The geometric superpotential–FLTZ comparison

Proof blueprint

The separate appendix gives auxiliary coordinate and extension-space presentations and their foundational lemmas. These do not replace the original intrinsic definitions.

Abstract

We specify a category-free geometric theorem whose conclusion suffices for the stopped comparison in *Fukaya categories of toric superpotentials and the FLTZ stop*. The conclusion records compact Liouville pairs, their witnessing contact forms, a finite chain of smooth deformations and marked completion identifications, and a cotangent ribbon whose full embedded core is exactly the FLTZ link. We prove the conditional passage from these data to the stated categorical result using the general GPS theorems.

1 Input data and the two endpoints

Fix a based lattice $N \simeq \mathbb{Z}^n$, $n \geq 1$, its dual M, and

$$T = M_{\mathbb{R}}/2\pi M, \qquad Y = M_{\mathbb{R}} \times T, \qquad z = \exp(u + i\theta).$$

Let $Q \subset N_{\mathbb{R}}$ be a full-dimensional lattice polytope containing zero in its interior. Set $V = \mathrm{Vert}(Q)$, assume that the vertices of every facet of Q form a lattice basis of N, and let Σ consist of zero and the cones on all nonempty proper faces of Q. Thus Σ is the smooth complete Fano fan determined by these data and V is exactly its set of primitive ray generators. These are the combinatorial inputs of [4]. Fix $c = (c_v)_{v \in V} \in (\mathbb{C}^*)^V$ and no other monomials:

$$W = W_c = \sum_{v \in V} c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}, \quad \phi_0 = \tfrac{1}{2}|u|^2, \quad \lambda_0 = \sum_j u_j \, d\theta_j.$$

Use $\iota_Z d\lambda = \lambda$ and

$$\lambda_{\mathbb{C}} = \tfrac{1}{2}(x\,dy - y\,dx), \qquad \lambda_K = \lambda_0 + KW^*\lambda_{\mathbb{C}} \quad (K \geq 0).$$

The angle variables are quotient coordinates: the exponentials and the one-forms $d\theta_j$ descend from $M_{\mathbb{R}}$ to T. They do not require globally defined real-valued angles. With the standard complex structure $J\partial_{u_j} = \partial_{\theta_j}$, W is holomorphic and $d\lambda_K = \sum_j du_j \wedge d\theta_j + KW^*(dx \wedge dy)$ is symplectic: the first summand is positive on (v, Jv) for $v \neq 0$, and the second is nonnegative. Thus Z_K, defined by $\iota_{Z_K} d\lambda_K = \lambda_K$, is well defined for every $K \geq 0$; completeness is a separate conclusion. Completeness means completeness of this Liouville field in both time directions. Finite Liouville type means that inclusion of a compact Liouville domain extends to a primitive-preserving diffeomorphism from its completion onto the given manifold.

Identify $T^*T = T \times N_{\mathbb{R}}$ with primitive $\lambda_{\mathrm{can}} = \sum_j p_j \, d\theta_j$. Choose the Euclidean norm determined by the fixed basis and put

$$D^*T = \{|p| \leq 1\}, \quad S^*T = \{|p| = 1\}, \quad \alpha_{\mathrm{can}} = \lambda_{\mathrm{can}}|_{S^*T}.$$

The target link is the following literal closed subset of S^*T:

$$\Lambda_\Sigma^\infty = \bigcup_{\sigma\in\Sigma,\ \sigma\neq 0} T_\sigma \times (\sigma \cap S^{n-1}),$$
$$T_\sigma = \{\theta : \langle v, \theta\rangle \in 2\pi\mathbb{Z} \text{ for every } v \in \sigma \cap N\}. \tag{1}$$

This specifies the positive-cone, period-2π convention. The FLTZ reference [3, §3.1, equation (8)] provides the formula's name and opposite-cone convention, not a geometric or categorical input. The cotangent lift $(\theta, p) \mapsto (-\theta, -p)$ changes the cone sign. The map $(x, p) \mapsto (2\pi x, p/(2\pi))$ changes period one to period 2π while preserving the cotangent primitive. These maps compare the conic subsets before taking their links in the chosen boundary sections.

2 The geometric data carried by a comparison

Differential-geometric conventions. All manifolds are finite-dimensional, Hausdorff and second countable. Smooth means C^∞, with smooth extension across boundaries and parameter-interval endpoints. A symplectic form is a closed, pointwise nondegenerate two-form. A Liouville domain (D, λ) is a compact smooth manifold with boundary, with $d\lambda$ symplectic and the unique field $\iota_Z d\lambda = \lambda$ strictly outward. Outward means $d\rho(Z) > 0$ for a local defining function $D = \{\rho \le 0\}$ at each boundary point. In dimension zero these conditions hold for the zero form and zero field on a finite set, whose boundary is empty. A contact form on a $(2n-1)$-manifold means $\alpha \wedge (d\alpha)^{n-1}$ is nowhere zero, including $n = 1$.

Definition 2.1 (Witnessed compact pair)**.** *A pair consists of a compact smooth $2n$-manifold with boundary D, a one-form λ with $d\lambda$ symplectic and Z_λ strictly outward at ∂D, a compact smooth $(2n-2)$-manifold with boundary F, a smooth embedding $j : F \hookrightarrow \partial D$, and a smooth positive function $b : \partial D \to \mathbb{R}_{>0}$, such that*

$$\alpha = \lambda|_{\partial D}, \qquad \mu = j^*(b\alpha)$$

makes F a Liouville domain. All smooth data extend across the indicated boundaries. The completion is $\widehat{D} = D \cup_{\partial D} (\partial D \times [0, \infty))$ with end primitive $e^r\alpha$. Its boundary at infinity is represented by ∂D. When another completed ambient space is named, a marking is a specified primitive-preserving diffeomorphism from $\widehat{D}$ onto that space.

The full intrinsic core and its embedded image are

$$\mathrm{Core}(F, \mu) = \bigcap_{t\ge 0} \mathrm{Fl}_{Z_\mu}^{-t}(F), \qquad j(\mathrm{Core}(F, \mu)) \subset \partial D.$$

The flow is computed in the completion of F. Equivalently the core consists of those $x \in F$ for which $\mathrm{Fl}_{Z_\mu}^t(x) \in F$ for every real $t \ge 0$. Here $\mathrm{Fl}^{-t}(F)$ denotes the image under time $-t$. The Liouville collar supplies the smooth gluing in the completion; the field there is ∂_r. Compactness on the interior and this end model give a flow defined for all real times on the completion. When $n = 1$, a ribbon is a compact zero-dimensional manifold; its Liouville field is zero and its core is the entire ribbon.

Definition 2.2 (Permitted comparison steps)**.** *A* pair deformation *is a smooth family of all the data in Definition 2.1 over $[0, 1]$. It may be given on compact smooth domain and ribbon bundles; a chosen trivialization presents it on fixed D and F. Thus λ_t, j_t, b_t and $\mu_t = j_t^*(b_t\lambda_t|_{\partial D})$ are jointly smooth, and both Liouville fields are outward for every t.*

A completion identification *consists of a specified primitive-preserving diffeomorphism* $\Psi : \widehat{D} \to \widehat{D}'$ *and its induced contact map* $\psi : \partial D \to \partial D'$ *at infinity, together with a diffeomorphism* $h : F \to F'$ *satisfying*

$$\Psi^*\lambda' = \lambda, \qquad \psi \circ j = j' \circ h, \qquad h^*\mu' = c_{\mathrm{scale}}\mu \quad (c_{\mathrm{scale}} > 0 \text{ constant}).$$

The boundary map uses Liouville trajectories to compare the chosen boundary sections; it is part of the data, not an arbitrary identification of abstract cores. A primitive-preserving map intertwines the Liouville fields and hence induces this map at infinity. More explicitly, for $x \in \partial D$*, follow the escaping trajectory* $\Psi(\mathrm{Fl}^r_Z(x))$ *to the unique representative on* $\partial D'$*; this representative is* $\psi(x)$*. Equivalently, for large* r *its end coordinates are* $(\psi(x), r + s(x))$ *for a smooth real function* s*. Primitive preservation gives* $\psi^*\alpha' = e^{-s}\alpha$*.*

Write $\mathcal{P} \rightsquigarrow \mathcal{Q}$ *for a finite sequence of these steps and their reverses, with an explicit finite list of endpoint tuples. Consecutive steps share their endpoint tuple literally; any change of presentation must be recorded as a permitted step with its maps. If the witnessing forms do not match, an actual pair deformation must be inserted. Neither replacement by a singular subset nor exchange of two ribbons merely because their embedded cores agree is a permitted step. These are deliberate geometric requirements, stronger than the minimum needed for categorical sufficiency.*

Remark 2.3. The positive factor b is essential data: arbitrary multiplication of a ribbon primitive by a positive function need not preserve its symplectic differential. No assertion here replaces λ on the ambient manifold by $b\lambda$. A bare exact symplectomorphism with uncontrolled behavior at infinity would also be insufficient for Definition 2.2. Primitive preservation is the stronger condition available for the completion identifications intended here.

Definition 2.4 (Ambient Maslov datum and transport)**.** *For a symplectic manifold* (X, ω) *choose an* ω*-compatible almost complex structure* J *and write* $K_J = \Lambda^n_{\mathbb{C}}(T^{1,0}X)^*$ *for its complex canonical line bundle. A Maslov datum is a homotopy class of nowhere-vanishing sections of* $K_J^{\otimes 2}$*. All sections and homotopies in this definition are smooth; section homotopies remain nowhere vanishing and homotopies of complex structures remain compatible with the relevant symplectic form. Compatibility means* $\omega(Jv, Jw) = \omega(v, w)$ *and* $\omega(v, Jv) > 0$ *for* $v \neq 0$*. Changing* J *through compatible structures identifies these classes; the space of compatible structures is contractible.*

On a pair deformation the datum is carried on the completed ambient family: choose a smooth compatible J_t *and a nowhere-vanishing section of the resulting squared canonical line bundle over the whole family. Its restrictions specify the endpoint classes. For a completion identification* Ψ*, a graded lift is a homotopy from the source section to the pullback of the target section, after identifying the compatible complex structures by a homotopy. A graded chain consists of a chain and these extension and homotopy data, agreeing at junctions. This definition asserts neither existence of a datum on every symplectic manifold nor the prescribed endpoint class for every possible chain.*

On Y *the initial datum is* $[\Omega^{\otimes 2}]$*, where* $\Omega = \bigwedge_j (du_j + i\, d\theta_j)$ *and the standard* J *above is compatible with every* $d\lambda_K$*. On* T^*T *use* $J_{\mathrm{can}}\partial_{p_j} = \partial_{\theta_j}$ *and the class of* $\left(\bigwedge_j (dp_j + i\, d\theta_j)\right)^{\otimes 2}$*. Its squared phase on a vertical Lagrangian plane is one; we call this the vertical-phase-zero datum. No grading is assigned to the singular subset* Λ^∞_Σ *or to individual branes here.*

3 The specified initial ribbon and its admissible choices

The following end bounds specify admissibility without assuming fullness or the desired comparison. Put

$$m_v = c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}, \qquad G = \sum_v m_v v,$$

$$a = \overline{G}^T G, \qquad b_0 = u \cdot G, \qquad Q_W = |u|^2 a - |b_0|^2 + \tfrac{1}{2}\operatorname{Re}(\overline{W} b_0).$$

An ambient end bound $A_{\mathrm{amb}} > 0$ satisfies $a > 0$ and $Q_W > 0$ on $\{\phi_0 \geq A_{\mathrm{amb}}\}$. Write $M(A_{\mathrm{amb}}) = \max_{\{\phi_0 \leq A_{\mathrm{amb}}\}} |W|$; this maximum exists on a nonempty compact set. For $R > \max(1, M(A_{\mathrm{amb}}))$, define the ratio set

$$S_R = \left\{ \frac{|b_0(y)|}{a(y)|W(y)|} : y \in Y,\ |W(y)| \geq R \right\}.$$

Its denominators are positive, since $|W| \geq R$ implies $\phi_0 > A_{\mathrm{amb}}$. When S_R is nonempty and bounded above, put

$$K_0(R) = 1 + 2 \sup S_R. \tag{2}$$

Existence of an ambient end bound, nonemptiness of S_R, and boundedness of S_R are conclusions required of the target. In particular $K_0(R)$ is this specified real number, at least one, rather than an independently chosen threshold. A formalization using a total real supremum must retain the nonemptiness and boundedness assertions explicitly. A fiber end bound $A_{\mathrm{fib}}(R) \geq A_{\mathrm{amb}}$ satisfies

$$a > 0, \qquad |u|^2 - \frac{|b_0|^2}{a} > 0 \quad \text{on } \{\phi_0 \geq A_{\mathrm{fib}}(R),\ |W| \leq R\}. \tag{3}$$

Its existence is likewise part of the target. When $n = 1$, the expression $|u|^2 - |b_0|^2/a$ is zero wherever $a > 0$. Consequently a valid fiber end bound makes the set on which (3) is required empty. Such a bound is possible because $W = c_+ e^{u+i\theta} + c_- e^{-u-i\theta}$ is proper in this case: the triangle inequality shows $|W| \to \infty$ as $u \to \pm\infty$. Do not replace (3) by a positivity claim on a nonempty end in dimension one.

For such bounds and a fixed $K \geq K_0(R)$ choose

$$A_- > A_{\mathrm{fib}}(R), \qquad A > A_-.$$

Use a smooth convex function $m_\varepsilon : \mathbb{R}^2 \to \mathbb{R}$ with nonnegative partial derivatives whose sum is one, satisfying

$$\max(x,y) \leq m_\varepsilon(x,y) \leq \max(x,y) + \varepsilon, \qquad m_\varepsilon(x,y) = \max(x,y) \text{ if } |x-y| \geq \varepsilon.$$

Define

$$\begin{aligned} E_\varepsilon &= \{m_\varepsilon(\phi_0 - A,\ |W|^2 - R^2) \leq 0\}, \\ F_R^- &= \{W = R,\ \phi_0 \leq A_-\}, \qquad \mu_R = \lambda_0|_{F_R^-}. \end{aligned} \tag{4}$$

Definition 3.1 (Admissible rounding)**.** *For these fixed R, K, A_-, A, a rounding consists of a real ε and a profile m_ε satisfying all the smoothness, convexity, derivative and equality conditions above. It is admissible when $0 < \varepsilon < A - A_-$ and, at every point of the explicitly defined band*

$$B_\varepsilon = \{-\varepsilon < \phi_0 - A \leq 0, \quad -\varepsilon < |W|^2 - R^2 \leq 0\},$$

one has

$$d(|W|^2)(Z_K) > 0, \qquad d\phi_0 \wedge d(|W|^2) \neq 0.$$

The second condition means linear independence of the two real covectors. It is allowed to hold vacuously when B_ε is empty. These conditions do not assume a completion marking, core equality, or comparison chain. Existence of admissible roundings and the asserted regularity of their domains are conclusions of the target.

For orientation, set $E_0 = \{\phi_0 \le A,\ |W| \le R\}$. The profile conditions imply $E_\varepsilon \subset E_0$ and $E_0 \setminus E_\varepsilon \subset B_\varepsilon$. Indeed $m_\varepsilon(x+t, y+t) = m_\varepsilon(x,y) + t$, by the sum of the partial derivatives. If $x \le -\varepsilon$ and $y \le 0$, monotonicity gives $m_\varepsilon(x,y) \le m_\varepsilon(-\varepsilon, 0) = 0$; interchange x, y for the other case. The portion of the zero level on which $m_\varepsilon >$ max also lies in B_ε. To see the strict band bounds there, use $m_\varepsilon(x,y) = x + g(y-x)$, where g is nondecreasing, $g(\varepsilon) = \varepsilon$, and $g'(\varepsilon) = 1$. Hence $g(s) < \varepsilon$ for $s < \varepsilon$, whereas $g(s) = s$ for $s \ge \varepsilon$. If $x \le -\varepsilon$ and $y < 0$, these two cases both give $m_\varepsilon(x,y) < 0$; the other case is symmetric. Unrounded faces need not lie in B_ε. The band is disjoint from F_R^- because $\phi_0 > A - \varepsilon > A_-$ there. For $n = 1$ it is empty, $E_\varepsilon = \{|W| \le R\}$, and $F_R^- = W^{-1}(R)$. On the unrounded face containing the ribbon, its inclusion j_R has

$$j_R^*(\lambda_K|_{\partial E_\varepsilon}) = \mu_R,$$

since W is constant on F_R^-. The initial witnessing factor is one.

4 The geometric endpoint theorem

Target theorem 4.1 (Geometric endpoint with specified bounds and grading)**.** *For every based lattice, polytope and coefficient tuple of Section 1, the following assertions hold with the nested quantifiers stated below.*

(G1) *For every real $K \ge 0$, (Y, λ_K) is complete and of finite Liouville type. There is an ambient end bound A_{amb} as in Section 3.*

(G2) *The ambient bound in (G1) can be chosen together with $R_0 > \max(1, M(A_{\mathrm{amb}}))$ so that, for every real $R \ge R_0$, S_R is nonempty and bounded above, and at least one fiber end bound $A_{\mathrm{fib}} \ge A_{\mathrm{amb}}$ exists. For every such R, for every valid fiber end bound (relative to this A_{amb}), for every real $K \ge K_0(R)$, and for every pair of levels $A_- > A_{\mathrm{fib}}$ and $A > A_-$, there exists an admissible rounding as specified in Section 3. For every such choice of levels and rounding, (4) defines a witnessed pair with nonempty ribbon. In particular R is a regular value of W and the full fiber is smooth. Inclusion extends to specified primitive-preserving diffeomorphisms*

$$(\widehat{E_\varepsilon, \lambda_K}) \longrightarrow (Y, \lambda_K), \qquad (\widehat{F_R^-, \mu_R}) \longrightarrow (W^{-1}(R), \lambda_0|_{W^{-1}(R)}).$$

These markings are induced by the corresponding Liouville flows and the given inclusions. Thus the ribbon is a truncation of the whole fiber. The fiber marking preserves the source theorem's fullness assertion; the categorical deduction itself only uses the ambient marking.

(G3) *For each of those initial pairs there exist a compact smooth ribbon F_Σ, an embedding $j_\Sigma : F_\Sigma \hookrightarrow S^*T$, and a smooth positive $b_\Sigma : S^*T \to \mathbb{R}_{>0}$ such that*

$$\mu_\Sigma = j_\Sigma^*(b_\Sigma \alpha_{\mathrm{can}})$$

is Liouville, and there is a finite chain

$$(E_\varepsilon, \lambda_K, F_R^-, j_R, 1) \rightsquigarrow (D^*T, \lambda_{\mathrm{can}}, F_\Sigma, j_\Sigma, b_\Sigma)$$

in the sense of Definition 2.2. The endpoint data and this chain are chosen together with the grading data specified below. The source marking is the one in (G2); the terminal marking is the standard completion of D^*T with T^*T. The endpoint satisfies the equality of embedded subsets

$$j_\Sigma\big(\mathrm{Core}(F_\Sigma, \mu_\Sigma)\big) = \Lambda_\Sigma^\infty \subset S^*T. \tag{5}$$

The same chain *is equipped with ambient Maslov data and graded lifts in the sense of Definition 2.4. At its source the datum is the pullback of $[\Omega^{\otimes 2}]$ by the marking in (G2). At its endpoint, under the standard cotangent marking, the transported datum is homotopic to the vertical-phase-zero datum of Definition 2.4. This is an existential requirement on the chosen chain, not a claim about every chain between these endpoints.*

Quantifier order. For each input, all-K completeness and finite type are asserted first. There exist A_{amb} and R_0 satisfying the stated conditions. For every $R \geq R_0$, assert nonemptiness and boundedness of S_R and existence of a fiber end bound; then define $K_0(R)$ by (2). For *every* valid fiber end bound, every $K \geq K_0(R)$, and every $A_- > A_{\mathrm{fib}}$, $A > A_-$, assert existence of an admissible rounding. For *every* admissible rounding, assert the witnessed pair and both specified markings, followed by one existential package containing the endpoint ribbon, witness, chain, embedded-core equality, and grading extensions and endpoint homotopy. The existence clauses precede the universal admissibility clauses and are not replaced by them. The endpoint ribbon and the chain need not be independent of R, K or the choices; their embedded core is always the fixed set (1). Item (G1) is asserted also below the ribbon threshold. The target assumes neither a strongly compatible potential, a full-core identification, nor the desired chain as input. Their construction belongs to its proof.

Lemma 4.2 (Comparison of initial choices)**.** *Assume (G1)–(G2). Fix the input and R, K in their stated range. Any two admissible initial pairs, including choices using different valid fiber bounds, are joined by a finite chain inside the fixed (Y, λ_K). Domain primitives are restrictions of λ_K, the completion markings extend inclusion, and every completion identification is the identity on the marked Y. The chain preserves $[\Omega^{\otimes 2}]$ at both ends and transports any fixed ambient background class identically.*

Proof. Let the two fiber levels be A_-^0, A_-^1. The smaller of their valid fiber end bounds is itself a valid bound, and both levels exceed it. Choose $L > \max(A_-^0, A_-^1)$ above this bound, then $A'' > L$ and an admissible rounding for these levels. Its domain E'' has $A'' - \varepsilon'' > L$, so both ribbons, and every intermediate fiber sublevel, lie in its unrounded $|W| = R$ face.

For each endpoint level, identify its original ambient completion with $\widehat{E}''$ using the two inclusion markings. The boundary map fixes the ribbon pointwise, because that ribbon lies on both boundary sections. Its primitive is μ_R on both sides, so take $h = \mathrm{id}$ and $c_{\mathrm{scale}} = 1$. Inside E'' vary the ribbon level linearly from A_-^0 to A_-^1. On a regular holomorphic fiber, the intrinsic Liouville field is the orthogonal projection of $Z_0 = \sum_j u_j \partial_{u_j}$ onto the fiber tangent space. Therefore its derivative of ϕ_0 is $|u|^2 - |b_0|^2/a$. The fiber-end inequality makes $d\phi_0$ nonzero on that tangent space throughout this compact band and makes the field outward. A transverse vector field on a neighborhood of the band, cut off outside it, gives smooth identifications of these sublevel domains. Thus this is a pair deformation, with ambient primitive λ_K and witness one. The two completion identifications and this deformation give the required finite chain. If ribbon presentations change in the chosen trivialization, record the resulting relabeling as an identity ambient completion step. Pull back the fixed ambient datum and any background class along the inclusion markings throughout. □

5 Geometric hypotheses and constructions

The categorical references in the following table are to the companion FS–FLTZ statement.

Requirement of the original result	Explicit geometric provision	Consequence checked here
All finite $K \geq 0$	(G1), separately from the large-K ribbon construction	Completeness and finite type are not weakened to the comparison range.
Every large R and every $K \geq K_0(R)$	(2) and (G2)–(G3)	No single choice of K replaces the stated conclusion.
Specified full far-fiber stop	(4), both markings in (G2)	The source is the stated embedded ribbon, not an arbitrary copy of its fiber.
Varying ambient primitives	Definition 2.2 and Lemma 5.1	An actual sector family is supplied before deformation invariance is used.
Valid terminal ribbon	$b_\Sigma > 0$ and $\mu_\Sigma = j_\Sigma^*(b_\Sigma \alpha_{\mathrm{can}})$ Liouville	The primitive used to define the full core is specified.
Correct target stop	Equality (5) in S^*T	The endpoint core replacement gives the precise FLTZ stop.
Admissible-choice independence	Lemma 4.2, including its fixed ambient grading	The comparison of initial choices is supplied, not inferred from one example.
Integer gradings	The grading data on the same chain in (G3)	The endpoint datum is identified without grading the singular link or fixing named-object shifts.
No additional categorical theorem	Proposition 5.2 and Remark 5.3	The deduction uses general GPS statements, not a toric application, CCC, HMS, or a thimble-generation assertion.

Geometric constructions. The proof of the geometric target uses the finite input in Section 1. In dependency order, this requires construction of the compatible convex potential; the end bounds and completeness; the original marked pair and compact tailoring families; all critical points and the strict intrinsic Lyapunov function; the full embedded core; smooth parameter-dependent hitting maps and a common ribbon neighborhood; the finite chain and endpoint core equality; and the ambient grading transport. In particular, neither (5) nor chain existence may be hidden in an admissible-potential hypothesis.

The intended parameter order is: fix coefficient and potential paths and analytic constants; choose inner and outer radius constants; choose $\beta = \log R$ sufficiently large; then choose finite normalized K_*. For the fixed coefficient/potential path (f_t, ϕ_t), first establish the finite path-uniform threshold $\kappa_{\mathrm{path}}(R) = \sup_t \kappa(f_t, \phi_t; R)$ of [4, §4], including $\kappa_{\mathrm{path}}(R) \geq K_0(R)$, and choose $K_*/R^2 \geq \kappa_{\mathrm{path}}(R)$. First connect the supplied original K to K_*/R^2 while keeping the original

data fixed; this interval remains above $K_0(R)$. Then traverse the coefficient/potential path at that coefficient. Quadratic homogeneity gives $K_*(f/R)^*\lambda_{\mathbb{C}} = (K_*/R^2)f^*\lambda_{\mathbb{C}}$ for any f; apply this at the path endpoint polynomial when passing to normalized tailoring data. Only afterward choose collars, hitting neighborhoods, and shrinking times. These later choices may depend on the fixed compact parameter intervals. No estimate for the growth of $K_*(R)$ is part of the target.

6 Foundational conventions

Auxiliary structures may bundle geometric objects and proofs of their defining properties; the existence of the target's objects remains the conclusion, not a field of the input data.

Foundations in dependency order.

1. Encode the based finite-rank lattice, its real dual spaces, the finite polytope and face conditions, nonzero coefficients, the quotient torus and its descended forms. Define the fan, W, λ_K and the literal subset (1). A coordinate model is acceptable provided its periodicity and descent to these quotients are proved.

2. Define differential forms, symplectic and contact nondegeneracy, smooth manifolds with boundary, Liouville fields and outwardness, smooth flows and completions. Include dimension zero. Define the core by the all-forward-time condition and connect it to the nested-image formula in Definition 2.1.

3. Define witnessed pairs, inclusion markings, compact parameter families, strict completion identifications and their induced boundary maps. A finite chain records its intermediate pairs and steps with matching endpoints; it is not an arbitrary relation supplied as input. The definition of smooth family must cover the completions as well as the compact pieces used to build them.

4. Define compatible almost complex structures, squared canonical line bundles, homotopies of nonvanishing sections, and the transport data of Definition 2.4. The source and cotangent endpoint classes are the explicit ones specified there.

5. Define the end-bound and rounding predicates, S_R, its nonemptiness and boundedness requirements, and the prescribed $K_0(R)$. Assemble the target with exactly the quantifier order above, including both markings and one chain carrying both core equality and grading. Record Lemma 4.2 as a separate conditional consequence.

Geometric correspondences. The foundation layer needs the well-definedness of quotient formulas; the symplectic calculation for $d\lambda_K$; the Liouville collar and complete flow on a completion; flow equivariance and the boundary map of a strict identification; and the correspondence between geometric families and any chosen coordinate encoding. An orbit characterization of a completion marking requires a proof that it produces the specified smooth primitive-preserving diffeomorphism. It cannot replace the marking by an unrelated predicate. Homotopy invariance under a change of compatible complex structure likewise needs a foundation lemma.

The general existence of the objects asserted in (G1)–(G3) is the geometric theorem. In particular, end bounds, boundedness of S_R, admissible roundings, full-fiber markings, the full embedded core, the finite comparison chain, and its endpoint grading must not be hidden in an input assumption. The GPS categorical deduction remains external to the Lean formalization.

Boundary cases and conventions. Check the $n = 1$ zero-dimensional ribbon and empty corner band, the positive cone and period-2π conventions, the two time directions of completeness, and the formula-defined threshold for every $R \geq R_0$. Check that changing the fiber bound does not change the quantifier meaning and that the graded chain is exactly the chain furnishing (5). The smoothness notation means C^∞, rather than analytic regularity. Supremum conventions must not rely on default values for an empty or unbounded set.

A Separate auxiliary representations and foundational lemmas

This appendix adds mathematical presentations separate from the original intrinsic definitions above. Throughout this appendix, a compact domain means the extension-space domain presentation specified below; pairs, completions and families mean the corresponding chosen presentations. In particular, the Liouville-neighborhood, Liouville-flow and Liouville-collar results below quantify over those presentations. The converse passage from an arbitrary intrinsic compact manifold with boundary, and the correspondence of whole completed families and of the encoded target, are separate correspondence requirements. The six proved Liouville collar producers concern the chosen presentations; their proofs do not discharge those correspondence requirements. The intrinsic boundary atlas and the local coefficient-evolution and transverse-seam results have their own explicit intrinsic or Euclidean inputs.

A.1 Auxiliary presentations: Finite input and the literal endpoints

Definition A.1 (Auxiliary presentation: Based input)**.** *An input consists of a based lattice $N = \mathbb{Z}^n$, $n \geq 1$, a finite set $V \subset N$ which is exactly the vertices of $Q = \mathrm{conv}(V)$, with $0 \in \mathrm{int}(Q)$, an integral basis of N consisting of the vertices of each facet, and nonzero coefficients $c_v \in \mathbb{C}$ for $v \in V$. No additional monomial, potential, end bound, marking, chain, or core identification is part of the input.*

Definition A.2 (Auxiliary presentation: Quotient endpoints)**.** *Let $M = N^*$ and put $T = M_{\mathbb{R}}/(2\pi M)$, $Y = M_{\mathbb{R}} \times T$, and $T^*T = T \times N_{\mathbb{R}}$. The based coordinates present T as $(\mathbb{R}/2\pi\mathbb{Z})^n$. Set $D^*T = \{(\theta, p) : \sum_j p_j^2 \leq 1\}$ and $S^*T = \{(\theta, p) : \sum_j p_j^2 = 1\}$. These use the Euclidean norm of the specified basis.*

Definition A.3 (Auxiliary presentation: Superpotential)**.** *For $y = (u, \theta) \in Y$, define $W(y) = \sum_{v \in V} c_v \exp(\langle v, u\rangle + i\langle v, \theta\rangle)$. The exponential in the angle variable is the character of the quotient induced by the integral pairing; it requires no real-valued global angle.*

Lemma A.4 (Presentation foundation: Quotient formulas)**.** *The quotient map is surjective, with two representatives equal precisely when their difference lies in $2\pi\mathbb{Z}^n$. The displayed superpotential formula is independent of representatives and agrees with the character formula on T.*

Proof. In each coordinate the quotient identifies two reals exactly when their difference is an integral multiple of 2π; apply this in all n coordinates. Every tuple of quotient classes has a tuple of representatives, so the product quotient is surjective. Replacing θ by $\theta + 2\pi k$ changes $\langle v, \theta\rangle$ by $2\pi\langle v, k\rangle$. The latter pairing is an integer, so its exponential is one. Each summand, and hence their finite sum, is unchanged. Evaluation of the induced character on a representative is exactly that same exponential. □

Definition A.5 (Auxiliary presentation: Positive-cone FLTZ link)**.** *Let Σ consist of zero and the positive cones on the nonempty proper faces of Q. For a cone σ, let $T_\sigma = \{\theta : \langle v, \theta\rangle \in 2\pi\mathbb{Z}$ for all $v \in \sigma \cap N\}$. The literal link is*

$$\Lambda_\Sigma^\infty = \bigcup_{\sigma\in\Sigma,\,\sigma\neq 0} T_\sigma \times (\sigma \cap S^{n-1}) \subset S^*T. \tag{6}$$

Lemma A.6 (Presentation foundation: Annihilator and boundary conventions)**.** *For a real angle representative, membership in T_σ is precisely $\langle v, \theta\rangle = 2\pi k_v$, $k_v \in \mathbb{Z}$, for every $v \in \sigma \cap N$. The link is contained in the specified Euclidean unit cosphere.*

Proof. A class in $\mathbb{R}/2\pi\mathbb{Z}$ is zero precisely when its representative is an integral multiple of 2π. Apply this to each integral pairing. Every point of the displayed union has momentum in S^{n-1}, which is the defining condition of the chosen cosphere. □

Lemma A.7 (Presentation foundation: Intrinsic polytope faces)**.** *The finite supporting-face presentation of Σ equals zero together with the radial cones on all nonempty proper intrinsic convex faces of Q. Consequently its link formula is the formula using all such faces.*

Proof. A supporting face is convex, and equality in a supporting linear inequality forces equality in every positive-weight summand. It is therefore an intrinsic face. Conversely, if F is an intrinsic convex face, iterating the open-segment face condition on finite convex combinations shows that $F = \mathrm{conv}(V \cap F)$. For a proper nonempty face let x be the uniform barycenter of $V \cap F$. It cannot be an interior point of Q: if it were, for any $y \in Q$ a point $z = x + \epsilon(x - y)$ would remain in Q for sufficiently small $\epsilon > 0$, and x would be in the open segment from y to z. The face condition would imply $y \in F$, contradicting properness. Thus x lies on the frontier. A supporting hyperplane at x cuts a proper exposed face G containing x. All vertices in $V \cap F$ lie in G, since their positive equal barycentric weights make a strict supporting inequality at any one vertex incompatible with equality at x. Choose a containing facet H. Such a facet exists by taking a maximal proper supporting face in the finite face set. Its vertex set B is a basis by the input, and $s = V \cap F$ is a subset of B.

Let $\ell \le a$ expose H, with equality exactly on its vertices. Define a linear functional m by $m(v) = 0$ for $v \in s$ and $m(v) = -1$ for $v \in B \setminus s$; the basis makes this definition possible. For every vertex outside B, the gap $a - \ell(v)$ is positive. There are finitely many such vertices, so choose $\epsilon > 0$ sufficiently small that $\epsilon m(v) < a - \ell(v)$ at each one with $m(v) > 0$; no restriction is needed where $m(v) \le 0$. Then $\ell + \epsilon m \le a$ on all vertices, with equality precisely on s. It therefore exposes $\mathrm{conv}(s) = F$. The radial cone on this convex hull equals the positive cone on s: expand a point of the hull as a convex combination and multiply by its radial scale; conversely normalize a nonzero positive combination by its positive coefficient sum. Substituting these face and cone equalities into the two union formulas proves both assertions. □

Definition A.8 (Auxiliary presentation: Actual quotient primitives)**.** *Set $\phi_0 = |u|^2/2$, $\lambda_0 = \sum_j u_j d\theta_j$, $\lambda_{\mathbb{C}} = (x\,dy - y\,dx)/2$, $\lambda_K = \lambda_0 + KW^*\lambda_{\mathbb{C}}$, and $\lambda_{\mathrm{can}} = \sum_j p_j d\theta_j$. Put $\alpha_{\mathrm{can}} = \lambda_{\mathrm{can}}|_{S^*T}$. Here $d\theta_j$ is the angular form descended to T, equivalently $\mathrm{Im}(\bar{q}_j dq_j)$ for the unit-circle coordinate $q_j = e^{i\theta_j}$.*

Lemma A.9 (Presentation foundation: Representative formulas for the primitives)**.** *Pullback to the logarithmic-angle cover gives the displayed λ_K; pullback to the momentum-angle cover gives the displayed λ_{can}.*

Proof. For $q = e^{i\theta}$ one has $dq = ie^{i\theta}d\theta$, so $\mathrm{Im}(\bar{q}dq) = d\theta$. The quotient formula for W identifies its pullback with the displayed finite exponential sum. Pullback of $(x\,dy - y\,dx)/2$ is therefore $(\mathrm{Re}\,W\,d\,\mathrm{Im}\,W - \mathrm{Im}\,W\,d\,\mathrm{Re}\,W)/2$. Multiplying the descended angular forms by u_j, respectively p_j, and summing proves both identities. □

Lemma A.10 (Presentation foundation: Symplectic differential)**.** *For every $K \geq 0$, $d\lambda_K$ is symplectic, and $d\lambda_{\mathrm{can}}$ is symplectic.*

Proof. On the cover, differentiating the primitive gives $d\lambda_K = \sum_j du_j \wedge d\theta_j + KW^*(dx \wedge dy)$. The exponential sum is holomorphic for $J(\delta u, \delta\theta) = (-\delta\theta, \delta u)$. Hence $d\lambda_K(v, Jv) = \sum_j((\delta u_j)^2 + (\delta\theta_j)^2) + K|dW(v)|^2 > 0$ when $v \neq 0$. A vector in the kernel would make this expression zero, so the kernel is zero. The cotangent differential is $\sum_j dp_j \wedge d\theta_j$, with the same argument and no second term. Both forms are closed since they are differentials of smooth one-forms. The quotient maps are local smooth diffeomorphisms, so these properties descend. □

A.2 Auxiliary presentations: Domains, collars, completions and cores

Definition A.11 (Auxiliary presentation: Extension-space domain presentation)**.** *A presentation of a compact domain consists of a compact subset D of a smooth boundaryless extension manifold, local smooth defining functions $D = \{\rho \leq 0\}$ with $d\rho \neq 0$ at its frontier, and a smooth embedded boundary presentation whose image is that frontier. In dimension zero that boundary is empty. A Liouville structure is a smoothly extended one-form λ with $d\lambda$ symplectic on D, a smoothly extended field Z satisfying $\iota_Z d\lambda = \lambda$ on D, and $d\rho(Z) > 0$ at the frontier.*

Lemma A.12 (Presentation foundation: Boundary charts from a defining function)**.** *Every frontier point of an extension-space domain has a smooth half-space chart that identifies its frontier with the flat boundary. In particular the dimension at a frontier point is positive.*

Proof. In an extension chart, the derivative of ρ at the point is a nonzero linear functional. Choose a complement to its kernel. Keep coordinates on that kernel and use $-\rho$ as the remaining coordinate. The derivative of this coordinate map is invertible. The inverse function theorem gives a diffeomorphism on sufficiently small neighborhoods. The equation $\rho \leq 0$ becomes the standard nonnegative half-space coordinate, and $\rho = 0$ becomes its flat boundary. A nonzero linear functional cannot exist on a zero-dimensional tangent space. □

Lemma A.13 (The boundary atlas of an intrinsic manifold)**.** *Let $k \geq 0$, and let M be a smooth Hausdorff second-countable $(k+1)$-dimensional manifold with boundary, presented by its intrinsic half-space atlas. Its actual boundary ∂M, with the subspace topology, has a smooth boundaryless k-dimensional atlas. The inclusion $\partial M \hookrightarrow M$ is a smooth topological embedding with injective differential at every boundary point. If M is compact, then ∂M is compact. This construction requires no ambient extension manifold, defining function or supplied boundary presentation.*

Proof. Write half-space coordinates as (r, v) with $r \geq 0$ and $v \in \mathbb{R}^k$. An intrinsic boundary point has $r = 0$ in every atlas chart containing it: the native characterization of interior points in such a chart is $r > 0$, and the boundary is its complement. For any ambient chart c, restrict its source to the boundary and define

$$c_\partial(x) = \pi(c(x)), \qquad c_\partial^{-1}(v) = c^{-1}(0, v),$$

where $\pi(r, v) = v$. Its source is open in the boundary subspace, and its target is the open set of v for which $(0, v)$ belongs to the half-space chart target. The inverse formulas follow from the

ambient chart inverse identities. The maps are continuous by restriction, projection and insertion, so they are boundary charts. They cover ∂M because the ambient atlas covers M.

For two such charts, their transition on its overlap is

$$v \longmapsto \pi\big(c' \circ c^{-1}(0, v)\big).$$

The ambient extended chart transition is smooth by the intrinsic smooth atlas assumption. Composing it with the linear hyperplane insertion and coordinate projection gives the displayed smooth transition. Thus these charts give the claimed smooth boundaryless atlas, with its original subspace topology. Hausdorffness and second countability are inherited by that subspace.

In these source and target coordinates the inclusion is $v \mapsto (0, v)$, which is smooth. Projection back to v is a left inverse, so its differential is injective. The inclusion is a topological embedding by the subspace topology. Finally the intrinsic interior is open; its complement ∂M is closed. If M is compact, this closed subset is compact, as required. The argument permits $k = 0$. □

Lemma A.14 (Local half-space collar from boundary coefficients)**.** *Let $k \geq 0$, write $\iota(v) = (0, v) \in \mathbb{R}^{k+1}$, and let $f : \mathbb{R}^{k+1} \to \mathbb{R}^{k+1}$ be smooth on an open set U. Suppose $\iota(u) \in U$ and $f(\iota(u))_0 > 0$. There are a smooth map $c : \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}^{k+1}$, an open neighborhood V of u, and $\delta > 0$, such that $c(0, v) = \iota(v)$ for every v and, for $v \in V$ and $|t| < \delta$,*

$$c(t, v) \in U, \qquad \partial_t c(t, v) = f(c(t, v)), \qquad c(t, v)_0 \geq 0 \iff t \geq 0, \qquad c(t, v)_0 > 0 \iff t > 0.$$

On the same rectangle $(-\delta, \delta) \times V$, the map c is injective and is a smooth local diffeomorphism at every point. Its restriction to $[0, \delta) \times V$, regarded as a map into the half-space $H = \{x \in \mathbb{R}^{k+1} : x_0 \geq 0\}$, is an open embedding. The smooth extension of the coefficients is an explicit local hypothesis; no extension manifold or compact uniform collar is supplied.

Proof. The open set $P = \{x \in U : f(x)_0 > 0\}$ contains $\iota(u)$. Choose a smooth compactly supported cutoff χ whose support lies in P and which is one on a neighborhood W of $\iota(u)$. The field $g = \chi f$ is smooth on P and is locally zero outside its support, so it is smooth everywhere. It has compact support. Local smooth ODE existence on its compact support gives a uniform positive existence time by finite covering. Outside the support its curves are constant. Successive concatenations of these local solutions give a smooth all-real flow F, with uniqueness giving the flow law. Continuity at $(0, \iota(u))$ supplies a product of a two-sided time interval and an open initial-point neighborhood whose trajectories lie in $W \cap P$. On that product, $\partial_t F_t(x) = f(F_t(x))$. Restrict the initial points to $\iota(V)$ for a sufficiently small open neighborhood V of u, and choose $\delta > 0$ within that time interval. Set $c(t, v) = F_t(\iota(v))$. This map is smooth on the whole product, and the zero-time flow identity gives its seam identity. For $v \in V$ the function $t \mapsto c(t, v)_0$ has positive derivative on $(-\delta, \delta)$ and is zero at $t = 0$. The mean value theorem makes it strictly increasing there. Comparison with its value at zero gives both asserted sign equivalences, together with the stated ODE and containment in U. At $(0, u)$ its time derivative is $f(\iota(u))$, with positive first coordinate, and its seam identity holds in all boundary variables. The transverse-seam inverse lemma therefore gives an open neighborhood of $(0, u)$ on which c agrees with a smooth local diffeomorphism and is injective. Choose a smaller open neighborhood V of u and a smaller $\delta > 0$ so that their rectangle lies in this inverse neighborhood and the previous flow rectangle. All the ODE and side identities restrict to this same rectangle. The inverse neighborhood gives the claimed injectivity and local diffeomorphism at every point. An injective local diffeomorphism is an open embedding of the full rectangle onto its open image. The normal-sign identity identifies the inverse image of H with exactly $[0, \delta) \times V$. Restrict its image homeomorphism to these subspaces. Its image is the intersection of the original open image with H, hence is open in H. This proves the one-sided open embedding, including its seam. □

Lemma A.15 (Local inversion at a transverse boundary seam)**.** *Let $k \geq 0$, $u \in \mathbb{R}^k$, and let $c : \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}^{k+1}$ be smooth on an open neighborhood of $(0, u)$. Write $\iota(v) = (0, v)$. Suppose $c(0, v) = \iota(v)$ for all v sufficiently near u, and $\partial_t c(0, u) = w$ with $w_0 > 0$. Then*

$$dc_{(0,u)}(s, v) = sw + \iota(v)$$

is invertible, and c is a smooth local diffeomorphism at $(0, u)$.

Proof. Differentiating the seam identity in the boundary variables gives $dc_{(0,u)}(0, v) = \iota(v)$. The asserted time derivative gives $dc_{(0,u)}(s, 0) = sw$. Linearity and the decomposition $(s, v) = (s, 0) + (0, v)$ give the displayed differential. If $sw + \iota(v) = 0$, its first coordinate gives $sw_0 = 0$. Since $w_0 > 0$, this implies $s = 0$, and then $\iota(v) = 0$ gives $v = 0$. Thus the differential is injective. Its source and target both have dimension $k + 1$, so it is bijective. The smooth inverse function theorem on the given open neighborhood yields a smooth local inverse at $(0, u)$. The argument includes $k = 0$. □

Lemma A.16 (Uniqueness of native one-sided integral curves)**.** *Let M be a smooth $(k + 1)$-dimensional manifold with boundary, e one of its half-space atlas charts, and E that chart regarded as a map into $\mathbb{R}^{k+1}$. Let Z be an intrinsic tangent field. Suppose $U \subset \mathbb{R}^{k+1}$ is compact and convex, W is smooth on U, and*

$$W(E(x)) = dE_x(Z(x)) \quad \text{for } x \in \mathrm{dom}(e) \text{ with } E(x) \in U.$$

Let $T \in \mathbb{R}$ and let $\gamma, \gamma' : \mathbb{R} \to M$ be continuous on $[0, T]$, with both paths in $\mathrm{dom}(e)$ and both coordinate paths in U throughout that interval. Suppose their intrinsic right derivatives at every $t \in [0, T)$ equal $Z(\gamma(t))$ and $Z(\gamma'(t))$, respectively. If $\gamma(0) = \gamma'(0)$, then $\gamma(t) = \gamma'(t)$ for every $t \in [0, T]$.

Proof. Put $\eta = E \circ \gamma$ and $\eta' = E \circ \gamma'$. The chart is continuous on its source, so these are continuous on $[0, T]$. The intrinsic chain rule for right derivatives gives

$$\partial_+\eta(t) = dE_{\gamma(t)}Z(\gamma(t)) = W(\eta(t)), \qquad \partial_+\eta'(t) = W(\eta'(t)) \quad (0 \leq t < T).$$

Smoothness on the compact convex set U gives a finite Lipschitz constant for W there: bound its first differential on the compact set and apply the mean value inequality along the segment between any two points of U. The right-derivative uniqueness theorem for Lipschitz ODEs on a closed interval now applies to η, η'. Its hypotheses hold because both paths stay in U, are continuous on the closed interval, have the displayed right ODE derivatives before its terminal endpoint, and agree at zero. Thus $\eta = \eta'$ on $[0, T]$. Injectivity of the chart on its source gives the asserted native equality. For $T = 0$ this is the initial equality; for $T < 0$ the interval is empty. □

Lemma A.17 (Local uniqueness from a smooth native coefficient germ)**.** *Let M, e, E, Z be as in the preceding uniqueness lemma. Suppose instead that U is open, W is smooth on U, and $W(E(x)) = dE_x(Z(x))$ for all $x \in \mathrm{dom}(e)$ with $E(x) \in U$. Let $T > 0$ and let $\gamma, \gamma' : \mathbb{R} \to M$ be continuous on $[0, T]$ and solve the intrinsic right ODE of Z at every $t \in [0, T)$. If $\gamma(0) = \gamma'(0) \in \mathrm{dom}(e)$ and $E(\gamma(0)) \in U$, then there is $0 < \varepsilon < T$ such that $\gamma(t) = \gamma'(t)$ for every $t \in [0, \varepsilon]$. No common chart or coefficient-set containment of the trajectories is assumed beyond their shared initial point.*

Proof. Put $a = E(\gamma(0))$ and choose $r > 0$ so that the closed Euclidean ball $B = \overline{B}(a, r)$ lies in U. On this compact convex ball, the smooth coefficient field has a finite Lipschitz constant. The set $\mathrm{dom}(e) \cap E^{-1}(B)$ is a neighborhood of the shared initial point: the chart source is open and a lies

in the interior of B, while the chart is continuous on its source. Continuity of both curves at zero relative to $[0,T]$ gives $\delta > 0$ such that both curves belong to this neighborhood for $t \in [0,T]$ with $t < \delta$. Choose $\varepsilon = \frac{1}{2}\min(\delta, T)$. Then $0 < \varepsilon < T$, both restricted curves stay in the chart with coordinates in B on $[0,\varepsilon]$, and their original right ODE identities hold at every $t < \varepsilon$ in that interval. Apply the finite-interval native uniqueness lemma on B to obtain the asserted equality. □

Lemma A.18 (A native local inward collar with its integral-curve germ)**.** *Let M be a smooth $(k+1)$-dimensional manifold with boundary, e an intrinsic half-space atlas chart, and E its Euclidean-valued coordinate map. Write $H = \{y \in \mathbb{R}^{k+1} : y_0 \geq 0\}$ and $\iota(v) = (0,v)$. Let Z be an intrinsic tangent field. Suppose $U \subset \mathbb{R}^{k+1}$ is open, f is smooth on U, $\iota(u) \in U$, $U \cap H \subset E(\mathrm{dom}(e))$, and*

$$f(y) = dE_{E^{-1}(y)} Z(E^{-1}(y)) \quad (y \in U \cap H), \qquad f(\iota(u))_0 > 0.$$

There are an open neighborhood V of u, $\delta > 0$, and a map $\gamma : \mathbb{R} \times \mathbb{R}^k \to M$ with the following properties on $S = [0,\delta) \times V$. It is smooth on S and its restriction to S is an open embedding into M. For every $p \in S$ there is a map $g : M \to \mathbb{R} \times \mathbb{R}^k$ smooth at $\gamma(p)$ with $g(\gamma(q)) = q$ for all $q \in S$ sufficiently near p. For $v \in V$,

$$\gamma(0,v) = E^{-1}(\iota(v)), \qquad \partial_t \gamma(t,v) = Z(\gamma(t,v)) \quad (0 \leq t < \delta), \qquad \gamma(t,v) \in \partial M \iff t = 0.$$

The ODE at $t = 0$ is the intrinsic within derivative on $[0,\delta)$; all these trajectories remain in $\mathrm{dom}(e)$. Moreover, for every $v \in V$, every $T > 0$, and every intrinsic integral curve $\eta : \mathbb{R} \to M$ of Z on $[0,T)$ with $\eta(0) = \gamma(0,v)$, there is $0 < \varepsilon < \min(\delta, T)$ such that $\eta(t) = \gamma(t,v)$ for $t \in [0,\varepsilon]$. Only the local Euclidean coefficient extension is assumed; no ambient extension manifold, compact uniform collar or completed flow is an input.

Proof. Construct the half-space collar c from the coefficient field f. For $t \geq 0$ in its strip the normal-sign identity puts $c(t,v)$ in H, and the trajectory containment puts it in U. The chart-target hypothesis therefore allows the definition $\gamma(t,v) = E^{-1}(c(t,v))$ on S. Define it arbitrarily outside S, for example with the fixed value $\gamma(0,u)$. Composition with the smooth native chart inverse makes γ smooth on S. The relative half-space open embedding of c, composed with the chart inverse, gives its open embedding into M. At each $p \in S$, choose a smooth Euclidean local inverse d^{-1} of c near p. Set $g = d^{-1} \circ E$ near $\gamma(p)$, extending it as a map outside that neighborhood. The native chart is smooth at $\gamma(p)$, so g is smooth there. The chart inverse identities and the local inverse identity give $g(\gamma(q)) = q$ for $q \in S$ sufficiently near p.

The seam formula follows from $c(0,v) = \iota(v)$. Apply the intrinsic within chain rule to the coefficient ODE. The derivative of the inverse chart composed with that of the chart is the identity on the intrinsic tangent space, including at the boundary. The assumed coefficient identity consequently gives

$$\partial_t \gamma(t,v) = d(E^{-1})_{c(t,v)} f(c(t,v)) = Z(\gamma(t,v)).$$

In native atlas coordinates a point is on the intrinsic boundary exactly when its first coordinate is zero. The strict normal-sign identity for c makes this equivalent to $t = 0$ on the one-sided strip.

Finally fix v, T, η as in the statement and put $b = \frac{1}{2}\min(\delta, T) > 0$. Both integral curves are continuous on $[0,b]$, and their within ODE identities give their right derivatives at every $t < b$. Their shared initial point is in the chart with coordinates in U. The local native uniqueness corollary therefore gives an interval $[0,\varepsilon]$ on which they agree, with $0 < \varepsilon < b < \min(\delta, T)$. □

Lemma A.19 (A common coefficient neighborhood from scalar extensions)**.** *Let e be an intrinsic half-space atlas chart of a smooth $(k+1)$-manifold with boundary M, let E be its Euclidean-valued*

coordinate map, and let Z be an intrinsic tangent field. Fix $x \in \mathrm{dom}(e)$ and put $a = E(x)$. Suppose that, for each $i = 0, \ldots, k$, there are an open neighborhood V_i of a and a smooth scalar function f_i on V_i whose germ on the extended chart target equals the ith coordinate of $dE_{E^{-1}(y)}Z(E^{-1}(y))$. There are one open neighborhood U of a and one smooth $f : U \to \mathbb{R}^{k+1}$ such that $U \cap H$ lies in the extended chart target, where $H = \{y : y_0 \geq 0\}$, and

$$f(y) = dE_{E^{-1}(y)}Z(E^{-1}(y)) \qquad (y \in U \cap H).$$

Here each scalar extension is smooth on an actual open neighborhood; smoothness merely relative to the half-space is not its hypothesis.

Proof. For each i, germ equality supplies an open neighborhood W_i of a on which the stated scalar equality holds at every point in the extended chart target. That target is relatively open in H and contains a. Choose an ambient open neighborhood O of a with $O \cap H$ contained in the target. The finite intersection

$$U = O \cap \bigcap_{i=0}^{k} (V_i \cap W_i)$$

is open and contains a. Define $f(y) = (f_0(y), \ldots, f_k(y))$. A finite tuple of smooth scalar functions is smooth, so f is smooth on U. Every point of $U \cap H$ lies in the chart target and in each W_i, so equality holds in every coordinate. Equality of all coordinates is equality of the vectors, giving the required identity. □

Lemma A.20 (A boundary-neighborhood collar from scalar coefficient extensions)**.** *Let c be an intrinsic boundary chart induced by the ambient atlas chart e, and let $x \in \mathrm{dom}(c) \subset \partial M$. Suppose the scalar coefficient-extension hypotheses of the preceding lemma hold for e, Z, x, and the normal coordinate of $dE_x Z(x)$ is strictly positive. There are an open neighborhood U of x in the actual boundary and $\delta > 0$ with a smooth open embedding*

$$\gamma : [0, \delta) \times U \longrightarrow M.$$

It satisfies $\gamma(0, y) = y$, its time curves solve the intrinsic ODE of Z, and $\gamma(t, y) \in \partial M$ exactly when $t = 0$. At each strip point p there is a map $g : M \to \mathbb{R} \times \partial M$ smooth at $\gamma(p)$ and satisfying $g(\gamma(q)) = q$ for every strip point q sufficiently near p. No common coefficient neighborhood or vector-valued extension is an additional input.

Proof. The boundary characterization gives $a = (0, u)$ for some $u \in \mathbb{R}^k$. Construct U and f by the coefficient-neighborhood lemma. Its coefficient identity at a gives $f(a)_0 = (dE_x Z(x))_0 > 0$. The same lemma gives containment of $U \cap H$ in the chart target and the coefficient identity throughout that set. Thus all hypotheses of the native local collar lemma hold. Apply it at u to obtain its coordinate collar a on $[0, \delta) \times V$ for an open $V \ni u$. Put $U = \mathrm{dom}(c) \cap c^{-1}(V)$ and $\gamma(t, y) = a(t, c(y))$. The boundary chart is a smooth open embedding on U, so composition gives both joint smoothness and the open embedding of γ. The coordinate seam formula and the boundary-chart inverse identity give $\gamma(0, y) = y$. The coordinate collar's time ODE and its boundary criterion give the same assertions for this composition, since its boundary coordinate is fixed along each time curve. For the inverse germ, compose the coordinate collar's smooth local inverse with the boundary chart inverse in its boundary component. That inverse is smooth near the point's boundary coordinate, so the composition is smooth near $\gamma(p)$ and has the asserted local identity with γ. □

Lemma A.21 (Uniqueness on the whole common interval)**.** *Let M be a smooth Hausdorff manifold with boundary and let Z be an intrinsic tangent field. At every point of M, suppose there is an intrinsic atlas chart and a smooth Euclidean coefficient field on an open neighborhood of that point's coordinates, agreeing there with the pushforward of Z. If $\gamma, \eta : \mathbb{R} \to M$ are intrinsic integral curves of Z on $[0, \delta)$ and $[0, T)$ respectively, and $\gamma(0) = \eta(0)$, then*

$$\gamma(t) = \eta(t) \qquad (0 \le t < \min(\delta, T)).$$

No common chart for their entire paths is required. It suffices to require the coefficient germs only at their common-image points $\gamma(t) = \eta(t)$ in the asserted common interval.

Proof. If $\min(\delta, T) \le 0$, the claimed interval is empty. Otherwise fix b with $0 < b < \min(\delta, T)$. Both curves are continuous on $[0, b]$. At any $s \in [0, b)$ where they agree, shift both curves by s. Their right derivatives solve the same intrinsic ODE. Choose the coefficient germ at their common value and apply local native uniqueness on a positive interval shorter than $b - s$. The curves therefore agree on $[s, s + \varepsilon]$ for some $\varepsilon > 0$.

Let A consist of the $s \in [0, b]$ for which the curves agree at every time in $[0, s]$. It contains zero. Put $s_* = \sup A$. For every $t < s_*$, the definition of supremum supplies an $s \in A$ with $t < s$, so the curves agree at t. If $s_* > 0$, continuity and Hausdorffness give their equality also at s_* by taking times approaching s_* from below. For $s_* = 0$ this is the given initial equality. Hence $s_* \in A$. If $s_* < b$, the local uniqueness interval at s_* extends equality to some $s_* + \varepsilon \le b$, contradicting that s_* is an upper bound for A. Thus $s_* = b$ and the curves agree on $[0, b]$. For any time t in the asserted common interval, choose such a b with $t \le b$; equality at t follows. □

Lemma A.22 (Compact assembly of native local inward collars)**.** *Let M be a smooth Hausdorff second-countable manifold with compact intrinsic boundary $B = \partial M$, and let Z have the local coefficient-extension property of A.21. Suppose that every $x \in B$ has an open boundary neighborhood U_x and a native local collar $c_x : [0, \delta_x) \times U_x \to M$, where $\delta_x > 0$. Each c_x is smooth and an open embedding, has smooth local inverse germs, satisfies $c_x(0, y) = y$, solves the intrinsic ODE of Z, and takes a point to B exactly at time zero. Then for some $\delta > 0$ there is one smooth open embedding*

$$C : [0, \delta) \times B \longrightarrow M$$

with the same seam identity, ODE, boundary criterion and smooth local inverse germs.

Proof. If B is empty, take any positive width and the unique map from the empty strip. Otherwise choose a finite subcover $U_{x_1}, \dots, U_{x_m}$ and put $d = \min_j \delta_{x_j} > 0$. For y belonging to two of these neighborhoods, their collar trajectories start at the same native point and solve the same field. Whole-interval uniqueness gives $c_{x_i}(t, y) = c_{x_j}(t, y)$ for $0 \le t < d$. Define $C(t, y)$ using any covering neighborhood containing y; this equality makes it well defined. Locally it equals the corresponding c_{x_j}, so it is smooth, is locally an open embedding, and has the asserted inverse germs, seam identity, ODE and boundary criterion on $[0, d) \times B$.

Write $S = [0, d) \times B$ and $K = \{0\} \times B$. For each $p \in K$, choose an open neighborhood N_p in S on which C is injective. The equal-image relation $E = \{(q, r) \in S^2 : C(q) = C(r)\}$ is closed, since C is continuous and M is Hausdorff. The closed set

$$P = E \setminus \bigcup_{p \in K} (N_p \times N_p)$$

is disjoint from $K \times K$: equality on the seam forces the two points to be equal, and each diagonal seam point lies in its $N_p \times N_p$. Compactness and the product-neighborhood lemma give open

neighborhoods O_1, O_2 of K such that $(O_1 \times O_2) \cap P$ is empty. Put $W = O_1 \cap O_2$. Two points of W with the same image belong to some $N_p \times N_p$, since their pair cannot belong to P. Injectivity on N_p makes them equal. Thus C is injective on the open neighborhood W of the compact seam. The tube lemma for $\{0\} \times B \subset W$ gives a relative time neighborhood J of zero with $J \times B \subset W$. Choose $0 < \delta < d$ so that $[0, \delta) \subset J$. Then C is injective on the smaller strip.

The restriction of C to this smaller strip is an open map: each open set is the union of its intersections with the local collar neighborhoods, and each of those has open image. An injective continuous open map is an open embedding. All the smoothness, inverse-germ and trajectory statements already proved restrict to this strip. □

Lemma A.23 (A uniform native collar from scalar extension germs)**.** *Let M be a smooth Hausdorff manifold with compact intrinsic boundary B, and let Z be an intrinsic tangent field. In every intrinsic atlas chart, at every point of its source, suppose each scalar coefficient of the chart pushforward of Z has an actual smooth extension germ on a Euclidean open neighborhood. For each $x \in B$, choose its intrinsic boundary chart and suppose that the normal coefficient of Z in that chart's ambient coordinates is strictly positive. Then there are $\delta > 0$ and a uniform native collar $C : [0, \delta) \times B \to M$ with all the conclusions of A.22, including its native smooth inverse germs. No local collars, overlap identities or common vector coefficient neighborhoods are additional inputs.*

Proof. For every point $x \in M$, choose an atlas chart containing it and apply the scalar coefficient-extension lemma. It constructs an open coefficient neighborhood and a smooth vector field agreeing with the native pushforward. Thus the pointwise coefficient hypothesis for whole-interval uniqueness holds throughout M. For each $x \in B$, use the chosen intrinsic boundary chart, the scalar extension germs in its ambient chart, and its positive normal coefficient. The boundary-neighborhood collar lemma then constructs an open $U_x \ni x$ and a positive-width local collar with its seam identity, native ODE, exact boundary criterion, open embedding and smooth inverse germs. These are precisely the local hypotheses of compact assembly. Applying that lemma produces the uniform collar and all its asserted properties. □

Lemma A.24 (A two-sided coordinate germ of the same native collar)**.** *Let M be a smooth Hausdorff $(k+1)$-manifold with boundary B, let Z be an intrinsic tangent field, and let $\delta > 0$. Suppose $C : [0, \delta) \times B \to M$ satisfies $C(0, y) = y$ and each time curve is an intrinsic integral curve of Z. Fix $x \in B$ and an intrinsic boundary chart χ at x, induced by the ambient half-space chart E. Suppose the scalar coefficients of Z in E have actual smooth Euclidean extension germs at $E(x)$, and $(dE_x Z_x)_0 > 0$. There are $0 < \varepsilon \le \delta$, an open neighborhood $V \ni \chi(x)$ contained in the target of χ, and a map $c_0 : \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}^{k+1}$ smooth everywhere, whose restriction to the two-sided open rectangle $(-\varepsilon, \varepsilon) \times V$ is injective and a local diffeomorphism at every point, such that*

$$c_0(0, v) = (0, v), \qquad (c_0(t, v))_0 \ge 0 \iff t \ge 0, \qquad (c_0(t, v))_0 > 0 \iff t > 0.$$

The seam identity holds for every $v \in \mathbb{R}^k$; the sign equivalences hold throughout the stated rectangle. For every $v \in V$ and $0 \le t < \varepsilon$, the point $C(t, \chi^{-1}(v))$ belongs to the source of E, and

$$E\big(C(t, \chi^{-1}(v))\big) = c_0(t, v).$$

Only local coefficient germs are required; the conclusion concerns the specified C itself.

Proof. Write $u = \chi(x)$ and $H = \{z : z_0 \ge 0\}$. The scalar coefficient-extension lemma supplies an open neighborhood U of $(0, u)$ and a smooth vector field f on U whose restriction to $U \cap H$ is the chart pushforward of Z, with $U \cap H$ contained in the target of E. At $(0, u)$ its normal component is

positive. The half-space flow lemma gives a two-sided smooth coordinate collar c_0 on $(-a, a) \times V_0$, with $a > 0$, all the displayed seam, sign, injectivity and local diffeomorphism properties, and values in U. Shrink V_0 to its intersection with the open target of χ, and put $\varepsilon = \min(a, \delta) > 0$. All these properties persist on the smaller rectangle.

For $v \in V_0$ and $0 \le t < \varepsilon$ define $\gamma_v(t) = E^{-1}(c_0(t, v))$. The nonnegative normal-coordinate identity and containment in U make this definition valid. The native chart chain rule and the coefficient identity give $\gamma_v' = Z \circ \gamma_v$ within $[0, \varepsilon)$, including at zero. The seam identity gives $\gamma_v(0) = \chi^{-1}(v) = C(0, \chi^{-1}(v))$. At every common-image point of these two curves, γ_v lies in the source of E with coordinates in U. The same f, U, E therefore supply the local coefficient hypothesis at every such point. Whole-common-interval native uniqueness gives $\gamma_v(t) = C(t, \chi^{-1}(v))$ for $0 \le t < \varepsilon$. This proves source containment for the specified collar and, on applying E, the asserted coordinate identity. □

Lemma A.25 (A local scalar extension across the same collar seam)**.** *Under the preceding lemma's hypotheses, let $\varphi : M \to \mathbb{R}$. Suppose its coordinate expression in E has an actual smooth Euclidean extension germ at $E(x)$. There are an open neighborhood W of $(0, x)$ in $\mathbb{R} \times B$ and a function $a : \mathbb{R} \times B \to \mathbb{R}$ smooth on W such that*

$$a(t, y) = \varphi(C(t, y)) \qquad \big((t, y) \in W \cap ([0, \delta) \times B)\big).$$

This is a local scalar extension across the seam of the same collar; it does not assert a global extension of additional data or maps.

Proof. Choose a smooth scalar extension g on an open neighborhood O of $E(x)$ whose germ on the target of E agrees with $\varphi \circ E^{-1}$. Germ equality permits shrinking O so that this agreement holds on $O \cap \mathrm{target}(E)$. Apply the preceding lemma to obtain c_0. Since $c_0(0, \chi(x)) = E(x)$ and c_0 is continuous, its inverse image of O is an open neighborhood of $(0, \chi(x))$. Choose a smaller two-sided product rectangle contained in that neighborhood and in the original rectangle. On it put $h = g \circ c_0$. This function is smooth by composition. For nonnegative time the preceding lemma identifies $c_0(t, v)$ with $E(C(t, \chi^{-1}(v)))$, which lies in $O \cap \mathrm{target}(E)$. The germ agreement for g then gives $h(t, v) = \varphi(C(t, \chi^{-1}(v)))$ at every such (t, v). Let W be the inverse image of this smaller rectangle under $(t, y) \mapsto (t, \chi(y))$, restricted to the open set $\mathbb{R} \times \mathrm{source}(\chi)$. This is an open neighborhood of $(0, x)$. On W define $a(t, y) = h(t, \chi(y))$, and extend a arbitrarily outside W. Smoothness of the boundary chart makes a smooth on W. Any point of W with nonnegative time has time less than the chosen $\varepsilon \le \delta$, so the previous scalar identity and the inverse boundary-chart identity give $a(t, y) = \varphi(C(t, y))$ throughout the asserted intersection. □

Lemma A.26 (Patching scalar extension germs on a closed carrier)**.** *Let X be a finite-dimensional smooth boundaryless Hausdorff second-countable manifold, let $A \subset X$ be closed, and let $f : X \to \mathbb{R}$ be a function. Suppose that every $x \in A$ has a neighborhood U_x and a function $g_x : X \to \mathbb{R}$ smooth on U_x with $g_x = f$ on $U_x \cap A$. There is a smooth function $g : X \to \mathbb{R}$ with $g = f$ on A. No agreement of the g_x away from A is required.*

Proof. Shrink each U_x to an open neighborhood of x. The sets U_x, together with $X \setminus A$, form an open cover of X. The smooth partition-of-unity theorem for finite-dimensional Hausdorff second-countable manifolds gives a locally finite smooth partition (ρ_i) subordinate to a refinement of this cover. For each refined set contained in a U_x, choose the corresponding g_x; for a refined set contained in $X \setminus A$, choose zero. Denote these choices by h_i. The product $\rho_i h_i$, defined on the open set where h_i is smooth and extended by zero outside, is smooth: the support of ρ_i is contained in that open set, so every point outside it has a neighborhood where the product vanishes. Local

finiteness therefore makes $g = \sum_i \rho_i h_i$ a smooth function on X. At $y \in A$, any index with $\rho_i(y) \neq 0$ is supported in a set contained in some U_x, and hence $h_i(y) = f(y)$. Indices subordinate to $X \setminus A$ vanish at y. Since $\sum_i \rho_i(y) = 1$, it follows that $g(y) = f(y)$. □

Lemma A.27 (Scalar extension from seam germs on the actual collar quotient)**.** *Let M be a smooth Hausdorff second-countable $(k+1)$-manifold with boundary B, and let $C : [0,\delta) \times B \to M$ be a smooth open collar of positive width, with seam $C(0,y) = y$, boundary criterion $C(t,y) \in B$ exactly when $t = 0$, and smooth native inverse germs. Let X_C be its actual boundaryless collar quotient, with original inclusion $i : M \to X_C$ and two-sided cylinder inclusion $j : (-\delta,\delta) \times B \to X_C$. Let $f : M \to \mathbb{R}$ be intrinsically smooth. Suppose each $b \in B$ has an open neighborhood W_b of $(0,b)$ in $\mathbb{R} \times B$ and a function a_b smooth on W_b, with $a_b(t,y) = f(C(t,y))$ on $W_b \cap ([0,\delta) \times B)$. There is a smooth function $g : X_C \to \mathbb{R}$ with $g(i(x)) = f(x)$ for every $x \in M$.*

Proof. The original inclusion is injective with closed image $A = i(M)$. Define F on A by $F(i(x)) = f(x)$ and give it arbitrary values off A. The quotient's interior inclusion is a smooth open embedding with smooth inverse on its image, by the lifted interior atlas. Thus at every original interior point the function obtained by transporting f through this inverse is a smooth local function agreeing with F on its whole patch.

At a boundary point b, the seam relation gives $j(0,b) = i(b)$. The cylinder is a smooth open embedding with smooth inverse on its image. Shrink its image near $j(0,b)$ so that inverse cylinder coordinates lie in W_b. Compose a_b with this inverse to obtain a smooth local scalar function there. If a point in this patch also equals $i(x)$, its inverse cylinder coordinates (t,y) have $t \geq 0$ and $C(t,y) = x$, by the original-cylinder equality criterion. Their time is less than δ. The assumed half-strip agreement therefore identifies the transported scalar with $f(x) = F(i(x))$.

These interior and boundary patches supply smooth local extension germs of F at every point of the closed carrier A. Apply the closed-carrier scalar patching lemma on the Hausdorff second-countable boundaryless manifold X_C. It produces a global smooth g with $g|_A = F|_A$, which is precisely the asserted original-value identity. □

Lemma A.28 (A scalar seam germ from the native Liouville primitive)**.** *Let M be a smooth Hausdorff $(k+1)$-manifold with boundary B. Let λ be a locally extendible intrinsic one-form whose actual exterior derivative is nondegenerate, and let Z be its contraction field. Fix $x \in B$, use its preferred boundary chart χ with ambient coordinates E, and put $\rho = -E_0$. Suppose $d\rho_x(Z_x) > 0$. Let $\delta > 0$ and suppose the specified $C : [0,\delta) \times B \to M$ has seam $C(0,y) = y$ and intrinsic time curves solving $-Z$. If $\varphi : M \to \mathbb{R}$ has an actual smooth scalar extension germ in E at $E(x)$, then there are a function $a : \mathbb{R} \times B \to \mathbb{R}$ and an open neighborhood $W \ni (0,x)$ on which a is smooth, with*

$$a(t,y) = \varphi(C(t,y)) \qquad \big((t,y) \in W \cap ([0,\delta) \times B)\big).$$

The field coefficient extensions are consequences of the primitive; the only scalar extension hypothesis concerns the additional function.

Proof. The native Liouville field-germ lemma applied to λ and its actual nondegenerate exterior derivative produces smooth scalar coefficient extension germs for $-Z$ in E. The canonical defining-function identity gives $d\rho_x(Z_x) = -(dE_x Z_x)_0 = (dE_x(-Z_x))_0$. Its assumed positivity is consequently the inward normal condition in the scalar seam lemma. Apply that lemma to this very C, the negative field, the derived coefficient germs and the given scalar germ of φ. Its open neighborhood, smooth scalar function and whole-half-strip agreement are exactly the required data. □

Lemma A.29 (One native compact presentation extending the primitive and every scalar)**.** *Let M be a compact smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Let λ be locally extendible with its actual exterior derivative nondegenerate, and suppose its contraction field Z satisfies the regular local defining-function outwardness hypotheses of Lemma A.42. There are a compact domain D in a smooth boundaryless $(k+1)$-manifold X, a smooth closed embedding $i : M \to X$ with bijective differential at every point, and a smooth one-form η on X, such that*

$$i(M) = D, \qquad \mathrm{Fr}(D) = i(\partial M), \qquad i^*\eta = \lambda.$$

There is a diffeomorphism $e : M \to D$ for the intrinsic manifold-with-boundary atlas of D, whose composite with $D \hookrightarrow X$ is i. For every $f : M \to \mathbb{R}$ whose coordinate expression in every native atlas chart has an actual smooth Euclidean extension germ at every chart point, there is a smooth $g : X \to \mathbb{R}$ with $g \circ i = f$. The objects X, D, i, η, e are chosen before f. This assertion concerns scalar data; arbitrary tensor data, maps and families require their own extension results.

Proof. The source primitive and outwardness construct one positive-width native collar C with the full primitive law and its smooth native inverse germs. Form its actual quotient X_C, its compact original carrier $D = i(M)$, and its glued primitive η. The compact-presentation theorem gives the smooth boundaryless manifold and compact domain, with the literal original carrier and boundary frontier. The original-inclusion differential lemma makes di bijective at every native point, including the seam. The primitive gluing and original-restriction lemmas give smoothness of η and $i^*\eta = \lambda$. The same-collar atlas identification gives the native diffeomorphism $e : M \to D$, with the stated original ambient composite. Fix these choices once.

At $b \in \partial M$, the defining-function sign lemma converts the supplied regular defining-function outwardness into strict outwardness for the preferred canonical function $\rho = -E_0$. Thus the hypotheses of the primitive-derived scalar seam lemma hold at every boundary point for this very C. Now fix f as stated. Its actual chart extension germs imply intrinsic smoothness: near each chart point, its coordinate expression agrees within the chart target with a smooth open extension, hence is smooth within that target. Its germ in the preferred boundary chart and the primitive-derived seam lemma give an open two-sided scalar collar germ at every b. Apply the actual-quotient scalar extension lemma to these germs and the established intrinsic smoothness. It gives a smooth function on the fixed X_C agreeing with f at every original point. This is the required g. Since the collar, quotient, carrier, inclusion and primitive were fixed before choosing f, the same X, D, i, η, e work for every such scalar. □

Definition A.30 (Intrinsic forms and their local extension coefficients)**.** *Let M be a smooth $(k+1)$-manifold with boundary. An intrinsic r-form assigns to each $x \in M$ an alternating covariant r-tensor on the actual tangent space T_xM. In an atlas chart with extended coordinate map E, its coordinate expression at y is*

$$\alpha^E_y(v_1, \dots, v_r) = \alpha_{E^{-1}(y)}(dE^{-1}_y v_1, \dots, dE^{-1}_y v_r).$$

The inverse derivative is taken within the nonnegative half-space. Local extendibility means that, at every point of every atlas chart, this coordinate expression agrees as a germ on the chart target with a smooth form-valued function on a Euclidean open neighborhood. For intrinsic forms α and ω of degrees r and $r+1$, the exterior-derivative relation requires, in every atlas chart,

$$\omega^E_y = d_{\mathrm{target}\, E}\alpha^E_y \qquad (y \in \mathrm{target}\, E),$$

where the right side is the alternating within-target derivative. A two-form is nondegenerate when $\omega_x(v, w) = 0$ for every w forces $v = 0$. The contraction equation is $\omega_x(Z_x, v) = \alpha_x(v)$ for every x, v.

The exterior-derivative relation specifies the coordinate equation; its existence and compatibility across charts are separate assertions.

Lemma A.31 (Recovery of intrinsic forms from native chart coefficients)**.** *For an intrinsic atlas chart E and x in its source, the actual chart differential dE_x and inverse within-half-space differential $dE^{-1}_{E(x)}$ are mutually inverse. Every intrinsic form is therefore recovered from its coordinate expression:*

$$\alpha^E_{E(x)}(dE_x v_1, \dots, dE_x v_r) = \alpha_x(v_1, \dots, v_r).$$

Proof. The chart and its inverse compose to the identity on their respective source and target. Differentiating these identities gives the two inverse identities. At a boundary point the chart target is a relative neighborhood in a half-space, whose tangent directions span the full Euclidean space; its within derivative consequently differentiates the identity as the identity linear map. Thus these identities include boundary points. Substitute the definition of α^E into the displayed expression. Each argument becomes $dE^{-1}_{E(x)} dE_x v_i = v_i$, giving the asserted form identity. □

Lemma A.32 (Native contraction and nondegeneracy in chart coordinates)**.** *Let E be an intrinsic atlas chart and y a point of its target, with $x = E^{-1}(y)$. If an intrinsic two-form ω is nondegenerate at x, then ω^E_y is nondegenerate. If intrinsic forms and a field satisfy $\omega(Z, \cdot) = \alpha$, their coordinate expressions satisfy*

$$\omega^E_y\big(dE_x Z_x, v\big) = \alpha^E_y(v) \qquad (v \in \mathbb{R}^{k+1}).$$

Proof. If $\omega^E_y(v, w) = 0$ for every coordinate vector w, insert $w = dE_x u$ for arbitrary $u \in T_x M$. The inverse differential identity gives $\omega_x(dE_y^{-1} v, u) = 0$ for every u. Native nondegeneracy implies $dE_y^{-1} v = 0$; applying dE_x gives $v = 0$. For the contraction formula, substitute the coordinate form definitions and use $dE_y^{-1} dE_x Z_x = Z_x$:

$$\omega^E_y(dE_x Z_x, v) = \omega_x(Z_x, dE_y^{-1} v) = \alpha_x(dE_y^{-1} v) = \alpha^E_y(v).$$

□

Lemma A.33 (Differentiating a matching local primitive extension)**.** *Let α, ω satisfy the intrinsic exterior-derivative relation, and let y belong to the target of an intrinsic atlas chart E. If a form-valued function β is differentiable at y and agrees with α^E as a germ within that target, then*

$$d\beta_y = \omega^E_y.$$

This assertion presupposes the intrinsic exterior-derivative relation; it does not assert the existence of ω.

Proof. The chart target is a relatively open subset of the nonnegative half-space. At each of its points the tangent directions span the full Euclidean space, so differentiation within that target is unique. The ordinary derivative of the differentiable function β is therefore also its within-target derivative. Germ equality gives equality of the within-target derivatives of β and α^E. Alternating these derivative identities and applying the assumed exterior-derivative relation yields $d\beta_y = d_{\text{target } E}\alpha^E_y = \omega^E_y$. □

Lemma A.34 (Exterior differentiation in overlapping native charts)**.** *Let α be a locally extendible intrinsic r-form, let E, E' be intrinsic atlas charts containing x, and put $T = E' \circ E^{-1}$ on their coordinate overlap. The coordinate form and its within-target exterior derivative obey*

$$\alpha^E = T^* \alpha^{E'}, \qquad d_{\text{target } E}\alpha^E = T^*(d_{\text{target } E'}\alpha^{E'})$$

at $E(x)$, including when x is a boundary point. The differential in the pullbacks is the actual within-half-space differential of T.

Proof. The chain rule and the inverse chart differential identities give $dT_{E(x)} = dE'_x \circ dE^{-1}_{E(x)}$. Substitution into the definition of the coordinate forms proves the first equality. Choose a smooth open extension β of $\alpha^{E'}$ near $E'(x)$. On a sufficiently small relative neighborhood of $E(x)$, the first equality identifies α^E with the pullback of β by T. Differentiation within a chart target agrees with differentiation within the half-space, since the target is relatively open there. It is unique there, and the ordinary derivative of β restricts to that same within derivative. Naturality of exterior differentiation now gives $d(T^*\beta) = T^*(d\beta)$ on this relative neighborhood. The within version follows from the same chain and product rules: second derivatives of the coordinate functions of T occur symmetrically and vanish after alternation. Those rules apply at the boundary because the half-space is the closure of its interior and its within differential is unique. Replacing $d\beta$ by the within exterior derivative of $\alpha^{E'}$ proves the second equality. □

Lemma A.35 (Existence and uniqueness of the native exterior derivative)**.** *Every locally extendible intrinsic r-form α has a unique intrinsic $(r+1)$-form $d\alpha$ satisfying the exterior-derivative relation in every atlas chart. It is determined by α alone. At x, any atlas chart E containing x gives*

$$(d\alpha)_x(v_1, \ldots, v_{r+1}) = \left(d_{\text{target } E}\alpha^E\right)_{E(x)}(dE_x v_1, \ldots, dE_x v_{r+1}).$$

Proof. At each x choose an atlas chart and use the displayed formula to define an alternating covariant tensor on T_xM. The chart-covariance lemma and the chain rule show that a second chart gives the same tensor: the transition differential followed by dE_x is dE'_x. Thus these pointwise tensors define an intrinsic form independent of the chosen chart. Composing its tensor at x with the inverse chart differential recovers $d_{\text{target } E}\alpha^E$ by the inverse identities. Hence the exterior-derivative relation holds in every chart. Conversely, if an intrinsic form ω satisfies that relation, recovering it from its chart coefficients gives exactly the displayed formula at each point. Therefore $\omega = d\alpha$, proving uniqueness. □

Lemma A.36 (Local smooth extension of the native exterior derivative)**.** *If α is a locally extendible intrinsic r-form, then $d\alpha$ is a locally extendible intrinsic $(r+1)$-form.*

Proof. Fix any atlas chart and a point of its source. Choose a smooth open extension β of α's coordinate expression there. The coordinate expression of $d\alpha$ agrees as a germ with $d\beta$: at every sufficiently nearby native point, germ equality and uniqueness of the within derivative give that identity. The function $d\beta$ is smooth on the same open neighborhood, because differentiation of a smooth function followed by the fixed linear alternation operation is smooth. Thus $d\beta$ supplies the required local extension in every chart. □

Lemma A.37 (The native exterior derivative squares to zero)**.** *For every locally extendible intrinsic form α,*

$$d(d\alpha) = 0.$$

In particular the actual exterior derivative of a locally extendible one-form is closed, including at boundary points.

Proof. Near a point in an atlas chart choose a smooth open extension β of the coordinate form. The preceding lemma identifies $d\beta$ with an extension of the coordinate form of $d\alpha$. Computing the

second derivative in this same chart therefore gives $d(d\beta)$. Write $\beta = \sum_I b_I\, dy_{i_1} \wedge \cdots \wedge dy_{i_r}$. In $d(d\beta)$ the second-derivative terms for each coefficient are

$$\sum_{j,l} \partial_l \partial_j b_I\, dy_l \wedge dy_j \wedge dy_{i_1} \wedge \cdots \wedge dy_{i_r}.$$

Terms with $j = l$ vanish. Terms with $j \neq l$ cancel in pairs, because mixed second derivatives agree and exchanging dy_l, dy_j changes the sign. Hence $d(d\beta) = 0$ on its open neighborhood. The matching within derivatives and form recovery carry this zero tensor to $d(d\alpha)$ at the original point. Every point admits such a chart, proving the identity everywhere. □

Lemma A.38 (The intrinsic contraction field and its smooth coefficient germs)**.** *Let α be a locally extendible intrinsic one-form and let ω satisfy its exterior-derivative relation. Suppose ω is nondegenerate at every point. There is a unique intrinsic field Z satisfying $\omega(Z, \cdot) = \alpha$. In every atlas chart, its coordinate expression has a smooth vector-valued extension germ on a Euclidean open neighborhood. Each scalar coefficient of the negative field $-Z$ has such a smooth extension germ as well. No two-form extension or field extension is an additional hypothesis.*

Proof. At x, the linear map $v \mapsto \omega_x(v, \cdot)$ is injective by nondegeneracy. Its domain and codomain have the same finite dimension, so it is bijective. Its inverse applied to α_x defines Z_x. If two fields satisfy the equation, their difference contracts to zero, hence is zero at every point. This proves uniqueness.

Fix a chart and a point of its source. Choose a smooth open extension β of the coordinate primitive near that point. Its derivative $d\beta$ is smooth there. On a smaller relative chart neighborhood, the exterior-extension agreement gives $d\beta = \omega^E$. Chart transport preserves nondegeneracy, so at the chosen point the matrix $A(y) : v \mapsto d\beta_y(v, \cdot)$ is invertible. Its entries are smooth. The determinant remains nonzero on a smaller Euclidean open neighborhood, and the adjugate formula makes A^{-1} smooth there. Thus $f(y) = A(y)^{-1}\beta_y$ is a smooth vector-valued function there. At native chart points it satisfies the transported contraction equation. By uniqueness it equals $dE(Z)$, proving the extension-germ assertion. Negating each scalar component of f gives the required scalar germs for $-Z$. □

Lemma A.39 (Canonical native boundary defining functions)**.** *Let E be an intrinsic atlas chart with inward nonnegative normal coordinate r. On its source put $\rho_E = -r \circ E$. This is a smooth intrinsic function, is nonpositive, and vanishes exactly at the boundary points of its chart source. Its differential is nonzero and satisfies*

$$d\rho_E(v) = -(dE(v))_0.$$

Consequently $d\rho_E(Z) > 0$ implies $(dE(-Z))_0 > 0$.

Proof. Smoothness follows by composing the smooth chart coordinate map with the negative normal projection. The half-space coordinate is nonnegative, and the boundary characterization is exactly $r = 0$. The chain rule gives the displayed differential formula. At a chart point let $v = dE^{-1}(e_0)$, where e_0 is the unit normal vector. The inverse differential identity gives $d\rho_E(v) = -1$, proving that $d\rho_E$ is nonzero. Finally linearity gives $(dE(-Z))_0 = -(dE(Z))_0 = d\rho_E(Z)$, so the stated positive inequality transfers to the inward field. □

Lemma A.40 (The sign of an arbitrary local defining function)**.** *Let x be a native boundary point in an atlas chart E with inward normal coordinate r. Suppose ρ vanishes at x, is nonpositive on a neighborhood of x in the manifold, and has nonzero differential there. Suppose its coordinate*

expression has a local scalar extension differentiable to all orders at $E(x)$ and agreeing as a germ on the chart target. There is a constant $c < 0$ such that

$$d\rho_x(v) = c\,(dE_x v)_0 \qquad (v \in T_x M).$$

In particular $d\rho_x(v) > 0$ if and only if $(dE_x v)_0 < 0$. Thus defining-function outwardness gives positive normal coefficient for the negative field in every native chart.

Proof. Put $a = E(x)$, so $a_0 = 0$, and let β be the coordinate extension. Its value at a is zero. Since the chart target is relatively open in the half-space, germ agreement and the local nonpositivity of ρ imply $\beta \le 0$ on a half-space neighborhood of a. Let $L = d\beta_a$. For any vector w with $w_0 \ge 0$, the points $a + tw$ lie in that neighborhood for all sufficiently small $t > 0$. The difference quotients are nonpositive, so $L(w) \le 0$. If $w_0 = 0$, applying this to both w and $-w$ gives $L(w) = 0$.

Let e_0 be the unit normal vector and put $c = L(e_0) \le 0$. The vector $w - w_0 e_0$ is tangential, so linearity gives $L(w) = cw_0$. If $c = 0$, this formula would make L zero, and the chain rule would make $d\rho_x$ zero, contrary to regularity. Thus $c < 0$. Germ agreement and uniqueness of differentiation on the half-space give the actual chart chain rule $d\rho_x = L \circ dE_x$, proving the displayed formula. Multiplication by $c < 0$ gives the stated sign equivalence, and $dE_x(-Z_x) = -dE_x(Z_x)$ gives its inward-field consequence. □

Lemma A.41 (Native Liouville coefficients from the original primitive)**.** *Let M be a smooth $2n$-manifold with boundary, $n \ge 1$, and let λ be a smooth one-form extending smoothly across its boundary in the source sense. Suppose $d\lambda$ is pointwise nondegenerate. The unique intrinsic field Z defined by $\iota_Z d\lambda = \lambda$ has scalar smooth extension germs in every native atlas chart. If Z is strictly outward, the field $-Z$ has a strictly positive normal coefficient in every boundary chart with inward nonnegative normal coordinate. Consequently $-Z$ satisfies the scalar-extension and inwardness hypotheses of A.23 whenever the intrinsic boundary is compact.*

Proof. Fix an atlas chart and a point of its source. Write its ambient coordinates as $y_1, \dots, y_m$. The source extension convention gives smooth scalar extensions a_j of the coefficients in $\lambda = \sum_j a_j\,dy_j$ on an actual Euclidean neighborhood of the coordinate point. Intersect their finitely many neighborhoods. On that neighborhood form

$$\omega = \sum_{i<j} (\partial_i a_j - \partial_j a_i)\,dy_i \wedge dy_j, \qquad L_y(v) = \omega_y(v, \cdot).$$

On native chart points this is the coordinate expression for $d\lambda$. The formula is independent of the coefficient extensions: two extensions agree on the relative half-space neighborhood, hence their first derivatives agree on its open interior; continuity gives agreement of those derivatives at its boundary as well. Nondegeneracy at the chosen point says L_y is invertible there. Its matrix depends smoothly on y. Nonvanishing of its determinant therefore persists on a smaller open neighborhood, and the adjugate formula divided by that nonzero determinant makes L_y^{-1} smooth there. The vector field

$$v(y) = L_y^{-1}\Big(\sum_j a_j(y)\,dy_j\Big)$$

is consequently smooth on that neighborhood. At its native points it solves the same contraction equation as the coordinate pushforward of Z. Nondegeneracy makes the solution unique, so these vectors agree. Each scalar coordinate of v is the required smooth extension germ. This argument applies in every atlas chart.

At a boundary point write the inward normal coordinate as $r \geq 0$ and choose a source-smooth regular local defining function ρ with $d\rho(Z) > 0$. The defining-function sign lemma gives $d\rho = c\,dr$ for a constant $c < 0$. Thus $dr(Z) < 0$ and $dr(-Z) > 0$. This applies in every native boundary chart. Negating the constructed scalar extensions gives the corresponding extensions for $-Z$. These are exactly the claimed collar hypotheses. □

Lemma A.42 (A compact native collar from the original Liouville primitive)**.** *Let M be a smooth Hausdorff manifold with compact intrinsic boundary B. Let λ be locally extendible, with its actual exterior derivative pointwise nondegenerate, and let Z be its unique contraction field. At every $x \in B$, suppose there is a regular local scalar function ρ_x, with a local scalar extension in the preferred boundary chart, satisfying*

$$\rho_x(x) = 0, \qquad \rho_x \leq 0 \text{ near } x \text{ in } M, \qquad d\rho_x(Z_x) > 0.$$

It suffices that the extension is differentiable to all orders at the coordinate point and agrees as a germ on the chart target. These data construct $\delta > 0$ and a smooth open embedding

$$C : [0, \delta) \times B \longrightarrow M$$

with $C(0, y) = y$, whose time curves solve the actual intrinsic ODE of $-Z$. Its image point lies in B exactly at time zero. At every strip point it has a native smooth local inverse germ.

Proof. The source's regular defining functions satisfy the listed local scalar conditions, which are the only properties used here. The native exterior-derivative construction determines $d\lambda$ from λ. The field-germ lemma constructs its unique contraction field and smooth scalar extension germs for $-Z$ in every atlas chart. At a boundary point choose the stated defining function. Its regularity, local sublevel inequality and scalar extension are precisely the hypotheses of the defining-function sign lemma. That lemma gives a strictly positive normal coefficient for $-Z$ in the preferred chart. The uniform collar from scalar extension germs therefore applies: it constructs local collars, proves their overlap agreement by ODE uniqueness, and assembles one positive-width collar using compactness of B. Its conclusions give the stated seam identity, open embedding, time ODE, boundary criterion and smooth inverse germs. If B is empty, that construction gives the unique map from an empty strip. □

Lemma A.43 (The inward spatial primitive equation in open coordinates)**.** *Let E, F be real normed vector spaces, let $U \subset F$ and $P \subset \mathbb{R} \times E$ be open, and let α be a smooth one-form on U. Suppose $C : P \to U$ is smooth and $f : F \to F$ satisfies, at every $(t, y) \in P$,*

$$\partial_t C(t, y) = f(C(t, y)), \qquad d\alpha_{C(t,y)}(f(C(t, y)), w) = -\alpha_{C(t,y)}(w) \quad (w \in F).$$

Then $\beta = C^*\alpha$ has zero time coefficient. For each fixed $y \in E$ and $v \in E$, the spatial coefficient

$$q(t) = \alpha_{C(t,y)}\big(dC_{(t,y)}(0, v)\big)$$

satisfies $q'(t) = -q(t)$ wherever $(t, y) \in P$.

Proof. Substituting $w = f(C(t, y))$ into the contraction equation and using alternation gives $\alpha_{C(t,y)}(f(C(t, y))) = 0$. The time ODE therefore gives $\beta(\partial_t) = 0$ throughout P. The constant product vector fields ∂_t and $(0, v)$ commute. The coordinate formula for the exterior derivative consequently gives

$$d\beta\big(\partial_t, (0, v)\big) = \partial_t\big(\beta(0, v)\big) - (0, v)\big(\beta(\partial_t)\big) = q'(t).$$

The spatial derivative of the identically zero time coefficient is zero. Pullback naturality and the time ODE give

$$d\beta\big(\partial_t,(0,v)\big) = d\alpha\big(f(C(t,y)),dC_{(t,y)}(0,v)\big) = -q(t),$$

proving the claimed equation. This is a local coordinate calculation; no intrinsic collar or global extension of its Liouville field is constructed by it. □

Lemma A.44 (The full inward primitive law in strip coordinates)**.** *Let E, F be real normed vector spaces, let $U \subset F$ and $V \subset E$ be open, and let $\delta > 0$. Let α be a smooth one-form on U, and let $C : (-\delta,\delta)\times V \to F$ be smooth with $C([0,\delta)\times V) \subset U$. Suppose that for every $0 \le t < \delta$ and $y \in V$,*

$$\partial_t C(t,y) = f(C(t,y)), \qquad d\alpha_{C(t,y)}(f(C(t,y)),w) = -\alpha_{C(t,y)}(w) \quad (w \in F).$$

Write $C_0(y) = C(0,y)$ and let π be the boundary-variable projection. On the whole nonnegative strip, including its seam,

$$C^*\alpha = e^{-t}\pi^*(C_0^*\alpha).$$

The identity holds on arbitrary product tangent vectors (r,v); the contraction hypothesis is imposed only at nonnegative times.

Proof. Fix $y \in V$ and $v \in E$, and let $q(t) = \alpha_{C(t,y)}(dC_{(t,y)}(0,v))$. The spatial primitive equation applies on $(0,\delta)\times V$, so $q' = -q$ there. Thus $(e^t q(t))' = 0$ on the connected interval $(0,\delta)$, making $e^t q(t)$ constant. Smoothness of C across zero and smoothness of α near $C(0,y)$ give continuity of q at zero. The constant is consequently its limit $q(0)$. Therefore $q(t) = e^{-t}q(0)$ for all $0 \le t < \delta$. The contraction equation and alternation give $\alpha_{C(t,y)}(f(C(t,y))) = 0$, including at $t = 0$. Using $(r,v) = r(1,0) + (0,v)$, linearity of dC and of α, and the time ODE, yields

$$\alpha_{C(t,y)}(dC_{(t,y)}(r,v)) = r\alpha_{C(t,y)}(f(C(t,y))) + q(t) = e^{-t}q(0).$$

The last expression is the value of $e^{-t}\pi^*(C_0^*\alpha)$ on (r,v), proving the full form identity. □

Lemma A.45 (Native pullback calculus on a one-sided strip)**.** *Let α be a locally extendible native one-form on M, and let ω satisfy its exterior-derivative relation. Let E be a real normed space, $V \subset E$ open and $\delta > 0$. Put $S = [0,\delta)\times V$, and let $C : S \to M$ be smooth in its intrinsic one-sided coordinates. The actual pullback*

$$\beta_p = \alpha_{C(p)} \circ d_S C_p$$

is smooth within S, and its within exterior derivative satisfies

$$d_S\beta = C^*\omega \qquad \text{on } S.$$

All differentials use this same one-sided source, including at time zero; no extension of C to negative times is required.

Proof. Near a strip point choose a native target chart E_M and a smooth open coefficient extension a of α there. Continuity of C keeps its nearby strip values in that chart. Chart recovery and the chain rule identify the actual pullback as a germ within S with G^*a, where $G = E_M \circ C$ uses the actual within differential. The derivatives of G are smooth within S; composition with a, followed by differentiation and contraction, therefore makes this pullback smooth within S. The strip has unique full-dimensional within differentials and is contained in the closure of its interior. Within pullback naturality gives $d_S(G^*a) = G^*(da)$, including at the seam. The extension agreement identifies da with ω's chart coefficients. The chart chain rule and form recovery identify the resulting pullback with $C^*\omega$. Germ equality then proves the asserted identity for the actual native pullback at each strip point. □

Lemma A.46 (The native primitive law with normed boundary variables)**.** *Let λ be a locally extendible native one-form, and let Z satisfy its actual Liouville equation $d\lambda(Z,\cdot)=\lambda$. Let E be a real normed space, $V\subset E$ open and $\delta>0$. Suppose $C:[0,\delta)\times V\to M$ is intrinsically smooth on this one-sided strip and its time curves solve the native ODE of $-Z$. With $C_0(y)=C(0,y)$ and boundary-variable projection π, the full native one-form identity is*

$$C^*\lambda = e^{-t}\pi^*(C_0^*\lambda) \qquad (0\le t<\delta).$$

It holds on every product tangent (r,v), using the actual within differential of this same C at the seam and everywhere else in the strip.

Proof. Put $\beta=C^*\lambda$ on $S=[0,\delta)\times V$. The time ODE gives $d_SC(\partial_t)=-Z\circ C$. Alternation and the Liouville equation give $\lambda(Z)=0$, hence $\beta(\partial_t)=0$. The strip pullback calculus gives $d_S\beta=C^*(d\lambda)$. For fixed y,v, the spatial coefficient $q(t)=\beta_{(t,y)}(0,v)$ consequently has within derivative

$$q'(t)=d_S\beta\big(\partial_t,(0,v)\big)=d\lambda\big(-Z,d_SC(0,v)\big)=-q(t).$$

Here the spatial derivative of the identically zero time component vanishes, and the constant product directions commute. Thus $e^tq(t)$ has zero within derivative throughout the convex interval $[0,\delta)$. The mean-value theorem on every compact subinterval shows it is constant, equal to $q(0)$. Therefore $q(t)=e^{-t}q(0)$. Finally write $(r,v)=r(1,0)+(0,v)$. The time component contributes zero and the spatial component gives the displayed full pullback identity, including at $t=0$. □

Lemma A.47 (The original primitive on the same inward collar)**.** *Assume the hypotheses of Lemma A.41 with Z strictly outward. Suppose the boundary B is compact, put $\alpha=\lambda|_B$, and construct the uniform collar $C:[0,\delta)\times B\to M$ using the field $-Z$. For the boundary projection $\pi:[0,\delta)\times B\to B$, the original primitive satisfies the full product-form identity*

$$C^*\lambda = e^{-t}\pi^*\alpha.$$

Thus the outward coordinate $r=-t$ has primitive $e^r\alpha$ on this very collar.

Proof. Construct C,δ by the compact native collar theorem, retaining its seam identity, open embedding, ODE, boundary criterion and smooth inverse germs. Fix $y\in B$, choose a boundary chart $\chi:U\to V\subset\mathbb{R}^k$ containing y, and reparameterize this same collar by

$$Q(t,u)=(t,\chi^{-1}(u)), \qquad D=C\circ Q.$$

The inverse chart identity and chain rule give $dQ_{(t,\chi(y))}(r,d\chi_y v)=(r,v)$, including at time zero. Thus the actual within differential of D on the one-sided strip has the same value as $dC(r,v)$ after this tangent reparameterization. The map D has the same negative-Liouville time curves, so the native normed-variable primitive law applies to it. It gives

$$\lambda_{C(t,y)}\big(dC_{(t,y)}(r,v)\big)=e^{-t}\lambda_{C(0,y)}\big(dC_{(0,y)}(0,v)\big).$$

Write $\iota:B\to M$ for the actual boundary inclusion. The identity $C\circ(y\mapsto(0,y))=\iota$, differentiated within the same strip, gives $dC_{(0,y)}(0,v)=d\iota_y(v)$. Substitution and $C(0,y)=y$ identify the right side with $e^{-t}\alpha_y(v)$. This holds for every y, every spatial tangent v and every time component r, proving the full product-form identity on precisely the constructed C and its original width. Substitution of the outward coordinate $r=-t$ gives $e^r\alpha$. □

Lemma A.48 (The Euclidean atlas of the native interior)**.** *Let $k \geq 0$, and let M be a smooth Hausdorff second-countable $(k+1)$-dimensional manifold with boundary with its intrinsic half-space atlas. The actual intrinsic interior M°, with its subspace topology, has a smooth boundaryless $(k+1)$-dimensional Euclidean atlas. Its inclusion into M is a smooth open embedding, and its differential is a linear isomorphism at every interior point. No Euclidean atlas on the interior is additional data.*

Proof. In a half-space chart c, write its inward normal coordinate as r. The interior points in the chart are exactly those with $r > 0$. Restrict the source to those points and regard the chart values as vectors in $\mathbb{R}^{k+1}$. The target is the intersection of the relative chart target with $\{r > 0\}$, which is Euclidean open: if the relative chart target is $O \cap \{r \geq 0\}$ with O open, this intersection is $O \cap \{r > 0\}$. The ambient chart and its inverse restrict to mutually inverse continuous maps on these sets. These charts cover M°, since every interior point lies in an ambient chart and has positive normal coordinate there.

On an overlap, the Euclidean transition is the restriction of the extended ambient chart transition. At every point in this overlap, the positive normal coordinate gives a Euclidean neighborhood contained in the half-space. Smoothness within the half-space consequently is ordinary smoothness on this neighborhood. The same argument applies to the inverse transition. Thus the restricted charts form a smooth Euclidean atlas. The native interior is open in M, and its subspace is Hausdorff and second countable.

In the resulting source chart and corresponding ambient target chart, the inclusion is the identity coordinate map into the half-space. At a positive normal coordinate this map is smooth and has identity differential. The differentials of the source and ambient charts are linear isomorphisms, so the inclusion differential is also an isomorphism. The inclusion is an open topological embedding by the open subspace topology, giving all assertions. □

Lemma A.49 (Smooth restriction of a native form to its actual boundary)**.** *Let M be a smooth Hausdorff second-countable $(k+1)$-dimensional manifold with boundary, $k \geq 0$. Write $\iota : B = \partial M \to M$ for its actual boundary inclusion. For every locally extendible intrinsic r-form α, the actual boundary pullback*

$$(\iota^*\alpha)_b(v_1, \ldots, v_r) = \alpha_{\iota(b)}(d\iota_b v_1, \ldots, d\iota_b v_r)$$

is a smooth r-form on the boundaryless manifold B. In particular the boundary primitive $\lambda|_B$ is smooth whenever the native primitive λ is locally extendible. This statement includes $k = 0$ and does not assert contact nondegeneracy.

Proof. Fix a boundary point b and a boundary chart induced by an ambient half-space chart E. In these charts the inclusion is the constant linear insertion $L(u) = (0, u)$. The chart chain rule therefore gives

$$dE_{\iota(b)} \circ d\iota_b \circ d\chi_b^{-1} = L.$$

Chart recovery of α and this identity express the boundary pullback coefficients as

$$u \longmapsto \alpha^E_{L(u)} \circ L^{\otimes r}$$

on the boundary chart target. Choose a smooth open coefficient extension β of α^E at $L(\chi(b))$. Germ agreement on the ambient chart target restricts to germ agreement along the boundary hyperplane. On a sufficiently small open neighborhood of $\chi(b)$ in the boundary chart target, the displayed coefficients therefore equal $\beta_{L(u)} \circ L^{\otimes r}$. Composition with L is smooth, and composition

of an alternating form with the fixed linear map L is a continuous linear operation. Thus these coefficient functions are smooth near $\chi(b)$.

This applies in every boundary chart near every boundary point. Evaluating the coefficient functions on tangent coordinates is a continuous multilinear operation, so the corresponding scalar function of the base point and r tangent vectors is smooth as well. This is precisely smoothness of the boundary form. The argument uses only hyperplane insertion, including when its source is the zero-dimensional vector space, so it covers $k = 0$. □

Lemma A.50 (Smooth pullback of a native primitive to a boundaryless source)**.** *Let N be a smooth boundaryless manifold, let M be a smooth intrinsic manifold with boundary, and let $f : N \to M$ be smooth. If λ is a locally extendible intrinsic one-form on M, its actual pullback*

$$(f^*\lambda)_x(v) = \lambda_{f(x)}(df_x v)$$

is a smooth one-form on N.

Proof. Fix $x \in N$. Choose a source chart χ near x and a native target chart E near $f(x)$. Continuity of f permits restriction of the source chart neighborhood so all its images lie in E. The coordinate map $g = E \circ f \circ \chi^{-1}$ is smooth there. Chart recovery and the chain rule identify the coefficients of the actual pullback with

$$u \longmapsto \lambda^E_{g(u)} \circ dg_u.$$

Choose a smooth open extension β of the native coefficients at $g(\chi(x))$. Germ agreement on the target chart and continuity of g make the displayed function agree near $\chi(x)$ with $u \mapsto \beta_{g(u)} \circ dg_u$. Both $\beta \circ g$ and dg are smooth. Composition of a covector and a linear map is a continuous bilinear operation, so this coefficient function is smooth. Every point of every source chart has such a neighborhood; thus the pullback has smooth coefficients throughout all source charts, which proves smoothness of the actual one-form. □

Lemma A.51 (The Euclidean cylinder atlas with time first)**.** *Let B be a smooth boundaryless k-manifold, $k \geq 0$, and let $\delta \in \mathbb{R}$. The open cylinder $Y = (-\delta, \delta) \times B$ has a smooth boundaryless $(k+1)$-dimensional Euclidean atlas obtained from product charts by*

$$L(t, u) = (t, u_1, \ldots, u_k).$$

Its inclusion into $\mathbb{R} \times B$ is smooth. A map into Y is smooth at a point exactly when its composite with this inclusion is smooth there. Restricting a map from $\mathbb{R} \times B$ to Y preserves smoothness at every cylinder point in both directions. These assertions include an empty B or an empty band.

Proof. The cylinder is an open subset of the product. Restrict the product charts to it and compose each coordinate map with the linear isomorphism L. Their Euclidean targets remain open. Their transition maps are the product transitions conjugated by L and L^{-1}; both linear maps are smooth, so all transitions are smooth. The subspace topology is Hausdorff and second countable, as inherited from the product. Thus these charts give the claimed boundaryless smooth manifold.

In the original product charts the inclusion is the identity on an open coordinate set. In the recharted source its expression is L^{-1}, proving smoothness. For a map with values in Y, passing between its product and recharted coordinate expressions applies L or L^{-1}, so smoothness is equivalent. For a map defined on the whole product, the band is a neighborhood of each of its points; restriction to that neighborhood preserves and reflects local smoothness. Recharting again preserves it by the same linear isomorphisms. Every assertion is local at an actual cylinder point, so it remains valid when there are no such points. □

Lemma A.52 (The regular time coordinate on the actual cylinder)**.** *For the Euclidean cylinder atlas, the actual time coordinate $T(t,b) = t$ is smooth and its differential is nonzero at every cylinder point. No positivity assumption on the band width or nonemptiness assumption on B is needed for this assertion.*

Proof. The time coordinate is the first projection after the smooth cylinder inclusion into $\mathbb{R} \times B$, so it is smooth. Fix $p = (t,b)$ in the band. Because $(-\delta, \delta)$ is open and contains t, the local curve $\gamma(s) = (s,b)$ is defined for all s near t and has values in the band. It is smooth in ordinary product coordinates, hence smooth in the recharted cylinder atlas. The identity $T \circ \gamma(s) = s$ gives $dT_p \circ d\gamma_t = \mathrm{id}_{\mathbb{R}}$ by the chain rule. The zero differential could not satisfy this equation, since evaluation at $1 \in \mathbb{R}$ would give $0 = 1$. Thus $dT_p \neq 0$. The proof is at an arbitrary actual cylinder point; if the band is empty the assertion holds without choosing any such point. □

Definition A.53 (The two-sided space of an intrinsic collar)**.** *Let M be a smooth manifold with boundary B, let $\delta > 0$, and let $C : [0,\delta) \times B \to M$ be an open embedding such that $C(0,b) = b$ and $C(t,b) \in B$ exactly when $t = 0$. Put $Y = (-\delta, \delta) \times B$. Form X_C from $M \sqcup Y$ by identifying $(t,b) \in Y$ with $C(t,b) \in M$ whenever $t \geq 0$, and give it the quotient topology. Equivalently, every equivalence class has a unique representative either in M or at a negative-time point of Y. Write $i : M \to X_C$ and $j : Y \to X_C$ for the two canonical maps. Define the folding map $F : X_C \to M$ by*

$$F(i(x)) = x, \qquad F(j(t,b)) = C(\max(t,0), b).$$

This is a finite two-sided collar extension, rather than the infinite cylindrical completion of Definition 2.1.

Lemma A.54 (The open pieces of the intrinsic collar extension)**.** *The maps $j : Y \to X_C$ and $i|_{M^\circ} : M^\circ \to X_C$ are open embeddings whose images cover X_C. Their overlap consists exactly of the points $j(t,b)$ with $0 < t < \delta$, and on the overlap*

$$j(t,b) = i(C(t,b)).$$

The folding map F is continuous and is a left inverse of i. Consequently i is a topological embedding. If M is compact, its image under i is compact.

Proof. The normal representatives in the definition show that i is injective. For j, negative-time representatives are distinct and cannot agree with nonnegative-time representatives. The latter agree only when $C(t,b) = C(t',b')$; injectivity of C then gives $(t,b) = (t',b')$. Thus j is injective as well. Both maps are continuous as the restrictions of the quotient projection.

Let $U \subset Y$ be open. The preimage of $j(U)$ in $M \sqcup Y$ is $C(U \cap ([0,\delta) \times B))$ on the M summand and U on the Y summand. The first set is open in M because the intersection is relatively open in the collar strip and C is an open embedding. Hence $j(U)$ is quotient-open, proving that j is an open embedding. If $V \subset M^\circ$ is open, it is open in M. The preimage of $i(V)$ is V on the M summand, and on Y it is

$$\{(t,b) : 0 < t < \delta,\ C(t,b) \in V\}.$$

The boundary criterion excludes time zero, and continuity of C on the positive strip makes this set open in Y. Thus $i(V)$ is open and $i|_{M^\circ}$ is an open embedding.

Every native point is either interior or boundary, and a boundary point b equals $C(0,b)$. Every other class has a negative cylinder representative. This proves the covering assertion. The boundary criterion and the normal representatives give exactly the stated overlap and its formula.

On the two summands the proposed folding map is the identity and $(t, b) \mapsto C(\max(t, 0), b)$, respectively. The latter is continuous because the maximum function is continuous and takes its values in $[0, \delta)$. These maps agree on every identified pair, so the quotient topology gives a continuous map F. The identity $F \circ i = \mathrm{id}_M$ makes the inverse of i on its image continuous; hence i is a topological embedding. Its image is compact when M is compact, as the continuous image of a compact space. □

Lemma A.55 (Separation and countability of the intrinsic collar extension)**.** *If M is Hausdorff, then X_C is Hausdorff. If M is second countable, then X_C is second countable. Its two open pieces have the actual subspace topologies of the native interior and of the boundary cylinder.*

Proof. Let $p \neq q$ be points of X_C. If $F(p) \neq F(q)$, choose disjoint open neighborhoods of these two points in the Hausdorff space M. Their inverse images under the continuous F are disjoint open neighborhoods of p, q. If $F(p) = F(q)$, at least one of p, q has a negative cylinder representative, because F is injective on $i(M)$. Its folded value is a boundary point b. A native point with that folded value is $i(b) = j(0, b)$; a point with a negative representative already lies in $j(Y)$. Thus both p and q belong to $j(Y)$. The cylinder Y is Hausdorff, so choose disjoint open neighborhoods of their distinct cylinder representatives. The open embedding j carries these to disjoint open neighborhoods in X_C. This proves Hausdorffness in every case.

The boundary B and interior M° inherit second countability from M. The open interval $(-\delta, \delta)$ is second countable, so its product Y with B is second countable. Choose countable bases on Y and M°. Their images under the two open embeddings are open in X_C and form a countable family. For any open set in X_C and any point in that set, one of the two pieces contains that point; the pullback of the set to that piece contains a basis neighborhood of the point. Its image is contained in the original open set. The union of the two image bases is therefore a countable basis for X_C, as asserted. □

Lemma A.56 (The original carrier's interior and frontier)**.** *Assume M is Hausdorff, and put $A = i(M) \subset X_C$. Then*

$$\mathrm{int}(A) = i(M^\circ), \qquad \mathrm{Fr}(A) = i(\partial M).$$

In the cylinder coordinates the first equality means $t > 0$, and the second means $t = 0$. These are equalities for the literal original carrier; no compactness hypothesis is needed.

Proof. The quotient's normal representatives give $j^{-1}(A) = \{(t, b) : t \geq 0\}$. In the open band its interior is $\{t > 0\}$ and its frontier is $\{t = 0\}$: positive and negative times have neighborhoods of the same sign, while at $(0, b)$ every product neighborhood contains both signs of time. As j is a homeomorphism onto an open subset, these computations are exactly the cylinder coordinates of the interior and frontier of A.

The native interior image is an open subset contained in A, hence contained in $\mathrm{int}(A)$. Conversely every point of $\mathrm{int}(A)$ is $i(x)$ for some $x \in M$. If x were a boundary point b, its cylinder representative would be $(0, b)$, which belongs to the frontier rather than the interior by the preceding calculation. Thus x is native interior, proving the first equality.

The quotient is Hausdorff, and i and F are continuous with $F \circ i = \mathrm{id}_M$. Thus A is the equality locus of the two continuous maps $i \circ F$ and id_{X_C}: a point of A is fixed, and a fixed point equals its own image $i(F(x))$. The diagonal of a Hausdorff space is closed, so this equality locus is closed. Therefore $\mathrm{Fr}(A) = A \setminus \mathrm{int}(A)$. Injectivity of i and the decomposition $M = M^\circ \sqcup \partial M$ identify this difference with $i(\partial M)$, proving the second equality and its seam formula. □

Lemma A.57 (The topological collar extension from the original primitive)**.** *Let $n \geq 1$, and let M be a compact smooth $2n$-dimensional Hausdorff second-countable manifold with its intrinsic half-space atlas. Let λ be locally extendible with its actual exterior derivative nondegenerate, and suppose its contraction field satisfies the local regular-defining-function outwardness hypotheses of A.42. These original data construct C, δ with all the properties of that native collar, including its smooth inverse germs and the full primitive identity*

$$C^*\lambda = e^{-t}\pi^*(\lambda|_{\partial M}).$$

For this same C and δ, the finite quotient extension X_C is Hausdorff and second countable. Its original carrier inclusion $i : M \to X_C$ is a closed embedding with compact image, and its cylinder and native interior inclusions are open embeddings. The interior of this carrier is $i(M^\circ)$, and its frontier is exactly $i(\partial M)$.

Proof. The actual boundary is a closed subset of compact M, hence compact. The native collar theorem with its full primitive law constructs C, δ from λ, its nondegenerate exterior derivative and the stated outwardness. Retain its positive width, one-sided smoothness, open embedding, seam identity, negative-Liouville time curves, exact boundary criterion, native inverse germs and primitive identity. Apply the finite quotient construction to this collar, without changing either witness. The open-piece lemma supplies the cylinder and native interior open embeddings, and the separation and countability lemma supplies the asserted topology on their quotient. The original inclusion is a continuous topological embedding, and its image is compact because M is compact. A compact subset of the Hausdorff quotient is closed, so this same inclusion is a closed embedding. The carrier lemma identifies its interior and frontier with the native interior and boundary images. Every conclusion uses precisely the initially constructed collar and its original width and primitive identity. □

Lemma A.58 (The smooth atlas on the finite intrinsic collar extension)**.** *Let $k \geq 0$, and let M be a smooth Hausdorff second-countable $(k+1)$-dimensional manifold with its intrinsic half-space atlas. Assume the hypotheses of its intrinsic collar extension, and suppose C is smooth on its one-sided strip. Suppose also that at every strip point p there is a native smooth local map g near $C(p)$ with $g(C(q)) = q$ for all q in a neighborhood of p within the strip. Then the Euclidean atlas of M° and the product atlas of $(-\delta, \delta) \times B$, carried through their two open embeddings, form a compatible smooth boundaryless $(k+1)$-dimensional atlas on X_C. These hypotheses hold for the native collar constructed from the original Liouville primitive.*

Proof. The two open embeddings identify their sources homeomorphically with open subsets covering X_C. Carry the charts of each source through these homeomorphisms. Identify $\mathbb{R} \times \mathbb{R}^k$ with $\mathbb{R}^{k+1}$ by $(t, u) \mapsto (t, u_1, \ldots, u_k)$. The resulting charts have open Euclidean targets and cover X_C. Transitions between charts from the same piece are smooth by the native interior atlas or the boundary product atlas.

For a mixed transition, its overlap on the cylinder side has $t > 0$. In these positive-time coordinates the map to the native interior is $(t, b) \mapsto C(t, b)$. One-sided smoothness is ordinary smoothness there, because every positive-time point has a neighborhood contained in the strip. Composing with the native interior charts proves smoothness of this direction of every mixed transition.

For the reverse direction fix a positive strip point p and a local map g as in the hypotheses. The collar is a homeomorphism onto its open image, so the image of a sufficiently small relative neighborhood of p is a neighborhood of $C(p)$ in M. On that neighborhood, the identity $g(C(q)) = q$

makes g equal to the collar inverse. Restricting further to the native interior preserves this equality and smoothness of g, since the native interior inclusion is smooth. Composing with a product chart therefore proves smoothness of the reverse mixed transition near the original point. Every overlap point has such a neighborhood, so all reverse mixed transitions are smooth. Thus the carried atlas is compatible and boundaryless. Its Hausdorff and second-countable topology is the quotient topology already established. The constructed Liouville collar has both one-sided smoothness and precisely the stated native inverse germs, so it satisfies these hypotheses without an additional inverse assumption. □

Lemma A.59 (Smooth canonical inclusions in the constructed extension)**.** *Under the hypotheses of the constructed mixed atlas, the canonical maps $i : M \to X_C$, $i|_{M^\circ} : M^\circ \to X_C$ and $j : Y \to X_C$ are smooth, including the full original inclusion at its boundary seam. At each $p \in Y$, the cylinder map j has a smooth local inverse germ at $j(p)$, agreeing there with its actual topological inverse.*

Proof. Each interior or cylinder chart is carried to X_C by its open embedding. In this chart and its carried chart, the corresponding embedding has identity coordinate expression. Hence both open-piece inclusions are smooth. A cylinder chart χ containing p gives a local inverse at $j(p)$ by composing its carried coordinate map on X_C with χ^{-1}. Both maps are smooth on their chart neighborhoods. Their chart inverse identities give value p at $j(p)$ and $j(g(z)) = z$ for all sufficiently nearby z. Thus this is an actual smooth inverse germ of the cylinder.

At a native interior point, lifting a sufficiently small neighborhood in M to M° is smooth by the restricted native interior charts. Locally i is the composite of this lift with the smooth interior-piece inclusion, so i is smooth there. At a boundary point $b = C(0, b)$, use the native inverse germ of the same collar at $(0, b)$. Because the collar is a homeomorphism onto an open subset of M, its inverse germ g agrees with its actual inverse at all sufficiently nearby native points. Continuity keeps g in the open cylinder band near $(0, b)$, and the cylinder atlas makes this band-valued map smooth. The quotient identity then gives $i(z) = j(g(z))$ near b. This composite is smooth at b. Every native point is interior or boundary, proving smoothness of the entire original inclusion. □

Lemma A.60 (The differential of a native collar inverse)**.** *Let M have its smooth intrinsic $(k+1)$-dimensional half-space atlas, let $B = \partial M$, and let $C : [0, \delta) \times B \to M$ be smooth, $\delta > 0$. At p in this strip, let $g : M \to \mathbb{R} \times B$ be smooth at $C(p)$ and satisfy $g(C(q)) = q$ near p within the strip. Then $dg_{C(p)}$ is a linear isomorphism, including when p is at time zero.*

Proof. The strip has a unique full differential at each of its points. At positive time this follows from openness. At time zero choose boundary-variable coordinates. The allowed directions (a, v) with $a \geq 0$ span $\mathbb{R} \times \mathbb{R}^k$: a linear map vanishing on those directions also vanishes on their negatives, thus on every direction. Directional difference quotients within the half-strip consequently determine a unique linear differential. The identity $g \circ C = \mathrm{id}$ near p within that strip may therefore be differentiated, giving

$$dg_{C(p)} \circ d_{[0,\delta)\times B} C_p = \mathrm{id}_{\mathbb{R}\times T_b B} .$$

Thus $dg_{C(p)}$ is surjective. Its domain and codomain both have finite dimension $k+1$, so surjectivity implies injectivity as well. This proves that it is an isomorphism without a separate rank hypothesis on the collar. □

Lemma A.61 (The actual differentials of the canonical inclusions)**.** *Under the mixed-atlas hypotheses, the differentials of the native interior inclusion, the cylinder inclusion and the full original inclusion $i : M \to X_C$ are linear isomorphisms everywhere. This includes every native boundary point of the original inclusion.*

Proof. In each native interior chart and its carried chart on X_C, the interior-piece inclusion has identity expression. Its differential is consequently injective. For the cylinder inclusion, its actual smooth inverse germ g satisfies $g \circ j = \mathrm{id}$ near the cylinder point by injectivity of j. The chain rule gives $dg \circ dj = \mathrm{id}$, so dj is injective as well. The equal finite tangent dimensions make both differentials bijective.

At a native interior point, the original inclusion composed with the native interior inclusion is the interior-piece inclusion. The latter two maps have bijective differentials, so the chain rule makes the original inclusion differential injective there. At a collar point $C(p)$, use the native inverse germ to write $i = j \circ h$ near $C(p)$, where h is smooth into the band. Let f be h followed by the smooth band inclusion into $\mathbb{R} \times B$. The quotient collar identity and injectivity of j give $f \circ C = \mathrm{id}$ near p within the strip. The preceding lemma makes df bijective, hence dh injective by the chain rule. As dj is injective, $di = dj \circ dh$ is injective. Every boundary point is $C(0, b)$, so this covers the entire boundary. Finally the original and extension tangent dimensions both equal $k + 1$; the proved injectivity therefore makes di bijective at every native point. □

Lemma A.62 (The smooth interior and cylinder primitive pieces)**.** *Let M be smooth, Hausdorff and second countable, with its intrinsic $(k+1)$-dimensional half-space atlas. Let λ be a locally extendible native one-form. On its native interior and on the actual cylinder $Y = (-\delta, \delta) \times \partial M$, define*

$$\eta_I = \iota_I^* \lambda, \qquad \eta_Y = e^{-T} \pi_B^* (\iota_B^* \lambda),$$

where ι_I and ι_B are the actual native inclusions, $T(t, b) = t$, and $\pi_B(t, b) = b$. Both pieces are smooth in their constructed Euclidean atlases. The cylinder piece is defined on both sides of its seam, for any real δ.

Proof. The native interior inclusion is smooth, so the smooth native pullback theorem gives smoothness of η_I. The actual boundary restriction $\iota_B^* \lambda$ is smooth. The cylinder inclusion into the product is smooth, and composition with the second projection makes π_B smooth. Its pullback of this boundary form is consequently smooth. The actual time coordinate T is smooth, so e^{-T} is smooth. Scalar multiplication of the smooth pullback by this smooth function gives the asserted smooth one-form η_Y on the whole band. These arguments do not require symplectic or contact nondegeneracy. □

Lemma A.63 (Exact primitive agreement on the positive collar overlap)**.** *Let λ be locally extendible, and let Z satisfy its actual Liouville equation $d\lambda(Z, \cdot) = \lambda$. Let $\delta > 0$, and suppose $C : [0, \delta) \times \partial M \to M$ is smooth, has negative-Z time curves, and satisfies $C(0, b) = b$. On $Y_+ = (0, \delta) \times \partial M$, the cylinder primitive from the preceding lemma satisfies*

$$C^* \lambda = \eta_Y.$$

This is equality on every tangent vector in the actual Euclidean cylinder model, including its arbitrary time component.

Proof. Fix $p = (t, b) \in Y_+$ and a tangent $w \in T_p Y$. The smooth band inclusion sends it to a product tangent (a, v). Choose a boundary chart χ near b and reparameterize the same collar by $(s, u) \mapsto C(s, \chi^{-1}(u))$. The chart inverse differential sends $(a, d\chi_b v)$ to (a, v), so the actual collar differential is preserved by this reparameterization. Its time curves still solve $-Z$. The normed-boundary-variable primitive law therefore gives

$$\lambda_{C(t,b)}(dC_{(t,b)}(a, v)) = e^{-t} \lambda_{C(0,b)}(dC_{(0,b)}(0, v)).$$

The seam differential in this identity is the one-sided strip differential. Differentiate $C(0,b) = \iota_B(b)$ in its spatial variable to identify $dC_{(0,b)}(0,v) = d\iota_{Bb}(v)$. At positive time the half-strip is a neighborhood, so its actual within differential equals the ordinary differential used here. The chain rule through the smooth band inclusion then identifies the left side with $(C^*\lambda)_p(w)$. The chain rule for the boundary projection identifies v with $d\pi_{Bp}(w)$, so the right side is exactly $(\eta_Y)_p(w)$. As w was arbitrary, this proves the asserted full one-form equality. □

Lemma A.64 (Regular defining functions on the actual cylinder patch)**.** *Assume the mixed-atlas hypotheses with M Hausdorff and second countable, and write $A = i(M)$ and $U = j(Y)$ in the resulting boundaryless smooth manifold X_C. For each cylinder point $p \in Y$, extend the actual inverse $j^{-1} : U \to Y$ arbitrarily outside U and put $\rho_p = -T \circ j^{-1}$ on U. The resulting scalar function is smooth throughout the genuine open set U, has nonzero differential at every point of U, and satisfies*

$$A \cap U = \{x \in U : \rho_p(x) \le 0\}, \qquad \mathrm{Fr}(A) \cap U = \{x \in U : \rho_p(x) = 0\}.$$

Every actual frontier point of A therefore has a regular local sublevel defining function constructed from the same collar. No nonemptiness assumption on the boundary is imposed.

Proof. The cylinder map is a smooth open embedding and has an actual smooth inverse germ at each point of its image. Injectivity makes every such germ agree with j^{-1} wherever both are defined. Thus this inverse is smooth at each point of the open set U, hence smooth on U. Composition with the smooth time coordinate and negation proves smoothness of ρ_p throughout this entire open neighborhood. For $q \in Y$ one has $\rho_p(j(q)) = -T(q)$. Differentiating this identity gives

$$d\rho_{pj(q)} \circ dj_q = -dT_q.$$

The right side is nonzero by time-coordinate regularity. Therefore $d\rho_{pj(q)}$ cannot be zero. This proves regularity at every point of U without an additional rank assumption.

The quotient's normal representatives say that $j(q)$ belongs to A exactly when $T(q) \ge 0$. Its frontier in cylinder coordinates is exactly $T(q) = 0$ by the carrier lemma. Substituting $\rho_p(j(q)) = -T(q)$ proves the two displayed equalities on U. Every original frontier point equals $i(b) = j(0,b)$ for a native boundary point b. Taking $p = (0,b)$ provides the stated defining function on the common open cylinder patch containing that point. If the boundary is empty there are no frontier points, so no cylinder point needs to be chosen. □

Lemma A.65 (A compact domain from the same native collar)**.** *Under the mixed-atlas hypotheses, suppose M is compact. The same finite extension X_C gives a compact domain presentation of dimension $k+1$, with carrier $A = i(M)$, actual boundary manifold $B = \partial M$ and boundary inclusion $b \mapsto i(b)$. This inclusion is a smooth topological embedding with injective differential and has image exactly $\mathrm{Fr}(A)$. Every frontier point has the regular local sublevel defining function constructed from the same cylinder time coordinate. The original inclusion preserves the topology and identifies the actual tangent spaces, including at its boundary seam.*

Proof. The constructed mixed atlas and quotient topology make X_C a boundaryless Hausdorff second-countable smooth manifold. The original inclusion is continuous, so its carrier A is compact. The defining-function lemma supplies at every frontier point a smooth scalar function on an open neighborhood, whose nonpositive sublevel is exactly the carrier there, whose zero level is exactly its frontier there, and whose differential is nonzero.

The zero-time map $s : B \to Y$, $s(b) = (0,b)$, is smooth in the cylinder's product atlas. Its boundary projection is a continuous left inverse, so it is a topological embedding. Differentiating

$\pi_B \circ s = \mathrm{id}_B$ gives $d\pi_B \circ ds = \mathrm{id}$, making ds injective. The cylinder inclusion j is a smooth topological embedding with injective differential. Therefore $j \circ s$ is a smooth topological embedding with injective differential. The quotient seam identity gives $j(s(b)) = i(b)$, and the carrier lemma identifies its image with $\mathrm{Fr}(A)$. These are exactly the carrier, regularity and embedded-boundary requirements for the stated compact domain presentation. The original inclusion is a topological embedding and has bijective differential by the inclusion lemmas, giving the final preservation assertion. The dimension here is $k+1 \geq 1$; the finite zero-dimensional construction is separate. □

Lemma A.66 (A single primitive with the exact two patch restrictions)**.** *Under the mixed-atlas hypotheses, let λ be locally extendible and let Z satisfy $d\lambda(Z, \cdot) = \lambda$. Suppose the same collar C has negative-Z time curves. The actual interior and cylinder primitives prescribe a single one-form η on X_C with the exact whole-patch identities*

$$i_I^* \eta = \eta_I, \qquad j^* \eta = \eta_Y,$$

where $i_I = i|_{M^\circ}$. At $z = i_I(x)$ its value on a tangent v is $\eta_{Ix}((di_{Ix})^{-1}v)$; at $z = j(p)$ it is $\eta_{Yp}((dj_p)^{-1}v)$.

Proof. Both patch differentials are linear isomorphisms, so each displayed formula defines an actual covector on the extension tangent space. Each patch map is injective, making its representative unique within that piece. Their images cover X_C. If $i_I(x) = j(p)$, the overlap criterion gives positive time. On a neighborhood of p the actual transition $g : Y \to M^\circ$ is induced by this same collar and satisfies $g(p) = x$ and $i_I \circ g = j$. The full-tangent primitive agreement says

$$\eta_{Ix} \circ dg_p = \eta_{Yp}, \qquad di_{Ix} \circ dg_p = dj_p.$$

For any extension tangent v, put $w = (dj_p)^{-1}v$. The second equality gives $(di_{Ix})^{-1}v = dg_p(w)$, and the first consequently gives

$$\eta_{Ix}((di_{Ix})^{-1}v) = \eta_{Yp}((dj_p)^{-1}v).$$

Thus the two covectors agree on every overlap tangent. Define η using the interior representative wherever it exists and the cylinder representative otherwise. The preceding equality proves that it has both prescribed values throughout both pieces. Pulling those values back through di_I or dj cancels the corresponding inverse linear map, proving both exact whole-patch form identities. □

Lemma A.67 (Smoothness of the actual globally assembled primitive)**.** *Under the preceding hypotheses, the actual one-form η on the constructed finite boundaryless extension X_C is smooth.*

Proof. Fix an extension point z. One of the two open patches contains it. Write that patch inclusion as k and its smooth primitive as β. Its carried charts give a smooth inverse germ g at z with $k \circ g = \mathrm{id}$ near z. On a smaller neighborhood g is continuously differentiable, so differentiating this identity there gives $dk \circ dg = \mathrm{id}$. The exact patch identity $k^* \eta = \beta$ consequently implies

$$\eta_{z'}(v) = \beta_{g(z')}(dg_{z'} v)$$

for every z' sufficiently near z and every tangent v. In charts the coefficients therefore agree as a germ with the coefficients of this actual pullback. The coefficient functions of β are smooth, g is smooth at z, and the derivative of a smooth coordinate map is smooth at that point. Composition and the continuous bilinear pairing of covectors with linear maps make these pullback coefficients smooth at z. Thus η has smooth coefficients at every point of every extension chart, proving smoothness on all of X_C. □

Lemma A.68 (The global primitive preserves the entire original manifold)**.** *Under the hypotheses of the actual global primitive construction, the entire original inclusion $i : M \to X_C$ satisfies*

$$i^*\eta = \lambda.$$

This identity uses the full actual native tangent space at every point, including the original boundary seam.

Proof. At an interior point the exact interior patch restriction gives $(i \circ \iota_I)^*\eta = \iota_I^*\lambda$. The differential of ι_I is bijective, so this equality tests every native tangent and gives $i^*\eta = \lambda$ there.

Fix a strip point $p = (t, b)$, allowing $t = 0$, and put $x = C(p)$. The native inverse germ gives a smooth band-valued map h near x with $h(x) = p$ and $i = j \circ h$ near x. Follow h by the band inclusion to obtain f into $\mathbb{R} \times B$. The quotient collar identity and injectivity of j give $f \circ C = \mathrm{id}$ near p within the half-strip. Writing $d_S C$ for the unique within differential on that strip, differentiation gives

$$df_x \circ d_S C_p = \mathrm{id}\,.$$

The native collar inverse differential lemma makes df_x bijective. Thus $d_S C_p = (df_x)^{-1}$ is surjective. For any native tangent $v \in T_x M$, choose $w = (a, u)$ with $d_S C_p(w) = v$. The chain rule for $i = j \circ h$, followed by the exact whole-cylinder primitive restriction, gives

$$\eta_{i(x)}(di_x v) = \eta_{Y\,p}(dh_x v).$$

The identity $df_x v = w$ says that the actual product tangent of $dh_x v$ has spatial component u. The definition of η_Y therefore makes the right side equal to $e^{-t}(\iota_B^*\lambda)_b(u)$.

Choose a boundary chart around b and reparameterize this same collar by its inverse chart. The full normed-variable primitive law, including time zero and arbitrary time component a, gives

$$\lambda_{C(t,b)}(d_S C_{(t,b)}(a, u)) = e^{-t}(\iota_B^*\lambda)_b(u).$$

Here the spatial derivative of $C(0, b) = \iota_B(b)$ identifies its seam covector with the actual boundary pullback. Substituting $d_S C_p(w) = v$ proves $\eta_{i(x)}(di_x v) = \lambda_x(v)$. Every boundary point is $C(0, b)$, so this covers the whole boundary; together with the interior calculation it proves the stated form identity on all of M. □

Lemma A.69 (A smooth primitive extension from the original compact source)**.** *Let $n \geq 1$, and let M be a compact smooth Hausdorff second-countable $2n$-dimensional manifold with its intrinsic half-space atlas. Let λ be locally extendible, suppose its actual exterior derivative is nondegenerate, and impose the local regular-defining-function outwardness hypotheses of A.42 on its contraction field. These original data produce a compact domain presentation D with carrier A in a boundaryless smooth $2n$-dimensional manifold X, a smooth closed embedding $i : M \to X$ and a globally smooth one-form η on X such that*

$$i(M) = A, \qquad \mathrm{Fr}(A) = i(\partial M), \qquad i^*\eta = \lambda.$$

The actual differential di_x is bijective at every native point, including its boundary. No collar, atlas, matching primitive or extension-domain presentation is additional input. This is the positive even-dimensional source specialization of the primitive extension construction.

Proof. The source-derived topological extension theorem constructs one collar C and one positive width δ from the given primitive, its nondegenerate exterior derivative and the stated outwardness.

Retain its open embedding, one-sided smoothness, exact boundary criterion, native inverse germs, seam identity and negative-time Liouville ODE. The same collar therefore meets all the mixed-atlas and compact-domain hypotheses. Apply the compact domain presentation lemma without changing C or δ, obtaining the boundaryless extension X_C and carrier $A = i(M)$. Its frontier is exactly the native boundary image. The canonical original inclusion is smooth, is a closed embedding by the source-derived quotient topology, and has bijective differential by the actual inclusion differential lemma.

The contraction field of the original primitive satisfies $d\lambda(Z, \cdot) = \lambda$ by nondegeneracy and its definition. Its same-collar ODE now supplies the actual global primitive construction and smoothness theorem. Let η be that constructed form on X_C. The full original restriction theorem gives $i^*\eta = \lambda$, including at the seam. Thus $X = X_C$, this actual compact domain and the same canonical inclusion satisfy all the asserted conclusions from the original source data alone. The dimension is positive here; the independent finite-carrier construction treats dimension zero. □

Lemma A.70 (Exterior differentiation of a boundaryless pullback to the native domain)**.** *Let M be a smooth positive-dimensional manifold with its intrinsic half-space atlas, let X be a smooth boundaryless manifold, let $f : M \to X$ be smooth, and let η be a smooth r-form on X. The actual native pullback, defined using df on the full native tangent spaces, satisfies*

$$d(f^*\eta) = f^*(d\eta).$$

Both exterior derivatives are the actual derivatives on their respective manifolds. The identity includes every native boundary point and requires no supplied exterior-derivative witness for the pullback.

Proof. Fix a native chart E around $x \in M$ and a boundaryless chart H around $f(x)$. By continuity shrink the native chart neighborhood so its image under f lies in the target chart's source. Put $F = H \circ f \circ E^{-1}$ on this relative half-space neighborhood, and write the smooth coordinate form as $\eta^H = \sum_I a_I \, dy_{i_1} \wedge \cdots \wedge dy_{i_r}$. The actual chart chain rule and inverse differential identities give

$$(f^*\eta)^E = \sum_I (a_I \circ F) \, dF_{i_1} \wedge \cdots \wedge dF_{i_r}.$$

All derivatives here are within the native chart target. That target is relatively open in the half-space and has unique full-dimensional within derivatives. Its interior is dense in it. The smooth coordinate functions of F have continuous second within derivatives; their mixed second derivatives commute on the interior by ordinary smooth calculus, and hence at the boundary by continuity. Consequently $d(dF_i) = 0$ throughout the target. The product rule for the displayed pullback therefore leaves only the terms

$$\sum_I d(a_I \circ F) \wedge dF_{i_1} \wedge \cdots \wedge dF_{i_r}.$$

The within chain rule identifies $d(a_I \circ F) = F^*(da_I)$. These remaining terms are exactly $F^*(d\eta^H)$. The coordinate characterization of the native derivative and recovery of intrinsic forms from their coefficients now give the asserted full-tangent identity at x. The argument applies to every native point. □

Lemma A.71 (A primitive extension with its actual symplectic derivative on the carrier)**.** *Under the positive even-dimensional source hypotheses of A.69, there are a compact domain presentation*

D with carrier A in a boundaryless smooth manifold X, a smooth closed embedding $i : M \to X$ and a globally smooth one-form η with all the conclusions of that construction, and additionally

$$i^*(d\eta) = d\lambda, \qquad d\eta \text{ is nondegenerate at every point of } A.$$

In particular $d\eta$ is the actual symplectic derivative on the carrier, rather than a separately prescribed two-form. The first identity holds on the full native tangent spaces, including the boundary.

Proof. Apply the smooth primitive extension construction to the original λ, its nondegenerate native derivative and its outwardness. It gives the actual D, X, i, η with $A = i(M)$, bijective di_x for every x, and $i^*\eta = \lambda$ on every native tangent. The native pullback naturality lemma gives

$$i^*(d\eta) = d(i^*\eta) = d\lambda.$$

Fix $z \in A$ and write $z = i(x)$. Suppose $v \in T_zX$ satisfies $d\eta_z(v, w) = 0$ for every $w \in T_zX$. Bijectivity of di_x gives $u \in T_xM$ with $di_x u = v$. For every $a \in T_xM$, the pullback identity gives

$$d\lambda_x(u, a) = d\eta_z(di_x u, di_x a) = 0.$$

Native nondegeneracy forces $u = 0$, hence $v = 0$. This proves nondegeneracy at every carrier point. Finally $d\eta$ is closed: in a boundaryless chart its second exterior derivative consists of mixed second derivatives of the smooth coefficients, and those terms cancel in pairs by symmetry and alternation. Thus the actual $d\eta$ is symplectic at every point of the carrier. □

Lemma A.72 (Restricting a compact domain to its symplectic locus)**.** *Let D be a compact domain presentation with carrier A in a finite-dimensional boundaryless smooth manifold X, and let η be a globally smooth one-form on X. Suppose its actual derivative $d\eta$ is nondegenerate at every point of A. The set*

$$U = \{z \in X : d\eta_z \text{ is nondegenerate}\}$$

is an open neighborhood of A. Restricting D to U gives a compact domain presentation with the same carrier and boundary under the open inclusion $q : U \hookrightarrow X$. The restricted primitive $\eta_U = q^*\eta$ is smooth, its actual derivative is symplectic on all of U, and there is a smooth field Z_U on U satisfying

$$\iota_{Z_U} d\eta_U = \eta_U.$$

This field is the unique contraction field at every point of U. No outwardness condition or Liouville-domain structure is an input.

Proof. In a tangent chart, contraction with $d\eta$ has a square matrix with smooth entries. Nondegeneracy is equivalent to its determinant being nonzero, an open condition. Hence U is open; the assumed carrier nondegeneracy gives $A \subset U$. The open inclusion identifies A with $q^{-1}(A)$ and its boundary with $q^{-1}(\mathrm{Fr}_X A)$. This carrier remains compact because the inverse of q on A is continuous. Restrict the original regular defining functions and boundary inclusion to U. Their smoothness, nonzero differentials and local sublevel identities are preserved by an open restriction, so they give the stated compact domain presentation.

Pullback along q preserves smoothness, and exterior naturality gives $d\eta_U = q^*(d\eta)$. The differential of q is a tangent isomorphism, so $d\eta_U$ is nondegenerate everywhere on U. It is closed by cancellation of mixed second derivatives of the smooth coefficients of η_U; thus it is symplectic throughout U. At each $z \in U$, the map $L_z : v \mapsto d\eta_{U\,z}(v, \cdot)$ is injective by nondegeneracy and hence bijective by equality of its finite domain and codomain dimensions. Define $Z_U(z) = L_z^{-1}(\eta_{U\,z})$.

In a tangent chart the inverse matrix is its adjugate divided by its nonzero determinant, so these coefficients are smooth. This proves smoothness of Z_U and its contraction equation. If another field satisfies the equation, the difference is annihilated by L_z at every z, hence is zero. This proves uniqueness. □

Lemma A.73 (A source-derived Liouville extension preserving the native field)**.** *Under the positive even-dimensional source hypotheses of A.69, the compact domain and its primitive can be extended to a boundaryless symplectic neighborhood with a smooth Liouville field, so that the resulting compact domain is a Liouville-domain presentation. Its original inclusion i preserves the whole primitive and its actual exterior derivative, has bijective differential on every native tangent space, and transports the original contraction field Z_M to the extension field Z_X:*

$$i^*\eta = \lambda, \qquad i^*(d\eta) = d\lambda, \qquad di_x(Z_M(x)) = Z_X(i(x)).$$

The compact carrier and frontier remain respectively $i(M)$ and $i(\partial M)$, and Z_X is strictly outward along that frontier. The same original inclusion is the ambient composite of a native diffeomorphism from M to this final compact domain's intrinsic manifold with boundary. Thus the atlas identification and the preserved primitive, derivative and field belong to the same domain and inclusion. No extension field or extension outwardness condition is supplied as additional input.

Proof. Construct the symplectic primitive extension on the carrier and restrict its compact domain to the open nondegeneracy locus. The restriction lemma gives a smooth primitive η, a symplectic actual derivative throughout the restricted extension, and its smooth canonical contraction field Z_X there. Lift the original inclusion to this open neighborhood. Its composite with the open inclusion is the original embedding, so its differential is still bijective. The carrier, frontier and primitive identities are preserved by this restriction, and exterior naturality preserves $i^*(d\eta) = d\lambda$ as well. Retain the actual same-collar compact domain from the source construction. Its original inclusion gives the native diffeomorphism of A.74; composing it with that lemma's native open-restriction identification gives a diffeomorphism onto this final compact domain. By the identities of these maps its ambient composite is the lifted original inclusion i.

For $x \in M$ and any $a \in T_xM$, the two primitive identities and the original contraction equation give

$$d\eta_{i(x)}(di_xZ_M(x), di_xa) = d\lambda_x(Z_M(x), a) = \lambda_x(a) = \eta_{i(x)}(di_xa).$$

Surjectivity of di_x makes this the extension contraction equation on every tangent at $i(x)$. Its unique solution is $Z_X(i(x))$, proving the field identity.

Fix a frontier point $i(x)$ and a regular local defining function ρ for the constructed carrier, with sublevel convention $\rho \le 0$. Its pullback $\rho \circ i$ vanishes at x, is nonpositive on a native neighborhood, and has nonzero differential: the chain rule and bijectivity of di_x preserve the nonzero differential of ρ. In a native boundary chart E, put $a = E(x)$ and let L be the actual within-half-space differential of $\rho \circ i \circ E^{-1}$ at a. For every w with $w_0 \ge 0$, the points $a + tw$ lie in the chart target for small $t > 0$. The nonpositive difference quotients give $L(w) \le 0$. When $w_0 = 0$, applying this to w and $-w$ gives $L(w) = 0$. Consequently $L(w) = cw_0$ with $c = L(e_0) \le 0$. If $c = 0$, the chart chain rule would give $d(\rho \circ i)_x = 0$, contradicting regularity. Thus $c < 0$, and

$$d(\rho \circ i)_x(v) = c\,(dE_xv)_0.$$

This argument uses only the actual native differential and local nonpositivity, without a separate scalar extension of $\rho \circ i$. Apply the native defining-function sign lemma to a source defining function

with positive derivative on $Z_M(x)$, supplied by the original outwardness hypothesis. It follows that $(dE_x Z_M(x))_0 < 0$. Hence

$$d\rho_{i(x)}(Z_X(i(x))) = d(\rho \circ i)_x(Z_M(x)) = c\,(dE_x Z_M(x))_0 > 0.$$

Thus the smooth canonical extension field is outward at every frontier point. The restricted compact domain, its smooth primitive and smooth field now satisfy all the Liouville-presentation axioms. □

Lemma A.74 (The native manifold and its constructed compact-domain atlas)**.** *Under the mixed-atlas and compactness hypotheses of A.65, let D be the compact domain constructed from that same collar, with carrier $A = i(M)$. Equip A with the intrinsic manifold-with-boundary atlas of D. The original inclusion, regarded as a map*

$$e : M \longrightarrow A, \qquad e(x) = i(x),$$

is a diffeomorphism of the two intrinsic manifolds with boundary. Its composite with $A \hookrightarrow X_C$ is literally the original inclusion. This identifies the original native atlas with the compact domain's actual intrinsic atlas, not merely their underlying spaces. For any open neighborhood $U \supset A$ in X_C, the canonical identification of A with the carrier of the open restriction $D|_U$ is also a native diffeomorphism. Its composite with e therefore gives a native diffeomorphism from M to $(D|_U)$'s intrinsic manifold, with the same original ambient inclusion after forgetting the restriction.

Proof. The original inclusion is a closed embedding with image A. It therefore gives a homeomorphism $e : M \to A$. To prove smoothness of its forward map, use interior target charts and the half-space charts supplied by regular defining functions at frontier points. Their coordinate maps are restrictions of smooth ambient coordinate maps on X_C. Since i is smooth and its image lies in A, the native coordinate expression of e is their smooth composition with i, with values in the appropriate half-space. Thus e is smooth in the target's actual intrinsic atlas.

At a point $e(x)$ with $x \in M^\circ$, the open interior patch has a smooth local inverse into M°. On the compact carrier near $e(x)$, compose its smooth inclusion into X_C with that inverse and then with $M^\circ \hookrightarrow M$. The resulting smooth map is e^{-1} there, by injectivity of the original inclusion. Thus the inverse is smooth at every such point.

If $x \in \partial M$, its image lies in the open cylinder patch $j(Y)$. The cylinder inverse is smooth on that patch. Its restriction to $A \cap j(Y)$ has nonnegative cylinder time: the same-collar quotient criterion identifies precisely those cylinder points with original points. Write its coordinates as (t, b), with $0 \le t < \delta$. The inverse is therefore

$$e^{-1}(z) = C(t, b), \qquad (t, b) = j^{-1}(z).$$

Indeed the quotient identity gives $i(C(t, b)) = j(t, b) = z$, and injectivity identifies $C(t, b)$ with the original representative. The restricted cylinder inverse is smooth as a map into the nonnegative half-strip: its ambient cylinder coordinates are smooth, and its values lie in that half-strip, whose native atlas is the restriction of these same product coordinates. The given one-sided collar C is smooth there, including at $t = 0$. Their composite is therefore native smooth at each original boundary point. Interior and boundary points exhaust M, proving smoothness of e^{-1} everywhere. Hence e is the claimed intrinsic diffeomorphism, and its stated ambient composite follows directly from its definition.

Now let U be an open neighborhood of A. On underlying carriers the map sends $z \in A$ to the same point regarded in U, and its inverse forgets this open restriction. Both maps are continuous because the open inclusion identifies their subspace topologies. Near a carrier point, restrict the

ambient chart and its defining function to U; this gives identical coordinate expressions for the original and restricted native charts on their common neighborhood. The two carrier maps are therefore represented there by the identity in the same Euclidean or half-space coordinates. They are consequently smooth in both directions, giving the claimed native restriction diffeomorphism. Composing it with e yields the final diffeomorphism, and forgetting the restriction preserves its original ambient composite pointwise. □

Lemma A.75 (One native symplectic compact presentation extending every scalar)**.** *Under the source hypotheses of Lemma A.29, one can choose its X, D, i, η, e so that also*

$$i^*(d\eta) = d\lambda, \qquad d\eta \text{ is symplectic at every point of } D.$$

All primitive and derivative identities use the full native tangent spaces, including at the boundary. For every scalar $f : M \to \mathbb{R}$ with actual smooth extension germs in every native atlas chart, there is a smooth $g : X \to \mathbb{R}$ with $g \circ i = f$. The same native diffeomorphism $e : M \to D$ has ambient composite i, and these choices are made before f.

Proof. Take the single compact presentation, primitive, inclusion and native diffeomorphism of the primitive-and-scalar extension lemma, retaining its universal scalar extension property. Exterior naturality on the full native tangent spaces gives $i^*(d\eta) = d(i^*\eta) = d\lambda$. Fix $z \in D$ and write $z = i(x)$. If $v \in T_zX$ satisfies $d\eta_z(v, w) = 0$ for every $w \in T_zX$, bijectivity of di_x gives $u \in T_xM$ with $di_x u = v$. For every $a \in T_xM$,

$$d\lambda_x(u, a) = d\eta_z(di_x u, di_x a) = 0.$$

The assumed nondegeneracy of $d\lambda_x$ gives $u = 0$, hence $v = 0$. Thus $d\eta$ is nondegenerate on D. It is closed because the second exterior derivative of a smooth form on a boundaryless manifold is zero: in each coordinate chart, mixed second derivatives cancel in pairs under alternation. Hence $d\eta$ is symplectic on D. These conclusions add no new choices to X, D, i, η, e, so their universal scalar extension property remains. □

Lemma A.76 (One native Liouville presentation extending every scalar)**.** *Under the source hypotheses of Lemma A.29, there are a boundaryless smooth manifold U, a compact Liouville domain $A \subset U$ with smooth primitive η_U, a smooth closed embedding $j : M \to U$ with bijective differential, and a native diffeomorphism $e_U : M \to A$ whose ambient composite is j, such that*

$$j(M) = A, \qquad \mathrm{Fr}_U(A) = j(\partial M), \qquad j^*\eta_U = \lambda, \qquad j^*(d\eta_U) = d\lambda, \qquad dj_x(Z_M(x)) = Z_U(j(x)).$$

Here Z_M is the contraction field of λ, and Z_U is the smooth contraction field of η_U; $d\eta_U$ is symplectic throughout U and Z_U is strictly outward along the frontier of A. For every $f : M \to \mathbb{R}$ with actual smooth extension germs in every native atlas chart there is a smooth $g_U : U \to \mathbb{R}$ with $g_U \circ j = f$. The objects U, A, j, η_U, e_U are chosen before f. If additionally $f(x) > 0$ at every source point, one can choose $g_U(y) > 0$ at every point of U. This conclusion concerns positive-dimensional compact domains and scalar data; general maps, tensor data and families require their own correspondence results.

Proof. Choose X, D, i, η, e from the native symplectic-and-scalar presentation lemma, keeping its universal scalar extension property. Let U be the open nondegeneracy locus of $d\eta$, containing D, and write $q : U \hookrightarrow X$ for its open inclusion. The symplectic restriction lemma gives the compact carrier $A = q^{-1}(D)$, primitive $\eta_U = q^*\eta$, symplectic actual derivative and smooth unique contraction field Z_U throughout U. Lift i to $j : M \to U$, so $q \circ j = i$. The open inclusion has bijective differential, and

hence dj_x is bijective because di_x is. The lift is smooth; it is a closed embedding since i is a closed embedding and its image is contained in U. Its image is A, whose frontier under q is the original frontier, giving $\mathrm{Fr}_U(A) = j(\partial M)$. Compose e with the native open-restriction diffeomorphism to obtain $e_U : M \to A$. Its ambient composite is j. The two pullback identities follow from $q \circ j = i$ and the actual exterior derivative identity for the restricted primitive.

For every $x \in M$ and $a \in T_xM$, these identities give

$$d\eta_{U\,j(x)}(dj_x Z_M(x), dj_x a) = d\lambda_x(Z_M(x), a) = \lambda_x(a) = \eta_{U\,j(x)}(dj_x a).$$

Surjectivity of dj_x and uniqueness of the contraction field give $dj_x Z_M(x) = Z_U(j(x))$. To verify outwardness, fix $x \in \partial M$ and a regular local defining function ρ for A at $j(x)$, with sublevel convention $\rho \leq 0$. Its native pullback $\rho \circ j$ is zero at x, locally nonpositive and has nonzero differential, by bijectivity of dj_x. In a native boundary chart E, its within-half-space differential L at $E(x)$ is nonpositive on every vector w with $w_0 \geq 0$: the values on $E(x) + tw$ for small $t > 0$ are nonpositive. On $w_0 = 0$, apply this to w and $-w$ to get $L(w) = 0$. Thus $L(w) = cw_0$ with $c \leq 0$. Regularity forces $c \neq 0$, so $c < 0$. The defining-function sign lemma applied to the source outwardness hypothesis gives $(dE_x Z_M(x))_0 < 0$. Consequently

$$d\rho_{j(x)}(Z_U(j(x))) = d(\rho \circ j)_x(Z_M(x)) = c(dE_x Z_M(x))_0 > 0.$$

This proves the Liouville-domain outwardness for the actual carrier.

Fix these objects, and now let f satisfy the stated germ hypothesis. The retained scalar extension property supplies a smooth $g : X \to \mathbb{R}$ with $g \circ i = f$. Set $g_U = g \circ q$. The open inclusion is smooth, so g_U is smooth, and $g_U \circ j = g \circ q \circ j = g \circ i = f$. Thus every scalar extends on this one Liouville presentation.

Suppose moreover that $f > 0$ on M. Fix a native chart E and a source point x in its domain. A smooth local coordinate extension β of f satisfies $\beta(E(x)) = f(x) > 0$. By continuity shrink its open neighborhood so that $\beta > 0$ there. Then $\log \beta$ is smooth there and agrees, on the native chart target near $E(x)$, with the coordinate expression of $\log f$. Thus $\log f$ has actual smooth extension germs in every native chart. Apply the scalar extension property on the already fixed U to obtain a smooth $h : U \to \mathbb{R}$ with $h \circ j = \log f$. The function $\exp h$ is smooth and strictly positive on all of U, and $(\exp h)(j(x)) = \exp(\log f(x)) = f(x)$ at every source point. It is the required positive extension on this same presentation. □

Lemma A.77 (Shrinking an old map domain into a target chart on the overlap)**.** *Let X be a locally compact regular space and Y a topological space. Let $U, V \subset X$ be open, let $L \subset V$ be compact, and let $K \subset U$ be any subset. Suppose $f : V \to Y$ is continuous and $T \subset Y$ is open, with $f(V \cap K) \subset T$. There are open $W, O \subset X$ such that*

$$L \subset W, \qquad \overline{W} \subset V, \qquad \overline{W} \text{ is compact}, \qquad K \subset O \subset U, \qquad f(W \cap O) \subset T.$$

The map is defined only on V; no nonempty-target hypothesis is required.

Proof. Local compactness and regularity give, for every $x \in L$, an open neighborhood W_x whose closure is compact and contained in V. Finitely many W_x cover L. Their union W is open, contains L, and its closure is the finite union of their compact closures, hence is compact and contained in V. If L is empty, take W empty. Restrict f to $\overline{W}$ and set

$$B = \{x \in \overline{W} : f(x) \notin T\}.$$

Continuity of the restriction makes B closed in $\overline{W}$. Since $\overline{W}$ is closed in X, B is closed in X. If $x \in B \cap K$, then $x \in V \cap K$, contradicting the assumed inclusion $f(V \cap K) \subset T$. Thus $K \cap B$ is empty. Put $O = U \setminus B$. This set is open with $K \subset O \subset U$. For $x \in W \cap O$, membership in $\overline{W}$ and exclusion from B imply $f(x) \in T$, as required. □

Lemma A.78 (Extending a manifold-valued map through one convex chart patch)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff and second countable. Let $A \subset X$ and $a : A \to N$. Let $U, V \subset X$ be open, and let $f : V \to N$ and $g : U \to N$ be smooth, with $f = a$ on $A \cap V$ and $g = a$ on $A \cap U$. Let e be a smooth target chart and let C be a convex subset of its coordinate image. Suppose $g(U)$ and $f(V \cap U)$ lie in the source of e, and their e-coordinates lie in C. Let $K \subset U$ be compact. There are an open set P and a smooth map $h : W \to N$ on $W = V \cup P$ such that*

$$K \subset P \subset U, \qquad h|_P = g|_P, \qquad h|_{A \cap W} = a|_{A \cap W}.$$

There is also a compact set $S \subset U$ with $h|_{V \setminus S} = f|_{V \setminus S}$. The target-chart conditions concern the patch alone; the rest of $f(V)$ need not lie in that chart. No inclusion $K \subset A$ is required.

Proof. Compactness of K inside U and smooth cutoffs on finite-dimensional Hausdorff second-countable manifolds give a smooth $\chi : X \to [0, 1]$ with compact closed support $S \subset U$, equal to one on an open neighborhood P of K contained in U. If K is empty, take P empty and $\chi = 0$. On $V \cap U$ define

$$H(x) = e^{-1}\big((1 - \chi(x))e(f(x)) + \chi(x)e(g(x))\big).$$

Both coordinate values lie in C. Convexity and $0 \le \chi(x) \le 1$ put their convex combination in C, hence in target(e). Composition, multiplication and addition of smooth coordinate functions show that H is smooth on $V \cap U$.

Define $r : V \to N$ to be H on $V \cap U$ and f on $V \setminus U$. The open sets $V \cap U$ and $V \setminus S$ cover V. Where they overlap, $\chi = 0$, so $H = f$ there. Moreover $r = f$ on all of $V \setminus S$, including its part in U. These local descriptions prove that r is smooth on V. On $V \cap P$, $\chi = 1$, giving $r = g$. Thus r on V and g on P agree on their overlap and glue to a smooth h on $W = V \cup P$. It equals g on P and f on $V \setminus S$.

For $x \in A \cap V \cap U$, both $f(x)$ and $g(x)$ equal $a(x)$. Their convex combination in coordinates equals $e(a(x))$, so $r(x) = a(x)$. For $x \in A \cap V$ outside U, $r(x) = f(x) = a(x)$. On $A \cap P$, $h = g = a$. These cases cover $A \cap W$ and prove preservation of every carrier value. □

Lemma A.79 (Adjoining a compact chart patch while retaining a covered compact set)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff and second countable, and let $a : A \to N$ for $A \subset X$. Suppose $g : U \to N$ is smooth on an open set U, equals a on $A \cap U$, and takes its values in a smooth target chart e with coordinates in an open convex set $C \subset$ target(e). Let $K \subset A \cap U$ be compact. Suppose $f : V \to N$ is smooth on an open set V, equals a on $A \cap V$, and $L \subset V$ is compact. There are an open $W \supset L \cup K$ and a smooth map $h : W \to N$ with $h = a$ on $A \cap W$.*

Proof. Set $T = \text{source}(e) \cap e^{-1}(C)$. This set is open, since C is open and the chart is continuous on its source. For $x \in K \cap V$, $f(x) = a(x) = g(x)$, so $f(x) \in T$. Apply the overlap-shrinking lemma to f, L, K, V, U, T. It gives open $V' \supset L$ with $V' \subset V$, and open $O \supset K$ with $O \subset U$, such that $f(V' \cap O) \subset T$. The restriction $g|_O$ still has its values in the source of e with coordinates in C. The restrictions of f and g agree with a on their respective carrier portions. The chart-patch extension lemma applies to $f|_{V'}$, $g|_O$ and K. It gives an open $P \supset K$ in O and a smooth map on $W = V' \cup P$ preserving a throughout $A \cap W$. Since $L \subset V'$ and $K \subset P$, this W contains $L \cup K$. □

Lemma A.80 (Finite assembly of manifold-valued chart patches)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff and second countable. Let $A \subset X$ and $a : A \to N$. For $j = 1, \dots, m$, let $U_j \subset X$ be open, let $K_j \subset A \cap U_j$ be compact, and let $g_j : U_j \to N$ be smooth with $g_j = a$ on $A \cap U_j$. Suppose each $g_j(U_j)$ lies in the source of a smooth target chart e_j, with coordinates in an open convex set $C_j \subset \mathrm{target}(e_j)$. There are an open set $W \subset X$ containing $\bigcup_{j=1}^{m} K_j$ and a smooth map $h : W \to N$ with $h = a$ on $A \cap W$.*

Proof. Put $L_i = \bigcup_{j=1}^{i} K_j$. These sets are compact, including $L_0 = \varnothing$. Begin with $V_0 = \varnothing$ and its empty map $h_0 : V_0 \to N$. Suppose an open $V_i \supset L_i$ and a smooth $h_i : V_i \to N$ with $h_i = a$ on $A \cap V_i$ have been constructed. Apply the compact-patch adjunction lemma with old compact set L_i, old map h_i, new compact set K_{i+1} and new patch $(U_{i+1}, g_{i+1}, e_{i+1}, C_{i+1})$. It supplies an open $V_{i+1} \supset L_i \cup K_{i+1} = L_{i+1}$ and a smooth $h_{i+1} : V_{i+1} \to N$ agreeing with a throughout $A \cap V_{i+1}$. This proves the induction step. After m steps, $W = V_m$ and $h = h_m$ have the required properties. When $m = 0$, the initial empty map supplies the conclusion. □

Lemma A.81 (Refining a smooth local map germ into a convex target chart)**.** *Let X, N be smooth boundaryless manifolds, let $A \subset X$, let $a : A \to N$, and let $x \in A$. Suppose x has an open neighborhood U in X and a smooth map $g : U \to N$ agreeing with a on $A \cap U$. There are an open neighborhood U' of x, a smooth map $g' : U' \to N$ agreeing with a on $A \cap U'$, a smooth target chart e, and an open convex set $C \subset \mathrm{target}(e)$, such that $g'(U') \subset \mathrm{source}(e)$ and $e(g'(U')) \subset C$.*

Proof. Choose a target chart e at $a(x)$ and a positive-radius open coordinate ball C centered at $e(a(x))$, contained in $\mathrm{target}(e)$. The set $T = \mathrm{source}(e) \cap e^{-1}(C)$ is open. Restrict g to its inverse image of T in U. This is an open neighborhood U' of x because $g(x) = a(x)$. The restricted map g' is smooth, retains all carrier values, and has coordinates in the open convex set C, as required. □

Lemma A.82 (A finite compact chart cover from smooth local map germs)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff. Let $A \subset X$ be compact and $a : A \to N$ a map. Suppose each $x \in A$ has an open neighborhood U_x in X and a smooth map $g_x : U_x \to N$ agreeing with a on $A \cap U_x$. There are finitely many open sets U_j, smooth maps $g_j : U_j \to N$ agreeing with a on $A \cap U_j$, smooth target charts e_j, open convex coordinate sets $C_j \subset \mathrm{target}(e_j)$, and compact sets $K_j \subset A \cap U_j$, such that*

$$A = \bigcup_{j=1}^{m} K_j, \qquad g_j(U_j) \subset \mathrm{source}(e_j), \qquad e_j(g_j(U_j)) \subset C_j.$$

Proof. Apply the convex-chart refinement lemma to each supplied local germ. It gives a smooth patch at each $x \in A$ agreeing with a on its carrier portion, with all its coordinates in one open convex target-chart subset. The resulting patches cover A. Choose a finite subcover, denoted $U_1, \dots, U_m$.

A compact Hausdorff space is normal. Applied to the finite open cover $(A \cap U_j)_{j=1}^{m}$ of A, normality gives a closed shrinking: closed sets K_j in A with $K_j \subset A \cap U_j$ and $\bigcup_j K_j = A$. For completeness, the shrinking follows by replacing one open member at a time by an open set whose closure stays in that member and still covers the portion omitted by the other members. At each replacement, that omitted portion is closed in A, lies in the member being replaced, and normality places it inside such an open set. Previously replaced members stay fixed, so the cover persists. After finitely many replacements their closures cover A and stay in their respective original members; take those closures for K_j. Each K_j is compact as a closed subset of compact A and is therefore compact in X. They provide the asserted compact cover. If A is empty, the empty family has all the stated properties. □

Lemma A.83 (Extending local manifold-valued germs around a compact carrier)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff and second countable. Let $A \subset X$ be compact and $a : A \to N$ a map. Suppose each $x \in A$ has an open neighborhood U_x in X and a smooth map $g_x : U_x \to N$ satisfying $g_x = a$ on $A \cap U_x$. There are an open neighborhood W of A and a smooth map $h : W \to N$ with $h|_A = a$. Agreement of different g_x away from A, a common target chart for $a(A)$, and nonemptiness of A or N are not required.*

Proof. The finite compact chart-cover lemma supplies patches (U_j, g_j, e_j, C_j) and compact sets $K_j \subset A \cap U_j$ with $\bigcup_j K_j = A$. Apply the finite chart-patch assembly lemma to those patches. Its open set W contains their compact union A, and its smooth map agrees with a on $A \cap W = A$. This proves the conclusion, including the empty-carrier case. □

Lemma A.84 (Adjoining a compact map patch while preserving an exterior region)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff and second countable, and let $a : A \to N$ for $A \subset X$. Let $g : U \to N$ be a smooth open patch agreeing with a on $A \cap U$, whose image lies in one smooth target chart with coordinates in an open convex subset of that chart's target. Suppose $K \subset A \cap U$ is compact and $U \cap B = \varnothing$ for a set $B \subset X$. Let $f : V \to N$ be smooth on an open V, agree with a on $A \cap V$, and let $L \subset V$ be compact. There exist an open $W \supset L \cup K$ and a smooth $h : W \to N$ such that*

$$h = a \text{ on } A \cap W, \qquad W \cap B \subset V, \qquad h = f \text{ on } W \cap B.$$

Proof. On $K \cap V$, $f = a = g$, so the overlap-shrinking lemma restricts V to an open V' still containing L and U to an open U' still containing K, with both maps in the same convex target chart on $V' \cap U'$. Apply the chart-patch extension lemma to these restrictions. It gives $W = V' \cup P$ with $K \subset P \subset U'$ and a smooth h agreeing with a on $A \cap W$. It agrees with f on $V' \setminus S$ for a compact support $S \subset U'$. A point in $W \cap B$ cannot lie in $U' \subset U$, and hence cannot lie in P or S. It therefore lies in $V' \subset V$, where $h = f$. This proves both exterior assertions. □

Lemma A.85 (Finite map patching retaining a prescribed exterior map)**.** *In the setting of the preceding lemma, supply finitely many smooth convex target-chart patches $g_j : U_j \to N$ agreeing with a on $A \cap U_j$, with compact $K_j \subset A \cap U_j$ and $U_j \cap B = \varnothing$. Given the same initial $f : V \to N$ and compact $L \subset V$, there exist an open $W \supset L \cup \bigcup_j K_j$ and a smooth $h : W \to N$ agreeing with a on $A \cap W$, with $W \cap B \subset V$ and $h = f$ on $W \cap B$.*

Proof. Begin with V and f. After adjoining the first j patches, retain the compact set $L \cup K_1 \cup \cdots \cup K_j$ in the current domain. The single-patch lemma adjoins K_{j+1} while retaining this compact set and the prescribed carrier values. At a point in the new domain and in B, that lemma puts the point in the previous domain and gives the previous map value. The inductive exterior assertion then puts it in the original V and identifies its value with f. This proves the exterior assertion after each step. Finite induction gives the result; the empty list leaves V, f unchanged. □

Lemma A.86 (Compact carrier germs extend while retaining a closed exterior region)**.** *Let X, N be finite-dimensional smooth boundaryless manifolds, with X Hausdorff and second countable. Let $a : A \to N$, let $K \subset A$ be compact, and let $B \subset X$ be closed and disjoint from K. Suppose each $x \in K$ has an open neighborhood with a smooth map agreeing with a on its intersection with A. Let $f : V \to N$ be smooth on an open V, agree with a on $A \cap V$, and let $L \subset V$ be compact. There exist an open $W \supset L \cup K$ and a smooth $h : W \to N$ such that*

$$h = a \text{ on } A \cap W, \qquad W \cap B \subset V, \qquad h = f \text{ on } W \cap B.$$

Neither A nor B needs to be compact. The local germs need agree only on A, and no agreement of their extensions away from A is required.

Proof. Refine each supplied germ at $x \in K$ to a convex target-chart patch. Intersect its source with the open complement of B. This still contains x, retains its smoothness and all its original A-values, and makes its source disjoint from B. Compactness gives a finite subcover $U_1, \dots, U_m$ of K. There are compact sets $K_j \subset K \cap U_j$ covering K: apply normality of the compact Hausdorff space K to shrink this finite relative open cover. Specifically, for each member the closed portion omitted by all the other members lies inside it; normality places that portion in an open set whose closure stays inside the member. Replacing the member by this smaller set preserves the cover. Repeating finitely many times gives closures contained in their original members that still cover K. These closures are the required compact K_j.

Apply the finite exterior-preserving patch lemma to these same patches, compact sets, and the initial V, f, L. It produces a smooth map on an open neighborhood of $L \cup \bigcup_j K_j = L \cup K$, retaining all prescribed values on A and the original map on the new domain's intersection with B. Empty compact sets are covered by the empty family, so no nonemptiness assumption is used. □

Definition A.87 (Native smooth extension germs for a manifold-valued map)**.** *Let M have its native $(k+1)$-dimensional half-space atlas, and let N be a smooth boundaryless manifold with a chosen chart e_y at each $y \in N$. A map $f : M \to N$ has native smooth extension germs if it is continuous and, for every native chart E and every $x \in \mathrm{source}(E)$, the coordinate expression $e_{f(x)} \circ f \circ E^{-1}$ agrees near $E(x)$ within $\mathrm{target}(E)$ with a smooth function on an actual open neighborhood of $E(x)$ in $\mathbb{R}^{k+1}$. The coordinate extension takes values in the vector space modeling N. Continuity places f in the source of $e_{f(x)}$ near x.*

Lemma A.88 (Native map extension germs imply intrinsic smoothness)**.** *A map with the native smooth extension germs of the preceding definition is smooth from the native manifold with boundary to its target manifold.*

Proof. Fix a source point x and a native chart E at x. Continuity of f and openness of the target chart source place f in that source near x. The coordinate expression in these two charts agrees, on a relative neighborhood of $E(x)$ in the half-space chart target, with the supplied smooth open coordinate extension. Its restriction is therefore differentiable to every order within that target. Together with continuity, this is the chart criterion for intrinsic smoothness at x. Since x was arbitrary, f is intrinsically smooth. □

Lemma A.89 (A manifold-valued germ across the same primitive-derived collar seam)**.** *Let M be a smooth Hausdorff $(k+1)$-manifold with boundary B, let λ be locally extendible with its actual derivative nondegenerate, and let Z be its contraction field. Fix $b \in B$ and the preferred boundary chart E there, and assume $d(-E_0)_b(Z_b) > 0$. Let $\delta > 0$, and suppose $C : [0, \delta) \times B \to M$ is jointly smooth, satisfies $C(0, y) = y$, and has time curves solving $\partial_t C = -Z \circ C$. Let $f : M \to N$ be continuous with an actual smooth coordinate extension germ for $e_{f(b)} \circ f \circ E^{-1}$ at $E(b)$, where $e_{f(b)}$ is the chosen smooth target chart at $f(b)$. There are an open neighborhood W of $(0, b)$ in $\mathbb{R} \times B$ and a smooth map $a : W \to N$ with*

$$a(t, y) = f(C(t, y)) \qquad ((t, y) \in W \cap ([0, \delta) \times B)).$$

The germ extends this specified collar map, including its whole local half-strip; no target-chart range condition is an additional hypothesis.

Proof. The primitive-derived coefficient lemma supplies smooth scalar extension germs for the coefficients of $-Z$ in E. The assumed normal defining-function inequality makes the normal coefficient of $-Z$ positive there. Apply the same-collar two-sided coordinate lemma to these germs and the specified C. In boundary coordinates $v = \chi(y)$ it gives a smooth two-sided map $c_0(t, v)$, centered at $E(b)$, agreeing with $E(C(t, y))$ for nonnegative time on a sufficiently small rectangle.

Let β be the supplied open coordinate extension of $e_{f(b)} \circ f \circ E^{-1}$. Shrink the rectangle so that c_0 lies in the smoothness neighborhood of β and, on its nonnegative half, in the neighborhood where the coordinate germ equality holds. Then $v(t, y) = \beta(c_0(t, \chi(y)))$ is smooth on a two-sided neighborhood of $(0, b)$ and satisfies

$$v(t, y) = e_{f(b)}(f(C(t, y)))$$

on its local half-strip. At the center its value is $e_{f(b)}(f(b)) \in \text{target}(e_{f(b)})$. Shrink the two-sided neighborhood so that all values of v lie in this open chart target. The smooth map there is $a = e_{f(b)}^{-1} \circ v$.

Continuity of f and continuity of C within the half-strip imply that $f(C(t, y))$ lies in $\text{source}(e_{f(b)})$ on a relative neighborhood of $(0, b)$. Intersect the two-sided neighborhood with an open set representing this relative neighborhood. For every remaining half-strip point the chart's inverse identity therefore gives $a(t, y) = f(C(t, y))$. The resulting open W contains $(0, b)$ and has every stated property. □

Lemma A.90 (Map extension from seam germs on the actual compact collar quotient)**.** *Let M be a compact smooth Hausdorff second-countable $(k+1)$-manifold with boundary B. Let $C : [0, \delta) \times B \to M$ be a smooth open collar of positive width, with seam $C(0, y) = y$, boundary criterion $C(t, y) \in B$ exactly when $t = 0$, and smooth native inverse germs. Let X_C be its actual boundaryless collar quotient, with original inclusion $i : M \to X_C$ and two-sided cylinder inclusion $j : (-\delta, \delta) \times B \to X_C$. Suppose $f : M \to N$ is intrinsically smooth and that each $b \in B$ has an open neighborhood W_b of $(0, b)$ in $\mathbb{R} \times B$ and a smooth partial map $a_b : W_b \to N$, agreeing with $f \circ C$ on $W_b \cap ([0, \delta) \times B)$. There are an open neighborhood W of $i(M)$ in this same X_C and a smooth map $g : W \to N$ with $g(i(x)) = f(x)$ for every $x \in M$.*

Proof. The original inclusion i is injective and continuous, so $A = i(M)$ is compact. Define $a : A \to N$ by $a(i(x)) = f(x)$. At an original interior point, transport $f|_{M^\circ}$ through the inverse of the smooth open interior inclusion. This inverse is smooth in the carried interior atlas. The transported map is a smooth partial map on the open interior image and agrees with a throughout its carrier portion.

At a boundary point b, the seam relation gives $j(0, b) = i(b)$. The cylinder inclusion is an open embedding with smooth local inverse germs. Restrict its inverse near $i(b)$ so that its coordinates lie in W_b, and compose with a_b. This is a smooth partial map on an open neighborhood of $i(b)$. If a point there also equals $i(x)$, its inverse cylinder coordinates (t, y) satisfy $t \geq 0$ and $C(t, y) = x$, by the original-cylinder equality criterion. The assumed half-strip equality gives $a_b(t, y) = f(x) = a(i(x))$.

These interior and boundary patches provide actual smooth local extensions of a at every point of the compact carrier A. Apply the compact-germ map extension lemma on the constructed Hausdorff second-countable boundaryless manifold X_C. It supplies an open $W \supset A$ and a smooth $g : W \to N$ with $g|_A = a$, which is exactly the required original-value identity. The construction also covers an empty carrier without choosing a point of N. □

Lemma A.91 (Native map extension on one fixed primitive collar quotient)**.** *Under the actual compact collar quotient hypotheses of the preceding lemma, let λ be locally extendible with nondegenerate actual derivative and contraction field Z. Suppose the chosen C solves $\partial_t C = -Z \circ C$, and $d(-E_0)_b(Z_b) > 0$ in the preferred boundary chart at every $b \in B$. For every $f : M \to N$ with native smooth extension germs, there are an open neighborhood W of $i(M)$ in this same X_C and a smooth $g : W \to N$ satisfying $g(i(x)) = f(x)$ for every $x \in M$. The collar quotient is fixed before f; its neighborhood W may depend on f. No extension on all of X_C is asserted.*

Proof. Native map extension germs imply intrinsic smoothness of f. At each boundary point b they also supply the actual smooth coordinate germ in its preferred native chart, and f is continuous. Apply the primitive-derived same-collar map seam lemma at b to the specified C, its integral equation, and the assumed outwardness. It constructs a smooth partial two-sided map germ agreeing with $f \circ C$ on the local half-strip. These germs at all boundary points, together with intrinsic smoothness of f, satisfy the actual-quotient map extension lemma. Its conclusion gives the required W and g on this very X_C, preserving every original value. □

Lemma A.92 (One native compact primitive presentation extending every source map)**.** *Let M be a compact smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Let λ be locally extendible with its actual exterior derivative nondegenerate, and suppose its contraction field satisfies the regular local defining-function outwardness hypotheses of Lemma A.42. There are a compact domain D in a boundaryless smooth manifold X, a smooth closed embedding $i : M \to X$ with bijective differential, a smooth one-form η on X, and a native diffeomorphism $e : M \to D$, such that*

$$i(M) = D, \qquad \mathrm{Fr}(D) = i(\partial M), \qquad i^*\eta = \lambda, \qquad (D \hookrightarrow X) \circ e = i.$$

The objects X, D, i, η, e are chosen before any additional scalar or map. Every source scalar with actual smooth native coordinate extension germs has a smooth extension on all of X. For every finite-dimensional smooth boundaryless target N and every $f : M \to N$ with native smooth extension germs, there are an open neighborhood W of $i(M)$ in this same X and a smooth map $g : W \to N$ satisfying $g(i(x)) = f(x)$ for all $x \in M$. The neighborhood W may depend on N and f. This assertion supplies a compact primitive presentation; it does not assert global symplecticity on X, a completion or a compact family.

Proof. The source primitive and outwardness construct one positive-width native Liouville collar C with smooth native inverse germs, its actual negative contraction-field equation and its full primitive law. Form its actual boundaryless quotient X_C and original compact carrier $D = i(M)$. The compact-presentation theorem gives the smooth boundaryless manifold and the frontier identity. The original-inclusion differential lemma makes di bijective at every point, including the seam. Gluing the interior primitive and the exponential cylinder primitive produces the smooth η, whose original pullback is λ. The same-collar native atlas identification supplies $e : M \to D$ with ambient composite i. Fix all these objects.

At each boundary point, the defining-function sign lemma converts the source's regular local defining-function outwardness into strict outwardness for the preferred canonical function $-E_0$. For a source scalar with actual native coordinate extension germs, these inequalities and the primitive-derived scalar seam lemma give two-sided smooth scalar germs on this very collar. The scalar's native coordinate germs imply its intrinsic smoothness by restricting each smooth open extension to the half-space chart target. The actual-quotient scalar extension lemma therefore extends it on all of this fixed X_C, preserving every original value.

Now choose a target N and a map f as stated. Apply the fixed primitive-collar map extension lemma to this same C, its native inverse germs, its negative contraction-field equation, the derived

canonical outwardness and the native map extension germs of f. It produces an open $W \supset i(M)$ in this very X_C and a smooth $g : W \to N$ with $g \circ i = f$. No choice of X, D, i, η, e depended on this map or its target, giving the asserted quantifier order. □

Lemma A.93 (One native symplectic primitive presentation extending every source map)**.** *Under the source hypotheses of Lemma A.92, the same native compact primitive presentation can be chosen with*

$$i^*(d\eta) = d\lambda, \qquad d\eta \text{ symplectic at every point of } D.$$

It retains the native diffeomorphism, bijective original-inclusion differential, global scalar extension property, and the universal manifold-valued map-neighborhood extension property. All these objects are chosen before the additional scalar or map.

Proof. Take X, D, i, η, e and both universal extension properties from the primitive-and-map presentation lemma. Exterior naturality on the full native tangent spaces gives $i^*(d\eta) = d(i^*\eta) = d\lambda$, including at the source boundary. At $z = i(x) \in D$, suppose $d\eta_z(v, w) = 0$ for all $w \in T_zX$. Bijectivity of di_x gives $u \in T_xM$ with $di_x u = v$. For every $a \in T_xM$ the pullback identity gives $d\lambda_x(u, a) = d\eta_z(di_x u, di_x a) = 0$. Nondegeneracy of $d\lambda_x$ forces $u = 0$ and hence $v = 0$. Thus $d\eta$ is nondegenerate throughout D. It is closed because the second exterior derivative of a smooth form vanishes, as follows in every coordinate chart by cancellation of mixed second derivatives under alternation. Hence it is symplectic on D. No object was changed, so both universal extension properties remain. □

Lemma A.94 (One native Liouville presentation extending every source map)**.** *Under the source hypotheses of Lemma A.92, there are a boundaryless smooth manifold U, a compact Liouville domain $A \subset U$ with smooth primitive η_U, a smooth closed embedding $j : M \to U$ with bijective differential, and a native diffeomorphism $e_U : M \to A$ whose ambient composite is j, such that*

$$j(M) = A, \qquad \mathrm{Fr}_U(A) = j(\partial M), \qquad j^*\eta_U = \lambda, \qquad j^*(d\eta_U) = d\lambda, \qquad dj_x(Z_M(x)) = Z_U(j(x)).$$

Here Z_M is the source contraction field, Z_U is the smooth contraction field of η_U, and $d\eta_U$ is symplectic throughout U. Every source scalar with actual native smooth extension germs has a smooth extension on all of U. If the source scalar is positive, its extension can be chosen positive on all of U. For every finite-dimensional smooth boundaryless target N and every $f : M \to N$ with native smooth extension germs, there are an open $W \subset U$ containing $j(M)$ and a smooth $g : W \to N$ such that $g(j(x)) = f(x)$ for all $x \in M$. The objects U, A, j, η_U, e_U are chosen before the scalar, target and map; the map neighborhood W may depend on the target and map. This is a positive-dimensional compact-domain result, without a compact-bundle, completed-family or pair correspondence assertion.

Proof. Choose X, D, i, η, e from the native symplectic-and-map presentation lemma, keeping both universal extension properties. Let U be the open nondegeneracy locus of $d\eta$, containing D, and let $q : U \hookrightarrow X$ be its open inclusion. The symplectic restriction lemma gives the compact carrier $A = q^{-1}(D)$, primitive $\eta_U = q^*\eta$ and smooth contraction field Z_U throughout U. Lift i to $j : M \to U$, so $q \circ j = i$. Open inclusion has bijective differential; therefore dj_x is bijective, and j is smooth and a closed embedding. The actual compact restriction preserves the frontier, giving $\mathrm{Fr}_U(A) = j(\partial M)$. Compose e with the native open-restriction diffeomorphism to obtain e_U, whose ambient composite is j. Pullback composition through $q \circ j = i$ gives both stated primitive and exterior-derivative identities.

For $x \in M$ and $a \in T_x M$, these identities yield

$$d\eta_{U\,j(x)}(dj_x Z_M(x), dj_x a) = d\lambda_x(Z_M(x), a) = \lambda_x(a) = \eta_{U\,j(x)}(dj_x a).$$

Surjectivity of dj_x and uniqueness of the contraction field give $dj_x Z_M(x) = Z_U(j(x))$. At a boundary point x, pull back a regular local defining function ρ of A through j. Its differential is nonzero, and in a native half-space chart E its differential annihilates all tangential directions: evaluate the locally nonpositive sublevel function in both signs of each such direction. On the inward normal it is nonpositive. Thus its differential has the form $c\,dE_0$ with $c < 0$, the strict sign following from regularity. The source defining-function sign lemma gives $(dE_x Z_M(x))_0 < 0$. Consequently

$$d\rho_{j(x)}(Z_U(j(x))) = d(\rho \circ j)_x(Z_M(x)) = c(dE_x Z_M(x))_0 > 0.$$

This proves outwardness for the actual restricted carrier.

Fix these objects. Restrict any smooth global scalar extension on X to U to retain its original values. For a positive source scalar s, continuity of each actual coordinate extension lets it be restricted to its positive locus; the logarithm is smooth there and extends the native coordinate germ of $\log s$. Extend $\log s$ by the universal scalar property on U, and exponentiate. The result is positive throughout U and equals s on the source.

Finally fix N and f as stated. The retained universal map property on X gives an open $O \supset i(M)$ and a smooth $h : O \to N$ with $h \circ i = f$. Let $W = q^{-1}(O)$, an open neighborhood of $j(M)$ in U, and set $g = h \circ q|_W$. This restriction is smooth and $g(j(x)) = h(q(j(x))) = h(i(x)) = f(x)$ for every $x \in M$. All presentation choices preceded N and f, proving the quantifier order. □

Remark A.95 (Scope of the manifold-valued patch extension). The compact-germ extension lemma concerns maps on a compact subset of an already chosen boundaryless presentation, with actual smooth extension germs there. Its finite assembly preserves all original carrier values and requires neither agreement of the germs away from the carrier nor one target chart containing the entire image. The native primitive-and-map presentation additionally transfers the source's actual native map germs through the same primitive-derived collar and quotient, choosing the compact primitive presentation before the map. The map is extended only on a neighborhood of the compact carrier. The native Liouville-and-map presentation retains this universal map property through the actual symplectic restriction, with the same native primitive, derivative and field. It concerns positive-dimensional source domains; the finite-carrier and zero-dimensional ribbon constructions retain their separate scope. Native maps with injective differential and injective compact restriction still require immersion and primitive-pullback transport on the same ribbon presentation and restriction to an embedding neighborhood. Recovering a fully intrinsic ambient pair also requires its native-boundary identification and contact-form transport. Intrinsic families and the converse presentation identifications retain their separate correspondence obligations.

Remark A.96 (Componentwise compact-domain presentations). The compact-domain correspondences preserve the original intrinsic Liouville domain, actual boundary, full primitive and exterior derivative, canonical Liouville field and strict outwardness. The source-smooth scalar, form, field, ribbon and map data used for witnessed pairs are retained on the same compact carrier through their native identifications. In positive dimension the original primitive constructs its inward collar, boundaryless extension and exact native compact presentation; the scalar and map extension results use that same presentation. In dimension zero the domain is its actual finite discrete carrier, with empty boundary and zero primitive and field.

Conversely, the actual compact restriction of a presented Liouville domain has all the native laws by A.326, and its canonical field has the original-chart germs of A.327. A prescribed change

of compact presentation retains the exact pulled primitive and defining functions by A.331. These component equations recover the compact witnessed-pair data in both directions. They concern the source-required data; they make no classification claim for arbitrary additional geometric structures.

Lemma A.97 (Presentation foundation: A symplectic neighborhood)**.** *An extension-space compact Liouville domain presentation has an open neighborhood on which the restricted primitive is symplectic and its canonical Liouville field satisfies the contraction equation. On the compact carrier this is the original field.*

Proof. In a tangent chart nondegeneracy is nonvanishing of the determinant of the two-form matrix, so it is an open condition. The nondegeneracy locus contains the compact carrier. Inverting that matrix gives a smooth field on the locus satisfying the contraction equation. On the carrier, nondegeneracy and the original contraction equation give equality of the two fields. □

Lemma A.98 (Presentation foundation: A constructed local Liouville flow)**.** *There is an open neighborhood of the compact domain and a common $\delta > 0$ on which a jointly smooth all-real auxiliary flow solves the actual neighborhood Liouville equation for $|t| < \delta$. The auxiliary field agrees with the Liouville field throughout these trajectories; completeness of the original ambient field is not asserted.*

Proof. Choose a compactly supported smooth cutoff equal to one on a neighborhood of the compact domain inside the symplectic neighborhood, and multiply the canonical field by it. Local integral curves on the compact support have a uniform positive existence time by finite covering. Off the support the field is zero and constant curves solve the equation. Thus the same uniform time works at every initial point. Concatenating these local curves in successive intervals, with uniqueness at every overlap, gives all-real integral curves. Uniqueness gives their action law. Local smooth ODE dependence gives joint smoothness near each time and point; the action law transports this to all times. At time zero the compact domain lies where the cutoff is one. Continuity and compactness give a common two-sided time strip and an open neighborhood of the compact domain whose trajectories stay in that locus. There the auxiliary equation is the Liouville equation. □

Lemma A.99 (Presentation foundation: Local collar inversion and domain side)**.** *The local flow from the boundary is injective and a local diffeomorphism on a common two-sided strip. On that strip its point lies in the domain if and only if its time is nonpositive.*

Proof. At time zero the differential is $(v, s) \mapsto v + sZ$. The tangent space of the boundary is $\ker d\rho$ and $d\rho(Z) > 0$, so this map is invertible. The inverse function theorem and compactness give a common strip of local invertibility. If no smaller strip were injective, choose pairs with times tending to zero and equal images. Compactness gives boundary subsequential limits; equality of images forces the limits equal. Both pairs eventually lie in one injective inverse-function neighborhood, a contradiction.

An orbit in this injective strip cannot hit the frontier at nonzero time: such a frontier point is also the time-zero image of another boundary point, contradicting injectivity. At a fixed starting point the derivative of its defining function at zero is strictly positive. Hence the orbit is inside the domain for sufficiently small negative times and outside for sufficiently small positive times. On each connected half interval it avoids the frontier. The inverse images of the interior and the open complement are disjoint open sets covering that half interval; connectedness forces membership to be constant. The pointwise signs therefore propagate to both entire half intervals. At time zero the boundary lies in the closed domain. □

Lemma A.100 (Presentation foundation: Naturality of exterior differentiation)**.** *For a smooth map f and smooth differential form η, $d(f^*\eta) = f^*(d\eta)$.*

Proof. This is local, so write in coordinates $\eta = \sum_I a_I\, dy_{i_1} \wedge \cdots \wedge dy_{i_k}$. Its pullback is $\sum_I (a_I \circ f)\, df_{i_1} \wedge \cdots \wedge df_{i_k}$. Differentiate this expression. Since mixed second derivatives of each smooth scalar f_i agree, $d(df_i) = 0$. The product rule leaves precisely $\sum_I d(a_I \circ f) \wedge df_{i_1} \wedge \cdots \wedge df_{i_k}$. The scalar chain rule identifies $d(a_I \circ f)$ with $f^*(da_I)$, which gives the claimed equality term by term. □

Lemma A.101 (Presentation foundation: The primitive pullback equation)**.** *Let λ be a smooth one-form. Suppose $F : \mathbb{R} \times N \to X$ is jointly smooth and I is a connected open real interval on which $\partial_t F = Z \circ F$, and $\iota_Z d\lambda = \lambda$ on X. Then $F_t^*\lambda = e^{t-t_0} F_{t_0}^*\lambda$ for $t, t_0 \in I$.*

Proof. Global contraction implies the contraction identity at every point of the trajectory image. Apply the local primitive-pullback lemma with $V = N$ and the given connected interval. Its pointwise equality holds at every $x \in N$, which gives the asserted equality of one-forms. □

Lemma A.102 (Presentation foundation: Liouville collar with its primitive)**.** *A Liouville domain presentation admits a symplectic open neighborhood of its compact domain and a two-sided smooth open collar $c : \partial D \times (-\delta, \delta) \longrightarrow X$, $\delta > 0$, given by the local Liouville flow from the boundary. On this same collar,*

$$c(x, 0) = x, \quad c(x, r) \in D \iff r \leq 0, \qquad c^*\lambda = e^r \alpha, \quad \alpha = \lambda|_{\partial D}.$$

The collar is an embedding and a local diffeomorphism; its time derivative is the actual Liouville field throughout the indicated strip.

Proof. The local collar with its domain side is supplied by the preceding construction. An injective local diffeomorphism is an open embedding. Apply the primitive pullback equation to its boundary flow and compare with time zero to obtain the spatial identity $e^r\alpha$. The time component of the collar derivative is sZ, and $\lambda(sZ) = s\, d\lambda(Z, Z) = 0$. The spatial and time components together therefore give the displayed full primitive identity on this same collar. □

Definition A.103 (Auxiliary presentation: A realized completion)**.** *A realized completion consists of a smooth exact symplectic ambient manifold with its smooth all-real-time Liouville flow, an embedding of a neighborhood of D preserving the primitive on D, and a smooth end parametrization by the same flow from the embedded boundary. On $\partial D \times [0, \infty)$ that parametrization is an embedding, has primitive $e^r\alpha$, covers the complement of the compact domain, and is disjoint from the compact domain when $r > 0$. An extension-space realization specifies its entire chosen extension (neighborhood) in which the compact inclusion is represented. Completeness means flow existence for every real time, in both directions. Finite Liouville type of a specified ambient means that the inclusion of some compact Liouville domain extends to a primitive-preserving diffeomorphism from this completion onto that very ambient.*

Lemma A.104 (Presentation foundation: Localized completion and complete flow)**.** *For every Liouville domain presentation, after restricting its extension manifold to an open neighborhood of the compact carrier, the collar gluing constructs a realized completion with all-real-time smooth Liouville flow. It has the specified compact inclusion and exponential cylindrical primitive. The open inclusions identify its compact carrier homeomorphically with the original carrier, preserving the original primitive and field; its end coefficient is the original boundary primitive. No realized completion or complete flow is an input to this construction.*

Proof. Construct a Liouville flow collar of the given presentation. Its interior and collar localization is an open neighborhood of the compact carrier in the symplectic neighborhood, and hence in the original extension manifold. The original compact carrier is identified with the localized carrier by the open inclusions; the restricted primitive and field and the boundary form are preserved by this identification. Apply the realized localized-completion construction to this constructed collar. Its identity compact inclusion, constructed complete flow, and all-real end with primitive $e^r\alpha$ give the required realization of the restricted presentation. The preservation identities identify these compact and boundary data with those of the supplied presentation. The collar and localized-completion constructions apply in every dimension, so no separate zero-dimensional case or supplied completion is needed. □

Definition A.105 (Auxiliary presentation: Full intrinsic core)**.** *For a realized completion and its actual flow Φ, set* $\mathrm{Core}(D,\lambda) = \{x \in D : \Phi^t(x) \in D \text{ for every } t \geq 0\}$.

Lemma A.106 (Presentation foundation: Nested-image core formula)**.** *In the completed ambient, the image of the full core equals $\bigcap_{t\geq 0} \Phi^{-t}(D)$, where $\Phi^{-t}(D)$ is the image under time $-t$ of this same all-real-time flow.*

Proof. The time maps are mutually inverse. Thus $x \in \Phi^{-t}(D)$ if and only if $\Phi^t(x) \in D$. Intersect over $t \geq 0$; time zero supplies $x \in D$. Apply the compact inclusion to identify the domain core with its ambient image. □

Lemma A.107 (Presentation foundation: Zero-dimensional core)**.** *The full core of a zero-dimensional realized Liouville domain is the entire domain.*

Proof. A zero-dimensional smooth manifold has zero tangent spaces and discrete point space. Every smooth orbit from the connected real line to this space is constant. Its value at time zero is its starting point, so every point of the domain remains there for every time. □

A.3 Auxiliary presentations: Pairs, comparison steps and ambient grading

Definition A.108 (Auxiliary presentation: Witnessed compact pair)**.** *A witnessed pair consists of Liouville domains (D,λ) of dimension $2n$ and (F,μ) of dimension $2n-2$, a smooth embedding $j : F \hookrightarrow \partial D$, and a smooth positive function b on ∂D such that $\mu = j^*(b\lambda|_{\partial D})$. Its extension-space realization includes realizations of the completions of both domains. The embedded full ribbon core is $j(\mathrm{Core}(F,\mu))$. An inclusion marking is a specified primitive-preserving smooth completion diffeomorphism onto a named completed ambient, extending the given compact inclusion and carrying the end to its Liouville orbits.*

Definition A.109 (Auxiliary presentation: Smooth families on fixed presentations)**.** *A presented pair deformation is an interval-indexed family of witnessed pairs with compact-domain, boundary and ribbon trivializations, and completed-ambient and completed-ribbon trivializations respecting the compact inclusions and cylinder coordinates. All pulled-back primitives, Liouville fields, ribbon embeddings and positive witnesses are jointly smooth with open extensions across the parameter endpoints. Smoothness of the completed primitives and fields is required on the whole completed spaces. Geometric inequalities are required on* $[0,1]$.

Lemma A.110 (A common nondegeneracy neighborhood for varying primitives)**.** *Let $D \subset M$ be one extension-space compact domain presentation. For each $a \in [0,1]$ let (λ_a, Z_a) be a Liouville structure on that presentation. Suppose only that the primitives form a jointly smooth family with smooth extensions across the parameter endpoints. There are an extension $\widetilde{\lambda}_a$ of the primitives to*

real parameters and an open neighborhood $\Omega \subset \mathbb{R} \times M$ *of* $[0,1] \times D$ *on which* $d_x \widetilde{\lambda}_a$ *is nondegenerate. On* Ω *its canonical Liouville field* $\widetilde{Z}_a(x)$ *is jointly smooth and satisfies*

$$\iota_{\widetilde{Z}_a(x)} d_x \widetilde{\lambda}_a = \widetilde{\lambda}_a \quad \text{at } x.$$

For $a \in [0,1]$ *and* $x \in D$*, this field is the original* $Z_a(x)$*. Neither symplecticity outside the interval nor joint smoothness of the given fields is an additional hypothesis.*

Proof. Choose the smooth extension of the primitive coefficients afforded by the hypothesis. In a fixed spatial chart write the extended primitive as $\sum_i l_i(a,y)\, dy_i$. Its spatial exterior derivative has coefficients $\partial_{y_i} l_j - \partial_{y_j} l_i$, which are jointly smooth on the same open coefficient neighborhood. The contraction map from tangent vectors to covectors therefore has a jointly smooth matrix $A(a,y)$. At every point of $[0,1] \times D$ this matrix is invertible by the original symplecticity. Intersect each coefficient neighborhood with the open locus $\det A \neq 0$ and take the union of the resulting chart neighborhoods. This gives the asserted open Ω containing the entire compact seam.

On each such neighborhood the vector of coefficients of the field is $A(a,y)^{-1}(l_i(a,y))_i$. Matrix inversion is smooth where the determinant is nonzero, so these coefficients are jointly smooth. They satisfy the displayed contraction equation. On chart overlaps the two fields solve the same equation for the same nondegenerate two-form; subtracting the equations gives zero contraction, hence equality of the fields. They consequently define a jointly smooth field on Ω. On $[0,1] \times D$, the original field satisfies that same contraction equation, and the same uniqueness proves its asserted equality with the constructed field. In dimension zero all these maps act between zero vector spaces; the contraction map is invertible and its unique solution is zero, so the argument includes that case. □

Lemma A.111 (Smooth replacement of compact-family field extensions)**.** *Let* $D \subset M$ *be a fixed compact domain presentation carrying Liouville structures* (λ_t, Z_t) *for* $t \in [0,1]$*, with jointly smooth primitives and open extensions across the parameter endpoints. There are replacement Liouville structures* (λ'_t, Z'_t) *on the same presentation and a real-parameter field* $\widetilde{Z}_t$ *on* M*, jointly smooth on all of* $\mathbb{R} \times M$*, such that*

$$\lambda'_t = \lambda_t, \qquad Z'_t(x) = Z_t(x) \quad (x \in D), \qquad \widetilde{Z}_t = Z'_t \quad (0 \le t \le 1).$$

In particular the replacement fields form a jointly smooth interval family with endpoint extensions. The contraction equation is required on the compact carrier; no such equation is asserted elsewhere.

Proof. The common nondegeneracy-neighborhood lemma gives a real extension $\widetilde{\lambda}_t$ and an open neighborhood Ω of $[0,1] \times D$ on which its canonical contraction field $Y_t(x)$ is jointly smooth. On the compact carrier it equals $Z_t(x)$. Choose a smooth function χ with compact support contained in Ω, equal to one on an open neighborhood of $[0,1] \times D$. Define $\widetilde{Z}_t(x) = \chi(t,x) Y_t(x)$ on Ω and set it equal to zero outside Ω. This is globally jointly smooth: on Ω it is a product of smooth data, and each point outside the closed support of χ has a neighborhood on which it vanishes. On $[0,1] \times D$ it is exactly the given field.

Set $\lambda'_t = \lambda_t$ and $Z'_t = \widetilde{Z}_t$ for interval parameters. Each field slice is smooth, and the primitive and its symplectic differential on D are unchanged. The carrier equality transfers the contraction equation. Since the boundary lies in the closed carrier, it also transfers strict outwardness. These data are therefore Liouville structures on the same compact presentation. The global smooth family $\widetilde{Z}$ supplies their asserted interval regularity and endpoint extensions. □

Lemma A.112 (A parameter-preserving uniform local flow)**.** *Let M be a smooth boundaryless finite-dimensional manifold, $J \subset \mathbb{R}$ and $K \subset M$ compact, and Ω an open neighborhood of $J \times K$. Let $Z_a(x) \in T_xM$ be a jointly smooth field on Ω. There are an open $U \subset \Omega$ containing $J \times K$, a single $\delta > 0$, and a jointly smooth all-real auxiliary flow on $\mathbb{R} \times M$,*

$$\Phi^s(a, x) = (a, \phi^s(a, x)),$$

whose first coordinate equals its initial parameter for every real s. If $(a, x) \in U$ and $|s| < \delta$, then $(a, \phi^s(a, x)) \in \Omega$ and the actual spatial equation is

$$\frac{d}{ds}\phi^s(a, x) = Z_a(\phi^s(a, x)), \qquad \phi^0(a, x) = x.$$

The same width works over the entire compact seam $J \times K$. The all-real auxiliary action does not assert completeness of the original field.

Proof. Put $C = J \times K$, which is compact. Choose a smooth function χ on $\mathbb{R} \times M$ whose support is compact and contained in Ω, and which equals one on an open neighborhood V of C. On Ω put $W(a, x) = (0, \chi(a, x)Z_a(x))$ and set $W = 0$ outside Ω. This extension is smooth: the support of χ is a closed subset contained in Ω, so every point outside that support has an open neighborhood on which the extension is zero. The field W has compact support.

Apply the compact-support flow construction to this field on the product manifold. Local smooth existence on its compact support gives one positive time width by a finite cover; off that support the constant curves solve the equation on the same interval. Restarting these solutions with increments smaller than that width constructs integral curves for all positive and negative times. Uniqueness makes the restarted solutions agree on overlaps and gives the flow action law. Local smooth dependence, followed by finite compositions of local time maps, gives joint smoothness of the resulting all-real action Φ. The derivative of its first coordinate is zero everywhere, since the first component of W vanishes. This real-valued function of time is constant on $\mathbb{R}$; its value at zero is a. Thus write $\Phi^s(a, x) = (a, \phi^s(a, x))$ for all real times.

The inverse image of V under the continuous map $(s, p) \mapsto \Phi^s(p)$ is open and contains $\{0\} \times C$, because Φ^0 is the identity. The tube lemma for this compact seam gives an open time neighborhood T of zero and an open spatial neighborhood A of C such that $\Phi^s(p) \in V$ whenever $s \in T$ and $p \in A$. Choose $\delta > 0$ with $(-\delta, \delta) \subset T$ and put $U = A \cap \Omega$. Every specified trajectory then lies in $V \subset \Omega$, where $\chi = 1$. Its equation for W becomes exactly the displayed equation for Z after taking the second coordinate. The initial-value identity follows from $\Phi^0 = \mathrm{id}$. No continuation of that original equation beyond this time strip was used or asserted. □

Lemma A.113 (A supported parameter-preserving local flow)**.** *Let M be a smooth boundaryless finite-dimensional manifold, let $J \subset \mathbb{R}$ and $K \subset M$ be compact, and let Ω be an open neighborhood of $J \times K$. Suppose the spatial field $Z_a(x)$ is jointly smooth on Ω. There are an open $U \subset \Omega$ containing $J \times K$, a positive δ, and a jointly smooth all-real flow $\Phi^s(a, x) = (a, \phi^s(a, x))$ on $\mathbb{R} \times M$ such that*

$$\Phi^s(p) = p \quad (p \notin \Omega,\ s \in \mathbb{R}).$$

For $p = (a, x) \in U$ and $|s| < \delta$, its value lies in Ω and $\partial_s \phi^s(a, x) = Z_a(\phi^s(a, x))$.

Proof. Choose a compactly supported smooth cutoff χ with support in Ω, equal to one on an open neighborhood V of $J \times K$. Extend $(0, \chi(a, x)Z_a(x))$ by zero outside Ω. It is smooth and compactly supported, so the compact-support flow construction gives a jointly smooth all-real flow Φ. Its parameter component has zero derivative and hence remains a. For $p \notin \Omega$ the extended field is zero

at p, so the constant curve at p is an integral curve. Uniqueness identifies it with $\Phi^s(p)$ for all real s.

Continuity of Φ and compactness of $J \times K$ give a neighborhood U of that set and one positive δ for which $\Phi^s(U) \subset V \cap \Omega$ whenever $|s| < \delta$. Taking $s = 0$ shows $U \subset \Omega$. On these trajectories $\chi = 1$, so the spatial equation of the extended field is precisely the displayed equation of the original field. Its initial condition is the identity because Φ is a flow. □

Lemma A.114 (A uniform local Liouville flow for varying primitives)**.** *Let $D \subset M$ and (λ_a, Z_a), $a \in [0,1]$, satisfy the hypotheses of the common nondegeneracy-neighborhood lemma. There are an open $U \subset \mathbb{R} \times M$ containing $[0,1] \times D$, a single $\delta > 0$, and a jointly smooth all-real auxiliary flow $\Phi^s(a,x) = (a, \phi^s(a,x))$ whose first coordinate is fixed for every real time. For every $a \in [0,1]$, $(a,x) \in U$ and $|s| < \delta$, put $y = \phi^s(a,x)$. Then $d\lambda_a$ is nondegenerate at y, its canonical Liouville field Z_{λ_a} satisfies $\iota_{Z_{\lambda_a}(y)} d\lambda_a = \lambda_a$ at y, and*

$$\frac{d}{ds}\phi^s(a,x) = Z_{\lambda_a}(\phi^s(a,x)), \qquad \phi^0(a,x) = x.$$

Thus the same width solves the actual primitive's equation over the entire compact parameter interval and compact domain. The all-real action solves an auxiliary cutoff field; completeness of the original Liouville fields is not asserted.

Proof. Construct $\widetilde{\lambda}$ and Ω by the common nondegeneracy-neighborhood lemma. Its canonical field is a jointly smooth section along the spatial projection of Ω. Apply the parameter-preserving local-flow lemma with $J = [0,1]$, $K = D$ and this field. It gives the stated U, δ and action, and every specified trajectory remains in Ω. At its point (a,y) the preceding neighborhood lemma supplies nondegeneracy and the contraction equation for $\widetilde{\lambda}_a$. For $a \in [0,1]$ the equality $\widetilde{\lambda}_a = \lambda_a$ holds as an equality of one-forms on the original extension space. Their differentials and their canonical contraction solutions are consequently equal there. Substitute these equalities into the local spatial equation and contraction identity. Parameter preservation and the initial-value identity are those of the same constructed action. □

Lemma A.115 (A family local flow preserving symplectic neighborhoods)**.** *Let $D \subset M$ and (λ_t, Z_t) satisfy the common nondegeneracy-neighborhood hypotheses. Put $X_t = \{x \in M : d\lambda_t \text{ is nondegenerate at } x\}$. There are an open $U \supset [0,1] \times D$, a positive δ, and a jointly smooth all-real auxiliary flow $\Phi^s(a,x) = (a, \phi^s(a,x))$ such that*

$$x \in X_t \quad \Longleftrightarrow \quad \phi^s(t,x) \in X_t \qquad (0 \le t \le 1,\ s \in \mathbb{R}).$$

For $(t,x) \in U$, $0 \le t \le 1$ and $|s| < \delta$, the point $y = \phi^s(t,x)$ lies in X_t, and

$$\partial_s \phi^s(t,x) = Z_{\lambda_t}(y), \qquad \iota_{Z_{\lambda_t}(y)} d\lambda_t = \lambda_t \quad \text{at } y.$$

The all-real flow is auxiliary; the original Liouville equation is asserted only on the indicated local strip.

Proof. Apply the supported local-flow lemma to the canonical contraction field on the common open nondegeneracy neighborhood Ω of $[0,1] \times D$. On each interval slice, every point of Ω belongs to X_t, since the extended primitive there is λ_t. Thus $x \notin X_t$ implies $(t,x) \notin \Omega$, and support gives $\phi^s(t,x) = x$ for every real s. Conversely, if $x \in X_t$ and $y = \phi^s(t,x) \notin X_t$, the same stationarity at y gives $\phi^{-s}(t,y) = y$. The inverse-flow identity also gives $\phi^{-s}(t,y) = x$, contradicting $x \in X_t$. This proves the equivalence.

The supported-flow local equation puts each specified trajectory in Ω. On an interval slice the extended primitive is exactly λ_t, so its nondegenerate differential, canonical contraction field and contraction equation agree with those for λ_t. Substituting these equalities into that equation proves the remaining assertions for this same auxiliary flow. □

Lemma A.116 (Primitive scaling from contraction along the actual trajectories)**.** *Let λ be a smooth one-form on M, let $H : \mathbb{R} \times N \to M$ be jointly smooth, let I be a connected open real time interval, and let $V \subset N$ be open. Suppose, only at the points $H(t,x)$ with $(t,x) \in I \times V$, that*

$$\partial_t H(t,x) = Z(H(t,x)), \qquad \iota_{Z(H(t,x))} d\lambda = \lambda \quad \text{at } H(t,x).$$

Then for $t, t_0 \in I$ and $x \in V$, $(H_t^*\lambda)_x = e^{t-t_0}(H_{t_0}^*\lambda)_x$. Contraction outside this trajectory image is not a hypothesis.

Proof. On $I \times V$ put $\beta = H^*\lambda$. The time derivative equation and alternation give $\beta(\partial_t) = \lambda(Z) = d\lambda(Z,Z) = 0$ there. Fix a spatial chart near x and extend any chosen tangent vector $v \in T_xN$ to the constant coordinate vector field in that chart. Write $b(t) = \beta_{(t,x)}(0,v)$. This field commutes with ∂_t, and the spatial derivative of the identically zero function $\beta(\partial_t)$ is zero. The coordinate formula for the exterior derivative therefore gives $b'(t) = d\beta_{(t,x)}(\partial_t, (0,v))$. Naturality identifies this expression with $d\lambda_{H(t,x)}(Z(H(t,x)), dH_{(t,x)}(0,v))$. The contraction identity at this very trajectory point makes it $\lambda_{H(t,x)}(dH_{(t,x)}(0,v)) = b(t)$. Consequently $(e^{-t}b(t))' = 0$ throughout I. Connectedness makes this function constant, so $b(t) = e^{t-t_0}b(t_0)$. As v was arbitrary, this is the asserted equality of pulled-back one-forms at x. □

Lemma A.117 (The primitive on the boundary map of the same family flow)**.** *Let (λ_a, Z_a), $a \in [0,1]$, be Liouville structures on one fixed compact domain presentation with boundary inclusion j. Let Φ be a jointly smooth all-real auxiliary action on $\mathbb{R} \times M$, and define $c_a(x,r) = (\Phi^r(a, j(x)))_2$. Suppose on one strip $|r| < \delta$, $\delta > 0$, that these curves solve the canonical λ_a-Liouville equation and that its contraction identity holds at every point of those same curves. Then, for each $a \in [0,1]$,*

$$c_a^*\lambda_a = e^r \alpha_a \quad \text{on } \partial D \times (-\delta, \delta), \qquad \alpha_a = j^*\lambda_a.$$

Explicitly, for $v \in T_x\partial D$ and $s \in \mathbb{R}$,

$$\lambda_a\big(dc_{a,(x,r)}(v,s)\big) = e^r \alpha_a(v).$$

This is the full spatial and radial pullback with the family parameter held fixed, and uses the given action without choosing another one.

Proof. Fix $a \in [0,1]$. Apply the local primitive-pullback lemma to $H(r,x) = c_a(x,r)$, $I = (-\delta, \delta)$ and $V = \partial D$. This is jointly smooth by smoothness of the given action and boundary inclusion. Its contraction and time-derivative hypotheses are exactly the assumed identities on these curves. At time zero the action is the identity, so $H_0 = j$. Thus its spatial pullback at radius r is $e^r j^*\lambda_a$. The radial derivative is the canonical field at $c_a(x,r)$. Contraction there gives $\lambda_a(Z_{\lambda_a}) = d\lambda_a(Z_{\lambda_a}, Z_{\lambda_a}) = 0$. Linearity of the differential and of the one-form now gives the displayed evaluation on (v,s): its spatial contribution is $e^r\alpha_a(v)$ and its radial contribution is zero. This holds at every point of the strip and for every tangent vector, which is the asserted full pullback equality. □

Lemma A.118 (A common Liouville collar with its primitive)**.** *Let $D \subset M$ and (λ_a, Z_a), $a \in [0,1]$, satisfy the hypotheses of the local primitive-family flow lemma, and write $j : \partial D \hookrightarrow M$ for the fixed*

boundary inclusion. The constructed parameter-preserving auxiliary action and its open initial neighborhood $U \supset [0,1] \times D$ can be chosen together with an open $J \subset \mathbb{R}$ containing $[0,1]$ and a single $\epsilon > 0$ such that all the preceding local-flow conclusions hold on U for $|r| < \epsilon$, and

$$c(a,x,r) = \phi^r(a, j(x)), \qquad q(a,x,r) = (a, c(a,x,r)),$$

are jointly smooth, and q is an open embedding and a local diffeomorphism on $J \times \partial D \times (-\epsilon, \epsilon)$. The seam satisfies $c(a,x,0) = j(x)$. For $a \in [0,1]$ every curve $r \mapsto c(a,x,r)$ on this strip solves the actual canonical λ_a-Liouville equation; $d\lambda_a$ is nondegenerate there and the canonical contraction identity holds along the curve. For that same interval parameter and collar,

$$c(a,x,r) \in D \iff r \le 0, \qquad c_a^* \lambda_a = e^r \alpha_a, \quad \alpha_a = j^* \lambda_a.$$

The primitive identity includes both spatial and radial tangent vectors and holds with the family parameter fixed. Only regularity and embedding are asserted for parameters in $J \setminus [0,1]$.

Proof. Use the auxiliary action, initial neighborhood U and width δ from the local primitive-family flow lemma. Composition with the fixed boundary inclusion gives the joint smoothness of c and q on their whole sources, and the time-zero identity gives the seam equation. Every boundary point belongs to the compact domain, so for $a \in [0,1]$ all these curves solve the original canonical equation on $(-\delta, \delta)$, with the stated nondegeneracy and contraction.

Suppose first that the boundary is nonempty. At a seam point $(a,x,0)$ with $a \in [0,1]$, differentiation gives

$$dq_{(a,x,0)}(u,v,s) = \big(u, dj_x(v) + sZ_a(j(x))\big).$$

Indeed the spatial seam map is independent of a, its boundary derivative is dj_x, and its time derivative is the canonical field, which equals Z_a on the carrier. Differentiating the local identity $\rho \circ j = 0$ gives $\operatorname{im} dj_x \subset \ker d\rho$. The boundary half-space chart gives dimension $\dim M - 1$ for this boundary tangent space. Since dj_x is injective and $d\rho \neq 0$, both subspaces have that same dimension, so they are equal. Outwardness gives $d\rho(Z_a) > 0$. Given a target spatial vector w, set $s = d\rho(w)/d\rho(Z_a)$. Then $w - sZ_a$ belongs to $\operatorname{im} dj_x$, and the injective boundary differential gives its unique preimage v. The displayed differential is consequently invertible. The inverse function theorem gives an injective local diffeomorphism neighborhood at every such seam point.

The local-diffeomorphism locus of q is open and contains $[0,1] \times \partial D \times \{0\}$. Compactness of $[0,1] \times \partial D$ and the tube lemma give an open neighborhood A of $[0,1] \times \partial D$ and a width $\epsilon_0 > 0$ for which $A \times (-\epsilon_0, \epsilon_0)$ is in that locus. Apply the tube lemma once more to the compact factors $[0,1]$ and ∂D inside A. It gives an open $J_0 \supset [0,1]$ with $J_0 \times \partial D \subset A$.

We claim that a smaller parameter neighborhood and time strip make q injective. If this failed, choose integers $m \to \infty$ and distinct points (a_m, x_m, r_m), (a'_m, x'_m, r'_m) in $(-1/m, 1+1/m) \times \partial D \times (-1/m, 1/m)$ with equal images. The first coordinate of q is its parameter, so $a_m = a'_m$. Pass to a subsequence for which $a_m \to a \in [0,1]$, $x_m \to x$ and $x'_m \to x'$; this is possible because the boundary is a compact metrizable manifold and the parameters are bounded. Both time coordinates tend to zero. Continuity and equality of images give $j(x) = j(x')$. Injectivity of j gives $x = x'$. Both source sequences therefore converge to the same seam point $(a,x,0)$ and eventually lie in its injective inverse-function neighborhood. This contradicts their distinctness and equal images.

Thus for some sufficiently large m, injectivity holds on that product strip. Choose this m also so that $(-1/m, 1+1/m) \subset J_0$, and choose $0 < \epsilon \le \min(1/m, \epsilon_0, \delta)$. Put $J = (-1/m, 1+1/m)$. The retaining map is injective and a local diffeomorphism on the asserted open strip. A local diffeomorphism is open there; an injective continuous open map has continuous inverse onto its image, proving the open-embedding assertion. Shrinking the time interval preserves the already

derived original equation and contraction identities for parameters in $[0,1]$. If the boundary is empty, take $J = \mathbb{R}$ and any $0 < \epsilon \le \delta$. The source of the retaining map is empty, so embedding, local invertibility and all boundary-curve assertions hold, and the same argument supplies its smoothness.

For the domain-side identity, fix an interval parameter a and a boundary point x. If $c(a,x,r)$ lay in the frontier at a nonzero radius in the strip, it would equal $j(y) = c(a,y,0)$ for some boundary point y. Injectivity of the retaining map would then force $r = 0$, a contradiction. Thus the two open half intervals of the orbit avoid the frontier. In a defining-function neighborhood at $j(x)$ the time derivative of $\rho(c(a,x,r))$ at zero is $d\rho(Z_a) > 0$. Consequently sufficiently small negative radii are inside the domain and sufficiently small positive radii are outside. On each connected half interval the inverse images of the interior of D and of its open complement are disjoint open sets covering that interval, because the frontier is avoided. The corresponding local sign therefore persists throughout each half interval. At radius zero the boundary inclusion lies in the closed domain. These statements prove membership if and only if $r \le 0$.

For every interval parameter the constructed boundary curves on this same strip already satisfy the actual canonical equation and contraction identity. Apply the boundary primitive lemma to this very action and width, without changing any choice. It yields $c_a^*\lambda_a = e^r j^*\lambda_a$, including every spatial tangent and radial speed. When the boundary is empty both final assertions have empty source. Thus the single constructed action supplies all the claimed collar, side and primitive properties. □

Lemma A.119 (Pullback of simultaneously varying maps and primitives)**.** *Let M, N be smooth boundaryless finite-dimensional manifolds. Suppose $f_t : M \to N$ and the one-forms α_t on N depend jointly smoothly on $t \in [0,1]$ with open extensions across the parameter endpoints. Then $f_t^*\alpha_t$ is a jointly smooth one-form family with such endpoint extensions. The open extension neighborhoods in parameter and spatial coordinates need not contain one common larger time interval when the manifolds are noncompact.*

Proof. First take real-parameter extensions of the maps and forms having the specified local smoothness. Fix $t_0 \in [0,1]$ and source coordinates y near any point $x \in M$. Choose target coordinates z near $f_{t_0}(x)$. On an open neighborhood of $(t_0, y(x))$ the map family has coordinate expression $g(t,y)$ in this target chart, because the map extension is smooth and hence continuous there. Intersect this neighborhood with the inverse image, under $(t,y) \mapsto (t, g(t,y))$, of an open neighborhood on which the target one-form coefficients $a(t,z)$ are smooth. This intersection remains open and contains $(t_0, y(x))$.

In these coordinates the pulled-back covector is

$$a(t, g(t,y)) \circ D_y g(t,y), \qquad D_y g(t,y) = Dg(t,y) \circ \big(v \mapsto (0,v)\big).$$

The chain rule proves this formula for the actual spatial pullback. Smoothness of g gives smoothness of Dg, and restriction along the fixed linear inclusion $v \mapsto (0,v)$ gives smoothness of $D_y g$. The coefficients $a(t, g(t,y))$ are smooth by composition. Composition of a covector and a linear map is bilinear in finite-dimensional spaces, so the displayed covector coefficients are smooth on this same open neighborhood.

For each fixed source chart, take the union of these neighborhoods over all $t_0 \in [0,1]$ and all points of its coordinate domain. It is open, contains the whole required parameter interval times that coordinate domain, and the actual pullback coefficients are smooth on it, since smoothness is local and the local formulas describe the same pullback. This proves the real-extension formulation. For interval-indexed families choose their specified real extensions and pull those back. At each interval parameter both extensions equal the prescribed data, so their pullback equals $f_t^*\alpha_t$. This supplies the required endpoint extension of the interval family without a uniform time margin. □

Lemma A.120 (Extending a smooth boundary-collar family)**.** *Let $D \subset M$ be a compact domain presentation, and let J be an open real neighborhood of $[0,1]$. Suppose $\delta > 0$ and $c_t : \partial D \times \mathbb{R} \to M$ depend jointly smoothly on (t,x,r), with $c_t(x,0) = j(x)$ for all real t. Suppose the map $(t,x,r) \mapsto (t, c_t(x,r))$ is an open embedding and a local diffeomorphism on $J \times \partial D \times (-\delta, \delta)$. There exist $\eta > 0$ and a real-parameter family of diffeomorphisms $h_t : M \to M$ such that $h_0 = \mathrm{id}$, both h_t and h_t^{-1} depend jointly smoothly on parameter and point, and*

$$h_t(j(x)) = j(x), \qquad h_t(D) = D, \qquad h_t(c_0(x,r)) = c_t(x,r) \quad (0 \le t \le 1,\ |r| < \eta).$$

Proof. The retained-parameter collar has a smooth inverse on its open image: its local inverses agree by injectivity. The compact seam $[0,1] \times j(\partial D)$ lies in that open image. Choose a jointly smooth function $\chi(t,y)$ which is one on an open neighborhood V of the seam and has compact support contained in the collar image. In its inverse coordinates define the time-dependent spatial field

$$X_t(c_t(x,r)) = \chi(t, c_t(x,r)) \frac{\partial c_t(x,r)}{\partial t},$$

and put it equal to zero outside the collar image. The inverse coordinates show smoothness on the image. The support of χ is a closed subset contained in that open image; extension by zero is consequently smooth also at points outside the image. Its spatial projection lies in one compact set $K \subset M$ for all times; enlarging K to a compact neighborhood if necessary makes the field zero near its frontier. Differentiating $c_t(x,0) = j(x)$ in t gives $X_t(j(x)) = 0$.

Local smooth existence and uniqueness give a smooth evolution $H(s,t)y$ for this nonautonomous field. It exists for every finite time interval: outside K the constant curves solve the equation, and an integral curve starting in K cannot reach its frontier and cross it, since the field vanishes there at all times and uniqueness would identify that curve with the constant frontier trajectory. On each compact time interval the resulting curves therefore remain in a compact spatial set. A finite cover of its product with that time interval by local existence neighborhoods supplies continuation to both endpoints. Repeating on successively larger time intervals supplies all real times. Uniqueness gives $H(t,u)H(s,t) = H(s,u)$ and $H(t,t) = \mathrm{id}$. Local smooth dependence and finite compositions give joint smoothness in (s,t,y). Thus $h_t = H(0,t)$ is a diffeomorphism with inverse $H(t,0)$, and both families are jointly smooth.

The zero field at each boundary point makes its trajectory constant, so h_t fixes $j(\partial D)$. An interior trajectory cannot reach the boundary: at such a time uniqueness, used backwards, would identify it with the constant boundary trajectory. Its connected time interval therefore remains in the interior, since exiting the closed domain would cross its frontier. This proves $h_t(D) \subset D$; applying the same argument to the inverse gives equality. The inverse image of V under $(t,x,r) \mapsto (t, c_t(x,r))$ is an open neighborhood of the compact seam $[0,1] \times \partial D \times \{0\}$. The tube lemma gives $\eta > 0$, with $\eta \le \delta$, such that $(t, c_t(x,r)) \in V$ for $t \in [0,1]$ and $|r| < \eta$. For fixed (x,r) in that strip, $c_t(x,r)$ therefore solves the equation for X_t on $[0,1]$, since $\chi(t, c_t(x,r)) = 1$. Its initial point is $c_0(x,r)$; uniqueness gives the asserted collar identity. If the boundary is empty, take $X = 0$ and $h_t = \mathrm{id}$; the collar identity then has empty domain. □

Lemma A.121 (Compact presentations adapted to varying Liouville primitives)**.** *Let $D \subset M$ be a fixed compact domain presentation with Liouville structures (λ_t, Z_t) for $t \in [0,1]$, whose primitives depend jointly smoothly with extension across the parameter endpoints. There are $\eta > 0$, jointly smooth boundary evolutions $c_t : \partial D \times \mathbb{R} \to M$ for all real t, and real-parameter compact presentation diffeomorphisms $h_t : M \to M$ such that $h_0 = \mathrm{id}$, both h_t and h_t^{-1} are jointly smooth, h_t fixes the boundary pointwise, $h_t(D) = D$, and*

$$c_t(x,0) = j(x), \qquad h_t(c_0(x,r)) = c_t(x,r) \quad (0 \le t \le 1,\ |r| < \eta).$$

For $t \in [0,1]$ and $|r| < \eta$, the same boundary evolutions obey $c_t(x,r) \in D$ exactly when $r \le 0$. On the negative part of this strip, including $r = 0$, they solve the original spatial Liouville ODE

$$\frac{\partial c_t(x,r)}{\partial r} = Z_t(c_t(x,r)).$$

Proof. Apply the common Liouville collar lemma to the supplied primitive family. It produces one jointly smooth auxiliary action, an open neighborhood J of the parameter interval and a positive collar width. Its boundary evolution is jointly smooth for all real parameters and satisfies $c_t(x,0) = j(x)$ also outside the interval. On J times the common strip its retained-parameter map is an open embedding and a local diffeomorphism. Apply the boundary-collar extension lemma to this map. The resulting diffeomorphisms preserve the compact carrier and fix its boundary, so they are compact presentation identifications with identity boundary map. Their value at parameter zero is the identity. Restrict to the smaller positive strip supplied by that lemma. The collar square and the domain-side law hold there for these same maps. The generated evolution satisfies the canonical Liouville equation on its local time strip. When $r \le 0$, the domain-side law puts its value in D, where nondegeneracy and the contraction equation identify the canonical field with the supplied field Z_t. Substituting that equality into the generated ODE gives the displayed original-field ODE. No completion is an input to this construction. □

Lemma A.122 (Common realized flow collars and their compact presentations)**.** *Let $D \subset M$ be a fixed compact domain presentation with interval Liouville structures (λ_t, Z_t) whose primitives are jointly smooth with open endpoint extensions. There are a positive δ, realized flow collars $c_t : \partial D \times \mathbb{R} \to X_t$ of width δ in the symplectic neighborhoods $X_t = \{x : d\lambda_t \text{ is nondegenerate at } x\}$, and compact presentation diffeomorphisms $h_t : M \to M$ for all real t such that both h_t and h_t^{-1} are jointly smooth, $h_0 = \mathrm{id}$, $h_t(D) = D$, h_t fixes the boundary pointwise, and*

$$h_t(c_0(x,r)) = c_t(x,r) \quad (0 \le t \le 1,\ |r| < \delta).$$

Each collar is obtained from a smooth all-real auxiliary flow on X_t; on the indicated strip it is an injective local diffeomorphism, obeys $c_t(x,r) \in D$ exactly when $r \le 0$, solves the canonical Liouville equation, and has primitive $c_t^*\lambda_t = e^r \lambda_t|_{\partial D}$.

Proof. Carry out the common Liouville collar construction with the invariant auxiliary flow. It gives one jointly smooth boundary family on a strip of width $\delta_0 > 0$, with the seam, injectivity, side and primitive identities of the common collar lemma. Each parameter slice of the auxiliary flow preserves X_t for all real flow times, so it restricts to a smooth all-real flow there. The open inclusion into M identifies its boundary map with the constructed common boundary family. The local equation in X_t is its canonical contraction equation: restriction of the primitive and its differential preserves contraction, and nondegeneracy makes its solution unique.

The restricted collar remains injective. Its spatial derivative is invertible because the retained-parameter collar derivative is invertible: the first component of that derivative is the parameter variation, so restricting to zero parameter variation identifies its kernel and image with those of the spatial derivative. Injectivity and surjectivity of the full derivative therefore give injectivity and surjectivity of the spatial derivative. The inverse function theorem makes the spatial collar a local diffeomorphism. The side identity is unchanged by the open inclusion. The primitive pullback identity follows from the local contraction equation and the boundary-flow primitive calculation, with the same original boundary form.

Apply the boundary-collar presentation isotopy to this same boundary family. It gives h_t and a common smaller positive width $\delta \le \delta_0$. Restrict the realized collars to that width. All their

asserted strip identities persist, and the isotopy gives the displayed square, the boundary and carrier preservation, identity at zero, and joint smoothness of the forward and inverse maps. □

Lemma A.123 (Recovering a supplied completion from a negative collar equation)**.** *Let (D, λ, Z) be a Liouville domain presentation with a supplied realized completion C, compact inclusion i, and end map e. Suppose $\delta > 0$ and $c : \partial D \times \mathbb{R} \to M$ satisfies $c(x, 0) = j(x)$, $c(x, r) \in D$ for $-\delta < r \le 0$, and*

$$\frac{\partial c(x,r)}{\partial r} = Z(c(x,r)) \quad (-\delta < r \le 0).$$

Then $e(x, r) = i(c(x, r))$ for every $x \in \partial D$ and $-\delta < r \le 0$. Only the field equation on the compact carrier is required; no field identification outside it is assumed.

Proof. Fix x and put $a = \sqrt{\delta}$. For $|u| < a$, the radius $r = -u^2$ belongs to $(-\delta, 0]$. Consider the two curves

$$\gamma(u) = \Phi^{-u^2}(i(j(x))), \qquad \widetilde{\gamma}(u) = i(c(x, -u^2)),$$

where Φ is the supplied completed Liouville flow with field $\widehat{Z}$. The chain rule gives $\gamma'(u) = -2u\widehat{Z}(\gamma(u))$. For the second curve the radial ODE and the chain rule give

$$\widetilde{\gamma}'(u) = -2u\, di_{c(x,-u^2)} Z(c(x, -u^2)) = -2u\widehat{Z}(\widetilde{\gamma}(u)).$$

The last equality is precisely the completion's field equation on D; its use is valid because the negative-radius curve lies in that carrier. Both curves are equal to $i(j(x))$ at $u = 0$. The smooth nonautonomous field $(u, y) \mapsto -2u\widehat{Z}(y)$ has unique local integral curves. Its uniqueness identifies these two curves throughout $(-a, a)$: the equality set is closed by continuity and is open by uniqueness with the common value at any time, and the interval is connected. For $-\delta < r \le 0$, take $u = \sqrt{-r}$, which belongs to $(-a, a)$ and satisfies $-u^2 = r$. Equality of the curves at this time gives $\Phi^r(i(j(x))) = i(c(x, r))$. The left side is $e(x, r)$ by the completion's end-orbit identity. This proves the assertion also at $r = 0$, without requiring a positive-radius equation for c. □

Lemma A.124 (Whole-completion regularity from a common compact collar)**.** *Let C and C_t be supplied realized completions of Liouville domain presentations D and D_t, respectively. Suppose compact presentation identifications $h_t : D \to D_t$, with boundary maps k_t, are supplied for $t \in [0, 1]$ and that their pulled-back compact primitives $\alpha_t = h_t^* \lambda_t$ form a jointly smooth family with open endpoint extensions. Suppose $\delta > 0$ is common to all parameters and the compact and end maps satisfy*

$$i(x) = e(y, r), \quad x \in D, \quad |r| < \delta \quad \Longrightarrow \quad i_t(h_t(x)) = e_t(k_t(y), r).$$

There are completion diffeomorphisms P_t agreeing with h_t on the compact carriers and with $(y, r) \mapsto (k_t(y), r)$ on their nonnegative cylindrical ends. On the fixed completed manifold of C, the forms $P_t^* \widehat{\lambda}_t$ and the pulled-back completed Liouville fields are jointly smooth with open endpoint extensions. No regularity of the supplied completions as a parameter family is assumed.

Proof. Use the compact interior and the cylindrical patch $r > -\delta$, which contains the seam and the whole nonnegative end. On their overlap one has $-\delta < r \le 0$, since any end point lying in the compact carrier has nonpositive radius. The assumed square on precisely this negative collar makes the compact and cylindrical formulas agree. On the interiors they are diffeomorphisms; on the open collar crossing the boundary they are the smooth coordinate change $(y, r) \mapsto (k_t(y), r)$. Local inverse coordinates exist because the end-orbit differential is invertible. The inverse compact and cylindrical formulas agree on the same overlap by the assumed square. These maps therefore glue to a diffeomorphism P_t of the whole completions, with the two asserted squares.

It remains to prove regularity across the compact-end seam. Choose a real extension a_t of α_t having its prescribed joint smooth coefficients, and write $\beta_t = j^* a_t$. In fixed two-sided compact collar coordinates around the boundary of C, put $b_t = e^r \beta_t$, with zero radial component. Such coordinates exist after shrinking the width, since the compact inclusion is a local diffeomorphism near its compact boundary and the end-orbit differential is invertible there. For interval parameters, the compact square and the exponential primitive identities of both completions imply $a_t = b_t$ on this collar inside the compact interior. Choose one smooth spatial cutoff χ supported in this collar and equal to one on a smaller neighborhood of the compact boundary. Replace the compact extension there by

$$a'_t = a_t + \chi(b_t - a_t),$$

and leave it equal to a_t elsewhere. For interval parameters it remains a_t on the compact interior. For every real parameter it equals b_t on the smaller collar, and its boundary restriction remains β_t: at the seam the tangential restriction of b_t is exactly β_t. The correction is jointly smooth on open neighborhoods of the required parameter interval in each chart, since its coefficient operations are smooth and its support is confined to the coordinate collar.

Transport a'_t through the fixed compact inclusion on its open interior, and use $e^r \beta_t$ in the fixed open cylindrical collar, including the whole nonnegative end. They agree on the smaller negative collar for all real parameters and hence define a real extension of $P_t^* \widehat{\lambda}_t$ over the entire fixed completion. In either kind of chart its coefficients are jointly smooth: the first formula is a pullback through a fixed smooth inverse; the second uses that inverse, the smooth factor e^r, and the jointly smooth boundary coefficients. Around each required time and point these formulas give an open neighborhood of joint smoothness. In any fixed spatial chart their union is an open neighborhood of the whole interval and coordinate domain; smoothness on this union follows locally. Thus the completed primitive family has the required endpoint extension, including on its noncompact end.

Take spatial exterior derivatives of these jointly smooth coefficients. Their contraction matrices are jointly smooth and invertible at every interval parameter because P_t pulls back a symplectic differential. The invertibility locus is open. Inverting on that locus and applying the inverse matrix to the primitive coefficients gives a jointly smooth extension of the canonical Liouville field. Naturality of contraction identifies this canonical field with the pulled-back completed field at every interval parameter. This proves its regularity without any regularity assumption on the chosen pointwise completions. □

Definition A.125 (Gluing compact and cylindrical map extensions)**.** *Let C, C' be realized completions of compact domain presentations $D \subset X$, $D' \subset X'$, with compact inclusions i, i' and end maps e, e'. For real-parameter maps $g_t : X \to X'$ and $b_t : \partial D \to \partial D'$, define a map $G_t : C \to C'$ by*

$$G_t(i(x)) = i'(g_t(x)) \quad (x \in D), \qquad G_t(e(y,r)) = e'(b_t(y), r) \quad (e(y,r) \notin i(D),\ r \geq 0).$$

The compact inclusion is injective, the end map is injective, and their images cover C, so these clauses specify one value at every point. The compact clause is used on their intersection. No carrier preservation, primitive identity or invertibility is required of these real-parameter maps.

Lemma A.126 (The glued map agrees on an open cylindrical collar)**.** *Fix a parameter t and a real number δ. Suppose that*

$$i(x) = e(y,r), \quad x \in D, \quad -\delta < r \quad \Longrightarrow \quad i'(g_t(x)) = e'(b_t(y), r).$$

Then the glued map satisfies

$$G_t(i(x)) = i'(g_t(x)) \quad (x \in D), \qquad G_t(e(y,r)) = e'(b_t(y), r) \quad (-\delta < r).$$

Proof. The compact clause follows from the definition and injectivity of i. If $e(y,r)$ lies in $i(D)$, write it as $i(x)$ and apply the assumed overlap equality to this same point. If it does not lie in $i(D)$, coverage gives a nonnegative end representation of that point. Injectivity of e identifies its boundary point and radius with (y,r). The end clause of the definition therefore gives the asserted equality. □

Lemma A.127 (Joint smoothness of the completed map from coherent extensions)**.** *Let $V \subset \mathbb{R}$ be open and $\delta > 0$. Suppose $(t,x) \mapsto g_t(x)$ and $(t,y) \mapsto b_t(y)$ are smooth on $V \times X$ and $V \times \partial D$, respectively, and the overlap equality of the preceding lemma holds for every $t \in V$. Then $(t,z) \mapsto G_t(z)$ is smooth on $V \times C$. In particular, if $[0,1] \subset V$, this is a smooth extension across both parameter endpoints and across the compact-end seam. The compatibility hypothesis concerns the real-parameter extensions on an open time neighborhood; interval-only compatibility does not by itself supply that hypothesis.*

Proof. Fix $(t,z) \in V \times C$. If $z = i(x)$ with x in the interior of D, the compact inclusion has a smooth local inverse a near z. Shrink its source so that a takes values in the interior of D. On this neighborhood the compact identity gives $G_s(w) = i'(g_s(a(w)))$. This composition is jointly smooth for $s \in V$.

Otherwise completion coverage gives $z = e(y,r)$ with $r \geq 0$. The end map has a smooth local inverse c near z. Since $r > -\delta$, continuity permits restriction to a neighborhood where the radius component of $c(w)$ is greater than $-\delta$. The open-collar identity gives

$$G_s(w) = e'\big(b_s(\mathrm{pr}_1\, c(w)), \mathrm{pr}_2\, c(w)\big) \qquad (s \in V).$$

Every factor is smooth on this neighborhood. This formula also applies at radius zero and on its negative-radius side. The two cases cover all completed points, so joint smoothness follows locally everywhere on the stated open product. If the parameter interval lies in V, the same open product supplies its endpoint extensions. □

Lemma A.128 (Prescribed interval maps retain a smooth completed extension)**.** *Under the hypotheses of the preceding lemma, suppose $[0,1] \subset V$ and maps $f_t : C \to C'$ are prescribed for $t \in [0,1]$, with*

$$f_t(i(x)) = i'(g_t(x)) \quad (x \in D), \qquad f_t(e(y,r)) = e'(b_t(y), r) \quad (r \geq 0).$$

Then $G_t = f_t$ for every interval parameter, and G gives a jointly smooth extension of this same prescribed map family to $V \times C$.

Proof. At a compact-image point the compact identities for G_t and f_t give equality. Outside that image, coverage supplies a nonnegative end representation, and both end identities give equality there. Thus the maps agree everywhere, including both interval endpoints. The preceding joint-smoothness lemma supplies the asserted extension. □

Lemma A.129 (A strict completed comparison family from coherent compact extensions)**.** *Fix two compact Liouville domain presentations and their realized completions C, C'. For each $t \in [0,1]$, let $h_t : X \to X'$ be a smooth presentation diffeomorphism carrying D onto D' and inducing a boundary diffeomorphism k_t. Suppose $h_t^* \lambda' = \lambda$ on the entire compact carrier D. Let $V \supset [0,1]$ be open and $\delta > 0$. Supply jointly smooth maps $g, \bar{g}, b, \bar{b}$ on, respectively, $V \times X$, $V \times X'$, $V \times \partial D$ and $V \times \partial D'$. At interval parameters these maps agree with h_t, h_t^{-1} on their compact carriers and with k_t, k_t^{-1} on their boundaries. Suppose for every $t \in V$ that*

$$i(x) = e(y,r),\ x \in D,\ -\delta < r \implies i'(g_t(x)) = e'(b_t(y), r),$$

$$i'(x') = e'(y', r),\ x' \in D',\ -\delta < r \implies i(\bar{g}_t(x')) = e(\bar{b}_t(y'), r).$$

There are primitive-preserving diffeomorphisms $H_t : C \to C'$ such that both H_t and H_t^{-1} have jointly smooth extensions across the interval endpoints. They satisfy, for every interval t,

$$H_t(i(x)) = i'(h_t(x)), \qquad H_t^{-1}(i'(x')) = i(h_t^{-1}(x')) \quad (x \in D,\ x' \in D'),$$

and, for every real radius r,

$$H_t(e(y, r)) = e'(k_t(y), r), \qquad H_t^{-1}(e'(y', r)) = e(k_t^{-1}(y'), r).$$

The completions in this assertion are fixed. The supplied open-time collar compatibility is an additional extension hypothesis; it is not asserted to follow from the original compact interval data.

Proof. At each interval parameter apply the compact-presentation completion extension theorem with scale one to h_t. It constructs a strict diffeomorphism H_t with the prescribed compact square. Primitive preservation intertwines the Liouville fields, and uniqueness of their all-real integral curves gives the stated end square at every real radius. Apply H_t^{-1} to the compact square at $h_t^{-1}(x')$ to obtain the inverse compact square. Applying it to the end square at $k_t^{-1}(y')$ gives the inverse end square.

The real maps g, b agree with these compact and end restrictions on the interval. Their open-time collar compatibility therefore lets the prescribed-interval extension lemma extend the same H_t jointly smoothly. Apply that lemma in the reverse direction to $\bar{g}, \bar{b}$ and the inverse squares to extend the same H_t^{-1} jointly smoothly. No inverse or primitive-preservation equation for these extensions outside the interval is used. □

Lemma A.130 (Joint completed map extensions in dimension zero)**.** *For zero-dimensional compact domains and their fixed realized completions, a jointly smooth compact map extension g on $V \times X$ makes the glued completed map jointly smooth on $V \times C$. No boundary-map regularity or collar compatibility is needed. If $[0, 1] \subset V$ and a prescribed interval map family f_t satisfies $f_t(i(x)) = i'(g_t(x))$ on D, this same glued map extends f_t.*

Proof. The boundary of a zero-dimensional compact domain is empty. All boundary regularity and overlap conditions in the completed-map smoothness lemma therefore have empty source; use any positive collar width. That lemma gives joint smoothness. Completion coverage, with empty cylindrical end, says $i(D) = C$. At every such compact point the definition of the glued map and the prescribed compact square agree. Thus the extension retains f_t everywhere on the interval. □

Lemma A.131 (Strict completed comparison families in dimension zero)**.** *Fix two zero-dimensional compact Liouville presentations and their realized completions. Let h_t be prescribed presentation diffeomorphisms for $t \in [0, 1]$. Suppose jointly smooth maps $g, \bar{g}$ on $V \times X, V \times X'$, for an open $V \supset [0, 1]$, agree on the interval with h_t, h_t^{-1} on their whole extension spaces. There exist strict completed diffeomorphisms $H_t : C \to C'$ with jointly smooth forward and inverse extensions and the exact squares*

$$H_t(i(x)) = i'(h_t(x)), \qquad H_t^{-1}(i'(x')) = i(h_t^{-1}(x')) \quad (x \in D,\ x' \in D').$$

No primitive equation, boundary extension, collar compatibility or nonemptiness assumption is required.

Proof. Every one-form on a zero-dimensional manifold vanishes, so $h_t^*\lambda' = \lambda$ holds automatically. The compact-presentation completion extension theorem with scale one constructs H_t and its compact square. Apply its inverse to this square at $h_t^{-1}(x')$ to obtain the inverse square. The zero-dimensional completed-map extension lemma applied to g and $\bar{g}$ proves joint smoothness of these same maps in both directions. Its proof uses only compact coverage and does not choose a point, so it includes empty domains. □

Lemma A.132 (Closed end cutoffs and open outward neighborhoods)**.** *For a fixed realized completion with end map e, and every real a, $e(\partial D \times [a, \infty))$ is closed and $e(\partial D \times (a, \infty))$ is open in the completed ambient.*

Proof. The nonnegative end is the complement of the compact interior image. The compact inclusion is a local diffeomorphism, so that interior image is open. Thus the nonnegative end is closed. The complete flow at time a is a homeomorphism and sends $e(y, r)$ to $e(y, a + r)$. Its image of the nonnegative end is exactly $e(\partial D \times [a, \infty))$, proving closedness. The all-real end map is a local diffeomorphism, hence an open map. Its image of the open set $\partial D \times (a, \infty)$ is open. □

Lemma A.133 (Joint cylindrical formulas on the actual escaping region)**.** *Let C be a fixed completion, N a smooth manifold, and $\beta(t, y, r)$ a jointly smooth map on $V \times \partial D \times \mathbb{R}$ for an open $V \subset \mathbb{R}$. On the actual region $\mathcal{E} = e(\partial D \times \mathbb{R}) \subset C$, the formula $F(t, e(y, r)) = \beta(t, y, r)$ defines a jointly smooth map on $V \times \mathcal{E}$.*

Proof. The injective all-real end map is a local diffeomorphism. Its image is open, and its local smooth inverses agree by injectivity. They give one smooth inverse $q : \mathcal{E} \to \partial D \times \mathbb{R}$. The formula is $F(t, z) = \beta(t, q(z))$, a composition of smooth maps. □

Lemma A.134 (Compact germs and a small inward end strip extend the whole interval map)**.** *Fix a completion C and a smooth target manifold N. Let $f_t : C \to N$ be prescribed for $t \in [0, 1]$. Suppose $a < 0$, $V \supset [0, 1]$ is open, and a jointly smooth $\beta : V \times \partial D \times (a, \infty) \to N$ satisfies*

$$f_t(e(y, r)) = \beta(t, y, r) \quad (t \in [0, 1],\ r > a).$$

At each $(t, i(x))$ with $t \in [0, 1]$ and $x \in D$, suppose there is a smooth target-valued map on an open neighborhood in $\mathbb{R} \times C$ agreeing with $(s, z) \mapsto f_s(z)$ at all interval points of that neighborhood. Then f has a jointly smooth extension on an open neighborhood of $[0, 1] \times C$. The compact germs and cylindrical extension need agree only at actual interval parameters. The inward end formula is required only for the finite band $a < r \leq 0$.

Proof. In $\mathbb{R} \times C$ put $A = [0, 1] \times C$ and prescribe $u(t, z) = f_t(z)$ on A. The compact set $K = [0, 1] \times i(D)$ is covered by the supplied local germs. On $O = V \times e(\partial D \times (a, \infty))$ use the cylindrical map $b(t, e(y, r)) = \beta(t, y, r)$. The end coordinates give smooth local inverses taking values in $r > a$, so this map is smooth on O. Its interval restriction agrees with the prescribed carrier map. The sets

$$L = [0, 1] \times e(\partial D \times [0, 2]), \qquad B = \mathbb{R} \times e(\partial D \times [1, \infty))$$

are compact and closed, respectively. The end-cutoff lemma gives closedness of B; compactness of the boundary gives compactness of L. Moreover $L \subset O$ because $a < 0$, and $K \cap B = \varnothing$, since positive end radii lie outside the compact carrier.

Apply the exterior-preserving compact-germ lemma to u, K, L, B, O, b. It gives a smooth h on an open $W \supset K \cup L$, agreeing with u on $A \cap W$ and with b on $W \cap B$. The open set $P = V \times e(\partial D \times (1, \infty))$ lies in $O \cap B$. Hence h and $b|_P$ agree on $W \cap P$ and glue to a smooth map

on $W \cup P$. Every interval compact point lies in $K \subset W$. An interval end point of radius in $[0,2]$ lies in $L \subset W$, and one of radius greater than 2 lies in P. Completion coverage therefore gives $[0,1] \times C \subset W \cup P$. Both glued clauses equal u at interval points, so this is an extension of the same prescribed f. If a map on all real parameters is desired, give it the value $f_0(z)$ outside $W \cup P$; this does not change the asserted smoothness on that open neighborhood and requires no additional target point. □

Lemma A.135 (Independent compact germs and all-real cylindrical formulas extend the interval map)**.** *Fix a completion C and a smooth target manifold N. Let $f_t : C \to N$ be prescribed for $t \in [0,1]$. Suppose an open $V \supset [0,1]$ and a jointly smooth $\beta : V \times \partial D \times \mathbb{R} \to N$ satisfy $f_t(e(y,r)) = \beta(t,y,r)$ at every interval parameter and every real radius. Suppose every interval compact point has a smooth target-valued local extension agreeing with f on the interval. Then this same f has a jointly smooth extension on an open neighborhood of $[0,1] \times C$.*

Proof. Restrict the cylindrical formula to $r > -1$. Its smoothness and interval equality are retained. Apply the preceding strip extension lemma with $a = -1$ and the supplied compact germs. Its extension agrees with f at every interval point, as required. □

Lemma A.136 (Independent compact and end formulas give joint completed regularity)**.** *Fix C, N, V, f_t as in the preceding lemma. Suppose jointly smooth maps $g : V \times X \to N$ and $\beta : V \times \partial D \times \mathbb{R} \to N$ satisfy*

$$f_t(i(x)) = g(t,x) \quad (x \in D), \qquad f_t(e(y,r)) = \beta(t,y,r) \quad (r \in \mathbb{R})$$

for every interval t. Then this same f has a jointly smooth extension on an open neighborhood of $[0,1] \times C$. No off-interval matching condition between g and β is imposed.

Proof. At a compact interior point choose a smooth local inverse q of i taking values in the interior of D. The formula $g(s, q(z))$ is a smooth target-valued germ and equals $f_s(z)$ at every interval point of that neighborhood by the compact square. A compact point outside the interior image belongs to the nonnegative end by completion coverage. It therefore lies in the open escaping region $\mathcal{E}$, where the preceding cylindrical formula gives a smooth germ agreeing with f throughout the interval. These two cases supply the required germs at every point of $[0,1] \times i(D)$. Apply the independent-germ extension lemma to them and the supplied cylindrical formula. It retains all prescribed interval values. □

Lemma A.137 (Strict completed comparison families from independent interval extensions)**.** *For two fixed compact Liouville presentations and their fixed realized completions, supply prescribed interval presentation diffeomorphisms h_t, with boundary maps k_t, satisfying $h_t^* \lambda' = \lambda$ on the entire source compact carrier. Suppose h_t, h_t^{-1} on the compact carriers and k_t, k_t^{-1} on the boundaries have independent jointly smooth extensions $g, \bar{g}, b, \bar{b}$ on $V \times X, V \times X', V \times \partial D, V \times \partial D'$ for an open $V \supset [0,1]$. Then there are strict completed diffeomorphisms $H_t : C \to C'$ with jointly smooth forward and inverse interval extensions, the exact compact forward and inverse squares, and the end squares*

$$H_t(e(y,r)) = e'(k_t(y), r), \qquad H_t^{-1}(e'(y',r)) = e(k_t^{-1}(y'), r) \quad (r \in \mathbb{R}).$$

Neither open-time collar compatibility nor inverse or primitive laws outside $[0,1]$ are required. This assertion compares fixed completions; it does not assert a correspondence for arbitrary moving completed families.

Proof. At each interval parameter the compact primitive identity constructs the strict completed diffeomorphism H_t with its compact square. Strict flow transport extends its boundary square to every real radius. Applying its inverse to these squares gives the inverse compact and end squares, as in the unique completion-extension lemma.

For the forward family take the independent compact formula $i' \circ g_t$ and the end formula $e'(b_t(y), r)$. These are jointly smooth because the target inclusion and its all-real end map are fixed smooth maps. On the actual interval they coincide with the prescribed compact and end restrictions of this same H_t. The independent-formula extension lemma therefore proves joint regularity of H_t. Apply it in the reverse direction to $i \circ \bar{g}_t$ and $e(\bar{b}_t(y'), r)$ to prove joint regularity of the same H_t^{-1}. The all-real end regularity here is derived from fixed end maps and the supplied boundary extensions, and is not an additional hypothesis on a moving completed family. □

Lemma A.138 (Coherent presentations of supplied varying completions)**.** *Let D be one fixed compact domain presentation with Liouville structures (λ_t, Z_t) for $t \in [0,1]$, whose primitives are jointly smooth with open endpoint extensions. For each parameter let C_t be a supplied realized completion of this original presentation. Then there exist compact presentation diffeomorphisms h_t for all real t, with identity boundary maps and $h_0 = \mathrm{id}$, such that h_t and h_t^{-1} are jointly smooth, and completion presentation diffeomorphisms $P_t : C_0 \to C_t$ for $t \in [0,1]$ agreeing with h_t on the compact carriers and with the identity boundary and radial coordinates on the ends. The pulled-back completed primitives and Liouville fields on C_0 are jointly smooth with open endpoint extensions. The pointwise completions are hypotheses; their family regularity and compact collar compatibility are conclusions.*

Proof. The compact presentation lemma constructs h_t and boundary evolutions c_t on a common strip of width $\eta > 0$. They have stationary seam, the exact domain-side law and the original-field ODE for negative radii. Apply the negative-collar recovery lemma at each parameter, including zero. For $-\eta < r \le 0$ it gives

$$e_t(y, r) = i_t(c_t(y, r)), \qquad e_0(y, r) = i_0(c_0(y, r)).$$

If $i_0(x) = e_0(y, r)$ with $x \in D$ and $|r| < \eta$, the completion's domain-side law gives $r \le 0$. The second displayed equality and injectivity of i_0 then give $x = c_0(y, r)$. The constructed compact square gives $h_t(x) = c_t(y, r)$, and the first displayed equality yields $i_t(h_t(x)) = e_t(y, r)$. Thus the common compact-end square is derived for the supplied completions, using their field equations only on the compact carrier.

The simultaneous pullback lemma applied to the jointly smooth h_t and the given primitive family makes $h_t^* \lambda_t$ jointly smooth with open endpoint extensions. The whole-completion regularity lemma, applied to this family and the derived square, now produces P_t and the stated completed primitive and field families. The boundary maps are the identity because those of h_t are the identity. The smooth forward and inverse compact maps and $h_0 = \mathrm{id}$ are retained from their original construction. No pointwise completion is constructed by this argument. □

Lemma A.139 (Localizing a collar-compatible presentation)**.** *Let c, c' be realized flow collars of equal positive width δ for Liouville domain presentations (D, λ) and (D', λ'). Let h be a compact presentation diffeomorphism with boundary map k, and suppose*

$$h(c(x, r)) = c'(k(x), r) \quad (|r| < \delta).$$

Let U, U' be their interior-and-collar localized extensions, with open inclusions a, a' into the original extensions. Then h restricts to a compact presentation diffeomorphism $H : U \to U'$ with boundary

map k, satisfying

$$a' \circ H = h \circ a, \qquad H^*(a'^*\lambda') = a^*(h^*\lambda').$$

For the cylindrical completions constructed from these collars, the same H is a completion presentation diffeomorphism. It satisfies the compact square and the end square $H(e(x,r)) = e'(k(x),r)$ for all $r \geq 0$.

Proof. As subsets of the original extensions, the localized neighborhoods are

$$U = \operatorname{int}(D) \cup c(\partial D \times (-\delta,\delta)), \quad U' = \operatorname{int}(D') \cup c'(\partial D' \times (-\delta,\delta)).$$

A homeomorphism carries interiors to interiors; hence carrier preservation gives $h(\operatorname{int}(D)) = \operatorname{int}(D')$. The collar square and bijectivity of k give equality of the two collar images under h. Therefore $h(U) = U'$. Restrict h and its smooth inverse to obtain H. Its carrier and boundary squares are the restrictions of those of h. The first displayed identity is the defining restriction identity. Pullback composition gives the second, since the localized primitives are exactly the pullbacks through a and a'.

For every $r > -\delta$, the constructed cylindrical end map has the formula $e(x,r) = c(x, \ell_\delta^{-1}(r))$, where ℓ_δ is the scalar collar stretch and its inverse lies in $(-\delta,\delta)$. Both collars use the same stretch because their widths agree. Applying the collar square at that inverse radius gives $H(e(x,r)) = e'(k(x),r)$. In particular this holds for every nonnegative radius. Both constructed completions have the identity compact inclusion in their localized extensions, so the compact square is immediate. Thus H with these two squares is the claimed completion presentation diffeomorphism. □

Lemma A.140 (Smooth cylindrical completion families from compatible collars)**.** *Let $D \subset M$ carry interval Liouville structures (λ_t, Z_t) with jointly smooth primitives and open endpoint extensions. Suppose realized flow collars c_t of one common positive width δ and real-parameter compact presentation diffeomorphisms h_t are given, with jointly smooth forward maps and boundary maps k_t, satisfying*

$$h_t(c_0(x,r)) = c_t(k_t(x),r) \quad (0 \leq t \leq 1,\ |r| < \delta).$$

Let U_t be their localized extensions and C_t their constructed cylindrical completions. There are localized compact presentation maps $H_t : U_0 \to U_t$ with boundary maps k_t and $a_t \circ H_t = h_t \circ a_0$, where $a_t : U_t \to M$ is the open inclusion. Their transported localized primitives are jointly smooth with open endpoint extensions. There are completion presentation diffeomorphisms $P_t : C_0 \to C_t$, with $P_t = H_t$ on these constructed ambient manifolds, whose transported completed primitives and Liouville fields are jointly smooth with open endpoint extensions. The collars and their compatibility are hypotheses; the pointwise completions and the completed regularity are constructed.

Proof. Localize each collar-compatible h_t by the preceding lemma. It gives H_t, its boundary map and open-inclusion square, and the exact formula

$$H_t^*(a_t^*\lambda_t) = a_0^*(h_t^*\lambda_t).$$

Simultaneous pullback makes $h_t^*\lambda_t$ jointly smooth with endpoint extensions on M. Pullback through the fixed smooth open inclusion a_0 preserves this regularity, so the displayed equality gives the asserted localized primitive family.

Construct C_t from each supplied collar. The preceding lemma gives its completion presentation map $P_t = H_t$. Its end square holds on the larger strip $r > -\delta$, including the negative compact overlap. Consequently, if the compact inclusion in C_0 represents an end point of radius $|r| < \delta$, applying H_t gives exactly the corresponding end point in C_t. This is the common compact-end

square required by the whole-completion regularity lemma. Apply that lemma's primitive gluing and canonical-field construction to these presentation maps and the localized smooth primitive family. It proves joint smoothness of their transported completed primitives and fields on all of C_0, with open endpoint extensions. This uses the constructed maps $P_t = H_t$ throughout, and assumes no pointwise completion or completed-family regularity. □

Lemma A.141 (Constructed localized completion families for varying primitives)**.** *Let $D \subset M$ be a fixed compact domain presentation carrying Liouville structures (λ_t, Z_t) for $t \in [0,1]$, with jointly smooth primitives and open endpoint extensions. There exist realized flow collars c_t of one common positive width, their localized extensions U_t with open inclusions $a_t : U_t \to M$, and the cylindrical completions C_t constructed from those collars. There are real-parameter compact presentation diffeomorphisms $h_t : M \to M$ with jointly smooth forward and inverse maps, identity boundary maps and $h_0 = \mathrm{id}$, and localized compact presentation diffeomorphisms $H_t : U_0 \to U_t$ satisfying*

$$a_t \circ H_t = h_t \circ a_0, \qquad H_t|_{\partial D} = h_t|_{\partial D}, \qquad H_0 = \mathrm{id}.$$

Both the transported localized primitives and their Liouville fields are jointly smooth with open endpoint extensions. There are completion presentation diffeomorphisms $P_t : C_0 \to C_t$ with $P_t = H_t$ on their constructed ambient manifolds and $P_0 = \mathrm{id}$. Their transported completed primitives and Liouville fields are jointly smooth on the entire fixed completion, with open endpoint extensions. Only the original compact primitive family is assumed jointly smooth; the collars, pointwise completions, presentation compatibility and completed regularity are conclusions.

Proof. The common realized-collar construction produces the collars and h_t, including their common width, collar square, boundary identity, joint smoothness of the forward and inverse maps, and $h_0 = \mathrm{id}$. Apply the cylindrical-family construction from compatible collars to these produced data. It constructs U_t, C_t, H_t, P_t and proves the original-inclusion and boundary squares, joint smoothness of the transported localized primitives, and both completed regularity assertions. Its equality $P_t = H_t$ concerns these same constructed maps. At $t = 0$ the inclusion square gives $a_0(H_0(z)) = a_0(z)$; injectivity of a_0 gives $H_0(z) = z$. Consequently $P_0 = \mathrm{id}$ as well.

For localized field regularity put $\alpha_t = H_t^*(a_t^*\lambda_t)$ and let Y_t be the pulled-back localized Liouville field. The differential of the localized primitive is nondegenerate everywhere on U_t, and its field satisfies contraction everywhere there, by the symplectic-neighborhood restriction construction. Naturality of exterior differentiation and contraction under H_t therefore give

$$d\alpha_t = H_t^* d(a_t^*\lambda_t), \qquad \iota_{Y_t} d\alpha_t = \alpha_t.$$

The first form is nondegenerate because H_t is a diffeomorphism. In a spatial chart, a smooth endpoint extension of α_t has jointly smooth coefficients, and its spatial exterior derivative has jointly smooth contraction matrices. At every interval parameter and every point of U_0 these matrices are invertible. Their invertibility loci are open. On those loci the inverse matrices applied to the primitive coefficients give jointly smooth field coefficients. Uniqueness of the contraction solution identifies them with Y_t at interval parameters and makes the local constructions agree on chart overlaps. The unions of these open coefficient neighborhoods cover $[0,1] \times U_0$, giving the required endpoint extensions. No common larger parameter interval on the noncompact U_0 is needed. □

Lemma A.142 (Constructing a localized witnessed pair deformation)**.** *Let n be a nonnegative integer, and let D, F be fixed compact domain presentations of dimensions $2n$ and $\max(2n-2, 0)$, respectively. Let (λ_t, Z_t) and (μ_t, Y_t) be interval Liouville structures on them, with jointly smooth*

primitives and open endpoint extensions. Suppose embeddings j_t of the whole chosen ribbon extension into ∂D and functions $b_t : \partial D \to \mathbb{R}$ form jointly smooth interval families with open endpoint extensions. Assume $b_t > 0$ everywhere and

$$\mu_t = j_t^*(b_t \alpha_t) \quad \text{at every point of } F, \qquad \alpha_t = \lambda_t|_{\partial D}.$$

There are realized ambient and ribbon flow collars, their localized domains D_t, F_t with open inclusions a_t^D, a_t^F, and a presented pair deformation whose parameter tuple is exactly

$$\big(D_t, (a_t^D)^*\lambda_t, F_t, j_t \circ a_t^F, b_t\big),$$

with ambient and ribbon completions given by the cylindrical constructions from those same collars. No collar, completion or presentation compatibility is an input. For $n \geq 1$ the dimensions are precisely those of the source's witnessed pairs. This result starts from chosen extension presentations and embeddings of their whole ribbon extensions.

Proof. Apply the localized-completion family construction to D with λ_t and to F with μ_t. It supplies both collar families, their localized compact presentations H_t^D, H_t^F, their original presentation maps h_t^D, h_t^F, and their completed presentations. All four transported primitive and field families, on the localized ambient and ribbon and on both entire completions, have the required joint smoothness and endpoint extensions.

Compose each j_t with the smooth open inclusion a_t^F. This is a smooth embedding. The ambient boundary form is still α_t, and the witness remains the given smooth positive b_t. Pullback composition and literal restriction through the open inclusion give, at every point of the localized ribbon carrier,

$$(j_t \circ a_t^F)^*(b_t\alpha_t) = (a_t^F)^* j_t^*(b_t\alpha_t) = (a_t^F)^*\mu_t.$$

The second equality uses the assumed primitive identity at the corresponding original carrier point. Thus each displayed tuple is a witnessed pair with exactly the constructed ambient and ribbon completions.

In the fixed presentations at parameter zero, its transported ribbon embedding is

$$(H_t^D|_{\partial D})^{-1} \circ j_t \circ a_t^F \circ H_t^F = j_t \circ h_t^F \circ a_0^F.$$

Here the ambient boundary map is the identity, and the ribbon open-inclusion square gives $a_t^F \circ H_t^F = h_t^F \circ a_0^F$. The right side is jointly smooth with open endpoint extensions by composition of the given embedding family with the globally smooth original ribbon presentation family and the fixed inclusion. The transported witness is exactly b_t, again because the ambient boundary map is the identity, so its given endpoint extensions suffice. Together with the constructed compact and completed presentations and their primitive and field regularity, these are all the data of a presented pair deformation. Its parameter tuples are the specified localized pairs throughout. □

Lemma A.143 (Local injectivity and immersion from an injective differential)**.** *Let $f : M \to N$ be a map between smooth boundaryless manifolds, smooth on an open subset $W \subset M$. If df_a is injective at $a \in W$, there is an open neighborhood $U \subset W$ of a on which f is injective and df is injective at every point.*

Proof. Choose source and target charts and a linear left inverse L of the coordinate derivative of f at a. The coordinate map for f followed by L has identity derivative at a. The inverse function theorem gives a neighborhood U on which this composition is a diffeomorphism. Shrink U inside W and the chosen chart domains. Equality of two f-images gives equality of their images under

the composition, hence equality of the original points. At every point of U, the chain rule expresses the invertible derivative of the composition as L times the coordinate derivative of f. Thus the latter derivative is injective, as asserted. □

Lemma A.144 (A compact embedding neighborhood)**.** *Let X be locally compact and Hausdorff, let Y be Hausdorff, and let $K \subset X$ be compact. Suppose $f : X \to Y$ is continuous on an open neighborhood W of K, injective on K, and injective on some neighborhood of each point of K. There is an open neighborhood U of K with compact closure contained in W such that $f|_U$ is a topological embedding.*

Proof. For each $(a, a) \in K \times K$, local injectivity gives a product neighborhood on which $f(x) = f(y)$ implies $x = y$. For distinct $a, b \in K$, their images are distinct; disjoint open neighborhoods of these images and continuity give a product neighborhood on which $f(x) \neq f(y)$. The union of all these neighborhoods is an open set P containing $K \times K$ on which the implication holds.

For each $a \in K$, finitely many product neighborhoods in P cover $\{a\} \times K$. Intersect their first factors and unite their second factors to obtain $A_a \times B_a \subset P$ with $a \in A_a$ and $K \subset B_a$. Finitely many A_a cover K. Their union A and the intersection B of the corresponding B_a satisfy $A \times B \subset P$. Thus f is injective on the open neighborhood $O = A \cap B \cap W$ of K.

Every point of O has an open neighborhood with compact closure in O. Indeed choose a compact neighborhood C of the point. Separate that point from the compact set $C \setminus O$ by disjoint open sets G, H with $C \setminus O \subset H$. Then the closure of $G \cap \operatorname{int} C$ is contained in $C \setminus H \subset O$ and is compact. Finitely many such neighborhoods cover K; let U be their union. Its closure is compact and contained in O. The continuous injection $f|_{\overline{U}}$ is a homeomorphism onto its image by compactness and Hausdorffness. Its restriction to U is the required topological embedding. If K is empty, take U empty. □

Lemma A.145 (A uniform compact embedding neighborhood)**.** *Let T, X be locally compact Hausdorff spaces, let Y be Hausdorff, and let $P \subset T$ and $K \subset X$ be compact. Suppose $f : T \times X \to Y$ is continuous on an open neighborhood W of $P \times K$, each $f_t|_K$ is injective for $t \in P$, and the graph map $J(t, x) = (t, f(t, x))$ is injective on a neighborhood of each point of $P \times K$. There are open neighborhoods V of P and U of K with $V \times U \subset W$ such that $f_t|_U$ is a topological embedding for every $t \in V$.*

Proof. The graph J is injective on $P \times K$, because equality of graph images forces equality of times and then equality of the compact slice points. Apply the compact embedding-neighborhood lemma to J to obtain an open neighborhood $Z \subset W$ of $P \times K$ on which J is an embedding. For each $t \in P$, finitely many product neighborhoods in Z cover $\{t\} \times K$. Their time-factor intersection and spatial-factor union give $V_t \times U_t \subset Z$ with $t \in V_t$ and $K \subset U_t$. Finitely many V_t cover P. Let V be their union and U the intersection of the corresponding U_t. Then $V \times U \subset Z$. For fixed $t \in V$, restriction of J to $\{t\} \times U$ is an embedding into $\{t\} \times Y$; the projection homeomorphism from this target slice to Y gives exactly the embedding $f_t|_U$. □

Lemma A.146 (Immersion of the parameter graph)**.** *For a map $j : \mathbb{R} \times M \to N$ between smooth boundaryless manifolds, differentiable at (t, x), the differential of $J(t, x) = (t, j(t, x))$ at (t, x) is injective if and only if the spatial differential of j_t at x is injective.*

Proof. The differential has the triangular form

$$dJ_{(t,x)}(s, v) = \big(s, (\partial_t j)_{(t,x)} s + d(j_t)_x v\big).$$

If the spatial differential is injective, a vector in the kernel has $s = 0$ from its first coordinate and then $v = 0$ from its second coordinate. Conversely a spatial vector v in the kernel of $d(j_t)_x$ gives $(0, v)$ in the kernel of dJ, so injectivity of dJ forces $v = 0$. □

Lemma A.147 (An embedding neighborhood of a compact injective immersion)**.** *Let M, N be smooth boundaryless manifolds, let $K \subset M$ be compact, and let $f : M \to N$ be smooth on an open neighborhood W of K. Suppose $f|_K$ is injective and df_x is injective for every $x \in K$. There is an open neighborhood U of K, contained in W, such that $f|_U$ is a smooth embedding.*

Proof. The local immersion-neighborhood lemma supplies local injectivity and an open neighborhood $O \subset W$ of K on which df remains injective. Smooth finite-dimensional manifolds are locally compact and Hausdorff. Apply the compact embedding-neighborhood lemma on O. It gives an open neighborhood U of K contained in O on which f is a topological embedding. Smoothness is inherited from W, and injectivity of the differential from O. Thus $f|_U$ is a smooth embedding. This also covers an empty K. □

Lemma A.148 (A smooth equivalence near a compact injective immersion)**.** *Let X, Y be smooth boundaryless manifolds of the same finite dimension, let $K \subset X$ be compact, and let $f : X \to Y$ be smooth on an open neighborhood W of K. Suppose $f|_K$ is injective and df_x is injective for every $x \in K$. There are open sets $U \subset W$ and $V \subset Y$, with $K \subset U$, and a smooth diffeomorphism $g : U \to V$ satisfying $g(x) = f(x)$ for every $x \in U$.*

Proof. The compact embedding-neighborhood lemma gives an open neighborhood $U \subset W$ of K on which f is a smooth embedding. At every $x \in U$ its derivative is injective between vector spaces of the same finite dimension, hence bijective. The inverse function theorem therefore supplies an open neighborhood $O_x \subset U$ on which f has a smooth local inverse and whose image V_x is open in Y. Consequently $V = f(U) = \bigcup_{x \in U} V_x$ is open. Injectivity of $f|_U$ gives a unique set-theoretic inverse $V \to U$. On each V_x this inverse is the smooth local inverse just obtained, because both take each image point to its unique preimage in U. It is therefore smooth on V. The restriction $g = f|_U$ is smooth and has this smooth inverse, proving the claim. If K is empty, one may take both neighborhoods empty; this also yields a smooth diffeomorphism with the stated identity. □

Lemma A.149 (Restricted extension presentations for a prescribed compact map)**.** *Let $D \subset X$ and $E \subset Y$ be compact smooth domain presentations of the same dimension. Suppose prescribed maps $h : X \to Y$ and $k : Y \to X$ are smooth on open neighborhoods of D and E, respectively, satisfy*

$$h(D) \subset E, \qquad k(E) \subset D, \qquad k \circ h|_D = \mathrm{id}_D, \qquad h \circ k|_E = \mathrm{id}_E,$$

and dh_x is injective for every $x \in D$. There are open neighborhoods U of D and V of E and a smooth presentation equivalence between the open restrictions $D \subset U$ and $E \subset V$, whose ambient map is $h|_U$. Its compact carrier map retains h at every point of D, and its inverse retains k at every point of E. Its smooth boundary equivalence is the restriction of that same ambient map and commutes with both boundary inclusions; no separate boundary map is required as a hypothesis. This assertion concerns supplied extension presentations and a supplied compact map; it does not construct such presentations from arbitrary native domain diffeomorphisms.

Proof. The identity $k \circ h = \mathrm{id}$ on D implies injectivity of $h|_D$. The carrier inclusions and $h \circ k = \mathrm{id}$ on E give $h(D) = E$. Apply the preceding lemma to $K = D$ and the prescribed smooth neighborhood extension of h. It gives $g : U \to V$ with $g = h|_U$. Since $g(D) = h(D) = E$, the open set V contains E, and its inverse maps E onto D. For $y \in E$, the point $k(y)$ belongs to $D \subset U$ and

$g(k(y)) = h(k(y)) = y$. Uniqueness of the inverse therefore gives $g^{-1}(y) = k(y)$, retaining the prescribed compact inverse. An ambient homeomorphism carries the frontier of a subset to the frontier of its image. Because U, V are open neighborhoods of the entire compact carriers, their relative frontiers are the original frontiers in X, Y. Thus g maps ∂D onto ∂E. Restrict g and its smooth inverse to these smooth boundary submanifolds. In boundary charts these restrictions are smooth, since the ambient maps are smooth and preserve the boundary submanifolds. They are inverse boundary diffeomorphisms. Their inclusion square is the pointwise equality $g(x) = h(x)$ for $x \in \partial D$. The ambient diffeomorphism, carrier equality and this boundary restriction constitute the required presentation equivalence and retain the prescribed compact map everywhere. □

Lemma A.150 (A uniform embedding neighborhood for a ribbon family)**.** *Let M, N be smooth boundaryless manifolds, let $K \subset M$ be compact, and let $j : \mathbb{R} \times M \to N$ be a map smooth on an open neighborhood W of $[0,1] \times K$ in $\mathbb{R} \times M$. For every $t \in [0,1]$, suppose $j_t|_K$ is injective and the spatial derivative $d(j_t)_x$ is injective for every $x \in K$. There are open neighborhoods V of $[0,1]$ and U of K with $V \times U \subset W$ such that every $j_t|_U$, $t \in [0,1]$, is a smooth embedding. The restrictions form a jointly smooth interval family, including open extensions across both endpoints.*

Proof. On W define $J(t,x) = (t, j(t,x))$. It is injective on the compact set $[0,1] \times K$: equality of the first coordinates forces equal times, and injectivity of the corresponding $j_t|_K$ then forces equal points. The parameter-graph immersion lemma makes dJ injective on $[0,1] \times K$. The local immersion-neighborhood lemma makes its injective-differential locus $O \subset W$ open and containing this compact product, and supplies local injectivity there. Apply the uniform compact embedding-neighborhood lemma to j on O. Its graph is J, so all hypotheses hold. It gives V, U with $V \times U \subset O$ and every $j_t|_U$ a topological embedding. The graph differential is injective on this rectangle; the graph-immersion equivalence makes the spatial derivative injective there as well. For interval times these are smooth embeddings, and the original smooth map restricted to $V \times U$ gives their joint smoothness and open endpoint extensions. □

Lemma A.151 (Constructing a witnessed pair family from compact ribbon germs)**.** *Let n be a nonnegative integer, and let D, F be fixed compact domain presentations of dimensions $2n$ and $\max(2n-2, 0)$, with interval Liouville structures (λ_t, Z_t) and (μ_t, Y_t) whose primitives are jointly smooth with open endpoint extensions. Let j be a map from the whole $\mathbb{R} \times F_{\mathrm{ext}}$ to ∂D, where F_{ext} is the chosen ribbon extension. Suppose j is jointly smooth on an open neighborhood W of $[0,1] \times F$, and suppose each restriction to the compact carrier F is injective with injective spatial derivative there. Let $b_t : \partial D \to \mathbb{R}_{>0}$ be jointly smooth with open endpoint extensions and satisfy*

$$\mu_t = j_t^*(b_t \alpha_t) \quad \text{on } F, \qquad \alpha_t = \lambda_t|_{\partial D}.$$

After restricting the ribbon extension to an open neighborhood of F, these data construct a presented pair deformation with both actual smooth completed families. The chosen ribbon neighborhood U and an open neighborhood V of $[0,1]$ satisfy $V \times U \subset W$. Under the native compact inclusions, its ribbon embeddings, witness and compact primitives are the prescribed j_t, b_t, λ_t and μ_t. The restricted embedding k_t satisfies the literal full-image equality $k_t(F|_U) = j_t(F)$, where $F|_U$ denotes the same compact carrier in the restricted extension. For $n \geq 1$ the dimensions are exactly those of the source's witnessed pair, including its zero-dimensional ribbon at $n = 1$.

Proof. Apply the uniform embedding-neighborhood lemma to the compact ribbon carrier. It gives one open neighborhood U of that carrier on which each j_t is a smooth embedding, jointly smooth on a neighborhood of the whole interval. Restrict the ribbon extension manifold to U. The carrier and boundary are unchanged. Restrict its local defining functions, primitive and field to U;

their regularity, symplecticity, contraction equation and outwardness on the carrier are unchanged. The primitive family and the embedding family retain their joint smoothness and open endpoint extensions. Their pullback identity on the carrier is the same identity through the open inclusion. Every point of the original compact carrier lies in U, so its unique lift to the restriction has the same image under k_t as under j_t. Conversely every point of the restricted carrier projects to an original carrier point with that same image. This proves the full-image equality.

The localized pair-family construction now applies to this restricted ribbon presentation and the original ambient presentation. It constructs both collar families, both actual completions and their smooth presentations. Its ribbon map is the restricted j_t composed with the native ribbon inclusion, its witness is b_t, and its primitives are the pullbacks of the original primitives through the native inclusions. Their composition with the inclusion of U into the original ribbon extension gives the stated compact identities. The same constructed compact and completed presentations supply the whole completed-family regularity, so all these data belong to one presented pair deformation. □

Lemma A.152 (A rectangular representative of a partial smooth family)**.** *Let M, N be smooth boundaryless manifolds, let $K \subset M$ be compact, and let $W \subset \mathbb{R} \times M$ be open with $[0,1] \times K \subset W$. Suppose $j : W \to N$ is smooth. There are open neighborhoods U of K and V of $[0,1]$ with $V \times U \subset W$, and a map $g : \mathbb{R} \times U \to N$ such that*

$$g(t,x) = j(t,x) \quad (t \in V,\ x \in U).$$

The map g is smooth on $V \times U$. No nonemptiness of K or N is assumed, and no regularity of g outside this rectangle is asserted.

Proof. If K is empty, take U empty and $V = \mathbb{R}$; the map with empty source has all the stated properties. Otherwise, for each $t \in [0,1]$ and $x \in K$, choose a product neighborhood $A_{t,x} \times B_{t,x}$ of (t,x) contained in W. For fixed t, finitely many $B_{t,x}$ cover K. Intersect their time neighborhoods and unite their spatial neighborhoods to obtain $A_t \times B_t \subset W$ with $t \in A_t$ and $K \subset B_t$. Finitely many A_t cover $[0,1]$. Their union V and the intersection U of the corresponding B_t satisfy $V \times U \subset W$, with the required inclusions. Since $0 \in V$, define

$$g(t,x) = \begin{cases} j(t,x), & t \in V, \\ j(0,x), & t \notin V. \end{cases}$$

Both evaluations are in the domain of j. On the open rectangle the map is the restriction of j along the smooth inclusion $V \times U \hookrightarrow W$, so it is smooth there and has the asserted equality. The value outside V uses this same partial map on the same spatial neighborhood and requires no additional point of N. □

Lemma A.153 (Constructing a witnessed pair family from partial ribbon germs)**.** *Let n be a nonnegative integer, and let D, F be fixed compact domain presentations of dimensions $2n$ and $\max(2n-2, 0)$, with interval Liouville structures (λ_t, Z_t) and (μ_t, Y_t) whose primitives are jointly smooth with open endpoint extensions. Let $W \subset \mathbb{R} \times F_{\mathrm{ext}}$ be open, containing $[0,1] \times F$, and let $j : W \to \partial D$ be smooth. For each real t put*

$$W_t = \{x \in F_{\mathrm{ext}} : (t,x) \in W\}, \qquad j_t : W_t \to \partial D, \quad j_t(x) = j(t,x).$$

For $t \in [0,1]$, suppose j_t is injective on F and its spatial differential is injective at every point of F. Let $b_t : \partial D \to \mathbb{R}_{>0}$ be jointly smooth with open endpoint extensions, and suppose the pullback on the native slice satisfies

$$j_t^*(b_t \alpha_t) = \mu_t \quad \text{on } F, \qquad \alpha_t = \lambda_t|_{\partial D}.$$

There are open $U_0 \supset F$ and $V_0 \supset [0,1]$ with $V_0 \times U_0 \subset W$, and an open neighborhood U_1 of the compact carrier in $F|_{U_0}$, such that the twice restricted ribbon $F' = (F|_{U_0})|_{U_1}$ admits a jointly smooth interval family of smooth embeddings k_t into ∂D. If $a : F'_{\mathrm{ext}} \to F_{\mathrm{ext}}$ is the composite open inclusion, then

$$k_t(x) = j(t, a(x)), \qquad k_t(F') = j_t(F), \qquad k_t^*(b_t\alpha_t) = a^*\mu_t \quad \text{on } F'.$$

These data construct a presented witnessed pair deformation with the prescribed witness b_t and both actual smooth completed families. Its parameter tuples are the localized pairs from the constructed ambient and ribbon collars, with primitives restricted from λ_t and $a^*\mu_t$ and ribbon maps induced by k_t. No total map on $\mathbb{R} \times F_{\mathrm{ext}}$ is an input. The result includes empty carriers and zero-dimensional ribbons. For $n \geq 1$ its dimensions agree with the intrinsic witnessed-pair definition; recovery of the initial extension presentations remains separate.

Proof. Apply the rectangular-representative lemma to $F \subset F_{\mathrm{ext}}$. It gives U_0, V_0 and a map $g : \mathbb{R} \times U_0 \to \partial D$ smooth on $V_0 \times U_0$, agreeing with j there. For interval times, the open inclusion $i_t : U_0 \hookrightarrow W_t$ satisfies $g_t = j_t \circ i_t$. In the inherited manifold charts, the differential of this open inclusion is the identity. Consequently the differential of g_t is the differential of j_t at the same point. The compact injectivity and injective-differential hypotheses therefore hold for g_t on the carrier of $F|_{U_0}$.

Restrict the ribbon primitives and fields to U_0. Restriction preserves their joint smoothness and endpoint extensions, their contraction equations, symplecticity and outwardness on the compact carrier. Pullback composition through i_t gives

$$g_t^*(b_t\alpha_t) = i_t^* j_t^*(b_t\alpha_t) = \mu_t|_{U_0} \quad \text{on } F|_{U_0}.$$

Apply the compact-germ pair-family construction to this restricted presentation and g, on $V_0 \times U_0$. Its additional open restriction U_1 yields the smooth embeddings k_t, their jointly smooth interval extensions, the restricted primitive identity, and both constructed collar and completion families. Its literal parameter-tuple identity gives the asserted pair deformation and leaves the supplied witness b_t.

Each original compact point belongs to U_0 and its lift belongs to U_1, because both neighborhoods contain the corresponding entire compact carriers. It therefore has a unique lift to F' with the same j_t-image. Every point of F' projects conversely to an original compact point. These two inclusions prove $k_t(F') = j_t(F)$, while $k_t(x) = g_t(x) = j(t, a(x))$ proves the pointwise identity. Thus the constructed deformation preserves the whole compact embedded ribbon and its primitive, including when that carrier is empty. □

Remark A.154 (Scope of the compact-family flow construction). The common neighborhood and bounded Liouville time strip concern one fixed extension-space domain presentation. They supply local integral curves for varying primitives without a supplied completed family. The common boundary collar, domain-side law and exponential primitive identity use the same action in A.118. The compact presentation diffeomorphisms also identify these collars on a common smaller strip and give jointly smooth pulled-back primitives. A.141 constructs their compatible localized completions and proves joint regularity of the transported localized and completed primitives and fields. Its completed maps are the same maps as the localized compact presentations. This does not supply a completion embedding on an arbitrary original extension presentation. For supplied completions of the original presentations, A.138 constructs coherent compact and completion presentations and proves joint regularity of the completed primitives and fields. It assumes those pointwise completions, without assuming their family regularity. The localized construction already proves its

original-inclusion squares and preservation of compact data. Passing from arbitrary intrinsic domain and compact-bundle families to these extension-space data remains a separate correspondence.

A given Liouville field is constrained by its contraction equation only on the compact carrier. The canonical field on the symplectic neighborhood agrees with it there, but arbitrary off-carrier field extensions need not vary smoothly. A jointly smooth field family on a whole chosen extension is furnished by A.111, preserving the primitive everywhere and the original field on the compact carrier. The supported auxiliary flow in A.115 also preserves each interval primitive's symplectic neighborhood for every real flow time. The replacement-field result is independent of the collar-based localized completion construction. The common realized-collar construction A.122 gives actual equal-width collars and their compatible compact presentations. A.140 constructs localized cylindrical completions and their smooth transported primitives and fields from such collar-compatible presentations. Neither result asserts the intrinsic compact-bundle correspondence. A.142 applies the constructed localized-completion family to both ambient and ribbon presentations. It retains their prescribed embeddings, positive witnesses and compact primitive identities, and supplies a presented pair deformation with both smooth completed families. A.151 obtains the required embedding on an extension neighborhood from a supplied total map whose germ is jointly smooth, injective and immersive along the compact ribbon. It retains the whole compact embedded ribbon image and both compact primitives, and supplies one common spatial neighborhood with genuine open parameter-endpoint extensions. A.153 instead starts with a map defined only on an open neighborhood of the compact parameter-ribbon product, whose slices are injective on the compact carrier for every interval parameter and have injective spatial differentials there. Restriction to a common spatial neighborhood gives the required rectangular representative without a nonempty-target assumption. A further embedding-neighborhood restriction preserves the whole compact image and primitive and constructs both completed families with the original witness. Intrinsic domain and bundle presentations and their full source correspondence remain separate.

Definition A.155 (Joint extension germs for native form families)**.** *Let M have its native $(k+1)$-dimensional half-space atlas, and let λ_a be native r-forms for $a \in [0,1]$. The family has joint extension germs if, for every native atlas chart E, every $a \in [0,1]$ and every x in its source, there are an open neighborhood U of $(a, E(x))$ in $\mathbb{R} \times \mathbb{R}^{k+1}$ and a smooth map β on U with values in alternating r-covectors such that*

$$\beta(s,y) = \lambda_s^E(y) \quad \text{for } s \in [0,1],\ y \in \operatorname{target}(E),\ (s,y) \in U.$$

Thus the same coefficient germ extends across the spatial boundary and the parameter endpoints. This condition imposes no symplectic or outwardness requirement outside the parameter interval.

Lemma A.156 (Native coordinate extensions on the fixed collar quotient)**.** *Let M be a smooth Hausdorff second-countable $(k+1)$-manifold with boundary and let C satisfy the smooth collar and inverse-germ hypotheses of its fixed boundaryless quotient X_C, with original inclusion i. Let a locally extendible reference primitive λ have nondegenerate exterior derivative. Suppose C solves $-Z_\lambda$ and Z_λ is outward for the preferred boundary defining functions. For every $x \in M$ there are a native atlas chart E at x, an open neighborhood O of $i(x)$ and a smooth map $Q : O \to \mathbb{R}^{k+1}$ such that every y with $i(y) \in O$ lies in the source of E and satisfies $Q(i(y)) = E(y)$. The boundary case uses the preferred ambient chart of the native boundary atlas; the interior case uses its native atlas chart.*

Proof. At an interior point the original interior inclusion is an open embedding with smooth inverse on its image. Restrict that image so its inverse lies in the source of the native chart E, and compose the inverse with E. This gives the required O, Q and all the asserted identities there.

At a boundary point x, let χ be its native boundary chart and E its preferred ambient chart. Smooth coefficient germs of the reference contraction field follow from its locally extendible primitive and nondegenerate derivative. Outwardness makes the normal coefficient of $-Z_\lambda$ positive. The same-collar two-sided coordinate lemma therefore gives $\varepsilon > 0$, a boundary coordinate neighborhood V and a smooth map c_0 with $c_0(t, v) = E(C(t, \chi^{-1}(v)))$ for $0 \le t < \varepsilon$, $v \in V$. Use the smooth inverse of the actual cylinder inclusion near $i(x)$; write its coordinates as (t, b). Restrict its open image so $|t| < \varepsilon$ and $b \in \mathrm{source}(\chi)$ with $\chi(b) \in V$, and set $Q = c_0(t, \chi(b))$ there. This map is smooth as a composite of smooth maps. An original point $i(y)$ in that image has $t \ge 0$ and $y = C(t, b)$ by the quotient overlap identity. The coordinate lemma gives y in the source of E and $Q(i(y)) = E(y)$, proving every boundary assertion. □

Lemma A.157 (Joint spatial pullback through a fixed local coordinate map)**.** *Let X be a smooth boundaryless manifold, let $O \subset X$ be open and let $Q : O \to \mathbb{R}^m$ be smooth. Let $U \subset \mathbb{R} \times \mathbb{R}^m$ be open and let $\beta(a, y)$ be a smooth map on U with values in one-covectors on $\mathbb{R}^m$. On*

$$W = \{(a, x) : x \in O,\ (a, Q(x)) \in U\}$$

the spatial covectors $\eta_a(x) = \beta(a, Q(x)) \circ dQ_x$ are jointly smooth.

Proof. Fix a spatial chart with inverse h and put $q = Q \circ h$ on its open coordinate domain over O. The chain rule identifies the chart coefficients of η with $\beta(a, q(y)) \circ dq_y$. The map $(a, y) \mapsto \beta(a, q(y))$ is smooth wherever $(a, q(y)) \in U$, by composition. The differential $y \mapsto dq_y$ is smooth on this open domain, since q is smooth. Composition of covectors and linear maps is continuous bilinear, so these chart coefficients are jointly smooth. This holds in every spatial chart, proving the assertion on W. □

Lemma A.158 (Patching spatial form families on a closed product carrier)**.** *Let X be a finite-dimensional smooth boundaryless Hausdorff second-countable manifold, let $A \subset \mathbb{R} \times X$ be closed, and let $f(a, x)$ be an alternating r-covector on T_xX for every (a, x). Suppose every $p \in A$ has an open neighborhood U_p and a spatial form family $g_p(a)$ whose coefficients are jointly smooth on U_p and whose actual covectors equal f on $U_p \cap A$. There is a spatial form family η_a jointly smooth for all real parameters with $\eta_a(x) = f(a, x)$ on A. In particular its coefficients extend smoothly across the endpoints of $[0, 1]$.*

Proof. The sets U_p, together with $(\mathbb{R} \times X) \setminus A$, form an open cover of the finite-dimensional Hausdorff second-countable product manifold. Choose a smooth locally finite partition of unity (ρ_i) subordinate to this cover. For a set contained in some U_p, choose its family g_p; for a set contained in the complement of A, choose the zero family. Denote these choices by g_i and set

$$\eta_a(x) = \sum_i \rho_i(a, x) g_i(a)_x.$$

Each sum takes place in the single actual covector space at x. In a fixed spatial chart, the coefficients of each summand are the product of the smooth scalar coefficient of ρ_i with the jointly smooth coefficients of g_i. The support of ρ_i is contained in the open set where these coefficients are smooth. Every point outside that set has a neighborhood disjoint from the support, on which the summand is zero. Thus every summand has jointly smooth coefficients everywhere. Local finiteness makes the sum finite on a neighborhood of every product point, proving joint smoothness of η. At $(a, x) \in A$, every nonzero summand has $g_i(a)_x = f(a, x)$; the terms subordinate to the complement vanish. The partition sums to one, so $\eta_a(x) = f(a, x)$, as required. □

Lemma A.159 (One native compact presentation extending a joint primitive family)**.** *Let M be a compact native smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Let λ_a, $a \in [0,1]$, be native one-forms with the joint extension germs of Definition A.155. Suppose $d\lambda_0$ is nondegenerate and Z_{λ_0} is outward for the regular local defining functions of the native domain. There are a single compact domain presentation $D \subset X$, a smooth closed embedding $i : M \to X$ with bijective differential, a native diffeomorphism $e : M \to D$ whose ambient composite is i, and jointly smooth one-forms η_a on X with smooth parameter-endpoint extensions, such that*

$$i(M) = D, \qquad i(\partial M) = \mathrm{Fr}(D), \qquad i^*\eta_a = \lambda_a \quad (a \in [0,1]).$$

At every original point $i(x)$, the same presentation has a native chart E, an open neighborhood O of $i(x)$ and a smooth coordinate extension $Q : O \to \mathbb{R}^{k+1}$, with $Q(i(y)) = E(y)$ for every original point in O. The pullback equality holds on every native tangent, including the boundary. Only the reference slice supplies the spatial collar; no symplecticity of the other slices or completion family is asserted here.

Proof. Restrict each joint coefficient germ to parameter zero to obtain local smooth extension germs of λ_0. Its nondegenerate derivative and outwardness construct one native inward collar C with positive width and smooth native inverse germs. Form its fixed boundaryless quotient X_C, original inclusion i and compact domain $D = i(M)$. The compact-presentation and inclusion-differential lemmas give the frontier identity and bijectivity of di, and the same-collar carrier identification gives e. Keep these choices fixed for the whole family.

For every $x \in M$, the coordinate-extension lemma gives a native chart E, a neighborhood O of $i(x)$ and a smooth $Q : O \to \mathbb{R}^{k+1}$ with $Q(i(y)) = E(y)$ at every original point in O. At a boundary point its hypotheses follow from the reference primitive and its outwardness; the defining-function sign lemma supplies the preferred normal sign. Now fix $a \in [0,1]$ and a joint coefficient germ β near $(a, E(x))$. On the open neighborhood where $z \in O$ and $(s, Q(z))$ is in this germ's domain define

$$g_s(z) = \beta(s, Q(z)) \circ dQ_z.$$

The joint coordinate-pullback lemma proves its joint smoothness there. At an original point $i(y)$ in this neighborhood, differentiation of $Q \circ i = E$ on a native neighborhood gives $dQ_{i(y)} \circ di_y = dE_y$ on the full native tangent space. The coefficient germ agrees with the chart coefficients of λ_s for $s \in [0,1]$. Recovery of the native form from those coefficients therefore gives $g_s(i(y)) \circ di_y = \lambda_s(y)$.

Define a covector on the closed set $A = [0,1] \times i(M)$ by $f(s, i(y)) = \lambda_s(y) \circ (di_y)^{-1}$, assigning zero elsewhere. Injectivity of i makes the definition unambiguous. The preceding local pullback identity and surjectivity of di_y imply $g_s(i(y)) = f(s, i(y))$ throughout each local intersection with A. Thus these germs meet the closed-product form-family patching hypotheses. Patching gives a global jointly smooth real-parameter family on X_C equal to f on A. Restrict its parameter to $[0,1]$ to obtain η_a. Its equality to f proves $i^*\eta_a = \lambda_a$ on every original tangent. All spatial choices preceded the parameter germs, so this is one common presentation; the coordinate maps used to construct the local forms are retained as its claimed native coordinate extensions. □

Lemma A.160 (One spatial symplectic neighborhood for a compact primitive family)**.** *Let $D \subset X$ be a compact domain presentation in a boundaryless smooth manifold, and let η_a be jointly smooth one-forms for $a \in [0,1]$ with smooth coefficient extensions across the parameter endpoints. Suppose $d\eta_a$ is nondegenerate at every point of D for every such a. There is one open spatial neighborhood $U \supset D$ on which every $d\eta_a$, $a \in [0,1]$, is nondegenerate.*

Proof. Choose the given real-parameter coefficient extension. In a spatial chart its spatial exterior derivative has jointly smooth matrix coefficients. At every $(a, x) \in [0,1] \times D$ the contraction matrix is invertible. Nonvanishing of its determinant is an open condition, giving an open product neighborhood of (a, x) where it remains invertible. The union of these neighborhoods is an open set V containing $[0,1] \times D$. The tube lemma for the two compact sets $[0,1]$ and D supplies open neighborhoods $J \supset [0,1]$ and $U \supset D$ with $J \times U \subset V$. Restricting to $a \in [0,1]$ identifies the coefficient extension with the supplied η_a, proving the asserted nondegeneracy on U. □

Lemma A.161 (A native Liouville family from one supplied primitive presentation)**.** *Let M be a compact native smooth Hausdorff second-countable $(k+1)$-manifold with boundary, carrying one-forms λ_a for $a \in [0,1]$. Suppose their native exterior derivatives are nondegenerate and their contraction fields are outward for regular local defining functions. Suppose a single compact domain presentation $D \subset X$ is supplied with a smooth closed embedding $i : M \to X$, bijective di, $i(M) = D$, $i(\partial M) = \mathrm{Fr}(D)$, and a native diffeomorphism $e : M \to D$ whose ambient composite is i. Suppose also that jointly smooth one-forms η_a on X, with smooth endpoint extensions, are supplied with $i^*\eta_a = \lambda_a$. There is one open spatial restriction of this presentation carrying Liouville structures (η'_a, Z'_a), a retained native inclusion j and carrier diffeomorphism e', with*

$$j^*\eta'_a = \lambda_a, \qquad j^*(d\eta'_a) = d\lambda_a, \qquad Z'_a(j(x)) = dj_x(Z_{\lambda_a}(x)).$$

The compact carrier and frontier are $j(M)$ and $j(\partial M)$. Both primitives and fields are jointly smooth with endpoint extensions, and $d\eta'_a$ is symplectic throughout the restricted extension for $a \in [0,1]$. These structures have constructed localized completion families with the whole-completion regularity and inclusion squares of Lemma A.141. If the supplied presentation has the native coordinate extensions of the preceding lemma, its open restriction retains them by composition with the open inclusion. The common presentation and joint primitive extension are hypotheses here.

Proof. Exterior naturality gives $i^*d\eta_a = d\lambda_a$. Since di_x is bijective, a vector in the contraction kernel of $d\eta_a$ at $i(x)$ pulls back to a vector in the contraction kernel of $d\lambda_a$ at x. The latter is zero, proving nondegeneracy on D for every parameter. The common-neighborhood lemma supplies one open $U \supset D$ on which all these derivatives are nondegenerate. Restrict X, D, i, e to U; the native open-restriction diffeomorphism retains the carrier and frontier, and the resulting inclusion j still has bijective differential. Pullback through the fixed open inclusion makes η'_a jointly smooth and preserves both displayed form identities. Its derivative is closed because the exterior derivative squares to zero, so it is symplectic on U.

Let Z'_a be the unique contraction field of η'_a on U. The two form identities show that $dj_x Z_{\lambda_a}(x)$ solves its contraction equation at $j(x)$ on every tangent, because dj_x is surjective. Nondegeneracy gives the displayed field identity. In local spatial coordinates the contraction field is the inverse matrix of $d\eta'_a$ applied to the coefficients of η'_a. These are jointly smooth wherever that matrix is invertible. Open invertibility neighborhoods of the interval product therefore give the required joint field regularity and endpoint extensions.

For outwardness, pull back a regular carrier defining function ρ at $j(x)$ through j. It is nonpositive on a native neighborhood, vanishes at x and has nonzero differential. In half-space coordinates that differential vanishes on the boundary tangent directions: each such direction and its negative has a nonpositive one-sided difference quotient. Its value on the positive normal is nonpositive, hence strictly negative by the nonzero differential. Thus it is $c\,dE_0$ with $c < 0$. Source outwardness gives $(dE_x Z_{\lambda_a}(x))_0 < 0$ by the defining-function sign lemma. The field identity consequently yields $d\rho_{j(x)}(Z'_a(j(x))) > 0$. This proves a Liouville structure at every interval parameter on the same restricted presentation. Finally apply the localized-completion-family construction to these

structures and their proved joint primitive regularity. It constructs the collars and completions with their inclusion squares and jointly smooth transported completed primitives and fields; none were supplied. A native coordinate extension on the original presentation restricts by composition with the fixed open inclusion, retaining its original pointwise coordinate identity. This proves the additional conditional coordinate-extension assertion. □

Lemma A.162 (A constructed Liouville completion family from native joint germs)**.** *Let M be a compact native smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Let λ_a, $a \in [0,1]$, have joint native extension germs, nondegenerate native exterior derivatives and outward contraction fields for regular local defining functions. These native data produce one fixed compact domain presentation and Liouville structures on it with all the conclusions of Lemma A.161: the same native carrier identification preserves every compact primitive, its full exterior derivative and its contraction field, both primitives and fields have joint endpoint extensions, and localized completion families are constructed with regularity on the whole completions. The same presentation also retains actual smooth local extensions of the coordinates of a suitable native chart at each original point. No spatial extension, coordinate extension, compact presentation or pointwise completion is supplied as input. The dimension is $k+1 \geq 1$; this assertion concerns one fixed native manifold and does not supply compact bundle trivializations or a varying witnessed ribbon embedding.*

Proof. Apply the native primitive-family extension lemma, using nondegeneracy and outwardness at parameter zero for its reference slice. It constructs one D, X, i, e and one jointly smooth family η_a with the native carrier and frontier identities, bijective inclusion differential and $i^*\eta_a = \lambda_a$ on all native tangents. These are exactly the presentation and joint-extension hypotheses of the supplied-presentation Liouville-family lemma. Its remaining hypotheses are the given native nondegeneracy and outwardness at each $a \in [0,1]$. Apply that lemma to the constructed presentation and family. It makes one common spatial restriction, proves all primitive, derivative and field identities through its retained native inclusion, and constructs the localized completion family. Thus every output uses the same selected compact presentation and the resulting family; no independently chosen pointwise extensions or completions enter the construction. The primitive-family construction retained the native coordinate maps, and the spatial restriction preserves them by the preceding lemma, so these maps are available on the same final presentation. □

Definition A.163 (Joint native extension germs for manifold-valued families)**.** *Let M have its native $(k+1)$-dimensional half-space atlas, let N be a smooth boundaryless manifold, and let $f_t : M \to N$ for $t \in [0,1]$. The family has joint native extension germs if for every native chart E, every $t \in [0,1]$ and every x in its source, there are an open neighborhood U of $(t, E(x))$ in $\mathbb{R} \times \mathbb{R}^{k+1}$ and a smooth map $\beta : U \to N$ such that*

$$\beta(s, E(y)) = f_s(y) \quad (s \in [0,1],\ y \in \mathrm{source}(E),\ (s, E(y)) \in U).$$

The germ takes values in N itself and extends across both the native spatial boundary and the parameter endpoints. No agreement between separate germs away from the native interval product is required.

Lemma A.164 (Joint native map germs on one spatial presentation)**.** *Let M be a compact native $(k+1)$-manifold, let X be a boundaryless smooth r-manifold, and let $i : M \to X$ be a closed embedding. Suppose that at every original point $i(x)$ there are a native chart E, an open neighborhood O in X and a smooth map $Q : O \to \mathbb{R}^{k+1}$ such that every y with $i(y) \in O$ lies in the source of E and*

satisfies $Q(i(y)) = E(y)$. Let $f_t : M \to N$ have joint native extension germs. There are an open neighborhood W of $[0,1] \times i(M)$ in $\mathbb{R} \times X$ and a smooth map $g : W \to N$ with

$$g(t, i(x)) = f_t(x) \qquad (t \in [0,1],\ x \in M).$$

In particular the construction needs neither a total extended map family nor compatible off-carrier germs, and allows empty carriers and empty targets.

Proof. Write $A = [0,1] \times i(M)$ and define $a(t, i(x)) = f_t(x)$ on A. Injectivity of i makes this well defined. The set A is compact because M and $[0,1]$ are compact and i is continuous. Fix $(t, i(x)) \in A$ and choose E, O, Q as in the hypothesis. The native germ at $(t, E(x))$ gives U, β. On the open set

$$V = \{(s, z) : z \in O,\ (s, Q(z)) \in U\}$$

put $h(s, z) = \beta(s, Q(z))$. This is smooth as a composite of smooth maps. It is defined at $(t, i(x))$ because $Q(i(x)) = E(x)$. If $(s, i(y)) \in V \cap A$, then y belongs to the source of E, $Q(i(y)) = E(y)$ and the germ identity gives $h(s, i(y)) = \beta(s, E(y)) = f_s(y) = a(s, i(y))$. Thus every point of the compact carrier A has an actual smooth manifold-valued extension on an open neighborhood. Apply the compact map-germ extension lemma in the boundaryless manifold $\mathbb{R} \times X$. It gives the claimed W, g, retaining the map a everywhere on A. That lemma includes the empty case and does not require agreement of these local extensions outside A. □

Lemma A.165 (One native boundary presentation for the whole primitive family)**.** *Let M be a compact native positive-dimensional smooth manifold with boundary, and let λ_t have joint native extension germs, nondegenerate exterior derivatives and outward contraction fields at every $t \in [0,1]$. The native family construction can be chosen with one compact presentation D, native inclusion i and carrier identification e, one Liouville family L_t, and one fixed smooth boundary diffeomorphism $e_B : \partial M \to \partial D$, satisfying*

$$j_D \circ e_B = i|_{\partial M}, \qquad e_B^*(\lambda_{L_t}|_{\partial D}) = \lambda_t|_{\partial M} \quad (t \in [0,1]).$$

All native primitive, exterior derivative and field identities, joint regularity, coordinate extensions and constructed completion families of the native family lemma are retained. The boundary map and its entire varying primitive identity are conclusions, rather than supplied family-presentation data.

Proof. Construct the fixed native Liouville presentation and its whole family from the native primitive germs. The result has a smooth injective inclusion i with bijective differential, carrier $i(M)$, frontier $i(\partial M)$ and $i^*\lambda_{L_t} = \lambda_t$ for every parameter. Apply the native boundary presentation equivalence to this same D, i. It constructs e_B and the inclusion square. These choices depend only on that fixed spatial presentation and do not vary with t. At each parameter apply the boundary primitive-transport lemma to the already proved equality $i^*\lambda_{L_t} = \lambda_t$. The inclusion square and pullback composition give precisely the displayed boundary primitive equality. The construction alters no primitive, field, coordinate extension or completion choice, so every other conclusion of the original family producer remains on the same D, i, e, L_t. □

Lemma A.166 (One uniform joint collar from the original native primitive family)**.** *Let M be a compact native smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Let λ_a, $a \in [0,1]$, have joint native extension germs and nondegenerate native exterior derivatives. At each parameter and boundary point suppose there is a smooth local real function ρ with $\rho = 0$ at that point, $\rho \le 0$ on a native neighborhood, $d\rho \neq 0$, and $d\rho(Z_{\lambda_a}) > 0$. These original data construct*

$\delta > 0$ and native inward collars $C_a : [0,\delta) \times \partial M \to M$ such that the actual parameter-preserving map

$$(s, a, b) \longmapsto (a, C_a(s, b))$$

is an open embedding into $[0,1] \times M$. Every slice is smooth and an open embedding, has smooth local inverse germs into $\mathbb{R} \times \partial M$, satisfies $C_a(0, b) = b$, meets the native boundary exactly at $s = 0$, and solves

$$\partial_s C_a(s, b) = -Z_{\lambda_a}(C_a(s, b)).$$

The construction retains one fixed compact presentation $D \subset Y$, a smooth closed native inclusion $i : M \to Y$ with bijective full differential and native coordinate extension germs, a fixed smooth boundary diffeomorphism $e_B : \partial M \to \partial D$, one jointly smooth real-parameter collar representative c, and an open $J \subset \mathbb{R}$ containing $[0,1]$ such that $(a, y, r) \mapsto (a, c(a, y, r))$ is a local diffeomorphism on $J \times \partial D \times (-\delta, \delta)$ and

$$i(C_a(s, b)) = c(a, e_B(b), -s) \qquad (a \in [0,1],\ 0 \le s < \delta).$$

No compact presentation, extended primitive, collar or joint open-embedding assertion is supplied as an input.

Proof. The native boundary-family construction gives one compact presentation, native carrier diffeomorphism $e : M \to D$, its ambient composite i, fixed boundary diffeomorphism e_B, and a jointly smooth primitive family on a neighborhood of D. It retains the full identities $i^*\widetilde{\lambda}_a = \lambda_a$ and $di\, Z_{\lambda_a} = Z_{\widetilde{\lambda}_a} \circ i$. Apply the common Liouville-collar lemma to this same presented family. It gives $\delta > 0$, a smooth real-parameter representative $c(a, y, r)$, an open parameter neighborhood $J \supset [0,1]$, and a joint open embedding

$$F(a, y, r) = (a, c(a, y, r)) \quad \text{on } J \times \partial D \times (-\delta, \delta),$$

which is a local diffeomorphism there. At interval parameters it has $c(a, y, 0) = j_D(y)$, solves the actual presented Liouville equation, and lies in D exactly when $r \le 0$.

For $0 \le s < \delta$, define $C_a(s, b) = e^{-1}(c(a, e_B(b), -s))$, regarding the collar value as a point of D. The side criterion makes this well defined. Restrict F to the preimage of $[0,1] \times D$. Restricting an open embedding to the preimage of any target subspace is an open embedding for the corresponding relative topologies. The side criterion identifies this preimage with the inward half collar after the homeomorphic rearrangement $(s, a, b) \mapsto (a, e_B(b), -s)$. The homeomorphism $[0,1] \times D \to [0,1] \times M$ induced by e^{-1} therefore gives exactly the claimed joint native open embedding. Restricting this parameter-preserving embedding to a parameter fiber proves the slice open-embedding property.

Smoothness of a map into the native compact carrier is equivalent, in its adapted half-space charts, to smoothness of its ambient composite. That composite is $c(a, e_B(b), -s)$, so every C_a is smooth up to $s = 0$. The boundary inclusion square gives $C_a(0, b) = b$. If $C_a(s, b)$ is a native boundary point b', then $c(a, e_B(b), -s) = i(b') = c(a, e_B(b'), 0)$. Injectivity of the same joint collar implies $s = 0$; the converse follows from the seam equation.

At a collar point, choose a smooth local inverse of F. Compose it with $x \mapsto (a, i(x))$, negate its radial coordinate, and apply e_B^{-1} to its boundary coordinate. This gives a smooth native local inverse germ of C_a into $\mathbb{R} \times \partial M$. The two inverse identities of F give its actual inverse identity on a relative neighborhood in the half collar, including the seam. Finally differentiate $i(C_a(s, b)) = c(a, e_B(b), -s)$ in s. The presented flow equation and field identity give $di(\partial_s C_a) = -di(Z_{\lambda_a})$. Injectivity of di proves the native time equation, with the one-sided derivative at $s = 0$ supplied by the smooth half-space extension. All maps and identities use the same chosen presentation and the same real-parameter collar representative. □

Definition A.167 (Joint scalar extension germs on the actual boundary)**.** *Let B be a fixed smooth boundaryless manifold and let $b_t : B \to \mathbb{R}$ for $t \in [0,1]$. These functions have joint local extension germs if every $(t,x) \in [0,1] \times B$ has an open neighborhood $U \subset \mathbb{R} \times B$ and a smooth scalar function g on U with $g(s,y) = b_s(y)$ for all $(s,y) \in U \cap ([0,1] \times B)$. Separate germs need only agree on the prescribed interval data.*

Lemma A.168 (Joint witness smoothness from germs on the actual boundary)**.** *A scalar family b_t on the fixed actual boundary B with the preceding joint local extension germs admits a real-parameter extension, jointly smooth on open neighborhoods of $[0,1] \times B$, retaining every value $b_t(x)$. In particular any positivity requirement on $[0,1] \times B$ is preserved; positivity outside that interval is not required.*

Proof. The set $A = [0,1] \times B$ is closed in $\mathbb{R} \times B$. Define $f(t,x) = b_t(x)$ on A and set it equal to zero elsewhere. At every point of A, the given scalar germ is smooth on an open neighborhood and agrees with f on its intersection with A. The closed-carrier scalar extension lemma gives a smooth extension equal to f on all of A. Its restrictions to the prescribed parameters are therefore exactly the original b_t. Composition with every spatial chart inverse gives jointly smooth coefficient functions near the required interval product, including both endpoint parameters. Equality of the interval values preserves positivity there. This constructs the asserted interval-family extension without any off-interval positivity hypothesis. □

Lemma A.169 (Joint smoothness on a native finite carrier)**.** *Let M be a compact native zero-dimensional smooth manifold and let μ_t be arbitrary one-forms on M, indexed by $t \in [0,1]$. On its native finite-carrier presentation, these primitives and their Liouville fields are jointly smooth with open parameter-endpoint extensions. Each primitive and each field is zero.*

Proof. Every tangent space of M is the zero vector space. A one-covector on that space and a vector in it are therefore zero. Hence $\mu_t = 0$ and the Liouville field is zero for every t. Extend both families by zero for all real parameters. In every native chart their coefficients are constant zero, and thus smooth in the parameter and spatial variables. These extensions agree with every interval value and give the asserted open endpoint extensions. No regularity assumption on the original indexing of the one-forms is needed. □

Lemma A.170 (Constructed completed families on a native finite carrier)**.** *For M and μ_t as above, the native finite-carrier presentation supports the Liouville family with primitive exactly μ_t. The localized completion-family construction produces both its compact presentations and its actual completed presentations, with jointly smooth transported completed primitives and fields and open endpoint extensions. At every interval parameter its completed flow fixes every point for every real time, and the full core is the whole compact ribbon. The original-point identification carries this core onto all of M. Empty carriers are allowed.*

Proof. The native zero-dimensional domain construction uses M itself, zero field, empty boundary and the supplied primitive $\mu_t = 0$. Its Liouville structures therefore retain μ_t literally at each parameter. The preceding lemma supplies the required joint primitive regularity. Apply the localized completion-family construction to this same fixed presentation and these structures. It constructs the collars, compact and completed presentations and their stated regularity.

The constructed completed manifold still has dimension zero, so its Liouville field is zero. The constant curve through any point solves its flow equation with that initial point. Uniqueness of integral curves therefore identifies its flow at every real time with the identity. A compact-carrier point stays in that carrier for every nonnegative time, so every such point belongs to the full core;

the core definition already restricts it to the compact carrier. This proves equality. The native carrier diffeomorphism is surjective and its original-point map is the identity on M, so the core image under that same identification is all of M. All constructions and equalities remain valid when M is empty. □

Lemma A.171 (Varying finite ribbons on a fixed ambient presentation)**.** *Let D be a fixed compact two-dimensional domain presentation carrying interval Liouville structures λ_t with jointly smooth primitives and open endpoint extensions. Let M be a compact native zero-dimensional manifold with arbitrary one-forms μ_t. Suppose $j_t : M \to \partial D$ is injective at every interval parameter, and j_t and a witness $b_t : \partial D \to \mathbb{R}$ are jointly smooth with open endpoint extensions, with $b_t > 0$ on the interval. These data construct a witnessed pair deformation using the localized ambient and finite-ribbon collars, with both actual smooth completed families. At every parameter its localized compact pair has the prescribed ribbon map j_t, witness b_t, ambient primitive restricted from λ_t, and ribbon primitive restricted from $\mu_t = 0$. Its full embedded core at every parameter is exactly $j_t(M)$. The ambient presentation is an input; no ribbon primitive regularity, pointwise ribbon completion or completed-family regularity is an input.*

Proof. The native zero-dimensional manifold is discrete and compact, hence finite. Each j_t is smooth on this native source. Its continuous injectivity and compactness make it a topological embedding into the Hausdorff boundary. Its derivative is injective because the source tangent space is zero. It is consequently a smooth embedding. For every parameter the pullback $j_t^*(b_t\alpha_t)$ is a one-form on a zero-dimensional manifold and is zero; it therefore equals μ_t on the entire native carrier.

Use the preceding lemma's fixed native ribbon presentation and jointly smooth zero primitive family. Apply the localized witnessed-pair-family construction to the given ambient family, these ribbon structures, the actual embeddings j_t and the given positive witness b_t. Its hypotheses on primitive regularity hold by the given ambient regularity and the zero-family lemma. Its map and witness hypotheses are exactly the assumed joint endpoint extensions, and its primitive identity is the zero-form equality just proved. The construction produces the two actual completion families and a pair deformation whose parameter tuples are these localized witnessed pairs. Localization only restricts the extension neighborhoods: every original compact point remains, the ambient boundary identification is the identity, and the ribbon map is j_t composed with its original-point inclusion. Thus the resulting maps, witnesses and compact primitive restrictions have precisely the claimed values at every parameter. The completed finite ribbon has stationary flow by the zero-dimensional core result, so its core is its entire compact carrier. Surjectivity of the native carrier diffeomorphism and the retained formula for the embedding give its full embedded core as exactly $j_t(M)$, at each parameter. □

Definition A.172 (Joint map germs on a native finite ribbon)**.** *Let M be a compact native zero-dimensional smooth manifold, let N be boundaryless and smooth, and let $j_t : M \to N$ for $t \in [0,1]$. Joint native map germs mean that every (t,x) has an open neighborhood $U \subset \mathbb{R} \times M$ and a smooth $g : U \to N$ agreeing with $j_s(y)$ whenever $(s,y) \in U$ and $s \in [0,1]$.*

Lemma A.173 (Extending joint map germs on a native finite ribbon)**.** *A finite-ribbon map family with these joint germs has a real-parameter extension that is jointly smooth on an open neighborhood of $[0,1] \times M$, retaining $j_t(x)$ at every interval point. No nonempty-target hypothesis or total extended family is needed.*

Proof. The set $A = [0,1] \times M$ is compact. Define its map by $(t,x) \mapsto j_t(x)$. The given germs satisfy the hypotheses of the compact manifold-valued germ-extension lemma on the smooth manifold

$\mathbb{R} \times M$. It gives an open $W \supset A$ and a smooth $g : W \to N$ agreeing with that map on A. Define the real-parameter extension to equal $g(t, x)$ for $(t, x) \in W$ and $j_0(x)$ elsewhere. It is smooth on the required open neighborhood W and agrees with all interval values. Its outside values use only the already prescribed reference map, so they also make sense with empty source or target. No smoothness outside W is asserted. □

Lemma A.174 (Finite-ribbon pair deformations from local native germs)**.** *In the finite-ribbon pair-family lemma on a fixed ambient presentation, replace the supplied joint map and witness extensions by their local native joint germs. The same conclusions hold: one constructed pair deformation retains every j_t, b_t, λ_t and μ_t, with both actual smooth completed families and full embedded core $j_t(M)$ at every parameter. Positivity and injectivity are required only on $[0, 1]$. The fixed ambient presentation and its Liouville family remain inputs.*

Proof. The finite-ribbon map-germ extension lemma constructs joint smoothness with open endpoint extensions for the original j_t. The scalar-germ extension lemma on the given actual boundary does the same for the original b_t, without changing its interval values and hence without changing their positivity. Apply the fixed-ambient finite-ribbon pair-family lemma to these two derived regularity statements and the original primitives and injective maps. It constructs the collars, both completed families and the pair deformation from these very interval data. Its exact parameter-tuple identity and its full-core equality give all the conclusions. The two extension steps leave every interval value unchanged, so neither the ribbon image nor the witnessing function has been replaced. □

Lemma A.175 (Joint native map germs under fixed target composition)**.** *For either positive-dimensional native ribbon charts or a native finite ribbon, joint native map germs of $j_t : M \to N$ are preserved by postcomposition with a fixed smooth map $f : N \to P$.*

Proof. At each required native interval point, let $g : U \to N$ be its actual joint extension germ. The composite $f \circ g : U \to P$ is smooth. At every native interval point of U it equals $f \circ j_t$, by the given germ identity. The same open set U extends through the parameter endpoints and, in positive dimension, the native spatial boundary. Thus these composites satisfy exactly the required joint germ identities in either dimension case. □

Lemma A.176 (Joint scalar germs under fixed source composition)**.** *Let $b_t : N \to \mathbb{R}$ have joint local scalar germs on a fixed smooth boundaryless manifold N. For any fixed smooth $f : P \to N$, the family $b_t \circ f$ has joint local scalar germs on P.*

Proof. At $(t, x) \in [0, 1] \times P$, take a scalar germ g on an open neighborhood U of $(t, f(x))$ extending b_t on interval points. The smooth map $p(s, y) = (s, f(y))$ has open inverse image $p^{-1}(U)$ containing (t, x). On this inverse image the composite $g \circ p$ is smooth and satisfies $(g \circ p)(s, y) = b_s(f(y))$ for every interval point. This supplies the required joint extension germ with its open parameter-endpoint domain. □

Lemma A.177 (Fully native pair families with finite ribbons)**.** *Let A be a compact native smooth surface with boundary and let λ_t have joint native extension germs, nondegenerate exterior derivatives and outward contraction fields at each $t \in [0, 1]$. Let M be a compact native zero-dimensional manifold with arbitrary one-forms μ_t. Suppose $j_t : M \to \partial A$ is injective and has joint native map germs, and $b_t : \partial A \to \mathbb{R}$ has joint local scalar germs with $b_t > 0$ on the interval. These original native data construct a fixed compact ambient presentation D, one fixed boundary diffeomorphism $e_B : \partial A \to \partial D$, and a pair deformation $\mathcal{Q}_t$ with both actual smooth completion families. At every*

parameter there are native carrier diffeomorphisms $e_{A,t} : A \to D_{\mathcal{Q}_t}$ and $e_{F,t} : M \to F_{\mathcal{Q}_t}$ preserving the original inclusions, with

$$j_{\mathcal{Q}_t} \circ e_{F,t} = e_B \circ j_t, \qquad b_{\mathcal{Q}_t} \circ e_B = b_t.$$

The ambient and ribbon primitives pull back through these same carrier diffeomorphisms to λ_t and μ_t, and the boundary primitive pulls back through e_B to $\lambda_t|_{\partial A}$. The full embedded core is exactly $e_B(j_t(M))$ at every parameter. No compact presentation, boundary equivalence, collar or completed family is supplied as input; compact bundle trivialization is separate.

Proof. Apply the native ambient boundary-family producer to A, λ_t. It constructs D, i, e, L_t, e_B, the native compact primitive identities, the original boundary inclusion square and $e_B^*(\lambda_{L_t}|_{\partial D}) = \lambda_t|_{\partial A}$ at every parameter. Define $j_t' = e_B \circ j_t$ and $b_t' = b_t \circ e_B^{-1}$. The two fixed-composition germ lemmas preserve their joint native germs. Bijectivity of e_B preserves injectivity of j_t and positivity of the original witness values.

Apply the finite-ribbon local-germ pair-family construction to $D, L_t, M, \mu_t, j_t', b_t'$. It constructs the ambient and ribbon collars, both actual completion families and the pair deformation with these exact parameter tuples. Its compact ribbon identification retains every point of M; compose the original ambient carrier diffeomorphism e with the native diffeomorphism of the ambient collar localization to obtain $e_{A,t}$. The two open restrictions of that localization retain the original ambient inclusion and its full primitive pullback, so $e_{A,t}^* \lambda_{\mathcal{Q}_t} = \lambda_t$. The zero ribbon primitive identity similarly gives $e_{F,t}^* \mu_{\mathcal{Q}_t} = \mu_t = 0$ on every tangent direction.

The inherited ambient boundary is the same ∂D and its primitive is the original boundary primitive of L_t. Thus the boundary pullback through e_B remains $\lambda_t|_{\partial A}$, and its inclusion agrees with the original native boundary inclusion through $e_{A,t}$. The prescribed embedding equality is $j_{\mathcal{Q}_t} \circ e_{F,t} = j_t' = e_B \circ j_t$. The witness equality follows from $b_t'(e_B(x)) = b_t(e_B^{-1}(e_B(x))) = b_t(x)$. Finally the finite-ribbon pair-family result identifies its full embedded core with $j_t'(M) = e_B(j_t(M))$. Every identification refers to the same constructed parameter tuple and completion family. □

Lemma A.178 (A native ribbon family from its fixed primitive presentation)**.** *Let $n \geq 2$, let D, F be fixed compact domain presentations of dimensions $2n$ and $2n-2$, and let λ_t, η_t be Liouville structures with jointly smooth primitives and open endpoint extensions. Let M be a compact native ribbon manifold, with a native carrier diffeomorphism $e : M \to F$ and smooth ambient inclusion $i : M \to F_{\mathrm{ext}}$. Suppose i is a closed embedding with bijective differential, $i(M) = F$, and has the native coordinate extensions of the joint map extension lemma. Let native one-forms μ_t satisfy $i^*\eta_t = \mu_t$. Suppose $j_t : M \to \partial D$ has joint native extension germs and is injective with injective differential at every interval parameter. Let b_t be a jointly smooth witness on the actual ∂D, with open endpoint extensions, positive on the interval, and suppose*

$$j_t^*(b_t \alpha_t) = \mu_t, \qquad \alpha_t = \lambda_t|_{\partial D}.$$

These data construct a pair deformation $\mathcal{Q}_t$ with both actual smooth completed families. For every parameter there are an ambient boundary identification $a_t : \partial D_{\mathcal{Q}_t} \to \partial D$ and a native ribbon carrier diffeomorphism $e_t : M \to F_{\mathcal{Q}_t}$ such that $a_t \circ j_{\mathcal{Q}_t} \circ e_t = j_t$, the whole compact ribbon image under a_t is $j_t(M)$, the witness transported through a_t is b_t, and the ribbon primitive pulls back through e_t to μ_t. The ambient is the exact collar localization of (D, λ_t) for a constructed collar C_t. Its boundary identification a_t is the identity of the inherited boundary presentation, and $(a_t^{-1})^* \alpha_{\mathcal{Q}_t} = \alpha_t$. If Φ_t and ι_t denote this same constructed ribbon completion's flow and compact inclusion, put

$$K_t = \{x \in M : \Phi_t^s(\iota_t(e_t(x))) \in \iota_t(e_t(M)) \text{ for every real } s \geq 0\}.$$

Then the image under a_t of its full embedded core is exactly $j_t(K_t)$. The ambient presentation and the native ribbon primitive presentation are hypotheses here; neither completed family is supplied.

Proof. Extend the native map germs on the same F, i to a smooth partial map g on an open neighborhood of $[0,1] \times F$, retaining every $j_t(x)$ at $i(x)$. The partial-family rectangle lemma restricts to one open spatial neighborhood of F and one open parameter neighborhood, giving maps f_t smooth jointly there with $f_t \circ i = j_t$. These maps are injective on the compact carrier, because $i(M) = F$ and every j_t is injective. Differentiation gives $df_{ti(x)} \circ di_x = dj_{tx}$. For a vector in the kernel of $df_{ti(x)}$, lift it through the surjective di_x. Its lift lies in the kernel of dj_{tx} and hence is zero. Thus df_t is injective on the whole compact carrier.

The two native primitive identities and pullback composition give $i^*(f_t^*(b_t\alpha_t)) = \mu_t = i^*\eta_t$. For any tangent vector at $i(x)$, lift it through di_x and evaluate this equality. It proves $f_t^*(b_t\alpha_t) = \eta_t$ on the full compact carrier, including at its boundary. Restricting the ribbon presentation to the common map neighborhood preserves the joint primitive regularity, the Liouville structures and these compact identities.

Apply the compact-germ witnessed-pair-family construction to these maps, the two original primitive families and the supplied b_t. Its further uniform embedding-neighborhood restriction gives genuine smooth embeddings, and it constructs the actual ambient and ribbon completion families and their pair deformation. The ambient boundary presentation is unchanged by its localization, so take its identity as a_t; the localized boundary primitive retains the original covector values. Compose the original e with the native carrier diffeomorphisms of the successive open restrictions and the collar localization. This is e_t. Every original compact point survives each restriction, and the map formulas agree on those points, so the resulting embedding satisfies $j_{\mathcal{Q}_t} \circ e_t = j_t$ at every $x \in M$. Surjectivity of e_t gives the full image equality. Localization retains the ambient boundary identity and b_t, while the witnessing primitive identity pulls back through e_t to the original μ_t.

Finally e_t bijects M with the compact ribbon of this very completed pair. The all-forward-time definition therefore says that a native point belongs to K_t precisely when its carrier image belongs to that ribbon's full core. Taking its embedded image and using $j_{\mathcal{Q}_t} \circ e_t = j_t$ gives exactly $j_t(K_t)$. Thus the primitive, embedded ribbon and full core identities all use the same produced pair and completion family. □

Lemma A.179 (Native positive-dimensional ribbons on a fixed ambient family)**.** *Let $n \geq 2$ and let D be a fixed compact $2n$-dimensional domain presentation with jointly smooth interval Liouville primitives λ_t and open endpoint extensions. Let M be a compact native $(2n-2)$-dimensional manifold with one-forms μ_t having joint native extension germs, nondegenerate exterior derivatives and outward contraction fields. Suppose $j_t : M \to \partial D$ has joint native map germs, is injective with injective differential at each parameter, and a witness b_t on the actual boundary has joint local scalar germs, is positive on the interval, and satisfies $j_t^*(b_t\alpha_t) = \mu_t$. These original native ribbon data construct the pair deformation and all preservation identities of the preceding lemma, including its native carrier maps and its full core computed in the same constructed ribbon completion. The ambient is the exact localization of (D, λ_t) for a constructed collar. No ribbon presentation, extended ribbon family, collar or completed family is supplied; the fixed ambient presentation remains an input.*

Proof. Construct the native ribbon Liouville family from M, μ_t and its joint coefficient germs, using the given interval nondegeneracy and outwardness. Its fixed native carrier presentation has bijective inclusion differential and actual local source-coordinate extensions. It retains every μ_t through the same inclusion and has the required joint primitive regularity. The boundary scalar-germ lemma makes the original b_t jointly smooth with open endpoint extensions, without changing

its positive interval values. Apply the native ribbon pair-family lemma to this constructed ribbon presentation, the supplied ambient family, the original map germs, injectivity and immersion, and the full original witnessing identity. It constructs the two collars and completion families, the actual pair deformation and all its native carrier, ribbon image, witness, primitive and full-core identities. Its ambient collar equality retains the actual (D, λ_t) localization, and its inherited identity boundary map retains the actual boundary primitive. All outputs use the same constructed native ribbon family and the same pair tuples. □

Lemma A.180 (Completed pair families from original native positive-dimensional data)**.** *Let $n \geq 2$, let A, M be compact native smooth manifolds with boundary of dimensions $2n$ and $2n-2$, and let λ_t, μ_t have joint native extension germs, nondegenerate exterior derivatives and outward contraction fields for every $t \in [0,1]$. Let $j_t : M \to \partial A$ have joint native map germs, be injective with injective differential, and let $b_t : \partial A \to \mathbb{R}$ have joint local scalar germs, be positive on the interval, and satisfy*

$$j_t^*(b_t \lambda_t|_{\partial A}) = \mu_t.$$

These original native data construct a presented pair deformation $\mathcal{Q}_t$ with both actual smooth completed families. For each parameter there are native carrier diffeomorphisms $e_{A,t} : A \to D_{\mathcal{Q}_t}$ and $e_{M,t} : M \to F_{\mathcal{Q}_t}$ and a boundary diffeomorphism $e_{B,t} : \partial A \to \partial D_{\mathcal{Q}_t}$ with

$$\iota_{\partial D_{\mathcal{Q}_t}} \circ e_{B,t} = e_{A,t}|_{\partial A}, \qquad j_{\mathcal{Q}_t} \circ e_{M,t} = e_{B,t} \circ j_t, \qquad b_{\mathcal{Q}_t} \circ e_{B,t} = b_t.$$

The ambient and ribbon primitives pull back through $e_{A,t}$ and $e_{M,t}$ to λ_t and μ_t on every native tangent, and the boundary primitive pulls back through $e_{B,t}$ to $\lambda_t|_{\partial A}$. The whole compact ribbon image is $e_{B,t}(j_t(M))$. With K_t defined by the all-forward-time condition on the same constructed ribbon completion through $e_{M,t}$, as in Lemma A.178, the full embedded core is $e_{B,t}(j_t(K_t))$. No compact presentation, boundary equivalence, extended map, witness extension or completed family is an input. The carrier and boundary identifications are asserted separately at each parameter; joint smoothness of these chosen identifications is not part of this conclusion. This is a construction on fixed native manifolds; arbitrary compact bundle trivializations and the full source-family correspondence remain separate.

Proof. Construct the native ambient Liouville family and fixed boundary presentation from A, λ_t. Write its compact inclusion and carrier map as i_A, e_A, its presented primitives as L_t, and its fixed boundary map as e_B. It has $i_A^* \lambda_{L_t} = \lambda_t$ and $e_B^*(\lambda_{L_t}|_{\partial D}) = \lambda_t|_{\partial A}$. Set $j_t' = e_B \circ j_t$ and $b_t' = b_t \circ e_B^{-1}$. The two germ-composition lemmas retain their joint native extensions. Bijectivity of e_B preserves injectivity, and its invertible differential composed with dj_t preserves immersion. Positivity of b_t' is exactly the original interval positivity. Pullback composition and the boundary primitive identity give

$$(j_t')^*(b_t' \lambda_{L_t}|_{\partial D}) = j_t^* e_B^*((b_t \circ e_B^{-1})\lambda_{L_t}|_{\partial D}) = j_t^*(b_t \lambda_t|_{\partial A}) = \mu_t.$$

Thus all hypotheses of the native fixed-ambient ribbon-family construction hold for these same transported data.

Apply that construction to $D, L_t, M, \mu_t, j_t', b_t'$. It gives $\mathcal{Q}_t$, its exact ambient collar localization C_t, its inherited identity boundary map and its native ribbon carrier diffeomorphism $e_{M,t}$. Compose e_A with the native carrier diffeomorphism of this very ambient localization to obtain $e_{A,t}$. If a_t is the composite open inclusion from its localized extension into D_{ext}, then $a_t \circ e_{A,t} = i_A$ on the native carrier and $\lambda_{\mathcal{Q}_t} = a_t^* \lambda_{L_t}$ there. Pullback composition therefore gives $e_{A,t}^* \lambda_{\mathcal{Q}_t} = i_A^* \lambda_{L_t} = \lambda_t$ on every native tangent. The inherited boundary primitive is unchanged, so the original e_B gives $e_{B,t}$ with its native inclusion square and full boundary primitive pullback.

The fixed-ambient consumer gives $j_{\mathcal{Q}_t} \circ e_{M,t} = j'_t = e_{B,t} \circ j_t$ and retains $e_{M,t}^* \mu_{\mathcal{Q}_t} = \mu_t$. Its witness is b'_t, so $b_{\mathcal{Q}_t}(e_{B,t}(x)) = b_t(x)$. Its whole compact image is $j'_t(M) = e_{B,t}(j_t(M))$. Finally the same consumer's all-forward-time core identity is $j'_t(K_t) = e_{B,t}(j_t(K_t))$, computed in its actual constructed ribbon completion. All carrier maps and primitive equalities therefore belong to the same constructed parameter tuple and completion family. □

Lemma A.181 (Joint native identifications in a localized completion family)**.** *Let $D \subset X$ be a fixed positive-dimensional smooth compact domain presentation, let N be a native compact manifold with boundary, and let $e : N \to D$ be a smooth diffeomorphism with ambient composite $i : N \to X$. Take the localized completion family of Lemma A.141, with open inclusions $a_t : U_t \to X$, compact presentation maps $H_t : U_0 \to U_t$ and real-parameter presentation maps $h_t : X \to X$. Let $D_t = a_t^{-1}(D)$ and let $e_t : N \to D_t$ be the original-point identification, characterized by $a_t \circ e_t = i$. The identifications with the reference compact carrier*

$$R_t = e_0^{-1} \circ H_t^{-1} \circ e_t : N \to N, \qquad R_t^{-1} = e_t^{-1} \circ H_t \circ e_0$$

are jointly smooth together with their inverses, with smooth parameter endpoint extensions in native coordinates. They satisfy

$$i \circ R_t = h_t^{-1} \circ i, \qquad i \circ R_t^{-1} = h_t \circ i.$$

The induced boundary maps are the identity on the native boundary. These identifications use the compact presentation maps of the same constructed completed family.

Proof. The localization inclusion square gives $a_t H_t = h_t a_0$. Since $a_t e_t = i$, applying this square to $H_t^{-1} e_t(x)$ gives $a_0 H_t^{-1} e_t(x) = h_t^{-1} i(x)$. Also $a_0 e_0 = i$. Injectivity of a_0 therefore proves $R_t = e^{-1} h_t^{-1} e$, where h_t is restricted to D. Inverting this equality gives $R_t^{-1} = e^{-1} h_t e$ and both displayed identities. Each h_t preserves D for every real parameter, so these two formulas define extensions for all real parameters.

The forward and inverse maps of h_t are jointly smooth on $\mathbb{R} \times X$. In an adapted domain chart, the compact carrier is a full chart at an interior point and a closed half-space at a boundary point. Its native chart coordinates are restrictions of smooth ambient coordinates. Restricting these joint maps to the carrier and composing with those coordinates consequently gives jointly smooth native coordinate functions. Composing with the fixed smooth diffeomorphisms e, e^{-1} gives joint native smoothness of both R_t and R_t^{-1}, including the parameter endpoints. The boundary map of every h_t is the identity; conjugating its restriction through $e|_{\partial N}$ proves the native boundary identity. Finally the same inclusion square identifies these maps with the given H_t, so no independent completed-family presentation has been introduced. □

Lemma A.182 (Coherent native identifications for positive-dimensional pair families)**.** *Under the hypotheses of Lemma A.180, its construction can choose one pair deformation $\mathcal{Q}_t$ and the same native carrier maps $e_{A,t}, e_{M,t}$ and boundary maps $e_{B,t}$ retaining every inclusion, primitive, ribbon, witness and full-core identity of that lemma, such that the native identifications are jointly smooth relative to the compact presentations of this very deformation. Explicitly, if $H_{A,t}, H_{M,t}$ are its ambient and ribbon compact presentation maps from parameter zero, then*

$$e_{A,0}^{-1} H_{A,t}^{-1} e_{A,t} : A \to A, \qquad e_{M,0}^{-1} H_{M,t}^{-1} e_{M,t} : M \to M$$

and their inverse families are jointly smooth in native coordinates, with smooth parameter endpoint extensions. The boundary identification transported through $H_{A,t}|_{\partial D_{\mathcal{Q}_0}}$ is $e_{B,0}$ for every parameter and is compatible with these same compact and completed presentations. The result concerns fixed native manifolds; trivialization of arbitrary compact domain and ribbon bundles is a separate assertion.

Proof. Keep the ambient primitive presentation, its native carrier map and its fixed boundary map from the native pair-family construction. Keep likewise the constructed ribbon primitive presentation and its native carrier map. The uniform map-neighborhood restrictions are fixed spatial open restrictions. They retain every compact point, and their native carrier identifications are fixed smooth diffeomorphisms.

Apply the localized completion-family construction to each of these two fixed presentations and its jointly smooth primitive family. Assemble the pair deformation using exactly the compact and completed presentation maps of these two constructed families. Choose $e_{A,t}, e_{M,t}$ by composing the fixed native carrier maps with the original-point identifications of the respective localizations. The preceding lemma applies to both localized families and proves the two claimed joint identifications and their inverse regularity. Their boundary maps are identities, so the original fixed e_B remains the boundary map at every parameter.

The chosen compact tuples and original-point maps are those in the native pair-family proof. Its pullback computations therefore retain both full native primitives, the native boundary inclusion square and its primitive, every ribbon point and every witness value. Its all-forward-time core is computed in the same constructed ribbon completion through the same $e_{M,t}$, so its embedded-core identity is retained as well. This adds joint identification regularity to that construction without changing any of its preservation maps. □

Lemma A.183 (Coherent native identifications for finite-ribbon pair families)**.** *Under the hypotheses of Lemma A.177, its construction can retain one pair deformation, both actual smooth completed families and every native inclusion, primitive, ribbon, witness and full-core identity, with jointly smooth native identifications and inverses relative to that deformation's ambient and ribbon compact presentations: composing a native carrier map with the inverse compact presentation and the inverse parameter-zero native carrier map gives a jointly smooth native self-map family, and so does its inverse. Both have smooth parameter endpoint extensions in native coordinates. Its boundary identification is constant after transport through the same ambient boundary presentation. The finite ribbon's full embedded core remains exactly $e_{B,t}(j_t(M))$. Empty finite carriers are allowed. Arbitrary compact bundle trivializations are a separate assertion.*

Proof. Retain the native ambient primitive presentation and fixed boundary map from the finite-ribbon pair-family construction. Apply the localized completion-family construction to its varying primitive and choose the native ambient carrier maps by its original-point identifications. The joint-identification lemma gives the ambient forward and inverse regularity and fixes its boundary map.

The native finite ribbon uses its fixed zero-dimensional carrier presentation and the identically zero primitive family. Its localized completion family has jointly smooth real-parameter ambient presentation maps h_t, preserving its finite carrier and satisfying $h_0 = \mathrm{id}$. Their restrictions to the finite carrier and the restrictions of their inverses are jointly continuous. Near each parameter-point pair, continuity keeps the image in one isolated native target chart. All its zero-dimensional coordinate functions are zero, hence smooth. Thus both restricted maps are jointly smooth in native coordinates for all real parameters. Conjugate them by the fixed original native carrier diffeomorphism. The localization inclusion square identifies the resulting native maps with the reference-transported finite carrier identifications and their inverses, exactly as in the positive-dimensional argument. This proves their joint smoothness and parameter endpoint extensions using that same localized family. Assemble the pair deformation with precisely these two families and their compact and completed presentation maps. The original-point identities retain every ambient and ribbon point, so the primitive pullbacks, boundary inclusion square, embedding and witness

formulas of the finite-ribbon construction remain valid on this same deformation. Its completed ribbon field is zero, hence every compact ribbon point is in the full core and no other point is in that core. Surjectivity of the native finite carrier map and the retained embedding identity give exactly $e_{B,t}(j_t(M))$. All maps and equalities are valid also for the empty carrier. □

Lemma A.184 (Native coordinate germs survive the constructed family endpoints)**.** *The coherent native pair-family constructions above retain smooth extensions of the original native coordinate functions on the actual ambient endpoint presentation at every parameter. For positive-dimensional ribbons they retain these extensions on the actual ribbon endpoint as well. All primitive, map, witness, full-core and two-way joint identification identities hold for those same pairs and the same native carrier maps.*

Proof. The original native primitive-family construction supplies smooth coordinate extensions on its fixed spatial presentation. If a native inclusion $a : M \to N$ has a coordinate extension q on an open set $O \subset N$, and $f : P \to N$ is smooth with $f \circ b = a$, then $q \circ f$ on the open set $f^{-1}(O)$ is a smooth extension of the same native coordinate function through $b : M \to P$. Its equality on native points follows by substituting $f(b(x)) = a(x)$.

Apply this observation to each fixed map-neighborhood restriction and to the actual collar-localization inclusion used by the pair-family construction. Those maps are smooth and their compact squares retain every original point. Thus the coordinate germs pull back to the very endpoint domains already used by the constructed compact and completed presentations. Keep these germs alongside the earlier construction's output. Nothing in its primitive, boundary, ribbon, witness, full-core or joint-identification proof changes. For a finite ribbon only the positive-dimensional ambient requires this coordinate argument; the finite native ribbon and its entire-core identity are retained as before. □

Lemma A.185 (Native presentation on a prescribed compact endpoint)**.** *Let D be a fixed positive-dimensional represented Liouville domain, let (M, μ) be a native manifold of the same dimension, and let $e : M \to D$ be a native diffeomorphism. Suppose the composite inclusion $a : M \to \widetilde{D}$ has the coordinate extension germs above and $a^*\lambda_D = \mu$. These data give a native Liouville presentation whose represented domain is exactly D and whose native inclusion is exactly a.*

Proof. The composite a is smooth because both e and the compact-domain inclusion are smooth. The compact-domain inclusion has injective full differential, while de is invertible. Their composite is injective between real tangent spaces of equal finite dimension, so it is bijective. The carrier map e is a homeomorphism and the inclusion of the compact carrier is a closed embedding; hence their composite is a closed embedding onto that carrier. Exterior naturality for a smooth map from a native manifold to a boundaryless ambient gives $a^*d\lambda_D = d\mu$, including boundary tangents. Together with the given primitive identity and coordinate germs these are exactly the defining properties of the claimed presentation. Each property uses the already fixed D and a, so neither is replaced. □

Lemma A.186 (Prescribed maps on the original positive-dimensional endpoint completions)**.** *Fix two positive-dimensional native carrier presentations on specified compact Liouville domains D, D', retaining their primitives and native coordinate germs. Let C, C' be their original specified completions. A prescribed native diffeomorphism h with native smooth extension germs and $h^*\mu' = c\mu$, for constant $c > 0$, extends to a smooth diffeomorphism $H : C \to C'$ preserving that primitive scale, both native compact squares, its boundary square and its same-radius end square for nonnegative radii. It intertwines the actual flows for every real time and identifies full native core membership.*

This applies to the literal ambient and positive-dimensional ribbon completions of the constructed native pair families; ambient maps use $c = 1$. It does not by itself supply the boundary–ribbon compatibility required for a permitted pair identification.

Proof. Construct native presentations on the fixed domains by the preceding endpoint-presentation lemma. Apply the native prescribed-map completion producer to these presentations and the original C, C'. That producer restricts extension neighborhoods only internally; its completed map has source C and target C', with the stated native forward and inverse squares and the boundary and end formulas. The scaled primitive identity gives all-real flow intertwining. Surjectivity of the two native carrier maps and of h identifies the image of the entire compact carrier under H with the target compact image. Hence a forward flow point lies in the source compact image exactly when its image lies in the target compact image. Requiring this for every nonnegative real time gives the full-core equivalence. For a constructed pair family, use the primitive and coordinate data retained on those very endpoint domains and its original completion records. No fresh domain or completion is selected by this application. □

Lemma A.187 (Prescribed finite maps on the original zero-dimensional completions)**.** *For two specified zero-dimensional compact domains with original completions C, C' and native carrier diffeomorphisms $a : M \to D$, $b : N \to D'$, every prescribed native diffeomorphism $h : M \to N$ extends to a diffeomorphism $H : C \to C'$ retaining both native compact squares. It preserves any specified constant primitive scale and intertwines the actual complete flows. In particular this applies to the original finite-ribbon completions of two constructed pair families. Empty native carriers are included.*

Proof. A zero-dimensional compact domain has empty boundary. Completion coverage therefore says that its compact inclusion is surjective; its embedding property gives injectivity. Compose this inclusion with a. This is a bijection $A : M \to C$. Both manifolds are discrete and zero-dimensional, so the map and its inverse are smooth. The same construction gives $B : N \to C'$. Define $H = B \circ h \circ A^{-1}$. Its forward and inverse native squares follow by cancellation. Every one-form on a zero-dimensional manifold is zero, so the scaled primitive identity holds for any real constant. Scaled primitive transport gives the actual all-real flow equation. The same bijections and cancellations apply to empty carriers without choosing a point. □

Lemma A.188 (Joint compact map germs on the constructed localized completion)**.** *Let a positive-dimensional constructed localized completion family have original compact carrier $D \subset X$, localized reference ambient C_0, open native inclusions $J_t : C_t \to X$, and jointly smooth real presentation maps $H_t : X \to X$ with jointly smooth inverses, preserving D. Write $P_t : C_0 \to C_t$ for its actual completed presentations; on these localized spaces their defining square is $J_t \circ P_t = H_t \circ J_0$ for $t \in [0, 1]$. There are open neighborhoods W^-, W^+ of $\mathbb{R} \times J_0^{-1}(D)$ in $\mathbb{R} \times C_0$ and jointly smooth maps $g^- : W^- \to C_0$, $g^+ : W^+ \to C_0$ satisfying*

$$J_0(g^-(t, z)) = H_t^{-1}(J_0(z)), \qquad J_0(g^+(t, z)) = H_t(J_0(z)).$$

For every original compact point $x \in D$, put $x_t = J_t^{-1}(x)$. At each interval parameter these maps retain

$$g^-(t, x_0) = P_t^{-1}(\iota_t(x_t)),$$

where ι_t is the actual compact inclusion into C_t. The completed ambients here are the actual collar-localized cylindrical constructions, whose compact inclusions are identities on their extension spaces.

Proof. Since J_0 is an open embedding, its image is open and its inverse on that image is smooth. Define

$$W^- = \{(t,z) : H_t^{-1}(J_0(z)) \in J_0(C_0)\}, \qquad W^+ = \{(t,z) : H_t(J_0(z)) \in J_0(C_0)\}.$$

Joint smoothness gives openness of these sets. For $z \in J_0^{-1}(D)$, carrier preservation puts both displayed images in $D \subset J_0(C_0)$ at every real parameter; hence both sets contain the required full real carrier product. On W^- put $g^- = J_0^{-1} \circ H_t^{-1} \circ J_0$, and on W^+ put $g^+ = J_0^{-1} \circ H_t \circ J_0$. These compositions are smooth and give the stated identities.

At an interval parameter, evaluate the actual presentation square at $P_t^{-1}(x_t)$ to obtain $H_t(J_0(P_t^{-1}(x_t))) = J_t(x_t) = x$. Thus $J_0(P_t^{-1}(x_t)) = H_t^{-1}(x) = J_0(g^-(t,x_0))$. Injectivity of J_0 and the identity compact inclusion prove the required actual completed-presentation square. □

Lemma A.189 (Native compact family germs retaining a prescribed reference completion)**.** *Let M be a compact native positive-dimensional smooth manifold with boundary. Let α_t, $t \in [0,1]$, have joint native smooth extension germs, nondegenerate exterior derivatives and outward Liouville fields for regular native defining functions. Let A be an independently prescribed native presentation of (M, α_0), with actual completion C and native compact inclusion $a : M \to C$. These data construct one fixed compact presentation of M with native carrier identification e, Liouville structures L_t satisfying $e^*\lambda_{L_t} = \alpha_t$, and its actual localized completion family C_t. Let $a_t : M \to C_t$ be its retained original compact inclusions, and $P_t : C_0 \to C_t$ its actual completed presentations. There is a fixed strict diffeomorphism $q : C \to C_0$ with $q \circ a = a_0$ and $q^*\widehat{\lambda}_0 = \widehat{\lambda}_C$. There are open neighborhoods W of $[0,1] \times a(M)$ in $\mathbb{R} \times C$ and W' of $[0,1] \times a_0(M)$ in $\mathbb{R} \times C_0$, with jointly smooth maps $g : W \to C_0$ and $\bar{g} : W' \to C$ retaining*

$$g(t, a(x)) = P_t^{-1}(a_t(x)), \qquad \bar{g}(t, a_0(x)) = a(A_t^{-1}(x)) \quad (t \in [0,1]),$$

where A_t is the native carrier diffeomorphism determined by $a_0(A_t(x)) = P_t^{-1}(a_t(x))$. The prescribed reference presentation and completion are retained. The same selected native carrier inclusion retains smooth local extensions of its native chart coordinates. These are compact comparison germs; a comparison on the whole moving completed family is not a conclusion.

Proof. Apply the native joint primitive-family construction. It gives one compact presentation and Liouville family, preserves every native primitive, and constructs its localized completed family with the same native coordinate extensions. Its time-zero localized domain therefore presents exactly (M, α_0). The native primitive and outward germs produce an inward collar of that native domain. Apply the strict inward-collar comparison to the prescribed presentation A, C and this constructed time-zero presentation and completion. It gives the fixed strict map q, its compact identity $q \circ a = a_0$, and the primitive equation.

Apply the preceding compact-germ lemma to the constructed family to get g^-, g^+ on open neighborhoods of the real compact carrier product. Pull the domain of g^- back through $(t,z) \mapsto (t, q(z))$ and put $g(t,z) = g^-(t,q(z))$ there. On the domain of g^+ put $\bar{g}(t,z) = q^{-1}(g^+(t,z))$. Both maps are smooth, and their domains contain the asserted interval carrier products because $q(a(x)) = a_0(x)$. The preceding lemma's actual presentation square gives $g(t,a(x)) = P_t^{-1}(a_t(x))$. The real original presentations preserve the carrier in both directions; their inverse restriction defines A_t through the stated native square, and their forward restriction gives $g^+(t, a_0(x)) = a_0(A_t^{-1}(x))$. Finally apply q^{-1} and its native compact square to obtain the formula for $\bar{g}$. Every choice uses the same constructed compact and completed family and the original prescribed reference completion. □

Lemma A.190 (Compact comparison extensions give full interior family germs)**.** *Let C be a realized completion of D, let N be a smooth manifold, and let $f_t : C \to N$ be prescribed maps for $t \in [0,1]$. Suppose a jointly smooth $g : W \to N$ on an open neighborhood W of $[0,1] \times \iota(D)$ satisfies $g(t, \iota(x)) = f_t(\iota(x))$ for every compact point $x \in D$. At each (t,z) with $t \in [0,1]$ and $z \in \iota(\operatorname{int} D)$, there is a smooth local germ agreeing with $f_s(y)$ at every interval point (s,y) of its open domain. No boundary agreement outside the compact image follows from this assertion.*

Proof. The compact inclusion is smooth with injective differential between manifolds of the same dimension. The inverse function theorem makes it a local diffeomorphism, hence an open map. Thus $\iota(\operatorname{int} D)$ is open in C. Intersect W with $\mathbb{R} \times \iota(\operatorname{int} D)$ and restrict g to this open intersection U. It contains (t,z) because $\operatorname{int} D \subset D$. Every interval point $(s,y) \in U$ has $y = \iota(x)$ for some $x \in \operatorname{int} D \subset D$. The assumed compact identity gives $g(s,y) = f_s(y)$. Hence this restriction of g is the required full local germ. □

Lemma A.191 (The transported completed field is smooth with the primitive)**.** *Let C be a realized completion of D, let C_t be realized completions of compact Liouville domains D_t, and let $P_t : C \to C_t$ be specified completed presentation diffeomorphisms for $t \in [0,1]$. Suppose the one-forms $\eta_t = P_t^* \widehat{\lambda}_t$ form a jointly smooth family with open extensions across the interval endpoints. Then the actual transported fields*

$$X_t(z) = d(P_t^{-1})_{P_t(z)}\big(\widehat{Z}_t(P_t(z))\big)$$

form a jointly smooth family with open endpoint extensions on C. No joint regularity of P_t as a family of maps is assumed.

Proof. Naturality of the exterior derivative gives $d\eta_t = P_t^* d\widehat{\lambda}_t$, which is nondegenerate because P_t is a diffeomorphism. Evaluating the Liouville equation on $dP_t(v)$ gives $\iota_{X_t} d\eta_t = \eta_t$. In each fixed coordinate chart on C, extend the coefficients of η_t jointly smoothly around the required parameter and spatial points. Their spatial derivatives form a smooth matrix $A(t,y)$ for the contraction map of $d\eta_t$. Restrict to the open locus where A is invertible; this contains every required interval point. The coefficients $A(t,y)^{-1}\eta(t,y)$ are jointly smooth there. At interval parameters they equal the actual X_t by uniqueness of the contraction equation. Thus the actual fields have the claimed chartwise smooth extensions. This argument is local on the completed manifold and requires no compactness of that manifold or common larger parameter interval. In dimension zero the unique field is zero. □

Lemma A.192 (A common two-sided germ of the prescribed completion collars)**.** *Let C, C_t be realized completions of compact Liouville domains D, D_t. Let h_t be compact presentation diffeomorphisms with boundary maps B_t, and let $P_t : C \to C_t$ be specified completed presentation diffeomorphisms, agreeing with h_t on the compact carriers and satisfying*

$$P_t(e(b,r)) = e_t(B_t(b), r) \qquad (r \geq 0).$$

Suppose $P_t^* \widehat{\lambda}_t$ is jointly smooth with open endpoint extensions on C. There are a single $\delta > 0$ and a jointly smooth map $\Gamma : \mathbb{R} \times \partial D \times \mathbb{R} \to C$ such that

$$\Gamma(t,b,r) = P_t^{-1}\big(e_t(B_t(b), r)\big) \qquad (t \in [0,1],\ -\delta < r < \delta).$$

The prescribed collar is evaluated using the actual all-real end trajectories. No negative-radius presentation square or joint regularity of those trajectories is an additional hypothesis.

Proof. The preceding lemma supplies smooth open extensions X_t of the actual transported completed fields. Their chart expressions give a smooth section on an open neighborhood Ω of $[0,1]\times C$: for each interval point choose a coordinate coefficient neighborhood, transfer it back to the manifold by that chart, and take the union of all open sets on which this section is smooth. The seam $s(b) = e(b,0)$ is a fixed smooth map, and $s(\partial D)$ is compact. Apply the supported parameter-preserving local-flow lemma to this compact seam. It gives one positive δ and a globally smooth auxiliary flow $\Phi^r(t,z) = (t,\phi^r(t,z))$ whose spatial equation is $\partial_r\phi^r(t,z) = X_t(\phi^r(t,z))$ for every seam point and $|r| < \delta$. Put $\Gamma(t,b,r) = \phi^r(t,s(b))$. It is jointly smooth by composition.

Fix an interval parameter t and $b \in \partial D$. The actual curve $q(r) = P_t^{-1}(e_t(B_t(b),r))$ solves X_t at every real r, by the target end-orbit equation and the chain rule. The nonnegative presentation square at radius zero gives $q(0) = s(b) = \Gamma(t,b,0)$. Both curves solve the same smooth field on $(-\delta,\delta)$. Local uniqueness makes their equality set open in this interval; continuity and the Hausdorff property make it closed. It contains zero, and the interval is connected, so the curves agree throughout it. The common width was chosen over the whole compact seam, including both interval endpoints. An empty boundary has no seam conditions and is covered by the same construction. □

Lemma A.193 (The prescribed collar germ extends over the outward cylinder)**.** *Under the hypotheses of the preceding collar lemma, there are $a < 0$ and a jointly smooth map $\beta : \mathbb{R}\times\partial D\times\mathbb{R} \to C$ such that*

$$\beta(t,b,r) = P_t^{-1}\big(e_t(B_t(b),r)\big) \qquad (t\in[0,1],\ r > a).$$

Agreement is required only above this negative cutoff; the auxiliary map need not satisfy a presentation square outside the parameter interval.

Proof. Choose δ,Γ from the collar lemma. Let ϑ be a smooth function with values in $[0,1]$, zero on $(-\infty,0]$ and one on $[1,\infty)$. Put

$$\theta(r) = \vartheta(4r/\delta - 1), \qquad s(r) = (1-\theta(r))r + \theta(r)\delta/2.$$

Then $s(r) = r$ for $r \le \delta/4$. For $0 \le r \le \delta/2$, $s(r)$ is a convex combination of two numbers in $[0,\delta/2]$; for $r \ge \delta/2$ it equals $\delta/2$. Hence $0 \le s(r) < \delta$ for every $r \ge 0$. Using the fixed reference completed flow Φ_C, define

$$\beta(t,b,r) = \Phi_C^{\,r-s(r)}\big(\Gamma(t,b,s(r))\big).$$

This is jointly smooth by smoothness of s,Γ and the fixed flow. Take $a = -\delta$. If $a < r \le \delta/4$, then $s(r) = r$ and the flow time is zero, so the collar equality gives the required value. If $r > \delta/4$, then $r > 0$ and $s(r) \in [0,\delta)$. The collar equality and the nonnegative presentation square give $\Gamma(t,b,s(r)) = e(b,s(r))$. The reference end-orbit equation gives $\Phi_C^{\,r-s(r)}(e(b,s(r))) = e(b,r)$. The nonnegative presentation square identifies this last point with $P_t^{-1}(e_t(B_t(b),r))$, completing the proof. □

Lemma A.194 (Positive cylindrical formulas and full compact germs extend completed maps)**.** *Let C be a realized completion of D, let N be a smooth manifold, and let $f_t : C \to N$ be prescribed maps for $t \in [0,1]$. Let $V \supset [0,1]$ be open and let $\beta(t,b,r)$ be jointly smooth on $V\times\partial D\times(0,\infty)$, with*

$$f_t(e(b,r)) = \beta(t,b,r) \qquad (t\in[0,1],\ r>0).$$

Suppose every $(t,\iota(x))$ with $t\in[0,1]$ and $x\in D$ has an open neighborhood U in $\mathbb{R}\times C$ and a smooth map $g_U : U \to N$ agreeing with $f_s(z)$ at every interval point $(s,z)\in U$. Then the same f_t has a jointly smooth extension on an open neighborhood of $[0,1]\times C$. No cylindrical formula at negative radius is required.

Proof. Write $A = [0,1] \times C$. Smooth end coordinates define a smooth map G on $O = V \times e(\partial D \times (0,\infty))$ by $G(t, e(b,r)) = \beta(t,b,r)$. It agrees with f on $A \cap O$. Put

$$K = [0,1] \times \big(\iota(D) \cup e(\partial D \times [0,1/2])\big), \quad L = [0,1] \times e(\partial D \times [1/2,2]), \quad B = \mathbb{R} \times e(\partial D \times [1,\infty)).$$

The sets K, L are compact; $L \subset O$, and B is closed. Moreover $K \cap B$ is empty: positive end points are outside the compact image, and injectivity of the end coordinates separates radii at most $1/2$ from radii at least one. Every point of K has a smooth germ agreeing with f on its entire interval neighborhood. At compact points use the supplied germ. At radius zero use $e(b,0) = \iota(b)$ and that same germ; at positive radius use G on O.

Apply exterior-preserving compact germ extension to f on A, the compact set K, closed set B, and initial map $G : O \to N$ with compact $L \subset O$. It gives a smooth map H on an open $W \supset K \cup L$, agreeing with f on $A \cap W$ and retaining G on $W \cap B$. Set $P = V \times e(\partial D \times (1,\infty))$. Since $P \subset O \cap B$, the maps H and G agree on $W \cap P$. They therefore glue to a smooth map on $W \cup P$. Completion coverage puts every interval point either in the compact image, in an end radius in $[0,1/2]$, in one in $[1/2,2]$, or in one greater than two. The first two cases lie in $K \subset W$; the third lies in $L \subset W$; the fourth lies in P. Thus $W \cup P$ contains A, and the glued map equals f at every point of A. This is the claimed extension of the prescribed family. □

Lemma A.195 (A spatial isomorphism gives a parameter-preserving local inverse)**.** *Let M, N be finite-dimensional smooth boundaryless manifolds and $g : \mathbb{R} \times M \to N$ smooth. If $d_x g_{(t,x)} : T_x M \to T_{g(t,x)} N$ is an isomorphism, then $G(s,y) = (s, g(s,y))$ is a local diffeomorphism at (t,x).*

Proof. The derivative of G sends (a,v) to $(a, d_t g_{(t,x)}(a) + d_x g_{(t,x)}(v))$. Given (a,w), its unique inverse image is

$$\big(a, (d_x g_{(t,x)})^{-1}\big(w - d_t g_{(t,x)}(a)\big)\big).$$

Thus $dG_{(t,x)}$ is an isomorphism. The smooth inverse function theorem gives a diffeomorphism between open neighborhoods of (t,x) and $G(t,x)$, which is the required local inverse. □

Lemma A.196 (Actual moving collar squares give full seam comparison germs)**.** *Let C, C_t, h_t, P_t, B_t satisfy the hypotheses of the prescribed-family collar lemma, including joint smoothness of $P_t^* \widehat{\lambda}_t$ on C. Put $\gamma_t(b,r) = P_t^{-1}(e_t(B_t(b), r))$ using the actual end orbits. Let N be a smooth manifold, let $f_t : C \to N$ be prescribed maps, and let $\beta : \mathbb{R} \times \partial D \times \mathbb{R} \to N$ be jointly smooth. Suppose one $\delta > 0$ satisfies*

$$f_t(\gamma_t(b,r)) = \beta(t,b,r) \qquad (t \in [0,1],\ |r| < \delta).$$

For every $t \in [0,1]$ and $b \in \partial D$, there is an open neighborhood U of $(t, e(b,0))$ in $\mathbb{R} \times C$ and a smooth $a : U \to N$ such that $a(s,z) = f_s(z)$ at every interval point $(s,z) \in U$. The neighborhood includes both sides of the seam; no negative-radius square for the fixed reference end map is assumed.

Proof. Choose the joint source collar extension Γ and its positive width ε from the prescribed-family collar lemma. Fix t, b. In a spatial neighborhood of $(b,0)$, $\Gamma(t,\cdot) = \gamma_t$. The actual γ_t is the composition of the diffeomorphism P_t^{-1}, the local diffeomorphism e_t, and the boundary product diffeomorphism $(c,r) \mapsto (B_t(c), r)$. Its spatial differential is therefore invertible, and equality on that spatial neighborhood gives the same differential for $\Gamma(t,\cdot)$. The preceding parameter inverse lemma applies to $G(s,c,r) = (s, \Gamma(s,c,r))$ at $(t,b,0)$. It gives smooth inverse coordinates v on an open neighborhood of $(t, e(b,0))$, since $\gamma_t(b,0) = e(b,0)$ by the nonnegative presentation square at the seam.

Shrink the inverse neighborhood so that its inverse radius lies in $(-\rho, \rho)$, where $\rho = \min(\varepsilon, \delta) > 0$, and put $a = \beta \circ v$ there. This map is smooth by composition. For an interval point (s,z) in this

shrunken neighborhood, write $v(s,z)=(u,c,r)$. The inverse identity gives $G(u,c,r)=(s,z)$, hence $u=s$ and $\Gamma(s,c,r)=z$. The radius restriction and the actual source collar equality give $\gamma_s(c,r)=z$. The assumed moving collar square then gives $a(s,z)=\beta(s,c,r)=f_s(\gamma_s(c,r))=f_s(z)$. This holds at every interval point of the open neighborhood, including at parameter endpoints, rather than only on its compact portion. □

Lemma A.197 (Compact extensions and actual moving collar squares give whole-completion regularity)**.** *Let C,K be realized completions of compact Liouville domains D,F. Supply actual moving completions C_t,K_t, compact presentations h_t,k_t, and completed presentations $P_t:C\to C_t$ and $Q_t:K\to K_t$. Suppose both transported primitive families $P_t^*\widehat{\lambda}_{C_t}$ and $Q_t^*\widehat{\lambda}_{K_t}$ are jointly smooth with open endpoint extensions. Let $B:\partial D\to\partial F$ be a fixed smooth diffeomorphism and let $f_t:C\to K$ be prescribed maps. Suppose there are an open neighborhood W of $[0,1]\times\iota_C(D)$ and a smooth $g:W\to K$ satisfying $g(t,\iota_C(x))=f_t(\iota_C(x))$ for every $x\in D$. Suppose also that the actual moving collars satisfy*

$$f_t\big(P_t^{-1}(e_{C_t}(B_{h_t}(b),r))\big)=Q_t^{-1}(e_{K_t}(B_{k_t}(B(b)),r))\qquad(t\in[0,1],\ r\in\mathbb{R}).$$

Then these same f_t have a jointly smooth extension on an open neighborhood of $[0,1]\times C$.

Proof. The target prescribed-family collar lemma supplies a globally smooth $\Gamma_K(t,c,r)$ agreeing with the actual target moving collar for $|r|<\delta$, where $\delta>0$. Put $\beta(t,b,r)=\Gamma_K(t,B(b),r)$. The assumed moving collar square identifies $f_t(P_t^{-1}(e_{C_t}(B_{h_t}(b),r)))$ with $\beta(t,b,r)$ for $|r|<\delta$. Apply the seam-comparison germ lemma to the source moving completion and this β. It yields full interval-neighborhood germs of f at each $(t,e_C(b,0))$.

At a compact point inside $\iota_C(\operatorname{int}D)$, the interior-germ lemma applied to g gives such a full germ. Every remaining compact point lies on the radius-zero seam: completion coverage outside $\iota_C(\operatorname{int}D)$ writes it as $e_C(b,r)$ with $r\geq 0$, while positive end points are outside the compact image. Thus $r=0$, and the preceding seam germ applies. Consequently every interval compact point has a smooth germ agreeing with $f_s(z)$ at every interval point of its open neighborhood.

For $r>0$, both completed presentation end squares apply and turn the assumed moving collar square into $f_t(e_C(b,r))=e_K(B(b),r)$. The right side is a jointly smooth cylindrical formula, independent of t. Positive-end gluing now extends the same f_t over the whole completion with open parameter endpoint extensions. The argument uses the fixed reference end squares only at positive radius; the two-sided seam germs use the actual moving collar square. □

Lemma A.198 (Strict moving slices retain the original native compact and end points)**.** *Let α_t, $t\in[0,1]$, be a compact native positive-dimensional Liouville primitive family with joint extension germs, nondegenerate exterior derivatives and outward defining-function germs. Let its localized completion family F_t have one original compact presentation D, native carrier diffeomorphism $e:M\to D$, native coordinate extensions, and compact primitives pulling back to α_t through this same e. Write $a_t:M\to F_t$ for the retained native compact inclusions and $B_{F,t}$ for their native boundary identifications. Independently let R be a native presentation of a primitive η on the same M, let $h_t:R\to E_t$ be compact presentation identifications, and let K_t be actual completions of E_t. Suppose $(h_t\circ i_R)^*\lambda_{E_t}=\alpha_t$ on all native tangents. There are strict diffeomorphisms $q_t:F_t\to K_t$ satisfying*

$$q_t(a_t(x))=\iota_{K_t}(h_t(i_R(x))),\qquad q_t^{-1}(\iota_{K_t}(h_t(i_R(x))))=a_t(x),$$
$$q_t(e_{F_t}(B_{F,t}(b),r))=e_{K_t}(B_{h_t}(B_R(b)),r)\qquad(r\in\mathbb{R}).$$

The constructed native boundary coordinates are literally stationary: $B_{F,t}=B_{F,0}$ in the original boundary of D. No joint regularity of q_t or primitive equality $\eta=\alpha_0$ is an assumption or conclusion here.

Proof. At each interval parameter, the original compact map h_t transports R to a native presentation of α_t on E_t. Its native carrier and coordinate extensions transport through that same h_t and its smooth inverse. The constructed localized domain also presents α_t, using its retained inclusion of $e(x)$ and the compact primitive equation. The native primitive and outward germs supply one inward collar for α_t. Apply the strict inward-collar comparison to these two presentations and their actual completions. It gives a strict q_t with both native compact squares. Choose such a map for each t.

Strict primitive preservation transports the complete Liouville flows. The end at $B_{F,t}(b)$ is the flow from $a_t(b)$; the target end at $B_{h_t}(B_R(b))$ is the flow from $\iota_{K_t}(h_t(i_R(b)))$. The compact square and flow transport therefore give the displayed end equality at every real radius. Finally, the open inclusion of each localized slice into the original compact presentation sends its boundary inclusion to the same boundary inclusion of D, and sends its native compact point to $e(x)$. Apply these two inclusion squares to $b \in \partial M$ at parameters t and zero. Both boundary images are $e(b)$ in D. Injectivity of the original boundary inclusion gives $B_{F,t}(b) = B_{F,0}(b)$. □

Lemma A.199 (Relative completion comparisons preserve the transported forms)**.** *Let C, K, F, F' be realized completions, let $P : C \to K$ and $S : F \to F'$ be specified completed presentation diffeomorphisms, and let $q : F' \to K$ be primitive-preserving. Put $A = P^{-1} \circ q \circ S$. Then*

$$A^*(P^*\widehat{\lambda}_K) = S^*\widehat{\lambda}_{F'}.$$

This compares the actual transported primitive $P^*\widehat{\lambda}_K$; it requires no primitive-preservation identity for P or S.

Proof. Cancellation gives $P \circ A = q \circ S$. Pullback composition and the strict primitive identity of q give

$$A^*(P^*\widehat{\lambda}_K) = (P \circ A)^*\widehat{\lambda}_K = (q \circ S)^*\widehat{\lambda}_K = S^*(q^*\widehat{\lambda}_K) = S^*\widehat{\lambda}_{F'}.$$

Every identity uses the specified maps themselves. □

Lemma A.200 (Original compact presentations give germs of the actual relative comparisons)**.** *Let F_t be a positive-dimensional localized completed family with its actual completed presentations $S_t : F_0 \to F_t$ and one fixed native carrier identification. Let $a_t : M \to F_t$ be its native compact inclusions. Independently let R be a native compact presentation, let C be its completion with native inclusion $a : M \to C$, and supply compact presentations $h_t : R \to E_t$, completions K_t, and completed presentations $P_t : C \to K_t$. Suppose smooth slice diffeomorphisms $q_t : F_t \to K_t$ satisfy $q_t(a_t(x)) = \iota_{K_t}(h_t(i_R(x)))$ for all $x \in M$. Put $A_t = P_t^{-1} \circ q_t \circ S_t$. There are jointly smooth maps on open neighborhoods of $[0,1] \times a_0(M)$ and $[0,1] \times a(M)$ agreeing, respectively, with these same A_t and A_t^{-1} on the entire compact carriers. Neither primitive preservation nor joint regularity of q_t is needed, and P_0 is not normalized.*

Proof. Let H_t be the constructed real original compact presentation maps, and define the real native diffeomorphism T_t by $S_t(a_0(x)) = a_t(T_t(x))$ at interval parameters. The actual compact squares of P_t and the assumed q_t give $A_t(a_0(x)) = a(T_t(x))$ and $A_t^{-1}(a(x)) = S_t^{-1}(a_t(x))$. The time-zero presentation S_0 is the identity, so the fixed smooth map A_0 satisfies $A_0(a_0(x)) = a(x)$.

The localized compact-germ lemma supplies smooth partial maps g^-, g^+ around $\mathbb{R} \times a_0(M)$, with $g^-(t, a_0(x)) = S_t^{-1}(a_t(x))$ and $g^+(t, a_0(x)) = a_0(T_t(x))$. The smooth map $A_0 \circ g^+$ therefore agrees with A_t on the whole compact carrier. Pull the domain of g^- back by $(t, z) \mapsto (t, A_0^{-1}(z))$ and compose g^- with this map. The resulting smooth map agrees with A_t^{-1} on the whole reference compact carrier, by $A_0^{-1}(a(x)) = a_0(x)$. Both domains contain the required compact products,

because the localized germs contain all real compact products and A_0 retains all native compact points. This proves the assertions without any normalization or joint smoothness of the prescribed moving maps. □

Lemma A.201 (Native compact and orbit squares make both relative comparisons smooth)**.** *Let F_t, a_t, S_t be a positive-dimensional localized native completed family of primitives α_t. Supply a reference native presentation R, its completion C with native inclusion a, original compact presentations h_t, completions K_t, and completed presentations $P_t : C \to K_t$. Suppose $P_t^*\widehat{\lambda}_{K_t}$ is jointly smooth with open endpoint extensions. Let $q_t : F_t \to K_t$ be smooth slice diffeomorphisms retaining the native compact points and satisfying*

$$q_t(e_{F_t}(B_{F,t}(b), r)) = e_{K_t}(B_{h_t}(B_R(b)), r) \qquad (r \in \mathbb{R}).$$

Put $A_t = P_t^{-1} \circ q_t \circ S_t$. There is a fixed boundary diffeomorphism $B : \partial F_0 \to \partial R$ such that $q_t(e_{F_t}(c, r)) = e_{K_t}(B_{h_t}(B(c)), r)$ for all real r, and the actual moving collars satisfy

$$A_t\big(S_t^{-1}(e_{F_t}(c, r))\big) = P_t^{-1}(e_{K_t}(B_{h_t}(B(c)), r)) \qquad (r \in \mathbb{R}).$$

Both these same A_t and their actual inverses have jointly smooth extensions around their entire interval products with F_0 and C.

Proof. The localized family's native boundary identification is stationary, $B_{F,t} = B_{F,0}$. Set $B = B_R \circ B_{F,0}^{-1}$. For each c write $c = B_{F,0}(b) = B_{F,t}(b)$. The assumed native end square then gives the asserted end square in c. The constructed compact presentation boundary map is the identity in this fixed original boundary, so its actual moving collar is $S_t^{-1}(e_{F_t}(c, r))$. Cancel S_t in the definition of A_t and apply the end square. This gives precisely the displayed actual collar square, at every real radius. Applying A_t^{-1} to it and replacing c by $B^{-1}(d)$ gives the inverse moving collar square.

The original compact-presentation extension lemma gives joint smooth partial maps agreeing with A_t and A_t^{-1} on their whole compact carriers. The constructed transported primitive family $S_t^*\widehat{\lambda}_{F_t}$ is jointly smooth; the original target transported primitive family is smooth by hypothesis. Apply the moving-comparison regularity lemma with these two primitive families, the forward compact extension, and the forward actual collar square. It gives joint smoothness of A_t on the whole reference completion. Apply it again with source and target exchanged, boundary map B^{-1}, the inverse compact extension and inverse collar square. This gives the same assertion for the actual A_t^{-1}. No joint regularity of the slice family q_t is assumed. □

Lemma A.202 (Native data give smooth comparisons with supplied positive-dimensional completed families)**.** *Let M be a compact native smooth manifold of dimension $k + 1 > 0$. Let α_t, $t \in [0, 1]$, have joint native one-form extension germs, nondegenerate exterior derivatives and outward boundary defining-function germs. Independently supply a native presentation R of a primitive η on M, a completion C of R, compact presentations $h_t : R \to E_t$, actual completions K_t, and completed presentations $P_t : C \to K_t$. Suppose $(h_t \circ i_R)^*\lambda_{E_t} = \alpha_t$ on all native tangents and the actual transported completed forms $\eta_t = P_t^*\widehat{\lambda}_{K_t}$ are jointly smooth with open endpoint extensions. The reference primitive η need not equal α_0, and P_0 is not normalized.*

There exist one compact native presentation D, e, a jointly smooth compact primitive family L_t pulling back to α_t through e, native coordinate extensions, and its localized completed family F_t, with actual completed presentations $S_t : F_0 \to F_t$ and native compact inclusions $a_t : M \to F_t$. There are strict diffeomorphisms $q_t : F_t \to K_t$ retaining the original compact points in both directions:

$$q_t(a_t(x)) = \iota_{K_t}(h_t(i_R(x))), \qquad q_t^{-1}(\iota_{K_t}(h_t(i_R(x)))) = a_t(x).$$

There is a fixed $B : \partial D \to \partial R$ such that

$$q_t(e_{F_t}(b,r)) = e_{K_t}(B_{h_t}(B(b)),r) \qquad (r \in \mathbb{R}).$$

For these same q_t, the relative diffeomorphisms $A_t = P_t^{-1} \circ q_t \circ S_t : F_0 \to C$ and their actual inverses have jointly smooth extensions around their entire interval products. They retain the actual moving collars at every real radius:

$$A_t\big(S_t^{-1}(e_{F_t}(b,r))\big) = P_t^{-1}(e_{K_t}(B_{h_t}(B(b)),r)).$$

They satisfy

$$A_t^*\eta_t = S_t^*\widehat{\lambda}_{F_t}, \qquad A_t(e_{F_0}(b,r)) = e_C(B(b),r) \quad (r \geq 0),$$

with the inverse nonnegative end square. Writing $a : M \to C$ for the native compact inclusion and $u_t(x) = S_t^{-1}(a_t(x))$, their compact squares are $A_t(u_t(x)) = a(x)$ and $A_t^{-1}(a(x)) = u_t(x)$. This statement compares the supplied original positive-dimensional moving completions. It does not assert the corresponding zero-dimensional moving result or matching of arbitrary witnessed pairs, chain junctions or prescribed gradings.

Proof. Apply the native primitive-family construction to α_t and its stated regularity, nondegeneracy and outward germs. It produces one compact D, native carrier diffeomorphism e, jointly smooth L_t, the required native coordinate extensions, and the actual localized completed family F_t. The compact primitives pull back to α_t through this same e. The strict moving slice comparison lemma, applied to this constructed family and the independently supplied R, h_t, E_t, K_t, produces strict q_t with both compact squares and their all-real native end squares. Its stationary native boundary identification compares all these end squares using one fixed boundary map B.

Apply the native moving comparison regularity lemma to these same q_t, their compact and end squares, and the assumed smooth original transported primitives. It supplies the actual all-real moving collar square and joint smoothness of A_t and its actual inverse. The relative-primitive lemma, applied at each interval parameter, gives $A_t^*(P_t^*\widehat{\lambda}_{K_t}) = S_t^*\widehat{\lambda}_{F_t}$. For $r \geq 0$, the two actual completed presentation end squares turn the moving collar square into $A_t(e_{F_0}(b,r)) = e_C(B(b),r)$. Applying A_t^{-1} and replacing b by $B^{-1}(c)$ gives the inverse nonnegative end square. No fixed negative-radius end square is used. Finally the definition of u_t, the compact q_t square, and the original compact P_t square give

$$A_t(u_t(x)) = P_t^{-1}(q_t(a_t(x))) = P_t^{-1}(\iota_{K_t}(h_t(i_R(x)))) = a(x).$$

Applying the actual inverse gives the other compact identity. All conclusions use the constructed family and the same slice maps throughout; no comparison map or its extension germs were supplied as additional data. □

Lemma A.203 (Smooth comparisons with prescribed moving finite completions)**.** *Let M be a compact native zero-dimensional smooth manifold and let μ_t, $t \in [0,1]$, be arbitrary native one-forms. Independently supply a zero-dimensional compact Liouville domain R, a native carrier diffeomorphism $e_R : M \to R$, its completion C, compact presentation maps $h_t : R \to E_t$, completions K_t of E_t, and completed presentation maps $P_t : C \to K_t$ retaining these compact maps. No normalization of P_0 is required.*

The original finite-carrier primitive family constructs a localized completed family F_t with native compact inclusions $a_t : M \to F_t$ and actual completed presentation maps $S_t : F_0 \to F_t$. There are primitive-preserving diffeomorphisms $q_t : F_t \to K_t$ with

$$q_t(a_t(x)) = \iota_{K_t}(h_t(e_R(x))), \qquad q_t^{-1}(\iota_{K_t}(h_t(e_R(x)))) = a_t(x).$$

For these same maps the relative comparisons $A_t = P_t^{-1} \circ q_t \circ S_t : F_0 \to C$ and their actual inverses have jointly smooth extensions around the entire closed parameter interval. Put $a(x) = \iota_C(e_R(x))$ and $u_t(x) = S_t^{-1}(a_t(x))$. Then

$$A_t(u_t(x)) = a(x), \qquad A_t^{-1}(a(x)) = u_t(x), \qquad A_t^*(P_t^* \widehat{\lambda}_{K_t}) = S_t^* \widehat{\lambda}_{F_t}.$$

Neither regularity of the prescribed transported primitives nor comparison-map germs are additional hypotheses. Empty carriers are allowed. The assertion concerns these supplied finite completions; it does not furnish the full witnessed-pair or grading correspondence.

Proof. Every tangent space of M is zero, so each μ_t vanishes. The finite-carrier primitive-family construction therefore gives F_t and S_t with genuine joint regularity and retained original native points. At each parameter the prescribed finite-endpoint comparison lemma applied to the identity of M gives a strict q_t with the displayed native compact square. Applying its inverse gives the other square. The completed presentation square for P_t then gives $A_t(u_t(x)) = a(x)$ and its inverse identity.

A zero-dimensional domain has empty boundary. Completion coverage therefore says that every completed point lies in its compact image. Its compact inclusion is an injective immersion and hence a local diffeomorphism in dimension zero. With this surjectivity it is a smooth diffeomorphism from the native compact carrier onto the entire completion. Consequently $a : M \to C$ and $a_0 : M \to F_0$ are diffeomorphisms.

The actual compact presentations of F_t and their inverses extend jointly smoothly in the real parameter. Restrict them to their finite carriers and conjugate by the fixed native carrier diffeomorphism. The resulting native diffeomorphisms $T_t : M \to M$ and their inverses are jointly smooth and satisfy the retained localization square $S_t(a_0(T_t(x))) = a_t(x)$. Thus $u_t = a_0 \circ T_t$. The preceding compact square and surjectivity of u_t identify the actual relative maps everywhere as

$$A_t = a \circ T_t^{-1} \circ a_0^{-1}, \qquad A_t^{-1} = a_0 \circ T_t \circ a^{-1}.$$

These formulas give jointly smooth real-parameter extensions for both actual directions. No regularity of the chosen slice maps q_t is needed. Finally pullback composition, cancellation of P_t with its inverse and primitive preservation of q_t give the stated varying primitive identity. Empty carriers satisfy the same coverage and map identities with empty source and target, so the proof includes them. □

Remark A.204 (Componentwise smooth completed-family correspondence). The family correspondences concern the smooth compact domain and ribbon bundles of the source, their local source-smooth trivializations, and the completions obtained by the specified Liouville collar gluing. The original bundle constructions retain the native carriers, varying full primitives, whole ribbon embeddings and positive boundary witnesses. Their coherent compact presentation maps construct the completed family, including smooth original interior and full signed-cylinder patches, parameter projection and strict original-slice comparisons.

The parameter-preserving total comparisons and their inverses are smooth in those actual original completion atlases. Their exact compact and cylinder equations preserve each original primitive, spatial derivative and Liouville field. The full total covector, two-covector and vertical field have real open coefficient germs on the original interior and seam-outward patches. The positive-dimensional ribbon has its own canonical completed family with these same identities, all-real flows and full intrinsic cores. In dimension zero the finite ribbon and its empty boundary give the exact discrete compact comparison, also when its carrier is empty.

In the reverse direction, A.332 retains the actual pulled ambient and positive-dimensional ribbon primitive families on their fixed native carriers, with joint primitive germs and every native nondegeneracy and strict actual-boundary law. A.346 retains the same full witnessing equation and positive witness; A.347 retains its positive-dimensional and finite native map germs. The native restriction and constant-scale identification laws preserve both prescribed compact map directions, actual boundary primitives, the whole ribbon square and the same eventual end map.

Supplied original ambient and ribbon inclusion markings transport through the same strict slice comparisons and are recovered on the whole original completed spaces. Independently supplied endpoint completions are joined by the original endpoint comparisons and equal-tuple normalizations. The section and grading correspondences preserve the original compatible complex structures, genuine nonzero canonical-square sections, their endpoint classes, literal finite-chain junctions and grading on that same chain. These statements are componentwise correspondences for the source's collar-generated completions, rather than a classification of unrelated total atlases. The existence of the toric itinerary, full-fiber markings, endpoint embedded-core equality and prescribed toric grading remains the geometric target.

Lemma A.205 (Presentation foundation: Induced boundary map of a strict identification)**.** *A primitive-preserving diffeomorphism $\Psi : \widehat{D} \to \widehat{D}'$ between realized completions induces a smooth boundary diffeomorphism ψ and a smooth real function s such that $\Psi(\Phi^r(x)) = \Phi'^{r+s(x)}(\psi(x))$ for every real r, and $\psi^*\alpha' = e^{-s}\alpha$.*

Proof. Primitive preservation and naturality of the exterior derivative imply $\Psi_* Z = Z'$ by the uniqueness of the Liouville equation. Uniqueness of integral curves implies flow equivariance at every real time. Positive orbits from the boundary escape every compact subset of the end. A diffeomorphism is proper, so their images eventually leave the target compact piece and lie in its cylindrical end. Write the end coordinates there as $(y, r+s)$. Equivariance shows that y and s do not depend on the sufficiently large r used. The cylinder parametrization is injective, giving uniqueness. Its smooth local inverse gives smooth coordinates near each boundary point; the unique local definitions glue. Applying the same construction to Ψ^{-1} gives the inverse boundary map. Equivariance extends the identity to every real r. Pulling back $e^{r+s}\alpha'$ by $(x, r) \mapsto (\psi(x), r + s(x))$ gives $e^{r+s}\psi^*\alpha'$: the primitive has no radial component. Compare with $e^r\alpha$ and cancel the positive exponential factor. □

Lemma A.206 (Boundary coordinates under strict native completion comparisons)**.** *Let X, X' be smooth manifolds with smooth one-forms λ, λ', and let K, K' be completions of compact Liouville domains with primitives η, η', end parametrizations $E_K, E_{K'}$, and boundary forms $\alpha_K, \alpha_{K'}$. Let B, B' be smooth native boundary manifolds, with maps $E : B \times \mathbb{R} \to X$ and $E' : B' \times \mathbb{R} \to X'$. Suppose there are smooth diffeomorphisms $S : X \to K$, $S' : X' \to K'$, $b : B \to \partial K$ and $b' : B' \to \partial K'$ satisfying*

$$S^*\eta = \lambda, \qquad (S')^*\eta' = \lambda',$$
$$S(E(x,r)) = E_K(b(x), r), \qquad S'(E'(x', r)) = E_{K'}(b'(x'), r) \quad (r \geq 0).$$

Let $\Psi : X \to X'$ be a smooth diffeomorphism with $\Psi^*\lambda' = \lambda$. Suppose a smooth diffeomorphism $\psi : B \to B'$ and a smooth real function $s : B \to \mathbb{R}$ satisfy, for every $x \in B$ and all sufficiently large real r,

$$\Psi(E(x,r)) = E'(\psi(x), r + s(x)).$$

Then $f = S' \circ \Psi \circ S^{-1}$ is primitive-preserving and its actual induced boundary diffeomorphism and shift are

$$\psi_K = b' \circ \psi \circ b^{-1}, \qquad s_K = s \circ b^{-1}.$$

They satisfy the all-real end and boundary primitive identities

$$f(E_K(y,r)) = E_{K'}(\psi_K(y), r + s_K(y)) \quad (r \in \mathbb{R}), \qquad \psi_K^* \alpha_{K'} = e^{-s_K} \alpha_K.$$

No sign restriction is imposed on s.

Proof. Pullback composition and the three primitive equations give

$$f^*\eta' = (S^{-1})^* \Psi^* (S')^* \eta' = (S^{-1})^* \lambda = \eta.$$

The last equality follows by pulling $S^*\eta = \lambda$ back by S^{-1}. The two displayed boundary formulas define a smooth diffeomorphism and a smooth function. Fix $y \in \partial K$ and put $x = b^{-1}(y)$. Choose r_0 such that the original end equation holds for $r \geq r_0$, and set $R = \max(r_0, 0, -s(x))$. Then $R \geq 0$ and $R + s(x) \geq 0$. The same-radius comparison equations and the original equation give

$$\begin{aligned} f(E_K(y,R)) &= S'(\Psi(E(x,R))) \\ &= S'(E'(\psi(x), R + s(x))) = E_{K'}(b'(\psi(x)), R + s(x)). \end{aligned}$$

Primitive preservation intertwines the complete Liouville flows of K, K'. For any real r, apply their time $r - R$ maps to this equality. The complete boundary-orbit identities then give

$$f(E_K(y,r)) = E_{K'}(b'(\psi(x)), r + s(x)),$$

which is exactly the asserted all-real formula. The boundary primitive calculation for a strict map, applied to these smooth coordinates and this all-real end equation, gives the displayed primitive law. Their end equation already makes them induced coordinates at infinity. Any other induced coordinates have the same all-real end law by strict flow equivariance. Comparing at radius zero, injectivity of $E_{K'}$ on $\partial K' \times \mathbb{R}$ identifies both the boundary point and the real shift. Thus the actual induced coordinates are precisely ψ_K, s_K. □

Lemma A.207 (A full native ribbon square yields the conjugated pair identification)**.** *Under the preceding strict comparison hypotheses, let $\mathcal{P}, \mathcal{Q}$ be witnessed compact pairs with completed ambients K, K', compact ribbons F, F', ribbon embeddings J, J' and primitives $\mu_F, \mu_{F'}$. Let R, R' be sets, with a surjection $e : R \to F$, maps $e' : R' \to F'$, $j : R \to B$, $j' : R' \to B'$ and $h : R \to R'$. Suppose the whole native ribbon squares are*

$$J(e(x)) = b(j(x)), \qquad J'(e'(x')) = b'(j'(x')), \qquad \psi(j(x)) = j'(h(x)).$$

Let $g : F \to F'$ be a specified smooth ribbon diffeomorphism with

$$g(e(x)) = e'(h(x)), \qquad g^*\mu_{F'} = c\mu_F \quad (c > 0).$$

Then the conjugated map $f = S' \circ \Psi \circ S^{-1}$, this same g and this same scale c define a permitted completion identification from $\mathcal{P}$ to $\mathcal{Q}$. Its induced boundary map and shift are exactly $\psi_K = b' \circ \psi \circ b^{-1}$ and $s_K = s \circ b^{-1}$. The bare carrier maps impose no smoothness claim on R, R'; the smooth compact map and its primitive scale are the specified g and its displayed identity.

Proof. The preceding lemma gives primitive preservation of f, the smooth boundary map ψ_K, smooth shift s_K, all-real end law and boundary primitive law. Fix $z \in F$ and choose $x \in R$ with $e(x) = z$, using surjectivity. The original native squares give

$$\begin{aligned} \psi_K(J(z)) &= b'(\psi(b^{-1}(b(j(x))))) = b'(\psi(j(x))) \\ &= b'(j'(h(x))) = J'(e'(h(x))) = J'(g(z)). \end{aligned}$$

Thus the actual induced boundary map carries the entire ribbon through the specified g. Use f as the ambient diffeomorphism, ψ_K as its induced boundary diffeomorphism and s_K as its shift. The preceding lemma gives their primitive, end and boundary primitive identities and the smoothness of the shift. Use the given g as the compact ribbon diffeomorphism and the given positive c as its scale. Its primitive equation is the assumed one, and its whole ribbon square is the equality just proved. These data satisfy every condition of a permitted completion identification, retaining f, g, c, ψ_K, s_K exactly. □

Lemma A.208 (Original endpoint identifications and markings on one native pair family)**.** *Under the original positive-dimensional native pair-family hypotheses of Lemma A.469, one construction retains its single deformation Q, original completed ambient family X, slice primitives L_t, strict slice comparisons $S_t : X_t \to K_t$, native carrier maps $e_{M,t}$ and boundary maps b_t, with all the compact, cylinder and total coefficient conclusions of that lemma. These very data have the following additional property.*

Fix $t, u \in [0,1]$. Let $\Psi : X_t \to X_u$ be an original smooth diffeomorphism with $\Psi^* L_u = L_t$. Suppose a smooth native boundary diffeomorphism ψ and smooth real function s satisfy the eventual original end law

$$\Psi(E_t(y,r)) = E_u(\psi(y), r + s(y)) \quad \text{for every } y \text{ and all sufficiently large } r.$$

Here only the original outward cylinders are used: the equation is required when $r \geq 0$ and $r + s(y) \geq 0$. Let $h : M \to M$ be an original smooth ribbon diffeomorphism with smooth native extension germs, and let $c > 0$ be constant. Require

$$h^* \mu_u = c\mu_t, \qquad \psi(j_t(x)) = j_u(h(x)) \quad (x \in M).$$

Then there is a permitted completion identification $I : Q_t \to Q_u$ whose ambient map, boundary map, shift and scale are exactly

$$S_u \circ \Psi \circ S_t^{-1}, \qquad b_u \circ \psi \circ b_t^{-1}, \qquad s \circ b_t^{-1}, \qquad c.$$

Its compact ribbon diffeomorphism g is constructed on the original ribbon domains of these same Q_t, Q_u and satisfies

$$\begin{aligned} g(e_{M,t}(x)) &= e_{M,u}(h(x)), \\ g^{-1}(e_{M,u}(x)) &= e_{M,t}(h^{-1}(x)), \\ g^* \mu_{Q_u} &= c\mu_{Q_t}. \end{aligned}$$

The same retained construction also transports every supplied original slice marking. Namely, fix t, a specified complete Liouville manifold (Z, λ_Z), a compact inclusion $i : A \to Z$ and an actual smooth diffeomorphism $q_0 : X_t \to Z$ satisfying

$$q_0^* \lambda_Z = L_t, \qquad q_0(o_t(x)) = i(x) \quad (x \in A).$$

There is an actual inclusion marking $m_t : K_t \to Z$ with compact inclusion $i \circ e_{A,t}^{-1}$, retaining

$$\begin{aligned} m_t = q_0 \circ S_t^{-1}, &\qquad m_t \circ S_t = q_0, \\ m_t(\iota_{K_t}(e_{A,t}(x))) = i(x), &\qquad m_t(E_{K_t}(b_t(y), r)) = \Phi_Z^r(i(y)) \quad (r \in \mathbb{R}). \end{aligned}$$

Here o_t is the original compact insertion and the boundary point y on the right is regarded as its original point in A. On the retained original ribbon completion Y_t, its very strict slice comparison

$S_t^F : Y_t \to K_t^F$ also transports every supplied original ribbon marking $q_F : Y_t \to Z_F$ with $q_F^* \lambda_{Z_F} = L_t^F$ and $q_F(o_t^F(x)) = i_F(x)$. The resulting actual inclusion marking is $m_t^F = q_F \circ (S_t^F)^{-1}$, with compact inclusion $i_F \circ e_{M,t}^{-1}$. It retains

$$\begin{aligned} m_t^F \circ S_t^F &= q_F, \\ m_t^F(\iota_{K_t^F}(e_{M,t}(x))) &= i_F(x), \\ m_t^F(E_{K_t^F}(b_t^F(y), r)) &= \Phi_{Z_F}^r(i_F(y)) \quad (y \in \partial M,\ r \in \mathbb{R}), \end{aligned}$$

where b_t^F is the native boundary map of that same ribbon comparison. There is no sign restriction on s. The original Ψ, h and their primitive and whole-ribbon equations are hypotheses; the compact map g and the represented whole-ribbon compatibility are conclusions. This statement concerns positive-dimensional ribbons. It asserts neither existence of the original endpoint maps and markings nor a prescribed grading transport.

Proof. Apply the original completed native pair-family construction once to the supplied native family. Retain exactly its $Q, X, L_t, S_t, e_{M,t}, b_t$ and every compact, cylinder and coefficient conclusion. In particular, the native ribbon carrier maps preserve the same μ_t and retain their smooth native-coordinate extension germs on these very ribbon domains.

Fix t, u and the stated original maps. Apply the fixed-endpoint native compact-map lemma to these retained ribbon domains and their original completions, the retained carrier maps, h and $h^*\mu_u = c\mu_t$. It constructs g on those exact domains, with both displayed carrier equations and the full compact primitive scaling equation. No domain, completion or native carrier map is changed by that application.

The original slice comparisons preserve L_t, L_u and satisfy their same-radius native end equations for all nonnegative radii. Thus they and the given eventual end equation satisfy the strict native conjugation hypotheses. The retained native ribbon squares are

$$J_t(e_{M,t}(x)) = b_t(j_t(x)), \qquad J_u(e_{M,u}(x)) = b_u(j_u(x)).$$

Together with $\psi \circ j_t = j_u \circ h$ and the constructed carrier equation for g, they are exactly the whole native ribbon squares in the conjugated pair-identification lemma. Its source carrier map is surjective because $e_{M,t}$ is a diffeomorphism. Apply that lemma with these very $S_t, S_u, \Psi, \psi, s, g, c$. It gives I retaining the stated ambient map, boundary map, shift and scale, and derives the represented whole-ribbon square and the all-real end and contact primitive laws. Since the original family was selected before t, u and the endpoint maps, this argument applies to every such choice with the same Q and the same original completed family and comparisons.

For the marking assertion retain this same t, Q, X_t, L_t, S_t and $e_{A,t}$. Their original compact equation is $S_t(o_t(x)) = \iota_{K_t}(e_{A,t}(x))$ and their primitive equation is $S_t^* \eta_t = L_t$. Apply the slice marking transport lemma to these equations and the supplied q_0, i. It constructs the actual marking $m_t = q_0 \circ S_t^{-1}$, with the entire-original-map recovery and native compact identities stated above. At an original boundary point y, the retained native boundary inclusion square identifies $\iota_{K_t}(b_t(y))$ with $\iota_{K_t}(e_{A,t}(y))$. Its all-real flow-induced equation therefore gives $m_t(E_{K_t}(b_t(y), r)) = \Phi_Z^r(i(y))$. This uses the same slice comparison and native maps as the endpoint identification construction, so both capabilities hold together on the single retained family. Finally apply the original ribbon-slice marking lemma to the retained S_t^F, its primitive and native compact squares and the supplied q_F. It constructs exactly the stated m_t^F, with its whole-original recovery, native compact equation and all-real native boundary trajectory. The application changes no retained ribbon completion or comparison, so the ribbon and ambient marking conclusions and the endpoint identification hold on the same selected family. □

Lemma A.209 (Original endpoint identifications and markings for a finite ribbon family)**.** *Under the original native surface-pair hypotheses of Lemma A.470, one construction retains its single deformation Q, original completed ambient family and strict slice comparisons $S_t : X_t \to K_t$, and its actual original finite ribbon comparisons $S_t^F : M \to K_t^F$. For $t, u \in [0,1]$, a supplied original strict smooth map $\Psi : X_t \to X_u$ with smooth boundary diffeomorphism ψ, smooth real shift s and eventual outward end law $\Psi(E_t(y,r)) = E_u(\psi(y), r + s(y))$, when both radii are nonnegative, gives a permitted identification $I : Q_t \to Q_u$ with exact data*

$$S_u \circ \Psi \circ S_t^{-1}, \qquad b_u \circ \psi \circ b_t^{-1}, \qquad s \circ b_t^{-1}, \qquad c,$$

provided $h : M \to M$ is an original finite smooth diffeomorphism, $c > 0$ is constant, and $\psi \circ j_t = j_u \circ h$ on the entire finite ribbon. Its compact ribbon map has both native forward and inverse equations through $e_{M,t}, e_{M,u}$ and this same h. There is no ribbon extension-germ or primitive scaling hypothesis: all ribbon one-forms vanish. The radial shift is unrestricted.

Every supplied original ambient marking $q_0 : X_t \to Z$, with $q_0^* \lambda_Z = L_t$ and $q_0(o_t(x)) = i(x)$ on the original compact ambient, gives an actual inclusion marking $m_t = q_0 \circ S_t^{-1} : K_t \to Z$ with compact inclusion $i \circ e_{A,t}^{-1}$. It retains $m_t \circ S_t = q_0$ on all of X_t and $m_t(\iota_{K_t}(e_{A,t}(x))) = i(x)$.

In addition, every supplied original ribbon marking $q_F : M \to Z_F$ onto a specified complete zero-dimensional Liouville manifold, with prescribed compact inclusion $q_F(x) = i_F(x)$, gives an actual inclusion marking

$$m_t^F = q_F \circ (S_t^F)^{-1} : K_t^F \to Z_F$$

with compact inclusion $i_F \circ e_{M,t}^{-1}$, entire-original-map recovery $m_t^F \circ S_t^F = q_F$ and native compact equation $m_t^F(\iota_{K_t^F}(e_{M,t}(x))) = i_F(x)$. Empty finite ribbons are allowed. The original endpoint maps and both original markings remain hypotheses; no geometric marking or chain existence or grading conclusion is asserted.

Proof. Apply the original completed finite-ribbon pair-family construction once and retain its Q, original completed ambient and ribbon comparisons, native maps and all its regularity conclusions. For given t, u, h, c, the finite native compact-map lemma constructs g on the very ribbon domains of Q_t, Q_u, with forward equation $g(e_{M,t}(x)) = e_{M,u}(h(x))$ and inverse equation $g^{-1}(e_{M,u}(x)) = e_{M,t}(h^{-1}(x))$. Every tangent space of these zero-dimensional domains is zero. Consequently $g^* \mu_{Q_u} = c \mu_{Q_t}$ on their full compact domains. The retained ambient strict slice comparisons and original eventual end law satisfy the native conjugation hypotheses. Their native ribbon squares, the given $\psi \circ j_t = j_u \circ h$, and this constructed g satisfy its full-ribbon consumer. That lemma gives the claimed permitted identification, retaining the exact conjugated ambient map, transported boundary map and arbitrary shift, constant c and both native ribbon carrier equations.

For a supplied original ambient marking, apply the slice marking transport lemma to the retained ambient strict comparison, original compact insertion and native carrier map. For a supplied original ribbon marking, apply the retained ribbon-slice marking lemma to this same S_t^F, whose retained primitive equation and compact square are $(S_t^F)^* \eta_t^F = \mu_t$ and $S_t^F(x) = \iota_{K_t^F}(e_{M,t}(x))$. It gives the two actual markings, their full recovery and compact equations, and their all-real flow-induced laws. The finite ribbon has empty boundary, so its boundary flow law is vacuous. All these constructions preserve the one initially selected Q and its maps; empty carriers require no choice of a native point. □

Definition A.210 (Auxiliary presentation: Finite geometric comparison chain)**.** *A permitted step is a presented pair deformation, a strict completion identification, or the reversal of a permitted*

step. A strict identification carries the ribbon by $\psi \circ j = j' \circ h$ for a compact ribbon diffeomorphism h and satisfies $h^*\mu' = c_{\mathrm{scale}}\mu$ for one positive constant. A finite chain is a finite path of such actual steps with the same endpoint tuple at each junction. Core equality is not a permitted-step constructor.

Definition A.211 (Auxiliary presentation: Ambient Maslov representatives and graded chains)**.** *For a compatible smooth almost complex structure J, put $K_J = \Lambda^n_{\mathbb{C}}(T^{1,0}X)^*$. A representative is a nowhere-zero smooth section of $K_J^{\otimes 2}$; a homotopy allows compatible J to vary and keeps the section nowhere zero. In tangent coordinates a square section is a complex-valued tensor alternating separately in two blocks of n arguments and complex-linear in each argument for J. A graded chain equips each deformation with such a section over its completed ambient family and each strict identification with a homotopy to the pulled-back target section. The chain is the same specified geometric path, and representatives agree at every junction.*

Lemma A.212 (Presentation foundation: Reconstructing the prescribed interval grading family)**.** *On a fixed presented smooth manifold let the prescribed symplectic forms vary arbitrarily. Suppose the interval's compatible complex structures have local smooth extensions as endomorphisms, and a prescribed nowhere-zero section of the intrinsic squared canonical line has local smooth scalar coefficients in its canonical charts. These data produce a presented Maslov family retaining the given complex structure and tensor section at every parameter of $[0,1]$, with the same prescribed two-form family. Only smooth extensions are required off the interval; compatibility, canonicality and nonvanishing are required on the interval.*

Proof. Choose the local smooth extensions of the endomorphisms and tensor coefficients. A smooth partition of unity on the product parameter-space neighborhood patches the endomorphism extensions. On the closed interval each summand equals the original endomorphism, so their weighted sum is that same endomorphism there. First apply the canonical-square map to view the intrinsic interval section as a vertical covariant tensor. Patch its local smooth extensions in that ambient tensor bundle, using its smooth transition maps; on the interval every extension equals the original tensor, so its weighted sum is unchanged. No off-interval canonical line or compatible complex structure is needed. Both patched families are smooth. Since their interval values are unchanged, all compatibility, canonicality and nonvanishing conditions, as well as both endpoints, are unchanged. No condition on the original two-form is changed and no geometric equation is required off the interval. □

Lemma A.213 (Joint smoothness of pulled tangent covectors)**.** *Let M, N be smooth manifolds of dimension $2n$. Suppose $f(t,x)$ is smooth on an open $R \subset \mathbb{R} \times M$, and $A(t,y) : T_yN \to \mathbb{C}^n$ is a jointly smooth family of real-linear maps on an open $U \subset \mathbb{R} \times N$. Then*

$$B(t,x) = A(t, f(t,x)) \circ d_x f_t$$

is a jointly smooth family of vertical real-linear maps on the open set $R \cap \{(t,x) : (t, f(t,x)) \in U\}$. Invertibility of A or B is not required for this regularity assertion.

Proof. The map $(t,x) \mapsto (t, f(t,x))$ is smooth, so the stated preimage domain is open. In source and target spatial charts, write the joint map as $a(t,u)$. Its spatial derivative is the restriction of its joint derivative to vectors $(0,v)$, and is therefore jointly smooth. The coordinate expression for B is the product of this derivative with the target coframe coefficients at $(t, a(t,u))$. Composition and multiplication of finite-dimensional linear maps preserve smoothness. This proves the assertion in each source chart, and the chain rule identifies these expressions on chart overlaps. □

Lemma A.214 (Reconstructing a smooth complex endomorphism from a coframe)**.** *Let $B(t,x) : T_xM \to \mathbb{C}^n$ be a jointly smooth family of invertible real-linear maps on an open set. Then*

$$L(t,x) = B(t,x)^{-1} \circ i \circ B(t,x)$$

is a jointly smooth vertical endomorphism. Wherever $BJ = iB$, one has $L = J$.

Proof. In a spatial chart, the coframe coefficient matrix is invertible and jointly smooth. Matrix inversion is smooth on the open set of invertible matrices, so its conjugation of multiplication by i is jointly smooth. Under a change of spatial tangent coordinates the coframe matrix is multiplied on the right by the tangent transition matrix. Its conjugation consequently changes by that matrix's usual endomorphism conjugation, so these are the coordinates of one vertical endomorphism. Multiplying $BJ = iB$ on the left by B^{-1} gives $L = J$. □

Lemma A.215 (Local interval canonical coframes and scalar coefficients)**.** *For a smooth Maslov family (J_t, s_t) on a fixed manifold N, every point of $[0,1] \times N$ has an open real-parameter neighborhood with a smooth invertible coframe $A : TN \to \mathbb{C}^n$ satisfying $AJ_t = iA$ at the interval parameters. In the resulting squared canonical-line coordinates, the prescribed section s_t has scalar coefficients with smooth extensions on such neighborhoods.*

Proof. At the chosen point take a real-linear coframe A_0 intertwining its complex structure with multiplication by i. In fixed spatial coordinates, use a smooth local extension L of the prescribed complex endomorphism and put

$$A = \tfrac{1}{2}(A_0 - iA_0L).$$

At the chosen point $A = A_0$. Invertibility is open, so restrict the neighborhood to make A invertible throughout. At interval parameters $L = J_t$ and $J_t^2 = -1$, whence $AJ_t = \frac{1}{2}(A_0J_t + iA_0) = iA$. The coefficient of a canonical n-form in this coframe is its evaluation on the inverse coframe's first n real standard basis vectors. The squared coefficient is the analogous evaluation on these vectors in both blocks. The inverse coframe is smooth. Evaluating the smooth local tensor extension of s_t on these vectors gives a smooth scalar extension whose interval values are precisely the required squared-line coefficients. When $n = 0$, the coframe is the unique map between zero spaces and the coefficient is the scalar tensor itself; the same argument applies. □

Lemma A.216 (Canonical interval charts pull back through a smooth map family)**.** *Let $e_t : M \to N$ be diffeomorphisms for $t \in [0,1]$ with a jointly smooth forward extension f on an open neighborhood of $[0,1] \times M$. Let A be an interval canonical coframe for J_t on N, and let V be an open subset of its domain containing $(t, e_t(x))$. There is an interval canonical coframe near (t,x) for $e_t^*J_t$, equal to*

$$B(s,y) = A(s, f(s,y)) \circ d_y f_s.$$

Its open domain is contained in the extension domain of f and in the preimage of V. No diffeomorphism condition is imposed on f_s outside the prescribed interval.

Proof. The vertical pullback lemma gives a smooth B on that open preimage. At (t,x), its first factor is invertible and its second factor is d_xe_t, also invertible. In fixed spatial coordinates the invertible matrices form an open set. Restrict to this invertibility locus near (t,x), giving the required open domain. On interval parameters, $f_s = e_s$ as maps, so their spatial derivatives agree. The definition $e_s^*J_s = (de_s)^{-1}J_sde_s$ gives

$$B(e_s^*J_s) = AJ_sde_s = iAde_s = iB.$$

Thus this same smooth invertible coframe is canonical for the prescribed pulled complex structure on the interval. □

Lemma A.217 (Local complex and section extensions for the pulled Maslov family)**.** *Let (J_t, s_t) be a smooth Maslov family on N, and let $e_t : M \to N$ be interval diffeomorphisms admitting a jointly smooth forward extension on a neighborhood of $[0,1] \times M$. Then the prescribed pulled complex structure $e_t^* J_t$ has genuine local smooth endomorphism extensions. The tensor $e_t^* s_t$ determines a nowhere-zero section of the actual squared canonical line for $e_t^* J_t$, and its scalar coefficients have local smooth extensions. The tensor image of this section is exactly the derivative pullback $e_t^* s_t$ at every interval point. Joint smoothness of the inverse maps is not required.*

Proof. Choose a target canonical coframe and a smooth scalar extension q of the target section coefficient near $(t, e_t(x))$. Pull that coframe back by the interval chart lemma to obtain a smooth invertible B near (t, x). The reconstruction lemma gives a smooth endomorphism $B^{-1} i B$. Its interval values equal $e_t^* J_t$ because B intertwines these complex structures.

The invertible complex-linear derivative de_t induces an isomorphism of canonical lines and of their tensor squares. Pull the original section back through that isomorphism. Its tensor image is the ordinary covariant derivative pullback, and its nonvanishing follows from injectivity of the isomorphism. In the pulled coframe $B = A \circ de_t$ the section coefficient is the original coefficient at $(t, e_t(x))$: the inverse frame is $(de_t)^{-1} A^{-1}$, so evaluation of the pulled tensor in each block cancels the derivative with its inverse. Consequently $q(s, f(s, y))$ is a smooth local scalar extension of that same section coefficient. These constructions apply at every interval point, including both endpoints, and preserve all original tensor values. □

Lemma A.218 (Transported grading on the specified completed deformation)**.** *Fix a completed pair deformation $\mathcal{D}$, with reference completed ambient M and its prescribed endpoint presentation maps P_0, P_1. Let ω_t be its completed symplectic forms in these presentations. Let (J_t, s_t) be a smooth Maslov family for forms η_t on a fixed manifold N, and let $e_t : M \to N$ be prescribed interval diffeomorphisms admitting a jointly smooth forward extension. Suppose $e_t^* \eta_t = \omega_t$ at every interval parameter. For prescribed endpoint Maslov representatives $(J_-, s_-), (J_+, s_+)$, assume the exact endpoint equations*

$$e_0^* J_0 = P_0^* J_-, \qquad e_1^* J_1 = P_1^* J_+, \qquad e_0^* s_0 = P_0^* s_-, \qquad e_1^* s_1 = P_1^* s_+.$$

There is a grading of this same completed deformation whose complex structure and tensor are exactly $e_t^* J_t, e_t^* s_t$ for every interval parameter, with these prescribed endpoint representatives.

Proof. Pullback through a diffeomorphism preserves the compatibility equations: invariance of the two-form under J_t follows by applying de_t to both arguments, and its positivity follows because de_t sends a nonzero vector to a nonzero vector. The form identity therefore makes $e_t^* J_t$ compatible with the actual ω_t. The preceding lemma supplies local smooth complex extensions and a nowhere-zero canonical-square section with local scalar extensions, retaining exactly $e_t^* s_t$. Apply the interval grading reconstruction lemma to these data and this completed deformation. Its reconstruction retains every interval value. At zero and one, the stated endpoint equations identify those values with the pullbacks through the actual presentation maps P_0, P_1. Thus the reconstructed grading has precisely the required endpoint representatives on the same deformation. □

Lemma A.219 (Local real map germs transport a supplied grading on the same deformation)**.** *Fix a completed pair deformation $\mathcal{D}$ with reference completed ambient M, completed symplectic forms ω_t, and actual endpoint presentation maps P_0, P_1. Let (J_t, s_t) be a supplied smooth Maslov family for forms η_t on a fixed manifold N. Let $e_t : M \to N$ be prescribed diffeomorphisms for $t \in [0,1]$. Suppose that for every (t, x) there is a smooth N-valued map on a real open neighborhood of (t, x)*

agreeing with $e_u(y)$ at all interval points in that neighborhood. Assume $e_t^*\eta_t = \omega_t$ for every interval parameter and the exact endpoint equations

$$e_0^*J_0 = P_0^*J_-, \qquad e_1^*J_1 = P_1^*J_+, \qquad e_0^*s_0 = P_0^*s_-, \qquad e_1^*s_1 = P_1^*s_+.$$

Then this same completed deformation has a grading retaining exactly $e_t^*J_t$ and the full tensor $e_t^*s_t$ at every interval parameter, with the prescribed endpoint representatives $(J_-, s_-), (J_+, s_+)$. The real germ may be chosen independently at each point. No common real smooth extension of the maps is required, and the assertion presupposes the supplied Maslov family and endpoint equations.

Proof. Fix (t, x) and choose one supplied real germ f near it. Choose a target canonical coframe A and a smooth scalar coefficient q for the prescribed target section near $(t, e_t(x))$. Intersect the domain of f with the preimage of this target neighborhood under $(u, y) \mapsto (u, f(u, y))$. On that open set the vertical pullback lemma makes

$$B(u, y) = A(u, f(u, y)) \circ d_y f_u$$

jointly smooth. At the center it is invertible because both A and $d_x e_t$ are invertible. The invertibility locus of its matrix in a spatial chart is open, so shrink the neighborhood to that locus. On its interval points f_u and e_u agree on a spatial neighborhood, and therefore $d_y f_u = d_y e_u$ there. Consequently $B(e_u^*J_u) = iB$. The coframe reconstruction lemma supplies a smooth local complex endomorphism $B^{-1}iB$ whose interval values are exactly $e_u^*J_u$.

The invertible complex-linear derivative de_u identifies the canonical lines and their squares. Pulling back s_u through this isomorphism gives a nowhere-zero section with full tensor $e_u^*s_u$. In the pulled coframe its scalar coefficient equals the original coefficient at $(u, e_u(y))$: each inverse frame vector is $(de_u)^{-1}A^{-1}$, so evaluation of the pulled tensor cancels de_u against its inverse in both tensor blocks. Thus $q(u, f(u, y))$ is a smooth local scalar extension for this exact pulled section. All these constructions are local at the chosen point and include both real parameter endpoints.

For a nonzero vector v, the derivative $de_t(v)$ is nonzero. The compatibility identities and positivity for J_t, η_t therefore pull back to those for $e_t^*J_t, e_t^*\eta_t = \omega_t$. Apply the interval grading reconstruction lemma to the local endomorphism and section-coefficient extensions just proved, the nonvanishing pulled section and this compatibility. It retains every interval complex structure and full tensor on $\mathcal{D}$. The four assumed endpoint equations give its exact required endpoint representatives, completing the proof. □

Definition A.220 (Auxiliary presentation: The prescribed endpoint grading classes)**.** *On Y put $\Omega = \bigwedge_j (du_j + i\,d\theta_j)$ and prescribe $[\Omega^{\otimes 2}]$. On T^*T prescribe $[(\bigwedge_j (dp_j + i\,d\theta_j))^{\otimes 2}]$. The latter squared form evaluates to one on the ordered vertical Lagrangian frame; this is the vertical-phase-zero convention.*

Remark A.221 (Coordinate presentation and intrinsic scope). The extension-space descriptions of domains, completions and families are auxiliary encodings; their full correspondence with the intrinsic objects requires the correspondence lemmas below. Requiring a global extension embedding is stronger than requiring an embedding near the compact piece; restriction to a suitable open neighborhood must therefore precede the completion construction. Fixed-completion results do not construct that completion. The interval grading reconstruction applies to varying prescribed forms on a fixed presented ambient. It does not construct the completed geometric family or its presentation; that obligation belongs to the completed-family correspondence.

A.4 Auxiliary presentations: Bounds, initial markings and the target

Definition A.222 (Auxiliary presentation: Specified end bounds and threshold)**.** *Write* $m_v = c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}$, $G = \sum_v m_v v$, $a = \bar{G}^T G$, $b_0 = u \cdot G$, *and* $Q_W = |u|^2 a - |b_0|^2 + \frac{1}{2}\operatorname{Re}(\bar{W} b_0)$. *An ambient bound* $A_{\mathrm{amb}} > 0$ *requires* $a, Q_W > 0$ *when* $\phi_0 \geq A_{\mathrm{amb}}$. *Set* $M(A) = \sup_{\phi_0 \leq A} |W|$. *For* $R > \max(1, M(A_{\mathrm{amb}}))$, *put* $S_R = \{|b_0|/(a|W|) : |W| \geq R\}$ *and*

$$K_0(R) = 1 + 2 \sup S_R. \tag{7}$$

Its nonemptiness and boundedness are separate assertions. A fiber bound $A_{\mathrm{fib}} \geq A_{\mathrm{amb}}$ *requires* $a > 0$ *and* $|u|^2 - |b_0|^2/a > 0$ *on* $\{\phi_0 \geq A_{\mathrm{fib}}, |W| \leq R\}$.

Lemma A.223 (Presentation foundation: The compact magnitude maximum)**.** *For every* $A \geq 0$, $M(A)$ *is an attained maximum on the nonempty compact quadratic sublevel in* Y.

Proof. The zero logarithmic coordinate belongs to the sublevel. If $\sum_j u_j^2 \leq 2A$, each coordinate satisfies $|u_j| \leq \sqrt{2A}$. Thus the sublevel is contained in a compact finite box times the compact quotient torus. It is closed by continuity of ϕ_0, so it is compact. The finite exponential sum defining W is continuous. Its magnitude image is consequently a nonempty compact subset of $\mathbb{R}$. Such a set contains its supremum, which therefore is the magnitude of a point in the sublevel and bounds every magnitude. □

Definition A.224 (Auxiliary presentation: Initial subsets and admissible rounding)**.** *Choose* $A_- > A_{\mathrm{fib}}$ *and* $A > A_-$. *A rounding profile is a smooth convex* $m_\varepsilon : \mathbb{R}^2 \to \mathbb{R}$ *with nonnegative partial derivatives summing to one,* $\max \leq m_\varepsilon \leq \max + \varepsilon$, *and equality with* max *when* $|x - y| \geq \varepsilon$. *Put*

$$\begin{gathered} E_\varepsilon = \{m_\varepsilon(\phi_0 - A, |W|^2 - R^2) \leq 0\}, \quad F_R^- = \{W = R, \phi_0 \leq A_-\}, \\ \mu_R = \lambda_0|_{F_R^-}. \end{gathered} \tag{8}$$

Admissibility requires $0 < \varepsilon < A - A_-$ *and, at every point of* $B_\varepsilon = \{-\varepsilon < \phi_0 - A \leq 0, -\varepsilon < |W|^2 - R^2 \leq 0\}$, $d(|W|^2)(Z_K) > 0$ *and linear independence of* $d\phi_0, d(|W|^2)$. *The band may be empty. Markings, chains and full-core equality are not admissibility conditions.*

Definition A.225 (Auxiliary presentation: Specified initial conclusion data)**.** *An initial realization is a witnessed pair whose compact ambient image is exactly* E_ε *and whose ribbon image is exactly* F_R^-, *with witness one and commuting ribbon inclusion. The whole regular fiber is presented smoothly with its restricted* λ_0, *and both inclusion markings are specified primitive-preserving completion diffeomorphisms onto* (Y, λ_K) *and* $(W^{-1}(R), \lambda_0|_{W^{-1}(R)})$, *induced by their Liouville flows. The ribbon is nonempty. These are conclusion data whose existence is to be proved.*

Definition A.226 (Auxiliary presentation: The one endpoint comparison package)**.** *The endpoint is a witnessed pair whose ambient compact domain and boundary are exactly* D^*T *and* S^*T *under its standard radial cotangent completion marking. It is joined to the specified initial pair by an actual finite comparison chain. Its embedded full ribbon core equals* Λ_Σ^∞ *literally. The same chain is graded, with source class pulled back from* $[\Omega^{\otimes 2}]$ *by the initial marking and endpoint homotopic under the cotangent marking to the vertical-phase-zero class. No grading is assigned to the singular link.*

Definition A.227 (Auxiliary presentation: The encoded geometric target proposition)**.** *The encoded target for an input asserts completeness and finite type for all* $K \geq 0$, *and existence of common* A_{amb}, R_0 *such that: for every* $R \geq R_0$, *the ratio set is nonempty and bounded above and*

a valid fiber bound exists; for every valid fiber bound, every $K \geq K_0(R)$, and every allowed pair of levels, an admissible rounding exists; and for every admissible rounding an initial realization and one endpoint comparison package exist together. The global encoded proposition quantifies over every positive-rank based input.

Lemma A.228 (Original pair-bundle realizations retain their complete marked and graded data)**.** *Let one original compact ambient and ribbon bundle satisfy the original geometric, primitive, embedding, positive-witness and Maslov coframe-germ hypotheses of Corollary A.760. In positive ribbon dimension take ambient dimension $2n$, $n > 1$, and ribbon dimension $2n-2$; in ambient dimension two allow any finite ribbon, including the empty ribbon. The original source data admit one retained realization consisting of the corollary's entire geometric deformation, its compatible global compact charts, chart covariance, coherent presentations, Maslov family and actual deformation grade. It retains all compact primitive, boundary, witness, whole ribbon image and full-core equations, and exact recovery of the original complex fields and full tensors in every original chart and at both endpoints.*

On this same chosen realization, every independently supplied original ambient or ribbon marking datum gives its specified inclusion marking, with original compact inclusion, full flow equation and whole original chart recovery. These marking data do not require another choice of the deformation. A choice of such a retained realization can be fixed for each source bundle. This statement retains the existing one-bundle construction; it does not assert a comparison between different bundles.

Proof. For positive ribbon dimension, apply the cited realization corollary to the supplied original ambient and ribbon primitives, geometric laws, Maslov data and coframe germs. Keep together its chosen global charts, their covariance identities, deformation, compact and coherent presentation maps, all geometric equations, Maslov family, actual grade and original endpoint recovery equations. Its ambient and ribbon marking conclusions quantify over independently supplied marking data after all these choices; retain those conclusions on the very same completions. This is a retained realization with every stated property.

In ambient dimension two, apply the corollary's finite-ribbon branch to the supplied original bundle and its full embedding data. Retain its global charts, coherent deformation, original primitive and whole-ribbon equations, Maslov family, grade and original endpoint recovery in the same way. Retain its two marking conclusions, whose original marking inputs are again quantified after the deformation is chosen. This construction allows the empty ribbon and uses no selected ribbon point. Thus a retained realization exists in that case also. Choosing an element of the nonempty set of retained realizations for each original source bundle fixes the stated choice without altering any of these properties. □

Lemma A.229 (Original junctions act on the same retained endpoint realizations)**.** *Fix the retained realization of every original pair bundle by Lemma A.228. An endpoint is an original bundle with any parameter $s \in I$; its pair and Maslov representative are the actual slice of that fixed realization. Take two endpoints with equal dimensions, possibly with different original ambient and ribbon manifolds. In local original ambient charts containing their parameters, supply a strict completed ambient map $\Psi : X \to Y$, its induced boundary map ψ, and its arbitrary real shift σ satisfying the eventual end law. In local original ribbon charts containing the same parameters, supply a full ribbon diffeomorphism h, a constant $c > 0$, and the whole embedding square. In positive ribbon dimension require real open native map germs for h and $h^*\mu_Y = c\mu_X$. In zero ribbon dimension these additional ribbon requirements are unnecessary and empty ribbons are allowed. Supply original Maslov representatives equal in complex field and full tensor to the original atlas*

data, and a smooth compatible nonvanishing homotopy H from the source representative to the pullback of the target by Ψ.

These original data construct one permitted identification I between the two retained endpoint pairs and a grade on that exact I between their actual retained slice representatives. If S, S' are the original local-to-retained endpoint comparisons and $b_S, b_{S'}$ their boundary maps, its maps are exactly

$$I_{\mathrm{amb}} = S' \circ \Psi \circ S^{-1}, \qquad I_{\partial} = b_{S'} \circ \psi \circ b_S^{-1}, \qquad \sigma_I = \sigma \circ b_S^{-1}.$$

At every real homotopy parameter τ, its fields are $(S^{-1})^* J_{H,\tau}$ and $(S^{-1})^* \mathbf{T}_{H,\tau}$. The realization attached to either endpoint is unchanged by this junction construction. The original maps and homotopy remain supplied geometric data.

Proof. In positive ribbon dimension, use the fixed realizations' compact primitive and embedding identities and their original ribbon covariance in the original positive-ribbon junction construction. They identify the actual original ribbon primitives and embeddings with the chosen endpoint pair data, so that construction gives a permitted identification on those very pairs. It preserves the given full ribbon map and its positive scale, and gives the three displayed conjugation formulas.

Original Maslov endpoint recovery gives the four identities

$$\nu_X . J = S^* \rho_L . J, \quad \nu_X . \mathbf{T} = S^* \rho_L . \mathbf{T}, \qquad \nu_Y . J = (S')^* \rho_R . J, \quad \nu_Y . \mathbf{T} = (S')^* \rho_R . \mathbf{T}.$$

Exterior differentiation of the strict primitive equations gives the corresponding symplectic pullback equations. Pull back H by S^{-1}. These four identities make its source exactly ρ_L and its target exactly $(S'\Psi S^{-1})^* \rho_R = I_{\mathrm{amb}}^* \rho_R$. The original homotopy lift lemma preserves smooth extension germs, compatibility, canonicality and nonvanishing under this pullback, so it gives a grade on the exact identification and the stated fields for all real τ.

For a zero-dimensional ribbon, original chart covariance and the retained whole embedding identity give the same construction by the original finite-ribbon junction lemma. Every ribbon one-form is zero, so the given positive constant needs no additional primitive identity, and no ribbon map extension is required. The original finite-ribbon graded junction lemma uses the same endpoint recovery and the same pullback of H to grade this exact identification, with the identical ambient, boundary, shift and homotopy-field formulas. The argument selects no ribbon point and therefore includes empty ribbons. Both constructions use the endpoint realizations already fixed and choose none afresh. □

Lemma A.230 (Every supplied finite original itinerary gives a grading on its exact comparison chain)**.** *Fix common ambient and ribbon dimensions, either $2n, 2n-2$ with $n > 1$, or $2, 0$. Original endpoints are original pair bundles and parameters, with the single retained realization and slice representative assigned by Lemma A.228. An original arrow is one entire supplied bundle deformation from 0 to 1, an original junction at arbitrary slice parameters satisfying the hypotheses of Lemma A.229, or a reverse of an original arrow. Let C be any finite ordered sequence of these arrows with literally matching original endpoints. The sequence may be empty, may contain consecutive junctions or arbitrary reversals, and may revisit a bundle. Original carriers may differ from one endpoint to another. Matching means equality of the original bundle-and-parameter vertices. Equal geometric endpoint tuples of two distinct bundles alone do not concatenate these sequences; an original identity or presentation junction between them must be supplied.*

There is a comparison chain $\widehat{C}$ obtained by replacing each original arrow by its constructed permitted comparison step in the same direction, and there is a grading on this very $\widehat{C}$ between its actual retained endpoint representatives. Its length equals that of C, and its literal vertex list is the original vertex list with each vertex replaced by its fixed retained slice pair. Repetitions are retained.

At every visited vertex the full retained realization is the same one used by these arrows; it keeps all original geometric, chart-covariance, coherent-presentation and Maslov-recovery equations and both independent completion-marking recovery conclusions. Finite ribbons, including empty ribbons, are allowed. This is conditional on the supplied original itinerary, its geometric laws, local regularity and homotopies; it does not construct the itinerary or the target's endpoint core or grading.

Proof. First assign to every original endpoint the slice pair and representative of its fixed bundle realization. For a deformation arrow use that realization's actual deformation and its actual grade. For a junction arrow use the identification and its grade produced by the retained junction lemma on those same endpoint choices. For a reverse arrow, reverse the already assigned comparison step and its grade. Induction on the construction of an arrow gives a permitted step and its grade with the prescribed retained representatives at both ends.

Construct the chain by induction on the number of arrows in C. For the empty sequence use the empty chain at its retained pair with the empty grading at its retained representative. Its length is zero and its vertex list is the singleton retained pair, as required. For a sequence obtained by appending one arrow e to a sequence C_0, use the inductively constructed chain and append the comparison step assigned to e. The endpoint of C_0 and the source of e name the same original vertex, hence the same fixed retained pair and the same representative literally. Append the assigned grade of e to the inductive grading. This grades precisely the appended comparison chain. Appending adds one to both lengths. It also appends exactly the retained pair assigned to the terminal original vertex to both vertex lists; the induction hypothesis therefore proves the length and list identities.

Every bundle realization was fixed before this arrow and sequence induction. At a visited vertex return that same full realization; its slice pair is the pair used by the chain by definition, and all its geometric and marking conclusions remain unchanged. Neither induction chooses a replacement presentation or inserts an arrow. The finite-ribbon producers used in the $2,0$ case allow empty ribbons, so the same induction includes them. This proves all the assertions. □

Lemma A.231 (Smooth original interior sections have smooth coefficients in every original chart)**.** *Assume the original collar, its smooth slice maps and local inverse germs, and the original complex-field extension hypotheses of Lemma A.674. Let σ be a section of the original canonical-square fibers whose restriction to $I \times M^\circ$ is smooth in that independent line bundle. For every original interior canonical chart c, the actual coefficient $q_c(\sigma)$ is smooth on the chart's entire base domain, with smoothness relative to the closed parameter interval at its endpoints. At a point in that domain where the original section is nonzero, $q_c(\sigma)$ is nonzero. This assertion gives intrinsic coefficient smoothness; real open extensions across the parameter endpoints are a separate requirement.*

Proof. Fix one original canonical chart c and a point p in its base domain. The chart supplies a smooth local bundle trivialization whose fiber map is exactly q_c. In this trivialization the given smooth section is the map $p \mapsto (p, q_c(\sigma(p)))$. Projection to the complex factor is smooth, so its coefficient is smooth at p. This argument applies to every point of the base domain, including the parameter endpoints in the closed interval model, and proves the claimed smoothness there. The fiber coordinate at p is a complex-linear isomorphism. If its value on $\sigma(p)$ were zero, injectivity and its value zero on the zero vector would imply $\sigma(p) = 0$. This proves nonvanishing where the original section is nonzero. □

Lemma A.232 (Smooth original cylindrical sections have smooth coefficients in every original chart)**.** *Assume the original collar, its smooth slice maps and local inverse germs, and the original complex-field extension hypotheses of Lemma A.687. Let σ be a section of the original canonical-square fibers whose restriction to the full signed cylinder $I \times B_\delta$ is smooth in that independent line*

bundle. For every original cylindrical canonical chart c, the actual coefficient $q_c(\sigma)$ is smooth on its entire base domain. This includes the radius-zero seam and the whole signed band, with smoothness relative to the closed parameter interval at its endpoints. Wherever the original section is nonzero, its coefficient is nonzero. Real open extensions across the parameter endpoints are not asserted by this intrinsic smoothness statement.

Proof. Fix a point in the base domain of c. The independently constructed smooth line-bundle trivialization has q_c as its fiber coordinate. The given smooth section therefore has smooth coordinate expression $p \mapsto (p, q_c(\sigma(p)))$. Composing with projection to $\mathbb{C}$ proves coefficient smoothness at that point. The base domain contains the actual cylinder coordinates, so this argument applies at radius zero, at every allowed signed radius, and at both interval endpoints in their closed parameter model. It proves smoothness on the whole base domain. At each point the fiber coordinate is a linear isomorphism; a zero coefficient would therefore force the original section value to be zero. This proves the nonvanishing assertion. □

Lemma A.233 (Intrinsic original section smoothness gives real open coefficient germs at interior times)**.** *Under the hypotheses of the two coefficient-smoothness lemmas, take any point $p = (t, x)$ in the base domain of an original interior or cylindrical canonical chart, with $0 < t < 1$. Intrinsic smoothness of the original section on that branch gives a real open smooth extension of its actual scalar coefficient near the chart coordinates of p. The extension agrees with the original coefficient at every nearby retained interval point in that chart. No coefficient extension germ at an interior time is an input. The assertion on the cylinder uses its full signed band and actual radial sign convention. It asserts no extension across $t = 0$ or $t = 1$.*

Proof. Let q be the actual coefficient in the chosen chart. It is smooth on its open base domain by the appropriate coefficient-smoothness lemma. In the interior branch, let a be the spatial chart and $\psi = a^{-1}$ on its open target. On $(0, 1) \times \mathrm{target}(a)$ the map $r(s, y) = (s, \psi(y))$ is smooth into $I \times M^\circ$, because the interval inclusion at strict interior times and the inverse chart are smooth. The inverse image of the coefficient's base domain under r is an open neighborhood of $(t, a(x))$. Intersect it with the real open domain of the original canonical chart. On this neighborhood set $\widetilde{q} = q \circ r$. Smooth composition gives a real smooth scalar, and the inverse chart identity gives $\widetilde{q}(s, a(w)) = q(s, w)$ at every retained point in the neighborhood.

For the cylindrical branch the spatial chart is taken on boundary and signed radius coordinates $(b, -\rho)$. Its inverse, followed by $(b, u) \mapsto (-u, b)$, is a smooth map to the signed band on the open set where $-u < \delta$. Restrict the inverse chart target to this open set; it contains the center by the original band condition. Use this map as ψ in the preceding construction and again intersect with the original canonical chart's real domain. The inverse chart and radial sign identities give $\widetilde{q}(s, a(b, -\rho)) = q(s, (\rho, b))$ throughout the retained locus. Thus the same construction gives the required full-band coefficient germ. All chosen neighborhoods have time coordinate in $(0, 1)$, as claimed. □

Corollary A.234 (Endpoint coefficient germs suffice for original section grading germs)*.* Under the original collar, complex-field and local inverse hypotheses, suppose the restriction of the same original section is intrinsically smooth on either original canonical-square branch. Suppose that in every original canonical chart of that branch and at every base point with $t = 0$ or $t = 1$ its scalar coefficient has a real open smooth extension. Then such coefficient extensions exist in every chart at every interval parameter. If the section is nowhere zero on the original completed fibers, its actual full tensor has the original smooth nowhere-vanishing native grading section germs on that branch. For the cylindrical branch, the matching identities use the entire signed band at every

nonnegative outward-radius center. Applied to both branches, these conclusions retain the same original section and full tensor on both. Endpoint extensions and restriction from a separately specified whole completed bundle remain hypotheses or separate correspondence obligations.

Proof. Fix any original chart and base point (t, x). If $t = 0$ or $t = 1$, use the supplied endpoint coefficient extension. Otherwise $t \in I$ implies $0 < t < 1$, and the preceding lemma constructs the required extension from intrinsic smoothness. Hence the coefficient-extension condition holds in every chart on the entire interval.

At an arbitrary point choose one chart from the original canonical coframe cover and use its coefficient extension just obtained. If the section is nowhere zero, the original interior grading-section germ lemma produces a smooth nowhere-vanishing canonical-square germ with the full original tensor and actual spatial differential identities. The original cylindrical grading-section germ lemma gives the same conclusion for the cylinder, matching at all nearby full-band points including the negative radii. Both lemmas use natural square evaluation of the given original section, so the tensor retained in either branch is exactly that same original tensor. Applying this argument at every center gives the stated branchwise grading germs. □

Lemma A.235 (Original canonical charts include their fibers in an independent real open bundle)**.** *Fix an original interior or full signed-cylinder canonical chart c with real open domain U, its smooth invertible coframe $C(z)$, and the corresponding fixed normalization N from $\mathbb{R}^{2n}$ to its spatial coordinate model. The normalized coframe $C(z)N$ determines a corrected complex structure and its canonical-square bundle on all of U, independently of a section. For a retained original point $p = (s, w)$ in the chart, put $z = (s, a(w))$ in the interior branch and $z = (s, a(b, -\rho))$ for $w = (\rho, b)$ in the cylinder branch. The original canonical-square fiber at p has a complex-linear inclusion ι_p into this extending bundle's fiber at z, determined by equality of the original line coordinate and the extending coframe coordinate. Thus*

$$q_z^{\mathrm{open}}(\iota_p \tau) = q_p^c(\tau)$$

for every original fiber element τ. If σ is any original section, let T be its natural square tensor on the original spatial tangent and let D_p be the actual original spatial chart differential, including the signed-radius differential in the cylinder branch. Natural square evaluation in the extending bundle satisfies the full tensor identity

$$m_z(\iota_p \sigma(p)) = T_s(\kappa_s(w)) \circ (D_p^{-1} N)^{\otimes 2n},$$

where κ_s is the actual original interior or cylinder insertion. All tensor slots and the cylinder radius-zero seam are retained. This construction does not assert that a section graph extends across either interval endpoint.

Proof. The invertible coframe defines the corrected complex field by conjugating the standard complex structure with CN. Its smoothness and invertibility give the independent canonical-square bundle on U by the native patch lemma. Let q_p^c be the original line coordinate and q_z^{open} the extending coframe coordinate. Both are complex-linear isomorphisms. Define $\iota_p = (q_z^{\mathrm{open}})^{-1} \circ q_p^c$. This is a complex-linear inclusion and its defining coordinate identity follows by cancelling q_z^{open} with its inverse.

Extending coframe reconstruction gives $m_z(\iota_p \sigma(p)) = q_z^{\mathrm{open}}(\iota_p \sigma(p)) \Gamma_{C(z)N}$. Substitute the coordinate identity to replace this scalar by $q_p^c(\sigma(p))$. The original full coordinate reconstruction lemma for the chosen branch identifies $q_p^c(\sigma(p)) \Gamma_{C(z)N}$ with $T_s(\kappa_s(w)) \circ (D_p^{-1} N)^{\otimes 2n}$. Combining these two tensor equalities proves the full identity in every slot. The bundle and the fiber inclusion were chosen from the chart and its coframe before any section, proving the asserted independence. □

Lemma A.236 (A supplied original native graph extension gives an actual coefficient extension)**.** *Fix either original chart and independent extending bundle of the preceding lemma. For an original section σ and a retained center p, suppose there is a real open neighborhood $V \subset U$ of its chart coordinates and a smooth map G into that bundle's total space. Require that at every retained interval point in V,*

$$G(z(p)) = (z(p), \iota_p \sigma(p)),$$

where $z(p)$ uses the original chart and signed-radius coordinates of the chosen branch. This is an equality of actual native section graphs; no scalar or full tensor matching germ is supplied separately. Then the actual original coefficient $q^c(\sigma)$ has a real open smooth extension near p, including when its parameter is 0 or 1. Existence of the supplied native graph extension remains an assumption.

Proof. For $z \in V$, write $G(z) = (b(z), v(z))$ and define

$$\widetilde{q}(z) = \mathrm{coeff}_{C(z)N}(m_{b(z)}(v(z))).$$

Natural tensor evaluation of the smooth map G is smooth by the native bundle-map tensor lemma. The coframe $C(z)N$ is smooth and invertible throughout V. Its coefficient operation evaluates the full tensor on the two blocks of inverse-coframe vectors, so smooth inversion and multilinear evaluation make $\widetilde{q}$ smooth on V. At a retained interval point the graph equality gives $b(z) = z$ and $v(z) = \iota_p \sigma(p)$. The inclusion's coefficient identity therefore gives $\widetilde{q}(z) = q_p^c(\sigma(p))$. This is precisely the original coefficient extension, on the given real open neighborhood. The argument makes no restriction to strict interior times, so applies also at both parameter endpoints when their graph extensions are supplied. □

Lemma A.237 (An intrinsically smooth original section has a smooth retained graph in the extending bundle)**.** *Assume the original collar, complex-field and inverse-germ hypotheses. For either original canonical-square branch, let σ have an intrinsically smooth restriction to its independent branch line bundle. In every original canonical chart the actual graph $p \mapsto (z(p), \iota_p \sigma(p))$ is smooth from the chart's retained base domain into its independently extending canonical-square bundle. This includes both interval endpoints in the closed parameter model and, in the cylinder branch, the full signed band and radius-zero seam. This is the smooth graph on the retained locus; a graph defined on a real open neighborhood across an interval endpoint is not asserted.*

Proof. The map $p \mapsto z(p)$ is smooth: the interval coordinate inclusion and original interior chart are smooth, and on the cylinder the boundary chart and prescribed radial sign change are smooth as well. The coefficient-smoothness lemma for the branch makes $p \mapsto q_p^c(\sigma(p))$ smooth on its base domain. In the extending coframe trivialization, the displayed graph is exactly

$$p \longmapsto (z(p), q_p^c(\sigma(p))),$$

by the inclusion's coordinate identity. Both components are smooth, and the inverse coframe trivialization is smooth. Their composition is therefore the claimed smooth native bundle map on the entire retained base domain. The argument uses the closed interval model there and all allowed cylinder radii, and proves precisely that retained-locus assertion. □

Corollary A.238 (Endpoint native graph extensions in one chart yield original grading germs). Assume the original collar, complex-field and local inverse hypotheses. Let the restriction of the same original section be intrinsically smooth on either original canonical-square branch. At every point of that branch with parameter 0 or 1, suppose that some original canonical chart containing the point has a supplied real open smooth extension of the actual native graph in its independently

extending bundle. Then every original canonical chart on that branch has actual scalar coefficient extension germs at all interval parameters. If the original section is nowhere zero on every completed fiber, its full tensor has the native smooth nowhere-vanishing grading section germs on the branch. For the cylinder these retain every full signed-band matching identity. Applied to both branches, all tensors are those of the same original section. No scalar or full-tensor endpoint germ is supplied separately; existence of the endpoint graph extensions and whole-bundle restriction remain separate obligations.

Proof. Fix an endpoint point and a chart whose native graph has the supplied extension. The graph-to-coefficient lemma gives its actual real open coefficient germ. Given any other original chart containing that point, the appropriate original chart-independence lemma transfers this germ, preserving the original section. Thus all charts have the endpoint coefficient extensions required by the smoothness-and-endpoints corollary. Apply that corollary to the intrinsically smooth branch section: its strict interior-time producer supplies the remaining parameters, and hence every chart has coefficient extension germs throughout the interval. When the original section is nowhere zero, the corollary's original native grading consumers give the claimed nonvanishing section germs and full tensor identities. The original signed-cylinder chart transfer and grading consumer use the actual full band, so preserve its signed-radius matching identities as well. This argument applies to each branch with the same original section. □

Lemma A.239 (The original completed total family has its actual canonical-square coordinates)**.** *Let X be the original completed total quotient with the total atlas of Lemma A.461. Retain the original slice complex fields J_s and one fixed invertible real normalization $N : \mathbb{R}^{2n} \to \mathbb{R}^{k+1}$ used on both patches. For $x \in X$, let s be its original parameter and let $z \in X_s$ be its point in the actual original completed slice. Put*

$$\widetilde{J}_x = N^{-1} J_s(z) N, \qquad E_x = K_{\widetilde{J}_x}^{\otimes 2}.$$

Here the canonical line is formed on the spatial tangent model of the completed slice, rather than on the tangent space of the parameterized base. On each actual original interior or full signed-cylinder inclusion, $\widetilde{J}$ equals the corresponding original normalized branch field. Consequently E is identified with that branch's actual tensor square.

Every original interior or cylindrical canonical coframe chart c defines linear local product coordinates on E over the image of its branch domain in X. That image is open. At an original branch point p, these coordinates send $v \in E_{i(p)}$ to

$$\big(i(p), q_c(p)(\iota_p v)\big),$$

where i is the actual branch inclusion and ι_p is the fiber identification just described. The coordinates and their fiber inverses preserve the original fiber topology. This construction uses no section and asserts no transition smoothness by itself.

Proof. The original parameter projection and fiber equivalence send $i(s,p)$ to parameter s and the original slice inclusion of p. Substitution in the definition of $\widetilde{J}$ gives the claimed branch field identities. Equality of these endomorphisms identifies their canonical lines and their tensor squares, giving the complex-linear continuous fiber isomorphisms ι_p.

Fix a branch chart c with open domain B_c. The total-atlas lemma makes its branch inclusion i an open embedding, so $i(B_c)$ is open. For $x \in i(B_c)$ there is a unique $p \in B_c$ with $i(p) = x$. Define its fiber coordinate by $q_c(p) \circ \iota_p$ and its inverse by $\iota_p^{-1} \circ q_c(p)^{-1}$. These are mutually inverse complex-linear continuous maps; the original branch square-coordinate lemma gives continuity of

q_c on each fiber and of its inverse. Together with the unchanged base coordinate they give local product coordinates over $i(B_c)$, with precisely the displayed formula. Restricting to any fiber therefore induces its original topology. All choices depend only on the original complex field and its coframe charts, so the construction precedes any choice of section. □

Lemma A.240 (Equal original compact tuples give strict full-ribbon identity transports)**.** *Take original positive-dimensional ribbon bundles and original compact ambient and ribbon charts containing the specified slice parameters. Their original compact geometric tuples consist of the native ambient and ribbon manifolds with their topologies and smooth structures, their full primitives λ, μ, the whole ribbon embedding j, and the witness b on the whole ambient boundary. No completed presentation is part of such a tuple. The identity maps of both native manifolds have local smooth real extensions across their boundaries, derived from the original bundle charts.*

If two such full original tuples are equal, equality transports give ambient, boundary and ribbon diffeomorphisms g, β, h. Their ambient and ribbon maps have those across-boundary extensions, and

$$\beta(x) = g(x) \quad (x \in \partial D), \qquad g^*\lambda' = \lambda, \qquad h^*\mu' = \mu, \qquad \beta \circ j = j' \circ h, \qquad b' \circ \beta = b.$$

Thus the primitive scale is exactly one, and every original ribbon point and every boundary witness value is retained.

Proof. The original coordinate change from a compact bundle chart to itself is the identity on its native compact fiber. The original smooth transition theorem gives local real extensions for this same coordinate change, so both identity maps have the asserted germs without any additional hypothesis.

Use the given tuple equality to identify the two manifolds, their structures, primitives, embedding and witness. Define g, β, h as the identity maps in these identifications. They are smooth diffeomorphisms with inverse themselves; the just-derived identity germs give the two across-boundary extensions. The boundary restriction of the ambient identity is the boundary identity. Its derivative and the ribbon derivative are identities, so pullback of either full primitive equals that primitive. Equality of the embedding gives $\beta j = j'h$ at every ribbon point, and equality of the witness gives $b'\beta = b$ at every boundary point. Transporting these conclusions back along the tuple equality proves all the displayed equations. □

Lemma A.241 (An original strict compact map constructs its completed ambient junction)**.** *Let U, V be independently constructed original local completion families of compact ambient bundle data with the same even ambient dimension. Take parameters s, s' in their original retained bases, with native compact primitives λ, λ', original compact insertions $o_s, o'_{s'}$ and completed primitives $L_s, L'_{s'}$. Supply a native compact diffeomorphism g with local real smooth germs across the compact boundary and $g^*\lambda' = \lambda$. Supply its native boundary restriction β, namely a boundary diffeomorphism with underlying compact map g at every boundary point. Then the original local completions admit an ambient junction with strict map Ψ, boundary map exactly β and shift exactly zero. The construction retains the equations*

$$\Psi^* L'_{s'} = L_s, \qquad \Psi(o_s(x)) = o'_{s'}(g(x)), \qquad \Psi(E_s(b, r)) = E'_{s'}(\beta(b), r) \quad (b \in \partial D,\ r \in \mathbb{R}).$$

Here E_s and $E'_{s'}$ are the original all-real boundary flow maps. Neither a completed map nor its end law is supplied as input.

Proof. At the retained base parameters the local family's compact primitives are precisely the original ones. Apply the prescribed-map theorem to the two local native presentations, g, its real extension germs, and the full primitive equality. It gives a strict diffeomorphism Ψ of these same chosen completions and the compact equation in the statement. In particular, at a native boundary point b it gives

$$\Psi(o_s(b)) = o'_{s'}(g(b)) = o'_{s'}(\beta(b)).$$

Strict primitive preservation intertwines the actual Liouville fields and their complete flows by the flow-transport lemma. Since each original end map is the full boundary orbit, for every real r,

$$\Psi(E_s(b,r)) = \Psi(\Phi^r_s(o_s(b))) = \Phi^r_{s'}(\Psi(o_s(b))) = \Phi^r_{s'}(o'_{s'}(\beta(b))) = E'_{s'}(\beta(b),r).$$

Thus β is the induced boundary map and the shift is the smooth constant zero. This all-real equation implies the required eventual end equation. Retaining these maps and equations gives the asserted original ambient junction on the independently chosen completions. □

Corollary A.242 (Equal original endpoint tuples construct their own completed ambient normalization)*.* Take two original positive-dimensional ribbon endpoints in independently supplied bundles, and original compact charts containing their actual parameters. If their full original compact tuples are equal, there is a specified choice of original ambient junction between the two corresponding local completed slices. Its boundary map is exactly the equality transport of the original compact boundary; its shift is zero. Its strict ambient map extends the original compact equality transport and obeys the same boundary orbit equation for every real time. The choice is made from the original compact equality before any represented endpoint pair or represented grading lift is used.

Proof. The full tuple equality gives the native ambient equality transport, its across-boundary germs, the full strict primitive identity and its specified boundary restriction by the equal-tuple transport lemma. Apply the preceding completed-junction lemma to the two original local families at their endpoint parameters, which lie in their retained bases by construction. It gives a nonempty collection of junctions with all the stated boundary, shift, compact and all-real end equations. Choose an element of that collection and retain those same equations. No represented pair or grading occurs in these inputs or in this choice, which proves the final assertion. □

Lemma A.243 (Equality of smooth Maslov classes has a single smooth homotopy witness)**.** *Fix a symplectic manifold and let ρ_0, ρ_1 be smooth Maslov representatives, allowing their compatible complex structures to differ. Their classes in the quotient by smooth compatible nowhere-vanishing homotopy are equal if and only if a single such smooth homotopy joins them. Smoothness includes real open extension across the time endpoints.*

Proof. The constant homotopy proves reflexivity, and replacing its time parameter by $1-t$ proves symmetry. To prove transitivity, take homotopies H from ρ_0 to ρ_1 and G from ρ_1 to ρ_2. Let $\vartheta : \mathbb{R} \to [0,1]$ be a smooth transition function that is zero on $(-\infty, 0]$ and one on $[1,\infty)$. On $t \le \frac{1}{2}$ use $H_{\vartheta(4t)}$, and on $t \ge \frac{1}{2}$ use $G_{\vartheta(4t-3)}$. For $\frac{1}{4} < t < \frac{3}{4}$ both expressions equal ρ_1 wherever the indicated expression is used: the first has time one and the second has time zero. Thus they agree on a neighborhood of the joining time and define a smooth family. Its endpoint values are ρ_0 and ρ_2. Both reparameterizations remain in $[0,1]$, so every complex structure remains compatible and every section remains canonical and nowhere vanishing. Composing their real open extension germs with these smooth time maps gives real open extension germs for the concatenation, including both endpoints. Hence the relation is transitive.

Quotient equality is generated by the original homotopy relation using reflexivity, symmetry and finite transitivity. Each of those operations preserves existence of one smooth homotopy by the preceding constructions. Induction on that generated relation therefore turns quotient equality into one such homotopy. Conversely a homotopy is one of the defining identifications in the quotient and hence implies equality of the classes. This proves both directions. □

Lemma A.244 (Equal original tuples and original Maslov classes produce a graded junction)**.** *Take two independently described original positive-ribbon bundle endpoints with equal full original compact tuples, and let Ψ be the original completed normalization constructed from that equality. The original smooth representatives ν_L, ν_R on their local completed ambients recover exactly their independently specified original complex fields and full canonical-square tensors. Any independently supplied smooth representatives with those original fields equal these representatives. Thus the following original class equality may equivalently use any such representatives. Suppose their original Maslov classes agree under this constructed map:*

$$[\nu_L] = [\Psi^* \nu_R].$$

Then there is an original graded junction between those endpoints, with this same completed ambient normalization, its zero shift, the full original ribbon equality transport and primitive scale one. Its smooth homotopy is constructed from the class equality; a junction or homotopy need not be supplied separately. The retained-junction construction turns this arrow into a permitted comparison step graded between the actual slice representatives of the two already fixed bundle realizations. Equality of compact tuples alone does not assert the displayed equality of grading classes.

Proof. Pull the retained endpoint representative back by the actual strict original local-to-retained comparison. Strictness identifies the symplectic forms, so the pullback is a smooth compatible nowhere-zero Maslov representative on the original local completion. Original endpoint recovery identifies its complex field and every tensor slot with the independently specified original atlas data. Perform this construction for both endpoints to obtain ν_L, ν_R. A representative is determined by its complex field and full tensor, so any independently supplied representative with those same fields equals it. Substituting those equal representatives preserves both classes, proving the claimed equivalence of class premises.

The original completed normalization is strict by the ambient normalization corollary. The assumed class equality and the preceding smooth-homotopy lemma therefore give one smooth compatible nonvanishing homotopy from ν_L to $\Psi^* \nu_R$. The full original tuple equality gives a ribbon equality transport h with its real native extension germs, $h^* \mu_R = \mu_L$, and the whole embedding square by the equal-tuple transport lemma. Use this h, scale one, the constructed Ψ and its boundary map, and the just-constructed homotopy as the original junction data. Every required geometric and Maslov endpoint equation has now been proved from the original tuple and class equalities. The retained junction lemma constructs a permitted identification and its grade on the same fixed endpoint realizations. Treating this junction as an original arrow gives the stated permitted comparison step. □

Corollary A.245 (An equal independently described endpoint appends to the same finite graded chain)*.* Let C be any supplied finite original positive-ribbon itinerary from A to L. Let R be an independently described original endpoint whose full compact tuple equals that of L in chosen original charts, and whose original Maslov class agrees under the ambient normalization constructed from this equality. Then C followed by the constructed normalization arrow has a permitted comparison chain $\widehat{C}'$, with a grade on that very chain between the retained representatives at A

and R. Its length is $\mathrm{length}(C)+1$, and its literal vertex list is the list of original vertices of C, followed by R, with each vertex replaced by its fixed retained slice pair. No original junction is an additional input.

Proof. The equal-endpoint normalization lemma constructs an original arrow from L to R, with all its strict, full-ribbon and grading data. Append this arrow to the given original itinerary C. The preceding endpoint L matches its source literally, so this is an original finite itinerary. Apply the finite itinerary theorem to that appended sequence. It returns precisely its compiled chain and a grade on that chain, without replacing the fixed realizations. Appending one arrow adds one to the original length and appends exactly R to the original vertex list. The itinerary theorem's exact length and vertex identities give both claimed formulas for the same compiled chain. □

Lemma A.246 (The whole original canonical square has smooth actual coordinate changes)**.** *Retain the original completed total atlas, its real two-way mixed collar germs, and its original interior and signed-cylinder canonical coframe charts with the common spatial tangent normalization N. Write $D_c \subset X$ for a promoted chart's open base domain and $q_c(x) : E_x \to \mathbb{C}$ for its actual fiber coordinate. For any two charts c, d of either branch, the actual coordinate change is multiplication by the original coframe scalar:*

$$q_d(x)q_c(x)^{-1}(z) = \Delta(F_c(x), F_d(x))z \quad (x \in D_c \cap D_d,\ z \in \mathbb{C}).$$

The scalar $x \mapsto \Delta(F_c(x), F_d(x))$ and the resulting real continuous-linear-map-valued multiplication map are smooth on the entire overlap in the original completed total atlas, including its parameter endpoints. No section is supplied in this assertion.

Proof. The branch fiber identifications preserve natural square multiplication. Original coframe reconstruction therefore gives, for $v \in E_x$,

$$m_x(v) = q_c(x)(v)\Gamma_{F_c(x)}.$$

Apply this to $v = q_c(x)^{-1}(z)$ and evaluate its full two-block tensor in the inverse $F_d(x)$ frame. The result is $z\Delta(F_c(x), F_d(x))$, which equals $q_d(x)(v)$ by the original coefficient formula. This proves the exact coordinate change whether the two charts are interior, cylindrical or of different branches.

For two interior charts, their unique interior preimages coincide; in the original interior patch the scalar is exactly the already proved smooth original interior transition. The local inverse of that open patch embedding is smooth in the original total atlas, so the scalar is smooth on its entire total overlap. For two cylindrical charts use their unique cylinder preimages and the original smooth cylindrical transition; the smooth inverse cylinder patch again gives smoothness on the whole total overlap.

For an interior and a cylindrical chart, fix any overlap point. The original real mixed collar germs give local spatial coordinate changes with unchanged parameter. The forward mixed canonical-square lemma supplies a smooth real extension of the actual scalar in the end coordinates, and the reverse mixed lemma supplies its corresponding extension in the interior coordinates. Composing either extension with the actual smooth local inverse patch and native spatial charts gives smoothness of the same scalar near the selected total point. These extensions include parameter endpoints and retain the genuine spatial collar derivative. Since the point was arbitrary, both directed mixed scalars are smooth on their full overlap.

Finally $a \mapsto (z \mapsto az)$ is a continuous real-linear map from $\mathbb{C}$ to the space of real continuous-linear maps $\mathbb{C} \to \mathbb{C}$. Composing it with the just-proved scalar smoothness proves smoothness of the actual linear coordinate-change map. All four cases use only the original field and coframes, proving independence from a section. □

Lemma A.247 (The canonical square is a smooth bundle on the whole original completed family)**.** *Retain the original completed total atlas, the slice collars and their smooth local inverses, the actual two-way real mixed collar germs, and the fixed spatial normalization N. Suppose $J_s^2 = -\mathrm{id}$ and that its original interior and cylindrical coordinate fields have local real smooth extensions as in the two original coframe-cover lemmas, including parameter endpoints. Then the actual fibers $E_x = K_{\widetilde{J}_x}^{\otimes 2}$ over the entire original completed quotient form a smooth complex line bundle with fiber model $\mathbb{C}$. In particular every fiber has complex dimension one. Its atlas consists of all original interior and full signed-cylinder canonical-square coordinates promoted through their actual inclusions. Its coordinate changes are exactly the scalar multiplications of the preceding overlap lemma, and its topology on each fiber is the original one. The construction covers the compact seam and both parameter endpoints and is made before any section is chosen.*

Proof. Every total point lies in an actual interior or signed-cylinder patch by the original quotient cover. If it lies in the interior patch, the interior complex-field extension and square identity give an original canonical coframe chart containing its unique branch preimage. If it lies in the cylindrical patch, the end coframe-cover lemma gives such a chart for that preimage, including the radius-zero seam. In either case the promoted chart contains the total point. Thus these promoted open domains cover the entire original total quotient.

The coordinates lemma gives mutually inverse linear fiber coordinates on every domain and identifies their original fiber topologies. Give the bundle total space the topology induced by all these local product coordinates. The overlap lemma proves continuity of their actual linear transition maps on every intersection. Hence the product coordinates are compatible local bundle trivializations for that topology. Their transitions are also smooth on the original total base, so the resulting vector bundle is smooth. The same fiber coordinates are complex-linear and each scalar transition is complex-linear, so these same charts also define its complex vector-bundle structure. Its fiber model is $\mathbb{C}$ by the same independent coordinate isomorphisms; every fiber therefore has complex dimension one. Restriction to a fiber preserves its original topology. Coverage used the full signed cylinder and the closed parameter interval, so includes the seam and endpoints. Every chart and transition was constructed from the original complex field and collars, with no section among the inputs; the resulting bundle has the asserted independence. □

Lemma A.248 (Equal original surface tuples normalize their entire finite ribbons)**.** *Take two original compact surface bundle endpoints with finite zero-dimensional ribbons, in original charts containing their respective parameters. Suppose their full original compact tuples are equal, including the native ambient and ribbon manifolds, their full primitives, the whole ribbon embedding and the entire boundary witness. Then the ambient equality transport is a native strict diffeomorphism with derived across-boundary germs; its boundary restriction β and the ribbon equality transport h satisfy*

$$\beta \circ j = j' \circ h, \qquad b' \circ \beta = b.$$

The two original local completed ambients have a specified strict normalization extending that ambient compact transport, with boundary map exactly β, shift zero, and the exact boundary orbit equation for every real time. The ribbon map is defined on the entire finite ribbon, and its primitive scale is one. Empty ribbons are allowed.

Proof. The original ambient chart self-transition is the identity, so its proved real transition germs supply the ambient identity's across-boundary extensions. Transport these germs and the identity map along the tuple equality. Its derivative is the identity after the same equality, so the full ambient primitive pulls back strictly. Equality of the whole embedding and witness gives both displayed

equations. The ribbon equality transport is an ordinary smooth zero-dimensional diffeomorphism. Every tangent vector of such a ribbon is zero; thus both ribbon one-forms and their pullbacks are zero, giving the primitive equation with scale one.

Apply the original strict compact-map completed-junction lemma to the two surface local families, their endpoint parameters and the ambient map just obtained. It constructs a nonempty collection of strict normalizations with this boundary map, shift zero, and the full compact and all-real end equations. Choose a member of this collection and retain those equations. No step chooses a ribbon point or requires the ribbon to be nonempty, so the construction also applies to empty ribbons. □

Lemma A.249 (Equal original finite-ribbon endpoints and classes give their own graded junction)**.** *Take two independently described original surface bundle endpoints with equal full compact tuples and the ambient normalization Ψ constructed in the preceding lemma. Their original local smooth Maslov representatives ν_L, ν_R recover the original complex fields and full tensors exactly. Any independently supplied representatives with those same original fields equal these representatives. If*

$$[\nu_L] = [\Psi^* \nu_R],$$

then an original graded junction is constructed with this same Ψ, its zero shift, the full finite-ribbon equality map and scale one. It gives a permitted step graded on the fixed retained endpoint realizations, including empty ribbons. The class equality can equivalently be stated using any independently supplied representatives with the original fields. No junction or homotopy is supplied as an input.

Proof. Pull each fixed retained representative back along its actual strict original local slice comparison. Original Maslov endpoint recovery identifies its complex field and full tensor with the original data. A representative is determined by these two fields, so any independently supplied representative with those same fields equals this one. Replacing either representative by that equal one changes neither its class nor the class of its strict pullback; this proves the asserted equivalence of class premises.

The ambient normalization is strict. The assumed original class equality therefore gives one smooth compatible nowhere-zero homotopy from ν_L to $\Psi^* \nu_R$ by the smooth class-equality lemma. Retain the original finite-ribbon equality map, its whole embedding square and scale one from the preceding normalization lemma. Together with Ψ, its boundary map and the constructed homotopy these give all original finite-ribbon junction data. The retained-junction theorem constructs the corresponding permitted identification and grade on the same fixed endpoint pairs and representatives. Both the finite-ribbon junction construction and all its original equations permit the empty ribbon, proving the stated generality. □

Corollary A.250 (An equal original finite-ribbon endpoint appends to the literal graded chain)*.* Let C be a supplied finite original surface-bundle itinerary from A to L, allowing finite empty ribbons. Let R be independently described with equal full original compact endpoint tuple and with equal original Maslov class under the ambient normalization constructed from this tuple equality. Appending that normalization gives a permitted comparison chain with a grade on that exact chain from the retained representative at A to the retained representative at R. Its length is $\mathrm{length}(C) + 1$; its literal vertex list is the original vertex list of C followed by R, mapped to the fixed retained endpoint pairs. No original junction is an extra input.

Proof. The preceding lemma constructs an original graded arrow from L to R. Append it to C at the identical original endpoint L. The original finite-itinerary theorem returns the compiled comparison chain with a grade on that very chain. Its exact length identity gives one more than

the original length of C; its exact vertex identity gives the original list followed by R, replaced pointwise by the fixed retained pairs. The normalization and itinerary constructions allow empty finite ribbons and empty input paths, so the same formulas hold in those cases as well. □

Lemma A.251 (A section of the whole original canonical square restricts to the exact original branch sections)**.** *Under the hypotheses of the whole-bundle construction, let σ_s be one section of the actual original slice canonical-square fibers. Regard it as the section σ^{tot} of that constructed whole bundle by sending an original total point x to $\sigma_s(z)$, where s and $z \in X_s$ are its actual original parameter and slice point. Through the proved fiber identifications its restrictions to the actual interior and full signed-cylinder inclusions are exactly the original branch sections of σ.*

If σ^{tot} is intrinsically smooth in this whole bundle, both these exact original restrictions are intrinsically smooth in the independently constructed branch bundles. This includes the parameter endpoints in the closed-interval model and the full signed cylinder, including its compact seam. It does not by itself assert real open section extensions across the parameter endpoints.

Proof. The original parameter and fiber equivalence send an interior insertion to its given parameter and original interior slice point. Evaluating σ there, and using the equality-induced canonical-square fiber identification, gives precisely its original interior branch value. For a cylinder insertion the original fiber-equivalence end equation gives the given parameter and original cylindrical slice point; the same fiber identification gives precisely the original end branch value. These are pointwise equalities in the actual fibers, rather than just identities of scalar coefficients.

Fix an original interior branch point and a canonical chart c containing it. Its fiber coordinate in the branch bundle is exactly the coordinate of the promoted chart in the whole bundle after composing with the actual interior inclusion, by the preceding fiber identities and the promoted-coordinate formula. Smoothness of the whole section implies smoothness of this scalar coordinate on the promoted chart domain. The total-atlas theorem makes the interior inclusion smooth, so its composition with that scalar is smooth. It equals the branch section's scalar on a neighborhood of the chosen point. A local bundle trivialization identifies smoothness of a section with smoothness of its scalar and unchanged base coordinate; hence the exact branch section is smooth there. Original canonical chart coverage proves smoothness at every interior branch point.

At any cylindrical branch point use an original end canonical chart and its promoted whole-bundle chart. Their scalar coordinates obey the same equality after composition with the smooth actual cylinder inclusion. The whole scalar is smooth, so the branch scalar is smooth, and the local bundle criterion proves smoothness of that exact end section at the point. End-chart coverage gives the result on the entire signed cylinder. Both arguments use the original closed-interval atlas and thus include its endpoints; no real open endpoint extension is constructed by these intrinsic smoothness arguments. □

Corollary A.252 (Whole original section smoothness reaches the full-tensor grading germs with endpoint graph extensions). Under the whole-bundle hypotheses, let one nowhere-zero original section σ be intrinsically smooth in that whole bundle. For every original interior branch point at parameter 0 or 1, supply one original canonical chart containing it and a real open native bundle-graph extension of that exact section there. Supply the corresponding one-chart graph extension at every original signed-cylinder branch point at parameter 0 or 1. Then the same original full tensor $m(\sigma)$ has both original native interior grading-section germs and original native signed-cylinder grading-section germs throughout the interval. Their matching equations recover every slot of this full tensor, including the seam and endpoint parameters. Existence of the supplied endpoint graph extensions remains a separate requirement.

Proof. The restriction lemma gives intrinsic smoothness of the exact original interior and signed-cylinder branch sections. Apply the endpoint-graph corollary to those two smooth restrictions, the supplied one-chart endpoint graph extensions and the original section's nonvanishing. Its strict interior-time coefficient construction supplies the remaining parameters. At endpoints it extracts coefficients from the supplied native graphs, transfers them to every original chart by the actual canonical-square transitions, and reconstructs the original nonzero section. Thus its two branch conclusions are grading-section germs for this same original full tensor, with their exact matching equations. The cylindrical conclusion uses the full signed branch domain, so includes the seam. All these conclusions retain the original section; the only endpoint extensions used are the ones explicitly supplied. □

Lemma A.253 (A nonzero scalar lifts a real open base change)**.** *Let $B : U_B \to V_B$ be a smooth diffeomorphism between real open sets in real normed vector spaces. Let U be a real open set in its source space and $a : U \to \mathbb{C}$ a smooth function. Put*

$$S = U_B \cap U \cap \{u : a(u) \neq 0\}, \qquad T = B(S).$$

Then S, T are open and

$$A : S \times \mathbb{C} \longrightarrow T \times \mathbb{C}, \qquad A(u, q) = (B(u), a(u)q)$$

is a smooth diffeomorphism with inverse $A^{-1}(v, r) = (B^{-1}(v), a(B^{-1}(v))^{-1}r)$. In particular its source contains (u_0, q_0) whenever $u_0 \in U_B \cap U$ and $a(u_0) \neq 0$. This is a local coordinate result; no extension of a canonical bundle or of a chosen section is assumed or asserted.

Proof. Continuity of a on U makes its nonzero locus relatively open there; since U and U_B are open, S is open. The base diffeomorphism is an open map, so $T = B(S)$ is open. Its restrictions identify S with T. The base diffeomorphism and smooth complex multiplication make A smooth. For $v \in T$, its inverse base point lies in S, where a is nonzero. Complex inversion is smooth away from zero, so the displayed inverse is smooth on $T \times \mathbb{C}$. Substituting it into A gives $(v, a(B^{-1}(v))a(B^{-1}(v))^{-1}r) = (v, r)$; substituting A into it gives $(u, a(u)^{-1}a(u)q) = (u, q)$. Thus the displayed maps are mutual inverses. The asserted source membership is exactly the defining condition $u_0 \in S$, with arbitrary scalar q_0. □

Lemma A.254 (Actual whole-bundle coordinates have reciprocal nonzero scalar transitions)**.** *Retain the original completed total space X and its canonical-square fibers E_x. For an original interior or signed-cylinder canonical chart c, let i_c be its actual branch inclusion, B_c its branch domain and $q_c(x) : E_x \to \mathbb{C}$ its promoted actual fiber coordinate. The image $D_c = i_c(B_c)$ is exactly the base domain of that promoted trivialization. Every $x \in D_c$ has a unique preimage $p \in B_c$. Define the raw coordinate of $v \in E_x$ by*

$$\Phi_c(v) = (u_c(p), q_c(x)(v)),$$

where $u_c(p)$ is the original real parameter together with its actual interior spatial chart, or its end spatial chart with the prescribed signed-radius convention. At a branch insertion $x = i_c(p)$ this is exactly the coordinate obtained using that p and its actual promoted fiber coordinate.

For $x \in D_c \cap D_d$, write $a_{cd}(x)$ for the original transition scalar from c to d. For every $v \in E_x$,

$$q_d(x)(v) = a_{cd}(x)q_c(x)(v), \qquad a_{dc}(x)a_{cd}(x) = 1, \qquad a_{cd}(x) \neq 0.$$

These are identities in every actual fiber, independent of a section. They do not assert a real open change germ across an interval endpoint.

Proof. The definition of each promoted trivialization gives its base domain as the actual branch image $i_c(B_c)$, proving the domain identity. Each original branch inclusion is an open embedding, hence injective. Thus a point in that image has exactly one preimage, and substituting an insertion $i_c(p)$ into the raw coordinate returns the displayed coordinate of p itself.

Fix $x \in D_c \cap D_d$ and $v \in E_x$. The whole-bundle overlap identity applied to $z = q_c(x)(v)$ says $q_d(x)q_c(x)^{-1}(z) = a_{cd}(x)z$. The fiber coordinates are mutually inverse, so this is precisely $q_d(x)(v) = a_{cd}(x)q_c(x)(v)$. Now choose the actual fiber vector $v_0 = q_c(x)^{-1}(1)$. Applying this identity in both directions gives

$$1 = q_c(x)(v_0) = a_{dc}(x)q_d(x)(v_0) = a_{dc}(x)a_{cd}(x).$$

If $a_{cd}(x)$ were zero, this product would be zero, contradicting $1 \neq 0$. The reciprocal and nonzero conclusions therefore use the actual fiber coordinate inverses and hold without any complex law outside the retained parameter locus. □

Lemma A.255 (Every actual whole-bundle overlap has a real open inverse scalar germ)**.** *Retain the original collar, actual total inclusions and their real mixed germs, with the original canonical coframe charts and their spatial normalizations. Fix charts c, d and retained branch points $p \in B_c$, $q \in B_d$ with $i_c(p) = i_d(q) = x$. There is a real open neighborhood U of $u_c(p)$ contained in the extending coframe domain of c and a smooth complex function a on U such that, for all retained $r \in B_c$, $s \in B_d$,*

$$i_c(r) = i_d(s), \quad u_c(r) \in U \quad \Longrightarrow \quad a(u_c(r)) = a_{dc}(i_c(r)).$$

Thus a extends the actual inverse fiber-coordinate scalar in the source chart's real coordinates, including at interval endpoints. Its value at the center is nonzero. Neither a section nor a nonzero scalar away from the retained locus is assumed.

Proof. The actual coframe formula for the whole-bundle transition identifies $a_{dc}(x)$ with the squared-coframe transition from the native frame of d at q to that of c at p.

If both charts are interior charts, injectivity of the interior inclusion gives $p = q$. Intersect the extending domain of c with the real chart-overlap domain and the inverse image of the extending domain of d under its actual spatial change, keeping the real parameter fixed. This open set contains $u_c(p)$ because the point belongs to both retained chart domains. The interior transition-smoothness lemma gives a smooth scalar there. At any two retained points with the same total insertion, injectivity again makes the branch points equal. The original spatial derivative identity then identifies that scalar with the displayed actual transition from d to c.

If both charts are cylindrical, use the original signed-cylinder spatial change and the same intersection of coframe domains. It is open and contains $u_c(p)$, and the end transition-smoothness lemma gives its smooth scalar. Injectivity of the full cylinder inclusion makes retained branch points with equal insertion identical; the actual end differential formula identifies this scalar with a_{dc}, throughout the full signed band.

If c is interior and d is cylindrical, the inverse mixed-scalar lemma supplies a smooth scalar in the real interior coordinates near $u_c(p)$, with its domain contained in c's extending coframe domain. At any retained interior point there, that lemma produces a cylinder point with the same total insertion and the prescribed scalar identity. If a cylinder point $s \in B_d$ with that insertion has already been specified, injectivity of the cylinder inclusion identifies it with the produced point. The mixed coframe formula therefore gives exactly the stated matching equation.

If c is cylindrical and d is interior, the forward mixed-scalar lemma gives the corresponding real cylinder neighborhood and smooth scalar, within c's extending coframe domain. At any retained cylinder point in it, its produced interior point has the same total insertion. Injectivity of the

interior inclusion identifies that point with every specified $s \in B_d$ having this insertion. The other mixed coframe formula gives the required inverse transition scalar in these source coordinates. These four cases cover all chart types. At the center, matching gives $a(u_c(p)) = a_{dc}(x)$, which is nonzero by the reciprocal transition lemma. □

Lemma A.256 (Actual whole-bundle coordinates have real open diffeomorphism germs)**.** *Retain the original whole-bundle coordinate, collar and real mixed-germ hypotheses. At any point belonging to two original chart domains, the actual raw base-coordinate change has a smooth diffeomorphism germ between real open sets. Its forward and inverse agree with the two actual base-coordinate changes on their retained common locus.*

For the raw bundle coordinates Φ_c of the preceding coordinate lemma, fix a vector $z \in E_x$ with $x \in D_c \cap D_d$. There is a smooth diffeomorphism $A : U_A \to V_A$ between real open sets in the full real bundle models of c, d, with

$$\Phi_c(z) \in U_A, \qquad A(\Phi_c(z)) = \Phi_d(z).$$

For every actual fiber vector w over a point of $D_c \cap D_d$,

$$\Phi_c(w) \in U_A \Longrightarrow A(\Phi_c(w)) = \Phi_d(w), \qquad \Phi_d(w) \in V_A \Longrightarrow A^{-1}(\Phi_d(w)) = \Phi_c(w).$$

Both directed changes thus use one pair of smooth mutual inverses. The construction covers all four interior/cylinder chart combinations, including parameter endpoints, and is independent of any section. It does not impose a common extension away from the retained interval.

Proof. For two interior charts, use the actual smooth interior spatial-chart change and its inverse, together with the real parameter. Injectivity of the interior insertion makes retained branch points with equal total image identical, so these product changes agree with both actual base-coordinate changes. For two cylindrical charts use the actual smooth boundary-product spatial changes. The signed-radius coordinate is a smooth sign change, and injectivity of the cylinder insertion again gives both retained matching identities. For an end-to-interior change, conjugate the original real mixed collar diffeomorphism by the two real parameter–spatial charts, using the prescribed radial sign. Its source contains the overlap point, and the original total insertion equation identifies its value with that point's interior coordinate. The two matching identities of the original collar germ give both directed base matching identities. For an interior-to-end change use the inverse of this same conjugated map. These four cases supply a real open base diffeomorphism B at the selected point, with both retained matching identities.

Let a be the preceding lemma's smooth extension of the inverse scalar a_{dc} in the real c base coordinates. Its center value is nonzero. Restrict its domain to its open nonzero locus and intersect with the source of B. There a^{-1} is smooth, so the skew-product lemma supplies one real open diffeomorphism

$$A(u, q) = (B(u), a(u)^{-1}q), \qquad A^{-1}(v, r) = (B^{-1}(v), a(B^{-1}(v))r).$$

Its source contains $\Phi_c(z)$, since both original base domains contain the center and the center scalar is nonzero. At any retained common point in this source, the base matching identity gives its actual d base coordinate. The inverse-scalar matching and reciprocal identity give $a(u)^{-1} = a_{cd}(x)$ there. The actual arbitrary-fiber transition formula therefore gives exactly $A(\Phi_c(w)) = \Phi_d(w)$, including the specified center.

If $\Phi_d(w)$ lies in the target, the retained inverse base matching identifies B^{-1} of its base with its actual c base coordinate. The target definition places that base in the same restricted nonzero source domain. Hence $\Phi_c(w)$ lies in the source of A. Apply the forward matching identity just

proved and then the inverse identity of A: this yields $A^{-1}(\Phi_d(w)) = \Phi_c(w)$. All matching equalities concern arbitrary actual fiber vectors. The base maps, coframe scalars and restricted skew product were chosen without a section. □

Definition A.257 (Local real extensions in the actual canonical atlas)**.** *Let $f : X \to E$ be any map from the original completed total base into its actual canonical-square total space. Write $\pi : E \to X$ for the bundle projection and E_d^{real} for the real base model of d times $\mathbb{C}$. The local extension condition means that, for every $x \in X$, there are original charts c, d, a point $p \in B_c$ with $i_c(p) = x$, and a real open neighborhood U of $u_c(p)$ in the extending coframe domain of c, such that $\pi f(x) \in D_d$ and a smooth map $G : U \to E_d^{\mathrm{real}}$ satisfies*

$$r \in B_c, \quad u_c(r) \in U \quad \Longrightarrow \quad \pi f(i_c(r)) \in D_d, \qquad G(u_c(r)) = \Phi_d(f(i_c(r))).$$

The all-chart extension condition requires such U, G for every pair c, d and every $p \in B_c$ with $\pi f(i_c(p)) \in D_d$. These conditions concern the entire coordinate map in the actual atlas, before specializing to a section. They impose no separately selected scalar or tensor-coordinate germ. The extension condition is supplied regularity across interval endpoints; bare smoothness on the retained closed-interval model is a separate condition.

Lemma A.258 (Real open extensions can be composed after shrinking)**.** *Let $A : E_0 \supset U_A \to V_A \subset E_1$ and $B : F_1 \supset U_B \to V_B \subset F_0$ be smooth diffeomorphisms between real open subsets of real normed vector spaces. Let $G : U \to F_1$ be smooth, with $U \subset E_1$ open. If $x \in U_A$, $A(x) \in U$ and $G(A(x)) \in U_B$, there is a real open neighborhood V of x with $V \subset U_A$ such that $A(y) \in U$ and $G(A(y)) \in U_B$ for every $y \in V$, and $B \circ G \circ A$ is smooth on V.*

Proof. The set $S = U_A \cap A^{-1}(U)$ is open, contains x, and is a domain on which $G \circ A$ is smooth. Its continuity makes $V = S \cap (G \circ A)^{-1}(U_B)$ open; this set contains x by hypothesis. Its defining conditions give both required inclusions. On V, the maps A, G, B are smooth and all their intermediate values belong to the stated smoothness domains. Their composition is therefore smooth there. □

Lemma A.259 (Actual chart neighborhoods admit real open restrictions)**.** *Fix an original chart c and $p \in B_c$. For every open neighborhood $O \subset X$ of $i_c(p)$, there is a real open neighborhood V of $u_c(p)$ in the extending coframe domain of c such that*

$$r \in B_c, \quad u_c(r) \in V \quad \Longrightarrow \quad i_c(r) \in O.$$

This is a topological restriction statement at every interval parameter, including its endpoints and the full signed-cylinder branch.

Proof. The actual inclusion i_c is continuous, so $i_c^{-1}(O)$ is an open neighborhood of p in the parameter–spatial branch. Let $\chi(t) = \min(1, \max(0, t))$. This continuous map fixes $[0, 1]$. On the product of the real parameter and the open target of the actual spatial chart, apply its inverse spatial chart and χ to obtain a continuous map into that branch. For a cylindrical chart, use the inverse of the signed-cylinder spatial chart, whose source is open in the full band. This map takes $u_c(p)$ to p and takes each retained coordinate $u_c(r)$ to r. The inverse image of $i_c^{-1}(O)$ under it is therefore real open, contains $u_c(p)$ and has the claimed retained-point inclusion. Intersect it with the extending coframe domain of c, which is real open and contains $u_c(p)$. The resulting set is the required V. Only continuity of χ is used. □

Lemma A.260 (Local full-map extensions transfer to all original charts)**.** *Let $f : X \to E$ satisfy the local extension condition in the actual canonical atlas. If $\pi \circ f : X \to X$ is continuous, then this same map satisfies the all-chart extension condition. Under the whole-bundle coframe-cover hypotheses, the converse holds for every map, and these two conditions are therefore equivalent whenever $\pi \circ f$ is continuous.*

Proof. Fix requested charts c, d and a point $p \in B_c$ with $\pi f(i_c(p)) \in D_d$. Choose local extension charts e, j, a branch point $r \in B_e$ with $i_e(r) = i_c(p)$, and a smooth full-map extension G on a real open neighborhood U of $u_e(r)$. The actual real base-change lemma gives a diffeomorphism germ A from c coordinates to e coordinates at this point, with $A(u_c(p)) = u_e(r)$. The actual whole-bundle change lemma gives a diffeomorphism germ B from j coordinates to d coordinates at $f(i_c(p))$. The local matching identity at r says $G(u_e(r)) = \Phi_j(f(i_c(p)))$, which belongs to the source of B. The composition lemma supplies a real open neighborhood of $u_c(p)$ on which $H = B \circ G \circ A$ is smooth and all its factors are defined.

Both D_e and D_d are open actual chart domains. Apply the neighborhood-restriction lemma to D_e and to $(\pi \circ f)^{-1}(D_d)$, the latter open by the continuity assumption. Intersect their real open restrictions with the just-chosen composition domain. This leaves a real open neighborhood W of $u_c(p)$ within c's extending coframe domain.

For any retained $q \in B_c$ with $u_c(q) \in W$, its total insertion belongs to D_e and its image under $\pi \circ f$ belongs to D_d. Let $s \in B_e$ be the unique preimage of $i_c(q)$. Actual base matching gives $A(u_c(q)) = u_e(s)$. The composition-domain restrictions place this point in U, so local full-map matching gives $G(u_e(s)) = \Phi_j(f(i_c(q)))$, and its bundle base lies in D_j. That value belongs to the source of B. Actual bundle matching therefore gives

$$H(u_c(q)) = B(\Phi_j(f(i_c(q)))) = \Phi_d(f(i_c(q))).$$

Together with $\pi f(i_c(q)) \in D_d$, these are all the required retained matching identities for W, H in the requested charts. The choices c, d, p were arbitrary, proving the all-chart condition.

Conversely assume the all-chart condition and fix $x \in X$. The section-independent whole-bundle coframe cover supplies one chart c with $x \in D_c$ and another chart d with $\pi f(x) \in D_d$. Choose the unique $p \in B_c$ with $i_c(p) = x$ and apply the all-chart condition to c, d, p. Its neighborhood and extension are exactly the local witnesses at x. This proves the converse without a continuity assumption. Combining the two implications gives the stated equivalence when $\pi \circ f$ is continuous. □

Lemma A.261 (Local full-map regularity extracts the exact original section graphs)**.** *Under the original whole-bundle and coframe-cover hypotheses, let σ be one section of the actual original slice canonical-square fibers. Suppose its actual whole-bundle graph*

$$f_\sigma(x) = (x, \sigma^{\mathrm{tot}}(x))$$

has the local full-map extension property in the actual atlas. Then every original interior and full signed-cylinder canonical chart has a smooth real open coefficient extension for the exact original restriction of σ at every retained branch point. These coefficients reconstruct smooth native graph extensions in the independently extending branch bundles, with exact equality to the original fiber inclusions of σ on the retained locus. In particular, endpoint graph extensions are produced at both interval endpoints, and the one-chart-at-each-total-point coefficient cover holds. Neither a coefficient germ nor an endpoint graph is supplied separately.

Proof. The projection of f_σ is the identity map of X, hence is continuous. The chart-independence lemma transfers its local extension property to every covering source and target chart, preserving this same graph map.

Fix any original interior chart c and a point $p \in B_c$. Apply that all-chart condition with both source and target equal to c. Their actual domains contain the graph's base $i_c(p)$. It gives a real open neighborhood U of $u_c(p)$ within the extending coframe domain and a smooth full coordinate map G on U. Put $q(z) = \mathrm{pr}_{\mathbb{C}}\, G(z)$, which is smooth. At every retained point of this chart with coordinate in U, the full map matching identifies q with the actual promoted tensor coordinate of σ^{tot}. The exact whole-section restriction identity identifies this coordinate with the original interior coefficient of σ. This proves the required coefficient extension throughout U. For a full signed-cylinder chart, use that chart for both source and target. The same last-coordinate projection and exact cylindrical restriction identity give the original cylinder coefficient extension, with the prescribed signed-radius coordinates. Both arguments apply at every interval parameter, including its two endpoints.

For either chart, use its independent extending canonical-square bundle and its extending coframe trivialization. Define the local graph in that trivialization by $z \mapsto (z, q(z))$ and apply the inverse bundle trivialization. Its base coordinate and scalar are smooth, and that inverse trivialization is smooth, so the resulting graph is smooth on U. At a retained point, the coefficient identity just proved and the original fiber-inclusion coordinate identity identify its value with the exact included original section, in the actual fiber of this independently extending bundle. Thus it is the required native graph extension, including every radius of the original signed cylinder.

Finally each total point has an original interior or cylindrical preimage, and the coframe-cover lemma supplies a chart containing that preimage. The corresponding coefficient extension just produced is one witness for the total-point cover. The cover-equivalence lemma therefore identifies these as the established original total local coefficient interface, still for the same supplied section. □

Corollary A.262 (Local full-map extensions reach the original whole-section grading consumer). Under the same hypotheses, let the same original section σ be intrinsically smooth in the whole original canonical-square bundle, nowhere zero on its original fibers, and locally extendible as a full map in the actual atlas. Then its original full tensor has both native interior and full signed-cylinder grading-section germ families throughout the parameter interval. Their matching equations retain that exact section and full tensor, including the compact seam and both parameter endpoints. An all-chart full-map extension premise also suffices. No endpoint graph is assumed separately.

Proof. The extraction lemma produces an actual native open graph extension at every original branch point. In particular, choose a covering interior canonical chart at each interior point with parameter 0 or 1, and choose a covering cylindrical chart at each such cylinder point. Their constructed graphs are the endpoint hypotheses of the original whole-section grading lemma. That lemma uses intrinsic whole-section smoothness to give smoothness of the exact original branch restrictions. It then combines these restrictions, the just-constructed endpoint graphs and the original section's nonvanishing to produce both full-tensor grading-section germ families. The extraction graphs agree in the actual fibers with this same original section, so the conclusion retains every slot of its original full tensor. The cylinder construction used the full signed band and thus also retains the seam. If the all-chart extension premise is supplied directly, the coefficient extraction and graph construction in the extraction proof apply directly, and the same whole-section grading lemma gives the same conclusions. □

Definition A.263 (Real open extensions on a retained part of the original base)**.** *Let X be the original completed total base, E its actual canonical-square total space, and $\pi : E \to X$ its projection. Retain the notation $i_c : B_c \to X$, $u_c : B_c \to E_c^{\mathrm{real}}$ and Φ_c for an original chart's inclusion, real base coordinates and full bundle coordinates. Fix a subset $A \subset X$ independently of a map $f : X \to E$. The local extension condition on A requires, at each $x \in A$, original charts c, d, a point $p \in B_c$ with $i_c(p) = x$ and $\pi f(x) \in D_d$, a real open neighborhood U of $u_c(p)$ within c's extending coframe domain, and a smooth full coordinate map $G : U \to E_d^{\mathrm{real}} \times \mathbb{C}$ such that*

$$r \in B_c, \quad i_c(r) \in A, \quad u_c(r) \in U \quad \Longrightarrow \quad \pi f(i_c(r)) \in D_d, \qquad G(u_c(r)) = \Phi_d(f(i_c(r))).$$

The all-chart condition requires these neighborhood and extension data for every c, d, p satisfying $i_c(p) \in A$ and $\pi f(i_c(p)) \in D_d$. The extensions are smooth on real open neighborhoods, including at parameter endpoints, but their matching is required only on A. No openness of A or matching outside A is imposed.

Lemma A.264 (Real open full-map extensions transfer on the same retained subset)**.** *Under the actual original atlas hypotheses, suppose $\pi \circ f$ is continuous and f has local full-map extensions on A. Then the same map has all-chart full-map extensions on that same subset A.*

Proof. Fix requested charts c, d and $p \in B_c$ with $i_c(p) \in A$ and $\pi f(i_c(p)) \in D_d$. Choose local witnesses e, j, r, U, G at this point, so $i_e(r) = i_c(p)$ and $G(u_e(r)) = \Phi_j(f(i_c(p)))$. The actual real base-change germ B_1 takes $u_c(p)$ to $u_e(r)$. The actual real full-bundle change germ B_2 takes the latter full map value from j coordinates to d coordinates. The composition-neighborhood lemma therefore gives a real open neighborhood on which $B_2 \circ G \circ B_1$ is smooth and its intermediate values lie in the domains of G and B_2.

The actual chart domains D_e, D_d are open. Continuity of $\pi \circ f$ makes $(\pi \circ f)^{-1}(D_d)$ open. Restrict the chosen real neighborhood by the raw-neighborhood lemma for these two open sets and intersect it with c's real coframe domain. The resulting neighborhood W contains $u_c(p)$ and ensures $i_c(q) \in D_e$ and $\pi f(i_c(q)) \in D_d$ for every retained chart point q with $u_c(q) \in W$.

For such q with $i_c(q) \in A$, let $v \in B_e$ be its actual preimage, so $i_e(v) = i_c(q)$. Then $i_e(v) \in A$ and actual base matching gives $B_1(u_c(q)) = u_e(v) \in U$. The local extension equation on A gives $G(u_e(v)) = \Phi_j(f(i_c(q)))$. Actual full-bundle matching consequently gives

$$(B_2 \circ G \circ B_1)(u_c(q)) = \Phi_d(f(i_c(q))).$$

Together with the target-domain membership already obtained, this is the required extension in the requested charts. All matching uses points in A, and c, d, p were arbitrary, proving the claim. □

Lemma A.265 (The retained section graph gives its exact branch coefficients)**.** *Fix a parameter subset $B \subset [0, 1]$, one original canonical-square section σ, and its actual whole-bundle graph $f_\sigma(x) = (x, \sigma^{\mathrm{tot}}(x))$. Put $A_B = \{x \in X : t(x) \in B\}$, where $t(x)$ is the actual original parameter. Suppose that graph has all-chart full-map extensions on A_B. At every original interior or signed-cylinder canonical chart point $p = (t, w)$ with $t \in B$, its exact original scalar coefficient has a smooth extension to a real open neighborhood of its coordinates within that chart's extending coframe domain. Matching holds at every chart point (s, v) in that neighborhood with $s \in B$. For the cylinder these are the prescribed coordinates of $(b, -\rho)$, throughout its entire signed band, including radius zero. No scalar extension is an input, and no matching is asserted outside B.*

Proof. Use the promoted original chart as both source and target for the all-chart condition at the insertion of p. The graph's projection is the identity, so its base belongs to that same target

chart. It gives one smooth full coordinate map G on a real open neighborhood U. Set $q = \mathrm{pr}_{\mathbb{C}} G$; projection makes q smooth on U.

At a retained interior point (s, v) with $s \in B$ and chart coordinates in U, its insertion belongs to A_B. Full-map matching identifies q with the actual promoted scalar coordinate of σ^{tot}. The exact whole-section restriction identity identifies this scalar with the original interior section coefficient. The inclusion U in the extending chart domain ensures that each such point is in the original chart's base domain. Hence q has every required interior matching equation.

For a signed-cylinder chart use its promoted chart in both positions instead. Its insertion also has parameter s, so belongs to A_B precisely when $s \in B$. Projecting its full-map matching equation and using the exact cylindrical restriction identity gives the original cylindrical coefficient. Its base coordinates are exactly the original chart applied to $(b, -\rho)$; no radial restriction is added. This proves all the cylindrical matching equations on the full signed band. Both constructions use real open neighborhoods and apply at either parameter endpoint whenever it belongs to B. □

Lemma A.266 (One prescribed representative gives covariant original local Maslov data)**.** *Retain the original compact bundle, its Liouville laws and the prescribed inclusion markings $q_s^a : X_s^a \to Z_s$ of the original completed atlas. Assume the ambient dimension is $2n$. Independently prescribe at each parameter s one compatible Maslov representative $(J_s^Z, \mathbf{T}_s^Z)$ on Z_s. Pullback along those same markings defines*

$$J_s^a = (q_s^a)^* J_s^Z, \qquad \mathbf{T}_s^a = (q_s^a)^* \mathbf{T}_s^Z \qquad (s \in U_a).$$

These are original completed-atlas Maslov data: the pulled field is compatible, the full canonical-square tensor is nowhere zero, and covariance holds under every actual completed overlap $G_{a,b}(s)$. Inverting any retained marking recovers exactly $J_s^Z, \mathbf{T}_s^Z$ on the whole prescribed ambient. This is a pointwise and slice statement; no joint parameter regularity follows from the slice markings alone.

Proof. Each q_s^a is a smooth primitive-preserving diffeomorphism. Exterior derivative commutes with its pullback, so it preserves the symplectic forms as well. Conjugating J_s^Z by its invertible differential preserves the complex law and compatibility: symplectic pairing of pulled vectors is the original pairing of their images, and a nonzero vector has a nonzero image. Tensor pullback preserves complex linearity and alternation in both blocks. Its invertible differential in every argument also preserves nonvanishing. Thus the pulled data have all required slice laws.

On an overlap the previously proved marking covariance gives $q_s^b \circ G_{a,b}(s) = q_s^a$. Pullback composition therefore gives

$$G_{a,b}(s)^* J_s^b = (q_s^b \circ G_{a,b}(s))^* J_s^Z = J_s^a,$$

and the same calculation in every tensor argument gives $G_{a,b}(s)^* \mathbf{T}_s^b = \mathbf{T}_s^a$. These are precisely the completed-atlas covariance equations. Finally $q_s^a \circ (q_s^a)^{-1} = \mathrm{id}$; pullback along $(q_s^a)^{-1}$ cancels both preceding definitions and recovers the prescribed field and full tensor on every point of Z_s. Every calculation is on an actual retained slice, including either interval endpoint. □

Lemma A.267 (The prescribed representative is recovered on each exact endpoint completion)**.** *Under the preceding hypotheses, assume $n \geq 1$. Fix an actual represented endpoint completion K_s with its original strict slice comparisons $A_s^a : X_s^a \to K_s$ and descended inclusion marking $q_s : K_s \to Z_s$. Pull back the prescribed representative along this exact q_s. The resulting representative on K_s recovers the original local field and full tensor through every actual A_s^a with $s \in U_a$. It is independent of the retained local family and comparison used to construct q_s. This assertion holds at every allowed parameter, in particular zero and one, and involves no endpoint homotopy or assertion of a jointly smooth Maslov family.*

Proof. The marking-descent lemma gives $q_s \circ A_s^a = q_s^a$ for every retained local family. Primitive preservation makes q_s symplectic, so pullback of the prescribed representative is a compatible nowhere-zero representative on K_s. Pullback composition gives

$$(A_s^a)^*(q_s^* J_s^Z) = (q_s^a)^* J_s^Z = J_s^a, \qquad (A_s^a)^*(q_s^* \mathbf{T}_s^Z) = (q_s^a)^* \mathbf{T}_s^Z = \mathbf{T}_s^a.$$

The tensor equality applies to every argument and every completed point. Two retained local choices give the same actual marking q_s by the marking-descent lemma. Their pulled fields and tensors consequently agree, so their Maslov representatives agree. None of these equations requires an interior parameter, proving the endpoint instances as well. □

Lemma A.268 (Each retained original full tensor has its exact canonical-square section)**.** *Fix original completed-atlas Maslov data S and an original local completion X_s^a over its retained base U_a. In its actual original native coordinates let J_s^a and $\mathbf{T}_s^a$ be the prescribed field and full canonical-square tensor. There is a pointwise original canonical-square section σ_s^a on these actual completions with*

$$m_{J_s^a}(\sigma_s^a) = \mathbf{T}_s^a \qquad (s \in U_a).$$

This equality recovers every tensor argument at every original completed point. The field in this construction is the exact original field on U_a and obeys the complex law there. The assertion requires no section on parameters outside U_a and proves no joint smoothness of this section.

Proof. At each retained s and original completed point, the native dimension identification conjugates the given field to a complex structure on $\mathbb{R}^{2n}$. Its complex law follows from the original field's complex law and cancellation of that linear identification. The same identification pulls the full original tensor into the squared canonical line for this complex structure.

Natural square evaluation is a complex linear isomorphism between the canonical tensor-product square and that full squared canonical tensor line, as proved in the canonical-square recovery lemma. Apply its inverse to the given full tensor at each point to define σ_s^a. Applying natural evaluation cancels that inverse and gives exactly the pulled original tensor. Evaluating this equality on the inverse images of arbitrary original tangent arguments cancels the dimension identification in every slot, proving the displayed original full-tensor equality. Every construction was at a retained parameter, so no off-base section regularity has been used or asserted. □

Lemma A.269 (A retained complex-field germ constructs the prescribed original chart)**.** *Retain the original collar and its smooth slice maps and local inverse hypotheses. Let $B \subset [0,1]$ be relatively open, and require the actual original field J_s to obey $J_s^2 = -\mathrm{id}$ only for $s \in B$. Fix an interior spatial chart with any prescribed anchor, or a boundary–radius chart with any prescribed anchor on the full signed cylinder. At a point (s,w) with $s \in B$ in that spatial chart, suppose there is a real open smooth germ K matching the chart-conjugated original field at every retained point of this same spatial chart. Then there is an original canonical coframe chart with exactly that anchor, containing (s,w) in its base domain, whose real open domain contains actual interval parameters only from B. The conditions requiring such raw complex-field germs at every retained point and every spatial anchor impose no coframes or section germs. The conclusion applies to the entire signed cylinder with its actual radius sign and to both parameter endpoints lying in B.*

Proof. Relative openness gives a real open set $H \subset \mathbb{R}$ with $H \cap [0,1] = B$. In the interior case intersect the given germ domain with H times the open spatial chart target. This is real open and contains the center. Every actual interval parameter in it belongs to B. Each point of its retained interval locus has a unique spatial chart inverse. Germ matching identifies K there with

conjugation of J_s by the actual original spatial differential. That differential is invertible by the original interior atlas theorem. Conjugation therefore preserves the complex law, so $K^2 = -\mathrm{id}$ on this retained locus. The native canonical-patch lemma gives a smooth invertible coframe on a smaller real open neighborhood, intertwining K on that locus. Substituting the original germ matching gives the original chart's field-intertwining equation. Its anchor was never changed, its domain contains the center, and its domain is contained in the preceding product, proving every interior assertion.

For the cylinder intersect instead with H times the real open spatial target consisting of points whose inverse chart satisfies $-r < \delta$ in the prescribed boundary–radius coordinates. This is exactly the signed-band condition; it contains the given cylinder point. The inverse spatial chart followed by $(b, r) \mapsto (-r, b)$ recovers a point of the original signed band. Its original cylinder insertion is a local diffeomorphism. Hence its spatial differential, including this sign reversal, is invertible, and the given matching again implies $K^2 = -\mathrm{id}$ at every retained point. Apply the same native canonical-patch lemma there and substitute germ matching to obtain the original cylindrical coframe chart. Its anchor and the entire signed-band convention are preserved, and its domain again has actual parameters only in B. Both domains are real open at the center, so these arguments also apply at either parameter endpoint when it belongs to B. □

Lemma A.270 (Regularity of the retained original section produces its prescribed coframe germs)**.** *Fix the original completed-atlas Maslov data S. On every actual local completion family a with retained base U_a, use the exact pointwise section σ^a of its given full tensor from the retained-section lemma. At every retained original interior and full signed-cylinder spatial chart point, suppose the actual chart-conjugated field has a real open smooth extension. Suppose also that the exact graph of σ^a has local full-map extensions on $A_a = \{x : t(x) \in U_a\}$ in the actual original canonical atlas. Then S has all the original smooth invertible coframe and scalar germs of Definition A.749. Their retained matching recovers the same original field and full tensor on each U_a, in every tensor argument and on the entire signed band. Coframe, coefficient and full-tensor extension germs are conclusions; no such germ is supplied separately. Real open field and graph regularity on these retained bases remains an explicit hypothesis.*

Proof. Fix any original local family and an interior chart center with any prescribed covering spatial anchor. Its retained base U_a is relatively open. The original complex law on that base and the supplied field germ therefore give, by the retained-chart lemma, a canonical coframe chart with exactly that anchor, containing the center. Denote its smooth invertible coframe by C. Its real domain has actual parameters only in U_a.

The exact section graph projects to the identity, so its projection is continuous. The relative full-map transfer lemma gives all-chart extensions on that same A_a. The relative extraction lemma applied in the constructed canonical chart gives a smooth scalar q on a real open neighborhood of the center within that chart domain. At every retained nearby point it is the exact coefficient of σ^a in C.

Let D be the actual original spatial differential from native tangent coordinates to this chart's coordinates, and let N be its fixed real normalization from $\mathbb{R}^{2n}$. The chart's intertwining equation is $C(DJD^{-1}) = iC$. Composing on the right with D gives $(CD)J = i(CD)$. The full coordinate reconstruction lemma and exact section-tensor recovery give

$$q\Gamma_{CN} = \mathbf{T}^a \circ (D^{-1}N)^{\otimes 2n}.$$

To recover the unnormalized original tensor, evaluate this identity on $N^{-1}Dv_j$ in every argument. The right side becomes $\mathbf{T}^a(v_1, \ldots, v_{2n})$ and the left side becomes $q\Gamma_{CD}(v_1, \ldots, v_{2n})$. Since the

arguments were arbitrary,

$$\mathbf{T}^a = q\Gamma_{CD}$$

in every slot. These are exactly the required original coframe and scalar matching equations. Smoothness and invertibility of C restrict to the extraction neighborhood, and q is smooth there.

At a cylindrical center use its prescribed boundary–radius anchor and supplied cylindrical field germ. The retained-chart lemma constructs its original cylindrical coframe chart. Relative graph transfer and coefficient extraction again give a smooth q for the same original section. Use now the original signed-cylinder differential for D and the cylindrical normalization for N. The end full-coordinate reconstruction lemma gives the displayed normalized tensor equality, and inserting $N^{-1}Dv_j$ proves the same unnormalized equality. This differential includes the actual radius sign, and neither chart construction nor coefficient extraction restricts the signed band. Thus the argument applies at the seam, at every allowed radius, and at either interval endpoint in the retained base. The local family, anchor and center were arbitrary, proving all original coframe germs. □

Corollary A.271 (The prescribed section realizes the same original marked and graded pair bundle). Supply the original compact ambient and ribbon bundles, primitive, embedding and positive-witness data and geometric laws of the retained pair-bundle realization theorem. In positive ribbon dimension take ambient dimension $2n$ with $n > 1$ and ribbon dimension $2n - 2$; in ambient dimension two allow any finite ribbon, including the empty ribbon. Independently supply original ambient inclusion markings q_s^a into complete ambients Z_s and one Maslov representative σ_s^Z on each Z_s.

Use the original local fields and full tensors obtained by pulling that same σ_s^Z back through these markings. Assume, on every retained original local base, the real open complex-field germs and local full-map regularity of the exact recovered section graph required by the coframe-recovery lemma. These data admit one retained original pair-bundle realization with its entire geometric deformation, coherent presentations, smooth Maslov family and actual deformation grade. It recovers those very original pulled fields and full tensors in every original local chart and at both parameter endpoints. On its same completed ambients and ribbons it retains both original marking capabilities and all the compact, witness, primitive, whole-ribbon and full-core equations of the retained realization theorem. Original Maslov covariance and coframe germs are derived before choosing this realization. The supplied marked geometry and retained-base field and section regularity remain hypotheses.

Proof. Pull back the independently prescribed representative through the original inclusion markings. The prescribed-pullback lemma proves that these are covariant original completed-atlas Maslov data S, with their exact original fields and full tensors. Construct the retained native section of those tensors. The given field germs and local full-map regularity are precisely the hypotheses of the retained-section coframe-recovery lemma, which constructs all the original coframe and scalar germs for that same S.

In positive ribbon dimension, combine the originally supplied geometric bundle data with this S and these constructed germs. The retained realization theorem applies and gives its entire positive-dimensional realization. Keep its chosen deformation, compatible charts, coherent presentations, all geometric equations, smooth Maslov family, grade and original chart recovery. The theorem's ambient and ribbon marking capabilities act on these same completed slices; in particular they apply to the original marking datum already supplied here.

For a finite ribbon in ambient dimension two, combine its originally supplied geometric and embedding data with the same pulled ambient Maslov data and their constructed coframe germs. The finite-ribbon branch of the retained realization theorem gives its complete realization, with all the corresponding equations, grade and both marking capabilities. Its full ribbon core is its whole embedded ribbon. That theorem permits an empty ribbon and chooses no ribbon point, so

this argument includes the empty case. In both cases the Maslov data used by the realization are exactly the S constructed first; no represented-family grading or replacement coframe germs were assumed. □

Lemma A.272 (Joint smoothness of the spatial derivative)**.** *Let F be a real normed vector space, $E = \mathbb{R}^{2n}$, and $\varphi : \mathbb{R} \times F \to E$. At each point $z = (t, x)$ where φ is differentiable, the spatial derivative $D(z) = d_x\varphi(t, x)$ is the restriction of the joint derivative to $\{0\} \times F$:*

$$D(z)v = d\varphi(z)(0, v).$$

If φ is smooth on a real open set U, then D is smooth on U.

Proof. For fixed t, the map $x \mapsto (t, x)$ has derivative $v \mapsto (0, v)$. The chain rule for its composition with φ gives the displayed identity. On a real open set, the derivative of a smooth map is smooth as a map into the space of continuous linear maps. Restricting that derivative by the fixed continuous linear inclusion $v \mapsto (0, v)$ is smooth, and the displayed identity identifies the resulting map with D throughout U. □

Lemma A.273 (Smooth pullback of local endomorphism and full tensor coefficients)**.** *Let F be a complete real normed vector space and $E = \mathbb{R}^{2n}$. Let $U \subset \mathbb{R} \times F$ and $W \subset \mathbb{R} \times E$ be real open. Suppose $\varphi : U \to E$ is smooth, and let $A : W \to \mathrm{End}_{\mathbb{R}}(E)$ and $T : W \to \mathrm{Mult}_{\mathbb{R}}(E^{2n}, \mathbb{C})$ be smooth coefficient fields. Put $q(t, x) = (t, \varphi(t, x))$ and $D(t, x) = d_x\varphi(t, x)$. If $p \in U$, $q(p) \in W$ and $D(p)$ is invertible, there is a real open neighborhood V of p, contained in U, such that $q(V) \subset W$, $D(z)$ is invertible for every $z \in V$, and both fields*

$$K(z) = D(z)^{-1}A(q(z))D(z), \qquad L(z)(v_1, \dots, v_{2n}) = T(q(z))(D(z)v_1, \dots, D(z)v_{2n})$$

are smooth on V. No complex-structure or canonicality equation is required for A or T.

Proof. The preceding lemma makes D smooth on U. Invertible continuous linear maps form an open set, so $C = \{z \in U : D(z) \text{ is invertible}\}$ is open and contains p. The map q is smooth on U. Hence $V = C \cap q^{-1}(W)$ is open, contains p, is contained in U, and has both asserted range properties. Inversion is smooth on the invertible locus, so $z \mapsto D(z)^{-1}$ is smooth on V. Composition gives smooth $A \circ q$ and $T \circ q$ there. Continuous bilinear composition of linear maps makes $D^{-1}(A \circ q)D$ smooth. The operation composing a continuous multilinear map with one continuous linear map in each of its finitely many arguments is continuous multilinear in those linear maps and linear in the tensor. It therefore preserves smoothness and gives smooth L, with exactly the displayed values in all $2n$ arguments. □

Corollary A.274 (The same smooth Maslov family has smooth coordinate pullbacks). Independently give a smooth Maslov family $(J_s, \mathbf{T}_s)$ on a fixed smooth $2n$-manifold N for its prescribed varying two-forms η_s. Choose $a \in N$ and its selected spatial chart c, and let $\varphi : U \subset \mathbb{R} \times F \to \mathbb{R}^{2n}$ be smooth on a real open set, where F is a complete real normed vector space. Suppose $p = (s, x) \in U$ has $s \in [0, 1]$, $\varphi(p)$ in the target of c, and invertible spatial derivative $D(p)$. There is a smaller real open neighborhood V of p on which φ stays in that chart target, D stays invertible, and the pullbacks

$$D^{-1}J_c(s, \varphi(s, x))D, \qquad \mathbf{T}_c(s, \varphi(s, x)) \circ D^{\otimes 2n}$$

are smooth. Here J_c and $\mathbf{T}_c$ are the actual spatial chart coefficients of this same Maslov family. This conclusion includes both parameter endpoints and requires no compatibility or canonicality off $[0, 1]$.

Proof. The given smooth Maslov family provides real open coefficient domains W_J, W_T containing every interval point of the chart target, with smooth actual coefficients J_c and $\mathbf{T}_c$ there. Intersect U with the preimage of that spatial chart target; this is open and still contains p. Apply the coefficient-pullback lemma on this intersection with $W = W_J \cap W_T$ and $A(s,y) = J_c(s,y)$, $T(s,y) = \mathbf{T}_c(s,y)$. Its center belongs to W because $s \in [0,1]$. The resulting V retains chart-target membership, invertibility and both smooth pullback fields. Its formulas are exactly the displayed actual coefficient pullbacks. All neighborhoods are real open at p, including when $s = 0$ or 1, and no field law outside the interval was used. □

Lemma A.275 (Compact target-map germs restrict to the original interior and boundary)**.** *Let M be a smooth $(k+1)$-manifold with boundary, N a smooth d-manifold, and $B \subset [0,1]$. Suppose $v_s : M \to N$ has joint open target-map germs on B in the actual compact atlas. For every selected native interior chart a, every $s \in B$ and x in its source, the map $v_s(x)$ has a smooth N-valued extension on a real open neighborhood of $(s, a(x))$, matching v_t at every nearby chart point with $t \in B$. The same assertion holds for every selected native boundary chart and every point of its source.*

Proof. The native interior chart is the restriction of its actual compact half-space chart, with the same coordinate values. Choose the supplied compact target-map germ in that chart. It is smooth on an open subtype; extend its underlying map arbitrarily outside that open set, using its value at the center. On the open set the map and all its local restrictions are unchanged, so it remains smooth there. The equality of compact and interior chart coordinates gives exactly the required interior matching.

For a boundary chart, its compact ambient coordinates are obtained by inserting zero as the normal coordinate into its boundary coordinates. This insertion is smooth. Pull the compact germ's domain back by $(t,y) \mapsto (t,(0,y))$ and compose the germ with that map. The preimage is real open and contains the boundary center. Smooth composition gives the required boundary extension. The actual boundary-coordinate identity and compact germ matching identify it with v_t at every indicated boundary point and retained parameter. □

Definition A.276 (Real local regularity of an interval family of flows)**.** *Let N be a fixed smooth manifold, $B \subset [0,1]$, and $\Phi^r_s : N \to N$ a given family of maps. Real local flow regularity on B means that for every $s \in B$, every finite real r and $x \in N$, there is a real open neighborhood U of (s,r,x) in $\mathbb{R} \times \mathbb{R} \times N$ and a smooth map $G : U \to N$ satisfying*

$$G(t,a,y) = \Phi^a_t(y) \quad \text{whenever } t \in B \text{ and } (t,a,y) \in U.$$

The extension is local in parameter, time and space. This condition asserts neither an off-interval flow law nor an off-interval complete flow. When applied to a specified complete source flow, its existence is an additional joint regularity requirement.

Lemma A.277 (Compact inclusion and source-flow germs give every boundary–radius germ)**.** *Under the compact target-map hypotheses of Lemma A.275, let Φ^r_s have real local flow regularity on the same B. For every selected boundary–radius product chart a, every $s \in B$ and (b,r) in its source, the actual map*

$$(t,z,\rho) \longmapsto \Phi^\rho_t(v_t(z))$$

has a smooth N-valued germ at $(s, a(b,r))$, agreeing with this map at every nearby retained chart point. The radius r may be any real number.

Proof. The preceding compact restriction lemma gives a smooth boundary inclusion germ β on a real open neighborhood U of $(s, a_\partial(b))$. Choose a flow germ G on a real open neighborhood W of $(s, r, v_s(b))$. Write $\rho(t, y, r') = (t, y)$ and $q(t, y, r') = (t, r', \beta(t, y))$. The map q is smooth on $\rho^{-1}(U)$, and at the center its value is $(s, r, v_s(b))$ by inclusion-germ matching. Restrict to the real open neighborhood $\rho^{-1}(U) \cap q^{-1}(W)$ and use $G \circ q$. It is smooth by composition. At any retained nearby point, $\beta(t, y) = v_t(a_\partial^{-1}(y))$, and substituting this identity into flow-germ matching gives $G(q(t, y, r')) = \Phi_t^{r'}(v_t(a_\partial^{-1}(y)))$. Product-chart coordinates retain the actual radius, proving the assertion for every real r, including zero. □

Lemma A.278 (The actual prescribed marking has real germs on its entire native atlas)**.** *Give original inclusion markings $q_s^a : X_s^a \to Z_s$ and retain one original local completion family a over U_a. Independently choose a fixed smooth manifold N and smooth slice diffeomorphisms $R_s : Z_s \to N$. Let $i_s^a : M \to X_s^a$ be its actual compact insertion and set $v_s(x) = R_s(q_s^a(i_s^a(x)))$. Suppose this original compact inclusion has joint open target-map germs on U_a, and the conjugated actual source flows*

$$\widetilde{\Phi}_s^r(y) = R_s\big(\Phi_{Z_s}^r(R_s^{-1}(y))\big)$$

have real local flow regularity there. Then $R_s \circ q_s^a$ has real open smooth target-map germs in every selected original interior chart and every selected original full signed-cylinder chart. Matching holds on the same retained base U_a and the whole original signed band, including its seam and all outward radii. No marking-map germ or pulled Maslov tensor regularity is supplied as an additional premise.

Proof. On the original interior the inclusion equation of the marking gives $R_s(q_s^a(i_s^a(x))) = v_s(x)$. Apply the compact chart restriction lemma to v. Its germ matching is consequently the literal marking matching at every retained interior point.

For a signed-band point with original coordinates (ρ, b), put $r = -\rho$. The original slice-completion equation identifies its cylindrical insertion with the actual end map at (b, r). The marking's all-real end equation therefore gives

$$R_s(q_s^a(\mathrm{cyl}_s(\rho, b))) = R_s\big(\Phi_{Z_s}^{-\rho}(q_s^a(i_s^a(b)))\big) = \widetilde{\Phi}_s^{-\rho}(v_s(b)).$$

The boundary–radius composition lemma constructs real open smooth germs of the last expression in exactly the original coordinates $(b, -\rho)$. Substitution of this literal marking equation proves the required full-cylinder matching. That composition lemma allows all real radii, so it covers the signed collar, the seam and every outward radius without adding a radial restriction. Both arguments use real open neighborhoods and the same retained base, so also apply at either parameter endpoint in U_a. □

Lemma A.279 (Ordinary field extensions give real germs of the actual finite-time flows)**.** *Let Q be a finite-dimensional Hausdorff second-countable smooth boundaryless manifold and $B \subset [0, 1]$. For every $s \in B$ give a smooth vector field ζ_s and curves $\Phi_s^r(x)$ defined for all real r, with $\Phi_s^0(x) = x$ and $\partial_r \Phi_s^r(x) = \zeta_s(\Phi_s^r(x))$. Give a real open $\Omega \subset \mathbb{R} \times Q$ and a jointly smooth vertical field Z there, with*

$$Z(s, x) = \zeta_s(x) \quad (s \in B,\ (s, x) \in \Omega).$$

Fix $s_0 \in B$, $x_0 \in Q$, and real r_0, R with $|r_0| < R$. Suppose $(s_0, \Phi_{s_0}^r(x_0)) \in \Omega$ for every $r \in [-R, R]$. There is a real open neighborhood U of (s_0, r_0, x_0) and a smooth $G : U \to Q$ such that

$$G(s, r, x) = \Phi_s^r(x) \quad (s \in B,\ (s, r, x) \in U).$$

Thus ordinary local field extensions near the compact center orbit produce a jointly smooth germ of the actual given flow. The field matching and the actual complete curves are required only on B.

Proof. The set $K = \{(s_0, \Phi^r_{s_0}(x_0)) : r \in [-R, R]\}$ is compact, since integral curves are continuous and $[-R, R]$ is compact. Choose a smooth compactly supported cutoff χ on $\mathbb{R} \times Q$, with support contained in Ω, equal to one on an open neighborhood V of K. Such a cutoff is obtained by finitely many smooth chart bumps around this compact subset of the open set. Multiply the parameter-constant lift $(0, Z(s, x))$ by χ and extend by zero outside Ω. It is smooth: outside the cutoff's closed support it vanishes on a neighborhood. Its compact support makes its smooth flow F^r exist for all real times. Indeed local smooth ODE existence has a uniform positive time on the compact support by a finite cover, and constant curves give that same time off the support. Successive restarts construct both time directions; uniqueness gives the flow law, and local smooth dependence and finite compositions give joint smoothness. The parameter derivative is zero, so the first coordinate of $F^r(s, x)$ is always s.

The center curve $(s_0, \Phi^r_{s_0}(x_0))$ solves this cutoff equation on $(-R, R)$ because $\chi = 1$ on K and Z matches ζ_{s_0} there. It has the same initial value as $F^r(s_0, x_0)$. Uniqueness on this common connected time interval identifies the two curves. Put $R' = (|r_0| + R)/2$, so $|r_0| < R' < R$. The inverse image of $V \cap \Omega$ under the continuous map $(r, p) \mapsto F^r(p)$ is open and contains $[-R', R'] \times \{(s_0, x_0)\}$. Compactness of $[-R', R']$ gives an open neighborhood A of (s_0, x_0) such that $F^r(A) \subset V \cap \Omega$ for every $r \in [-R', R']$.

For $(s, x) \in A$ with $s \in B$, the spatial curve $\mathrm{pr}_Q F^r(s, x)$ therefore solves ζ_s on $(-R', R')$: the cutoff is one, the parameter is s, and the original field matching holds there. It starts at x, so uniqueness identifies it with $\Phi^r_s(x)$ throughout this interval. Take $U = \{(s, r, x) : (s, x) \in A,\ |r| < R'\}$ and $G(s, r, x) = \mathrm{pr}_Q F^r(s, x)$. This U is real open and contains the required center; G is smooth by the joint smoothness of F and permutation of product coordinates. The preceding equality gives its matching for every retained input in U, including either parameter endpoint in B. The auxiliary cutoff field has no asserted source meaning outside B. □

Lemma A.280 (The prescribed source coordinates transport the actual field and its complete curves)**.** *Let (Z, λ_Z) be a complete Liouville manifold, N a fixed smooth manifold of the same dimension, and $R : Z \to N$ a smooth diffeomorphism. The actual transported field and curves are*

$$X(y) = dR_{R^{-1}(y)} Z_{\lambda_Z}(R^{-1}(y)), \qquad \widetilde{\Phi}^r(y) = R\big(\Phi^r_Z(R^{-1}(y))\big).$$

The field X is smooth, and each displayed all-real curve is an integral curve of X with initial value y.

Proof. The tangent map of the smooth diffeomorphism R is smooth. Composing it with the smooth source-field section and R^{-1} gives a smooth tangent section based at $R(R^{-1}(y)) = y$, namely X. Differentiate the displayed curve by the chain rule. Its derivative is $dR_{\Phi^r_Z(R^{-1}(y))} Z_{\lambda_Z}(\Phi^r_Z(R^{-1}(y)))$. Substituting $R^{-1}(R(\Phi^r_Z(R^{-1}(y))))$ identifies this expression with $X(\widetilde{\Phi}^r(y))$. At zero the source flow is the identity, so the initial value is y. Completeness of the supplied source flow gives every real time. □

Lemma A.281 (Local extensions of the actual source field produce its real flow germs)**.** *Give complete source Liouville manifolds Z_s, a fixed smooth N, and smooth slice diffeomorphisms $R_s : Z_s \to N$. Let $X_s, \widetilde{\Phi}_s$ be their actual transported fields and curves, as in Lemma A.280. Fix $B \subset [0, 1]$. Suppose that for every $s \in B$, finite real r and $x \in N$, there are $T > |r|$, a real open Ω, and a smooth vertical field W on Ω, matching X_t at retained parameters $t \in B$, such that*

$$(s, \widetilde{\Phi}^a_s(x)) \in \Omega \quad (-T \le a \le T).$$

Then $\widetilde{\Phi}$ has real local flow regularity on B.

Proof. The preceding transport lemma gives the smooth actual slice fields, the all-real integral-curve equations and the zero-time identity. For any retained center (s, r, x) choose the given T, W, Ω. These are exactly the hypotheses of the finite-orbit flow-germ lemma with radius T. Its smooth germ agrees with the same actual transported curves at every retained input. Applying it at all centers gives precisely the real local flow regularity condition. □

Lemma A.282 (Smooth actual primitives produce all source-field and finite-time flow germs)**.** *Under the same independently prescribed Z_s, N, R_s, suppose the actual one-forms $\alpha_s = (R_s^{-1})^* \lambda_{Z_s}$ form a smooth interval family on N, with real open coefficient extensions. The actual transported fields X_s have smooth real extensions on one open neighborhood Ω of $[0,1] \times N$. Consequently the actual transported complete flows $\widetilde{\Phi}_s$ have real local flow regularity on every $B \subset [0,1]$. No joint smoothness of R_s and no off-interval complete flow is assumed.*

Proof. Naturality gives $d\alpha_s = (R_s^{-1})^* d\lambda_{Z_s}$, which is nondegenerate. The source Liouville equation and the chain rule give $\iota_{X_s} d\alpha_s = \alpha_s$. In a fixed chart, the smooth extended coefficients of α_s and their spatial derivatives give a smooth contraction matrix for $d\alpha_s$. Its matrix is invertible at each interval point. Shrink locally to its invertible locus. Smooth inversion solves the contraction equation with smooth coefficients there, agreeing with the actual X_s by uniqueness. Thus X_s is a smooth interval field family. The defining real extension of that family has a smooth tangent section on the union of its native chart neighborhoods; this union is real open and contains every point of $[0,1] \times N$. Call that neighborhood Ω and the extended field W.

For any retained s, time r and point x, take $T = |r| + 1$. Every point $(s, \widetilde{\Phi}_s^a(x))$ of its finite orbit belongs to Ω because Ω contains $[0,1] \times N$. The extension matches X_t on all retained parameters. The preceding field-extension lemma therefore constructs real flow germs at every center. All original flow equations were required only on the interval, as asserted. □

Corollary A.283 (Smooth source primitives and compact inclusions give the literal marking germs)*.* Give the original inclusion markings and fixed source coordinates $R_s : Z_s \to N$ of Lemma A.278. Assume joint open target-map germs of the original compact inclusion on its retained local base, and ordinary smooth interval regularity of the actual pulled primitives $(R_s^{-1})^* \lambda_{Z_s}$. Then the actual $R_s \circ q_s^a$ has all the real open marking-map germs of that lemma on the same base, in every original interior chart and throughout the full signed cylinder. Joint flow germs, field germs and marking-map germs are conclusions here.

Proof. The smooth-primitive lemma constructs real local regularity of the actual conjugated source flows on the retained base. Apply the inclusion-and-flow marking lemma with these constructed flow germs and the supplied compact inclusion germs. Its literal marking matching and full signed-cylinder conclusion are unchanged. □

Lemma A.284 (A literal marking germ has the derivative of the actual marked insertion)**.** *Let X, Y be manifolds, N a smooth $2n$-manifold, and $e : Y \to N$ a smooth diffeomorphism. Let $A \subset X$ be open, $z \in A$, and let $\chi : X \to F$ and $j : X \to Y$ be differentiable at z, with $d\chi_z$ invertible. Suppose a smooth map $\beta : U \subset \mathbb{R} \times F \to N$ on a real open set matches*

$$\beta(s, \chi(x)) = e(j(x)) \quad (x \in A,\ (s, \chi(x)) \in U)$$

at a fixed retained s, and $(s, \chi(z)) \in U$. Choose a selected target chart c containing $e(j(z))$ in its source. Then

$$d_y\big(c(\beta(s,y))\big)_{\chi(z)} = dc_{e(j(z))}\, de_{j(z)}\, dj_z\, (d\chi_z)^{-1}.$$

Proof. Differentiability of χ makes it continuous at z. The open conditions $x \in A$ and $(s, \chi(x)) \in U$ consequently hold throughout a neighborhood of z. On that neighborhood, $c(\beta(s, \chi(x))) = c(e(j(x)))$ by the literal matching equation. The target chart is differentiable at the common center, so differentiating this equality and applying the chain rule gives

$$d_y(c \circ \beta_s)_{\chi(z)}\, d\chi_z = dc_{e(j(z))}\, de_{j(z)}\, dj_z.$$

Compose on the right by $(d\chi_z)^{-1}$ and cancel to obtain the claimed derivative in the actual source coordinates. □

Lemma A.285 (Actual marking and chart differentials cancel in every tensor argument)**.** *Under the preceding hypotheses also require dj_z invertible. Put $E = de_{j(z)}$, $B = dc_{e(j(z))}$, $D = d\chi_z(dj_z)^{-1}$ and $L = d_y(c \circ \beta_s)_{\chi(z)}$. Then L is invertible. For any real endomorphism J of the actual target tangent space and any complex-valued real multilinear tensor T on its $2n$ arguments, actual chart pullback satisfies*

$$L^{-1}(BJB^{-1})L = D(E^{-1}JE)D^{-1},$$
$$(T \circ (B^{-1})^{\otimes 2n}) \circ L^{\otimes 2n} = (T \circ E^{\otimes 2n}) \circ (D^{-1})^{\otimes 2n}.$$

These are precisely the coefficient pullbacks of the literal germ $\varphi(t, y) = c(\beta(t, y))$ at $(s, \chi(z))$.

Proof. The marking and selected chart are diffeomorphisms on their indicated local domains, so E and B are invertible. The hypotheses make D invertible. The derivative lemma gives $L = BED^{-1}$; hence $L^{-1} = DE^{-1}B^{-1}$. Substituting these formulas into the first expression and cancelling $B^{-1}B$ gives the first displayed equality. In each argument of the second expression, $B^{-1}Lv = ED^{-1}v$. Substitution in all $2n$ arguments gives the second equality, including every full tensor slot. The coordinate coefficient of J is BJB^{-1}, and that of T is $T \circ (B^{-1})^{\otimes 2n}$, because the inverse target chart differential is B^{-1}. These are the actual selected-chart coefficients at the center by the literal germ matching, proving the last assertion as well. □

Lemma A.286 (Literal marking germs produce the original field and full tensor germs)**.** *Give original completed-atlas Maslov data S, a fixed smooth $2n$-manifold N, its varying two-forms η_s, and an independently smooth Maslov family $f = (J_s^f, \mathbf{T}_s^f)$ for those forms. For every original retained completion X_s^a, give an actual smooth slice diffeomorphism $e_s^a : X_s^a \to N$. Require exactly*

$$J_s^a = (e_s^a)^* J_s^f, \qquad \mathbf{T}_s^a = (e_s^a)^* \mathbf{T}_s^f \quad (s \in U_a),$$

with the actual native tangent-dimension identification in these equations. Suppose the actual maps e_s^a have real open target-map germs in every original interior and full signed-cylinder spatial chart, matching on the retained bases U_a. Then the same S has real open smooth germs of its actual chart-conjugated field and full chart-pulled tensor in all those charts, with exact matching in every tensor argument. No field, full-tensor, coframe or scalar-coefficient germ is an input.

Proof. Fix an original interior chart center. Its original insertion into the completed slice is a smooth local diffeomorphism by the native interior atlas, and its selected spatial chart has invertible differential. Choose the supplied literal marking germ β. Restrict its real open domain to the preimage of the selected target chart based at its value at the center. The coordinate map $\varphi = c \circ \beta$ is smooth there. The literal derivative and cancellation lemma makes its spatial derivative invertible at the center. Apply the Maslov coordinate-pullback corollary to this same f and φ. It gives smooth real coefficient pullbacks on a smaller real open neighborhood. At each retained point of that neighborhood apply the literal cancellation lemma again. The supplied field and full-tensor

identities identify the results with the actual original chart pullbacks of S. Native dimension identification cancels in every argument, so the tensor equality is the full original one.

At a signed-cylinder center use its original boundary–radius chart and insertion instead. The original cylinder atlas gives the same local-diffeomorphism and invertibility facts, with its actual radius sign. Choose its supplied germ, restrict to the target chart, and repeat the identical derivative and coordinate-pullback calculation. Every matching point belongs to the same retained base and signed band; no radial restriction is added. Thus the resulting germs retain the seam and the whole signed cylinder. The argument applies to all original chart centers and either retained parameter endpoint, giving all required original field and full-tensor germs. □

Lemma A.287 (One smooth source family determines the exact original prescribed representatives)**.** *Give original inclusion markings $q_s^a : X_s^a \to Z_s$ into independently prescribed complete source manifolds, fixed smooth coordinates $R_s : Z_s \to N$, and a smooth Maslov family $f = (J_s^f, \mathbf{T}_s^f)$ on N for varying η_s. Require the actual symplectic equation*

$$R_s^* \eta_s = d\lambda_{Z_s} \quad (s \in [0,1]).$$

Pullback along R_s gives one prescribed Maslov representative on each Z_s. Pullback of those representatives along the original markings gives covariant original data S, whose actual fields and full tensors satisfy

$$J_s^a = (R_s \circ q_s^a)^* J_s^f, \qquad \mathbf{T}_s^a = (R_s \circ q_s^a)^* \mathbf{T}_s^f \quad (s \in U_a).$$

The equations retain the actual native tangent-dimension identifications and all full tensor arguments.

Proof. The displayed symplectic equation makes each R_s a symplectic smooth diffeomorphism for the indicated forms. Its invertible differential pulls the compatible complex structure and nowhere-zero canonical-square representative of f_s back to a compatible nowhere-zero representative on Z_s. The original prescribed-pullback lemma then constructs covariant original data S from these representatives and the same markings. Pullback composition gives $(q_s^a)^* R_s^* = (R_s \circ q_s^a)^*$ for the complex field and in every argument of the full tensor. The native dimension identification commutes with composition of these actual diffeomorphisms, so the same equalities hold in the prescribed original native coordinates. This proves both equations without assuming them separately. □

Corollary A.288 (Ordinary smooth source data produce all original Maslov coefficient germs). Under the preceding original source and symplectic hypotheses, assume the actual primitives $(R_s^{-1})^* \lambda_{Z_s}$ form an ordinary smooth interval family on N. In every original compact chart assume joint open target-map germs of the actual compact inclusion $R_s \circ i_s$ in those original coordinates, on its retained local base. Then the original data S pulled from this same smooth f have all real open smooth field and full-tensor germs. Consequently they have all smooth invertible original coframe and scalar germs. Their matching retains the actual original fields and full tensors, all original chart anchors, both parameter endpoints and the whole signed cylinder. Flow, marking-map, coefficient and coframe germs are derived from these ordinary source hypotheses.

Proof. The preceding pullback lemma constructs S and proves its exact field and full-tensor pullback equations through $R_s \circ q_s^a$. For every original local completion, the smooth-primitive marking corollary constructs all literal real open map germs of this same composite from its supplied compact inclusion regularity and ordinary primitive smoothness. Use these derived maps and the just-proved pullback equations in the original tensor-germ lemma. It constructs all original field and full-tensor germs of S on their retained bases. The original smooth-coefficient lemma then reconstructs smooth

invertible coframes and their actual scalar coefficients from these germs. Every step matches the same original fields and tensors on the same retained locus, so all full tensor slots, chart anchors, endpoints and signed-band conditions remain exactly those stated. □

Corollary A.289 (The ordinary smooth source realizes the same full marked and graded pair bundle). Supply the original compact pair-bundle geometry and laws of the retained graded realization theorem. In positive ribbon dimension use ambient dimension $2n$ with $n > 1$ and ribbon dimension $2n - 2$; in ambient dimension two allow any finite ribbon, including the empty one. Independently supply complete source manifolds Z_s, original inclusion markings, fixed smooth coordinates $R_s : Z_s \to N$, and one smooth Maslov family f for varying η_s, with $R_s^* \eta_s = d\lambda_{Z_s}$. Assume ordinary smooth interval regularity of the actual pulled source primitives and joint open target-map germs of every original compact inclusion as in the preceding corollary.

These data produce one full original pair-bundle realization. It retains its complete original geometric deformation, actual grade, coherent native presentations, original local primitive, ribbon and positive-witness recovery, all compact and full-core equations, and both ambient and ribbon marking-recovery capabilities on those same completed spaces. Its smooth Maslov family recovers the very original fields and full tensors pulled from f through the supplied source markings, in every original local chart and at both endpoints. In the finite-ribbon case the full embedded core is the whole ribbon. General correspondence with an arbitrary completed source family is not asserted: the fixed smooth source presentation and its ordinary primitive and compact-inclusion regularity remain supplied.

Proof. The smooth-source germ corollary first constructs the original Maslov data S and all their coframe and scalar germs from the independently supplied source geometry and f. For a positive-dimensional ribbon, combine these same S and derived germs with the original geometric pair-bundle data. They are exactly the inputs of the retained graded realization theorem. Apply its positive-dimensional construction once and keep the entire resulting realization: its deformation, grade, coherent presentations, original recovery equations and both marking capabilities. The prescribed Maslov fields and full tensors in those equations are this very S by construction.

For a finite ribbon in dimension two, use the same construction of ambient S and germs in rank one and combine them with the original finite-ribbon data. Its supplied ribbon one-form is zero on the zero-dimensional tangent spaces. The finite-ribbon realization therefore applies and retains the same complete package, including its full ribbon-core equality and both marking capabilities. It requires no ribbon point, so includes the empty ribbon. In both cases the coframe germs were produced before choosing the realization and are consumed by its original construction. No represented grading, replacement section or desired tensor germ was assumed. The conditional source inputs stay precisely those listed in the statement. □

A.5 Auxiliary foundations: source data on retained local bases

The local source presentations below are defined only for the actual retained parameters. Their ordinary smooth extensions carry no symplectic, compatible-complex or canonical-section conditions outside that locus. The prescribed section on each original source is unchanged when another local presentation is chosen.

Definition A.290 (Ordinary source primitive extensions on a retained base)**.** *Give complete Liouville manifolds (Z_s, λ_s), a smooth manifold N, a retained base $B \subset [0,1]$ and smooth diffeomorphisms $R_s : Z_s \to N$ only for $s \in B$. Ordinary primitive regularity on B means that there*

are real-parameter one-forms α_r on N and a real open neighborhood Ω of $B \times N$ on which their coefficients are jointly smooth, such that

$$\alpha_s = (R_s^{-1})^* \lambda_s \quad (s \in B).$$

No diffeomorphism R_s or geometric law is required outside B.

Lemma A.291 (A smooth primitive extension has an open smooth contraction-field neighborhood)**.** *Let N be a smooth finite-dimensional manifold, $A \subset \mathbb{R}$ and α_r a one-form family with jointly smooth coefficients on a real open neighborhood Ω of $A \times N$. Assume $d\alpha_r$ is nondegenerate at every point for $r \in A$. There is a real open neighborhood V of $A \times N$ on which the form-determined field X_r, defined by $\iota_{X_r} d\alpha_r = \alpha_r$ wherever $d\alpha_r$ is nondegenerate, is a jointly smooth vertical tangent section.*

Proof. Fix $(r, x) \in A \times N$ and choose a smooth chart near x. The coefficients of $d\alpha$ are spatial exterior derivatives of the smooth coefficients of α, hence are jointly smooth on this open chart neighborhood. Contraction with $d\alpha$ is therefore a smooth matrix family. At (r, x) its matrix is invertible by nondegeneracy. Its invertible locus is open; there smooth matrix inversion applied to the one-form coefficients gives the smooth solution of the contraction equation. Uniqueness of that solution identifies it with the form-determined field, and the inverse chart differential gives its smooth tangent section in native coordinates. Take the union of all open neighborhoods where this very tangent section is smooth. It contains $A \times N$, is open, and the section is smooth there because smoothness is local. □

Lemma A.292 (Retained source primitives give both signs of actual finite-time flow germs)**.** *Under the retained source hypotheses and ordinary primitive regularity above, the actual conjugated source flow*

$$\widetilde{\Phi}_s^r(y) = R_s\big(\Phi_{Z_s}^r(R_s^{-1}(y))\big) \quad (s \in B)$$

has a jointly smooth real open germ at every (s, r, y), for every finite real time r. Each germ agrees with this same actual flow at every retained input in its neighborhood. Both parameter endpoints and both signs of time are included; no source flow outside B is assumed.

Proof. On B, exterior differentiation and pullback give $d\alpha_s = (R_s^{-1})^* d\lambda_s$, which is nondegenerate. The preceding lemma therefore gives a jointly smooth contraction field on an open neighborhood V of $B \times N$. The transported source Liouville field satisfies its contraction equation by pullback naturality, so uniqueness identifies it with this field on B. The source-coordinate flow lemma identifies the displayed all-real curves as its actual integral curves there. Fix a retained center (s, r, y) and choose $T = |r| + 1$. Every point of its compact orbit for time in $[-T, T]$ lies in V, since V contains $B \times N$. Apply the finite-time real-germ lemma to this open smooth field neighborhood and these actual curves. It gives the required germ and matching on B. This argument has no restriction on the sign of r or on whether s is an endpoint. □

Corollary A.293 (Retained primitive and compact-inclusion smoothness give the actual marking germs). Give original inclusion markings $q_s : X_s \to Z_s$ and one original local completion family with retained base B. Let N be smooth and give smooth diffeomorphisms $R_s : Z_s \to N$ only for $s \in B$. Assume ordinary source primitive regularity on B and, in every original compact chart, real open smooth target-map germs of the actual R_s composed with the compact inclusion, matching only on B. Then the actual $R_s \circ q_s$ composed with each original interior or signed-cylinder insertion has a real open smooth map germ at every retained center. Each germ matches all actual retained-base and signed-cylinder inputs lying in its neighborhood, including the seam and both parameter endpoints.

Proof. Restrict a compact-inclusion germ to the native interior chart. The original marking extends that same inclusion, so the restricted germ equals its actual marked interior insertion on B. For a cylinder center, restrict the compact-inclusion germ to the native boundary chart. Compose it with the actual finite-time source flow germs supplied by the preceding lemma, taking the time to be the negative of the original cylinder-radius coordinate. The boundary flow-germ lemma gives a real open smooth germ of this composite. The original marking's all-real end identity identifies it with $R_s \circ q_s$ on the cylinder: $R_s^{-1} R_s$ cancels on the compact boundary inclusion. This identity uses the actual signed radius, so it retains the whole original band and the seam. All matching conditions use only $s \in B$, including either retained endpoint. □

Definition A.294 (Ordinary local field and canonical-square coefficient extensions)**.** *Let N be a smooth $2n$-manifold, $B \subset [0,1]$, and give fields J_s and full complex-valued $2n$-argument tensors T_s on N. Ordinary local coefficient extensions mean the following: for every $s \in B$, every selected chart c of N and every y in its target, there are a real open neighborhood W of (s,y) and smooth matrix and multilinear-tensor functions A, Q on W such that*

$$A(v,z) = c_* J_v(z), \qquad Q(v,z) = c_* T_v(z) \quad (v \in B,\ z \in \mathrm{target}(c),\ (v,z) \in W).$$

Here $c_* J = dc\, J\, (dc)^{-1}$ and $c_* T$ evaluates T on the inverse chart differential in every argument. Only these equalities on B are required of the extensions.

Lemma A.295 (Local source coefficients pull back through a smooth spatial map)**.** *Assume these coefficient extensions on N and B. Let V_0 be a real open subset of $\mathbb{R} \times E$, where E is a complete normed real vector space, and let $\varphi : V_0 \to \mathbb{R}^{2n}$ be smooth. Fix $s \in B$, $y \in E$ and a selected target chart c such that $(s,y) \in V_0$, $\varphi(s,y) \in \mathrm{target}(c)$ and $L(s,y) = d_y \varphi_s$ is invertible. There is a real open neighborhood $V \subset V_0$ of (s,y) on which φ remains in this chart target and L remains invertible, and smooth fields K and full tensors Q' on V satisfying*

$$K(v,z) = L(v,z)^{-1} (c_* J_v)(\varphi(v,z)) L(v,z), \qquad Q'(v,z) = (c_* T_v)(\varphi(v,z)) \circ L(v,z)^{\otimes 2n}$$

for every $v \in B$ and $(v,z) \in V$.

Proof. Choose the coefficient extensions A, Q at $(s, \varphi(s,y))$. Intersect V_0 with the inverse image of the open chart target. Apply the smooth coefficient-pullback lemma there to A, Q, φ. It gives an open neighborhood V whose image lies in the chosen coefficient neighborhood and on which L is invertible, together with the smooth functions $L^{-1}(A \circ \varphi) L$ and $(Q \circ \varphi) \circ L^{\otimes 2n}$. At retained parameters the defining matching of A, Q identifies these functions with the two displayed expressions. The intersection and neighborhood restrictions retain the center and give $V \subset V_0$. □

Lemma A.296 (Retained local source coefficients give the full original tensor germs)**.** *Let S be covariant original completed-atlas Maslov data. For each original local completion X_s^a with retained base B_a, give a smooth $2n$-manifold N_a, fields J_s^a, T_s^a with the ordinary coefficient extensions just defined, and smooth slice diffeomorphisms $e_s^a : X_s^a \to N_a$ only for $s \in B_a$. Require that the original fields and full tensors of S are exactly $(e_s^a)^* J_s^a$ and $(e_s^a)^* T_s^a$ under the native tangent-dimension identification. In every original interior and signed-cylinder chart, assume real open smooth map germs of the actual e_s^a composed with the original insertion, with matching on B_a. Then S has real open smooth germs for all its original complex fields and full tensors, retaining every original chart, both parameter endpoints and the whole original signed cylinder.*

Proof. At an interior chart center choose its actual smooth map germ β. Select a target chart containing its center value and restrict the germ to the inverse image of that chart. The original

interior insertion and the slice map e_s^a are local diffeomorphisms. The literal marking-germ differential lemma therefore makes the spatial derivative of $\varphi = c \circ \beta$ invertible at the center. Apply the local source coefficient-pullback lemma. It gives smooth pulled matrix and tensor functions on a real open neighborhood. At each retained point the literal marking pullback identities cancel the selected target-chart differential in every tensor slot. The assumed exact pullback equations for S then identify these functions with its actual original field and full tensor in that interior chart.

At a signed-cylinder center use its original boundary–radius chart and insertion. The original cylinder atlas gives the same local invertibility, including at radius zero and at negative radii in the original band. Repeat the preceding construction with its actual map germ. The same cancellation identifies every tensor argument and the complex field with the original signed-cylinder coefficients. The germs match only at retained parameters, so the construction also applies at either parameter endpoint without any off-base equations. □

Definition A.297 (Ordinary local presentations of the original completed sources)**.** *Give original inclusion markings into complete source manifolds Z_s. For each original local completion with retained base B_a, an ordinary local source presentation consists of a smooth manifold N_a and smooth diffeomorphisms $R_s^a : Z_s \to N_a$ for $s \in B_a$, together with ordinary primitive regularity on that base and real open smooth target-map germs of R_s^a composed with the original compact inclusion in every compact chart, matching only at retained parameters. The model N_a may vary between local bases. This definition does not assert existence of these presentations from intrinsic family smoothness.*

Lemma A.298 (The local source coordinates cancel from the prescribed original Maslov pullback)**.** *Give one prescribed Maslov representative $\sigma_s = (J_s, T_s)$ on each complete source Z_s and original inclusion markings $q_s^a : X_s^a \to Z_s$. Pull these same representatives back to covariant original completed-atlas data S. For any retained local smooth coordinates $R_s^a : Z_s \to N_a$, put $\widetilde{J}_s^a = (R_s^{a-1})^* J_s$ and $\widetilde{T}_s^a = (R_s^{a-1})^* T_s$ on that base. Then*

$$J_S^a = (R_s^a \circ q_s^a)^* \widetilde{J}_s^a, \qquad T_S^a = (R_s^a \circ q_s^a)^* \widetilde{T}_s^a \quad (s \in B_a).$$

Both equalities use the actual native tangent-dimension identification and the full tensor, with every argument retained.

Proof. The original prescribed-pullback construction defines $J_S^a = (q_s^a)^* J_s$ and $T_S^a = (q_s^a)^* T_s$. Composition of field pullbacks gives $(R_s^a \circ q_s^a)^* (R_s^{a-1})^* = (q_s^a)^*$, because $R_s^{a-1} \circ R_s^a$ is the identity. The same cancellation holds independently in every full tensor argument. The native dimension identification composes with the actual differentials of these maps, so neither equality changes when expressed on the original native tangent spaces. □

Corollary A.299 (Ordinary local source data give the same original tensor and coframe germs)*.* Give the original markings, one prescribed σ_s on each source, and an ordinary local source presentation for every original local base. Require ordinary local smooth coefficient extensions of the actual transported $\widetilde{J}_s^a, \widetilde{T}_s^a$ on N_a for $s \in B_a$. Then the covariant original data S pulled from these same σ_s have real open smooth field and full-tensor germs, and hence smooth invertible original coframe and scalar germs. All matching retains the original local charts, both parameter endpoints and the entire signed cylinder. No smooth Maslov family on a full auxiliary cutoff interval is required.

Proof. For each original local base the retained primitive and inclusion smoothness produces the actual marking-map germs on its interior and signed cylinder. The preceding cancellation lemma proves the field and full-tensor pullback identities of S through these very maps from the very

transported σ_s. Apply the local source tensor-germ lemma using the supplied ordinary coefficient extensions. It gives every original field and full-tensor germ of S, with its full retained matching. The original coefficient-to-coframe construction applies to these derived germs and gives smooth invertible coframes and the actual scalar coefficients of the same tensors. None of these steps changes the prescribed representatives or the original signed cylinder. □

Corollary A.300 (Local ordinary source data realize the full original marked and graded pair bundle). Supply the original compact pair-bundle geometry and laws of the retained graded realization theorem. In positive ribbon dimension use ambient dimension $2n$, $n > 1$, and ribbon dimension $2n - 2$; in dimension two allow a finite ribbon, including the empty one. Give complete source manifolds Z_s, original inclusion markings, one prescribed Maslov representative σ_s on each source, and an ordinary local source presentation for every original local completion. Require ordinary local coefficient extensions of the actual transported complex fields and full tensors of these same σ_s on each retained base.

These data produce one full original pair-bundle realization, with its actual deformation, grade, coherent native presentations, original primitive, ribbon, positive-witness, compact and full-core recovery equations, and both ambient and ribbon marking-recovery capabilities on those same chosen completions. Its smooth Maslov family recovers precisely the original fields and full tensors pulled from σ_s in every local chart and at both endpoints. For a finite ribbon its full core is the entire ribbon. Existence of the local source presentations and their ordinary regularity from arbitrary intrinsic completed-family data is not asserted by this conditional result.

Proof. The preceding germ corollary constructs the original covariant Maslov data from σ_s and produces all their original coframe and scalar germs. In positive ribbon dimension combine these same data and derived germs with the original compact pair-bundle geometry. Apply its retained graded realization once and retain the complete result, including its actual deformation, grade, native recovery identities and universal independent marking capabilities. Its Maslov inputs are exactly the pullbacks of the prescribed σ_s, so its field and full-tensor recovery retains them.

In dimension two the identical ambient construction provides the original Maslov data and coframe germs in rank one. Combine them with the supplied finite-ribbon geometry and apply the finite-ribbon retained realization. Its tangent spaces are zero-dimensional, so its ribbon primitive is zero and its full core is the whole ribbon. The construction uses no ribbon point and therefore includes the empty ribbon. It retains the same deformation, grade and both original marking-recovery capabilities on its chosen completions. In both cases the coframe germs are derived before and used by the realization; no prescribed data are replaced during that construction. □

A.6 Auxiliary foundations: the canonical square on the retained completion

The following constructions restrict the actual collar-generated completed total space to its retained parameter base. The complex laws and smooth extension matching are required only on that base; no section is used to construct the canonical-square bundle.

Definition A.301 (The actual canonical square over the retained completed base)**.** *Let $\mathcal{X}$ be the actual collar-generated completed total space, $p : \mathcal{X} \to [0,1]$ its parameter map and $B \subset [0,1]$ relatively open. Fix the prescribed real normalization of each vertical slice tangent space with $\mathbb{R}^{2n}$. The retained completed base is the open subset $\mathcal{X}_B = p^{-1}(B)$. For a given complex field J on its slices, using these fixed vertical-tangent identifications, the retained canonical-square fiber at $x \in \mathcal{X}_B$ is the literal complex tensor product*

$$K_{J,x} \otimes_{\mathbb{C}} K_{J,x}, \qquad K_{J,x} = \Lambda^n_{\mathbb{C}}(T^{1,0}_x \mathcal{X}_{p(x)})^*.$$

It is the original canonical-square fiber at the same completed point, with its original fiber topology and operations.

Lemma A.302 (Retained complex-field germs give covering original canonical charts)**.** *Assume the original collar has its smooth slice maps, native inverse germs and joint open total embedding. Give the original native complex field J with $J_s^2 = -\mathrm{id}$ only for $s \in B$. In every original interior and boundary–radius chart require ordinary real open smooth coefficient germs of this actual field, matching only for $s \in B$ and including the full original signed cylinder. The restricted original canonical-square charts cover the whole $\mathcal{X}_B$, including the seam and full end.*

Proof. The native interior and cylinder patches cover the original completed total space. Fix a retained point and choose one of these patches containing it. Its parameter is in B, since the point belongs to $\mathcal{X}_B$. In a spatial chart containing its patch representative, the assumed raw field germ and complex law satisfy the hypotheses of the retained canonical-chart producer. Apply that producer in the interior or boundary–radius case. It gives an original canonical coframe chart containing the representative, hence its original tensor-coordinate chart contains the given completed point. Restrict that chart to $\mathcal{X}_B$. The fiber coordinate and center point do not change, so this is a covering restricted chart. The interior–cylinder cover includes the seam and entire original end, which therefore are also covered. □

Lemma A.303 (The retained actual canonical square is a smooth complex line bundle)**.** *Under the retained canonical-chart hypotheses above, assume also the ordinary real mixed-germ regularity of the original collar transitions. The literal fibers $K_{J,x} \otimes_{\mathbb{C}} K_{J,x}$ over $\mathcal{X}_B$ form a smooth complex line bundle. Its local tensor coordinates are the original canonical-square coordinates restricted to this retained open base; its coordinate changes are the same original ones restricted there. Its fiber topology is the original canonical-square topology, and each fiber has complex dimension one. Neither a section nor a complex-field law outside B is an input to this construction.*

Proof. Restrict every original tensor-coordinate map and its inverse to $\mathcal{X}_B$. The base coordinate is restricted by inclusion and the fiber coordinate is unchanged. The original inverse identities therefore prove the restricted inverse identities. The maps remain complex linear on each fiber. The retained chart-cover lemma gives one such coordinate chart at every point of $\mathcal{X}_B$.

On an overlap, the inverse fiber coordinate of a restricted chart is exactly the inverse coordinate of the original chart at the same point. Thus the restricted transition is the original transition, a multiplication by its original complex scalar. The original overlap lemma proves this scalar is smooth, including the mixed interior–end cases, from the stipulated mixed germs. Composition with the smooth inclusion of the retained open base preserves smoothness. The original fiber-coordinate maps induce the given fiber topology; restriction of the base does not change those maps on a fixed fiber, so the same assertion holds for every restricted chart. These covering linear coordinates and their smooth compatible changes define the vector-bundle topology and smooth vector-bundle atlas with the prescribed fiber topologies. Their complex linearity gives the complex vector-bundle structure on this same atlas. Finally a covering fiber coordinate is a complex linear isomorphism with $\mathbb{C}$, so every fiber has complex dimension one. All choices and overlap calculations used only retained charts and their original coordinates. □

Definition A.304 (A section of the actual retained canonical-square bundle)**.** *A section σ of the independently constructed retained canonical-square bundle assigns to every $x \in \mathcal{X}_B$ an element $\sigma(x) \in K_{J,x} \otimes_{\mathbb{C}} K_{J,x}$. No smoothness or nonvanishing is included in this definition.*

Lemma A.305 (Actual slice restriction recovers the same retained section and graph)**.** *Restrict any retained section σ to each actual slice by the original slice inclusion and its exact canonical-square fiber identification. Its reconstructed total section agrees with σ at every point of $\mathcal{X}_B$. The reconstructed total graph is exactly the graph of this same section after forgetting the retained open-base inclusion. These identities retain the full original cylinder and the seam. When functions on the whole auxiliary parameter interval are used, their unused values are zero.*

Proof. The original slice equivalence sends a slice point to its actual total point and has inverse the corresponding fiber equivalence. The complex field in total coordinates at this included point is exactly the same normalized slice field. Thus the slice restriction uses the canonical-square fiber identification for that equality. For a total point x, let $s = p(x)$ and take its unique slice representative under the actual fiber equivalence. Including this representative gives x again. The two canonical-square fiber identifications are therefore inverse, so reconstruction returns $\sigma(x)$ when $s \in B$. Outside B both total functions are zero. Pairing these identical values with the identical base points gives the graph identity. The fiber equivalence and original slice inclusion are defined on the whole completion, so the identities also hold on the full cylinder and at the seam. □

Lemma A.306 (Actual retained bundle coordinates agree with the original tensor coordinates)**.** *For a point of the retained canonical-square total space in a covering original tensor chart, its actual restricted bundle coordinates are exactly the original base and tensor coordinates at that same point after forgetting the retained open-base inclusion. The full fiber element is unchanged.*

Proof. The restricted tensor chart sends the retained base point to itself and uses the very original tensor-coordinate map on its fiber. Consequently its fiber coordinate is the original one. The base coordinate is chosen using the same original chart-domain representative before and after forgetting the retained subtype. Thus both coordinates agree, with the same full fiber element. □

Definition A.307 (Real open extension regularity in the retained bundle atlas)**.** *Let f be any map from $\mathcal{X}_B$ into the total space of its retained canonical-square bundle. Its ordinary real open extension regularity means that at every retained source point there are a covering original source chart and a covering original target bundle chart such that the full coordinate map of f has a smooth extension to a real open neighborhood of the source coordinates. The extension agrees with f at all retained inputs in that neighborhood, including both the target base coordinates and its complex fiber coordinate. This condition applies to arbitrary maps; for a section it is applied to its actual graph. Matching outside $\mathcal{X}_B$ is not required.*

Lemma A.308 (Actual retained bundle map extensions give original full-map extensions)**.** *For any map into the actual retained canonical-square bundle, ordinary real open extension regularity in its actual restricted coordinates gives the original retained-subset full-map extension property after forgetting the retained open-base inclusion. The same real open neighborhoods and extension functions apply; only the coordinate identification changes.*

Proof. At a retained center choose the source chart, target chart, real open neighborhood and smooth extension supplied by the actual restricted bundle map condition. Forgetting the retained subtype changes neither source nor target base point. The coordinate-recovery lemma identifies the actual restricted target coordinates with the original target coordinates for every retained matching input. Substitute this identity into the supplied matching equation. It gives the required original full-map extension using those same charts, neighborhood and extension function. Every input still lies on the same retained subset. □

Lemma A.309 (Ordinary retained section graphs transfer to every original bundle chart)**.** *Assume the original collar-transition mixed germs, and let the actual graph of a retained section σ have the preceding real open extension regularity. The identical native slice section then has such full-map extensions in every pair of covering original source and target bundle charts, with matching confined to the retained base. Both parameter endpoints and the full original signed cylinder are retained. No scalar coefficient or coframe germ of the section is a premise.*

Proof. The actual slice-restriction graph identity identifies the graph map used by the retained extension condition with the graph of the native slice section. Use the actual retained-map lowering lemma to obtain the original local full-map extensions in the selected covering charts. The base projection of a section graph is the identity, hence continuous. Apply the original retained-subset map-transfer lemma, using the actual real source and target bundle changes established from the collar-transition mixed germs. It transfers these local extensions to every pair of covering charts. That transfer matches only on the same retained subset and changes neither graph values nor source points, giving all the stated retention properties. □

Lemma A.310 (A nonzero retained section gives its nonzero full original canonical tensor)**.** *Suppose $J_s^2 = -\mathrm{id}$ on the retained base and the actual retained section σ is nowhere zero. Its native slice restriction σ_s is nowhere zero at every retained completed point. Natural canonical-square evaluation m_J gives the full native tensor T_s, with the exact normalized identity*

$$T_s \circ N^{\otimes 2n} = m_{N^{-1}J_sN}(\sigma_s),$$

where N is the prescribed real vertical-tangent normalization. This is a squared canonical tensor for that very complex field, and it is nonzero. Every original tensor argument is retained.

Proof. The slice restriction uses an invertible canonical-square fiber identification, so it cannot send a nonzero section value to zero. The native tensor is defined by natural square evaluation followed in every argument by the inverse normalization. Composing each argument with N cancels this inverse and gives the displayed identity. The image of natural square evaluation lies in the full squared canonical tensor line by its definition, proving canonicity. The normalized field satisfies the complex law by conjugation of $J_s^2 = -\mathrm{id}$. Natural square evaluation is then a complex linear isomorphism onto this full tensor line. If T_s were zero, the displayed identity would make its natural square evaluation zero, and injectivity would force $\sigma_s = 0$, a contradiction. □

Definition A.311 (Smooth compatible complex data on the original retained completions)**.** *Compatible original complex data consist of complex fields J_s^a on the actual original local completions X_s^a for their retained parameters. Each is compatible with the actual completed symplectic form, and on overlaps they transform by the original completion comparison pullbacks. Smooth regularity means ordinary real open coefficient extensions of these very fields in every original interior and boundary–radius chart, matching only on each retained base and including the full signed cylinder. No section is part of these data.*

Definition A.312 (A nowhere-zero section on the original retained canonical-square bundles)**.** *For the preceding compatible complex data, a prescribed original retained section assigns a nowhere-zero section σ^a of the actual canonical-square bundle on each original retained completed base. Its full tensors, obtained by natural square evaluation and the original tangent normalizations, transform under the original completion overlaps by their actual tensor pullbacks. The actual graph of each σ^a has ordinary real open extensions in the constructed restricted bundle coordinates, with matching only on that retained base. These are the overlap and smoothness conditions on the section itself; no original tensor or coframe germ is included.*

Lemma A.313 (The retained section constructs the exact original Maslov data)**.** *A prescribed original retained section produces covariant original completed-atlas Maslov data S whose complex fields are the very given J_s^a and whose full tensors are exactly those obtained from the given σ^a by natural square evaluation on every actual slice. Compatibility, canonicity and nonvanishing hold on all retained completed points, and the original overlap identities hold.*

Proof. Use the given compatible complex field for the field component of S. Restrict each σ^a through the actual slice inclusion and apply natural canonical-square evaluation with the same original tangent normalization to define its full tensor component. The retained section full-tensor lemma proves canonicity and nonvanishing of this very tensor. Compatibility is the given compatibility of J_s^a. Its complex-field overlap identity is the one in the original complex data, and its tensor overlap identity is the prescribed section's geometric covariance under the same original comparisons. These are exactly all the defining laws of covariant original completed-atlas Maslov data, with no replacement of a section value or tensor argument. □

Lemma A.314 (Reconstruction from the produced tensors recovers the originally chosen section)**.** *Let S be the original Maslov data constructed from the prescribed retained section σ. Its full native tensor is exactly the natural square evaluation of the actual slice restriction of σ, with the original normalization in every argument. The canonical-square section reconstructed from this full tensor is exactly that original slice restriction, and its reconstructed total section equals the original σ at every retained completed point. Thus the construction recovers the section itself, not only a scalar coordinate of its tensor.*

Proof. The tensor component of S is defined by natural square evaluation of the original slice restriction, followed by dimension reindexing. Returning to native coordinates cancels the reindexing. Composing all its arguments with the prescribed normalization cancels its inverse and gives precisely natural square evaluation of the original section. The section reconstructed from S has this same natural square evaluation by the original full-tensor recovery lemma. The compatible field satisfies the complex law at each retained point, so natural square evaluation is injective there. Applying injectivity to this full tensor equality identifies the two slice section values. At unused parameters both totalized sections are zero. Their slice restrictions therefore agree as functions. The actual total/native restriction recovery lemma now gives the stated total-section equality at every retained completed point, including the seam and full end. □

Lemma A.315 (The actual retained section produces all required original coframe germs)**.** *Assume ordinary raw complex-field regularity for the independently given compatible original fields, and let σ be the prescribed nowhere-zero retained section with ordinary real open graph extensions in the actual restricted bundle coordinates. The constructed original data S have smooth invertible coframe and scalar germs in every original interior and signed-cylinder chart. These germs recover the full tensors of the very σ, with the original chart anchors, retained parameter endpoints and entire signed band. No original coframe or scalar germ is supplied.*

Proof. Use the raw complex-field extensions in the interior and cylinder as the field inputs of the original retained-section coframe producer. The exact section-reconstruction lemma identifies its required native section with the actual slice restriction of σ. Ordinary extensions in the actual restricted bundle coordinates transfer, through the proved coordinate identification and graph recovery, to local full-map extensions of this identical native section graph in the original bundle atlas. These are exactly the remaining inputs of the retained-section coframe producer. It constructs the covering coframes from the raw field germs, extracts smooth scalar coordinates of this very graph, and recovers the original full tensors in every required chart. All matching is confined to the

same retained base and uses the actual signed cylinder, so the asserted endpoints and whole-band recovery are retained. □

Corollary A.316 (The prescribed retained section realizes the same full marked graded pair bundle). Supply the original compact pair-bundle geometry and laws of the retained graded realization theorem. In positive ribbon dimension use ambient dimension $2n$, $n > 1$, and ribbon dimension $2n-2$; in dimension two allow a finite ribbon, including the empty one. Give compatible covariant original complex data with ordinary raw field regularity, and one prescribed nowhere-zero section on its actual retained canonical-square bundles, with its defining overlap and ordinary graph-extension conditions.

These data produce one full original realization, retaining its actual geometric deformation, grade, coherent native presentations, original primitive, ribbon, witness, compact and full-core equations, and both ambient and ribbon marking-recovery capabilities on those same chosen completions. Under the actual original completion comparisons, both its smooth Maslov family and its completed-slice representatives recover precisely the given complex fields and the full tensors of the original prescribed section, for all retained parameters and in particular at both endpoints. The finite ribbon's full core is the whole ribbon. Existence of the target's prescribed toric data, itinerary and endpoint class is not a conclusion here.

Proof. Construct S from the actual prescribed retained section by natural square evaluation. The preceding lemma derives all its coframe and scalar germs from raw field regularity and ordinary section-graph extensions. In positive ribbon dimension combine these derived data with the supplied original pair-bundle geometry and apply its full retained graded realization once. Keep its actual deformation, grade, native recovery equations and both universal independent marking capabilities on that chosen realization. Its field and tensor recovery equations concern exactly this S. By its construction they are therefore the given complex fields and the natural full tensors of the originally prescribed section. Composing with its actual ambient completion presentation gives the same identities for its completed-slice representatives, including both parameter endpoints; no new endpoint representative is chosen.

In dimension two construct the identical ambient S and germs in rank one and combine them with the finite-ribbon geometry. Its full retained realization applies, since every ribbon tangent space is zero-dimensional and its primitive is zero. It retains the whole ribbon core, including for the empty ribbon, and all the same grading, native recovery and independent marking capabilities. The same two recovery identities again give the original prescribed fields and section tensors throughout the retained parameters. □

Definition A.317 (Restriction to the actual native compact domain). *Let D be a compact domain of dimension $k+1$ in its boundaryless extension X, equipped with its actual extended half-space atlas, and let $i : D \hookrightarrow X$ be the inclusion. For an ambient r-form η, its native restriction is the actual differential pullback*

$$(i^*\eta)_x(v_1, \dots, v_r) = \eta_{i(x)}(di_x v_1, \dots, di_x v_r).$$

The vectors here lie in the full tangent space of the native manifold with boundary, including when $x \in \partial D$.

Lemma A.318 (Actual restriction coefficients from the extended inverse chart). *Let c be an extended native half-space chart of D, with smooth ambient inverse extension h on an open real target,*

and let b be an ambient chart. Put $g = b \circ h$. For every y in the native chart target for which $h(y)$ lies in the source of b,

$$(i^*\eta)^c_y(v_1, \ldots, v_r) = \eta^b_{g(y)}(dg_y v_1, \ldots, dg_y v_r).$$

The derivative dg_y is the ordinary real derivative of the ambient extension, including at native boundary points.

Proof. The native extended chart target is the intersection of the ambient extension target with the closed half-space. On that intersection $i \circ c^{-1} = h$. Thus the actual coordinate representative $b \circ i \circ c^{-1}$ agrees with g on a relative neighborhood of y in the half-space. Its within derivative consequently equals the within derivative of g. A half-space has unique full-dimensional within derivatives, so this last derivative is the ordinary derivative of the smooth g. The differential chain rule for the inclusion and inverse native chart now identifies each argument of the pullback with the ambient inverse chart differential applied to $dg_y v_j$. Substituting these identities into the definition of the restriction gives the displayed coefficient formula, for every argument. □

Lemma A.319 (Ambient one-form families produce native endpoint and boundary germs)**.** *Suppose η_s, $s \in [0,1]$, is a smooth family of ambient one-forms on X with real open coefficient extensions across the parameter endpoints. Then the actual native restrictions $i^*\eta_s$ have the joint extension germs of A.155. In particular, a smooth ambient one-form has a locally extendible native restriction. No geometric law outside $[0,1]$ is required.*

Proof. Fix a native chart c, a point x in its source and $t \in [0,1]$. Choose an ambient chart b at $i(x)$, and let h be the smooth ambient inverse extension of c. On the open set $S = \mathrm{dom}(h) \cap h^{-1}(\mathrm{source}(b))$, the map $g = b \circ h$ is smooth and maps into the ambient chart target. The center $c(x)$ belongs to S by the inverse chart identities. Choose a real extension $\widetilde{\eta}_s$ of the supplied interval family. Its coefficients in b are jointly smooth on an open set V containing $[0,1] \times \mathrm{target}(b)$. The map $q(s,y) = (s, g(y))$ is smooth on $\mathbb{R} \times S$. Hence

$$U = (\mathbb{R} \times S) \cap q^{-1}(V)$$

is open and contains $(t, c(x))$. On U define the one-covector

$$\beta(s,y) = \widetilde{\eta}^b_s(g(y)) \circ dg_y.$$

The coefficient map and the spatial derivative of g are smooth; continuous bilinear composition of linear covectors with linear maps therefore makes β jointly smooth. At every actual interval parameter and native chart point in U, the coefficient formula of the preceding lemma gives $\beta(s,y) = (i^*\eta_s)^c_y$. These are the required germs at every center and chart, including parameter endpoints and spatial boundary points. For a single ambient one-form use the constant real family, whose coefficients are smooth on the product with each ambient chart target, and restrict any one of the resulting joint germs to a time slice. □

Corollary A.320 (Presented deformation primitives restrict to the native source). Let Q be a presented pair deformation in ambient dimension $2n$. Use its actual compact-domain presentation maps to pull every ambient primitive back to the extension of the compact ambient domain at time zero. If $2n = k + 1$, the actual differential restrictions of this family to that fixed native domain have joint real open coefficient germs. Likewise the ribbon primitives, pulled back by the actual ribbon presentations to their time-zero extension and then restricted to its native domain, have such germs when $2n - 2 = k + 1$. These conclusions concern the actual primitives. They do not assemble an intrinsic pair deformation or treat a zero-dimensional ribbon by a positive-dimensional half-space chart.

Proof. The definition of Q supplies joint smoothness with parameter endpoint extensions for precisely the ambient primitive family pulled back by its actual ambient presentations to the fixed time-zero extension. After identifying its dimension by $2n = k + 1$, apply the native family restriction lemma to that family and domain. The resulting forms are exactly their differential restrictions by definition. For the ribbon, the supplied ribbon-primitive smoothness is the same hypothesis on the fixed time-zero ribbon extension. The equality $2n - 2 = k + 1$ permits the identical restriction lemma there. In both cases only the stated dimension equality reindexes the tangent model; no primitive or presentation map is replaced. □

Lemma A.321 (Native coordinates of the actual compact inclusion)**.** *Let D be a positive-dimensional compact domain in a boundaryless extension X, with its extended half-space atlas. The actual inclusion $i : D \hookrightarrow X$ has smooth real open native-coordinate extensions at every point, including its boundary. For the selected native chart c, the ambient coordinate extension Q satisfies $Q(i(y)) = c(y)$ on the entire corresponding compact chart patch.*

Proof. Choose the extended half-space chart at the point. Its ambient coordinate map is smooth on an open neighborhood of that point in X. On the compact carrier it agrees exactly with the native chart, by the defining forward-chart identity. This ambient coordinate map is therefore the required extension of the coordinate expression of i. The identity holds at every point in the native chart source, giving the asserted whole-patch agreement. □

Lemma A.322 (Native scalar germs from ambient smooth functions)**.** *Let ρ be smooth on an open subset $V \subset X$, let c be an extended native chart of D, and let x belong to its source with $i(x) \in V$. The coordinate expression of $\rho \circ i$ has a smooth real open extension near $c(x)$, agreeing as a germ on the native chart target, including when $x \in \partial D$.*

Proof. Write $\widetilde{c}$ for the ambient chart extending c, and $h = \widetilde{c}^{-1}$ for its smooth inverse. The set $U = \mathrm{target}(\widetilde{c}) \cap h^{-1}(V)$ is open and contains $c(x)$: the inverse chart identity gives $h(c(x)) = i(x)$. The function $\beta = \rho \circ h$ is smooth on U. At every native chart target point y, the inverse-chart identity is $h(y) = i(c^{-1}(y))$. Consequently $\beta(y)$ equals the native coordinate expression of $\rho \circ i$ throughout their common relative neighborhood. This gives the claimed germ. □

Definition A.323 (The same Liouville domain as a native presentation)**.** *For a positive-dimensional presented Liouville domain (D, λ), its native restriction presentation uses the actual native compact carrier, primitive $\alpha = i^*\lambda$, identity carrier identification, the same represented domain D, and the literal inclusion $i : D \hookrightarrow X$. Its native coordinate germs are the ones supplied by the compact inclusion lemma. No compact domain, primitive or inclusion is changed.*

Lemma A.324 (Full native symplectic and Liouville restriction laws)**.** *For this same presentation, $d\alpha = i^*(d\lambda)$ is nondegenerate on every full native tangent space, and*

$$di_x Z_\alpha(x) = Z_\lambda(i(x)) \qquad (x \in D).$$

The identities include all boundary tangent directions.

Proof. Exterior naturality for the smooth inclusion gives $d(i^*\lambda) = i^*(d\lambda)$. Its full native differential is bijective by the exact endpoint presentation. If $d\alpha_x(v, w) = 0$ for all w, surjectivity of di_x and ambient nondegeneracy give $di_x v = 0$; injectivity then gives $v = 0$. For any ambient vector $u = di_x w$,

$$d\lambda_{i(x)}(di_x Z_\alpha(x), u) = d\alpha_x(Z_\alpha(x), w) = \alpha_x(w) = \lambda_{i(x)}(u).$$

The ambient contraction equation has a unique solution, so $di_x Z_\alpha(x) = Z_\lambda(i(x))$. □

Lemma A.325 (Actual defining functions retain strict native outwardness)**.** *At each actual native boundary point b, let ρ be the stored ambient local defining function, with convention $D = \{\rho \leq 0\}$. Its restriction $\rho_D = \rho \circ i$ vanishes at b, is nonpositive near b in D, has local zero set exactly the actual native boundary, has nonzero full native differential, and has a smooth real open germ in the native boundary chart. Moreover*

$$d\rho_D(Z_\alpha)(b) > 0.$$

Proof. The native boundary corresponds to the ambient frontier, so the defining function vanishes at $i(b)$. On its defining neighborhood every point of the compact carrier satisfies $\rho \leq 0$; continuity of i gives the asserted native neighborhood. On that neighborhood the ambient zero-level law identifies $\rho_D(y) = 0$ with the ambient frontier; the native boundary/frontier identity identifies this with $y \in \partial D$. The scalar restriction lemma supplies the real open germ of ρ_D. By the chain rule, $d\rho_D = d\rho \circ di$. If this covector were zero, surjectivity of di_b would force $d\rho_{i(b)} = 0$, contradicting the ambient defining function's regularity. Finally the field restriction law gives $d\rho_D(Z_\alpha)(b) = d\rho(Z_\lambda)(i(b))$, which is strictly positive by the original outwardness law. □

Corollary A.326 (All native Liouville laws on the same compact carrier)*.* The actual native restriction $\alpha = i^*\lambda$ is locally smoothly extendible. Its actual exterior derivative is locally smoothly extendible, satisfies the defining derivative relation, is closed and nondegenerate, and its canonical field satisfies $\iota_{Z_\alpha} d\alpha = \alpha$. At every native boundary point this field is strictly outward for a regular local defining function whose local zero set is exactly the actual boundary and whose coordinates have a smooth real open germ. Thus the same compact carrier retains all the native Liouville laws.

Proof. The ambient primitive is smooth, so the single-form specialization of native family restriction gives local smooth extendibility of α. The native exterior-derivative construction gives its actual defining derivative relation; derivative regularity gives local extendibility of $d\alpha$, and the square-zero theorem gives $d(d\alpha) = 0$. The differential-field restriction lemma gives nondegeneracy. The canonical Liouville field is the unique solution of contraction with this nondegenerate derivative, hence satisfies the stated equation. The defining-function restriction lemma supplies, at every boundary point, its exact native zero level, local nonpositive side, regular differential, coordinate germ and strict outward derivative. These are all the asserted laws on the originally fixed carrier and primitive. □

Corollary A.327 (Smooth native coefficients of the restricted Liouville field)*.* In every original native chart of the same compact Liouville domain, the canonical field Z_α has a smooth vector-valued extension on a real open neighborhood of each chart point, agreeing as a germ with its full native chart expression. This includes the boundary.

Proof. The native restriction laws give a locally extendible primitive, its actual exterior-derivative relation and nondegeneracy at every point. Apply the intrinsic contraction-field germ lemma to these data in the chosen original chart. Its unique contraction solution is exactly Z_α, so the resulting smooth extension agrees with this canonical field. The lemma is formulated in full half-space tangent coordinates, and therefore applies equally to boundary chart points. □

Lemma A.328 (A prescribed presentation retains its exact native compact primitive)**.** *Let S be a fixed positive-dimensional compact presentation and let $A : X_S \to X_D$ be a prescribed extension diffeomorphism identifying its carrier with a Liouville domain (D, λ). Put $i = A \circ i_S$ and $\alpha = i_S^*(A^*\lambda)$. The actual induced native carrier diffeomorphism $e : S \to D$ preserves interior and boundary. These data give a native Liouville presentation of exactly α on S, with represented domain exactly D and inclusion exactly i.*

Proof. The smooth ambient diffeomorphism and its inverse restrict to the native carrier diffeomorphism e. A native diffeomorphism preserves interior and boundary, by the local inverse function theorem in the interior and application of the same argument to its inverse. Differentiating $i = A \circ i_S$ and applying pullback composition gives $i^*\lambda = i_S^*(A^*\lambda) = \alpha$ on every full native tangent. The inclusion-coordinate germs for i_S transfer through the smooth square with A^{-1}, giving the actual native coordinate germs of i. Apply the exact endpoint presentation lemma with this fixed D, e, i and primitive equation. Its output retains D and i exactly, as asserted. □

Lemma A.329 (The transported native primitive retains nondegeneracy and field)**.** *The actual derivative $d\alpha$ of the same fixed-carrier primitive is nondegenerate at every native point, and its canonical field satisfies*

$$di_x Z_\alpha(x) = Z_\lambda(i(x)).$$

These conclusions use the full actual native tangent spaces, including the boundary of the fixed source carrier.

Proof. The transported presentation has the actual primitive and exterior pullback identities and bijective full inclusion differential. Apply the native presentation field lemma to this exact presentation. Its nondegeneracy argument uses surjectivity and injectivity of di, and its contraction argument identifies the pushforward with the unique ambient Liouville field. Thus both conclusions concern the prescribed i and the actual primitive α on the fixed source carrier. □

Lemma A.330 (Stored target defining functions define the fixed source boundary)**.** *At a native boundary point b of S, let ρ be the stored local defining function of the target D at $i(b)$. Then $\rho_S = \rho \circ A \circ i_S$ vanishes at b, is nonpositive near b in S, and its local zero set is exactly the actual native boundary of S. It has nonzero full differential and a smooth real open germ in the original native boundary chart. Moreover $d\rho_S(Z_\alpha)(b) > 0$.*

Proof. The native carrier map preserves boundary, so $i(b)$ lies on the target frontier. Pull the stored defining neighborhood back by the smooth A and the continuous i_S. The carrier and zero-level laws of ρ give the nonpositive side there. They identify its zero set with the target frontier, hence with the source native boundary by boundary preservation. The scalar restriction lemma applied to $\rho \circ A$ gives the stated real open coordinate germ. The chain rule gives $d\rho_S = d\rho \circ di$. If the former were zero, surjectivity of di_b would force $d\rho = 0$, contrary to the target defining function's regularity. The field equation gives $d\rho_S(Z_\alpha)(b) = d\rho(Z_\lambda)(i(b)) > 0$ by the stored outwardness law. □

Corollary A.331 (All native Liouville laws for the exact transported primitive)*.* The exact fixed-carrier primitive $\alpha = i_S^*(A^*\lambda)$ is locally smoothly extendible. Its actual exterior derivative is locally smoothly extendible, satisfies the defining derivative relation, is closed and nondegenerate, and its canonical field satisfies $\iota_{Z_\alpha} d\alpha = \alpha$. At every actual native boundary point it has a regular defining function with the exact local boundary zero set, original-chart real open germ and strict outward derivative.

Proof. Smooth pullback makes $A^*\lambda$ a smooth ambient one-form. Native form restriction gives local extendibility of α. The native derivative construction, regularity and square-zero theorems give respectively its derivative relation, local smooth derivative and closedness. The transported differential-field lemma gives nondegeneracy, so the canonical contraction solution has the stated contraction equation. The transported defining-function lemma supplies all the asserted actual boundary laws. Every step uses the same α and fixed compact carrier. □

Corollary A.332 (Actual fixed deformation primitives retain every native slice law). Let Q be a presented deformation. Restrict its actual ambient primitive family pulled onto the time-zero extension by its prescribed ambient presentations to the fixed native carrier. When $2n = k + 1$ this family has joint real open primitive germs and, at every original parameter, a nondegenerate actual exterior derivative and the strict native outwardness laws. In particular the local defining functions have exactly the actual fixed boundary as zero set, nonzero full differential and smooth real open original-chart germs. The same conclusion holds for the exact pulled ribbon primitive family when $2n - 2 = k + 1$, on that fixed positive-dimensional ribbon carrier.

Proof. The original deformation's primitive restriction corollary gives joint native family germs for precisely these pulled primitives, using the stated dimension identification. At a fixed original parameter apply the transported Liouville restriction laws to that parameter's actual ambient presentation and Liouville domain. Its primitive is exactly the family member already restricted, so it gives its nondegeneracy and all the stated actual boundary laws without a new hypothesis. Apply the same argument to the actual ribbon presentation and its Liouville domain after $2n-2 = k+1$. This treats every parameter and retains the same fixed ambient or ribbon carrier throughout. □

Lemma A.333 (Smooth ambient map families restrict through native boundaries). *Let $D \subset X$ be a positive-dimensional compact domain with its extended native atlas, and let N be a smooth boundaryless manifold. A jointly smooth family $f_t : X \to N$, $t \in [0,1]$, with smooth open extensions across parameter endpoints restricts to joint native manifold-valued germs for $f_t \circ i : D \to N$. The values agree exactly at all original compact points.*

Proof. Choose a real-parameter extension $\widetilde{f}$ smooth on an open $V \supset [0,1] \times X$. At a native chart center use its smooth ambient inverse extension h, and put $q(s,y) = (s, h(y))$. This is smooth on the product of $\mathbb{R}$ with the ambient chart target. Intersect that product with $q^{-1}(V)$ to obtain an open neighborhood U of the chosen parameter and chart point. The map $\beta(s,y) = \widetilde{f}(s, h(y))$ is smooth on U and takes values in N itself. For original parameters and native chart points, $h(c(x)) = i(x)$ and $\widetilde{f}(t, i(x)) = f_t(i(x))$. Thus $\beta(t, c(x)) = f_t(i(x))$ on the required relative product, including both parameter endpoints and every spatial boundary point. □

Lemma A.334 (Native map germs retain an ambient compact-carrier square). *Let $D \subset X$ and $E \subset Y$ be positive-dimensional compact domains of the same dimension. Suppose f is smooth on an open neighborhood of D in X, and $h : D \to E$ has the exact compact square $i_E \circ h = f \circ i_D$. Then h has smooth real open coordinate extensions in every pair of original source and target native charts at a compact point. The same assertion applies separately to a specified inverse map with its own smooth ambient neighborhood.*

Proof. For source and target extended charts c, a, write $g = f \circ \widetilde{c}^{-1}$. On the intersection of the ambient source target with the preimage of the supplied neighborhood, g is smooth. Intersect further with $g^{-1}(\mathrm{source}(\widetilde{a}))$. This is open and contains the selected source chart point, because the compact square puts its image in the target native chart source. On this open set $\widetilde{a} \circ g$ is smooth. At native points the inverse source chart identity and the compact square give $\widetilde{a}(g(c(x))) = \widetilde{a}(i_E(h(x))) = a(h(x))$. The same image condition ensures that $h(x)$ belongs to the target native chart source there. These equations are the required exact coordinate germs. Applying this construction to the separately supplied inverse proves the inverse assertion. □

Lemma A.335 (Smooth boundary scalar families give their actual local germs). *A smooth interval scalar family b_t on a fixed boundaryless manifold B, with real open parameter extensions, has joint*

local smooth germs on $\mathbb{R} \times B$ retaining every original value $b_t(x)$. No positivity away from the original interval is asserted.

Proof. Fix (t, x) and a smooth chart c at x. Choose the real-parameter scalar extension and its smooth coefficients on an open set V in $\mathbb{R} \times \text{target}(c)$ containing the original interval chart product. The map $q(s, y) = (s, c(y))$ is smooth on $\mathbb{R} \times \text{source}(c)$, so its intersection with $q^{-1}(V)$ is an open neighborhood of (t, x). Composing the smooth coefficients with q gives a smooth scalar there. At every original parameter and point in this neighborhood the inverse chart identity $c^{-1}(c(y)) = y$ and the scalar extension identity recover $b_s(y)$. This gives the claimed local germs at every center. □

Lemma A.336 (Prescribed compact equivalences retain both native maps)**.** *A prescribed compact-domain equivalence, smooth in both directions on open carrier neighborhoods, restricts to an equivalence of the actual compact carriers with exactly the original forward and inverse values. In positive dimension these maps form a native diffeomorphism through the boundary. For zero-dimensional compact carriers they form a native diffeomorphism for the discrete smooth atlases, including empty carriers.*

Proof. The forward and inverse maps send their respective carriers into each other by the given carrier laws. The supplied inverse identities restrict to the carriers and make these restrictions inverse bijections. In positive dimension the compact-domain smoothness criterion says that a carrier-valued map is smooth when its ambient inclusion is smooth. The ambient inclusions of these restrictions are exactly the prescribed neighborhood maps composed with the smooth compact inclusions, so both are smooth. In dimension zero every map from the discrete carrier is smooth; hence the same inverse bijections are diffeomorphisms there. The definitions use the original maps throughout, giving both asserted value identities without changing a compact point. □

Lemma A.337 (Both prescribed native compact maps retain real chart extensions)**.** *The exact native forward and inverse maps of a prescribed compact-domain equivalence have smooth real open extensions in every original pair of native charts. In positive dimension these extensions use the original forward and inverse neighborhood maps separately. In dimension zero both coordinate maps have globally smooth extensions in the zero real model, with exact agreement at every compact point.*

Proof. For the forward map its literal carrier square is $i_E \circ h = f \circ i_D$. Apply the ambient compact-carrier square germ lemma to the prescribed smooth forward neighborhood map f. The inverse has its own square and smooth neighborhood map, so the same lemma applied to those data gives its germs separately. In dimension zero the source and target real coordinate spaces each have only the zero vector. The constant map $\beta = 0$ is globally smooth, and both original chart expressions equal zero at every carrier point. It therefore extends each of the exact coordinate maps, without changing either carrier map. □

Lemma A.338 (Compact restriction preserves full differential pullback and scale)**.** *Let $h : D \to E$ be the native restriction of a prescribed positive-dimensional compact equivalence f. For every ambient form η on the target extension,*

$$h^*(i_E^*\eta) = i_D^*(f^*\eta).$$

If $f^*\beta = c\alpha$ on the entire original compact carrier, then $h^*(i_E^*\beta) = c\, i_D^*\alpha$ on every full native tangent space. Here c is any real constant; a completion step has the prescribed $c > 0$. In dimension zero the native primitive scaling equation holds for the same carrier map and any c, since every one-form is zero.

Proof. The literal compact square is $i_E \circ h = f \circ i_D$. Differentiating it gives $di_E \circ dh = df \circ di_D$. Substituting this identity into every argument of η in the differential pullback proves the first formula. If the carrier primitive equation holds, substitute it at $i_D(x)$ in this formula. Composition of a covector with di_D commutes with multiplication by c, giving the second formula at every x and all full native tangent directions, including boundary directions. In dimension zero every tangent vector is zero and a one-form vanishes on zero, so both sides of the scale equation are zero for the same map and constant. □

Lemma A.339 (Boundary conjugation retains its actual primitive factor)**.** *Let $u_D : B_D \to \partial D$ and $u_E : B_E \to \partial E$ be the actual boundary presentation diffeomorphisms. A prescribed boundary map $\psi : B_D \to B_E$ induces $\psi_N = u_E \circ \psi \circ u_D^{-1}$ on the genuine native boundaries. If $\psi^*\beta = c\alpha$ for a pointwise scalar function c on B_D, then, for the actual boundary restrictions α_N, β_N,*

$$\psi_N^*\beta_N = (c \circ u_D^{-1})\alpha_N.$$

The conjugated map satisfies $\psi_N \circ u_D = u_E \circ \psi$ on the whole boundary.

Proof. The displayed map is a composition of smooth boundary diffeomorphisms, and its square follows by canceling $u_D^{-1}u_D$. The boundary primitive transport identities are $u_D^*\alpha_N = \alpha$ and $u_E^*\beta_N = \beta$. Pulling the first identity back by u_D^{-1} gives $(u_D^{-1})^*\alpha = \alpha_N$. Pullback composition now yields

$$\psi_N^*\beta_N = (u_D^{-1})^*(\psi^* u_E^*\beta_N) = (u_D^{-1})^*(c\alpha) = (c \circ u_D^{-1})\alpha_N.$$

All pullbacks use the differentials of the stated maps, giving the assertion on every actual boundary tangent space. □

Lemma A.340 (The same completion step retains its native infinity laws)**.** *For a supplied completion identification Ψ of positive-dimensional ambient pairs, express its prescribed boundary map and shift in the actual native boundaries by the preceding conjugation. Writing $s_N = s \circ u_D^{-1}$, its boundary map satisfies $\psi_N^*\alpha'_N = e^{-s_N}\alpha_N$. For every native boundary point x, there is a real r_0 such that for every real $r \geq r_0$,*

$$\Psi(E_D(u_D^{-1}(x), r)) = E_{D'}(u_{D'}^{-1}(\psi_N(x)), r + s_N(x)).$$

The ambient completion map itself and its signed end coordinates are unchanged.

Proof. Apply boundary primitive conjugation to the supplied equation $\psi^*\alpha' = e^{-s}\alpha$. This gives the first formula with exactly s_N. At $u_D^{-1}(x)$ the original induced-at-infinity law supplies r_0 and the stated equation with target boundary point $\psi(u_D^{-1}(x))$. The conjugation square identifies this point with $u_{D'}^{-1}(\psi_N(x))$. Substitution gives the displayed equation for all real $r \geq r_0$ using the same Ψ, shift and end maps. □

Lemma A.341 (A supplied completion step preserves the whole native ribbon)**.** *Let h_N be the actual compact-carrier restriction of the ribbon map of the same supplied completion step. At every compact ribbon point x, including in a zero-dimensional or empty ribbon, its original boundary square and its actual native-frontier square hold:*

$$\psi(j(x)) = j'(h_N(x)), \qquad \psi_N(u_D(j(x))) = u_{D'}(j'(h_N(x))).$$

In positive or zero ribbon dimension the full native primitives satisfy $h_N^*\mu'_N = c_{\mathrm{scale}}\mu_N$ with exactly the step's prescribed positive constant.

Proof. Restrict the supplied whole-ribbon square to a point of the actual carrier. Its ribbon equivalence has exactly the prescribed value there, so this gives the first equality. Apply $u_{D'}$ and use $\psi_N \circ u_D = u_{D'} \circ \psi$ to obtain the second equality. For a positive-dimensional ribbon, apply the compact primitive restriction lemma to the given carrier equation $h^*\mu' = c_{\mathrm{scale}}\mu$. This proves the equation on every full native tangent direction. In dimension zero every native one-form is zero, so the same equation holds with exactly the prescribed scale and native carrier diffeomorphism. The square proof uses only carrier membership and the supplied square, so it also covers dimension zero and the empty case. □

Definition A.342 (The exact ribbon map and witness on native carriers)**.** *Let D be a positive-dimensional compact ambient domain with actual boundary presentation $u : B_D \to \partial D$, let F be a compact ribbon domain, and let $j : F \to B_D$ and $b : B_D \to \mathbb{R}$ be the prescribed maps. Their native restrictions are*

$$j_N(x) = u(j(i_F(x))), \qquad b_N = b \circ u^{-1}.$$

For a zero-dimensional ribbon the same pointwise formula defines j_N, and the native restriction of a form is its actual differential pullback along i_F in the discrete zero-dimensional atlas. All the original compact points and the boundary witness values are retained.

Lemma A.343 (Native ribbon and positive witness families retain joint germs)**.** *For fixed positive-dimensional D, F, jointly smooth original families j_t and b_t with parameter endpoint extensions give joint native germs for $j_{N,t}$ and local boundary germs for $b_{N,t}$. If $b_t > 0$ on the original boundary for every $t \in [0,1]$, then $b_{N,t} > 0$ on the whole actual native boundary for those parameters.*

Proof. Apply native map-family restriction to $j_t \circ i_F$, and compose its manifold-valued germs with the smooth boundary diffeomorphism u. This gives germs with exactly the values $j_{N,t}$. Apply the boundary scalar-family restriction lemma to b_t, and pull its local germs back by the smooth map u^{-1}. Their values are exactly $b_{N,t}$. Finally at a native boundary point y, the value is $b_{N,t}(y) = b_t(u^{-1}(y)) > 0$. Neither construction alters any prescribed interval value or imposes positivity outside the interval. □

Lemma A.344 (Finite ribbons retain real-time native map germs)**.** *A jointly smooth map family on the extension of a compact zero-dimensional domain restricts to actual local real-time manifold-valued germs on its discrete carrier, including at the interval endpoints. In particular a finite ribbon family j_t has these germs with exactly its prescribed native-frontier values $j_{N,t}$. An empty ribbon is allowed.*

Proof. The product map $q(s,x) = (s, i_F(x))$ is smooth, since the zero-dimensional compact inclusion is smooth on its discrete carrier. Choose the real open smooth extension $\widetilde{f}$ of the original family, with smoothness domain $V \supset [0,1] \times X_F$. Its preimage $q^{-1}(V)$ is an open neighborhood of every original parameter-carrier point. The map $\widetilde{f} \circ q$ is smooth there and agrees with the original restricted family on all interval points by the extension identity. For j_t, compose these germs with u to obtain exactly $j_{N,t}$. This construction uses no selected point of F, so also applies to an empty carrier. □

Lemma A.345 (The full witnessing equation survives native restriction)**.** *Suppose in positive ribbon dimension that the actual ambient ribbon primitive satisfies $j^*(b\alpha) = \mu$ on its entire compact carrier, where α is the boundary restriction of the ambient primitive. Then the exact native maps and primitives satisfy*

$$j_N^*(b_N \alpha_N) = \mu_N$$

at every compact ribbon point and every full native tangent direction, including boundary points. In dimension zero the same equation holds for the actual restricted forms and prescribed map and witness.

Proof. The boundary transport identity is $u^*\alpha_N = \alpha$, and $b_N \circ u = b$. Differentiating $j_N = u \circ j \circ i_F$ gives the actual differential chain rule. Therefore

$$j_N^*(b_N\alpha_N) = i_F^* j^* u^*(b_N\alpha_N) = i_F^* j^*(b\alpha) = i_F^*\mu = \mu_N.$$

Each differential is the full native differential, so the equation holds also at ribbon boundary points. In dimension zero each tangent vector is zero; a one-form evaluated on zero is zero. Both actual pullback forms consequently vanish for the same prescribed j_N and b_N, proving the equation there as well. □

Lemma A.346 (A presented deformation retains its fixed full ribbon equation)**.** *Let Q be a presented deformation. Write A_t, R_t, B_t for its actual ambient, ribbon and boundary presentation maps from time zero, and let $\lambda_t, \mu_t, j_t, b_t$ be its original data. On the fixed extensions put*

$$\widetilde{j}_t = B_t^{-1} \circ j_t \circ R_t, \quad \widetilde{b}_t = b_t \circ B_t, \quad \widetilde{\lambda}_t = A_t^*\lambda_t, \quad \widetilde{\mu}_t = R_t^*\mu_t.$$

The fixed witness is positive and has real local parameter germs. On every point of the fixed compact ribbon carrier its full primitive satisfies $\widetilde{j}_t^*(\widetilde{b}_t\widetilde{\alpha}_t) = \widetilde{\mu}_t$, where $\widetilde{\alpha}_t$ is the actual boundary restriction of $\widetilde{\lambda}_t$.

Proof. Positivity is $\widetilde{b}_t(x) = b_t(B_t(x)) > 0$. The supplied fixed witness smoothness and the scalar restriction lemma give its local parameter germs. The actual boundary presentation square gives $\widetilde{\alpha}_t = B_t^*\alpha_t$. Consequently $\widetilde{b}_t\widetilde{\alpha}_t = B_t^*(b_t\alpha_t)$. Pullback composition and $B_t \circ \widetilde{j}_t = j_t \circ R_t$ then yield

$$\widetilde{j}_t^*(\widetilde{b}_t\widetilde{\alpha}_t) = R_t^* j_t^*(b_t\alpha_t) = R_t^*\mu_t = \widetilde{\mu}_t$$

on the fixed compact carrier. The middle equality uses the original whole-ribbon equation at $R_t(x)$, which belongs to the original compact carrier by the ribbon presentation law. Thus every covector argument is retained, rather than only a subset of ribbon points. □

Corollary A.347 (The same fixed ribbon family has exact native parameter germs)*.* The exact family $\widetilde{j}_t$ on the fixed compact ribbon of Q has joint native target-valued germs through its boundary and the parameter endpoints when $2n - 2 = k + 1$. For $n = 1$ the exact map on the actual finite ribbon has real-time native germs, including when the ribbon is empty. These germs take values in the original fixed smooth boundary presentation and retain every prescribed compact ribbon value.

Proof. The presented deformation supplies joint smoothness of exactly $\widetilde{j}_t$ on its fixed extension. In positive dimension identify the tangent model using $2n - 2 = k + 1$ and apply the compact map-family restriction lemma. For $n = 1$ apply its discrete-carrier counterpart. These procedures restrict the prescribed family without replacing any value and keep its fixed smooth boundary target. The constructions require no selected point of the finite ribbon, so the empty case is included. □

Remark A.348 (Componentwise source correspondence)*.* The geometric target retains the following order of quantifiers. For each finite input, it asserts completeness and finite Liouville type for every $K \geq 0$. An ambient bound and R_0 are chosen; for every $R \geq R_0$ the ratio set is nonempty and bounded and a fiber bound exists. For every valid fiber bound, every $K \geq K_0(R)$ and every allowed pair of levels, a rounding exists. Every admissible rounding then has the specified whole-ribbon pair

and both inclusion markings. One existential endpoint package contains its finite geometric chain, literal embedded full-core equality and ambient grading on that same chain with the prescribed endpoint class.

The component correspondences represent and recover the actual intrinsic compact pairs and smooth interval families, including their canonical collar-glued completions. They retain full ribbon maps, positive boundary witnesses, constant-scale ribbon diffeomorphisms and the prescribed strict ambient completion maps. The marking correspondences preserve both supplied inclusion markings. The grading correspondences retain compatible almost complex structures and genuine nowhere-zero sections of the squared canonical line bundles. The finite-chain constructions retain literal junctions and the grading on the same geometric chain.

The reverse restriction laws use the same compact carriers, original parameter values, whole ribbons and witnesses, full tangent differentials and prescribed maps at infinity. Together with the forward component correspondences they account for the defining source data in both directions. They are applied to each object under the quantifiers just stated; neither a comparison chain nor its endpoint core equality or grading is moved into the input hypotheses. These are componentwise correspondences, with their individual mathematical statements and attachments. This synthesis introduces no additional global equivalence theorem.

Definition A.349 (Auxiliary presentation: The conditional initial-choice comparison proposition)**.** *Assume G1–G2 with specified A_{amb}, R_0. For each fixed R, K in their stated range and every two admissible initial realizations, allowing different valid fiber bounds, the conditional proposition asserts a finite comparison chain inside the fixed (Y, λ_K). Its intermediate compact primitives are restrictions of λ_K, its markings extend inclusion, and its strict identifications are the identity on marked Y. The same chain carries the prescribed logarithmic ambient grading at both ends. This is a proposition definition, not a proof of the conditional assertion. Transport of additional background classes requires its separate functoriality foundation.*

A.7 Auxiliary presentations: Local completion foundations

The following results concern extension-space domain presentations. A zero-dimensional completion is constructed on the finite carrier itself. In positive dimension, localization and scalar reparameterization are separate prerequisites for constructing the cylindrical primitive and its complete flow; neither prerequisite asserts that construction.

Lemma A.350 (Smooth inverse coordinates for the flow collar)**.** *The constructed Liouville collar $c : \partial D \times (-\delta, \delta) \longrightarrow X_0$, in the symplectic neighborhood X_0 of the compact domain, is a diffeomorphism onto its open image. Each point of this image has unique boundary and radial coordinates (x, r), depending smoothly on the point, with $c(x, r)$ equal to that point and $-\delta < r < \delta$.*

Proof. The collar construction gives an injective local diffeomorphism on the open strip. Its image is open because every local diffeomorphism is open. Injectivity and the definition of image give a bijection onto that image. At every source point the inverse function theorem gives a smooth inverse on an open neighborhood of its image. This inverse agrees there with the set-theoretic inverse: both return a point in the strip with the specified image, and injectivity makes those points equal. These local inverses therefore give a smooth global inverse. The inverse takes values in the strip, and both inverse identities follow from the bijection. □

Lemma A.351 (Interior and collar localization)**.** *Let D_0 be the compact domain with its extension restricted to the symplectic neighborhood X_0 used by the collar construction. With its constructed*

collar c, set

$$U = \operatorname{int}_{X_0}(D_0) \cup c\big(\partial D_0 \times (-\delta, \delta)\big).$$

Then U is an open neighborhood of D_0, and

$$U \setminus D_0 = c\big(\partial D_0 \times (0, \delta)\big).$$

Restriction to U preserves the compact domain by its open inclusion, giving a homeomorphism on the compact carriers, and its boundary primitive is exactly $\alpha = \lambda|_{\partial D}$ from the original domain. Every exterior point has the unique smooth positive collar coordinates furnished above.

Proof. The interior and the collar image are open, so their union is open. The compact carrier is closed. Every one of its points is either an interior point or a frontier point. The frontier equals the image of the boundary inclusion, and each boundary point is the time-zero collar image. Thus the carrier is contained in U. An exterior point of U cannot lie in the interior of the carrier, so it lies in the collar image. On the collar the domain-side identity is $c(x, r) \in D_0$ if and only if $r \leq 0$. Hence its unique radial coordinate is positive. Conversely each positive collar point belongs to U and lies outside the carrier. This proves the displayed equality. The restricted carrier is the inverse image of D_0 under the open inclusion. That inclusion and the map sending $x \in D_0$ to the same point in U are inverse continuous maps, giving the homeomorphism. The boundary inclusion still sends each boundary point to the same point, and the restricted primitive evaluates on the same tangent vectors as before. Its boundary pullback is therefore unchanged. The earlier restriction from the original extension to X_0 has the same property, so the boundary form is the original α. □

Definition A.352 (Scalar collar stretch)**.** *Fix $\delta > 0$ and the standard smooth transition $S : \mathbb{R} \to [0, 1]$ used in this construction. It is smooth and nondecreasing, equal to zero on $(-\infty, 0]$ and to one on $[1, \infty)$. Put*

$$q_\delta(r) = S(3r/\delta - 1), \qquad h_\delta(r) = r + \frac{q_\delta(r)}{\delta - r} \quad (r < \delta).$$

This scalar function reparameterizes a finite collar; its definition does not construct a cylindrical primitive or a completion.

Lemma A.353 (Smoothness, seam and positivity of the stretch)**.** *The function h_δ is smooth and strictly increasing on $(-\infty, \delta)$, has positive derivative there, and satisfies*

$$h_\delta(r) = r \quad (r \leq \delta/3), \qquad h_\delta(r) = r + (\delta - r)^{-1} \quad (2\delta/3 \leq r < \delta).$$

Proof. The denominator is positive on the specified interval, so the quotient is smooth there. The transition is nonnegative and nondecreasing, and its argument has positive derivative $3/\delta$. Therefore $q_\delta \geq 0$ and $q'_\delta \geq 0$. Direct differentiation gives

$$h'_\delta(r) = 1 + \frac{q'_\delta(r)(\delta - r) + q_\delta(r)}{(\delta - r)^2} > 0.$$

For any two points in the interval, the mean value theorem now makes the difference of function values strictly positive when their ordered difference is positive. Thus h_δ is strictly increasing. If $r \leq \delta/3$, the transition argument is at most zero, so $q_\delta(r) = 0$. If $r \geq 2\delta/3$, it is at least one, so $q_\delta(r) = 1$. Substitution gives both formulas. □

Lemma A.354 (The finite collar stretches onto the whole line)**.** *As $r \to \delta$ from below, $h_\delta(r) \to +\infty$. The restriction $h_\delta : (-\infty, \delta) \to \mathbb{R}$ is a bijection, and its restriction to $[0, \delta)$ is a bijection onto $[0, \infty)$.*

Proof. Near δ the preceding formula is $h_\delta(r) = r + (\delta - r)^{-1}$. The first summand tends to the finite number δ and the second to $+\infty$, proving the limit. For $y \in \mathbb{R}$, set $a = \min(y, 0)$. Then $a < \delta$ and $a \leq \delta/3$, so $h_\delta(a) = a \leq y$. The limit gives $b < \delta$ with $h_\delta(b) \geq y$; it can be chosen larger than a. Continuity and the intermediate value theorem on $[a, b]$ give a preimage of y. Strict increase gives uniqueness. Finally $h_\delta(0) = 0$, and strict increase implies $h_\delta(r) \geq 0$ if and only if $r \geq 0$. This identifies the restricted nonnegative domain and image. □

Lemma A.355 (Smooth inverse and restricted collar coordinates)**.** *The inverse $k_\delta : \mathbb{R} \to (-\infty, \delta)$ of h_δ is smooth and is the identity for $s \leq \delta/3$. The stretch restricts to a smooth bijection $(-\delta, \delta) \to (-\delta, \infty)$ with smooth inverse, and retains its identity restriction near $r = 0$. Moreover $k_\delta(s) \geq 0$ if and only if $s \geq 0$.*

Proof. The preceding bijection is continuous and strictly increasing. Its inverse is continuous: at any $r_0 < \delta$, choose points on either side of r_0 as close as desired; their ordered images form a neighborhood of $h_\delta(r_0)$ whose preimage lies between them. The derivative of h_δ is nonzero everywhere. The inverse function theorem at r_0 gives a smooth local inverse, which agrees with the global inverse by injectivity. Thus the global inverse is smooth. For $s \leq \delta/3$, the seam formula gives $h_\delta(s) = s$; the inverse identity then gives $k_\delta(s) = s$. Since $h_\delta(-\delta) = -\delta$ and the stretch is strictly increasing, its inverse satisfies $k_\delta(s) > -\delta$ exactly when $s > -\delta$. Its values always lie below δ. The inverse identities now prove the asserted restricted bijection, and the seam formula gives the identity on a neighborhood of zero. Strict increase and $k_\delta(0) = 0$ give the final equivalence. □

Lemma A.356 (A smoothly glued stretched primitive)**.** *On the localized extension U, write r for the finite collar radius. Define a function f to be $\exp(h_\delta(r) - r)$ on the collar image and one outside that image. Then f is smooth and positive, $f = 1$ on the compact carrier, and the one-form $\eta = f\lambda$ is smooth on U and equals λ on that carrier.*

Proof. The inverse collar coordinates are smooth. Composing their radius with the smooth scalar stretch and exponential therefore makes f smooth on the collar image. For a collar point in the carrier the domain-side identity gives $r \leq 0$, so the seam identity $h_\delta(r) = r$ gives $f = 1$. Off the collar image this equality holds by definition. Thus $f = 1$ on the entire carrier. On its open interior it is consequently locally constant. The interior and the collar image cover U, so these two smoothness assertions prove smoothness everywhere, including their overlap. Exponentials are positive, and the other defining value is one. Multiplying a smooth one-form by this smooth function gives a smooth one-form; its equality with λ on the carrier follows from $f = 1$. □

Lemma A.357 (Smooth coordinates on the stretched end)**.** *Let k_δ be the inverse scalar stretch. The map*

$$e : \partial D \times (-\delta, \infty) \longrightarrow U, \qquad e(x, s) = c(x, k_\delta(s)),$$

is smooth and an open embedding with injective differential. Its restriction to $\partial D \times [0, \infty)$ is an embedding with the subspace topology. The reparameterization $(x, s) \mapsto (x, k_\delta(s))$ is a diffeomorphism onto the original finite collar strip.

Proof. The scalar inverse and its forward stretch give mutually inverse smooth maps between $(-\delta, \infty)$ and $(-\delta, \delta)$. Taking their product with the boundary identity gives the asserted reparameterization diffeomorphism. The original collar is smooth and an open embedding onto its image. That image is contained in U, and its map into the open submanifold U retains these properties. Composing with the reparameterization gives e and proves its smoothness and open embedding property. The differentials of the collar diffeomorphism and of the reparameterization are invertible,

as follows by differentiating their inverse identities. The intervening open-submanifold inclusions have invertible differentials as well. The chain rule therefore gives an injective differential for e. The inclusion of the nonnegative subspace into the open source is an embedding; composing the two embeddings proves the last restriction assertion. □

Lemma A.358 (The stretched end is the whole exterior)**.** *For the constructed end coordinates,*

$$e(x,0) = x, \qquad e(x,s) \in D \iff s \le 0 \quad (s > -\delta).$$

Moreover $U \setminus D = e(\partial D \times (0,\infty))$, and $D \cup e(\partial D \times [0,\infty)) = U$.

Proof. The scalar inverse fixes zero, and the collar at zero is the boundary inclusion, proving the first identity. On the original collar the domain-side identity is membership in D precisely when the finite radius is nonpositive. Strict increase of k_δ and $k_\delta(0) = 0$ identify that condition with $s \le 0$. Localization identifies the whole exterior with original collar points having finite radius $0 < r < \delta$. For each such point set $s = h_\delta(r) > 0$; the inverse identity then gives $e(x,s) = c(x,r)$. Conversely a positive s has positive inverse radius and hence lies outside D. This proves the exterior identity. Every point of U either belongs to D or to that exterior, which proves the covering identity with the nonnegative end. □

Lemma A.359 (Compact radial slabs and a compact exhaustion step)**.** *For $-\delta < a$ and every real b, the set $e(\partial D \times [a,b])$ is compact. Consequently*

$$K = D \cup e(\partial D \times [0,1])$$

is compact, and every $z \in U \setminus K$ has end coordinates $z = e(x,s)$ with $s \ge 1$.

Proof. The boundary is compact, and the closed bounded interval $[a,b]$ is compact, also when it is empty. Their product is compact. Its points belong to the open source of the smooth end map since $s \ge a > -\delta$. The continuous image of this product is therefore compact. Apply this with $a = 0, b = 1$ and unite it with the compact carrier to obtain compactness of K. A point outside K lies outside D, so the exterior identity gives $z = e(x,s)$ with $s > 0$. If $s < 1$, this same point belongs to the slab in K, a contradiction. Hence $s \ge 1$. □

Lemma A.360 (Exterior derivative of a scalar multiple)**.** *For a smooth real function f and smooth one-form α on a smooth manifold,*

$$d(f\alpha)_x(v,w) = f(x)d\alpha_x(v,w) + df_x(v)\alpha_x(w) - df_x(w)\alpha_x(v).$$

Thus $d(f\alpha) = f\,d\alpha + df \wedge \alpha$ for the actual exterior derivatives of these forms.

Proof. Work in a smooth chart at x, writing $\alpha = \sum_i A_i\,dy_i$. The coefficients of $f\alpha$ are fA_i, whose differentials are $f\,dA_i + A_i\,df$ by the scalar product rule. Hence

$$d(f\alpha) = \sum_i (f\,dA_i + A_i\,df) \wedge dy_i = f\sum_i dA_i \wedge dy_i + df \wedge \sum_i A_i\,dy_i.$$

Evaluate on the chart images of v, w. The chain rule for the chart inverse identifies the derivative of the chart expression of f on these vectors with $df_x(v), df_x(w)$; the inverse chart differential cancels the chart differential in each one-form evaluation. The two alternating evaluations of $df \wedge \alpha$ are precisely the final two terms displayed in the statement. This gives the intrinsic formula. □

Lemma A.361 (Nondegeneracy of a cylindrical alternating-form deformation)**.** *Let E be a real vector space and ω a nondegenerate alternating bilinear form. Suppose $Z \in E$, $\rho \in E^*$ and $\ell : E \to \mathbb{R}$ satisfy $\omega(Z, w) = \ell(w)$ for every w and $\rho(Z) = 1$. For nonzero real numbers a, f, the form*

$$B(v, w) = f\big(\omega(v, w) + (a-1)(\rho(v)\ell(w) - \rho(w)\ell(v))\big)$$

has trivial kernel. The same conclusion holds without the scalar factor f. No nondegeneracy of the new form is assumed.

Proof. The contraction identity makes ℓ linear. Alternation gives $\ell(Z) = \omega(Z, Z) = 0$. Let v belong to the kernel of B. Since $f \neq 0$, remove the scalar factor. Evaluating the resulting identity at $w = Z$ gives

$$0 = -\ell(v) - (a-1)\ell(v) = -a\ell(v),$$

so $\ell(v) = 0$. The kernel identity now reads $\omega(v, w) + (a-1)\rho(v)\omega(Z, w) = 0$ for every w. Nondegeneracy of ω yields $v + (a-1)\rho(v)Z = 0$. Apply ρ to obtain $a\rho(v) = 0$, whence $\rho(v) = 0$ and then $v = 0$. The argument with no scalar factor is the same argument before removing f, or is the case $f = 1$. □

Lemma A.362 (Actual differential with a cylindrical evaluation formula)**.** *Let α, η be one-forms on a smooth manifold. At a point x, suppose $d\alpha_x$ is nondegenerate, $d\alpha_x(Z, w) = \alpha_x(w)$ for every tangent vector w, and $\rho(Z) = 1$ for a linear covector ρ. If $a, f \neq 0$ and*

$$d\eta_x(v, w) = f\big(d\alpha_x(v, w) + (a-1)(\rho(v)\alpha_x(w) - \rho(w)\alpha_x(v))\big)$$

for every v, w, then the actual exterior derivative $d\eta_x$ is nondegenerate. The displayed evaluation identity is a hypothesis; construction and smooth gluing of η are separate assertions.

Proof. Apply the preceding algebraic result on T_xM with $\omega = d\alpha_x$ and $\ell = \alpha_x$. Each of its hypotheses is precisely one of the stated assumptions. The displayed identity identifies $d\eta_x$ with its deformed form B. Thus a tangent vector annihilated by $d\eta_x$ is zero, which is nondegeneracy. □

Lemma A.363 (The derived differential of the stretched primitive)**.** *Let Z be the original Liouville field restricted to U. On the collar image the smooth inverse radius r satisfies $dr(Z) = 1$. Writing $f = \exp(h_\delta(r) - r)$ and $a = h'_\delta(r)$ there, the actual exterior derivative of the constructed $\eta = f\lambda$ is*

$$d\eta = f\big(d\lambda + (a-1)dr \wedge \lambda\big).$$

On the interior of the compact carrier it is $d\lambda$.

Proof. For fixed $x \in \partial D$, inverse collar coordinates give $r(c(x, t)) = t$ throughout the finite open strip. Differentiating and using $\partial_t c(x, t) = Z(c(x, t))$ gives $dr(Z) = 1$ at each collar point. The scalar chain rule applied to f gives $df = f(h'_\delta(r) - 1)dr$. The product rule for the exterior derivative now yields $d\eta = f\, d\lambda + df \wedge \lambda$, which is the stated formula. On the open interior of the carrier f is identically one, so $df = 0$ there and this same product rule gives $d\eta = d\lambda$. □

Lemma A.364 (The constructed primitive is symplectic with a smooth Liouville field)**.** *The actual two-form $d\eta$ is symplectic everywhere on U. Consequently it determines a unique smooth field $\widetilde{Z}$ there satisfying $\iota_{\widetilde{Z}} d\eta = \eta$. No nondegeneracy assumption on $d\eta$ is needed.*

Proof. The original $d\lambda$ is nondegenerate on its symplectic neighborhood, hence on U, and its field satisfies $d\lambda(Z, w) = \lambda(w)$. On the carrier interior the preceding differential formula identifies $d\eta$ with $d\lambda$. At each collar point apply the cylindrical differential nondegeneracy lemma with $\rho = dr$, the original field Z, scalar $a = h'_\delta(r)$ and multiplier f. Its contraction hypothesis is the original Liouville equation, and $\rho(Z) = 1$ was just proved. The two required nonzero scalars are positive by stretch derivative positivity and exponential positivity. The derived differential identity supplies its remaining hypothesis. Thus $d\eta$ is nondegenerate on the collar image as well. These two open sets cover U, proving nondegeneracy everywhere. Smoothness of η gives smoothness of its exterior derivative, and $d(d\eta) = 0$ gives closedness, so it is symplectic. In each tangent chart its invertible smooth matrix sends a unique vector to the smooth covector η. Matrix inversion is smooth on invertible matrices, so these vectors vary smoothly. Uniqueness makes the local fields agree on overlaps, giving the stated global field and contraction equation. □

Lemma A.365 (The actual stretched field and its compact restriction)**.** *On the finite collar image the canonical field of η is*

$$\widetilde{Z} = (h'_\delta(r))^{-1} Z.$$

On the entire compact carrier, including its boundary, it agrees with the original domain's Liouville field under the open inclusions.

Proof. Set $a = h'_\delta(r) > 0$. Alternation and the original contraction equation give $\lambda(Z) = d\lambda(Z, Z) = 0$. Since $dr(Z) = 1$, evaluation of the derived differential at $a^{-1}Z$ and an arbitrary vector w gives

$$d\eta(a^{-1}Z, w) = f\big(a^{-1}\lambda(w) + (a-1)a^{-1}\lambda(w)\big) = f\lambda(w) = \eta(w).$$

Nondegeneracy makes the solution of this contraction equation unique, so $\widetilde{Z} = a^{-1}Z$ there. On the compact interior $\eta = \lambda$ and $d\eta = d\lambda$, so uniqueness gives equality with Z. At a collar point of the carrier $r \le 0$. Because $\delta > 0$, a neighborhood of this r lies below $\delta/3$, where h_δ is the identity. Hence $h'_\delta(r) = 1$, and the collar formula again gives equality with Z. These cases cover the compact carrier. The neighborhood Liouville field agrees on that carrier with the original domain field, and restriction by the open inclusions preserves this equality. □

Lemma A.366 (The constructed end curves solve the canonical field)**.** *For each boundary point x, the actual constructed end curve satisfies*

$$\frac{d}{ds} e(x, s) = \frac{Z(e(x, s))}{h'_\delta(k_\delta(s))} = \widetilde{Z}(e(x, s)) \qquad (s > -\delta).$$

Thus it is an integral curve of the canonical field of η throughout this entire open interval, including all nonnegative radii.

Proof. Differentiate the identity $h_\delta(k_\delta(s)) = s$. The positive derivative of h_δ gives $k'_\delta(s) = (h'_\delta(k_\delta(s)))^{-1}$. Since $e(x, s) = c(x, k_\delta(s))$ and the collar time derivative is the original field Z, the chain rule gives the first equality. The inverse collar radius at this point is exactly $k_\delta(s)$. Substitution into the canonical stretched-field formula gives the second equality. Both equalities hold at each point of the specified open interval, which is precisely the integral-curve assertion there. □

Lemma A.367 (Boundary orbit calculus for a supplied complete flow)**.** *Let $D \subset M$ be a compact Liouville domain presentation with boundary inclusion j and original outward field Z_D. Suppose a*

smooth field Z on M is supplied with its smooth all-real-time flow Φ, and $Z(j(x)) = Z_D(j(x))$ on the boundary. Let η be a smooth one-form on M satisfying $\iota_Z d\eta = \eta$ everywhere. Then

$$q(x,r) = \Phi^r(j(x)), \qquad (x,r) \in \partial D \times \mathbb{R},$$

is smooth, is an immersion at every real radius, satisfies $q(x,0) = j(x)$, and has $q^*\eta = e^r j^*\eta$, with zero component in the radial direction. This assertion assumes the complete flow and does not construct it.

Proof. Smoothness follows by composing the smooth action with the boundary inclusion, and the time-zero identity is the flow identity. Apply the primitive pullback equation on the connected time interval $\mathbb{R}$ to obtain the spatial pullback $e^r j^*\eta$. The time derivative of q is $Z(q)$, and alternation gives $\eta(Z) = d\eta(Z,Z) = 0$, so the radial component vanishes as asserted. At radius zero the differential is $(v,s) \mapsto dj(v) + sZ_D(j(x))$. A defining function ρ vanishes on the boundary, so $d\rho(dj(v)) = 0$, while outwardness gives $d\rho(Z_D) > 0$. A vector in the kernel therefore has $s = 0$, and then $v = 0$ by injectivity of the boundary embedding's differential. For any fixed real r the flow action law gives, near (x,r), $q(y,t) = \Phi^r(q(y,t-r))$. The time map Φ^r has smooth inverse Φ^{-r}, so its differential is invertible. Translation of the radial variable has identity differential. The chain rule transports the injective differential at radius zero to that at (x,r). □

Lemma A.368 (Complete flow from a compact part and radial integral curves)**.** *Let M be a smooth boundaryless finite-dimensional manifold, Z a smooth vector field, and $K \subset M$ compact. Suppose a family of curves $e_b : \mathbb{R} \to M$, indexed by an arbitrary set B, satisfies $e_b'(r) = Z(e_b(r))$ for every $r > 0$, and for each $x \notin K$ there exist $b \in B$ and $r \geq 1$ with $e_b(r) = x$. Then Z has a smooth flow defined for all real times. No complete flow is assumed, and no smoothness, injectivity or manifold structure of the index set B is required.*

Proof. Local existence and smooth dependence for the differential equation of a smooth vector field give, near each point of K, integral curves through every nearby point on a common open time interval about zero. A finite subcover gives $\epsilon_K > 0$ working at every point of K. If K is empty, take $\epsilon_K = 1$. Set $\epsilon = \min(\epsilon_K, 1/2) > 0$. Points of K have integral curves for $|t| < \epsilon$. If $x \notin K$, choose b, r as in the hypothesis. The curve $t \mapsto e_b(t+r)$ starts at x and solves the equation on that same interval, because $r \geq 1$ and $|t| < \epsilon \leq 1/2$ imply $t + r > 0$. Thus every initial point has an integral curve on one uniform interval. Uniqueness follows from local Lipschitz continuity of the smooth field in charts. Extend each curve in successive intervals by restarting at its current endpoint, with increments $\epsilon/2$. The intervals overlap, and uniqueness makes the solutions agree there, so this constructs an integral curve for every positive and negative real time. Uniqueness makes the curve starting at its time-t point equal to its time translate. Hence $\Phi^{s+t}(x) = \Phi^s(\Phi^t(x))$ and $\Phi^0(x) = x$. Local smooth dependence makes the action jointly smooth near time zero. Near any prescribed finite time, cover the compact orbit segment by finitely many local smooth flow neighborhoods and compose those local flows. Their composition agrees with the constructed action by uniqueness and is jointly smooth there. The all-real action therefore is smooth everywhere and solves the specified vector field. □

Lemma A.369 (A complete flow of the constructed stretched field)**.** *The actual canonical field $\widetilde{Z}$ of the constructed primitive η on U admits a jointly smooth flow Φ for all real times. Together these give a complete exact symplectic ambient $(U, \eta, \widetilde{Z}, \Phi)$, without a supplied complete flow.*

Proof. The canonical field is smooth by the symplectic primitive construction. Take the compact set $K = D \cup e(\partial D \times [0,1])$ furnished by the compact-slab lemma. Its exterior points all have the

form $e(x, s)$ with $s \geq 1$. The derived radial ODE says that these end curves solve the actual field equation for every positive radius. The compact-part and radial-curve continuation lemma therefore supplies the smooth all-real-time flow of this very field. The constructed primitive is smooth, its exterior derivative is symplectic, and the field satisfies its contraction equation. Together with that flow, these are precisely the asserted complete exact symplectic ambient data. □

Lemma A.370 (A realized completion of the localized domain)**.** *For every constructed collar c of a Liouville domain presentation, the localized domain on U has a realized completion with ambient (U, η) and identity compact inclusion. Its end map*

$$q(x, s) = \Phi^s(x), \qquad s \in \mathbb{R},$$

is smooth and an immersion for every real radius, agrees with the constructed $e(x, s)$ whenever $s > -\delta$, and has primitive $q^*\eta = e^s\alpha$, where α is the original domain's boundary primitive. Its nonnegative end is embedded, covers the complement of the compact carrier, and is disjoint from that carrier for $s > 0$. This construction applies in every dimension; its output is a completion of the localized presentation.

Proof. Use the complete flow of the actual stretched field just constructed. The identity inclusion is smooth, an embedding and an immersion. The primitive and field preservation assertions on its compact carrier are exactly the compact restriction identities for η and $\widetilde{Z}$. Restricting the primitive identity to the boundary shows that its boundary form is the localized original form, which localization identifies with the original α. Define q by the displayed all-real flow formula. The boundary orbit calculus gives joint smoothness, immersion at every real radius, and the primitive identity with this same boundary form, since the field agrees with the original outward field on the boundary.

For fixed x, the curves $s \mapsto q(x, s)$ and $s \mapsto e(x, s)$ both solve the smooth stretched-field equation on $(-\delta, \infty)$ and agree at zero. Local uniqueness for the smooth differential equation makes the equality set open in this interval. Continuity makes it closed, since the ambient is Hausdorff. It is nonempty, and the interval is connected, so the curves agree on the whole interval. The nonnegative end of q is consequently the embedded nonnegative end of e. The covering identity for e gives the ambient covering by the compact carrier and this end. Its domain-side identity makes every positive-radius end point exterior to the carrier. These facts verify every inclusion, end, primitive and complete-flow property of the realized completion. No positive-dimensional assumption was used; when the boundary is empty the end assertions have empty source. □

Lemma A.371 (Zero-dimensional smooth geometry)**.** *A smooth zero-dimensional manifold is discrete, and every tangent vector and every one-form on it is zero.*

Proof. Each chart takes values in the zero-dimensional real vector space, which has a single point. Its open chart domain is therefore a singleton, so every point is open. Each tangent space is isomorphic to that same zero vector space. A linear functional sends the zero vector to zero, and there are no other tangent vectors. Thus every one-form vanishes. □

Lemma A.372 (A native zero-dimensional Liouville presentation)**.** *Let M be a compact Hausdorff second-countable smooth manifold modeled on the zero-dimensional real vector space, with its given atlas, and let λ be a one-form on M. Then M is finite and admits a Liouville-domain presentation whose extension is this same smooth manifold and whose compact carrier is the whole extension. The original inclusion gives a native diffeomorphism onto that compact carrier, preserves λ and its actual exterior derivative, has bijective differential, and transports the original zero Liouville field*

to the presentation's field. Both the intrinsic boundary and the frontier of the carrier are empty. The exterior derivative of the primitive is symplectic on the entire extension. The construction also applies when M is empty.

Proof. A chart has singleton target because its model space has one point. Its open source containing the chart center is therefore a singleton. Thus M is discrete. Compactness applied to the open singleton cover makes M finite. Use M itself, with its given smooth structure, as the extension and use all of M as the compact carrier. Its frontier is empty, so take the empty smooth manifold as its boundary. The boundary map and all local defining-function requirements have empty source. The carrier is compact by hypothesis.

Every tangent vector is zero, hence every alternating form of positive degree is zero. In particular $\lambda = 0$ and its actual exterior derivative is zero. These forms are smooth, the two-form is closed, and it is nondegenerate: a vector annihilating it must be the only vector, zero. Choose the zero vector field. It is smooth and satisfies the contraction equation; that equation has no other solution because each tangent space has one point. Outwardness has no boundary points at which it must be checked.

The map $x \mapsto x$ into the extension is the identity and is a closed embedding. The map $x \mapsto (x, x \in M)$ to the compact carrier is a bijection with inverse the subtype inclusion. Both spaces have discrete zero-dimensional smooth atlases, so both maps are smooth in their given charts. Their differentials are maps between zero vector spaces and are bijective. The primitive, actual exterior derivative and field identities follow from their vanishing. A zero-dimensional Euclidean model has no boundary; the carrier is the whole extension, so its frontier is empty as asserted. None of these arguments chooses a point of M, and hence they include the empty manifold. □

Lemma A.373 (Completion on the zero-dimensional carrier)**.** *Let D be a zero-dimensional Liouville domain presentation. Its carrier is an open finite subset of the extension manifold. Restricting the extension and its geometric data to that carrier constructs a realized completion whose ambient manifold is the carrier, whose primitive and Liouville field are zero, and whose compact inclusion is the identity. The compact carrier is identified diffeomorphically with the original compact carrier by the open inclusion, preserving the original primitive and field. Its boundary and cylindrical end are empty. No completion is assumed.*

Proof. The extension is discrete by the preceding lemma, so its carrier is open. The open singleton cover of this compact carrier has a finite subcover, so the carrier is finite. Restriction to this open subset preserves the smooth structure and the given primitive and field. All tangent vectors and one-forms vanish. The zero two-form is nondegenerate because the only tangent vector is zero; its exterior derivative is zero. The contraction equation is consequently satisfied everywhere on the restricted extension. The compact inclusion and its inverse between the two finite carriers are smooth: every map between discrete zero-dimensional smooth manifolds is locally a map between singleton charts, hence smooth. The restrictions agree with the original primitive and field pointwise. The identity action $(t, x) \mapsto x$ is smooth and solves the zero field equation for every real t. The boundary is empty by the zero-dimensional domain convention, so the end map has empty source. All end identities, immersion conditions and disjointness conditions are vacuous. The identity compact inclusion preserves the primitive and field and covers the whole ambient manifold. These data give the claimed realized completion. □

Lemma A.374 (Flow and core of the constructed finite completion)**.** *For the completion just constructed, $\Phi^t(x) = x$ for every real time t, and its full intrinsic core is its entire finite ambient carrier. Under the open inclusion into the original extension, the image of this core is exactly the original compact carrier.*

Proof. The constructed flow is the identity action for every real time. Every ambient point belongs to the compact domain after the restriction, and its trajectory therefore remains in that domain at every nonnegative time. The all-forward-time definition of the core gives the assertion. The open inclusion maps the restricted carrier bijectively onto the original carrier, giving the stated image equality. □

Lemma A.375 (The full completed core of a native zero-dimensional source)**.** *For the native zero-dimensional source M and primitive λ of A.372, the carrier completion of its constructed presentation is primitively diffeomorphic to M. The identification retains the original native smooth structure. Its flow fixes every point for every real time, and its full intrinsic core corresponds to all of M. Consequently, for any map $j : M \to B$, the image of this core under the induced map to B is exactly $j(M)$.*

Proof. Apply the carrier completion construction to the Liouville presentation just obtained from M. Its carrier is the entire original extension M. The open restriction to that carrier therefore has points $(x, x \in M)$; forgetting this membership proof is a bijection onto M, inverse to the map supplying that proof. These maps are smooth because the two zero-dimensional manifolds are discrete, and they preserve the primitive because both primitives vanish. The constructed completion has precisely this restricted carrier as its whole ambient manifold. Its flow is the identity action for every real time. Thus every compact point stays in the compact carrier at every nonnegative time, and the full core is the whole carrier. Forgetting the membership proof identifies it with all of M. Composing with j gives image $j(M)$, since every $x \in M$ has exactly that compact representative. The argument also holds for the empty source. □

Lemma A.376 (Native finite ribbons in a presented ambient)**.** *Let D be a two-dimensional Liouville-domain presentation with boundary form α, let M be a compact Hausdorff second-countable native zero-dimensional smooth manifold, let $j : M \to \partial D$ be injective, and let $b : \partial D \to \mathbb{R}_{>0}$ be smooth. There is a witnessed pair whose compact ambient is the localization of D used in its constructed completion, whose boundary and boundary form are the original ∂D and α, and whose ribbon is the native finite source M, under the carrier identification above. Its ribbon map is literally j under that identification and its witness is b. Both completions are constructed. Its embedded full core is exactly $j(M)$.*

Proof. Construct the ambient's localized domain and completion. Localization restricts its extension to an open neighborhood of the compact carrier and retains the same boundary and boundary primitive. Construct the zero-dimensional Liouville presentation of M with zero primitive and apply its carrier completion. The carrier identification makes its extension points precisely the points of M with a membership proof. Compose the inverse identification with j to obtain the ribbon map. The source is discrete, so this map is smooth. It is injective; because the source is compact and the boundary is Hausdorff, the map is a closed embedding. Its differential is injective because every source tangent space is zero. It is therefore a smooth embedding.

Pulling back $b\alpha$ along this map gives a one-form on a zero-dimensional source and hence the zero one-form. This is exactly the ribbon primitive of the constructed Liouville domain. Smoothness and positivity of the unchanged witness are the given hypotheses, so these data define a witnessed pair. The full ribbon core corresponds to all of M by the preceding completion result. Applying the actual ribbon map therefore identifies its embedded image with $j(M)$. □

A.8 Native boundary and ribbon presentations

The following compact-data correspondences preserve the actual maps and one-forms of Definition 2.1. Their conclusions concern presentations of supplied domains and ribbons; they do not assert the existence of the initial or terminal pairs of Target Theorem 4.1.

Lemma A.377 (The native boundary of a compact presentation)**.** *Let M be a smooth $(k+1)$-dimensional manifold with boundary, $k \geq 0$, and let $A \subset X$ be a compact domain presentation of that dimension. Suppose $i : M \to X$ is a smooth injective map with injective differential and*

$$\mathrm{Fr}_X(A) = i(\partial M).$$

There is a diffeomorphism $e_\partial : \partial M \to B_A$ onto the presentation's supplied boundary, satisfying

$$\iota_A \circ e_\partial = i \circ \iota_M,$$

where ι_M and ι_A are the native and supplied boundary inclusions. Thus the boundary map is the restriction of the same i.

Proof. Give $\mathrm{Fr}_X(A)$ the boundary atlas obtained by restricting the domain's extended half-space charts to their zero normal coordinate. The map $f : \partial M \to \mathrm{Fr}_X(A)$ with ambient value $i(\iota_M(x))$ is well defined by the frontier equality. It is injective because i and ι_M are injective, and surjective by that equality. In a target boundary chart its coordinate expression is the tangential projection of the ambient chart applied to $i \circ \iota_M$. This expression is smooth, so f is smooth.

Write $q : \mathrm{Fr}_X(A) \to X$ for the actual frontier inclusion. The chain rule applied to $q \circ f = i \circ \iota_M$ gives $dq \circ df = di \circ d\iota_M$. Both maps on the right are injective, so df is injective. Its source and target have dimension k; therefore it is bijective. The inverse function theorem supplies a smooth local inverse near every point. Since f is bijective, these local inverses are restrictions of its set-theoretic inverse and agree on every overlap. Consequently f is a diffeomorphism.

The supplied boundary inclusion similarly gives a bijection $s : B_A \to \mathrm{Fr}_X(A)$. Its smoothness follows from the same target boundary charts, and its differential is injective because $dq \circ ds = d\iota_A$ is injective. Equal dimensions and the inverse function theorem give its smooth inverse. Set $e_\partial = s^{-1} \circ f$. Then $\iota_A \circ e_\partial = q \circ f = i \circ \iota_M$, as required. For $k = 0$ the same argument applies to the zero-dimensional tangent spaces and boundary charts; empty boundaries give the unique empty map. □

Lemma A.378 (Transport of the actual boundary primitive)**.** *Under the preceding hypotheses, let η be a one-form on X and λ a native one-form on M satisfying $i^*\eta = \lambda$. Write $\alpha_A = \iota_A^*\eta$ and $\alpha_M = \iota_M^*\lambda$. Then*

$$e_\partial^* \alpha_A = \alpha_M.$$

In particular the boundary identification preserves the actual restricted primitive, rather than selecting an unrelated boundary form.

Proof. Pullback composition and the boundary inclusion identity give

$$e_\partial^*(\iota_A^*\eta) = (\iota_A \circ e_\partial)^*\eta = (i \circ \iota_M)^*\eta = \iota_M^*(i^*\eta) = \iota_M^*\lambda.$$

These equalities can also be checked on a tangent vector v at a boundary point x: both sides evaluate to $\eta_{i(\iota_M(x))}(di_{\iota_M(x)} d\iota_{Mx} v)$. □

Lemma A.379 (A native domain and its actual boundary presentation)**.** *Let (A, λ) be a compact positive-dimensional native Liouville domain whose primitive has smooth native extension germs, whose actual exterior derivative is nondegenerate, and whose contraction field is outward. There are a Liouville domain presentation D, a native carrier diffeomorphism $e : A \to D$, its ambient inclusion i, and a boundary diffeomorphism $e_B : \partial A \to B_D$ such that i is smooth with bijective differential, $i(A) = D$, and*

$$i^*\lambda_D = \lambda, \qquad \iota_D \circ e_B = i \circ \iota_A, \qquad e_B^*(\lambda_D|_{B_D}) = \lambda|_{\partial A}.$$

The compact and boundary identifications are therefore constructed on one presentation from the original native Liouville data alone.

Proof. The native Liouville presentation with scalar extensions gives D, i, e, the primitive equality, and bijectivity of di, with i injective and $\mathrm{Fr}(D) = i(\partial A)$. Apply the native boundary-presentation lemma to these same objects. It gives e_B and the literal inclusion square. Applying the boundary primitive-transport lemma to $i^*\lambda_D = \lambda$ gives the last displayed identity. The chosen domain, primitive and carrier map have not changed during either application, so all assertions hold simultaneously on this single presentation. □

Lemma A.380 (A presentation of a native ribbon embedding)**.** *Let (M, μ) be a compact positive-dimensional native Liouville domain of dimension $k+1$, with smooth extension germs for its primitive in native half-space coordinates, nondegenerate $d\mu$, and outward contraction field. Let N be a finite-dimensional smooth boundaryless manifold, let $j : M \to N$ be injective with injective native differential, and suppose j has smooth extension germs across its native boundary. Let η be a one-form on N satisfying $j^*\eta = \mu$. There are a Liouville domain presentation $(F, \widetilde{\mu})$ in a boundaryless manifold U, a native diffeomorphism $e : M \to F$, and a smooth embedding $q : U \to N$ such that, writing i for the ambient composite of e,*

$$q \circ i = j, \qquad i(M) = F, \qquad q^*\eta|_F = \widetilde{\mu}|_F.$$

The last equality holds on the full tangent spaces at every compact carrier point, including its boundary.

Proof. Choose the native Liouville presentation with its universal map extension property. Write $F_0 \subset U_0$ for its compact carrier, $\widetilde{\mu}_0$ for its primitive, and $i_0 : M \to U_0$ for the ambient inclusion of its native diffeomorphism. Then $i_0^*\widetilde{\mu}_0 = \mu$ and di_0 is bijective everywhere. Extend j to a smooth map $g : W \to N$ on an open neighborhood $W \subset U_0$ of F_0, retaining $g \circ i_0 = j$.

Differentiating this identity gives $dg_{i_0(x)} \circ di_{0x} = dj_x$. Since dj_x is injective and di_{0x} is surjective, $dg_{i_0(x)}$ is injective. Indeed, lift any vector in its kernel through di_{0x}; its lift lies in the kernel of dj_x and is therefore zero. The map g is injective on F_0: every point there is $i_0(x)$, and equality of two g-images gives equality of the corresponding j-images and hence of the native points. The compact embedding neighborhood lemma now gives an open $U \subset W$ containing F_0 such that $q = g|_U$ is a smooth embedding.

Restrict the presentation and its primitive to U. The compact carrier, boundary, outward field and native atlas are preserved under this open restriction. Let $i : M \to U$ and $e : M \to F$ be the restricted native inclusion and carrier diffeomorphism. Then $q \circ i = j$. Pullback composition gives $i^*(q^*\eta) = j^*\eta = \mu = i^*\widetilde{\mu}$. For $z = i(x)$ and any $v \in T_zU$, choose $a \in T_xM$ with $di_x a = v$. Evaluating the pullback equality on a proves $(q^*\eta)_z(v) = \widetilde{\mu}_z(v)$. This uses the full tangent isomorphism, so it applies equally to boundary points and proves the claimed carrier equality. □

Lemma A.381 (A witnessed pair retaining its native compact ribbon)**.** *Let D be a Liouville domain presentation of dimension $2n$, with $n \geq 2$, and put $\alpha = \lambda_D|_{\partial D}$. Let (M, μ) be a compact native Liouville domain of dimension $2n-2$ satisfying the preceding primitive extension, nondegeneracy and outwardness hypotheses. Suppose $j : M \to \partial D$ has native smooth extension germs, is injective with injective differential, and let $b : \partial D \to \mathbb{R}_{>0}$ be smooth with*

$$j^*(b\alpha) = \mu.$$

After open localization of the ambient and ribbon presentations, there is a witnessed pair with constructed completions of both domains. The ambient compact domain and its boundary primitive retain their original values; its witness is exactly b. Its ribbon compact carrier is natively diffeomorphic to M, and its ribbon embedding under that identification is exactly j, on every point of M. Its compact primitive pulls back to μ under that same native diffeomorphism, and its embedded ribbon image is exactly $j(M)$.

Proof. Apply the native ribbon embedding presentation to $N = \partial D$ and $\eta = b\alpha$. It gives a presented ribbon F and an embedding q of its extension into ∂D, with $q \circ i = j$ and $q^*(b\alpha) = \widetilde{\mu}$ throughout its compact carrier. Construct Liouville collars for D and F, and the corresponding localized cylindrical completions. The localization inclusions a_D and a_F preserve the original compact primitives, fields and boundary forms. The localized ambient boundary presentation is the same ∂D, with boundary primitive α.

Use $q \circ a_F$ as the localized ribbon embedding and keep b as its witness. On the localized compact ribbon, pullback composition gives

$$(q \circ a_F)^*(b\alpha) = a_F^*(q^*(b\alpha)) = a_F^*\widetilde{\mu},$$

which is precisely the localized ribbon primitive. The compact domains remain Liouville, and both completions were constructed from their collars. Hence these data satisfy the witnessed-pair definition. The native open-restriction carrier diffeomorphism identifies M with the localized ribbon and has ambient composite $a_F^{-1} \circ i$. Its composite with $q \circ a_F$ is $q \circ i = j$. The witnessing function was kept literally equal to b. Pulling the localized ribbon primitive back through this native identification gives $j^*(b\alpha) = \mu$, by the witnessed primitive identity already proved. Surjectivity of the native carrier diffeomorphism gives the embedded image $j(M)$, proving all assertions. □

Lemma A.382 (Native smooth map germs under target composition)**.** *Let M be a native manifold with boundary and let N, P be finite-dimensional smooth boundaryless manifolds. If $f : M \to N$ has actual smooth open extension germs in native source coordinates and $g : N \to P$ is smooth, then $g \circ f$ has such extension germs in those same source coordinates.*

Proof. Continuity follows by composition. Fix a native chart at $x \in M$, a target chart a at $f(x)$, and a target chart c at $g(f(x))$. Write U_a and U_c for the domains of these target charts. Let β be a smooth open extension of the coordinate expression of f, on an open neighborhood U of the source coordinate of x. The equality of the extension germ with that coordinate expression gives $\beta(x) = a(f(x))$, with x here expressed in its source chart. The coordinate map $q = c \circ g \circ a^{-1}$ is smooth on the open set

$$V = a(U_a) \cap (g \circ a^{-1})^{-1}(U_c),$$

which contains $a(f(x))$. By continuity of β, the set $U' = U \cap \beta^{-1}(V)$ is an open neighborhood of the source coordinate of x. The map $q \circ \beta$ is smooth on U'. On a sufficiently small native source-chart neighborhood, f takes values in the domain of a and its coordinate expression agrees with β. There

$q \circ \beta = c \circ g \circ f$, using the inverse chart identity $a^{-1} \circ a = \mathrm{id}$. Thus this map is the required actual open extension germ of $g \circ f$. The source chart was arbitrary, proving the assertion in every native chart. □

Lemma A.383 (The embedded core through a native carrier)**.** *Let P be a witnessed pair with actual ribbon completion C and complete flow Φ, and let $e : M \to F$ be a bijection onto its compact ribbon carrier. Suppose the ribbon embedding q satisfies $q \circ e = j$. Put $I(x) = \iota_C(e(x))$, where ι_C is the compact inclusion into this same completion, and define*

$$K_M = \{x \in M : \Phi^t(I(x)) \in I(M) \text{ for every } t \geq 0\}.$$

Then the embedded full ribbon core of P is exactly $j(K_M)$.

Proof. Surjectivity of e gives $I(M) = \iota_C(F)$. Therefore $x \in K_M$ if and only if $\Phi^t(\iota_C(e(x))) \in \iota_C(F)$ for every $t \geq 0$, which is precisely the condition that $e(x)$ belong to the full ribbon core of P. Thus this core is $e(K_M)$. Applying its actual ribbon embedding gives $q(e(K_M)) = j(K_M)$, as claimed. Every forward time is tested in the given completion; no different flow or singular stop is substituted. □

Lemma A.384 (A presentation of a fully native witnessed pair)**.** *Let (A, λ) and (M, μ) be compact native Liouville domains of dimensions $2n$ and $2n - 2$, respectively, with $n \geq 2$. Their primitives have smooth native extension germs, their exterior derivatives are nondegenerate, and their contraction fields are outward. Write $\alpha_A = \lambda|_{\partial A}$. Suppose $j : M \to \partial A$ has native smooth extension germs, is injective with injective differential, and $b : \partial A \to \mathbb{R}_{>0}$ is smooth with $j^*(b\alpha_A) = \mu$. These data admit a presented witnessed pair with constructed ambient and ribbon completions, native compact-carrier diffeomorphisms, and a derived boundary diffeomorphism $e_B : \partial A \to B_D$. The presented compact ribbon map under its native identification is $e_B \circ j$, and its witness is $b_D = b \circ e_B^{-1}$. In particular*

$$b_D(e_B(x)) = b(x), \qquad e_B^* \alpha_D = \alpha_A, \qquad \iota_D \circ e_B = i_A \circ \iota_A,$$

where i_A is the same native ambient inclusion of the chosen compact presentation, retained under open localization. The native ambient and ribbon carrier diffeomorphisms pull the final compact primitives back to λ and μ, respectively, and the compact embedded ribbon image is $e_B(j(M))$. For K_M computed in the same constructed ribbon completion as in the preceding core transport lemma, the full embedded core is $(e_B \circ j)(K_M)$.

Proof. Choose a native Liouville presentation of (A, λ), with compact carrier inclusion i_A, primitive pullback identity $i_A^* \lambda_D = \lambda$, and the native carrier diffeomorphism. The inclusion is smooth and injective with bijective differential, and its frontier is $i_A(\partial A)$. The boundary-presentation lemma therefore constructs e_B from this same inclusion. Boundary primitive transport gives $e_B^* \alpha_D = \alpha_A$, and its inclusion square is the stated one.

Set $j_D = e_B \circ j$ and $b_D = b \circ e_B^{-1}$. The witness b_D is smooth and positive. Each native extension germ of j can be composed with the globally smooth target map e_B, so j_D has native smooth extension germs. It is injective because both factors are injective. The chain rule gives $dj_D = de_B \circ dj$; the first factor is a tangent isomorphism and the second is injective, so dj_D is injective. Pullback of a scalar multiple and composition give

$$j_D^*(b_D \alpha_D) = j^*\big((b_D \circ e_B) e_B^* \alpha_D\big) = j^*(b\alpha_A) = \mu.$$

The native compact ribbon consumer now constructs a witnessed pair with this presented ambient, ribbon map j_D on its native carrier, witness b_D, and both completions. Its open localizations

preserve the original compact ambient values, boundary primitive and boundary inclusion. Composing the original ambient carrier diffeomorphism with the native open-restriction identification retains i_A as its ambient value. The ribbon consumer similarly preserves every native ribbon point. Thus its actual map on the original ribbon is $e_B \circ j$, and the witness and boundary identities remain the displayed ones. Pullback through the native open-restriction identifications preserves both original compact primitives, and surjectivity of the native ribbon carrier diffeomorphism gives its compact image $e_B(j(M))$. The core transport lemma applied to that same diffeomorphism and actual completion gives the stated full embedded core formula. □

Lemma A.385 (A native surface pair with a finite ribbon)**.** *Let (A, λ) be a compact native two-dimensional Liouville domain whose primitive has smooth native extension germs, with nondegenerate $d\lambda$ and outward contraction field. Let M be a compact native zero-dimensional manifold with any native one-form μ, let $j : M \to \partial A$ be injective, and let $b : \partial A \to \mathbb{R}_{>0}$ be smooth. These data admit a presented witnessed pair with constructed ambient and ribbon completions. Its native ambient carrier and boundary identifications retain the original inclusion and primitive; writing $e_B : \partial A \to B_D$ for the boundary identification, its witness is $b_D = b \circ e_B^{-1}$. Under the native identification of its ribbon with M, the ribbon embedding is $e_B \circ j$, its primitive pulls back to $\mu = 0$, its field is zero, and its full embedded core is exactly $e_B(j(M))$.*

Proof. Choose the native surface Liouville presentation and its native carrier diffeomorphism. Derive the boundary diffeomorphism e_B through the same inclusion, and transport its actual boundary primitive by the boundary primitive lemma. Put $j_D = e_B \circ j$ and $b_D = b \circ e_B^{-1}$. The witness is smooth and positive, and j_D is injective.

A native zero-dimensional manifold is discrete: each chart takes values in the one-point zero-dimensional Euclidean space and therefore has a singleton source neighborhood. Its compactness makes it finite. Thus j_D is smooth. A continuous injection from this compact Hausdorff source into the Hausdorff boundary is a topological embedding. Its derivative is injective because its source tangent space is zero. Every one-form and two-form on M is zero. Consequently the pullback $j_D^*(b_D \alpha_D)$ is its zero primitive; the zero field satisfies the contraction equation, and the empty boundary makes outwardness vacuous.

Apply the native finite ribbon pair construction to this presented ambient, j_D and b_D. It constructs the ambient completion and the finite ribbon completion. The ribbon completion has identity flow for all real times, so its entire carrier is its full core. Its actual embedded image under the native identification is $j_D(M) = e_B(j(M))$. The ambient localization preserves the compact inclusion, boundary primitive and witness values. Composing its native open-restriction carrier diffeomorphism with the original ambient carrier diffeomorphism retains the same native ambient atlas and inclusion, proving all the asserted preservation properties. □

Lemma A.386 (Compact-carrier equality determines exterior derivatives)**.** *Let $D \subset X$ be a compact domain presentation in a boundaryless smooth d-manifold. If smooth one-forms α, β on X agree on D, then $d\alpha$ and $d\beta$ agree on D as two-forms on the full ambient tangent spaces, including at its boundary. Equality outside D is not required.*

Proof. In dimension zero every two-covector is zero, so the assertion holds. For $d \geq 1$, let $i : D \to X$ be the native compact inclusion. The hypothesis gives $i^*\alpha = i^*\beta$ on the native domain. Taking native exterior derivatives and using pullback naturality gives $i^*d\alpha = i^*d\beta$. At every $x \in D$, including its boundary, the native half-space inclusion has bijective differential $di_x : T_xD \to T_{i(x)}X$. For any $v, w \in T_{i(x)}X$, choose their inverse images $a, b \in T_xD$. The pullback equality then evaluates to

$$d\alpha_{i(x)}(v, w) = (i^*d\alpha)_x(a, b) = (i^*d\beta)_x(a, b) = d\beta_{i(x)}(v, w).$$

Thus equality holds on the full ambient tangent space at every point of the carrier. □

Lemma A.387 (Constant-scale exterior transport on the compact carrier)**.** *Let (D,λ) and (D',λ') be Liouville domain presentations, and let h be a prescribed diffeomorphism of their extension manifolds preserving the compact carriers and boundary presentations. For a real constant c, suppose $h^*\lambda' = c\lambda$ on D. Then*

$$h^* d\lambda' = c\, d\lambda \quad \text{on } D$$

on all ambient tangent directions. The primitive scaling identity is required only on the compact carrier.

Proof. The forms $h^*\lambda'$ and $c\lambda$ are smooth and agree on D. The preceding carrier-equality lemma identifies their exterior derivatives there. Exterior naturality gives $d(h^*\lambda') = h^* d\lambda'$. The scalar-multiple formula gives $d(c\lambda) = dc \wedge \lambda + c\, d\lambda = c\, d\lambda$, because c is constant and $dc = 0$. Combining these identities proves the claimed equality at every carrier point, including the boundary. □

Lemma A.388 (The prescribed compact map transports Liouville fields)**.** *Under the hypotheses of the preceding lemma, the prescribed compact presentation map transports the actual Liouville fields:*

$$dh_x(Z_\lambda(x)) = Z_{\lambda'}(h(x)) \quad (x \in D).$$

The assertion also holds in dimension zero and requires no scaling identity outside the compact carrier.

Proof. Carrier preservation puts $h(x)$ in D'. For any tangent vector $w \in T_{h(x)}X'$, bijectivity of dh_x gives $v \in T_xX$ with $dh_x v = w$. The exterior and primitive identities imply

$$\begin{aligned} d\lambda'_{h(x)}(dh_x Z_\lambda(x), w) &= c\, d\lambda_x(Z_\lambda(x), v) \\ &= c\lambda_x(v) = \lambda'_{h(x)}(w). \end{aligned}$$

Thus $dh_x Z_\lambda(x)$ solves the target Liouville contraction equation. Nondegeneracy of $d\lambda'$ on the target carrier makes that solution unique, proving the equality. The argument does not cancel c and applies in dimension zero as well. □

Lemma A.389 (The original compact map has a compatible negative collar)**.** *Let C, C' be realized completions of Liouville domain presentations (D,λ) and (D',λ'). Let h be a prescribed compact presentation diffeomorphism with boundary map h_∂ and $h^*\lambda' = c\lambda$ on D, for a real constant c. There is an open neighborhood U of $\partial D \times [0,\infty)$ in $\partial D \times \mathbb{R}$ such that*

$$\iota(x) = E(y,r), \quad x \in D, \quad (y,r) \in U \quad \Longrightarrow \quad \iota'(h(x)) = E'(h_\partial(y), r).$$

Here ι, ι' are the supplied compact inclusions and E, E' their actual end maps. The compact map is the original h.

Proof. Construct an inward Liouville collar $f(y,r)$ of D of width $\delta > 0$, represented in its original extension space. For $-\delta < r \le 0$ its values lie in D, its initial value is y, and it solves the actual compact Liouville equation. The prescribed field-transport lemma shows that

$$g(y',r) = h(f(h_\partial^{-1}(y'), r))$$

lies in D' and solves its Liouville equation on this same negative strip, with initial value y'. The chain rule and carrier-only field transport suffice; no equation outside either carrier is used. Negative-collar recovery in the two supplied completions gives

$$E(y,r) = \iota(f(y,r)), \qquad E'(h_\partial(y),r) = \iota'(h(f(y,r))) \quad (-\delta < r \le 0).$$

Take $U = \partial D \times (-\delta, \infty)$. If $\iota(x) = E(y,r)$ with $(y,r) \in U$, the completion's positive-end disjointness forces $r \le 0$. The first displayed recovery identity and injectivity of ι give $x = f(y,r)$. The second identity then gives exactly the required square. Empty boundaries cause no exception, since the assertion then has empty source. □

Lemma A.390 (Completed primitive scaling from the compact and end squares)**.** *Let $H : C \to C'$ be a supplied completed presentation diffeomorphism for a compact presentation map h, so that*

$$H \circ \iota = \iota' \circ h, \qquad H(E(y,r)) = E'(h_\partial(y),r) \quad (r \ge 0).$$

If $h^*\lambda' = c\lambda$ on D for a real constant c, then $H^*\widehat{\lambda}' = c\widehat{\lambda}$ on the entire completion.

Proof. On the compact interior, pull the two forms back through ι. The compact square and the completion primitive identities make their pullbacks $h^*\lambda'$ and $c\lambda$, which agree. The compact inclusion is a local diffeomorphism there, so the forms agree on its interior image. Restricting the compact primitive equality to the boundary gives $h_\partial^*\alpha' = c\alpha$. On a positive end, the end square therefore gives

$$E^*H^*\widehat{\lambda}' = e^r h_\partial^*\alpha' = ce^r\alpha = E^*(c\widehat{\lambda}).$$

There is no radial term in either exponential end primitive. The end coordinates are local diffeomorphisms on this open positive end, so this proves form equality there as well. The compact interior image and positive end cover the completion except for its compact boundary. In every signed collar chart, each seam point is a limit of interior or positive-end points. Both forms have continuous coefficients there, so the equality extends to the seam. Thus it holds everywhere; an empty boundary leaves just the compact interior case. □

Lemma A.391 (Completion extension of the prescribed compact presentation)**.** *Let C, C' be arbitrary supplied realized completions of Liouville domain presentations (D,λ) and (D',λ'). Suppose a prescribed compact presentation diffeomorphism h satisfies $h^*\lambda' = c\lambda$ on D, for a constant $c > 0$. There is a completed presentation diffeomorphism $H : C \to C'$ with*

$$H \circ \iota = \iota' \circ h, \qquad H(E(y,r)) = E'(h_\partial(y),r) \quad (r \ge 0), \qquad H^*\widehat{\lambda}' = c\widehat{\lambda}.$$

In particular the strict case $c = 1$ preserves the primitive and the prescribed compact map. An ambient completion-identification step in Definition 2.2 uses this scale-one case; general positive scaling is an auxiliary Liouville-map statement. This result starts with an extension-space compact presentation map; it does not construct a presentation of an arbitrary native compact diffeomorphism.

Proof. The negative-collar square lemma gives an open cylinder neighborhood U containing the seam and the nonnegative end. On the compact interior define $H = \iota' \circ h \circ \iota^{-1}$, and on this cylindrical neighborhood define $H(E(y,r)) = E'(h_\partial(y),r)$. The former is smooth on the interior, where the compact inclusion is a local diffeomorphism. The latter is smooth in the signed end charts. At every overlap point the derived negative-collar square makes these formulas agree. They therefore glue to a smooth map of the whole completion. The inverse formulas use h^{-1} on the target

compact piece and $(h_\partial^{-1}(y'), r)$ on its end. The same overlap square, read backwards, makes them agree, and the compact and end identities show that these two glued maps are inverse. This gives the required completed diffeomorphism and both squares, retaining the original h. The preceding primitive-scaling lemma applied to these squares proves $H^*\widehat{\lambda'} = c\widehat{\lambda}$ everywhere. Setting $c = 1$ gives the stated strict case without changing the compact map. □

Lemma A.392 (Completion of a prescribed locally represented compact map)**.** *Let C, C' be supplied realized completions of Liouville domain presentations (D, λ) and (E, λ'), in any dimension. Let h, k be the prescribed locally represented compact inverse maps of Lemma A.149, and suppose $h^*\lambda' = c\lambda$ on the full compact carrier D, for a constant $c > 0$. There are a smooth diffeomorphism $H : C \to C'$ and a smooth boundary diffeomorphism $h_\partial : \partial D \to \partial E$ satisfying*

$$H^*\widehat{\lambda'} = c\widehat{\lambda}, \qquad H(\iota(x)) = \iota'(h(x)) \quad (x \in D), \qquad h|_{\partial D} = \iota_{\partial E} \circ h_\partial,$$

and

$$H(E_C(y, r)) = E_{C'}(h_\partial(y), r) \quad (r \geq 0).$$

Thus scale one gives a strict completed identification retaining the prescribed compact map, its derived boundary restriction and the same nonnegative end-radius coordinate. This result assumes the compact map's actual local representation; producing it from arbitrary native diffeomorphisms is separate.

Proof. Restrict the source and target extension manifolds to the open neighborhoods U, V of the compact-map presentation lemma. Its presentation equivalence p agrees with h everywhere in U and carries D onto E. The open inclusions have identity differentials in the original tangent coordinates. Hence the restricted primitives and fields have the original values and the restricted primitive identity is $p^*\lambda' = c\lambda$ on D. Restriction preserves the full compact pieces, their boundary forms and outwardness. It also presents the supplied completions on the same completed manifolds with the same primitives, flows and end maps: compose each compact inclusion with its open restriction inclusion, and keep its end map. Their compact images are unchanged, so their covering and overlap identities remain valid.

Apply the prescribed compact-presentation completion lemma to these two restricted presentations and their retained completions. It gives H, the primitive identity and the compact and end squares. Substituting $p = h$ on U turns its compact square into the asserted formula for every original $x \in D$. Its boundary square identifies the induced boundary diffeomorphism with the restriction of the same h, and its end square retains exactly the original end maps and radius. When $c = 1$ its primitive identity is strict, with all these maps and squares unchanged. □

Lemma A.393 (Completion of a prescribed native finite-carrier diffeomorphism)**.** *Let M, N be compact native zero-dimensional smooth manifolds, let α, β be arbitrary native one-forms, and let $h : M \to N$ be a prescribed smooth diffeomorphism. Use the independently constructed native finite-carrier Liouville presentations of M, α and N, β. For any supplied realized completions C, C' of these presentations and every constant $c > 0$, there is a smooth diffeomorphism $H : C \to C'$ with*

$$H^*\widehat{\beta} = c\widehat{\alpha}, \qquad H(\iota(x)) = \iota'(h(x)), \qquad H^{-1}(\iota'(y)) = \iota(h^{-1}(y)) \quad (x \in M,\ y \in N).$$

At scale one the map is primitive-preserving. No local map representation or coupled choice of the two compact presentations is an input. Completions of both original presentations can also be constructed from these native data, giving the strict conclusion without supplied completions. Empty native carriers are allowed.

Proof. The native zero-dimensional Liouville construction takes the original finite manifolds as their own boundaryless extensions, with full carriers and empty boundaries. Every tangent vector and every one-form is zero. Thus the global maps h, h^{-1} are smooth on these extensions, preserve their entire carriers and satisfy both inverse identities. Their differentials are injective because their source tangent spaces are zero. The full compact primitive identity $h^*\beta = c\alpha$ holds for every real c, since both sides vanish.

These very maps therefore meet the locally represented compact-map completion lemma's hypotheses. Apply it to the original supplied completions and the chosen $c > 0$. It gives the asserted H and its prescribed forward compact formula. For $y \in N$, apply that formula to $h^{-1}(y)$ and then apply H^{-1}. The identity $h(h^{-1}(y)) = y$ gives the prescribed inverse compact formula. At $c = 1$ the primitive equality is strict. Every construction uses the independent original native presentations, and the argument remains valid when both carriers are empty.

For the final existence assertion, take the finite carrier completion and its original native primitive-preserving diffeomorphism from each source, supplied by the native zero-dimensional completion lemma. Use that diffeomorphism as the compact inclusion of the original whole native domain. Its compact image is the entire completed manifold, all primitive and field identities hold on zero tangent spaces, and all end conditions have empty source. This constructs a completion of each original presentation while retaining the completed ambient and its flow. Apply the scale-one conclusion to these two constructed completions. □

Lemma A.394 (Native maps on two independently fixed compact presentations)**.** *Let M, N be compact native positive-dimensional Liouville manifolds with boundary of the same dimension. Fix independently two native Liouville presentations, with boundaryless extensions X, Y, compact inclusions $i : M \to X$, $j : N \to Y$, native carrier diffeomorphisms, bijective inclusion differentials and actual local native-coordinate extensions, as in the native Liouville-family construction. Let $f : M \to N$ be continuous and have smooth local native-coordinate extension germs across the source boundary. There is an open neighborhood $W \subset X$ of $i(M)$ and a smooth map $g : W \to Y$ satisfying*

$$g(i(x)) = j(f(x)) \quad (x \in M).$$

Neither presentation depends on f, and no ambient map extension is an input. The map g is only asserted on this neighborhood; no global ambient diffeomorphism or completed identification is asserted by this lemma.

Proof. Fix $x \in M$. At $j(f(x))$ choose a native chart a of N and its smooth ambient coordinate extension R on an open set in Y, with $R \circ j = a$ at native points of that set. Differentiating this equality on the full native tangent space gives $dR_{j(f(x))} \circ dj_{f(x)} = da_{f(x)}$. Both $dj_{f(x)}$ and the chart differential are bijective, so $dR_{j(f(x))}$ is bijective. The inverse function theorem therefore gives a smooth local inverse of R near $a(f(x))$.

Choose a native chart e of M at x and its smooth ambient coordinate extension S near $i(x)$, with $S \circ i = e$. The given map germ extends $a \circ f \circ e^{-1}$ to a smooth map v on an ambient neighborhood of $e(x)$. Compose S, this germ v and the local inverse of R. Shrink its open domain around $i(x)$ so these compositions are defined. Continuity of f and the two native inclusions permits a further shrinking so that all relevant native image points lie in the chosen target chart and inverse neighborhood. For such a native point y, the composite equals $R^{-1}(v(S(i(y)))) = R^{-1}(a(f(y))) = j(f(y))$. It is thus a smooth manifold-valued extension germ of the prescribed map on the source carrier.

The carrier $i(M)$ is compact and i is injective, so $i(x) \mapsto j(f(x))$ is well defined there. Apply the compact manifold-valued germ-extension lemma to the germs just constructed on the independently

fixed X, Y. It produces one smooth map on an open neighborhood of the whole carrier, retaining this prescribed map at every carrier point. This is the required W, g; no off-carrier agreement between the individual coordinate germs was required. □

Lemma A.395 (Full tangent and primitive transport through a native map square)**.** *Let M, N be native manifolds with boundary of the same positive dimension, let X, Y be boundaryless smooth manifolds of that dimension, and let $i : M \to X$, $j : N \to Y$ be smooth maps with bijective full native tangent differentials. Let $h : M \to N$ be a smooth diffeomorphism and let $f : X \to Y$ be differentiable at each $i(x)$, satisfying $f \circ i = j \circ h$. Then*

$$df_{i(x)} \circ di_x = dj_{h(x)} \circ dh_x$$

on every native tangent, including boundary tangents, and $df_{i(x)}$ is injective. If one-forms λ on X and μ on Y satisfy $i^*\lambda = \alpha$, $j^*\mu = \beta$ and $h^*\beta = c\alpha$ for a real constant c, then

$$(f^*\mu)_{i(x)} = c\lambda_{i(x)} \quad (x \in M).$$

The assertions concern the actual f in this prescribed native square; they construct neither the inclusions nor an ambient map extension.

Proof. Differentiate the equality $f \circ i = j \circ h$ at x using the native chain rule. This gives the displayed differential square on the full native tangent space. For a vector v in the kernel of $df_{i(x)}$, choose w with $di_x(w) = v$ by surjectivity of di_x. The square gives $dj_{h(x)}(dh_x(w)) = 0$. Injectivity of $dj_{h(x)}$ and of the derivative of the diffeomorphism h gives $w = 0$, and therefore $v = 0$.

For an arbitrary tangent vector $v = di_x(w)$ at $i(x)$, the same square and the three primitive identities give

$$(f^*\mu)_{i(x)}(v) = \mu_{j(h(x))}(dj_{h(x)}(dh_x(w))) = \beta_{h(x)}(dh_x(w)) = c\alpha_x(w) = c\lambda_{i(x)}(v).$$

Every ambient tangent at $i(x)$ has such a lift w, so this proves equality of the full ambient covectors there. No boundary tangent direction has been omitted. □

Lemma A.396 (Independent presentations of a prescribed native compact diffeomorphism)**.** *Let (M, α) and (N, β) be compact native positive-dimensional Liouville domains of the same dimension, whose primitives have smooth native extension germs. Let $h : M \to N$ be a prescribed smooth diffeomorphism with smooth native-coordinate extension germs, and suppose $h^*\beta = c\alpha$ for a real constant c. The two native primitive constructions independently produce compact Liouville presentations. After restriction of only their extension neighborhoods, they give presentations (D, λ), (E, μ) with native inclusions i, j and a locally represented compact diffeomorphism g satisfying*

$$g(i(x)) = j(h(x)), \qquad g^{-1}(j(y)) = i(h^{-1}(y)), \qquad g^*\mu = c\lambda \text{ on } D.$$

The same inclusions retain the full original native primitives and exterior derivatives, with native carrier diffeomorphisms and bijective full inclusion differentials. No compact presentation or ambient map extension is an input. More generally, any two independently fixed native Liouville presentations with these inclusion and coordinate extension properties admit such simultaneous open restrictions and the same prescribed-map formulas.

Proof. Apply the native primitive-family construction separately to the constant families α and β. It produces two fixed native Liouville presentations with the asserted primitive and derivative identities, native carrier diffeomorphisms, bijective inclusion differentials and native-coordinate

extensions. The fixed-presentation map-extension lemma gives a smooth map f on an open neighborhood W of $i(M)$ with $f \circ i = j \circ h$. Work in this open restriction of the source extension, keeping every native source point. Thus f is an actual map there, including when the carrier is empty.

The native tangent-square lemma proves injectivity of df at every native carrier point. The map is injective on that carrier: from $f(i(x)) = f(i(x'))$ one obtains $j(h(x)) = j(h(x'))$, and the two injectivities give $x = x'$. Apply the compact smooth-equivalence neighborhood lemma to $i(M)$ and f on W. It gives an ambient diffeomorphism on an open source neighborhood onto an open target neighborhood, with forward map exactly f. Its target neighborhood contains all of $j(N)$ because h is surjective and $f(i(M)) = j(N)$. Restrict the two presentations to these neighborhoods. Their carriers remain precisely $i(M), j(N)$. The forward native formula is retained. Applying the ambient inverse to that formula at $h^{-1}(y)$ proves the prescribed native inverse formula.

The tangent-square primitive-transport lemma now applies to these same maps and the full native relation $h^*\beta = c\alpha$. It gives $g^*\mu = c\lambda$ on every carrier tangent. Open inclusions have identity differentials in the original tangent coordinates, so all restrictions retain the native primitive and exterior derivative identities, the carrier diffeomorphisms and bijective inclusion differentials. The ambient map and its smooth inverse, with their carrier formulas, form the required locally represented compact diffeomorphism. If the two presentations are already fixed, begin with their map-extension lemma and perform the same restrictions; none of the subsequent choices modifies their original native inclusions or compact data. □

Lemma A.397 (The prescribed native map on original chosen completions)**.** *Fix two independent native positive-dimensional Liouville presentations of (M, α) and (N, β), with compact domains $(D, \lambda), (E, \mu)$ and native inclusions i, j, as in the preceding lemma. Let $h : M \to N$ be a prescribed smooth diffeomorphism with smooth native-coordinate extension germs and $h^*\beta = c\alpha$, where $c > 0$ is constant. For arbitrary supplied realized completions C, C' of these original presentations, there are smooth diffeomorphisms $H : C \to C'$ and $b : \partial D \to \partial E$ satisfying*

$$H^*\widehat{\mu} = c\widehat{\lambda}, \qquad H(\iota(i(x))) = \iota'(j(h(x))), \qquad H^{-1}(\iota'(j(y))) = \iota(i(h^{-1}(y))),$$

for every native $x \in M$, $y \in N$, together with

$$H(\iota(\iota_{\partial D}(z))) = \iota'(\iota_{\partial E}(b(z))), \qquad H(E_C(z, r)) = E_{C'}(b(z), r) \quad (r \geq 0).$$

The case $c = 1$ is primitive-preserving. All formulas concern the original supplied completed manifolds, primitives, inclusions and end maps. No ambient map extension or compact-map representation is supplied. This assertion still assumes the independently fixed native presentations and their chosen completions.

Proof. Apply the fixed-presentation variant of the native compact-map presentation lemma. It restricts the source extension first to the map-extension neighborhood and then to its diffeomorphism neighborhood, and restricts the target to the corresponding image neighborhood. It gives a compact represented map g retaining both native directions and the full carrier primitive equality. Restrict the original completions along these same open inclusions. As in the local compact-map completion lemma, the restrictions retain the entire original completed manifolds, their primitives and flows, every native compact inclusion and the original end maps.

Apply that completion lemma to g and these restricted completions. It gives H, b, the primitive equality and the end square on the original completed spaces. Its compact formula at the retained native point $i(x)$ and $g(i(x)) = j(h(x))$ give the prescribed native forward formula. Apply H^{-1} to that equality at $h^{-1}(y)$ to obtain the prescribed native inverse formula. At $r = 0$, the end identity is exactly the boundary formula, because each end map there equals its original boundary

compact inclusion. Finally $c = 1$ makes the same primitive identity strict. Thus no alteration of the prescribed native map or of the original chosen completions has occurred. □

Lemma A.398 (Constructed completions of fixed native presentations)**.** *Under the native map and primitive hypotheses of A.397, two independently fixed native presentations have open restrictions, retaining their entire original native carriers and tensors, whose constructed completions admit the prescribed scaled comparison. Its native forward and inverse formulas, boundary square and same-radius end square are those of that lemma on these actual constructed completions. No completion is supplied. The completion domains are the localized presentations, rather than the initial unrestricted extension neighborhoods. Scale one gives the strict comparison.*

Proof. Restrict each extension to its symplectic neighborhood and then to the neighborhood of its constructed flow collar. Both neighborhoods contain the whole compact carrier. Open restriction retains the native inclusion, carrier diffeomorphism, full tangent isomorphism, primitive and exterior derivative identities and native coordinate extension germs. Thus the localized presentations still represent the same native domains and forms. The realized localized completion lemma supplies a completion of each, with its actual stretched primitive, complete flow, compact inclusion and end map.

Apply the fixed-presentation completed-map lemma to these two localized presentations and the two completions just constructed. The native diffeomorphism and its primitive identity are unchanged by restriction. Consequently that lemma gives the prescribed native forward and inverse equations, completed primitive scale, boundary square and nonnegative same-radius end square on these same constructed completions. Setting the scale to one gives the strict assertion. □

Lemma A.399 (Constructed completion comparison from original native data)**.** *Let (M, α) and (N, β) be compact positive-dimensional native Liouville domains: their native primitives have smooth local extension germs, their exterior derivatives are nondegenerate, and their native Liouville fields are strictly outward along the boundary, with smooth local defining-function germs. Let $h : M \to N$ be a prescribed native smooth diffeomorphism with smooth native-coordinate extension germs, and suppose $h^*\beta = c\alpha$ for a constant $c > 0$. There exist independent native presentations with inclusions i, j, completions C, C' of their compact domains, and completed and boundary diffeomorphisms H, b satisfying*

$$H^*\widehat{\mu} = c\widehat{\lambda}, \qquad H(\iota(i(x))) = \iota'(j(h(x))), \qquad H^{-1}(\iota'(j(y))) = \iota(i(h^{-1}(y))),$$

for all native x, y, and

$$H(\iota(\iota_{\partial D}(z))) = \iota'(\iota_{\partial E}(b(z))), \qquad H(E_C(z, r)) = E_{C'}(b(z), r) \quad (r \geq 0).$$

The same H transports the actual completed Liouville fields and all-real-time flows, and the same prescribed h transports the full native cores:

$$dH(Z_C) = Z_{C'} \circ H, \qquad H(\Phi_C^t(w)) = \Phi_{C'}^t(H(w)) \quad (t \in \mathbb{R}),$$

$$i(x) \in \mathrm{Core}_C(D) \iff j(h(x)) \in \mathrm{Core}_{C'}(E).$$

Each presentation retains its original native primitive, full exterior derivative and tangent identification. Neither presentation, completion nor ambient extension of h is an input. Scale one gives a strict comparison. This is an existence assertion for the constructed presentations and completions; it does not identify arbitrary intrinsic completions or assert uniqueness.

Proof. Apply the native Liouville presentation construction independently to the constant primitive families α and β. Nondegeneracy and outwardness supply its hypotheses, and each result retains the original native carrier, primitive, exterior derivative and full tangent identification. These two constructions do not use h or a common choice of extension manifold. Apply the preceding lemma to the resulting presentations and the prescribed h. It constructs localized completions and their scaled comparison, preserving the native forward and inverse, boundary and end equations. Take its localized presentations as the asserted presentations. Open restriction retains every listed native identity, so these completions and equations have precisely the claimed original native data.

Apply the scaled complete-flow transport lemma to the completed primitive equation of this very H. It gives its field identity and flow identity for every real time. The compact images are $\iota(i(M))$ and $\iota'(j(N))$, because each native inclusion parametrizes the entire carrier. The compact square and surjectivity of h give $H(\iota(i(M))) = \iota'(j(N))$. Since H is bijective, membership in these two compact images corresponds in both directions. Apply this equivalence to $\Phi^t_C(\iota(i(x)))$, and use the flow and native compact squares. For each $t \geq 0$, its membership in the source compact image is equivalent to membership of $\Phi^t_{C'}(\iota'(j(h(x))))$ in the target compact image. Requiring these conditions for every nonnegative real time is exactly the two full-core definitions. This proves the native core equivalence without any supplied flow or core transport. Scale one reduces the completed primitive equation to strict preservation. □

Lemma A.400 (Scaled primitives transport the actual complete flows)**.** *Let (X, λ, Z, Φ) and (Y, μ, Z', Ψ) be complete Liouville manifolds with their actual contraction fields and complete flows. If a smooth diffeomorphism $H : X \to Y$ satisfies $H^*\mu = c\lambda$ for a real constant c, then*

$$dH_x Z_x = Z'_{H(x)}, \qquad H(\Phi^t(x)) = \Psi^t(H(x)) \quad (x \in X,\ t \in \mathbb{R}).$$

Neither field nor flow equivariance is an assumption. Positivity of c is needed in the preceding completion construction, but not in these consequences of its primitive equation.

Proof. Exterior naturality and the constant-scalar derivative formula give $H^*d\mu = c\,d\lambda$. Fix x and a target tangent vector w. Since H is a diffeomorphism, write $w = dH_x v$. The contraction equation and the two pullback identities yield

$$d\mu_{H(x)}(dH_x Z_x, w) = c\,d\lambda_x(Z_x, v) = c\lambda_x(v) = \mu_{H(x)}(w).$$

Nondegeneracy of $d\mu$ makes the solution of the target contraction equation unique, so $dH_x Z_x = Z'_{H(x)}$. This argument never divides by c.

The chain rule now shows that $t \mapsto H(\Phi^t(x))$ is an integral curve of the actual smooth field Z'. The curve $t \mapsto \Psi^t(H(x))$ solves the same equation and has the same value at zero. Local uniqueness for a smooth vector field makes their equality set open in $\mathbb{R}$; continuity and the Hausdorff property make it closed. It contains zero, and both complete curves are defined on the connected real line. They therefore agree for every real time, as asserted. □

Lemma A.401 (Strict comparison of chosen completions)**.** *Any two supplied realized completions C, C' of the same Liouville domain presentation have a strict completed comparison $H : C \to C'$ with*

$$H^*\widehat{\lambda}' = \widehat{\lambda}, \qquad H(\iota(x)) = \iota'(x) \quad (x \in D), \qquad H(E(y,r)) = E'(y,r) \quad (r \in \mathbb{R}).$$

Thus it preserves the compact inclusion pointwise and the actual end flow at every real radius, including negative radii.

Proof. Use the identity compact and boundary presentation in the prescribed completion-extension lemma at scale one. Its primitive hypothesis is the identity pullback of λ. The result is a completed comparison H preserving the primitive and the compact inclusion, with the end identity for $r \geq 0$. Exterior naturality and primitive preservation give $H^* d\widehat{\lambda}' = d\widehat{\lambda}$. Evaluating this identity on Z and any tangent vector shows that $dH(Z)$ solves the target contraction equation. Nondegeneracy makes its solution the target Liouville field. Hence the image of each source flow curve solves the target smooth field equation with the corresponding initial point. Uniqueness of integral curves intertwines the complete flows for every real time. Starting at a boundary point and using the compact identity there gives

$$H(E(y,r)) = H(\widehat{\Phi}^r(\iota(y))) = \widehat{\Phi}'^r(\iota'(y)) = E'(y,r)$$

for all real r. This proves the asserted extension of the end square. □

Lemma A.402 (A scaled completed map is determined on the compact domain)**.** *Let C be a realized completion of a compact Liouville domain D, with inclusion ι, and let X be a complete Liouville manifold of the same dimension. Suppose $H_1, H_2 : C \to X$ are smooth diffeomorphisms and*

$$H_i^* \lambda_X = c_i \widehat{\lambda}, \qquad c_i > 0 \quad (i = 1, 2).$$

If $H_1(\iota(x)) = H_2(\iota(x))$ for every $x \in D$, then $H_1 = H_2$. The constants need not be equal. In particular, the strict comparison of two chosen completions and an inclusion marking into a fixed complete ambient are each uniquely determined by their compact restriction.

Proof. Constant primitive scaling intertwines the actual complete Liouville flows by A.400. Every point of C is either in $\iota(D)$ or has the form $E(y,r) = \widehat{\Phi}^r(\iota(y))$ for some $y \in \partial D$ and $r \geq 0$. The two maps agree in the first case by hypothesis. In the second case, since y lies in D, flow intertwining gives

$$H_1(E(y,r)) = \Phi_X^r(H_1(\iota(y))) = \Phi_X^r(H_2(\iota(y))) = H_2(E(y,r)).$$

Thus the maps agree everywhere. This also covers empty boundary and dimension zero, when the compact inclusion alone covers the completion. Apply the result with $c_1 = c_2 = 1$ to two strict comparisons or two inclusion markings having the same compact restriction. □

Lemma A.403 (Unique scaled extension of a prescribed compact presentation)**.** *Let D, D' be represented compact Liouville domains, let h be a smooth diffeomorphism of their extension spaces preserving their compact carriers, with a boundary diffeomorphism h_∂ satisfying the boundary inclusion square. Suppose $c > 0$ is constant and $h^*\lambda' = c\lambda$ on the compact carrier of D. For any two realized completions C, C' there is a unique smooth diffeomorphism $H : C \to C'$ such that*

$$H^* \widehat{\lambda}' = c\widehat{\lambda}, \qquad H(\iota(x)) = \iota'(h(x)) \quad (x \in D).$$

It necessarily satisfies $H(E(y,r)) = E'(h_\partial(y), r)$ for every real r.

Proof. The prescribed compact-presentation extension supplies a completed map with the primitive and compact identities. Any two such maps agree on $\iota(D)$, so compact-domain uniqueness proves they are equal. For a boundary point y, the boundary square gives $H(\iota(y)) = \iota'(h_\partial(y))$. Flow intertwining and the end flow formula, valid for all real times, then give

$$H(E(y,r)) = \Phi'^r(H(\iota(y))) = \Phi'^r(\iota'(h_\partial(y))) = E'(h_\partial(y), r).$$

This derives the end formula from the primitive and compact identities. □

Lemma A.404 (The full core is independent of the chosen completion)**.** *For two realized completions of the same Liouville domain presentation, their full cores are equal as subsets of the original compact carrier.*

Proof. Let H be the strict comparison above. For any $x \in D$ and any real t, flow intertwining and the compact identity give $H(\widehat{\Phi}^t(\iota(x))) = \widehat{\Phi}'^t(\iota'(x))$. Moreover $H(\iota(D)) = \iota'(D)$, with inverse equality, because $H(\iota(y)) = \iota'(y)$ for every compact point y and H is bijective. Consequently the source flow point lies in $\iota(D)$ if and only if the target flow point lies in $\iota'(D)$. Requiring this for every real $t \geq 0$ is exactly the full-core condition in the two completions. Thus the same compact points satisfy it in both, proving equality of the original carrier subsets. □

Lemma A.405 (Transporting an inclusion marking between chosen completions)**.** *Let C, C' be realized completions of one domain presentation, let X be a specified complete Liouville manifold, and let $i : D \to X$ be a fixed compact inclusion. A supplied primitive-preserving inclusion marking $q' : C' \to X$ extending i induces one $q : C \to X$ extending the same i. The induced marking retains the Liouville-flow equation on the original boundary for every real radius.*

Proof. Let $H : C \to C'$ be the strict comparison and put $q = q' \circ H$. It is a diffeomorphism and $q^*\lambda_X = H^*(q'^*\lambda_X) = H^*\widehat{\lambda}' = \widehat{\lambda}$, so it preserves the primitive. For each compact point, $q(\iota(x)) = q'(\iota'(x)) = i(x)$. For every original boundary point y and every real r, the all-real end identity gives

$$q(E(y,r)) = q'(E'(y,r)) = \Phi_X^r(q'(\iota'(y))) = \Phi_X^r(q(\iota(y))).$$

These are the required inclusion and flow-induced identities for the same supplied compact map i. The marking q' remains an input; this argument asserts no target-specific marking existence. □

Lemma A.406 (An original slice marking on the retained represented completion)**.** *Let C be a realized completion of a compact Liouville domain D, with completed primitive η and compact inclusion ι_C. Let X be a smooth manifold with a one-form L, let M be a set with a bijection $e : M \to D$, and let $\nu : M \to X$ be its original compact inclusion. Suppose an actual smooth diffeomorphism $S : X \to C$ satisfies*

$$S^*\eta = L, \qquad S(\nu(x)) = \iota_C(e(x)).$$

Let (Z, λ_Z) be a specified complete Liouville manifold and let $q : X \to Z$ be a supplied smooth diffeomorphism with

$$q^*\lambda_Z = L, \qquad q(\nu(x)) = i(x) \quad (x \in M).$$

Then $q_C = q \circ S^{-1}$ is an actual inclusion marking of C onto Z, with compact inclusion $i \circ e^{-1}$, and $q_C \circ S = q$ on all of X. It retains the native compact equation $q_C(\iota_C(e(x))) = i(x)$ and satisfies

$$q_C(E_C(y,r)) = \Phi_Z^r\big(q_C(\iota_C(y))\big) \quad (y \in \partial D,\ r \in \mathbb{R}).$$

The original marking q is a hypothesis. This transports its prescribed compact inclusion through the actual strict comparison S; it does not infer a marking from an orbit predicate or assert geometric marking existence.

Proof. The composite q_C is a smooth diffeomorphism. Pullback composition and the given primitive identities imply

$$q_C^*\lambda_Z = (S^{-1})^*(q^*\lambda_Z) = (S^{-1})^*L = (S^{-1})^*(S^*\eta) = \eta.$$

For every $z \in X$, cancellation gives $q_C(S(z)) = q(S^{-1}(S(z))) = q(z)$, retaining the entire original map. For $x \in M$, the original compact square gives $S^{-1}(\iota_C(e(x))) = \nu(x)$, whence $q_C(\iota_C(e(x))) =$

$q(\nu(x)) = i(x)$. Every point of D is $e(x)$ for a unique x, so this is the compact inclusion law for $i \circ e^{-1}$ on the entire domain.

Primitive preservation makes q_C intertwine the complete Liouville flows. The complete boundary-orbit formula is $E_C(y, r) = \Phi_C^r(\iota_C(y))$ for every real r. Apply q_C and flow intertwining to obtain the displayed all-real flow-induced equation. Primitive preservation, the compact inclusion law and this flow equation give exactly the required inclusion marking. □

Lemma A.407 (Original markings on the same retained ribbon completion slices)**.** *Fix the actual native ribbon slice comparisons of one pair deformation, as in the positive-dimensional or finite-ribbon slice comparison lemmas. At a parameter t, write Y_t for the original ribbon completion, K_t^F for the ribbon completion of this same pair, $\nu_t : M \to Y_t$ for the original compact insertion and $e_{M,t}$ for the retained native ribbon carrier map. Let $S_t^F : Y_t \to K_t^F$ be the retained strict slice comparison, with original primitive L_t^F. Every supplied smooth diffeomorphism $q_F : Y_t \to Z_F$ onto a specified complete Liouville manifold with*

$$q_F^* \lambda_{Z_F} = L_t^F, \qquad q_F(\nu_t(x)) = i_F(x)$$

gives an actual inclusion marking $m_t^F = q_F \circ (S_t^F)^{-1} : K_t^F \to Z_F$, with compact inclusion $i_F \circ e_{M,t}^{-1}$. It recovers the entire original map by $m_t^F \circ S_t^F = q_F$ and retains every native compact value. In the positive-dimensional case, its actual native boundary map b_t^F satisfies

$$m_t^F(E_{K_t^F}(b_t^F(y), r)) = \Phi_{Z_F}^r(i_F(y)) \quad (y \in \partial M,\ r \in \mathbb{R}).$$

For a finite ribbon the primitive hypothesis is automatic, and the marking intertwines the actual complete flows at all points and real times. Empty finite ribbons are allowed. No new slice comparisons, native maps or original marking are selected.

Proof. The retained comparison satisfies $(S_t^F)^* \eta_t^F = L_t^F$ and $S_t^F(\nu_t(x)) = \iota_{K_t^F}(e_{M,t}(x))$. Apply the original slice marking transport lemma to these exact equations, the given q_F and its compact inclusion. Its marking is exactly $q_F \circ (S_t^F)^{-1}$, and it gives both entire-original-map recovery and the native compact equation. For a positive-dimensional native boundary point y, the retained boundary inclusion square identifies the compact boundary point $b_t^F(y)$ with the compact carrier point $e_{M,t}(y)$. Substitute this square and the native compact equation into the marking's all-real flow-induced law; this yields the displayed boundary trajectory. For a finite ribbon, all one-forms are zero, so the original primitive equation requires no hypothesis. Primitive preservation of the constructed marking gives complete flow intertwining on the entire completion. The construction uses only the retained carrier bijections and their equations and therefore also applies when the finite ribbon is empty. □

Lemma A.408 (Extending a native inclusion into a prescribed smooth manifold)**.** *Let (M, μ) be a positive-dimensional compact native Liouville domain with a fixed presentation P, native inclusion $a : M \to \widetilde{D}$, carrier $a(M)$, bijective full inclusion differential and smooth native coordinate extensions. Let X be a specified smooth boundaryless manifold, let η be a one-form on X, and let $i : M \to X$ have smooth native-chart extension germs. Suppose i is injective, di_x is injective for every x, and $i^*\eta = \mu = a^*\lambda_P$. There are an open neighborhood U of $a(M)$ and a smooth embedding $J : U \to X$ with*

$$J \circ a = i, \qquad dJ_{a(x)} \circ da_x = di_x, \qquad (J^*\eta)_{a(x)} = (\lambda_P)_{a(x)} \quad (x \in M).$$

Neither a complete field nor a completion of X is required.

Proof. Apply the fixed-presentation native map-extension lemma to the constant family i. Its smooth map on a neighborhood of $[0,1] \times a(M)$ restricts at parameter zero to a smooth map f on an open neighborhood of $a(M)$, with $f \circ a = i$. The chain rule gives $df_{a(x)} \circ da_x = di_x$. If $df_{a(x)}(v) = 0$, write $v = da_x(w)$ using surjectivity of da_x. Injectivity of di_x gives $w = 0$, hence $v = 0$. Thus df is injective on the carrier. Also f is injective there, because a bijects M with $a(M)$ and i is injective. The compact injective-immersion neighborhood lemma restricts f to a smooth embedding $J : U \to X$ containing that whole carrier. Its value and differential squares remain unchanged. For any $v = da_x(w)$, the two native primitive identities imply

$$(J^*\eta)_{a(x)}(v) = \eta_{i(x)}(di_x(w)) = \mu_x(w) = (\lambda_P)_{a(x)}(da_x(w)).$$

Every full tangent vector has this form, proving the primitive identity including at native boundary points. No property of η away from $i(M)$ has been used. □

Lemma A.409 (Primitive inclusion determines the full carrier field)**.** *Let (D, λ) be a represented compact Liouville domain with smooth extension manifold $\widetilde{D}$, and let X be a specified smooth boundaryless manifold of the same dimension. Let η be a smooth one-form on X and $J : \widetilde{D} \to X$ a smooth embedding satisfying $J^*\eta = \lambda$ on the entire compact carrier D. For every $z \in D$,*

$$(J^*d\eta)_z = (d\lambda)_z, \qquad (d\eta)_{J(z)} \text{ is nondegenerate}, \qquad dJ_z(Z_\lambda(z)) = Z_\eta(J(z)).$$

Here Z_η is the canonical contraction vector, characterized where $d\eta$ is nondegenerate by $\iota_{Z_\eta} d\eta = \eta$. The identities use every tangent direction, including the boundary. Nondegeneracy away from $J(D)$ and completeness are not hypotheses.

Proof. The two smooth forms $J^*\eta$ and λ agree on the full compact carrier. The compact-carrier exterior equality lemma gives equality of their exterior derivatives there. Exterior naturality gives $d(J^*\eta) = J^*(d\eta)$, proving the first identity. At z, the embedding differential $A = dJ_z$ is injective. Equal finite dimensions make it bijective. If an ambient tangent vector $v = A(u)$ satisfies $(d\eta)_{J(z)}(v, w) = 0$ for every ambient w, then testing against $w = A(q)$ and using the first identity gives $(d\lambda)_z(u, q) = 0$ for every q. Source nondegeneracy gives $u = 0$, hence $v = 0$. Thus $d\eta$ is nondegenerate at $J(z)$. For every ambient $w = A(q)$, the same differential identity and the primitive equality imply

$$(d\eta)_{J(z)}(A(Z_\lambda(z)), w) = (d\lambda)_z(Z_\lambda(z), q) = \lambda_z(q) = \eta_{J(z)}(w).$$

Nondegeneracy identifies this vector with $Z_\eta(J(z))$, proving the field equation. These arguments apply also in dimension zero. □

Lemma A.410 (A prescribed native inclusion with its derived carrier field)**.** *Under the hypotheses of A.408, assume X has the same dimension as M and η is smooth. The embedding J can be chosen with all the value, differential and primitive identities of that lemma, and on its whole original compact carrier it satisfies*

$$J^*d\eta = d\lambda_P, \qquad d\eta \text{ is nondegenerate on } J(a(M)), \qquad dJ(Z_{\lambda_P}) = Z_\eta \circ J.$$

These are identities on the actual restricted presentation and the specified ambient manifold. Neither a supplied ambient field nor a complete flow, end equation or completion is required.

Proof. Construct U, J by the prescribed smooth inclusion extension lemma. Restrict P to U; its whole original compact carrier, primitive and Liouville field are retained. The primitive square from that lemma holds on every carrier tangent. Apply the preceding carrier-field lemma to this

same restricted presentation, the smooth form η and the constructed smooth embedding J. Equal dimensions and the carrier primitive identity give exactly the three displayed laws. The prescribed native value and differential squares are those of the same J, so they remain part of the conclusion. □

Lemma A.411 (Extending a native inclusion into its prescribed ambient)**.** *Let M be a positive-dimensional compact native Liouville domain and let P be a fixed presentation, with native inclusion $a : M \to \widetilde{D}$, carrier $a(M)$, bijective full differential and smooth extensions of native coordinate functions. Let (X, λ_X) be a specified complete Liouville ambient. Suppose $i : M \to X$ has smooth target-valued germs across native boundary charts, is injective and has injective differential, and satisfies $i^*\lambda_X = \mu$, where $a^*\lambda_P = \mu$. There are an open neighborhood U of $a(M)$ in $\widetilde{D}$ and a smooth embedding $J : U \to X$ such that*

$$J(a(x)) = i(x), \qquad dJ_{a(x)} \circ da_x = di_x, \qquad (J^*\lambda_X)_{a(x)} = (\lambda_P)_{a(x)}.$$

The prescribed ambient X and every original inclusion point are retained.

Proof. Regard the target-valued spatial germs as constant parameter families. The fixed-presentation native map-extension theorem supplies a smooth partial map on a neighborhood of $[0,1] \times a(M)$, agreeing with i at every native point. Restricting to one parameter and an open spatial neighborhood gives a smooth f with $f \circ a = i$. Differentiating this equality proves $df_{a(x)} \circ da_x = di_x$. Since da_x is surjective and di_x injective, $df_{a(x)}$ is injective. Also f is injective on $a(M)$, since i is injective and a bijects M with that carrier. The compact injective-immersion neighborhood theorem now restricts f to a smooth embedding J on an open neighborhood U of $a(M)$. The point and differential identities survive this restriction. For a full tangent vector v at $a(x)$, choose w with $da_x(w) = v$. Then

$$\lambda_X(dJ(v)) = \lambda_X(di_x(w)) = \mu_x(w) = \lambda_P(da_x(w)) = \lambda_P(v).$$

Thus the equality holds on every full tangent vector, including boundary points, and establishes the primitive identity on the whole carrier. □

Lemma A.412 (Native compact maps on the original completed endpoint domains)**.** *Fix two positive-dimensional native Liouville presentations with domains D, D' and supplied completions C, C'. A prescribed native diffeomorphism h with smooth native extension germs and constant primitive scale has a compact-domain map represented by total forward and inverse extensions on the original extension spaces, each smooth on a neighborhood of its compact carrier. These extensions retain every original native value and the primitive scale on all compact tangent vectors. Neither original domain is replaced and no nonempty-carrier hypothesis is required.*

Proof. First, a point z in an extension space maps into its completion under the completion inclusion. Coverage puts its image either in the compact image or on a cylindrical trajectory. The first case provides a compact point; the second provides a boundary base point, which is also compact. Thus a nonempty extension space has nonempty compact carrier. Composing a compact point with the native carrier inverse and h supplies a point in the other compact carrier. The inverse construction uses h^{-1}. These choices supply forward and reverse fallback values without an inhabitedness assumption, even when both native carriers are empty.

The native fixed-presentation map theorem supplies a compact diffeomorphism on open restrictions of the two original extension spaces. Compose its maps with the target open inclusions. On the images of its open source and target neighborhoods retain these maps; outside those neighborhoods use the fallback values just constructed. Open-subspace smoothness proves the resulting

total functions smooth on the same original neighborhoods. Successive restrictions may be removed in turn by this construction. Their native compact squares are unchanged, hence they preserve the compact carriers and their two compositions are identity there. Differentiating the native forward square, using the native tangent bijections and the invertibility of dh, gives injectivity of the full forward differential at every compact point. Finally pull back the native primitive scaling identity through that same square and lift each ambient tangent vector through the native inclusion differential. This gives the full original-carrier primitive equation, including the boundary. These are all the required properties of the original-domain compact map. □

Lemma A.413 (Finite native maps on the original compact domains)**.** *Let D, D' be fixed zero-dimensional compact Liouville domain presentations with supplied completions. Let $a : M \to D$ and $b : N \to D'$ be native carrier diffeomorphisms, and let $h : M \to N$ be a prescribed native diffeomorphism. There are forward and inverse maps on the original extension spaces, smooth on neighborhoods of the compact carriers, that restrict to $b \circ h \circ a^{-1}$ and $a \circ h^{-1} \circ b^{-1}$, respectively. They represent a compact domain diffeomorphism on exactly D, D' and satisfy the full compact primitive scaling equation for every real constant c. Empty carriers are allowed; neither the domains nor their completions are replaced.*

Proof. For z in the source compact carrier set $g(z) = b(h(a^{-1}(z)))$, and on the target compact carrier set $k(z') = a(h^{-1}(b^{-1}(z')))$. To extend these functions to the original extension spaces, completion coverage supplies a compact source point whenever the source extension has a point: its completed image is either the image of a compact point or an end point with a compact boundary base point. Applying a^{-1}, h and b to such a point supplies a target compact fallback value. The reverse construction supplies a source compact fallback value for each target extension point. Use these fallback values outside the respective carriers. If an extension space is empty its map has empty source, so no choice of a point is required there.

The zero-dimensional extension manifolds are discrete, so both extended functions are smooth on their whole original spaces. Their carrier restrictions have the asserted formulas, map each carrier onto the other, and are inverse by cancellation of a, b, h with their inverses. Their differentials are injective because the source tangent spaces are zero. Every one-form on a zero-dimensional extension is zero; thus $g^*\lambda_{D'} = c\lambda_D$ holds for every real c, in particular on the whole compact carrier. These properties give the required compact domain diffeomorphism on the original domains. □

Lemma A.414 (The completed finite map retains its original compact map)**.** *Under the hypotheses of the preceding finite compact-map lemma, its specific original compact map g has a smooth completed diffeomorphism $H : C \to C'$ on the same supplied completions. For every real c, it satisfies $H^*\widehat{\lambda}_{D'} = c\widehat{\lambda}_D$, both native compact squares, and*

$$H(\iota(z)) = \iota'(g(z)), \qquad H^{-1}(\iota'(z')) = \iota(g^{-1}(z')) \quad (z \in D,\ z' \in D').$$

It intertwines the actual completed flows for every real time.

Proof. Apply the fixed zero-dimensional endpoint completion lemma with a, b, h, c to obtain H with both native compact squares, primitive scaling and all-real-time flow transport. For $z \in D$, surjectivity of a gives $z = a(x)$ for some $x \in M$. The native forward square gives $H(\iota(z)) = \iota'(b(h(x)))$. The specific original compact map has $g(a(x)) = b(h(x))$, so this is the asserted forward equality. For $z' \in D'$ choose $y \in N$ with $z' = b(y)$; the native inverse square and $g^{-1}(b(y)) = a(h^{-1}(y))$ give the inverse equality. All other conclusions belong to the same completed map already constructed. Surjectivity has empty source obligations when the carriers are empty, so this proof covers that case as well. □

Lemma A.415 (Native compact maps on prescribed positive-dimensional domains)**.** *Let D, D' be fixed positive-dimensional compact Liouville domains with supplied completions, and let $a : M \to D$, $b : N \to D'$ be native carrier diffeomorphisms retaining the native primitive pullbacks and smooth native-coordinate extension germs. A prescribed native diffeomorphism $h : M \to N$ with smooth native extension germs and constant primitive scale $c \in \mathbb{R}$ has a compact diffeomorphism g on exactly D, D' with $g(a(x)) = b(h(x))$, $g^{-1}(b(y)) = a(h^{-1}(y))$, and the full compact primitive equation $g^*\lambda_{D'} = c\lambda_D$. Neither supplied completion nor original domain is replaced.*

Proof. The exact-endpoint presentation lemma constructs native Liouville presentations on the already fixed D, D' from a, b, their primitive identities and coordinate germs. Apply the original-domain native compact-map lemma to those presentations, the originally supplied completions and h. Its two native squares and full compact primitive equation give precisely the stated conclusion. □

Lemma A.416 (Boundary representatives and their actual trajectories)**.** *Suppose a smooth completed diffeomorphism Ψ has boundary coordinates ψ, s satisfying $\Psi(E(x,t)) = E'(\psi(x), t + s(x))$ for all real t. For boundary points x, y,*

$$\psi(x) = y \quad \Longleftrightarrow \quad \exists r \in \mathbb{R} : \ \Psi(E(x,0)) = E'(y,r).$$

Proof. If $\psi(x) = y$, take $r = s(x)$ and evaluate the end equation at time zero. Conversely, compare that zero-time equation with the asserted equality at r. The end parametrization is injective at all real radii: flow both points forward by a common time making their radii nonnegative, use the nonnegative-end embedding's injectivity, and cancel the common radial increment. Equality of the first end coordinates then gives $\psi(x) = y$. □

Lemma A.417 (A full ribbon orbit square assembles a pair identification)**.** *Let P, Q be fixed presented witnessed pairs, Ψ a specified primitive-preserving smooth diffeomorphism between their original ambient completions, and $g : F_P \to F_Q$ a compact domain diffeomorphism with $g^*\mu_Q = c\mu_P$ for one constant $c > 0$. Suppose that, for every $z \in F_P$, there is a real r such that*

$$\Psi(E_P(j_P(z), 0)) = E_Q(j_Q(g(z)), r).$$

There is a permitted completion identification whose ambient map, compact ribbon map and scale are exactly Ψ, g, c. Its boundary map is induced by Ψ, and satisfies the full boundary–ribbon square. There is no restriction on the real radial shifts.

Proof. The induced boundary-map lemma supplies a smooth boundary diffeomorphism ψ, a smooth shift s, the all-real end equation and $\psi^*\alpha_Q = e^{-s}\alpha_P$. Fix $z \in F_P$ and choose the r in the hypothesis. The end equation at $j_P(z)$ and time zero gives

$$E_Q(\psi(j_P(z)), s(j_P(z))) = E_Q(j_Q(g(z)), r).$$

The boundary-representative equivalence applied to the supplied orbit equality gives $\psi(j_P(z)) = j_Q(g(z))$. This holds on the whole compact ribbon. The specified Ψ, g, c, their primitive equations, and the derived boundary data and square satisfy every condition of a completion identification. No endpoint or specified map has changed. □

Lemma A.418 (Assembling a native pair identification on the original endpoints)**.** *Let P, Q be fixed presented witnessed pairs with positive-dimensional ribbons, and let Ψ be a specified primitive-preserving smooth diffeomorphism between their original ambient completions. Let $a : M \to F_P$,*

$b : N \to F_Q$ be native carrier diffeomorphisms on their exact original ribbon domains, retaining native primitive pullbacks and smooth native-coordinate extension germs. Suppose a prescribed native diffeomorphism $h : M \to N$ has smooth native extension germs, scales these ribbon primitives by one constant $c > 0$, and satisfies

$$\forall x \in M \quad \exists r \in \mathbb{R} : \qquad \Psi(E_P(j_P(a(x)), 0)) = E_Q(j_Q(b(h(x))), r),$$

where E_P, E_Q are the specified ambient end maps. These data yield a permitted completion identification whose ambient map is exactly Ψ, whose ribbon scale is exactly c, and whose compact ribbon map and inverse retain every native value of h, h^{-1} on the original domains. Its boundary map is induced by Ψ and satisfies the full boundary–ribbon square. The real shifts in the hypothesis are unrestricted.

Proof. Construct native Liouville presentations on the exact original ribbon domains using a, b and their primitive and coordinate data. The original-domain compact-map lemma gives a compact diffeomorphism g with its inverse, the native forward and inverse squares, and $g^*\mu_Q = c\mu_P$ on every compact tangent vector.

For an arbitrary compact ribbon point z, surjectivity of a gives $z = a(x)$ for some $x \in M$. The native forward square for g changes the supplied orbit equation to $\Psi(E_P(j_P(z), 0)) = E_Q(j_Q(g(z)), r)$ with the same real r. Apply the preceding full ribbon orbit-square lemma to this g and the supplied Ψ, c. It produces the asserted identification and retains those three choices. Thus the native forward and inverse squares for its ribbon map are exactly those of the constructed g. □

Lemma A.419 (A strict compact native pair square gives a completion identification)**.** *Let P, Q be specified witnessed pairs with positive-dimensional ribbons and their specified completions. Let native carrier diffeomorphisms $e_A : A \to D_P$, $e'_A : A' \to D_Q$, $e_M : M \to F_P$, $e'_M : M' \to F_Q$ retain the full compact primitive pullbacks $\alpha, \alpha', \mu, \mu'$ and native coordinate extension germs. Let $e_B : \partial A \to \partial D_P$ and $e'_B : \partial A' \to \partial D_Q$ be smooth boundary diffeomorphisms whose inclusion composites equal those of e_A, e'_A. Suppose maps $j : M \to \partial A$, $j' : M' \to \partial A'$ retain the literal whole-ribbon squares $j_P \circ e_M = e_B \circ j$ and $j_Q \circ e'_M = e'_B \circ j'$. Let $h_A : A \to A'$ and $h_M : M \to M'$ be native diffeomorphisms with smooth extension germs, satisfying*

$$h_A^*\alpha' = \alpha, \qquad h_M^*\mu' = c\mu \quad (c > 0), \qquad h_A \circ i_{\partial A} \circ j = i_{\partial A'} \circ j' \circ h_M.$$

There is a permitted completion identification $I : P \to Q$ of scale c retaining both original completed endpoints, the native ambient map h_A and its inverse, the native ribbon map h_M and its inverse, and the full boundary equation

$$\psi(e_B(j(x))) = e'_B(j'(h_M(x))) \qquad (x \in M).$$

The compact-square hypothesis derives the ribbon trajectory condition at radius zero. This statement concerns strict maps of the compact domains; arbitrary moved boundary sections and their radial shifts are separate data.

Proof. Apply the fixed-native-endpoint map theorem to the prescribed ambient domains and completions, with h_A and scale one. It constructs a primitive-preserving diffeomorphism Ψ between those very completions, retaining the native ambient forward and inverse squares and the same-radius end square. In particular its boundary section shift is zero.

For each $x \in M$, the seam equations of the two specified completions, the whole-ribbon squares, the boundary inclusion squares, the native ambient square for Ψ and the given native pair equation give

$$\begin{aligned}\Psi(E_P(j_P(e_M(x)),0)) &= \Psi(\iota_P(e_A(i_{\partial A}(j(x)))))\\ &= \iota_Q(e'_A(h_A(i_{\partial A}(j(x)))))\\ &= \iota_Q(e'_A(i_{\partial A'}(j'(h_M(x)))))\\ &= E_Q(j_Q(e'_M(h_M(x))),0).\end{aligned}$$

Thus the required full ribbon trajectory condition holds with radius exactly zero at every compact ribbon point. Apply the native pair-identification assembly theorem with this Ψ, the given h_M, c and the retained ribbon primitive and coordinate germs. It constructs I on the exact given pairs with ambient map Ψ, scale c and the forward and inverse native ribbon squares. Its defining full ribbon square, followed by those native squares, is the displayed boundary equation. The ambient forward and inverse native squares remain those of Ψ. Every trajectory condition used in this application follows from the literal compact pair equation. □

Lemma A.420 (Assembling a finite native ribbon identification)**.** *Let P, Q be fixed presented surface pairs, with zero-dimensional ribbons. Supply a primitive-preserving smooth ambient completed diffeomorphism Ψ, native carrier diffeomorphisms $a : M \to F_P$, $b : N \to F_Q$, a native diffeomorphism $h : M \to N$, and a constant $c > 0$. Suppose, for every $x \in M$, that*

$$\exists r \in \mathbb{R} : \qquad \Psi(E_P(j_P(a(x)),0)) = E_Q(j_Q(b(h(x))),r).$$

There is a permitted completion identification with ambient map exactly Ψ, scale exactly c, and both prescribed native compact ribbon squares. It retains the original endpoint domains and completions, and includes empty ribbons and unrestricted radial shifts. No ribbon primitive scaling hypothesis is needed.

Proof. Apply the original-domain finite native map lemma to the fixed ribbon domains, their supplied completions, a, b, h, c. Its compact map g retains both native squares and satisfies the full primitive scaling equation because the ribbon one-forms vanish. Given $z \in F_P$, choose $x \in M$ with $a(x) = z$. The supplied orbit equation and the forward square $g(a(x)) = b(h(x))$ give the full compact orbit-square hypothesis. The orbit-square assembly lemma therefore supplies a completion identification retaining Ψ, g, c. In particular its ribbon map and inverse have the original native values. If the ribbon is empty the orbit-square argument has empty source, while all ambient boundary data are still derived from the given Ψ; the same conclusion follows. □

Lemma A.421 (A strict native ambient comparison with its prescribed boundary square)**.** *Let (A, λ) be a positive-dimensional compact native Liouville domain with native primitive extension germs, nondegenerate exterior derivative and outward boundary defining-function germs. Let D, E be independently prescribed compact presentations, with native carrier diffeomorphisms $e_D : A \to D$, $e_E : A \to E$, retaining the full pullbacks of λ and native coordinate extension germs. Supply their actual completions C, K and smooth boundary identifications $b_D : \partial A \to \partial D$, $b_E : \partial A \to \partial E$ satisfying*

$$\iota_{\partial D}(b_D(x)) = e_D(x), \qquad \iota_{\partial E}(b_E(x)) = e_E(x) \quad (x \in \partial A).$$

There is a primitive-preserving diffeomorphism $H : C \to K$ with

$$H(\iota_C(e_D(x))) = \iota_K(e_E(x)), \qquad H^{-1}(\iota_K(e_E(x))) = \iota_C(e_D(x)) \quad (x \in A),$$

$$H(E_C(b_D(x), r)) = E_K(b_E(x), r) \quad (x \in \partial A,\ r \in \mathbb{R}).$$

Neither a completed comparison nor its boundary or orbit square is supplied as an input.

Proof. The two native carrier diffeomorphisms, full primitive identities and coordinate germs make D, E presentations of the same (A, λ). Each canonical native boundary map has the displayed inclusion square. Injectivity of the respective boundary inclusion therefore identifies it with the supplied b_D or b_E, point by point. The primitive and outward germs construct one native inward collar. Apply the strict inward-collar completion comparison theorem to that collar and these two independently supplied completions. Its compact and inverse squares are the stated native point equations, and strict primitive preservation intertwines their all-real Liouville flows. The end maps are the flows from the actual boundary sections. Substituting the identified boundary maps into the resulting all-real end square proves the last equation. □

Lemma A.422 (A completion identification of two presentations of the same whole native pair)**.** *Let P, Q be witnessed pairs of rank $n \geq 2$ whose ambient domains and actual completions present the same native (A, λ) as in Lemma A.421. Write $e_{A,P}, e_{A,Q}$ and b_P, b_Q for those native ambient and boundary identifications. Let M be one compact native ribbon with primitive μ with native extension germs, nondegenerate exterior derivative and outward boundary defining-function germs, and a native map $j : M \to \partial A$. Supply native carrier diffeomorphisms $e_{M,P} : M \to F_P$, $e_{M,Q} : M \to F_Q$ with full ribbon primitive pullbacks equal to μ and native coordinate extension germs. Assume the whole ribbon squares*

$$j_P \circ e_{M,P} = b_P \circ j, \qquad j_Q \circ e_{M,Q} = b_Q \circ j.$$

Supply one native witness $b : \partial A \to \mathbb{R}$ with $b_P^{\mathrm{wit}} \circ b_P = b = b_Q^{\mathrm{wit}} \circ b_Q$, where $b_P^{\mathrm{wit}}, b_Q^{\mathrm{wit}}$ are the original pair witnesses. There is a permitted completion identification $I : P \to Q$ with scale one, retaining every native compact ambient point and every native compact ribbon point in both directions. Its boundary map is

$$\psi = b_Q \circ b_P^{-1},$$

and its ambient map satisfies the same-radius end square for every real radius. Its radial shift is identically zero, and $b_Q^{\mathrm{wit}} \circ \psi = b_P^{\mathrm{wit}}$ on the entire ambient boundary. Its full boundary–ribbon square is derived from these native data; it is not an assumed orbit comparison.

Proof. The preceding ambient comparison lemma constructs a strict H with both ambient native point squares and its all-real boundary orbit square. For $x \in M$, the two supplied whole ribbon squares turn the radius-zero instance of that orbit square into

$$H(E_P(j_P(e_{M,P}(x)), 0)) = E_Q(j_Q(e_{M,Q}(x)), 0).$$

The native ribbon primitive and outward germs construct an inward collar for (M, μ). Apply the strict native completion comparison to its two prescribed ribbon presentations and their actual completions. This gives a strict ribbon completed diffeomorphism G retaining every native compact ribbon point in both directions.

The completed compact inclusions are smooth embeddings between boundaryless manifolds of equal dimension, hence have open images and smooth inverses on those images. Restrict the source extension to the open inverse image under G of the target inclusion image; it contains the entire compact source ribbon by the native point square. On that open neighborhood compose G with the source inclusion and the target inclusion's smooth inverse. Perform the inverse construction using G^{-1} and restrict to these corresponding open neighborhoods. The resulting maps are inverse smooth diffeomorphisms, carrying the entire compact source ribbon onto the entire target ribbon and retaining both native point squares. Strict primitive preservation by G and the two inclusion primitive identities give $g^*\mu_Q = \mu_P$ on every compact ribbon tangent.

Surjectivity of $e_{M,P}$ turns the displayed native orbit equation into the whole compact ribbon orbit square for this very g and H, with shift zero. The orbit-square pair assembly lemma therefore constructs I with ambient map H, compact ribbon map g and scale one. Its ambient and ribbon squares are exactly those just established.

The boundary map induced by H must equal $b_Q \circ b_P^{-1}$. Indeed, at a point $b_P(y)$ the constructed same-radius end equation and the defining end equation for the induced boundary map give equal points of E_Q. Injectivity of that end parametrization identifies their boundary coordinates with $b_Q(y)$. Surjectivity of b_P then proves the asserted formula at every boundary point. Equality of the radius coordinates in that same injective end map gives $r + s(b_P(y)) = r$, so the shift vanishes everywhere. The common witness squares give $b_Q^{\mathrm{wit}}(\psi(b_P(y))) = b_Q^{\mathrm{wit}}(b_Q(y)) = b(y) = b_P^{\mathrm{wit}}(b_P(y))$; surjectivity of b_P proves witness preservation on the entire boundary. The full boundary–ribbon square is precisely the one supplied by the assembly lemma using these same maps. □

Lemma A.423 (One native deformation identified with every prescribed whole-pair slice)**.** *Assume the native joint primitive, ribbon and positive witness hypotheses of Lemma A.182, with positive-dimensional ribbon. At every parameter independently supply a witnessed pair O_t on its original compact domains and completions, with native ambient, ribbon and boundary identifications retaining the same full primitives λ_t, μ_t, whole ribbon j_t and witness b_t. Require the exact compact boundary and ribbon squares and native ambient and ribbon coordinate extension germs of Lemma A.422.*

There exists one pair deformation Q retaining all primitive, boundary, whole ribbon, witness, full-core, coordinate-germ and jointly smooth native presentation conclusions of the coherent native construction. For every t there is a permitted completion identification $I_t : O_t \to Q_t$ with scale one, zero shift, both exact native compact ambient and ribbon squares, the exact prescribed boundary square, whole-boundary witness preservation and the all-real same-radius ambient end square. Every identification refers to this same Q and its retained native maps. The original endpoint pairs are joined by the literal three-step chain

$$O_0 \xrightarrow{I_0} Q_0 \xrightarrow{Q} Q_1 \xrightarrow{I_1^{-1}} O_1.$$

Its endpoint identifications have scale one and preserve their whole boundary witnesses. The slice assertion is pointwise in t; it does not assert joint completed-family regularity of the I_t or identify a prescribed original completed atlas or grading family.

Proof. Apply the coherent native pair-family construction once to the given joint native data, retaining its produced Q, its native carrier and boundary maps, the jointly smooth ambient and ribbon relabelings, and their literal compact presentation squares. Retain its native coordinate extensions on these same slices, including at both interval endpoints. Its full primitive, whole ribbon and witness identities hold for exactly $\lambda_t, \mu_t, j_t, b_t$; its embedded core is computed in the ribbon completion of this same Q_t.

Fix t. Freezing the supplied joint primitive germs gives the native slice germs. The original pairs O_t and the already selected Q_t thus satisfy the same-native whole-pair identification theorem: the primitive, coordinate, compact boundary, ribbon and witness squares on the Q_t side are precisely those retained by the construction, and the corresponding original squares are hypotheses. That theorem gives I_t with all the stated exact native and end equations. Doing this for every t changes neither Q nor any of its native presentation maps or coherence data. Choose the identifications at $t = 0$ and $t = 1$. Append the first identification, the deformation Q and the reverse of the second identification, in that order. Their consecutive endpoint tuples agree literally with Q_0 and Q_1, so this is a permitted finite chain of length three. The scale and witness conclusions are exactly those

of the selected endpoint identifications. No parameter regularity of the chosen I_t is needed for the pointwise or finite-chain conclusions. □

Lemma A.424 (A completion identification of two presentations of the same finite native ribbon)**.** *The preceding whole-pair conclusion holds for witnessed surface pairs and one native zero-dimensional ribbon M, including $M = \varnothing$. The ambient primitive, native coordinate, boundary and common witness hypotheses are unchanged. Supply its two native ribbon carrier diffeomorphisms and both whole ribbon squares. No ribbon primitive equality or ribbon coordinate extension hypothesis is required. The resulting identification has scale one, boundary map $b_Q \circ b_P^{-1}$, zero radial shift, exact boundary witness preservation, both native ambient and ribbon point squares, and its ambient all-real same-radius end square.*

Proof. Construct the ambient strict comparison H from the same native ambient data by the ambient comparison lemma. At every finite native ribbon point the full ribbon squares give exactly the radius-zero orbit equation displayed in the preceding proof. The original-domain finite native map lemma applied to the identity of M supplies the compact ribbon diffeomorphism g retaining both native point squares. Every ribbon primitive vanishes on its zero tangent spaces, so $g^*\mu_Q = \mu_P$ automatically. Surjectivity of the source native carrier map turns the native radius-zero orbit equation into the whole compact ribbon orbit square for H, g. Apply the orbit-square pair assembly lemma with these maps and scale one. It retains H and both native ribbon point squares. The same injectivity of the target end map identifies its induced boundary map with $b_Q \circ b_P^{-1}$ and its shift with zero at every ambient boundary point. The two common witness squares then prove witness preservation there, exactly by substituting $b_P(y)$ and using surjectivity of b_P. If M is empty, the ribbon equations have empty source; the ambient comparison and its boundary equation still follow from the unchanged ambient hypotheses. □

Lemma A.425 (One finite-ribbon deformation identified with every prescribed whole-pair slice)**.** *Assume the native joint ambient primitive, finite ribbon map and positive witness hypotheses of Lemma A.183. The native finite ribbon primitive family μ_t is arbitrary. At every parameter independently supply a witnessed surface pair O_t on its original domains and completions, with native ambient and finite ribbon carrier diffeomorphisms and a native boundary identification. Require the ambient full primitive pullback and coordinate extension germs, actual compact boundary inclusion square, whole ribbon square and common witness values of Lemma A.424. No finite ribbon primitive or coordinate extension condition is required.*

There is one pair deformation Q retaining all primitive, boundary, whole ribbon, witness, coordinate-germ and jointly smooth native presentation conclusions of the coherent finite-ribbon construction. Its full embedded core at each parameter is the whole native ribbon image $e_{B,t}(j_t(M))$. For every t there is a permitted identification $I_t : O_t \to Q_t$ with scale one, zero shift, exact native compact ambient and ribbon squares in both directions, the prescribed boundary square, whole-boundary witness preservation and the all-real same-radius ambient end square. The exact original endpoint tuples are joined by

$$O_0 \xrightarrow{I_0} Q_0 \xrightarrow{Q} Q_1 \xrightarrow{I_1^{-1}} O_1,$$

a three-step chain whose endpoint identifications have scale one and preserve their full boundary witnesses. The slice identifications are pointwise; no joint original completed atlas or grading correspondence is asserted. Empty ribbons are allowed.

Proof. Apply the coherent native finite-ribbon producer once, retaining its same Q, native ambient, ribbon and boundary maps, jointly smooth native relabelings, exact compact presentation

squares and ambient coordinate extensions. Its primitives, full ribbon and positive witness are the prescribed $\lambda_t, \mu_t, j_t, b_t$. Since the completed finite ribbon flow is stationary, its full embedded core is exactly the retained whole native ribbon image at every parameter.

Fix t. The slice of the ambient primitive germs and its outward and nondegeneracy assumptions give the native ambient hypotheses of the same-native finite pair identification theorem. Its native compact, boundary, ribbon and witness squares on the Q_t side are the retained producer conclusions; the corresponding O_t squares are hypotheses. Apply that theorem to obtain I_t with every stated exact square, scale, shift and witness identity. This application requires no finite ribbon primitive or coordinate hypotheses and changes none of the selected Q or its coherence data.

Choose these identifications at zero and one. Append I_0, the same deformation Q, and the reverse of I_1. Their endpoint tuples match literally, so these are three consecutive permitted steps from O_0 to O_1. The chosen endpoint identifications retain scale one and their full witness identities. All finite ribbon statements have empty source when M is empty, while the ambient constructions and endpoint equations are unchanged, so the same argument covers that case. □

Lemma A.426 (Completion of a prescribed compact Liouville map)**.** *Let (D, λ) and (D', λ') be compact native Liouville domains and let $h : D \to D'$ be a prescribed native diffeomorphism with*

$$h^*\lambda' = c\lambda, \qquad c > 0 \text{ constant.}$$

Smoothness here includes real open local native map extensions across the boundary, as in the source smoothness convention. Let C, C' be their Liouville completions, with compact inclusions ι, ι', completed primitives $\widehat{\lambda}, \widehat{\lambda}'$, and end maps E, E'. There is a unique smooth diffeomorphism $H : C \to C'$ satisfying

$$H \circ \iota = \iota' \circ h, \qquad H(E(x, r)) = E'(h_\partial(x), r) \quad (r \geq 0), \qquad H^*\widehat{\lambda}' = c\widehat{\lambda},$$

where $h_\partial : \partial D \to \partial D'$ is the restriction of h. In particular, when $c = 1$ this is a primitive-preserving completed identification retaining the prescribed compact map. Taking h to be the identity compares two completions of the same compact Liouville domain while preserving its compact inclusion and cylindrical ends.

Proof. The positive-dimensional existence construction retains arbitrary supplied completions and the prescribed native map. Its finite-carrier counterpart provides the zero-dimensional construction. Compact-map uniqueness applies in all dimensions: surjectivity of the native carrier identifications turns agreement on native points into agreement on the whole compact carrier. The following argument gives the stated formulas and uniqueness.

First h preserves the intrinsic interior and boundary. At an interior point its full tangent derivative is invertible, since the derivative of its smooth inverse is an inverse. In native charts a smooth local extension of h therefore has invertible derivative. The inverse function theorem maps a sufficiently small open interior neighborhood onto an ambient open neighborhood contained in the target half-space. Its image point must have positive normal coordinate: an ambient open neighborhood of a zero normal coordinate contains points outside that half-space. Applying the same argument to h^{-1} proves equality of the interior images, hence equality of boundary images. The native boundary charts express h_∂ as the tangential projection of the smooth ambient coordinate expression of h restricted to the zero normal hyperplane. This proves smoothness; the same argument for h^{-1} gives its smooth inverse.

On the intrinsic interior, ordinary exterior naturality and differentiation of $h^*\lambda' = c\lambda$ give $h^*d\lambda' = c\, d\lambda$. In every native half-space chart both sides have continuous coefficients, because the map and forms have smooth extension germs. The positive normal-coordinate interior is dense in

that half-space. Hence equality of the coefficients on the interior extends to the zero-coordinate boundary. This proves the exterior identity everywhere without requiring arbitrary open extensions to obey the scaling identity outside the source domain. Write Z, Z' for the two contraction fields. For each $x \in D$ and $v \in T_xD$ the two primitive identities give

$$d\lambda'_{h(x)}(dh_x Z_x, dh_x v) = c\, d\lambda_x(Z_x, v) = c\lambda_x(v) = \lambda'_{h(x)}(dh_x v).$$

Surjectivity of dh_x and nondegeneracy of $d\lambda'$ therefore give $dh_x Z_x = Z'_{h(x)}$. The constant c cancels in this equation.

Choose the native inward flow collars of the two compact domains, and shrink their widths to one common $\delta > 0$. Smooth integral-curve uniqueness and the field identity show

$$h(\mathrm{Fl}^r_Z(x)) = \mathrm{Fl}^r_{Z'}(h_\partial(x)) \quad (-\delta < r \le 0).$$

Indeed both sides start at $h_\partial(x)$ and solve the same smooth field equation. Local uniqueness makes their equality set open in the time interval, continuity makes it closed, and connectedness gives the whole interval. The negative-collar recovery identity identifies these curves in the two completions with their signed end coordinates.

Define H on the compact piece by $\iota' \circ h \circ \iota^{-1}$, and on the nonnegative end by $E(x, r) \mapsto E'(h_\partial(x), r)$. At radius zero these definitions agree by the compact boundary inclusion identities. In the signed collar chart across the seam, the preceding negative-time identity shows that this combined map is exactly $(x, r) \mapsto (h_\partial(x), r)$ on both sides. Hence it is smooth across the seam as well as on the compact interior and positive end. Using h^{-1} and h_∂^{-1} constructs a smooth inverse with the two analogous squares. On every compact or end point, the two compositions are the identity, so H is a diffeomorphism.

Put $\alpha = \lambda|_{\partial D}$ and $\alpha' = \lambda'|_{\partial D'}$. Boundary restriction of the compact primitive identity gives $h_\partial^* \alpha' = c\alpha$. On the end, the primitive identity for H is consequently

$$(h_\partial \times \mathrm{id})^*(e^r \alpha') = e^r h_\partial^* \alpha' = c e^r \alpha.$$

On the compact piece it is the prescribed equality $h^*\lambda' = c\lambda$. These pieces and their signed seam collars cover the completion, giving the stated equality everywhere. They also determine H at every point, proving uniqueness. If the boundary is empty, the compact piece alone covers its completion and the same compact formula suffices. In dimension zero the finite-carrier completion has zero primitive and no end, so the given finite native diffeomorphism already gives this construction. □

Lemma A.427 (Contraction of the actual outward cylindrical primitive)**.** *Let E be a real normed vector space and let β be a one-form on E, differentiable at $y \in E$. Define the one-form η on $\mathbb{R} \times E$ by*

$$\eta_{(r,y)}(s, v) = e^r \beta_y(v).$$

Its actual exterior derivative at (r, y) satisfies

$$(d\eta)_{(r,y)}((1, 0), (s, v)) = \eta_{(r,y)}(s, v).$$

Proof. The directional derivative of the coefficients of η in direction $(1, 0)$ is η itself, because the spatial argument is fixed and $\frac{d}{dr} e^r = e^r$. In every direction (s, v), the derivative of $\eta(1, 0)$ is zero, because this scalar function is identically zero. Exterior differentiation of a one-form is the antisymmetrization of these two coefficient derivatives. Therefore

$$(d\eta)((1, 0), (s, v)) = D\eta(1, 0)(s, v) - D\eta(s, v)(1, 0) = \eta(s, v) - 0.$$

Differentiability of β at y supplies the coefficient derivative used in this calculation. □

Lemma A.428 (The radial contraction equation in a prescribed ambient)**.** *Let X be a specified smooth boundaryless manifold with a smooth one-form λ, let $U \subset \mathbb{R}\times E$ be open, and let $F : U \to X$ be smooth. Suppose $F^*\lambda = \eta$ on U, where η is the cylindrical primitive of the preceding lemma. At $p = (r, y) \in U$, suppose β is differentiable at y and dF_p is surjective. Then the actual vector $dF_p(1, 0)$ satisfies*

$$(d\lambda)_{F(p)}(dF_p(1, 0), w) = \lambda_{F(p)}(w) \quad (w \in T_{F(p)}X).$$

Proof. Given w, choose $v \in \mathbb{R} \times E$ with $dF_p(v) = w$. Exterior naturality and the equality $F^*\lambda = \eta$ on an open neighborhood of p give

$$(d\lambda)_{F(p)}(dF_p(1, 0), w) = (F^*d\lambda)_p((1, 0), v) = (d\eta)_p((1, 0), v) = \eta_p(v) = \lambda_{F(p)}(w).$$

The middle contraction identity is the preceding lemma; the last identity is the original full primitive pullback evaluated at v. Surjectivity ensures that every ambient tangent is tested. □

Lemma A.429 (The actual canonical field is outward radial)**.** *Under the hypotheses of A.428, if $d\lambda$ is nondegenerate at $F(p)$, its canonical contraction field satisfies*

$$Z_\lambda(F(p)) = dF_p(1, 0).$$

Proof. The radial equation states that the vector on the right solves $\iota_v d\lambda = \lambda$. Nondegeneracy makes this solution unique: if v, v' solve it, then $d\lambda(v - v', w) = 0$ for every w, whence $v - v' = 0$. The canonical contraction field is that unique solution. □

Lemma A.430 (Radial coordinate curves solve the actual ambient equation)**.** *Let $a \in \mathbb{R}$, let $V \subset E$ be open, and let F be smooth on $(a, \infty) \times V$ with values in the specified ambient X. Suppose λ is smooth, β is smooth on V, $F^*\lambda = e^r\beta$ on this product, dF is surjective there, and $d\lambda$ is nondegenerate at every image point. For every $y \in V$, the actual curve $\gamma_y(r) = F(r, y)$ satisfies*

$$\gamma_y'(r) = Z_\lambda(\gamma_y(r)) \quad (r > a).$$

Proof. At each $r > a$, the chain rule applied to $r \mapsto (r, y)$ gives $\gamma_y'(r) = dF_{(r,y)}(1, 0)$. Smoothness of β near y implies its differentiability there. All hypotheses of the canonical radial-field lemma therefore hold at (r, y), and that lemma identifies this derivative with $Z_\lambda(F(r, y))$. This is exactly the asserted integral-curve equation throughout (a, ∞). □

Lemma A.431 (Complete flow from prescribed cylindrical primitive charts)**.** *Let X be a specified smooth boundaryless manifold and λ a smooth one-form whose actual exterior derivative is nondegenerate everywhere. For an arbitrary index set I, suppose $\delta_i > 0$, $V_i \subset E$ is open, β_i is smooth on V_i, and $F_i : (-\delta_i, \infty) \times V_i \to X$ is smooth with surjective differential and full primitive identity*

$$F_i^*\lambda = e^r\beta_i.$$

Suppose $K \subset X$ is compact and every $x \notin K$ has a representation $x = F_i(r, y)$ with $y \in V_i$ and $r \geq 1$. Then the canonical Liouville field of this same λ has a jointly smooth flow for every real time on this same manifold X. The chart and compact-coverage hypotheses are explicit inputs; no complete flow, radial equation or completed ambient is an input.

Proof. In each tangent chart, contraction with $d\lambda$ is an invertible matrix whose entries are smooth. Its inverse is the adjugate divided by its nowhere-zero determinant, so the solution of $\iota_Z d\lambda = \lambda$ has smooth coefficients. Thus the canonical field Z_λ is smooth everywhere. Index curves by all pairs (i, y) with $y \in V_i$, and put $e_{i,y}(r) = F_i(r, y)$. The radial integral-curve lemma gives

$e'_{i,y}(r) = Z_\lambda(e_{i,y}(r))$ whenever $r > -\delta_i$. Because $\delta_i > 0$, this holds throughout $r > 0$. The supplied compact-coverage assertion says precisely that these curves cover $X \setminus K$ at radii at least one. The compact radial-end complete-flow lemma applied to X, Z_λ, K and this curve family therefore constructs the asserted all-real smooth flow on X. No change of the ambient manifold or primitive occurs. □

Definition A.432 (The half-infinite native collar attachment)**.** *Let M, B be topological spaces, let $\delta > 0$, and let $C : [0, \delta) \times B \to M$ be a prescribed inward collar. Put $A = (-\infty, \delta) \times B$. Normalize the disjoint union $M \sqcup A$ by retaining $x \in M$, sending (t, b) with $t \geq 0$ to $C(t, b) \in M$, and retaining (t, b) with $t < 0$ in a separate negative-cylinder summand. Let X_C be the quotient identifying two points exactly when their normalizations agree, with its quotient topology. Write $o : M \to X_C$ and $q : A \to X_C$ for its two quotient maps.*

Lemma A.433 (Literal points and overlap in the half-infinite attachment)**.** *In the attachment of the preceding definition, o is injective and*

$$o(x) = q(t, b) \quad \Longleftrightarrow \quad 0 \leq t < \delta \text{ and } C(t, b) = x.$$

If C is injective, then q is injective. The images of o and q cover X_C.

Proof. The normalization of an original point is that same original point. Thus equal quotient classes of original points have equal original representatives, proving injectivity of o. An original representative and a cylinder representative have the same normalization exactly when the cylinder time is nonnegative and its collar value equals the original point. This proves the displayed equivalence. For two cylinder representatives with nonnegative times, equality of normalizations gives equality of collar values; injectivity of C gives equality of representatives. With negative times their normalizations retain the representatives themselves. A negative and a nonnegative representative normalize to different summands and cannot agree. This proves injectivity of q whenever C is injective. Every quotient class has a representative in one of the two summands, proving coverage. □

Lemma A.434 (The topology of the half-infinite attachment)**.** *If C is an open embedding, then q is an open embedding and o is an embedding. If M and B are Hausdorff, then X_C is Hausdorff and o is a closed embedding.*

Proof. Both quotient maps are continuous. A subset W of X_C is open exactly when $o^{-1}(W)$ and $q^{-1}(W)$ are open in their respective summands. For open $V \subset A$, injectivity of q gives $q^{-1}(q(V)) = V$. The overlap formula gives

$$o^{-1}(q(V)) = C\big(V \cap ([0, \delta) \times B)\big),$$

where the intersection is relatively open in the half collar. The open embedding C maps it to an open set in M. Thus $q(V)$ is open, so the continuous injective map q is an open embedding. Define $f : X_C \to M$ by $f(o(x)) = x$ and $f(q(t, b)) = C(\max(t, 0), b)$. The normalization makes these values well defined. Both expressions are continuous on the summands, so the quotient property gives continuity of f. The identity $f \circ o = \mathrm{id}$ shows that the inverse of o on its image is continuous, hence o is an embedding.

For distinct $z, w \in X_C$ with $f(z) \neq f(w)$, disjoint open neighborhoods of their fold images in M pull back to disjoint open neighborhoods of z, w. If their fold images agree, both points lie in the cylinder image: when one point is $o(x)$ and the other $q(t, b)$, the fold equality gives

$x = C(\max(t,0),b)$, hence $o(x) = q(\max(t,0),b)$; two original points with the same fold would be equal. The Hausdorff space A therefore separates the two cylinder representatives; the open embedding q gives disjoint open neighborhoods in X_C. Thus X_C is Hausdorff. Finally $o(M)$ is the equalizer of the continuous maps $o \circ f$ and id_{X_C}, since $o(f(z)) = z$ holds exactly on $o(M)$. The diagonal of the Hausdorff space X_C is closed, so this equalizer is closed. Therefore o is a closed embedding. □

Definition A.435 (The original varying-collar quotient)**.** *Let T, M, B be topological spaces, let $\delta > 0$, and let $C_a : [0,\delta) \times B \to M$ be a prescribed map for each $a \in T$. Define the total collar by*

$$C_{\mathrm{tot}}(s,a,b) = (a, C_a(s,b)).$$

The shared attachment X is the original half-infinite quotient for this total collar: it identifies (s,a,b) with $(a,C_a(s,b))$ when $0 \le s < \delta$, retaining all of $T \times M$ and all strictly negative cylinder points. Write o_{tot} for the original-point map and $q_{\mathrm{tot}}(a,s,b)$ for the cylinder map. Its parameter projection $\pi : X \to T$ sends either representative to a. This is well defined because normalization preserves the parameter in both its original and negative-cylinder summands. When M has its native manifold-with-boundary structure and $B = \partial M$, write $i_{\mathrm{tot}}(a,x) = o_{\mathrm{tot}}(a,x)$ for the original interior map on $T \times \operatorname{int} M$.

Lemma A.436 (The shared quotient retains every original completed slice)**.** *The parameter projection of the preceding definition is continuous. For every $a \in T$, the original slice attachment X_{C_a} is canonically in bijection with the full fiber $\pi^{-1}(a)$, with*

$$o_a(x) \longmapsto o_{\mathrm{tot}}(a,x), \qquad q_a(s,b) \longmapsto q_{\mathrm{tot}}(a,s,b).$$

The resulting map from X_{C_a} into X is continuous. This is a bijection of the actual quotient fibers; no homeomorphism or smooth equivalence of those fibers is asserted.

Proof. On the original summand, the parameter function is the first product projection; on the cylinder summand it is the boundary parameter projection. Both are continuous. They agree on the quotient relation, so the defining property of the quotient topology makes π continuous.

Every slice quotient class has a unique normalized representative, either an original point $x \in M$ or a strictly negative cylinder point (s,b). Send these representatives respectively to (a,x) and (s,a,b) in the normalized total quotient. Distinct representatives remain distinct, so this defines an injective map into $\pi^{-1}(a)$. Conversely every class in that fiber has an original or strictly negative normalized representative whose parameter is necessarily a. Removing that parameter gives a normalized slice representative. These insertion and removal maps are mutually inverse. For a nonnegative cylinder point, its slice normalization is $C_a(s,b)$, whose total normalization is $(a,C_a(s,b))$; this is exactly the normalization of the total cylinder point. Hence both displayed formulas hold at every cylinder radius as well as every original point. The displayed original-point formula is continuous in x, and the cylinder formula is continuous in (s,b), because both total quotient maps and insertion of the fixed parameter are continuous. The defining property of the slice quotient topology therefore makes this same slice map into X continuous. □

Lemma A.437 (The actual shared cylinder is an open chart across the seam)**.** *If the actual total collar C_{tot} is an open embedding, then*

$$q_{\mathrm{tot}} : T \times \big((-\infty,\delta) \times B\big) \to X$$

is an open embedding. In particular its image contains the attachment seam $q_{\mathrm{tot}}(a,0,b)$ and the whole outward end, in these same parameter coordinates. The total collar's open-embedding property is an explicit hypothesis; this statement supplies no smooth atlas on all of X.

Proof. The coordinate rearrangement $(a,(s,b)) \mapsto (s,(a,b))$ is a homeomorphism from the displayed source to $(-\infty,\delta) \times (T \times B)$: its inverse is $(s,(a,b)) \mapsto (a,(s,b))$, and both maps are continuous product coordinate projections. The half-infinite topology lemma applied to C_{tot} makes its cylinder map an open embedding. Composing with this homeomorphism gives precisely q_{tot}, so it too is an open embedding. Since $\delta > 0$, its source contains $s = 0$ and every $s \le 0$, proving the stated seam and outward-end coverage. □

Lemma A.438 (The original interiors and shared cylinders give an actual open cover)**.** *Let M be a native smooth manifold with boundary, let T be a topological parameter space, and let $C_a : [0,\delta) \times \partial M \to M$ have $\delta > 0$ and $C_a(0,b) = b$. Suppose the actual total collar C_{tot} is an open embedding. In the original shared quotient,*

$$i_{\mathrm{tot}} : T \times \operatorname{int} M \to X, \qquad i_{\mathrm{tot}}(a,x) = o_{\mathrm{tot}}(a,x)$$

is an open embedding, and its image together with the image of q_{tot} covers X. Thus both patches have the original quotient topology; smooth compatibility of their transitions is a separate assertion.

Proof. The total collar is continuous, so the original-point map is an embedding by the half-infinite topology lemma. The native interior inclusion is an open embedding. Taking its product with the identity of T and composing with the original-point map proves that i_{tot} is an embedding.

Put $V = T \times \operatorname{int} M$, which is open in $T \times M$. Its image under o_{tot} has original-summand preimage exactly V. Its cylinder-summand preimage consists of those (s,a,b) with $s \ge 0$ and $(a, C_a(s,b)) \in V$. At $s = 0$ this last condition cannot hold, because $C_a(0,b) = b \in \partial M$. The preimage is relatively open in the half collar by continuity of C_{tot}, and contains only $s > 0$; hence it is open in the full cylinder. The quotient topology therefore makes $o_{\mathrm{tot}}(V)$ open in X. Thus i_{tot} is an open embedding.

Every quotient point has an original or cylinder representative. A cylinder representative already lies in the second patch. An original representative (a,x) with x interior lies in the first. Otherwise x is a native boundary point b, and the seam equation identifies $o_{\mathrm{tot}}(a,b)$ with $q_{\mathrm{tot}}(a,0,b)$. This lies in the second patch, since $0 < \delta$. These cases exhaust all representatives and prove the asserted cover. □

Lemma A.439 (The exact overlap of the original interior and shared cylinder)**.** *Let M be a native smooth manifold with boundary and let C_a be prescribed half-collar maps with $C_a(s,b) \in \partial M$ exactly when $s = 0$. For original interior points $x \in \operatorname{int} M$ and full cylinder points (s,b), the equality*

$$i_{\mathrm{tot}}(a,x) = q_{\mathrm{tot}}(a',s,b)$$

holds exactly when $a = a'$, $s > 0$, and $C_{a'}(s,b) = x$. The equality condition evaluates $C_{a'}$ only on its half-collar domain; $s < \delta$ is already part of the cylinder point's domain.

Proof. If the two points agree, applying the parameter projection gives $a = a'$. The exact original-cylinder overlap formula then gives $s \ge 0$ and $C_{a'}(s,b) = x$. If $s = 0$, the boundary condition would make x a native boundary point, contradicting its interior membership. Thus $s > 0$. Conversely, these three conditions and the original-cylinder overlap formula identify the original representative (a,x) with the cylinder representative (s,a',b). This proves the equivalence on the same actual quotient. □

Lemma A.440 (Joint smooth germs of the literal mixed quotient transitions)**.** *Let M be a native smooth Hausdorff second-countable manifold with boundary, and let C_a be half-collar maps which*

meet its boundary exactly at inward time zero. Let $D \subset Y$ be a fixed compact domain presentation, let $i : M \to Y$ be a smooth injective map with bijective full differential, and let $e_B : \partial M \to \partial D$ be a smooth diffeomorphism. Suppose $J \subset \mathbb{R}$ is open and contains $[0,1]$, and that

$$F(a, y, r) = (a, c(a, y, r))$$

is a local diffeomorphism on $J \times \partial D \times (-\delta, \delta)$, with the exact native square $i(C_a(s,b)) = c(a, e_B(b), -s)$ for $a \in [0,1]$ and $0 \le s < \delta$. At every interval parameter and positive overlap point (a, s, b), $0 < s < \delta$, there is a smooth diffeomorphism

$$A : U \longrightarrow V, \qquad U \subset \mathbb{R} \times \big((0,\delta) \times \partial M\big), \quad V \subset \mathbb{R} \times \operatorname{int} M,$$

between open neighborhoods, with $(a, s, b) \in U$. Both A and A^{-1} preserve the real parameter. At every interval point of U,

$$A(a', s', b') = (a', C_{a'}(s', b')), \qquad i_{\mathrm{tot}}(a', C_{a'}(s', b')) = q_{\mathrm{tot}}(a', s', b').$$

At every interval point $(a', x) \in V$, its inverse is (a', s', b') with $0 < s' < \delta$ and $q_{\mathrm{tot}}(a', s', b') = i_{\mathrm{tot}}(a', x)$. Thus both directions are germs of the literal mixed transitions of the original quotient patches, with real-parameter endpoint extensions.

Proof. Fix (a, s, b) with $0 < s < \delta$. The boundary condition makes $x = C_a(s, b)$ an interior point. The restriction $j = i|_{\operatorname{int} M}$ is smooth; its differential is the composition of the bijective native interior inclusion differential with di, so it is bijective at x. The inverse function theorem gives a smooth diffeomorphism q from an open neighborhood of x in the native interior to an open neighborhood of $i(x)$ in Y, agreeing there with j.

The coordinate map $E(a', s', b') = (a', e_B(b'), -s')$ is a diffeomorphism. At $z = (a, e_B(b), -s)$, the assumed local diffeomorphism F has a smooth local diffeomorphism representative d, agreeing with F on an open neighborhood of z. The native square gives $F(z) = (a, i(x))$, which belongs to the target of $\mathrm{id}_{\mathbb{R}} \times q$. Consequently the local composition

$$A = (\mathrm{id}_{\mathbb{R}} \times q)^{-1} \circ d \circ E$$

is defined on an open neighborhood of (a, s, b). Restrict its source to $0 < s' < \delta$ and take its corresponding open target. It remains a diffeomorphism, and its source contains the chosen point. Each of E, d and the product map preserves parameter, so A does too. Applying this identity to A^{-1} and using $A \circ A^{-1} = \mathrm{id}$ proves parameter preservation in the reverse direction.

At an interval source point, write $A(a', s', b') = (a', y)$. The definition of A and the native square give $i(y) = c(a', e_B(b'), -s') = i(C_{a'}(s', b'))$. Injectivity of i gives $y = C_{a'}(s', b')$. The exact original interior-cylinder overlap lemma then proves the forward quotient identity. At an interval target point (a', x), the inverse lies in the positive source and has parameter a'. Apply the forward identity there and cancel $A \circ A^{-1}$; this gives precisely the reverse quotient identity. All neighborhoods are open in real parameter times the indicated boundaryless spatial manifolds, so these are endpoint-extension germs rather than one-sided parameter identities. □

Lemma A.441 (A genuine open cover by native interior and cylinder)**.** *Suppose M is a second-countable native smooth manifold with boundary, $B = \partial M$, and C is an open embedding satisfying*

$$C(0, b) = b, \qquad C(t, b) \in \partial M \iff t = 0.$$

Then $o|_{\operatorname{int} M}$ and q are open embeddings whose images cover X_C. Their overlap is precisely $0 < t < \delta$ under C. The space X_C is second countable, including when B is empty.

Proof. The native interior is open in M, and its inclusion followed by o is an embedding. The inverse image of its image under o is the interior. Its inverse image under q is exactly $\{(t,b) : 0 < t < \delta\}$: the overlap formula requires $t \geq 0$, and the boundary characterization excludes exactly $t = 0$. Both inverse images are open, so the quotient topology makes the interior image open. This proves it is an open embedding. The cylinder map is already an open embedding by the preceding lemma. An original point is either interior or boundary; a boundary point b has $o(b) = q(0,b)$ by the zero-time identity. Every cylinder point is already in the other open piece. Thus the two pieces cover X_C, and the overlap formula just proved identifies their overlap exactly. The interior of M and the boundary subspace B are second countable. So is $A = (-\infty, \delta) \times B$. Transport countable bases to the two open images by their embeddings. The union of these two countable bases is a countable basis for X_C: each point of an open set lies in one open piece and hence in a basis member contained in that set. This proves second countability also for an empty boundary. □

Lemma A.442 (The literal full outward end and compact enlargements)**.** *Let C be an open embedding as above. The outward end map*

$$e : [0,\infty) \times B \to X_C, \qquad e(r,b) = q(-r,b),$$

is an embedding. Its image and $o(M)$ cover X_C, and

$$o(x) = e(r,b) \iff r = 0 \text{ and } C(0,b) = x.$$

Thus the positive-radius end is disjoint from $o(M)$. If M and B are compact, the sets

$$K_R = o(M) \cup e([0,R] \times B) \quad (R \in \mathbb{R})$$

are compact, increase with R, and every point of X_C belongs to some K_R with $R \geq 0$; interpret $[0,R]$ as empty when $R < 0$.

Proof. The map $(r,b) \mapsto (-r,b)$ embeds the closed nonnegative cylinder into A, with its inverse on the image given by $(t,b) \mapsto (-t,b)$. Composing it with the cylinder embedding q gives the end embedding. The original-cylinder overlap requires $-r \geq 0$. Together with $r \geq 0$ this is equivalent to $r = 0$, proving the seam formula and positive-end disjointness. Each cylinder representative with $t \geq 0$ is an original point $o(C(t,b))$; for $t < 0$ it is $e(-t,b)$. This and the original representatives prove coverage by the original carrier and end. The image $o(M)$ is compact by continuity. For any R, the product $[0,R] \times B$ is compact, including the empty case, and its continuous image under e is compact. A finite union gives compactness of K_R. For $R \leq S$, interval inclusion gives $K_R \subset K_S$. An original point belongs to K_0, and an end point $e(r,b)$ belongs to K_r, where $r \geq 0$. Coverage therefore proves the final assertion. □

Lemma A.443 (The finite native extension is the actual open collar window)**.** *Let $\delta > 0$ and let $C : [0,\delta) \times B \to M$ be injective. Let X_C^{fin} be the same normalization quotient as X_C, but using the finite band $(-\delta,\delta) \times B$. There is an open embedding $j : X_C^{\mathrm{fin}} \to X_C$ which fixes every original point and every finite cylinder coordinate, with image*

$$o(M) \cup q\big((-\delta,\delta) \times B\big).$$

It is a homeomorphism onto that actual open image. For $r \geq 0$,

$$e(r,b) \in j(X_C^{\mathrm{fin}}) \quad \iff \quad r < \delta.$$

No smooth structure on the whole half-infinite attachment is asserted.

Proof. Include the finite band into A by leaving both coordinates fixed, and retain the identity on the original summand. This map commutes with normalization: nonnegative coordinates have the same collar value in both quotients, while negative coordinates remain themselves. It therefore defines j on quotient classes and gives the stated original and cylinder formulas. The induced inclusion of normalized representatives is injective on both summands, so j is injective. Continuity follows from the quotient property and continuity on each summand.

For open $W \subset X_C^{\mathrm{fin}}$, the inverse image of $j(W)$ under o is the original-summand inverse image of W. Its inverse image under q is the finite-band inclusion of the finite-cylinder inverse image of W: injectivity of C gives injectivity of each cylinder quotient map, and the exact overlap formula identifies an original representative in the cylinder image with its unique nonnegative collar coordinates. Both finite inverse images are open by the finite quotient topology. The finite band is open in A, so its inclusion maps open sets to open sets. Hence both full inverse images are open, and the quotient topology of X_C makes $j(W)$ open. Thus j is an open embedding and is a homeomorphism onto its image. Every finite quotient point is represented by an original point or a finite cylinder point. Conversely those representatives all occur. This gives the exact displayed image. If $r = 0$, the end point is original and $r < \delta$. If $r > 0$, it is not original by the seam formula; injectivity of q puts $q(-r, b)$ in the finite-cylinder image exactly when $-\delta < -r$, equivalently $r < \delta$. This proves the end-radius characterization. □

Lemma A.444 (The full cylinder in the native Euclidean model)**.** *Let B be a smooth boundaryless k-manifold and let $\delta \in \mathbb{R}$. The open half-infinite cylinder $A = (-\infty, \delta) \times B$ has a smooth boundaryless $(k+1)$-dimensional atlas. Its product coordinates are smooth, and the finite band $(-\delta, \delta) \times B$ includes smoothly into A without changing either coordinate.*

Proof. The cylinder is an open subset of $\mathbb{R} \times B$. Restrict its product charts and identify $\mathbb{R} \times \mathbb{R}^k$ with $\mathbb{R}^{k+1}$ by $(t, u) \mapsto (t, u_1, \ldots, u_k)$. The restricted targets are open. Their transition maps retain the time coordinate and apply a boundary-chart transition to the spatial coordinate, so they and their inverses are smooth. The two product coordinates are smooth in these charts. The finite-band inclusion is the identity in these same coordinates, proving its smoothness. □

Lemma A.445 (The smooth atlas on the actual half-infinite native attachment)**.** *Suppose M is a smooth Hausdorff second-countable $(k+1)$-manifold with boundary and $C : [0, \delta) \times \partial M \to M$, $\delta > 0$, is its native inward collar. Suppose C satisfies the smoothness and local inverse hypotheses of the finite collar-extension atlas. The finite native extension and the full cylinder, carried through their open embeddings into X_C, give a compatible smooth boundaryless $(k+1)$-dimensional atlas on that actual half-infinite attachment.*

Proof. Write $P = X_C^{\mathrm{fin}}$ for the finite smooth extension, $j : P \to X_C$ for its open inclusion, and $q : A \to X_C$ for the full-cylinder open inclusion. Their images cover X_C: all original points belong to $j(P)$ and all remaining points belong to $q(A)$. Carry the smooth charts of P and A through these homeomorphisms onto their open images. They have open Euclidean targets and cover the whole attachment.

The exact quotient formulas give $q^{-1}(j(P)) = \{(t, b) : -\delta < t < \delta\}$. On this overlap, the map from full-cylinder coordinates to the finite extension is its finite cylinder inclusion, leaving t, b unchanged. It is smooth by the finite atlas theorem and the preceding coordinate identification. Near each overlap point the inverse of this finite cylinder inclusion is smooth, because its carried product charts are smooth charts of the finite extension. Composing with the smooth finite-to-full band inclusion gives smoothness in the reverse direction. Transitions within either piece are already smooth in that piece's atlas. Thus all carried transitions and their inverses are smooth.

The existing quotient topology is Hausdorff and second countable, so these charts give the claimed smooth boundaryless manifold on X_C. □

Lemma A.446 (Smooth patch maps and the native inclusion on the full attachment)**.** *Under the hypotheses of the full-attachment atlas, the finite-patch inclusion $j : P \to X_C$ and full-cylinder inclusion $q : A \to X_C$ are smooth open embeddings with smooth local inverses. Their actual differentials are linear isomorphisms everywhere. The original native inclusion $o : M \to X_C$ is also smooth and has bijective full differential, including at every boundary point.*

Proof. The carried charts of each open piece belong to the constructed atlas of X_C. They make its inclusion smooth and its local inverse smooth. For either inclusion a, choose a local inverse g at a point of its image. The identities $a \circ g = \mathrm{id}$ and $g \circ a = \mathrm{id}$ hold on neighborhoods of the corresponding points, because a is injective and an open embedding. Differentiating the two identities shows that da and dg are inverse linear maps. This proves bijectivity of the differential for both pieces. The original inclusion factors as $o = j \circ o_{\mathrm{fin}}$. The finite-extension native inclusion o_{fin} is smooth with bijective full differential, including at its native boundary. Composition proves smoothness of o, and the chain rule expresses do as the composition of its two bijective differentials. Hence do is bijective at every native point. □

Lemma A.447 (The primitive on the entire half-infinite cylinder)**.** *Let λ be a locally extendible one-form on the native smooth manifold with boundary M, and put $\alpha = \lambda|_{\partial M}$. For $\delta > 0$, on the entire cylinder $A = (-\infty, \delta) \times \partial M$,*

$$\eta_A = e^{-t}\pi_{\partial M}^{*}\alpha$$

is a smooth one-form. Its pullback to the finite band $(-\delta, \delta) \times \partial M$ is exactly the finite cylinder primitive, on every tangent vector. Thus for the outward radius $r = -t$ its primitive is $e^r\alpha$ for every $r \geq 0$.

Proof. The native boundary restriction α is smooth. The projection $\pi_{\partial M}$ and time coordinate t are smooth in the full cylinder atlas. Pullback by this projection and multiplication by the smooth function e^{-t} give the asserted smoothness on all of A. The finite-band inclusion retains both coordinates. Composing it with the boundary projection gives the finite boundary projection, and its time coordinate is still t. The chain rule therefore gives the same boundary covector on each tangent vector, multiplied by the same scalar e^{-t}. This proves exact pullback equality with the finite primitive. Substituting $t = -r$ gives the prescribed outward formula $e^r\alpha$ for every nonnegative radius. □

Lemma A.448 (Joint native boundary restriction)**.** *Let M be a smooth Hausdorff second-countable $(k+1)$-manifold with its native half-space atlas. Let λ_a, $a \in [0, 1]$, be native one-forms with joint extension germs. Their actual native boundary restrictions*

$$\alpha_a = \iota_{\partial M}^{*}\lambda_a$$

form a jointly smooth family on the original boundary, with smooth extensions across the parameter endpoints. No nondegeneracy or Liouville field is required for this conclusion.

Proof. Fix a parameter a and a boundary point b. Choose a native chart E at b and a boundary chart χ whose source maps into that chart. Joint extension germs supply smooth covector coefficients $\beta(s, y)$ near $(a, E(b))$ which agree with those of λ_s at native chart points for interval parameters. In boundary coordinates put $q = E \circ \iota_{\partial M} \circ \chi^{-1}$. The native boundary atlas makes q

smooth. After restricting to an open parameter-coordinate neighborhood, the actual coefficients of α_s are

$$\beta(s, q(v)) \circ dq_v.$$

Composition and differentiation of the smooth map q make these coefficients jointly smooth. The chain rule identifies them with the actual boundary pullback at every interval point of the neighborhood. Transfer this coefficient germ through χ to an actual spatial one-form germ on the boundary. These germs cover the closed subset $[0,1] \times \partial M$ of $\mathbb{R} \times \partial M$. The closed-carrier spatial-form extension lemma patches them to one jointly smooth real-parameter family, preserving every prescribed boundary covector. Restricting that family to the interval proves the assertion, including the parameter endpoint extensions. If the boundary is empty, the unique family of forms on it satisfies the assertion. □

Lemma A.449 (Joint primitive on the full original boundary cylinder)**.** *Under the hypotheses of the preceding lemma, fix $\delta > 0$ and put $A = (-\infty, \delta) \times \partial M$, with its actual smooth cylinder atlas. The spatial one-forms*

$$\eta_a = e^{-s} \pi_{\partial M}^* \alpha_a$$

are jointly smooth on A, with smooth parameter endpoint extensions. Their pullbacks to $(-\delta, \delta) \times \partial M$ are exactly the finite-band primitives for λ_a, on every spatial tangent. In the outward coordinate $r = -s$, the formula is $e^r \alpha_a$; it includes the seam $r = 0$ and every outward radius $r \geq 0$. These statements concern the actual cylinder and its primitive family; they do not assert a smooth atlas on a whole varying quotient attachment.

Proof. Choose a jointly smooth extension $\widetilde{\alpha}_a$ of the original boundary family, supplied by the preceding lemma. On A set $\widetilde{\eta}_a = e^{-s} \pi_{\partial M}^* \widetilde{\alpha}_a$. The boundary projection is a fixed smooth map. In any cylinder chart, the coefficients of its pullback are the boundary coefficients composed with that projection and applied to its differential. Both composition and this covector evaluation preserve joint smoothness. Multiplication by the smooth scalar e^{-s} therefore gives jointly smooth coefficients of $\widetilde{\eta}$ near every interval parameter and cylinder point. At interval parameters they are exactly η_a.

The finite-band inclusion preserves the time and boundary coordinates. Its pullback thus preserves e^{-s} and the boundary projection; pullback composition gives precisely the finite cylinder primitive. This is equality of covectors on all tangents, including arbitrary time components. Substituting $s = -r$ gives the outward formula. Since $0 < \delta$, every $r \geq 0$ corresponds to $s = -r < \delta$, so this formula covers the seam and the entire attached end. □

Lemma A.450 (Full covector agreement on the actual completion overlap)**.** *Under the native full-attachment hypotheses, let λ be locally extendible, let Z satisfy $d\lambda(Z, \cdot) = \lambda$, and suppose the same collar has negative-Z time curves. Let β be its finite glued primitive and η_A its full cylinder primitive. At every overlap point $j(u) = q(p)$, the ambient covectors*

$$\beta_u \circ (dj_u)^{-1}, \qquad (\eta_A)_p \circ (dq_p)^{-1}$$

on $T_{j(u)} X_C$ agree.

Proof. Let F be the finite-cylinder inclusion into the finite extension, and let L be the coordinate inclusion of the finite band into A. The exact overlap formula supplies a finite-band point v with $u = F(v)$ and $p = L(v)$. The quotient identities give $j \circ F = q \circ L$, so

$$dj_u \circ dF_v = dq_p \circ dL_v.$$

The finite primitive law and full-cylinder restriction give $F^*\beta = \eta_{\mathrm{fin}} = L^*\eta_A$. Fix an arbitrary ambient tangent vector w. Both dj_u and dF_v are bijective, so choose a with $dj_u(dF_v(a)) = w$. The finite transported covector evaluates on w as $\beta_u(dF_v(a)) = \eta_{\mathrm{fin},v}(a)$. The differential identity gives $w = dq_p(dL_v(a))$, and the full transported covector therefore evaluates as $(\eta_A)_p(dL_v(a)) = \eta_{\mathrm{fin},v}(a)$. The two evaluations agree for every w, proving covector equality. □

Lemma A.451 (One smooth primitive on the actual half-infinite attachment)**.** *Under the hypotheses of the preceding lemma, the actual smooth half-infinite attachment X_C carries a smooth one-form η with the exact whole-patch identities*

$$j^*\eta = \beta, \qquad q^*\eta = \eta_A.$$

Proof. At a point of $j(P)$ define the covector using $\beta \circ (dj)^{-1}$, and at a point of $q(A)$ use $\eta_A \circ (dq)^{-1}$. Each representative is unique because the patch inclusions are injective. The two open pieces cover X_C, and the preceding lemma proves the two prescriptions agree wherever both apply. They therefore define one actual one-form η. Pulling it back through either patch differential cancels that differential's inverse and gives the displayed whole-patch identities.

For smoothness fix a point in either open image and choose the smooth local inverse g of that patch inclusion. On a neighborhood of the point, the constructed one-form is the pullback by g of the smooth primitive on its source patch: differentiating the local inverse identities identifies dg with the inverse patch differential. The source coefficients, coordinate expression of g and derivative of g are smooth. Thus their covector composition has smooth coefficients in every local chart. Such neighborhoods cover X_C, proving smoothness on the whole attachment. □

Lemma A.452 (Nondegeneracy through a bijective differential)**.** *Let $f : N \to X$ be a smooth map between smooth boundaryless manifolds, let γ be a smooth one-form on X, and suppose df_u is bijective at $u \in N$. Then $d\gamma$ is nondegenerate at $f(u)$ if and only if $d(f^*\gamma)$ is nondegenerate at u.*

Proof. Exterior naturality gives $d(f^*\gamma)_u(v, w) = d\gamma_{f(u)}(df_u(v), df_u(w))$. Suppose the target form is nondegenerate and v annihilates every w under the pulled-back form. Surjectivity of df_u shows $df_u(v)$ annihilates every target tangent, so it is zero. Injectivity of df_u gives $v = 0$. Conversely, suppose the pulled-back form is nondegenerate and a target tangent a annihilates every target tangent. Choose v with $df_u(v) = a$. The displayed identity then makes v annihilate every source tangent, so $v = 0$ and $a = df_u(0) = 0$. These are precisely the two nondegeneracy implications. □

Lemma A.453 (Nonzero constant scaling of a symplectic differential)**.** *If β is a smooth one-form, $c \in \mathbb{R} \setminus \{0\}$, and $d\beta$ is nondegenerate at u, then $d(c\beta)$ is nondegenerate at u.*

Proof. The exterior scalar product rule gives $d(c\beta) = c\, d\beta$, because the derivative of the constant c vanishes. If a tangent v satisfies $d(c\beta)_u(v, w) = 0$ for every w, division by the nonzero c gives $d\beta_u(v, w) = 0$ for every w. Nondegeneracy of $d\beta$ gives $v = 0$, proving the assertion. □

Lemma A.454 (Joint seam coefficients of the same original quotient primitives)**.** *Let M be a native smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Let λ_a, $a \in [0, 1]$, have joint native extension germs and nondegenerate exterior derivatives, and let $\iota_{Z_a} d\lambda_a = \lambda_a$. Fix $\delta > 0$ and actual varying inward collars $C_a : [0, \delta) \times \partial M \to M$. For every a, suppose C_a is smooth and an open embedding, satisfies $C_a(0, b) = b$, meets the native boundary exactly at $s = 0$, and has smooth local inverse germs into $\mathbb{R} \times \partial M$ whose compositions with C_a are the identity near each point of the half collar. Suppose its time curves solve $-Z_a$. Assume also that the actual total collar $(s, a, b) \mapsto (a, C_a(s, b))$ is an open embedding.*

On these same original slice quotient attachments $X_a = X_{C_a}$, with their native completion atlases, there are one-forms L_a such that each L_a is smooth and dL_a is nondegenerate everywhere, with

$$o_a^* L_a = \lambda_a, \qquad q_a^* L_a = e^{-s}\pi_{\partial M}^*(\lambda_a|_{\partial M}).$$

The first equality holds on every native tangent, including those based at the native boundary. The actual spatial pullbacks $q_a^* L_a$ form a jointly smooth family on $(-\infty, \delta) \times \partial M$, with smooth parameter endpoint extensions. In the shared quotient, their cylinder points are exactly $q_{\mathrm{tot}}(a, s, b)$ under the canonical original fiber bijection. The same q_{tot} and the original interior map i_{tot} are open embeddings whose images cover this shared quotient. Thus these are the spatial coefficients of the same chosen original primitives in that actual open cylinder patch, including at the seam. A smooth atlas on the entire total quotient requires further compatibility and assembly beyond this local coefficient assertion.

Proof. Fix an interval parameter a. Joint native extension germs restrict to local extension germs for λ_a. The assumed collar properties, its negative-Liouville time equation and nondegeneracy therefore satisfy the symplectic-primitive construction on the original half-infinite attachment. Apply that construction to this very C_a and λ_a, and choose its primitive L_a for every a. It gives smoothness and nondegeneracy on the same X_a, full native primitive preservation, and the exact full-cylinder pullback above.

The joint-cylinder lemma constructs smooth parameter-endpoint extensions of $e^{-s}\pi_{\partial M}^*(\lambda_a|_{\partial M})$ on the whole cylinder. The exact pullback equation identifies this family with the actual $q_a^* L_a$ for every interval parameter. Consequently those same selected primitives have the claimed jointly smooth spatial pullbacks; no completed-family regularity has been assumed. The total-collar open-embedding hypothesis gives the actual open cylinder chart of the shared quotient. Together with the original seam equations, the open-cover lemma makes the native interior map an open embedding and gives the two-patch cover of this same quotient. The original-fiber identification sends $q_a(s, b)$ exactly to $q_{\mathrm{tot}}(a, s, b)$, so this joint coefficient family belongs to that same open patch and the same original quotient points. □

Lemma A.455 (Original native data construct the same joint quotient primitive family)**.** *Under the original compact native primitive hypotheses of Lemma A.166, those data construct collars C_a and primitives L_a on their same original slice quotient attachments, with every conclusion of Lemma A.454. In particular, each L_a is smooth with nondegenerate exterior derivative, preserves the entire original native primitive, and has the exact full-cylinder pullback whose spatial coefficients extend jointly smoothly across the parameter endpoints and every seam. The shared quotient has its original native-interior and full-cylinder open cover, retaining the exact cylinder-point fiber identification. The same chosen δ, C_a retain the fixed D, i, e_B, real-parameter collar representative c, open parameter neighborhood J, local diffeomorphism law and native collar square of the native collar producer. No collar, completed space, completed primitive regularity or comparison map is an input. This constructs the specified local charts and primitive coefficients on one fixed compact native domain; the smooth atlas on the entire varying quotient and the full source-family correspondence remain separate.*

Proof. Apply the source-only native collar lemma to the given primitive family. It constructs one δ, C_a with all slice smoothness, native seam, boundary, local inverse and joint open-embedding laws, and the actual inward time equation for the native canonical Liouville fields. Keep its same D, i, e_B, c, J and their extension identities.

Nondegeneracy of $d\lambda_a$ gives the actual contraction equation $\iota_{Z_{\lambda_a}} d\lambda_a = \lambda_a$ by the intrinsic contraction-field lemma. Thus the constructed collars and these same fields satisfy every hypothesis

of the conditional joint-seam lemma. Apply it without changing any collar or extension witness. Its chosen primitives live on the original quotient attachments for precisely these C_a, and it gives all the displayed primitive, nondegeneracy, joint coefficient, open-cover and fiber-point identities there. The collar producer's real-parameter representative remains attached to that same C_a and δ, so all of its fixed-presentation and native collar equations are retained by this construction. □

Lemma A.456 (One original quotient family with its actual mixed-transition germs)**.** *Under the original compact native primitive hypotheses of Lemma A.166, one choice of $\delta, C_a, L_a, D, i, e_B, c, J$ has every conclusion of Lemma A.455 and, at each interval parameter and positive overlap point, the mutually inverse smooth real-parameter germs of Lemma A.440. Both directions preserve parameter and restrict to the literal transitions between the original interior and full cylinder patches of that same quotient family. All original primitive, nondegeneracy, joint cylinder coefficient, open-cover, fiber-point and native real-collar extension identities are retained by this one choice. The canonical Liouville field of each same L_a has a smooth flow for all real times on its original quotient X_a; joint regularity of these flows in a is not asserted. The joint germs are conclusions from the original primitive family; neither transition germs nor a completed family are supplied as inputs. A global smooth atlas and completed bundle trivialization, and their correspondence with independently prescribed completions, remain separate assertions.*

Proof. Apply the source-only joint-seam construction once. Keep its same $\delta, C_a, L_a, D, i, e_B, c, J$ and every primitive and quotient identity. The native collar meets the boundary exactly at time zero. The map i is smooth and a closed embedding, hence injective, and has bijective full differential. The same e_B is a smooth diffeomorphism, the same J contains the parameter interval, and the same real collar map $(a, y, r) \mapsto (a, c(a, y, r))$ is a local diffeomorphism on its stated strip. Its native square still relates this very C_a to c. These are exactly the hypotheses of the mixed-transition germ lemma. Apply that lemma at each interval parameter and each $0 < s < \delta$. It gives both smooth inverse germs and their actual forward and reverse quotient identities for the same original collars and patches. No witness from the source-only construction is replaced, so its primitive, nondegeneracy, joint coefficient, open-cover, fiber and extension equations are all retained together with the new germs. For each interval parameter, the same L_a is smooth and has nondegenerate exterior derivative, and its cylinder pullback is the full exponential primitive. The original native carrier is compact. The actual outward charts pull this same primitive back to $e^r \pi^* \alpha_{a,B}$ and cover the complement of a compact truncation. Nondegeneracy identifies its canonical contraction field with the radial field in these charts. The prescribed-cylinder complete-flow theorem, as applied in the intrinsic half-infinite complete-flow construction, therefore gives an all-real smooth flow of that field on the same X_a. This changes neither L_a nor any collar or transition germ and makes no joint-in-parameter flow assertion. □

Lemma A.457 (Hausdorffness of the original shared completion quotient)**.** *Let T, M, B be Hausdorff spaces, let $\delta > 0$, and suppose the actual total collar $C_{\mathrm{tot}} : [0, \delta) \times (T \times B) \to T \times M$ is an open embedding. The same original shared quotient X is Hausdorff.*

Proof. Products of Hausdorff spaces are Hausdorff, so both $T \times M$ and $T \times B$ are Hausdorff. The shared quotient is exactly the half-infinite attachment with original carrier $T \times M$, boundary carrier $T \times B$, and collar C_{tot}. The half-infinite topology lemma applied to this collar gives Hausdorffness of that same quotient. □

Lemma A.458 (Second countability of the original shared completion quotient)**.** *Let T be second countable, let M be a second-countable native smooth manifold with boundary, and suppose the*

original collars have $\delta > 0$, $C_a(0, b) = b$, and an open-embedding total collar. Their same original shared completion quotient X is second countable.

Proof. The native interior and native boundary are subspaces of M, hence are second countable. The interval $(-\infty, \delta)$ is second countable, and finite products preserve second countability. Thus both patch sources $T \times \operatorname{int} M$ and $T \times ((-\infty, \delta) \times \partial M)$ are second countable. Choose a countable base in each. Their open embeddings into X send these bases to countable collections of open subsets of X. Those collections together form a base: given x in an open set U, choose a patch containing x using the original open-cover lemma. The preimage of U in that patch is open and contains its representative of x. A member of the patch base contains this representative and lies in the preimage; its image contains x and lies in U. The union of the two countable collections is countable, proving the claim. □

Lemma A.459 (The original completed slices are the actual topological fibers)**.** *Let T be a topological parameter space and M a native smooth manifold with boundary. Suppose the original collars have $\delta > 0$, $C_a(0, b) = b$, and an open-embedding total collar. For every $a \in T$, the same canonical insertion $X_{C_a} \to X$ is an embedding and gives a homeomorphism onto the full fiber $\pi^{-1}(a)$ with its subspace topology. Its underlying bijection is exactly the original fiber bijection:*

$$o_a(x) \longmapsto o_{\mathrm{tot}}(a, x), \qquad q_a(s, b) \longmapsto q_{\mathrm{tot}}(a, s, b).$$

Proof. The canonical insertion $f_a : X_{C_a} \to X$ is continuous and injective by the original-fiber lemma. The total interior and total cylinder are open embeddings and their images cover X. The slice interior parametrization u_a and cylinder parametrization v_a are continuous, by continuity of the slice quotient maps and native interior inclusion. Their composites with f_a are respectively $x \mapsto i_{\mathrm{tot}}(a, x)$ and $(s, b) \mapsto q_{\mathrm{tot}}(a, s, b)$, by the original-fiber identities. These are embeddings: insertion of the fixed parameter is an embedding, and so are both total patch maps. Consequently u_a and v_a are embeddings too. Indeed their inverse maps on their images are restrictions of the continuous inverse maps of $f_a \circ u_a$ and $f_a \circ v_a$, composed with f_a.

The preimage of the total interior patch is precisely the slice interior patch. Indeed, any equality with a total interior representative forces its parameter to be a; removing that parameter and using injectivity of the canonical slice insertion gives the slice interior representative. The same argument with the cylinder representatives proves that the preimage of the total cylinder patch is precisely the slice cylinder patch. The inverse on the image of the canonical insertion is therefore continuous on its intersections with each of these two open patches, by the patch embedding statements. Those intersections form an open cover of the image, so this inverse is continuous everywhere. The insertion itself is continuous and injective by the original-fiber lemma; it is consequently an embedding. That lemma identifies its image with the entire parameter fiber by precisely the displayed bijection. Restricting the codomain to that image proves the asserted homeomorphism with the fiber's subspace topology. □

Lemma A.460 (Continuous maps descend through the actual shared attachment)**.** *Let T, M, B, N be topological spaces and let C_a be prescribed half-collar maps. Suppose continuous maps*

$$f : T \times M \to N, \qquad g : T \times ((-\infty, \delta) \times B) \to N$$

satisfy the exact original half-collar square $f(a, C_a(s, b)) = g(a, s, b)$ for $0 \le s < \delta$. There is a unique continuous map $F : X \to N$ on their original shared quotient such that

$$F(o_{\mathrm{tot}}(a, x)) = f(a, x), \qquad F(q_{\mathrm{tot}}(a, s, b)) = g(a, s, b).$$

For every a, its restriction through the canonical slice insertion is exactly the analogous slice descent of $f(a,\cdot)$ and $g(a,\cdot,\cdot)$ on X_{C_a}. No collar-embedding hypothesis is needed for this universal property.

Proof. Each class of the total quotient has a unique normalized representative, either an original point (a,x) or a strictly negative cylinder point (s,a,b). Assign it respectively $f(a,x)$ or $g(a,s,b)$. A nonnegative cylinder representative normalizes to $(a,C_a(s,b))$, whose assigned value equals $g(a,s,b)$ by the half-collar square. Thus the resulting map F has both required restrictions. The original and full cylinder summands map continuously to N by f and g, with the latter summand differing only by the homeomorphic reordering of its parameter and radius coordinates. Since the topology on X is the quotient topology of those summands, F is continuous. Every quotient point has a representative in one of the two summands, so any map with those restrictions has the same value there; this proves uniqueness even without assuming continuity of the competing map.

For the slice quotient at a, use the same normalized-representative construction with $f(a,\cdot)$ and $g(a,\cdot,\cdot)$. The canonical insertion sends its original points and cylinder points to their total counterparts, preserving every displayed coordinate. The two maps obtained by slice descent and by F after insertion therefore agree on every original and full cylinder representative. Those representatives cover the slice quotient, so the maps are equal on that entire original completed slice. □

Lemma A.461 (A smooth atlas on the same original completed total quotient)**.** *Let the parameter space be $[0,1]$ and let M be a smooth Hausdorff second-countable $(k+1)$-manifold with boundary. Suppose its collars have $\delta > 0$, the original seam values, the boundary criterion of Lemma A.439, and an open-embedding total collar. Suppose the literal positive-collar transitions have mutually inverse smooth real-parameter germs preserving parameter, as in Lemma A.440. Then the two actual open patches*

$$[0,1] \times \operatorname{int} M, \qquad [0,1] \times ((-\infty,\delta) \times \partial M)$$

supply a smooth $(k+2)$-dimensional atlas with parameter boundary on the same original total quotient X, using its original quotient topology. Its local model is $[0,\infty) \times \mathbb{R}^{k+1}$: the only boundary comes from the parameter endpoints, while the original compact seam is an interior spatial locus. Both actual patch maps are jointly smooth in this atlas, including at the parameter endpoints.

Proof. The interval has smooth boundary charts: near 0 use the coordinate a, near 1 use $1-a$, and in its interior use an affine coordinate with positive values. Their changes of coordinates are affine maps with nonzero linear part, hence are smooth with smooth inverses. Give each patch the product of this interval atlas with its native boundaryless spatial atlas. Both patch maps are open embeddings by the actual open-cover lemma. Pushing a product chart through either embedding gives a partial homeomorphism on X, with an open source and an open target in the stated half-space model. Those charts cover X.

Two charts from the same patch have precisely the smooth product transition of that patch. For charts from different patches, the exact overlap lemma says their common points have $0 < s < \delta$ and $x = C_a(s,b)$, with the parameter unchanged. At any such point choose the assumed forward and reverse real-parameter germs. Their restrictions to interval parameters agree with precisely these two patch transitions. Compose them with the spatial charts and the interval charts. All those maps have smooth local extensions; the interval chart extensions are affine. Thus the extended mixed coordinate transition and its inverse are smooth on a neighborhood of every overlap point, including points at $a = 0$ and $a = 1$. Smoothness is local, so every mixed transition is smooth on its full overlap. The pushed charts consequently form the claimed smooth atlas. In these charts each patch map is represented either by its own product-chart transition or by one of the smooth mixed transitions just proved. Hence both patch maps are smooth.

The Hausdorff and second-countability lemmas apply to this same open-embedding total collar and give the required topological properties for X. The spatial patches are boundaryless, and each seam point lies in the cylinder patch with $s = 0 \in (-\infty, \delta)$. It therefore has an ordinary spatial neighborhood. Only the interval factor contributes the boundary of the total atlas. □

Lemma A.462 (Smooth maps are determined by the two actual open patches)**.** *Under the hypotheses of the total-atlas lemma, let N be a smooth manifold with corners. A map $F : X \to N$ is smooth in that same original total atlas if and only if $F \circ i_{\mathrm{tot}}$ and $F \circ q_{\mathrm{tot}}$ are smooth on their product patch manifolds.*

Proof. If F is smooth, both composites are smooth because the patch maps are smooth. Conversely suppose both composites are smooth. Every point of X lies in at least one actual open patch. In any chart lifted from that patch, the coordinate expression of F is precisely the coordinate expression of the corresponding composite in its product source chart. It is therefore smooth there. All charts of the total atlas are compatible with these lifted charts, so the ordinary change of coordinates preserves that local smoothness. These neighborhoods cover X, proving that F is smooth everywhere. □

Lemma A.463 (Original native primitive data construct that same smooth total family)**.** *Under the original compact native primitive hypotheses of Lemma A.166, there is one choice of δ, C_a, L_a on the original native domain with the smooth total atlas of Lemma A.461 on its same original shared quotient. This quotient is Hausdorff and second countable. Its actual interior and full-cylinder maps and its parameter projection are smooth, including at the interval endpoints. The chosen collars retain their negative native Liouville equation, seam values, slice open embeddings, boundary criterion and smooth local inverse germs. Their literal mixed transitions have the two-way smooth real-parameter germs on that same quotient.*

The original completed slices carry those same L_a, each smooth with nondegenerate exterior derivative and complete canonical Liouville flow for all real times. Their native pullbacks equal the full λ_a, their full-cylinder pullbacks equal $e^{-s}\pi_{\partial M}^{*}(\lambda_a|_{\partial M})$, and those same spatial cylinder coefficients form a jointly smooth family with parameter-endpoint extensions. The same original slices are homeomorphic to the full parameter fibers by the original point and cylinder insertion formulas. This is a construction on one fixed native compact domain; completed bundle trivializations, transported forms on the whole total family, whole pair identifications and independently prescribed endpoint markings remain additional obligations.

Proof. Apply the source-only mixed-germ construction once and retain its chosen δ, C_a, L_a. It gives $\delta > 0$, the original seam equations, the precise boundary criterion, the actual open-embedding total collar and the two-way smooth real-parameter germs for every positive overlap. It also gives exactly the claimed primitive restrictions, cylinder coefficients and all-real slice flows on the original completion atlases for those C_a.

The total-atlas lemma applies to this same collar and these same mixed germs, giving the atlas and smooth patch maps on its original shared quotient. The parameter projection composed with either patch map is the first product projection, hence is smooth. The smooth-patch criterion therefore proves smoothness of this same parameter map. The Hausdorff and second-countability conclusions are those of the total-atlas lemma. The fiber-homeomorphism lemma identifies each original slice with the full parameter fiber by the same insertion map. These operations change neither the collars nor the chosen primitives, so all the retained primitive, coefficient and all-real slice-flow conclusions hold on this one family together with the new smooth total atlas and topological fiber identifications. □

Lemma A.464 (An end-strip formula suffices on the entire original quotient)**.** *Let T, M, B, N be topological spaces and let $\delta > 0$. Suppose the original total collar is continuous, $f : T \times M \to N$ is continuous, and $g : T \times ((-\infty, \delta) \times B) \to N$ satisfies $f(a, C_a(s, b)) = g(a, s, b)$ throughout $0 \le s < \delta$. If g is continuous on $\{s < \varepsilon\}$ for some $\varepsilon > 0$, then g is continuous on its whole original cylinder, and its same quotient descent is continuous on the entire original shared attachment. The collar width δ is retained.*

Proof. For $s < \delta$ put

$$H(a, s, b) = f(a, C_a(\max(s, 0), b)).$$

Since $\delta > 0$, one has $0 \le \max(s, 0) < \delta$, so this is defined on the whole original cylinder. The maximum function, product-coordinate reordering, total collar and f are continuous, hence so is H. On $\{s \ge 0\}$, the half-collar square gives $H = g$. The subsets $\{s < \varepsilon\}$ and $\{s > 0\}$ are open in the full cylinder and cover it, because $\varepsilon > 0$. The restriction of g to the first is continuous by hypothesis; on the second it agrees with the continuous H. Continuity is local, so g is continuous on the whole cylinder. The continuous-descent lemma now gives continuity of the same descended map on the entire shared quotient, using precisely the original half-collar square and the unchanged δ. □

Lemma A.465 (The native inward collar square on that same completed deformation)**.** *Let Q be a pair deformation of ambient dimension $2n$, $n \ge 1$, with compact ambients D_t, their completions K_t, and its specified compact and completed presentation maps $h_t : D_0 \to D_t$, $P_t : K_0 \to K_t$. Let (M, λ_t) be a native $(2n)$-manifold with boundary and let $e_t : M \to D_t$ be smooth diffeomorphisms retaining the full native primitive and having native coordinate extension germs for their ambient composites. Let $b_t : \partial M \to \partial D_t$ be smooth diffeomorphisms satisfying the exact boundary inclusion squares and $h_{t,\partial} \circ b_0 = b_t$. Define the actual reference compact maps by*

$$f_t(x) = P_t^{-1}(\iota_{K_t}(e_t(x))).$$

Suppose $C_t(s, b)$ solves the negative native Liouville equation for $0 \le s < \delta$ and $C_t(0, b) = b$. Then, on these same specified completions,

$$f_t(C_t(s, b)) = P_t^{-1}(E_{K_t}(b_t(b), -s)) \qquad (0 \le s < \delta),$$

where E_{K_t} is the actual all-real end orbit map. In particular, the negative-radius square is a consequence of the primitive and boundary squares and the native trajectory equation.

Proof. Fix t. The prescribed compact-endpoint lemma gives a native Liouville presentation with exactly D_t and the ambient composite of e_t. Its native boundary map and b_t have the same composite with the boundary inclusion, namely the ambient composite of e_t on each native boundary point. That inclusion is injective, so the two boundary maps agree. The native inward-trajectory lemma for this presentation and the specified K_t gives

$$\iota_{K_t}(e_t(C_t(s, b))) = E_{K_t}(b_t(b), -s) \qquad (0 \le s < \delta).$$

Apply P_t^{-1} to this equality and use the definition of f_t. The exact boundary coherence identifies $b_t(b)$ with $h_{t,\partial}(b_0(b))$, so the right side is precisely the prescribed end map of the same completed presentation evaluated at $(b_0(b), -s)$. No completion or compact map is replaced. □

Lemma A.466 (One real smooth extension of the same native cylinder family)**.** *Under the hypotheses of the preceding native inward-square lemma, there are $a < 0$ and a jointly smooth map $\beta : \mathbb{R} \times \partial M \times \mathbb{R} \to K_0$ such that*

$$\beta(t, b, r) = P_t^{-1}(E_{K_t}(b_t(b), r)) \qquad (t \in [0, 1],\ r > a).$$

Consequently

$$\beta(t, b, -s) = f_t(C_t(s, b)) \qquad (0 \le s < \delta,\ a < -s),$$
$$\beta(t, b, r) = E_{K_0}(b_0(b), r) \quad (r \ge 0).$$

The map is smooth on the whole real parameter product, so its restrictions supply actual extensions at both parameter endpoints. Its required agreement uses the same completed family Q and is imposed only on interval parameters and radii above the displayed cutoff.

Proof. The definition of a pair deformation gives joint smoothness, with real parameter-endpoint extensions, of the completed ambient primitives transported to the reference completion by these very P_t. The prescribed end-strip lemma therefore gives one $a < 0$ and a smooth map $\Gamma : \mathbb{R} \times \partial D_0 \times \mathbb{R} \to K_0$ with

$$\Gamma(t, y, r) = P_t^{-1}(E_{K_t}(h_{t,\partial}(y), r)) \quad (t \in [0, 1],\ r > a).$$

Set $\beta(t, b, r) = \Gamma(t, b_0(b), r)$. It is smooth because Γ and b_0 are smooth. Boundary coherence gives the first required formula. At an inward radius $r = -s > a$, the preceding native collar-square lemma identifies this formula with $f_t(C_t(s, b))$. At $r \ge 0$, the specified completion presentation satisfies $P_t(E_{K_0}(y, r)) = E_{K_t}(h_{t,\partial}(y), r)$. Applying its inverse and taking $y = b_0(b)$ gives the last formula. These arguments use the same a, β, P_t throughout. □

Lemma A.467 (A continuous original quotient map into the same completed deformation)**.** *Under the hypotheses of the native inward-square lemma, suppose $\delta > 0$ and the actual total collar of the same C_t is an open embedding. In addition, let $R_t : M \to M$ be a jointly smooth real-parameter family of native diffeomorphisms, with jointly smooth inverses, satisfying the actual compact coherence square*

$$h_t(e_0(R_t(x))) = e_t(x) \qquad (t \in [0, 1],\ x \in M).$$

On these collars' original shared quotient X, there is a continuous map $q : X \to K_0$ with the exact formulas

$$q(o_{\mathrm{tot}}(t, x)) = f_t(x),$$
$$q(q_{\mathrm{tot}}(t, s, b)) = P_t^{-1}(E_{K_t}(b_t(b), -s)).$$

This map retains the original collar width δ, every original native point and the full cylinder of this same completed deformation Q.

Proof. The completion presentation's compact square and the native compact coherence square give

$$f_t(x) = \iota_{K_0}(e_0(R_t(x))).$$

The right side is jointly continuous in (t, x): $R_t(x)$ is jointly smooth, and $\iota_{K_0} \circ e_0$ is smooth. Define

$$g(t, s, b) = P_t^{-1}(E_{K_t}(b_t(b), -s)).$$

The full native inward square is exactly the required agreement $f_t(C_t(s, b)) = g(t, s, b)$ on the whole original half collar. The cylinder-extension lemma gives $a < 0$ and a smooth β for these same

collars and completed presentations. For $s < -a$, its formula identifies $g(t, s, b)$ with $\beta(t, b, -s)$, so g is continuous on that open strip. Its width $-a$ is positive. The total collar is continuous because it is an open embedding. The end-strip descent lemma therefore constructs a continuous map on the entire original shared quotient with precisely the two displayed formulas, without changing δ, any collar or any completed presentation. □

Lemma A.468 (A smooth end strip controls the whole original total atlas)**.** *Under the hypotheses of the original total-atlas lemma, let N be a smooth manifold with corners and let $F : X \to N$ be a map. Suppose $F \circ i_{\mathrm{tot}}$ is smooth and $F \circ q_{\mathrm{tot}}$ is smooth on $\{s < \varepsilon\}$ for some $\varepsilon > 0$. Then F is smooth on the whole same original total atlas, with its collar width δ retained.*

Proof. Consider a cylinder point (t, s, b). If $s < \varepsilon$, its open strip neighborhood gives smoothness of $F \circ q_{\mathrm{tot}}$ there by hypothesis. Otherwise $s \geq \varepsilon > 0$, so the original boundary criterion makes $x = C_t(s, b)$ an interior point. The exact overlap identity gives $q_{\mathrm{tot}}(t, s, b) = i_{\mathrm{tot}}(t, x)$. The real mixed-germ hypotheses restrict to a smooth local map A from this cylinder neighborhood to the original interior patch, with $q_{\mathrm{tot}} = i_{\mathrm{tot}} \circ A$ there, including at the parameter endpoints. Thus $F \circ q_{\mathrm{tot}}$ locally equals the smooth map $(F \circ i_{\mathrm{tot}}) \circ A$. These two cases prove smoothness of $F \circ q_{\mathrm{tot}}$ everywhere. The smooth-patch criterion now gives smoothness of F on the whole original X. Every argument used the original collar and overlap, so no collar width was changed. □

Lemma A.469 (One native pair construction with its original completed ambient family)**.** *Under the original native ambient and positive-dimensional ribbon family hypotheses of Lemma A.182, one construction gives both a pair deformation Q and an original completed ambient total quotient X for the same given native ambient primitive family λ_t. The deformation retains every native compact primitive, boundary, whole ribbon, positive witness, full-core, coordinate-extension and joint relabeling conclusion of that coherent native construction. The same X retains the collars, real mixed germs, smooth total atlas, Hausdorffness, second countability, smooth patch and parameter maps, original slice primitives, joint cylinder coefficients and all-real slice flows of the source-only native total-atlas construction.*

Writing K_t for these very Q ambient completions, $P_t : K_0 \to K_t$ for their specified completed presentation maps, e_t and b_t for these very native carrier and boundary identifications, there is a jointly smooth map $q : X \to K_0$ satisfying

$$q(o_{\mathrm{tot}}(t, x)) = P_t^{-1}(\iota_{K_t}(e_t(x))),$$
$$q(q_{\mathrm{tot}}(t, s, b)) = P_t^{-1}(E_{K_t}(b_t(b), -s)).$$

Both formulas use the exact collars defining this same X and the same constructed Q, with the original collar width retained. The parameter-preserving comparison $H(z) = (\pi(z), q(z))$ is a smooth diffeomorphism in this original total atlas. For the very slice primitives L_t constructed on this same X, writing j_t for its canonical original slice insertion, η_t for the primitive of K_t and $\ell_t = P_t^* \eta_t$, one has

$$(q \circ j_t)^* \ell_t = L_t \qquad (t \in [0, 1]).$$

There are actual smooth slice diffeomorphisms $S_t = P_t \circ q \circ j_t : X_{C_t} \to K_t$ with

$$S_t(o_t(x)) = \iota_{K_t}(e_t(x)),$$
$$S_t(q_t^{\mathrm{cyl}}(s, b)) = E_{K_t}(b_t(b), -s) \qquad (s < \delta),$$
$$S_t^* \eta_t = L_t,$$
$$dS_{t,x}(Z_{L_t}(x)) = Z_{K_t}(S_t(x)).$$

Each slice uses its original atlas and its same chosen primitive L_t. In addition, there are a total covector Ω, an alternating total two-covector Ξ and a total tangent field $\mathcal{Z}$ on this same original X such that their tangent evaluation, tangent-pair evaluation and tangent section are C^∞ relative to its closed parameter model, and

$$\Omega_{j_t(x)} \circ d(j_t)_x = (L_t)_x,$$
$$\Xi_{j_t(x)}(d(j_t)_x(v), d(j_t)_x(w)) = (dL_t)_x(v,w),$$
$$d(j_t)_x(Z_{L_t}(x)) = \mathcal{Z}_{j_t(x)}.$$

The full pullback of this very Ω by the original interior insertion has real open smooth coefficient germs on the entire native interior product. There exists $a < 0$ such that its full pullback by the original cylinder insertion has such germs on $s < \min(\delta, -a)$. Both properties include the parameter component at both endpoints. Every point of the same original quotient has an interior representative or a cylinder representative in this very strip. These conclusions retain the same Q, collars, primitives, q and slice diffeomorphisms. The covector has real open coefficients in one actual smooth chart at every original point, with its full tangent identity preserved. The two-covector and field extension assertions below concern the original product patches. The same canonical Ξ, constructed from the spatial derivative $d\ell_t$ and this same q, has real open full two-covector coefficient germs on the entire original interior product and, for some $a_\Xi < 0$, on $s < \min(\delta, -a_\Xi)$ in the original cylinder. Both tangent arguments are retained, and these two patches cover the same original quotient. The same transported $\mathcal{Z}$ has real spatial coefficient extensions in the original interior product patch and, for some $a_\mathcal{Z} < 0$, in its original cylinder patch on $s < \min(\delta, -a_\mathcal{Z})$. On interval points the full differential of each insertion sends $(0, W)$ to this very $\mathcal{Z}$, where W is the respective real spatial extension. These patches also cover X. In addition, the original positive-dimensional ribbon primitive family constructs one original completed ribbon total quotient Y, with its own common collar width $\delta_F > 0$, collars C_t^F and selected slice primitives L_t^F. The original native atlas and cylinder primitive conclusions hold on this same Y. Its actual smooth slice comparisons $f_t : Y_t \to K_t^F$ target the ribbon completions of this very Q, retaining the already constructed native ribbon carrier maps e_t^F. They have both exact native inclusion squares, the native boundary and entire cylinder squares, strict primitive preservation, all-real flow conjugacy, the full native core equivalence, and the actual embedded-core identity of Lemma A.474. The same retained native ribbon relabelings and compact square also construct an all-real reference boundary diffeomorphism family, jointly smooth in both directions. Its interval forward and inverse maps are exactly the actual reference boundary comparisons, and its full reference cylinder equation is that of Lemma A.475. These same comparisons construct a parameter-preserving smooth ribbon total diffeomorphism $H_F : Y \to [0,1] \times K_0^F$ with

$$H_F(\iota_t(z)) = (t, P_t^{-1}(f_t(z))), \qquad H_F^{-1}(t,y) = \iota_t(f_t^{-1}(P_t(y))).$$

It retains every original native point and the full original cylinder, including all $s < \delta_F$, through these exact slice equations. The spatial interior formula has a jointly smooth real-parameter extension. At every reference compact interior point its actual inverse has a real smooth coordinate germ preserving parameter and agreeing, after interval restriction and original interior insertion, with H_F^{-1}. There are also a negative cutoff a_F and a real smooth cylinder formula $\Gamma_F(t,b,r)$ agreeing with the actual spatial coordinate of H_F whenever $a_F < r$ and $-r < \delta_F$. At nonnegative radii it equals the reference end map on the retained moving boundary point, and $(t,b,r) \mapsto (t, \Gamma_F(t,b,r))$ has a real smooth local inverse there. All these conclusions use this same Q, ribbon quotient, selected primitives and slice comparisons. The same original Y carries the canonical total covector

$$(\Omega_F)_z(v) = (\ell^F_{\pi_F(z)})_{q_F(z)}(d(q_F)_z(v)), \qquad \ell_t^F = P_t^* \eta_t^F.$$

Its tangent evaluation is C^∞ in the original closed parameter model, and its pullback by every original slice insertion is exactly L_t^F. This same covector has real open full coefficient germs in an actual smooth chart at every point of Y. The same chosen reference family and comparison define

$$(\Xi_F)_z(v,w) = (d\ell^F_{\pi_F(z)})_{q_F(z)}(dq_{F,z}v, dq_{F,z}w),$$
$$(\mathcal{Z}_F)_z = d(H_F^{-1})_{H_F(z)}(0, Z_{\ell^F_{\pi_F(z)}}(q_F(z))).$$

Their tangent-pair evaluation and tangent section are jointly smooth in the original closed parameter model. Their exact original slice restrictions are dL_t^F and $Z_{L_t^F}$, respectively. The full pullbacks of this very Ξ_F have real open coefficient germs on the entire original interior product and, for some $a_\Xi < 0$, on $s < \min(\delta_F, -a_\Xi)$ in the original cylinder. The same $\mathcal{Z}_F$ has real spatial field extensions on the interior and, for some $a_Z < 0$, on $s < \min(\delta_F, -a_Z)$ in the original cylinder; the full differential of each insertion sends $(0, W)$ to this field at every interval point. Each interior-and-strip pair covers Y. All these conclusions retain the very same Q, original ribbon quotient, collars, primitives, comparisons and native relabelings.

Proof. Apply the coherent native pair-family construction with its native coordinate conclusions once, to the given ambient and ribbon primitives, ribbon maps and positive witnesses. Keep its Q, native carrier and boundary maps, joint native relabelings and all compact coherence squares. It supplies every stated primitive, ribbon, witness, full-core and coordinate-extension conclusion on that Q.

Apply the source-only native total-atlas construction once to the same original ambient λ_t. Keep its δ, C_t, original quotient X and original slice primitives, and every atlas, topology, coefficient and complete-flow conclusion. Its collars solve the negative native Liouville equation for precisely that λ_t. The Q native maps preserve precisely that same primitive and retain the required coordinate and boundary squares. Their joint native relabelings give the compact coherence square required by the original-quotient map lemma. Apply that lemma to these very C_t, Q and native maps. It gives a continuous q with both displayed formulas on the actual original quotient, without changing its collar width or any witness of either construction. Therefore all of the two constructions' retained conclusions and this same map hold together on the claimed tuple. The same collars retain their boundary criterion and real mixed germs. The same-quotient smoothness lemma applied to this very q and its two exact formulas gives joint smoothness in that original total atlas, retaining all of these witnesses. The exact same formulas and compact coherence square satisfy the comparison-bijection lemma, giving bijectivity of H. The constructed boundary primitive square follows from the retained full native primitive and boundary inclusion squares. Apply the slice primitive transport lemma to those squares, this same q and these very chosen L_t. It gives the displayed equality with ℓ_t, preserving the entire original primitive family and every witness of the construction. The global comparison diffeomorphism lemma supplies this same H with its smooth inverse. Restrict it to the original slices and compose with the same P_t, using the slice completion diffeomorphism lemma. This constructs the actual S_t with both native and full-cylinder equations. The strict slice comparison lemma gives $S_t^*\eta_t = L_t$. Finally each chosen dL_t is nondegenerate by the original primitive construction. Apply the original slice field-transport lemma to the proved primitive equality and this same q; it gives the displayed field equation for $S_t = P_t \circ q \circ j_t$. Apply the total-covector slice lemma to this same jointly smooth q, its proved pullback identity and these same original L_t. It gives one Ω with the displayed exact restrictions and closed-model smooth tangent evaluation. The total-exterior lemma, applied to the same varying reference primitive and its proved strict original slice pullback, gives the canonical Ξ with smooth closed-model tangent-pair evaluation and exact restriction to dL_t. Keep the already constructed exact total diffeomorphism

$H(z) = (\pi(z), q(z))$ and the already proved nondegeneracy of dL_t. The original total-field lemma applies to these very witnesses and gives the canonical $\mathcal{Z}$ with closed-model smooth tangent section and the displayed original Liouville-field restriction. Choose Ω to be exactly the covector defined by that same varying reference primitive and q, as in the total-covector slice lemma. Apply the original cylinder coefficient lemma to this same covector and the retained native, boundary and full-cylinder equations. It gives the stated real cutoff and full coefficient germs for this very Ω, without replacing its slice restrictions. Apply the original interior coefficient lemma to the same covector, native relabeling family and compact coherence square. It supplies the full real coefficient germs on the entire original interior product. The interior-and-strip cover lemma, with the cutoff just obtained, gives the asserted representatives of every point of the same original quotient. All these constructions use the given Q completed primitive and field families and introduce no new premise on the native input. Apply the original two-form patch coefficient lemma to this same canonical Ξ, using the retained actual q formulas, native relabelings and varying reference primitive. It gives the claimed entire-interior germs and its negative-cutoff cylinder germs. The strip-cover lemma with this cutoff gives coverage of the same X, with both full tangent arguments preserved. Apply the selected-chart covector lemma to the same Ω and its proved interior and cylinder germs. It gives its stated real open coefficient identity in an actual chart at every original point. Keep the canonical transported field furnished by the total-field construction. Apply the original field patch lemma using this same H, native relabelings, boundary square and full-cylinder formula. It supplies its two real spatial extensions and their exact full differential identities. The strip-cover lemma with the field cutoff gives the asserted coverage of the unchanged original quotient. Keep the original ribbon primitive family and the already chosen native ribbon carrier identifications of this same Q. Their full primitive identities and coordinate germs are retained by the initial coherent pair construction. Apply the original ribbon-slice comparison lemma to those very data. It constructs the original ribbon collar, total atlas and slice primitives once, and compares their exact slices with the already selected Q ribbon completions. All primitive, native, boundary, cylinder, flow, full-core and embedded-core conclusions are the conclusions of that lemma. The previously constructed ambient quotient, its comparisons and every witness of Q remain the same. The initial coherent pair construction also retains the original ribbon relabeling family and its inverse with all-real joint smoothness and the exact native compact presentation square. Apply the joint reference boundary lemma to those relabelings and the very ribbon slice comparisons just constructed. It supplies the all-real smooth boundary family and inverse with exact interval reference equations and the full original reference cylinder identity. This application preserves every collar, primitive, completed slice and compact carrier map already selected. Apply the original ribbon total diffeomorphism theorem to the same slice comparisons and retained relabelings. It gives H_F with its exact original slice and inverse formulas, hence the stated native and entire-cylinder equations. Its forward real interior formula is $\iota_{K_0^F}(e_0(R_s(x)))$. The inverse construction uses a local inverse of this fixed compact interior inclusion followed by R_s^{-1}; its real coordinate germ preserves parameter, and the exact compact square identifies its restriction with H_F^{-1} in the original interior patch. Finally retain the real cylinder extension from the moving-boundary lemma. Its exact actual cylinder equation and nonnegative-radius local inverse properties give the stated Γ_F, a_F on this same H_F. None of these operations selects another pair, slice primitive, collar or completed presentation. Finally apply the original ribbon total-covector theorem to these same slice comparisons, Q and retained native relabelings. Its reference pullback law retains precisely L_t^F, and its canonical covector is defined from this very q_F and ℓ_t^F. Retain that covector and its proved smooth tangent evaluation and exact original slice restrictions in the same constructed tuple. Apply the ribbon actual-chart coefficient lemma to this very covector and the retained real comparison extensions. It supplies its

full real endpoint germs in actual smooth charts. Apply the ribbon total-exterior and total-field lemmas to the same reference primitive, comparison and strict slice identities. Choose exactly their canonical Ξ_F and $\mathcal{Z}_F$, retaining their smooth evaluations and exact slice restrictions. The ribbon exterior-tensor and field patch germ lemmas apply to those same canonical tensors and the same real comparison extensions; they supply the asserted coefficient germs and full insertion differential identities. Apply the original strip-cover lemma with each resulting negative cutoff. This proves both cover assertions on the unchanged Y. No new premise is required on the original compact input, and no primitive or completed slice is reselected. □

Lemma A.470 (The same original ambient family for a finite native ribbon)**.** *Under the original native surface and finite-ribbon family hypotheses of Lemma A.183, including an empty finite ribbon, there is one pair deformation Q and one original completed ambient total quotient X with all the ambient atlas and jointly smooth same-Q comparison conclusions of Lemma A.469. The finite-ribbon construction retains its actual native ambient and ribbon identifications, joint relabelings, compact primitive and boundary squares, whole ribbon and positive witness. Its ribbon primitive is the given native one-form, necessarily zero, and its full embedded core is the whole native ribbon image. The ambient original native points and full cylinders satisfy exactly the same two comparison formulas, with the same constructed collars and Q. The parameter-preserving comparison is a smooth diffeomorphism, and its slice pullbacks preserve the very original primitives L_t of this construction: $(q \circ j_t)^*(P_t^*\eta_t) = L_t$ for every interval parameter. The actual smooth slice maps $S_t = P_t \circ q \circ j_t$ have both native and full-cylinder equations of the preceding lemma, satisfy $S_t^*\eta_t = L_t$, and transport the original slice Liouville fields to the same constructed ambient completion fields. In addition, there are a total covector Ω, an alternating total two-covector Ξ and a total tangent field $\mathcal{Z}$ on this same original X such that their tangent evaluation, tangent-pair evaluation and tangent section are C^∞ relative to its closed parameter model, and*

$$\begin{aligned} \Omega_{j_t(x)} \circ d(j_t)_x &= (L_t)_x, \\ \Xi_{j_t(x)}(d(j_t)_x(v), d(j_t)_x(w)) &= (dL_t)_x(v,w), \\ d(j_t)_x(Z_{L_t}(x)) &= \mathcal{Z}_{j_t(x)}. \end{aligned}$$

The full pullback of this very Ω by the original interior insertion has real open smooth coefficient germs on the entire native interior product. There exists $a < 0$ such that its full pullback by the original cylinder insertion has such germs on $s < \min(\delta, -a)$. Both properties include the parameter component at both endpoints. Every point of the same original quotient has an interior representative or a cylinder representative in this very strip. These conclusions retain the same Q, collars, primitives, q and slice diffeomorphisms. The covector has real open coefficients in one actual smooth chart at every original point, with its full tangent identity preserved. The two-covector and field extension assertions below concern the original product patches. The same canonical Ξ, constructed from the spatial derivative $d\ell_t$ and this same q, has real open full two-covector coefficient germs on the entire original interior product and, for some $a_\Xi < 0$, on $s < \min(\delta, -a_\Xi)$ in the original cylinder. Both tangent arguments are retained, and these two patches cover the same original quotient. The same transported $\mathcal{Z}$ has real spatial coefficient extensions in the original interior product patch and, for some $a_\mathcal{Z} < 0$, in its original cylinder patch on $s < \min(\delta, -a_\mathcal{Z})$. On interval points the full differential of each insertion sends $(0, W)$ to this very $\mathcal{Z}$, where W is the respective real spatial extension. These patches also cover X. The original finite ribbon is itself its completed native space. Its canonical slice diffeomorphisms to these same Q ribbon completions retain both native inclusion squares, its exact zero primitive, all-real flows and the whole embedded core as in Lemma A.488. For their actual reference maps $A_t = (P_t^F)^{-1} \circ f_t$, the same retained

ribbon relabelings give one smooth parameter-preserving diffeomorphism $\mathbb{R} \times M \to \mathbb{R} \times K_0^F$. Its interval restrictions are precisely $(t, x) \mapsto (t, A_t(x))$ and $(t, y) \mapsto (t, A_t^{-1}(y))$ in the two directions. Both directions extend smoothly for every real parameter, including the original interval endpoints and the empty-ribbon case.

Proof. Apply the coherent finite native pair-family construction with its ambient coordinate conclusions once. Retain its Q, ambient and finite-ribbon identifications, boundary maps, joint native relabelings and all compact coherence squares. Every one-form on a zero-dimensional manifold is zero, since every tangent vector is zero. Its Liouville field is zero, so its all-forward-time core is the whole finite ribbon. The construction retains that entire core, the given ribbon map and the positive witness, including for an empty ribbon.

Apply the source-only native total-atlas construction once to the same original surface primitive family. It supplies its original δ, C_t, X, smooth total atlas, topology, original slice primitives, joint cylinder coefficients and all-real slice flows. The ambient native maps of the already chosen Q retain that same primitive, its coordinate germs and boundary squares, and the smooth ambient relabeling family gives the required compact coherence square. The original-quotient map lemma therefore applies to those same C_t and that same Q in ambient dimension two. It gives the continuous map with both original-point and full-cylinder formulas, retaining the collar width and every witness of both constructions. The same collars have their boundary criterion and real mixed germs. The same-quotient smoothness lemma therefore proves that this same comparison is jointly smooth in the original total atlas. The exact native and full-cylinder formulas and compact coherence square give bijectivity by the comparison-bijection lemma. The retained native primitive and boundary inclusion squares give the boundary primitive square. The slice primitive transport lemma then identifies the pullback of $P_t^* \eta_t$ with the same chosen original L_t at every parameter. The comparison diffeomorphism lemma gives the same total H with its smooth inverse. The slice completion diffeomorphism lemma then gives the actual $S_t = P_t \circ q \circ j_t$ with both native and full-cylinder equations, and the strict slice comparison lemma gives $S_t^* \eta_t = L_t$. Apply the original slice field-transport lemma to this same q, its proved primitive pullback and the constructed nondegeneracy of dL_t. It gives the asserted transport of these same original fields to the constructed completion fields. These arguments use only the ambient boundary and collars; the zero-dimensional ribbon retains its own boundaryless native atlas. Apply the total-covector slice lemma to this same jointly smooth q, its proved pullback identity and these same original L_t. It gives one Ω with the displayed exact restrictions and closed-model smooth tangent evaluation. The total-exterior lemma, applied to the same varying reference primitive and its proved strict original slice pullback, gives the canonical Ξ with smooth closed-model tangent-pair evaluation and exact restriction to dL_t. Keep the already constructed exact total diffeomorphism $H(z) = (\pi(z), q(z))$ and the already proved nondegeneracy of dL_t. The original total-field lemma applies to these very witnesses and gives the canonical $\mathcal{Z}$ with closed-model smooth tangent section and the displayed original Liouville-field restriction. Choose Ω to be exactly the covector defined by that same varying reference primitive and q, as in the total-covector slice lemma. Apply the original cylinder coefficient lemma to this same covector and the retained native, boundary and full-cylinder equations. It gives the stated real cutoff and full coefficient germs for this very Ω, without replacing its slice restrictions. Apply the original interior coefficient lemma to the same covector, native relabeling family and compact coherence square. It supplies the full real coefficient germs on the entire original interior product. The interior-and-strip cover lemma, with the cutoff just obtained, gives the asserted representatives of every point of the same original quotient. All these constructions use the given Q completed primitive and field families and introduce no new premise on the native input. Apply the original two-form patch coefficient lemma to this same canonical Ξ, using the retained actual q formulas,

native relabelings and varying reference primitive. It gives the claimed entire-interior germs and its negative-cutoff cylinder germs. The strip-cover lemma with this cutoff gives coverage of the same X, with both full tangent arguments preserved. Apply the selected-chart covector lemma to the same Ω and its proved interior and cylinder germs. It gives its stated real open coefficient identity in an actual chart at every original point. Keep the canonical transported field furnished by the total-field construction. Apply the original field patch lemma using this same H, native relabelings, boundary square and full-cylinder formula. It supplies its two real spatial extensions and their exact full differential identities. The strip-cover lemma with the field cutoff gives the asserted coverage of the unchanged original quotient. Apply the original finite-ribbon slice comparison lemma to the already retained Q, native ribbon one-forms and carrier maps. It gives their canonical completed inclusion diffeomorphisms, strict primitive equality, both native squares, all-real flow equation and full embedded-core identity on the original native ribbon itself. The same initial coherent pair construction gives jointly smooth all-real ribbon relabelings and inverses, with their exact compact presentation square. Apply the finite-ribbon total comparison lemma with those very relabelings and that square. Its reference equality identifies the actual Q completed reference maps with the fixed native time-zero completion diffeomorphism composed with those relabelings. Its displayed all-real maps and inverses therefore give the required joint total diffeomorphism with the exact interval restrictions. These applications preserve the original Q, its native ribbon points, and every earlier ambient family witness. □

Lemma A.471 (Strict comparison of a selected original completion primitive)**.** *Let M be a compact smooth native manifold with boundary of positive dimension, and let α be a locally extendible one-form with nondegenerate differential and outward Liouville field. Fix a width $\delta > 0$ and an inward collar C satisfying the original attachment hypotheses, the zero-time boundary identity, smooth native inverse germs, and $\partial_s C(s,b) = -Z_\alpha(C(s,b))$ for $0 \le s < \delta$. On its original full quotient X_C, fix a smooth primitive L with globally nondegenerate differential and the full identities*

$$o^*L = \alpha, \qquad q_C^*L = e^{-s}\pi_{\partial M}^*(\alpha|_{\partial M}).$$

Let D be an independently specified Liouville domain, K a specified completion of D, and $e : M \to D$ a native carrier diffeomorphism whose full primitive pullback is α and whose native coordinates have smooth extension germs. There exist a smooth diffeomorphism $f : X_C \to K$, a native boundary diffeomorphism $b : \partial M \to \partial D$, and a complete flow Φ of Z_L such that

$$\begin{aligned} f^*\lambda_K &= L, \\ f(o(x)) &= \iota_K(e(x)), & f^{-1}(\iota_K(e(x))) &= o(x), \\ \iota_{\partial D}(b(y)) &= e(y), & f(q_C(s,y)) &= E_K(b(y),-s) \quad (s<\delta), \\ f(\Phi^r(z)) &= \Phi_K^r(f(z)) \quad (r \in \mathbb{R}). \end{aligned}$$

For every $x \in M$ the full core correspondence is

$$\big(\forall r \ge 0,\ \Phi^r(o(x)) \in o(M)\big) \quad \Longleftrightarrow \quad e(x) \in \mathrm{Core}(K).$$

The chosen C and L are retained throughout this construction.

Proof. The outward defining-function germs give a strictly negative inward normal component of Z_α at every native boundary point. The same-quotient Liouville-domain theorem therefore equips the compact carrier $o(M)$ with the already chosen primitive L. Its native carrier diffeomorphism is the original o with its compact target atlas. The collar equation and native coordinate-extension

theorem give its required coordinate germs. Thus the fixed-endpoint presentation construction makes this very carrier a native presentation P of α. Apply the same fixed-endpoint construction to e, obtaining a native presentation N on the prescribed D.

The quotient-completion theorem supplies a completion K_0 of P on exactly X_C, with exactly L and its canonical complete flow Φ. Apply the strict inward-collar comparison to P, N, K_0, K. It supplies a strict diffeomorphism f with both displayed native squares. Let b be the boundary diffeomorphism of N; its inclusion square is the displayed boundary equation. The boundary diffeomorphism of P is the identity on the original native boundary: both its boundary inclusion and o send y to $o(y)$, and that inclusion is injective. The comparison's all-real end identity consequently reads $f(E_{K_0}(y,r)) = E_K(b(y),r)$. The signed-cylinder orbit theorem gives $E_{K_0}(y,-s) = q_C(s,y)$ for every $s < \delta$, which proves the entire cylinder equation. Strict primitive preservation transports the canonical complete flows, proving their equality for every real time.

The native square and surjectivity of e give $f(o(M)) = \iota_K(D)$. Since f is bijective, a point belongs to $o(M)$ exactly when its image belongs to $\iota_K(D)$. Apply this equivalence to $\Phi^r(o(x))$, use the flow equation, and require it for every nonnegative real r. This is precisely the displayed full-core equivalence. Every step used the fixed collar, its fixed quotient, and its already selected primitive. □

Lemma A.472 (Smoothness into the intrinsic boundary is reflected by its inclusion)**.** *Let M be a smooth native manifold with boundary, with its actual intrinsic boundary B and inclusion $i : B \to M$. For every map $f : N \to B$ from a smooth manifold with corners and every $x \in N$, f is smooth at x if and only if $i \circ f$ is smooth at x. Consequently the same equivalence holds for global smoothness.*

Proof. The boundary inclusion is smooth, so composition proves the forward implication. Conversely suppose $i \circ f$ is smooth at x. It is continuous there. Since i induces the boundary subspace topology, f is continuous at x as a map into B. Choose a native half-space chart c at $i(f(x))$ and its actual boundary restriction c_B. Continuity permits restricting near x so that f takes values in the source of c_B. On this neighborhood the literal boundary chart formula is

$$c_B \circ f = \mathrm{drop} \circ c \circ i \circ f,$$

where deletion removes the zero normal coordinate of the native chart. The native extended chart is smooth near the given point, and deletion is a linear smooth map. The displayed coordinate expression is therefore smooth at x. Composing with the smooth boundary chart inverse proves that f is smooth at x. Applying this local equivalence at every point proves the global equivalence. □

Lemma A.473 (Joint native relabelings restrict to a joint intrinsic boundary family)**.** *Let $R_s : M \to M$, $s \in \mathbb{R}$, be native diffeomorphisms of a smooth manifold with boundary, and suppose both R_s and R_s^{-1} are jointly smooth. Their literal restrictions $R_s^B : B \to B$ are boundary diffeomorphisms satisfying*

$$i \circ R_s^B = R_s \circ i, \qquad i \circ (R_s^B)^{-1} = R_s^{-1} \circ i.$$

Both boundary directions are jointly smooth for every real parameter.

Proof. A native diffeomorphism preserves the intrinsic interior. Indeed its differential at an interior point is invertible by the chain rule for its smooth inverse. In native target coordinates, the inverse function theorem maps an open source neighborhood onto an open ambient neighborhood of the image coordinate. This image lies in the target half-space, so its center is an interior point of that half-space. Applying the same argument to the inverse identifies the two intrinsic interiors. Taking complements shows that R_s and R_s^{-1} preserve B.

Restrict these two maps to B. Their inverse identities remain true, and their inclusion composites are the two displayed smooth native maps. The smoothness reflection lemma makes both restrictions smooth, so they are boundary diffeomorphisms. For joint smoothness, compose the given native joint map with the smooth product insertion $(s,b) \mapsto (s, i(b))$. This is the inclusion composite of $(s,b) \mapsto R_s^B(b)$. Apply the global smoothness reflection lemma to obtain smoothness of that actual boundary-valued family. The same product insertion and reflection applied to the given joint inverse map prove joint smoothness of the boundary inverse family. Every construction is on the entire real parameter line. □

Lemma A.474 (Original positive-dimensional ribbon slices compared with the same deformation)**.** *Let Q be a pair deformation with positive-dimensional ribbons, and let M be a fixed compact native manifold of that ribbon dimension. Let μ_t have jointly locally extendible coefficients, nondegenerate spatial exterior derivatives and outward native Liouville fields. Suppose $e_t : M \to F_t$ are carrier diffeomorphisms to the literal ribbon domains of Q, retain the full primitive identities $e_t^* \mu_{F_t} = \mu_t$, and have native coordinate extension germs on those same domains. These data construct one original ribbon completion family with collars C_t, common width $\delta > 0$, original quotient slices Y_t, and selected primitives L_t^F. Its original total quotient has the smooth atlas, real mixed transition germs, original fiber homeomorphisms, and native and full-cylinder primitive conclusions of Lemma A.463.*

For every interval parameter there are smooth diffeomorphisms $f_t : Y_t \to K_t^F$ and $b_t : \partial M \to \partial F_t$, and a complete flow Φ_t of $Z_{L_t^F}$, with

$$\begin{aligned} f_t^* \mu_{K_t^F} &= L_t^F, \\ f_t(o_t(x)) &= \iota_{K_t^F}(e_t(x)), \qquad & f_t^{-1}(\iota_{K_t^F}(e_t(x))) &= o_t(x), \\ \iota_{\partial F_t}(b_t(y)) &= e_t(y), & f_t(q_t^F(s,y)) &= E_{K_t^F}(b_t(y), -s) \quad (s < \delta), \\ f_t(\Phi_t^r(z)) &= \Phi_{K_t^F}^r(f_t(z)) \quad (r \in \mathbb{R}). \end{aligned}$$

Here K_t^F is the ribbon completion of this same Q. For every native x,

$$\big(\forall r \geq 0,\ \Phi_t^r(o_t(x)) \in o_t(M)\big) \quad \Longleftrightarrow \quad e_t(x) \in \mathrm{Core}(K_t^F).$$

Writing j_t^Q for the actual ribbon embedding of Q, its full embedded core is exactly

$$j_t^Q(\mathrm{Core}(K_t^F)) = (j_t^Q \circ e_t)\big(\{x \in M : \forall r \geq 0,\ \Phi_t^r(o_t(x)) \in o_t(M)\}\big).$$

The chosen quotient primitives and the prescribed Q, e_t are unchanged. No joint smoothness of the maps f_t or their inverses is asserted.

Proof. Apply the source-only native completed-family atlas construction once to the given μ_t, its joint coefficient germs, nondegeneracy and outwardness. Retain its common δ, original collars C_t, original quotient slices, original total atlas, primitives L_t^F, native and full-cylinder primitive equations and global nondegeneracy. In particular each chosen C_t satisfies the native collar equation for $-Z_{\mu_t}$ and every attachment hypothesis of the selected-primitive strict comparison lemma.

Fix an interval parameter t. Its given e_t presents μ_t on the actual ribbon domain F_t of Q, with the required native coordinate extensions. Apply that strict comparison lemma with the already selected C_t, L_t^F and the actual completion K_t^F. It supplies the displayed f_t, b_t, Φ_t, all native and cylinder squares, strict primitive preservation, complete flow equality, and the full-core equivalence. Choose such witnesses for every t. Surjectivity of e_t and that equivalence identify the native set on

the right of the displayed embedded-core equation with the inverse image of the entire completed ribbon core. Applying j_t^Q therefore gives exactly the displayed set equality, in both directions. This last choice adds no joint regularity assertion. The quotient family was chosen once before these applications, and every target domain and completion is that of the fixed Q. All retained original atlas and primitive conclusions therefore coexist with these same pointwise comparisons. □

Lemma A.475 (The same ribbon comparisons have a joint reference boundary family)**.** *Retain the original positive-dimensional ribbon slice comparisons $f_t : Y_t \to K_t^F$, carrier maps e_t and boundary maps $b_t : B \to \partial F_t$ of the same deformation Q. Let $h_t : \widetilde{F}_0 \to \widetilde{F}_t$ be its actual compact ribbon presentation on the extension spaces, $\bar{h}_t : \partial F_0 \to \partial F_t$ its boundary map, and $P_t : K_0^F \to K_t^F$ its actual completed presentation. Suppose a native diffeomorphism family $R_s : M \to M$ and its inverse are jointly smooth for all real s, and retain the exact compact square*

$$h_t(e_0(R_t(x))) = e_t(x) \qquad (t \in [0,1]).$$

Then the actual boundary square is

$$\bar{h}_t(b_0(R_t^B(y))) = b_t(y).$$

There is an all-real family of boundary diffeomorphisms $\beta_s : B \to \partial F_0$, jointly smooth in both directions, with

$$\beta_t = \bar{h}_t^{-1} \circ b_t, \qquad \beta_t^{-1} = b_t^{-1} \circ \bar{h}_t \qquad (t \in [0,1]).$$

For the actual prescribed reference end map

$$\widetilde{E}_t(v,r) = P_t^{-1}(E_{K_t^F}(\bar{h}_t(v), r)),$$

the unchanged slice comparisons retain the full cylinder equation

$$P_t^{-1}(f_t(q_t^F(s,y))) = \widetilde{E}_t(\beta_t(y), -s) \qquad (s < \delta_F).$$

Proof. Let $i_t : \partial F_t \to F_t^{\mathrm{ext}}$ be the actual boundary inclusion. Its embedding property makes it injective. The compact presentation boundary square and the native boundary squares of the retained comparisons give

$$\begin{aligned} i_t(\bar{h}_t(b_0(R_t^B(y)))) &= h_t(i_0(b_0(R_t^B(y)))) \\ &= h_t(e_0(R_t(y))) \\ &= e_t(y) = i_t(b_t(y)). \end{aligned}$$

In the middle line the native boundary point is regarded as its original point of M; the restriction identity $i \circ R_t^B = R_t \circ i$ justifies this equality. Injectivity of i_t proves the displayed actual boundary square.

For every real s set $\beta_s = b_0 \circ R_s^B$. The preceding joint-boundary restriction lemma gives smoothness of $(s,y) \mapsto R_s^B(y)$ and its inverse family. The fixed b_0 and b_0^{-1} are smooth, so β_s and $\beta_s^{-1} = (R_s^B)^{-1} \circ b_0^{-1}$ are jointly smooth. At interval parameters the proved boundary square gives $\bar{h}_t \circ \beta_t = b_t$. Applying $\bar{h}_t^{-1}$ proves the reference equation; taking its inverse proves the inverse equation. Finally the retained cylinder square gives $f_t(q_t^F(s,y)) = E_{K_t^F}(b_t(y), -s)$. Substitute $b_t(y) = \bar{h}_t(\beta_t(y))$ and apply P_t^{-1}. The definition of $\widetilde{E}_t$ yields the asserted equation at every allowed original cylinder radius, with no change of slice map, collar, primitive, or completed presentation. □

Lemma A.476 (The original ribbon quotient has an exact total reference bijection)**.** *Retain the original positive-dimensional ribbon family, its selected collars C_t, width δ, primitives L_t^F, slice comparisons $f_t : Y_t \to K_t^F$ and total quotient X^F. Let the jointly smooth native relabelings R_s and their inverses satisfy the exact compact square of Lemma A.475. Write $P_t : K_0^F \to K_t^F$ for the actual completed presentation of the same pair deformation Q, and set $A_t = P_t^{-1} \circ f_t$. There is a bijection*

$$H : X^F \longrightarrow [0,1] \times K_0^F, \qquad H(\iota_t(z)) = (t, A_t(z)),$$

preserving the parameter in both directions. It retains the exact equations

$$\begin{aligned} A_t(o_t(x)) &= \iota_{K_0^F}(e_0(R_t(x))), \\ A_t(q_t(s,b)) &= P_t^{-1}(E_{K_t^F}(b_t(b), -s)) \quad (s < \delta), \\ H^{-1}(t,y) &= \iota_t(f_t^{-1}(P_t(y))). \end{aligned}$$

Here $\iota_t : Y_t \to X^F$ is the original slice insertion. In particular these are equations for the unchanged original quotient, not for a replacement family of completions.

Proof. First descend the spatial maps from the compact and full-cylinder summands of the original attachment. Their values on an overlap agree, because the original collar point and its cylinder representative are the same point of Y_t and are evaluated by the same A_t. Thus their quotient descent restricts to A_t on each original slice. Pair that descended map with the original parameter projection to define H. The original fiber theorem gives, for each t, a bijection from Y_t onto the entire fiber of the total parameter map. Every A_t is a bijection because P_t and f_t are diffeomorphisms. Assign to a total point its parameter t and apply A_t to its unique original slice representative. Equality of two images first gives equality of parameters and then equality of slice representatives by injectivity of A_t and of the original slice insertion. Thus H is injective. The point $\iota_t(f_t^{-1}(P_t(y)))$ maps to (t,y), proving surjectivity and the displayed inverse formula. The parameter identities follow from the original slice insertion formula in both directions.

The compact presentation square gives $P_t(\iota_{K_0^F}(e_0(R_t(x)))) = \iota_{K_t^F}(h_t(e_0(R_t(x)))) = \iota_{K_t^F}(e_t(x))$. The retained native equation for f_t, followed by P_t^{-1}, proves the first displayed equation for A_t. Applying P_t^{-1} to the retained full-cylinder equation for f_t proves the second, at every radius $s < \delta$. No collar, selected primitive, slice map or completed presentation is changed in these constructions. □

Lemma A.477 (A real cylinder extension for the same moving ribbon boundary)**.** *Under the hypotheses of the preceding total-reference lemma, let $\beta_s : B \to \partial F_0$, $s \in \mathbb{R}$, be the two-way jointly smooth moving boundary family already constructed from those same R_s, b_0. There are $a < 0$ and a smooth map*

$$\Gamma : \mathbb{R} \times B \times \mathbb{R} \longrightarrow K_0^F$$

such that for interval parameters, $r > a$ and $-r < \delta$,

$$\Gamma(t,b,r) = P_t^{-1}(E_{K_t^F}(b_t(b), r)).$$

This real extension uses the actual moving boundary maps and the prescribed completed presentation of Q. For every $r \geq 0$ it also satisfies $\Gamma(t,b,r) = E_{K_0^F}(\beta_t(b), r)$, and the real map $(s,b,r) \mapsto (s, \Gamma(s,b,r))$ is a local smooth diffeomorphism at every (t,b,r) with $t \in [0,1]$ and $r \geq 0$.

Proof. The prescribed-family end-strip lemma gives $a < 0$ and a real jointly smooth map $\gamma(s,v,r)$ extending the actual reference end map $\widetilde{E}_t(v,r)$ for every interval parameter and $r > a$. Define

$\Gamma(s,b,r)=\gamma(s,\beta_s(b),r)$. The insertion $(s,b,r)\mapsto(s,\beta_s(b),r)$ is smooth by the proved joint boundary smoothness; composition with γ proves smoothness of Γ on the entire real product. For $t\in[0,1]$ and $r>a$, the reference boundary square gives $\bar{h}_t(\beta_t(b))=b_t(b)$. Consequently

$$\Gamma(t,b,r)=\widetilde{E}_t(\beta_t(b),r)=P_t^{-1}(E_{K_t^F}(\bar{h}_t(\beta_t(b)),r))=P_t^{-1}(E_{K_t^F}(b_t(b),r)),$$

which is precisely the required actual cylinder identity. At nonnegative radii the actual end-presentation square identifies this value with $E_{K_0^F}(\beta_t(b),r)$. Near each such radius the strict inequality $a<r$ makes the fixed-parameter spatial germ equal to the prescribed end map composed with the boundary diffeomorphism β_t. The prescribed end map is a local diffeomorphism by boundary orbit calculus and the completed presentation, so this spatial differential is bijective. The parameter-preserving inverse theorem applied to the real smooth Γ gives the claimed real local diffeomorphism, including at interval endpoints. □

Lemma A.478 (The exact ribbon total comparison is jointly smooth)**.** *The preceding exact total comparison H is smooth from the unchanged original ribbon total atlas to $[0,1]\times K_0^F$, including at both parameter endpoints. Its spatial coordinate retains the same native and full-cylinder formulas, with the original inward collar width δ.*

Proof. The spatial coordinate of H on the actual total interior patch is $\iota_{K_0^F}(e_0(R_t(x)))$. The fixed compact map $\iota_{K_0^F}\circ e_0$ is smooth, and $R_s(x)$ is jointly smooth on the entire real parameter line. Restriction to the native interior and the parameter interval therefore proves smoothness of this patch composite, including at the endpoints.

On the original full cylinder, at every radius $s<-a$, its spatial coordinate is $\Gamma(t,b,-s)$ by the preceding real extension and the exact cylinder equation. Radius negation, coordinate insertion and Γ are smooth. Hence the cylinder composite is smooth on this end strip of positive width $-a$. The original family already has its actual smooth total atlas and real mixed transition germs. The smooth end-strip criterion uses those mixed germs to extend this patch smoothness through the remaining positive inward collar, and proves smoothness of the same spatial coordinate on all of X^F. The total parameter map is smooth in that atlas. Taking the product of that parameter map and the spatial coordinate proves smoothness of H. □

Lemma A.479 (The exact ribbon comparison has a jointly smooth inverse)**.** *The same bijection $H:X^F\to[0,1]\times K_0^F$ is a smooth diffeomorphism in the original total atlas. Both directions preserve parameter and retain the exact slice and inverse equations of Lemma A.476. Every original strict slice primitive, complete flow and full-core identity remains the one in Lemma A.474.*

Proof. Put $a_0(x)=\iota_{K_0^F}(e_0(x))$. On the native interior its differential is injective: the native interior inclusion is immersive, e_0 is a diffeomorphism, and the specified compact completion inclusion is immersive. Its source and target have the same spatial dimension, so that differential is bijective. Both manifolds there are boundaryless. The inverse function theorem supplies a smooth inverse germ l of a_0 at every native interior point. Near $(t,a_0(x))$ the formula

$$(u,y)\longmapsto i_{\mathrm{tot}}(u,R_u^{-1}(l(y)))$$

is smooth. Diffeomorphisms of native manifolds with boundary preserve the interior, so this formula takes its values in the actual total interior patch. Its image under H is (u,y) by the exact compact formula and the inverse identities for R_u. Bijectivity of H therefore identifies it with H^{-1} near the chosen point. This proves inverse smoothness over the whole reference compact interior.

Now take a point $(t,E_{K_0^F}(v,r))$ with $r\geq 0$. Write $b=\beta_t^{-1}(v)$. The actual nonnegative end-presentation square identifies this point with $(t,\Gamma(t,b,r))$. Since $a<0\leq r$, the real cylinder

extension agrees near (b, r) with $P_t^{-1} \circ E_{K_t^F} \circ (b_t \times \mathrm{id})$ at this fixed t. Boundary orbit calculus makes the spatial differential of $E_{K_t^F}$ injective. Its domain and target have equal dimension, so it is bijective. The other two maps are diffeomorphisms, giving a bijective spatial differential for $\Gamma(t, \cdot, \cdot)$. The parameter-preserving inverse theorem thus gives a smooth real local inverse for $(u, c, \rho) \mapsto (u, \Gamma(u, c, \rho))$ at (t, b, r). The exact total cylinder equation is $H(q_{\mathrm{tot}}(u, s, c)) = (u, \Gamma(u, c, -s))$ whenever $a < -s$. The strict inequalities $a < r$ and $-r < \delta$ permit the original real-cylinder inverse lemma to apply. It proves smoothness of H^{-1} at this actual nonnegative-end point in the unchanged total atlas, including the seam $r = 0$ and either parameter endpoint.

The completion coverage places every reference point either in the compact interior or in the nonnegative end, including the compact boundary by its seam equation. These two arguments prove smoothness of H^{-1} everywhere. The forward direction was already proved smooth. Their unchanged inverse identities give the asserted diffeomorphism. None of these operations replaces a slice comparison, so its strict primitive, all-real flow and full-core conclusions remain those already proved for the selected original slices. □

Lemma A.480 (The same reference ribbon primitive pulls back to the selected original primitive)**.** *Retain the original positive-dimensional ribbon slices Y_t, their selected primitives L_t^F, comparisons $f_t : Y_t \to K_t^F$, and the actual reference presentations $P_t : K_0^F \to K_t^F$ of the same deformation Q. Write η_t^F for the actual primitive of K_t^F and put $\ell_t^F = P_t^* \eta_t^F$ on K_0^F. Then the same reference comparison $A_t = P_t^{-1} \circ f_t$ and original total spatial map q_F satisfy*

$$A_t^* \ell_t^F = L_t^F, \qquad (q_F \circ \iota_t)^* \ell_t^F = L_t^F \qquad (t \in [0, 1]).$$

Both identities retain the full selected primitive on every original slice tangent space, including the compact seam.

Proof. Pullback composition and the inverse identities of P_t give

$$A_t^* \ell_t^F = f_t^* (P_t^{-1})^* (P_t^* \eta_t^F) = f_t^* \eta_t^F = L_t^F.$$

The last equality is the retained strict primitive identity for the original slice comparison. The exact original total restriction is $q_F \circ \iota_t = A_t$. Substitution proves the second equality. Every map in these pullbacks is the already selected smooth slice map or prescribed presentation, so no primitive or slice atlas is changed. □

Lemma A.481 (One canonical covector on the original ribbon total tangent bundle)**.** *For the same original ribbon total quotient Y, smooth parameter projection π_F, original spatial map q_F and reference primitive family ℓ_t^F, define the actual total tangent covector by*

$$(\Omega_F)_z(v) = (\ell^F_{\pi_F(z)})_{q_F(z)}(d(q_F)_z(v)), \qquad v \in T_z Y.$$

Its tangent evaluation $TY \to \mathbb{R}$ is C^∞ relative to the closed parameter model of the original total atlas, and

$$(\Omega_F)_{\iota_t(x)} \circ d(\iota_t)_x = (L_t^F)_x \qquad (t \in [0, 1],\ x \in Y_t).$$

This canonical covector allows parameter components. Its real open coefficient extensions and the spatial exterior and field realizations are described separately below.

Proof. The original total map q_F and parameter projection are smooth, and Q supplies joint interval smoothness of the reference primitive family ℓ_t^F. Apply the total-covector evaluation lemma to these

same maps and forms. It gives exactly the displayed covector, with smooth tangent evaluation in the closed parameter model. For every original slice, $\pi_F \circ \iota_t = t$ and the chain rule give

$$\begin{aligned}(\Omega_F)_{\iota_t(x)} \circ d(\iota_t)_x &= (\ell_t^F)_{q_F(\iota_t(x))} \circ d(q_F \circ \iota_t)_x \\ &= \big((q_F \circ \iota_t)^* \ell_t^F\big)_x = (L_t^F)_x,\end{aligned}$$

where the last equality is the preceding exact primitive identity. Thus the smooth total covector and all its spatial restrictions use the same original quotient, selected primitives and given Q. □

Lemma A.482 (The exact spatial exterior tensor on the original ribbon total family)**.** *Retain the original positive-dimensional ribbon total quotient Y, its parameter projection π_F, comparison q_F, selected slice primitives L_t^F, and the reference family ℓ_t^F of the same deformation. Define the alternating total two-covector*

$$(\Xi_F)_z(v, w) = (d\ell^F_{\pi_F(z)})_{q_F(z)}(dq_{F,z}v, dq_{F,z}w).$$

Its evaluation on pairs of total tangent vectors is C^∞ relative to the original closed parameter model, and

$$(\Xi_F)_{\iota_t(x)}(d\iota_{t,x}v, d\iota_{t,x}w) = (dL_t^F)_x(v, w).$$

The derivative $d\ell_t^F$ is spatial. This construction retains both full tangent arguments and does not identify Ξ_F with $d\Omega_F$.

Proof. The reference primitive family has jointly smooth spatial exterior derivatives by the exterior-family lemma. The tangent-pair evaluation lemma gives smooth evaluation of that family in the reference product. The map $(z, v, w) \mapsto (\pi_F(z), q_F(z), dq_{F,z}v, dq_{F,z}w)$ is smooth because π_F, q_F and the tangent map of q_F are smooth. Composition gives exactly the displayed evaluation of Ξ_F. Continuity and alternation follow from those of $d\ell_t^F$. On a slice, $\pi_F \circ \iota_t = t$ and the chain rule identify the restriction with $(q_F \circ \iota_t)^* d\ell_t^F$. Naturality of the spatial exterior derivative and the exact reference primitive identity give

$$(q_F \circ \iota_t)^* d\ell_t^F = d\big((q_F \circ \iota_t)^* \ell_t^F\big) = dL_t^F.$$

Thus every original slice restriction is the specified exterior tensor. □

Lemma A.483 (The exact vertical Liouville field on the original ribbon total family)**.** *Retain the same ribbon comparison diffeomorphism $H_F(z) = (\pi_F(z), q_F(z))$ and write Z_t^F for the reference Liouville field of ℓ_t^F. The total field*

$$(\mathcal{Z}_F)_z = d(H_F^{-1})_{H_F(z)}(0, Z^F_{\pi_F(z)}(q_F(z)))$$

has a C^∞ tangent section relative to the original closed parameter model and satisfies

$$d\iota_{t,x}(Z_{L_t^F}(x)) = (\mathcal{Z}_F)_{\iota_t(x)}.$$

Proof. The reference completed field family is jointly smooth. Apply the inverse-differential transport lemma to this family and the same H_F; it gives the displayed smooth total section. The strict slice primitive identity and nondegeneracy imply

$$d(q_F \circ \iota_t)_x(Z_{L_t^F}(x)) = Z_t^F(q_F(\iota_t(x))).$$

Indeed pull back the reference contraction equation along the slice diffeomorphism; injectivity of pullback on covectors and uniqueness of the contraction solution identify the two fields. Since $\pi_F \circ \iota_t = t$, the parameter component of $d(H_F \circ \iota_t)$ is zero. Its evaluation on $Z_{L_t^F}$ is therefore $(0, Z_t^F(q_F(\iota_t(x))))$. Differentiating $H_F^{-1} \circ H_F \circ \iota_t = \iota_t$ gives precisely the claimed original slice identity. □

Lemma A.484 (Real full covector germs on the original ribbon patches)**.** *The full pullback of the same Ω_F by the original interior insertion $o_F : [0,1] \times \operatorname{int} F \to Y$ has real open C^∞ coefficient germs on the entire interior product. There is $a_\Omega < 0$ such that its full pullback by the original cylinder insertion $c_F(t,s,b)$ has such germs on $s < \min(\delta_F, -a_\Omega)$. Agreement includes the parameter component at both endpoints. These two patches cover Y.*

Proof. The same native relabelings give the real smooth interior extension $f(t,x) = \iota_{K_0^F}(e_0^F(R_t(x)))$ of $q_F \circ o_F$. The original cylinder-extension lemma gives $a_\Omega < 0$ and a real smooth map $g(t,s,b) = \Gamma_F(t,b,-s)$ extending $q_F \circ c_F$ on the stated strip. The reference primitive family has real smooth coefficient extensions. Apply the real covector pullback lemma to f and g with that same family. At interval points the insertion parameter is t and the exact comparison squares hold. Differentiate these squares: the chain rule identifies the resulting full covector pullbacks with those of Ω_F. In the natural tangent coordinates the interval inclusion in the real line has identity differential, including at both endpoints. Thus all parameter components are retained. The two real extensions supply the asserted germs. The original interior-and-strip cover lemma, applied with $a_\Omega < 0$, proves coverage of the same Y. □

Lemma A.485 (Real full covector germs in actual original ribbon charts)**.** *At every point z of the original ribbon total atlas, there is an actual smooth chart in which the full coefficient function of this same Ω_F extends smoothly to a real open neighborhood of the chart point. At interval points this extension agrees on the entire actual tangent space, including the parameter direction.*

Proof. Choose an original interior or cylinder-strip representative of z using the preceding cover. The insertion is a diffeomorphism onto an open subset of the original total atlas. Choose a spatial chart at its representative and combine it with the parameter chart. The preceding patch lemma supplies a real open smooth covector coefficient extension there. Transport this chart by the insertion; its transition maps are smooth in the original atlas, so it is an actual smooth chart. The tangent maps of the insertion and its inverse are inverse linear maps. Consequently the coefficient identity for the patch pullback gives exactly the coefficient identity of Ω_F in this transported chart. It holds on every tangent vector. This is the asserted real open germ at z. □

Lemma A.486 (Real full exterior-tensor germs on the original ribbon patches)**.** *The full pullback of the same Ξ_F by o_F has real open C^∞ coefficient germs on the entire original interior product. For some $a_\Xi < 0$, its full pullback by c_F has such germs on $s < \min(\delta_F, -a_\Xi)$. Both full tangent arguments are retained at every interval point, including the endpoints, and these patches cover Y.*

Proof. The spatial exterior-family lemma supplies real smooth coefficient extensions of $d\ell_t^F$. Use the retained smooth real interior extension f and cylinder extension g of the comparison, choosing $a_\Xi < 0$ from the cylinder-extension lemma. Apply the real two-form pullback coefficient lemma to these two maps and the same $d\ell_t^F$. On interval points the exact squares $q_F \circ o_F = f$ and $q_F \circ c_F = g$ hold on the respective patches, and both insertions have parameter t. Differentiating and substituting in the formula for Ξ_F identifies these pullbacks with the real two-form pullbacks just constructed. The interval inclusion has identity differential at both endpoints; therefore neither full tangent argument loses its parameter component. The interior-and-strip cover lemma with this a_Ξ proves coverage. □

Lemma A.487 (Real spatial Liouville-field germs on the original ribbon patches)**.** *The same $\mathcal{Z}_F$ has real spatial field extensions with C^∞ coefficient germs on the entire original interior product*

and, for some $a_Z < 0$, on the original cylinder strip $s < \min(\delta_F, -a_Z)$. If W_{int} and W_{cyl} are these real spatial extensions, their exact interval identities are

$$do_{F,(t,x)}(0, W_{\mathrm{int}}(t,x)) = (\mathcal{Z}_F)_{o_F(t,x)},$$
$$dc_{F,(t,s,b)}(0, W_{\mathrm{cyl}}(t,s,b)) = (\mathcal{Z}_F)_{c_F(t,s,b)}.$$

The two patches cover the same original Y.

Proof. Use the same real comparison extensions on the interior and cylinder patches. At every interval parameter their spatial restrictions are local diffeomorphisms: each insertion is a local diffeomorphism onto an open original slice patch, and the reference slice comparison is a diffeomorphism. Their spatial differentials are therefore bijective. The reference exterior derivatives are nondegenerate, so their pullbacks are nondegenerate on these interval patches. Apply the real transported-field germ lemma to both comparison squares, the same H_F and the reference primitive family. It constructs real spatial fields with smooth coefficient germs and the two displayed identities on the full insertion differentials. Nondegeneracy is required only at interval points; its openness supplies the local real neighborhoods used to solve the contraction equation. The unchanged slice-field identities therefore remain valid. The interior-and-strip cover lemma with the cylinder cutoff proves coverage of the unchanged original quotient. □

Lemma A.488 (Original finite ribbon completions compared with the same deformation)**.** *Let Q be a pair deformation with zero-dimensional ribbons, M a compact smooth native zero-dimensional manifold, and μ_t arbitrary one-forms on M. Let $e_t : M \to F_t$ be native carrier diffeomorphisms to the literal ribbon domains of this same Q. The original completion of (M, μ_t) is M itself. Its canonical map to the actual Q ribbon completion K_t^F is the smooth diffeomorphism $f_t(x) = \iota_{K_t^F}(e_t(x))$. These maps satisfy*

$$f_t^* \mu_{K_t^F} = \mu_t, \qquad f_t^{-1}(\iota_{K_t^F}(e_t(x))) = x,$$
$$f_t(\Phi_t^r(x)) = \Phi_{K_t^F}^r(f_t(x)) \quad (r \in \mathbb{R}), \quad x \in \mathrm{Core}(M, \mu_t) \iff e_t(x) \in \mathrm{Core}(K_t^F).$$

Both cores are the whole compact ribbons. For the actual embedding j_t^Q, the full embedded core equals $(j_t^Q \circ e_t)(\mathrm{Core}(M, \mu_t)) = (j_t^Q \circ e_t)(M)$. This construction includes empty ribbons and imposes no joint regularity on the supplied e_t.

Proof. Every tangent vector on M and on the ribbon completion is zero, so both primitives are zero. The original zero-dimensional completion theorem gives the completion on exactly M, with identity inclusion, constant complete flow and full core M. The target compact boundary is empty, so target completion coverage makes its compact inclusion surjective. Its embedding property makes it injective. Composing that inclusion with e_t gives the canonical bijection f_t. Source and target are discrete zero-dimensional smooth manifolds, so this bijection and its inverse are smooth. Its primitive pullback is zero, hence is exactly μ_t. The inverse native identity follows by applying f_t^{-1} to its defining equation. Strict primitive preservation gives the stated all-real flow equation by complete Liouville-flow transport.

The zero-dimensional full-core theorem identifies the target core with its entire compact carrier. Every $e_t(x)$ lies in that carrier, and every x lies in the original full core, proving the core equivalence. Surjectivity of e_t identifies the two whole carriers. Applying the actual j_t^Q gives exactly the asserted full embedded-core equation. The proof chooses no point and uses no nonemptiness assertion, so it also applies to the empty ribbon. Each construction is at its specified parameter, with no additional regularity assumption on e_t. □

Lemma A.489 (A joint original finite-ribbon comparison on the same deformation)**.** *Let Q be a pair deformation whose ribbon has dimension zero. Let M be a smooth zero-dimensional manifold and let $e_t : M \to F_t$ be native carrier diffeomorphisms to the ribbon domains of this very Q. Suppose $R_s : M \to M$, $s \in \mathbb{R}$, and R_s^{-1} are jointly smooth and retain the compact square*

$$h_t(e_0(R_t(x))) = e_t(x) \qquad (t \in [0,1]),$$

where h_t is the actual compact ribbon presentation of Q. Write K_t for its actual ribbon completions, $P_t : K_0 \to K_t$ for their completed presentation maps, and $a_t(x) = \iota_{K_t}(e_t(x))$. Then the actual reference comparison $A_t = P_t^{-1} \circ a_t$ satisfies

$$A_t = a_0 \circ R_t \qquad (t \in [0,1]).$$

There is a smooth diffeomorphism $H : \mathbb{R} \times M \to \mathbb{R} \times K_0$ such that both directions preserve the real parameter and, at every interval parameter,

$$H(t,x) = (t, A_t(x)), \qquad H^{-1}(t,y) = (t, A_t^{-1}(y)).$$

Both actual reference directions thus have smooth real-parameter extensions, including at the two interval endpoints. Empty ribbons are included.

Proof. Every compact zero-dimensional domain has empty boundary. Completion coverage and injectivity of the compact inclusion therefore make a_t a bijection onto K_t. The source and target are discrete smooth zero-dimensional manifolds, so a_t and its inverse are smooth. The completed compact presentation square gives $P_t(a_0(R_t(x))) = \iota_{K_t}(h_t(e_0(R_t(x))))$. Substitution of the retained native compact square turns the right side into $a_t(x)$. Applying P_t^{-1} proves $A_t = a_0 \circ R_t$.

For every real s define

$$H(s,x) = (s, a_0(R_s(x))), \qquad G(s,y) = (s, R_s^{-1}(a_0^{-1}(y))).$$

Joint smoothness of R_s and R_s^{-1}, and smoothness of the fixed a_0 and its inverse, prove that H and G are smooth on their entire real products. The two inverse identities for a_0 and R_s give $G(H(s,x)) = (s,x)$ and $H(G(s,y)) = (s,y)$. Hence H is a diffeomorphism with inverse G and both directions preserve parameter. On the interval the formula for H is the proved reference equality. Taking inverses in $A_t = a_0 \circ R_t$ gives the claimed formula for G. All identities and smoothness assertions hold on empty products as well, so no nonemptiness assumption is needed. □

Lemma A.490 (Smoothness on the original completion slice is detected by its native patches)**.** *Under the hypotheses of the original half-infinite completion atlas, let N be a smooth manifold with corners. A map $F : X_C \to N$ is smooth in that original completion atlas if and only if its composites with the actual native interior and full-cylinder inclusions are smooth.*

Proof. The native interior inclusion into X_C is the composite of its smooth inclusion into M with the smooth original inclusion o. Both constituent full differentials are bijective, so its differential is bijective by the chain rule. The inverse function theorem gives a smooth inverse germ at each interior-patch point. The actual full cylinder inclusion already has smooth inverse germs by the full-attachment inclusion lemma. Both inclusions are smooth, so smoothness of F implies smoothness of both composites.

Conversely suppose both composites are smooth. At any original interior-patch point, compose its smooth composite with that patch's smooth inverse germ. The inverse identity makes this composition agree with F near the point, proving smoothness of F there. At any cylinder-patch point the same argument uses the cylinder's smooth inverse germ and the given cylinder composite. The original interior and full cylinder cover X_C, so these cases prove smoothness everywhere in its unchanged original atlas. □

Lemma A.491 (Smooth local retractions to the unchanged original slice atlas)**.** *Suppose the same C_t satisfy the total-atlas hypotheses and each slice's original smooth-collar and local inverse hypotheses. Let $j_t : X_{C_t} \to X$ be the canonical original slice insertion. For every $x \in X_{C_t}$ there is a map $r : X \to X_{C_t}$, smooth at $j_t(x)$ in these original atlases, such that $r(j_t(x)) = x$ and, near $j_t(x)$,*

$$\pi(z) = t \quad \Longrightarrow \quad j_t(r(z)) = z.$$

This includes the two endpoint parameters.

Proof. If x has a native interior representative u, choose an actual total-interior chart about $j_t(x) = i_{\mathrm{tot}}(t, u)$. Its smooth local inverse G takes values in $[0, 1] \times \operatorname{int} M$ and satisfies $G(j_t(x)) = (t, u)$ and $i_{\mathrm{tot}}(G(z)) = z$ near that point. Set $r(z) = i_t(\mathrm{pr}_2\, G(z))$ there, with arbitrary values outside that neighborhood. The original slice interior map is smooth, so r is smooth at $j_t(x)$ and $r(j_t(x)) = x$. Applying π to the inverse identity gives $\mathrm{pr}_1\, G(z) = \pi(z)$. If $\pi(z) = t$, the original-point insertion identity consequently gives $j_t(r(z)) = i_{\mathrm{tot}}(t, \mathrm{pr}_2\, G(z)) = z$.

If x has a full-cylinder representative p, use instead an actual total-cylinder chart and its smooth inverse G, with $G(j_t(x)) = (t, p)$ and $q_{\mathrm{tot}}(G(z)) = z$. Define $r(z) = q_t(\mathrm{pr}_2\, G(z))$ near that point. The original slice cylinder map is smooth. The cylinder insertion identity and the equation $\mathrm{pr}_1\, G(z) = \pi(z)$ give both required inverse formulas by the same displayed product-coordinate argument. These two cases exhaust the original slice. All total charts use the interval's boundary model, so the construction applies equally at $t = 0$ and $t = 1$. □

Lemma A.492 (The canonical original slice insertion is a smooth embedding)**.** *Under the same total and slice collar hypotheses, the canonical $j_t : X_{C_t} \to X$ is smooth in the original slice and total atlases, has injective differential everywhere, and is a topological embedding onto the full parameter fiber. Thus it is a smooth embedding, including at $t = 0$ and $t = 1$, retaining the original-point and cylinder formulas.*

Proof. On the original slice interior, j_t is $i_{\mathrm{tot}} \circ (u \mapsto (t, u))$. On its full cylinder it is $q_{\mathrm{tot}} \circ (p \mapsto (t, p))$. Both expressions are smooth: fixed-parameter insertion and both total patch maps are smooth. The original-slice smooth-patch criterion therefore proves smoothness of j_t in its unchanged completion atlas.

At any x, take the preceding smooth retraction r. Continuity of j_t pulls its local fiber-inverse identity back to a neighborhood of x. Since $\pi \circ j_t = t$ and j_t is injective, that identity gives $r \circ j_t = \mathrm{id}$ near x. Differentiating yields $dr_{j_t(x)} \circ dj_{t,x} = \mathrm{id}$. A linear map with a left inverse is injective, so $dj_{t,x}$ is injective. The original fiber-homeomorphism lemma already proves that this same j_t is a topological embedding onto the full fiber with its subspace topology. Together these are the claimed smooth-embedding properties. □

Lemma A.493 (The total atlas reflects smoothness into every original slice)**.** *Under the same collar hypotheses, let N be a smooth manifold with corners and $F : N \to X_{C_t}$ a map. At every $y \in N$, F is smooth at y in the original completion slice atlas if and only if $j_t \circ F$ is smooth at y in the original total atlas. Consequently F is smooth everywhere if and only if $j_t \circ F$ is smooth everywhere. The assertion uses the original slice atlas, without choosing a replacement atlas on the fiber.*

Proof. Smoothness of F implies that of $j_t \circ F$ by composition with the smooth j_t. Conversely suppose $j_t \circ F$ is smooth at y. Take the preceding retraction r at $j_t(F(y))$. Continuity of $j_t \circ F$ pulls the retraction's fiber-inverse identity back to a neighborhood of y. The image of $j_t \circ F$ always has parameter t, so that identity gives $j_t(r(j_t(F(z)))) = j_t(F(z))$ near y. Injectivity of j_t yields

$r(j_t(F(z))) = F(z)$ there. The left side is smooth at y by composition with r, proving smoothness of F at y. Applying this equivalence at every point proves the global equivalence. □

Lemma A.494 (The same original quotient comparison is jointly smooth)**.** *Under the hypotheses of the original-quotient map lemma, suppose those same collars satisfy the boundary criterion and real mixed-germ hypotheses of the original total-atlas lemma. Any map $q : X \to K_0$ with the same original-point and full-cylinder formulas is jointly smooth in that original total atlas, including at the interval endpoints. The same map and collar width are retained.*

Proof. Put $f_0 = \iota_{K_0} \circ e_0$, a smooth map on the native compact domain. The actual original-point formula and compact coherence give

$$(q \circ i_{\mathrm{tot}})(t, x) = f_0(R_t(x)).$$

The map $(t, x) \mapsto R_t(x)$ is jointly smooth with real-parameter extensions. Restricting x to the native interior and t to the interval therefore proves smoothness of this interior expression in its product boundary atlas.

Choose the smooth cylinder extension β and $a < 0$ for these same collars and the same completed presentations. The prescribed full-cylinder formula gives

$$(q \circ q_{\mathrm{tot}})(t, s, b) = \beta(t, b, -s) \qquad (s < -a).$$

The right side is smooth, including at the parameter endpoints, by composition of the real smooth β with parameter inclusion, product-coordinate reordering and radius negation. The strip width $-a$ is positive. The smooth end-strip criterion now proves smoothness of this very q on the whole original total atlas, using its actual mixed germs on the remaining inward overlap and without changing any collar width. □

Lemma A.495 (The original completed-family comparison is a bijection)**.** *Under the compact coherence and exact native and full-cylinder formulas of the original-quotient map lemma, the same map*

$$H : X \longrightarrow [0, 1] \times K_0, \qquad H(z) = (\pi(z), q(z))$$

is a bijection. This assertion needs no assumed inverse or topological identification of the original completed family with the presentation.

Proof. Each point of the original attachment has a unique normalized representative: either an original point $o_{\mathrm{tot}}(t, x)$ or a cylinder point $q_{\mathrm{tot}}(t, s, b)$ with $s < 0$. The compact coherence square gives

$$q(o_{\mathrm{tot}}(t, x)) = \iota_{K_0}(e_0(R_t(x))).$$

For $s < 0$, the prescribed nonnegative-radius completion square gives

$$q(q_{\mathrm{tot}}(t, s, b)) = E_{K_0}(b_0(b), -s).$$

In both cases π is t. Suppose two points have the same image under H. Their parameters agree. If both representatives are native, injectivity of ι_{K_0}, e_0 and R_t makes their native points agree. If both are strictly outward, injectivity of the reference end coordinates and b_0 makes their radius and boundary point agree. The two different cases cannot have the same image: a positive-radius end point lies outside the compact image of K_0. Thus H is injective.

Now take $(t, y) \in [0, 1] \times K_0$. The reference completion's coverage places y either in its compact image or in its end at a radius $r \geq 0$. In the compact case write $y = \iota_{K_0}(e_0(u))$ and set $x = R_t^{-1}(u)$.

The displayed native formula gives $H(o_{\mathrm{tot}}(t,x)) = (t,y)$. In the end case write $y = E_{K_0}(b_0(b), r)$. Since $-r \le 0 < \delta$, $(-r, b)$ is a point of the full original cylinder, and its exact prescribed formula gives $H(q_{\mathrm{tot}}(t,-r,b)) = (t,y)$, also when $r = 0$. Hence H is surjective. □

Lemma A.496 (The constructed inverse retains all original native and cylinder points)**.** *Let H be the preceding bijection. Its inverse satisfies*

$$\begin{aligned} H^{-1}(t, f_t(x)) &= o_{\mathrm{tot}}(t,x), \\ H^{-1}\big(t, P_t^{-1}(E_{K_t}(b_t(b), -s))\big) &= q_{\mathrm{tot}}(t,s,b) \qquad (s < \delta), \\ H^{-1}(t, E_{K_0}(b_0(b), r)) &= q_{\mathrm{tot}}(t,-r,b) \qquad (r \ge 0), \\ \pi(H^{-1}(t,y)) &= t. \end{aligned}$$

In particular the full inward collar, seam and outward end retain their actual points; these are equations for the inverse of the same H.

Proof. The actual native formula gives $H(o_{\mathrm{tot}}(t,x)) = (t, f_t(x))$. Apply H^{-1} to this identity to obtain the first equation. The actual full-cylinder formula similarly gives $H(q_{\mathrm{tot}}(t,s,b)) = (t, P_t^{-1}(E_{K_t}(b_t(b), -s)))$; applying H^{-1} gives the second equation at every allowed radius. For $r \ge 0$ use this equation with $s = -r$ and the specified nonnegative-radius completion square, which identifies its second coordinate with $E_{K_0}(b_0(b), r)$. This proves the third equation. Finally $H(H^{-1}(t,y)) = (t,y)$; comparing its first coordinates gives the last equation. □

Lemma A.497 (The actual reference inclusion has smooth inverse germs on the native interior)**.** *Let M have dimension $k+1 = 2n$, and let $e_0 : M \to D_0$ be the actual native compact diffeomorphism for Q. The map*

$$f_0 : \operatorname{int} M \longrightarrow K_0, \qquad f_0(u) = \iota_{K_0}(e_0(u))$$

is a local smooth diffeomorphism into the given reference completion.

Proof. The native interior inclusion, e_0, the compact carrier inclusion into its extension manifold and ι_{K_0} are smooth, so their composite f_0 is smooth. The native interior differential is injective; the differential of e_0 is invertible; and both the compact carrier inclusion and the specified completion inclusion have injective full differentials. The chain rule therefore makes $df_{0,u}$ injective at every $u \in \operatorname{int} M$. Its source and target have the same finite dimension $k+1 = 2n$, so it is also surjective. Both source and target here are boundaryless manifolds. The inverse function theorem applied in their smooth charts gives a smooth local inverse at every $f_0(u)$, proving the assertion for this same f_0. □

Lemma A.498 (The constructed comparison inverse is smooth over the compact interior)**.** *Under the original total-atlas and comparison-bijection hypotheses, suppose $k+1 = 2n$ and the actual compact coherence family $R_t : M \to M$ and its inverse are jointly smooth with real-parameter extensions. Then the constructed H^{-1} is smooth at every $(t, f_0(u))$ with $u \in \operatorname{int} M$. Equivalently this holds over every point of the actual reference compact interior, using its given native presentation. Smoothness is in the unchanged original total atlas, including at both parameter endpoints.*

Proof. Fix $(t, f_0(u))$ and choose the preceding smooth inverse germ l of f_0 there. A diffeomorphism of manifolds with boundary preserves their interiors, so R_s^{-1} maps $\operatorname{int} M$ to itself. Joint smoothness of the inverse family, restricted to the native interior, makes $(s,v) \mapsto R_s^{-1}(v)$ smooth in its interior atlas. Thus near $(t, f_0(u))$ the formula

$$G(s,y) = i_{\mathrm{tot}}(s, R_s^{-1}(l(y)))$$

is smooth, by smoothness of the actual total interior patch. On the inverse germ's neighborhood $f_0(l(y)) = y$. The exact native formula and compact coherence then give

$$H(G(s,y)) = (s, f_0(R_s(R_s^{-1}(l(y))))) = (s,y).$$

Injectivity of H implies $G(s,y) = H^{-1}(s,y)$ on that neighborhood. Hence the actual inverse is smooth at the chosen point. The smooth parameter extension of R_s^{-1} and the original product boundary atlas make this argument apply at $s = 0$ and $s = 1$ as well.

The compact carrier's intrinsic manifold interior equals its interior in the chosen extension manifold. The given diffeomorphism e_0 preserves interior points, so every reference compact interior point is $e_0(u)$ for some $u \in \operatorname{int} M$. Applying the assertion just proved gives smoothness over every point of its image under ι_{K_0}, as claimed. □

Lemma A.499 (A real cylinder inverse germ gives an inverse in the original family atlas)**.** *Under the original total-atlas hypotheses, let N be a smooth manifold and let $A : X \to [0,1] \times N$ be a bijection. Let $\beta : \mathbb{R} \times \partial M \times \mathbb{R} \to N$ and $a \in \mathbb{R}$ satisfy*

$$A(q_{\mathrm{tot}}(t,s,b)) = (t, \beta(t,b,-s)) \qquad (t \in [0,1],\ s < \delta,\ a < -s).$$

Suppose $a < r$ and $-r < \delta$, and the actual map $G(u,b,v) = (u, \beta(u,b,v))$ has a smooth local inverse at (t,b,r). Then A^{-1} is smooth at $(t, \beta(t,b,r))$ in the original total atlas, including when t is a parameter endpoint.

Proof. Let v be the smooth local inverse of G at (t,b,r). Restrict its domain to interval parameters and write $v(u,y) = (w,c,\rho)$. The inverse identity $G(v(u,y)) = (u,y)$ implies $w = u$. Thus $\beta(u,c,\rho) = y$. At the base point $(c,\rho) = (b,r)$. The strict inequalities $a < r$ and $-r < \delta$ and continuity of v allow restriction to a neighborhood where $a < \rho$ and $-\rho < \delta$. On this neighborhood define

$$B(u,y) = q_{\mathrm{tot}}(u, -\rho, c).$$

The radius and boundary coordinates of v are smooth after interval restriction, and the pair $(-\rho, c)$ lies in the open full cylinder. Consequently B is smooth in the original total atlas. The prescribed cylinder square gives $A(B(u,y)) = (u, \beta(u,c,\rho)) = (u,y)$. Since A is bijective, $B = A^{-1}$ on this neighborhood, proving smoothness of the actual inverse. All restrictions are of smooth real-parameter germs to the interval product boundary atlas, so the argument also holds at $t = 0$ and $t = 1$. □

Lemma A.500 (The actual completed-family inverse is smooth across the seam and outward end)**.** *Under the native inward-square and original total-atlas hypotheses, with $\dim M = 2n$, let H be the constructed comparison bijection. Its inverse is smooth at every*

$$(t, E_{K_0}(b_0(b), r)), \qquad t \in [0,1], \quad r \geq 0,$$

in the unchanged original family atlas. This includes the seam $r = 0$ and both parameter endpoints.

Proof. Choose the actual real smooth cylinder extension β and $a < 0$ of the same completed presentation maps. For interval parameters and every $v > a$ it satisfies

$$\beta(t,c,v) = P_t^{-1}(E_{K_t}(b_t(c), v)).$$

At (b,r) with $r \geq 0$, the inequality $r > a$ persists on a spatial neighborhood. Thus the spatial derivative of $\beta(t,\cdot,\cdot)$ there is the derivative of the displayed actual prescribed end map. The

boundary product map $b_t \times \mathrm{id}$ and P_t^{-1} are smooth diffeomorphisms. The all-real end orbit map has injective differential by boundary orbit calculus; its source and target both have dimension $2n$, so that differential is bijective. Therefore the displayed spatial derivative of β is bijective. The parameter inverse lemma now gives a smooth local inverse for $G(u,c,v) = (u, \beta(u,c,v))$ at (t,b,r).

The exact full-cylinder formula for H and the extension equality give $H(q_{\mathrm{tot}}(u,s,c)) = (u, \beta(u,c,-s))$ whenever $a < -s$. Since $a < r$ and $-r \le 0 < \delta$, the preceding real cylinder inverse lemma applies. It proves smoothness at $(t, \beta(t,b,r))$. The actual nonnegative-radius presentation square identifies $\beta(t,b,r)$ with $E_{K_0}(b_0(b), r)$. This proves precisely the assertion at every seam and outward-end point, using the original collars and the same inverse H^{-1}. □

Lemma A.501 (The same parameter-preserving comparison is a smooth diffeomorphism)**.** *Under the hypotheses of the original smooth quotient-map lemma, including its jointly smooth native compact coherence family and inverse, the bijection*

$$H : X \longrightarrow [0,1] \times K_0, \qquad H(z) = (\pi(z), q(z))$$

and its constructed inverse are smooth in the actual original total atlas and the reference product atlas. They are consequently a smooth diffeomorphism retaining the same underlying bijection, all original native and full-cylinder equations, and the actual parameter projection.

Proof. The original total atlas makes π smooth, and the original quotient-map lemma makes this same q smooth. Their product H is therefore smooth. The comparison-bijection lemma gives its inverse. At every reference compact interior point, the interior inverse lemma proves smoothness of this inverse. Every reference point outside that interior image belongs to the nonnegative end, by the completion's compact and end coverage and its boundary seam equation. The end inverse lemma proves smoothness there as well. These two cases exhaust K_0. Thus H^{-1} is smooth at every point of the reference product, including its parameter boundary. The inverse identities are those of the already proved bijection, so these same two maps define the claimed smooth diffeomorphism. □

Lemma A.502 (A parameter-preserving total diffeomorphism restricts to the original smooth slices)**.** *Under the original total and slice atlas hypotheses, let N be a smooth manifold and let $A : X \to [0,1] \times N$ be a smooth diffeomorphism preserving parameter: $\mathrm{pr}_1(A(z)) = \pi(z)$. For every t, the map*

$$A_t : X_{C_t} \longrightarrow N, \qquad A_t(x) = \mathrm{pr}_2(A(j_t(x)))$$

is a smooth diffeomorphism in the unchanged original slice atlas. Its inverse satisfies $j_t(A_t^{-1}(y)) = A^{-1}(t,y)$.

Proof. Since A preserves parameter, $A^{-1}(t,y)$ lies in the full fiber $\pi^{-1}(t)$. The original fiber bijection therefore gives a unique $x \in X_{C_t}$ with $j_t(x) = A^{-1}(t,y)$; define $B_t(y) = x$. The inverse identities of A and injectivity of j_t give $B_t(A_t(x)) = x$ and $A_t(B_t(y)) = y$. Thus A_t is bijective with inverse B_t, satisfying the displayed inclusion formula.

The actual slice insertion, A, and the second product projection are smooth, so A_t is smooth. The inclusion formula writes $j_t \circ B_t$ as the smooth map $y \mapsto A^{-1}(t,y)$. The original slice smoothness-reflection lemma consequently makes B_t smooth in that same original slice atlas. These two smooth maps and their inverse identities give the required diffeomorphism. □

Lemma A.503 (The exact original slice comparison is a diffeomorphism to the reference completion)**.** *Under the original smooth comparison and slice-atlas hypotheses, the literal spatial restriction*

$q_t = q \circ j_t : X_{C_t} \to K_0$ *is a smooth diffeomorphism in the unchanged original slice atlas. Its inverse has the exact relation*

$$j_t(q_t^{-1}(y)) = H^{-1}(t, y)$$

with the already constructed inverse of that same original total comparison $H(z) = (\pi(z), q(z))$.

Proof. The total comparison diffeomorphism preserves parameter by its exact formula. Apply the preceding slice restriction lemma with $A = H$. Its spatial restriction is $\mathrm{pr}_2(H(j_t(x))) = q(j_t(x)) = q_t(x)$. The resulting diffeomorphism therefore has this exact forward map, and the preceding lemma's inverse inclusion formula is precisely the displayed equation. No original slice atlas or total map is changed. □

Lemma A.504 (The original slice maps to its same constructed completion by the prescribed diffeomorphism)**.** *Under the same original slice comparison hypotheses, the map*

$$F_t = P_t \circ q \circ j_t : X_{C_t} \longrightarrow K_t$$

is a smooth diffeomorphism in the unchanged original slice atlas. It satisfies the exact native and full-cylinder equations

$$F_t(o_t(x)) = \iota_{K_t}(e_t(x)),$$
$$F_t(q_t^{\mathrm{cyl}}(s, b)) = E_{K_t}(b_t(b), -s) \qquad (s < \delta).$$

All maps refer to the same constructed Q*, original collars and completed slices; no new completion is chosen.*

Proof. The reference slice comparison $q \circ j_t$ is a smooth diffeomorphism, and P_t is the specified smooth diffeomorphism to K_t. Their composite is therefore a smooth diffeomorphism. The original native formula gives $q(j_t(o_t(x))) = P_t^{-1}(\iota_{K_t}(e_t(x)))$. Apply P_t to obtain the first displayed equation. The full-cylinder formula gives $q(j_t(q_t^{\mathrm{cyl}}(s, b))) = P_t^{-1}(E_{K_t}(h_{t,\partial}(b_0(b)), -s))$. Apply P_t and use the exact boundary coherence $h_{t,\partial} \circ b_0 = b_t$ to obtain the second equation at every allowed original cylinder radius. □

Lemma A.505 (The same slice diffeomorphism preserves the actual original primitive)**.** *Under the preceding slice diffeomorphism hypotheses, let* L_t *satisfy the prescribed full native and cylinder primitive restrictions, and let the native boundary identifications retain the boundary primitive. Then the same* $F_t = P_t \circ q \circ j_t$ *satisfies*

$$F_t^* \eta_t = L_t \qquad (t \in [0, 1]).$$

It is consequently a primitive-preserving diffeomorphism of these same ambient completions. This statement supplies no ribbon identification or independent endpoint marking.

Proof. The original slice primitive lemma, applied to these very patch restrictions and native boundary primitive squares, gives $(q \circ j_t)^*(P_t^* \eta_t) = L_t$. The two maps in this pullback composition are smooth by the preceding slice diffeomorphism lemma. Hence $F_t^* \eta_t = (q \circ j_t)^*(P_t^* \eta_t) = L_t$. The preceding lemma already proves that this very F_t is a smooth diffeomorphism, so the displayed equality gives the claimed primitive preservation without selecting another map or completion. □

Lemma A.506 (The native and full-cylinder restrictions determine the completed one-form)**.** *Under the original half-infinite completion atlas hypotheses, two one-forms* A, B *on* X_C *are equal if their pullbacks by the actual original inclusion* $o : M \to X_C$ *and the actual full-cylinder map* $q_C : (-\infty, \delta) \times \partial M \to X_C$ *are equal. The native equality is on the full native tangent spaces, including those based at the native boundary.*

Proof. Every point $z \in X_C$ has an original or full-cylinder representative. If $z = o(x)$, the differential do_x is surjective by the original inclusion lemma. Given $v \in T_z X_C$, choose $u \in T_x M$ with $do_x(u) = v$. Equality of the native pullbacks gives $A_z(v) = A_z(do_x(u)) = B_z(do_x(u)) = B_z(v)$. If $z = q_C(p)$, choose instead u with $dq_{C,p}(u) = v$, using surjectivity of the full-cylinder differential. Equality of the cylinder pullbacks gives the same equality on v. Both cases hold for every v and exhaust X_C, so $A = B$. □

Lemma A.507 (The actual native and cylinder formulas force primitive preservation)**.** *Let K be a specified completion of a compact Liouville domain D, with primitive λ_K, compact inclusion ι_K and actual all-real end map E_K. Let λ be a native one-form on M. Suppose $i : M \to D$ is smooth and preserves this full native form, and $b : \partial M \to \partial D$ is a smooth diffeomorphism with $b^*(\lambda_D|_{\partial D}) = \lambda|_{\partial M}$. On the original half-infinite completion X_C, let L satisfy*

$$o^* L = \lambda, \qquad q_C^* L = e^{-s} \pi_{\partial M}^* (\lambda|_{\partial M}).$$

Every smooth map $f : X_C \to K$ with the exact formulas

$$f(o(x)) = \iota_K(i(x)), \qquad f(q_C(s, x)) = E_K(b(x), -s)$$

satisfies $f^* \lambda_K = L$. The conclusion is on the entire original completion, with no primitive-preservation assumption on f.

Proof. Pullback composition and the compact formula give

$$o^*(f^* \lambda_K) = i^*(\iota_K^* \lambda_K) = i^* \lambda_D = \lambda = o^* L.$$

All differentials in this identity use full native tangent spaces, so it also holds at native boundary points. For the full cylinder put $J(s, x) = (b(x), -s)$. The prescribed all-real end primitive is $E_K^* \lambda_K = e^r \pi_{\partial D}^* (\lambda_D|_{\partial D})$; it has zero radial component. Hence the cylinder formula and the boundary pullback identity give

$$q_C^*(f^* \lambda_K) = J^*(E_K^* \lambda_K) = e^{-s} \pi_{\partial M}^*(b^*(\lambda_D|_{\partial D})) = e^{-s} \pi_{\partial M}^*(\lambda|_{\partial M}) = q_C^* L.$$

The preceding patch uniqueness lemma now gives $f^* \lambda_K = L$ on the whole same X_C. □

Lemma A.508 (The original slice transports the same varying completed primitive)**.** *Under the original slice and total atlas hypotheses, let Q be the same pair deformation and let $q : X \to K_0$ be jointly smooth with its exact original-point and full-cylinder formulas. Suppose the native identifications $e_t : M \to D_t$ preserve λ_t, and the boundary identifications b_t preserve its restriction and satisfy $h_{t,\partial} \circ b_0 = b_t$. Let L_t be one-forms on the original completed slices with the exact native and full-cylinder pullbacks of the joint primitive lemma. Writing $j_t : X_{C_t} \to X$ for the canonical original slice insertion, η_t for the primitive on K_t and $\ell_t = P_t^* \eta_t$ for the varying primitive on the reference K_0, one has*

$$(q \circ j_t)^* \ell_t = L_t \qquad (t \in [0, 1]).$$

This is equality of spatial one-forms on every entire original completed slice, including the seam. It specifies no component in the parameter direction of the total space.

Proof. Fix t and put $q_t = q \circ j_t$ and $f_t = P_t \circ q_t$. The original slice insertion and q are smooth in their actual atlases, so both q_t and f_t are smooth. On a native point, their exact formulas cancel P_t with its inverse and give

$$f_t(o_t(x)) = \iota_{K_t}(e_t(x)).$$

On every full-cylinder point, the same cancellation and the exact boundary square give

$$f_t(q_t^{\mathrm{cyl}}(s,b)) = E_{K_t}(h_{t,\partial}(b_0(b)), -s) = E_{K_t}(b_t(b), -s).$$

The preceding primitive pullback lemma applies to these very native and cylinder formulas, their native and boundary primitive identities, and the prescribed patch pullbacks of L_t. It yields $f_t^*\eta_t = L_t$. By pullback composition,

$$(q \circ j_t)^*\ell_t = q_t^*(P_t^*\eta_t) = (P_t \circ q_t)^*\eta_t = f_t^*\eta_t = L_t.$$

The parameter was arbitrary, so this proves the assertion on every interval slice without changing any collar, presentation or primitive. □

Lemma A.509 (A nondegenerate pulled-back two-form forces an injective differential)**.** *Let $f : M \to N$ be a map with differential at x, and let η be a two-form on N. If $(f^*\eta)_x$ is nondegenerate, then df_x is injective.*

Proof. If $df_x(v) = 0$, then for every $w \in T_xM$,

$$(f^*\eta)_x(v,w) = \eta_{f(x)}(df_x(v), df_x(w)) = 0.$$

Nondegeneracy gives $v = 0$. Thus the kernel of df_x is zero, which proves injectivity. □

Lemma A.510 (Equal-dimensional primitive transport forces an invertible differential)**.** *Let M, N be smooth manifolds of the same finite dimension, let $f : M \to N$ be smooth, and let β be a smooth one-form on N. Suppose $f^*\beta = \alpha$. If $d\alpha_x$ is nondegenerate, then $df_x : T_xM \to T_{f(x)}N$ is an isomorphism.*

Proof. Naturality of exterior differentiation gives $d\alpha = d(f^*\beta) = f^*(d\beta)$. The preceding lemma therefore makes df_x injective. Its source and target have the same finite dimension, so its image has the full target dimension; hence it is surjective as well. An injective and surjective linear map is an isomorphism. □

Lemma A.511 (Primitive transport sends the actual Liouville vectors to each other)**.** *Let $f : M \to N$ be a smooth map between equal-dimensional smooth manifolds, and let β be a smooth one-form on N with $f^*\beta = \alpha$. At x, suppose $d\alpha_x$ and $d\beta_{f(x)}$ are nondegenerate. The unique vectors satisfying their Liouville equations obey*

$$df_x(Z_\alpha(x)) = Z_\beta(f(x)).$$

In particular, when N is a specified complete Liouville ambient, the vector on the right is its actual given Liouville field.

Proof. The preceding lemma makes df_x surjective. For any $w \in T_{f(x)}N$, choose $v \in T_xM$ with $df_x(v) = w$. Naturality and the source Liouville equation give

$$\begin{aligned} d\beta_{f(x)}(df_x(Z_\alpha(x)), w) &= d\alpha_x(Z_\alpha(x), v) \\ &= \alpha_x(v) = \beta_{f(x)}(df_x(v)) = \beta_{f(x)}(w). \end{aligned}$$

Thus $df_x(Z_\alpha(x))$ satisfies the target Liouville equation. Nondegeneracy of $d\beta_{f(x)}$ makes its solution unique, proving the equality. In a specified complete ambient its given field satisfies that same equation and is consequently that same unique vector. □

Lemma A.512 (The same original completed slices transport their actual fields)**.** *Under the original slice and total atlas hypotheses, suppose* $\dim M = 2n$ *and* $q : X \to K_0$ *is smooth for the same pair deformation* Q*. Let* L_t *have nondegenerate* dL_t *on each original completed slice, and suppose the proved spatial identity* $(q \circ j_t)^*(P_t^*\eta_t) = L_t$ *holds. Put* $F_t = P_t \circ q \circ j_t : X_{C_t} \to K_t$*. Then*

$$dF_{t,x}(Z_{L_t}(x)) = Z_{K_t}(F_t(x)) \qquad (t \in [0,1],\ x \in X_{C_t}),$$

where Z_{K_t} *is the field of that same specified completion. No completed inverse or additional field-transport equation is assumed.*

Proof. The original slice insertion is smooth, so smoothness of q and P_t gives smoothness of F_t. Pullback composition gives

$$F_t^*\eta_t = (q \circ j_t)^*(P_t^*\eta_t) = L_t.$$

The actual completion K_t has smooth primitive η_t with nondegenerate differential. The original slice and K_t both have dimension $2n$. Apply the preceding field-transport lemma to this very F_t, L_t and η_t, using the stated nondegeneracy of dL_t. It identifies the derivative image of Z_{L_t} with the given actual field of K_t, exactly as displayed. □

Lemma A.513 (Smooth tangent evaluation of an interval family of one-forms)**.** *Let* N *be a smooth finite-dimensional manifold and let* ℓ_a *be a jointly smooth family of one-forms on* N*,* $a \in [0,1]$*, with real smooth coefficient germs through both parameter endpoints. Then*

$$[0,1] \times TN \longrightarrow \mathbb{R}, \qquad (a,(y,w)) \longmapsto (\ell_a)_y(w)$$

is C^∞ *relative to the product's closed parameter model, including at both parameter endpoints.*

Proof. Fix a spatial chart χ on N near y. Its induced tangent chart has coordinates (u,v), where $u = \chi(y)$ and $v = d\chi_y(w)$. Let $A(a,u)$ be a real smooth germ for the covector coefficients of ℓ_a in this chart. At interval parameters the displayed map is $A(a,u)(v)$. Evaluation of a covector on a vector is bilinear, so this expression is smooth on the same real open neighborhood, together with all fiber coordinates v. It is the actual evaluation at every interval parameter there. Such tangent charts cover TN, proving the stated C^∞ regularity at every point in the closed parameter model. □

Lemma A.514 (A smooth total covector from the same varying reference primitive)**.** *Let* X *have the actual original completed-family total atlas, with its smooth parameter projection* π*. Let* $q : X \to K_0$ *be the same smooth original comparison map for a pair deformation* Q*, and let* $\ell_a = P_a^*\eta_a$ *be its jointly smooth varying primitive on* K_0*. The covectors on the actual total tangent spaces defined by*

$$\Omega_z(v) = (\ell_{\pi(z)})_{q(z)}(dq_z(v)), \qquad v \in T_zX,$$

have C^∞ *tangent evaluation* $TX \to \mathbb{R}$ *relative to the closed parameter model of the original total atlas, including its parameter boundary. This assertion does not prescribe a real open extension of the total covector through that boundary. No vanishing on parameter-direction tangents is asserted.*

Proof. Smoothness of q makes its tangent map $(z,v) \mapsto (q(z), dq_z(v))$ smooth. The tangent-bundle projection $(z,v) \mapsto z$ and π are smooth, so

$$(z,v) \longmapsto \big(\pi(z), (q(z), dq_z(v))\big)$$

is smooth into $[0,1] \times TK_0$. Compose it with the smooth tangent evaluation of the varying one-form family from the preceding lemma. The composite is exactly the displayed evaluation of Ω.

The chain rule in their actual manifold charts gives derivatives of every finite order continuous up to the parameter endpoints, so the same holds for this composite in the closed parameter model. Linearity and continuity in v follow from those of dq_z and $(\ell_{\pi(z)})_{q(z)}$, giving a covector on each actual T_zX. □

Lemma A.515 (The total covector restricts to the exact original slice primitives)**.** *Under the preceding hypotheses, let L_a be the very original slice primitives with the proved strict spatial identity $(q \circ j_a)^*\ell_a = L_a$. The same total covector Ω restricts to L_a on every original slice insertion $j_a : X_a \to X$:*

$$\Omega_{j_a(x)}(d(j_a)_x(v)) = (L_a)_x(v) \qquad (a \in [0,1],\ x \in X_a,\ v \in T_xX_a).$$

Thus one total tangent covector, with C^∞ evaluation in the closed parameter model, restricts to all the original primitives on their unchanged original total atlas.

Proof. The original slice insertion satisfies $\pi(j_a(x)) = a$ and is smooth in its unchanged original slice atlas. By the chain rule, $dq_{j_a(x)} \circ d(j_a)_x = d(q \circ j_a)_x$. Substitute these two identities in the definition of Ω:

$$\begin{aligned}\Omega_{j_a(x)}(d(j_a)_x(v)) &= (\ell_a)_{q(j_a(x))}(d(q \circ j_a)_x(v)) \\ &= \big((q \circ j_a)^*\ell_a\big)_x(v) = (L_a)_x(v).\end{aligned}$$

The strict slice identity concerns the same chosen L_a at every original point, including the seam. Hence all displayed restrictions belong to this one smooth total covector without changing the collars, completion spaces, slice atlases or primitive choices. □

Lemma A.516 (Real open coefficients for a full parameter-dependent covector pullback)**.** *Let P, N be smooth boundaryless finite-dimensional manifolds, let ℓ_t be a family of one-forms on N, $t \in [0,1]$, with real smooth coefficient extensions, and let $\beta : \mathbb{R} \times P \to N$ be smooth. Choose a real extension $\widetilde{\ell}_t$ witnessing the assumed real smooth coefficient extensions of that same family. The covectors*

$$A_{(t,x)} = \big(\widetilde{\ell}_t\big)_{\beta(t,x)} \circ d\beta_{(t,x)}$$

have smooth coefficients on a real open neighborhood of each (t_0, x_0) with $t_0 \in [0,1]$. At interval parameters these are the full differential pullbacks of the given ℓ_t, including their values on the parameter direction.

Proof. Fix (t_0, x_0) and choose charts χ on P near x_0 and ψ on N near $\beta(t_0, x_0)$. By continuity of β, restrict the source chart to a real open neighborhood V of $(t_0, \chi(x_0))$ on which $g(t,u) = \psi(\beta(t, \chi^{-1}(u)))$ is defined and smooth. The chosen extension of ℓ has a smooth coefficient covector $L(t,v)$ on a real open neighborhood of $(t_0, \psi(\beta(t_0, x_0)))$. Shrink V so that $(t, g(t,u))$ lies in that coefficient neighborhood. In these source coordinates the displayed covector is

$$L(t, g(t,u)) \circ Dg_{(t,u)}.$$

Both factors are smooth: the first by composition, and the second because differentiation of a smooth map between finite-dimensional spaces is smooth. Composition of a covector and a linear map is bilinear, so this coefficient covector is smooth on V. The chain rule identifies it with A in the actual source chart. At interval parameters $\widetilde{\ell}_t = \ell_t$, which proves the stated agreement on every tangent vector, including the parameter direction. □

Lemma A.517 (Real endpoint coefficients of the same total covector on the original cylinder strip)**.** *Keep the original completed total quotient X, its original cylinder insertion $q_{\mathrm{tot}}(t,s,b)$, its same smooth comparison $q : X \to K_0$, and the total covector Ω constructed from the same reference primitive $\ell_t = P_t^*\eta_t$. There exists $a < 0$ such that, on the original strip $s < \min(\delta, -a)$, the full pullback $q_{\mathrm{tot}}^*\Omega$ has real open smooth coefficient germs at every interval parameter, including both endpoints. The germs are obtained from a real cylinder extension $\beta : \mathbb{R} \times \partial M \times \mathbb{R} \to K_0$ as in Lemma A.466, by taking the coefficients of*

$$(\widetilde{\ell}_t)_{\beta(t,b,-s)} \circ d\big((t,s,b) \mapsto \beta(t,b,-s)\big),$$

where $\widetilde{\ell}$ is a real coefficient extension of that same reference primitive family. Agreement holds on the full cylinder tangent space, including its parameter direction.

Proof. Choose $a < 0$ and β from the real cylinder-extension lemma. Set $f(t,s,b) = \beta(t,b,-s)$. This real cylinder extension is smooth, so f is smooth. The reference primitive family has real smooth coefficient extensions. Apply the preceding coefficient lemma to this f and the same ℓ.

At every interval parameter and every $s < \min(\delta, -a)$, the exact original cylinder formula gives $q(q_{\mathrm{tot}}(t,s,b)) = f(t,s,b)$. This equality holds on an open cylinder strip relative to the parameter interval, so the two maps have the same differential there. In the natural parameter tangent coordinates the inclusion $[0,1] \hookrightarrow \mathbb{R}$ has derivative the identity, also at zero and one. The chain rule therefore identifies the differential of the interval restriction of f with its full real differential. Applying the chain rule to the actual cylinder insertion gives $dq \circ dq_{\mathrm{tot}} = df$ on the full tangent space at every interval parameter, including both endpoints. The parameter projection of $q_{\mathrm{tot}}(t,s,b)$ is t. Substituting these identities into the defining formula for Ω therefore identifies its full cylinder pullback with the displayed real-extension covector at interval parameters. The preceding lemma supplies real open smooth coefficients for exactly this covector, which proves the claimed germs. The strip contains the seam $s = 0$ and the entire outward end $s \le 0$ because $\delta > 0$ and $-a > 0$. □

Lemma A.518 (The original interior and any seam-outward strip cover the total family)**.** *Let $\delta > 0$ and let the original native collars satisfy $C_t(0,b) = b$ and the boundary criterion $C_t(s,b) \in \partial M$ if and only if $s = 0$ for $0 \le s < \delta$. Let $a < 0$. Every point of the same total quotient X is either $o_{\mathrm{int}}(t,x)$ for an original native interior point x, or $q_{\mathrm{tot}}(t,s,b)$ with $s < \min(\delta, -a)$.*

Proof. The original interior and full cylinder cover X. An interior representative already has the required form. For a cylinder representative $q_{\mathrm{tot}}(t,s,b)$, one has $s < \delta$. If $s < -a$, it belongs to the required strip. Otherwise $s \ge -a > 0$. The native boundary criterion makes $C_t(s,b)$ an interior point, and the exact original overlap formula gives

$$q_{\mathrm{tot}}(t,s,b) = o_{\mathrm{int}}(t, C_t(s,b)).$$

This is an original interior representative of that very point, which proves the claim in the remaining case. □

Lemma A.519 (Real endpoint coefficients of the same total covector on the original interior)**.** *Keep the same original completed quotient X, comparison q, reference primitive family ℓ_t and total covector Ω. Let $o_{\mathrm{int}} : [0,1] \times \operatorname{int} M \to X$ be its original interior insertion. Under the native compact coherence hypotheses, keep the jointly smooth real family of native diffeomorphisms R_t and the fixed map $f_0(x) = \iota_{K_0}(e_0(x))$ from that same construction. The full pullback $o_{\mathrm{int}}^*\Omega$ has real open smooth coefficient germs on the entire original interior product, including the parameter component at both endpoints. They are obtained from*

$$(\widetilde{\ell}_t)_{f_0(R_t(x))} \circ d\big((t,x) \mapsto f_0(R_t(x))\big), \qquad x \in \operatorname{int} M,$$

where $\widetilde{\ell}$ witnesses the real smooth coefficient extensions of that same reference primitive family.

Proof. The fixed native inclusion map f_0 is smooth, and the retained family $(t,x) \mapsto R_t(x)$ is smooth on the real parameter product with the original native manifold. Restrict its source to the open native interior and compose with f_0. This gives a smooth real map $f(t,x) = f_0(R_t(x))$ on $\mathbb{R} \times \operatorname{int} M$. The actual compact coherence square and original quotient formula give

$$q(o_{\mathrm{int}}(t,x)) = f(t,x) \qquad (t \in [0,1],\ x \in \operatorname{int} M).$$

The parameter projection of $o_{\mathrm{int}}(t,x)$ is t. The chain rule for this actual smooth insertion identifies the full pullback of Ω with the interval covector $(\ell_t)_{f(t,x)} \circ df_{(t,x)}$. Here the differential of the interval inclusion in the real parameter line is the identity in natural tangent coordinates, also at zero and one, so the displayed formula uses the full real differential at those endpoints. Apply the real pullback coefficient lemma to f and the same reference primitive family. It supplies the required real open smooth coefficients and their agreement with this exact pullback at every original interior point and every interval parameter. □

Lemma A.520 (Smooth inverse-differential transport of the reference field)**.** *Let $H : X \to [0,1] \times N$ be a smooth diffeomorphism in the original total atlas, and let Z_a be a jointly smooth interval family of fields on N with real smooth coefficient germs. Define a field on that actual X by*

$$\mathcal{Z}_z = d(H^{-1})_{H(z)}(0, Z_{\mathrm{pr}_1 H(z)}(\mathrm{pr}_2 H(z))).$$

Its tangent section $z \mapsto (z, \mathcal{Z}_z)$ is C^∞ relative to the closed parameter model of the original total atlas.

Proof. In a spatial tangent chart the family Z_a has jointly smooth vector coefficients, with real smooth germs. Thus $(a,y) \mapsto (y, Z_a(y))$ is a smooth tangent section on $[0,1] \times N$. Combine it with the zero tangent section of $[0,1]$ and the tangent-bundle identification for a product. This gives the smooth product tangent section $(a,y) \mapsto ((a,y),(0,Z_a(y)))$. Compose this section with H and then with the tangent map of the smooth inverse H^{-1}. The basepoint of the resulting tangent vector is $H^{-1}(H(z)) = z$, and its vector is exactly the displayed $\mathcal{Z}_z$. Both tangent maps and their compositions are smooth in the actual atlases, giving the asserted closed-model regularity. □

Lemma A.521 (The same total field retains an exact original slice field)**.** *Under the preceding hypotheses let $j_a : X_a \to X$ be a smooth original slice insertion and let V_a be a field on X_a. If*

$$d(H \circ j_a)_x(V_a(x)) = (0, Z_{\mathrm{pr}_1 H(j_a(x))}(\mathrm{pr}_2 H(j_a(x))))$$

for every x, then the same total field has the exact slice identity $d(j_a)_x(V_a(x)) = \mathcal{Z}_{j_a(x)}$.

Proof. The inverse identity gives $H^{-1} \circ H \circ j_a = j_a$. Differentiate it and evaluate at $V_a(x)$ to obtain

$$d(j_a)_x(V_a(x)) = d(H^{-1})_{H(j_a(x))}\big(d(H \circ j_a)_x(V_a(x))\big).$$

Substitution of the assumed derivative equation makes the right-hand side exactly the defining formula for $\mathcal{Z}_{j_a(x)}$. This proves the claimed equality on the actual original tangent spaces. □

Lemma A.522 (A total tangent section for the same original Liouville fields)**.** *Under the original completed-family and strict primitive hypotheses, suppose $\dim X_a = \dim K_a = 2n$ and dL_a is nondegenerate on every original slice. Keep the same diffeomorphism $H(z) = (\pi(z), q(z))$ and the*

same original L_a. There is one total tangent field $\mathcal{Z}$ on this actual original total atlas whose tangent section is C^∞ relative to the closed parameter model and which satisfies

$$d(j_a)_x(Z_{L_a}(x)) = \mathcal{Z}_{j_a(x)} \qquad (a \in [0,1],\ x \in X_a).$$

No real open extension of this total field through the parameter boundary is prescribed.

Proof. Let Z_a^{ref} be the pullback to K_0 of the actual field of K_a by the same P_a. This reference family is jointly smooth by the given pair deformation. Use it in the smooth inverse-differential transport lemma with the same actual H to construct $\mathcal{Z}$. That lemma gives the asserted tangent-section regularity.

Put $q_a = q \circ j_a$. The proved original field-transport identity is $d(P_a \circ q_a)_x(Z_{L_a}(x)) = Z_{K_a}(P_a(q_a(x)))$. Apply $d(P_a^{-1})$ at the image point. Differentiation of $P_a^{-1} \circ P_a = \mathrm{id}$ and the chain rule give

$$d(q_a)_x(Z_{L_a}(x)) = Z_a^{\mathrm{ref}}(q_a(x)).$$

Since $H \circ j_a(x) = (a, q_a(x))$, its derivative at $Z_{L_a}(x)$ is $(0, Z_a^{\mathrm{ref}}(q_a(x)))$: the first component is the constant parameter a. The exact slice-field lemma now identifies $d(j_a)_x(Z_{L_a}(x))$ with the very $\mathcal{Z}_{j_a(x)}$ already constructed. This proves the statement for every original slice, using the unchanged total diffeomorphism, primitives and atlases. □

Lemma A.523 (Spatial exterior derivatives retain joint coefficient regularity)**.** *Let ℓ_a be a jointly smooth interval family of one-forms on a smooth manifold N, with real smooth coefficient germs across the parameter endpoints. The spatial exterior derivatives $d\ell_a$ form a jointly smooth interval family of two-forms, with such real smooth coefficient germs.*

Proof. In a spatial chart write $\ell_a = \sum_i A_i(a,u)\,du_i$ with real smooth coefficient germs A_i. At every interval parameter the exterior derivative has coefficient matrix $B_{ij}(a,u) = \partial_{u_i} A_j(a,u) - \partial_{u_j} A_i(a,u)$. Spatial partial differentiation preserves real smoothness, so B gives real smooth coefficient germs for the actual $d\ell_a$ throughout the interval, including both endpoints. Naturality of the spatial exterior derivative identifies these expressions on overlapping spatial charts. Thus they are the coefficients of the same actual two-form family in all charts, proving its asserted joint regularity. □

Lemma A.524 (Joint evaluation on pairs of spatial tangent vectors)**.** *Let ω_a be a jointly smooth interval family of two-forms on a smooth finite-dimensional manifold N with real smooth coefficient germs. On the fiberwise tangent-pair bundle, the function*

$$(a, y, w_0, w_1) \longmapsto (\omega_a)_y(w_0, w_1), \qquad w_0, w_1 \in T_yN,$$

is C^∞ relative to the product's closed parameter model.

Proof. An induced tangent-pair chart has coordinates (a, u, v_0, v_1), where u is a spatial coordinate and v_i is the coordinate vector of w_i. If $B(a,u)$ is a real smooth coefficient germ for the alternating bilinear form, the displayed function is $B(a,u)(v_0, v_1)$. Evaluation is trilinear in B, v_0, v_1, so this function is smooth on the same real coefficient neighborhood with arbitrary fiber coordinates. Such charts cover the tangent-pair bundle, proving the stated closed-model regularity at all points. □

Lemma A.525 (A total two-covector for the exact original spatial derivatives)**.** *Under the original completed-family hypotheses, keep the same X, π, q, reference primitives ℓ_a and original slice primitives L_a with $(q \circ j_a)^* \ell_a = L_a$. There is one alternating total two-covector given by*

$$\Xi_z(v, w) = (d\ell_{\pi(z)})_{q(z)}(dq_z(v), dq_z(w)),$$

whose evaluation on pairs of actual total tangent vectors is C^∞ relative to the original closed parameter model. Its original slice restrictions are exactly the spatial derivatives:

$$\Xi_{j_a(x)}(d(j_a)_x(v), d(j_a)_x(w)) = (dL_a)_x(v, w).$$

The two-covector is defined using the spatial derivatives $d\ell_a$; no identity with a total exterior derivative of Ω is asserted.

Proof. The spatial exterior-family lemma gives joint regularity of $d\ell_a$. Apply the tangent-pair evaluation lemma to that same family. The map on total tangent pairs sending (z, v, w) to $(\pi(z), q(z), dq_z(v), dq_z(w))$ is smooth, since it is assembled from the smooth parameter projection, q and its tangent map. Composing it with the reference tangent-pair evaluation gives exactly the displayed evaluation of Ξ, proving its regularity. Alternation and continuity follow from those of $d\ell_a$ and dq_z.

Fix a. The chain rule and $\pi \circ j_a = a$ identify the slice restriction with $(q \circ j_a)^*(d\ell_a)$. The reference primitive ℓ_a and the original comparison slice are smooth. Naturality of the spatial exterior derivative therefore gives

$$(q \circ j_a)^*(d\ell_a) = d\big((q \circ j_a)^*\ell_a\big) = dL_a.$$

This is the claimed exact restriction on every original slice tangent pair, with all spaces and primitive choices unchanged. □

Lemma A.526 (Real open local coefficients of the same total covector in its actual atlas)**.** *Under the original native completed-family hypotheses, keep its unchanged total smooth structure and the same covector Ω. For every original point z, there exist a chart e in the maximal atlas of this smooth structure with z in its source, a real open neighborhood U of $e(z)$ in the full coordinate space, and a smooth covector-valued function B on U, such that*

$$\Omega_y = B(e(y)) \circ de_y$$

for every y in the chart source with $e(y) \in U$. The equality retains the complete tangent covector, including parameter components at both endpoints. It requires one such chart at each point.

Proof. Use the original interior-and-strip cover to choose an actual open product patch containing z. Choose its product coordinates from a spatial chart and a parameter chart. The given real open patch coefficient germ extends the full pullback of Ω in the natural real time coordinate t and these spatial coordinates.

The parameter charts may be chosen with real affine expression $r = t$ or $r = 1-t$. Their inverses have the same corresponding affine expressions, so both extend smoothly to real open intervals, with nonzero derivative equal to 1 or -1. Write $A(t, u)$ for the chosen patch coefficient germ and let $t = \tau(r)$ be that real affine inverse. Define on a sufficiently small real open neighborhood of the chart coordinate of z

$$B(r, u) = A(\tau(r), u) \circ \big((v, w) \mapsto (\tau'(r)v, w)\big).$$

Composition and covector evaluation are smooth, so B is smooth. At every interval point in the chosen neighborhood the chain rule identifies B with the full covector in these product chart coordinates, including the reflected time derivative when it occurs. Lift this product chart through the actual smooth open patch insertion. The lifted chart belongs to the maximal atlas of the unchanged total smooth structure; differentiating its inverse and the insertion gives exactly the displayed covector identity. Its source contains z. Thus e, U, B have all required properties at this original point, which was arbitrary. □

Lemma A.527 (Real open coefficients of a parameter-dependent two-form pullback)**.** *Let P, N be smooth boundaryless finite-dimensional manifolds, and let ω_t be a family of two-forms on N, $t \in [0,1]$, with real smooth coefficient extensions. Let $f : \mathbb{R} \times P \to N$ be smooth and choose a real coefficient extension $\widetilde{\omega}$ of that same family. The alternating two-covectors*

$$B_{(t,x)}(v,w) = (\widetilde{\omega}_t)_{f(t,x)}(df_{(t,x)}v, df_{(t,x)}w)$$

have real open smooth coefficient germs at every (t,x) with $t \in [0,1]$. Their interval restrictions retain both full tangent arguments, including the parameter direction.

Proof. Fix (t_0, x_0). Choose a chart χ near x_0 and a chart ψ near $f(t_0, x_0)$. On a sufficiently small real open neighborhood of $(t_0, \chi(x_0))$, continuity of f ensures that $g(t,u) = \psi(f(t, \chi^{-1}(u)))$ is defined. Shrink this neighborhood so that $(t, g(t,u))$ belongs to a real smooth coefficient neighborhood for $\widetilde{\omega}$. Write its alternating coefficient form as $C(t,y)$. The coefficient form of the displayed pullback is

$$(v,w) \longmapsto C(t, g(t,u))(Dg_{(t,u)}v, Dg_{(t,u)}w).$$

The function g, its differential, and C are smooth there. Evaluation of an alternating form on two linear images is a polynomial operation between finite-dimensional coefficient spaces, so this coefficient form is smooth. The chain rule identifies it with the actual pullback. At interval parameters the chosen extension equals ω_t, giving the claimed equality on every pair of full tangent vectors, without deleting either parameter component. □

Lemma A.528 (Real endpoint coefficients of the original total two-covector on both patches)**.** *Keep the original completed-family quotient X, parameter π, comparison q, reference primitive ℓ_t, and the same total alternating two-covector*

$$\Xi_z(v,w) = (d\ell_{\pi(z)})_{q(z)}(dq_z v, dq_z w).$$

Under the native coherence hypotheses there exists $a < 0$ such that $o_{\mathrm{int}}^{*}\Xi$ has real open smooth coefficient germs on the whole original interior product, and $q_{\mathrm{tot}}^{*}\Xi$ has such germs on $s < \min(\delta, -a)$. These germs retain both full tangent arguments and cover every original quotient point.

Proof. The spatial exterior-family lemma gives real smooth coefficient extensions of $d\ell_t$. On the original native interior the comparison has the retained real extension $f(t,x) = f_0(R_t(x))$. On the cylinder strip the real extension is $g(t,s,b) = \beta(t,b,-s)$ for the $a < 0$ and β supplied by the cylinder-extension lemma. Apply the preceding two-form coefficient lemma to f and to g, using the same reference exterior family.

At interval parameters the exact original comparison formulas say $q \circ o_{\mathrm{int}} = f$ and $q \circ q_{\mathrm{tot}} = g$ on these respective patches. Both insertions have parameter projection t. Differentiate these identities using the chain rule. The interval inclusion has identity differential in the real parameter coordinate, also at its two endpoints. Substitution in the definition of Ξ therefore identifies the full patch pullbacks with the real two-form pullbacks just constructed. This establishes agreement on both tangent arguments at every interval point, including the endpoints. The original interior and any such seam-outward strip cover X by the strip-cover lemma, proving coverage without changing any quotient, primitive, comparison or slice atlas. □

Lemma A.529 (Real endpoint germs for a field transported through an actual product patch)**.** *Let $H : X \to [0,1] \times N$ be a smooth diffeomorphism and let ℓ_t have real smooth coefficient extensions and nondegenerate spatial derivatives for all interval parameters. Let an actual open product patch*

$i : [0,1] \times P \to X$ satisfy $H(i(t,x)) = (t, f(t,x))$, where f has a jointly smooth real extension and each spatial differential $d(f_t)_x$ is bijective on this patch. For the same transported field

$$\mathcal{Z}_z = d(H^{-1})_{H(z)}(0, Z_{\ell_t}(y)), \qquad (t,y) = H(z),$$

there is a real spatial field family $W(t,x)$ whose coefficients have real open smooth germs near the entire interval product and which satisfies

$$di_{(t,x)}(0, W(t,x)) = \mathcal{Z}_{i(t,x)} \qquad (t \in [0,1],\ x \in P).$$

The neighborhoods and germs are constructed from the interval data; no nondegeneracy or field is assumed at off-interval parameters.

Proof. The spatial pullback forms $\alpha_t = f_t^* \ell_t$ have a real extension $\widetilde{\alpha}_t$ with smooth coefficient germs by the chain rule. Define $W(t,x)$ to be the contraction solution for this real extension where its spatial derivative is nondegenerate, and zero elsewhere. At an interval point their spatial exterior derivatives satisfy $d\alpha_t = f_t^* d\ell_t$ and are nondegenerate because $d(f_t)$ is bijective. Fix one such point and use the same real extension near it. Nondegeneracy is an open condition on alternating matrices, so its spatial exterior derivative stays nondegenerate on a smaller real open neighborhood. On that neighborhood solve $\iota_Y d\alpha = \alpha$ by inversion of this matrix. Matrix inversion is smooth on the open set of invertible matrices, hence Y has real smooth vector coefficients there. At every interval point in this neighborhood the equation is the original one, so uniqueness gives $W = Y = Z_{\alpha_t}$ there.

Naturality of pullback and the Liouville equation give $d(f_t) Z_{\alpha_t} = Z_{\ell_t} \circ f_t$ at those interval points. Differentiate the retained product square and evaluate at $(0, Z_{\alpha_t})$. Its image under $d(H \circ i)$ is $(0, Z_{\ell_t})$. Applying dH^{-1} identifies $di(0, Z_{\alpha_t})$ with the same $\mathcal{Z}$. Thus the same real family W has smooth coefficients near every interval point and the displayed exact identity, proving the assertion. □

Lemma A.530 (Real endpoint field germs on the original interior and seam-outward patches)**.** *Under the native coherence, strict primitive and nondegeneracy hypotheses, keep the same original completed total quotient and its transported field $\mathcal{Z}$. There exists $a < 0$ such that this field has real open smooth coefficient germs in the original interior product patch and in the original cylinder product patch on $s < \min(\delta, -a)$. These patches cover every original quotient point, and the interval restrictions remain the exact original slice Liouville fields.*

Proof. The original interior and full cylinder insertions are open smooth patches. Their product comparison squares have the retained real extensions $f_0(R_t(x))$ and $\beta(t, b, -s)$, respectively, with $a < 0$ from the real cylinder-extension construction. For each interval parameter the spatial restrictions are local diffeomorphisms: the insertions are local diffeomorphisms onto open patches of the original slice, and the original comparison on that slice is a diffeomorphism. Their spatial differentials are therefore bijective. Apply the preceding transported-field lemma to these two actual squares and the same varying reference primitive. It gives real smooth coefficient germs for the unchanged $\mathcal{Z}$ on the stated two product patches. The strip-cover lemma covers every point of the same original quotient. Finally the original slice-field restriction identity was proved for this very transported field, so it still gives the original Liouville fields on all slices. □

Lemma A.531 (Time translation of the whole cylinder primitive)**.** *For $a \le 0$, the map $s_a : A \to A$, $s_a(t,b) = (t + a, b)$, is smooth, has bijective differential at every point, and satisfies*

$$s_a^* \eta_A = e^{-a} \eta_A.$$

Proof. Since $t + a \le t < \delta$, the map is defined throughout A. In product coordinates it translates time and fixes the spatial coordinate, so it is smooth. At an image point $(t + a, b)$ the inequality $(t + a) - a < \delta$ persists on a neighborhood. There s_a has the smooth inverse $(v, b) \mapsto (v - a, b)$. Differentiating the two local inverse identities makes its differential bijective. The boundary projection is unchanged by s_a, while its time coordinate is $t+a$. The chain rule and $e^{-(t+a)} = e^{-a}e^{-t}$ give the displayed full one-form identity. □

Lemma A.532 (Exact original native restriction of the full primitive)**.** *Under the native primitive-gluing hypotheses, any one-form η on X_C satisfying $j^*\eta = \beta$ preserves the entire original native primitive:*

$$o^*\eta = \lambda$$

on every full tangent space, including the native boundary.

Proof. The original inclusion factors as $o = j \circ o_{\mathrm{fin}}$. The finite primitive already satisfies $o_{\mathrm{fin}}^*\beta = \lambda$ on the full native tangent spaces. Both inclusions are smooth at every native point, including its boundary. The chain rule for their composite gives

$$o^*\eta = o_{\mathrm{fin}}^*(j^*\eta) = o_{\mathrm{fin}}^*\beta = \lambda.$$

Thus the asserted equality holds on every original full tangent. □

Lemma A.533 (Global nondegeneracy of the prescribed completion differential)**.** *Let λ be a locally extendible native one-form with nondegenerate $d\lambda$. Under the full-attachment atlas hypotheses, suppose a smooth one-form η on X_C satisfies $o^*\eta = \lambda$ and $q^*\eta = \eta_A$. Then $d\eta$ is nondegenerate at every point of X_C.*

Proof. At every original point $o(x)$, native exterior naturality gives $o^*d\eta = d\lambda$, including at the native boundary. The differential do_x is bijective, so the same tangent-space argument as in the pullback-nondegeneracy lemma makes $d\eta$ nondegenerate at $o(x)$. Every point of X_C is original or has a cylinder representative $q(t, b)$. If $t \ge 0$, this representative equals $o(C(t, b))$, so its nondegeneracy is already proved.

If $t < 0$, put $v = (0, b) \in A$. Its image is the original boundary point $q(v) = o(b)$. Pullback by the bijective differential dq_v therefore gives nondegeneracy of $d\eta_A$ at v. The time translation s_t satisfies $s_t(v) = (t, b)$ and $s_t^*\eta_A = e^{-t}\eta_A$. The constant e^{-t} is nonzero, so $d(e^{-t}\eta_A)$ is nondegenerate at v. The bijective differential of s_t and its pullback identity give nondegeneracy of $d\eta_A$ at (t, b). Finally, the cylinder inclusion has bijective differential and $q^*\eta = \eta_A$, so nondegeneracy transfers to $d\eta$ at $q(t, b)$. These cases cover the whole attachment. □

Lemma A.534 (A symplectic primitive on the same actual half-infinite attachment)**.** *Under the native primitive-gluing hypotheses, suppose also that $d\lambda$ is nondegenerate. Then the same actual full attachment has a smooth one-form η with nondegenerate differential and*

$$j^*\eta = \beta, \qquad o^*\eta = \lambda, \qquad q^*\eta = \eta_A.$$

Proof. The smooth primitive-gluing lemma constructs a smooth η with the two whole-patch identities. Its original native restriction is λ by the native-restriction lemma. The global nondegeneracy lemma now applies to this same smooth one-form and gives nondegeneracy of $d\eta$ everywhere. All three identities are retained by these deductions. □

Lemma A.535 (Actual outward charts and compact-complement coverage)**.** *Under the full-attachment hypotheses, let $\chi_b : U_b \to V_b$ be a native boundary chart containing $b \in B$, with $V_b \subset \mathbb{R}^k$. The maps*

$$H_b(r,u) = q(-r, \chi_b^{-1}(u)), \qquad (r,u) \in (-\delta, \infty) \times V_b,$$

are smooth and have surjective differential throughout their domains. If $q^*\eta = \eta_A$, their full primitive identity is $H_b^*\eta = e^r \pi^* \alpha_b$, where $\alpha_b = (\chi_b^{-1})^*(\lambda|_B)$ and $\pi(r,u) = u$. Every point outside $K_1 = o(M) \cup e([0,1] \times B)$ is $H_b(r,u)$ for some such chart and $r \geq 1$. When M is compact, K_1 is compact.

Proof. The map $(r,u) \mapsto (-r, \chi_b^{-1}(u))$ is smooth and takes the displayed domain into A. Its derivative is bijective: sign reversal of time is invertible, and the boundary-chart inverse has invertible differential. Composition with the smooth cylinder inclusion, whose differential is bijective, proves smoothness and surjectivity for H_b. The chain rule for its boundary projection sends an arbitrary tangent (a,v) to $d\chi_b^{-1}(v)$. Its cylinder time is $-r$. Using $q^*\eta = e^{-t}\pi_B^*(\lambda|_B)$ therefore evaluates the pulled-back covector as $e^r(\lambda|_B)_{\chi_b^{-1}(u)}(d\chi_b^{-1}(v))$. This is exactly $e^r\pi^*\alpha_b$, including the zero radial component. An original point belongs to K_1. By original-and-end coverage, any remaining point is $e(r,b)$ with $r > 1$; otherwise it too lies in K_1. Choose the chart containing b and put $u = \chi_b(b)$. Then $H_b(r,u) = e(r,b)$. Compactness of K_1 follows from the compact-enlargement lemma, since a compact manifold has compact boundary. □

Lemma A.536 (The all-real Liouville flow on the actual intrinsic attachment)**.** *Under the native primitive-gluing hypotheses, suppose M is compact and $d\lambda$ is nondegenerate. There is a single smooth primitive η on the actual attachment X_C, with globally nondegenerate differential and the finite, native and full-cylinder pullback identities, whose canonical Liouville field has a smooth flow defined for every real time.*

Proof. Construct η by the symplectic-primitive lemma. The preceding outward-chart lemma supplies actual smooth charts with surjective differentials, the full exponential primitive formula and coverage outside the compact set K_1. The boundary coefficient forms α_b are smooth because the native boundary restriction is smooth. Thus every hypothesis of the complete-flow theorem for prescribed cylindrical primitive charts holds on this very X_C and for this very η. That theorem derives the radial field from the primitive law and proves two-sided compact continuation. Applying it gives the smooth all-real flow of the canonical contraction field. The primitive and its three pullback identities were fixed before this application and remain the same throughout. □

Lemma A.537 (The signed native cylinder curve solves the actual Liouville equation)**.** *Under the native full-attachment hypotheses, let λ be locally extendible and let η be a smooth one-form on the attachment with globally nondegenerate differential and $q^*\eta = \eta_A$. For each $b \in B$, the signed cylinder curve*

$$\gamma_b(r) = q(-r, b), \qquad -\delta < r < \infty,$$

solves the canonical Liouville vector-field equation of η on that entire interval.

Proof. Take the native boundary chart at b and fix $u = \chi_b(b)$. Then $H_b(r,u) = q(-r,b)$ throughout $r > -\delta$. The outward-chart lemma supplies its full exponential primitive identity and surjective actual differential. The radial-integral-curve theorem derived from that identity gives

$$\frac{d}{dr} H_b(r,u) = Z_\eta(H_b(r,u))$$

throughout the same signed chart domain. Substitution of the fixed u identifies this equation with the asserted equation for γ_b. □

Lemma A.538 (Transport of a regular compact domain through an open smooth embedding)**.** *Let $D \subset X$ be a regular compact domain in a smooth boundaryless d-manifold, with a specified smooth boundary manifold and inclusion. If $f : X \to Y$ is an open smooth embedding into a smooth boundaryless d-manifold, then $f(D)$ is a regular compact domain in Y. Its boundary is the same specified manifold with inclusion composed with f, and $f(\mathrm{Fr}_X D) = \mathrm{Fr}_Y f(D)$.*

Proof. Continuity makes $f(D)$ compact and hence closed in the Hausdorff space Y. Because f is a homeomorphism onto an open subset, frontiers inside that subset correspond under f. Every frontier point of $f(D)$ lies in $f(D)$, so no frontier point lies outside $f(X)$. This proves the stated frontier equality. At a source boundary point, let ρ be a smooth regular local defining function. The differential of f is injective and its source and target have equal finite dimension, so it is bijective. The smooth inverse function theorem gives a smooth local inverse g near the image point. Restrict that neighborhood so g takes values where ρ is defined. Injectivity of f then gives

$$y \in f(D) \iff \rho(g(y)) \le 0, \qquad y \in \mathrm{Fr}_Y f(D) \iff \rho(g(y)) = 0.$$

The function $\rho \circ g$ is smooth. Its differential is nonzero: if it vanished, the chain rule for $g \circ f = \mathrm{id}$ would force $d\rho = 0$, contrary to regularity. These are regular local defining functions for $f(D)$. Composing the original boundary inclusion with f gives a smooth topological embedding with injective differential and, by frontier equality, exactly the required image. Thus it is the stated boundary presentation of the transported domain. □

Lemma A.539 (Outwardness through the transported defining functions)**.** *Let $D \subset X$, $f : X \to Y$ and $f(D)$ be as in the domain transport lemma. Let Z, W be vector fields on X, Y satisfying $df_x(Z_x) = W_{f(x)}$ for every $x \in D$. For a source boundary point x, let ρ_x be its chosen defining function and $\rho'_{f(x)}$ the transported function obtained with the smooth local inverse of f. Then*

$$d\rho'_{f(x)}(W_{f(x)}) = d\rho_x(Z_x).$$

If Z is strictly outward for these source defining functions, W is strictly outward for the defining functions of the transported domain $f(D)$.

Proof. The source boundary is contained in D, since D is closed. Thus the vector-field transport equality holds at every source boundary point. Near such a point the smooth local inverse g satisfies $g \circ f = \mathrm{id}$, and the transported function is $\rho_x \circ g$. Differentiating $(\rho_x \circ g) \circ f = \rho_x$ at x gives

$$d\rho'_{f(x)} \circ df_x = d\rho_x.$$

Evaluate this equality at Z_x and use $df_x(Z_x) = W_{f(x)}$ to obtain the asserted scalar equality. Every frontier point of $f(D)$ is the image of a source frontier point by the preceding domain-transport lemma. Its chosen defining function is precisely one of these transported functions. If the source scalar is positive, the equal target scalar is positive. This proves strict outwardness at every target frontier point. □

Lemma A.540 (The original compact native domain inside its actual full attachment)**.** *Under the native full-attachment hypotheses, suppose M is compact. Its literal original image $o(M)$ is*

a regular compact domain whose extension is the constructed smooth manifold X_C. Its boundary manifold is the original native $B = \partial M$, with boundary inclusion

$$b \longmapsto o(b) = q(0, b).$$

Thus this inclusion has image exactly the frontier of $o(M)$ in the full attachment, and all frontier points have regular smooth local sublevel defining functions there.

Proof. The finite native extension P already contains its original image $o_{\mathrm{fin}}(M)$ as a regular compact domain with boundary B and inclusion $o_{\mathrm{fin}}|_B$. Its inclusion $j : P \to X_C$ is a smooth open embedding. Apply the preceding transport lemma to this domain and j. The coordinate-preserving quotient identities give $j \circ o_{\mathrm{fin}} = o$. Thus the transported carrier is exactly $o(M)$ and its boundary inclusion is $o|_B$. The zero-time identity $C(0, b) = b$ and the quotient overlap formula give $o(b) = q(0, b)$. Regular local defining functions and the exact frontier image follow from the same transport lemma. □

Lemma A.541 (The canonical field on every original native point)**.** *Let λ be a native one-form with nondegenerate differential. Under the full-attachment atlas hypotheses, suppose a smooth one-form η on X_C has nondegenerate differential and $o^*\eta = \lambda$. Its canonical Liouville field satisfies*

$$Z_\eta(o(x)) = do_x(Z_\lambda(x))$$

at every original native point, including the native boundary.

Proof. Native exterior naturality gives $o^*d\eta = d\lambda$ on every full source tangent space. For any source tangent v, the two primitive identities and the native contraction equation give

$$d\eta_{o(x)}(do_x Z_\lambda, do_x v) = d\lambda_x(Z_\lambda, v) = \lambda_x(v) = \eta_{o(x)}(do_x v).$$

Surjectivity of do_x makes this the contraction equation against every target tangent. Nondegeneracy of $d\eta$ makes its solution unique, so $do_x Z_\lambda = Z_\eta(o(x))$ as asserted. □

Lemma A.542 (The actual quotient compact domain is Liouville)**.** *Under the native primitive-gluing hypotheses, suppose M is compact, $d\lambda$ is nondegenerate, and the canonical native Liouville field has negative inward normal coefficient at every boundary point. The same actual compact domain $o(M) \subset X_C$ carries a Liouville structure whose primitive η has the finite, native and entire cylinder pullback identities and globally nondegenerate differential. Its field is the canonical field of η and preserves the native field by the original inclusion.*

Proof. Construct η by the full symplectic-primitive theorem. Its smooth nondegenerate differential gives the unique canonical contraction field Z_η. In each smooth coordinate chart this field is obtained by inversion of the smooth invertible matrix of $d\eta$, so it is smooth. The differential $d\eta$ is closed because $d^2\eta = 0$. The original field identity follows from the preceding lemma.

It remains to check outwardness for the actual compact-domain defining functions. Its frontier is $o(\partial M)$ by its boundary inclusion theorem. At $o(x)$ on this frontier, pull back a regular target defining function ρ through o. In native half-space coordinates its smooth local expression vanishes on the zero normal hyperplane and is nonpositive on the positive side. Thus its tangential derivatives vanish. Its normal derivative is nonpositive by its values on the inward normal ray, and is nonzero because $d\rho \neq 0$ and do_x is surjective. The normal derivative is therefore strictly negative. The native field has strictly negative normal coefficient, so the derivative of $\rho \circ o$ on that field is the product of two negative numbers and is positive. The field identity and chain rule give $d\rho(Z_\eta) > 0$

at $o(x)$. This holds at every frontier point, proving outwardness. Compactness, regular boundary, smooth primitive, symplectic differential and the contraction equation now give the asserted Liouville structure on this very domain. All primitive and field identities were retained. □

Lemma A.543 (A full native quotient Liouville domain from the original primitive)**.** *Let M be a compact smooth native manifold with boundary of dimension $k+1$, and let λ be a locally extendible native one-form with nondegenerate differential. Suppose at each boundary point x there is a real function ρ with a smooth extension germ in its native chart, satisfying*

$$\rho(x) = 0, \qquad \rho \le 0 \text{ near } x \text{ within } M, \qquad d\rho_x \neq 0, \qquad d\rho_x(Z_\lambda(x)) > 0.$$

These source data construct a positive-width native inward collar and a Liouville structure on its actual full quotient compact domain. The same primitive preserves the original native primitive and field, the finite-patch primitive and full exponential cylinder law, and has globally nondegenerate differential. The chosen collar's time curves solve $-Z_\lambda$.

Proof. Apply the native Liouville collar construction to the original primitive, its nondegenerate differential and the stated local outward germs. It supplies a single positive width and native collar with its zero-time identity, exact boundary characterization, one-sided smoothness, smooth inverse germs and negative-field time curves. The native defining-function sign lemma turns each given strictly positive $d\rho(Z_\lambda)$ into a strictly negative inward normal coefficient of that field. Thus all hypotheses of the preceding same-collar quotient Liouville theorem hold for this constructed collar. Apply it to obtain the stated structure and all primitive and field identities on the same full quotient. The selected collar already has the asserted negative-field time curves, and this choice is retained in the conclusion. □

Lemma A.544 (The complete boundary orbit equals the signed native cylinder)**.** *For the same actual quotient compact Liouville domain and primitive, let Φ be a complete flow of its canonical Liouville field. Then its all-real boundary orbit satisfies*

$$\Phi^r(o(b)) = q(-r, b) \qquad (b \in B,\ r > -\delta).$$

Proof. Fix b. The curve $r \mapsto \Phi^r(o(b))$ solves the canonical field equation for every real time. The signed cylinder curve $r \mapsto q(-r, b)$ solves the same equation on $(-\delta, \infty)$. At $r = 0$ their common value is $o(b)$ by the actual zero-time seam identity. The canonical field is smooth, so local integral-curve uniqueness identifies the two curves near any time at which they agree. Their equality set in $(-\delta, \infty)$ is therefore open. It is closed by continuity and the Hausdorff property, and contains zero. The interval is connected, so the equality set is the whole interval, proving the asserted equality at every signed radius. □

Lemma A.545 (The actual quotient with its all-real orbit end is a completion)**.** *For a locally extendible native λ, let L be a Liouville structure on the actual quotient compact domain $o(M) \subset X_C$. Suppose its primitive η has globally nondegenerate differential and the prescribed full-cylinder law. Then X_C, with this primitive and canonical complete field, is a completion of that compact domain. Its compact inclusion is the identity on their common extension, and its end map is*

$$E(b, r) = \Phi^r(o(b)) \qquad (b \in B,\ r \in \mathbb{R}).$$

At every $r \ge 0$ this end is the literal attached outward end.

Proof. The actual primitive gives a smooth canonical contraction field, a symplectic differential on all of X_C and the complete flow already constructed there. Take this as the completed ambient. Take the identity as the compact inclusion. It preserves the primitive. On the compact carrier, the field of L solves the same contraction equation as the canonical ambient field, so uniqueness from nondegeneracy proves field preservation.

Define E by the complete boundary orbit. It is smooth and is an immersion at every real radius by boundary-orbit calculus, using the compact field identity and outwardness. The same calculus gives its exponential primitive identity for every real radius and every tangent, including its zero radial component. Its orbit identity is the definition of E. By the preceding lemma, at every nonnegative radius it equals the attached end $e(r, b) = q(-r, b)$. The actual attached nonnegative end is an embedding, so this proves the required end embedding. The original image and this end cover X_C by the original-and-end coverage theorem. Its positive-radius image is disjoint from $o(M)$ by the literal overlap theorem. These are all compact, end, primitive, field and flow requirements for a completion of the same Liouville domain. □

Lemma A.546 (The all-real completion end preserves the original boundary primitive)**.** *If the same compact quotient primitive satisfies $o^*\eta = \lambda$, its compact boundary form is exactly the original native restriction $\lambda|_B$. The completed end map of the preceding lemma satisfies*

$$E^*\eta = e^r \pi_B^*(\lambda|_B)$$

for every real r, including radii below the local signed collar.

Proof. The compact boundary inclusion is $o \circ \iota_B$. The chain rule and full native primitive identity give its boundary form as $\iota_B^* o^* \eta = \iota_B^* \lambda = \lambda|_B$. The all-real complete boundary-orbit primitive theorem gives $E^*\eta = e^r \pi_B^* \alpha_D$ for the compact boundary form α_D. Substitute the boundary equality just proved. This theorem applies on the whole real orbit and therefore proves the formula at every real radius, without extending the signed cylinder coordinates beyond their domain. □

Lemma A.547 (A full completion produced from the original native Liouville data)**.** *Under the original native primitive and outward-germ hypotheses of A.543, there exist a Liouville domain presentation D, a completion Q of D and a smooth closed embedding $i : M \to D^{\mathrm{ext}}$ with bijective full differential at every native point, such that*

$$i(M) = D, \qquad \mathrm{Fr}(D) = i(\partial M).$$

Both the compact primitive and completed primitive pull back to the original λ on all native tangent spaces. The completed field at every original point is the differential image of Z_λ under the native embedding followed by the completion inclusion.

Proof. The source-native Liouville theorem constructs a single native collar, its actual full quotient compact domain and a Liouville structure with the prescribed primitive and field identities. Use that domain as D, and apply the actual quotient-completion construction to this very primitive and its global nondegeneracy and full-cylinder law to obtain Q. Let $i = o$ be the original native inclusion. It is a closed embedding by quotient topology, is smooth with bijective full differential by the inclusion theorem, and its range and boundary image are the domain carrier and frontier by the compact-domain theorem. The compact primitive identity is the constructed full native pullback. The completion inclusion is identity on the common extension, so composing with it preserves that primitive identity. The compact native field identity and the identity completion differential likewise give the claimed completed field equation. Thus every original point, full tangent, primitive and field is retained in the produced domain and completion. □

Lemma A.548 (The original native atlas on the full quotient carrier)**.** *Under the full native attachment hypotheses, the original inclusion $o : M \to X_C$ induces a diffeomorphism from M to the actual compact carrier $o(M)$ with its regular compact-domain native atlas. Its composite with the carrier inclusion is literally o.*

Proof. The original inclusion is a closed embedding onto the actual carrier, so it induces a homeomorphism $e : M \to o(M)$. Its ambient composite is o, which is smooth. The compact-domain atlas characterizes native smoothness by smoothness of the ambient composite, so e is smooth. To check its inverse at $y = e(x)$, write $a : M \to A$ for the native carrier diffeomorphism in the finite extension P, and $j : P \to X_C$ for the finite-patch embedding. The full derivative of j is bijective. The inverse function theorem supplies a smooth local inverse g on an open neighborhood of y, with $j(g(z)) = z$ there. For z in the full compact carrier in this neighborhood, write $z = o(x') = j(a(x'))$. Injectivity of j gives $g(z) = a(x') \in A$. Thus g restricts to a map from the full native carrier to A. Its ambient composite is smooth, so this restriction is native smooth by the same compact-domain atlas criterion. Composing with the finite native diffeomorphism a^{-1} gives e^{-1} near y. The point was arbitrary, so the inverse is smooth everywhere. The equality of its ambient composite with o is its definition. □

Lemma A.549 (Native coordinate extensions in the full completion quotient)**.** *Native coordinate extension germs transport through a smooth embedding of boundaryless manifolds of equal dimension. In particular, if the native collar solves $-Z_\lambda$, its locally extendible primitive has nondegenerate differential and its field is outward for the preferred boundary defining functions, every native coordinate germ along the original inclusion extends smoothly on the actual full quotient X_C, preserving its native source chart.*

Proof. Let $f : N \to P$ be such a smooth embedding and let native maps $i : M \to N$, $h : M \to P$ satisfy $f \circ i = h$. Fix $x \in M$ and choose its source chart a, open neighborhood O of $i(x)$ and smooth coordinate extension $Q : O \to \mathbb{R}^{k+1}$. Injectivity and equal dimension make $df_{i(x)}$ bijective. The inverse function theorem gives a smooth local inverse g of f near $h(x)$. Shrink its open source by $g^{-1}(O)$ and define $Q' = Q \circ g$ there. For any $y \in M$ whose image $h(y)$ lies there, the local inverse identity gives $f(g(h(y))) = h(y) = f(i(y))$. Injectivity gives $g(h(y)) = i(y) \in O$. The original extension identity therefore gives y in the source of a and $Q'(h(y)) = a(y)$. This proves every asserted germ property. For the actual quotient take N to be the finite extension, $P = X_C$ and $f = j$. The fixed-collar coordinate extension theorem supplies Q at interior and boundary points under the primitive, outwardness and collar equation hypotheses. The square $j \circ o_{\mathrm{fin}} = o$ is literal, so the preceding transport applies. □

Lemma A.550 (A native Liouville presentation on the same full quotient)**.** *The original native primitive and outward-germ hypotheses of A.543 produce a native Liouville presentation on that same full quotient compact domain. Its native inclusion is exactly o, and its carrier diffeomorphism is the preceding one. The same selected quotient primitive and canonical field retain their finite, native and full cylinder identities and global symplectic nondegeneracy.*

Proof. Choose the collar and Liouville structure by the source-native Liouville theorem. It retains the collar equation for $-Z_\lambda$. The native defining-function sign theorem turns the given outward germs into outwardness for the preferred defining functions. Consequently the preceding coordinate extension theorem applies to this very collar. Its full quotient carrier diffeomorphism and its native primitive identity now satisfy the fixed-endpoint native presentation theorem. Apply that theorem to the actual quotient compact domain with this Liouville structure. Its domain and native inclusion are unchanged by construction. The finite, native and cylinder primitive identities, global

nondegeneracy and canonical and native field identities are those already proved for the chosen Liouville structure; packaging the presentation retains all of them. □

Lemma A.551 (A completion of the same produced native Liouville presentation)**.** *Under the original native primitive and outward-germ hypotheses, there exist a native Liouville presentation P of (M, λ) and a completion K of that very represented domain. The completed ambient is its same extension manifold, with its same primitive, and its inclusion on the entire extension is the identity. The completed native inclusion $a : M \to \widehat{P}$ satisfies $a^*\lambda_{\widehat{P}} = \lambda$ on every native tangent space.*

Proof. Choose the collar, quotient Liouville structure and native presentation once by the preceding source-native presentation theorem. The selected structure has the full-cylinder primitive law and global symplectic nondegeneracy. Apply the actual quotient completion theorem to that same collar and structure. Its ambient is the structure's extension, its primitive is unchanged and its full extension inclusion is the identity. The presentation's domain is exactly this selected quotient Liouville domain, so this completion is a completion of P's domain. Its native composite inclusion therefore equals P's native inclusion. The native presentation's primitive identity, followed by the identity completion inclusion, gives the asserted completed native pullback on all tangent directions. All choices in this argument are the single collar and structure already selected for P. □

Lemma A.552 (The full native contraction field of any Liouville presentation)**.** *Let P be a native Liouville presentation of a positive-dimensional (M, μ), with inclusion $i : M \to X$, primitive λ and bijective full native differential. Then $d\mu$ is nondegenerate at every native point and*

$$Z_\lambda(i(x)) = di_x(Z_\mu(x)) \qquad (x \in M).$$

These conclusions use the native primitive and exterior-derivative pullback identities, including all boundary tangent directions.

Proof. If $v \in T_xM$ has $d\mu_x(v, u) = 0$ for every $u \in T_xM$, the derivative pullback identity gives $d\lambda_{i(x)}(di_x v, di_x u) = 0$ for every u. Surjectivity of di_x and nondegeneracy of $d\lambda_{i(x)}$ imply $di_x v = 0$. Injectivity of di_x gives $v = 0$. Hence $d\mu_x$ is nondegenerate. For any $w \in T_{i(x)}X$, write $w = di_x u$. The defining contraction equation and the two pullback identities give

$$d\lambda_{i(x)}(di_x Z_\mu(x), w) = d\mu_x(Z_\mu(x), u) = \mu_x(u) = \lambda_{i(x)}(w).$$

Thus $di_x Z_\mu(x)$ solves the ambient contraction equation. Its unique solution is $Z_\lambda(i(x))$, proving the field identity. □

Lemma A.553 (Native boundary comparison and its actual collar seam)**.** *For two native Liouville presentations P, Q of the same positive-dimensional (M, μ), the frontier of each carrier is the image of the native boundary. There are native boundary diffeomorphisms $a_P : \partial M \to B_P$ and $a_Q : \partial M \to B_Q$ with*

$$\iota_{B_P} a_P(x) = i_P(x), \qquad \iota_{B_Q} a_Q(x) = i_Q(x).$$

Their induced boundary comparison is $b = a_Q \circ a_P^{-1}$. For supplied actual completions C, K, let A be the radius-preserving comparison of their compact flow-collar neighborhoods using b. At every original native boundary point in its domain, $A(i_P(x)) = i_Q(x)$.

Proof. The native carrier diffeomorphism carries the native boundary onto the intrinsic boundary of the carrier. The latter boundary maps under carrier inclusion exactly onto its frontier, so this gives the asserted frontier equality for each presentation. The native inclusion is smooth, injective and immersive. Apply the native boundary presentation lemma to obtain a_P, a_Q and their displayed inclusion squares. Their composition b is a diffeomorphism and $b(a_P(x)) = a_Q(x)$.

The compact flow collar at radius zero is the actual boundary inclusion. The radius-preserving collar comparison sends its point with coordinates $(0, a_P(x))$ to the point with coordinates $(0, b(a_P(x)))$. Using the preceding equality and the two inclusion squares identifies these points with $i_P(x)$ and $i_Q(x)$, respectively. This proves the exact original native boundary seam identity wherever A is defined. □

Lemma A.554 (Uniqueness on a right half-open interval in a smooth manifold)**.** *Let X be a smooth boundaryless manifold and Z a smooth vector field. Two curves $\mathbb{R} \to X$ solving the integral-curve equation on $[0, \delta)$ with the same value at zero agree throughout that interval. Their equations include the right derivative at zero; no negative-time extension of either curve is required.*

Proof. If $[0, \delta)$ is empty the assertion is vacuous. Otherwise, at any common-image point choose a chart and a small closed coordinate ball inside its open target. The smooth coordinate field has bounded derivative on a slightly larger closed ball and is therefore Lipschitz on the smaller ball. Continuity of the two curves keeps both coordinate paths in that ball for a short right interval starting at their common value. For coordinate paths u, v with $u(0) = v(0)$, their differential equations and the fundamental theorem of calculus give

$$|u(s) - v(s)| \le L \int_0^s |u(r) - v(r)|\, dr.$$

Put $D(s) = \int_0^s |u(r) - v(r)|\, dr$. Then $D(0) = 0$, $D \ge 0$, and $D' \le LD$ on this interval. Thus $e^{-Ls} D(s)$ is nonincreasing and nonnegative with initial value zero; it vanishes. Consequently $u = v$, and injectivity of the chart gives local right agreement. The same proof applies after translating any common time to zero.

Fix $b \in [0, \delta)$. The agreement set is closed in $[0, b]$ by continuity and the Hausdorff property, contains zero, and has local right agreement at each of its points before b by the preceding argument. To see that it contains the whole interval, let c be the supremum of times through which all earlier times agree. Closedness gives agreement at c. If $c < b$, local right agreement extends that initial interval beyond c, contradicting the supremum. Hence $c = b$ and agreement holds on $[0, b]$. Since b was arbitrary, the curves agree on all of $[0, \delta)$. □

Lemma A.555 (Actual completion collars retain native inward trajectories)**.** *Let P, Q present the same positive-dimensional native (M, μ), and let C, K be actual completions of their respective domains. Let $\gamma : \mathbb{R} \to M$ solve $\dot\gamma = -Z_\mu$ on $[0, \delta)$ with $\gamma(0) = x \in \partial M$. If a_P, a_Q are the native boundary identifications and E_C, E_K the all-real boundary orbit maps, then*

$$\iota_C(i_P(\gamma(t))) = E_C(a_P(x), -t), \qquad \iota_K(i_Q(\gamma(t))) = E_K(a_Q(x), -t).$$

The radius-preserving compact collar comparison A induced by $a_Q a_P^{-1}$ satisfies

$$A(i_P(\gamma(t))) = i_Q(\gamma(t))$$

at every such point in its domain.

Proof. Put $f = \iota_C \circ i_P$. The native field identity and the completion's field identity on the compact carrier, combined with the full chain rule, give $df(Z_\mu) = Z_C \circ f$. Thus $f \circ \gamma$ is an integral curve of $-Z_C$ on $[0, \delta)$. The curve $t \mapsto \Phi_C^{-t}(f(x))$ is another such curve with the same initial value. The completed field is smooth on its boundaryless ambient, so right-interval uniqueness identifies these curves throughout that interval. The native boundary square identifies $f(x)$ with the actual completed boundary point at $a_P(x)$, and the definition of the orbit map gives the first displayed identity. The identical argument for Q, K proves the second.

If $i_P(\gamma(t))$ lies in the compact collar comparison domain, its completed image equals $E_C(a_P(x), -t)$ by the first identity. Injectivity of the all-real end parametrization therefore identifies its collar coordinates with $(a_P(x), -t)$. The comparison preserves that radius and replaces the boundary point by $a_Q(x)$. Its completed image is consequently $E_K(a_Q(x), -t) = \iota_K(i_Q(\gamma(t)))$ by the second identity. Injectivity of the target completion inclusion gives the asserted original-point equality in its compact extension manifold. □

Lemma A.556 (Comparing two compact presentations through the original inclusion)**.** *Let P, Q be two Liouville presentations of the same positive-dimensional compact native manifold (M, μ). Write $a : M \to X_P$ and $i : M \to X_Q$ for their original inclusions, and λ_P, λ_Q for their primitives. Assume that i has smooth open extension germs in the original native source charts. There are restricted open neighborhoods of the entire carriers in X_P, X_Q and a smooth diffeomorphism h between them such that*

$$h \circ a = i, \qquad h(a(M)) = i(M), \qquad (h^*\lambda_Q)_{a(x)} = (\lambda_P)_{a(x)} \quad (x \in M).$$

It induces a smooth diffeomorphism b between the actual boundary presentations with $h \circ \iota_{\partial P} = \iota_{\partial Q} \circ b$. Neither h nor germs of its inverse are supplied as hypotheses.

Proof. Extend the specified original inclusion i through P by the native inclusion-extension lemma. This gives a smooth embedding J on an open neighborhood of $a(M)$ with $J \circ a = i$. The two extension manifolds have the same dimension. Apply the compact equivalence-neighborhood lemma to J, restricting its domain further to an open neighborhood U of $a(M)$. It is a diffeomorphism $h : U \to V$ onto an open set $V \subset X_Q$. For every $x \in M$, $h(a(x)) = i(x)$, so $i(M) \subset V$ and $h(a(M)) = i(M)$. An open-neighborhood diffeomorphism preserves frontiers of these carriers. The frontiers are exactly the embedded boundaries. Restricting h and h^{-1} to these smooth embedded submanifolds gives inverse smooth boundary maps, defining b and its stated square.

Fix $x \in M$ and $v \in T_{a(x)}X_P$. The native presentation gives a surjective full differential da_x, so $v = da_x(w)$ for some $w \in T_xM$. The chain rule and the two native primitive identities give

$$(h^*\lambda_Q)_{a(x)}(v) = \lambda_{Q\,i(x)}(di_x(w)) = \mu_x(w) = \lambda_{P\,a(x)}(da_x(w)).$$

This proves the full covector equality, including at boundary points. All the restrictions contain the entire original carriers. □

Lemma A.557 (Strict comparison onto a prescribed completion from original inclusion germs)**.** *Under the preceding hypotheses, let C, K be supplied actual completions of P, Q, respectively. There exist a primitive-preserving smooth diffeomorphism $H : C \to K$ and a smooth boundary diffeomorphism b with*

$$H(\iota_C(a(x))) = \iota_K(i(x)), \qquad H^{-1}(\iota_K(i(x))) = \iota_C(a(x)) \quad (x \in M),$$

$$H(\iota_C(\iota_{\partial P}(y))) = \iota_K(\iota_{\partial Q}(b(y))), \qquad H(E_C(y, r)) = E_K(b(y), r) \quad (r \in \mathbb{R}),$$

and $H(\Phi_C^t(z)) = \Phi_K^t(H(z))$ for every real t and $z \in C$. In particular H is an inclusion marking onto the prescribed ambient K, with compact inclusion determined by the original map i. Its underlying strict comparison is uniquely determined by the native point equations; this uniqueness requires no original inclusion extension germs. The same-radius end identity here comes from matching the actual compact boundary sections of the two presentations of (M, μ).

Proof. Choose the compact neighborhood comparison h, b from the preceding lemma. Restrict the two compact presentations to these neighborhoods. Their carriers and completed ambients are unchanged. The full compact primitive identity is $h^*\lambda_Q = \lambda_P$, so the compact presentation completion theorem with scale one gives a strict smooth diffeomorphism $H : C \to K$. Its compact square gives $H(\iota_C(a(x))) = \iota_K(i(x))$. Apply H^{-1} to obtain the inverse native square. Its nonnegative end square at radius zero, together with $E_C(y, 0) = \iota_C(\iota_{\partial P}(y))$ and the corresponding identity for K, gives the displayed boundary square.

Primitive preservation transports the actual Liouville fields and, by complete-flow uniqueness, their flows for all real times. Since $E_C(y, r) = \Phi_C^r(\iota_C(\iota_{\partial P}(y)))$ and likewise for K, flow transport and the boundary square give the asserted end identity for every real radius. Every point of the source compact carrier is $a(x)$ for a unique native x. Define its prescribed target inclusion to be $\iota_K(i(x))$. The native square is precisely the compact inclusion identity for H, and the all-real end and flow identities give its flow-induced marking condition. Thus H with these identities is the required inclusion marking. If another strict comparison has the same native point values, carrier coverage makes its compact restriction identical. The compact uniqueness theorem for completed maps then makes the two maps equal everywhere. □

Lemma A.558 (Extending the original native identity by interior and collar germs)**.** *Let P, Q present the same positive-dimensional native (M, μ), with actual completions C, K. Suppose $\delta > 0$ and $c : [0, \delta) \times \partial M \to M$ is an open embedding with $c(0, x) = x$ and each curve $t \mapsto c(t, x)$ solving $-Z_\mu$. There are an open neighborhood U of $i_P(M)$ in X_P and a smooth map $g : U \to X_Q$ with $g(i_P(y)) = i_Q(y)$ for every native y. No open source-chart extensions of the original inclusions are supplied.*

Proof. At an interior point y, restrict i_P to the boundaryless native interior. Its differential is bijective, so the smooth inverse function theorem supplies a local smooth inverse q on an open neighborhood of $i_P(y)$. Compose q with the restriction of i_Q to the native interior. If $i_P(z)$ belongs to this open neighborhood, the inverse identity gives $i_P(q(i_P(z))) = i_P(z)$; injectivity of i_P implies $q(i_P(z)) = z$. Thus the resulting smooth local map has the prescribed native values at every carrier point of its domain.

At a boundary point x, the native collar image A is open in M and contains x. Since i_P is an embedding, there is an open $T \subset X_P$ with $i_P^{-1}(T) = A$. Intersect T with the open compact collar comparison neighborhood W. This intersection contains $i_P(x)$ because the comparison is defined at the actual boundary seam. Its comparison map is smooth. If $i_P(y)$ belongs to this intersection, then $y = c(t, z)$ for some $0 \le t < \delta$ and $z \in \partial M$. The native inward collar identity proves that the comparison sends $i_P(y)$ to $i_Q(y)$. This gives the required boundary local germ.

Every point of the compact carrier is the image of a unique native point, which is either interior or boundary. The preceding germs therefore extend the single carrier map $i_P(y) \mapsto i_Q(y)$ at all its points. Apply the compact manifold-valued germ extension lemma to obtain g on one open neighborhood of the entire carrier, retaining exactly these native values. Agreement away from the carrier is not required by that lemma, and has not been assumed. □

Lemma A.559 (Compact primitive-preserving comparison from the native inward collar)**.** *Under the preceding hypotheses, open restrictions of P, Q admit a smooth neighborhood diffeomorphism h and an induced smooth boundary map b satisfying*

$$h(i_P(x)) = i_Q(x), \qquad h(i_P(M)) = i_Q(M), \qquad (h^*\lambda_Q)_{i_P(x)} = (\lambda_P)_{i_P(x)}.$$

The primitive equality holds on all ambient tangent directions at every carrier point. The maps h, b are constructed from the native identity and collar, rather than supplied.

Proof. Choose g on an open carrier neighborhood from the preceding lemma. Its carrier restriction is injective because its native values are $i_Q(x)$ and i_Q is injective. Differentiating $g \circ i_P = i_Q$ gives $dg_{i_P(x)} \circ di_{Px} = di_{Qx}$ on the full native tangent space. Both native inclusion differentials are bijective, hence $dg_{i_P(x)}$ is injective. The compact equivalence-neighborhood lemma now restricts g to a smooth diffeomorphism $h : U' \to V$ on open neighborhoods of the carriers. Every $i_Q(x) = h(i_P(x))$ lies in V, and the original native square gives exact carrier preservation. Frontiers are preserved by this diffeomorphism and equal the actual embedded boundaries, so the boundary restriction induces b.

For $v \in T_{i_P(x)}X_P$, write $v = di_{Px}(w)$. The native primitive identities and the chain rule give

$$(h^*\lambda_Q)_{i_P(x)}(v) = \lambda_{Qi_Q(x)}(di_{Qx}(w)) = \mu_x(w) = \lambda_{Pi_P(x)}(v).$$

This proves the full carrier covector equality, including at its boundary. □

Lemma A.560 (Strict completed comparison from the native inward collar)**.** *Under these hypotheses there are a primitive-preserving diffeomorphism $H : C \to K$ and a smooth boundary diffeomorphism b such that*

$$H(\iota_C(i_P(x))) = \iota_K(i_Q(x)), \qquad H^{-1}(\iota_K(i_Q(x))) = \iota_C(i_P(x)),$$

$$H(E_C(y, r)) = E_K(b(y), r) \quad (r \in \mathbb{R}), \qquad H(\Phi_C^t(z)) = \Phi_K^t(H(z)) \quad (t \in \mathbb{R}).$$

In particular H is an inclusion marking onto the prescribed ambient K, retaining the original native points and their all-real trajectories.

Proof. Apply the compact comparison lemma and restrict P, Q to its open neighborhoods. Their completed ambients are unchanged. The full compact primitive identity permits the prescribed compact presentation completion theorem at scale one. It supplies a primitive-preserving smooth completed diffeomorphism H, the induced smooth boundary map b, and the compact square. The native compact comparison square gives the first displayed identity; application of H^{-1} gives the second.

At radius zero the completed end square identifies the two actual boundary sections. Primitive preservation transports the completed Liouville fields and their all-real flows. Expressing the orbit end map as the flow from its actual boundary section therefore proves the same-radius end square for every real radius, and gives the displayed all-real flow equation. Every source carrier point is $i_P(x)$ for a unique native point x. The compact native square gives the prescribed inclusion $\iota_K(i_Q(x))$, while the flow equation gives its flow-induced condition. Thus the constructed H is the asserted inclusion marking onto K. □

Lemma A.561 (The constructed native completion marked onto a prescribed completion)**.** *Let (M, μ) have the original native primitive extension, nondegeneracy and outward defining-function germs of the native completion theorem. Let Q be an independently specified native presentation of*

this same primitive, with original inclusion $i : M \to X_Q$, and let K be a supplied actual completion of Q. There are a constructed native presentation P, a completion C of that very P, and an inclusion marking $q : C \to K$ with

$$q(\iota_C(a(x))) = \iota_K(i(x)) \quad (x \in M).$$

The ambient of C is the extension manifold of P, its primitive is the same constructed primitive, and its inclusion on that entire extension is the identity. On the full native tangent space the completed inclusion pulls back the primitive to μ. The independently prescribed completion remains an explicit input; open extension germs of its original inclusion are not inputs.

Proof. Apply the native completion theorem once to the supplied primitive, nondegeneracy and outward germs. It constructs P together with a completion C of that same presentation, with the stated ambient, primitive, identity inclusion and full native primitive pullback. The same original primitive and outward germs produce a uniform native inward collar: it is an open embedding, its seam is the original native boundary, and its time curves solve $-Z_\mu$. Use that produced collar in the preceding strict completed comparison theorem for P, C and the independently supplied Q, K. It gives an inclusion marking $q : C \to K$ with the displayed native square. The comparison changes none of the already selected P, C, their primitive or inclusion identities. Thus all these conclusions hold simultaneously without supplying original-inclusion extension germs. □

Lemma A.562 (The zero-dimensional completion on the original native space)**.** *Let M be a compact Hausdorff second-countable smooth zero-dimensional manifold, with an arbitrary one-form λ. Its Liouville completion can be taken on the original space M with its original smooth structure, primitive λ, identity compact inclusion and empty cylindrical end. The flow fixes every point for every real time. The full core is literally M, and for every map $j : M \to A$ its core image is $j(M)$.*

Proof. Every tangent space of M is zero. Thus every positive-degree form, including λ and $d\lambda$, is zero, and the Liouville field is zero. The zero two-form is nondegenerate on the zero vector space. The intrinsic boundary is empty. Use M itself as the completed ambient, with identity compact inclusion and the unique map from the empty end. The identity inclusion preserves the primitive and field, has invertible differential, and covers every ambient point. All seam and end conditions hold because their domains are empty. The action $(t, x) \mapsto x$ is smooth and solves the zero field for every real time, so it gives a complete flow. Every point stays in the compact carrier for all nonnegative times. Hence the full core equals M, and taking its image under j gives exactly $j(M)$. This also covers the empty manifold. □

Lemma A.563 (Marking the carrier completion onto the native zero-dimensional source)**.** *The carrier completion of a native zero-dimensional source has a primitive-preserving inclusion marking onto the complete ambient M of the preceding lemma. It sends the compact representative of each $x \in M$ to that same native point x.*

Proof. The carrier completion consists of the points of M carrying their membership in the whole compact carrier. Forgetting that membership is a bijection onto M, whose inverse supplies it. Both spaces are discrete zero-dimensional smooth manifolds, so the bijection and its inverse are smooth. Both primitives vanish, proving primitive preservation. The compact inclusion is sent to its original point by construction. The boundary is empty, so the flow-induced end condition has no instances. These data give the asserted inclusion marking. □

Definition A.564 (Intrinsic compact smooth bundles over the parameter interval)**.** *Fix a finite-dimensional real model with corners and a compact, Hausdorff, second-countable smooth model fibre*

F. An intrinsic compact smooth bundle over $[0,1]$ is a topological bundle with actual fibres E_t and local trivializations modelled on F, with the following joint extension condition for every ordered pair of local trivializations. Write $g_t : F \to F$ for their transition at an overlap parameter t. For every native source chart a and x in its source, use a's extended source coordinates and the native target chart at $g_t(x)$. The coordinate expression of g_s admits a C^∞ extension on an open subset of $\mathbb{R}\times\mathbb{R}^d$ containing $(t, a(x))$, agreeing with the actual transition for interval parameters s in the overlap and native source coordinates there. This condition includes both parameter endpoints and spatial boundaries. The source models include the native half-space for positive-dimensional domains and ribbons, and the boundaryless zero-dimensional model for zero-dimensional ribbons.

Lemma A.565 (Smooth native fibre transition diffeomorphisms)**.** *At each overlap parameter of an intrinsic compact smooth bundle, the actual fibre transition is a smooth diffeomorphism of F. Its inverse is the transition for the reversed ordered pair of trivializations.*

Proof. Fix the parameter t and a native source point x. The joint extension condition gives a smooth coordinate map $\beta(s,z)$ on an open neighborhood of $(t, a(x))$. Restricting it along $z \mapsto (t,z)$ gives a smooth extension of the actual fibre transition's coordinate expression near x. The topological bundle transition is continuous, so these local coordinate expressions prove its native smoothness at every x. The same argument for the reversed ordered pair gives smoothness of the reversed transition. The bundle trivialization identities make these actual maps inverse to one another. Thus the original transition, with that smooth inverse, is the asserted diffeomorphism. □

Lemma A.566 (Boundary preservation by native bundle transitions)**.** *Every actual transition g_t of an intrinsic compact smooth bundle preserves the intrinsic boundary: $g_t^{-1}(\partial F) = \partial F$.*

Proof. The preceding lemma gives a smooth inverse for g_t. At an interior point the derivative of g_t is invertible, since the chain rule for its smooth inverse gives its inverse linear map. In native coordinates at that interior point, the smooth inverse function theorem makes its coordinate map take an open neighborhood onto an open ambient neighborhood of the image. This neighborhood lies in the target model range. Hence the image coordinate is an interior point of that range, so the image point is intrinsically interior. Applying the same argument to g_t^{-1} gives equality of the interior images. Taking complements proves the displayed boundary equality. In the zero-dimensional boundaryless model both boundaries are empty, and the same conclusion holds. □

Lemma A.567 (Joint smoothness under a smooth overlap reparameterization)**.** *For an ordered bundle transition g_t, let $\sigma : \mathbb{R} \to [0,1]$ have smooth real-valued coordinate and have image in the two trivializations' overlap. Then*

$$(s,x) \longmapsto g_{\sigma(s)}(x)$$

is jointly smooth on $\mathbb{R}\times F$ in the native product model.

Proof. The scalar coordinate of σ is smooth and therefore continuous; the induced interval topology makes σ continuous. Continuity of the original bundle transition gives continuity of the displayed map. Fix (s_0, x_0) and apply the joint extension condition at $(\sigma(s_0), x_0)$, obtaining an open smooth coordinate extension $\beta(t,z)$ in a fixed target chart near the image point. Compose it with the smooth map $(s,z) \mapsto (\sigma(s), z)$. On a neighborhood of the chosen point this composition agrees with the actual reparameterized transition at native source coordinates: every $\sigma(s)$ lies in the overlap by hypothesis. The composition is a smooth open extension of those coordinate expressions, including source-boundary points. Together with continuity this proves native joint smoothness at (s_0, x_0). The point was arbitrary, so joint smoothness holds everywhere. □

Lemma A.568 (A smooth overlap reparameterization fixed near an interior seam)**.** *If S is an open subset of $[0,1]$ and $c \in S \cap (0,1)$, there is $\sigma : \mathbb{R} \to [0,1]$ whose scalar coordinate is smooth, whose image lies in S, which equals the identity near c and equals c outside a bounded neighborhood of c.*

Proof. Choose $u < c < v$ with $(u, v) \subset S \cap (0,1)$, using the relative openness of S and the interior condition on c. Choose $r > 0$ smaller than both $c - u$ and $v - c$. Let b be a smooth bump function equal to one near c, zero whenever $|s - c| \geq r$, and satisfying $0 \leq b \leq 1$. Set $\sigma(s) = c + (s - c)b(s)$. Its scalar coordinate is smooth. If $|s - c| < r$, then $|\sigma(s) - c| \leq |s - c| < r$; otherwise $\sigma(s) = c$. Thus $u < \sigma(s) < v$ for every s, proving both interval-valuedness and membership in S. Where $b = 1$ it equals s, and where $|s - c| \geq r$ it equals c. These are the required neighborhood and bounded-support identities. □

Lemma A.569 (A smooth actual transition correction preserving the base)**.** *Let e, e' be two atlas trivializations and write g_t for the actual change from the e fibre coordinate to the e' fibre coordinate. If $\sigma : \mathbb{R} \to [0,1]$ has smooth scalar coordinate and image in their overlap, then*

$$G(s, x) = (s, g_{\sigma(s)}(x))$$

is a smooth diffeomorphism of $\mathbb{R} \times F$ preserving time. Its restriction to $[0,1] \times F$ corrects e to an actual bundle trivialization $\bar{e}$ with the same base open set. At every base point t where $\sigma(t) = t$, this corrected trivialization equals e' on the entire fibre over t.

Proof. Joint reparameterized transition smoothness makes G smooth. The reversed actual transition gives the map $(s, x) \mapsto (s, g^{-1}_{\sigma(s)}(x))$, which is smooth by the same theorem. The ordered transition identities make these maps inverse at each s, so G is the asserted diffeomorphism. Both maps preserve their first coordinate, hence restrict to inverse continuous maps of $[0,1] \times F$. Composing the open partial homeomorphism e with this product homeomorphism gives an actual trivialization $\bar{e}$. Since the first coordinate is unchanged, its target is the same base open set times F and it still projects to the bundle base. For a point p over t in that base set its formula is $\bar{e}(p) = (t, g_{\sigma(t)}(\mathrm{pr}_F\, e(p)))$. If $\sigma(t) = t$, membership of its value in the overlap puts t in both base sets, and the actual transition identity identifies this expression with $e'(p)$, for every such p. □

Lemma A.570 (An actual bundle-chart splice with whole-fibre agreement near the seam)**.** *Let e_L, e_R be atlas trivializations meeting at $c \in (0,1)$. There is a smooth overlap reparameterization σ and an actual topological bundle trivialization T whose base open set is*

$$(\mathrm{base}(e_L) \cap [0, c]) \ \cup \ (\mathrm{base}(e_R) \cap (c, 1]).$$

Its formula is e_L at base parameters at most c, and at parameters greater than c it is the correction of e_R by the actual transition to e_L evaluated at σ. On an open base neighborhood of c, T equals e_L on every point of every fibre. The transition correction is a smooth product diffeomorphism; no membership of T in the designated smooth bundle atlas is asserted here.

Proof. Apply the cutoff lemma to the open overlap to construct σ with $\sigma(t) = t$ near c. Correct e_R toward e_L using the preceding lemma. At c the corrected chart equals e_L on the entire fibre. Paste e_L on the side $t \leq c$ with the corrected chart on the side $t \geq c$. The equal seam values give a continuous map by the closed-set pasting lemma; the corresponding inverses have the same seam values and paste continuously in the same way. Both maps preserve the base and are inverse fibrewise, so they form an open partial homeomorphism onto the displayed base open set times F. That set is open: away from c this follows from openness of the selected chart base, and near c both

chart bases contain a common open neighborhood. Thus the paste is an actual topological bundle trivialization with the stated piecewise formula. Shrink a base neighborhood of c so it lies in both base sets and so $\sigma(t) = t$ there. At $t \le c$ its value is e_L by definition, and at $t > c$ the corrected right chart equals e_L on the whole fibre by the preceding lemma. This proves literal whole-fibre agreement throughout that open neighborhood. The correction's product smoothness was already established. □

Definition A.571 (Joint open coordinate germs of a native map family)**.** *For $S \subset [0,1]$, a family $f_t : F \to F$ has joint open coordinate germs on S if it satisfies the coordinate-extension condition of the intrinsic smooth bundle definition, with $t \in S$ and with f_t in place of a transition. Thus every source chart a, $t \in S$ and x in its source admit a smooth map on an open neighborhood of $(t, a(x))$ in $\mathbb{R} \times \mathbb{R}^d$, agreeing with f_s in the fixed target chart at $f_t(x)$ for actual $s \in S$ and native source coordinates in that neighborhood. Continuity of the actual family is a separate condition; the open extension need not take values in the native model away from those actual coordinates.*

Lemma A.572 (Composition of actual joint open coordinate germs)**.** *Let $S \subset [0,1]$ be relatively open. Suppose $f_t, g_t : F \to F$ have joint open coordinate germs on S, and $(t,x) \mapsto f_t(x)$ is continuous on $S \times F$. Then $g_t \circ f_t$ has joint open coordinate germs on S.*

Proof. Fix a native source chart a, $t \in S$, and x in its source. Use the native chart b at $f_t(x)$ as intermediate chart. Let $\beta(s,z)$ on U be the given open coordinate extension of f_s from a to b, and let $\gamma(s,w)$ on V be the extension of g_s from b to the native chart at $g_t(f_t(x))$. Put $p = (t, a(x))$ and $q(s,z) = (s, \beta(s,z))$. The map q is smooth on U, and $q(p) = (t, b(f_t(x))) \in V$.

The extended source-coordinate inverse is continuous at $a(x)$. Together with the continuous retraction of $\mathbb{R}$ to $[0,1]$ and continuity of the actual inner family on the relatively open $S \times F$, this gives an open ambient neighborhood N of p on which the corresponding actual inner values lie in the source of b. For actual interval parameters and native coordinates, this retraction leaves the inputs unchanged. On the open neighborhood $W = U \cap q^{-1}(V) \cap N$, the composition $\gamma \circ q$ is smooth. At actual $(s, a(y)) \in W$ with $s \in S$, the inner germ identity gives $\beta(s, a(y)) = b(f_s(y))$. By the choice of N this is a native coordinate in b's target, so the outer germ identity gives

$$\gamma(s, \beta(s, a(y))) = \text{the fixed target coordinate of } g_s(f_s(y)).$$

This is exactly the required open germ. The continuous retraction was used only to select N; the smooth extension itself is the composition of the two supplied smooth germs. □

Lemma A.573 (Locality of joint open coordinate germs in the parameter)**.** *Suppose that for each $t \in S \subset [0,1]$ there is a relatively open neighborhood T of t on which f_s has joint open coordinate germs for $s \in S \cap T$. Then it has such germs on S.*

Proof. Fix a source chart and a point at parameter $t \in S$. Choose T as in the hypothesis, and write $T = O \cap [0,1]$ with O open in $\mathbb{R}$ and containing t. Take its supplied coordinate germ on an open set U and restrict it to $U \cap (O \times \mathbb{R}^d)$. Every actual parameter $s \in S$ in this restricted neighborhood also belongs to T, so the local germ's agreement identity applies. Smoothness and the distinguished point are preserved by this open restriction, proving the result. □

Lemma A.574 (Reparameterization of joint open coordinate germs)**.** *If f_t has joint open coordinate germs on S, and $\sigma : \mathbb{R} \to [0,1]$ has smooth scalar coordinate and image in S, then $t \mapsto f_{\sigma(t)}$ has joint open coordinate germs for all $t \in [0,1]$.*

Proof. At a chosen (t, x), take the germ β on U for the original family at $(\sigma(t), x)$. Compose it with the smooth ambient map $q(s, z) = (\sigma(s), z)$ and use the open domain $q^{-1}(U)$. This domain contains the chosen point. Its actual inputs have $\sigma(s) \in S$, so the original agreement identity gives precisely the required coordinate expression for $f_{\sigma(s)}$. □

Definition A.575 (Charts smoothly compatible with the original bundle atlas)**.** *An actual topological bundle trivialization e is compatible with the original smooth bundle atlas if, for every original chart a, both ordered actual transitions $e \to a$ and $a \to e$ satisfy the joint open coordinate-extension condition on their base overlap.*

Lemma A.576 (Joint smooth transitions between compatible charts)**.** *Every ordered pair of compatible charts has joint open coordinate germs for its actual transition, including self transitions.*

Proof. Fix an overlap parameter t of two compatible charts e, e'. Choose an original atlas chart a with t in its base set. On the open triple overlap the actual transition $e \to e'$ is the composition of $e \to a$ and $a \to e'$, by the trivialization inverse identities. Both factors have joint open germs by compatibility. The inner actual transition is jointly continuous: it is the second coordinate of the composition of the inverse bundle chart e^{-1} with a on this overlap. The germ composition lemma therefore gives germs of $e \to e'$ there. Such an original chart exists at every overlap parameter, so parameter locality supplies the desired germs on the whole overlap. The argument also applies when $e = e'$. □

Lemma A.577 (A smooth bundle atlas containing all compatible charts)**.** *The compatible charts form a smooth compact bundle atlas containing the original atlas, on the same dependent fibres, model fibre and total-space topology. The original chosen local chart at each base point can be retained.*

Proof. Every original chart is compatible, because the original atlas supplies joint open germs for each ordered pair of its charts. Thus the compatible charts contain the original atlas and still cover every base point; its original chosen chart remains available there. All charts are actual topological trivializations on the given total space. The preceding lemma supplies the joint open extension condition for every ordered pair of compatible charts. The compact, Hausdorff, second-countable smooth model fibre is the same F. These facts give all the required smooth compact bundle conditions for the enlarged atlas. □

Lemma A.578 (Compatibility of an actual smooth transition correction)**.** *Let e, e' be original atlas charts, and let $\sigma : \mathbb{R} \to [0, 1]$ have smooth scalar coordinate and image in their overlap. The corrected chart $\bar{e}$ of the smooth transition-correction lemma is compatible with the original smooth atlas. If $\tau_{a,b}(t)$ denotes the actual transition from chart a to chart b, then for every original chart a and every t in its overlap with e the two transitions are*

$$\tau_{a,\bar{e}}(t) = \tau_{e,e'}(\sigma(t)) \circ \tau_{a,e}(t), \qquad \tau_{\bar{e},a}(t) = \tau_{e,a}(t) \circ \tau_{e',e}(\sigma(t)).$$

Proof. The correction formula for $\bar{e}$ preserves the base coordinate and applies $\tau_{e,e'}(\sigma(t))$ to e's fibre coordinate. Composing it with a^{-1} gives the first displayed identity. The inverse correction applies $\tau_{e',e}(\sigma(t))$ before e^{-1}; composing with a gives the second identity.

The original transitions have joint open germs, and germ reparameterization gives such germs for both reparameterized corrections on the whole interval. Restrict to the open base overlap of a and e. In the first composition the inner transition $\tau_{a,e}(t)$ is jointly continuous by the topological bundle chart identities. In the second composition the inner reparameterized reversed correction

is jointly continuous because it is the restriction of the smooth product correction's inverse. The germ composition lemma applies to both displayed identities, giving both required ordered extension conditions. Since a was an arbitrary original chart, $\bar{e}$ is compatible. □

Lemma A.579 (Compatibility from local whole-fibre chart agreement)**.** *Let e be an actual bundle chart. Suppose every t in its base set has an open neighborhood Q and a compatible chart b such that, for each $s \in Q \cap \mathrm{base}(e)$, the chart b is defined over s and e, b agree on every point of the entire fibre E_s. Then e is compatible.*

Proof. Fix an original atlas chart a and an overlap parameter t. Choose Q, b as in the hypothesis. For each relevant $s \in Q$, whole-fibre agreement gives identical transitions from a into e and b, since those transitions evaluate the two charts on the same point $a^{-1}(s, x)$. Their reverse transitions also agree: for any $x \in F$, the point $p = e^{-1}(s, x)$ lies over s, and whole-fibre agreement gives $b(p) = e(p) = (s, x)$. The inverse-chart identity therefore implies $b^{-1}(s, x) = p$. Applying a gives equality of the two reverse transitions. Both ordered transitions of b against a have joint open germs by compatibility. Their restrictions hence supply germs for e against a on Q intersected with the overlap. Parameter locality supplies these germs on the entire overlap, in both orders. As a was arbitrary, this proves compatibility of e. □

Lemma A.580 (Compatible smooth splicing of actual bundle charts)**.** *Let e_L, e_R be compatible charts meeting at $c \in (0, 1)$. There are a smooth overlap reparameterization σ and a compatible actual trivialization T with base open set*

$$(\mathrm{base}(e_L) \cap [0, c]) \ \cup \ (\mathrm{base}(e_R) \cap (c, 1]).$$

At parameters at most c its formula is e_L, and at parameters greater than c it is e_R followed by the transition from e_R to e_L evaluated at σ. It equals e_L on whole fibres throughout an open neighborhood of c. More generally, for original atlas charts the gauge splice for any smooth σ into the overlap equal to the identity near c is compatible.

Proof. First consider original atlas charts and such a σ. The topological gauge splice is an actual chart with the displayed base set and formulas. At a base parameter less than c, it agrees with the compatible original chart e_L on a whole-fibre open neighborhood on that side. At a parameter greater than c, it similarly agrees with the correction of e_R toward e_L. The preceding correction lemma makes that chart compatible. At c, the near-identity condition on σ gives whole-fibre agreement with e_L on an open neighborhood containing c. The whole-fibre locality lemma now proves that the splice is compatible with the original atlas.

For arbitrary compatible e_L, e_R, use the enlarged smooth atlas of all compatible charts. They are atlas charts there. Construct σ from their open overlap by the cutoff lemma and form the same topological gauge splice. The first part of the proof gives its compatibility with the enlarged atlas. Every original chart belongs to that enlarged atlas, so both ordered germ conditions hold against the original atlas as well. The construction retains the literal base-set and piecewise formulas, and its near-seam whole-fibre equality follows from $\sigma(t) = t$ there. This proves all the stated properties. □

Lemma A.581 (Global interval charts from local charts and interior splicing)**.** *Let $E \to [0, 1]$ have local bundle trivializations with model fibre F, and let P be a property of these actual trivializations. Suppose every $t \in [0, 1]$ belongs to the base open set of a trivialization satisfying P. Suppose also*

that whenever two trivializations e_L, e_R satisfy P and meet at $c \in (0,1)$, there is a trivialization T satisfying P whose base open set is

$$(\mathrm{base}(e_L) \cap [0,c]) \ \cup \ (\mathrm{base}(e_R) \cap (c,1]).$$

Then an actual trivialization satisfying P has base open set $[0,1]$. This covering argument imposes no smoothness on P; any required smoothness must be preserved by the stated splice construction.

Proof. Let S be the set of $x \in [0,1]$ for which one trivialization satisfying P covers $[0,x]$. A local chart at zero proves $0 \in S$, and 1 is an upper bound. Put $c = \sup S$, so $0 \le c \le 1$. We first prove $c \in S$. Suppose otherwise. Then $c > 0$, since $0 \in S$. Choose a local chart e_c satisfying P at c. Its base open set contains $(u,c]$ for some $0 \le u < c$. The supremum property supplies $d \in S$ with $u < d \le c$. Since $c \notin S$, we have $d < c$, and consequently $0 < d < 1$. Choose a chart e_d satisfying P and covering $[0,d]$. Both e_d and e_c contain d in their base sets. Their splice at d covers $[0,d]$ by its left part and $(d,c]$ by its right part. It therefore covers $[0,c]$ and satisfies P, contradicting $c \notin S$.

Choose a chart e satisfying P and covering $[0,c]$, using $c \in S$. If $c < 1$, openness of its base set at c supplies $v \in (c,1]$ with $[c,v)$ in that base set. Choose $c < d < v$. Then e covers $[0,d]$, so $d \in S$, contradicting $c = \sup S$. Thus $c = 1$, and e covers the entire parameter interval. Its base set is a subset of that interval, so it is exactly $[0,1]$. □

Lemma A.582 (An actual global smooth chart for an intrinsic compact interval bundle)**.** *Every intrinsic smooth compact bundle over $[0,1]$ has an actual global bundle trivialization e whose ordered transitions against every original atlas chart satisfy the joint open coordinate-extension condition in both directions. Its base set is the entire interval. Thus these transitions extend smoothly across parameter endpoints and native spatial boundaries.*

Proof. Take $P(e)$ to be compatibility with the original smooth atlas. At every parameter the original chosen atlas chart is defined and is compatible, by the compatible-atlas lemma. The compatible splice theorem produces a chart satisfying P with the required literal piecewise base set for every two compatible charts meeting at an interior seam. Apply the global interval covering theorem to this P, these local charts and this actual splice construction. It gives a compatible actual chart with base set $[0,1]$. Compatibility is exactly the two asserted ordered joint extension conditions. □

Definition A.583 (Joint open germs of value and target-map families)**.** *Let F be the native model fibre of dimension d, let $S \subset [0,1]$, and let V be a real normed space. A family $v_t : F \to V$ has joint open value germs on S if, for every native atlas chart a, every $t \in S$ and every $x \in \mathrm{source}(a)$, there are an open neighborhood U of $(t, a(x))$ in $\mathbb{R} \times \mathbb{R}^d$ and a smooth $\beta : U \to V$ such that*

$$\beta(s,z) = v_s(a^{-1}(z)) \quad (s \in S,\ z \in \mathrm{target}(a),\ (s,z) \in U).$$

For a smooth boundaryless target manifold N, a family $f_t : F \to N$ has joint open target-map germs if the same condition holds with a smooth map $\beta : U \to N$ taking values in that actual target manifold and agreeing with $f_s(a^{-1}(z))$ at the indicated inputs. No separate continuity condition is included in either definition. The target-map condition includes an actual manifold-valued extension; it is stronger than requiring only extensions of target coordinates.

Lemma A.584 (Transport of joint value germs by an actual fibre family)**.** *Let $S \subset [0,1]$ be relatively open. Suppose $v_t : F \to V$ has joint open value germs on S, $g_t : F \to F$ has joint open coordinate germs on S, and $(t,x) \mapsto g_t(x)$ is continuous on $S \times F$. Then $v_t \circ g_t$ has joint open value germs on S.*

Proof. Fix a source chart a, $t \in S$ and x in its source. Choose the native chart b at $g_t(x)$. The inner family has a smooth coordinate extension $\gamma(s,z)$ on an open neighborhood U of $(t,a(x))$; the value family has a smooth extension $\beta(s,w)$ on an open neighborhood W of $(t,b(g_t(x)))$. Put $q(s,z)=(s,\gamma(s,z))$. It is smooth on U and its value at $(t,a(x))$ belongs to W.

Continuity of the actual inner family lets us shrink to an open ambient neighborhood U' of $(t,a(x))$ on which actual inputs (s,z) with $s \in S$ and $z \in \mathrm{target}(a)$ have $g_s(a^{-1}(z)) \in \mathrm{source}(b)$. To obtain an ambient neighborhood, use the continuous interval retraction and the continuous extended source-coordinate inverse near $a(x)$; they agree with the actual parameter and point on these inputs. Intersect U' with U and $q^{-1}(W)$. This is open and contains $(t,a(x))$. On this intersection $\beta \circ q$ is smooth. For an actual input, $\gamma(s,z)=b(g_s(a^{-1}(z)))$ belongs to $\mathrm{target}(b)$, so the outer germ identity gives $\beta(q(s,z))=v_s(g_s(a^{-1}(z)))$. This is the required value germ. □

Lemma A.585 (Transport of joint target-map germs by an actual fibre family)**.** *Under the hypotheses of the preceding lemma, replace $v_t : F \to V$ by a family $f_t : F \to N$ having joint open target-map germs. Then $f_t \circ g_t$ has joint open target-map germs on S.*

Proof. At the chosen (t,x) use the intermediate native chart b at $g_t(x)$. Choose the inner extension γ and the manifold-valued extension β of $f_s \circ b^{-1}$ supplied by the two germ conditions. The preceding construction shrinks the open source neighborhood so that actual inner values lie in the source of b and $(s,\gamma(s,z))$ lies in the domain of β. On that neighborhood $\beta(s,\gamma(s,z))$ is a smooth map to N, by the smooth chain rule. At actual native inputs the two agreement identities identify it with $f_s(g_s(a^{-1}(z)))$. This is the required manifold-valued extension of the composite family. □

Lemma A.586 (Parameter locality of value and target-map germs)**.** *Suppose every $t \in S$ has a relatively open parameter neighborhood T on which a value or target-map family has joint open germs for parameters in $S \cap T$. Then the family has such germs on S.*

Proof. Fix a native source chart, $t \in S$ and a point x in that chart. Choose the supplied relatively open T at t, and write it as $[0,1] \cap O$ with $O \subset \mathbb{R}$ open. Take the supplied germ on an open neighborhood U of $(t,a(x))$ for parameters in $S \cap T$. Restrict its domain to $U \cap (O \times \mathbb{R}^d)$. This remains open, contains $(t,a(x))$, and preserves smoothness. Any actual input there with parameter $s \in S$ also has $s \in T$, so the supplied agreement identity applies. This proves the required germ for both value and actual target-manifold extensions. □

Definition A.587 (Local germs of original total-space data)**.** *Let $E \to [0,1]$ be an intrinsic smooth compact bundle with model fibre F. For an actual function $v : E \to V$ and a bundle chart e, write $v_t^e(x)=v(e^{-1}(t,x))$ when t lies in the base set of e. The original total-space data have local value germs if v^e has joint open value germs on that base set for every designated atlas chart e. For a map $j : E \to N$, define local target-map germs by the corresponding condition for j^e. These conditions are imposed on the original total-space functions in local atlas charts.*

Lemma A.588 (Original data germs in any compatible bundle chart)**.** *Original total-space values or target maps with local germs have joint open germs in every bundle chart compatible with the original atlas. Their coordinate expressions retain the actual total-space values.*

Proof. Let e be compatible and fix t in its base set. Choose an original atlas chart c whose base set contains t. On the relatively open overlap the actual transition $g_s = e_s \to c_s$ has joint open coordinate germs by compatibility and is jointly continuous by the bundle chart identities. For every s in this overlap and every $x \in F$,

$$c^{-1}(s,g_s(x)) = e^{-1}(s,x).$$

Indeed, apply c to $e^{-1}(s,x)$ and use the definition of the actual transition; applying c^{-1} gives the identity. Consequently $v_s^c(g_s(x)) = v_s^e(x)$ and $j_s^c(g_s(x)) = j_s^e(x)$. Restrict the supplied original data germs to this overlap and apply the respective composition lemma. The displayed identities identify the resulting germs with the actual expressions in e. Such an original chart exists near every t in the base set of e, so parameter locality gives the asserted germs on the entire base set. All identities evaluate the original functions at the same actual total-space point. □

Lemma A.589 (One global chart transporting original values and target maps)**.** *Given an original value function $v : E \to V$ and target map $j : E \to N$ with their local germs, there is one actual global compatible bundle chart e on $[0,1]$ in which both v^e and j^e have joint open germs throughout the interval, including its endpoints and the native spatial boundary. The transported expressions are obtained by evaluating v and j on the actual inverse of this same chart.*

Proof. Construct a global compatible chart by the global smooth trivialization lemma. Apply the compatible-chart data lemma separately to v and j using that very chart. Its base set is $[0,1]$, so both conclusions hold throughout the whole parameter interval. The definition of their chart expressions gives the asserted literal inverse-chart evaluation. □

Definition A.590 (Joint native form germs and covariant pullback)**.** *Let M be a positive-dimensional native manifold with boundary, $S \subset [0,1]$, and α_t a family of intrinsic r-forms. The family has joint open form germs on S if, for every native atlas chart a, $t \in S$ and $x \in \mathrm{source}(a)$, its coefficients in a agree with a smooth alternating-covector-valued function on an open ambient neighborhood of $(t, a(x))$, at all actual parameters in S and native chart coordinates in that neighborhood. For a native map $f : M \to M$, its covariant pullback is*

$$(f^*\alpha)_x(v_1, \dots, v_r) = \alpha_{f(x)}(df_x v_1, \dots, df_x v_r),$$

using the full native differential, also at boundary points.

Lemma A.591 (The full native coordinate formula for covariant pullback)**.** *Let a, b be native atlas charts, let $x \in \mathrm{source}(a)$, assume f is differentiable at x and $f(x) \in \mathrm{source}(b)$, and put $y = a(x)$ and $h = b \circ f \circ a^{-1}$. The coefficients of $f^*\alpha$ in a at y are*

$$(\alpha^b)_{h(y)} \circ (dh_y)^{\otimes r},$$

where dh_y is computed within the native half-space. This formula holds on all boundary tangent directions as well.

Proof. The within-half-space chain rule gives $dh_y = db_{f(x)} \circ df_x \circ da_y^{-1}$. The native half-space has a unique derivative at each of its points, including its boundary, so this is its full within-half-space differential. For vectors $w_1, \dots, w_r$ in the model space, the displayed covector therefore evaluates as

$$\alpha_{f(x)}\big(db_{h(y)}^{-1} db_{f(x)} df_x da_y^{-1} w_1, \dots, db_{h(y)}^{-1} db_{f(x)} df_x da_y^{-1} w_r\big).$$

The inverse-chart differential identity cancels each adjacent pair $db_{h(y)}^{-1} db_{f(x)}$. What remains is exactly $(f^*\alpha)_x(da_y^{-1}w_1, \dots, da_y^{-1}w_r)$, the coefficient of the native pullback in a. This proves the formula. □

Lemma A.592 (Joint open germs of the actual varying native one-form pullback)**.** *Let $S \subset [0,1]$ be relatively open. Suppose $f_t : M \to M$ has joint open native coordinate germs on S and is jointly continuous on $S \times M$, and α_t is a native one-form family with joint open form germs on S. Then the actual covariant pullbacks $f_t^*\alpha_t$ have joint open form germs on S. No spatial differential extension is supplied separately.*

Proof. Fix a native chart a, $t \in S$ and x in its source. Use the native chart b at $f_t(x)$ and a smooth coordinate extension $\beta(s,z)$ of $b \circ f_s \circ a^{-1}$ near $(t, a(x))$. Let $\gamma(s,w)$ be a joint open coefficient extension of α_s in b near $(t, b(f_t(x)))$, and put $q(s,z) = (s, \beta(s,z))$. By continuity of the actual inner family, shrink the source open neighborhood so that actual $f_s(a^{-1}(z))$ lies in the source of b. Intersect further with the inverse image under q of the open domain of γ. This is still an open neighborhood of $(t, a(x))$. On it define the covector

$$\theta(s,z) = \gamma(s, \beta(s,z)) \circ D_z\beta(s,z).$$

The spatial derivative $D_z\beta$ is smooth, since β is smooth on this open set. Covector composition with a linear map is a continuous bilinear operation, so θ is smooth there.

At an actual native input, the inner germ equality holds on a relative neighborhood in the half-space. Differentiating this equality within the half-space identifies $D_z\beta(s,z)$ with the within-half-space differential of $b \circ f_s \circ a^{-1}$: uniqueness of derivatives on the half-space makes the derivative of the open extension equal to that relative differential, including at its boundary. The outer germ identity identifies $\gamma(s, \beta(s,z))$ with the actual coefficients of α_s in b. The preceding coordinate pullback formula now identifies $\theta(s,z)$ with the coefficients of $f_s^*\alpha_s$ in a. Thus θ is the required joint open germ of that actual covariant pullback. □

Lemma A.593 (Actual boundary-valued joint germs from ambient family germs)**.** *Let M be a positive-dimensional smooth native manifold with boundary, $S \subset [0,1]$ relatively open, and $f_t : M \to M$ a jointly continuous family on $S \times M$ with joint open native coordinate germs. Suppose $f_t(\partial M) \subset \partial M$ for $t \in S$. The actual restrictions $f_t|_{\partial M}$ have joint open manifold-valued germs with target the original intrinsic boundary. Boundary preservation is required only at these actual parameters.*

Proof. Fix $t \in S$, $x \in \partial M$ and a native boundary source chart a, induced by an ambient half-space atlas chart A. Choose the ambient native target chart at $f_t(x)$ and restrict that very chart to a boundary chart b. Let $\beta(s,z)$ be the supplied ambient coordinate extension for f_s from A to that ambient target chart. Continuity of the actual family permits shrinking its open domain so that actual $f_s(y)$ lies in the source of the chosen target chart. Native boundary coordinates insert the zero normal coordinate literally. Consequently the coordinate expression of the restricted family is

$$q(s,v) = \mathrm{drop}\big(\beta(s, \mathrm{insert}(v))\big),$$

where insertion adds the zero normal component and deletion removes it. Both linear operations are smooth, so q is smooth wherever the inserted coordinates lie in the domain of β.

At $(t, a(x))$, q equals $b(f_t(x))$, which lies in the open target of b. Intersect its open domain with $q^{-1}(\mathrm{target}(b))$. On this open neighborhood $b^{-1} \circ q$ is a smooth map taking values in the actual intrinsic boundary. For an actual parameter $s \in S$ and a native boundary point in the source neighborhood, ambient germ agreement and actual boundary preservation imply that deletion of the ambient output coordinates equals its boundary coordinate in b. The boundary chart inverse identity therefore gives $b^{-1}(q(s, a(y))) = f_s(y)$. This is the required actual boundary-valued germ, including extension across parameter endpoints. □

Lemma A.594 (Actual boundary transition germs between compatible bundle charts)**.** *Every ordered transition between compatible bundle charts of a positive-dimensional native model fibre is an actual smooth fibre diffeomorphism. On the overlap its restrictions to the native boundary have joint open germs with values in that original boundary. At every fixed parameter they are smooth boundary diffeomorphisms, whose full boundary tangent differentials are bijective. No boundary regularity is an additional hypothesis.*

Proof. Both ordered transitions have joint open coordinate germs by the compatible-transition lemma. Freezing a parameter therefore gives smooth maps in both directions. The actual bundle transition inverse identities make them inverse, so the transition is a native diffeomorphism. A diffeomorphism of manifolds with boundary preserves their native boundary, as it preserves the interior and its complement. The overlap is relatively open and the actual transition is jointly continuous by the bundle chart identities. Apply the boundary-family germ lemma to this transition and its derived boundary preservation. This gives the required manifold-valued boundary germs on the entire overlap without requiring boundary-preserving off-interval extensions. Freezing the germs gives smooth boundary maps in both ordered directions. The ambient inverse transition identities restrict to the actual boundary, so these maps are inverse. Differentiating both inverse identities makes their full boundary tangent differentials mutual inverses, proving bijectivity. □

Definition A.595 (Covariant local primitive data in the original bundle atlas)**.** *Let $E \to [0,1]$ be an intrinsic smooth compact bundle with a positive-dimensional native model fibre M. A covariant atlas presentation of fibre primitives assigns a native one-form family α_t^a to each original bundle atlas chart a, with joint open form germs on its base set, satisfying on every overlap*

$$\alpha_t^a = \tau_{a,b}(t)^* \alpha_t^b.$$

Here $\tau_{a,b}$ is the actual fibre coordinate change from a to b. For an arbitrary bundle chart e at t, choose the designated original chart c_t at that parameter and define

$$\alpha_t^e = \tau_{e,c_t}(t)^* \alpha_t^{c_t}.$$

Only the original atlas families and their actual overlap covariance are source data; no global-chart form family is supplied.

Lemma A.596 (The primitive in an arbitrary chart using any original local chart)**.** *For t in the overlap of a compatible bundle chart e and an original atlas chart c, the primitive defined above satisfies*

$$\alpha_t^e = \tau_{e,c}(t)^* \alpha_t^c.$$

Thus it retains the covariant original primitive in any original chart that is defined at that parameter.

Proof. The chosen original chart c_t is defined at t. Covariance of the original families gives $\alpha_t^c = \tau_{c,c_t}(t)^* \alpha_t^{c_t}$. Compatibility makes $\tau_{e,c}(t)$ smooth, and the original smooth atlas makes $\tau_{c,c_t}(t)$ smooth. The native pullback chain rule gives

$$\tau_{e,c}(t)^* \alpha_t^c = \tau_{e,c}(t)^* \tau_{c,c_t}(t)^* \alpha_t^{c_t} = \big(\tau_{c,c_t}(t) \circ \tau_{e,c}(t)\big)^* \alpha_t^{c_t}.$$

The actual bundle inverse identities give the fibre cocycle $\tau_{c,c_t}(t) \circ \tau_{e,c}(t) = \tau_{e,c_t}(t)$. The last expression is therefore the defining value of α_t^e, proving the asserted equality. If $e = c$ is itself an original chart, its self transition is the identity. Pullback by the identity fixes all native covectors, so α_t^c is recovered exactly. □

Lemma A.597 (Covariance of original primitives between compatible charts)**.** *For compatible charts e, e' and t in their overlap,*

$$\alpha_t^e = \tau_{e,e'}(t)^* \alpha_t^{e'}.$$

This identity uses the full native tangent differential of the actual transition, including boundary tangent directions.

Proof. Use the same designated original chart c_t at t in the defining formulas for both sides. Both $\tau_{e,e'}(t)$ and $\tau_{e',c_t}(t)$ are smooth by compatibility. The native pullback chain rule gives

$$\tau_{e,e'}(t)^*\alpha_t^{e'} = \big(\tau_{e',c_t}(t) \circ \tau_{e,e'}(t)\big)^*\alpha_t^{c_t}.$$

The bundle cocycle identifies the composite with $\tau_{e,c_t}(t)$, so the result is the defining value of α_t^e. Since these are equal covectors on the full native tangent spaces, the identity retains every boundary tangent evaluation. □

Lemma A.598 (Local primitive germs in a compatible global bundle chart)**.** *Covariant local primitive data have joint open form germs in every compatible bundle chart. In particular there is an actual compatible global chart on* $[0,1]$ *whose primitive family has these germs throughout the interval, retaining the stated overlap pullbacks of the original atlas primitives.*

Proof. Fix a compatible chart e and t in its base set. Choose an original chart c at t. On their relatively open overlap the preceding lemma expresses the actual transported primitive as $\tau_{e,c}(s)^*\alpha_s^c$. The transition has joint open coordinate germs by compatibility and is jointly continuous, and α_s^c has the supplied joint open form germs. The varying native one-form pullback lemma therefore gives joint open form germs of α^e on this overlap.

To pass from such local parameter neighborhoods to the base set of e, fix any distinguished parameter and native point and take its supplied coefficient germ on U. Write the chosen relatively open parameter neighborhood as $[0,1] \cap O$ with $O \subset \mathbb{R}$ open, and restrict the germ domain to $U \cap (O \times \mathbb{R}^{k+1})$. This is an open neighborhood of the distinguished point. Every actual input there has parameter in the chosen overlap, so its coefficient agreement holds. Hence the germs hold on the entire base set. Finally construct a global compatible chart by the global smooth trivialization lemma and apply this result to that chart. Its base set is the whole interval, giving the asserted global primitive family. □

Lemma A.599 (Postcomposition of actual target and value germs)**.** *Let N and P be smooth boundaryless manifolds. Suppose $j_t : F \to N$ has actual target-map germs on S, and $g_t : N \to P$ has local smooth extensions on open subsets of $\mathbb{R} \times N$ at every actual input (t, y) with $t \in S$. Then $g_t \circ j_t$ has actual target-map germs on S. The same assertion holds for normed-space-valued g_t, with value germs as its conclusion.*

Proof. Fix a parameter $t \in S$, a source chart, and a native source point x. Choose an inner germ $\beta : U \to N$ at its coordinate input z_0 and an outer germ $\gamma : V \to P$ at $(t, j_t(x))$. The smooth map $q(s,z) = (s, \beta(s,z))$ takes z_0 into V. Thus $W = U \cap q^{-1}(V)$ is an open neighborhood of z_0, and $\gamma \circ q$ is smooth on W. At every actual input with parameter in S, inner germ agreement gives $\beta(s,z) = j_s(y)$, and outer agreement then gives $\gamma(s, \beta(s,z)) = g_s(j_s(y))$. This is the required actual target germ. The proof for normed-space-valued γ is identical. □

Definition A.600 (Covariant original-atlas ribbon maps)**.** *Let E_R and E_A be compact smooth bundles over $[0,1]$, with ribbon fibre F and ambient fibre A, respectively. The ribbon fibre may have either the positive-dimensional half-space model or the zero-dimensional model; the ambient fibre has its half-space model. Original-atlas ribbon data consist of maps $j_t^{c_R,c_A} : F \to \partial A$, specified for pairs of original bundle charts, with actual boundary-target germs on their base overlaps and with the covariance equation*

$$j_t^{d_R,d_A} \circ \tau_{c_R,d_R}(t) = \tau^{\partial}_{c_A,d_A}(t) \circ j_t^{c_R,c_A}$$

on every fourfold base overlap. Here τ^{∂} is the actual induced boundary map. For any ribbon chart e_R and compatible ambient chart e_A, choose original charts $c_R(t), c_A(t)$ over t and define, on e_R's and e_A's base overlap,

$$j_t^{e_R,e_A} = \tau^{\partial}_{c_A(t),e_A}(t) \circ j_t^{c_R(t),c_A(t)} \circ \tau_{e_R,c_R(t)}(t).$$

These data describe the full local ribbon maps. No Liouville primitive identity or smooth immersion condition is part of this definition.

Lemma A.601 (Exact ribbon coordinate squares)**.** *For fixed original charts c_R, c_A and a parameter in their overlap with e_R, e_A, the constructed map is exactly*

$$j_t^{e_R,e_A} = \tau^{\partial}_{c_A,e_A}(t) \circ j_t^{c_R,c_A} \circ \tau_{e_R,c_R}(t).$$

In the original charts it recovers the original map. Changing both presentations gives the same covariance square for every ribbon point, provided the two ambient charts are compatible. The ribbon charts need not be smooth for these exact identities.

Proof. Write d_R, d_A for the original charts chosen in the definition at t. The source cocycle gives $\tau_{e_R,d_R} = \tau_{c_R,d_R} \circ \tau_{e_R,c_R}$. Apply original covariance to $j^{d_R,d_A} \circ \tau_{c_R,d_R}$, obtaining $\tau^{\partial}_{c_A,d_A} \circ j^{c_R,c_A}$. The ambient cocycle then gives $\tau^{\partial}_{d_A,e_A} \circ \tau^{\partial}_{c_A,d_A} = \tau^{\partial}_{c_A,e_A}$, proving the local formula. For original e_R, e_A, use this formula with $c_R = e_R, c_A = e_A$: both self transitions are identities. For another presentation f_R, f_A, use the same designated d_R, d_A in both definitions. Composing the f_R, f_A formula on the right with τ_{e_R,f_R} and applying both cocycles gives the e_R, e_A formula followed by $\tau^{\partial}_{e_A,f_A}$. This proves the full covariance square. □

Lemma A.602 (Transport of the entire ribbon image and its topology)**.** *On the overlap of the preceding lemma, the entire ribbon image satisfies*

$$j_t^{e_R,e_A}(F) = \tau^{\partial}_{c_A,e_A}(t)\big(j_t^{c_R,c_A}(F)\big).$$

If the original local map is a topological embedding, so is the constructed map. If it is injective, so is the constructed map.

Proof. The exact local formula expresses each image point as the stated ambient transition of an original image point. Conversely, the source transition $\tau_{e_R,c_R}(t)$ is a homeomorphism of F, hence surjective. For any original image point choose a preimage of its source point under this transition, and the local formula supplies its constructed image point. This proves both inclusions of the displayed equality. A composition of a topological embedding with homeomorphisms on its source and ambient target is a topological embedding. The actual boundary is a subspace of the ambient manifold, so the same conclusion holds for the boundary-valued map. The analogous composition of injective maps is injective. □

Lemma A.603 (Full ribbon differential and immersion transport)**.** *Suppose both constructed charts e_R, e_A are compatible and c_R, c_A are original charts on the fourfold overlap. Write $f = \tau_{e_R,c_R}(t)$, $j = j_t^{c_R,c_A}$ and $b = \tau^{\partial}_{c_A,e_A}(t)$. At every native source point,*

$$d(j_t^{e_R,e_A})_x = db_{j(f(x))} \circ dj_{f(x)} \circ df_x.$$

These are full native tangent maps, including at a source boundary. If dj_y is injective at every source point y, then $d(j_t^{e_R,e_A})_x$ is injective at every x.

Proof. The compatible source transition is a smooth diffeomorphism. The original ribbon map is smooth by freezing its supplied target germs, and the actual boundary transition is a smooth boundary diffeomorphism. The exact local map identity is $j_t^{e_R,e_A} = b \circ j \circ f$. Applying the native chain rule twice gives the displayed differential on the full source tangent space. The differentials of f and b are bijective because they are differentials of diffeomorphisms. If $dj_{f(x)}$ is injective, the displayed composition of injective linear maps is injective. This proves the immersion conclusion everywhere. □

Lemma A.604 (Original ribbon germs in two constructed global charts)**.** *Original-atlas ribbon data have joint actual boundary-target germs in any pair of compatible charts, on their base overlap. There exist compatible global charts for both bundles in which the constructed full ribbon family has these germs on the whole interval. This applies to both ribbon model cases of the preceding definition.*

Proof. Fix compatible charts e_R, e_A and a parameter t in their overlap. Choose original charts c_R, c_A over t. Their joint base overlap is an open parameter neighborhood. There the exact local formula expresses the constructed ribbon family as the original local family precomposed with the actual ribbon transition and postcomposed with the actual ambient boundary transition. Compatibility supplies joint smooth germs and joint continuity of the source transition, so the target-family composition lemma transports the original target germs through it. The boundary transition has joint actual boundary-valued germs. In a boundary chart, these germs also give smooth native product germs: compose their coordinate representatives with the smooth chart map and restrict to its open source. The actual-target postcomposition lemma therefore supplies the germs of the full composite. Parameter locality passes from these neighborhoods to the entire base overlap. Finally construct one compatible global chart for each bundle and apply the result to that pair. Both base sets are $[0, 1]$, proving the assertion. □

Definition A.605 (Covariant positive local boundary witnesses)**.** *Positive witness data on the original ambient atlas consist of functions $g_t^c : \partial A \to \mathbb{R}$ with joint value germs on each chart's base set, positive at every actual boundary point, and satisfying $g_t^c = g_t^d \circ \tau_{c,d}^{\partial}(t)$ on base overlaps. For a compatible chart e and original chart c_t over t, set*

$$g_t^e = g_t^{c_t} \circ \tau_{e,c_t}^{\partial}(t)$$

for t in the base set of e. Positivity is required on actual interval values; no sign condition is imposed on an off-interval extension.

Lemma A.606 (Exact recovery and positivity of transported witnesses)**.** *For any fixed original chart c on its overlap with compatible e, $g_t^e = g_t^c \circ \tau_{e,c}^{\partial}(t)$. An original chart recovers its original witness exactly. The transported witness is positive at every boundary point throughout the base set of e.*

Proof. Original covariance gives $g_t^c \circ \tau_{e,c}^{\partial}(t) = g_t^{c_t} \circ \tau_{c,c_t}^{\partial}(t) \circ \tau_{e,c}^{\partial}(t)$. The actual boundary cocycle identifies the composite transition with $\tau_{e,c_t}^{\partial}(t)$, which is the defining formula for g_t^e. For $e = c$ the self transition is the identity, proving recovery. Every transported value is a value of the positive original witness $g_t^{c_t}$ at an actual boundary point. The original chart c_t contains t, so positivity follows from its supplied hypothesis. □

Lemma A.607 (Original positive witness germs in a constructed global chart)**.** *The actual transported witness has joint value germs in every compatible chart. There is a constructed compatible global chart in which it is a positive native scalar family with local smooth extensions across all parameter endpoints, on the entire boundary.*

Proof. At any parameter in a compatible chart e, choose a fixed original chart c containing that parameter. The exact local formula is $g_s^c \circ \tau_{e,c}^{\partial}(s)$ on their base overlap. The original value germs become native product value germs in boundary charts by composing with the smooth native chart map and restricting to its open source. The actual boundary transition has target germs. Postcomposition for value germs therefore proves the germs of this local formula. Parameter locality gives the result throughout e's base set. Construct a compatible global chart and apply the result there. Native boundary charts identify the coordinate value germs with local smooth extensions on open subsets of $\mathbb{R} \times \partial A$. Positivity follows from the preceding lemma at every actual interval value. □

Lemma A.608 (Boundary restriction of the same transported primitive)**.** *If a smooth ambient map $f : A \to A$ and a smooth boundary map $h : \partial A \to \partial A$ satisfy $f \circ \iota_\partial = \iota_\partial \circ h$, then for every native differential form α,*

$$\iota_\partial^*(f^*\alpha) = h^*(\iota_\partial^*\alpha).$$

In particular, for original local primitive data and a compatible chart e, the boundary restriction of α_t^e is the pullback of the original boundary primitive through the actual transition $\tau_{e,c}^{\partial}(t)$ on every overlap with an original chart c.

Proof. Differentiate the given point square. The native chain rule gives $df \circ d\iota_\partial = d\iota_\partial \circ dh$ at every boundary point. Evaluating each side of the asserted form identity on any list of full boundary tangent vectors gives α at the same ambient point on these two equal lists of ambient tangent vectors. Thus the forms agree in every degree. Apply this identity to the compatible ambient transition $\tau_{e,c}(t)$ and its actual smooth boundary restriction. The exact local primitive identity $\alpha_t^e = \tau_{e,c}(t)^* \alpha_t^c$ proves the stated specialization. □

Lemma A.609 (The original witnessed ribbon primitive identity in compatible charts)**.** *Let the ribbon be positive-dimensional. Supply covariant original-atlas ambient primitives λ, ribbon primitives μ, ribbon maps j, and positive boundary witnesses g. Fix compatible charts e_R, e_A and original charts c_R, c_A on their fourfold base overlap. If the original local maps satisfy*

$$(g_t^{c_A} \circ j_t^{c_R,c_A})\,(j_t^{c_R,c_A})^*(\lambda_t^{c_A}|_{\partial A}) = \mu_t^{c_R},$$

then the actual data in these same compatible charts satisfy

$$(g_t^{e_A} \circ j_t^{e_R,e_A})\,(j_t^{e_R,e_A})^*(\lambda_t^{e_A}|_{\partial A}) = \mu_t^{e_R}.$$

Both equations are equalities on the full native ribbon tangent spaces. No primitive-matching equation in the transported charts is supplied.

Proof. Write $R = \tau_{e_R,c_R}(t)$, $h = \tau_{e_A,c_A}^{\partial}(t)$, $j_G = j_t^{e_R,e_A}$ and $j_L = j_t^{c_R,c_A}$. Exact ribbon coherence, witness coherence, and primitive covariance give

$$h \circ j_G = j_L \circ R, \qquad g_G = g_L \circ h, \qquad \lambda_G|_{\partial A} = h^*(\lambda_L|_{\partial A}), \qquad \mu_G = R^*\mu_L.$$

All maps in this square are smooth; the native chain rule applies on full tangent spaces. Therefore the transported left-hand side equals

$$(g_L \circ h \circ j_G)\, j_G^* h^*(\lambda_L|_{\partial A}) = R^*\big((g_L \circ j_L)\, j_L^*(\lambda_L|_{\partial A})\big) = R^*\mu_L = \mu_G.$$

The first equality uses the exact point square and the pullback chain rule; multiplication by the scalar witness commutes with pullback by pointwise evaluation. The second equality is precisely the supplied original local primitive identity. This proves the full transported identity, including at native boundary points of the ribbon. □

Definition A.610 (Geometric laws on the original bundle atlas)**.** *The original local Liouville laws require, at every parameter in each original chart's base set, that $d\alpha_t^c$ be nondegenerate on every full native tangent space and its Liouville field be outward. Concretely, at each native boundary point there is a local defining function ρ with value zero, $\rho \leq 0$ near that point, nonzero full differential, a smooth open coordinate extension agreeing with ρ on the native half-space, and $d\rho(Z_{\alpha_t^c}) > 0$. For positive-dimensional ribbons, the original local pair laws require, on every original chart-pair overlap, that j_t be a topological embedding with injective full native tangent differential and that*

$$(g_t \circ j_t)\, j_t^*(\lambda_t|_{\partial A}) = \mu_t.$$

Every law is imposed on original local data; none is imposed on data in a subsequently constructed global chart.

Lemma A.611 (Native exterior derivative and Liouville field under pullback)**.** *Let P present the native primitive α, and let $f : M \to M$ be smooth. Then $d(f^*\alpha) = f^*(d\alpha)$ on full native tangent spaces. If every df_x is bijective, $d(f^*\alpha)$ is nondegenerate and*

$$df_x(Z_{f^*\alpha}(x)) = Z_\alpha(f(x)).$$

Proof. Write $i : M \to X_P$ for the presentation inclusion and λ_P for its smooth boundaryless primitive. The native primitive identity and the chain rule give $f^*\alpha = (i \circ f)^*\lambda_P$. Exterior naturality for this smooth map to the boundaryless ambient gives $d(f^*\alpha) = (i \circ f)^* d\lambda_P = f^* d\alpha$. If v pairs to zero under this pulled-back two-form with every native vector u, then $df_x(v)$ pairs to zero with every target vector, because df_x is surjective. Nondegeneracy of $d\alpha$ gives $df_x(v) = 0$, and injectivity gives $v = 0$.

For $Z = Z_{f^*\alpha}(x)$ and any target vector w, choose u with $df_x(u) = w$. The contraction equation and the established exterior identity give $d\alpha_{f(x)}(df_x(Z), w) = \alpha_{f(x)}(w)$. Thus $df_x(Z)$ satisfies the defining contraction equation of $Z_\alpha(f(x))$. Nondegeneracy makes that solution unique, proving the claimed field identity. □

Lemma A.612 (Original local Liouville laws in every compatible chart)**.** *Covariant original primitive data satisfying the original local Liouville laws satisfy nondegeneracy and the complete outward defining-function condition in every compatible chart. These transported conditions are conclusions, including their open coordinate extension clauses.*

Proof. At a parameter t in compatible chart e, choose an original chart c over t. Its local smooth primitive and original local laws construct one native Liouville presentation P of α_t^c. The exact local formula is $\alpha_t^e = f^*\alpha_t^c$, where $f = \tau_{e,c}(t)$ is an actual native diffeomorphism. The preceding lemma gives nondegeneracy and the full Liouville field square.

For a boundary point x, the point $f(x)$ is native boundary and its image under P lies on the actual carrier frontier. Pull back a regular ambient defining function of this frontier along $i_P \circ f$. Its native value is zero at x, its native values are nonpositive near x, its differential is nonzero because $d(i_P \circ f)$ is bijective, and its derivative on the transported Liouville field is positive by the field square and the presentation's outwardness. The defining-function sign lemma identifies this differential as a positive multiple of the negative normal-coordinate differential in the source boundary chart. Hence the canonical negative normal coordinate has positive derivative on the transported field. Use this canonical coordinate as the source defining function: it is nonpositive on the native half-space, vanishes at its boundary, has nonzero full differential, and extends smoothly to the open ambient coordinates by $z \mapsto -z_0$. These facts supply every clause of the transported outward condition. □

Lemma A.613 (Completed positive-dimensional pair family from original bundle data)**.** *Let $n > 1$. Suppose the ambient and ribbon compact smooth bundles have native model dimensions $2n$ and $2n-2$, respectively, and supply the covariant original primitive, full ribbon and positive witness data, with the original local Liouville and pair laws above. There is one compatible global chart for each bundle and an actual completed pair family Q_t over $[0,1]$ with native diffeomorphisms $e_{A,t}, e_{M,t}$ to its compact ambient and ribbon domains and smooth boundary identifications $e_{B,t}$. In these same charts write the actual transported data as $\lambda_t, \mu_t, j_t, g_t$. They recover every original local value and full covector by the exact coordinate formulas above. For every parameter, including both endpoints,*

$$e_{A,t}^*\lambda_{Q_t,A} = \lambda_t, \qquad e_{M,t}^*\lambda_{Q_t,M} = \mu_t, \qquad e_{B,t}^*(\lambda_{Q_t,A}|_\partial) = \lambda_t|_\partial,$$
$$\iota_{\partial Q_t} \circ e_{B,t} = \iota_{A,t} \circ e_{A,t}|_\partial, \qquad j_{Q_t} \circ e_{M,t} = e_{B,t} \circ j_t, \qquad g_{Q_t} \circ e_{B,t} = g_t.$$

The entire embedded ribbon is $e_{B,t}(j_t(M))$. If

$$K_t = \{x \in M : \Phi^s_{Q_t,M}(\iota_{Q_t,M}(e_{M,t}(x))) \in \iota_{Q_t,M}(e_{M,t}(M)) \text{ for every } s \geq 0\},$$

then the full embedded core is exactly $e_{B,t}(j_t(K_t))$. There are jointly smooth native diffeomorphism families R_A, R_M, defined also across the parameter endpoints, such that the constructed ambient and ribbon presentation maps satisfy

$$F_{A,t} \circ \iota_{A,0} \circ e_{A,0} \circ R_A(t) = \iota_{A,t} \circ e_{A,t}, \qquad F_{M,t} \circ \iota_{M,0} \circ e_{M,0} \circ R_M(t) = \iota_{M,t} \circ e_{M,t},$$

and $F_{B,t} \circ e_{B,0} = e_{B,t}$. Both native carrier inclusions have their original coordinate extension germs on these same actual presentations.

Proof. Construct the two compatible global charts once, using the original ribbon-data global-chart lemma. In that same chart pair, transport both primitive families and the positive witness by their universal compatible chart theorems. These give all joint form, scalar and actual target-map germs on the entire interval. The compatible Liouville law theorem derives nondegeneracy and the full outward clauses from the original local laws. The whole-ribbon and immersion theorems retain the supplied local embedding and immersion conditions. The witnessed primitive matching theorem derives the native ribbon primitive equation in these same charts from its original local equation.

The native coherent positive-dimensional pair-family constructor now applies to precisely these four transported families and the derived laws. Its conclusions give Q, the native domain and boundary maps, the primitive and map squares, the jointly smooth R_A, R_M and their presentation identities. The endpoint coordinate-germ theorem retains the original coordinates on these same domains. Exact local primitive, witness and ribbon coherence give all the original-atlas recovery identities, without changing the chart choices or Q. The whole-ribbon square and surjectivity of $e_{M,t}$ identify its embedded full ribbon with $e_{B,t}(j_t(M))$. The native ribbon carrier maps onto the actual compact ribbon carrier; thus its completed forward trajectories define exactly the inverse image K_t of the full core. The same ribbon square then identifies the embedded core with $e_{B,t}(j_t(K_t))$. Every construction uses the whole closed parameter interval, proving the endpoint assertions as well. □

Lemma A.614 (Original bundle data retain endpoint identifications and markings on one completed family)**.** *Let $n > 1$. Supply the original compact ambient and ribbon bundle data and local geometric laws of Lemma A.613. There exist compatible global charts and one completed pair family Q, with all the original local covector, witness and full ribbon recovery identities of that lemma. On these same charts and this same family, there are original completed ambient slices X_t, their primitives L_t and strict slice comparisons $S_t : X_t \to K_t$ retaining the original native inclusion and signed cylinder formulas, together with all compact, cylinder and total coefficient conclusions of*

Lemma A.208. For any two parameters, any supplied original strict ambient and ribbon identification with its eventual shifted-end law, constant positive ribbon scale and embedding square gives a permitted identification of these very Q_t, Q_u with ambient map $S_u \circ \Psi \circ S_t^{-1}$ and native boundary map $b_u \circ \psi \circ b_t^{-1}$. Every supplied original ambient or ribbon inclusion marking transports through these same S_t and the corresponding ribbon slice comparison, retaining its compact inclusion pointwise. All original identification and marking hypotheses are exactly those in Lemma A.208. Their existence is not asserted for arbitrary geometric endpoints.

Proof. Choose the compatible global ambient and ribbon charts furnished by the original bundle data. Transport the original primitive, ribbon and witness into these chosen charts, retaining their local recovery identities. The compatible Liouville laws give nondegeneracy and outwardness; the whole ribbon and immersion laws give the original embedding and differential conditions. The witnessed primitive law gives the required ribbon pullback equation in these same charts. Apply the native pair-family construction retaining original endpoint identifications and markings to these transported data and the derived laws. It constructs one family Q and its original completed slices and strict comparisons, with the native and cylinder formulas. Its identification conclusion applies to each supplied strict ambient and ribbon identification, and its marking conclusions apply to each supplied ambient and ribbon marking. Thus all additional conclusions hold for the same constructed Q. The coordinate recovery identities precede this construction and are unchanged by it; the compact ribbon square and surjectivity of the native ribbon diffeomorphism identify the entire embedded ribbon. The same native carrier square identifies all its forward trajectories with those used to define its full core. Hence the original covector, witness, ribbon and core identities hold together with the endpoint and marking conclusions, as claimed. □

Lemma A.615 (Completed zero-dimensional ribbon family from original bundle data)**.** *Let the ambient model dimension be two and the ribbon model dimension zero. Supply the actual compact smooth bundles, original covariant ambient primitive data satisfying the original local Liouville laws, original boundary-target ribbon data injective on every original chart overlap, and original covariant positive witnesses. An arbitrary native ribbon one-form family μ_t requires no additional regularity assumption. There are compatible global charts and an actual completed pair family Q_t of rank one with the native ambient, ribbon and boundary maps, primitive pullback identities, whole ribbon square, witness values, and joint presentation coherence of the preceding lemma. It retains exact original local primitive, ribbon and witness values. At every parameter its full embedded core is exactly*

$$\mathrm{Core}_{\mathrm{emb}}(Q_t) = e_{B,t}(j_t(M)).$$

The actual ambient native inclusion retains its coordinate extension germs at every parameter, including both endpoints.

Proof. Construct both compatible global charts from the original ribbon data. Transport the ambient primitive and positive witness through that same pair of charts. Their local germs give the required native ambient form and scalar families, and the compatible Liouville theorem derives the ambient nondegeneracy and outward clauses. The actual ribbon target germs convert to native product target germs on the zero-dimensional source. Its source transition is a homeomorphism, so original overlap injectivity implies injectivity of the constructed full ribbon map.

Every tangent space of the zero-dimensional native ribbon is zero. Consequently every native one-form μ_t is the zero form, and its smoothness is automatic for every parameter family. Apply the coherent native zero-dimensional pair-family constructor to these actual families. It gives the primitive, boundary, ribbon and witness squares and the joint native presentation identities,

retaining the ambient coordinate germs. The original-atlas recovery identities follow from the same compatible chart formulas already used to build the family. The native zero-dimensional completed ribbon flow fixes every point, so its full core is its entire ribbon carrier. Surjectivity of its native carrier map and the full ribbon square therefore give exactly the displayed whole-image equality. This proof also applies to an empty ribbon. □

Lemma A.616 (Original finite ribbon bundle data retain endpoint identifications and markings)**.** *Supply the rank-one original compact bundle data of Lemma A.615, including the original overlap injectivity of the full ribbon and the arbitrary native ribbon one-form family. There are compatible global charts and one completed pair family Q retaining all that lemma's local primitive, ribbon, witness, presentation and whole-core identities. This same construction has original completed ambient slices and strict slice comparisons with their native and signed-cylinder formulas and all compact, cylinder and total coefficient conclusions of the native finite ribbon endpoint construction, and transports every supplied original endpoint identification and every supplied ambient or finite ribbon inclusion marking by the exact conjugation formulas of Lemma A.209. The original smooth maps, primitive identities, eventual shifted-end law, ribbon square and constant positive scale required by that lemma remain the explicit conditions for these conditional transports.*

Proof. Choose the compatible global ambient and finite ribbon charts from the original bundle ribbon data. Transport the original ambient primitive, positive witness and entire ribbon through these same charts. Compatible ambient Liouville laws give nondegeneracy and outwardness; original overlap injectivity gives full ribbon injectivity. Every ribbon tangent space is zero, so the arbitrary ribbon one-form family is the zero family and automatically smooth. Apply the native finite ribbon construction retaining original endpoint identifications and markings to these transported data. It produces one Q, its original completed ambient slices and strict comparisons, and all conditional identification and marking formulas. The original local recovery identities are unchanged. The finite ribbon flow fixes every point, so its full core is the entire ribbon; the retained ribbon square gives the same whole-image core formula. Thus all claims hold on the same charts and the same completed family, including the empty ribbon case. □

Intrinsic compact bundle scope. The local compact smooth-bundle data and their transition theorems in A.564 supply intrinsic fibre coordinate changes with smooth extensions across parameter endpoints and spatial boundaries. The base-dependent correction and topological splice in A.570 agree with a chosen atlas chart on a whole open neighborhood of the seam. The compatible splice in A.580 preserves the full joint extension condition, and A.582 supplies an actual global compact bundle chart compatible in both directions with the original atlas. Original total-space values and maps with local germs transport through that same chart by A.589. Covariant original-atlas primitive families transport by A.598, with full native tangent identities on all chart overlaps. The compatible transitions have actual boundary-valued germs by A.594. The full original ribbon maps and their entire images transport through both constructed bundle charts by A.604 and A.602. Their original immersion condition is retained by A.603; the original positive witness values and germs transport by A.607. The original positive-dimensional witnessed primitive equation is retained in those same charts by A.609. These are covariant local representations. The original local Liouville and pair laws produce actual completed pair families in both ribbon dimension cases by A.613 and A.615. Their stronger native completion conclusions, conditional original endpoint identifications and marking transports hold on one and the same family by A.614 and A.616. Identification with every intrinsic source pair, matching independently prescribed endpoint completions throughout a finite chain, and transport of the prescribed grading along that same chain remain separate requirements of A.204 and A.212.

The native primitive family constructs joint target-valued compact comparison germs on its actual localized completed family by A.188. Their transfer to a prescribed reference completion presenting α_0 is A.189. For an arbitrary reference primitive and unnormalized time-zero presentation, A.200 instead retains both actual relative comparison maps on their entire compact carriers through one fixed time-zero diffeomorphism. The interior lemma A.190 gives full neighborhood agreement inside the compact interior.

For original moving completions with smooth actual transported primitives, A.191 derives their transported field regularity, and A.192 gives joint germs of $P_t^{-1}(e_t(B_t(b), r))$ on one common two-sided strip. Its outward extension is A.193. The slice comparisons of A.198 retain the original native compact points and all-real boundary orbits. Their actual moving collar squares give full seam-neighborhood agreement by A.196. These seam germs and the compact interior germs give whole-completion regularity through A.197, whose cylindrical formula is used only at positive radius. Both directions are assembled for the same maps by A.201.

The source-data construction A.202 therefore compares original positive-dimensional moving completions with a family built from the native primitive germs. The same relative maps and their actual inverses are jointly smooth, retain the native compact squares and nonnegative reference end squares, and compare the actual varying transported primitives by A.199. It assumes original completed presentation data with their specified nonnegative same-radius end squares and smooth actual transported primitives. For these component comparison lemmas the coherent collar-adapted presentation maps are supplied data. The original compact-bundle constructors in A.613 and A.615 realize the full witnessed families, retaining the original ambient and ribbon relabelings, whole ribbons, positive witnesses and primitive equations. Their endpoint and marking consumers retain the independently supplied completed endpoints and inclusion markings on those same families. For the finite native ribbon, A.203 constructs comparisons with prescribed moving completions and proves both actual relative directions jointly smooth from their retained compact squares. Its empty boundary makes each completed ambient equal to its compact image; no regularity of the prescribed transported primitives is an additional input. These moving comparison lemmas isolate the conditional comparison maps; the full completed-family correspondence is accounted for by the component statements of A.204. The retained-section realization in A.316 and the original finite-chain constructions preserve the supplied grading data and literal junctions on the same geometric chain. The prescribed toric endpoint class and geometric itinerary remain existence conclusions of the target. A.218 transports a specified grading once the actual completed deformation, pulled two-forms and exact endpoint equations have been supplied; the full original bundle realization supplies those same identifications for the retained source data.

Prescribed completion realization boundary. The native compact presentation uses a finite signed collar band. Such a band gives an extension across the seam of the compact domain; it does not contain the whole half-infinite cylindrical end of Definition 2.1. If t is the inward collar coordinate, the outward completion radius is $r = -t$: the primitive $e^{-t}\alpha$ is $e^r\alpha$, and the outward radial field is $\partial_r = -\partial_t$. The prescribed smooth inclusion extension in A.408 likewise constructs only a neighborhood of the compact carrier. Its derivative and contraction field are derived on that entire carrier in A.410. The half-infinite topological attachment in A.442 contains the entire outgoing end, retains the original compact points and proves their coverage and compact enlargements. Its genuine open cover is A.441. The coordinate-preserving open window in A.443 identifies the existing finite native extension with the actual finite neighborhood inside this same attachment. The full-cylinder atlas and the smooth atlas on the whole attachment are constructed in A.444 and A.445; the smooth patch and native inclusions with bijective full differentials are

A.446. The regular compact domain with literal original carrier and native boundary is A.540. The actual whole-patch primitive gluing, exact native restriction and global nondegeneracy are proved in A.534. The compact domain is Liouville with its actual canonical field in A.542, and the original primitive and outward germs construct its collar and structure in A.543. The canonical field of the actual primitive has an all-real smooth flow by A.536. Its completion package, identity compact inclusion and all-real boundary orbit end are assembled in A.545; the original source data produce such a completion in A.547. The signed cylinder curve in A.537 is defined on $r > -\delta$. The all-real orbit end agrees with it there by A.544, and its primitive is the prescribed original native exponential form at every real radius by A.546. The full compact carrier has its original native atlas and coordinate extensions by A.548 and A.549; these give the native presentation on the same quotient in A.550, with a completion of that very presentation in A.551. The original native primitive and outward germs produce an inward collar. The actual boundary and field equations, followed by the inward collar trajectory identities, give smooth local comparison germs at the native boundary. Interior inverse-function germs and compact manifold-valued extension assemble them in A.558. Consequently A.561 constructs a strict marking of this same completion onto an independently supplied completion, retaining every original native point without requiring open extension germs of the prescribed original inclusion. The supplied native presentation and its actual completion remain explicit inputs. Once actual signed cylinder charts with the full exponential primitive identity, surjective differentials and compact coverage are supplied, A.431 constructs the complete flow on that specified ambient. These chart and primitive hypotheses are not consequences of the topological attachment alone. In particular, compact inclusion identities alone assert neither a diffeomorphism onto a prescribed completed ambient nor either marking required in (G2).

A.9 Independent endpoint identifications and their literal junctions

Lemma A.617 (Independent original completions give a full ribbon identification)**.** *Let P, Q be fixed witnessed pairs with positive-dimensional ribbons F_P, F_Q and completed ambients K_P, K_Q. Independently choose smooth completed manifolds X, Y with primitives L_X, L_Y, native boundary manifolds B_X, B_Y, outward cylinder maps E_X, E_Y, and smooth boundary diffeomorphisms $b_X : B_X \to \partial P$ and $b_Y : B_Y \to \partial Q$. Suppose specified smooth diffeomorphisms $S_X : X \to K_P$ and $S_Y : Y \to K_Q$ satisfy*

$$S_X^* \lambda_{K_P} = L_X, \qquad S_Y^* \lambda_{K_Q} = L_Y,$$
$$S_X(E_X(y, r)) = E_{K_P}(b_X(y), r),$$
$$S_Y(E_Y(z, r)) = E_{K_Q}(b_Y(z), r) \quad (r \geq 0).$$

Let M, N be native compact ribbon manifolds, with carrier diffeomorphisms $a : M \to F_P$, $a' : N \to F_Q$ preserving their specified native primitives μ_M, μ_N, and with the smooth coordinate extension germs of Lemma A.415. Let $j_X : M \to B_X$ and $j_Y : N \to B_Y$ satisfy the whole ribbon squares

$$j_P \circ a = b_X \circ j_X, \qquad j_Q \circ a' = b_Y \circ j_Y.$$

Suppose a specified smooth diffeomorphism $\Psi : X \to Y$ is strict, $\Psi^* L_Y = L_X$, and has smooth boundary coordinates $\psi : B_X \to B_Y$ and $s : B_X \to \mathbb{R}$ with the eventual end law

$$\Psi(E_X(y, r)) = E_Y(\psi(y), r + s(y))$$

for every y and all sufficiently large r with both displayed outward radii nonnegative. Finally, let $h : M \to N$ be a native smooth diffeomorphism with smooth coordinate extension germs, satisfying

$$h^* \mu_N = c \mu_M \quad (c > 0), \qquad \psi \circ j_X = j_Y \circ h.$$

Then there is a permitted completion identification $I : P \to Q$ whose ambient map, induced boundary map, shift and scale are exactly

$$f = S_Y \circ \Psi \circ S_X^{-1}, \qquad \psi_K = b_Y \circ \psi \circ b_X^{-1}, \qquad s_K = s \circ b_X^{-1}, \qquad c.$$

Its compact ribbon diffeomorphism g satisfies

$$g(a(x)) = a'(h(x)), \qquad g^{-1}(a'(z)) = a(h^{-1}(z)), \qquad g^*\mu_{F_Q} = c\mu_{F_P}.$$

Both fixed endpoint pairs and their chosen completions are retained. No sign restriction is imposed on s.

Proof. The fixed-endpoint native compact-map lemma constructs g on the already specified domains F_P, F_Q. Its conclusions are both native point equations and the full primitive scale equation above. The strict maps S_X, S_Y, their outward end squares, Ψ, ψ, s and its eventual end law satisfy the boundary-conjugation lemma. It gives primitive preservation of f and the stated actual induced boundary map and shift, with their all-real represented end equation and boundary primitive equation. The whole native ribbon equations and surjectivity of a show that $\psi_K \circ j_P = j_Q \circ g$: for $a(x) \in F_P$ their two values are respectively $b_Y(\psi(j_X(x)))$ and $b_Y(j_Y(h(x)))$, which agree by hypothesis. Use f, ψ_K, s_K, g, c in the full-ribbon conjugation construction. Every defining equation is the one just derived, and so it gives the required permitted identification with exactly these maps and scale. □

Lemma A.618 (Independent original completions with finite ribbons)**.** *The preceding independent-completion assertion holds for witnessed surface pairs and finite native ribbon manifolds M, N, including empty ribbons. The ambient strict maps, outward cylinders, eventual end law and whole ribbon squares are unchanged. Supply the native carrier diffeomorphisms a, a' and a native diffeomorphism $h : M \to N$. For any constant $c > 0$, no native ribbon primitive identity or smooth ribbon coordinate extension germs are required. The resulting compact ribbon map has the same two native point equations, and the permitted identification retains the same displayed ambient map, induced boundary map, arbitrary real shift and chosen scale c.*

Proof. The native finite-domain compact-map lemma constructs the compact ribbon diffeomorphism g with both prescribed native point equations. Every ribbon tangent space is zero-dimensional. Both $g^*\mu_{F_Q}$ and $c\mu_{F_P}$ therefore vanish on every tangent vector, proving the full primitive identity. Boundary conjugation uses only the ambient hypotheses, so gives the same f, ψ_K, s_K and their strict and end identities. The whole native ribbon square and surjectivity of a imply $\psi_K \circ j_P = j_Q \circ g$ at every point, by the calculation in the positive-dimensional proof. The full-ribbon conjugation construction then gives the asserted permitted identification. If the ribbons are empty, their point and primitive equations have empty domain; the ambient construction and all its boundary conclusions still apply. □

Lemma A.619 (A derived identification joins two independent deformations literally)**.** *Let $\mathcal{D}, \mathcal{E}$ be two specified witnessed pair deformations. Their endpoint tuples need not coincide. Suppose the independent original completion and full native ribbon hypotheses of one of the preceding two lemmas hold for exactly $P = \mathcal{D}_1$ and $Q = \mathcal{E}_0$. Then these hypotheses construct a permitted identification $I : \mathcal{D}_1 \to \mathcal{E}_0$ and a finite comparison chain of exactly three steps*

$$\mathcal{D}_0 \xrightarrow{\mathcal{D}} \mathcal{D}_1 \xrightarrow{I} \mathcal{E}_0 \xrightarrow{\mathcal{E}} \mathcal{E}_1.$$

The original deformations, all four endpoint tuples and the derived identification's prescribed maps and scale are retained.

Proof. Apply the appropriate independent-endpoint lemma to the already fixed $\mathcal{D}_1$ and $\mathcal{E}_0$ and the supplied original data. It constructs I between exactly those tuples. Form the finite path with first edge the deformation $\mathcal{D}$, second edge this I, and third edge the deformation $\mathcal{E}$. The first junction is literally $\mathcal{D}_1$ on both edges, and the second is literally $\mathcal{E}_0$ on both edges. Each edge is permitted and the list has three entries, hence is the claimed chain of length three. Neither appending the edges nor counting them changes their maps or endpoints. □

A.10 Real inverse germs of the original completed comparison

Lemma A.620 (A real cylinder inverse retains every original interval point)**.** *Let X be an original completed total space constructed from collars C_t of width δ, with cylinder insertion $q_{\mathrm{cyl}}(t,s,b)$ for $s<\delta$. Let $A : X \to [0,1]\times N$ be a specified bijection. Suppose a map $\beta : \mathbb{R}\times B\times\mathbb{R}\to N$ and real a satisfy*

$$A(q_{\mathrm{cyl}}(t,s,b)) = (t,\beta(t,b,-s)) \quad (t\in[0,1],\ s<\delta,\ a<-s).$$

Fix $t\in[0,1]$, $x\in B$, and r with $a<r$ and $-r<\delta$. Suppose $G(u,b,v)=(u,\beta(u,b,v))$ is a local smooth diffeomorphism at (t,x,r). Then there is a real open neighborhood U of $(t,\beta(t,x,r))$ and a smooth map $g : U\to B\times\mathbb{R}$ with $g(t,\beta(t,x,r))=(x,r)$ such that

$$a<g_2(u,y), \qquad -g_2(u,y)<\delta, \qquad \beta(u,g_1(u,y),g_2(u,y))=y \quad ((u,y)\in U).$$

The map $(u,y)\mapsto(u,g(u,y))$ is the restriction of a smooth partial diffeomorphism. At every interval point $(u,y)\in U$, the actual original inverse is exactly

$$A^{-1}(u,y) = q_{\mathrm{cyl}}(u,-g_2(u,y),g_1(u,y)).$$

Proof. Choose the smooth local inverse L of G near $G(t,x,r)$. Shrink its open source to the inverse image under L of the open conditions $a<v$ and $-v<\delta$. This remains an open neighborhood U of the distinguished point. Write $g=\mathrm{pr}_{B\times\mathbb{R}}\circ L$. The equation $G(L(u,y))=(u,y)$ gives $\mathrm{pr}_{\mathbb{R}}(L(u,y))=u$ and $\beta(u,g(u,y))=y$. Thus $L(u,y)=(u,g(u,y))$, giving both the parameter-preserving partial diffeomorphism and the stated inequalities. The inverse equation at the distinguished point gives $g=(x,r)$ there. For any interval $(u,y)\in U$, its inequalities make $q_{\mathrm{cyl}}(u,-g_2(u,y),g_1(u,y))$ a valid original cylinder point. The assumed square sends that point under A to $(u,\beta(u,g(u,y)))=(u,y)$. Applying the inverse of the same bijection A proves the exact original inverse equation throughout the interval part of U. □

Lemma A.621 (The actual original comparison has real inverse germs on its full reference end)**.** *Under the native inward-square and original-comparison hypotheses of Lemma A.466, let*

$$H : X\to[0,1]\times K_0, \qquad H(z)=(\pi(z),q(z))$$

be the actual original comparison bijection, with original cylinder insertion q_{cyl} and native reference boundary map b_0. For every $t\in[0,1]$, native boundary point x and $r\ge 0$, there is a real open neighborhood U of $(t,E_{K_0}(b_0(x),r))$ and a smooth $g : U\to\partial M\times\mathbb{R}$ such that

$$g(t,E_{K_0}(b_0(x),r))=(x,r), \qquad -g_2(u,y)<\delta.$$

The parameter-preserving map $(u,y)\mapsto(u,g(u,y))$ is the restriction of a smooth partial diffeomorphism, and at every interval point of U one has the exact original equation

$$H^{-1}(u,y) = q_{\mathrm{cyl}}(u,-g_2(u,y),g_1(u,y)).$$

The assertion includes the original seam $r=0$ and both parameter endpoints; no larger interval of globally invertible maps is assumed.

Proof. Retain the real smooth cylinder extension β and its negative cutoff $a < 0$ from the cylinder-extension lemma. On interval parameters and radii above a, it equals $P_u^{-1}(E_{K_u}(b_u(b), v))$. At the chosen $r \geq 0$ the inequality $r > a$ persists on a spatial neighborhood. Its spatial derivative therefore agrees with that of this same prescribed end map. The maps b_u and P_u^{-1} are diffeomorphisms; the all-real boundary orbit map has an invertible spatial differential by complete boundary orbit calculus. Consequently this spatial derivative is an isomorphism. The parameter inverse lemma makes $(u, b, v) \mapsto (u, \beta(u, b, v))$ a local smooth diffeomorphism at (t, x, r). The exact original cylinder square identifies H on its cylinder with $(u, \beta(u, b, -s))$ when $a < -s$. Apply the preceding real germ lemma, using $a < r$ and $-r < \delta$, to this very H. It gives the smooth g, its parameter-preserving partial inverse and the original inverse equality at every interval point of U. The nonnegative-radius presentation square gives $\beta(t, x, r) = E_{K_0}(b_0(x), r)$, identifying the distinguished point and the value of g exactly as stated. □

Lemma A.622 (The actual original comparison has real inverse germs on the compact interior)**.** *Let H be the original comparison bijection with its actual native compact square and a jointly smooth real-parameter family of native relabelings R_u and their inverses. Write $f_0 : \operatorname{int} M \to K_0$ for the actual reference native interior inclusion. For every $t \in [0, 1]$ and $v \in \operatorname{int} M$, there is a real open neighborhood U of $(t, f_0(v))$ and a smooth map $g : U \to \operatorname{int} M$ such that*

$$g(t, f_0(v)) = R_t^{-1}(v), \qquad H^{-1}(u, y) = i_{\mathrm{tot}}(u, g(u, y)) \quad (u \in [0, 1],\ (u, y) \in U).$$

Here i_{tot} is the same original interior insertion. The assertion retains all interval values of the actual original inverse, including at both parameter endpoints.

Proof. The reference native inclusion is a local diffeomorphism at v. Choose its smooth local inverse l near $f_0(v)$, with $l(f_0(v)) = v$ and $f_0(l(y)) = y$. Take $U = \mathbb{R} \times \operatorname{dom}(l)$ and set $g(u, y) = R_u^{-1}(l(y))$. Native diffeomorphisms preserve interior points, so this is interior-valued. Joint smoothness of R_u^{-1} and smoothness of l make g smooth on this real open set. Its value at the distinguished point is $R_t^{-1}(v)$. For interval parameters the exact native square gives

$$H(i_{\mathrm{tot}}(u, g(u, y))) = (u, f_0(R_u(R_u^{-1}(l(y))))) = (u, y).$$

Applying the inverse of the same comparison bijection H proves the stated inverse equation at every interval point of U. □

A.11 Supplied native grading lifts on the same endpoint identification

Lemma A.623 (A supplied native grading homotopy conjugates through the specified slice maps)**.** *Let M, N, X, Y be smooth $2n$-manifolds, with two-forms η on M and η' on N. Fix smooth diffeomorphisms $S : X \to M$, $S' : Y \to N$, and $\Psi : X \to Y$. Let ρ_M, ρ_N be specified Maslov representatives for η, η'. Suppose a supplied native grading lift of Ψ is a compatible nowhere-vanishing homotopy from $S^*\rho_M$ to $\Psi^*(S')^*\rho_N$, for the fixed two-form $S^*\eta$ on X. Write (J_τ, s_τ) for its real-indexed endomorphism and tensor representatives, with smooth coefficient germs near the closed interval; compatibility, canonicality and nonvanishing hold on $[0, 1]$. Then the specified conjugated diffeomorphism*

$$f = S' \circ \Psi \circ S^{-1} : M \to N$$

has a grading lift from ρ_M to $f^*\rho_N$. Its actual family is exactly

$$\widetilde{J}_\tau = (S^{-1})^* J_\tau, \qquad \widetilde{s}_\tau = (S^{-1})^* s_\tau \quad (\tau \in \mathbb{R}).$$

The lift is compatible with η and canonical and nowhere zero on $[0, 1]$, with smooth coefficient germs across its endpoints.

Proof. Pull the supplied family back along the fixed diffeomorphism S^{-1}. In spatial coordinates an endomorphism pulls back by conjugating with the forward and inverse derivative matrices; a covariant tensor pulls back by inserting the forward derivative in every argument. These matrices are smooth, and composition with the original coefficient germs gives smooth real-parameter germs of both pulled fields. The invertible differential identifies the canonical lines and their squares, so preserves canonicality and nonvanishing. It also preserves the compatibility equations for the pulled two-form. That form is $(S^{-1})^*(S^*\eta) = \eta$, proving compatibility with exactly the specified η on the interval. At zero, composition of pullbacks gives $(S^{-1})^*S^*\rho_M = \rho_M$. At one it gives

$$(S^{-1})^*\Psi^*(S')^*\rho_N = f^*\rho_N.$$

These identities apply to both the complex endomorphism and the squared tensor. They prove the required exact endpoint equations. The construction is the stated derivative pullback at every real parameter, so retains both displayed all-real family formulas. □

Lemma A.624 (The native homotopy grades the exact constructed endpoint identification)**.** *Fix a permitted identification $I : P \to Q$, specified native completed ambients X, Y, and smooth slice comparisons $S : X \to K_P$, $S' : Y \to K_Q$. Suppose an original smooth map $\Psi : X \to Y$ satisfies the exact retained ambient equation*

$$I_{\mathrm{amb}} = S' \circ \Psi \circ S^{-1}.$$

Fix endpoint Maslov representatives ρ_P, ρ_Q and a supplied native grading lift from $S^*\rho_P$ to $\Psi^*(S')^*\rho_Q$, compatible with $S^*\omega_P$. Then the same I has a graded lift from ρ_P to $I^*_{\mathrm{amb}}\rho_Q$. If the supplied lift's real-indexed fields are J_τ, s_τ, the constructed graded lift retains exactly $(S^{-1})^*J_\tau, (S^{-1})^*s_\tau$ at every real parameter. In particular this applies to the ambient equality retained by either independent-endpoint identification lemma.

Proof. The preceding conjugation lemma gives a grading lift for $S' \circ \Psi \circ S^{-1}$ with exactly these endpoint representatives and the stated all-real family. The retained ambient equality identifies this map with I_{amb}. Substituting the equality in the target endpoint equations makes this same family a graded lift of exactly I. Its source equations and its family are unchanged. Thus both actual pulled field formulas hold at every real parameter as claimed. □

Lemma A.625 (Supplied grading lifts decorate the literal three-step junction chain)**.** *Let $\mathcal{D}, \mathcal{E}$ be specified pair deformations and $I : \mathcal{D}_1 \to \mathcal{E}_0$ a permitted identification. Let $\rho_0, \rho_1, \rho_2, \rho_3$ be specified Maslov representatives on their four literal endpoint tuples. Supply a grading of $\mathcal{D}$ from ρ_0 to ρ_1 and a grading of $\mathcal{E}$ from ρ_2 to ρ_3. Suppose I retains the native conjugation equation and supplied native grading lift of the preceding lemma for ρ_1, ρ_2. Then the exact path*

$$\mathcal{D}_0 \xrightarrow{\mathcal{D}} \mathcal{D}_1 \xrightarrow{I} \mathcal{E}_0 \xrightarrow{\mathcal{E}} \mathcal{E}_1$$

has a grading from ρ_0 to ρ_3, retaining the supplied two deformation gradings and the derived native conjugation grading of the same middle identification. The representatives at both junctions agree literally. This is conditional on the supplied native homotopy and two deformation gradings.

Proof. Construct the middle graded lift by the preceding lemma, with source ρ_1 and target ρ_2. Start with the zero-length graded path at ρ_0, append the supplied grading of $\mathcal{D}$, then this middle graded lift of I, then the supplied grading of $\mathcal{E}$. The first append ends at ρ_1, the literal source of the middle lift. The middle append ends at ρ_2, the literal source of the last lift. Thus both junctions share exactly one representative, and the final endpoint is ρ_3. Forgetting these grading data leaves precisely the displayed original three-edge path. No edge, family or endpoint representative has been replaced. □

A.12 Descent of original value coefficients through the actual inverse

Lemma A.626 (The spatial derivative of a real parameter-preserving inverse is invertible)**.** *Let P, N be finite-dimensional smooth boundaryless manifolds and let L be a smooth partial diffeomorphism from $\mathbb{R} \times P$ to $\mathbb{R} \times N$. On an open subset U of its source, suppose*

$$L(u, x) = (u, g(u, x))$$

for a smooth $g : U \to N$. At every $(t, x) \in U$, the spatial differential $d_x g_t : T_x P \to T_{g(t,x)} N$ is an isomorphism.

Proof. Since L is a local diffeomorphism, its full differential is an isomorphism. The displayed parameter identity makes that differential

$$(a, v) \longmapsto (a, d_t g(a) + d_x g_t(v)).$$

If $d_x g_t(v) = d_x g_t(w)$, the images of $(0, v)$ and $(0, w)$ under this isomorphism agree, so $v = w$. For any $z \in T_{g(t,x)} N$, surjectivity of the full differential gives (a, v) mapping to $(0, z)$. The first coordinate forces $a = 0$, and the second then gives $d_x g_t(v) = z$. Thus the spatial differential is bijective; in finite dimension its inverse is continuous, so it is a linear isomorphism as asserted. □

Lemma A.627 (The actual original end inverse has invertible spatial derivatives)**.** *Under the actual original end inverse-germ hypotheses, every $(t, E_{K_0}(b_0(x), r))$ with $r \geq 0$ has a real open neighborhood U and a smooth $g : U \to \partial M \times \mathbb{R}$ retaining*

$$g(t, E_{K_0}(b_0(x), r)) = (x, r), \qquad -g_2(u, y) < \delta,$$
$$H^{-1}(u, y) = q_{\mathrm{cyl}}(u, -g_2(u, y), g_1(u, y)) \quad (u \in [0, 1],\ (u, y) \in U).$$

Moreover the spatial differential $d_y g_u$ is an isomorphism at every $(u, y) \in U$, including its real off-interval points.

Proof. Retain the U, g and the parameter-preserving smooth partial diffeomorphism $L(u, y) = (u, g(u, y))$ produced by the actual end germ lemma. Their distinguished value, band inequality and every interval inverse equation are already the displayed ones. Apply the preceding vertical-derivative lemma to L, its same open subset U, and this same smooth g. It makes $d_y g_u$ an isomorphism at every point of U. This adds the claimed differential conclusion without changing any original inverse value or neighborhood. □

Lemma A.628 (An original end coefficient descends through the retained real inverse germ)**.** *Let $H : X \to [0, 1] \times K_0$ be the actual original comparison of Lemma A.621, and let E be a normed real vector space. Fix a function $f : X \to E$, a parameter t, native boundary point x, and $r \geq 0$. Suppose $a : V \to E$ is smooth on a real open neighborhood V of (t, x, r) in $\mathbb{R} \times \partial M \times \mathbb{R}$, and satisfies*

$$a(u, b, v) = f(q_{\mathrm{cyl}}(u, -v, b))$$

for every interval parameter and original cylinder point in V. Then on a real open neighborhood W of $(t, E_{K_0}(b_0(x), r))$ there is a smooth $a' : W \to E$ such that

$$a'(u, y) = f(H^{-1}(u, y)) \quad (u \in [0, 1],\ (u, y) \in W).$$

The extension preserves every such interval value, including at the seam and both parameter endpoints.

Proof. Retain the actual real inverse germ g on its neighborhood U from the end inverse-germ lemma. Its distinguished value is (x, r). The map $G(u, y) = (u, g(u, y))$ is smooth on U, so $W = U \cap G^{-1}(V)$ is a real open neighborhood of the distinguished reference point. Set $a' = a \circ G$ on W; it is smooth. At every interval $(u, y) \in W$, the retained germ equation is $H^{-1}(u, y) = q_{\mathrm{cyl}}(u, -g_2(u, y), g_1(u, y))$. The source coefficient identity at $G(u, y) \in V$ therefore gives $a'(u, y) = f(H^{-1}(u, y))$, proving the required equality throughout the interval part of W. □

Lemma A.629 (Original interior coefficients descend through the retained real inverse germ)**.** *Let H be the actual original comparison of Lemma A.622, E a normed real vector space, and $f : X \to E$ a function. Suppose $f \circ i_{\mathrm{tot}}$ has smooth real open extensions near every $(u, v) \in [0, 1] \times \operatorname{int} M$, agreeing with this original interior coefficient at every nearby interval point. Then near every $(t, f_0(v))$ there is a real open neighborhood W and a smooth $a' : W \to E$ satisfying*

$$a'(u, y) = f(H^{-1}(u, y)) \quad (u \in [0, 1],\ (u, y) \in W).$$

Proof. Retain the real interior inverse germ g near $(t, f_0(v))$. Choose a supplied smooth source extension a on a real open neighborhood V of $(t, g(t, f_0(v)))$. The smooth map $G(u, y) = (u, g(u, y))$ makes $W = U \cap G^{-1}(V)$ a real open neighborhood of the distinguished point. Define $a' = a \circ G$ there. It is smooth, and for every interval point of W the retained inverse identity and the source coefficient identity give

$$a'(u, y) = f(i_{\mathrm{tot}}(u, g(u, y))) = f(H^{-1}(u, y)).$$

This proves every required interval value using the same original inverse. □

Lemma A.630 (Original local value coefficients descend on the whole reference completion)**.** *Let H be the same original completed comparison, E a normed real vector space, and $f : X \to E$. Suppose its original interior coefficient $f \circ i_{\mathrm{tot}}$ has smooth real open extension germs at every interval interior point. Suppose also its original end coefficient $f(q_{\mathrm{cyl}}(u, -r, b))$ has smooth real open extension germs at every interval point with $r \geq 0$, with the source agreement on all valid nearby interval cylinder points. Then for every $(t, y) \in [0, 1] \times K_0$ there is a real open neighborhood W and smooth $a' : W \to E$ with*

$$a'(u, z) = f(H^{-1}(u, z)) \quad (u \in [0, 1],\ (u, z) \in W).$$

This assertion concerns values in a fixed vector space. Transport of tangent endomorphisms or covariant canonical tensors additionally requires their spatial differential transformation laws.

Proof. The reference completion is covered by its compact interior and its nonnegative cylindrical end. If $y = f_0(v)$ is in the compact interior, the preceding interior coefficient lemma supplies W, a'. Otherwise completion coverage writes $y = E_{K_0}(b_0(x), r)$ for some native boundary x and $r \geq 0$; the seam $r = 0$ is included in this case. Choose the supplied source end extension at (t, x, r) and apply the end coefficient lemma. It supplies W, a' with the same required interval equality. Both cases use the same actual comparison inverse H^{-1} and exhaust every reference point, proving the assertion. □

Lemma A.631 (The original interior inverse has invertible spatial derivatives)**.** *Under the actual original interior inverse-germ hypotheses, let $f_0 : \operatorname{int} M \to K_0$ be the reference native inclusion and R_u the real smooth family of native relabelings. For every $t \in [0, 1]$ and $v \in \operatorname{int} M$ there is a real open neighborhood U of $(t, f_0(v))$ and a smooth $g : U \to \operatorname{int} M$ such that*

$$g(t, f_0(v)) = R_t^{-1}(v),$$
$$H^{-1}(u, y) = i_{\mathrm{tot}}(u, g(u, y)) \quad (u \in [0, 1],\ (u, y) \in U).$$

At every real point $(u, y) \in U$, the spatial differential $d_y g_u : T_y K_0 \to T_{g(u,y)} \operatorname{int} M$ is invertible. This retains the original inverse values at both parameter endpoints.

Proof. A native diffeomorphism and its inverse preserve the intrinsic interior: a local diffeomorphism preserves whether a point is interior. Their restrictions are mutually inverse and smooth in the interior charts, so are interior diffeomorphisms. The joint smooth native inverse family consequently restricts to a joint smooth interior family. The reference native inclusion is a local diffeomorphism at v; choose its smooth local inverse l on an open neighborhood O of $f_0(v)$ with $l(f_0(v)) = v$ and $f_0(l(y)) = y$ throughout O. Set $U = \mathbb{R} \times O$ and $g(u, y) = R_u^{-1}(l(y))$. It is jointly smooth and has the stated distinguished value. For fixed u, both l and the restriction of R_u^{-1} are local diffeomorphisms at every point of this domain. The chain rule expresses $d_y g_u$ as the composition of their two invertible differentials, so it is invertible everywhere on U. For an interval point, the actual native square gives

$$H(i_{\mathrm{tot}}(u, g(u, y))) = (u, f_0(R_u(R_u^{-1}(l(y))))) = (u, y).$$

Applying the inverse of the same original comparison bijection H proves the asserted interval equation. Neither the construction nor this equation changes u or the original interior insertion. □

A.13 Local coframe transport through the original end inverse

Lemma A.632 (Smooth spatial pullback of coordinate covectors)**.** *Let P be a smooth $2n$-manifold and F a normed real vector space. Let U be an open subset of $\mathbb{R} \times P$, and let $f : U \to F$ be smooth. Suppose $A(u, z) : F \to \mathbb{C}^n$ is a smooth family of continuous real-linear maps on an open $V \subset \mathbb{R} \times F$. On $W = U \cap \{(u, y) : (u, f(u, y)) \in V\}$ put*

$$B(u, y) = A(u, f(u, y)) \circ d_y f_u.$$

Then B is jointly smooth as a family of vertical covectors. Only the spatial differential occurs in this formula; neither its invertibility nor invertibility of A is assumed.

Proof. The map $G(u, y) = (u, f(u, y))$ is smooth, so $W = U \cap G^{-1}(V)$ is open. In a spatial chart c on P, put $a(u, v) = f(u, c^{-1}(v))$. The spatial derivative of a is its joint derivative restricted to vectors $(0, w)$. The joint derivative is smooth on its open domain, and restriction to this fixed spatial subspace is a continuous linear operation, so the spatial derivative varies smoothly. The coefficients of B in this chart are $A(u, a(u, v)) \circ d_v a_u$. Both factors are smooth, hence their product is smooth. The chain rule identifies this coordinate product with $A(u, f(u, y)) \circ d_y f_u \circ d_v c^{-1}$, which is precisely the chart expression of the displayed vertical covectors. Thus the expressions agree on overlapping charts and prove joint smoothness. □

Lemma A.633 (The determinant square of a smooth vertical coframe is smooth)**.** *Let $A(u, y) : T_y P \to \mathbb{C}^n$ be a jointly smooth family of real-linear maps on a subset of $\mathbb{R} \times P$. Then*

$$\Gamma_A(u, y)(v_1, \ldots, v_n; w_1, \ldots, w_n) = \det(A(u, y)v_1, \ldots, A(u, y)v_n) \det(A(u, y)w_1, \ldots, A(u, y)w_n)$$

is jointly smooth as a vertical covariant tensor, alternating in both blocks. This regularity statement requires no invertibility.

Proof. In any spatial tangent chart, the coefficients of the displayed tensor are products of minors of the coefficient matrix of A. Each minor is a finite sum of signed products of entries of that matrix. Since these entries are jointly smooth, all such products and sums are jointly smooth. The determinant's alternating multilinearity gives the asserted alternation in each block. Changing spatial charts inserts the tangent transition in every argument, exactly the covariant-tensor transformation law. The chart coefficients therefore describe one jointly smooth vertical tensor. □

Lemma A.634 (Original end coframes and scalar coordinates give local canonical-square tensors)**.** *Let $H : X \to [0,1] \times K_0$ be the actual original comparison of Lemma A.627. Fix $t \in [0,1]$, a native boundary point x, and $r \geq 0$. Let c be the spatial product chart of $\partial M \times \mathbb{R}$ at (x,r), with coordinate model F. Suppose $A(u,v) : F \to \mathbb{C}^n$ is a smooth invertible real-linear coframe on a real open neighborhood V of $(t, c(x,r))$. Let $a : V \to \mathbb{C}$ be smooth with $a(t,c(x,r)) \neq 0$, and let $f : X \to \mathbb{C}$ be a specified original coefficient. Assume that, for every valid interval cylinder point in the chart whose coordinates lie in V,*

$$a(u, c(b,v)) = f(q_{\mathrm{cyl}}(u, -v, b)).$$

Then a real open neighborhood W of $(t, E_{K_0}(b_0(x), r))$ has a smooth $g : W \to \partial M \times \mathbb{R}$ with

$$g(t, E_{K_0}(b_0(x), r)) = (x, r), \qquad g(u,y) \in \mathrm{dom}(c),$$
$$H^{-1}(u,y) = q_{\mathrm{cyl}}(u, -g_2(u,y), g_1(u,y)) \quad (u \in [0,1],\ (u,y) \in W).$$

For $\varphi_u = c \circ g_u$, the local fields

$$B(u,y) = A(u, \varphi_u(y)) \circ d_y\varphi_u, \qquad a'(u,y) = a(u, \varphi_u(y)),$$
$$J_B = B^{-1} i B, \qquad s_B = a' \Gamma_B$$

are jointly smooth, B is invertible, and s_B is a nowhere-zero squared canonical tensor for J_B throughout W. With $D = d_y\varphi_u$ and $A_0 = A(u, \varphi_u(y))$, their exact spatial transport laws are

$$J_B = D^{-1}(A_0^{-1} i A_0) D, \qquad s_B = D^*(a' \Gamma_{A_0}),$$

where D^* inserts D in every tensor argument. At every interval point of W one has $a'(u,y) = f(H^{-1}(u,y))$. This is a local construction from supplied original coframes and scalar coordinates. It asserts neither compatibility with a prescribed symplectic form nor existence of a global ambient grading.

Proof. Retain the smooth inverse germ g on U from the original inverse vertical-derivative lemma. Its spatial differential is invertible everywhere. Shrink first to $U_1 = U \cap g^{-1}(\mathrm{dom}(c))$; the distinguished value (x,r) belongs to this chart, so U_1 is a real open neighborhood of the distinguished point. Set $G(u,y) = (u, c(g(u,y)))$. It is smooth on U_1. The set $V' = V \cap a^{-1}(\mathbb{C} \setminus \{0\})$ is open and contains $(t, c(x,r))$. Hence $W = U_1 \cap G^{-1}(V')$ is open and contains the distinguished reference point.

The coordinate coframe pullback lemma gives joint smoothness of $B = A \circ d\varphi$ on W. The spatial chain rule factors $d\varphi$ as $dc \circ dg$. The chart differential, the retained inverse spatial differential, and the supplied A are all invertible there, so B is invertible. The coframe complex-reconstruction lemma gives joint smoothness of $J_B = B^{-1} i B$. It satisfies $J_B^2 = -1$ and $B J_B = iB$ directly by conjugation. The determinant-square lemma makes Γ_B smooth, and composition makes $a' = a \circ G$ smooth; therefore $s_B = a' \Gamma_B$ is smooth. Each determinant defining Γ_B is complex-linear for J_B because $B J_B = iB$, and alternating in its block. Thus s_B is a squared canonical tensor for J_B. Substituting $B = A_0 D$ and $(A_0 D)^{-1} = D^{-1} A_0^{-1}$ proves the displayed conjugation law. Inserting D in every argument of the two determinants proves the full tensor pullback law, including its scalar factor. It is nonzero: a' never vanishes on W, and invertibility of B lets one insert preimages of a complex basis in both determinant blocks, giving Γ_B value one.

Restricting the original inverse germ to W preserves its distinguished value and every interval inverse equation; the corresponding cylinder point is valid by that germ's band condition. Its source point lies in the chart and its coordinates lie in V. The supplied original coefficient equation consequently gives $a'(u,y) = f(q_{\mathrm{cyl}}(u, -g_2(u,y), g_1(u,y))) = f(H^{-1}(u,y))$ throughout the interval part of W, proving the last assertion. □

Lemma A.635 (Original interior coframes give local canonical-square tensors)**.** *Let H be the actual original comparison, and let f_0 and R_u be the original interior inclusion and native relabelings of Lemma A.631. Fix $t \in [0,1]$ and $v \in \operatorname{int} M$. Put $y_0 = f_0(R_t(v))$, the actual reference image of the original native point at time t, and choose the spatial chart c at v, with coordinate model F. Suppose $A(u,z) : F \to \mathbb{C}^n$ is a smooth invertible real-linear coframe on a real open neighborhood V of $(t, c(v))$. Let $a : V \to \mathbb{C}$ be smooth with $a(t,c(v)) \neq 0$, and let $f : X \to \mathbb{C}$ be a specified original coefficient. Assume, at every interval interior point in this chart with coordinates in V, that*

$$a(u, c(z)) = f(i_{\mathrm{tot}}(u,z)).$$

Then a real open neighborhood W of (t, y_0) has a smooth $g : W \to \operatorname{int} M$ such that

$$g(t,y_0) = v, \qquad g(u,y) \in \operatorname{dom}(c),$$
$$H^{-1}(u,y) = i_{\mathrm{tot}}(u, g(u,y)) \quad (u \in [0,1],\ (u,y) \in W).$$

For $\varphi_u = c \circ g_u$, define $B = A(u, \varphi_u(y)) d_y \varphi_u$, $a' = a(u, \varphi_u(y))$, $J_B = B^{-1} i B$, and $s_B = a' \Gamma_B$. These fields are jointly smooth, B is invertible, and s_B is a nowhere-zero squared canonical tensor for J_B on W. For $D = d_y \varphi_u$ and $A_0 = A(u, \varphi_u(y))$, one has

$$J_B = D^{-1}(A_0^{-1} i A_0) D, \qquad s_B = D^*(a' \Gamma_{A_0}).$$

At every interval point, $a'(u,y) = f(H^{-1}(u,y))$. The assertion is local and does not assert compatibility with a prescribed symplectic form or a global ambient grading.

Proof. Apply the original interior spatial-invertibility lemma at the reference interior point $R_t(v)$. Its distinguished reference image is y_0, and its inverse germ g takes value $R_t^{-1}(R_t(v)) = v$ there. It has an invertible spatial differential throughout its real open domain U and the exact original interval inverse equation. Restrict to $U_1 = U \cap g^{-1}(\operatorname{dom}(c))$, and put $G(u,y) = (u, c(g(u,y)))$. Smoothness of g and c makes G smooth. The open set $V' = V \cap a^{-1}(\mathbb{C} \setminus \{0\})$ contains $(t, c(v))$, so $W = U_1 \cap G^{-1}(V')$ is an open neighborhood of (t, y_0). The same restricted g retains its distinguished value and all interval inverse equations. The coordinate pullback lemma makes B smooth. Its factors A, dc and dg are invertible on W, so B is invertible. Complex reconstruction makes J_B smooth. Conjugation gives $J_B^2 = -1$ and $B J_B = iB$. The determinant-square lemma and smoothness of $a' = a \circ G$ make s_B smooth. Its two determinant blocks are alternating and complex-linear for J_B, so it is squared canonical. Since a' is nonzero and inserting preimages of a complex basis under B gives Γ_B value one, s_B is nowhere zero. Substituting $B = A_0 D$ and its inverse proves the displayed conjugation identity; inserting D in each determinant argument proves the tensor pullback identity. Finally, at an interval point the supplied source coefficient identity and the retained original inverse equation give $a'(u,y) = f(i_{\mathrm{tot}}(u, g(u,y))) = f(H^{-1}(u,y))$. □

A.14 Reconstructing the prescribed grading from original coframes

A canonical square in the original coordinate model is recovered from its computed inverse-coframe coefficient. This local algebraic reconstruction is a prerequisite for retaining a prescribed grading under spatial pullback. It does not construct a compatible grading from the toric input or determine the endpoint class required by the target.

Lemma A.636 (Inverse-coframe reconstruction in the original coordinate model)**.** *Let F be a real vector space of dimension $2n$, identified by a specified real-linear isomorphism with the tangent coordinate model. Let $C : F \to \mathbb{C}^n$ be an invertible real-linear coframe, and let $J : F \to F$ satisfy*

$CJ = iC$. For a covariant tensor τ which is alternating separately in two blocks of n arguments and complex-linear in every argument for J, define

$$a(C,\tau) = \tau(C^{-1}e_1,\dots,C^{-1}e_n;C^{-1}e_1,\dots,C^{-1}e_n).$$

Here $e_1,\dots,e_n$ are the complex standard basis vectors, regarded as real vectors. Then the full tensor satisfies

$$\tau = a(C,\tau)\Gamma_C.$$

In particular $\tau \neq 0$ implies $a(C,\tau) \neq 0$. For jointly smooth coframes C and covariant tensors τ on an open parameter-coordinate domain, with C invertible throughout that domain, the computed scalar $a(C,\tau)$ is smooth. This regularity assertion requires no canonicality equation off the original interval. The specified linear identification serves only to express canonicality; the coefficient and reconstruction use the original vectors of F.

Proof. Since $CJ = iC$, the map C is complex-linear from (F,J) to $\mathbb{C}^n$. Its inverse sends the complex standard basis to a complex basis of (F,J). Expand each of the n arguments in the first block in this basis. Multilinearity expands the value into a sum indexed by n-tuples of basis indices. Alternation makes every term with a repeated index zero, and the terms with distinct indices differ by their permutation sign. Their sum is therefore the determinant of the first block's coordinate matrix times the tensor's value on the ordered basis in that block. Repeating this expansion in the second block gives

$$\tau(v_1,\dots,v_n;w_1,\dots,w_n) = a(C,\tau)\det(Cv_1,\dots,Cv_n)\det(Cw_1,\dots,Cw_n).$$

This is the asserted full reconstruction. If its coefficient were zero, every value of τ would vanish, proving the nonvanishing implication. When $n = 0$, the two empty determinants are one and the identity is the identity for a scalar, so the assertion includes that dimension.

On the open invertibility locus, inversion of finite-dimensional linear maps is smooth. Thus all vectors $C^{-1}e_j$ depend smoothly on the parameter and coordinates. Evaluation of a continuously multilinear tensor on finitely many vectors is smooth, so evaluation of the smooth τ on these inverse-frame vectors is smooth. This proves the scalar regularity without imposing canonicality away from the interval. Changing the auxiliary linear identification inserts that isomorphism and its inverse together in every argument; they cancel. Consequently neither the coefficient nor the displayed reconstruction depends on it. □

Lemma A.637 (Smooth local coframes from the original complex coefficients)**.** *Let F be a real vector space of dimension $2n$ with a specified real-linear identification with the tangent coordinate model. Let V be an open subset of a normed real parameter-coordinate space, let A be a prescribed subset, and let $p \in V \cap A$. Suppose $K : V \to \mathrm{End}_{\mathbb{R}}(F)$ is smooth and $K(z)^2 = -\mathrm{id}$ for $z \in V \cap A$. Then some open neighborhood $W \subset V$ of p admits a smooth invertible coframe $L(z) : F \to \mathbb{C}^n$ with*

$$L(z)K(z) = iL(z) \qquad (z \in W \cap A).$$

For the native grading application, A is the original parameter interval locus. Neither a complex equation nor a symplectic compatibility equation is required away from A.

Proof. The equation $K(p)^2 = -\mathrm{id}$ makes F a complex vector space by $(a+ib)v = av + bK(p)v$. The real dimension of a complex vector space is twice its complex dimension: a complex basis and its multiples by i give a real basis. Hence F has complex dimension n. Choose a complex basis, giving an invertible real-linear coframe $L_0 : F \to \mathbb{C}^n$ with $L_0K(p) = iL_0$. Define on V

$$L(z) = \tfrac{1}{2}\big(L_0 - iL_0K(z)\big).$$

This is smooth and $L(p) = L_0$. Invertibility of finite-dimensional linear maps is open, so $W = V \cap \{z : L(z) \text{ is invertible}\}$ is an open neighborhood of p. On $W \cap A$, use $K(z)^2 = -\operatorname{id}$ to compute

$$L(z)K(z) = \tfrac{1}{2}(L_0K(z) + iL_0) = iL(z).$$

Restricting the same smooth L to W proves every assertion. When $n = 0$, the unique linear map between zero spaces is invertible, and the same formula and identities hold. □

A.15 Local native canonical-square section germs

Lemma A.638 (A native complex extension gives a genuine local canonical bundle)**.** *Let F be a real vector space of dimension $2n$, and choose a real linear isomorphism $P : \mathbb{R}^{2n} \to F$. Let $A \subset E$ be a prescribed locus in a finite-dimensional real vector space. Suppose $p \in U \cap A$, U is open, and $K : U \to \operatorname{End}_{\mathbb{R}}(F)$ is smooth with $K(z)^2 = -\operatorname{id}$ for $z \in U \cap A$. There is an open neighborhood $W \subset U$ of p and a smooth invertible coframe $B : W \to \operatorname{Iso}_{\mathbb{R}}(F, \mathbb{C}^n)$ whose normalized complex structure*

$$L(z) = (B(z)P)^{-1}i(B(z)P)$$

is smooth and squares to $-\operatorname{id}$ throughout W, and satisfies $L(z) = P^{-1}K(z)P$ on $W \cap A$. Thus L defines its genuine canonical-square bundle on the whole open patch, without any complex-structure equation on the given K outside A.

Proof. The local native coframe existence lemma gives, after shrinking U, a smooth invertible B with $B(z)K(z) = iB(z)$ on $W \cap A$. The constant map P is invertible, so BP is smooth and invertible. Matrix inversion is smooth on the invertible matrices; hence its conjugation $L = (BP)^{-1}i(BP)$ is smooth on W. The identity $i^2 = -\operatorname{id}$ gives $L^2 = -\operatorname{id}$ there. At a point of $W \cap A$, multiplying $BK = iB$ by B^{-1} and conjugating by P gives $L = P^{-1}KP$. These identities define a complex vector bundle on the trivial real bundle over W and hence its canonical line and canonical square. □

Lemma A.639 (A smooth native canonical-square section gives full tensor coordinates)**.** *Use the patch W, identification P and corrected complex structure L of the preceding lemma. Supply an actual smooth section σ of the canonical square for L over W. Let m_L be the natural multiplication map from that canonical square to the full two-block covariant tensor space, and define a tensor on the original tangent model F by*

$$T(z) = (P^{-1})^* m_{L(z)}(\sigma(z)).$$

Then T is smooth on W, $P^*T = m_L(\sigma)$ everywhere there, and T is squared canonical for the original K on $W \cap A$. If $\sigma(p) \neq 0$, then $T(p) \neq 0$. Smoothness of the tensor image is a conclusion from smoothness of the bundle section.

Proof. Choose a local complex coframe for L. Its top exterior power is a smooth local basis of the canonical line; the square is a smooth local basis of the canonical square. In that basis, the smooth section is a smooth scalar times the squared determinant tensor. The determinant tensor is smooth, since it is a polynomial in the coframe entries. Consequently $m_L(\sigma)$ is smooth in full tensor coordinates. Composition of every tensor argument with the fixed map P^{-1} is linear and continuous, so T is smooth as well. Composing in turn with P gives $P^*T = m_L(\sigma)$, because $P^{-1}P = \operatorname{id}$ in every argument. The right side is squared canonical for L, and $L = P^{-1}KP$ on $W \cap A$, proving the asserted canonicality there. The natural multiplication map is an isomorphism between two one-dimensional complex spaces: it sends the square of a nonzero top exterior form to its nonzero two-block determinant tensor. Therefore it is injective. If $T(p) = 0$, its pullback

by P would be zero, so injectivity would give $\sigma(p) = 0$, contrary to the hypothesis. This proves nonvanishing. □

Lemma A.640 (Original local bundle section germs supply both tensor-coordinate branches)**.** *Use the original completed slices with native interior inclusions j_s and full-cylinder inclusions q_s. Supply native fields $J_s^{\mathrm{nat}}, T_s^{\mathrm{nat}}$. At each original interior point and at each original boundary–radial point of nonnegative radius, choose its original spatial chart c, a fixed real linear identification $P : \mathbb{R}^{2n} \to F$ of its coordinate model, and a smooth real complex endomorphism germ K about the corresponding parameter–coordinate point. Require $K^2 = -\mathrm{id}$ only at interval parameters in that open germ. On a smaller open neighborhood, supply an actual smooth canonical-square section σ for the corrected complex structure L of Lemma A.638, nonzero at the chosen center.*

At every valid original interval chart point, require

$$K = EJ_s^{\mathrm{nat}}E^{-1}, \qquad m_L(\sigma) = P^*((E^{-1})^*T_s^{\mathrm{nat}}),$$

where $E = dc \circ (dj_s)^{-1}$ for the interior patch. For the end patch, use $e(a, b) = (b, -a)$ on the original band $a < \delta$, and $E = dc \circ de \circ (dq_s)^{-1}$. All fields on the right are evaluated at the corresponding original included point. Then the tensor-coordinate germs required by the original interior and end grading extension predicates exist, with exactly these same source chart equations, interval complex and canonical laws, and nonzero center. The extending local section and the complex germ are supplied here; no extension existence theorem for arbitrary interval-only intrinsic bundle sections is asserted.

Proof. Fix one of the chosen source chart centers. Restrict the given smooth complex germ to the smaller open domain of σ; its complex square law still holds at every interval point. Apply the full tensor coordinate lemma to obtain $T = (P^{-1})^* m_L(\sigma)$, smooth on this same open domain, squared canonical for K on the interval locus, and nonzero at the center. The given original tensor equation implies

$$T = (P^{-1})^* P^*((E^{-1})^* T_s^{\mathrm{nat}}) = (E^{-1})^* T_s^{\mathrm{nat}}$$

at every valid original interval chart point, because $PP^{-1} = \mathrm{id}$ in every original-coordinate tangent argument. The given complex equation is unchanged by restriction. Thus K, T are precisely the required smooth source coefficient germs with all original matching identities. This argument uses the interior expression for E at interior centers and the stated cylinder expression at end centers, including radius zero. Applying it at every supplied center proves both coordinate extension predicates. □

A.16 Rectifying native bundle graph extensions

Lemma A.641 (A native bundle-valued map has a smooth full tensor image)**.** *Let B be a smooth manifold with a smooth field of complex structures L on a fixed real vector space of dimension $2n$, and equip its canonical-square bundle $\mathcal{K} \to B$ with its genuine smooth bundle atlas. If X is a smooth manifold and $G : X \to \mathrm{Tot}(\mathcal{K})$ is smooth, its full tensor image*

$$x \longmapsto m_{L(\pi G(x))}(G(x))$$

is smooth in the fixed two-block covariant tensor space. No condition on the bundle base map $\pi \circ G$ is required.

Proof. Fix $x_0 \in X$ and choose a local smooth complex coframe around $\pi G(x_0)$. It trivializes the canonical-square bundle and gives a smooth squared determinant generator $\gamma(y)$. Continuity of

the smooth base map restricts G near x_0 into this trivialization. There, write its bundle scalar coordinate as $a(x)$. Smoothness of a map into a smooth bundle says that both $a(x)$ and its base coordinate $\pi G(x)$ are smooth. The tensor image is

$$a(x)\gamma(\pi G(x)).$$

The generator is polynomial in the smooth coframe entries, hence smooth, and composition and scalar multiplication preserve smoothness. This proves smoothness near each x_0, hence on X. □

Lemma A.642 (A smooth bundle graph extension produces a genuine local section)**.** *Let W be an open native canonical-square patch, with its smooth complex structure L and genuine canonical-square bundle $\mathcal{K} \to W$. Let $A \subset W$ contain p and suppose a smooth map $G : U \to \mathrm{Tot}(\mathcal{K})$ is defined on an open neighborhood $U \subset W$ of p such that*

$$\pi G(z) = z \qquad (z \in U \cap A).$$

On $U \cap A$ let $\sigma(z)$ denote the fiber component of $G(z)$, which lies in $\mathcal{K}_z$ by this projection identity, and assume $\sigma(p) \neq 0$. The base point of $G(z)$ may differ from z outside A. Then an open neighborhood $V \subset U$ of p carries a genuine smooth section $\widetilde{\sigma}(z) \in \mathcal{K}_z$ which is nowhere zero on V and equals $\sigma(z)$ at every $z \in V \cap A$. Its full tensor image is smooth and retains every prescribed canonical-square tensor on $V \cap A$.

Proof. Apply the preceding bundle-map lemma to obtain the smooth full tensor $T(z) = m_{L(\pi G(z))}(G(z))$. On $U \cap A$ this tensor is exactly $m_{L(z)}(\sigma(z))$, so it is squared canonical for $L(z)$ there. Choose a smooth invertible complex coframe $C(z)$ for $L(z)$ near p, and let $\gamma_C(z)$ be its squared determinant generator. Evaluate $T(z)$ on the inverse-coframe standard vectors in both blocks to obtain its smooth scalar coefficient $a_C(z)$. Define

$$Q(z) = a_C(z)\gamma_C(z).$$

Both factors are smooth. The generator is squared canonical for $L(z)$, so Q is squared canonical for $L(z)$ throughout this neighborhood. The inverse-coframe reconstruction lemma gives $Q(z) = T(z)$ at each point of A there, since $T(z)$ is canonical at those points.

The canonical-square multiplication is an isomorphism onto the squared canonical tensor line. Its inverse gives a section $\widetilde{\sigma}(z)$ with image $Q(z)$, in the fiber over the original point z. In coframe coordinates its scalar is precisely the smooth $a_C(z)$, so it is a smooth native bundle section. Injectivity of multiplication and $Q = T = m_L(\sigma)$ on A imply $\widetilde{\sigma} = \sigma$ there. At p this section is nonzero, hence $a_C(p) \neq 0$. Intersect the current neighborhood with the open set $\{z : a_C(z) \neq 0\}$ to obtain V. The section is nowhere zero on V, and every prescribed locus value and full tensor value is retained. This constructs the section through correction of the smooth tensor image, without any assumption on the base projection of G outside A. □

Corollary A.643 (Graph rectification retains the original complex germ). Fix an original smooth real endomorphism germ K about $p \in A$, with $K^2 = -\mathrm{id}$ on its prescribed locus, and its produced native canonical-square patch. Let W be any smaller real open patch containing p. Supply a smooth bundle-valued map G on W whose base projection is the identity on $W \cap A$ and whose fiber value at p is nonzero. Then some open $V \subset W$ containing p carries an actual smooth nowhere-zero canonical-square section. The original endomorphism germ K is retained unchanged, and the section's full tensor image equals the full tensor image of G at every point of $V \cap A$.

Proof. Restrict the produced canonical-square patch to W. On $W \cap A$, the projection identity makes the fiber component of $G(z)$ a section in the fiber over z. Apply graph rectification to this prescribed section and to G itself. It produces V and a smooth section which is nonzero throughout V, retains all the prescribed locus values and hence retains their full tensor images. Only the patch has been restricted; the original K defining that patch has not been replaced. Thus the same original endomorphism germ, with the smaller open domain and the constructed section, gives the asserted local section data. □

Lemma A.644 (Original interior and end graph extensions produce their section germs)**.** *Use the original completed slices with interior inclusions j_s, full-cylinder inclusions q_s, and native fields J_s^{nat}, T_s^{nat}. At every original interior chart center and every original boundary–radial center of nonnegative radius, supply a smooth endomorphism germ K, with $K^2 = -\mathrm{id}$ at interval parameters, and its produced canonical-square patch with fixed tangent-model identification P. On a real open neighborhood of that center, supply a smooth map G into this genuine canonical-square bundle. Its base projection is the identity only on the original interval locus, and its fiber at the center is nonzero. For every valid original interval chart point require*

$$K = EJ_s^{\mathrm{nat}}E^{-1}, \qquad m_{L(\pi G)}(G) = P^*((E^{-1})^*T_s^{\mathrm{nat}}),$$

where $E = dc \circ (dj_s)^{-1}$ on the original interior chart c. For the end chart use $E = dc \circ de \circ (dq_s)^{-1}$ with $e(a,b) = (b,-a)$ on the original band $a < \delta$. Then the original interior and end extending section germs exist, retaining both source chart equations and every full tensor value. Each constructed section is nonzero at its prescribed center. No extending section, smooth scalar coordinate or full tensor regularity is supplied as input.

Proof. At each given interior center apply graph rectification with the original complex germ. It keeps that germ K, restricts the given open graph domain and produces an actual smooth nonzero section whose full tensor image equals that of G on the interval locus. The original complex equation is therefore unchanged, and the given tensor equation becomes the required tensor equation for this very section. These are exactly the original interior section-germ data.

At each end center, including radius zero, apply the same rectification to its boundary–radial graph domain and original complex germ. Restriction keeps the original end chart and the specified differential $dc \circ de \circ (dq_s)^{-1}$ unchanged. The retained graph tensor image consequently gives the original end section equation with every tangent slot and the radial sign intact. Performing these two constructions at every supplied center gives both families of section germs, with all interval values, including both parameter endpoints, retained. Nonvanishing at each prescribed center is retained by graph rectification. □

Lemma A.645 (Original graph extensions grade the same completed deformation)**.** *Under the original completion comparison, primitive and compatibility hypotheses of Lemma A.653, supply the original interior and end graph extensions of the preceding lemma. For prescribed endpoint Maslov representatives, assume the native endpoint complex intertwining and full tensor equations. Then the same completed pair deformation has a grading with those endpoints and a graded step whose underlying ungraded step is literally that deformation. Its interval complex structure and full squared canonical tensor retain exactly $(S_t^{-1})^*J_t^{\mathrm{nat}}$ and $(S_t^{-1})^*T_t^{\mathrm{nat}}$. Extending sections and represented coordinate or endpoint equations are conclusions derived from the original graph and native data.*

Proof. Apply the original graph-extension lemma at every interior center and every nonnegative-radius end center. It produces both families of original extending section germs with exactly the

complex and full tensor source equations. Use these two produced families, the given original primitive and compatibility identities and the native endpoint equations in the original section-germ grading lemma. That lemma constructs a grading on the exact given completed deformation with the stated prescribed interval fields and endpoint representatives. Its graded step forgets to that very deformation, so passing from graphs to sections has changed neither the geometric step nor any prescribed interval value. □

The graph-extension hypothesis is local data in the genuine bundle; it is weaker than supplying an extending section, since it places no condition on the graph's base projection away from the prescribed locus. This rectification does not itself construct a smooth graph extension from arbitrary smooth intrinsic interval sections. That correspondence for the original completed family remains a separate obligation.

Lemma A.646 (Local extensions retaining the original grading on the completed slices)**.** *Let $S_u : X_{C_u} \to K_0$ be the actual original completed-slice comparisons, under the original native comparison hypotheses. Choose an original interior point v at parameter t, with reference image $y_0 = f_0(R_t(v))$, or an original boundary–radial point (x, r) with $r \geq 0$ and reference image $y_0 = E_{K_0}(b_0(x), r)$. Let c be the respective original spatial chart at that point, with real coordinate model F of dimension $2n$. Let $J_u^{\mathrm{nat}}, T_u^{\mathrm{nat}}$ be prescribed endomorphism and covariant tensor fields on the original completed slices. Suppose that on a real open neighborhood V of the chosen source coordinate point there are smooth coefficient fields K, T satisfying, at every interval point of V,*

$$K^2 = -\operatorname{id}, \qquad T \text{ is a squared canonical tensor for } K.$$

Assume T is nonzero at the chosen point. For every interval parameter u and every valid original patch point whose chart coordinates lie in V, require the actual source coordinate equations

$$K(u, c(p)) = E_u J_u^{\mathrm{nat}}(z) E_u^{-1}, \qquad T(u, c(p)) = (E_u^{-1})^* T_u^{\mathrm{nat}}(z).$$

Here in the interior case $z = j_u(p)$ and $E_u = d_p c \circ (d_p j_u)^{-1}$. In the boundary–radial case, write $p = e(a, b) = (b, -a)$ with $a < \delta$, set $z = q_u(a, b)$, and put $E_u = d_p c \circ d_{(a,b)} e \circ (d_{(a,b)} q_u)^{-1}$. Thus these are source chart expressions of the original native fields, with the original cylinder sign and band retained.

Then a real open neighborhood U of (t, y_0) admits a jointly smooth invertible vertical coframe B, its smooth induced endomorphism $L = B^{-1} i B$, and a smooth nowhere-zero covariant tensor Q squared canonical for L throughout U. For every interval point $(u, y) \in U$, with $D = d_y(S_u^{-1})$, these fields retain the entire original grading:

$$L(u, y) = D^{-1} J_u^{\mathrm{nat}}(S_u^{-1}(y)) D, \qquad Q(u, y) = D^* T_u^{\mathrm{nat}}(S_u^{-1}(y)).$$

No source coframe or descended tensor representation is supplied. The smooth source coefficient germs and their original chart equations are the local grading inputs; no compatibility or prescribed endpoint class is asserted by this local statement.

Proof. Apply the source-coordinate coframe existence lemma to K at the chosen point, with the original interval as prescribed locus. After shrinking V, it supplies a smooth invertible source coframe A with $AK = iA$ at every interval point. Compute its scalar $a = a(A, T)$ by inverse-coframe evaluation. The reconstruction lemma makes a smooth; canonicality and nonvanishing of T at the chosen point make a nonzero there.

The original interior insertion is injective, since equality of inserted points recovers their parameter and native point. For the cylinder insertion, apply the actual parameter-preserving comparison

to two coincident inserted points. Its prescribed end formula and injectivity of the reference end map and native boundary identification recover both the radial and boundary coordinates. Thus this insertion is also injective. Define the source coefficient on the chosen insertion image to equal $a(u, c(p))$ at valid chart points with coordinates in this smaller V, and set it to zero elsewhere. Injectivity makes this single valued. Apply the original interior coframe-descent lemma in the interior case, or the original end coframe-descent lemma in the boundary–radial case. Each produces a real open reference neighborhood and a smooth original inverse germ g, retaining the literal original total inverse at every nearby interval point. It also produces the jointly smooth invertible coframe $B = A(u, c(g(u, y)))d_y(c \circ g_u)$ and the smooth tensor $Q = a(u, c(g(u, y)))\Gamma_B$, canonical and nowhere zero for $L = B^{-1}iB$. Restrict the reference neighborhood so all source coordinates remain in the smaller V. This preserves every conclusion, and in the end case every interval inverse point remains in the original band $a < \delta$.

At each interval point, the original grading-identification lemma applies to this same g, B, the source-coordinate equations, and the computed coefficient reconstruction. It gives exactly the two displayed original slice pullback equations, with the actual derivative $d_y(S_u^{-1})$ in every place and every tensor argument. This proves the local existence statement in both native patches. All regularity and nonvanishing conclusions off the interval concern the produced fields B, L, Q; they impose no additional equations on the original data there. □

Lemma A.647 (The original inverse germs have the actual completed-slice differentials)**.** *Let $S_u : X_{C_u} \to K_0$ be the actual original completed-slice comparison. Let $W \subset \mathbb{R} \times K_0$ be real open, and let g be a smooth original interior or boundary–radial inverse germ on W. For an interval point (u, y), write $w = g(u, y)$. In the interior case suppose*

$$H^{-1}(u, y) = i_{\mathrm{tot}}(u, w).$$

In the boundary–radial case suppose $w = (b, \rho)$ with $\rho > -\delta$ and

$$H^{-1}(u, y) = q_{\mathrm{cyl}}(u, -\rho, b).$$

These equations must hold at every nearby interval point of W. Let j_u be the original interior inclusion, let q_u be the original full-cylinder inclusion, and put $e(s, b) = (b, -s)$ on the original cylinder band $s < \delta$. Then

$$S_u^{-1}(y) = j_u(w) \quad \text{or} \quad S_u^{-1}(y) = q_u(p), \quad p = e^{-1}(w),$$

respectively. For an original spatial chart c containing w, set

$$E = \begin{cases} d_w c \circ (d_w j_u)^{-1}, & \text{interior}, \\ d_w c \circ d_p e \circ (d_p q_u)^{-1}, & \text{boundary–radial}. \end{cases}$$

Both coordinate changes E are invertible, in the unchanged original completion atlases, and the actual spatial derivative satisfies

$$d_y(c \circ g_u) = E \circ d_y(S_u^{-1}).$$

The asserted value and derivative agreements are required only at original interval parameters.

Proof. The actual slice comparison satisfies $\iota_u(S_u^{-1}(y)) = H^{-1}(u, y)$, where ι_u inserts the original completed slice into its total quotient. The original patch insertions satisfy $\iota_u(j_u(w)) = i_{\mathrm{tot}}(u, w)$

and $\iota_u(q_u(p)) = q_{\mathrm{cyl}}(u,p)$. The respective given inverse equation makes the two slice insertions equal. Injectivity of ι_u gives the asserted original slice point identity in each case.

Fix the interval parameter u. Real openness of W gives an open spatial neighborhood of y on which the same point identity holds. The original interior and full-cylinder inclusions are smooth local diffeomorphisms in the original completion atlases. Their differential is therefore invertible. In the interior case, differentiation of $S_u^{-1} = j_u \circ g_u$ gives

$$d_y(S_u^{-1}) = d_w j_u \circ d_y g_u.$$

Multiply by $(d_w j_u)^{-1}$ and then compose with $d_w c$. The spatial chain rule gives the displayed derivative formula. In the boundary–radial case choose a smooth local inverse l of q_u at p. On a smaller spatial neighborhood, the point identity and injectivity of q_u give $g_u = e \circ l \circ S_u^{-1}$. Differentiating $l \circ q_u = \mathrm{id}$ gives $d_{q_u(p)} l = (d_p q_u)^{-1}$. Differentiating the preceding factorization and composing with $d_w c$ yields

$$d_y(c \circ g_u) = d_w c \circ d_p e \circ (d_p q_u)^{-1} \circ d_y(S_u^{-1}).$$

The map e is the interchange of factors and reversal of the radial sign, so its differential is invertible. Chart differentials are invertible as well. Thus every factor defining either E is invertible. All differentiations use equality on an open spatial neighborhood in the original slice atlas; no equation outside the interval is needed. □

Lemma A.648 (Native coordinate descent equals the prescribed original slice pullback)**.** *Under the actual original slice and inverse-germ hypotheses of Lemma A.647, fix an interval point (t,y), put $w = g(t,y)$, and let c be the respective original spatial chart containing w. Let $z = S_t^{-1}(y)$, and let E be the respective original slice-to-coordinate differential from that lemma. Put*

$$D = d_y(S_t^{-1}), \qquad A = d_y(c \circ g_t).$$

Let $J^{\mathrm{nat}}, T^{\mathrm{nat}}$ be prescribed endomorphism and covariant tensor fields on the original completed slice X_{C_t}. Suppose an invertible coframe L in this coordinate model intertwines K with i, and the squared canonical tensor T represents the original data by the source coordinate equations

$$K = E J^{\mathrm{nat}}(z) E^{-1}, \qquad T = (E^{-1})^* T^{\mathrm{nat}}(z).$$

Then the local descended fields equal the literal original slice pullbacks:

$$(LA)^{-1} i (LA) = D^{-1} J^{\mathrm{nat}}(S_t^{-1}(y)) D,$$
$$a(L,T)\Gamma_{LA} = D^* T^{\mathrm{nat}}(S_t^{-1}(y)).$$

The original-to-coordinate equations are hypotheses alongside the stated inverse-germ and coframe data; both descended matching equations are conclusions.

Proof. The original grading-derivative lemma gives the actual original point identity and $A = ED$ in each patch. The chart, patch-inclusion and boundary–radial coordinate-change differentials are invertible, so E is invertible. The actual slice diffeomorphism has invertible differential D. Multiplying $LK = iL$ by L^{-1} gives $L^{-1} i L = K$. Therefore

$$(LA)^{-1} i (LA) = A^{-1} K A = D^{-1} E^{-1} (E J^{\mathrm{nat}}(z) E^{-1}) E D = D^{-1} J^{\mathrm{nat}}(z) D.$$

Substitute $z = S_t^{-1}(y)$ to prove the first matching equation. The inverse-coframe reconstruction lemma gives $T = a(L,T)\Gamma_L$. Pulling this full tensor back by A gives

$$a(L,T)\Gamma_{LA} = A^*T = (ED)^*((E^{-1})^*T^{\mathrm{nat}}(z)) = D^*T^{\mathrm{nat}}(z).$$

In the last equality, $E^{-1}ED = D$ in every argument of both tensor blocks. Substitute the same original point identity to obtain the second matching equation. This proves both equations in the interior and boundary–radial patches, using their respective exact coordinate changes and the same original slice differential. □

Lemma A.649 (Original compatibility passes to the same completed presentation)**.** *Fix an interval parameter t and the actual original slice comparison $S_t : X_{C_t} \to K_0$. Assume the native and boundary primitive identifications and the original completed primitive restrictions of Lemma A.508. Let Λ_t be this smooth original primitive, and let ℓ_t be the same completed deformation's primitive in its reference presentation. Then*

$$S_t^*\ell_t = \Lambda_t.$$

If J_t^{nat} is compatible with $d\Lambda_t$, and a smooth field J_t on K_0 satisfies, for $D = d_y(S_t^{-1})$,

$$J_t(y) = D^{-1}J_t^{\mathrm{nat}}(S_t^{-1}(y))D,$$

then J_t is compatible with $d\ell_t$. Smoothness of the represented field and source native compatibility are hypotheses; represented compatibility follows for the actual completed primitive.

Proof. The actual original comparison has the original-point and full-cylinder formulas of the primitive transport lemma. Applying that lemma to these same formulas gives $S_t^*\ell_t = \Lambda_t$. Pull back by S_t^{-1} to obtain $\ell_t = (S_t^{-1})^*\Lambda_t$. Naturality of exterior differentiation for the smooth primitive Λ_t yields $d\ell_t = (S_t^{-1})^*d\Lambda_t$.

The differential D is invertible because S_t is a diffeomorphism. The prescribed conjugation identity gives $DJ_t = J_t^{\mathrm{nat}}D$. Thus $J_t^2 = -\mathrm{id}$ follows from the same identity for J_t^{nat} by conjugation. For any tangent vectors v, w at y, two-form pullback and source compatibility give

$$\begin{aligned} d\ell_t(J_tv, J_tw) &= d\Lambda_t(J_t^{\mathrm{nat}}Dv, J_t^{\mathrm{nat}}Dw) = d\Lambda_t(Dv, Dw) = d\ell_t(v,w), \\ d\ell_t(v, J_tv) &= d\Lambda_t(Dv, J_t^{\mathrm{nat}}Dv). \end{aligned}$$

If $v \neq 0$, injectivity of D gives $Dv \neq 0$, so the last expression is strictly positive by source compatibility. Together with the assumed smoothness of J_t, these are all the compatibility conditions. Every form and differential belongs to the actual original slice and the same completed deformation presentation. □

Lemma A.650 (Original source coordinate germs grade the same completed deformation)**.** *Fix a positive-dimensional completed pair deformation $\mathcal{D}$ under the original native comparison hypotheses, its actual original slice comparisons $S_t : X_{C_t} \to K_0$, and its completed presentation maps $P_t : K_0 \to K_t$. Supply original slice fields $J_t^{\mathrm{nat}}, T_t^{\mathrm{nat}}$ and smooth original primitives Λ_t satisfying the native and boundary primitive identifications and original completed primitive restrictions of Lemma A.649. Assume J_t^{nat} is compatible with $d\Lambda_t$ at every interval parameter. At every original interior point, and at every original boundary–radial point of radius $r \geq 0$, assume the smooth source coordinate extensions, interval complex and canonical equations, nonzero center, and exact original source chart equations of Lemma A.646. These are explicit original-coordinate regularity hypotheses. Supply a real-parameter two-form family η on K_0 satisfying $\eta_t = d\ell_t$ for $t \in [0,1]$,*

where ℓ_t is this same deformation's completed primitive. No regularity of η away from the interval is required.

Define the prescribed interval fields on the reference completion by

$$J_t^*(y) = D^{-1} J_t^{\mathrm{nat}}(S_t^{-1}(y)) D,$$
$$T_t^*(y) = D^* T_t^{\mathrm{nat}}(S_t^{-1}(y)), \qquad D = d_y(S_t^{-1}).$$

For specified endpoint Maslov representatives (J_-, T_-) and (J_+, T_+), assume the exact endpoint equations

$$J_0^* = P_0^* J_-, \quad J_1^* = P_1^* J_+, \qquad T_0^* = P_0^* T_-, \quad T_1^* = P_1^* T_+.$$

Then this same completed deformation admits a grading whose interval complex structure and full squared canonical tensor are exactly J_t^*, T_t^* at every point, with those endpoint representatives. This conditional result assumes original coordinate extension data; the passage from every arbitrary intrinsic bundle section to those coordinate data is a separate correspondence assertion.

Proof. The reference completion is the union of its embedded interior and its nonnegative cylindrical end. Fix (t, y) in its interval product. If y is an original reference interior point $f_0(v)$, apply the interior local existence lemma at the original point $R_t^{-1}(v)$; its reference center is $f_0(R_t(R_t^{-1}(v))) = y$. Otherwise completion coverage gives $y = E_{K_0}(b_0(x), r)$ with $r \geq 0$, including the seam. Apply the end local existence lemma at (x, r). Thus every interval point has a real open neighborhood with a smooth invertible coframe B, a smooth endomorphism L, and a smooth nonzero squared canonical tensor Q, whose interval values are exactly J^*, T^*.

In particular J_t^* is smooth on each slice. The original compatibility lemma, with the actual primitive identity, makes it compatible with $d\ell_t$, the same deformation's completed symplectic form. Each local coframe intertwines this prescribed interval complex structure with i. Its determinant defines a basis of the canonical line, so Q is the tensor image of one section of its square. Evaluation on the inverse-coframe complex standard basis in both blocks computes a smooth scalar coefficient. On the interval that coefficient is exactly the coefficient of T^*, since $Q = T^*$ there. The canonical-square tensor map is injective and its image is the squared canonical tensors; hence these local descriptions specify one intrinsic squared-canonical section for J^* on the whole interval product. It is nowhere zero, because every local Q is nowhere zero. They give genuine local smooth scalar extensions for this same section.

Apply the interval grading reconstruction lemma to these local smooth endomorphism extensions, the intrinsic section and its local scalar extensions, and η, whose interval values are the actual forms $d\ell_t$. Its partition-of-unity construction takes place in the endomorphism and covariant tensor bundles. At an interval point all contributing extensions equal the same J^*, T^*, so the weighted sums retain those values exactly. No compatibility or canonicality is needed outside the interval. The resulting smooth family therefore remains compatible, canonical and nowhere zero on the interval. The four assumed endpoint equations supply precisely the two endpoint representatives through this same P_0, P_1. This is a grading of the given completed deformation, with all prescribed interval values unchanged. □

A.17 Original native endpoint representatives

Lemma A.651 (Native endpoint representatives retain the four reference pullback equations)**.** *Under the original native comparison hypotheses, fix an interval parameter s. Write $S_s : X_{C_s} \to K_0$ for the actual original reference comparison, $P_s : K_0 \to K_s$ for the specified completion presentation,*

and $F_s = P_s \circ S_s$ for their prescribed native completion comparison. Let (J_s, T_s) be a Maslov representative on K_s. Suppose the native complex field and full squared canonical tensor obey

$$dF_s \circ J_s^{\mathrm{nat}} = J_s \circ dF_s, \qquad T_s^{\mathrm{nat}} = F_s^* T_s.$$

Then their actual inverse-slice pullbacks satisfy

$$(S_s^{-1})^* J_s^{\mathrm{nat}} = P_s^* J_s, \qquad (S_s^{-1})^* T_s^{\mathrm{nat}} = P_s^* T_s.$$

In particular, applying these two statements at $s = 0$ and $s = 1$ produces the four endpoint equations required by the grading construction on this same completed deformation. The native endpoint intertwining and full tensor equations are explicit hypotheses; no equality of endpoint classes for an arbitrary geometric chain is asserted.

Proof. Fix $y \in K_0$, and put $z = S_s^{-1}(y)$, $D = d_y(S_s^{-1})$, $A = d_y P_s$, and $B = d_z F_s$. The three maps are diffeomorphisms, so all three differentials are invertible. The equation $F_s \circ S_s^{-1} = P_s$ and the chain rule give $BD = A$. Evaluate the assumed intertwining equation at z, substitute $B = AD^{-1}$, and multiply by A^{-1} on the left and D on the right. This gives

$$D^{-1} J_s^{\mathrm{nat}}(z) D = A^{-1} J_s(P_s(y)) A,$$

which is exactly the complex pullback equation. For arbitrary tangent vectors $v_1, \ldots, v_{2n}$ at y, the full tensor hypothesis gives

$$\begin{aligned} ((S_s^{-1})^* T_s^{\mathrm{nat}})_y(v_1, \ldots, v_{2n}) &= T_s^{\mathrm{nat}}(z)(Dv_1, \ldots, Dv_{2n}) \\ &= T_s(P_s(y))(BDv_1, \ldots, BDv_{2n}) \\ &= T_s(P_s(y))(Av_1, \ldots, Av_{2n}) \\ &= (P_s^* T_s)_y(v_1, \ldots, v_{2n}). \end{aligned}$$

Equality for every tangent argument proves tensor equality. These arguments apply separately at each endpoint without any normalization of P_0, proving all four endpoint equations. □

Lemma A.652 (Native endpoint representatives grade the same completed deformation)**.** *Under the source coordinate extension, original primitive and native compatibility hypotheses of Lemma A.650, prescribe endpoint representatives (J_-, T_-) and (J_+, T_+). Assume the native endpoint intertwining and full tensor equations of Lemma A.651 for these representatives at parameters zero and one respectively. Then the same completed pair deformation admits a grading with these endpoints whose interval complex field and full squared canonical tensor retain exactly $(S_t^{-1})^* J_t^{\mathrm{nat}}$ and $(S_t^{-1})^* T_t^{\mathrm{nat}}$ for every interval parameter t. The resulting graded step has this very deformation as its underlying ungraded step.*

Proof. Apply the native endpoint lemma to (J_-, T_-) at parameter zero and to (J_+, T_+) at parameter one. It yields both complex equations and both full tensor equations in the reference presentation. Use these four conclusions, together with the given source coordinate extension, primitive and compatibility data, in the same-deformation grading lemma. Its grading has exactly the required interval values and the specified endpoint representatives. Regard this grading as a graded deformation step; its underlying ungraded step is precisely the given deformation, proving the final assertion. □

Lemma A.653 (Original bundle section germs grade the literal native deformation step)**.** *Under the original completion comparison, primitive and compatibility hypotheses of Lemma A.652, supply the original interior and end bundle section germs of Lemma A.640. For the prescribed endpoint representatives, require the native endpoint intertwining and full tensor equations. Then the same completed pair deformation has a grading with those endpoints and a graded deformation step whose underlying ungraded step is literally that deformation. Its complex structure and full squared canonical tensor retain exactly* $(S_t^{-1})^* J_t^{\mathrm{nat}}$ *and* $(S_t^{-1})^* T_t^{\mathrm{nat}}$ *at every interval parameter and point. No represented coordinate-extension data, represented endpoint equation or deformation grading is supplied.*

Proof. Apply the source section-germ conversion to the supplied interior germs and to the supplied end germs. Its conclusions give both original tensor-coordinate extension predicates, including every source chart matching equation, interval complex and canonical law, and nonzero center. Apply the native endpoint grading consumer to these two derived predicates, the given original primitive and compatibility data and the native endpoint identities. It gives the required grading with all interval values unchanged and the graded step on the exact given deformation. Thus both the coordinate inputs and the reference endpoint equations used by this construction have been derived from the supplied original data. □

A.18 Junctions derived from common native endpoint tuples

Lemma A.654 (Two fixed-source native families produce their own whole-pair junction)**.** *Let* $n \geq 2$*. On fixed compact smooth native manifolds* A *and* M *of dimensions* $2n$ *and* $2n-2$*, supply two jointly smooth, locally extendible families of witnessed pair data* $(\lambda_t^L, \mu_t^L, j_t^L, b_t^L)$ *and* $(\lambda_t^R, \mu_t^R, j_t^R, b_t^R)$*, with the intrinsic Liouville, embedding, positive-witness and pullback hypotheses of Lemma A.469. Assume equality of their native ambient primitives, embeddings and witnesses:*

$$(\lambda_1^L, j_1^L, b_1^L) = (\lambda_0^R, j_0^R, b_0^R).$$

There are completed pair deformations F, G *presenting these two families, retaining at every slice their native compact primitives, boundary and whole ribbon identities, full embedded cores, coordinate extensions, jointly smooth native transports and native ribbon completion slices, and a permitted completion identification* $I : F_1 \to G_0$*. If* $e_{A,L}, e_{M,L}, e_{B,L}$ *and* $e_{A,R}, e_{M,R}, e_{B,R}$ *denote the endpoint native identifications of these very* F, G*, then* I *has scale one and satisfies*

$$\begin{aligned}
I_{\mathrm{rib}}(e_{M,L}(x)) &= e_{M,R}(x), & I_{\mathrm{rib}}^{-1}(e_{M,R}(x)) &= e_{M,L}(x),\\
I_{\partial}(e_{B,L}(y)) &= e_{B,R}(y), & s_I &= 0,\\
I_{\mathrm{amb}}(\iota_{F_1}(e_{A,L}(z))) &= \iota_{G_0}(e_{A,R}(z)),\\
I_{\mathrm{amb}}^{-1}(\iota_{G_0}(e_{A,R}(z))) &= \iota_{F_1}(e_{A,L}(z)),\\
I_{\mathrm{amb}}(E_{F_1}(e_{B,L}(y), r)) &= E_{G_0}(e_{B,R}(y), r) \qquad (r \in \mathbb{R}).
\end{aligned}$$

Its boundary map preserves the whole witness function, and its boundary–ribbon square holds on the whole compact ribbon. No junction map or completed identification is supplied as input. The larger native constructor's whole completed-family atlas and inclusion-marking capability are not asserted as output here.

Proof. Apply the original native family construction of the endpoint identification lemma separately to the two given families. This produces F, G and their native carrier, boundary, primitive, ribbon and witness identities for every slice. The two defining ribbon pullback laws give

$$\mu_1^L = (j_1^L)^*(b_1^L \lambda_1^L|_{\partial A}) = (j_0^R)^*(b_0^R \lambda_0^R|_{\partial A}) = \mu_0^R.$$

Thus equality of the ribbon primitives is derived. At the junction, use this equality and the given tuple equalities to regard the two sets of endpoint identities as presentations of the single native pair $(A, \lambda_1^L, M, \mu_1^L, j_1^L, b_1^L)$. The endpoint primitive extension, nondegeneracy and outwardness hypotheses are restrictions of the corresponding left-family hypotheses at parameter one. Thus the two ambient primitive pullbacks, two ribbon primitive pullbacks, coordinate extensions, boundary inclusion squares, full ribbon squares and whole witness squares are exactly the hypotheses of the same-native-pair completion-identification lemma. Apply that lemma to these very F_1, G_0. It constructs I with scale one and every displayed forward, inverse and all-real end equation. Its conclusions also give shift zero, preservation of the entire witness and the full boundary–ribbon square. No alteration of either constructed family or its endpoint presentation occurs. □

Lemma A.655 (The derived junction gives the literal three-step comparison chain)**.** *Under the hypotheses of the preceding lemma, the same constructed F, G, I give a permitted chain with ordered steps*

$$\mathrm{deform}(F), \quad \mathrm{identify}(I), \quad \mathrm{deform}(G),$$

length exactly three and vertex list exactly (F_0, F_1, G_0, G_1). The middle step is the whole-pair junction derived from the native endpoint tuple equality, with all its original point, witness and end equations retained. These two families and this chain need not carry a grading or connect the prescribed geometric target endpoints.

Proof. Take the very F, G, I provided by the preceding lemma. The first deformation is a step from F_0 to F_1, the derived identification is a step from F_1 to G_0, and the second deformation is a step from G_0 to G_1. Successive endpoint tuples agree literally, so appending these three steps to the empty chain at F_0 constructs a permitted comparison chain. Each append increases the length by one and appends precisely its target tuple. Consequently its length is three and its ordered vertex list is (F_0, F_1, G_0, G_1), even if some tuples coincide. The middle step was included with the exact I just constructed, so every point, witness and end identity of that identification is retained. No grading has been used or asserted by this construction. □

Lemma A.656 (Two finite-ribbon native families produce their exact junction chain)**.** *Let A be a compact smooth native surface with boundary and let M be a compact zero-dimensional native manifold, allowing $M = \varnothing$. Supply two locally smoothly extendible families of ambient primitives λ_t^L, λ_t^R with nondegenerate exterior derivatives and outward Liouville fields; two injective ribbon maps $j_t^L, j_t^R : M \to \partial A$ with local smooth real-parameter extensions; and two positive boundary witnesses b_t^L, b_t^R with local smooth real-parameter extensions. Any prescribed ribbon one-form families μ_t^L, μ_t^R may be used. Assume*

$$\lambda_1^L = \lambda_0^R, \qquad j_1^L = j_0^R, \qquad b_1^L = b_0^R.$$

Then two completed rank-one pair deformations F, G present these original families and admit a derived identification $I : F_1 \to G_0$ with scale one, shift zero, preservation of the whole witness, forward and inverse native compact ambient and full ribbon squares, and the same-radius native end law for every real radius. At every slice their native compact primitives, boundary and full ribbon identities, smooth transports and native ribbon completion slices are retained. The embedded full core is the whole embedded native finite ribbon.

The chain with these very F, I, G has length three and vertex list (F_0, F_1, G_0, G_1). No completed map, orbit-square certificate, grade or target-specific endpoint is supplied or asserted. The possibility of an empty ribbon does not establish the nonemptiness required for the initial far fiber.

Proof. Every tangent space of M is zero, so every one-form on M is zero. Consequently both prescribed ribbon primitive families are identically zero, their smoothness follows from that equality, and their Liouville fields are zero. Their boundaries are empty. Thus their Liouville and outwardness conditions hold, and their cores equal all of M.

Apply the original finite-ribbon family construction separately to the two given families. It produces F, G with the stated compact native correspondence, smooth transport and ribbon-completion slice data. At the junction the equalities of the original ambient primitive, whole embedding and witness present one common native pair. The same-native finite-ribbon completion-identification lemma therefore constructs I between the exact chosen F_1, G_0. Its conclusions give scale one, shift zero, the whole witness identity, both compact ambient and full ribbon squares, and its all-real end law. The full finite ribbon equals its core in every slice because its flow is the zero-field flow.

Append the deformation of F, this very identification I and the deformation of G to the empty chain at F_0. Their successive endpoint tuples are literally F_1 and G_0, so the result is permitted. Exactly three steps were appended and precisely their targets were appended to the vertex list; hence its length is three and its vertices are (F_0, F_1, G_0, G_1). All arguments also apply when M is empty. □

A.19 Finite assembly from supplied original junction data

Definition A.657 (A finite itinerary of deformations and original native junctions)**.** *An original native junction between fixed witnessed pairs P, Q consists of the ambient and full ribbon hypotheses of Lemma A.617, or its finite-ribbon version in Lemma A.618. Its strict native map, boundary map, end shift, compact ribbon map, positive scale and all stated equations are supplied data. An itinerary edge is a specified pair deformation, such an original native junction, or the reversal of an itinerary edge. A finite itinerary is a finite list of these edges with literal common witnessed pair tuples at successive endpoints. The empty itinerary has one endpoint tuple and no edges. Its length is the number of edges; its vertex list contains the source followed by every successive target. No existence of these deformations or native junction data is asserted by this definition.*

Lemma A.658 (Every supplied finite itinerary gives a chain with the same vertices)**.** *Every finite itinerary $\mathcal{I}$ determines a permitted comparison chain $C(\mathcal{I})$ with exactly its original endpoints, length and vertex list. Each deformation edge is retained. Each native junction edge is replaced by the identification constructed from its supplied original data, retaining the exact conjugated ambient and boundary maps, end shift, positive scale and forward and inverse compact ribbon equations of the applicable independent-endpoint lemma. A reversed edge is replaced by the reversal of that same constructed step.*

Proof. First define the comparison step associated with one itinerary edge. For a deformation retain that specified deformation. For a native junction apply the corresponding positive-dimensional or finite-ribbon independent-endpoint lemma to its supplied original data; this gives an identification between the exact same two endpoint tuples with all listed equations. For a reversed edge take the reversal of the step already constructed for its underlying edge. This recursion terminates because a reversed edge has a finite number of reversal constructors. Every resulting step therefore has exactly the original edge endpoints.

For an empty itinerary at P, take the empty comparison chain at P. It has length zero and the single vertex P, as required. Inductively, suppose the assertion holds for an itinerary from P to Q, and append an edge from Q to R. Append its constructed comparison step to the chain already obtained. The common endpoint is literally Q, so the concatenation is permitted and ends at R.

Both lengths increase by one, and both vertex lists append exactly R. The inductive length and vertex equalities consequently persist. The construction of each edge retains all its stated data, proving the remaining edgewise assertions as well. □

Lemma A.659 (A supplied original grading lift grades the same native junction)**.** *Let a supplied original native junction from P to Q have strict native map $\Psi : X \to Y$ and specified slice comparisons $S : X \to K_P$, $S' : Y \to K_Q$. Fix endpoint Maslov representatives ρ_P, ρ_Q. Suppose a supplied native homotopy grades Ψ from $S^*\rho_P$ to $\Psi^*(S')^*\rho_Q$, compatible with $S^*\omega_P$. Then the identification selected for this same junction has a graded lift from ρ_P to ρ_Q. If the supplied homotopy is (J_τ, s_τ), its constructed grading family is exactly $((S^{-1})^*J_\tau, (S^{-1})^*s_\tau)$ at every real parameter. This is conditional on the supplied native homotopy.*

Proof. The independent-endpoint construction retains the ambient equation $I_{\mathrm{amb}} = S' \circ \Psi \circ S^{-1}$ for this very selected identification I. Apply the supplied-grading conjugation lemma with that retained equation and the given native homotopy. It gives the required compatible nowhere-vanishing homotopy for I, with the exact endpoint representatives and the stated pulled family at every real parameter. It applies in both ribbon dimension cases because the grading homotopy is on the ambient and the same retained ambient equation holds in each case. □

Lemma A.660 (Supplied grading data decorate the exact assembled finite chain)**.** *Fix a finite itinerary and one Maslov representative at each of its vertices, shared literally at each junction. Supply a grading of each specified deformation between these representatives and a native homotopy as in the preceding lemma for every native junction. For a reversed edge supply the grading of its underlying edge with the two endpoint representatives exchanged. These data give a grading of exactly the chain $C(\mathcal{I})$ constructed above, with the same source and target representatives. No edge, vertex or geometric family is changed by the decoration.*

Proof. For a deformation use its supplied grading. For a native junction use the preceding lemma to grade its exact constructed identification. For a reversed edge reverse the grading of its underlying edge; its source and target representatives are exchanged, matching the reversed step. This recursion follows the finite construction of each edge. For an empty itinerary at P, take the empty graded chain at its specified representative. Inductively, append the grading of the last constructed edge to the graded chain of the preceding itinerary. Their junction representatives are the one supplied representative at that literal common vertex, so they agree. Forgetting the grading therefore appends exactly the constructed comparison step to exactly the preceding comparison chain. Induction proves that forgetting all grading data gives $C(\mathcal{I})$ itself and that its final representative is the specified target representative. □

Corollary A.661 (An explicit finite list produces its exact graded chain). Let $m \geq 0$ be an integer, and specify witnessed pairs $P_0, \ldots, P_m$. For each $0 \leq i < m$, supply an itinerary edge $e_i : P_i \to P_{i+1}$. Choose one Maslov representative ρ_i on each P_i, and supply the grading data of the preceding lemma for e_i from ρ_i to ρ_{i+1}. Then the recursively assembled comparison chain from P_0 to P_m has length exactly m, vertex list exactly $(P_0, \ldots, P_m)$, and a grading from ρ_0 to ρ_m on that very chain. This includes $m = 0$ and lists with repeated pair tuples.

Proof. For $m = 0$, form the empty itinerary at P_0 and the empty grading at ρ_0. Its length is zero and its vertex list is (P_0). Inductively form the itinerary and grading from the first m edges, then append e_m and its grading. The preceding target is literally P_m with representative ρ_m, which is also the source of this last edge and its grading. The new itinerary has length $m+1$ and vertex list $(P_0, \ldots, P_m, P_{m+1})$, since one edge and precisely its target vertex were appended. Induction gives

an itinerary and its grading for any finite supplied list. Apply the itinerary assembly lemma and the exact-chain grading lemma. Assembly preserves this length and ordered vertex list, and decoration grades exactly its assembled chain with the supplied endpoints, proving the assertion. □

A.20 Native canonical sections under spatial coordinate changes

Lemma A.662 (A native canonical section produces its coordinate graph and section)**.** *Let U be an open subset of a real normed parameter space, let A be a prescribed locus with $p \in U \cap A$, and let a native canonical-square patch on U carry a smooth complex structure J. Let σ be a smooth section of its canonical-square bundle, nonzero at p. Write $m_J(\sigma)$ for its full two-block covariant tensor. Fix a real vector space F of dimension $2n$ with a based identification with the native tangent model. Suppose $D(z) : F \to \mathbb{R}^{2n}$ is smooth and invertible for $z \in U$, and $K(z) : F \to F$ is smooth on U, with*

$$D(z)K(z) = J(z)D(z) \qquad (z \in U \cap A).$$

Then the transported full tensor $S(z) = m_{J(z)}(\sigma(z)) \circ D(z)^{\otimes 2n}$ is smooth on U. On a smaller neighborhood of p there is a genuine corrected complex patch agreeing with K on A, and a smooth bundle graph whose full tensor equals S on A. The graph yields a nowhere-zero local canonical-square section with this same full, unnormalized tensor identity. The original patch, section and coordinate differential are inputs to this local assertion.

Proof. Native section smoothness gives smoothness of $m_J(\sigma)$ by the native section-to-tensor lemma. Composition in each of the finitely many tensor slots is multilinear and continuous, so the smoothness of D gives smoothness of S. On A, apply the intertwining identity twice and use $J^2 = -1$. Injectivity of D gives $K^2 = -1$ there. The local complex-extension construction therefore supplies an open patch and a smooth invertible coframe defining a genuine complex structure that agrees with K on A.

Intertwining implies that S is a squared canonical tensor for this corrected structure at every point of A. Contract S with the inverse coframe to obtain its canonical scalar coefficient, then multiply this coefficient by the squared determinant generator of the coframe. The local coframe-reconstruction lemma makes the result a smooth canonical tensor everywhere on the patch and leaves S unchanged on A. The native canonical-square isomorphism converts it to an actual smooth bundle section, hence a smooth graph over the identity base map. At p, composition through the invertible $D(p)$ cannot annihilate a nonzero tensor: composing a hypothetical zero tensor back through $D(p)^{-1}$ would annihilate $m_{J(p)}(\sigma(p))$, contradicting injectivity of canonical-square multiplication. Thus the graph is nonzero at p. Apply graph rectification and shrink to its nonzero neighborhood. Rectification retains the graph's tensor on A. Undoing the fixed based identification in each slot gives exactly S as stated. □

Corollary A.663 (The actual spatial differential supplies canonical section transport)*.* In the preceding lemma let $U \subset \mathbb{R} \times F$ be open and let $\varphi : U \to \mathbb{R}^{2n}$ be smooth. Suppose its spatial derivative $D_{(t,y)} = d_y\varphi(t, y)$ is invertible on U and satisfies the preceding intertwining law on A. Then both the graph and the nonvanishing section are constructed with this actual spatial derivative in every tensor slot. Smoothness of D is a conclusion; no extension of φ beyond U is required.

Proof. The derivative of $y \mapsto (t, y)$ is the inclusion $v \mapsto (0, v)$. The chain rule therefore identifies $d_y\varphi(t, y)$ with $d\varphi(t, y) \circ (v \mapsto (0, v))$. On the open set U, the derivative of a smooth map is smooth. Composition with this fixed linear inclusion proves smoothness of D on U. Apply the preceding graph construction with this D and then its exact graph-to-section construction. Substitution of

the chain-rule identity gives the claimed full tensor identities. All neighborhood restrictions occur inside U. □

A.21 Finite sequences of independently presented original families

Lemma A.664 (The exact alternating itinerary of finitely many pair families)**.** *Let $m \geq 0$, let $\mathcal{D}_0, \ldots, \mathcal{D}_m$ be pair deformations, and for $i < m$ specify a permitted completion identification from the target of $\mathcal{D}_i$ to the source of $\mathcal{D}_{i+1}$. There is a comparison chain alternating these exact deformations with these exact identifications, in their supplied order. Its length is $2m+1$, and its vertex list is the concatenation of the two-element lists consisting of the source and target of each $\mathcal{D}_i$. The two tuples on opposite sides of an identification are both retained, even when they present the same intrinsic pair.*

Proof. For $m = 0$, use the one deformation arrow $\mathcal{D}_0$. Its length is one and its vertices are its source and target. Suppose the construction has been made for $\mathcal{D}_0, \ldots, \mathcal{D}_m$. Append the specified identification from the target of $\mathcal{D}_m$ to the source of $\mathcal{D}_{m+1}$, then append the deformation $\mathcal{D}_{m+1}$. Both endpoints match literally by the specified source and target tuples of these two arrows. The length increases by two and the vertex list gains exactly the source and target of $\mathcal{D}_{m+1}$. Induction proves both formulas, and the recursive construction retains each supplied arrow. □

Lemma A.665 (Finite forward comparisons constructed from original native families)**.** *Fix compact native manifolds A and M of dimensions $2n$ and $2n-2 > 0$. Let $\mathcal{F}_0, \ldots, \mathcal{F}_m$, $m \geq 0$, be original smooth witnessed-pair families on these fixed manifolds, with the local smooth extensions, outwardness and defining pullback laws of Lemma A.208. Assume that at every adjacent junction the ambient primitives, full ribbon embeddings and positive boundary witnesses agree on the native manifolds. There are completed realizations Q_i of all these families and strict identifications $I_i : Q_i(1) \to Q_{i+1}(0)$, chosen together, with scale one and boundary shift zero. Their compact ambient, boundary and full ribbon maps are the native identity expressed in the two chosen presentations, and their cylindrical maps retain every real radial coordinate. The boundary witnesses agree under these maps.*

Each chosen Q_i retains the full original ambient and ribbon completion correspondences of the cited family producer, including their total atlases and transport of supplied inclusion markings. The exact alternating chain built from these Q_i and I_i has length $2m+1$ and ordered vertices $(Q_0(0), Q_0(1), \ldots, Q_m(0), Q_m(1))$. This assertion concerns forward families on fixed native manifolds; it supplies neither grading extensions nor arbitrary changes of native bundle trivialization.

Proof. Apply the original family construction once for each index i, retaining all its ambient and ribbon completion conclusions on the resulting Q_i. These choices are made before constructing any junction. At a junction the defining equations $\mu = j^*(b\lambda|_{\partial A})$ and the three assumed matches imply equality of the two native ribbon primitives as well. Thus the endpoint presentations represent the same whole native pair, with the same ambient and ribbon primitives and witnessing function. Apply the same-native-pair completion-identification lemma to these exact two presentations. It gives the forward and inverse compact-map equations, boundary-map equation, witness equation, scale one, shift zero and all-real cylindrical equation stated above. Choose its resulting identification at every adjacent index. The preceding finite-itinerary lemma now supplies the claimed alternating path and both formulas. Nothing in this assembly changes any Q_i or discards its retained completion correspondence. Its junction proof uses the compact presentation laws; the further total-family data remain available on the same choices. □

Lemma A.666 (Finite forward native comparisons with zero-dimensional ribbons)**.** *Let A be a fixed compact native surface with boundary and M a fixed compact native zero-dimensional manifold. Supply $m+1$ original families of ambient Liouville primitives, smooth injective ribbon maps $M \to \partial A$, positive boundary witnesses and ribbon one-forms, with the original local extension and ambient outwardness conditions of Lemma A.209. Suppose the ambient primitive, whole ribbon map and witness agree at each adjacent native endpoint. Then one can choose all completed realizations once and construct strict junctions and their alternating finite chain with the same scale-one, zero-shift, compact forward and inverse, witness, all-real end, length and vertex conclusions as in Lemma A.665. The full ambient completion correspondence and the all-real ribbon total comparison, including transport of supplied markings, remain on these same choices. Every ribbon point is retained; M may be empty.*

Proof. Every tangent space of M is zero, so every ribbon one-form and every pulled-back one-form is zero. Thus no additional ribbon primitive matching, nondegeneracy, outwardness or smoothness assumption is needed. For each index apply the original zero-ribbon family construction once, retaining its full ambient and ribbon completion conclusions. At each junction, the assumed native ambient, embedding and witness equalities, together with the automatic zero ribbon-form equality, allow the same-native-zero-pair identification lemma to be applied to these exact endpoint presentations. It provides all the strict-map equations, scale and shift claimed above. Choose these identifications and apply the finite-itinerary lemma. Its induction gives length $2m+1$ and precisely the ordered source and target tuples. The construction neither changes the realizations nor removes points of their native ribbon correspondences, so it also applies to the empty zero-dimensional manifold. □

A.22 Canonical coordinates on the actual original interior

Lemma A.667 (Original interior tangent coordinates have the intrinsic chart transition)**.** *Let M be a smooth $(k+1)$-manifold with boundary and let M° be its intrinsic interior. Use the actual full native attachment X_C of a collar $C : [0,\delta)\times\partial M \to M$, with $\delta > 0$, under the smooth collar and local inverse hypotheses of the full-attachment atlas. Let $i : M^\circ \to X_C$ be its original interior inclusion. Fix a point $w \in M^\circ$ and intrinsic interior charts a, b whose domains both contain w. Write $A_w = da_w$, $B_w = db_w$ and $L_w = di_w$. All three differentials are invertible, and $A_w^{-1} = d(a^{-1})_{a(w)}$. The original ambient tangent coordinates defined by these charts are*

$$H_a(w) = A_w L_w^{-1}, \qquad H_b(w) = B_w L_w^{-1}.$$

They are invertible and their transition is the actual intrinsic chart differential:

$$H_b(w) H_a(w)^{-1} = B_w A_w^{-1} = d(b \circ a^{-1})_{a(w)}.$$

This equality is independent of the collar inclusion differential. For a family of collars it applies to each original parameter slice; it does not assert joint smoothness of a family of canonical charts.

Proof. The native-inclusion theorem gives invertibility of L_w. The charts and their inverses are smooth on their respective open domains. Differentiating both inverse identities gives invertibility of A_w, B_w and $A_w^{-1} = d(a^{-1})_{a(w)}$. Composing the invertible maps A_w, L_w^{-1} gives invertibility of $H_a(w)$ and its inverse $L_w A_w^{-1}$; the same argument applies to $H_b(w)$. Substitution now gives

$$H_b(w) H_a(w)^{-1} = B_w L_w^{-1} L_w A_w^{-1} = B_w A_w^{-1}.$$

The chain rule on the chart overlap identifies the last composition with $d(b \circ a^{-1})_{a(w)}$. Only the original spatial differentials at the same point w occur. Applying this argument separately to each collar of a parameter family proves the final slice statement. □

Lemma A.668 (The squared canonical transition uses the actual interior chart derivative)**.** *For real-linear maps $E, F : \mathbb{R}^{2n} \to \mathbb{C}^n$, let Γ_E be the full two-block tensor obtained by taking the product of the coordinate determinants of E in the two blocks. Define $\Delta(E, F)$ by evaluating Γ_E on the inverse F-images of the complex standard basis in each block. Here the inverse of a noninvertible real-linear map is defined to be zero. For every invertible real-linear map $A : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$,*

$$\Delta(EA, FA) = \Delta(E, F).$$

Under the hypotheses and notation of the preceding lemma, fix a real-linear isomorphism $P : \mathbb{R}^{2n} \to \mathbb{R}^{k+1}$ and real-linear maps $C_a, C_b : \mathbb{R}^{k+1} \to \mathbb{C}^n$. Then

$$\Delta(C_b H_b P, C_a H_a P) = \Delta\big(C_b\, d(b \circ a^{-1})_{a(w)} P, C_a P\big).$$

Thus the scalar transition of the full canonical square, in genuine invertible canonical coframes, uses the literal intrinsic spatial chart derivative. No smooth canonical-bundle atlas is asserted by this pointwise identity.

Proof. Since A is invertible, FA is invertible precisely when F is. When F is invertible, $(FA)^{-1} = A^{-1}F^{-1}$. When it is not, both sides of this inverse identity are zero by the stated convention. Also Γ_{EA} is Γ_E with A inserted in each tensor slot. Substituting $(FA)^{-1} = A^{-1}F^{-1}$ in its basis evaluation cancels A and A^{-1} in every slot, proving the first formula. Write $L = d(b \circ a^{-1})_{a(w)}$. The preceding lemma gives $H_b = LH_a$. Apply the first formula with

$$E = C_b L P, \qquad F = C_a P, \qquad A = P^{-1} H_a P.$$

This A is invertible, $EA = C_b H_b P$, and $FA = C_a H_a P$. Substitution proves the second formula and its stated interpretation. □

Lemma A.669 (The actual interior transition derivative is smooth on its open overlap)**.** *For the intrinsic interior charts a, b, the coordinate overlap*

$$O_{ab} = a(\operatorname{dom} a) \cap (a^{-1})^{-1}(\operatorname{dom} b)$$

is open. The actual transition $b \circ a^{-1}$ and its spatial derivative are smooth on O_{ab}. These maps are defined on an open real spatial domain, with no additional parameter derivative.

Proof. The chart target $a(\operatorname{dom} a)$ and the domain of b are open. The inverse chart is continuous on that target, so its preimage of the domain of b is relatively open there. Intersection with the open target proves that O_{ab} is open. The inverse chart is smooth on its target and the chart b is smooth on its domain. Their composition is therefore smooth on O_{ab}. On an open subset of a finite-dimensional real vector space, the derivative of a smooth map is smooth. Applying this to the same composition proves smoothness of its actual spatial derivative on exactly O_{ab}. □

Lemma A.670 (Original coframe extensions have smooth squared-canonical transitions)**.** *Let $g = b \circ a^{-1}$ be the actual intrinsic interior chart change, and let $P : \mathbb{R}^{2n} \to \mathbb{R}^{k+1}$ be a real-linear isomorphism. Let $U, V \subset \mathbb{R} \times \mathbb{R}^{k+1}$, with smooth real-linear-map families C_a on U and C_b on V. Suppose $U \subset \mathbb{R} \times O_{ab}$, $(t, g(y)) \in V$ for $(t, y) \in U$, and $C_a(t, y)$ is invertible throughout U. Then*

$$(t, y) \longmapsto \Delta\big(C_b(t, g(y))\, dg_y P, C_a(t, y) P\big)$$

is smooth on U. In particular, for the original canonical coframe coordinates this gives smoothness of their actual squared-canonical scalar transition in real parameter coordinates.

Proof. The preceding overlap lemma gives smoothness of g and dg on O_{ab}. Projection to y and composition therefore give smoothness of $g(y)$ and dg_y on U. The map $(t,y)\mapsto(t,g(y))$ is smooth there and takes U into V; composing it with C_b gives smoothness of $C_b(t,g(y))$. Composition of linear maps is continuous bilinear, so $E(t,y)=C_b(t,g(y))\,dg_yP$ and $F(t,y)=C_a(t,y)P$ are smooth on U. The map F is invertible there. Inversion is smooth on the open set of invertible finite-dimensional real-linear maps, so each inverse-frame vector $F(t,y)^{-1}e_j$ is smooth. The determinant generator of E is polynomial in its coefficients. Evaluating its two-block tensor on the finitely many inverse-frame vectors therefore gives a smooth scalar, exactly $\Delta(E,F)$. The pointwise canonical-transition identity identifies this scalar with the original transition when the supplied maps are the original canonical coframes. □

Lemma A.671 (The original interval-interior complex field supplies a canonical coframe cover)**.** *Let $I=[0,1]$ and use the original collar completion X_s in each slice, under the preceding collar and inverse hypotheses. Let $J_s(x)$ be the original complex endomorphism, with $J_s^2=-\mathrm{id}$, and fix $P:\mathbb{R}^{2n}\to\mathbb{R}^{k+1}$ invertible. Suppose that, in each intrinsic interior chart a, the field*

$$K_a(s,w)=H_a(s,w)J_s(i_s(w))H_a(s,w)^{-1}$$

has a smooth local extension in the real coordinates $(s,a(w))$ near every interval point, including both endpoints. Then $I\times M^\circ$ has an open cover by canonical coframe charts. Each chart has an open real coordinate domain U_a, contained in $\mathbb{R}\times a(\operatorname{dom}a)$, and a smooth invertible $C_a:U_a\to\mathrm{Iso}_{\mathbb{R}}(\mathbb{R}^{k+1},\mathbb{C}^n)$ satisfying $C_aK_a=iC_a$ at original interval points. Its base domain is

$$B_a=\{(s,w):w\in\operatorname{dom}a,\ (s,a(w))\in U_a\}.$$

The actual native coframe $F_a=C_a(s,a(w))H_a(s,w)P$ is invertible on B_a and complex-linear for $\widetilde{J}(s,w)=P^{-1}J_s(i_s(w))P$. No bundle section is supplied.

Proof. At (s_0,w_0) use the chart centered at w_0 and its local endomorphism extension. Intersect its open extension domain with $\mathbb{R}\times a(\operatorname{dom}a)$. On the interval locus the extension squares to $-\mathrm{id}$, since K_a is conjugate to J_s. The local canonical-patch construction supplies, on a smaller open neighborhood containing $(s_0,a(w_0))$, the required smooth invertible C_a with its intertwining identity on the interval locus. The parameter inclusion $I\hookrightarrow\mathbb{R}$ is continuous and the chart is continuous on its open domain; hence the displayed preimage B_a is open in $I\times M^\circ$. Its distinguished point belongs to it. Choosing a chart at each point proves coverage. The preceding differential lemma makes H_a invertible; P and C_a are also invertible. Thus F_a is invertible. Finally

$$F_a\widetilde{J}=C_aH_aJ_sP=C_aK_aH_aP=iF_a,$$

which proves complex linearity for the actual normalized original field. □

Lemma A.672 (Original line coordinates give independent tensor-square coordinates)**.** *At $x=(s,w)\in B_a$, let L_x be the actual canonical line of $\widetilde{J}(x)$. Evaluation on the inverse $F_a(x)$-images of the complex standard basis defines a complex-linear isomorphism $\ell_a:L_x\to\mathbb{C}$. Its tensor square, followed by $\mathbb{C}\otimes_{\mathbb{C}}\mathbb{C}\simeq\mathbb{C}$, defines coordinates $q_a:L_x^{\otimes2}\to\mathbb{C}$ independently of natural square multiplication. These maps define linear local product coordinates over B_a and preserve the original topology on each fiber. If $m_x:L_x^{\otimes2}\to$ full two-block tensors is natural square multiplication, then*

$$q_a(\tau)=\operatorname{coeff}_{F_a}(m_x(\tau)),\qquad m_x(\tau)=q_a(\tau)\Gamma_{F_a}.$$

Proof. The coframe is an invertible complex-linear map. Its determinant form γ_a is a basis of L_x and evaluates to one on the inverse-frame basis. Expansion of an alternating complex-linear form in that basis gives $\eta = \ell_a(\eta)\gamma_a$. Consequently ℓ_a is a complex-linear isomorphism, and tensoring it with itself gives q_a with inverse $z \mapsto z(\gamma_a \otimes \gamma_a)$. These inverses give the local product coordinates. Linear maps between finite-dimensional complex vector spaces and their inverses are continuous, so the original fiber topology is preserved. For a pure tensor $\eta \otimes \zeta$, natural square multiplication has value $\eta(v_1, \dots, v_n)\zeta(w_1, \dots, w_n)$. Inverse-frame evaluation therefore gives $\ell_a(\eta)\ell_a(\zeta)$, exactly $q_a(\eta \otimes \zeta)$. Every tensor is a finite sum of pure tensors, so linearity proves the first identity for every τ. Expanding τ in the basis $\gamma_a \otimes \gamma_a$ and applying m_x gives the second identity, since $m_x(\gamma_a \otimes \gamma_a) = \Gamma_{F_a}$. □

Lemma A.673 (The original canonical-square coordinates have smooth actual overlaps)**.** *On $B_a \cap B_b$, the independent coordinates satisfy*

$$q_b(q_a^{-1}(z)) = \Delta(F_a, F_b)z.$$

The scalar factor has a smooth extension in original real parameter–spatial chart coordinates, including the parameter endpoints. Consequently the actual fiber coordinate-change maps are smooth over $I \times M^\circ$.

Proof. The preceding reconstruction gives $m_x(q_a^{-1}(z)) = z\Gamma_{F_a}$. Taking the coefficient in F_b gives $q_b(q_a^{-1}(z)) = z\Delta(F_a, F_b)$. In the b-chart coordinates (t, y) the exact cancellation lemma identifies this factor with

$$\Delta\big(C_a(t, g(y))\, dg_y P, C_b(t, y)P\big), \qquad g = a \circ b^{-1}.$$

Restrict to the real overlap where $y \in O_{ba}$, $(t, y) \in U_b$, and $(t, g(y)) \in U_a$. There C_b is invertible and both coframe extensions are smooth. The preceding transition-smoothness lemma proves smoothness of the displayed factor on this real overlap. Its restriction to original interval points equals the actual scalar factor, by the cancellation identity. Multiplication of a scalar and a complex coordinate is continuous bilinear and smooth, so the resulting fiber coordinate change is smooth, as asserted. □

Lemma A.674 (The original interval-interior canonical square is a smooth line bundle)**.** *Under the original-complex-field extension hypotheses above, the actual fibers $L_{(s,w)}^{\otimes 2}$ form a smooth complex line bundle over $I \times M^\circ$. Its charts are the independently constructed original line-square coordinates, its overlap maps are the actual scalar transitions, and its topology restricts to the original topology on every fiber. The construction precedes any choice of section. This assertion covers the interior patch; it does not construct its extension over the entire completed family.*

Proof. Take all the local product coordinates of the preceding coframe cover. Their base domains are open and cover every original interval-interior point. The coordinate maps are linear in each fiber, mutually invertible there, and identify each original fiber topology with the standard topology of $\mathbb{C}$. Give the total space the topology induced by these local product coordinates. The overlap lemma supplies continuous and smooth linear coordinate changes; hence these product coordinates are compatible local trivializations. Their smooth overlap maps give a smooth complex vector bundle on the interval-interior base. Every fiber is complex-linearly isomorphic to $\mathbb{C}$ by the same coordinate map, so has complex dimension one. All charts were constructed from the original J, collar inclusion and its actual spatial differentials before any section was selected; the construction therefore has the stated independence from a section. □

Lemma A.675 (One original section coefficient germ transfers to every original chart)**.** *Let σ_s be a section of the actual canonical-square fibers of the original completion X_s, normalized by the fixed P. Suppose that at each $(s,w) \in I \times M^\circ$ its restriction has, in some one original canonical chart containing that point, a scalar coefficient with a smooth local extension on an open real parameter–spatial neighborhood. Then the same assertion holds in every original canonical chart containing that point. This is the explicit local extension convention; smoothness of an unrelated full tensor field or a chosen target graph is not assumed.*

Proof. Fix a point, a source chart a with its supplied scalar germ q_a, and another original chart b containing the point. In b-chart coordinates, let $g = a \circ b^{-1}$. Restrict to the open real coordinate overlap and to the preimage under $(t,y) \mapsto (t, g(y))$ of the domain of q_a. This is an open neighborhood of the point. The original scalar transition has the smooth real extension δ_{ab} on this overlap. Set

$$q_b(t,y) = \delta_{ab}(t,y) q_a(t, g(y)).$$

Composition and multiplication preserve smoothness, so q_b is a smooth real germ. At an original interval point, the exact tensor coordinate-change identity gives $q_b = q_b^{\mathrm{chart}}(\sigma)$, because q_a is the given original coefficient there. Hence this germ agrees with the same original section in chart b. Applying the argument to each point and chart proves the assertion, with no change to the section. □

Lemma A.676 (Original section coefficients reconstruct every tensor slot in actual spatial coordinates)**.** *Let σ_s be an original native canonical-square section. At an original interval-interior point $x = (s,w)$ in a canonical chart a, let $q = q_a(\sigma_s(i_s(w)))$. Write T_s for natural square multiplication of σ_s, expressed on the original native spatial tangent space. The full tensor in chart coordinates is*

$$q\Gamma_{C_a P} = T_s(i_s(w)) \circ (H_a^{-1} P)^{\otimes 2n}.$$

Every slot uses the actual original inclusion and chart differential; no independently supplied tensor is used.

Proof. The independent coordinate reconstruction gives the normalized native tensor $q\Gamma_{F_a}$, where $F_a = C_a H_a P$. Undo its normalization by inserting P^{-1} in every slot, and then insert $H_a^{-1} P$ to pass to normalized chart coordinates. In each slot the resulting coframe is

$$C_a H_a P \, P^{-1} H_a^{-1} P = C_a P.$$

Invertibility of P and of the actual H_a justifies both cancellations. The determinant-square generators therefore become $\Gamma_{C_a P}$ in every slot, giving the displayed identity for the full unnormalized tensor. □

Lemma A.677 (Original local coefficient extensions give a smooth native interior section)**.** *Under the original-complex-field extension hypotheses and the local one-chart coefficient convention above, the restriction of σ to $I \times M^\circ$ is a smooth section of the independently constructed original canonical-square line bundle. This assertion is the forward passage from local open extensions to native section smoothness.*

Proof. Fix a point and use the distinguished original canonical chart chosen in the bundle cover. The chart-independence lemma gives a smooth scalar germ q in this chart. The base coordinate map $(s,w) \mapsto (s, a(w))$ is smooth on its domain, including the parameter endpoints. Its composition with q is therefore smooth near the point. On the original interval this composition equals the

actual fiber coordinate of σ, by the germ's matching identity. In a local product trivialization the section is thus the smooth map $x \mapsto (x, q(x))$. This proves smoothness at the chosen point, and the argument applies to every point. □

Lemma A.678 (The original native section produces its actual interior grading graphs)**.** *Suppose the original complex field has the local extensions above and $\sigma_s(x) \neq 0$ at every original completion point. Suppose its interior coefficients have the local one-chart open extensions. At each original interval-interior point there is an open real parameter–spatial neighborhood carrying a corrected genuine complex structure and a smooth canonical-square bundle-valued graph, nonzero at the center, whose projection is the identity on the original interval. On that interval its complex field is exactly $K_a = H_a J_s H_a^{-1}$ and its full tensor is exactly*

$$T_s(i_s(w)) \circ (H_a^{-1} P)^{\otimes 2n}.$$

The original native section and its actual spatial differential produce every interior grading graph required by the coordinate presentation; none of these graphs or matching equations is an input.

Proof. Choose the original coframe chart centered at the given interior point. Transfer the section's local coefficient germ to that chart. Conjugation by its smooth invertible coframe supplies a smooth real endomorphism extension which squares to $-\mathrm{id}$ everywhere on its domain and agrees with K_a on the interval. Apply the local canonical-patch construction to this extension, shrinking the coefficient-germ domain if necessary. The tensor $S = q\Gamma_{C_a P}$ is smooth on this smaller patch. It is squared canonical for the corrected complex structure at original interval points. Contract S with the inverse coframe of the constructed complex patch and multiply its resulting scalar by that coframe's squared determinant generator. The local coframe-reconstruction lemma makes the corrected tensor smooth and canonical throughout the patch and leaves S unchanged on the interval. Natural square multiplication is an isomorphism onto these squared canonical tensors; its inverse gives a smooth local bundle section and hence a smooth bundle-valued graph. The coefficient at the center is nonzero: the original independent fiber coordinate is an isomorphism and the original section is nonzero. Its determinant generator is nonzero as well, so S and the resulting graph are nonzero there. The full-coordinate reconstruction lemma identifies S on the interval with exactly the displayed full native tensor in every actual spatial slot. The coframe intertwining law gives exactly the original complex-field identity. Repeating this construction at each original interval-interior point gives all the asserted graphs with their exact matching equations. □

Lemma A.679 (The constructed interior graphs yield the original native section germs)**.** *Under the hypotheses of the preceding lemma, every original interval-interior point has a genuine smooth nowhere-vanishing canonical-square section germ on an open real coordinate patch. Its complex field and full tensor retain the same original chart identities and actual spatial slots. This supplies the interior section-germ data used by completed-family grading transport; it does not assert the cylindrical end germs or a grading of the whole completed deformation.*

Proof. Use the actual original graph constructed in the preceding lemma. The graph-to-section rectification lemma gives a local section on a smaller real patch and preserves its complex field and full tensor on the original interval. The graph is nonzero at the center. Continuity makes the nonzero locus open, so shrink once more to this locus. The section is now nowhere vanishing on its whole domain. Neither restriction changes its original interval identities, proving every asserted section-germ equation. □

A.23 Canonical coordinates on the actual original cylindrical end

Lemma A.680 (Original end tangent coordinates have the intrinsic product-chart transition)**.** *Let M be a smooth $(k+1)$-manifold with boundary and let X_C be its actual full collar completion under the smooth collar and local inverse hypotheses. Write*

$$B_\delta = \{(\tau, b) : \tau < \delta,\ b \in \partial M\}, \qquad e(\tau, b) = (b, -\tau), \qquad j : B_\delta \longrightarrow X_C$$

for the original cylinder, its boundary–outward-radius coordinates, and its inclusion. Thus the outward radius satisfies $r > -\delta$; the band includes the seam $r = 0$ and the whole nonnegative end. Fix $p \in B_\delta$ and two actual product charts a, b on $\partial M \times \mathbb{R}$ whose domains contain $e(p)$. Put $A_a = da_{e(p)}$, $R_p = de_p$, and $L_p = dj_p$. These maps are invertible. The actual original end tangent coordinates are

$$H_a(p) = A_a R_p L_p^{-1}, \qquad H_b(p) = A_b R_p L_p^{-1}.$$

They are invertible, and

$$H_b(p) H_a(p)^{-1} = A_b A_a^{-1} = d(b \circ a^{-1})_{a(e(p))}.$$

The original cylinder differential and radial sign change cancel from this transition, including at the seam.

Proof. The smooth full-cylinder inclusion and its local inverse give invertibility of L_p. The coordinate map e has smooth inverse $(b, r) \mapsto (-r, b)$ on $r > -\delta$, so differentiating both inverse identities gives invertibility of R_p. The charts and their inverses are smooth on their open domains; the same argument gives invertibility of A_a, A_b and $A_a^{-1} = d(a^{-1})_{a(e(p))}$. Consequently $H_a^{-1} = L_p R_p^{-1} A_a^{-1}$, and substitution gives $H_b H_a^{-1} = A_b R_p L_p^{-1} L_p R_p^{-1} A_a^{-1} = A_b A_a^{-1}$. The chain rule identifies the last map with the derivative of the literal product-chart transition. This calculation uses only invertible spatial differentials at p, so applies also when its radial coordinate is zero. □

Lemma A.681 (The squared canonical transition uses the actual end chart derivative)**.** *Write $V = \mathbb{R}^{2n}$, $N = \mathbb{R}^{k+1}$, and $E = \mathbb{R}^k \times \mathbb{R}$. Fix real-linear isomorphisms $P : V \to N$ and $Q : V \to E$. For a real-linear map $F : V \to \mathbb{C}^n$, let Γ_F denote its full two-block determinant tensor, and let $\Delta(F, G)$ be its coefficient in G, using zero inverse for a noninvertible map. For real-linear maps $C_a, C_b : E \to \mathbb{C}^n$, the preceding end coordinates satisfy, with $g = b \circ a^{-1}$,*

$$\Delta(C_b H_b P, C_a H_a P) = \Delta\big(C_b\, dg_{a(e(p))} Q, C_a Q\big).$$

For genuine canonical coframes this is the scalar transition of the same original canonical square, expressed through the actual boundary–radius chart derivative.

Proof. The preceding lemma gives $H_b = dg\, H_a$. The common invertible real-coordinate cancellation identity applies with $F = C_b\, dg\, Q$, $G = C_a Q$, and $A = Q^{-1} H_a P$. Then $FA = C_b H_b P$ and $GA = C_a H_a P$. Cancelling A proves the formula. In canonical coframes the coefficient is exactly the independent line-square coordinate change, by determinant evaluation. □

Lemma A.682 (The actual end chart transition and its derivative are smooth)**.** *For the original product charts a, b, the real coordinate overlap*

$$O_{ab} = a(\operatorname{dom} a) \cap (a^{-1})^{-1}(\operatorname{dom} b)$$

is open. The literal transition $g = b \circ a^{-1}$ and its spatial derivative are smooth on O_{ab}. Intersecting with the open image of $r > -\delta$ gives the same conclusions on the actual original cylinder-chart overlap.

Proof. The chart target is open and its inverse is continuous there. The inverse image of the open domain of b is therefore relatively open in that target, proving openness of O_{ab}. Both chart maps are smooth on the domains used in the composition, hence g is smooth on this overlap. The derivative of a smooth map on an open finite-dimensional real domain is smooth on that domain. The band image $r > -\delta$ is open, and restriction to an open intersection preserves each of these conclusions. □

Lemma A.683 (Original end coframe extensions have smooth squared-canonical transitions)**.** *Let $g = b \circ a^{-1}$ and $Q : V \to E$ be as above. Suppose C_a and C_b are smooth real-linear-map families on real parameter–spatial domains U and W, with $U \subset \mathbb{R} \times O_{ab}$, $(t, g(y)) \in W$ for $(t, y) \in U$, and $C_a(t, y)$ invertible throughout U. Then*

$$(t, y) \longmapsto \Delta\big(C_b(t, g(y))\, dg_y Q, C_a(t, y)Q\big)$$

is smooth on U. Its restriction to the original interval locus is the actual canonical-square transition whenever these are the original canonical coframe extensions.

Proof. Smoothness of g, dg gives smoothness of their compositions with the spatial projection. The map $(t, y) \mapsto (t, g(y))$ is smooth and maps U into W, so $C_b(t, g(y))$ is smooth on U. Composition of linear maps is continuous bilinear. Thus both $C_b(t, g(y))\, dg_y Q$ and $C_a(t, y)Q$ are smooth there. The second map is invertible; its inverse and its inverse-frame basis vectors are smooth because inversion is smooth on invertible finite-dimensional real-linear maps. The determinant-square generator is polynomial in its coframe coefficients. Evaluating it on the finitely many inverse-frame vectors proves smoothness of the displayed scalar. The pointwise transition identity gives its equality with the actual original transition on the interval. □

Lemma A.684 (The original cylindrical complex field supplies a canonical coframe cover)**.** *Use the original completion X_s for every $s \in I = [0, 1]$, with its original cylinder j_s and end coordinate $H_a(s, p)$. Let $J_s^2 = -\mathrm{id}$ on X_s, and retain $P : V \to N$, $Q : V \to E$. At each original $(s, p) \in I \times B_\delta$, use the product chart a centered at $e(p)$. Suppose the end-chart field*

$$K_a(s, p) = H_a(s, p) J_s(j_s(p)) H_a(s, p)^{-1}$$

has a smooth local real extension near $(s, a(e(p)))$, including the interval endpoints and the seam. No extension outside the actual band is required. Then $I \times B_\delta$ is covered by original canonical charts. Each chart has an open real coordinate domain U_a, contained in the actual product-chart image of $r > -\delta$, and a smooth invertible $C_a : U_a \to \mathrm{Iso}_{\mathbb{R}}(E, \mathbb{C}^n)$ with $C_a K_a = iC_a$ on original interval points. Its base domain is

$$D_a = \{(s, p) : e(p) \in \operatorname{dom} a,\ (s, a(e(p))) \in U_a\}.$$

The native coframe $F_a = C_a(s, a(e(p)))H_a(s, p)P$ is invertible and complex-linear for $\widetilde{J} = P^{-1} J_s(j_s(p)) P$.

Proof. At the given point choose its original product chart and the assumed complex-field extension. Intersect its real open domain with the product-chart target and the open image of the actual band. On the original interval locus the extension squares to $-\mathrm{id}$ because it is conjugate to J_s. The local canonical-patch construction, normalized by Q, supplies a smaller open neighborhood and the smooth invertible C_a satisfying the required intertwining law on this locus. The parameter inclusion, e, and the chart on its domain are continuous, so D_a is open in $I \times B_\delta$ and contains the original point. Choosing this construction at each point proves coverage. Invertibility of C_a, H_a, P proves invertibility of F_a. Finally $F_a \widetilde{J} = C_a H_a J_s P = C_a K_a H_a P = iF_a$, which gives the asserted native complex linearity. □

Lemma A.685 (Original cylindrical line coordinates give independent tensor-square coordinates)**.** *At $x = (s, p) \in D_a$, let L_x be the actual canonical line of $\widetilde{J}(x)$. Evaluation on the inverse $F_a(x)$-images of the complex standard basis gives a complex-linear isomorphism $\ell_a : L_x \to \mathbb{C}$. Its tensor square, followed by $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C} \simeq \mathbb{C}$, gives an independent coordinate $q_a : L_x^{\otimes 2} \to \mathbb{C}$. These give linear local product coordinates preserving the original topology on each fiber. For natural square multiplication m_x into full two-block tensors,*

$$q_a(\tau) = \operatorname{coeff}_{F_a}(m_x(\tau)), \qquad m_x(\tau) = q_a(\tau)\Gamma_{F_a}.$$

Proof. The determinant form γ_a of the invertible complex coframe F_a is a basis of L_x and evaluates to one on its inverse-frame basis. Every $\eta \in L_x$ therefore equals $\ell_a(\eta)\gamma_a$. Tensoring this linear isomorphism with itself gives q_a, with inverse $z \mapsto z(\gamma_a \otimes \gamma_a)$. These mutually inverse maps give the local product coordinates. Finite-dimensional linear isomorphisms and their inverses are continuous, so each original fiber topology is preserved. For $\eta \otimes \zeta$, inverse-frame evaluation of its natural square multiplication is $\ell_a(\eta)\ell_a(\zeta)$, exactly its tensor-square coordinate. Finite sums of pure tensors span the tensor product, so linearity gives the first identity for every τ. Expanding in $\gamma_a \otimes \gamma_a$ and applying m_x gives the second identity. □

Lemma A.686 (The original cylindrical canonical-square coordinates have smooth overlaps)**.** *On $D_a \cap D_b$ the independent coordinates obey*

$$q_b(q_a^{-1}(z)) = \Delta(F_a, F_b)z.$$

The scalar factor has a smooth extension in the original real parameter–boundary–radius coordinates, including the seam and the parameter endpoints. Hence the actual fiber coordinate-change maps are smooth over $I \times B_\delta$.

Proof. Reconstruction gives $m_x(q_a^{-1}(z)) = z\Gamma_{F_a}$. Taking its coefficient in F_b gives the displayed identity. In the b-chart coordinates (t, y), write $g = a \circ b^{-1}$. The pointwise cancellation identity expresses its factor as

$$\Delta\big(C_a(t, g(y))\, dg_y Q, C_b(t, y)Q\big).$$

Restrict to the real overlap where both coframe extensions are defined and $y \in O_{ba}$. This is an open neighborhood of every original overlap point. The denominator coframe is invertible there, so the preceding smooth-transition lemma proves smoothness of this scalar extension. Its interval restriction is the original scalar by the exact cancellation formula. Scalar multiplication is smooth, proving smoothness of the fiber coordinate changes. □

Lemma A.687 (The original cylindrical canonical square is a smooth line bundle)**.** *Under the original complex-field extension hypotheses, the actual canonical-square fibers $L_{(s,p)}^{\otimes 2}$ form a smooth complex line bundle over the whole $I \times B_\delta$. Its charts are the independently constructed original line-square coordinates, and its topology restricts to the original fiber topology. It is constructed before any section is chosen. This is the actual full cylinder patch; compatibility with the interior bundle remains a separate assertion.*

Proof. The canonical chart domains are open and cover the entire base. Their fiber coordinates are linear isomorphisms preserving each original fiber topology. Endow the total space with the topology specified by these local product coordinates. The actual overlap maps are continuous and smooth by the preceding lemma, so the coordinates are compatible local trivializations. Their smooth linear overlap maps define a smooth complex vector bundle. Each fiber is isomorphic to $\mathbb{C}$, hence has complex dimension one. The construction uses only the original complex field and spatial differentials and never selects a section, giving the final assertion. □

Lemma A.688 (One original cylindrical section coefficient germ transfers to every chart)**.** *Let σ_s be a section of the actual canonical-square fibers of X_s, normalized by P. Suppose that at every original $(s,p) \in I \times B_\delta$ some original canonical chart containing that point has a scalar coefficient with a smooth local extension on an open real parameter–boundary–radius neighborhood. Then the same assertion holds in every original canonical chart containing that point, for the same section σ.*

Proof. Fix a point, its supplied coefficient germ q_a in a chart a, and any other chart b containing it. Put $g = a \circ b^{-1}$. Intersect the real chart overlap with the preimage of the germ domain under $(t,y) \mapsto (t, g(y))$. This is an open neighborhood of the point. The actual scalar transition has a smooth real extension δ_{ab} there. Define $q_b(t,y) = \delta_{ab}(t,y) q_a(t, g(y))$. Composition and multiplication give a smooth germ. On the original interval, the exact coordinate-change equation makes its value equal to the coefficient of the same σ in chart b. This proves chart independence pointwise on the whole band. □

Lemma A.689 (Original cylindrical section coefficients reconstruct every actual tensor slot)**.** *At an original cylinder point in chart a, let $q = q_a(\sigma_s(j_s(p)))$ and let T_s be natural square multiplication of σ_s on the original spatial tangent. Then its full tensor in boundary–radius coordinates is*

$$q\Gamma_{C_aQ} = T_s(j_s(p)) \circ (H_a^{-1}Q)^{\otimes 2n}.$$

Every slot uses the actual original cylinder differential and the prescribed outward-radius sign change.

Proof. The independent coordinates reconstruct the normalized native tensor as $q\Gamma_{F_a}$, where $F_a = C_a H_a P$. Undo the native normalization by inserting P^{-1} in every slot, then insert $H_a^{-1}Q$ to pass to normalized end coordinates. The coframe in each slot becomes $C_a H_a P P^{-1} H_a^{-1} Q = C_a Q$. All cancellations are justified by invertibility of P, H_a. Thus every determinant slot becomes that of Γ_{C_aQ}, proving the full tensor identity. □

Lemma A.690 (Original cylindrical coefficient extensions give a smooth native section)**.** *Under the original complex-field extension and one-chart section coefficient hypotheses, the restriction of σ to $I \times B_\delta$ is a smooth section of the independently constructed original canonical-square line bundle.*

Proof. At a base point use the distinguished original canonical chart of the bundle cover. Chart independence supplies a smooth real coefficient germ in this chart. The original base coordinate map $(s,p) \mapsto (s, a(e(p)))$ is smooth, including at parameter endpoints and at radius zero. Compose it with the coefficient germ; on the original interval this composition equals the actual fiber coordinate of σ. In the local trivialization the section is therefore $x \mapsto (x, q(x))$, a smooth map. This holds at every base point and proves the assertion. □

Lemma A.691 (The original native section produces its actual cylindrical grading graphs)**.** *Suppose $\sigma_s \neq 0$ everywhere on the original completions, and retain the original end complex-field and one-chart coefficient extension hypotheses. At every original interval–cylinder point there is an open real parameter–boundary–radius patch with a corrected genuine complex structure and a smooth canonical-square bundle-valued graph, nonzero at the center, whose projection is the identity on the original interval. There its complex field is exactly $K_a = H_a J_s H_a^{-1}$ and its full tensor is exactly*

$$T_s(j_s(p)) \circ (H_a^{-1}Q)^{\otimes 2n}.$$

In particular this constructs the cylindrical grading graphs from the same original section and its actual spatial coordinates.

Proof. Choose the original canonical chart at the point and transfer the given coefficient germ to it. Conjugation by the smooth invertible coframe gives a smooth complex-field extension squaring to $-\mathrm{id}$ throughout its domain and equaling K_a on the interval. Apply the local canonical-patch construction to that corrected field, shrinking within the coefficient-germ domain. The full tensor $S = q\Gamma_{C_a Q}$ is smooth and is squared canonical for the corrected field at original interval points. Contract it against the inverse coframe of the constructed patch, then multiply the resulting scalar by that coframe's squared generator. The coframe-reconstruction lemma gives a smooth canonical tensor throughout the patch, equal to S on the original interval. Inverse natural square multiplication gives its smooth bundle-valued section and graph. The original independent coordinate is an isomorphism, so its coefficient is nonzero at the center because σ is nonzero. The determinant generator is also nonzero, making the constructed graph nonzero there. Full-coordinate reconstruction identifies its interval tensor with the displayed expression in every slot. The coframe intertwining equation gives its exact complex-field identity. Repeating at each point constructs all claimed graphs, including radius zero and both parameter endpoints. □

Lemma A.692 (The constructed cylindrical graphs yield the original section germs)**.** *Under the preceding hypotheses, every original interval parameter and every cylindrical center of nonnegative outward radius has a genuine smooth nowhere-vanishing canonical-square section germ on an open real patch. Its complex field and full tensor retain the same original end-chart identities and actual spatial slots at every nearby full-band point, including negative outward radii. These are the full cylindrical section germs used in grading transport. Interior/end compatibility and the prescribed endpoint identities remain separate assertions.*

Proof. Use the original graph produced by the preceding lemma. Graph-to-section rectification gives a section on a smaller real patch and preserves its complex field and full tensor on the original interval. Its nonzero value at the center and continuity give an open nonzero neighborhood. Restricting to that neighborhood gives a nowhere-vanishing section on its whole patch. Restriction does not change either original matching identity, proving the assertion at every nonnegative radial center, with all full-band matching identities retained. □

A.24 Canonical coordinates across the original interior–cylinder overlap

Lemma A.693 (The original mixed spatial differential comes from the literal collar transition)**.** *Retain an original collar C, its completion X_C, interior inclusion i, full-cylinder inclusion j, and $e(\tau, b) = (b, -\tau)$. Fix a cylinder point p, an end chart b containing $e(p)$, and a smooth spatial germ F from end chart coordinates into the native interior. Put $y = b(e(p))$, $w = F(y)$, and choose an interior chart a containing w. Suppose that on a neighborhood of p the actual attachment maps satisfy*

$$i(F(b(e(q)))) = j(q).$$

The original tangent coordinates $H_I = da_w(di_w)^{-1}$ and $H_E = db_{e(p)}de_p(dj_p)^{-1}$ then satisfy, for $g = a \circ F$,

$$H_I = dg_y H_E, \qquad dg_y = H_I H_E^{-1}.$$

The end differential includes the prescribed outward-radius sign. The smooth germ and its actual attachment equality are hypotheses of this local statement.

Proof. The two smooth spatial maps $i \circ F \circ b \circ e$ and j agree near p, so the chain rule gives $di_w\, dF_y\, db_{e(p)}\, de_p = dj_p$. All original inclusion and chart-coordinate differentials in H_I and H_E are invertible. Multiply this equality on the left by $da_w(di_w)^{-1}$ and on the right by $(dj_p)^{-1}$. The

spatial chain rule identifies $da_w dF_y$ with dg_y, yielding $H_I = dg_y H_E$. Multiplication on the right by H_E^{-1} gives the second equation. The map e retains $r = -\tau$ in its actual derivative throughout this calculation. □

Lemma A.694 (The mixed canonical-square transition retains both spatial normalizations)**.** *Let $V = \mathbb{R}^{2n}$, $N = \mathbb{R}^{k+1}$ and $E = \mathbb{R}^k \times \mathbb{R}$. Retain the preceding spatial germ, coordinates and attachment equality, and independent isomorphisms $P_I : V \to N$ and $P_E : V \to E$. For arbitrary real-linear maps $C_I : N \to \mathbb{C}^n$ and $C_E : E \to \mathbb{C}^n$, put $F_I = C_I H_I P_I$ and $F_E = C_E H_E P_I$. Then*

$$\Delta(F_I, F_E) = \Delta(C_I\, dg_y\, P_E, C_E P_E).$$

For genuine canonical coframes this is the original mixed canonical-square factor. The native and end normalizations remain separate in the formula.

Proof. Put $B = P_E^{-1} H_E P_I$. This is invertible, and the mixed spatial differential identity gives $(C_I\, dg_y\, P_E)B = F_I$ and $(C_E P_E)B = F_E$. The common-coordinate cancellation lemma gives the equation, including the stipulated zero-inverse convention for noninvertible coframes. For genuine canonical coframes, independent line-square reconstruction identifies $\Delta(F_I, F_E)$ with their actual scalar coordinate change. The identity explains the occurrence of the distinct end normalization P_E on the right. □

Lemma A.695 (The original mixed canonical-square factor has a smooth real extension)**.** *Retain the original collar family, its actual total interior and full-cylinder inclusions, and its source-derived real mixed-transition germs. Fix an original point represented both by (s, w) in an interior canonical chart c and by (s, p) in an end canonical chart d. There is an open real neighborhood U of $(s, b(e(p)))$ and smooth maps $g : U \to N$, $f : U \to \mathbb{C}$ such that*

$$g(s, b(e(p))) = a(w), \qquad U \subset U_d, \qquad (t, g(t, y)) \in U_c \quad ((t, y) \in U),$$

where a, b are the underlying spatial charts and U_c, U_d their coframe-extension domains. At every nearby original interval point (t, q) in the spatial end-chart domain whose coordinate lies in U, there is an original interior point v in the spatial chart a with

$$a(v) = g(t, b(e(q))), \qquad i_{\mathrm{tot}}(t, v) = j_{\mathrm{tot}}(t, q),$$

and

$$f(t, b(e(q))) = \Delta(F_c(t, v), F_d(t, q)).$$

Here F_c, F_d are the same original normalized native coframes. Thus both the actual spatial transition and its canonical-square factor have real smooth extensions, including parameter endpoints.

Proof. The actual quotient overlap criterion makes the inward time of p strictly positive. Choose its source-derived real mixed-transition germ A and retain its parameter preservation in both directions. In the stated chart coordinates put

$$g(t, y) = a(\pi_M A(t, e^{-1}(b^{-1}(y)))),$$

where π_M denotes spatial projection. Intersect the open source of this expression with U_d and the inverse image of U_c under $(t, y) \mapsto (t, g(t, y))$. The resulting open set contains the center: the actual quotient identity and injectivity of the interior inclusion make $A(s, p) = (s, w)$. It has the asserted

domain inclusions and center value, and g is smooth there. Its spatial derivative $d_y g$ is the total derivative restricted to the spatial summand, hence is smooth. Define

$$f(t,y) = \Delta\big(C_c(t,g(t,y))\, d_y g(t,y) P_E, C_d(t,y) P_E\big).$$

The denominator coframe is invertible throughout its domain. Smooth finite-dimensional inversion, the polynomial determinant and finite inverse-frame evaluation therefore make f smooth. At an original interval point the exact mixed identity of A produces v with both stated equations. Restricting to that fixed parameter gives a smooth spatial germ with actual attachment agreement. The preceding mixed differential and cancellation lemmas identify f with $\Delta(F_c, F_d)$ there. This proves all matching equations on the same open real neighborhood. □

Lemma A.696 (The inverse mixed canonical-square factor has a smooth real extension)**.** *Retain the original family and its real mixed germs. Fix the same original mixed point $(s,w),(s,p)$ in interior and end canonical charts c,d. There is an open real neighborhood V of $(s,a(w))$ and smooth maps $h : V \to E$, $f_{EI} : V \to \mathbb{C}$ with*

$$h(s,a(w)) = b(e(p)), \qquad V \subset U_c, \qquad (t,h(t,x)) \in U_d \quad ((t,x) \in V).$$

At every nearby original interval point (t,v) in the spatial interior-chart domain with coordinate in V, there is an original cylinder point q in the spatial end-chart domain with

$$b(e(q)) = h(t,a(v)), \qquad j_{\mathrm{tot}}(t,q) = i_{\mathrm{tot}}(t,v),$$

and

$$f_{EI}(t,a(v)) = \Delta(F_d(t,q), F_c(t,v)).$$

Its scalar uses the interior coordinate normalization P_I; its spatial map retains the actual outward-radius sign.

Proof. Use the actual real mixed germ A at the point, and its inverse which preserves parameter. Compose its spatial component with the native end coordinates e and the end chart b, and compose its input with the inverse interior chart a^{-1}. Restrict to the open set where this inverse lies in the positive collar and both coframe domains are defined. This contains the center by the exact original overlap identity and injectivity of the cylinder patch. The resulting h is smooth, with the stated center value and domain inclusions. At interval points, the inverse mixed identity gives q and both original equations. Locally this inverse is an actual smooth map into the open collar; any extension used outside that collar is irrelevant to this germ. At a fixed interval parameter its actual spatial equality is $j \circ F \circ a = i$ in the indicated chart coordinates. Differentiating this equality and cancelling the invertible original inclusions gives $H_E = d_x h\, H_I$. The common-coordinate cancellation identity, using the invertible map $P_I^{-1} H_I P_I$, identifies the reverse native factor with

$$\Delta\big(C_d(t,h(t,x))\, d_x h(t,x) P_I, C_c(t,x) P_I\big).$$

Define f_{EI} by this expression on V. The spatial derivative of the smooth h is smooth; the denominator coframe is invertible. Smooth inversion and determinant evaluation therefore make f_{EI} smooth and give precisely the claimed interval scalar identity. The definition of e retains the radial sign in the actual spatial derivative, proving the final assertion. □

Lemma A.697 (The same original section grades the retained completed deformation)**.** *Retain the original completed slices X_s, their actual reference comparisons $A_s : X_s \to K_0$, and the same*

completed pair deformation $\mathcal{Q}$. Write $p_s : K_0 \to K_s$ for its completed spatial presentation and $S_s = p_s \circ A_s : X_s \to K_s$ for its actual completed-slice comparison. Work under the completion-comparison, primitive and compatibility hypotheses of the original section-germ grading lemma. In particular the original smooth completed primitives Λ_s restrict to the given compact primitive and to its full exponential cylinder primitive, J_s is compatible with $d\Lambda_s$, and the represented spatial exterior forms have the stipulated real parameter extension. Supply one original canonical-square section σ_s which is nowhere zero on each X_s. Assume the original complex field has the local real extensions in both interior and end charts, and that the same σ has local one-chart coefficient extensions on both patches as above. Let T_s be its natural full tensor image, with the fixed native normalization undone. For the prescribed endpoint representatives (J^0, T^0) and (J^1, T^1) require, for $s = 0, 1$,

$$dS_s\, J_s = J^s\, dS_s, \qquad T_s = S_s^* T^s.$$

Then this very deformation $\mathcal{Q}$ has a grading with these endpoint representatives and a graded step whose underlying step is literally $\mathcal{Q}$. At every original interval parameter, its complex structure and full tensor are exactly $(A_s^{-1})^* J_s$ and $(A_s^{-1})^* T_s$ on the fixed reference completion K_0.

Proof. Compatibility implies $J_s^2 = -\mathrm{id}$, so the complex-square hypotheses of both extraction lemmas hold. Apply the interior section-extraction lemma to the given original J, σ. It produces actual interior section germs with the full tensor T_s in every original spatial slot. Apply the end section-extraction lemma to the same J, σ and the same native normalization. It produces actual cylindrical section germs with that same T_s, including at radius zero. Their original fields and tensors are therefore identical on every point where both branches apply; no separate tensor family is selected. The original section-germ grading lemma now applies to these two derived germ families, the retained completion and primitive data, compatibility, and the stated native endpoint equations through $S_s = p_s \circ A_s$. Transporting these equations through A_s^{-1} identifies the reference endpoint fields with $p_s^* J^s$ and $p_s^* T^s$ on K_0. The constructor gives the grading on the same supplied deformation, retains precisely both pullback fields through A_s^{-1} at every interval point, and gives its literal graded deformation step. This proves all the assertions. □

A.25 One local coefficient condition on the original total family

Lemma A.698 (Original section coefficient germs transfer across the mixed overlap)**.** *Retain the original total family, one original complex field J, and one section σ of its actual canonical-square fibers. Suppose $J^2 = -\mathrm{id}$ and its original interior and full-cylinder coordinate fields have the real local extensions used in the two canonical coframe covers. At an original mixed overlap point, a smooth real coefficient germ of σ in an interior canonical chart gives one in every end canonical chart containing that point. Conversely an end germ gives one in every interior canonical chart containing the point. The section and every original tensor slot are preserved.*

Proof. Injectivity of the original slice inclusion identifies the two representatives with the same point of X_s, so their coordinates evaluate the same original section. Use the actual real mixed-transition germ A at the point and choose the indicated interior and end charts. Shrink its real coordinate domain to where the supplied coefficient germ and both coframe extensions are defined. Write $x = g(t, y)$ for its spatial transition, and let $\delta(t, y)$ be the smooth scalar extension of the preceding lemma. If q_I is the supplied interior germ, set $q_E(t, y) = \delta(t, y) q_I(t, g(t, y))$. This is smooth and agrees with the coefficient of the same original σ on the interval, because independent line-square reconstruction writes the native tensor as $q_I \Gamma_{F_c}$, whose end coordinate is $\Delta(F_c, F_d) q_I$. For the reverse direction, use the smooth inverse coordinate germ h and its direct reverse scalar

germ δ_{EI} constructed in the inverse-transition lemma. Put

$$q_I(t,x) = \delta_{EI}(t,x)q_E(t,h(t,x)).$$

Composition and multiplication prove smoothness, and the exact reverse coefficient equation proves interval matching. Within either patch, the corresponding chart-independence lemma transfers the resulting germ to every other chart containing the point. Each matching identity is an identity of coefficients of the original section; independent coordinate reconstruction therefore preserves its full native tensor. □

Definition A.699 (An original total local canonical-section cover)**.** *For one section σ of the actual original canonical-square fibers, a total local coefficient cover means the following assertion. At every point of the same original total quotient family, there is either an original interior point and an interior canonical chart, or an original full-cylinder point and an end canonical chart, mapping to that quotient point, such that the point lies in the chart's base domain and the coefficient of σ has a smooth local extension in real parameter–spatial coordinates. The choice of patch may vary with the quotient point. This condition supplies neither a grading on a represented deformation nor a section on a different bundle.*

Lemma A.700 (The original total coefficient cover is equivalent to both patch conditions)**.** *Under the original complex-field and real-extension hypotheses of the mixed transfer lemma, an original section has a total local coefficient cover if and only if it has the local one-chart coefficient extension property at every original interior point and at every original full-cylinder point. Both properties refer to the same section on the same original completed family.*

Proof. Suppose both patch conditions hold. The original interior and full-cylinder images cover the quotient. At each quotient point choose either preimage from this cover and apply its corresponding coefficient condition. This supplies the total condition. Conversely, the original complex-field hypotheses give a canonical chart at every point of either patch. Fix an original interior point and apply the total condition to its quotient image. If the chosen preimage is interior, injectivity of the interior patch makes it the same point, so its germ supplies the interior property. If it is cylindrical, the original quotient overlap criterion identifies it with this interior point at positive inward collar time. The mixed transfer lemma therefore supplies an interior germ. Now fix a full-cylinder point. If the total condition chooses a cylindrical preimage, injectivity makes it the same point. If it chooses an interior preimage, the same overlap criterion gives positive inward time, and mixed transfer supplies an end germ. These arguments apply at every original point, including parameter endpoints. A cylinder point at inward time zero or negative inward time cannot have an interior preimage, so the argument also includes the seam and the entire outward end. □

Lemma A.701 (One original total section cover grades the retained deformation)**.** *Retain the original completion-comparison, primitive, compatibility, complex-field extension and native endpoint hypotheses of Lemma A.697. Instead of its two section-coefficient hypotheses, supply one nowhere-vanishing original canonical-square section with a total local coefficient cover. Then the same retained completed deformation has the grading and literal graded step of that lemma, with precisely its original interval complex-field and full-tensor pullbacks and its prescribed endpoint representatives.*

Proof. Compatibility makes the original complex field square to $-\mathrm{id}$. The total-cover equivalence therefore produces both original patch coefficient conditions for this very section. Apply the same-original- section grading lemma to these derived conditions and the retained completion-comparison,

primitive, compatibility and endpoint data. Its output grades that same deformation and has exactly its asserted interval pullbacks and endpoint representatives. No original section, completed slice or comparison map is changed in this application. □

A.26 Endpoint representatives from the original total section

Lemma A.702 (The original total section constructs all slice representatives and grades the same deformation)**.** *Fix the completed pair deformation $\mathcal{Q}$, original completed slices X_s, smooth primitives Λ_s, and actual comparisons $A_s : X_s \to K_0$ of Lemma A.701. Retain its compact and cylinder primitive equations, collar and completion-comparison hypotheses, original compatible complex field J_s, and real local complex-field extensions on both patches. Supply one nowhere-vanishing original canonical-square section σ with the total local coefficient cover. Write T_s for its natural full tensor, with the original native normalization undone, and set*

$$J_s^{\mathrm{ref}} = (A_s^{-1})^* J_s, \qquad T_s^{\mathrm{ref}} = (A_s^{-1})^* T_s.$$

Let $p_s : K_0 \to K_s$ be the retained completed spatial presentation and $S_s = p_s \circ A_s$. There exist Maslov representatives $\rho_s = (\mathcal{J}_s, \mathcal{T}_s)$ on every K_s, for $s \in [0,1]$, such that

$$dS_s\, J_s = \mathcal{J}_s\, dS_s, \qquad T_s = S_s^* \mathcal{T}_s,$$
$$J_s^{\mathrm{ref}} = p_s^* \mathcal{J}_s, \qquad T_s^{\mathrm{ref}} = p_s^* \mathcal{T}_s.$$

The same $\mathcal{Q}$ has a grading from ρ_0 to ρ_1 and a graded step with underlying deformation literally $\mathcal{Q}$. Its interval fields on K_0 are exactly $J_s^{\mathrm{ref}}, T_s^{\mathrm{ref}}$. These representatives are constructed from the supplied original section; no prescribed endpoint class or comparison chain is asserted.

Proof. Compatibility gives $J_s^2 = -\mathrm{id}$. The total-cover equivalence supplies both patch coefficient conditions for the same σ. The interior and end extraction lemmas give its original section germs, hence local smooth extensions of the reference complex field and full tensor. The compatibility lemma identifies the reference complex field as compatible with the exterior differential of the represented completed primitive. Apply the grading-family reconstruction lemma to these extensions and the nowhere-zero section. This gives one smooth family f on K_0 whose interval values are exactly $J_s^{\mathrm{ref}}, T_s^{\mathrm{ref}}$.

Restrict f to s. The restriction is smooth and its complex field is compatible, while its tensor is canonical and nowhere zero, because these conditions hold at every interval parameter. Define ρ_s by pulling this restriction back through p_s^{-1}. Exterior derivative commutes with pullback, so the primitive presentation equation identifies the resulting two-form with the actual completed symplectic form on K_s. Thus ρ_s is a Maslov representative on that actual slice. Inverse pullback cancellation gives both displayed reference equations.

At $x \in X_s$, the differentials of A_s and A_s^{-1} are inverse. The equation $J_s^{\mathrm{ref}} = (A_s^{-1})^* J_s$ therefore gives $dA_s J_s = J_s^{\mathrm{ref}} dA_s$. The reference equation gives $dp_s J_s^{\mathrm{ref}} = \mathcal{J}_s dp_s$. Composing yields $dS_s J_s = \mathcal{J}_s dS_s$. Similarly $T_s^{\mathrm{ref}} = (A_s^{-1})^* T_s$ implies $T_s = A_s^* T_s^{\mathrm{ref}}$ by inserting the inverse differentials in every tensor slot. Substitution of $T_s^{\mathrm{ref}} = p_s^* \mathcal{T}_s$ gives $T_s = S_s^* \mathcal{T}_s$. These identities hold for every s, so in particular furnish the four required equations at 0 and 1. The original-total-section grading lemma, applied to these constructed representatives and derived equations, grades this very $\mathcal{Q}$ with the stated reference interval values. Its corresponding graded step retains the same deformation and both constructed endpoints. □

A.27 Original endpoint homotopies on the constructed junction

Lemma A.703 (The original primitive equations determine the exact symplectic pullbacks)**.** *Let an original native ambient junction between witnessed pairs P, Q have smooth completed ambients X, Y, original primitives α, β, slice comparisons $S : X \to K_P$, $S' : Y \to K_Q$, and original strict map $\Psi : X \to Y$, as in Definition A.657. Write λ_P, λ_Q for the actual completed primitives and $\omega_P = d\lambda_P$, $\omega_Q = d\lambda_Q$. The retained primitive equations imply*

$$d\alpha = S^*\omega_P, \qquad d\beta = (S')^*\omega_Q, \qquad d\alpha = \Psi^* d\beta.$$

In particular the original strict map is symplectic for these actual original two-forms, without a separate symplectic pullback hypothesis.

Proof. The source and target comparison equations are $\alpha = S^*\lambda_P$ and $\beta = (S')^*\lambda_Q$. All completed primitives and comparison maps are smooth. Exterior derivative commutes with smooth pullback, so differentiating these two equations gives the first two identities. The target comparison also proves smoothness of β. The original strict equation $\alpha = \Psi^*\beta$ can therefore be differentiated, giving $d\alpha = \Psi^* d\beta$. The original two-forms are nondegenerate because they are pullbacks of the completed symplectic forms by diffeomorphisms. Together with the last identity, this proves that Ψ is symplectic for them. □

Lemma A.704 (Original differential and tensor equations identify the endpoint representatives)**.** *Retain the ambient junction and notation of the preceding lemma. Let $\widehat{\rho}_X = (J_X, T_X)$ and $\widehat{\rho}_Y = (J_Y, T_Y)$ be original Maslov representatives for $d\alpha, d\beta$, and let $\rho_P = (J_P, T_P)$ and $\rho_Q = (J_Q, T_Q)$ be specified representatives on K_P, K_Q. Suppose the original endpoint equations are*

$$\begin{aligned} dS\, J_X &= J_P\, dS, & T_X &= S^*T_P, \\ dS'\, J_Y &= J_Q\, dS', & T_Y &= (S')^*T_Q, \end{aligned}$$

where each tensor equation holds in every full tangent slot. Then

$$\widehat{\rho}_X = S^*\rho_P, \qquad \widehat{\rho}_Y = (S')^*\rho_Q.$$

The two-form identifications required by these representative pullbacks are precisely those derived from the primitive equations above.

Proof. At every $x \in X$, the differential dS_x is invertible. Its stated intertwinement equation therefore gives $J_X(x) = (dS_x)^{-1} J_P(Sx) dS_x$, the complex field of $S^*\rho_P$. The full tensor equation identifies the tensor of $\widehat{\rho}_X$ with the tensor of $S^*\rho_P$. A Maslov representative consists of these two fields with their compatibility, smoothness, canonicality and nonvanishing properties; equality of the fields identifies the representatives. Their indexing two-forms agree by $d\alpha = S^*\omega_P$. For $y \in Y$ the same argument uses the invertible differential dS'_y to obtain $J_Y(y) = (dS'_y)^{-1} J_Q(S'y) dS'_y$. The target tensor equation and $d\beta = (S')^*\omega_Q$ then give the second representative equality, including its full tensor. □

Lemma A.705 (The original homotopy grades the same constructed completion identification)**.** *Retain the original ambient junction and endpoint representatives of the preceding lemma, with its four original comparison equations. Supply either positive-dimensional or finite original whole-ribbon junction data of Definition A.657. Let $I : P \to Q$ be the permitted identification constructed from these very ambient and ribbon data, so $I_{\mathrm{amb}} = S' \circ \Psi \circ S^{-1}$. Suppose H is a smooth original Maslov homotopy from $\widehat{\rho}_X$ to $\Psi^*\widehat{\rho}_Y$, compatible with $d\alpha$. This pullback uses the strict symplectic*

identity derived above. Then the same I has a graded lift from ρ_P to ρ_Q. If J^H_τ, T^H_τ are the real-indexed fields of H, the constructed grading has precisely

$$J^I_\tau = (S^{-1})^* J^H_\tau, \qquad T^I_\tau = (S^{-1})^* T^H_\tau \quad (\tau \in \mathbb{R}).$$

The homotopy and original endpoint comparison equations are supplied data; existence of a prescribed target grading is not concluded.

Proof. The preceding representative lemma identifies the original endpoints as $\widehat{\rho}_X = S^*\rho_P$ and $\widehat{\rho}_Y = (S')^*\rho_Q$. Thus the original source equations of H become the complex-field and full-tensor equations for $S^*\rho_P$. Its target equations become those for $\Psi^*(S')^*\rho_Q$. The primitive-derived identity $d\alpha = S^*\omega_P$ identifies its compatibility form with the one required by the native supplied-grading lemma. Retain the actual real-indexed fields of H throughout these identifications. This gives that lemma exactly its native homotopy, including all four endpoint equations, without supplying a separately constructed grading of I.

Apply the native supplied-grading lemma to the same original ambient and whole-ribbon junction. It grades the exact identification I constructed from those data, by pulling the given family through S^{-1}. The pullback is defined by derivative conjugation for the complex field and by inserting the derivative in every tensor slot. It consequently has the two displayed formulas for every real parameter. Compatibility, canonicality, nonvanishing and endpoint extension hold on the interval as in that lemma, and its source and target representatives are exactly ρ_P, ρ_Q. Neither the geometric identification nor its ribbon data has been changed. □

A.28 Section capability of one chosen original bundle realization

Lemma A.706 (One original bundle realization grades every qualifying section on its chosen completion)**.** *Let $n > 1$ and supply the original compact ambient and ribbon bundle data and local geometric laws of Lemma A.614. There exist compatible global compact charts and one completed pair deformation $\mathcal{Q}$ with all of that lemma's original recovery, whole-ribbon, slice-comparison, endpoint-identification and marking conclusions. Write C_s for its chosen original collar, X_s for its original completed slices, Λ_s for their retained primitives, $S_s : X_s \to K_s$ for its actual completed comparisons, and $p_s : K_0 \to K_s$ for the retained completed presentation. These choices can be made before the section data below.*

For every original complex field J_s on these chosen X_s, with any fixed native tangent normalizations, and every original canonical-square section σ on this chosen completed family, assume J_s is compatible with $d\Lambda_s$, has the original real local extensions on both patches, and σ is nowhere zero and has the total local coefficient cover. Then there are Maslov representatives $\rho_s = (\mathcal{J}_s, \mathcal{T}_s)$ on the actual K_s such that, for every $s \in [0,1]$,

$$dS_s\, J_s = \mathcal{J}_s\, dS_s, \qquad T_s = S_s^* \mathcal{T}_s,$$

where T_s is the natural full tensor of the same σ. The very same $\mathcal{Q}$ has a grading from ρ_0 to ρ_1 and a graded step retaining that deformation. Its interval complex field and tensor on K_0 are precisely $p_s^*\mathcal{J}_s$ and $p_s^*\mathcal{T}_s$. The quantifier order chooses the completed family before admitting sections on its chosen collar; no correspondence with a section independently prescribed on an untrivialized completed bundle is asserted.

Proof. Choose the compatible compact charts and transport the covariant original primitive, witness and full ribbon data as in the original-bundle endpoint-and-marking lemma. Its original local geometric laws imply the required global nondegeneracy, outwardness, embedding and primitive

equations in those charts. Apply the native completed-family construction once to these data. Retain its $\mathcal{Q}, C_s, X_s, \Lambda_s$, total comparison, and actual maps S_s, together with its complete original endpoint and marking conclusions. No section was used in making these choices.

Now fix any of the stated J, σ on this chosen family. All the compact primitive, collar, completion and boundary equations needed by the original-total-section endpoint lemma are among the retained construction equations. The stipulated compatibility and local extension conditions give that lemma its remaining hypotheses. Apply it with this very $\mathcal{Q}$. It constructs all slice representatives and their native comparison equations, and grades this same deformation. Its completed-slice comparison is the canonical map obtained by composing the retained total slice map with p_s. The retained formula for S_s is exactly this same composite at every point. Thus these two smooth maps, and hence their differentials, agree. The lemma's complex intertwinement and full-tensor restriction equations are consequently the displayed equations through the actual S_s. Its reference equations identify the interval grading fields as $p_s^*\mathcal{J}_s, p_s^*\mathcal{T}_s$. This argument applies to every qualifying J, σ after the same geometric choices. The original endpoint and marking conclusions were retained before this application and are unchanged, which proves the asserted order of quantifiers and all conclusions together. □

A.29 The three-step chain from original sections and homotopy

Lemma A.707 (Original sections and one original homotopy grade the literal three-step chain)**.** *Supply two completed pair deformations $\mathcal{F}, \mathcal{G}$, each with the original completion, primitive, compatibility and total-section data of Lemma A.702. Write Z_F, Z_G for their original completed slices at parameters $1, 0$, respectively, and $S_F : Z_F \to K_{F_1}$, $S_G : Z_G \to K_{G_0}$ for their retained actual comparisons. Supply an original whole-ribbon native junction from F_1 to G_0, with completed ambients X, Y, comparisons $S : X \to K_{F_1}$, $S' : Y \to K_{G_0}$, original strict map $\Psi : X \to Y$ and constructed identification I. Let $\widehat{\rho}_X = (J_X, T_X)$ and $\widehat{\rho}_Y = (J_Y, T_Y)$ be original Maslov representatives, and supply an original homotopy H from $\widehat{\rho}_X$ to $\Psi^*\widehat{\rho}_Y$, compatible with the original source form. Suppose original smooth diffeomorphisms $U : Z_F \to X$ and $U' : Z_G \to Y$ satisfy*

$$S \circ U = S_F, \qquad S' \circ U' = S_G,$$

and compare the original section fields by

$$dU\, J_1^F = J_X\, dU, \qquad T_1^F = U^* T_X,$$
$$dU'\, J_0^G = J_Y\, dU', \qquad T_0^G = (U')^* T_Y.$$

Then there are slice representatives ρ_s^F, ρ_s^G, gradings g_F, g_G of these very two deformations, and a lift g_I of this very I, such that the exact chain

$$F_0 \xrightarrow{\mathcal{F}} F_1 \xrightarrow{I} G_0 \xrightarrow{\mathcal{G}} G_1$$

is graded by their ordered concatenation. The two deformation gradings retain their original reference complex fields and full tensors on $[0, 1]$. For every real τ, the middle lift retains $(S^{-1})^* J_\tau^H, (S^{-1})^* T_\tau^H$. The geometric chain has length three and literal vertex list $[F_0, F_1, G_0, G_1]$. This is conditional on the original realizations, comparisons and homotopy just supplied.

Proof. Apply the original-total-section endpoint lemma to $\mathcal{F}$ and its section. It produces ρ_s^F, g_F with the original reference interval values and the native endpoint equations through S_F. Apply it to $\mathcal{G}$ and its section to obtain ρ_s^G, g_G and the corresponding equations through S_G.

For either of the two original endpoint comparisons, write its maps as $U : Z \to X$ and $S : X \to K$, its original complex fields as J_Z, J_X, and the produced endpoint field as J_K. The original and produced equations are $dU J_Z = J_X dU$ and $d(S \circ U) J_Z = J_K d(S \circ U)$. Fix $y \in X$ and $v \in T_y X$. Set $z = U^{-1} y$ and $w = (dU_z)^{-1} v$. The chain rule in the second equation gives

$$dS_y \, dU_z \, J_Z w = J_K(Sy) \, dS_y \, dU_z w.$$

The first equation and $dU_z w = v$ give $dS_y J_X v = J_K(Sy) dS_y v$. For an arbitrary list of full tangent arguments v_i at y, insert $w_i = (dU_z)^{-1} v_i$ in both tensor identities $T_Z = U^* T_X = (S \circ U)^* T_K$. This gives $T_X(y)(v_i) = T_K(Sy)(dS_y v_i)$, hence $T_X = S^* T_K$. Use this calculation first for Z_F, U, S and then for Z_G, U', S'. It produces all four original-to-represented endpoint equations needed by the original-homotopy lift lemma, for the actual ρ_1^F, ρ_0^G constructed above.

That lemma gives g_I on the same constructed I, with the claimed all-real pulled family. Concatenate g_F, g_I, g_G in that order, starting with the empty graded chain at ρ_0^F. The first grading ends at ρ_1^F, the literal source of g_I, and g_I ends at ρ_0^G, the literal source of g_G. Thus the concatenation is a grading of precisely the displayed geometric chain, and equals the ordered concatenation of these three gradings. Its deformation fields and middle family are those already constructed. Appending three edges gives length three and successively appends F_1, G_0, G_1 to the initial vertex F_0, which proves the asserted vertex list. □

A.30 Ambient sections on the chosen finite-ribbon bundle realization

Lemma A.708 (One finite-ribbon bundle realization grades every qualifying section on its chosen completion)**.** *Supply the rank-one original compact ambient and finite ribbon bundle data of Lemma A.616, including the arbitrary native ribbon one-form family. There exist compatible global compact charts and one completed pair deformation $\mathcal{Q}$ retaining all of that lemma's original recovery, whole-ribbon, slice-comparison, endpoint-identification and marking conclusions. Write C_s for its chosen original ambient collar, X_s for its original completed slices, Λ_s for their retained primitives, $S_s : X_s \to K_s$ for their actual completed comparisons, and $p_s : K_0 \to K_s$ for the completed presentation. These choices precede the section data below.*

For every original complex field J_s on these chosen X_s, with any fixed native tangent normalizations, and every original canonical-square section σ on this chosen completed family, assume that J_s is compatible with $d\Lambda_s$ and has real local extensions on the interior and full-cylinder patches, and that σ is nowhere zero and has the total local coefficient cover. Then there are Maslov representatives $\rho_s = (\mathcal{J}_s, \mathcal{T}_s)$ on the actual K_s such that

$$dS_s \, J_s = \mathcal{J}_s \, dS_s, \qquad T_s = S_s^* \mathcal{T}_s \quad (s \in [0,1]),$$

where T_s is the natural full tensor of the same σ. This same $\mathcal{Q}$ has a grading from ρ_0 to ρ_1 and a graded deformation step retaining $\mathcal{Q}$ literally. Its interval complex field and tensor on K_0 are exactly $p_s^* \mathcal{J}_s$ and $p_s^* \mathcal{T}_s$. The finite ribbon may be empty. The section is admitted after the completed family is chosen; a section prescribed independently on an untrivialized completed bundle requires a separate comparison.

Proof. Apply the original finite-ribbon endpoint-and-marking lemma once. Retain its compatible global charts, $\mathcal{Q}, C_s, X_s, \Lambda_s$, total comparison and actual maps S_s, together with all original recovery identities and conditional endpoint and marking transports. Every ribbon tangent space is zero, so the arbitrary ribbon one-form is zero and its smoothness requires no additional hypothesis. Its completed flow fixes every ribbon point, hence its full core is the entire finite ribbon. This argument includes the empty ribbon. None of these choices uses an ambient section.

Fix any of the stated J, σ on this chosen ambient family. The retained compact and cylinder primitive equations, collar laws and completion-comparison identities are precisely the geometric hypotheses of the original-total-section endpoint lemma. The supplied compatibility, real complex-field extensions, total coefficient cover and nonvanishing provide its remaining hypotheses. Apply that lemma to this same $\mathcal{Q}$. It constructs every ρ_s and gives the complex intertwinement and full-tensor restriction through its completed-slice comparison. That comparison is the composite of the retained total slice map with p_s, which is exactly the formula for the retained S_s. Equality of these smooth maps identifies their differentials, giving the displayed equations in every tangent slot. The lemma also grades this very deformation with endpoints ρ_0, ρ_1, and identifies its interval reference fields as $p_s^*\mathcal{J}_s, p_s^*\mathcal{T}_s$. Thus every qualifying section is graded after the same geometric choices, with all the previously retained endpoint and marking conclusions. □

A.31 Transport of a section fixed on an independent completed ambient

Lemma A.709 (The derived slice comparisons preserve the original primitive and its differential)**.** *Let $\mathcal{Q}$ be a completed pair deformation, with actual completed slices K_s, primitives λ_s, and retained presentation maps $p_s : K_0 \to K_s$. Write $\lambda_s^{\mathrm{ref}} = p_s^*\lambda_s$. Let N be a smooth $2n$-manifold with smooth one-forms ℓ_s, and let $e_s : K_0 \to N$ be specified diffeomorphisms satisfying $e_s^*\ell_s = \lambda_s^{\mathrm{ref}}$. Define*

$$E_s = e_s \circ p_s^{-1} : K_s \longrightarrow N.$$

Then $E_s^*\ell_s = \lambda_s$ and $E_s^* d\ell_s = d\lambda_s$ for every $s \in [0,1]$.

Proof. Functoriality of pullback and the supplied primitive equation give

$$E_s^*\ell_s = (p_s^{-1})^* e_s^*\ell_s = (p_s^{-1})^* p_s^*\lambda_s = \lambda_s.$$

Both maps are smooth and ℓ_s is smooth. Exterior derivative commutes with smooth pullback, so differentiating this equality gives $E_s^* d\ell_s = d\lambda_s$. The same calculation with e_s gives $e_s^* d\ell_s = d\lambda_s^{\mathrm{ref}}$ on the reference completion. □

Lemma A.710 (A preselected original section grades the chosen completed deformation)**.** *Fix N, ℓ_s first, together with a complex field J_s compatible with $d\ell_s$ and a nowhere-vanishing section σ_s of its actual squared canonical line, for $s \in [0,1]$. Assume J has local smooth endomorphism extensions in real parameter coordinates and σ has local smooth scalar extensions in its canonical-line coordinates. Write T_s for the natural full tensor of this very section.*

Supply a completed pair deformation $\mathcal{Q}$ and the actual comparison diffeomorphisms e_s and primitive equations of the preceding lemma. Assume the maps e_s admit a jointly smooth forward extension on an open neighborhood of $[0,1] \times K_0$. There is a smooth Maslov family on N retaining exactly J_s, T_s at every interval parameter. The field $e_s^*J_s$ has local smooth real endomorphism extensions on K_0, and there is a nowhere-zero section τ_s of its actual squared canonical line with local smooth scalar extensions and full tensor exactly $e_s^*T_s$. There are Maslov representatives $\rho_s = (\mathcal{J}_s, \mathcal{T}_s)$ on every actual K_s, given by

$$\mathcal{J}_s = E_s^*J_s, \qquad \mathcal{T}_s = E_s^*T_s, \qquad E_s = e_s \circ p_s^{-1}.$$

The same $\mathcal{Q}$ has a grading from ρ_0 to ρ_1 and a graded deformation step retaining it literally. The grading's interval complex field and full tensor are precisely $e_s^*J_s$ and $e_s^*T_s$, with the latter equal to the tensor of τ_s. In particular the four required endpoint pullback equations are conclusions:

$$p_s^*\mathcal{J}_s = e_s^*J_s, \qquad p_s^*\mathcal{T}_s = e_s^*T_s \qquad (s = 0, 1).$$

The tensor equations hold in every tangent slot. This result assumes the geometric comparison maps and their forward extension; it does not construct them from an untrivialized original completed bundle or prescribe the transported endpoint class of the geometric target.

Proof. Reconstruct a Maslov family f on N from the supplied original endomorphism and section coefficient germs. The reconstruction lemma preserves J_s and the natural full tensor T_s at every interval parameter. Apply the interval Maslov pullback lemma to this f and the specified maps e_s. It gives the local endomorphism extensions of $e_s^* J_s$ and the actual nowhere-zero canonical-square section τ_s, with local scalar extensions and tensor image $e_s^* T_s$. Thus the pulled section is that of the original σ.

Restrict f to s and pull this Maslov representative back by the actual diffeomorphism E_s. The preceding lemma identifies $E_s^* d\ell_s$ with the actual completed symplectic form $d\lambda_s$, so the resulting ρ_s is a representative on K_s and has exactly the displayed field and tensor. Since $E_s \circ p_s = e_s$, functoriality of complex-field and tensor pullback gives

$$p_s^* \mathcal{J}_s = e_s^* J_s, \qquad p_s^* \mathcal{T}_s = e_s^* T_s$$

for every s, hence in particular at both endpoints. For tensors this functoriality follows by applying the differential chain rule in each of the two blocks of tangent arguments; no tensor slot is discarded.

The primitive calculation also gives $e_s^* d\ell_s = d\lambda_s^{\mathrm{ref}}$. Apply the interval grading transport lemma to f, e_s, this form identity, and the four endpoint equations just derived. It grades this very $\mathcal{Q}$ with endpoints ρ_0, ρ_1 and interval fields exactly $e_s^* J_s, e_s^* T_s$. Its graded deformation step retains $\mathcal{Q}$, and the pulled-section tensor identity identifies the last field with the tensor of τ_s throughout the interval. □

A.32 Strict completed transitions from the original compact bundle atlas

Lemma A.711 (Completions of the original local Liouville fibres)**.** *Let $E \to [0,1]$ be a smooth compact bundle with positive-dimensional native model fibre M. Supply covariant original-atlas primitives α_t^a and the original nondegeneracy and outwardness laws. For each original chart a and parameter t in its base set there is a native presentation of (M, α_t^a) and a localized completion $C_{a,t}$. Write $i_{a,t} : M \to C_{a,t}$ for its full compact inclusion and $\widehat{\alpha}_t^a$ for its completed primitive. These retain the original native primitive and its full exterior derivative. Each $C_{a,t}$ has its actual complete Liouville flow and cylindrical end.*

Proof. Fix a, t. Restrict the original chart's joint form extension germs to this parameter. They give smooth local extension germs for α_t^a. Its nondegenerate exterior derivative and outwardness are exactly the original local Liouville laws at a, t. The native Liouville presentation construction therefore gives a compact presentation with the same native carrier, primitive and full exterior derivative. Localize this presentation to its symplectic neighborhood and realized inward collar, then apply the localized completion construction. Both restrictions contain the full compact carrier, so every native inclusion and tensor identity is retained. That construction provides the stated completed primitive, complete flow and cylindrical end. Carry out these choices for each a, t. No assertion of joint smoothness of these independently constructed completions is made. □

Lemma A.712 (The original atlas germs give full native transition extensions)**.** *For a smooth compact bundle with positive-dimensional native model fibre, every original compact atlas transition at every overlap parameter has smooth native-coordinate extension germs for every requested source and target spatial chart. These follow from the bundle's original joint coordinate germs, including its self transitions; no additional compact-transition regularity is required.*

Proof. Fix an overlap parameter t, write f for the original compact transition there, and fix a native source chart a at x and a requested native target chart b at $y = f(x)$. Write c for the preferred native chart at y. Freezing the transition's original joint germ at t gives a smooth map γ on an open neighborhood of $a(x)$, agreeing with $c \circ f \circ a^{-1}$ on its native inputs. Freeze the original self-transition germ using b as source. Since that compact map is the identity, it gives a smooth map β on an open neighborhood of $b(y)$, agreeing with $c \circ b^{-1}$ on native inputs.

At y, the native chart differentials db_y and dc_y are linear isomorphisms on the full tangent space. Their chart-inverse differential identities hold also at boundary points. Differentiating the germ agreement of $\beta \circ b$ and c therefore gives

$$d\beta_{b(y)} \circ db_y = dc_y.$$

It follows that $d\beta_{b(y)} = dc_y \circ (db_y)^{-1}$ is invertible. The smooth inverse function theorem provides a smooth local inverse q near $c(y)$ with $q(\beta(z)) = z$ near $b(y)$.

The original transition is smooth, hence continuous. Shrink the open source neighborhood of $a(x)$ so that actual native inputs map into the source of b, their b-coordinates lie in the neighborhood where $q \circ \beta$ is the identity, and γ takes values in the open domain of q. Such an open neighborhood exists: continuity gives the first two restrictions near the native point, and smoothness of γ gives the third restriction. On this neighborhood $q \circ \gamma$ is smooth. For every native source input there, the two germ agreements and the inverse identity give

$$q(\gamma(a(z))) = q(c(f(z))) = q(\beta(b(f(z)))) = b(f(z)).$$

Thus $q \circ \gamma$ is the required genuine open extension in the requested target chart b. Since a, b, t, x were arbitrary, every original compact transition has the asserted extension germs. □

Lemma A.713 (The original compact coordinate change has a unique strict completed extension)**.** *Retain these original chart completions. At a parameter t in the overlap of original charts a, b, let $\tau_{a,b}(t) : M \to M$ be their actual compact coordinate change. There is a unique smooth diffeomorphism $H_{a,b}(t) : C_{a,t} \to C_{b,t}$ satisfying*

$$H_{a,b}(t)^* \widehat{\alpha}_t^b = \widehat{\alpha}_t^a, \qquad H_{a,b}(t) \circ i_{a,t} = i_{b,t} \circ \tau_{a,b}(t).$$

It has the prescribed inverse compact formula and intertwines the actual Liouville fields and complete flows:

$$dH_{a,b}(t)(Z_{a,t}) = Z_{b,t} \circ H_{a,b}(t), \qquad H_{a,b}(t)(\Phi_{a,t}^r(w)) = \Phi_{b,t}^r(H_{a,b}(t)(w)) \quad (r \in \mathbb{R}).$$

There is a boundary diffeomorphism $h_{a,b}(t)$ for which the all-real orbit end formula is

$$H_{a,b}(t)(E_{a,t}(x, r)) = E_{b,t}(h_{a,b}(t)(x), r) \qquad (r \in \mathbb{R}).$$

The native-coordinate extension germs needed in its construction are derived from the original compact atlas.

Proof. Original atlas compatibility makes $\tau_{a,b}(t)$ a native smooth diffeomorphism. The preceding lemma derives its full native-coordinate extension germs from the original transition and self-transition germs. Original primitive covariance gives $\tau_{a,b}(t)^* \alpha_t^b = \alpha_t^a$ on every full native tangent space. Apply the fixed-presentation completion-map lemma to the retained native presentations, their localized completions, this prescribed diffeomorphism and its extension germs, with scale one. It constructs $H_{a,b}(t)$ and $h_{a,b}(t)$ with the displayed primitive, compact and inverse compact

formulas and the same-radius end formula for $r \geq 0$. Primitive preservation gives the actual field and all-real-time flow equations by the complete-flow transport lemma. At radius zero the end equation identifies the boundary compact inclusions. For every real r, each end map is the time-r flow of that inclusion. Applying the flow equation to the radius-zero boundary identity therefore gives the asserted end formula for every real radius.

If another strict completed map has the same native compact formula, then it agrees with $H_{a,b}(t)$ on the full compact carrier, because the native inclusion parametrizes that carrier. The compact uniqueness lemma makes these two completed maps equal. Thus the two displayed conditions determine $H_{a,b}(t)$ uniquely. □

Lemma A.714 (Identity, inverse and cocycle laws of the strict completed transitions)**.** *Choose the unique maps above for all original transitions. At every parameter where the indicated original charts are defined,*

$$H_{a,a} = \mathrm{id}, \qquad H_{b,a} = H_{a,b}^{-1}, \qquad H_{b,c} \circ H_{a,b} = H_{a,c}.$$

Thus completed coordinate changes retain the literal compact atlas identities. These are fibrewise laws; joint smooth completed transition germs and a glued smooth completed bundle require further results.

Proof. The original self transition is the identity. Hence $H_{a,a}$ and the identity of $C_{a,t}$ preserve its primitive and agree on its full compact carrier. Compact uniqueness gives $H_{a,a} = \mathrm{id}$. Functoriality of smooth pullback shows that $H_{b,c} \circ H_{a,b}$ preserves the completed primitive. For every $x \in M$, its compact formula is

$$(H_{b,c} \circ H_{a,b})(i_{a,t}(x)) = i_{c,t}(\tau_{b,c}(t)(\tau_{a,b}(t)(x))) = i_{c,t}(\tau_{a,c}(t)(x)).$$

The second equality is the actual original bundle cocycle. Thus this composite and $H_{a,c}$ agree on the full compact carrier; compact uniqueness proves the cocycle equality. Taking $c = a$ gives $H_{b,a} \circ H_{a,b} = \mathrm{id}$. Since $H_{a,b}$ is a diffeomorphism, compose on the right with its inverse to obtain $H_{b,a} = H_{a,b}^{-1}$. □

Lemma A.715 (Original inclusion markings agree through the completed transition)**.** *Fix t in the overlap of original charts a, b. Let (X, λ_X) be a complete Liouville manifold and let $\nu : E \to X$ be one fixed map on the original total compact bundle. Supply inclusion markings $q_a : C_{a,t} \to X$ and $q_b : C_{b,t} \to X$ whose restrictions to the native compact pieces are the expressions of this same ν in charts a, b. Then the actual completed transition satisfies*

$$q_b \circ H_{a,b}(t) = q_a$$

on the whole completed ambient. The two markings are supplied hypotheses; their existence is a separate requirement.

Proof. For a native point x in chart a, the two compact chart expressions of ν satisfy $\nu_b(\tau_{a,b}(t)(x)) = \nu_a(x)$, since both evaluate ν at the same point of E_t. Thus the compact transition formula and the two marking restrictions give

$$(q_b \circ H_{a,b}(t))(i_{a,t}(x)) = q_b(i_{b,t}(\tau_{a,b}(t)(x))) = \nu_b(\tau_{a,b}(t)(x)) = \nu_a(x) = q_a(i_{a,t}(x)).$$

Both $q_b \circ H_{a,b}(t)$ and q_a are primitive-preserving diffeomorphisms to X, by pullback functoriality. The native inclusion covers the entire compact carrier. The completion compact-uniqueness lemma therefore identifies these two maps everywhere, including all cylindrical points. □

A.33 Joint local completion spaces in the original compact bundle atlas

Lemma A.716 (A chart-preserving parameter extension at every interval point)**.** *Let S be relatively open in $I = [0,1]$ and let $t \in S$. There exist a relatively open neighborhood U of t, contained in S, a map $\sigma : I \to I$ with image in S, and a smooth real function $f : \mathbb{R} \to \mathbb{R}$ such that $f(s) = \sigma(s)$ on I and $\sigma(s) = s$ on U. This includes $t = 0$ and $t = 1$.*

Proof. Relative openness gives $r > 0$ such that $I \cap (t - r, t + r) \subset S$. Choose a smooth function $\chi : \mathbb{R} \to [0,1]$, equal to one on a neighborhood $(t - a, t + a)$ with $a > 0$, and zero outside $(t - r, t + r)$. Set $f(s) = t + \chi(s)(s - t)$. This is smooth on $\mathbb{R}$. For $s \in I$, it is the convex combination $(1 - \chi(s))t + \chi(s)s$, so lies in I. If $|s - t| < r$, then $|f(s) - t| \leq |s - t| < r$, hence $f(s) \in S$. Otherwise $\chi(s) = 0$ and $f(s) = t \in S$. Thus its restriction defines the asserted σ. The relative open set $U = I \cap (t - \min(a, r), t + \min(a, r))$ contains t, lies in S, and has $\sigma(s) = s$ because $\chi = 1$ there. The convex-combination argument remains valid at both interval endpoints. □

Lemma A.717 (Real endpoint germs survive a chart-preserving interval reparameterization)**.** *Let α_u be a native differential-form family with joint smooth real coordinate extensions over a subset $S \subset I$. Suppose $\sigma : I \to I$ has image in S and has scalar coordinate equal on I to a smooth real function f. Then $\beta_s = \alpha_{\sigma(s)}$ has joint smooth real coordinate extensions over all of I, including its endpoints.*

Proof. Fix $s \in I$ and a native spatial point x in an arbitrary spatial chart. The original joint form-germ hypothesis at $(\sigma(s), x)$ gives a smooth coefficient extension $A(u, z)$ on an open real product neighborhood, agreeing with α_u on native inputs with $u \in S$. The smooth map $(v, z) \mapsto (f(v), z)$ takes (s, x) into that neighborhood. Its inverse image is open and contains (s, x); there $A(f(v), z)$ is smooth. For interval inputs v one has $f(v) = \sigma(v) \in S$, so these coefficients agree with β_v. This produces the required real open germ at every point and every spatial chart, including when s is an interval endpoint. □

Lemma A.718 (The original local primitives construct a joint total completion space)**.** *Let $E \to I$ carry the original positive-dimensional compact bundle primitives and local Liouville laws. For every original chart c and t in its base set, there exist U, σ, f as above and one joint smooth total completion space with patchwise primitive laws for $\beta_s = \alpha^c_{\sigma(s)}$ on the fixed native compact fibre M. It has a common collar width $\delta > 0$, original inward collars C_s, completed quotient slices X_s and primitives L_s, with*

$$o_s^* L_s = \beta_s, \qquad q_s^* L_s = e^{-r} \pi_{\partial M}^* (\beta_s|_{\partial M}).$$

Here $q_s(r, y)$ is the signed inward-cylinder insertion, $r < \delta$. The shared quotient has the smooth total atlas, smooth interior and cylinder insertions, smooth parameter projection, Hausdorffness, second countability and actual fibre homeomorphisms of Lemma A.463. Each L_s is smooth, has nondegenerate exterior derivative and its actual complete Liouville flow for every real time. The spatial full-cylinder primitive coefficients are jointly smooth with real parameter-endpoint extensions. On U the compact primitive is exactly the original α^c_s. Assembly of a smooth vertical one-form on the whole total space and joint cross-chart completed comparisons remain additional assertions. No completed family or smooth completed transition is supplied as input.

Proof. Apply the parameter-extension lemma to the base set of c at t. Apply the preceding reparameterization lemma to the original joint form germs in chart c, the produced σ, and its smooth real extension f. This gives joint native form germs for β at every parameter, including both interval endpoints.

The spatial exterior derivative and Liouville field at s are those of the original $\alpha^c_{\sigma(s)}$. Its nondegeneracy and outwardness are therefore the original local laws at a parameter in the base set. Apply the native completed-family total-atlas construction to β, these joint germs and these laws. Retain its one choice of collars, shared quotient, slice primitives, total atlas, actual fibre identifications and complete flows. Its compact and full-cylinder primitive formulas are the displayed identities. Finally $\sigma(s) = s$ on U, giving the exact original compact primitive there. Every retained conclusion belongs to this same construction. □

Lemma A.719 (The joint local slices retain the earlier original chart completions)**.** *Retain a local joint construction in chart c as above. Each quotient slice is itself a Liouville completion of its full compact native carrier, retaining exactly L_s and the chosen collar. For each $s \in U$ there is a strict diffeomorphism $F^c_s : X^c_s \to C_{c,s}$ and a boundary diffeomorphism h^c_s such that*

$$\begin{aligned} (F^c_s)^*\widehat{\alpha}^c_s &= L^c_s, \\ F^c_s(o^c_s(x)) &= i_{c,s}(x), & (F^c_s)^{-1}(i_{c,s}(x)) &= o^c_s(x), \\ F^c_s(E^c_s(b,r)) &= E_{c,s}(h^c_s(b), r) & & (r \in \mathbb{R}), \\ F^c_s(\Phi^c_s(r,z)) &= \Phi_{c,s}(r, F^c_s(z)) & & (r \in \mathbb{R}). \end{aligned}$$

Here $E^c_s, E_{c,s}$ are the actual boundary-orbit end maps of these same completions and $\Phi^c_s, \Phi_{c,s}$ their complete Liouville flows. For an arbitrary interval parameter the same assertion holds with the target original-chart parameter $\sigma(s)$. No joint smoothness of these comparison maps is asserted.

Proof. The original outwardness law implies the strictly negative inward normal component of the native Liouville field. The compact-carrier Liouville-domain construction therefore equips the compact carrier of the chosen quotient with the restriction of this very L_s. Its native presentation retains the actual native inclusion, full primitive equation and collar equation. The half-infinite completion construction gives a completion on the same entire quotient with identity ambient inclusion and primitive exactly L_s.

Use the earlier original chart native presentation at $\sigma(s)$ as the prescribed target, localized to its symplectic neighborhood. Both native presentations have primitive $\alpha^c_{\sigma(s)}$. The chosen collar obeys the inward native Liouville equation, so apply the strict inward-collar completion-comparison lemma to these two presentations and their completions. It constructs the asserted strict comparison, boundary map, both native compact squares, all-real boundary-orbit formula and all-real complete-flow equation. At $s \in U$, the identity $\sigma(s) = s$ makes its target exactly $C_{c,s}$. Every source object is the retained quotient and primitive, so this introduces no second local completed space. □

Lemma A.720 (The strict local overlaps are determined by the original compact transitions)**.** *On the overlap of two retained local constructions, define*

$$G_{c,d}(s) = (F^d_s)^{-1} \circ H_{c,d}(s) \circ F^c_s.$$

This map is strict and sends $o^c_s(x)$ to $o^d_s(\tau_{c,d}(s)(x))$. Any strict diffeomorphism of these same slices with that compact formula equals $G_{c,d}(s)$. It intertwines their actual complete flows for every real time:

$$G_{c,d}(s)(\Phi^c_s(r,z)) = \Phi^d_s(r, G_{c,d}(s)(z)) \qquad (r \in \mathbb{R}).$$

Let $\tau^{\partial}_{c,d}(s)$ be the actual boundary restriction of $\tau_{c,d}(s)$. The same map satisfies

$$\begin{aligned} G_{c,d}(s)(E^c_s(b,r)) &= E^d_s(\tau^{\partial}_{c,d}(s)(b), r) \quad (r \in \mathbb{R}), \\ G_{c,d}(s)(q^c_s(r,b)) &= q^d_s(r, \tau^{\partial}_{c,d}(s)(b)) \quad (r < \min(\delta_c, \delta_d)). \end{aligned}$$

The last formula retains both collar widths and includes every outward cylinder radius $r \leq 0$ and their common inward strip. This assertion is fibrewise. Joint smoothness of the overlap maps and correspondence with a whole completed bundle remain additional assertions.

Proof. Pullback functoriality and the three strict primitive identities give $G_{c,d}(s)^* L_s^d = L_s^c$. The two compact squares and the compact equation of $H_{c,d}(s)$ give

$$G_{c,d}(s)(o_s^c(x)) = (F_s^d)^{-1}(i_{d,s}(\tau_{c,d}(s)(x))) = o_s^d(\tau_{c,d}(s)(x)).$$

The retained source quotient is itself a completion of the compact carrier $o_s^c(M)$ by the preceding lemma. If another strict map has the same compact formula, it agrees with $G_{c,d}(s)$ on that entire carrier. Completion compact uniqueness therefore identifies the two maps on the whole source quotient. Finally strict primitive preservation gives Liouville-field intertwinement and hence the complete-flow equation for every real time.

The original compact transition preserves the native boundary, so its restriction $\tau_{c,d}^{\partial}(s)$ is a boundary diffeomorphism and has the same underlying native values as $\tau_{c,d}(s)$. At a native boundary point b, the compact square therefore gives

$$G_{c,d}(s)(o_s^c(b)) = o_s^d(\tau_{c,d}^{\partial}(s)(b)).$$

Each end map at radius r is the time-r Liouville flow of this native boundary inclusion. Apply the complete-flow equation to the last boundary identity to obtain the asserted end equation for every real radius. The original signed-cylinder orbit formula on a collar of width δ is $q_s(r,b) = E_s(b,-r)$ for every $r < \delta$. For $r < \min(\delta_c, \delta_d)$ it applies to both retained collars. Substitute both identities into the end equation at radius $-r$; this gives the displayed literal signed-cylinder formula, without changing either collar width. □

Lemma A.721 (Original local slices compare strictly with the chosen compatible completion)**.** *Let F be a retained positive-dimensional local completion family in an original compact bundle chart c, with base U_F. Let g be a compatible global chart of the same compact bundle, and write α_s^g for its transported original primitive. For each interval parameter s, fix a compact Liouville domain D_s, its completion K_s, and a native diffeomorphism $e_s : M \to D_s$ whose ambient inclusion has smooth spatial extension germs and pulls the compact primitive back to α_s^g. These may be the fixed compact domains, completions and native maps of one chosen completed pair deformation. Write $\kappa_s : D_s \to K_s$ for its compact inclusion.*

For every $s \in U_F$, there is a smooth diffeomorphism $A_s : X_s^F \to K_s$ such that

$$\begin{aligned} A_s^* \lambda_{K_s} &= L_s^F, \\ A_s(o_s^F(x)) &= \kappa_s(e_s(\tau_{c,g}(s)(x))), \\ A_s^{-1}(\kappa_s(e_s(y))) &= o_s^F(\tau_{g,c}(s)(y)) \end{aligned}$$

for every native $x, y \in M$. The map A_s is uniquely determined among strict completion diffeomorphisms by this compact square. There is a boundary diffeomorphism $h_s : \partial M \to \partial D_s$ such that

$$\begin{aligned} A_s(E_s^F(b,r)) &= E_{K_s}(h_s(b), r) \qquad (r \in \mathbb{R}), \\ A_s(\Phi_s^F(r,z)) &= \Phi_{K_s}(r, A_s(z)) \qquad (r \in \mathbb{R}), \\ A_s(q_s^F(a,b)) &= E_{K_s}(h_s(b), -a) \quad (a < \delta_F). \end{aligned}$$

If a native boundary presentation e_s^{∂} is specified with $\iota_{\partial D_s} \circ e_s^{\partial} = e_s \circ \iota_{\partial M}$, then $h_s = e_s^{\partial} \circ \tau_{c,g}^{\partial}(s)$, with the actual original bundle boundary transition. All objects in these equations are the retained

original local slices and the fixed chosen target completions. This assertion is fibrewise; joint real parameter extension germs are a separate assertion.

Proof. Fix $s \in U_F$. The retained local slice comparison gives a native Liouville presentation of its compact carrier with primitive α_s^c and a completion on the same quotient X_s^F, with primitive exactly L_s^F. The endpoint presentation lemma applied to the fixed D_s, e_s gives a target native presentation with primitive α_s^g and the specified completion K_s.

The actual compact transition $h = \tau_{c,g}(s)$ is a native smooth diffeomorphism with inverse $\tau_{g,c}(s)$. Compatibility gives its smooth coordinate extension germs in both directions, and original primitive covariance gives $h^*\alpha_s^g = \alpha_s^c$ on every full native tangent space. Apply the fixed-presentation completion extension lemma with scale one to these two presentations and this very h. It produces a strict smooth completion diffeomorphism A_s on the given completed manifolds and the asserted full compact square. Apply A_s^{-1} to that square at $x = h^{-1}(y)$; the compact transition inverse identity gives the displayed inverse square. The primitive equation is precisely the strict extension identity. Strict primitive preservation intertwines the Liouville fields, hence their complete flows for every real time. The boundary square produced by the same extension construction identifies the boundary point $h_s(b)$ with $e_s(\tau_{c,g}(s)(b))$ in the compact target. This also identifies it with $e_s^\partial(\tau_{c,g}^\partial(s)(b))$ when the stated native boundary presentation is given, since compact and boundary inclusions are injective. Each boundary end map is the all-real flow of its compact boundary inclusion. Flow intertwinement and the boundary square therefore give the all-real end identity. Substituting the original signed-cylinder identity $q_s^F(a,b) = E_s^F(b,-a)$ gives the full cylinder equation without reducing its original width. Finally compact uniqueness equates any two strict maps with the same compact square on the whole completion. No target completion or native compact map has been replaced. □

Lemma A.722 (The local comparison retains the same deformation's varying primitive)**.** *Let $\mathcal{Q}$ be a specified completed pair deformation, and let $p_s : K_0 \to K_s$ be its specified ambient completion presentations. Assume its fixed compact domains D_s, native maps e_s and completed primitives satisfy the hypotheses of the preceding lemma with the same compatible global chart g. Let $A_s : X_s^F \to K_s$ be that lemma's comparison, and set $B_s = p_s^{-1} \circ A_s$. Then*

$$B_s^*\ell_s = L_s^F, \qquad (B_s^{-1})^* L_s^F = \ell_s, \qquad \ell_s = p_s^*\lambda_{K_s}.$$

The full compact square and its inverse are the preceding lemma's squares composed with p_s^{-1} and p_s, respectively. With $\overline{E}_s(b,r) = p_s^{-1}(E_{K_s}(b,r))$, the all-real boundary and full signed-cylinder squares are

$$B_s(E_s^F(b,r)) = \overline{E}_s(h_s(b),r), \qquad B_s(q_s^F(a,b)) = \overline{E}_s(h_s(b),-a).$$

Here the boundary coordinate may equivalently be expressed in the fixed compact-boundary presentation of $\mathcal{Q}$. These maps use the original local completion and the fixed source completion of that very $\mathcal{Q}$, with its actual varying primitive ℓ_s. No joint regularity or new geometric deformation is asserted.

Proof. Apply the preceding lemma to the exact fixed compact domain and completion at s, using the given native endpoint primitive and extension equations. Compose its comparison with p_s^{-1}. Pullback functoriality and $p_s \circ p_s^{-1} = \mathrm{id}$ give

$$B_s^*\ell_s = A_s^*(p_s^{-1})^* p_s^*\lambda_{K_s} = A_s^*\lambda_{K_s} = L_s^F.$$

Pulling this equality back by B_s^{-1} gives the inverse primitive identity. Composing each forward square with p_s^{-1} gives the compact, all-real end and full signed-cylinder equations. The inverse

compact square follows from $A_s^{-1} \circ p_s = B_s^{-1}$. Changing the boundary coordinate through the fixed boundary presentation changes neither point nor radius. Thus every equation holds on the same specified domains, completions and presentation maps at each parameter. □

Lemma A.723 (Original compact transition germs restrict to the actual interior)**.** *Let $S \subset [0,1]$ be relatively open and let $f_s : M \to M$ be a jointly continuous family with the original joint real coordinate extension germs. Suppose f_s preserves the native interior for $s \in S$. Its literal interior restriction has actual interior-valued smooth extensions on real open neighborhoods of every (s,x), agreeing with $f_u(y)$ at every interval point in the neighborhood with $u \in S$. In particular, this holds for the transition of any ordered pair of compatible compact bundle charts on their common base, with no additional interior-preservation or extension input. Both parameter endpoints are included.*

Proof. Fix $s \in S$ and an interior point x. Restrict an original native chart containing x to the interior; its coordinate image has strictly positive normal coordinate and is open in the full Euclidean model. Do the same for an original target chart containing $f_s(x)$. The supplied coordinate germ, after restriction using joint continuity, has a smooth real extension β whose interval values lie in that target chart and are its actual coordinates of $f_u(y)$. At the center, β belongs to the target interior chart's open image. Intersect its domain with the inverse image of that open image. This is a real open neighborhood of the same center. Composing β there with the smooth inverse interior chart produces an actual interior-valued smooth germ. The chart inverse identity gives its exact agreement with $f_u(y)$ at every retained interval point. The real parameter has not been restricted to the interval in this construction.

For a compatible bundle transition, the original transition germs and joint continuity supply the preceding hypotheses. Its native smooth diffeomorphism carries interior to interior, as follows from invariance of the interior under a diffeomorphism of manifolds with boundary. Hence no preservation hypothesis remains for this specialization. The construction works at zero and one because its domains are open in the real parameter throughout. □

Lemma A.724 (Real patch germs give real germs in the retained total atlas)**.** *Let X be a retained native total completion, with parameter projection $\pi : X \to I$ and original collar width $\delta > 0$. Let $S \subset I$, $A = \pi^{-1}(S)$ and $f : A \to N$, where N is a boundaryless smooth manifold. Suppose its interior composite has actual target-valued smooth germs on real open neighborhoods of every (s,x), $s \in S$, agreeing with that composite at all retained interval points. Suppose the same holds for its cylinder composite on $a < \delta$, $-a > c$, where $c < 0$ is one fixed cutoff. Then at every $z \in A$ there exist a chart e in the original total atlas, with z in its source, a real open neighborhood U of its full Euclidean coordinates $e(z)$, and a smooth map $\beta : U \to N$, such that*

$$\beta(e(y)) = f(y) \qquad (y \in A \cap \operatorname{source}(e),\ e(y) \in U).$$

One such original atlas chart is required at each point. No map on $X \setminus A$ or equations at off-interval parameters are required.

Proof. The original interior-and-strip cover puts each $z \in A$ in either an interior patch or a cylinder patch with $-a > c$. Choose the product chart of that patch at the preimage of z. Write its spatial chart as χ and its parameter chart as $r = t$ or $r = 1 - t$; the inverse parameter expression $t = \tau(r)$ is real affine and smooth. Lift this product chart through the original open patch insertion. By the retained total atlas construction, the lifted chart e belongs to that original atlas and contains z.

Let γ be the supplied target-valued real germ of the patch composite at this center. On the real open set where $\chi^{-1}(u)$ is defined and $(\tau(r), \chi^{-1}(u))$ belongs to the domain of γ, set

$$\beta(r,u) = \gamma(\tau(r), \chi^{-1}(u)).$$

Intersect this set with the inverse image of the chosen spatial patch region; in the cylinder case retain the strict inequality $-a > c$. These are open conditions and hold at the center. Both τ and χ^{-1} are smooth, so the resulting β is smooth and takes actual values in N throughout its real domain. For y in the stated interval locus, the lifted chart inverse recovers exactly its original patch preimage, the parameter belongs to S, and the germ agreement gives $\beta(e(y)) = f(y)$. This argument includes both parameter endpoints and keeps the original collar width and all outward radii. □

Lemma A.725 (Original local compact comparisons have real interior germs)**.** *Let c, g be compatible compact bundle charts and let $\mathcal{Q}$ be a fixed completed pair deformation. Write $e_s : M \to D_s$ for its native compact ambient maps, h_s for its compact presentation and $p_s : K_0 \to K_s$ for its completed presentation. Suppose a jointly smooth real family of native diffeomorphisms R_s satisfies*

$$h_s(e_0(R_s(x))) = e_s(x) \qquad (s \in I,\ x \in M).$$

Then on $S = U_c \cap U_g$ the actual interior map

$$(s, x) \longmapsto p_s^{-1}\big(\kappa_s(e_s(\tau_{c,g}(s)(x)))\big), \qquad x \in M^\circ,$$

has target-valued smooth extension germs on real open neighborhoods of every original (s, x), retaining this exact interval formula. The target is the fixed source completion of that very $\mathcal{Q}$. The family R and the compact presentation square are explicit hypotheses, as supplied by the original compact bundle realization.

Proof. The compact presentation square and completion presentation compact square identify the displayed map with

$$\kappa_0(e_0(R_s(\tau_{c,g}(s)(x)))).$$

The original compatible transition has actual interior-valued real germs by the interior transition lemma. At a chosen center let γ be such a germ. The map $(u, y) \mapsto \kappa_0(e_0(R_u(y)))$ is jointly smooth for real u and interior y: R is jointly smooth and e_0, κ_0 are smooth. Compose this map with $(u, x) \mapsto (u, \gamma(u, x))$ on the real germ domain. The result is a smooth actual completion-valued germ. At every retained interval point, γ equals the literal compact transition, so the preceding identity gives precisely the desired reference compact map. The same argument applies at both endpoints, without requiring additional extension data for the original transition. □

Lemma A.726 (The retained slice comparisons define the actual local total map)**.** *Retain the local completion family F, compatible global chart g and fixed completed deformation $\mathcal{Q}$ of the local deformation comparison lemma. Write $A = \pi^{-1}(U_F)$ in the actual original total quotient. There is a map $\Phi : A \to K_0$ whose restriction to every completed slice is exactly that lemma's B_s:*

$$\Phi(\iota_s(z)) = B_s(z) \qquad (s \in U_F,\ z \in X_s^F).$$

Consequently its full slice restriction preserves the actual varying primitive ℓ_s. On every original compact point it has the formula

$$\Phi(o_{\mathrm{tot}}(s, x)) = p_s^{-1}\big(\kappa_s(e_s(\tau_{c,g}(s)(x)))\big).$$

The map is defined on its actual relative-open base only; no value outside that base is selected. Joint regularity remains an additional assertion.

Proof. Every total point z has its actual parameter $s = \pi(z)$. The retained fibre identification is a bijection from X_s^F onto the parameter fibre of z. Let z_s be its unique inverse image and define $\Phi(z) = B_s(z_s)$ for $z \in A$. For a slice insertion $z = \iota_s(w)$, the parameter is exactly s and the inverse fibre identity gives $z_s = w$. Thus $\Phi \circ \iota_s = B_s$ on the full slice, and its primitive pullback is L_s^F by the local deformation comparison lemma. The original total compact insertion is the slice insertion of its original compact point. Apply the same restriction identity and the retained compact square of B_s to obtain the displayed compact formula. Each definition and equation uses the same quotient, its actual fibre identification and the same chosen completed deformation. □

Lemma A.727 (The actual local comparison has real germs on the completed total space)**.** *Retain the actual local total map $\Phi : A \to K_0$ of the preceding lemma and the real family R with its compact presentation square from the reference interior-germ lemma. Supply the native boundary presentations e_s^∂ with their full compact boundary squares and the retained boundary presentation coherence $h_{s,\partial} \circ e_0^\partial = e_s^\partial$, where $h_{s,\partial}$ is the boundary map of this same deformation's compact presentation. No real end extension or cutoff is supplied. Then Φ has actual target-valued real extension germs in the original total atlas on A. Its interval values retain the full original slice comparison, the varying primitive and the original compact formula. On the full original signed cylinder its values are*

$$\Phi(q_{\mathrm{tot}}(s,a,b)) = p_s^{-1}(E_{K_s}(e_s^\partial(\tau_{c,g}^\partial(s)(b)), -a)) \qquad (s \in U_F,\ a < \delta_F).$$

No real inverse germs, descent over the whole original compact bundle, or correspondence with independently prescribed sections are asserted.

Proof. Joint smoothness of the completed primitives is part of the given completed pair deformation. The prescribed end-strip extension lemma therefore produces $c < 0$ and a smooth real family Γ extending its actual prescribed boundary orbits. Compose Γ with the fixed smooth native boundary presentation e_0^∂ to obtain a smooth real map $\beta : \mathbb{R} \times \partial M \times \mathbb{R} \to K_0$. The retained boundary presentation coherence gives

$$\beta(s,b,r) = p_s^{-1}(E_{K_s}(e_s^\partial(b), r)) \qquad (s \in I,\ r > c).$$

The full slice restriction of Φ and the retained signed-cylinder square of its slice comparison give the displayed cylinder formula for every original radius $a < \delta_F$. In particular, on $-a > c$ this is $\beta(s, \tau_{c,g}^\partial(s)(b), -a)$. The actual original boundary transition has target-valued real germs. At a chosen boundary center take one such germ γ and compose it, retaining the parameter and signed radius, with the smooth map β. This gives a smooth target-valued real germ on the inverse image of the domain of γ under $(u,a,b) \mapsto (u,b)$. Its interval restriction agrees exactly with the actual cylinder composite wherever $-a > c$.

The retained compact formula of Φ and the reference interior-germ lemma give actual target-valued real germs for its interior composite. Apply the original total real-map germ criterion to these interior germs, the constructed cylinder germs and the same cutoff $c < 0$. It gives one original atlas chart and a real open completion-valued germ at every point of A, including either parameter endpoint, the seam and every outward radius. Every germ has the same interval restriction Φ, so none of the original slice, compact or primitive identities is changed. □

Lemma A.728 (Original compact bundle data choose one completion for every local comparison)**.** *Let $n > 1$. Supply the original compact ambient and ribbon bundles of dimensions $2n$ and $2n-2$, their covariant primitive data, original full ribbon data and positive witness data, together with the original local Liouville and pair laws. There exist a compatible global ambient compact chart g, one*

completed pair deformation $\mathcal{Q}$, and its native ambient diffeomorphisms e_s retaining the original primitive pullbacks and spatial coordinate extensions. These choices precede every original local chart comparison below.

For every original ambient chart c and every $t \in U_c$, let F be the local completed family constructed from that original chart, parameter and ambient primitive. The actual comparison $\Phi_F : \pi_F^{-1}(U_F) \to K_0$ of the local total-map lemma has target-valued real extension germs in its original total atlas. For every $s \in U_F$, on the entire original completed slice,

$$(\Phi_F \circ \iota_s)^* \ell_s = L_s^F, \qquad \ell_s = p_s^* \lambda_{K_s}.$$

The global chart, completed deformation, native maps and primitive are the same for every c, t. No completed deformation, real inverse germs or grading sections are supplied as input. The assertion does not identify a completed bundle with independently prescribed sections.

Proof. Apply the original compact bundle completed-family construction once to the given primitive, full ribbon, witness and local geometric data. Retain its compatible global ambient chart g, completed deformation $\mathcal{Q}$, native maps and exact primitive and coordinate extension laws. Retain also its real jointly smooth native family R, compact presentation square and native boundary presentation with both boundary squares. All these are produced for the same choices and before any original local chart is selected.

Now fix any original ambient chart c and $t \in U_c$. The local completion construction from the original ambient primitive and Liouville laws gives F. Apply the local total-map lemma using the retained $g, \mathcal{Q}$ and their produced native endpoint laws. Its restriction to each full completed slice is the strict comparison, so its primitive pullback is exactly the displayed L_s^F. The retained R and compact and boundary presentation squares satisfy the local total real-germ lemma's remaining hypotheses. Apply it to this same total comparison to obtain the real original-atlas germs. The chart and parameter were arbitrary; the global choices were made before them, proving the stated quantifier order. □

Lemma A.729 (The actual local total comparison has a parameter-preserving inverse)**.** *Retain the local completion F, compatible global chart g, completed pair deformation $\mathcal{Q}$, and strict slice comparisons $B_s : X_s^F \to K_0$ of the local total-map lemma. Set $A = \pi_F^{-1}(U_F)$. The map*

$$T : A \longrightarrow U_F \times K_0, \qquad T(z) = (\pi_F(z), \Phi_F(z))$$

is a bijection with inverse

$$T^{-1}(s, y) = \iota_s(B_s^{-1}(y)).$$

It retains exactly the original parameter and the entire original completed slice. For the native reference map $a_s(x) = p_s^{-1}(\kappa_s(e_s(x)))$, its compact formula is

$$T^{-1}(s, a_s(x)) = o_{\mathrm{tot}}(s, \tau_{g,c}(s)(x)).$$

Here $\tau_{g,c}$ is the actual reverse original compact coordinate change. No joint regularity of this inverse is asserted here.

Proof. The slice insertion has parameter s, and the forward total map restricts to B_s. Thus $T(\iota_s(B_s^{-1}(y))) = (s, B_s(B_s^{-1}(y))) = (s, y)$. Conversely let $z \in A$ and set $s = \pi_F(z)$. The retained fibre bijection has a unique $z_s \in X_s^F$ with $\iota_s(z_s) = z$. Then $\Phi_F(z) = B_s(z_s)$, so $\iota_s(B_s^{-1}(\Phi_F(z))) = \iota_s(z_s) = z$. These two equations prove bijectivity and the displayed inverse. The compact inverse square of the strict slice comparison gives $B_s^{-1}(a_s(x)) = o_s(\tau_{g,c}(s)(x))$. Apply ι_s and its compact insertion identity to obtain the asserted total compact equation. □

Lemma A.730 (The exact local inverse has real interior germs)**.** *Retain the preceding inverse and a real jointly smooth native family of diffeomorphisms $R_u : M \to M$ satisfying $h_s(e_0(R_s(x))) = e_s(x)$ on the interval. Write $i_0(x) = \kappa_0(e_0(x))$ for the fixed source inclusion. For every $s \in U_F$ and $x \in M^\circ$, there exist a real open neighborhood V of $(s, i_0(x))$ in $\mathbb{R}\times K_0$ and a smooth map $\gamma : V \to M^\circ$ such that*

$$T^{-1}(v, y) = o_{\mathrm{tot}}(v, \gamma(v, y)) \qquad (v \in U_F,\ (v, y) \in V).$$

The equality holds for every retained interval point in that real neighborhood, including either parameter endpoint. The germ takes actual interior values, rather than merely coordinate values.

Proof. The fixed reference interior map $i_0 : M^\circ \to K_0$ is a local smooth diffeomorphism. Choose a smooth local inverse ℓ near $i_0(x)$, with $i_0(\ell(y)) = y$. The map $(u, y) \mapsto R_u^{-1}(\ell(y))$ is jointly smooth on that real open neighborhood and takes values in M°, since a smooth native diffeomorphism preserves the interior.

At the center choose an actual interior-valued real germ η of the reverse original transition $\tau_{g,c}$. Intersect the neighborhood with the inverse image of the domain of η under $(u, y) \mapsto (u, R_u^{-1}(\ell(y)))$ and define $\gamma(u, y) = \eta(u, R_u^{-1}(\ell(y)))$ there. This is smooth and interior-valued. On an interval input v, the compact presentation square gives

$$a_v(R_v^{-1}(\ell(y))) = i_0(\ell(y)) = y.$$

The compact inverse identity of the preceding lemma therefore gives

$$T^{-1}(v, y) = o_{\mathrm{tot}}(v, \tau_{g,c}(v)(R_v^{-1}(\ell(y)))).$$

This equals the asserted formula by the interval agreement of η. The germ domains are open in the real parameter, so this argument includes both endpoints without changing the inverse or its base. □

Lemma A.731 (The exact inverse retains the full original signed cylinder)**.** *Retain T^{-1} and native boundary presentations e_s^∂ with the full compact boundary squares and the coherence $h_{s,\partial} \circ e_0^\partial = e_s^\partial$. Write $\overline{E}_s(b, r) = p_s^{-1}(E_{K_s}(h_{s,\partial}(b), r))$ for $b \in \partial D_0$. For every $s \in U_F$, native boundary point b and real r with $-r < \delta_F$, the actual inverse satisfies*

$$T^{-1}(s, \overline{E}_s(e_0^\partial(b), r)) = q_{\mathrm{tot}}(s, -r, \tau_{g,c}^\partial(s)(b)).$$

The original collar width and every permitted radius are retained.

Proof. The original forward and reverse compact coordinate changes are inverse on their chart overlap. Their actual boundary restrictions have the same native values, so $\tau_{c,g}^\partial(s) \circ \tau_{g,c}^\partial(s) = \mathrm{id}$. Apply the retained full-cylinder forward formula to $q_{\mathrm{tot}}(s, -r, \tau_{g,c}^\partial(s)(b))$. Its boundary argument reduces to b, its outward radius is r, and the native boundary coherence makes its reference image $\overline{E}_s(e_0^\partial(b), r)$. The two-sided inverse identity for T now gives the displayed equation. The sole radial restriction is $-r < \delta_F$, exactly the original cylinder domain. □

Lemma A.732 (The exact local inverse has real end germs)**.** *Retain the preceding inverse and native boundary coherence. For every $s \in U_F$, native boundary point b and $r \geq 0$, there exist a real open neighborhood V of $(s, E_{K_0}(e_0^\partial(b), r))$ in $\mathbb{R} \times K_0$ and a smooth map $\gamma = (\gamma_\partial, \gamma_r) : V \to \partial M \times \mathbb{R}$ with $-\gamma_r < \delta_F$ throughout V, such that*

$$T^{-1}(v, y) = q_{\mathrm{tot}}(v, -\gamma_r(v, y), \gamma_\partial(v, y)) \qquad (v \in U_F,\ (v, y) \in V).$$

These are actual native boundary–radius germs of the same inverse. They include the seam $r = 0$, all outward radii and both interval endpoints without altering the collar width.

Proof. Joint regularity of the given completed primitive produces $a < 0$ and a jointly smooth real map $\beta(u, b, r)$ agreeing with $\overline{E}_u(e_0^\partial(b), r)$ on interval parameters and $r > a$. At the selected interval parameter, the spatial map $(b, r) \mapsto \beta(s, b, r)$ agrees near (b, r) with the prescribed end map composed with the fixed boundary diffeomorphism e_0^∂. Both factors are local smooth diffeomorphisms, so its derivative is invertible. The parameter-preserving inverse lemma applied to $(u, b, r) \mapsto (u, \beta(u, b, r))$ gives a smooth local inverse. Write its spatial components as $(b'(u, y), r'(u, y))$; its first component equals u by its inverse equation.

Choose an actual boundary-valued real germ η of the reverse original transition $\tau_{g,c}^\partial$ near (s, b). Shrink the inverse domain so that $(u, b'(u, y))$ lies in its domain, $r'(u, y) > a$, and $-r'(u, y) < \delta_F$. All conditions are open and hold at the center because $a < 0 \le r$ and $\delta_F > 0$. Set

$$\gamma(u, y) = (\eta(u, b'(u, y)), r'(u, y)).$$

This is smooth and obeys the stated radial bound. On an interval input v, the local inverse equation and the interval agreement of β give $y = \overline{E}_v(e_0^\partial(b'(v, y)), r'(v, y))$. The full original signed-cylinder inverse identity therefore gives

$$T^{-1}(v, y) = q_{\mathrm{tot}}(v, -r'(v, y), \tau_{g,c}^\partial(v)(b'(v, y))).$$

The interval agreement of η identifies this with the required γ formula. Each neighborhood is real open, so the argument also holds at both parameter endpoints and at $r = 0$. □

Lemma A.733 (The exact inverse has real patch germs at every completed point)**.** *Retain the actual inverse T^{-1}, the real native family R_u and the compact and boundary presentation squares of the preceding lemmas. At every $(s, y) \in U_F \times K_0$, this inverse has a real open spatial germ of one of the following two kinds: an actual interior map γ with $T^{-1}(v, z) = o_{\mathrm{tot}}(v, \gamma(v, z))$ on all retained interval inputs, or an actual boundary–radius map γ with $-\gamma_r < \delta_F$ and $T^{-1}(v, z) = q_{\mathrm{tot}}(v, -\gamma_r(v, z), \gamma_\partial(v, z))$ there. In either case the germ is smooth on its real open neighborhood and retains the exact original inverse values at every interval point.*

Proof. If y lies in the image of the compact interior of the reference completion, the native diffeomorphism e_0 has an interior preimage x. Apply the real interior inverse-germ lemma at $(s, i_0(x))$. If y is outside that image, the completion's nonnegative end covers this complement. Thus $y = E_{K_0}(b', r)$ for $r \ge 0$. The native boundary diffeomorphism e_0^∂ has a preimage b of b'. Apply the real end inverse-germ lemma at (s, b, r). These cases cover the whole reference completion, including its seam, and give exactly the two claimed spatial germs of the same inverse. □

Lemma A.734 (The actual local completion comparison is a joint native diffeomorphism)**.** *Retain the local completion F, the chosen completed deformation $\mathcal{Q}$, and the original compact and boundary presentation squares and real native family of the preceding germ lemmas. Equip $A = \pi_F^{-1}(U_F)$ with its original total-completion atlas and $U_F \times K_0$ with the relative-open product atlas. Then the actual parameter-preserving equivalence*

$$T(z) = (\pi_F(z), \Phi_F(z))$$

is a smooth diffeomorphism between these very spaces. Its inverse is exactly $T^{-1}(s, y) = \iota_s(B_s^{-1}(y))$. It retains the full original compact and cylinder formulas and both whole-slice primitive equations. This is joint native smoothness on the retained interval spaces, including their endpoints; it asserts neither a common real extension nor a whole original completed-bundle section correspondence.

Proof. At each $z \in A$, the forward real-germ theorem gives an original atlas chart e and a smooth completion-valued real map β agreeing with Φ_F on the interval locus. Restrict to the open intersection of A, the source of e and the inverse image of the domain of β. There $\Phi_F = \beta \circ e$, which is smooth in the native total atlas. These neighborhoods cover A, so Φ_F is smooth. Its parameter projection is smooth by the retained total-atlas construction. Their product is therefore smooth into $U_F \times K_0$.

At (s, y), choose one real spatial inverse germ. In the interior case it gives the original inverse as $(v, z) \mapsto o_{\mathrm{tot}}(v, \gamma(v, z))$ on a relative-open neighborhood. Both the restricted real germ and the retained total interior insertion are smooth, so this inverse is smooth there. In the cylindrical case write the inverse as $(v, z) \mapsto q_{\mathrm{tot}}(v, -\gamma_r(v, z), \gamma_\partial(v, z))$. The strict radial inequality keeps this smooth germ inside the original open cylinder band. The retained total cylinder insertion is smooth, so its composition is smooth on that neighborhood. The all-point inverse-germ theorem supplies one of these two cases everywhere. Thus the actual inverse is jointly smooth throughout the relative-open product. The two-sided inverse equations already proved give the claimed diffeomorphism. Its maps were never changed, so all their compact, cylinder and strict slice identities are retained. □

Lemma A.735 (Original positive-ribbon bundles choose bidirectional local completion comparisons)**.** *Let $n > 1$. Supply the original compact ambient and ribbon bundles of dimensions $2n$ and $2n-2$, their covariant primitives, original full ribbon data and positive witnesses, and the original local Liouville and pair laws. There exist a compatible global ambient chart g, one completed pair deformation $\mathcal{Q}$, and native ambient maps e_s with the exact original compact primitive pullbacks and spatial extension germs, chosen before every original chart c and $t \in U_c$.*

For each such c, t, the retained original local completion F has the actual forward total map Φ_F with real original-atlas germs and the actual inverse T_F^{-1} with real spatial patch germs at every point. There is a joint native smooth diffeomorphism whose underlying equivalence is exactly $z \mapsto (\pi_F(z), \Phi_F(z))$, with this same original inverse. Writing $B_s = \Phi_F \circ \iota_s$, on every whole slice

$$B_s^* \ell_s = L_s^F, \qquad (B_s^{-1})^* L_s^F = \ell_s, \qquad \ell_s = p_s^* \lambda_{K_s} \qquad (s \in U_F).$$

All comparisons use the same chosen deformation and varying primitive. The completed comparison maps and their regularity are conclusions; correspondence with independently prescribed sections is not asserted.

Proof. Apply the original positive-ribbon bundle construction once. Retain its compatible global chart, completed deformation, native maps, real jointly smooth native family R, compact and boundary presentation squares and original primitive laws. Now fix arbitrary c, t and construct the original local completion F from its ambient primitive and local Liouville laws. The forward real-germ theorem applies to these retained native data and gives the exact original-atlas germs of Φ_F. The all-point inverse-germ theorem applies to the same R and boundary squares and gives the real spatial germs of its actual two-sided inverse. Applying the native diffeomorphism lemma to these produced germs gives a jointly smooth equivalence whose underlying maps are unchanged.

The forward slice primitive identity is the retained local deformation comparison identity. Its strict inverse has the inverse primitive identity: pulling the forward identity back by B_s^{-1} gives $(B_s^{-1})^* L_s^F = (B_s \circ B_s^{-1})^* \ell_s = \ell_s$. All choices of $g, \mathcal{Q}, e_s$ preceded c, t and are unchanged through these applications. This proves the stated simultaneous quantifier order and both primitive equations. □

Lemma A.736 (Original finite-ribbon bundles choose bidirectional local completion comparisons)**.** *Let the original compact ambient bundle have dimension two and its ribbon bundle dimension zero. Supply the covariant ambient primitive and original local Liouville laws, original boundary-target*

ribbon maps injective on every chart overlap, and original positive witnesses. An arbitrary native ribbon one-form family is allowed. There exist one compatible global ambient chart g, one completed rank-one pair deformation $\mathcal{Q}$, and native ambient maps with the exact original primitive and spatial extension laws, chosen before every original local ambient chart and parameter. For every resulting retained local completion F, its actual forward total map has real original-atlas germs and its actual inverse has real spatial patch germs at every completed point. There is a joint native smooth diffeomorphism whose underlying equivalence is exactly $z \mapsto (\pi_F(z), \Phi_F(z))$, with the same original inverse. The same whole-slice identities

$$B_s^* \ell_s = L_s^F, \qquad (B_s^{-1})^* L_s^F = \ell_s \qquad (s \in U_F)$$

hold for that same deformation's varying primitive. No original-bundle section correspondence or endpoint class homotopy is part of this assertion.

Proof. Every ribbon tangent space is zero, so its arbitrary one-form family is zero. Apply the original finite-ribbon bundle construction once to the given ambient primitive, ribbon injectivity and witness data. Retain its compatible ambient chart, completed deformation, native ambient and boundary maps, jointly smooth real native family R, compact and boundary presentation squares and exact primitive laws. For any subsequently selected original ambient chart and parameter, construct F from the original ambient primitive and local Liouville laws. Apply the forward total real-germ theorem and the all-point inverse-germ theorem to these same retained maps and squares. The native diffeomorphism lemma applies to these produced germs and gives a jointly smooth equivalence with exactly these same maps. Their restrictions are the same strict slice comparisons B_s and B_s^{-1}. The local deformation comparison gives the first primitive identity, and its inverse primitive equation gives the second. The global choices were made before the local chart, establishing the claimed order. The zero-dimensional argument also permits an empty ribbon. □

Lemma A.737 (Original finite-ribbon bundle data choose one ambient completion for every local comparison)**.** *Let the ambient compact bundle have dimension two and the ribbon bundle dimension zero. Supply the original ambient primitive and its local Liouville laws, the original boundary-target ribbon maps with their injectivity on every original chart overlap, and the original positive witness. Allow any native one-form family on the ribbon. There exist a compatible global ambient chart g, one completed rank-one pair deformation $\mathcal{Q}$, and native ambient diffeomorphisms $e_s : M \to D_s$ with the original compact primitive pullback identities and spatial extension germs. These choices precede every original local ambient chart c and every $t \in U_c$.*

For each such c, t, retain the original local completion family F. Its actual total comparison $\Phi_F : \pi_F^{-1}(U_F) \to K_0$ has target-valued smooth real extension germs in the original total atlas, including both interval endpoints. On every entire completed slice,

$$(\Phi_F \circ \iota_s)^* \ell_s = L_s^F, \qquad \ell_s = p_s^* \lambda_{K_s} \qquad (s \in U_F).$$

The compatible chart, completed deformation and varying primitive are identical for all local chart choices. This assertion supplies forward comparison germs and the displayed whole-slice primitive identity; whole completed-bundle and prescribed-section correspondence are separate assertions.

Proof. Every tangent space of the zero-dimensional ribbon is zero, so its arbitrary one-form family is identically zero. Apply the original finite-ribbon bundle completed-family construction to the original ambient primitive, ribbon maps, injectivity and positive witness. Retain its compatible global ambient chart, completed deformation, native ambient and boundary maps, jointly smooth real native family R_s, compact presentation square and boundary presentation coherence. The

retained compact primitive identity and spatial extension germs are those asserted here. These choices are made once.

Now fix an original ambient chart c and $t \in U_c$. The original ambient primitive and its local Liouville laws construct the local completion family F. Apply the local total-map lemma to the same retained global chart and completed deformation. Its exact full-slice restriction is the strict local slice comparison, which proves the displayed primitive equation for every $s \in U_F$. The retained R_s and native compact and boundary squares satisfy the hypotheses of the total real-germ lemma, so that lemma gives the asserted real original-atlas germs of this actual map. No inverse or section extension is introduced. Since c, t were arbitrary after the global choices, the stated quantifier order follows. The argument also applies to an empty ribbon. □

Lemma A.738 (A positive end strip detects smoothness on the retained total space)**.** *Let X be a retained native total completion space, $S \subset I$ relatively open, and N a smooth manifold with corners. Let $f : X \to N$. If f is jointly smooth on the interior insertion over S and on the cylinder insertion over S for inward radii $r < \varepsilon$, where $\varepsilon > 0$, then f is smooth over the full parameter preimage of S. The original collar width is retained.*

Proof. The interior and cylinder insertions cover X. The interior composite is smooth by hypothesis. For a cylinder point with $r < \varepsilon$, the second hypothesis gives smoothness. Every remaining cylinder point has $r \geq \varepsilon > 0$, and hence lies in the positive collar overlap. On a neighborhood of that point its cylinder insertion is the interior insertion composed with the retained joint collar map. The latter has smooth mixed-coordinate germs by the native total-atlas construction. The interior composite is smooth on that neighborhood, so their composition is smooth there. Restricting to S preserves these neighborhoods because S is relatively open. Thus the full cylinder composite is smooth, and the two-patch criterion gives the conclusion. □

Lemma A.739 (Joint smoothness of the compact-determined local overlaps)**.** *Let F, G be retained local total completions with relatively open base sets U_F, U_G, and let $S = U_F \cap U_G$. Their canonical slice overlaps induce a parameter-preserving smooth bijection*

$$\mathcal{G}_{F,G} : X_F|_S \longrightarrow X_G|_S, \qquad (s, z) \longmapsto (s, G_{F,G}(s)(z)).$$

Its inverse is the reversed local overlap, and is smooth in the same native total atlases. The map preserves the original compact coordinate changes and the signed-cylinder formulas on their common strips. This applies in particular to the local families constructed directly from the original compact primitives and Liouville laws. No completed-overlap smoothness is an input.

Proof. The actual fiber identifications define the map from the already constructed canonical slice overlaps. Its parameter coordinate is s. On the interior patch its compact formula is the original native coordinate transition followed by the target interior insertion. The original transition germs make this composite jointly smooth. Take $\varepsilon = \delta_G > 0$. On the source cylinder strip $r < \varepsilon$ the retained overlap formula is

$$(s, r, b) \longmapsto q_s^G(r, \tau_{F,G}^{\partial}(s)(b)).$$

The boundary transition has the original joint smooth boundary germs, and the target cylinder insertion is smooth. Hence this composite is smooth. The end-strip criterion proves smoothness on the full source preimage of S. Apply the same argument with F, G reversed. Slice compact uniqueness identifies the two composites with the identity: each is strict and fixes every original compact point. The actual fiber identifications therefore give inverse identities on the total spaces. This proves all assertions. □

Lemma A.740 (A smooth total covector extending the exact slice primitives)**.** *For every retained local total completion X_F, there is a total one-form Ω with smooth tangent evaluation in the native total atlas, whose pullback by every full spatial slice insertion ι_s is exactly the retained primitive:*

$$\iota_s^*\Omega = L_s \qquad (s \in I).$$

For a family constructed from the original compact bundle primitives, its further pullback to the full native compact fibre equals α_s^c for every $s \in U_F$. The parameter component of Ω is unrestricted; no splitting of the total tangent bundle is prescribed.

Proof. Fix a point in the interior patch. The original joint native primitive germ gives smooth coefficients on a real product neighborhood of its parameter and spatial coordinates. Extend these coefficients to a total covector by assigning zero to the parameter direction. Transport it through the actual open interior patch and its smooth local inverse. The chain rule and the interior primitive equation show that its pullback to each spatial slice on this neighborhood is L_s. At a cylinder-patch point use instead the jointly smooth full-cylinder coefficients of $e^{-r}\pi_{\partial M}^*(\beta_s|_{\partial M})$. Assign zero to the parameter direction in cylinder coordinates and transport through that patch inverse. Its slice pullback is again L_s by the full-cylinder primitive equation. These constructions include parameter endpoints using the given real coefficient germs. Thus every point has a local smooth total covector with the required full slice restriction.

The finite-dimensional Hausdorff second-countable total manifold admits a smooth locally finite partition of unity (ρ_a) subordinate to these neighborhoods. Let Ω_a be the respective local covectors. Extend $\rho_a\Omega_a$ by zero off its neighborhood; it is smooth because the support of ρ_a is contained there. Define $\Omega = \sum_a \rho_a\Omega_a$. Local finiteness makes this a smooth total one-form. For x in the s-slice and v in its full spatial tangent space, every nonzero term satisfies $\Omega_a(d\iota_s(v)) = L_s(v)$. Therefore

$$\Omega(d\iota_s(v)) = \sum_a \rho_a(x)L_s(v) = L_s(v).$$

This proves the exact full slice pullback. On U_F, the retained native primitive equation is $o_s^*L_s = \alpha_s^c$. Applying that identity to the full slice restriction gives the asserted original compact pullback. No agreement between local parameter components is required by this calculation. □

Lemma A.741 (The reference slice comparisons agree on original overlaps)**.** *Retain one compatible global compact chart g, one completed pair deformation $\mathcal{Q}$, its native maps e_s, and its reference primitive $\ell_s = p_s^*\lambda_{K_s}$. Let F^c, F^d be two retained original local completions of that same compact bundle and primitive. For every $s \in U_c \cap U_d$, their canonical completed overlap $G_{c,d}(s) : X_s^c \to X_s^d$ and the strict reference comparisons $B_s^c : X_s^c \to K_0$, $B_s^d : X_s^d \to K_0$ satisfy*

$$B_s^d \circ G_{c,d}(s) = B_s^c, \qquad G_{c,d}(s) \circ (B_s^c)^{-1} = (B_s^d)^{-1}.$$

The identities use the original completed slices and their entire compact carriers, with the same reference primitive and deformation.

Proof. Set $A_s^c = p_s \circ B_s^c$ and $A_s^d = p_s \circ B_s^d$. Both $A_s^d \circ G_{c,d}(s)$ and A_s^c pull back the actual complete endpoint primitive λ_{K_s} to L_s^c, by the strict endpoint comparison, overlap identity and pullback functoriality. On a compact point $o_s^c(x)$ their values are respectively

$$\kappa_s e_s\big(\tau_{d,g}(s)(\tau_{c,d}(s)(x))\big), \qquad \kappa_s e_s(\tau_{c,g}(s)(x)).$$

The original compact chart transitions obey $\tau_{d,g}(s) \circ \tau_{c,d}(s) = \tau_{c,g}(s)$: each side sends the c-coordinate of a point to the g-coordinate of that same point. Thus these two strict maps agree on the

entire compact carrier. Completion compact uniqueness, applied with complete target (K_s, λ_{K_s}), identifies them on the whole completed slice. Compose with p_s^{-1} to obtain $B_s^d \circ G_{c,d}(s) = B_s^c$. Compose this equality with $(B_s^c)^{-1}$, then with $(B_s^d)^{-1}$, to obtain the inverse equality. □

Lemma A.742 (The actual total reference comparisons agree on original overlaps)**.** *Under the preceding hypotheses, write $\Omega_{c,d}$ for the canonical parameter-preserving total overlap, restricted to $U_c \cap U_d$. Write Φ_c, Φ_d for the actual total reference maps and $T_c(z) = (\pi_c(z), \Phi_c(z))$, $T_d(z) = (\pi_d(z), \Phi_d(z))$. On their actual common base they satisfy*

$$\Phi_d \circ \Omega_{c,d} = \Phi_c, \qquad T_d \circ \Omega_{c,d} = T_c, \qquad \Omega_{c,d} \circ T_c^{-1} = T_d^{-1}.$$

No comparison outside either original local base is required.

Proof. Let z be in the original c total space over the common base and set $s = \pi_c(z)$. The actual fibre bijection gives a unique x with $z = \iota_s^c(x)$. The retained total overlap is $\Omega_{c,d}(z) = \iota_s^d(G_{c,d}(s)(x))$. Consequently the two total slice restriction identities and slice coherence give

$$\Phi_d(\Omega_{c,d}(z)) = B_s^d(G_{c,d}(s)(x)) = B_s^c(x) = \Phi_c(z).$$

The overlap retains parameter s, giving the equality of T_d and T_c. For (s, y) in the common-base product, the original inverse formula and inverse slice coherence give

$$\Omega_{c,d}(T_c^{-1}(s, y)) = \iota_s^d(G_{c,d}(s)((B_s^c)^{-1}(y))) = \iota_s^d((B_s^d)^{-1}(y)) = T_d^{-1}(s, y).$$

All points used have their parameter in both original local bases. □

Definition A.743 (Preselected pointwise canonical-square data on the original completed atlas)**.** *Fix the original compact bundle, its primitive data and local Liouville laws, hence its retained local completed families X_s^a over U_a. Pointwise canonical-square data consist, for every a and $s \in U_a$, of a smooth almost complex field J_s^a compatible with dL_s^a and an everywhere nonzero full tensor $\mathbf{T}_s^a$ in the squared canonical line for J_s^a. The tensor records its value on every pair of n-tuples of real tangent vectors, rather than only a scalar coefficient in one coframe. For the canonical completed overlaps $G_{a,b}(s)$, require*

$$J_s^a = G_{a,b}(s)^* J_s^b, \qquad \mathbf{T}_s^a = G_{a,b}(s)^* \mathbf{T}_s^b \qquad (s \in U_a \cap U_b).$$

Here $f^*J = (df)^{-1} \circ J \circ df$, and tensor pullback applies df in every argument. These data are chosen before a compatible global chart or completed pair deformation. This pointwise definition imposes no spatial smoothness on the tensor and no joint parameter regularity; the smooth section and homotopy conditions of Definition 2.4 remain additional requirements.

Lemma A.744 (Chart-independent pointwise descent and exact recovery)**.** *Supply the preselected pointwise data of the preceding definition. Afterward retain a compatible global chart and one completed deformation $\mathcal{Q}$ with its strict reference comparisons $B_s^a : X_s^a \to K_0$. There exist uniquely determined pointwise fields J_s and full tensors $\mathbf{T}_s$ on K_0 such that, for every original local family a and every $s \in U_a$,*

$$J_s = ((B_s^a)^{-1})^* J_s^a, \qquad \mathbf{T}_s = ((B_s^a)^{-1})^* \mathbf{T}_s^a.$$

Their exact forward recovery equations are

$$(B_s^a)^* J_s = J_s^a, \qquad (B_s^a)^* \mathbf{T}_s = \mathbf{T}_s^a.$$

Every original family is retained on its own base. In particular the recovery equations hold at parameter 0 or 1 whenever that endpoint belongs to the original base. No smoothness or endpoint homotopy is asserted.

Proof. Each parameter s belongs to an original compact chart. Center the retained local construction at s in that chart; its base contains s. Choose one such family a and define the fields by the displayed inverse pullbacks. For another local family b containing s, inverse comparison coherence gives $G_{a,b}(s) \circ (B_s^a)^{-1} = (B_s^b)^{-1}$. The two original covariance equations and pullback functoriality give

$$((B_s^a)^{-1})^* J_s^a = ((B_s^a)^{-1})^* G_{a,b}(s)^* J_s^b = ((B_s^b)^{-1})^* J_s^b,$$
$$((B_s^a)^{-1})^* \mathbf{T}_s^a = ((B_s^a)^{-1})^* G_{a,b}(s)^* \mathbf{T}_s^b = ((B_s^b)^{-1})^* \mathbf{T}_s^b.$$

For complex fields, functoriality follows by composing the inverse differentials; for full tensors it follows by composing the differential in each argument. Thus the definitions are independent of the chosen family. Pulling them back by B_s^a and using $(B_s^a)^{-1} \circ B_s^a = \mathrm{id}$ proves exact recovery. Conversely, any fields obeying recovery for one covering local family are determined by its inverse pullback, proving uniqueness. Each argument uses only membership in the actual base, so it also applies to both endpoints. □

Lemma A.745 (Pointwise descent preserves compatibility and the nonzero full canonical tensor)**.** *The descended J_s is compatible with $d\ell_s$ on the same reference completion K_0, and $\mathbf{T}_s$ is an everywhere nonzero full tensor in its squared canonical line. These are pointwise properties; they do not assert a smooth Maslov family on the completed deformation.*

Proof. Fix s and any covering local family a, and set $f = (B_s^a)^{-1}$. The pullback $J_s = f^* J_s^a$ is spatially smooth: in smooth local coordinates its coefficients are the products of the smooth coefficients of J_s^a, the derivative of f, and the inverse derivative. The latter is smooth because f is a diffeomorphism and its derivative is invertible at every point. The strict primitive identity implies $f^* dL_s^a = d\ell_s$. At $y \in K_0$, df_y is a real linear isomorphism and $J_s(y) = (df_y)^{-1} J_s^a(f(y)) df_y$. Therefore $J_s^2 = -\mathrm{id}$. For tangent vectors v, w at y, compatibility on the original slice and the two-form pullback give

$$d\ell_s(J_s v, J_s w) = dL_s^a(J_s^a df(v), J_s^a df(w)) = dL_s^a(df(v), df(w)) = d\ell_s(v, w).$$

If $v \neq 0$, then $df(v) \neq 0$, and the same pullback gives $d\ell_s(v, J_s v) = dL_s^a(df(v), J_s^a df(v)) > 0$. Thus J_s is compatible. Since $df \circ J_s = J_s^a \circ df$, pullback preserves the complex linear and alternating identities in each canonical factor, so $\mathbf{T}_s = f^* \mathbf{T}_s^a$ belongs to the full squared canonical line of J_s. At any y, the nonzero tensor $\mathbf{T}_s^a(f(y))$ has some pair of n-tuples on which its value is nonzero. Apply $(df_y)^{-1}$ to every vector in these tuples. The pullback evaluation of $\mathbf{T}_s(y)$ is that same nonzero value. Hence the descended full tensor is everywhere nonzero. □

Corollary A.746 (Pointwise recovery in the actual canonical tensor-product square). For every s the descended full tensor is the natural evaluation of a unique pointwise nowhere-vanishing section τ_s of $K_{J_s} \otimes_{\mathbb{C}} K_{J_s}$ on the same reference completion. Write m_J for evaluation of a tensor-product canonical square as a full two-block tensor. Then

$$m_{J_s}(\tau_s) = \mathbf{T}_s, \qquad (B_s^a)^* m_{J_s}(\tau_s) = \mathbf{T}_s^a \qquad (s \in U_a).$$

No smoothness of τ_s or of its parameter dependence is asserted.

Proof. At each point choose a complex basis of the tangent space for J_s, and let η be its nonzero dual top exterior form. The canonical line has basis η, so its tensor-product square has basis $\eta \otimes \eta$. A full two-block tensor alternating and complex linear in each block is determined by its value on two copies of that basis: each block's value on any tuple is the determinant of its coordinates

times the value on the chosen basis. Hence the full squared canonical tensor space has basis $(v, w) \mapsto \eta(v)\eta(w)$. Natural evaluation sends $\eta \otimes \eta$ to this nonzero basis vector, and is therefore a complex linear isomorphism.

Define τ_s pointwise by the inverse of this natural evaluation applied to $\mathbf{T}_s$. The isomorphism makes its value unique and independent of the auxiliary complex basis. Since $\mathbf{T}_s$ is nonzero everywhere, so is τ_s. Its natural evaluation is exactly $\mathbf{T}_s$; applying the preceding exact recovery equation proves the displayed pullback on every original local base. □

Lemma A.747 (The actual inverse germs have invertible spatial derivatives)**.** *Retain an original local family F, its reference comparison B_s, and an actual real spatial inverse germ at (s, y), with $s \in U_F$. In the interior case let $\gamma(v, z) \in M^\circ$ satisfy $T_F^{-1}(v, z) = o_{\mathrm{tot}}(v, \gamma(v, z))$ on all retained inputs in its real open domain. In the end case let $\gamma(v, z) = (\gamma_\partial(v, z), \gamma_r(v, z))$ satisfy $-\gamma_r < \delta_F$ and*

$$T_F^{-1}(v, z) = q_{\mathrm{tot}}(v, -\gamma_r(v, z), \gamma_\partial(v, z))$$

there. Choose any smooth spatial chart at $\gamma(s, y)$, using a boundary chart and the real radius in the end case. The spatial differential at y of $z \mapsto \gamma(s, z)$ in that chart is a linear isomorphism. The assertion concerns the retained parameter; it does not assert regularity of prescribed sections.

Proof. Restrict the real germ domain to parameter s. This gives an open neighborhood of y in K_0. The actual inverse identity and the slice insertion formula show in the interior case that

$$(B_s)^{-1}(z) = o_s(\gamma(s, z))$$

on that neighborhood. The compact interior insertion o_s is a local smooth diffeomorphism onto its open image in the completed slice. Since B_s is a smooth diffeomorphism, after restricting to the chosen interior chart this equation identifies $\gamma(s, \cdot)$ with the local smooth inverse of $B_s \circ o_s$. Its differential is therefore a linear isomorphism, by differentiating both local inverse identities.

In the end case the same slice insertion formula gives $(B_s)^{-1}(z) = q_s(-\gamma_r(s, z), \gamma_\partial(s, z))$. The signed-cylinder insertion q_s is a local smooth diffeomorphism on its open band, including the seam and each outward radius. The map $(b, r) \mapsto (-r, b)$ is a linear coordinate isomorphism between the radius-last and signed-time-first charts. Hence the displayed identity identifies $\gamma(s, \cdot)$ in the chosen boundary–radius chart with the local smooth inverse of that coordinate change followed by $B_s \circ q_s$. Differentiating the two inverse equations again shows that its spatial differential is a linear isomorphism. □

Corollary A.748 (Real inverse patch germs with invertible spatial derivatives). Under the hypotheses of the actual all-point inverse-germ lemma, for every $s \in U_F$ and $y \in K_0$ there is a real open neighborhood U of (s, y) and a smooth actual spatial germ γ of T_F^{-1} of either the interior or boundary–radius kind, with one fixed original spatial chart whose source contains $\gamma(U)$. In that fixed chart the spatial differential of $z \mapsto \gamma(v, z)$ is a linear isomorphism at every real $(v, z) \in U$. The exact original inverse equation still holds at every retained interval parameter in U. In the boundary–radius case the original bound $-\gamma_r < \delta_F$ holds throughout U. These germs cover the seam, all outward radii and both interval endpoints. No prescribed section or smooth grading is an input or conclusion.

Proof. At (s, y) obtain an actual interior or boundary–radius inverse germ from the all-point inverse-germ lemma. Choose an original spatial chart at its center value, and restrict its real open domain to the inverse image of that chart's open source. This retains the center, all inverse equations and

the radial bound. By the preceding lemma, the spatial differential of its coordinate expression is invertible at the center.

Choose also a smooth source chart on K_0 around y, and restrict the domain to its source. In these two fixed spatial charts write the germ's coordinate expression as $f(v,\xi)$. Its joint smoothness makes $D_\xi f(v,\xi)$ continuous in (v,ξ). The set of invertible linear maps between these finite-dimensional coordinate spaces is open: after composing with the inverse of the center derivative, sufficiently small perturbations of the identity have an inverse by the convergent geometric series. Thus the inverse image of this open set under $D_\xi f$ gives a real open neighborhood of the center on which that derivative is invertible. On this neighborhood the derivative of the source chart is also an isomorphism, so the spatial differential of the original germ in the fixed target chart is invertible. Restrict the germ to this last neighborhood. All restrictions were made inside its original real open domain; therefore its literal inverse equation at retained parameters and its original signed-radius bound are unchanged. □

A.34 Smooth transport from retained original coframe germs

The following local regularity data express real open extensions of the preselected original completed-atlas fields in native coframes. Their transport is conditional on these source data. They do not construct a Maslov datum from the toric input or assert its prescribed endpoint class.

Definition A.749 (Retained original coframe and scalar germs)**.** *Fix the preselected original fields $(J_s^a, \mathbf{T}_s^a)$ and their retained local bases U_a. At every retained interior chart point and every retained signed-cylinder chart point, in every original spatial chart containing that point, require a real open neighborhood of its parameter and spatial coordinates, a smooth invertible coframe $C(v,\xi) : F \to \mathbb{C}^n$, and a smooth scalar $a(v,\xi)$. Here F is the original spatial coordinate model. In the cylinder case the entire band $-r < \delta_a$ is used, with its original insertion $q_s(-r,b)$; centers need not have nonnegative radius.*

Let c be the original spatial chart and let κ_s be its inverse chart followed by the original slice insertion. At every retained chart point in the real open neighborhood require

$$C(s,\xi)(d\kappa_s)^{-1} J_s^a(\kappa_s(\xi))d\kappa_s = iC(s,\xi),$$
$$\kappa_s^* \mathbf{T}_s^a(\xi) = a(s,\xi)\Gamma_{C(s,\xi)}.$$

The squared determinant generator is

$$\Gamma_C(v_1,\dots,v_n;w_1,\dots,w_n) = \det(Cv_1,\dots,Cv_n)\det(Cw_1,\dots,Cw_n).$$

Tensor matching retains every argument. Retained points mean $s \in U_a$ and the original interior or signed-cylinder chart conditions. The coframe is invertible throughout its real open domain; only the retained locus is required to satisfy the original field equations. At the center the scalar is necessarily nonzero: otherwise the tensor matching equation would contradict the original tensor's nonvanishing. No global chart, reference deformation, transported field or grading occurs in these source data.

Lemma A.750 (Original smooth chart coefficients produce coframe and scalar germs)**.** *Fix preselected original completed-atlas data $(J_s^a, \mathbf{T}_s^a)$. At every retained original interior chart point and every retained original signed-cylinder chart point, suppose there is a real open neighborhood of the parameter and spatial coordinates with smooth extensions $K(z)$ and $T(z)$ of the actual chart-conjugated complex field and the full chart pullback of the tensor. Matching is required only at retained chart points in the local base; in the cylinder it uses the entire original signed band and*

the original insertion. Explicitly, if κ_s is that insertion in spatial coordinates, then on the retained locus

$$K(s,\xi) = (d\kappa_s)^{-1} J_s^a(\kappa_s(\xi)) d\kappa_s, \qquad T(s,\xi) = \kappa_s^* \mathbf{T}_s^a(\xi).$$

Tensor equality holds in every argument. There are no complex-square or canonicality requirements outside this retained locus. Then these original data have smooth invertible coframe germs C and smooth scalar germs a satisfying Definition A.749. The scalar is the computed inverse-coframe coefficient of T. This assertion assumes original real open coefficient extensions; it does not extract them from a separately specified intrinsic section.

Proof. Fix one of the stated chart centers. On its retained locus, $K^2 = -\mathrm{id}$ because the differential of the insertion is invertible and the original J_s^a squares to $-\mathrm{id}$. The same differential pulls the original canonical-square tensor back to a tensor squared canonical for K. Apply the local canonical coframe patch lemma to K, this retained locus and the chosen center. It gives, after shrinking the real open neighborhood, a smooth coframe C invertible everywhere there and satisfying $CK = iC$ on the retained locus.

Define a by evaluating T on two blocks of the vectors $C^{-1}e_1, \ldots, C^{-1}e_n$. Smooth inversion of the invertible coframe and smooth tensor evaluation make a smooth on the whole open neighborhood. On each retained point, inverse-coframe reconstruction gives $T = a\Gamma_C$ in every tensor argument. Substituting the two original chart matching equations gives exactly the complex intertwinement and tensor matching required for the original coframe and scalar germs. At the center the original tensor is nonzero; its pullback by the invertible $d\kappa_s$ is nonzero, and reconstruction implies $a \neq 0$ there. Apply this construction at every retained interior and signed-cylinder chart center. The neighborhoods are real open, including at interval endpoints; only the retained-locus equations have been used. Thus all required original germs have been produced on their original bases and full signed bands. □

Lemma A.751 (Original local smooth canonical-square sections give the coefficient extensions)**.** *Fix the original completed-atlas Maslov data. At every retained original interior and full signed-cylinder chart center, supply a real open smooth extension K of the actual chart-conjugated complex field, with matching on the exact retained chart locus A. Supply a corrected native canonical-square patch, with a fixed real linear normalization $P : \mathbb{R}^{2n} \to F$, whose complex structure L agrees with $P^{-1}KP$ on A. Supply an actual smooth section σ of this patch's canonical square, nonzero at the center, satisfying the full original tensor matching on A:*

$$m_L(\sigma(s,\xi)) = P^*(\kappa_s^* \mathbf{T}_s^a)(\xi).$$

Here m_L is the natural evaluation of the canonical tensor-product square on two blocks of tangent arguments. The patch is the local correction of the supplied K; no equation $K^2 = -\mathrm{id}$ is required outside A. Then real open smooth full-tensor coefficient extensions of the original data exist, and hence their coframe and scalar germs exist. The assertion assumes original local native section germs and their matching. Extracting these germs from an independently specified section on a whole intrinsic completed bundle is a separate question.

Proof. At a fixed center put $T = (P^{-1})^* m_L(\sigma)$ on the given real open patch. The smooth native section evaluation lemma makes T smooth in all real parameter and spatial coordinates. Its full normalized identity is $P^*T = m_L(\sigma)$. At each retained chart point the assumed matching therefore gives $P^*T = P^*\kappa_s^* \mathbf{T}_s^a$. Pulling this equality back by P^{-1} yields $T = \kappa_s^* \mathbf{T}_s^a$ in every tensor argument. The supplied K already matches the original chart-conjugated complex field there. Thus K, T are precisely the real open smooth extensions required by the original coefficient-germ producer. Apply

that producer at every interior and full signed-cylinder center to construct the invertible coframes and their computed smooth scalars. The original retained chart loci, local bases, full signed bands and interval endpoints are preserved throughout these equations. □

Corollary A.752 (Original coefficient extensions grade the same realized pair bundle). Assume the original pair-bundle hypotheses and original preselected completed-atlas Maslov data of Lemma A.756, with real open smooth complex-field and full-tensor coefficient extensions as in the preceding lemma. The pair bundle admits the same simultaneous geometric and graded realization of Corollary A.760, including recovery in every original completed chart and retention of independently supplied ambient and ribbon markings on those same realized completions. Both positive-dimensional and finite ribbons are allowed, including empty finite ribbons. No separately supplied coframe or scalar germs are needed. The original coefficient extensions, markings and Maslov data remain hypotheses; their existence from toric input or from an independently given whole intrinsic completed bundle is not asserted.

Proof. The preceding lemma constructs the original coframe and scalar germs from the given smooth coefficient extensions. Insert those germs into the original pair-bundle realization theorem, retaining its one set of geometric choices and its resulting smooth Maslov family. The graded pair-family corollary then equips that same deformation with its grade and descends each independently supplied original marking to its same completed slices. Every chart recovery, compact inclusion and full boundary-orbit equation is the conclusion of that corollary for these very choices. Its positive-dimensional ribbon case applies when the ribbon dimension is positive, and its finite-ribbon case applies in dimension zero without using any ribbon point. Thus the claimed simultaneous realization and marking retention hold also for empty finite ribbons. □

Lemma A.753 (The actual inverse differential factors through the original chart). *Fix a retained original local family a, its comparison B_s^a and an actual real inverse germ γ on a real open domain W. In a fixed original spatial chart c, put $\varphi_s = c \circ \gamma_s$ and let κ_s be its inverse chart followed by the original slice insertion. At a retained $(s, y) \in W$ in this chart, put $H_s = (d\kappa_s)^{-1}$ at the corresponding original slice point. Then*

$$d_y\varphi_s = H_s \circ d_y(B_s^a)^{-1}.$$

For the signed cylinder, H_s includes the derivative of $(a, b) \mapsto (b, -a)$ and the inverse derivative of the original cylinder insertion. The formula uses the whole signed band and every retained interval parameter, including either endpoint.

Proof. Fix s in the retained local base. The real germ domain has an open spatial slice about y. Restrict this slice so its germ values remain in the fixed spatial chart. The literal inverse equation there reads $(B_s^a)^{-1} = \kappa_s \circ \varphi_s$. The original interior insertion is a local smooth diffeomorphism on the interior; the original cylinder insertion is one on its signed band. The chart is also a local smooth diffeomorphism. Thus $d\kappa_s$ is invertible. Differentiating the displayed local equation in the spatial variable gives $d(B_s^a)^{-1} = d\kappa_s \circ d\varphi_s$. Compose with $(d\kappa_s)^{-1}$ to obtain the assertion.

On the cylinder write $e(a, b) = (b, -a)$. Its inverse converts the boundary–radius coordinates to the original signed-time coordinates, so $\kappa_s = q_s \circ e^{-1} \circ c^{-1}$. Consequently $H_s = dc \circ de \circ (dq_s)^{-1}$ at the relevant points, retaining the radial sign. The same differentiation proves the cylinder formula. Only spatial derivatives at the fixed retained s were taken, so the argument applies without change to $s = 0$ or $s = 1$. □

Lemma A.754 (Actual inverse comparisons give the descended local smooth extensions)**.** *Supply the retained original coframe and scalar germs. Choose afterward a compatible global chart and the same completed deformation $\mathcal{Q}$ and comparisons B_s^a used in pointwise descent. At every (s,y) in $[0,1]\times K_0$, the specific descended fields J_s and $\mathbf{T}_s$ have jointly smooth real local extensions, including at both interval endpoints.*

More precisely, select any original family a containing s and an actual inverse germ in its interior or signed-cylinder chart. After shrinking its real open domain W, let $\varphi_v(z)=c(\gamma(v,z))$ and $D=d_z\varphi_v$. With the supplied source coframe and scalar germ, put

$$\begin{aligned} C_0(v,z)&=C(v,\varphi_v(z)), & B(v,z)&=C_0(v,z)D,\\ a'(v,z)&=a(v,\varphi_v(z)), & J_B(v,z)&=B(v,z)^{-1}iB(v,z),\\ T_B(v,z)&=a'(v,z)\Gamma_{B(v,z)}. \end{aligned}$$

The coframe B is invertible and these fields are smooth on W. The domain can be restricted so that every interval parameter in W belongs to U_a. At every such $(v,z)\in W$,

$$J_B(v,z)=J_v(z), \qquad T_B(v,z)=\mathbf{T}_v(z).$$

Tensor equality retains both complete blocks of tangent arguments, and T_B is nonzero at those interval points. No prescribed smooth reference family or inverse germ is supplied as an extra hypothesis.

Proof. A retained original family contains s by the original atlas cover. Obtain an actual inverse germ at (s,y) from the invertible-real-germ lemma. It supplies a fixed original spatial chart, an invertible spatial derivative on its real open neighborhood, and the literal inverse comparison equation at retained parameters. In that chart choose the supplied source coframe and scalar germ at the inverse image of y. Intersect the inverse-germ domain with the inverse image of this source germ domain under $(v,z)\mapsto(v,c(\gamma(v,z)))$. This is real open and contains (s,y). Since U_a is relatively open in $[0,1]$, choose a real open O with $O\cap[0,1]=U_a$ and further restrict the domain to $O\times K_0$. Call the result W.

The spatial coframe pullback lemma makes B jointly smooth on W. Its factors C_0 and D are invertible, so B is invertible. Smooth matrix inversion makes J_B smooth, and the determinant-square and scalar composition laws make T_B smooth.

At a retained parameter, the literal inverse equation identifies $(B_v^a)^{-1}=\kappa_v\circ\varphi_v$ on the spatial neighborhood. The inverse-differential lemma gives $d(B_v^a)^{-1}=d\kappa_v\circ D$. The original coordinate equations therefore give

$$\begin{aligned} ((B_v^a)^{-1})^*J_v^a&=D^{-1}(C_0^{-1}iC_0)D=J_B,\\ ((B_v^a)^{-1})^*\mathbf{T}_v^a&=D^*(a'\Gamma_{C_0})=a'\Gamma_{C_0D}=T_B. \end{aligned}$$

The first equality follows by inverse differential composition. The second follows by inserting D in every determinant argument. Pointwise descent identifies the left sides with the specific J_v and $\mathbf{T}_v$. In the cylinder case κ_v uses the original signed insertion, so differentiating the same equation retains its radial sign and band bound. Every interval parameter in W lies in the actual local base, so these equations hold also at $v=0$ or $v=1$. At such points the original tensor is nonzero, and the inverse comparison differential is an isomorphism. If its pulled tensor vanished, composing back with the inverse differential in every slot would force the original tensor to vanish. Thus T_B is nonzero at every interval point of W. □

Corollary A.755 (The descended data form a Maslov family on the same deformation). Under the same conditional original coframe-germ hypotheses, the specific descended complex fields and canonical squares form a smooth nowhere-vanishing Maslov family for $d\ell_s$ on the completed ambient of the very same deformation $\mathcal{Q}$. The family retains

$$J_s = ((B_s^a)^{-1})^* J_s^a, \qquad m_{J_s}(\tau_s) = \mathbf{T}_s, \qquad (B_s^a)^* m_{J_s}(\tau_s) = \mathbf{T}_s^a \quad (s \in U_a),$$

including all chart-recovery equations at both endpoints. This supplies smooth representative data; it asserts neither prescribed endpoint homotopy classes nor an arbitrary-source-section correspondence.

Proof. The preceding lemma gives jointly smooth real local extensions of J_s and $\mathbf{T}_s$ at every completed-family point. Exact equality with the descended fields means these germs agree on all retained interval overlaps. The pointwise Maslov laws give compatibility with the same $d\ell_s$, canonicality, and nonvanishing everywhere. In a local coframe the inverse of natural tensor-square evaluation sends a canonical tensor to its scalar coefficient times the square of the top exterior coframe. Both the coefficient and this local basis are smooth, so the unique section τ_s from pointwise canonical-square recovery is jointly smooth. Local sections agree on overlaps because they have the same full tensor image and evaluation is injective. The reconstructed sections are nonzero by the same injectivity. To obtain one real smooth presentation, apply the local-extension family reconstruction lemma to the endomorphism and full tensor germs. Its smooth partition of unity has weights summing to one on the closed interval slab. At each point of that slab every contributing germ has the same descended value, so the weighted sums equal J_s and $\mathbf{T}_s$ there. Compatibility, canonicality and nonvanishing on the interval are therefore unchanged. These real extending fields, with the recovered canonical-square sections, form a Maslov family on $\mathcal{Q}$. Its tensor image is exactly $\mathbf{T}_s$ at every interval parameter, and its all-chart recovery and endpoint equations are precisely the already proved pointwise recovery identities. No geometric deformation has been changed. □

Lemma A.756 (An original pair bundle realizes its smooth ambient Maslov data)**.** *Supply original pair-bundle data in either the positive-dimensional ribbon case of Lemma A.613 or the zero-dimensional ribbon case of Lemma A.615. Independently supply covariant original completed-atlas Maslov data and their retained coframe and scalar germs. There exist compatible global charts, one completed pair deformation $\mathcal{Q}$, its native ambient, ribbon and boundary identifications, and a smooth Maslov family $f = (J_s^f, \mathbf{T}_s^f)$ on its completed ambient.*

The chosen geometric data retain every conclusion of the applicable pair-bundle construction: all original local primitive, ribbon and positive-witness recovery equations, both primitive pullbacks and the boundary-form pullback, the entire embedded ribbon and full embedded core, jointly smooth native ambient and ribbon relabelings, their exact presentation squares, and the applicable native coordinate germs. In dimension zero the full embedded core is the entire embedded ribbon. For the strict comparisons B_s^a of this same $\mathcal{Q}$,

$$J_s^f = J_s, \qquad \mathbf{T}_s^f = \mathbf{T}_s, \qquad (B_s^a)^* J_s^f = J_s^a, \qquad (B_s^a)^* \mathbf{T}_s^f = \mathbf{T}_s^a \quad (s \in U_a).$$

These equations hold for every retained original chart and at both interval endpoints. The original smooth ambient data precede the choice of $\mathcal{Q}$; the conclusion does not take a represented deformation or a smooth reference grading as an extra input. No prescribed marking or endpoint homotopy class is asserted.

Proof. For a positive-dimensional ribbon, apply the original pair-bundle construction once and retain its entire conclusion. In particular keep the chosen global charts, $\mathcal{Q}$, native maps, primitive

equations, full ribbon and core equations, jointly smooth relabelings and presentation squares. Its ambient primitive equation supplies the strict slice comparison input, and its native coordinate germs and coherent ambient and boundary presentation squares supply the inverse-germ regularity input to the smooth Maslov-family corollary. Apply that corollary to the independently given original fields and coframe germs on this exact $\mathcal{Q}$. It constructs f with $J_s^f = J_s$ and $\mathbf{T}_s^f = \mathbf{T}_s$ at every interval parameter. Its all-chart recovery equations give the displayed pullbacks. None of the retained geometric data is replaced.

For a zero-dimensional ribbon, apply its original pair-bundle construction once and retain its entire conclusion, including its arbitrary ribbon one-form family. That family is zero because every ribbon tangent space is zero. The ambient primitive, native coordinate germs and coherent ambient and boundary presentation squares give the same hypotheses for the smooth Maslov-family corollary in ambient rank one. Apply it to this very $\mathcal{Q}$ and the supplied original ambient data. It produces the same all-parameter and all-chart equations, while every compact geometric conclusion remains the one already constructed. In particular the full ribbon-core equality is unchanged. Both constructions and the family recovery include the interval endpoints, completing both cases. □

Remark A.757 (Regularity needed for smooth original-section transport). Pointwise descent and exact recovery do not alone give a smooth section on the completed parameter family. Such transport also requires original native coframes and scalar coefficient germs on real open neighborhoods, agreeing with the prescribed section at the retained parameters of each original local base. Their pullbacks must use the actual inverse comparison and its spatial derivative. The real spatial inverse germs provide the local maps, and Lemma A.747 gives invertibility of their spatial derivatives at retained points. Corollary A.748 retains invertibility on shrunk real neighborhoods around every retained point. This local requirement does not demand a single smooth real inverse on the whole completed family.

Definition A.749 records this local real-extension requirement on the original data. The conditional smooth transport lemma retains their literal interval values and all tensor arguments, and its corollary assembles a Maslov family on the same deformation. Prescribed endpoint class homotopies remain separate data. Independently supplied original completed-atlas markings descend slice by slice by Lemma A.763, retaining their common compact inclusion. A globally glued marked target family requires its own geometric and regularity data.

Remark A.758 (Native total regularity and real parameter germs). The joint overlap and total covector assertions above concern smoothness in the native closed-interval total atlases. They do not assert smooth coefficient extensions on an open real neighborhood of every parameter endpoint. Such extensions for the assembled total covector and completed overlap inverse, as well as descent over a whole completed bundle, remain additional requirements for correspondence with the specified smooth completed-family data.

Lemma A.759 (A smooth Maslov family grades its exact pair deformation)**.** *Let $\mathcal{Q}$ be a represented completed pair deformation, with completion presentation $p_s : K_0 \to K_s$, completed symplectic form $\omega_s = p_s^* d\lambda_{K_s}$, and a smooth Maslov family $f = (J_s^f, \mathbf{T}_s^f)$ for that precise symplectic extension. For every $s \in [0,1]$, define the representative on K_s by*

$$\rho_s = (p_s^{-1})^*(J_s^f, \mathbf{T}_s^f).$$

These are smooth compatible, nowhere-vanishing canonical-square representatives, and they satisfy $p_s^* \rho_s = (J_s^f, \mathbf{T}_s^f)$ throughout the interval. The same $\mathcal{Q}$, its same symplectic extension, and the literal family f form a graded pair deformation from ρ_0 to ρ_1.

Proof. Restriction of f to a fixed s is a smooth compatible complex structure and a smooth nonzero canonical-square section for ω_s on K_0. Naturality of exterior differentiation gives $p_s^* d\lambda_{K_s} = \omega_s$ on the whole completed ambient. Pull this equation back by p_s^{-1} to obtain $(p_s^{-1})^* \omega_s = d\lambda_{K_s}$. Thus inverse presentation pullback transports the restricted complex structure to a compatible structure for $d\lambda_{K_s}$. The differential of p_s^{-1} is an isomorphism at every point, so its induced canonical-square map is an isomorphism as well. It therefore preserves the canonical-square type and nonvanishing of the section. Smoothness follows by smooth pullback of the restricted fields and section. This proves that ρ_s is a Maslov representative.

The chain rule and $p_s^{-1} \circ p_s = \mathrm{id}$ give $p_s^*(p_s^{-1})^* J_s^f = J_s^f$ and $p_s^*(p_s^{-1})^* \mathbf{T}_s^f = \mathbf{T}_s^f$, on every tangent argument of the full tensor. Taking $s = 0, 1$ supplies the two required endpoint equations. The given f is already smooth on the full completed family for the specified symplectic extension. Together with these endpoint equations it is exactly a graded pair deformation with the same underlying $\mathcal{Q}$. □

Corollary A.760 (Original pair-bundle data retain their grade and both supplied markings). Under the original pair-bundle and original smooth Maslov-data hypotheses of Lemma A.756, the same geometric choices and smooth family f admit the graded pair deformation just constructed. They retain every original compact, ribbon, witness, primitive, full-core and presentation conclusion of that lemma in both ribbon dimensions. If $B_s^a : X_s^a \to K_0$ is any retained original reference comparison, the actual endpoint representative ρ_s satisfies

$$(p_s \circ B_s^a)^* \rho_s = (J_s^a, \mathbf{T}_s^a)$$

on the entire original completed slice for every parameter in its base, including zero and one. After these same geometric and grading choices, every independently supplied original ambient completed-atlas marking datum into complete targets Z_s yields markings $q_s : K_s \to Z_s$ satisfying

$$q_s \circ (p_s \circ B_s^a) = q_s^a$$

for every original local slice and every parameter in its base. Their compact restrictions are the original common inclusion, expressed in the retained native compact coordinates, and their end equations use the actual complete Liouville flows at every real radius. Independently, every supplied original ribbon marking datum into complete targets Z_s^F yields markings $q_s^F : K_s^F \to Z_s^F$ on the ribbon completions of this same $\mathcal{Q}$. If $A_s^{F,a} : Y_s^{F,a} \to K_s^F$ is the retained original ribbon comparison and $r_s^{F,a}$ is its supplied marking, then on the entire original completed ribbon slice,

$$q_s^F \circ A_s^{F,a} = r_s^{F,a}.$$

Their compact restrictions recover the original common ribbon inclusion in the retained compact coordinates. They intertwine the actual complete flows at every real time, and in positive ribbon dimension recover every real boundary orbit. In dimension zero, the original completion is the finite ribbon itself, including when it is empty. The geometric deformation and its grade are chosen before either kind of marking datum is supplied. Neither marking construction changes their geometric, full-core or Maslov recovery equations. No joint smooth marked target family, prescribed toric endpoint homotopy class, or comparison between distinct original bundle families is asserted.

Proof. Apply the original pair-bundle Maslov realization once, in the applicable positive-dimensional or zero-dimensional ribbon case. Retain its chosen $\mathcal{Q}$, f, all geometric data and every recovery equation. Apply the preceding lemma to this very $\mathcal{Q}$ and f. It gives ρ_s and a grade

with precisely the retained deformation, symplectic extension and smooth family. For each original local base, pullback functoriality and that lemma's all-parameter recovery give

$$(p_s \circ B_s^a)^* \rho_s = (B_s^a)^* p_s^* \rho_s = (B_s^a)^* (J_s^f, \mathbf{T}_s^f) = (J_s^a, \mathbf{T}_s^a).$$

The final equality is the retained all-chart Maslov recovery. It includes every completed point and both parameter endpoints. The grading construction replaces none of the retained geometric choices, so all their compact, ribbon, witness, primitive and core equations remain exactly those already obtained in either case. Now supply any original ambient marking datum into Z_s. For every s apply the marking descent lemma to the already chosen ambient completion of this $\mathcal{Q}$, its native compact relabeling, and its retained primitive and coordinate-extension laws. It gives q_s with exact recovery through every actual endpoint comparison $p_s \circ B_s^a$. The compact and end recovery corollary gives the specified inclusion and all-real flow equations. The marking input is supplied only after the deformation and grade were chosen, and this construction changes neither. The ambient dimension is $2n > 0$ in both ribbon cases, so the ambient marking construction applies also when the ribbon itself has dimension zero.

For a positive-dimensional ribbon, the retained compact geometric conclusion gives both the full primitive pullback for its e_s and its smooth chart extensions. Apply the original ribbon marking descent to these exact laws and each independently supplied ribbon marking datum. Its all-chart recovery is precisely the displayed equation for $A_s^{\tilde{F},a}$, and its compact and end recovery gives the stated original inclusion and all-real orbit formulas. For a finite ribbon, apply the original zero-dimensional marking descent to the retained compact ribbon diffeomorphisms. Every original primitive is zero, so its strictness requires no added hypothesis. Its chart-independent marking recovers every original finite completion and compact inclusion, and intertwines the complete flows. This proof also applies to an empty ribbon because it makes no choice of a ribbon point. Both constructions use the same already chosen $\mathcal{Q}$ and its exact ribbon completions, and neither alters its grading or any earlier recovery law. □

A.35 Supplied original markings on the same represented completed fibres

Definition A.761 (Markings of the original completed atlas with one compact inclusion)**.** *Retain an original compact bundle $E \to [0,1]$, its original primitives and Liouville laws, and its original local completion families. Write C_s^c for an original chart completion, $\widehat{\alpha}_s^c$ for its primitive, $i_s^c : M \to C_s^c$ for its native compact inclusion, and $\gamma_s^c : M \to E_s$ for its original compact chart inverse. Independently prescribe a complete Liouville manifold (Z_s, λ_s^Z) and a map $i_s : E \to Z_s$ for each $s \in [0,1]$, whose restriction to E_s is the prescribed compact inclusion. Only that restriction is used below. Original completed-atlas markings consist of primitive-preserving smooth diffeomorphisms*

$$q_s^c : C_s^c \longrightarrow Z_s, \qquad (q_s^c)^* \lambda_s^Z = \widehat{\alpha}_s^c, \qquad q_s^c(i_s^c(x)) = i_s(\gamma_s^c(x))$$

for every original compact chart c, every parameter in its base, and every native compact point x. For a retained local completed slice X_s^a constructed in chart c, let $\sigma_s^a : X_s^a \to C_s^c$ be its strict base-slice comparison and define $q_s^a = q_s^c \circ \sigma_s^a$ and $\gamma_s^a = \gamma_s^c$. With L_s^a its primitive and o_s^a its native compact inclusion, this gives

$$(q_s^a)^* \lambda_s^Z = L_s^a, \qquad q_s^a(o_s^a(x)) = i_s(\gamma_s^c(x)).$$

The markings and their common compact inclusion precede any choice of represented pair deformation. This is slice data; it specifies no joint regularity of $s \mapsto q_s^a$ or of the target family.

Lemma A.762 (The common original compact inclusion forces strict marking covariance)**.** *Let $H_{c,d}(s) : C_s^c \to C_s^d$ be the strict original chart transition and $G_{a,b}(s) : X_s^a \to X_s^b$ the canonical strict completed local overlap. On every indicated common base, the markings satisfy, on the entire completed slices,*

$$q_s^d \circ H_{c,d}(s) = q_s^c, \qquad q_s^b \circ G_{a,b}(s) = q_s^a.$$

Thus completed marking covariance is a consequence of the common original compact inclusion, rather than an additional input.

Proof. The chart transition preserves the primitive and sends $i_s^c(x)$ to $i_s^d(\tau_{c,d}(s)(x))$, where $\tau_{c,d}$ is the actual compact coordinate change. Both chart inverses recover the same original point, so $\gamma_s^d(\tau_{c,d}(s)(x)) = \gamma_s^c(x)$. Consequently

$$q_s^d(H_{c,d}(s)(i_s^c(x))) = i_s(\gamma_s^d(\tau_{c,d}(s)(x))) = i_s(\gamma_s^c(x)) = q_s^c(i_s^c(x)).$$

Pullback functoriality makes the composite a primitive-preserving diffeomorphism to Z_s. It agrees with q_s^c on the full compact carrier. Compact uniqueness identifies these strict maps everywhere. The local overlap is exactly $(\sigma_s^b)^{-1} \circ H_{c,d}(s) \circ \sigma_s^a$. Substituting this into $q_s^b \circ G_{a,b}(s)$ and cancelling $\sigma_s^b \circ (\sigma_s^b)^{-1}$ reduces it to $q_s^d \circ H_{c,d}(s) \circ \sigma_s^a = q_s^c \circ \sigma_s^a = q_s^a$, proving the local equation. □

Lemma A.763 (Chart-independent marking descent to the retained pair completion)**.** *Retain one compatible global compact chart g, one represented pair deformation $\mathcal{Q}$, its compact ambient relabelings $e_s : M \to D_s$, and all its strict original slice comparisons $A_s^a : X_s^a \to K_s$, where K_s is the completion of D_s. These are positive-dimensional ambient completions, including when the ribbon has dimension zero. Write $\kappa_s : D_s \to K_s$ for its compact inclusion and set*

$$I_s(y) = i_s(\gamma_s^g(e_s^{-1}(y))) \qquad (y \in D_s).$$

There is a chart-independent inclusion marking $q_s : K_s \to Z_s$ extending this precise compact inclusion. At every parameter in every original local base,

$$q_s = q_s^a \circ (A_s^a)^{-1}, \qquad q_s \circ A_s^a = q_s^a, \qquad q_s^* \lambda_s^Z = \lambda_{K_s}.$$

These identities include all completed points and both parameter endpoints. They retain the same $\mathcal{Q}$ and its actual comparisons.

Proof. Fix s and choose any original local family containing s; such a family exists because the original compact atlas covers the base and each retained local family contains its chosen centre. Define $q_s = q_s^a \circ (A_s^a)^{-1}$. For another family b containing s, slice coherence gives $A_s^b \circ G_{a,b}(s) = A_s^a$. Hence $G_{a,b}(s) \circ (A_s^a)^{-1} = (A_s^b)^{-1}$, and covariance gives

$$q_s^b \circ (A_s^b)^{-1} = q_s^b \circ G_{a,b}(s) \circ (A_s^a)^{-1} = q_s^a \circ (A_s^a)^{-1}.$$

The definition is therefore independent of the local choice, and composition with each A_s^a proves exact whole-slice recovery. Strictness of A_s^a and q_s^a, followed by inverse pullback, gives $q_s^* \lambda_s^Z = \lambda_{K_s}$.

The comparison compact square sends $o_s^a(\tau_{g,a}(s)(x))$ to $\kappa_s(e_s(x))$. The original chart inverses satisfy $\gamma_s^a(\tau_{g,a}(s)(x)) = \gamma_s^g(x)$. Exact recovery and the original marking restriction thus give

$$q_s(\kappa_s(e_s(x))) = i_s(\gamma_s^g(x)) = I_s(e_s(x)).$$

Since e_s is onto D_s, this is the required inclusion equation on its full compact carrier. Primitive preservation intertwines the complete Liouville flows, so the marking is induced by this inclusion also on the cylindrical end. None of the construction restricts s to the interior of the parameter interval, proving the endpoint claims. □

Corollary A.764 (Exact native inclusion and all-real end recovery). For the same descended marking and every $s \in [0,1]$, native compact point x, represented boundary point b and real radius r,

$$q_s(\kappa_s(e_s(x))) = i_s(\gamma_s^g(x)), \qquad q_s(E_{K_s}(b,r)) = \Phi^r_{Z_s}(I_s(b)).$$

Here a boundary point is regarded as its actual compact inclusion in D_s. In particular these recover the supplied original markings at zero and one on the represented endpoint completions.

Proof. The native equation is the compact restriction just proved. The actual end point $E_{K_s}(b,r)$ is $\Phi^r_{K_s}(\kappa_s(b))$ for every real r. The descended marking preserves the primitive, so it intertwines these complete flows. Applying its compact restriction gives

$$q_s(E_{K_s}(b,r)) = \Phi^r_{Z_s}(q_s(\kappa_s(b))) = \Phi^r_{Z_s}(I_s(b)).$$

This uses the full real flow, and hence has no radial restriction. Specializing the same formulas to zero and one proves the endpoint assertion. □

Remark A.765 (Scope of original-atlas marking descent). These are conditional representation results for independently supplied original markings. Their strict overlap equality is derived, and the descended markings belong to the same represented deformation as its recovered geometric and Maslov data. Neither existence of the toric full-fibre markings nor joint smoothness of an independently supplied marked target family follows from this slice construction. Positive-dimensional endpoint descent applies to the ambient completions in both ribbon dimensions. The ribbon completions themselves have the separate positive-dimensional and finite-ribbon descent statements below.

A.36 Supplied original ribbon markings on the same pair deformation

Lemma A.766 (Original positive-dimensional ribbon slices retain their exact endpoint comparison). *Let the original positive-dimensional ribbon bundle have primitives μ_s^a, original Liouville laws, and a compatible global chart g. Retain one completed pair deformation $\mathcal{Q}$, its ribbon domains F_s, their completions K_s^F, and compact ribbon diffeomorphisms $e_s : M \to F_s$. Suppose $e_s^*\mu_{F_s} = \mu_s^g$ and the maps e_s have smooth extensions in the indicated boundary charts. For every original local completion X_s^a, with primitive L_s^a, compact insertion o_s^a and end map E_s^a, there are a strict comparison $A_s^a : X_s^a \to K_s^F$ and a boundary diffeomorphism $\beta_s^a : \partial M \to \partial F_s$ satisfying*

$$\begin{aligned}
(A_s^a)^*\mu_{K_s^F} &= L_s^a, \\
A_s^a(o_s^a(x)) &= \kappa_s^F(e_s(\tau_{a,g}(s)(x))), \\
(A_s^a)^{-1}(\kappa_s^F(e_s(x))) &= o_s^a(\tau_{g,a}(s)(x)), \\
A_s^a(E_s^a(b,r)) &= E_{K_s^F}(\beta_s^a(b), r) \qquad (r \in \mathbb{R}).
\end{aligned}$$

Here τ is the actual original compact chart transition and κ_s^F is the compact inclusion in this very ribbon completion. The comparisons intertwine the complete Liouville flows for all real times. No ribbon domain or completion of $\mathcal{Q}$ is replaced.

Proof. Fix a parameter in the original local base. The full primitive pullback and the assumed smooth chart extensions present μ_s^g on the actual compact domain F_s. Apply the compatible local slice comparison to that presentation and to its already specified completion K_s^F. Its compact equations are exactly the forward and inverse equations above; its boundary square provides β_s^a, and its end square gives the displayed equation at every real radius. Its strict primitive identity

is the first equation. Strict primitive preservation intertwines both complete Liouville fields, and uniqueness of their integral curves then intertwines their flows at every real time. Apply this construction at each retained parameter, including zero and one. It always uses the given F_s and K_s^F, proving the last assertion. □

Lemma A.767 (Original positive-dimensional ribbon markings descend independently of the local chart)**.** *Keep the original ribbon bundle and the exact $\mathcal{Q}$ of the preceding lemma. Independently supply original completed-atlas ribbon markings $q_s^a : X_s^a \to Z_s^F$ with a common compact inclusion $i_s^F : E_s^F \to Z_s^F$ as in Definition A.761. There is an inclusion marking $q_s^F : K_s^F \to Z_s^F$ at every parameter, extending*

$$I_s^F(y) = i_s^F(\gamma_s^g(e_s^{-1}(y))) \qquad (y \in F_s).$$

For every original local base containing s, on its entire completion,

$$q_s^F = q_s^a \circ (A_s^a)^{-1}, \qquad q_s^F \circ A_s^a = q_s^a, \qquad (q_s^F)^* \lambda_{Z_s^F} = \mu_{K_s^F}.$$

Thus the marking is independent of the original local chart. The conclusion holds at both parameter endpoints and assumes no joint smoothness of the family $s \mapsto q_s^F$.

Proof. Apply the chart-independent marking descent lemma to the original ribbon bundle, its strict comparisons from the preceding lemma, and the exact positive-dimensional target completion K_s^F. Its hypotheses are the supplied original marking datum, the compact primitive identity, and the smooth chart extensions already assumed. It gives the stated compact inclusion and the formula $q_s^a \circ (A_s^a)^{-1}$. For two local choices a, b, the strict maps $A_s^b \circ G_{a,b}(s)$ and A_s^a agree on every original compact point: their compact equations and the original coordinate cocycle both give $\kappa_s^F(e_s(\tau_{a,g}(s)(x)))$. Compact uniqueness therefore gives $A_s^b \circ G_{a,b}(s) = A_s^a$. The common original compact inclusion gives $q_s^b \circ G_{a,b}(s) = q_s^a$. Consequently

$$q_s^b \circ (A_s^b)^{-1} = q_s^b \circ G_{a,b}(s) \circ (A_s^a)^{-1} = q_s^a \circ (A_s^a)^{-1}.$$

This proves choice independence and recovery through every local comparison. Strictness and inverse pullback give the primitive equation. Every construction is at an arbitrary retained parameter, including the endpoints, and introduces no joint regularity assertion. □

Corollary A.768 (Original ribbon inclusions and all real boundary orbits are recovered). For the same marking, every native compact point x, every original local boundary point b, and every real r, one has

$$q_s^F(\kappa_s^F(e_s(x))) = i_s^F(\gamma_s^g(x)),$$
$$q_s^F(E_{K_s^F}(\beta_s^a(b), r)) = \Phi^r_{Z_s^F}(i_s^F(\gamma_s^a(b))).$$

Moreover $q_s^F \circ \Phi^r_{K_s^F} = \Phi^r_{Z_s^F} \circ q_s^F$ on the entire ribbon completion.

Proof. The first equation is the compact restriction of the descended marking. Primitive preservation intertwines its complete Liouville flows. The boundary square of A_s^a and its whole-local-marking recovery give $q_s^F(\kappa_s^F(\beta_s^a(b))) = i_s^F(\gamma_s^a(b))$, where the boundary point is regarded as its compact inclusion. The end map is the all-real trajectory through that point, so flow intertwining gives the second equation at every real radius. These arguments also prove the displayed whole-completion flow identity. □

Definition A.769 (Original finite-ribbon markings with a common compact inclusion)**.** *Let $E^F \to [0,1]$ be an original smooth compact zero-dimensional ribbon bundle, with finite model fibre M. Its original completion in any compact chart is M itself. Prescribe complete zero-dimensional Liouville targets Z_s^F and one compact inclusion $i_s^F : E_s^F \to Z_s^F$ at each parameter. Original finite-ribbon marking data are smooth diffeomorphisms $q_s^a : M \to Z_s^F$ in every original compact chart, with*

$$q_s^a(x) = i_s^F(\gamma_s^a(x)) \qquad (x \in M).$$

No primitive equation or overlap covariance is imposed as an additional condition. No joint smoothness of the parameterwise markings is imposed. Empty model fibres and empty targets are allowed.

Lemma A.770 (Original finite-ribbon marking covariance is forced by the compact inclusion)**.** *Every original finite-ribbon chart transition $\tau_{a,b}(s)$ is a smooth diffeomorphism of the entire original zero-dimensional completion, and*

$$q_s^b \circ \tau_{a,b}(s) = q_s^a.$$

In any retained global compact chart g there is a smooth marking $q_s^g : M \to Z_s^F$, independent of the local chart used to obtain it, with $q_s^g(x) = i_s^F(\gamma_s^g(x))$ at every point.

Proof. The compact chart transition is a homeomorphism. A zero-dimensional manifold is discrete, and each of its coordinate neighborhoods is a single point; on such a neighborhood every map is smooth. The transition and its inverse are therefore smooth. The original chart inverses satisfy $\gamma_s^b(\tau_{a,b}(s)(x)) = \gamma_s^a(x)$. Substituting this equation in the defining compact marking equations gives $q_s^b(\tau_{a,b}(s)(x)) = q_s^a(x)$ for every point of the entire completion. Choose any original chart a containing s and set $q_s^g = q_s^a \circ \tau_{g,a}(s)$. The same coordinate equation gives $q_s^g(x) = i_s^F(\gamma_s^g(x))$, so any other choice gives exactly the same map. This also proves its smoothness and bijectivity. The construction is valid on an empty fibre without choosing a point. □

Lemma A.771 (Original finite-ribbon markings descend to the same exact completion)**.** *Retain a completed pair deformation $\mathcal{Q}$ with zero-dimensional ribbons, a global original compact chart g, and its actual compact ribbon diffeomorphisms $e_s : M \to F_s$. Write K_s^F for the ribbon completion of this same $\mathcal{Q}$ and κ_s^F for its compact inclusion. The map $a_s = \kappa_s^F \circ e_s$ is a smooth diffeomorphism onto K_s^F, and each original chart has the exact comparison $A_s^{F,a} = a_s \circ \tau_{a,g}(s)$. Every original finite-ribbon marking datum gives an inclusion marking*

$$q_s^F = q_s^g \circ a_s^{-1} : K_s^F \longrightarrow Z_s^F$$

with, on each entire original completion,

$$q_s^F \circ A_s^{F,a} = q_s^a, \qquad q_s^F(\kappa_s^F(e_s(x))) = i_s^F(\gamma_s^g(x)).$$

It preserves the primitive and intertwines the complete Liouville flows at all real times. The assertions hold at both parameter endpoints and for empty ribbons, with no joint marking-family regularity claim.

Proof. Completion coverage and the empty boundary imply that every completed point is a compact point; the compact inclusion is an embedding and hence a bijection onto the completion. In dimension zero it and its inverse are smooth. Thus its composite with e_s is the required a_s. Define q_s^F by the displayed inverse composite. It is a smooth diffeomorphism, and all one-forms on the zero-dimensional source and target vanish. It therefore preserves their primitives. Its compact restriction is $q_s^g(x) = i_s^F(\gamma_s^g(x))$, which makes it an inclusion marking of the exact K_s^F. Finally,

$$q_s^F \circ A_s^{F,a} = q_s^g \circ a_s^{-1} \circ a_s \circ \tau_{a,g}(s) = q_s^g \circ \tau_{a,g}(s) = q_s^a.$$

The last equation is the compact inclusion covariance just proved. Strict primitive preservation gives complete flow intertwining for all real times. None of the argument uses a ribbon point or excludes either endpoint, proving the remaining assertions. □

Lemma A.772 (Original local bundle junctions retain two independently realized positive-dimensional endpoints)**.** *Independently retain two original positive-dimensional pair-bundle realizations $\mathcal{F},\mathcal{G}$ with their compatible global compact charts, ambient and ribbon carrier maps, full primitive pullbacks, smooth chart extensions, boundary inclusion squares and whole ribbon embedding squares, as in Lemma A.613. Choose parameters s,s' and original local ambient completions $(X,L_X),(Y,L_Y)$ at those parameters. Their boundary models are B_X,B_Y and their end maps are E_X,E_Y. Independently supply a strict smooth diffeomorphism $\Psi:X\to Y$, a smooth boundary diffeomorphism $\psi:B_X\to B_Y$, and a smooth real shift $\sigma:B_X\to\mathbb{R}$ such that*

$$\Psi^*L_Y=L_X,\qquad \Psi(E_X(b,r))=E_Y(\psi(b),r+\sigma(b))$$

for each b and all sufficiently large real r.

Choose original local ribbon charts with compact models M,N, primitives μ,ν and full embeddings $j_X:M\to B_X$, $j_Y:N\to B_Y$ given literally by those original bundle charts. Supply a compact smooth diffeomorphism $h:M\to N$ with local smooth extensions across the boundary, a constant $c>0$, and

$$h^*\nu=c\mu,\qquad \psi\circ j_X=j_Y\circ h.$$

The local ribbon completions and their bases contain the respective chosen parameters. None of these original strict or ribbon junction data refer to a represented endpoint.

Let $S:X\to K_{\mathcal{F},s}$ and $S':Y\to K_{\mathcal{G},s'}$ be the actual original ambient slice comparisons and let $b_S,b_{S'}$ be their actual boundary maps. If $e_{M,s},e_{N,s'}$ are the retained global ribbon carrier maps, put

$$a=e_{M,s}\circ\tau_{M,g_M}(s),\qquad a'=e_{N,s'}\circ\tau_{N,g_N}(s').$$

There is a permitted completion identification between the literal pairs $\mathcal{F}_s$ and $\mathcal{G}_{s'}$. Its ambient map, boundary map, shift and ribbon scale are exactly

$$f=S'\circ\Psi\circ S^{-1},\quad \psi_K=b_{S'}\circ\psi\circ b_S^{-1},$$
$$\sigma_K=\sigma\circ b_S^{-1},\qquad c_K=c.$$

Its compact ribbon diffeomorphism g satisfies

$$g(a(x))=a'(h(x)),\qquad g^{-1}(a'(y))=a(h^{-1}(y)),\qquad g^*\mu_{\mathcal{G},s'}=c\mu_{\mathcal{F},s}.$$

The original-chart primitive, smooth extension and full embedding laws for a,a' are consequences of the retained global realization and original bundle covariance. They are not additional endpoint-junction assumptions. The shift has no sign restriction. This conclusion does not supply the original Ψ,h, a zero-dimensional ribbon junction, a junction grading or a finite itinerary.

Proof. The ambient primitive pullbacks and smooth chart extensions of the retained realizations give the strict original slice comparisons S,S' with

$$S(E_X(b,r))=E_{K_{\mathcal{F},s}}(b_S(b),r),\qquad S'(E_Y(b',r))=E_{K_{\mathcal{G},s'}}(b_{S'}(b'),r)$$

for all real r. Their compact boundary squares identify $b_S,b_{S'}$ with the retained global boundary maps composed with the actual original boundary chart transitions.

The actual original compact ribbon transition followed by the retained carrier map is the smooth diffeomorphism a displayed above. By the chain rule and original primitive covariance,

$$a^*\mu_{\mathcal{F},s} = \tau_{M,g_M}(s)^* e_{M,s}^* \mu_{\mathcal{F},s} = \tau_{M,g_M}(s)^* \mu_s^{g_M} = \mu.$$

The same argument gives $(a')^*\mu_{\mathcal{G},s'} = \nu$. For its chart extensions, the actual strict comparison A_M from the original local ribbon completion satisfies $A_M(o_M(x)) = \kappa_s^F(a(x))$ for every original compact point. Set $v = A_M^{-1} \circ \kappa_s^F$, a smooth map on the retained ribbon extension space. Then $v(a(x)) = o_M(x)$. Fix any original native chart χ near x. Its original insertion has a smooth inverse-coordinate germ q on an open set U containing $o_M(x)$, with $q(o_M(y)) = \chi(y)$ wherever that chart identity is required. The open set $v^{-1}(U)$ contains $a(x)$, the function $q \circ v$ is smooth there, and $(q \circ v)(a(y)) = q(o_M(y)) = \chi(y)$. These are precisely the transported inverse-coordinate germs on the same retained domain. Apply this construction at every native point and then to a' as well.

Original ribbon-map covariance gives

$$j_{\mathcal{F},s}(a(x)) = e_{B,s}(\tau^{\partial}_{X,g_A}(s)(j_X(x))) = b_S(j_X(x)).$$

Indeed substitute the original compact ribbon transition into the retained global embedding equation. The two successive original ribbon coordinate changes cancel by their cocycle, leaving the original chart embedding and the displayed boundary transition. The compact boundary comparison identifies that transition with b_S. The same computation at the target gives $j_{\mathcal{G},s'} \circ a' = b_{S'} \circ j_Y$. These are the full ribbon equations, not restrictions to their cores.

The supplied Ψ, ψ, σ and its eventual end law now satisfy the independent original-completion identification lemma together with these derived compact primitive, chart and whole embedding laws and the supplied h, c. Apply that lemma to the two already fixed endpoint pairs and their exact completions. It constructs the required permitted identification and retains the displayed conjugated ambient and boundary maps, arbitrary real shift, scale, and both directions of the compact ribbon map. Every comparison used the existing $\mathcal{F}_s$ and $\mathcal{G}_{s'}$, so neither endpoint is replaced. □

Lemma A.773 (Original finite-ribbon bundle junctions retain two independently realized endpoints)**.** *Retain two independently realized original pair bundles $\mathcal{F}, \mathcal{G}$ in ambient dimension two, with their original finite ribbon bundles, global compact carrier maps, ambient primitive pullbacks and smooth chart extensions, boundary inclusion squares, and full global ribbon embedding equations, as in Lemma A.615. At chosen parameters s, s', let $(X, L_X), (Y, L_Y)$ be independently selected original local ambient completions, with boundary models B_X, B_Y and end maps E_X, E_Y. Supply a strict smooth diffeomorphism $\Psi : X \to Y$, a smooth boundary diffeomorphism $\psi : B_X \to B_Y$ and a smooth real shift σ with*

$$\Psi^* L_Y = L_X, \qquad \Psi(E_X(b, r)) = E_Y(\psi(b), r + \sigma(b))$$

for each b and all sufficiently large real r. Supply original finite ribbon charts M, N containing the respective parameters, their literal original full embeddings $j_X : M \to B_X$, $j_Y : N \to B_Y$, a smooth diffeomorphism $h : M \to N$, and a constant $c > 0$ with $\psi \circ j_X = j_Y \circ h$ on every original ribbon point. These junction data refer only to the original local completions and original bundle charts. No ribbon primitive or smooth extension germs are added as hypotheses.

Write S, S' for the actual original-to-endpoint ambient comparisons, $b_S, b_{S'}$ for their boundary maps, and put

$$a = e_{M,s} \circ \tau_{M,g_M}(s), \qquad a' = e_{N,s'} \circ \tau_{N,g_N}(s')$$

using the retained finite ribbon carrier maps and actual original compact transitions. There is a permitted completion identification between the literal endpoints $\mathcal{F}_s, \mathcal{G}_{s'}$ with

$$f = S' \circ \Psi \circ S^{-1}, \quad \psi_K = b_{S'} \circ \psi \circ b_S^{-1},$$
$$\sigma_K = \sigma \circ b_S^{-1}, \qquad c_K = c,$$

and with compact ribbon diffeomorphism g satisfying

$$g(a(x)) = a'(h(x)), \qquad g^{-1}(a'(y)) = a(h^{-1}(y)).$$

The full original-chart embedding equations for a, a' are derived from the existing global realization and original bundle covariance. The shift may have either sign. Empty ribbons are allowed; this result asserts no original junction existence, grading or finite itinerary.

Proof. The actual original compact ribbon transitions are homeomorphisms of zero-dimensional manifolds. Such manifolds are discrete, so the transitions and their inverses are smooth, as proved by finite-ribbon marking covariance. Their composites with the retained carrier maps are exactly the smooth diffeomorphisms a, a' above.

Substitute $\tau_{M,g_M}(s)(x)$ in the retained global source embedding equation. Original ribbon-map covariance expresses the result in the original local charts. The two successive source ribbon transitions cancel by their cocycle, giving

$$j_{\mathcal{F},s}(a(x)) = e_{B,s}(\tau^{\partial}_{X,g_A}(s)(j_X(x))).$$

The actual ambient endpoint comparison has boundary map $b_S = e_{B,s} \circ \tau^{\partial}_{X,g_A}(s)$, by its compact boundary square and the retained boundary inclusion law. Thus the last equation is $j_{\mathcal{F},s} \circ a = b_S \circ j_X$ on the whole finite ribbon. The target computation gives $j_{\mathcal{G},s'} \circ a' = b_{S'} \circ j_Y$.

The same ambient endpoint comparisons are strict and satisfy their outward end squares at every real radius. The supplied Ψ, ψ, σ and their eventual end law, together with the derived full ribbon equations and the supplied h, c, therefore satisfy the independent finite-ribbon endpoint identification lemma. Apply it to the two exact retained endpoints. It constructs the permitted identification with precisely the displayed ambient, boundary, shift, scale and both compact ribbon-map equations. Every ribbon tangent space is zero, so its one-form is zero and the required primitive scale identity is automatically $0 = c\,0$. There are no ribbon boundary or chart-extension conditions to add. The argument is valid at every ribbon point without choosing one, and so also proves the assertions when both finite ribbons are empty. □

Lemma A.774 (Original completed-atlas recovery determines the exact endpoint representatives)**.** *Fix one retained completed pair deformation $\mathcal{Q}$ from original pair-bundle data, with its specified presentation $p_s : K_0 \to K_s$. Let X^a_s be a local completed slice constructed from the original Liouville laws in an original compact chart, containing s in its actual local base. Write $A^a_s : X^a_s \to K_s$ for its direct endpoint comparison and $B^a_s : X^a_s \to K_0$ for its reference comparison. Suppose a smooth Maslov family $f = (J^f_s, \mathbf{T}^f_s)$ on this same $\mathcal{Q}$ recovers the preselected original fields:*

$$(B^a_s)^* J^f_s = J^a_s, \qquad (B^a_s)^* \mathbf{T}^f_s = \mathbf{T}^a_s.$$

Let $\rho_s = (p_s^{-1})^* f_s$ be its actual representative on K_s. Then

$$A^a_s = p_s \circ B^a_s, \qquad (A^a_s)^* \rho_s = (J^a_s, \mathbf{T}^a_s).$$

Consequently any original smooth Maslov representative ν on X^a_s whose complex field and full tensor are $J^a_s, \mathbf{T}^a_s$ is exactly $(A^a_s)^* \rho_s$. The conclusion holds for every original local base containing s, including $s = 0$ or $s = 1$.

Proof. The reference comparison is obtained from the direct comparison by $B_s^a = p_s^{-1} \circ A_s^a$, on the entire original completion. Composing with p_s gives the first equality. Pullback composition and the two inverse identities for p_s give

$$(A_s^a)^*(p_s^{-1})^* f_s = (p_s^{-1} \circ A_s^a)^* f_s = (B_s^a)^* f_s = (J_s^a, \mathbf{T}_s^a).$$

For complex fields these equalities conjugate by the differential and its inverse; for full tensors they hold on every pair of n-tuples of tangent vectors. The primitive comparison also gives $(A_s^a)^* d\lambda_{K_s} = dL_s^a$, by exterior differentiation, so both sides are representatives for the same original symplectic form. A Maslov representative is determined by its complex field and its full canonical-square tensor. The asserted equality with ν follows. All identities hold on each original local base, so apply equally at either interval endpoint. □

Lemma A.775 (Two original positive-dimensional pair bundles produce one graded three-step junction)**.** *Supply two original positive-dimensional pair bundles satisfying the original geometric, primitive, ribbon and witness hypotheses of Corollary A.760. Supply their preselected original completed-atlas Maslov data and coframe and scalar germs. In original ambient charts whose local bases contain, respectively,* 1 *and* 0*, supply the strict ambient junction $\Psi : X \to Y$, boundary map ψ and real shift σ of Lemma A.772. Supply its original compact full ribbon diffeomorphism h, constant scale $c > 0$ and whole embedding square; the original ribbon chart primitives and embeddings at* $1, 0$ *are exactly those in that junction, and their local ribbon bases contain the corresponding endpoints. All this data is chosen on the original bundles.*

Supply original Maslov representatives ν_X, ν_Y whose complex fields and full tensors equal the respective original atlas data on X, Y, and a smooth compatible nowhere-vanishing homotopy H from ν_X to $\Psi^*\nu_Y$ for dL_X. There exist two simultaneous graded realizations $\mathcal{F}, \mathcal{G}$ with every original chart covariance, compact primitive, ribbon, witness, full-core and coherent presentation conclusion of that corollary. Their actual families f, g and deformation grades are retained, with their whole-original-chart recovery at every parameter. For every independently supplied original ambient or ribbon marking datum, the corresponding markings exist on these same chosen completions and retain the original compact inclusions and full boundary-orbit equations. Neither realization is chosen again for the marking datum or the junction.

Let $S : X \to K_{\mathcal{F},1}$ and $S' : Y \to K_{\mathcal{G},0}$ be the actual direct comparisons. The exact permitted identification $I : \mathcal{F}_1 \to \mathcal{G}_0$ constructed from the original bundle ribbon data has a graded lift between the actual representatives $\rho_1^{\mathcal{F}}, \rho_0^{\mathcal{G}}$. If H has real-indexed fields $J_\tau, \mathbf{T}_\tau$, this lift's fields are exactly

$$(S^{-1})^* J_\tau, \qquad (S^{-1})^* \mathbf{T}_\tau \qquad (\tau \in \mathbb{R}).$$

Its ambient, boundary and shift maps are $S' \circ \Psi \circ S^{-1}$, $b_{S'} \circ \psi \circ b_S^{-1}$ and $\sigma \circ b_S^{-1}$. The ordered sequence consisting of $\mathcal{F}$, this exact I, and $\mathcal{G}$ is graded by their actual three grades. Its length is three and its vertex list is literally

$$[\mathcal{F}_0, \mathcal{F}_1, \mathcal{G}_0, \mathcal{G}_1].$$

This is a conditional construction from original bundles and an original junction homotopy; it asserts neither their toric existence nor the prescribed endpoint Maslov classes.

Proof. Apply the original graded pair-bundle realization corollary once to each original bundle. Retain both entire conclusions: choose the geometric presentations, pair deformations, Maslov families and their actual grades before constructing any junction. The corollary's marking conclusions already quantify over independently supplied original marking data on these fixed completions, so remain valid without making a further geometric choice.

At the junction parameters, their primitive and whole embedding conclusions agree with the specified original ribbon charts. The original bundle positive-junction lemma therefore constructs an identification I between the very $\mathcal{F}_1, \mathcal{G}_0$ just chosen, with the displayed conjugated maps and the supplied constant ribbon scale and original full ribbon maps. The endpoint recovery lemma gives $\nu_X = S^*\rho_1^{\mathcal{F}}$ and $\nu_Y = (S')^*\rho_0^{\mathcal{G}}$ on the entire original completions. Differentiating the strict primitive equations gives $S^*d\lambda_{K_{\mathcal{F},1}} = dL_X$, $(S')^*d\lambda_{K_{\mathcal{G},0}} = dL_Y$ and $\Psi^*dL_Y = dL_X$. Thus the original native homotopy lift lemma applies to H and this exact I. Pullback by S^{-1} starts at $\rho_1^{\mathcal{F}}$ and ends at $(S'\circ\Psi\circ S^{-1})^*\rho_0^{\mathcal{G}} = I_{\mathrm{amb}}^*\rho_0^{\mathcal{G}}$. It preserves smooth endpoint extension germs, compatibility, canonicality and nonvanishing and has the stated all-real fields.

Start with the empty graded chain at $\rho_0^{\mathcal{F}}$. Append the actual grade of $\mathcal{F}$, the just-constructed grade of I, and the actual grade of $\mathcal{G}$, in this order. The adjacent pairs and their representatives agree literally at $\mathcal{F}_1$ and $\mathcal{G}_0$ by their definitions. The resulting chain therefore has exactly those three arrows, the displayed four vertices and endpoints $\rho_0^{\mathcal{F}}, \rho_1^{\mathcal{G}}$. All original geometric and marking conclusions remain those of the two realizations already fixed. □

Lemma A.776 (Original homotopies grade the retained finite-ribbon bundle junction)**.** *Retain two independently realized original pair bundles in ambient dimension two, with their preselected original smooth Maslov data and resulting families f, g. Supply an original strict ambient junction between local completed slices at parameters s, s' and the original finite ribbon diffeomorphism, positive constant scale and full embedding square of Lemma A.773. Let I be its exact constructed permitted identification and S, S' its direct original ambient slice comparisons. Supply original Maslov representatives ν_X, ν_Y equal in both complex field and full tensor to the preselected original atlas data, and an original smooth compatible nonvanishing homotopy H from ν_X to $\Psi^*\nu_Y$ for dL_X. Then the same I has a graded lift from the actual representative ρ_s of f to the actual representative $\rho'_{s'}$ of g. Its real-indexed complex field and tensor are exactly the pullbacks of those of H by S^{-1}, at every real homotopy parameter. All original ambient, boundary, shift, scale and full ribbon formulas remain those of I. Empty ribbons are allowed; no positive-dimensional ribbon primitive, boundary-extension germs, original homotopy existence, prescribed toric endpoint class or finite itinerary is asserted.*

Proof. The original endpoint recovery lemma gives $\nu_X = S^*\rho_s$ and $\nu_Y = (S')^*\rho'_{s'}$ on the entire original ambient completions. Strict primitive preservation and exterior differentiation give all three symplectic pullback identities used in the positive-dimensional case. Apply the original native homotopy lift lemma to the exact zero-ribbon junction I and these recovered endpoints. Its pullback by S^{-1} starts at ρ_s and ends at $I_{\mathrm{amb}}^*\rho'_{s'}$, by pullback composition and the retained ambient conjugation formula. The invertible differential preserves compatibility and nonvanishing, and smooth pullback preserves the homotopy's endpoint extension germs. Its actual fields are the claimed all-real pullbacks. This construction uses only the ambient homotopy; it requires no ribbon point and therefore applies to empty ribbons. The underlying identification is exactly the previously constructed finite-ribbon identification, retaining its full map formulas. □

Corollary A.777 (Two original finite-ribbon pair bundles produce one graded three-step junction). Supply two original pair bundles in ambient dimension two with their finite ribbon bundles, original primitives, witnesses and full embedding data satisfying the original graded realization hypotheses. Supply their preselected original completed-atlas Maslov data and coframe and scalar germs. At 1 in an original source ambient chart and 0 in an original target ambient chart, supply a strict original junction with its smooth boundary map, arbitrary real shift and eventual end law. The local bases contain those parameters. Supply an original finite ribbon diffeomorphism, a constant $c > 0$ and its full embedding square in original ribbon charts containing $1, 0$. Supply original endpoint Maslov

representatives equal in complex field and full tensor to the original atlas data there, and an original smooth compatible nonvanishing homotopy H from the source representative to the pullback of the target by that strict original map.

There exist two simultaneous original graded realizations $\mathcal{F}, \mathcal{G}$ retaining every original chart covariance, compact, ribbon, witness, full-core, presentation, Maslov recovery and independent marking conclusion of Corollary A.760. Their exact original finite-ribbon junction $I : \mathcal{F}_1 \to \mathcal{G}_0$ is graded by the pullback of H along the inverse direct source comparison, with that same pullback family at every real parameter. The ambient, boundary and shift formulas are those of the original zero-junction lemma. The ordered deformations $\mathcal{F}, \mathcal{G}$ with this I between them form a graded chain of length three whose literal vertices are

$$[\mathcal{F}_0, \mathcal{F}_1, \mathcal{G}_0, \mathcal{G}_1].$$

The actual two deformation grades and junction grade are retained on these same arrows. Every independently supplied original ambient and ribbon marking is realized on these already chosen completions. No ribbon primitive-scale or ribbon boundary-extension assumption is added. Empty ribbons are allowed. This does not construct the original toric data, homotopy or prescribed endpoint class.

Proof. Apply the original graded pair-bundle corollary once to each original finite-ribbon bundle. Retain both full conclusions, including each actual Maslov family and grade and the universal independent marking conclusions on the chosen completions. Their whole embedding and boundary inclusion equations at $1, 0$ are the inputs for the original finite-ribbon junction construction. That construction therefore uses precisely the already selected $\mathcal{F}_1, \mathcal{G}_0$. Apply the retained finite-ribbon graded-junction lemma to these realizations, the recovered families, original endpoint representatives and H. It supplies the exact grade on this very I, with the all-real inverse-comparison pullback family and the original geometric map equations.

Append in order the actual grade of $\mathcal{F}$, this exact junction grade, and the actual grade of $\mathcal{G}$ to the empty graded chain at $\rho_0^{\mathcal{F}}$. The grades meet on their actual endpoint representatives $\rho_1^{\mathcal{F}}$ and $\rho_0^{\mathcal{G}}$; the underlying pair tuples meet literally at $\mathcal{F}_1$ and $\mathcal{G}_0$. Hence this chain has exactly three arrows and the displayed vertex list, ending at $\rho_1^{\mathcal{G}}$. Its original geometric and marking conclusions are still those of the two realizations fixed at the start. Both the finite-ribbon realization and its junction construction allow empty ribbons, and this proof uses no ribbon point, so that case is included. □

A.37 Foundations and intrinsic correspondences

The following list records foundations and correspondences, not additional hypotheses of Target Theorem 4.1. The encoded proposition is separately defined in A.227. Its component summaries are A.96, A.204 and A.348.

- Quotient and endpoint formulas: quotient descent, the literal fan and link, primitive descent and symplectic differential.
- Intrinsic manifolds with boundary: the actual intrinsic boundary atlas and its smooth immersive inclusion in A.13, the constructed local half-space collar in A.14, the local transverse-seam inverse in A.15, the native one-sided collar with smooth inverse germs and its actual integral-curve compatibility in A.18, and the local uniqueness producer A.17. Actual scalar smooth extension germs supply the common coefficient neighborhood in A.19 and the boundary-neighborhood collar with native inverse germs in A.20. Whole-interval uniqueness is A.21.

These results retain explicit smooth extension hypotheses. Compact assembly in A.22 and the direct scalar-extension construction in A.23 give one uniform inward collar with native inverse germs. The native forms of A.30 and their chart recovery in A.31 support the actual exterior derivative in A.35, its smooth local extension in A.36, and its square-zero identity in A.37. A.38 derives the intrinsic contraction field and smooth coefficient germs. The arbitrary defining-function sign lemma A.40 supplies source outwardness in every native chart. Together they give A.41 and the actual compact native collar in A.42. Native pullback calculus and its full one-sided primitive law with normed boundary variables are A.45 and A.46. The primitive law on the same collar with its actual boundary source is A.47. The actual boundary pullback of any locally extendible native form is smooth by A.49. The smooth pullback of the native primitive to an arbitrary boundaryless source is A.50; in particular this supplies its smooth restriction to the native interior. The native interior has its Euclidean atlas in A.48. The explicit finite collar extension is A.53; its cylinder and interior open pieces are A.54, and separation and countability are A.55. The source-derived quotient with its compact closed original carrier is A.57; it retains the same constructed collar, width and full primitive law. The carrier's actual interior and frontier are identified in A.56. These half-space collar statements have positive ambient dimension. The separate zero-dimensional source construction is A.372, retaining the original native atlas. Its completion and full-core consumer is A.375. The actual witnessed-pair consumer for a native finite ribbon in a presented ambient is A.376. The time-first Euclidean cylinder atlas is A.51, including the empty-band case. Its actual time coordinate has nonzero differential in A.52. The actual compatible atlas on the whole extension is A.58, with both mixed transitions derived from the same native collar and its inverse germs. The original carrier includes smoothly across the seam, and the cylinder has actual smooth inverse germs, in A.59. The actual tangent isomorphisms, including at the original seam, are A.61, using A.60 without a rank input. The actual smooth primitive pieces are A.62, and their full-tangent agreement on the positive collar overlap is A.63. Regular local sublevel defining functions on the entire open cylinder patch are A.64. The compact domain presentation itself is A.65, with the literal carrier and native embedded boundary. The actual global primitive with its exact whole-patch restrictions is A.66, and its smoothness is A.67. Preservation on the entire original manifold, including every boundary tangent, is A.68. The source-only construction of the actual compact domain and this smooth primitive extension is A.69. The native exterior naturality theorem is A.70; its application in A.71 preserves the actual derivative and gives nondegeneracy on the literal compact carrier. Restriction to the derived symplectic locus in A.72 gives a globally symplectic primitive and a smooth canonical contraction field there. The same-collar native atlas identification and its preservation under open restriction are A.74. The source-derived Liouville extension with the same native atlas, original primitive, actual derivative, field and outwardness is A.73. The same-collar two-sided coordinate germ is A.24; its local scalar consumer is A.25, with primitive-derived field hypotheses in A.28. The actual quotient consumer A.27 uses A.26 to patch those local extensions while preserving every original value. A.29 chooses one compact presentation and smooth primitive extension before the additional scalar, and extends every scalar with actual smooth native chart extension germs on that same presentation. The native atlas and actual derivative are retained together with this universal scalar property in A.75. Its open symplectic restriction gives A.76, where the same native diffeomorphism, primitive and field are preserved and every scalar extends on the resulting Liouville presentation. The compact-germ extension lemma A.83 extends manifold-valued maps on an open neighborhood of a compact carrier, from supplied smooth local germs on a fixed boundaryless presentation. Its route uses overlap shrinking, convex-chart patching and finite assembly, without a common target

chart for the whole compact image. Transfer of arbitrary source-smooth native map germs is supplied by A.89 and A.91, on the same fixed collar quotient. The source-only constructor A.92 chooses one native compact primitive presentation before every additional scalar and map. It extends each such map on a neighborhood of the whole original carrier. The actual native Liouville consumer A.94 preserves both universal extension properties through the same symplectic restriction, with the full native primitive, derivative and field identities. The native compact ribbon extension and its actual witnessed-pair consumer are A.380 and A.381. They preserve every original ribbon point and the full compact primitive through the selected native carrier diffeomorphism. The native boundary identification and actual primitive transport are A.377 and A.378. Their fully native ambient-and-ribbon consumer is A.384, with witness transport through the same derived boundary diffeomorphism. The rank-one native ambient consumer is separately A.385; its actual finite ribbon core is the whole ribbon. Contact nondegeneracy transport, the converse identification for arbitrary domain presentations, and the full intrinsic pair correspondence remain obligations of the open intrinsic-domain presentation correspondence A.96. The extension-space boundary charts are separate prerequisites of that correspondence.

- Completion and core: the proved localized-presentation construction A.104 and its actual completion A.370, with their collar, scalar, primitive and complete-flow producers. The arbitrary intrinsic-domain presentation correspondence is a separate requirement. A prescribed native compact Liouville diffeomorphism with actual smooth native-coordinate germs now has the constructive route A.399: independent presentations and their actual localized completions retain its native map, inverse, positive primitive scale, boundary and end squares. For independently fixed presentations and arbitrary supplied realized completions, A.397 retains those formulas on the original completed spaces. The independent zero-dimensional native construction is A.393. Its positive-scale primitive equation derives transport of the actual completed fields, all-real-time flows and full native cores along that same comparison. On realized completions, A.402 and A.403 determine the completed map from its compact restriction and constant primitive scaling. Correspondence with arbitrary independently specified intrinsic completions remains an obligation of A.426. The prescribed native inclusion extends into that original ambient by A.411; the whole intrinsic completion adapter remains separate. Using these maps at literal comparison-chain junctions uses the exact endpoint presentations of A.185 and the actual completed maps of A.186 and A.187. Full pair identifications use the original-domain compact maps of A.412 and their fixed-endpoint consumer A.415. The finite original-domain adapter and its completed compatibility are A.413 and A.414. For a supplied strict ambient completed map, the full ribbon orbit condition is equivalent to the boundary square by A.416; the actual permitted identification is assembled in A.417. The native positive-dimensional and finite-ribbon consumers are A.418 and A.420. They retain the original pairs and completions and permit arbitrary real radial shifts. These orbit-square adapters require the specified strict ambient map and its compatibility with the whole ribbon. When the original pairs present the same native compact ambient and whole native ribbon, A.422 and A.424 instead construct the ambient strict map and derive the whole ribbon orbit square at radius zero from the actual native boundary and ribbon squares. Their exact same-deformation consumer is A.423: the original joint native positive-dimensional pair data produce one coherent deformation Q, retaining all native relabelings and coordinate germs, together with permitted identifications from every prescribed original slice to the corresponding slice of that very Q. Its finite-ribbon counterpart is A.425, retaining the same whole-core equality and allowing empty ribbons. Both constructions give

literal three-step original endpoint chains through their respective single deformations. Their slice identifications are pointwise; joint original completed-family coherence and prescribed grading transport are separate obligations. Constructing the target-specific maps, markings and geometric chain remains a geometric obligation. The compatible compact bundle charts are already supplied by A.582; matching the original whole completed families and their prescribed endpoint markings with coherent collar-adapted presentations remains separate. For supplied represented compact maps, A.391 constructs the scaled completed extension. On the same represented domain, A.401, A.404 and A.405 compare arbitrary chosen completions and transport a supplied inclusion marking, retaining its compact inclusion and all-real-time flow equation. The separate zero-dimensional completion and core results retain their own scopes.

- Completed families: the open varying-primitive correspondence A.204, including the whole completed ambient family. For a strict identification compared through actual same-radius native completion maps, A.206 retains the original induced boundary map and arbitrary radial shift under conjugation, with the all-real end equation and contact primitive law. Its full native ribbon square consumer, A.207, retains a supplied compact ribbon diffeomorphism and primitive scale while deriving the represented boundary square and permitted identification. Lemma A.208 instead constructs the scaled compact ribbon map from an original native map between two slices of one constructed deformation, with the finite-ribbon counterpart A.209. Its original strict ambient map and native ribbon map, with their full primitive equations and native ribbon square, remain inputs. A supplied actual marking on an original completed slice transports to its represented completion by Lemma A.406, preserving its compact inclusion and all-real flow law. Its retained ribbon comparison transports supplied original ribbon markings by A.407, retaining whole-original recovery and native compact and flow equations. Full pair-family correspondence, comparison of independently constructed families or different carriers, existence of original geometric maps and markings, and their grading transport remain separate. Its compact-domain prerequisites are the common nondegeneracy neighborhood A.110 and the parameter-preserving bounded flow A.114; neither supplies the completed-family correspondence. The common boundary collar and primitive law A.118 now feed the compact presentation route A.121; transporting its primitives uses A.119. The supplied-completion theorem A.138 derives coherence and completed regularity from these producers when pointwise completions of the original presentations are supplied. The localized construction's native inclusions already preserve the original compact data and identify its presentation maps with the original maps; the remaining relation is the full intrinsic domain, bundle and source correspondence. The common realized flow collars of A.122 have compatible compact presentation maps. The conditional construction A.140 assembles the actual localized cylindrical completions from such maps and proves their transported completed regularity. A.141 constructs the collars, localized compact presentations and completions from the original compact primitive family alone, with jointly smooth transported fields and primitives on both localized and completed spaces. Its maps agree with the original compact presentations under the open inclusions. The intrinsic domain and bundle presentation correspondence remains separate. A.142 consumes two such constructed families to give a presented witnessed pair deformation from fixed ambient and ribbon domain data, jointly smooth embeddings and witnesses, and the compact ribbon primitive identity. It constructs both completed families. A.151 removes the whole-extension embedding hypothesis: a supplied total map whose germ along the compact ribbon is jointly smooth and immersive, and injective on the compact carrier for every interval parameter, gives a common

actual embedding neighborhood and the same localized pair-family construction, preserving the whole compact ribbon image. A.153 removes the supplied total-map requirement as well: the ribbon map is defined only on an open neighborhood of the compact parameter-ribbon product, is jointly smooth, and is injective on the compact carrier for every interval parameter, with injective spatial differential there. The rectangular representative and embedding restrictions preserve its compact image, primitive and prescribed witness. Intrinsic domain and bundle extensions remain separate requirements. The independent field-replacement lemma A.111 supplies globally smooth extensions preserving all compact fields; that lemma is not a prerequisite of the localized collar route. Native primitive coordinate germs construct one fixed spatial presentation and both completed primitive/field families in A.162. Its fixed actual boundary diffeomorphism transports every native boundary primitive by A.165. A.164 transfers joint native map germs to that same spatial presentation, while A.168 derives witness-family smoothness from actual local scalar germs. The finite-ribbon family and its whole original-native ambient consumer are A.170 and A.177, retaining every primitive, ribbon map, witness and full embedded core on the same constructed parameter tuple. The positive-dimensional native consumer A.180 likewise constructs the actual pair family from both original primitives, native ribbon maps and positive witnesses. Their coherent refinements A.182 and A.183 retain the same original data through jointly smooth native carrier identifications and inverse families, compatible with the compact and completed presentations of the same constructed deformation. Compatible global compact bundle charts and original-atlas data transport are supplied by the bundle producers. Joint native coefficient germs also give the actual original boundary and full exponential cylinder families in A.448 and A.449. The shared quotient of the original varying collars retains its exact original slice fibers by A.436. Its actual cylinder chart is supplied by A.437, conditional on the original total collar being an open embedding. For compact fixed native domains, the original joint primitive germs, nondegeneracy and outwardness construct that total collar embedding, its same real-parameter representative and local inverse germs by A.166. With the original seam values, these same cylinders and the original native interiors give the actual open cover of A.438. The exact shared quotient overlap is A.439. Under the explicit native slice-collar hypotheses, A.454 chooses the same original quotient primitives and proves joint regularity of their actual spatial pullbacks on that cylinder patch, including at every original seam. The source-only fixed-native-domain consumer A.455 constructs those same quotient primitives, charts and extension witnesses from the original primitive germs, nondegeneracy and outwardness alone. The same retained real collar supplies both literal mixed-transition germs in A.440. The source-only consumer A.456 retains the same primitive family and actual open cover together with these two-way germs and all-real canonical Liouville flows on each original slice; joint regularity of those flows is a separate assertion. The topology of that same original shared quotient is Hausdorff by A.457 and second countable by A.458. Its original slice-to-fiber bijections are actual homeomorphisms by A.459. The source-only consumer A.463 constructs the smooth total atlas of that same original varying attachment, with boundary only at the parameter endpoints, smooth original interior and full-cylinder patches, and smooth parameter projection. The exact criterion for maps out of this total atlas is A.462. These assertions retain the original primitive and all-real slice-flow conclusions, while joint transported forms on the whole family remain separate. The original completed slices insert as smooth embeddings by A.492; their local smooth retractions and exact smoothness reflection are A.491 and A.493. For original native positive-dimensional and finite-ribbon pair data, A.469 and A.470 construct the same original ambient total atlas and the same pair deformation together, with a jointly smooth forward map q retaining every original native point and full-cylinder

formula. The parameter-preserving comparison is $z \mapsto (\pi(z), q(z))$. Its smooth inverse in that original total atlas is A.501. The exact slice diffeomorphisms retain the original slice atlases by A.503 and A.504. They preserve the same chosen spatial primitives by A.505 and transport their actual Liouville fields by A.512. The same original primitives admit the total tangent-covector realization of A.515, with C^∞ evaluation relative to the closed parameter model and exact original slice restrictions. The original fields similarly admit the exact total tangent-section realization of A.522. The original spatial exterior derivatives admit the exact total two-covector realization of A.525. The same total covector has real open coefficient germs on the original cylinder seam and outward strip by A.517, and on the entire original native interior product by A.519. These patches cover the same actual quotient by A.518. The same covector has real open full coefficients in an actual smooth chart at every original point by A.526. The spatial two-covector has real open full patch coefficients by A.528; the same transported field has real spatial extensions with exact full patch differential identities by A.530. Their original interior and negative-cutoff cylinder strips cover the same quotient. The original positive-dimensional ribbon primitives also construct one original ribbon total atlas and strict same-deformation slice comparisons by A.474. Their actual primitives, native forward and inverse squares, full cylinders, all-real flows and full embedded cores are retained. The same native ribbon relabelings construct the all-real moving reference boundary family and its inverse, with their exact interval restrictions and the whole original reference cylinder formula, by A.475. Their exact original total comparison and inverse are jointly smooth by A.479, with real compact interior and seam/outward cylinder map and inverse germs. The canonical ribbon total covector now has smooth tangent evaluation in its closed parameter model and exact selected primitive restrictions by A.481. Its full real open coefficient germs on both original patches and in actual smooth charts are A.484 and A.485. The exact spatial exterior tensor and vertical Liouville field on that same total quotient are A.482 and A.483, with smooth tangent evaluations and exact original slice restrictions. Their real coefficient germs and exact insertion differential identities are A.486 and A.487. These are retained by the same native pair construction. For finite ribbons, the original native space is its completion, with canonical strict slice comparisons by A.488; the actual same-deformation reference maps and inverses have all-real parameter-preserving smooth extensions by A.489. The original endpoint comparison on one retained native family and transport of supplied ambient and ribbon inclusion markings are proved by A.208 and A.209. Their original strict endpoint maps, eventual end laws and whole native ribbon squares are hypotheses. Independent original completions now give permitted endpoint identifications by A.617 and A.618; the literal three-step junction of two retained deformations is A.619. These results require the prescribed original strict map, its eventual end law and the full native ribbon square. Constructing those data for the target remains work for the geometric existence theorem. The full intrinsic completed-family correspondence still requires the full-pair identities over the same completed families and descent of arbitrary supplied original grading families to those same completed presentations. For supplied native homotopies and two supplied deformation gradings, A.625 grades the literal three-step path through the exact retained middle identification. More generally, A.658 constructs an arbitrary supplied finite itinerary with exactly its original length and ordered pair tuples. Its supplied grading data decorate that same chain by A.660; the explicit finite-list assertion is A.661. These conditional constructions do not provide the original deformations, junction maps, grading families or their prescribed endpoint class homotopies.

- Grading: the representative and interval-family constructions and the prescribed endpoint

classes. The real inverse germs of A.622 and A.621 retain the actual original total inverse on every nearby interval point. They provide real local map representatives for transporting supplied grading coefficients. Lemma A.219 transports a supplied Maslov family along specified interval diffeomorphisms using individual real forward germs. Its actual completed-form pullback and four prescribed endpoint equations remain explicit hypotheses. Local value coefficients descend through that exact inverse by A.630, and A.626 proves the invertibility of a parameter-preserving real inverse's spatial differential. The actual original interior inverse retains this invertibility by A.631. The original end and interior inverses transport supplied coordinate coframes by their actual spatial differentials in A.634 and A.635. These give local coframe-induced complex structures and nonzero squared canonical tensors, with their scalar coordinates retaining the original inverse values. The computed inverse-coframe coefficient reconstructs an arbitrary original canonical tensor in its actual coordinate model by A.636. Smooth original complex coefficients produce their own local coframes by A.637. The original interior and boundary–radial inverse derivatives are the actual completed-slice inverse derivatives, including the coordinate sign at the seam, by A.647. Source coordinate expressions therefore identify both descended complex fields and whole tensors with the original slice pullbacks by A.648. Their local existence from smooth original coordinate germs is A.646; neither a source coframe nor a descended generator formula is an input there. Source compatibility passes through the actual native slice inverse to the same represented completed primitive by A.649, provided the represented field is smooth and retains that exact inverse pullback. The interior and nonnegative-end cover assembles these local fields into one grading over the same completed deformation by A.650, retaining every original interval complex structure and full tensor. Its original source coordinate extension interface, native compatibility and primitive restrictions are explicit hypotheses. The native endpoint intertwining and full tensor equations imply its four reference endpoint equations by A.651; A.652 then grades the same deformation step with those exact endpoint representatives. The local corrected bundle patch of A.638 requires complex square only on the prescribed locus. An actual smooth local bundle section on that patch produces full tensor coordinates by A.639. Original local section germs retain both source chart equations and supply the interior and end tensor-coordinate interfaces by A.640. Together with the native endpoint identities, these local section germs grade the literal original deformation step by A.653. Existence of such extending local sections for every arbitrary intrinsic completed family, and the prescribed native endpoint identities or class homotopies for the source-level grading, remain separate correspondence obligations. On the actual original interval-interior base, the spatial chart transition is A.667. Local extensions of the original complex coefficients construct the coframe cover and the original complex line bundle in A.671 and A.674. The source's one-chart local coefficient convention is independent of the chart by A.675 and gives native smoothness by A.677. For an original nowhere-vanishing section, this produces the actual interior grading graphs and section germs in A.678 and A.679, retaining all original spatial tensor slots. The corresponding original cylindrical bundle and section extraction are supplied by A.687, A.688, A.690 and A.692, retaining the same original section, its outward-radius sign and every full-band matching identity. Their interior and end germs grade the same supplied deformation by A.697, under explicit native compatibility, completion-comparison and endpoint equations. The actual mixed coordinate and scalar germs are A.695 and A.696. They retain both native tangent normalizations and transfer the same original section coefficient in both directions by A.698. On the same total quotient, the one-chart-at-each-point condition of A.699 is equivalent to both patch extension conditions by A.700. That one total cover grades the retained deformation by A.701, with the exact original field and full-tensor pullbacks and the prescribed

endpoint equations. Recovery of coefficient extensions from bare intrinsic smoothness and a whole mixed canonical bundle correspondence are optional stronger presentations of these local data. The original total section constructs all actual slice representatives and derives their native endpoint equations by A.702, while grading that same supplied deformation. Original junction primitive equations derive the symplectic pullbacks in A.703. The original endpoint differential and full-tensor equations identify its complete representatives by A.704. An original smooth homotopy with those endpoint equations grades the exact constructed junction by A.705, retaining its real-indexed pulled fields. The two supplied original total sections and their original comparison and homotopy data grade the literal three-step chain by A.707, with the same produced deformation gradings and middle lift. These are conditional representation producers, not existence of the target's prescribed grading. For positive-dimensional ribbons, A.706 constructs one original bundle realization before admitting every qualifying section on its chosen completed family. Its actual slice maps, whole-ribbon, endpoint-identification and marking conclusions are retained with that same deformation and grading capability. At the original compact-atlas level, A.711 constructs the local Liouville completions, and A.713 extends their actual compact coordinate changes strictly. The required native-coordinate germs follow from the original atlas by A.712. Their identity, inverse and cocycle equations hold on whole completed fibres by A.714. Two supplied inclusion markings extending the same original compact fibre map agree through that transition by A.715; existence of those markings remains a separate requirement. Lemma A.716 supplies chart-preserving local parameter extensions at both interval endpoints; A.717 retains the real joint primitive germs under those extensions. Lemma A.718 constructs one joint smooth local total completion space with its actual slice primitives and native and cylinder patch laws. Lemma A.719 strictly identifies each retained slice with its earlier original chart completion, preserving the whole compact fibre and all-real end and flow equations. The compact-determined overlap route is Lemma A.720. The total covector with exact full slice restriction is Lemma A.740. The positive end-strip criterion A.738 proves joint native smoothness of the canonical overlaps in A.739, with the reversed overlap as smooth inverse. Lemma A.721 strictly compares every retained local slice with the chosen compatible completion, using its actual original compact and boundary transition. Lemma A.722 retains the varying completed primitive of the same deformation in its fixed source completion. The original compact transition restricts to actual real interior germs by A.723; A.725 gives those germs for the literal compact comparison into that same source completion. Lemma A.724 turns supplied real interior and seam-outward germs into actual target-valued real germs in the original total atlas. Lemma A.726 binds the actual local total map to those same strict slice comparisons. Lemma A.727 derives its real original-atlas germs from the retained compact and boundary presentation squares, with the end extension produced from the same varying completed primitive. Lemma A.728 chooses one compatible chart and completed deformation from the original positive ribbon bundle data before all original local chart comparisons. Lemma A.737 gives the same forward ambient comparison and full-slice primitive identity from original dimension-zero ribbon bundle data. Lemma A.729 identifies the exact parameter-preserving two-sided inverse. Its real interior and signed-cylinder germs give real spatial patch germs at every reference completion point by A.733. Lemmas A.735 and A.736 produce these forward and inverse germs from the original compact data, with one chosen deformation and both whole-slice primitive identities. Lemma A.734 proves joint native regularity of that same total equivalence on its retained relative-open spaces. Both source constructions retain a native diffeomorphism with exactly this underlying equivalence. The strict compact coordinate cocycle and completion uniqueness give slice comparison coherence in A.741; A.742 retains that identity for the actual total

maps and their inverses on the original common base. Preselected pointwise full canonical-square data on that original atlas are defined in A.743; A.744 gives chart-independent inverse transport and exact recovery on every original local base. Compatibility, canonical type and nonzero tensor values are retained by A.745. The actual tensor-product canonical square and its exact all-chart evaluation are recovered pointwise in A.746. Conditional joint smooth reconstruction from retained original coframe and scalar germs is supplied by A.754 and A.755. Lemma A.756 constructs a completed pair deformation and this smooth family together from the original positive-dimensional or finite-ribbon bundle data, retaining the entire original geometric realization. The literal completed family gives an actual grade on its same deformation by Lemma A.759 and Corollary A.760, with exact recovery of the original data on every completed slice. For independently supplied original completion markings, Definition A.761 retains one original compact inclusion into the target at each parameter. Their strict overlap covariance is derived by A.762. Lemma A.763 descends them to the same retained ambient completion, independent of the local chart, with exact whole-slice recovery. Corollary A.764 retains the native compact inclusion and every real boundary orbit, including the source and terminal parameters. The positive-dimensional original ribbon completions have strict all-chart comparisons and chart-independent marking descent by Lemmas A.766 and A.767; their native inclusions and all real boundary orbits are recovered by Corollary A.768. The finite-ribbon marking data and covariance are A.769 and A.770; Lemma A.771 recovers the same exact ribbon completion, including empty ribbons. Corollary A.760 retains both independently supplied markings on its already chosen geometric deformation and grade. These results assume the original markings; they do not construct the toric full-fibre markings or a jointly smooth marked target family. Independently prescribed original completed-atlas Maslov data are recovered with the supplied real coframe and scalar germs. Original real open smooth coefficients for the complex field and full tensor yield the required coframe and scalar germs by Lemma A.750; Corollary A.752 uses those constructed germs in the same graded bundle realization. An independently supplied original local smooth native canonical-square section gives these coefficient extensions by Lemma A.751, with matching only on the retained source chart locus. For the source's collar-generated completed bundle, the retained canonical-square bundle and section correspondences extract these original coefficient and section germs from the ordinary source-local regularity of the actual section. Their full realization consumer is A.316. The original endpoint class homotopies and finite-chain integration for supplied source data are retained by A.245 and A.250. These constructions do not assert the prescribed toric endpoint classes or the existence of the toric geometric itinerary. The bounded positive-dimensional local transport in Lemma A.772 retains two independent endpoint realizations and derives their original-chart ribbon correspondence from bundle covariance. Its finite-ribbon companion is Lemma A.773, including empty ribbons. Both assume the original strict junction, its eventual end law and its full ribbon map, and supply neither a graded junction nor an original finite itinerary. Original completed-atlas recovery determines the exact endpoint representatives in Lemma A.774. Given independently supplied original endpoint representatives and an original smooth homotopy, the resulting grade retains the exact finite-ribbon junction, including empty ribbons, by Lemma A.776. Two original graded pair bundles and their original junction homotopy produce simultaneous realizations and a literal three-step chain by Lemma A.775 in positive ribbon dimension and Corollary A.777 for finite ribbons. These conditional transports do not extract smooth section germs from an independently given intrinsic section, construct original junctions or homotopies, or furnish the toric itinerary and endpoint classes of the geometric target. The conjugation of a supplied native grading lift and its attachment to the exact retained identification are A.623

and A.624. Their family is the literal derivative pullback at every real parameter. The conditional three-step assembly is A.625; it requires both supplied deformation gradings and the supplied original native lift. Compatible-complex-structure transport and fixed-structure class equivalence have proved component declarations. Their presence does not close the whole source-and-encoding correspondence. The retained one-bundle outputs of A.228 and original junction transport of A.229 assemble every supplied finite original itinerary by A.230, with the same grading and literal vertex list. Matching here is equality of original bundle-and-parameter vertices. Deriving an original presentation junction merely from geometric equality of endpoint tuples in distinct bundles is a separate correspondence obligation. The original geometric arrows, homotopies and marking data remain supplied. Intrinsic smoothness in the independent original canonical-square branch bundles gives every-chart coefficient smoothness by A.231 and A.232. Their actual real open coefficient germs at strict interior times are A.233. With endpoint coefficient extensions supplied, the same original full tensor has branchwise grading section germs by A.234. Independent open bundles and their actual original fiber inclusions are A.235; intrinsic section smoothness gives smooth graphs on their retained bases by A.237. A supplied real open native graph extension produces an actual scalar germ by A.236. Supplying such a graph in one chart at each endpoint point suffices for all original branch grading germs by A.238. Restriction from a separately specified whole completed canonical-square bundle and existence of its endpoint extension data remain separate. The actual canonical square on the whole original completed total family is constructed in Lemma A.247, with its original fibers, full mixed atlas and complex line structure. Intrinsic smoothness of a section in that bundle restricts to the exact original branch sections by Lemma A.251. With endpoint native graph extensions supplied, its same full tensor has both original grading-section germ families by Corollary A.252. The existence of those real open endpoint extensions remains a separate smoothness correspondence requirement when only intrinsic retained-locus smoothness is specified. The actual raw coordinate laws and real open chart changes are A.254 and A.256. The full-map local extension condition in this actual atlas is A.257; A.260 proves its equivalence to the all-chart condition under the actual coframe cover and continuous bundle projection. For the graph of one original section the projection is the identity. Its generic local full-map extensions produce exact native endpoint graphs by A.261 and reach the original whole-section grading consumer by A.262. Their matching retains the section itself and every full-tensor slot. Real open full-map extensions on each retained subset, with matching only on that subset, transfer to every original chart by A.264. For a section on a retained parameter base, the exact original branch coefficients follow from A.265. The prescribed representative on each marked ambient pulls back to covariant original pointwise Maslov data by A.266; its recovery on each exact endpoint completion is A.267. The pointwise section of each given original full tensor is A.268. Raw real open field germs on the relatively open retained bases construct original canonical charts by A.269. Those field germs and local full-map regularity of that exact section graph produce all original coframe and scalar germs by A.270. These regularity assumptions remain explicit; the coframe germs are conclusions for the same prescribed original tensors. They are consumed by the complete original pair-bundle realization in A.271, retaining its geometry, grade, all original chart recovery and both marking capabilities on that same realization. For independently prescribed complete sources with a fixed smooth presentation, ordinary smoothness of the actual pulled primitives produces all finite-time source-flow germs by A.282. Together with the original compact inclusion regularity, it produces literal marking-map germs on the original interior and full signed cylinder by A.283. One independent smooth Maslov family for those source forms determines the original representatives and their exact marking pullbacks by A.287.

The same source data produce the full original field, tensor and coframe germs by A.288, and are consumed by the complete positive-ribbon or finite-ribbon realization in A.289. This route assumes ordinary primitive and compact-inclusion smoothness in the independently supplied fixed source presentation; it assumes no field, flow, marking-map, tensor, coframe or scalar germ. Correspondence from an arbitrary intrinsically smooth marked completed family to that fixed smooth presentation and its ordinary source data remains a separate foundation. The prescribed toric endpoint classes and comparison itinerary remain geometric obligations. The retained local source route in A.300 requires source coordinates and ordinary primitive, compact-inclusion, complex-field and full-tensor coefficient regularity only on each original local base. The prescribed source representative is the same before all local coordinate choices. The coordinates may differ between local bases; no Maslov family on a whole auxiliary cutoff interval is assumed. This is a conditional adapter from supplied ordinary source presentations. Independently, A.303 constructs the actual smooth complex line bundle on the retained collar-generated completed base from the original complex-field germs, complex laws on that base and ordinary collar-transition regularity, without a section input. An independently selected section in these actual fibers, with ordinary full-map real open extensions in the constructed restricted bundle coordinates, gives the exact original Maslov data in A.313. The full original tensor and the originally chosen section itself are recovered in A.314. Its original coframe and scalar germs are produced in A.315, then consumed by A.316 on the same full positive-dimensional or finite-ribbon realization, with both marking capabilities and the actual grade. These results concern the canonical collar-generated source and do not require a separate fixed source manifold or Maslov family on its auxiliary cutoff interval. Conversely, A.319 derives joint native boundary and parameter-endpoint germs from actual smooth ambient one-form families by the extended inverse half-space charts. Corollary A.320 applies this construction to the actual ambient and positive-dimensional ribbon primitives of a presented deformation. The full actual compact Liouville laws, exact native boundary defining functions and field germs are supplied by A.326 and A.327. Their exact fixed-carrier deformation-family consumer is A.332. The same fixed whole-ribbon equation and witness are retained by A.346, and its positive-dimensional and finite native map germs by A.347. The original forward and inverse compact maps have the chart extensions of A.337. Their actual native boundary and whole-ribbon equations, original primitive scale and induced infinity law are A.340 and A.341. These reverse laws concern supplied geometric objects and steps; they construct no toric comparison itinerary or endpoint core equality. Equality of full original compact tuples in independently described bundles constructs their strict original completed normalization by Corollary A.242 and Lemma A.248. Under equality of the independent original Maslov classes, the resulting junction has a constructed smooth homotopy and appends to the exact finite graded chain by Corollaries A.245 and A.250. Full tuple equality retains the witness and whole ribbon; it does not supply the grading-class equality. These normalizations are conditional foundations using the independent source section data and class equality. The retained-section producer A.316 supplies its original section germs and full graded realization from the source-local smoothness data. Neither normalization nor that realization constructs the toric geometric itinerary, full-fiber markings, embedded-core equality or prescribed toric endpoint classes.

- Bounds and assembly: the compact magnitude maximum, explicitly defined threshold and rounding predicates, both markings, and the same chain carrying core equality and grading. Their source correspondence is part of A.348.

Componentwise source correspondence preserves the defining geometric data and their quanti-

fiers.

B Geometric proof route

This chapter follows the explicit geometric argument in [1]. All inputs and endpoint conventions are those of Section 1; the target is Theorem 4.1. The full intrinsic fiber estimates below are distinct from the estimates on a finite ambient band. The endpoint grading is attached to the same chain that supplies the embedded-core equality.

C Elementary flow tools and parameter order

Lemma C.1 (Local product coordinates at a regular level)**.** *Let M be a smooth d-manifold without boundary, F a finite-dimensional real vector space, and $f : M \to F$ smooth. If df_x is surjective and $f(x) = c$, there are local smooth product coordinates near x in which f is exactly the first coordinate. The second coordinate lies in a vector space of dimension $d - \dim F$.*

Proof. Choose an ambient coordinate chart sending x to $z_0 \in E$, and write $\widetilde{f}$ for f in that chart. Its derivative $L : E \to F$ at z_0 is surjective. Choose a linear right inverse $S : F \to E$ and set $P(v) = v - SL(v) \in \ker L$. The linear map $v \mapsto (Lv, Pv)$ is an isomorphism $E \to F \times \ker L$, with inverse $(a, k) \mapsto S(a) + k$. Thus the smooth map

$$z \longmapsto \big(\widetilde{f}(z),\ P(z - z_0)\big)$$

has invertible derivative at z_0. The inverse function theorem restricts it to a smooth diffeomorphism of open neighborhoods. Its first coordinate is exactly $\widetilde{f}$, by construction. Rank-nullity gives $\dim \ker L = d - \dim F$. Composing with the original ambient chart proves the assertion. □

Lemma C.2 (Restriction of product coordinates to a literal level)**.** *Fix a point of the literal level set $f^{-1}(c)$. If a local product homeomorphism $\Theta : U \to V \subset F \times G$ satisfies $\mathrm{pr}_F \Theta = f$, then it restricts to a local homeomorphism from the literal level set $f^{-1}(c)$ to $V_c = \{g : (c, g) \in V\} \subset G$. Its inverse is $g \mapsto \Theta^{-1}(c, g)$.*

Proof. The set V_c is open as the inverse image of the open set V under the continuous map $g \mapsto (c, g)$. The domain $U \cap f^{-1}(c)$ is open in the subspace topology. On this domain the first coordinate of Θ is c. Conversely the stated inverse lies in the level set because its first product coordinate is c. The inverse identities are restrictions of those of Θ. Both maps are continuous by their displayed formulas and the subspace universal property. □

Lemma C.3 (Smooth manifold and tangent space of a regular level)**.** *If df_x is surjective at every $x \in f^{-1}(c)$, the literal level set with its subspace topology is a smooth manifold of dimension $\max(d - \dim F, 0)$. Its inclusion in M is a smooth embedding whose differential identifies its tangent space with $\ker df_x$.*

Proof. At every point take the local product coordinates and restricted chart of the two preceding lemmas. At each point surjectivity forces $d \geq \dim F$. Choose a linear isomorphism from each second-coordinate space to a fixed real vector space of dimension $\max(d - \dim F, 0)$. If the level is empty there are no charts to choose, and its empty atlas gives the same assertion without a dimension restriction. These charts cover the level set and preserve its subspace topology. A transition map is the restriction of the second-coordinate projection of $\Theta_j \Theta_i^{-1}(c, g)$, with the chosen

linear isomorphisms inserted. Ambient chart transitions and both local product diffeomorphisms are smooth, so every such transition is smooth. Hausdorffness and second countability are inherited from the ambient subspace. This constructs the asserted smooth manifold. In its local chart the inclusion is $g \mapsto \Theta^{-1}(c, g)$, so it is smooth and has injective differential. It is a topological embedding by the chosen subspace topology. Differentiating $f\Theta^{-1}(a, g) = a$ shows that its tangent image lies in $\ker df_x$. Conversely, under the invertible derivative of Θ, a vector in this kernel has first coordinate zero and hence is the image of a unique second-coordinate vector. This proves the tangent-space equality. □

Lemma C.4 (The full original far fiber as a smooth embedded manifold)**.** *For every original ambient bound A_{amb} and $R > \max(1, M(A_{\mathrm{amb}}))$, the literal full fiber $W^{-1}(R)$ is the image of a smooth embedding of a $(2n-2)$-manifold. The image of its differential is exactly* $\ker dW$.

Proof. The original far-value estimate makes dW surjective at every point of $W^{-1}(R)$. Apply the regular-level construction with the real vector space $\mathbb{C}$, of dimension two. Its range is the literal fiber and its tangent-image identity gives the assertion. This construction also applies when $n = 1$, where the fiber manifold is zero-dimensional. □

Lemma C.5 (Smooth factorization through an embedded submanifold)**.** *Let $i : N \hookrightarrow M$ and $g : P \hookrightarrow M$ be smooth embeddings of finite-dimensional manifolds, with $g(P) \subset i(N)$. Then there is a smooth embedding $q : P \hookrightarrow N$ with $i \circ q = g$.*

Proof. Injectivity of i defines the unique lift q pointwise. The embedding property makes it continuous and a topological embedding. Fix $x \in P$ and choose manifold charts near $q(x)$ and $g(x)$. The injective derivative of i has a linear left inverse L. The derivative of the coordinate map $L \circ i$ is the identity at $q(x)$, so the inverse function theorem supplies a smooth local inverse. Continuity of q permits restricting to a neighborhood of x where $q = (L \circ i)^{-1} \circ L \circ g$ in these charts. This proves smoothness locally and hence everywhere. Differentiating $i \circ q = g$ shows that dq is injective because dg is. Together with the established topological embedding this proves the assertion. □

Lemma C.6 (The full ribbon in the unchanged rounded boundary)**.** *For every admissible original rounding, the whole compact fiber truncation at A^- admits an extension neighborhood smoothly embedded in the rounded boundary, with image on its compact carrier exactly the prescribed full ribbon. On that carrier the induced boundary primitive is the restricted λ_0. The inclusion square in the original ambient commutes literally.*

Proof. The proved unchanged open neighborhood contains the entire specified ribbon. Intersect its inverse image in the full fiber with the fiber-domain extension; restrict the extension to this open neighborhood, retaining its whole compact carrier. On the full fiber $W = R$ and the rounded scalar is identically zero in this neighborhood, so the original fiber inclusion lands in the actual rounded boundary. The embedding-factorization lemma lifts it to a smooth embedding into the native boundary manifold. Its ambient composite is the original inclusion by construction. Thus its compact image is exactly $W^{-1}(R) \cap \{\phi_0 \le A^-\}$, with all components included. The K-dependent base primitive restricts to zero on the fiber, since W is constant there. Hence the boundary primitive pulls back to the literal restriction of λ_0. Restriction of the extension preserves the existing completion and its onto-full-fiber inclusion marking. □

Lemma C.7 (Frontier of a regular scalar sublevel)**.** *Let $\rho : M \to \mathbb{R}$ be smooth on a manifold without boundary. If $d\rho_x \neq 0$ whenever $\rho(x) = 0$, then*

$$\operatorname{int}\{\rho \le 0\} = \{\rho < 0\}, \qquad \partial\{\rho \le 0\} = \{\rho = 0\}.$$

Proof. Continuity makes the strict sublevel open and the weak sublevel closed. If a zero-level point lay in the interior of the weak sublevel, it would be a local maximum of ρ. In a coordinate chart Fermat's theorem would give $d\rho_x = 0$, a contradiction. Thus the interior is exactly the strict sublevel. Subtracting the interior from the closed weak sublevel gives the frontier formula. □

Lemma C.8 (The literal rounded frontier)**.** *For every admissible original rounding under the initial parameter hypotheses, the frontier of the literal rounded domain is exactly the zero level of* $\rho = m_\varepsilon(\phi_0 - A, |W|^2 - R^2)$*. Its interior is exactly* $\{\rho < 0\}$*.*

Proof. The rounded defining function is smooth and its differential on the zero level is nonzero by the proved outward derivative inequality. Apply the preceding scalar-sublevel lemma to this defining function. □

Lemma C.9 (Compact Liouville domains from regular outward sublevels)**.** *Let* M *be a smooth manifold,* $\rho : M \to \mathbb{R}$ *smooth, and* $\{\rho \le 0\}$ *compact. If zero is a regular value, this literal sublevel is a compact domain with boundary* $\{\rho = 0\}$ *and defining function* ρ*. If a smooth primitive* λ *has symplectic derivative on* M *and* $d\rho(Z_\lambda) > 0$ *on* $\{\rho = 0\}$*, the same domain is a Liouville domain with primitive* λ*.*

Proof. A nonzero real-valued differential is surjective. Apply the regular-level construction to obtain the boundary manifold and its actual embedded inclusion, whose image equals the topological frontier by the preceding lemma. The global function ρ supplies a defining function at every boundary point, and compactness is the given sublevel compactness. For the second assertion, positivity implies that the differential is nonzero. Nondegeneracy of $d\lambda$ gives the smooth Liouville vector field, and the stated derivative inequality is precisely its outwardness for this defining function. These data supply the asserted Liouville domain. In dimension zero regularity forces the boundary to be empty, so no dimension restriction is needed. □

Lemma C.10 (The actual rounded Liouville domain)**.** *For every admissible original rounding in the initial parameter range, the literal rounded sublevel is a compact Liouville domain in the original ambient, with primitive* λ_K *and its entire rounded zero level as boundary.*

Proof. The rounded defining function is smooth and its weak sublevel is compact. The original primitive λ_K is smooth and symplectic for the prescribed $K \ge K_0 \ge 1$, and its actual Liouville vector has strictly positive derivative on the whole zero level. Apply the regular outward-sublevel construction with this defining function and the unchanged original ambient primitive. □

Lemma C.11 (Actual ambient marking of every rounded domain)**.** *For every admissible original rounding, inclusion of its literal rounded Liouville domain extends from any completion to a primitive-preserving diffeomorphism onto the original complete ambient with primitive* λ_K*, with the prescribed all-real boundary-flow formula.*

Proof. The original complete flow reaches $\{\phi_0 \le A_{\mathrm{amb}}\}$ in finite negative time. This entire sublevel lies in the strict interior of the rounded domain. Indeed $A_{\mathrm{amb}} \le A_{\mathrm{fib}} < A^- < A - \varepsilon$, so $x = \phi_0 - A < -\varepsilon$, whereas $y = |W|^2 - R^2 < 0$ by the strict far-radius choice. Monotonicity in the second coordinate and the exact flat tail give $m_\varepsilon(x, y) \le m_\varepsilon(x, 0) = 0$. Equality is impossible by the proved unchanged-face zero-level classification, since $x < -\varepsilon$ and $y < 0$. Thus every point enters the rounded domain in finite negative time. Its actual rounded defining function has the literal frontier as zero level and positive Liouville derivative there. Apply the boundary-outward absorption marking theorem to the constructed domain and original complete primitive. □

Lemma C.12 (Symplectic restriction to the superpotential kernel)**.** *Let $j : N \hookrightarrow X$ be a smooth embedding in the original toric ambient whose tangent image at every point is* $\ker dW$*. Then $d(j^*\lambda_0)$ is symplectic on N.*

Proof. Pullback commutes with exterior differentiation, so the form is the restriction of the original Kähler form $\omega_0 = d\lambda_0$. It is smooth and closed. Holomorphicity of W makes $\ker dW$ invariant under the standard complex structure J. If $v \in T_xN$ pairs to zero against every tangent vector, set $u = dj_x(v)$. The kernel-image equality supplies $w \in T_xN$ with $dj_x(w) = Ju$. Thus $0 = \omega_0(u, Ju)$. Kähler positivity forces $u = 0$, and injectivity of dj_x forces $v = 0$. This proves nondegeneracy, also in dimension zero. □

Lemma C.13 (The full far fiber with its actual restricted symplectic primitive)**.** *Under the original far-radius hypotheses, there exist a smooth $(2n-2)$-manifold N and smooth embedding $j : N \hookrightarrow X$ whose image is literally $W^{-1}(R)$ and for which $d(j^*\lambda_0)$ is symplectic everywhere.*

Proof. Choose the constructed far-fiber manifold and embedding. Its tangent-image theorem identifies every tangent image with $\ker dW$. Apply the preceding restriction lemma to this actual embedding. □

Lemma C.14 (Original intrinsic Liouville projection)**.** *At a regular point of an original fiber set $a = \sum_j |G_j|^2$, $b_0 = \sum_j u_j G_j$, and $q_j = u_j - \overline{G_j} b_0/a$ in logarithmic complex tangent coordinates. Then $dW(q) = 0$, and q represents the intrinsic Liouville vector for the restriction of λ_0. Its derivative on the quadratic potential is*

$$d\phi_0(q) = |u|^2 - \frac{|b_0|^2}{a}.$$

Proof. The formula for dW gives $dW(q) = b_0 - ab_0/a = 0$. For any tangent vector v with $dW(v) = 0$, the normal correction $u - q = \overline{G} b_0/a$ is symplectically orthogonal to v: its pairing is the imaginary part of $\overline{b_0/a}\, dW(v) = 0$. Thus $\omega_0(q, v) = \omega_0(u, v) = \lambda_0(v)$. The proved tangent-kernel identity lifts q uniquely to the fiber tangent space; the proved nondegeneracy identifies that lift with its Liouville vector. The quadratic differential is the real radial pairing, which gives $|u|^2 - \mathrm{Re}(\overline{b_0} b_0)/a$ and hence the displayed formula. This intrinsic defect differs from the ambient quantity Q_W. □

Lemma C.15 (Intrinsic exhaustion bounds and outwardness)**.** *On an original far fiber with its restricted primitive, $|Z_{\mathrm{fib}}\phi_0| \le 2(1 + \phi_0)$. For every valid original fiber-end bound A_{fib}, $Z_{\mathrm{fib}}\phi_0 > 0$ at every fiber point with $\phi_0 \ge A_{\mathrm{fib}}$.*

Proof. Cauchy–Schwarz gives $0 \le |u|^2 - |b_0|^2/a \le |u|^2 = 2\phi_0$, and $\phi_0 \ge 0$, proving the growth bound. On the prescribed end, every valid fiber-end bound asserts that the same defect is strictly positive. The preceding intrinsic derivative identity proves both assertions for the actual native field. In rank one the purported end is empty; no positive derivative is asserted on a nonempty zero-dimensional region. □

Lemma C.16 (Complete smooth flow from a controlled proper exhaustion)**.** *Let Z be a smooth vector field on a finite-dimensional smooth manifold M. Suppose $\rho : M \to \mathbb{R}$ is smooth, nonnegative, has compact sublevels, and $|d\rho(Z)| \le C(1 + \rho)$ everywhere for some $C > 0$. Then Z has a jointly smooth flow for all real times.*

Proof. Fix an initial point and a finite time horizon $T > 0$. Choose a smooth cutoff $0 \le \chi \le 1$ of ρ that is one on a sufficiently large sublevel and vanishes outside a larger compact sublevel.

The field χZ is smooth and compactly supported, hence has a complete smooth flow. Along its trajectories the scalar function $1+\rho$ satisfies the same differential bound with constant C. Gronwall, applied forward and backward, bounds it on $[-T,T]$ by $(1+\rho(x))e^{CT}$, independently of the cutoff. Choosing its one-region strictly above this bound makes the cutoff flow solve the original field equation throughout $[-T,T]$. Uniqueness identifies these solutions for different horizons on overlaps. They give a global integral curve through every point. Uniqueness supplies the flow law, and smooth local dependence on time and initial conditions gives joint smoothness of the global flow. □

Lemma C.17 (The original full far fiber is complete)**.** *For every original ambient bound and far radius, the entire literal fiber $W^{-1}(R)$, with the restriction of λ_0, is a complete Liouville manifold of real dimension $2n-2$. Its embedded image and primitive are the original full fiber and literal restricted primitive.*

Proof. Use the constructed smooth symplectic far-fiber embedding j. The fiber is a closed level of the continuous function W. Consequently every sublevel of $\phi_0 \circ j$ is compact: under the embedding it is the intersection of the closed fiber with a compact ambient quadratic sublevel. This exhaustion is smooth and nonnegative. The restricted symplectic primitive gives a smooth intrinsic Liouville field; the preceding bound and proper-exhaustion flow theorem make its flow complete in both time directions. Retain the original embedding and its proved literal range and differential-surjectivity identities. □

Lemma C.18 (Compact original fiber truncations and full-fiber markings)**.** *For every valid original fiber-end bound and $A^- > A_{\mathrm{fib}}$, the entire truncation $W^{-1}(R) \cap \{\phi_0 \le A^-\}$ is a compact Liouville domain in the constructed complete full fiber. Its primitive is the literal restriction of λ_0. Inclusion extends from any completion to a primitive-preserving diffeomorphism onto that entire full fiber, respecting its all-real boundary flow.*

Proof. The function $\eta = \phi_0|_{W^{-1}(R)}$ is smooth with compact sublevels. Its actual intrinsic derivative is strictly positive wherever $\eta \ge A^- > A_{\mathrm{fib}}$. Apply the regular outward-sublevel construction to $\eta - A^-$. The actual full-fiber flow is complete, and the compact-annulus absorption lemma gives finite negative entry into this truncation. The sublevel absorption marking theorem constructs the claimed strict onto-full-fiber completion map. Every construction retains all components of the literal truncation; no connected component or smaller neighborhood replaces it. □

Lemma C.19 (Finite negative-time entry into an exhaustion sublevel)**.** *Let Φ be a continuous real flow on a topological space X, and let $\rho, q : X \to \mathbb{R}$ be continuous. Suppose every sublevel of ρ is compact and $\frac{d}{dt}\rho(\Phi^t(x)) = q(\Phi^t(x))$ for all x,t. If $q > 0$ on $\{\rho \ge A\}$, then for every x there is $T \ge 0$ with $\rho(\Phi^{-T}(x)) \le A$.*

Proof. Suppose a negative orbit never enters the sublevel. Its potential stays strictly above A and is increasing as a function of the original time $t \le 0$, since its derivative is positive. It therefore stays in the compact annulus $A \le \rho \le \rho(x)$. The continuous positive function q has a positive minimum m there. The mean value inequality on $[-T,0]$ yields $mT \le \rho(x) - \rho(\Phi^{-T}(x))$ for every $T \ge 0$. Taking $T = (\rho(x) - A + 1)/m > 0$ contradicts $\rho(\Phi^{-T}(x)) > A$. Positivity at the boundary level is used to obtain the positive minimum on the closed annulus. □

Lemma C.20 (Inclusion marking from boundary outwardness and negative entry)**.** *Let $D = \{\rho \le A\}$ be a compact Liouville domain in X, with $\partial D = \{\rho = A\}$ for a smooth scalar ρ. Suppose its primitive extends to a complete Liouville primitive on X, $d\rho(Z) > 0$ on $\{\rho = A\}$, and every*

point enters D in finite negative time. Then inclusion extends from any completion of D to a primitive-preserving smooth diffeomorphism onto X, agreeing with the original flow at every all-real boundary-flow point.

Proof. The scalar boundary barrier makes $\{\rho \geq A\}$ forward invariant. A boundary trajectory cannot return to level A at a positive time: invariance would make that time a local minimum of its scalar trajectory, whose derivative would vanish by Fermat, contrary to boundary positivity. The derivative at the initial boundary point is positive, so the trajectory lies strictly outside at all positive times. The same barrier and no-return argument place it strictly inside at negative times. Thus each orbit has at most one boundary crossing, and the all-real boundary-flow map is injective. Outward transversality gives an invertible differential at time zero; flow transport gives this at every time. The inverse function theorem makes the map an open embedding. Finite negative entry and the intermediate value theorem express every exterior point as a nonnegative boundary-flow point. The Liouville flow identity supplies the primitive $e^r\alpha$ in these coordinates. Gluing with inclusion across the shared collar gives a smooth strict completion onto X. Comparison with any other completion preserves both inclusion and the all-real boundary-flow formula. Only boundary positivity, rather than monotonicity on the entire exterior, is needed in this argument. □

Lemma C.21 (Inclusion marking for an outward exhaustion sublevel)**.** *Let D be a compact Liouville domain in a manifold X, with its primitive extending to a complete Liouville primitive on X. Suppose a smooth function $\rho : X \to \mathbb{R}$ satisfies $D = \{\rho \leq A\}$, $\partial D = \{\rho = A\}$, and $d\rho(Z) > 0$ on $\{\rho \geq A\}$. Assume every point enters D in finite negative time. For any completion of D, inclusion extends to a primitive-preserving smooth diffeomorphism onto X, whose value on every boundary-flow point is $\mathrm{Fl}_Z^r(b)$, with the same boundary point b and real time r.*

Proof. For a scalar orbit function $f(t) = \rho(\mathrm{Fl}_Z^t(x))$, the derivative is positive whenever $f \geq A$. The scalar barrier inequality implies that the superlevel $f \geq A$ is forward invariant; on it the mean value inequality makes f strictly increasing. It follows that a boundary orbit is strictly outside D at positive times and strictly inside at negative times. In particular it has only one boundary crossing. Thus the map $(b, r) \mapsto \mathrm{Fl}_Z^r(b)$ on $\partial D \times \mathbb{R}$ is injective: any coincidence, after applying inverse flow, would give a second boundary crossing, and equality of times then gives equality of boundary points. Outwardness makes its differential invertible at $r = 0$; transport by the smooth flow diffeomorphisms makes it invertible at every r. The inverse function theorem therefore makes this injective map an open embedding. For any $x \notin D$, negative entry and the intermediate value theorem give a boundary hit at some nonpositive time; reversing that time expresses x as $\mathrm{Fl}_Z^r(b)$ with $r \geq 0$. These end coordinates cover precisely the exterior, with no positive-time overlap with D. The Liouville identity $\mathrm{Fl}_Z^{r*}\lambda = e^r\lambda$ and $\lambda(Z) = 0$ show that the end primitive is exactly $e^r\alpha$, including variable boundary and time directions. Consequently inclusion on D and this end map agree smoothly through their common Liouville collar, and define the required bijective smooth map with smooth inverse. For another completion of the same compact domain, its collar coordinates give the same formula, so the resulting strict comparison preserves the primitive and agrees on every all-real boundary-flow point. The extra global sublevel and positivity hypotheses specialize the source absorption lemma to the original exhaustion levels. □

Lemma C.22 (Flow absorption and the actual completion map)**.** *Let (X, λ) have a complete Liouville field, and let $D \subset X$ be a compact Liouville domain with restricted primitive. Suppose every point of X enters D in finite negative time. Then inclusion extends to a primitive-preserving diffeomorphism $\widehat{D} \to X$. More generally, if $E \subset D$ is another compact Liouville domain and every point of D enters E in finite negative time, inclusion gives $\widehat{E} \simeq \widehat{D}$.*

Proof. Cartan's identity gives $\lambda(Z) = 0$ and $\mathcal{L}_Z\lambda = \lambda$, hence $\mathrm{Fl}_Z^{t*}\lambda = e^t\lambda$. On the cylindrical end define the map by $(x,t) \mapsto \mathrm{Fl}_Z^t(x)$, $x \in \partial D$, $t \geq 0$; on D use inclusion. A Liouville collar identifies these formulas smoothly near their seam. Outwardness makes D negatively invariant. Every trajectory meeting its boundary crosses outwards and can never cross back in; uniqueness of the ODE gives injectivity of the end map. Negative absorption gives surjectivity. Transversality of the crossing and the implicit-function theorem make its inverse smooth. The displayed flow identity proves primitive preservation, also for varying time because $\lambda(Z) = 0$. For the second assertion work in $\widehat{D}$, whose Liouville field is complete; every cylindrical point first enters D. The first assertion then applies to E. □

Lemma C.23 (Finite intrinsic shrinking)**.** *Let (H,μ) be the marked completion of a compact Liouville domain F_0. Given any open neighborhood U of its full core, there is $T \geq 0$ such that $F_T = \mathrm{Fl}_{Z_H}^{-T}(F_0) \subset U$. If F_0 is a ribbon in an unchanged boundary face, these domains give a smooth ribbon deformation in that same face, with primitive $e^{-\tau}\mu|_{F_0}$ in the fixed F_0 presentation for $0 \leq \tau \leq T$.*

Proof. Outwardness makes F_0 negatively invariant. Its negative-flow images are nested compact domains with intersection the core. If no integer-time image lay in U, the nested nonempty compact sets $F_m \setminus U$ would have a common point outside U, contradicting the definition of the core. Smooth complete flow and its primitive identity give the family and the constant factor. This factor preserves the Liouville field, so every member is still a Liouville domain. All images lie in F_0, hence in its unchanged face. This argument includes dimension zero. □

Lemma C.24 (A smooth nonvanishing coefficient path with uniform bounds)**.** *Fix a complex logarithm $\ell_v = \log c_v$ for each input coefficient and put $c_v(t) = \exp((1-t)\ell_v)$. This coefficient tuple is smooth and nonvanishing for all real t, with $c_v(0) = c_v$ and $c_v(1) = 1$. Replacing only the coefficients by this tuple preserves the based polytope data, the literal fan, support height and FLTZ link. There are $A, B > 0$ such that $A \leq |c_v(t)| \leq B$ for every $t \in [0,1]$ and $v \in V$.*

Proof. The logarithms are fixed constants; smoothness follows by composing the affine real-parameter map with the complex exponential. The exponential never vanishes, and $\exp(\ell_v) = c_v$ gives the two endpoints. The fan, support height and link depend only on the unchanged geometric data and not on the coefficients, so those sets and functions are equal along the path. The positive continuous function $(t,v) \mapsto |c_v(t)|$ on the compact product $[0,1] \times V$ has a positive minimum and a finite maximum. Take that minimum for A and increase the maximum to a positive B. □

Lemma C.25 (The literal affine potential path)**.** *Let ϕ_0, ϕ_1 be smooth potentials on a finite-dimensional real inner-product space and put $\phi_t = (1-t)\phi_0 + t\phi_1$. This is jointly smooth in (t,x) with the stated endpoint values, and $D^2\phi_t = (1-t)D^2\phi_0 + tD^2\phi_1$. If $c_i|v|^2 \leq D^2\phi_i(x)[v,v] \leq C_i|v|^2$ for $i = 0,1$, then for $0 \leq t \leq 1$ the common lower and upper bounds are $\min(c_0,c_1)|v|^2$ and $\max(C_0,C_1)|v|^2$. If both endpoints are positively homogeneous of degree two, the whole path is positively homogeneous of degree two.*

Proof. Sums and products of smooth functions prove joint smoothness, and substitution gives the endpoints. Differentiation is linear, proving the Hessian identity. Its two coefficients are nonnegative and sum to one on the interval, so multiplying each endpoint bound by its coefficient and adding proves the claimed common bounds. Substitution of tx for x in both endpoint homogeneity identities proves the final assertion for every positive dilation. □

Lemma C.26 (Uniform Hessian bounds for the constructed potential path)**.** *There are a strongly compatible end ϕ_∞, a smooth global extension ϕ, a radius $R > 0$, and constants $c, C > 0$ such that $\phi_t = (1-t)|x|^2/2 + t\phi(x)$ is jointly smooth and*

$$c|v|^2 \le D^2\phi_t(x)[v,v] \le C|v|^2 \quad (0 \le t \le 1,\ x, v \in M_{\mathbb{R}}).$$

For this same radius, every ϕ_t agrees outside the ball with $\phi_{\infty,t} = (1-t)|x|^2/2 + t\phi_\infty$, which is positively homogeneous of degree two. All norms and Hessian bounds here use the fixed Euclidean inner product.

Proof. Construct the continuously differentiable strong end with convexity correction $\delta > 0$ and growth at least $\delta|x|^2$. The corrected-convex extension gives a smooth ϕ with Hessian bounded below by δ times the identity and exact equality to this end outside a ball of radius $R > 0$. The global Hessian-upper-bound lemma gives an operator bound $C_1 > 0$; Cauchy–Schwarz then gives $D^2\phi(x)[v,v] \le C_1|v|^2$. The Hessian of $|x|^2/2$ is the identity, so the affine-path lemma gives the common bounds $c = \min(1,\delta) > 0$ and $C = \max(1, C_1) > 0$. Both terms defining the end path are homogeneous of degree two. The same affine-path lemma proves its homogeneity, and the literal end equality of ϕ implies equality of each pair of path members for all parameters outside the same radius. Joint smoothness is the affine-path smoothness assertion. □

Choice dependencies. Choose complex logarithms ℓ_v of the finitely many nonzero c_v and use $c_v(t) = \exp((1-t)\ell_v)$ to join c to 1. Construct the potential below and set $\phi_t = (1-t)\phi_0 + t\phi$. Its Hessian is positive definite and its end is homogeneous of degree two; compactness of the parameter interval and the unit sphere gives uniform positive lower and finite upper Hessian bounds. No radial cutoff is used in this interpolation. All coefficient and potential paths are fixed before any large-radius choices. Their constants are uniform on the compact parameter interval. Choose fixed inner and outer radius constants, then $\beta = \log R$ above every threshold in the proof. These thresholds are independent of the later base coefficient. For the supplied K, finally choose a finite normalized K_* with $K_*/R^2 \ge K$ and above the path-uniform threshold, as well as the compact tailoring threshold. Collars, rounding widths, hitting neighborhoods and shrinking times may depend on this now fixed compact parameter range. Throughout the proof, "for large β" means that there is one finite β_0 such that the assertion holds for every $\beta \ge \beta_0$. Unless stated otherwise, its constants depend only on the finite input, the fixed paths and the fixed auxiliary cutoffs, not on K, K_* or the eventual admissible levels and profiles. Finitely many such thresholds are replaced by their maximum.

All unlabelled vector norms are Euclidean in the fixed basis. On the homogeneous end the metrics defined by $D^2\phi_t$ are uniformly equivalent to that norm, by the preceding Hessian bounds; comparisons between these norms include those fixed constants. The symbol M denotes the lattice in the input section and a maximal monomial magnitude in estimates; H denotes a Hessian matrix, whereas H_β is a hypersurface.

Lemma C.27 (Polar face and character facts)**.** *The polar $P = \{x : \langle v, x\rangle \le 1\ (v \in V)\}$ is bounded and simple. Incident normals are independent and extend to a lattice basis; for $\sigma = \mathrm{cone}(v_1, \dots, v_k)$, the torus T_σ is connected of dimension $n-k$ and is defined by the generator character conditions.*

Proof. The inclusion of a ball about zero in Q bounds every point of P: for $x \in P$, pairing x with the point of that ball in its direction gives a uniform upper bound for $|x|$. The inequalities defining P are therefore a bounded full-dimensional intersection of finitely many half-spaces. At a boundary point x, its active vertices lie in the proper exposed face $\{q \in Q : \langle q, x\rangle = 1\}$. Every proper face of Q lies in a facet. A facet's vertices form a lattice basis, so they are independent and

that facet is a simplex. Its faces are simplices as well. Thus the active vertices are independent and extend to that basis. At a vertex of P, the active normals must span the real vector space, since otherwise a small segment in a common annihilating direction would remain in P. There are therefore precisely n active normals at every vertex, proving that P is simple. For any nonempty common facet intersection, its active normals are independent; choose a point in its relative interior. The remaining inequalities are strict there, so a neighborhood in the affine solution space of the k independent equalities has dimension $n-k$. No additional facet can contain this neighborhood: its normal would belong to the span of the k normals, contradicting independence of all active normals at that point. For $\sigma = \operatorname{cone}(v_1, \dots, v_k)$, extend these generators to an integral basis. Every lattice point in their real span has integral coordinates in that basis; nonnegative cone coordinates thus give $\sigma \cap N = \sum_i \mathbb{Z}_{\geq 0} v_i$. The generator character equations are consequently equivalent to all the lattice character equations. In the dual quotient coordinates they fix the first k angles to zero and leave the others arbitrary, giving a connected torus of dimension $n-k$. Finally, a vector in $\operatorname{relint} \sigma$ has strictly positive coordinates on every generator. It cannot lie in the span of a proper face, whose missing generator coordinates are zero. □

D A strongly compatible potential, constructed directly

Write $h(x) = \max_{v \in V} \langle v, x \rangle$. All norms in this section may be any fixed norm on the finite-dimensional real spaces, with its corresponding dual norm. Constants include the comparison factors between this norm and the coordinate maximum norm.

Definition D.1 (The literal polar body and its vertex facets)**.** *Put* $P = \{x : h(x) \leq 1\} = \{x : \langle v, x \rangle \leq 1 \ (v \in V)\}$ *and* $F_v = \{x \in P : \langle v, x \rangle = 1\}$.

Lemma D.2 (Compact polar facets and active pairings)**.** *The polar* P *and every* F_v *are compact, and* $0 \notin F_v$. *If* $v \in V$ *and* $x \in F_v$, *then* $h(x) = \langle v, x \rangle = 1$.

Proof. The support height is continuous, so P is closed. The lower bound $h(x) \geq c|x|$ bounds P by the closed ball of radius $1/c$. In finite dimension P is therefore compact. The equation $\langle v, x \rangle = 1$ defines a closed set, whose intersection with P is the compact F_v. Zero cannot satisfy that equation. For an actual vertex v, the finite maximum defining h gives $1 = \langle v, x \rangle \leq h(x) \leq 1$, proving the last assertion. □

Lemma D.3 (Finite closed cones cover the real space)**.** *The family* Σ *of cones on proper faces of* Q, *together with zero, is finite. Each cone is closed, and their union is* $N_{\mathbb{R}}$.

Proof. A face is specified by a subset of the finite vertex set, so the cone family is finite. Every nonempty proper face lies in a facet, whose vertices form a lattice basis. Its real generators are therefore independent. The linear map from their coefficient space to their span is an isomorphism onto a closed finite-dimensional subspace. The cone is the image of the closed nonnegative orthant under this isomorphism, so is closed in that subspace and hence in $N_{\mathbb{R}}$. The zero cone is closed as well. For $p \neq 0$, the intersection of Q with its positive ray is a bounded closed interval starting at zero and containing a positive initial interval, since zero is interior and Q is compact. Its last point tp lies on the boundary and hence in a proper face of Q. That point is a convex combination of the face vertices; scaling by $t^{-1} > 0$ expresses p as their nonnegative combination. Zero belongs to the zero cone. This proves coverage. □

Lemma D.4 (The additive face law for native cones)**.** *If* $\sigma, \tau \in \Sigma$, $p, q \in \sigma$, *and* $p + q \in \tau$, *then* $p, q \in \tau$.

Proof. The assertion is immediate for $\sigma = \{0\}$. Write $\sigma = \mathrm{cone}(S)$ for a nonempty proper face and choose its supporting covector ℓ_S, normalized to equal one on S and to be at most one on Q. The normalization is positive since zero is interior. A vector in σ with nonnegative coefficients has ℓ_S value equal to their sum; it is zero only when the vector is zero. This proves the assertion if $\tau = \{0\}$. Otherwise write $\tau = \mathrm{cone}(T)$ and normalize its supporting covector ℓ_T the same way. On σ, $L = \ell_T - \ell_S$ is nonpositive. On τ, it is nonnegative, so $L(p+q) = 0$. Since $L(p), L(q) \le 0$, both vanish. For any nonnegative expression of p in the vertices of S, $L(p) = 0$ forces every vertex with positive coefficient to have ℓ_T value one. Such vertices lie in T, so $p \in \tau$. The same reasoning applies to q. □

Lemma D.5 (Open-star incidence)**.** *Define* $\mathrm{Star}(v) = \bigcup_{\sigma\in\Sigma,\ v\in\sigma} \sigma$. *For every p and v,*

$$p \in \operatorname{int}\mathrm{Star}(v) \quad \Longleftrightarrow \quad \text{every } \sigma \in \Sigma \text{ containing } p \text{ contains } v.$$

Proof. If p is interior to the star, then for sufficiently small $t > 0$, $p - tv$ belongs to a cone τ containing v. Both $p - tv$ and tv belong to τ. For any cone σ containing p, the additive face law applied to $p = (p - tv) + tv$ puts tv in σ. Positive scaling then puts v in σ. Conversely, the finite union of closed cones not containing v is closed and misses p by the stated condition. Its complement is an open neighborhood of p. Coverage by the fan puts every point of this neighborhood in a cone containing v, hence in the star. □

Lemma D.6 (Open stars of a complete fan)**.** *For a ray v, let* $\mathrm{Star}(v)$ *be the union of the closed cones containing it. If τ is the minimal cone whose relative interior contains p, then*

$$p \in \operatorname{int}\mathrm{Star}(v) \quad \Longleftrightarrow \quad v \text{ is a ray of } \tau. \tag{9}$$

Proof. Every cone containing p has τ as a face: its intersection with τ is a face meeting $\operatorname{relint}\tau$, so equals τ. The finitely many closed cones not containing p miss a neighborhood of p. Completeness covers that neighborhood by the remaining cones. If v is a ray of τ, each remaining cone contains v, proving membership in the interior of the star. Conversely, if v is not a ray of τ, no point $p - tv$ with $t > 0$ belongs to the star. Otherwise a cone σ contains $p - tv$ and v, so also p, and τ is a face of σ. The face property applied to $p = (p - tv) + tv \in \tau$ forces $v \in \tau$. The fan intersection axiom then makes the ray of v a face of τ, a contradiction. The points $p - tv$ converge to p, excluding interior membership. This also proves the statement when $p = 0$. □

Lemma D.7 (Positive face averages belong to an open star)**.** *Let S be the vertex set of a proper face of Q. For nonnegative numbers a_b on S and $v \in S$ with $a_v > 0$, the vector $p = \sum_{b\in S} a_b b$ belongs to* $\operatorname{int}\mathrm{Star}(v)$.

Proof. The summand $a_v v$ and the remainder $\sum_{b\neq v} a_b b$ belong to $\mathrm{cone}(S)$. For any fan cone containing their sum p, the additive face law puts $a_v v$ in that cone. Since $a_v > 0$, scaling puts v in the same cone. The open-star incidence equivalence now puts p in the interior of the star. □

Lemma D.8 (Near-maximal weighted normals lie in open stars)**.** *There is $\eta > 0$ such that, for every $u \neq 0$ and nonnegative weights a_w on V satisfying $h(u) - \langle w, u\rangle \le \eta\|u\|_\infty$ whenever $a_w > 0$, one has $\sum_{w\in V} a_w w \in \operatorname{int}\mathrm{Star}(v)$ for every $v \in V$ with $a_v > 0$.*

Proof. Take the near-maximal-face tolerance η. The set of vertices with positive weight is contained in one proper face with vertex set S. All weights outside S vanish, so the sum over V equals the sum over S. The positive-face-average lemma applies to this sum for each vertex with positive weight, proving the result. □

Definition D.9 (Strongly compatible end potential)**.** *A strongly compatible end potential is a function ϕ_∞ on $M_{\mathbb{R}}$, smooth away from zero and homogeneous of degree two under positive dilations, with positive-definite Hessian away from zero, whose gradient Φ_∞ satisfies*

$$\Phi_\infty(F_v) \subset \operatorname{int}\operatorname{Star}(v) \qquad (v \in V). \tag{10}$$

Remark D.10 (Zero extension of the end function). The source defines the end function on the punctured space. Here it is written on the whole space by assigning zero at the origin, which is forced by positive degree-two homogeneity. Restriction and this zero extension are inverse operations. Strong compatibility requires smoothness and Hessian positivity only away from zero; continuous differentiability at zero is a separate conclusion of the construction.

Lemma D.11 (Smoothing the actual support height)**.** *Use the coordinate maximum norm and put $C = 1 + \sum_{v\in V} \|\langle v, \cdot\rangle\|$, with the operator norm on the linear functionals. For $\epsilon > 0$ normalize a smooth bump centered at zero, with inner radius $\epsilon/2$ and outer radius ϵ, to obtain a nonnegative smooth density ρ_ϵ of integral one, supported in the closed radius-ϵ ball and positive in the open ball. The convolution*

$$q_\epsilon(x) = \int h(x - z)\rho_\epsilon(z)\, dz$$

is smooth and convex, and $|q_\epsilon(x) - h(x)| \le C\epsilon$ everywhere. Its unit sublevel is bounded, and for sufficiently small ϵ, $q_\epsilon(0) < 1$.

Proof. Each linear function defining h has Lipschitz constant at most C. Taking their finite maximum preserves that bound, so $|h(x - z) - h(x)| \le C\|z\|$. Integration against the probability density gives the asserted approximation. Convexity of h follows from its being a maximum of linear functions. For $0 \le t \le 1$, its convexity inequality at $x - z$ and $y - z$, integrated against ρ_ϵ, gives convexity of q_ϵ. To prove smoothness, write the integral as $\int h(w)\rho_\epsilon(x - w)\, dw$. On every compact set of x, the integrands and all their x derivatives are supported in one compact set of w. The continuous function h is bounded there, and each derivative of the smooth density is bounded there. Differentiation under the integral is therefore valid to every order, with continuous derivatives, proving smoothness. The lower bound $h(x) \ge c\|x\|$ gives $q_\epsilon(x) \ge c\|x\| - C\epsilon$, so its unit sublevel is bounded. Finally $h(0) = 0$ gives $q_\epsilon(0) \le C\epsilon < 1$ for small ϵ. □

Lemma D.12 (Positive measure near an active maximizing vertex)**.** *For $v \in V$, let U_v be the open set where v uniquely maximizes among all vertices. If $\langle v, y\rangle = h(y)$ and $\epsilon > 0$, then $B(y, \epsilon) \cap U_v$ has positive Lebesgue measure. Equivalently, $\{z : \|z\| < \epsilon,\ y - z \in U_v\}$ has positive Lebesgue measure.*

Proof. Strictly separate the vertex v from the compact convex hull of all the other vertices, obtaining w with $\langle v, w\rangle > \langle b, w\rangle$ for every $b \ne v$. Since v maximizes at y, it uniquely maximizes at $y + tw$ for every $t > 0$. Choose t with $\|tw\| < \epsilon$. The finitely many strict inequalities remain true on an open neighborhood of $y + tw$, so that neighborhood lies in U_v. A sufficiently small open ball about $y + tw$ lies also in $B(y, \epsilon)$. This ball has positive Lebesgue measure, proving the first assertion. The affine change of variable $z \mapsto y - z$ preserves Lebesgue measure and carries the second set onto the first, proving the equivalence. □

Lemma D.13 (The deficit of a nearby maximizer)**.** *If $v \in V$ maximizes at y and $|x - y| \le \epsilon$, in the coordinate maximum norm, then $0 \le h(x) - \langle v, x\rangle \le 2C\epsilon$.*

Proof. The lower bound is the defining maximum inequality. Since $h(y) = \langle v, y\rangle$,

$$h(x) - \langle v, x\rangle = h(x) - h(y) + \langle v, y - x\rangle.$$

The first term is at most $C|x-y|$ by Lipschitz continuity, and the second is at most $C|x-y|$ by the operator norm of the vertex pairing. Adding these bounds proves the result. □

Lemma D.14 (The gradient average and its deficits)**.** *There are nonnegative weights $a_b(x)$, summing to one over $b \in V$, such that*

$$dq_\epsilon(x) = \sum_{b \in V} a_b(x) b.$$

Every b with $a_b(x) > 0$ satisfies

$$0 \le h(x) - \langle b, x \rangle \le 2C\epsilon. \tag{11}$$

If v maximizes at x, then $a_v(x) > 0$.

Proof. Outside the finitely many hyperplanes $\langle b - b', w \rangle = 0$ for distinct vertices, the maximum defining $h(w)$ is unique. These hyperplanes have Lebesgue measure zero. On the open region U_b where b uniquely maximizes, $dh = b$. Fix a direction w. The difference quotient of $h(x + tw - z)$ at $t = 0$ is bounded in absolute value by $C\|w\|$, uniformly in z and nonzero t, by Lipschitz continuity. Multiplication by $\rho_\epsilon(z)$ gives an integrable dominating function. Away from the null tie hyperplanes the quotient converges to $dh(x-z)[w]$. Dominated convergence therefore gives the directional derivative $\int dh(x-z)[w]\rho_\epsilon(z)\,dz$. The convolution is already smooth, so these identities for every w identify its differential with $\int dh(x-z)\rho_\epsilon(z)\,dz$. Thus take $a_b(x) = \int_{x-z \in U_b} \rho_\epsilon(z)\,dz$. The partition outside a null set proves nonnegativity and sum one. If $a_b(x) > 0$, there is z with $\|z\| < \epsilon$ such that b maximizes at $x - z$. The nearby-maximizer deficit lemma applied at $x - z$ gives the bound $0 \le h(x) - \langle b, x \rangle \le 2C\epsilon$. If v maximizes at x, the active-maximizer positive-volume lemma shows that the open ball about x meets U_v in positive measure. The density $\rho_\epsilon(x - \cdot)$ is positive on that intersection, so its integral there is positive. This is $a_v(x) > 0$. □

Lemma D.15 (The smooth gauge of a bounded convex sublevel)**.** *Let $q : M_{\mathbb{R}} \to \mathbb{R}$ be smooth and convex, with $q(0) < 1$ and bounded sublevel $B = \{x : q(x) \le 1\}$. Its Minkowski gauge j is convex and smooth away from zero. For $x \neq 0$ put $y = x/j(x)$. Then $q(y) = 1$, $dq(y)[y] \ge 1 - q(0) > 0$, and*

$$dj(x) = \frac{dq(y)}{dq(y)[y]}.$$

Proof. Continuity and $q(0) < 1$ put a neighborhood of zero in B. The set B is closed and convex. On each ray through $x \neq 0$, its intersection is a closed interval containing an initial positive interval, and boundedness makes its far endpoint tx finite. Continuity gives $q(tx) = 1$. For any z with $q(z) = 1$, the first-order convexity inequality gives $q(0) \ge q(z) - dq(z)[z]$, so $dq(z)[z] \ge 1 - q(0) > 0$. A ray cannot meet $q = 1$ twice: if $0 < s < t$ and $q(tx) = 1$, convexity at $sx = (1 - s/t)0 + (s/t)tx$ gives $q(sx) < 1$. Thus tx is the unique unit-level point of the ray and $j(x) = 1/t$. This proves positivity and positive homogeneity. For convexity, if $j(x), j(z) > 0$, the points $x/j(x)$ and $z/j(z)$ belong to B, so their convex combination proves $j(x + z) \le j(x) + j(z)$. Cases with a zero vector follow directly, and subadditivity together with homogeneity proves convexity. For smoothness apply the implicit function theorem to $(x, s) \mapsto q(x/s) - 1$ at $s = j(x) > 0$. Its s derivative is $-dq(y)[y]/j(x) \neq 0$. Uniqueness of the ray intersection identifies the resulting local smooth function with j. Differentiating $q(x/j(x)) = 1$ gives $dq(y)[h]/j(x) - dq(y)[y]dj(x)[h]/j(x) = 0$, which is the stated differential formula. □

Lemma D.16 (The actual homogeneous gauge prepotential)**.** *For $\epsilon > 0$ with $C\epsilon < 1$, let j_ϵ be the gauge of $\{q_\epsilon \le 1\}$ and put $f_\epsilon = j_\epsilon^2$. This is a continuously differentiable convex function, zero at zero, positive and smooth away from zero, and homogeneous of degree two. For $x \neq 0$ put $y = x/j_\epsilon(x)$. Then $q_\epsilon(y) = 1$ and $df_\epsilon(x)$ is a positive multiple of $dq_\epsilon(y)$.*

Proof. The actual support smoothing is smooth and convex, its unit body is bounded, and $q_\epsilon(0) < 1$. The native sublevel-gauge result therefore gives smoothness and normalization away from zero. The convex body contains a neighborhood of zero, which bounds its gauge above by a constant times $|x|$; subadditivity applied in both directions then proves continuity of the gauge. Boundedness of the body bounds the gauge below by a positive constant times $|x|$, proving positivity away from zero. The squared-gauge result gives convexity and degree-two homogeneity. Its value at zero is zero, and composition with the square gives the continuity and punctured smoothness assertions. The Lipschitz gauge vanishes at zero, so the squared-Lipschitz origin lemma also proves continuous differentiability on the whole space. Finally the chain rule and gauge differential formula give

$$df_\epsilon(x) = \frac{2j_\epsilon(x)}{dq_\epsilon(y)[y]} dq_\epsilon(y).$$

Both numerator and denominator are positive, proving the last assertion. □

Lemma D.17 (Compatibility of the smoothed gauge)**.** *There is $\epsilon > 0$ with $C\epsilon < 1$ such that the gauge j_ϵ of $\{q_\epsilon \le 1\}$ satisfies $d(j_\epsilon^2)(F_v) \subset \operatorname{int}\operatorname{Star}(v)$ for every $v \in V$.*

Proof. Let $\eta > 0$ be the weighted near-maximal open-star constant and choose $\epsilon = \min(1/(4C), \eta/(8C^2)) > 0$. Then $C\epsilon \le 1/4$ and $4C^2\epsilon \le \eta$. At $y = x/j_\epsilon(x)$ normalization and approximation give $h(y) \ge 1 - C\epsilon \ge 1/2$. The Lipschitz bound $h(y) \le C\|y\|_\infty$ therefore yields $C\|y\|_\infty \ge 1/2$. In particular $y \ne 0$, and

$$2C\epsilon \le 4C^2\epsilon\|y\|_\infty \le \eta\|y\|_\infty.$$

If $x \in F_v$, then $h(x) = \langle v, x\rangle = 1$. Positive homogeneity makes v maximize also at y. Every positive coefficient in $dq_\epsilon(y)$ has deficit at most $2C\epsilon \le \eta\|y\|_\infty$, and v has deficit zero. The weight of v is positive, so the weighted near-maximal open-star lemma puts $dq_\epsilon(y)$ in $\operatorname{int}\operatorname{Star}(v)$. The gauge differential formula and positivity of j_ϵ show that $d(j_\epsilon^2)(x) = 2j_\epsilon(x)dq_\epsilon(y)/dq_\epsilon(y)[y]$ is a positive multiple of this vector. Positive dilations preserve the star and its interior, completing the proof. □

Lemma D.18 (Convexity and homogeneity of the squared gauge)**.** *If B is a convex neighborhood of zero in a real normed space, its squared Minkowski gauge is convex. For every $t \ge 0$, $j_B(tx)^2 = t^2 j_B(x)^2$.*

Proof. The gauge is nonnegative, subadditive and positively homogeneous. Subadditivity and homogeneity imply its convexity. For a convex combination of two points, apply this convexity inequality, then the increasing square function on nonnegative numbers, and then convexity of the square function. This proves convexity of the square. Squaring the positive-homogeneity identity gives the displayed equality. □

Lemma D.19 (Quadratic perturbation preserves compact gradient images)**.** *Let $K_i \subset M_{\mathbb{R}}$ be finitely many compact sets, let $U_i \subset N_{\mathbb{R}}$ be open, and let f be smooth on a neighborhood of each point of every K_i, with $\nabla f(K_i) \subset U_i$. There is $\delta > 0$ such that $\nabla(f + \delta|x|^2)(K_i) \subset U_i$ for every i.*

Proof. On each K_i the gradient is continuous and its image is a compact subset of U_i. Its distance from the closed complement of U_i is positive whenever that complement is nonempty; an empty complement imposes no condition. The sets K_i are bounded. The gradient perturbation equals $2\delta x$, so its norm tends to zero uniformly on each K_i as δ tends to zero. For each index choose a positive upper bound on δ making this perturbation smaller than the indicated distance; empty K_i impose no condition. Take one positive δ smaller than the finitely many bounds. This proves all inclusions at once. □

Lemma D.20 (Existence of the actual strongly compatible end)**.** *For every finite toric input there is a strongly compatible end potential ϕ_∞, continuously differentiable on the whole space and zero at zero, and there is $\delta > 0$ such that $\phi_\infty - \delta|x|^2$ is convex, $\phi_\infty(x) \geq \delta|x|^2$, and $D^2\phi_\infty(x)[v, v] \geq 2\delta|v|^2$ for every $x \neq 0$ and v.*

Proof. Take the actual gauge prepotential whose gradient maps every polar facet into its open star. These facets are compact and avoid zero, so the prepotential is smooth near every point of each facet. The compact-gradient-image perturbation lemma supplies one $\delta > 0$ preserving all the star inclusions after adding $\delta|x|^2$. Set $\phi_\infty = j_\epsilon^2 + \delta|x|^2$. The actual prepotential gives its punctured smoothness, continuous differentiability, zero value at zero and degree-two homogeneity. Its convexity gives the convex correction, and nonnegativity gives the quadratic lower bound. At every nonzero point, restrict the corrected convex function to a line. Its derivative is nondecreasing; differentiating there gives the Hessian bound $2\delta|v|^2$ by the same argument as the convexity-correction Hessian lemma. This bound is strictly positive on nonzero v, proving the required positive Hessian. All strong-compatibility conditions therefore hold for this constructed function. □

Lemma D.21 (A small quadratic makes the Hessian positive)**.** *For a sufficiently small $\delta > 0$, $\phi_\infty(x) = j_\epsilon(x)^2 + \delta|x|^2$ is a strongly compatible end potential. Its Hessian has uniform positive lower and finite upper bounds on the punctured space.*

Proof. The nonnegative convex gauge has convex square: apply convexity of the gauge, monotonicity of the square on nonnegative numbers, and convexity of the square. Its smooth Hessian away from zero is consequently positive semidefinite, as follows by taking the second derivative on every line. Adding $\delta|x|^2$ adds 2δ times the identity to the Hessian and makes it positive definite. Each compact F_v has continuous image under $d(j_\epsilon^2)$ contained in the open set $\operatorname{int}\operatorname{Star}(v)$. Its distance from the closed complement is positive; if that complement is empty there is no restriction. The perturbation of the gradient is $2\delta x$, bounded uniformly on the finitely many facets. Taking one positive δ sufficiently small preserves all star inclusions. Smoothness and degree-two homogeneity are immediate. Differentiating homogeneity twice makes the Hessian degree zero. Its least and greatest quadratic-form values on the product of two Euclidean unit spheres attain positive and finite bounds, respectively. Scaling the base point therefore gives these bounds everywhere away from zero. □

Lemma D.22 (Smooth extension with a common convexity correction)**.** *Let E be a real inner-product space, let $c \leq a < d$, and let $f : E \to \mathbb{R}$ be continuous, with $f(0) = 0$, and smooth away from zero. Suppose $f - c|x|^2$ is convex and $f(x) \geq d|x|^2$ everywhere. Then there is a smooth $g : E \to \mathbb{R}$ such that $g - c|x|^2$ is convex, $g(x) = a|x|^2 + 1$ on a neighborhood of zero, and there is $R > 0$ such that $g(x) = f(x)$ whenever $|x| \geq R$.*

Proof. Use the smooth convex maximum of width $1/2$ and set $g(x) = m_{1/2}(a|x|^2 + 1, f(x))$. Its inputs differ by one at zero, so continuity and the exact tails give the quadratic formula on a neighborhood of zero. Away from zero both inputs are smooth; this formula therefore proves global smoothness. Since the profile derivatives sum to one, integration along a simultaneous translation gives $m_{1/2}(r + t, s + t) = m_{1/2}(r, s) + t$. Consequently

$$g(x) - c|x|^2 = m_{1/2}((a - c)|x|^2 + 1, f(x) - c|x|^2).$$

Both arguments on the right are convex. For any convex combination of points, their convexity inequalities, monotonicity of the profile in both arguments, and convexity of the profile prove convexity of this composition. Choose $R = \sqrt{(3/2)/(d - a)} > 0$. If $|x| \geq R$, then $f(x) - (a|x|^2 + 1) \geq (d - a)|x|^2 - 1 \geq 1/2$. The exact tail gives $g(x) = f(x)$, as required. □

Lemma D.23 (A convexity correction gives a Hessian lower bound)**.** *Let E be a real Hilbert space and $f : E \to \mathbb{R}$ differentiable. If $f(x) - c|x|^2$ is convex and its gradient is differentiable at x, then for every v,*

$$D^2 f(x)[v, v] \geq 2c|v|^2.$$

Proof. The function $l(t) = f(x+tv) - c|x+tv|^2$ is differentiable and convex on $\mathbb{R}$. The secant-slope inequalities for a convex function show that l' is nondecreasing. For $h > 0$ the difference quotient $(l'(h) - l'(0))/h$ is therefore nonnegative. Its limit as h decreases to zero is $l''(0)$, so $l''(0) \geq 0$. The chain rule gives $l''(0) = D^2 f(x)[v, v] - 2c|v|^2$, proving the result. In particular, a globally smooth f satisfies the bound at every point. □

Lemma D.24 (A smooth extension with its actual Legendre inverse)**.** *Let E be a finite-dimensional real inner-product space, and let $0 < c \leq a < d$. Suppose $f : E \to \mathbb{R}$ is continuous with $f(0) = 0$, smooth away from zero, $f - c|x|^2$ convex, and $f(x) \geq d|x|^2$. There is a smooth g equal to $a|x|^2 + 1$ near zero and equal to f outside a finite-radius ball, with $D^2 g(x)[v, v] \geq 2c|v|^2$ for every x, v. Its gradient is a diffeomorphism of E onto itself.*

Proof. Take the smooth extension retaining the convexity correction. The convexity-correction Hessian lemma gives the stated uniform lower bound. Since $2c > 0$, the uniform-gradient-inverse lemma applies to its gradient and proves the final assertion. All end and near-zero equalities are those of the same constructed extension. □

Lemma D.25 (Smooth convex extension across the origin)**.** *The end potential $\phi_\infty = j_\epsilon^2 + \delta|x|^2$ constructed above has a smooth positive-Hessian extension ϕ to the whole space, equal to ϕ_∞ outside a compact set and to $a|x|^2 + 1$ near zero, for some $a > 0$. The Hessian of ϕ has uniform positive lower and finite upper bounds.*

Proof. Extend the squared gauge by its value zero at the origin. The gauge is continuous because its convex body contains a neighborhood of zero, so this gives a continuous function $f = \phi_\infty$ with $f(0) = 0$. It is smooth away from zero. Put $c = \delta/2$, $a = 3\delta/4$, and $d = \delta$. Then $0 < c \leq a < d$, and $f - c|x|^2 = j_\epsilon^2 + (\delta/2)|x|^2$ is convex. Also $f(x) \geq d|x|^2$ because the squared gauge is nonnegative. The corrected-convex extension with Legendre inverse supplies the same smooth ϕ with the required exact end and near-zero formulas and a uniform positive Hessian lower bound. Outside a ball its Hessian is the degree-zero Hessian of the end potential, with a finite upper bound from the compact unit sphere. On the intervening compact ball, continuity supplies a finite upper bound. Taking the larger of these bounds proves the final assertion. □

Lemma D.26 (Strong monotonicity gives a global smooth inverse)**.** *Let E be a finite-dimensional real inner-product space and $g : E \to E$ a smooth map. Suppose there is $c > 0$ such that $\langle Dg(x)v, v\rangle \geq c|v|^2$ for all x, v. Then*

$$\langle g(y) - g(x), y - x\rangle \geq c|y - x|^2, \qquad |y - x| \leq c^{-1}|g(y) - g(x)|,$$

and g is a smooth diffeomorphism from E onto itself.

Proof. Apply the mean-value inequality to $t \mapsto \langle g(x + t(y - x)), y - x\rangle$ on $[0, 1]$. Its derivative is at least $c|y - x|^2$, giving the first inequality. Cauchy–Schwarz gives the second after division by $c|y - x|$ if $x \neq y$; if $x = y$ it is immediate. This also proves injectivity. At every point the derivative is injective, since a vector in its kernel has $c|v|^2 \leq 0$. Equal finite dimensions make it invertible. The inverse function theorem makes the image open. For every compact K, the second inequality with $y = 0$ bounds $g^{-1}(K)$, and continuity makes this preimage closed. In finite dimension it is

compact, so g is proper. Its image is closed: a convergent sequence of image points and its limit form a compact set; the chosen preimages have a convergent subsequence in its compact preimage, whose limit maps to the given image limit. The image is nonempty, open and closed in the connected vector space, so it is the whole space. The local smooth inverses from the inverse function theorem agree with the unique global inverse, proving its smoothness. □

Lemma D.27 (Euler identities on a homogeneous end)**.** *Let ϕ be homogeneous of degree two under positive dilations, and differentiable away from zero. For $x \neq 0$ and $t > 0$,*

$$d\phi(x)[x] = 2\phi(x), \qquad d\phi(tx) = t\, d\phi(x).$$

In fixed real coordinates set $\Phi_j(x) = d\phi(x)[e_j]$. Then $\Phi(tx) = t\Phi(x)$, and whenever Φ is differentiable at x, its derivative $H(x) = D\Phi(x)$ satisfies $H(x)x = \Phi(x)$.

Proof. Differentiate $\phi(sx) = s^2\phi(x)$ with respect to s at $s = 1$; positive s form a neighborhood of one. The chain rule gives the first identity. For a fixed $t > 0$, differentiate $\phi(ty) = t^2\phi(y)$ with respect to y at x in direction v. This gives $t\, d\phi(tx)[v] = t^2 d\phi(x)[v]$. Division by t proves the second identity. Applying it to each coordinate basis vector gives $\Phi(tx) = t\Phi(x)$. If Φ is differentiable at x, differentiate this last identity with respect to t at one, obtaining $D\Phi(x)x = \Phi(x)$. □

Lemma D.28 (Uniform derivative bounds on a homogeneous end)**.** *Let E, F be finite-dimensional real normed spaces. A continuous degree-zero map $A : E \setminus \{0\} \to F$ is uniformly bounded. If $g : E \to F$ is smooth away from zero and $g(tx) = tg(x)$ for $t > 0$, then*

$$Dg(tx) = Dg(x) \quad (x \neq 0, t > 0), \qquad \sup_{x \neq 0} \|Dg(x)\| < \infty.$$

Proof. Continuity bounds A on the compact unit sphere. For every $x \neq 0$, degree-zero homogeneity gives $A(x) = A(x/\|x\|)$, proving the uniform bound. If the sphere is empty, the punctured space is empty and any positive bound works. For the second assertion differentiate $g(ty) = tg(y)$ at $y = x$, obtaining $tDg(tx) = tDg(x)$. Division by $t > 0$ gives the identity. Smoothness makes the derivative continuous away from zero, so the first assertion applied to Dg proves its bound. □

Lemma D.29 (A homogeneous end bounds the global extension Hessian)**.** *Let E be a finite-dimensional real inner-product space, $\phi : E \to \mathbb{R}$ smooth, and $\phi_\infty : E \to \mathbb{R}$ continuously differentiable and smooth away from zero. Suppose ϕ_∞ is positively homogeneous of degree two and $\phi(x) = \phi_\infty(x)$ whenever $|x| \geq R$, for some $R > 0$. There is $C > 0$ such that $\|D(\nabla\phi)(x)\| \leq C$ for every x.*

Proof. The gradient of ϕ_∞ is homogeneous of degree one. This holds away from zero by differentiation of degree-two homogeneity; at zero its derivative vanishes, since differentiating $\phi_\infty(tx) = t^2\phi_\infty(x)$ at $x = 0$ gives $t\, d\phi_\infty(0) = t^2 d\phi_\infty(0)$ for $t > 0$. Apply the homogeneous-derivative bound to this gradient to bound its Hessian at every nonzero point by a finite constant $C_1 > 0$. On the compact ball $\{|x| \leq R\}$, continuity of the smooth global Hessian supplies a finite bound C_2. When $|x| > R$, the two potentials agree on an open neighborhood of x, so their gradient derivatives agree at x and the first bound applies. Taking $\max(C_1, C_2) > 0$ proves the claim, including the sphere $|x| = R$ by the compact-ball bound. □

Lemma D.30 (A squared Lipschitz gauge is continuously differentiable at zero)**.** *Let j be a real-valued C-Lipschitz function on a real normed space, with $j(0) = 0$, smooth away from zero. Then j^2 is continuously differentiable on the whole space, with derivative zero at zero, and $\|d(j^2)(x)\| \leq 2C^2\|x\|$ everywhere.*

Proof. Lipschitz continuity gives $|j(x)| \le C\|x\|$. Thus $|j(x)^2| \le C^2\|x\|^2 = o(\|x\|)$ at zero, proving the zero derivative there. At a nonzero point, differentiating the Lipschitz inequality in each unit direction gives $\|dj(x)\| \le C$. The product rule now yields $\|d(j^2)(x)\| \le 2|j(x)|\|dj(x)\| \le 2C^2\|x\|$; the same bound holds at zero. The derivative is continuous away from zero by smoothness and tends to its zero value at zero by this estimate. This proves continuous differentiability everywhere. □

Lemma D.31 (A strongly convex punctured gradient has a smooth inverse)**.** *Let E be a finite-dimensional real inner-product space. Suppose $f : E \to \mathbb{R}$ is continuously differentiable, $c > 0$, $f - c|x|^2$ is convex, $f(x) \ge c|x|^2$, and $\nabla f(0) = 0$. Then its gradient is a homeomorphism of E onto itself, restricting to a homeomorphism of $E \setminus \{0\}$. If f is smooth away from zero and for some $d > 0$ its Hessian satisfies $D^2 f(x)[v, v] \ge d|v|^2$ whenever $x \ne 0$, this punctured homeomorphism and its inverse are smooth.*

Proof. The supporting inequality for the differentiable convex function $f - c|x|^2$ gives

$$f(y) \ge f(x) + df(x)[y - x] + c|y - x|^2.$$

Interchanging x, y and adding yields $\langle \nabla f(y) - \nabla f(x), y - x\rangle \ge 2c|y - x|^2$. Cauchy–Schwarz implies $|y - x| \le (2c)^{-1}|\nabla f(y) - \nabla f(x)|$, proving injectivity and continuity of the inverse on the image. For each p, the lower quadratic bound makes $f(x) - \langle p, x\rangle$ coercive. It attains a minimum on a compact sublevel, and differentiability gives $\nabla f(x) = p$ there. Thus the gradient is surjective. Continuous differentiability of f makes the gradient continuous, so it is a homeomorphism. Its injectivity and value zero at zero identify zero as the unique preimage of zero, giving the punctured restriction. Smoothness of f gives smoothness of its gradient at every nonzero point. The positive Hessian bound makes that derivative invertible in finite dimension. The inverse function theorem supplies local smooth inverses, which agree with the unique global inverse. This proves smoothness of the punctured inverse. □

Lemma D.32 (The punctured homogeneous gradient inverse)**.** *The gradient of the constructed degree-two end potential $\phi_\infty = j_\epsilon^2 + \delta|x|^2$ is a smooth diffeomorphism between the punctured real dual spaces.*

Proof. The gauge of the bounded convex unit body containing a neighborhood of zero is Lipschitz: its upper bound by a constant times $|x|$ and subadditivity in both directions bound $|j_\epsilon(x) - j_\epsilon(y)|$ by a constant times $|x - y|$. Its value at zero is zero, and it is smooth away from zero. The squared-Lipschitz origin lemma therefore gives a continuously differentiable extension of j_ϵ^2, with derivative zero at zero. Add the quadratic term to obtain the same properties for ϕ_∞. The function $\phi_\infty - \delta|x|^2 = j_\epsilon^2$ is convex, and $\phi_\infty(x) \ge \delta|x|^2$. Its gradient at zero is zero, and away from zero it is smooth with Hessian bounded below by 2δ times the identity. The strongly convex punctured-inverse lemma, after identifying the dual spaces by the fixed Euclidean basis, gives the claimed diffeomorphism. □

Lemma D.33 (Existence without an external potential theorem)**.** *A strongly compatible end potential exists. Its extension by zero is continuously differentiable. It has a smooth positive-Hessian extension ϕ to $M_\mathbb{R}$, equal to ϕ_∞ outside a compact set and equal to $a|x|^2 + 1$ near zero for some $a > 0$. The global gradient $\Phi = \nabla\phi$ is a diffeomorphism onto $N_\mathbb{R}$. The homogeneous gradient Φ_∞ is a diffeomorphism between the punctured real vector spaces.*

Proof. Choose the strongly compatible end from its existence lemma, together with $\delta > 0$ for which $\phi_\infty - \delta|x|^2$ is convex, $\phi_\infty(x) \ge \delta|x|^2$, and its punctured Hessian is bounded below by 2δ times the

identity. This same function is continuously differentiable on the whole space and has value zero at zero. Its difference with $(\delta/2)|x|^2$ is convex, being the sum of the first convex correction and $(\delta/2)|x|^2$. Apply the corrected-convex extension with Legendre inverse with $c = a = \delta/2$ and $d = \delta$. It produces a single smooth ϕ with the stated exact formulas, Hessian at least δ times the identity, and a smooth gradient inverse. The quadratic lower bound and value zero show that ϕ_∞ has its global minimum at zero. Differentiability gives $d\phi_\infty(0) = 0$. The strongly convex punctured-inverse lemma, with convexity correction δ and punctured Hessian bound 2δ, gives the other smooth inverse. Identifying the dual spaces using the fixed Euclidean basis transports both actual gradients and their inverses to $M_{\mathbb{R}}$ and $N_{\mathbb{R}}$; the coordinate components are precisely $d\phi(x)[e_j]$ and $d\phi_\infty(x)[e_j]$. □

Remark D.34 (How the construction uses the finite input). The common-face step uses the direction-uniform near-maximal-face estimate rather than a separate minimum on a compact annulus. Radial regularity of the smoothed unit level follows directly from convexity and $q_\epsilon(0) < 1$. The global inverse argument uses strong monotonicity and an open and closed image; the homogeneous punctured inverse uses the continuously differentiable squared gauge and retains the strictly convex minimization argument. These changes preserve the potential, facet conditions and gradient conclusions of the source.

Definition D.35 (The literal intersection of polar facets)**.** *For a finite set S of vertices, put $F_S = \{x \in P : \langle v, x\rangle = 1$ for every $v \in S\}$. For $S \neq \varnothing$ this is exactly $\bigcap_{v\in S} F_v$.*

Lemma D.36 (The support of a nonempty polar-facet intersection)**.** *Every F_S is compact and convex. If $S \subset V$ is nonempty and F_S is nonempty, S is the vertex set of a proper face of Q. Conversely, for the vertex set S of any proper face there is $y \in F_S$ with $\langle w, y\rangle < 1$ for every $w \in V \setminus S$.*

Proof. The defining hyperplanes are closed, so F_S is a closed subset of the compact polar body. Intersecting its convex body with affine hyperplanes proves convexity. For a point $x \in F_S$ with S nonempty, an imposed pairing equal to one makes $x \neq 0$ and $h(x) = 1$. Thus S lies in the proper exposed maximizing face at x. The input makes every proper face a simplex, and each subset of a simplex's vertices is a face; its subface is also a face of Q. Hence S is a proper face support. For the converse choose an exposing functional L for S with maximum a on Q. This level is positive because zero is interior and the face is proper. In the fixed dual basis write $L(v) = \langle v, u\rangle$ and put $y = u/a$. All vertices pair at most one with y, exactly the vertices of S pair to one, and the others pair strictly below one. This is the required point. □

Lemma D.37 (Intersection of native face cones)**.** *For proper face supports S, T,* $\operatorname{cone}(S) \cap \operatorname{cone}(T) = \operatorname{cone}(S \cap T)$*, with the empty cone interpreted as $\{0\}$. In particular, a nonzero vertex ray belongs to a face cone exactly when its vertex is among that face's generators.*

Proof. The inclusion from right to left follows by extending nonnegative coefficients by zero. Zero belongs to both sides. For a nonzero point on the left, normalize each of its nonnegative representations to write it as $rx = qy$, where $r, q > 0$, $x \in \operatorname{conv}(S)$, and $y \in \operatorname{conv}(T)$. Normalize exposing functionals L_S, L_T to have maximum one on Q. Then $r = qL_S(y) \leq q$ and $q = rL_T(x) \leq r$. Thus $r = q$ and $x = y$. The common point lies in both faces. Their intersection is the convex hull of their common vertices: in a convex combination of vertices of S, equality for the exposing functional of T forces every positive coefficient to lie in T. Consequently $x \in \operatorname{conv}(S \cap T)$, and the point lies in its cone. Apply this equality to the singleton vertex face and the other face. If their vertex intersection were empty, their cones could intersect only at zero, proving the final assertion. □

Lemma D.38 (Positive active normals at a face minimum)**.** *Let ϕ_∞ be strongly compatible, $S \subset V$ nonempty, and x a minimum of its restriction to F_S. With $A = \{v \in V : \langle v, x\rangle = 1\}$, there are coefficients $a_v > 0$ for $v \in A$ such that $\Phi_\infty(x) = \sum_{v\in A} a_v v$.*

Proof. An imposed pairing makes $x \neq 0$ and $h(x) = 1$, so A is a nonempty proper face support. A direction annihilating every active normal admits a two-sided line through x in F_S for sufficiently small times: active pairings remain one and each of the finitely many inactive strict inequalities persists. Differentiability and minimality therefore imply $d\phi_\infty(x)[d] = 0$ for every such direction. Choose a native face cone containing $p = \Phi_\infty(x)$ and write $p = \sum_{w\in T} a_w w$ with $a_w \geq 0$. Strong compatibility puts p in the interior of the star of each $v \in A$. The star incidence criterion and the native cone-intersection lemma give $A \subset T$. Extend T to a facet basis. For $w \in T \setminus A$, its dual basis direction has pairing zero with all active normals and extracts the coefficient a_w from p. Stationarity forces $a_w = 0$. For $v \in A$, if $a_v = 0$, the same representation would put p in the cone on $T \setminus \{v\}$, a native face cone lacking the ray of v. This contradicts its open-star incidence. Thus $a_v > 0$ on A, and removing the zero coefficients proves the formula. □

Lemma D.39 (Interior and prescribed normals of a face minimum)**.** *For a nonempty proper face support S, every minimum x of a strongly compatible end on F_S belongs to* relint F_S *and has $\Phi_\infty(x) = \sum_{v\in S} a_v v$ with all $a_v > 0$.*

Proof. Take the point $y \in F_S$ at which only S is active, supplied by the support lemma. The positive-active-normal formula at x gives

$$d\phi_\infty(x)[y - x] = \sum_{v\in A} a_v(\langle v, y\rangle - 1).$$

Every summand is nonpositive, and if $A \setminus S$ is nonempty at least one is strictly negative. But the segment from x to y stays in F_S, so its one-sided derivative at the minimum cannot be negative. Thus $A = S$, proving the prescribed-normal formula. All unprescribed inequalities are now strict at x. In the affine span of F_S the prescribed equations hold identically; the finitely many strict inequalities give an open neighborhood of x contained in F_S. This proves its actual relative-interior membership. □

Lemma D.40 (Uniqueness from a positive convexity correction)**.** *Suppose a strongly compatible end has a continuously differentiable zero extension and $\phi_\infty - c|x|^2$ is convex for $c > 0$ in the fixed Euclidean coordinates. For every nonempty proper face support S, its restriction to F_S has a unique minimum, with the relative interior and positive prescribed coefficients of the preceding lemma.*

Proof. The support lemma gives a nonempty compact face. Continuity gives a minimum, and the preceding lemma gives its interior and normal sum. For two minima x, y, each normal in that sum pairs to zero with $y - x$, since both points satisfy its equation with value one. Thus both gradients annihilate $y - x$. Strong gradient monotonicity for the positive convexity correction gives

$$2c|y - x|^2 \leq \langle \Phi_\infty(y) - \Phi_\infty(x), y - x\rangle = 0.$$

As $c > 0$, this forces $y = x$, proving uniqueness. □

Lemma D.41 (The minimum on every nonempty polar-facet intersection)**.** *The finite input supplies a strongly compatible end ϕ_∞, continuously differentiable on the whole space with value zero at zero, such that for every nonempty $S \subset V$ with $F_S \neq \varnothing$, the restriction has a unique minimum $x_S \in$ relint F_S and $\Phi_\infty(x_S) = \sum_{v\in S} a_v v$ with all $a_v > 0$. The same face conclusion holds for every end satisfying the quantitative properties in Lemma D.40.*

Proof. Choose the end from the strong-end-existence lemma, retaining its continuous differentiability and positive convexity correction. For each nonempty facet intersection the support lemma derives a nonempty proper face support from its actual points. Apply the quantitative minimum lemma to this same end and that support. Its assumptions are supplied by the construction, so no face-minimum or boundary hypothesis is added to the input. Applying it to any other end satisfying those explicit assumptions proves the final assertion. □

Lemma D.42 (Uniform angular exclusion)**.** *For any strongly compatible end potential there is $\delta_0 > 0$ such that for every native fan cone σ, every u, and every vertex $a \notin \sigma$,*

$$\Phi_\infty(u) \in \sigma \quad \Longrightarrow \quad h(u) - \langle a, u\rangle \geq \delta_0 |u|. \tag{12}$$

Proof. Suppose $u \neq 0$ and a maximizes at u. The positive support height gives $u/h(u) \in F_a$. Strong compatibility and gradient homogeneity give $\Phi_\infty(u) \in \operatorname{int} \operatorname{Star}(a)$. The open-star incidence criterion therefore puts a in every native cone containing $\Phi_\infty(u)$. Consequently the deficit is strictly positive whenever $a \notin \sigma$ and $\Phi_\infty(u) \in \sigma$. For a fixed such pair (σ, a), consider $K = \{|u| = 1 : \Phi_\infty(u) \in \sigma\}$. It is compact, because the gradient is continuous on the unit sphere and σ is closed. The continuous strictly positive deficit has a positive minimum on K when K is nonempty; when it is empty any positive bound works. There are finitely many cones and vertices, so the minimum of the resulting positive constants is a positive δ_0. For $u \neq 0$, apply this bound to $u/|u|$. Positive scaling preserves cone membership, and both the gradient and support deficit are homogeneous of degree one. Multiplication by $|u|$ proves the claim. At zero its two sides are zero. □

Lemma D.43 (Face minima and the angular exclusion gap)**.** *There is a single constructed strongly compatible end for which every nonempty proper polar-facet intersection has its unique relative interior minimum with positive prescribed normal coefficients, and for which one positive constant gives (12) for all native fan cones. These conclusions also hold for a fixed strongly compatible end with the continuous differentiability and positive convexity correction of Lemma D.40.*

Proof. Choose the end with all face minima from Lemma D.41. It is strongly compatible, so apply the angular-exclusion lemma to that same function; this gives one angular constant simultaneously for every cone and vertex. For a fixed end satisfying the stated quantitative assumptions, the quantitative minimum lemma gives each face conclusion and the angular lemma again applies to the same end. In particular the originally constructed end with its retained positive convexity correction satisfies both parts together. □

E The original total space and its full far-fiber ribbon

In this section ϕ is any smooth positive-Hessian potential with a degree-two positive-Hessian end. Set

$$\lambda_\phi = \sum_j \partial_j \phi \, d\theta_j, \quad \omega_0 = d\lambda_\phi, \quad H = D^2\phi, \quad r = \sqrt{2\phi} \quad \text{on the homogeneous end.}$$

There $Hu = \nabla\phi$, $u^T H u = r^2$, and the initial Liouville field is $Z_0 = u\partial_u$. For a holomorphic polynomial $f = \sum_{v \in V} m_v$, $m_v = c_v e^{\langle v, u + i\theta\rangle}$, write

$$G = \sum_v m_v v, \qquad M = \max_v |m_v|, \qquad a = \overline{G}^T H^{-1} G, \qquad b = u \cdot G.$$

All constants below can be chosen uniformly on the compact coefficient and potential paths fixed in Section C.

Definition E.1 (The adapted quotient primitive)**.** *For a based real potential ϕ on $M_{\mathbb{R}}$, its adapted primitive on $Y = M_{\mathbb{R}} \times T$ is the actual one-form $\lambda_\phi = \sum_j d\phi(u)[e_j]\, d\theta_j$, where the angular forms are the descended forms of Definition A.8.*

Lemma E.2 (The strict cotangent Legendre map)**.** *Suppose ϕ is smooth and its actual gradient Φ is a smooth diffeomorphism between the based real dual spaces. The quotient map $\mathcal{L}(u,\theta) = (\theta, \Phi(u))$ is a smooth diffeomorphism $Y \to T^*T$ with $\mathcal{L}^*\lambda_{\mathrm{can}} = \lambda_\phi$. In particular $d\lambda_\phi$ is symplectic.*

Proof. The displayed product map has inverse $(\theta, p) \mapsto (\Phi^{-1}(p), \theta)$, which is smooth because Φ^{-1} is smooth. Its torus component is the identity, so $\mathcal{L}^* d\theta_j = d\theta_j$. Its momentum component is $\Phi_j(u) = d\phi(u)[e_j]$. Substituting these two identities into $\lambda_{\mathrm{can}} = \sum_j p_j d\theta_j$ proves the primitive equality on the quotient itself. Exterior differentiation commutes with pullback, giving $d\lambda_\phi = \mathcal{L}^* d\lambda_{\mathrm{can}}$. The latter is closed; it is nondegenerate because $d\lambda_{\mathrm{can}}$ is nondegenerate and the derivative of the diffeomorphism is an isomorphism at each point. Thus it is symplectic. □

Lemma E.3 (A constructed adapted cotangent identification)**.** *The finite toric input supplies a strongly compatible homogeneous end and a smooth global potential with literal equality outside a ball, whose adapted primitive is symplectic and has the strict cotangent identification of Lemma E.2.*

Proof. Choose the potential and its actual smooth gradient inverse from Lemma D.33. Apply the strict cotangent-map lemma using that same gradient and potential. Its primitive equality and symplecticity belong to this constructed quotient map, while the literal equality outside a ball is the one supplied by the potential construction. □

Lemma E.4 (The adapted Liouville field on a homogeneous end)**.** *Let ϕ be smooth with smooth gradient diffeomorphism, and suppose it equals a degree-two homogeneous ϕ_∞, smooth away from zero, whenever $|u| \geq R > 0$. Throughout the open end $|u| > R$, its Hessian $H = D\Phi$ satisfies $Hu = \Phi(u)$, and the Liouville field of its literal primitive is $Z_0 = u\partial_u$. In local real-angle tangent coordinates,*

$$d\lambda_\phi((v,w),(v',w')) = \sum_j \big((Hv)_j w'_j - (Hv')_j w_j\big).$$

Proof. The derivative of $\mathcal{L}$ sends (v,w) to (w, Hv). Evaluate the canonical form $\sum_j dp_j \wedge d\theta_j$ on these images and use the strict pullback identity to obtain the displayed formula. On the open end the two potentials agree in a neighborhood of every point, so their first and second derivatives agree there. The homogeneous Euler identities therefore give $Hu = \Phi(u)$. Substitution of $(u,0)$ into the first slot of the displayed form gives $\sum_j \Phi_j(u) w'_j = \lambda_\phi(v',w')$. This is exactly the Liouville contraction equation. Nondegeneracy from the strict cotangent-map lemma makes its solution unique, so the actual Liouville field is radial there. □

Lemma E.5 (The constructed potential has a radial Liouville end)**.** *There is a strongly compatible end and a smooth global potential with symplectic adapted primitive, agreeing literally outside a ball, for which the actual Liouville field is $u\partial_u$ throughout the open homogeneous end.*

Proof. Take the constructed potential, end, radius and gradient inverse from Lemma D.33. The adapted radial-end lemma applies with these data and proves the field equality. Its primitive is symplectic by the same strict cotangent-map argument. □

Definition E.6 (Largest actual monomial magnitude)**.** *For $y = (u,\theta)$, put*

$$M(y) = \max_{v \in V} |c_v e^{\langle v,u\rangle + i\langle v,\theta\rangle}|.$$

This is the maximum of the actual finite support, with its prescribed coefficients.

Lemma E.7 (Attainment and positivity of the largest monomial)**.** *The maximum $M(y)$ is attained by a vertex, every actual monomial has magnitude at most $M(y)$, and $M(y) > 0$.*

Proof. The support is finite and nonempty: a polytope containing a neighborhood of zero in positive dimension cannot have empty vertex set. A finite nonempty set of real numbers has a maximum, obtained by successively comparing its entries. Each entry is positive because $c_v \neq 0$ and the complex exponential never vanishes. The attained entry is therefore positive and dominates all entries. □

Lemma E.8 (A vertex pairing controls the Euclidean radius)**.** *There exists $c > 0$ such that for every $u \in M_{\mathbb{R}}$ some $v \in V$ satisfies $\langle v, u\rangle \geq c|u|$.*

Proof. Choose $c > 0$ such that the Euclidean ball of radius c about zero lies in Q. For $u \neq 0$, the point $cu/|u|$ is a convex combination $\sum_v t_v v$, with $t_v \geq 0$ and $\sum_v t_v = 1$. Pairing with u gives $c|u| = \sum_v t_v \langle v, u\rangle \leq \max_v \langle v, u\rangle$. A maximizing vertex supplies the claim. At $u = 0$ every vertex supplies it. □

Lemma E.9 (Exponential growth of the actual finite support)**.** *There exist $A, c > 0$ such that $M(u, \theta) \geq Ae^{c|u|}$ for every $(u, \theta) \in Y$.*

Proof. Take c from Lemma E.8 and let $A = \min_{v\in V} |c_v| > 0$. Choose v with $\langle v, u\rangle \geq c|u|$. The quotient character has modulus one, hence its actual monomial has magnitude $|c_v|e^{\langle v,u\rangle} \geq Ae^{c|u|}$. The maximum dominates that monomial. This proof includes rank one. □

Lemma E.10 (Uniform complex independence on proper faces)**.** *There is $c > 0$ such that for every nonempty proper face with vertex set B and every complex coefficient tuple $\gamma \in \mathbb{C}^B$,*

$$c \max_{v\in B} |\gamma_v| \leq \max_j \left| \sum_{v\in B} \gamma_v v_j \right|.$$

Proof. The face vertices extend to the integral basis of a facet, so they are real linearly independent. If their complex linear combination is zero, its real and imaginary parts each vanish and real independence makes every coefficient zero. For each fixed nonempty face the map $\gamma \mapsto \sum_v \gamma_v v$ is therefore injective. Its norm has a positive minimum on the compact coefficient unit sphere for the maximum norm: a zero minimum would be attained and contradict injectivity. By homogeneity this minimum is a lower bound for every tuple, including zero. There are only finitely many proper faces. Taking the minimum of their positive constants gives one positive constant for all of them. □

Remark E.11 (Norm convention for the face estimate). The coefficient and image norms in Lemma E.10 are coordinate maximum norms. The source estimate is Euclidean. In finite dimension $\max_j |z_j| \leq |z| \leq \sqrt{n} \max_j |z_j|$, so this intermediate bound implies the required Euclidean lower bound, with its constant adjusted if necessary.

Remark E.12 (Coefficient bounds). The support geometry is unchanged by varying the coefficients. The uniform estimates below use only a common positive lower bound and a finite upper bound for their magnitudes. The discrepancy argument uses $te^{-t} \leq 1$ in place of $te^{-t} \leq 1/e$, which changes only an unspecified estimate constant.

Lemma E.13 (Relative exponential monomial tails)**.** *For each fixed input there is $C > 0$ such that every monomial satisfies $|m_v(y)| \leq Ce^{-(h(u)-\langle v,u\rangle)}M(y)$. Consequently, for every $\epsilon > 0$ there is $D > 0$ such that $h(u) - \langle v, u\rangle \geq D$ implies $|m_v(y)| \leq \epsilon M(y)$, uniformly in y and v.*

Proof. Let $A = \min_v |c_v| > 0$ and $S = \sum_v |c_v| > 0$. A vertex attaining h has monomial magnitude at least Ae^h, so $M \geq Ae^h$. For every v, $|m_v| = |c_v|e^{\langle v,u\rangle} \leq Se^{\langle v,u\rangle}$. Combining these bounds gives the first assertion with $C = S/A$. For a given $\epsilon > 0$ choose $D = \max(1, \log(C/\epsilon))$. Then $D > 0$ and a deficit at least D gives $Ce^{-(h-\langle v,u\rangle)} \leq Ce^{-\log(C/\epsilon)} = \epsilon$. Multiplication by the positive M proves the conclusion. □

Definition E.14 (Total pairing deficit)**.** *For $B \subset V$ set*

$$\Delta_B(u) = \sum_{v\in B} \sum_{w\in V} \max(\langle w, u\rangle - \langle v, u\rangle, 0).$$

In the following three lemmas write $\|u\|_\infty = \max_j |u_j|$.

Lemma E.15 (Uniform positive deficit for subsets outside proper faces)**.** *There is $\delta > 0$ such that every subset $B \subset V$ which is not contained in a nonempty proper face satisfies $\Delta_B(u) \geq \delta\|u\|_\infty$ for all u.*

Proof. Fix such a subset B and a nonzero u. If $\Delta_B(u) = 0$, each nonnegative summand is zero. Every $v \in B$ then maximizes the pairing among all vertices. The maximizing vertex set is a nonempty exposed face. It is proper: zero interior containment gives vertices pairing positively and negatively with the nonzero u. Thus it contains B, a contradiction. Hence $\Delta_B(u) > 0$. The finite sum Δ_B is continuous and positively homogeneous of degree one. It has a positive attained minimum δ_B on the compact sphere $\|u\|_\infty = 1$. For $u \neq 0$, scaling $u/\|u\|_\infty$ gives $\Delta_B(u) \geq \delta_B\|u\|_\infty$; at zero both sides vanish. There are finitely many subsets of V. Taking the minimum of the positive δ_B over the subsets in question gives one positive δ. If there are no such subsets, any positive δ works. □

Lemma E.16 (Near-maximal vertices lie in one proper face)**.** *There is $\eta > 0$ such that, for every $u \neq 0$ and $B \subset V$, the bounds $h(u) - \langle v, u\rangle \leq \eta\|u\|_\infty$ for all $v \in B$ imply that B is contained in a nonempty proper face.*

Proof. Take δ from Lemma E.15, let $N = |V|$, and choose $\eta = \delta/(2(N^2+1)) > 0$. Every summand of $\Delta_B(u)$ is at most $\eta\|u\|_\infty$, since $\langle w, u\rangle \leq h(u)$ and the positive-part operation preserves this nonnegative bound. There are at most N^2 summands, so $\Delta_B(u) \leq N^2\eta\|u\|_\infty < \delta\|u\|_\infty$. If B were not contained in a proper nonempty face, the last inequality would contradict the preceding lemma. □

Lemma E.17 (A proper face contains a largest monomial and its small complement)**.** *For every $\epsilon > 0$ there exists $L > 0$ such that whenever $\|u\|_\infty \geq L$, one nonempty proper face contains a vertex attaining $M(y)$ and every off-face monomial has magnitude at most $\epsilon M(y)$.*

Proof. Put $q = \min(\epsilon, 1/2) > 0$ and obtain a deficit cutoff $D > 0$ for tolerance q from Lemma E.13. Take η from Lemma E.16 and choose $L = 1 + D/\eta$. For $\|u\|_\infty \geq L$, let $B = \{v \in V : h(u) - \langle v, u\rangle \leq D\}$. Its deficits are at most $\eta\|u\|_\infty$, so it is contained in a nonempty proper face F. Every vertex outside F is outside B and has deficit greater than D. Its monomial magnitude is therefore at most $qM \leq \epsilon M$. A monomial attaining $M > 0$ cannot lie outside F, where the bound $qM \leq M/2$ would contradict attainment. Thus F has the required properties. □

Lemma E.18 (Native derivative bound from one face and its literal tail)**.** *There is $c > 0$ such that whenever a nonempty proper face F contains a vertex attaining M and $\|G - G_F\|_\infty \leq cM$, where $G_F = \sum_{v\in F} m_v v$, one has $cM \leq \sqrt{a}$.*

Proof. Let $c_0 > 0$ be the uniform face-independence constant and put $c = c_0/2$. Since the face contains a largest monomial, its coefficient maximum is M and $\|G_F\|_\infty \geq c_0 M$. The triangle inequality gives $\|G\|_\infty \geq \|G_F\|_\infty - \|G - G_F\|_\infty \geq cM$. The Euclidean length of G is $\sqrt{a}$ and dominates its coordinate maximum norm. □

Lemma E.19 (Original Euclidean dominant-gradient bound)**.** *For each fixed finite input there exist $c, L > 0$ such that $|u| \geq L$ implies $\sqrt{a(y)} \geq cM(y)$.*

Proof. Take c from Lemma E.18 and set $S = \sum_{v \in V} \|v\|_\infty$. Use Lemma E.17 with tolerance $\epsilon = c/(1+S) > 0$ to obtain an outer coordinate radius $L_\infty > 0$. On that end its face F contains a largest monomial and every complementary monomial has magnitude at most ϵM. Consequently $\|G - G_F\|_\infty \leq \sum_{v \notin F} |m_v| \|v\|_\infty \leq \epsilon M S \leq cM$. The face-tail lemma gives $\sqrt{a} \geq cM$ there. Finally $|u| \leq \sqrt{n}\|u\|_\infty \leq (n+1)\|u\|_\infty$. Thus the Euclidean radius $L = (n+1)L_\infty$ suffices. □

Remark E.20 (Quantitative adaptation of the dominant-face argument). The source proves noncancellation by an escaping-sequence argument. The chosen route above replaces its normalized-limit step by a simultaneous finite-subset deficit estimate on the coordinate unit sphere. The exponentially small monomial complement then supplies a quantitative proper face before applying its complex independence bound. The conversion to the prescribed Euclidean radius is explicit. All these intermediate bounds concern the actual finite input and its actual monomials.

Lemma E.21 (Uniform monomial bounds from coefficient bounds)**.** *Fix $A, B > 0$. For every coefficient tuple $d = (d_v)_{v \in V}$ with $A \leq |d_v| \leq B$, write $m_v = d_v e^{\langle v,u \rangle + i\langle v,\theta \rangle}$, $f = \sum_v m_v$, $G = \sum_v m_v v$, $M = \max_v |m_v|$, and $b = u \cdot G$. There is $C > 0$, depending only on A, B, V, such that for all these tuples and every (u, θ),*

$$M \geq Ae^{h(u)}, \qquad |m_v| \leq (B/A)e^{-(h(u)-\langle v,u \rangle)}M, \qquad |b - h(u)f| \leq CM.$$

Proof. A vertex maximizing the pairing gives $M \geq Ae^h$. Since $|m_v| \leq Be^{\langle v,u \rangle}$, division by Ae^h proves the relative monomial bound. Put $\delta_v = h - \langle v, u \rangle \geq 0$. The literal sum satisfies $b - hf = -\sum_v \delta_v m_v$. Thus

$$|b - hf| \leq Be^h \sum_v \delta_v e^{-\delta_v} \leq |V| Be^h \leq (|V|B/A)M.$$

Here $te^{-t} \leq 1$ for $t \geq 0$ follows from $e^t \geq 1 + t$. Increasing $|V|B/A$ to a positive C proves the claim. □

Lemma E.22 (Uniform dominant gradient from coefficient bounds)**.** *For fixed $A, B > 0$ there are $c, L > 0$, depending only on A, B, V, such that every coefficient tuple satisfying $A \leq |d_v| \leq B$ satisfies $|G(u, \theta)| \geq cM(u, \theta)$ whenever $|u| \geq L$.*

Proof. Let $c_0 > 0$ be the common proper-face independence constant, put $c = c_0/2$, and let $S_V = \sum_v \|v\|_\infty$. Set $q = \min(1/2, c/(1 + S_V)) > 0$. Choose $D > 0$ so that $(B/A)e^{-D} \leq q$; the coefficient-box monomial bound then makes every term with deficit at least D at most qM. Let $\eta > 0$ be the near-maximal-face tolerance. For $\|u\|_\infty \geq 1 + D/\eta$, the set $\{v : h - \langle v, u \rangle \leq D\}$ lies in one nonempty proper face F. Every complementary monomial has magnitude at most $qM \leq M/2$, so a monomial attaining $M > 0$ lies in F. Proper-face independence gives $\|G_F\|_\infty \geq c_0 M$, while

$$\|G - G_F\|_\infty \leq qMS_V \leq cM.$$

Consequently $|G| \geq \|G\|_\infty \geq cM$. The inequality $|u| \leq (n+1)\|u\|_\infty$ shows that $L = (n+1)(1+D/\eta)$ suffices. Every constant used is independent of the tuple d. □

Lemma E.23 (Uniform estimates on the prescribed coefficient path)**.** *For the prescribed path* $c_v(t) = \exp((1-t)\ell_v)$*, there are* $A, c_1, c_2, L, C > 0$ *such that, for every* $t \in [0,1]$ *and every* $y = (u, \theta)$*,*

$$M_t(y) \geq Ae^{c_1|u|}, \qquad |b_t - h(u)f_t| \leq CM_t(y), \qquad |u| \geq L \implies |G_t(y)| \geq c_2 M_t(y).$$

Proof. The coefficient-path lemma supplies a common $A > 0$ and $B > 0$ for all $t \in [0,1]$. Apply the two coefficient-box lemmas with those bounds. For the growth assertion, the vertex-pairing radius lemma gives $h(u) \geq c_1|u|$. The support and its height are unchanged along the path, so all these constants apply simultaneously to every member. □

Lemma E.24 (Uniform critical-value bound and far-fiber regularity)**.** *There are* $L, B_{\mathrm{ex}} > 0$ *such that every critical point of every* f_t*,* $t \in [0,1]$*, has* $|u| < L$ *and critical value of modulus at most* B_{ex}*. For each* t *every fiber over* $|w| > B_{\mathrm{ex}}$ *is regular.*

Proof. Choose the common outer radius L and dominant-gradient constant from the path-estimate lemma. At a critical point the logarithmic derivative G_t is zero, which contradicts $|G_t| \geq cM_t > 0$ if $|u| \geq L$. Thus every critical point satisfies $|u| < L$. Let $B > 0$ be the common coefficient upper bound and let $V_* = \max_{v \in V} |v|$. For $|u| \leq L$, each monomial has modulus at most $Be^{V_* L}$, by Cauchy–Schwarz. Hence $|f_t| \leq |V|Be^{V_* L}$ there, uniformly in the angles and t. Take $B_{\mathrm{ex}} = 1 + |V|Be^{V_* L}$. Finally the local real-angle differential is $df_t(v, w) = G_t \cdot (v + iw)$. If $G_t \neq 0$, choose an index j with $(G_t)_j \neq 0$. For any complex z, writing $z/(G_t)_j = a + ib$ gives $df_t(ae_j, be_j) = z$. Thus this real differential is surjective exactly when $G_t \neq 0$. A point over $|w| > B_{\mathrm{ex}}$ cannot be critical by the bound already proved, so every such fiber is regular. □

Lemma E.25 (Dominant-face estimate)**.** *Outside a common compact set,* $|G| \geq cM$ *and* $M \geq c'e^{c''|u|}$*. In particular all critical points and critical values of the polynomials in the chosen family lie in common compact sets.*

Proof. For a fixed tuple, the two estimates are Lemmas E.9 and E.19. For the prescribed compact coefficient path, its magnitudes have a common positive minimum A and a finite common sum bound S. The tail lemma therefore has one coefficient-ratio constant S/A and one cutoff for any prescribed tolerance, valid throughout the path. The total pairing deficit and proper-face independence constants depend only on the fixed vertices. The concentration and face-tail arguments consequently give one common outer radius and one positive derivative constant for the path. The growth argument uses the same common A and its fixed vertex-radius constant, so its exponential bound is uniform as well. At a critical point $G = 0$, the positive M makes the outer derivative estimate impossible. All critical points therefore lie in a common compact radial cylinder. Continuity of the polynomial on that cylinder times the compact coefficient path bounds all critical values. □

Lemma E.26 (The original quadratic Liouville differential identities)**.** *For the actual Liouville vector field of* λ_K *with* $K \geq 0$*, one has*

$$dW(Z_K) = \frac{b_0 + KaW/2}{1 + Ka}, \qquad d\phi_0(Z_K) = \frac{|u|^2 + KQ_W}{1 + Ka}, \qquad d|W|^2(Z_K) = \frac{2\,\mathrm{Re}(\overline{W}b_0) + Ka|W|^2}{1 + Ka}.$$

Proof. For the standard metric let $L = dW$ and let L^* be its real adjoint. Holomorphicity gives $LL^* = a\,\mathrm{id}_{\mathbb{C}}$ and $LZ_0 = b_0$, where $Z_0 = (u, 0)$. The Liouville contraction equation is $(\mathrm{id} + KL^*L)Z_K = Z_0 + (K/2)L^*W$. Applying L gives $(1 + Ka)dW(Z_K) = b_0 + KaW/2$. Moreover $Z_K = Z_0 + \frac{K}{1+Ka}L^*(W/2 - b_0)$. Since $d\phi_0(v) = g(Z_0, v)$, its evaluation is $|u|^2 + \frac{K}{1+Ka}\,\mathrm{Re}(\overline{b_0}(W/2 - b_0))$, which simplifies to the second displayed formula by the definition of Q_W. Finally $d|W|^2(v) = 2\,\mathrm{Re}(\overline{W}dW(v))$ gives the third formula. Here $a \geq 0$, so $1 + Ka > 0$ even when $a = 0$; all divisions are valid. □

Lemma E.27 (Inverse Hessian estimates and uniform path dominance)**.** *Let H be a symmetric invertible operator on Euclidean real space with $0 \le \langle Hv, v\rangle \le C|v|^2$ and $C > 0$. Put*

$$a_H = \langle H^{-1} \operatorname{Re} G, \operatorname{Re} G\rangle + \langle H^{-1} \operatorname{Im} G, \operatorname{Im} G\rangle, \qquad r_H = \sqrt{\langle Hu, u\rangle}.$$

Then $a/C \le a_H$ and $|b_0| \le r_H\sqrt{a_H}$. Now fix a smooth potential ϕ with global Hessian bounds $c|v|^2 \le \langle D\nabla\phi(x)v, v\rangle \le C|v|^2$, $c > 0$. For the actual Hessian H_s of $\phi_s(x) = (1-s)|x|^2/2 + s\phi(x)$ there are $d, L > 0$ such that

$$dM_t(y) \le \sqrt{a_{H_s}(W_t, y)} \quad (t, s \in [0,1],\ |u| \ge L).$$

Both path parameters are independent; d, L are chosen before them.

Proof. The symmetric nonnegative form $B(v, w) = \langle Hv, w\rangle$ satisfies Cauchy's inequality. With $w = H^{-1}z$ it gives $|\langle v, z\rangle|^2 \le B(v, v)\langle H^{-1}z, z\rangle$. Setting $v = z$ and using the upper bound yields $|z|^2/C \le \langle H^{-1}z, z\rangle$, first for $z \ne 0$ by cancellation and then for $z = 0$ directly. Summing over the real and imaginary parts of G proves the first assertion. Taking $v = u$ in Cauchy's inequality for those two parts and summing proves the second. Schwarz symmetry makes each actual smooth Hessian symmetric. The positive lower bound makes it injective and hence invertible in finite dimension. Along the affine path the lower bound is $\min(1, c) > 0$ and the upper bound is $C' = \max(1, C) > 0$. The common coefficient-path estimate gives $d_0 M_t \le \sqrt{a_t}$ outside one radius L. The first assertion therefore gives the claimed estimate with $d = d_0/\sqrt{C'}$. □

Remark E.28. These are estimates for the constructed inverse of the actual Hessian of the selected potential. Identification with the native symplectic Poisson pairing, the Euler radial identities, and the varying-potential ambient margin are additional obligations in the following nodes.

Lemma E.29 (Uniform inverse operator bound)**.** *If an invertible real operator H on a Euclidean space satisfies $c|v|^2 \le \langle Hv, v\rangle$ with $c > 0$, then $|H^{-1}v| \le c^{-1}|v|$ and $\|H^{-1}\| \le c^{-1}$. In particular these bounds apply to the actual potential Hessian with its previously constructed common lower bound.*

Proof. Put $w = H^{-1}v$. Coercivity and Euclidean Cauchy's inequality give $c|w|^2 \le \langle v, w\rangle \le |v||w|$. If $w = 0$ the claim follows from nonnegativity. Otherwise divide by $c|w| > 0$ to obtain the pointwise bound. Taking the operator supremum proves the norm bound. The constructed Hessian inverse has precisely this contraction identity and lower quadratic bound, so substitution gives the final assertion. □

Lemma E.30 (Native Hamiltonian fields in the potential metric)**.** *Let ϕ be smooth with positive definite Hessian H, and write $\lambda_\phi = \nabla\phi \cdot d\theta$ on the quotient ambient. For a smooth real function A, write its differential on the logarithmic cover as $dA(q_u, q_\theta) = A_u \cdot q_u + A_\theta \cdot q_\theta$. The native Hamiltonian field defined by $\iota_{X_A} d\lambda_\phi = dA$ has cover coordinates*

$$X_A = (H^{-1}A_\theta, -H^{-1}A_u).$$

Consequently, for $f = A + iB$,

$$\pi_f = A_u \cdot H^{-1}B_\theta - A_\theta \cdot H^{-1}B_u.$$

For the actual toric holomorphic polynomial this is exactly $\overline{G}^T H^{-1} G$.

Proof. The differential of the primitive evaluates on cover tangent vectors as $\omega(q,r) = (Hq_u)\cdot r_\theta - (Hr_u)\cdot q_\theta$. Substitution of the displayed X_A, symmetry of H, and both inverse identities give $\omega(X_A,r) = A_u\cdot r_u + A_\theta\cdot r_\theta$. The cover differential is bijective, so this proves the native contraction equation on every tangent vector. Nondegeneracy identifies its unique solution with the native Hamiltonian field. Evaluating dA on X_B gives the Poisson formula. For the holomorphic polynomial the actual differential has blocks $A_u = \operatorname{Re} G$, $A_\theta = -\operatorname{Im} G$, $B_u = \operatorname{Im} G$, $B_\theta = \operatorname{Re} G$. Substitution gives the sum of its two inverse-metric squares. □

Lemma E.31 (Native holomorphic ambient field identities)**.** *Let the selected smooth positive-Hessian potential agree outside a fixed ball with its smooth degree-two end. On that open end, for every $K \geq 0$, the actual Liouville field of $\lambda_\phi + KW^*\lambda_{\mathbb{C}}$ satisfies*

$$dW(Z_K) = \frac{b + Ka_H W/2}{1 + Ka_H}, \qquad d\phi(Z_K) = \frac{r_H^2 + KQ_H}{1 + Ka_H}, \quad Q_H = r_H^2 a_H - |b|^2 + \tfrac{1}{2}\operatorname{Re}(\overline{W}b).$$

Here H is the actual Hessian, $a_H = \overline{G}^T H^{-1} G$, $b = u\cdot G$, and $r_H^2 = u^T H u$.

Proof. Exterior agreement transfers the homogeneous Hessian identity to the chosen global potential. Thus its native base Liouville field is $(u,0)$, and $dW(Z_0) = b$, $d\phi(Z_0) = r_H^2$. The native Hamiltonian formulas give $\pi_W = a_H \geq 0$, $d\phi(X_{\operatorname{Re} W}) = -\operatorname{Im} b$, and $d\phi(X_{\operatorname{Im} W}) = \operatorname{Re} b$. The denominator is at least one. Substitution into the exact rank-two identities gives the first formula; in the second numerator the two products simplify to $-|b|^2 + \operatorname{Re}(\overline{W}b)/2$, giving the second formula. All fields and differentials are those of the native quotient primitive. □

Lemma E.32 (Holomorphic ambient and intrinsic field identities)**.** *For $\lambda_{\phi,K} = \lambda_\phi + Kf^*\lambda_{\mathbb{C}}$, $K \geq 0$, $d\lambda_{\phi,K}$ is symplectic, since its additional term is a semipositive $(1,1)$-form. On the homogeneous end, direct rank-two inversion gives*

$$df(Z_K) = \frac{b + Kaf/2}{1 + Ka}, \tag{13}$$

$$d\phi(Z_K) = \frac{r^2 + KQ_f}{1 + Ka}, \qquad Q_f = r^2 a - |b|^2 + \tfrac{1}{2}\operatorname{Re}(\overline{f}b). \tag{14}$$

These are identities for the new *Liouville field; no old-metric gradient is substituted for it.*

On a regular fiber, $Z^{\mathrm{fib}} = Z_0 - L^*b/a$ and $d\phi(Z^{\mathrm{fib}}) = r^2 - |b|^2/a$.

Proof. For clarity, if $L = df$ and L^* is its real adjoint for the Kähler metric of ω_0, then $LL^* = a\,\mathrm{id}_{\mathbb{C}}$. The Liouville equation is globally equivalent to the following identity, with $Z_0 = H^{-1}\nabla\phi$ in the radial directions:

$$(\mathrm{id} + KL^*L)Z_K = Z_0 + (K/2)L^*f.$$

The identity $(\mathrm{id} + KL^*L)L^* = (1 + Ka)L^*$ proves the rank-two inversion, including points with $a = 0$. On a regular holomorphic fiber, the intrinsic field is the orthogonal projection of Z_0 onto $\ker L$: this tangent space is J_0-invariant, so its symplectic and metric orthogonal complements coincide. On the homogeneous end this gives

$$Z^{\mathrm{fib}} = Z_0 - L^*b/a, \qquad d\phi(Z^{\mathrm{fib}}) = r^2 - |b|^2/a.$$

For the original ϕ_0, the homogeneous identities hold on all of Y; in particular $H = I$ gives exactly the scalar formulas in the target. For the interpolated potentials these identities are used only on their common homogeneous end.

□

Definition E.33 (Actual support height)**.** *Set* $h(u) = \max_{v \in V} \langle v, u \rangle$.

Lemma E.34 (Two-sided linear control of the actual support height)**.** *For each fixed finite input there are* $c, H > 0$ *such that* $c|u| \le h(u) \le H|u|$ *for every* $u \in M_{\mathbb{R}}$.

Proof. Choose c from Lemma E.8. A vertex with pairing at least $c|u|$ gives the lower bound for their maximum. Put $H = 1 + \sum_{v \in V} |v| > 0$. For every vertex, the Cauchy–Schwarz inequality gives $\langle v, u \rangle \le |v||u| \le H|u|$. Taking the finite maximum gives the upper bound, also at $u = 0$. □

Lemma E.35 (Uniform monomial discrepancy bound)**.** *For the prescribed coefficient tuple there exists* $C > 0$ *such that* $|b_0(y) - h(u)W(y)| \le CM(y)$ *for every* $y = (u, \theta)$.

Proof. Differentiating each monomial radially gives $b_0 = \sum_v \langle v, u \rangle m_v$. Write $\delta_v = h(u) - \langle v, u \rangle \ge 0$. Then $b_0 - hW = -\sum_v \delta_v m_v$ and $\delta_v |m_v| = |c_v| e^{h(u)} \delta_v e^{-\delta_v} \le |c_v| e^{h(u)}$. The last inequality follows from $e^t \ge 1 + t \ge t$ for $t \ge 0$. Choose a maximizing vertex for h, whose monomial has magnitude at least $c_{\min} e^h$, where $c_{\min} = \min_v |c_v| > 0$. Thus $e^h \le M / c_{\min}$, and the triangle inequality proves the bound with $C = (\sum_v |c_v|)/c_{\min} > 0$. □

Lemma E.36 (Sharp original quadratic margin from the constructed discrepancy)**.** *There exists* $C > 0$ *such that, with* $r = |u|$, $g = \sqrt{a}$ *and* $h = h(u)$, *every point with* $h \ge 1/2$ *satisfies*

$$Q_W \ge \frac{rg(rg - CM)}{2h}.$$

Proof. Take the actual discrepancy constant C and put $E = b_0 - hW$. Cauchy–Schwarz gives $|b_0| \le rg$ and $|E| \le CM$. Since $W = (b_0 - E)/h$ and $h > 0$,

$$Q_W \ge r^2 g^2 - \left(1 - \frac{1}{2h}\right) |b_0|^2 - \frac{CM}{2h} |b_0|.$$

For $h \ge 1/2$ the expression on the right is nonincreasing as $|b_0|$ ranges from zero to rg: its derivative is $-2(1 - 1/(2h))|b_0| - CM/(2h) \le 0$. Substituting rg gives exactly the asserted bound. □

Lemma E.37 (Original margin and ambient bound from an independent gradient estimate)**.** *Suppose* $c, L > 0$ *and* $g = \sqrt{a} \ge cM$ *whenever* $r = |u| \ge L$. *Then there exist* $q, R > 0$ *such that* $a > 0$ *and* $Q_W \ge qrM^2$ *for* $r \ge R$. *In particular* $A_{\text{amb}} = R^2/2$ *is an ambient end bound for the original quadratic potential.*

Proof. Choose $m, H > 0$ with $mr \le h \le Hr$ and the discrepancy constant $C > 0$. Take $R > \max(L, 1/(2m), 2C/c)$ and put $q = c^2/(4H) > 0$. At $r \ge R$, the gradient premise gives $g \ge cM > 0$, hence $a > 0$. Moreover $h \ge 1/2$ and $CM \le rg/2$. Lemma E.36 now gives

$$Q_W \ge \frac{(rg)^2}{4h} \ge \frac{c^2 r^2 M^2}{4Hr} = qrM^2 > 0.$$

The condition $\phi_0 = r^2/2 \ge R^2/2$ is equivalent to $r \ge R$ because both radii are nonnegative. Thus the required positivity of a and Q_W holds on that potential end. □

Lemma E.38 (Unconditional original quadratic ambient margin and end bound)**.** *For every original finite input there exist* $q, R > 0$ *such that* $r \ge R$ *implies* $a > 0$ *and* $Q_W \ge qrM^2$. *An ambient end bound therefore exists without an additional analytic hypothesis.*

Proof. Lemma E.19 constructs $c, L > 0$ and the independent native gradient estimate required by Lemma E.37. Applying that reduction gives the quantitative margin and the stated end bound. □

Lemma E.39 (Uniform quadratic margin along the coefficient path)**.** *Keep the quadratic potential $\phi_0 = |u|^2/2$ and vary the coefficients along the prescribed path. There are $q, L > 0$, independent of $t \in [0,1]$, such that $r = |u| \geq L$ implies $a_t > 0$ and $Q_{W_t} \geq qrM_t^2$. Consequently one common $A_{\mathrm{amb}} > 0$ is an ambient end bound for all these quadratic path members.*

Proof. The uniform coefficient bounds and dominant estimate supply $c, C, L_0 > 0$ with $\sqrt{a_t} \geq cM_t$ for $r \geq L_0$ and $|b_t - hW_t| \leq CM_t$ everywhere. The height, unchanged along the path, satisfies $mr \leq h \leq Hr$ for fixed $m, H > 0$. Choose $L > \max(L_0, 1/(2m), 2C/c)$ and $q = c^2/(4H)$. Writing $g = \sqrt{a_t}$, the sharp scalar margin gives

$$Q_{W_t} \geq \frac{rg(rg - CM_t)}{2h} \geq \frac{r^2 g^2}{4h} \geq qrM_t^2 > 0.$$

All constants precede t. Since $\phi_0 \geq L^2/2$ implies $r \geq L$, $A_{\mathrm{amb}} = L^2/2$ has the required property. This statement fixes the Hessian to the identity; the varying-potential margin below also requires its inverse-Hessian estimates. □

Lemma E.40 (Uniform margin for the actual potential-path Hessians)**.** *Fix a smooth potential ϕ with common positive lower and finite upper Hessian bounds, and let $H_s(u) = D^2\phi_s(u)$ for its prescribed affine path. Write $f = W_t$, $b = u \cdot G_t$, $a_H = \overline{G}_t^T H_s^{-1} G_t$, and $r_H^2 = u^T H_s u$. There are $q, L > 0$, independent of $s, t \in [0,1]$, such that on $|u| \geq L$ one has $a_H > 0$ and*

$$r_H^2 a_H - |b|^2 + \tfrac{1}{2}\operatorname{Re}(\overline{f}b) \geq qr_H M_t^2.$$

More generally the same constants work for every symmetric invertible metric satisfying the fixed positive lower and upper bounds.

Proof. Let $c > 0$ be the common lower metric bound, put $k = \sqrt{c}$, and let $h \leq D|u|$ be the support-height upper bound. Then $r_H \geq k|u|$ and $h \leq (D/k)r_H$. The common inverse-metric dominance gives $g = \sqrt{a_H} \geq dM_t > 0$, and the discrepancy satisfies $|b - hf| \leq BM_t$. Enlarge L until $h \geq 1/2$ and $dr_H \geq 2B$. Weighted Cauchy's inequality gives $|b| \leq r_H g$. Writing $f = (b - (b - hf))/h$, the real-part estimate gives

$$Q \geq r_H^2 g^2 - (1 - 1/(2h))|b|^2 - BM_t|b|/(2h) \geq \frac{r_H g(r_H g - BM_t)}{2h}.$$

The second inequality follows since the subtracted polynomial is nondecreasing for $0 \leq |b| \leq r_H g$. Also $r_H g - BM_t \geq r_H g/2$, so $Q \geq r_H^2 g^2/(4h) \geq d^2 r_H M_t^2/(4D/k)$. Choose $q = d^2/(4D/k) > 0$. Every bound was chosen before the metric and coefficient parameters. The affine Hessians satisfy these same bounds, which proves their stated specialization. □

Lemma E.41 (Uniform normalized holomorphic margin)**.** *Fix positive constants c, C. For the coefficient-one polynomial W_1 and its normalization $f_\beta = e^{-\beta} W_1$, let $G_\beta = e^{-\beta} G_1$, $b_\beta = u \cdot G_\beta$, $a_\beta = \overline{G}_\beta^T H^{-1} G_\beta$ and $r_H^2 = u^T H u$. For all symmetric invertible H with $cI \leq H \leq CI$ one has $Q_{f_\beta} = e^{-2\beta} Q_{W_1}$. There exist $q, L > 0$ such that for every real β and $|u| \geq L$,*

$$a_\beta > 0, \qquad Q_{f_\beta} \geq qr_H e^{2(h(u) - \beta)}.$$

Proof. The formulas for f, G, b are linear in the common coefficient scalar $e^{-\beta}$. Thus a, $|b|^2$, and $\operatorname{Re}(\overline{f}b)$ each acquire the factor $e^{-2\beta}$, giving the exact margin identity. At the coefficient-one endpoint the maximum monomial magnitude is $e^{h(u)}$. Apply the uniform metric margin there and multiply its two conclusions by the positive number $e^{-2\beta}$. The same q, L work for every β. □

Lemma E.42 (Euler radius on the retained common homogeneous end)**.** *Suppose the selected smooth ϕ agrees outside the radius $R > 0$ with a degree-two function ϕ_∞ which is continuously differentiable everywhere and smooth away from zero. For the same affine path ϕ_s, on $|u| > R$ its actual Hessian satisfies $u^T D^2\phi_s(u)u = 2\phi_s(u)$.*

Proof. The global function and its homogeneous end agree on a neighborhood of each such u, so their first and second derivatives there agree. Differentiating $\phi_\infty(tu) = t^2\phi_\infty(u)$ gives Euler's identity $d\phi_\infty(u)[u] = 2\phi_\infty(u)$. Differentiating its degree-one gradient gives $D^2\phi_\infty(u)u = \nabla\phi_\infty(u)$. Pairing with u proves the result. The affine combination with $|u|^2/2$ retains the same exterior agreement region and degree-two homogeneity, so the argument applies to every s. □

Lemma E.43 (Common literal logarithmic potential levels)**.** *Retain the smooth potential, homogeneous end, exterior agreement radius, and global Hessian bounds of the preceding lemmas. Given any A_{amb}, there are $F, \beta_0 > 0$ such that, for every $\beta \geq \beta_0$, the literal level $A_\beta = F^2\beta^2/2$ is at least A_{amb}. For all independent $s, t \in [0,1]$, on $\{\phi_s \geq A_\beta,\ |W_t| \leq e^\beta\}$ one has $|u| > R$, $a_H > 0$, and*

$$2\phi_s - |b|^2/a_H \geq \phi_s, \qquad H = D^2\phi_s(u), \quad b = u \cdot G_t.$$

Proof. Write $r_H = \sqrt{u^T H u}$. The metric bounds compare r_H with $|u|$, uniformly in s. The coefficient bounds therefore give constants $d, A, e, D, B > 0$ with $\sqrt{a_H} \geq dM$, $M \geq Ae^{er_H}$, $h \leq Dr_H$, and $|b| \leq h|W_t| + BM$ outside a fixed radius. Choose $F = 2/e$. For $r_H \geq F\beta$,

$$\frac{|b|}{r_H\sqrt{a_H}} \leq \frac{D}{dA}e^{\beta - er_H} + \frac{B}{dr_H} \leq \frac{D}{dA}e^{-\beta} + \frac{B}{dF\beta}.$$

Choose β_0 so each last term is at most $1/4$ and the fixed exceptional radius is excluded. The resulting bound is at most $1/2$; squaring proves $r_H^2 - |b|^2/a_H \geq r_H^2/2$. The potentials ϕ_s are uniformly bounded above on $|u| \leq R$: bound both endpoint potentials on that compact ball and take their maximum. Increase β_0 until A_β strictly exceeds that bound and is at least A_{amb}. Thus $\phi_s \geq A_\beta$ forces $|u| > R$. The Euler radius identity now gives $r_H^2 = 2\phi_s$ and $r_H \geq F\beta$, so the preceding metric estimate is applicable. □

Lemma E.44 (Empty logarithmic fiber ends in rank one)**.** *In rank one, for any invertible H one has $r_H^2 a_H = |b|^2$. Where $a_H \neq 0$, the defect $r_H^2 - |b|^2/a_H$ is zero. Consequently, under the hypotheses and with the constants of the common literal logarithmic level theorem, for every $\beta \geq \beta_0$ and $s, t \in [0,1]$ the set $\{\phi_s \geq F^2\beta^2/2,\ |W_t| \leq e^\beta\}$ is empty.*

Proof. Write H as multiplication by its nonzero one-dimensional coefficient h. Its inverse multiplies by h^{-1}, so $(hu^2)(|G|^2/h) = |uG|^2$. Division by nonzero a_H proves the vanishing defect. A point in the asserted set would have $a_H > 0$ and defect at least $\phi_s \geq F^2\beta^2/2 > 0$ by the common level theorem, contradicting that vanishing. This proves emptiness without imposing a nonempty positive-dimensional fiber end. □

F The intrinsic potential-fiber exhaustion

Lemma F.1 (The full potential truncation and its completion marking)**.** *Retain the chosen potential, its full regular fiber $j : F \hookrightarrow Y$, and the proved complete primitive $\mu = j^*\lambda_\phi$. Suppose at a chosen level A the actual inequality $d(\phi \circ j)(Z_F) > 0$ holds throughout $\phi \circ j \geq A$. Then $D = \{\phi \circ j \leq A\}$ is a nonempty compact native Liouville domain with frontier $\phi \circ j = A$ and image $\{W_c = w,\ \phi \leq A\}$. Its actual intrinsic flow constructs a completion, and every chosen completion of this same domain has a primitive-preserving full-fiber inclusion marking onto F.*

Proof. The defining function is $\phi\circ j - A$. Its differential is that of $\phi\circ j$, so the assumed strictly positive derivative makes zero a regular outward level. The properness lemma makes its sublevel compact; the regular-sublevel construction gives the native domain and the stated frontier. The exact full-fiber image of j gives the displayed image equality in both directions. The actual complete flow, compact exhaustion bands, and positive derivative make every point enter D in finite negative time by the exhaustion absorption theorem. Laurent surjectivity gives a point in the full fiber; negative entry therefore also proves $D \neq \varnothing$. The boundary-orbit map of this very flow and negative entry construct the completion and its inclusion marking. The strict comparison of completions transports that marking to every other chosen completion of this same domain. □

Lemma F.2 (Uniform actual markings for every large full-fiber truncation)**.** *Retain the smooth chosen potential with uniform positive Hessian bounds and a degree-two homogeneous end, and the prescribed coefficient path. For every A_{amb} there exist $B, F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$, every $t \in [0,1]$, every complex value $B < |w| \leq e^{\beta}$, every smooth realization of the literal full fiber $W_t^{-1}(w)$, and every $A \geq F^2\beta^2/2$, its actual full potential truncation is a nonempty compact Liouville domain and every chosen completion is marked onto that same full fiber. Moreover $F^2\beta^2/2 \geq A_{\mathrm{amb}}$.*

Proof. The coefficient-path critical-value bound supplies actual regularity for all $|w| > B$. Specialize the proved intrinsic potential-path outwardness to its retained endpoint $s = 1$; this endpoint is literally the chosen ϕ. The regular-level theorem supplies the tangent kernel for every supplied full-fiber realization. Since its polynomial value is w and $|w| \leq e^{\beta}$, the intrinsic estimate applies throughout $\phi \geq F^2\beta^2/2$, and hence throughout $\phi \geq A$ for every stated A. Apply the preceding domain and actual flow completion construction. The image remains the entire literal truncation, and comparison transports its inclusion marking to every chosen completion. All constants are selected before β, t, w, the fiber realization, the truncation level, and the completion. □

Lemma F.3 (Potential exhaustion on the projected holomorphic field)**.** *Let ϕ be smooth, let $H = D^2\phi(u)$ be invertible, and suppose $Hu = \nabla\phi(u)$ at the point in question. Write $f = W_c$, $G = \sum_v m_v v$, $b = u \cdot G$ and $a = \overline{G}^T H^{-1} G$. For the displayed rank-two formula Π_f applied to the radial Liouville vector one has, with real division also defined at zero,*

$$d\phi(\Pi_f Z_0) = u^T H u - |b|^2/a.$$

Proof. Set $p = d\operatorname{Re} f$, $q = d\operatorname{Im} f$. The Hamiltonian-coordinate formula gives $p(X_q) = a$, $d\phi(X_p) = -\operatorname{Im} b$, and $d\phi(X_q) = \operatorname{Re} b$. Euler's Hessian identity gives $Z_0 = u\partial_u$, so $p(Z_0) = \operatorname{Re} b$, $q(Z_0) = \operatorname{Im} b$, and $d\phi(Z_0) = u^T H u$. Substitute these into $\Pi_f Z_0 = Z_0 + (q(Z_0)/a)X_p - (p(Z_0)/a)X_q$. Linearity of $d\phi$ yields the displayed identity, since $|b|^2 = (\operatorname{Re} b)^2 + (\operatorname{Im} b)^2$. □

Lemma F.4 (The literal potential fiber is smooth and symplectic)**.** *Retain a smooth chosen potential with $cI \leq D^2\phi \leq UI$, where $c, U > 0$. If $G \neq 0$ everywhere on the literal level $W_c = w$, the whole level has a smooth embedded realization $j : F \hookrightarrow Y$ of dimension $2n-2$ with tangent image $\ker dW_c$. The literal form $j^* d\lambda_\phi$ is symplectic; this conclusion holds for every smooth realization with that same full image.*

Proof. The actual inverse-Hessian scalar satisfies $a_\phi \geq |G|^2/U > 0$ by the upper Hessian bound. The native Hamiltonian-coordinate identity makes the Poisson pairing equal to this scalar. The holomorphic differential is surjective when $G \neq 0$, so the regular-level manifold theorem constructs the whole level and identifies the tangent range. Nonzero pairing makes the intersection of the real and imaginary differential kernels symplectic; pullback along the embedding gives the asserted restricted form. □

Lemma F.5 (The actual full potential fiber has a complete intrinsic flow)**.** *For the retained potential with the preceding Hessian bounds and the degree-two homogeneous end hypotheses, every full regular fiber with its literal restricted primitive has an actual complete intrinsic Liouville flow, jointly smooth for all real times.*

Proof. On the homogeneous end, the intrinsic identity and weighted Cauchy bound give $0 \le d\phi(Z_F) \le 2\phi$. Choose the proved nonnegative shift $\rho = \phi + D$. The literal fiber is closed, hence its restricted ρ sublevels are compact by properness of the ambient potential and closedness of its embedding. The intrinsic field is smooth because its primitive is smooth and its exterior derivative is nondegenerate. On the compact radial remainder continuity bounds $|d\rho(Z_F)|$ by a finite B. On the end the preceding inequality gives $|d\rho(Z_F)| \le 2(1+\rho)$. Thus globally $|d\rho(Z_F)| \le \max(B, 2)(1+\rho)$. The proper-exhaustion flow theorem constructs its complete all-real smooth flow. Together with the actual restricted symplectic primitive and contraction equation this is a complete primitive on the same fiber. □

Lemma F.6 (Every far fiber on the chosen coefficient path is complete)**.** *For the retained potential and the prescribed coefficient path there is $B > 0$ such that, for every $t \in [0, 1]$ and every complex value $|w| > B$, the entire literal fiber $W_t^{-1}(w)$ has a smooth embedded realization with an actual complete primitive $j^*\lambda_\phi$.*

Proof. The uniform critical-value bound gives $G_t \neq 0$ at every point over $|w| > B$, before t and w are selected. Apply the preceding full regular-fiber construction and completeness theorem to W_t with the same retained potential. All other hypotheses concern only that potential and therefore remain unchanged along the coefficient path. □

Lemma F.7 (Properness and lower bound of the actual end potential)**.** *Let ϕ be smooth with $D^2\phi \ge cI$, $c > 0$, and agree outside $|u| \ge L > 0$ with a C^1 degree-two homogeneous function that is smooth away from zero. Then on $|u| > L$, $u^T D^2\phi(u)u = 2\phi(u)$ and $c|u|^2 \le 2\phi(u)$. Every sublevel of ϕ in the logarithmic quotient ambient is compact, and there is $D \ge 0$ with $\phi + D \ge 0$ everywhere.*

Proof. Differentiating the degree-two Euler identity gives $u^T D^2\phi(u)u = 2\phi(u)$ on the open region of agreement. The Hessian lower bound gives the quadratic inequality there. The sublevel $\phi \le A$ is closed and contained in the quadratic cylinder $|u|^2/2 \le \max(L^2/2, A/c)$, which is compact because the angular torus is compact. On the closed radial ball continuity gives a finite lower bound for ϕ; on its complement the quadratic inequality makes ϕ nonnegative. A nonnegative shift of that finite lower bound proves the last claim. □

Lemma F.8 (Cauchy bounds for the actual inverse-metric defect)**.** *For a symmetric positive invertible Hessian H and $a > 0$, the literal defect satisfies $0 \le u^T Hu - |u \cdot G|^2/a \le u^T Hu$.*

Proof. Weighted Cauchy–Schwarz gives $|u \cdot G|^2 \le (u^T Hu)(\overline{G}^T H^{-1} G)$. Divide by $a > 0$ for the lower bound; nonnegativity of $|u \cdot G|^2/a$ gives the upper bound. □

Lemma F.9 (The actual intrinsic potential-fiber identity)**.** *Under the preceding assumptions, let $j : F \hookrightarrow Y$ be a smooth embedding whose tangent image at the point is $\ker df$, and suppose $a > 0$. For the actual Liouville vector Z_F of $\mu = j^*\lambda_\phi$, one has*

$$d(\phi \circ j)(Z_F) = u^T Hu - |b|^2/a.$$

Proof. The complex kernel is the intersection of the real and imaginary kernels. The nonzero Poisson pairing makes the restricted symplectic form nondegenerate and identifies $dj(Z_F) = \Pi_f(Z_0)$ by the intrinsic projection lemma. Apply the chain rule to $\phi \circ j$ and then the preceding projected exhaustion formula. □

Lemma F.10 (Intrinsic exhaustion on the retained potential path)**.** *Retain a smooth potential with a uniform positive Hessian lower bound and a degree-two homogeneous end outside $|u| \geq L > 0$. For each $s \in [0,1]$, put $\phi_s = (1-s)|u|^2/2 + s\phi$. At every point with $|u| > L$ and $a_s > 0$ of a smooth fiber realization whose tangent image is $\ker df$, its actual restricted field satisfies*

$$d(\phi_s \circ j)(Z_{F,s}) = u^T D^2\phi_s(u)u - |u \cdot G|^2/a_s, \qquad a_s = \overline{G}^T (D^2\phi_s(u))^{-1} G.$$

Proof. The affine path remains smooth and has uniformly positive Hessian, so each actual Hessian is invertible. On the common open exterior region it agrees with the affine combination of the quadratic and homogeneous ends. That combination has degree two, so differentiating its gradient scaling gives $D^2\phi_s(u)u = \nabla\phi_s(u)$. Apply the actual intrinsic exhaustion identity with this Hessian. □

Lemma F.11 (One logarithmic level makes the actual fiber fields outward)**.** *Retain the common Hessian bounds, homogeneous end and chosen coefficient path. Given any ambient level A_{amb}, there exist $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$ the level $A_\beta = F^2\beta^2/2$ is at least A_{amb}. For every independent $s, t \in [0,1]$ and every smooth fiber realization with tangent image $\ker dW_t$, on $\{\phi_s \geq A_\beta,\ |W_t| \leq e^\beta\}$ one has*

$$d(\phi_s \circ j)(Z_{F,s}) \geq \phi_s > 0.$$

The statement includes zero-dimensional fibers whenever the stated end set is empty.

Proof. The common logarithmic-level lemma supplies $|u| > L$, $a_s > 0$, and $u^T D^2\phi_s(u)u - |u \cdot G_t|^2/a_s \geq \phi_s$ throughout this set. The preceding intrinsic identity turns this scalar estimate into the asserted inequality for the actual restricted field. Since $F > 0$ and $\beta \geq \beta_0 > 0$, the level A_β is strictly positive, proving strict outwardness. No dimension restriction is used. □

G The actual holomorphic potential ambient

The continuation argument below uses the actual potential as a proper exhaustion. This is an adaptation of the vector-growth argument of Lemma G.20: it controls the exhaustion derivative directly and applies the proper-exhaustion continuation theorem. It concerns the holomorphic polynomial ambient before tailoring.

Lemma G.1 (A uniform rank-two exhaustion bound)**.** *Let $R \geq 0$, $a > 0$, $D, K \geq 0$, and $f, b \in \mathbb{C}$ satisfy $|b| \leq \sqrt{R}\sqrt{a}$ and $|f| \leq D\sqrt{a}$. Then*

$$\left| \frac{R + K(Ra - |b|^2 + \mathrm{Re}(\bar{f}b)/2)}{1 + Ka} \right| \leq (1 + D/2)(1 + R).$$

Proof. The first bound gives $0 \leq Ra - |b|^2 \leq Ra$. Also $\sqrt{R} \leq 1 + R$, since $(1+R)^2 - R = 1 + R + R^2 > 0$. Consequently $|\mathrm{Re}(\bar{f}b)| \leq Da\sqrt{R} \leq Da(1+R)$. Writing the numerator as N, these inequalities give

$$-KDa(1+R)/2 \leq N \leq R(1+Ka) + KDa(1+R)/2.$$

Both bounds lie between the negatives and positives of $(1+D/2)(1+R)(1+Ka)$, since $Ka \leq 1+Ka$ and $R \leq 1+R$. Divide by the positive denominator. □

Lemma G.2 (Every holomorphic potential update is symplectic)**.** *Let ϕ be smooth with $D^2\phi \geq cI$, $c > 0$. For every finite $K \geq 0$, the literal primitive $\lambda_{\phi,K} = \lambda_\phi + KW_c^*\lambda_{\mathbb{C}}$ has symplectic exterior derivative on the whole quotient ambient.*

Proof. The actual Hessian is invertible by its positive lower bound. The native Hamiltonian identity identifies the rank-two Poisson scalar with $a_\phi = \overline{G}^T(D^2\phi)^{-1}G \geq 0$, including points with $G = 0$. The native rank-two nondegeneracy criterion therefore applies because $1 + Ka_\phi \geq 1$. Smoothness and closedness follow from the exterior calculus of the smooth literal primitive. □

Lemma G.3 (Both-time complete flow of the retained potential ambient)**.** *Retain a smooth potential with $cI \leq D^2\phi \leq UI$, where $c, U > 0$, agreeing outside $|u| \geq L > 0$ with a continuously differentiable degree-two function smooth away from zero. For every literal polynomial W_c with the prescribed finite support and nonzero coefficients, and every finite $K \geq 0$, its actual native Liouville field for $\lambda_{\phi,K}$ has an all-real flow jointly smooth in time and initial point. The primitive, contraction equation and complete field all concern this same fixed quotient ambient and literal primitive.*

Proof. The Hessian and holomorphic update give a smooth symplectic primitive, hence its native field is smooth. On a sufficiently large homogeneous end, inverse-metric dominance gives $\sqrt{a_\phi} \geq dM$, with $d > 0$ and M the maximal monomial magnitude. The finite sum gives $|W_c| \leq NM \leq (N/d)\sqrt{a_\phi}$, where N is the support cardinality. Weighted Cauchy gives $|b| \leq \sqrt{u^T D^2\phi u}\sqrt{a_\phi}$. Apply the scalar bound to the actual native exhaustion identity with $R = u^T D^2\phi u = 2\phi$ and $D = N/d$. The properness lemma supplies $S \geq 0$ with $\rho = \phi + S \geq 0$. Thus on this end $|d\rho(Z_K)| \leq 2(1 + D/2)(1 + \rho)$. On the remaining compact radial cylinder, continuity bounds $|d\rho(Z_K)|$ by a finite constant B. Consequently the global bound holds with $C = \max(B, 2(1 + D/2)) > 0$. Every sublevel of ρ is compact by the potential properness lemma. The proper-exhaustion continuation theorem constructs the complete flow, jointly smooth in time and initial point, in both directions. Combining it with the actual primitive and its native contraction equation gives the complete primitive. □

Lemma G.4 (Full potential sublevels with actual inclusion markings)**.** *For the same potential and primitive, suppose at a chosen level A that $d\phi(Z_K) > 0$ throughout $\phi \geq A$. Then the full literal sublevel $D_A = \{\phi \leq A\}$ is a compact native Liouville domain with frontier $\phi = A$. Its actual Liouville flow constructs a completion, and every chosen completion of this same domain has a primitive-preserving inclusion marking onto the whole given ambient, agreeing with its boundary flow.*

Proof. The function $\phi - A$ has compact sublevel by properness. Its differential equals $d\phi$, so its positive derivative at zero makes it a regular outward defining function. The native regular-sublevel construction gives the domain and exact frontier. Completeness and positive derivative on all exterior exhaustion bands give finite negative entry into the sublevel by the exhaustion absorption theorem. The boundary-orbit construction with this very flow therefore gives an actual completion and its inclusion marking. The strict comparison between completions of this same domain transports that marking to any other chosen completion, retaining inclusion and the native boundary flow. □

Lemma G.5 (One ambient truncation threshold for the coefficient path)**.** *For the retained potential above and the prescribed coefficient path, there is one $A_0 > 0$ such that for every $t \in [0, 1]$, every finite $K \geq 0$, and every $A \geq A_0$, the full sublevel $\{\phi \leq A\}$ has the actual completion and arbitrary-completion inclusion marking of the preceding lemma. Its actual derivative is strictly positive throughout $\phi \geq A_0$.*

Proof. The quantitative metric margin supplies one exterior radial threshold, independent of t and K, where $a_\phi > 0$ and $Q_{W_t} > 0$. Increase that threshold to include the nonhomogeneous region. Bound ϕ above by B on its compact radial cylinder and take $A_0 = \max(B, 0) + 1$. Thus $\phi \geq A_0$ forces entry into this homogeneous exterior region. There $r^2 = u^T D^2 \phi u > 0$ and the actual exhaustion identity has positive numerator $r^2 + KQ_{W_t}$ and denominator $1 + Ka_\phi$. Apply the full sublevel construction for every $A \geq A_0$. Every constant is chosen before t, K, A or the completion is selected. □

Lemma G.6 (The native retained potential interpolation)**.** *Retain the chosen endpoint ϕ and end ϕ_{end} of the preceding lemmas. Let $\phi_0(u) = |u|^2/2$ and $\phi_s = (1-s)\phi_0 + s\phi$, independently of the coefficient parameter. For $s \in [0, 1]$, the actual Hessian in the fixed Euclidean coordinates satisfies*

$$\min(1, c)I \leq D^2\phi_s \leq \max(1, U)I.$$

The path member is smooth and agrees at the same radius L with $(1-s)\phi_0 + s\phi_{\mathrm{end}}$. This end is continuously differentiable globally, smooth off zero, and homogeneous of degree two.

Proof. Pullback by the fixed coordinate equivalence commutes with the literal affine interpolation and sends ϕ_0 to the Euclidean norm square divided by two. Its Hessian is the identity. Differentiating the actual affine interpolation gives $D^2\phi_s = (1-s)I + sD^2\phi$; both weights are nonnegative and sum to one, giving the asserted lower and upper bounds. Multiplication by fixed scalars and addition preserve the stated smoothness and continuously differentiable regularity. The two endpoints are degree-two homogeneous on the prescribed end, so their combination has that same homogeneity. Their literal end agreement also persists at precisely the same threshold. □

Lemma G.7 (The complete native ambient at every potential parameter)**.** *For every retained $s \in [0, 1]$, every coefficient-path member W_t, and every finite $K \geq 0$, the literal primitive $\lambda_{\phi_s} + KW_t^* \lambda_{\mathbb{C}}$ on the given quotient ambient has an actual complete native Liouville flow for all real time.*

Proof. The preceding lemma supplies all hypotheses of the native ambient complete-flow lemma, with constants $\min(1, c) > 0$ and $\max(1, U) > 0$. Apply that lemma to the retained path member and its literal homogeneous end. Each coefficient-path member has the prescribed finite support and nonzero coefficients. The resulting complete primitive therefore consists of this actual literal primitive and its own native contraction field and both-time flow. □

Lemma G.8 (Common full ambient markings for independent parameters)**.** *There is one $A_0 > 0$, chosen before s, t, K, A, such that for all independent $s, t \in [0, 1]$, all finite $K \geq 0$, and all $A \geq A_0$, the literal full domain $D_{s,t,K,A} = \{\phi_s \leq A\}$ has its actual completion and every chosen completion has an inclusion-induced marking onto the given complete ambient with primitive $\lambda_{\phi_s} + KW_t^* \lambda_{\mathbb{C}}$. The derivative $d\phi_s(Z_{s,t,K})$ is strictly positive throughout $\phi_s \geq A_0$. The markings are asserted at each parameter separately; this statement does not assert joint smoothness of a selected family of markings.*

Proof. Use the quantitative inverse-metric margin with the common Hessian bounds $\min(1, c)$ and $\max(1, U)$. It supplies one radial exterior threshold, independent of s, t, K, where $a_s > 0$ and $Q_{s,t} > 0$. Enlarge this threshold to $L_{\mathrm{big}} \geq L$. On the compact cylinder $|u| \leq L_{\mathrm{big}}$, let B bound ϕ above and put

$$B_{\mathrm{path}} = \max(B, L_{\mathrm{big}}^2/2), \qquad A_0 = \max(B_{\mathrm{path}}, 0) + 1.$$

Because ϕ_s is a convex combination of ϕ and ϕ_0, it is at most B_{path} on that cylinder for every s. Thus $\phi_s \geq A_0$ forces $|u| > L_{\mathrm{big}}$. On this actual homogeneous end, the native derivative identity is

$$d\phi_s(Z_{s,t,K}) = \frac{u^T D^2\phi_s u + KQ_{s,t}}{1 + Ka_s} > 0.$$

The first numerator term is strictly positive by the common positive Hessian lower bound and nonzero radius; the second is nonnegative, and the denominator is positive. The complete primitive from the preceding lemma and the full-sublevel absorption construction now give the native domain, its exact full carrier, the flow-built completion, and the inclusion marking for every other completion of that same domain. □

Lemma G.9 (Joint smoothness of the literal moving rounded boundary)**.** *Let ϕ be any smooth retained potential, let m be any allowed smooth rounding profile, and fix R, A. The literal scalar*

$$\rho(s, y) = m\big(\phi_s(y) - A, |W_s(y)|^2 - R^2\big)$$

is jointly smooth on $\mathbb{R} \times Y$, in the native quotient atlas. At $s = 0$ it is exactly the original supplied rounded defining function $m(\phi_0 - A, |W_c|^2 - R^2)$.

Proof. The finite sum defining W_s is the sum of products of the smooth fixed coefficient paths and the quotient monomials with unit coefficient. Thus it is jointly smooth, and the fixed linear recharting to the native quotient atlas preserves this assertion. Its norm square is smooth. The literal affine expression for ϕ_s is jointly smooth by multiplication and addition. Composition with the given smooth profile proves the scalar assertion. The exact endpoint equalities $\phi_s|_{s=0} = \phi_0$ and $W_s|_{s=0} = W_c$ give the final identity. □

Lemma G.10 (Actual rounded-boundary motion at the initial parameter)**.** *Fix every valid original ambient and fiber end bound, far radius, base coefficient above the prescribed threshold, and strict truncation levels of the all-initial-pairs assertion. Retain any supplied admissible rounding profile m. At every y on its original rounded boundary, there is an open neighborhood O of $(0, (0, y))$ and a jointly smooth motion $\Psi : O \to \mathbb{R} \times Y$ such that*

$$\Psi(0, (s, z)) = (s, z), \qquad \Psi(u, (s, z))_s = s + u, \qquad \rho(\Psi(u, (s, z))) = \rho(s, z).$$

Here ρ is the literal moving scalar of the preceding lemma. This conclusion is local near the initial parameter. It does not assert regularity for all $s \in [0, 1]$, simultaneous ribbon preservation, or a global compact-family trivialization.

Proof. The original rounded-boundary outwardness theorem applies to this exact arbitrary supplied profile and gives $d\rho_0(Z_K) > 0$ at y. Consequently the actual spatial differential $D = d\rho_0$ is surjective onto $\mathbb{R}$: for target a, the vector $aZ_K/D(Z_K)$ maps to a. The endpoint identity of the preceding lemma identifies this differential with that of the actual moving scalar at $(0, y)$. Joint smoothness and surjectivity therefore give the native parameter-level local normal form. Translate its actual parameter coordinate and apply its inverse, leaving the scalar and kernel coordinates fixed. The local-motion theorem yields the asserted open domain, joint smoothness and literal preservation identities, including the identity at zero motion time. □

Lemma G.11 (Quantitative ambient margin)**.** *There are $c, r_0 > 0$ such that $Q_f \geq crM^2$ for $r \geq r_0$, uniformly on the fixed coefficient and potential paths. Multiplying all coefficients of f by one nonzero scalar does not change these constants when both Q_f and M^2 are scaled accordingly.*

Proof. Set $h = \max_v \langle v, u\rangle$, $g = \sqrt{a}$, and $E = b - hf$. With $\delta_v = h - \langle v, u\rangle \geq 0$, the inequality $\delta_v e^{-\delta_v} \leq 1/e$ and uniform coefficient bounds give $|E| \leq CM$. Here $h \asymp r$, $g \geq cM$, and $|b| \leq rg$. Since $f = (b - E)/h$,

$$Q_f \geq r^2 g^2 - \left(1 - \frac{1}{2h}\right)|b|^2 - \frac{CM}{2h}|b|.$$

For $h \geq 1/2$ the right side decreases as $|b|$ increases from 0 to rg. Thus

$$Q_f \geq \frac{rg(rg - CM)}{2h} \geq crM^2$$

once r is large. All constants are uniform on the fixed paths. □

Lemma G.12 (Original bounded-value fiber end bounds)**.** *For every original finite input and every pair of real numbers A_{amb}, R, there is $A_{\mathrm{fib}} \geq A_{\mathrm{amb}}$ such that on $\{\phi_0 \geq A_{\mathrm{fib}},\ |W| \leq R\}$ one has $a > 0$ and $|u|^2 - |b_0|^2/a > 0$.*

Proof. Take $c, L > 0$ giving $g = \sqrt{a} \geq cM$ on $r = |u| \geq L$. Choose the constructed constants $C, H, A, d > 0$ with $|b_0 - hW| \leq CM$, $0 \leq h \leq Hr$, and $M \geq Ae^{dr}$. Put $B = \max(R, 0)$ and choose $S > \max(L, 4C/c, 4HB/(cAd))$. Let $A_{\mathrm{fib}} = \max(A_{\mathrm{amb}}, S^2/2)$. On the indicated set, $r \geq S > 0$, $g \geq cM > 0$, and therefore $a > 0$. The inequality $e^{dr} \geq dr$ gives $M \geq Adr$ and hence $cM \geq 4HB$. Consequently $HrB \leq rg/4$, while $r \geq 4C/c$ gives $CM \leq rg/4$. Since $|W| \leq R \leq B$, $|b_0| \leq h|W| + CM \leq HrB + CM \leq rg/2$. Thus $|b_0|^2/a \leq r^2/4$ and the Cauchy defect is at least $3r^2/4 > 0$. The chosen level has $A_{\mathrm{fib}} \geq A_{\mathrm{amb}}$ as required. In rank one the defect is identically zero wherever $a > 0$, so these implications make the specified end set empty. □

Remark G.13 (Range of the original fiber-bound producer). The fiber-bound lemma allows arbitrary real R and A_{amb} by using $B = \max(R, 0)$ and enlarging the potential level. It strengthens the local analytic range of the source estimate; the fixed target retains its original large-positive-R and admissible-choice quantifiers. This lemma asserts the analytic end inequalities, not a smooth intrinsic field or a full-fiber completion map.

Lemma G.14 (Global Euclidean growth of the actual original Liouville field)**.** *For each original finite input and every $K \geq 0$ there is $C > 0$ such that the actual Liouville vector of λ_K, lifted at any logarithmic-coordinate representative (u, θ), has full Euclidean length at most $C(1 + |u|)$.*

Proof. Put $r = |u|$, $g = \sqrt{a}$ and $q = K/(1 + Ka)$. The actual Liouville equation, with $L = dW$, gives $Z_K = Z_0 + qL^*(W/2 - b_0)$. In the complex coordinates for the real tangent pair, its jth entry is $u_j + q\overline{G_j}(W/2 - b_0)$. Since $|u_j| \leq r$, $|G_j| \leq g$ and $|b_0| \leq rg$ by Cauchy–Schwarz, every entry has modulus at most $r + qg(|W|/2 + rg)$. The maximum norm of the real radial and angular coordinates is bounded by this complex maximum norm. Also $q \geq 0$ and $qg^2 \leq 1$, so

$$\|Z_K\|_\infty \leq r + qg(|W|/2 + rg) \leq 2r + qg|W|/2.$$

Choose $k, L_0 > 0$ with $g \geq kM$ on $r \geq L_0$. Writing $N = |V|$, the bound $|W| \leq NM \leq Ng/k$ on this end gives $qg|W|/2 \leq Nqg^2/(2k) \leq N/(2k)$. On $r \leq L_0$, take $H > 0$ with $h(u) \leq Hr$ and let $S = \sum_v |c_v|$. Then $|W| \leq Se^{h(u)} \leq Se^{HL_0} =: B$. The inequality $g \leq 1 + g^2$ gives $qg \leq K + 1$: multiply by the positive $1 + Kg^2$ and use $K(1 + g^2) \leq (K + 1)(1 + Kg^2)$. Thus $qg|W|/2 \leq (K + 1)B/2$ on the remaining region, including critical points where $g = 0$. Choose $F = N/(2k) + (K + 1)B/2 \geq 0$. The global estimate $\|Z_K\|_\infty \leq 2r + F \leq (2 + F)(1 + r)$ bounds the coordinate maximum norm. There are $2n$ real tangent coordinates, so their Euclidean length is at most $\sqrt{2n}\|Z_K\|_\infty \leq (2n + 1)\|Z_K\|_\infty$. The constant $C = (2n + 1)(2 + F) > 0$ therefore

gives the asserted Euclidean bound. The logarithmic quotient cover carries this native inverse-system solution to the actual vector by its derivative, so the same Euclidean estimate holds for its coordinate lift at every representative. □

Lemma G.15 (Linear growth gives complete smooth flow)**.** *Let E be a finite-dimensional real normed vector space and let $V : E \to E$ be smooth. If $C > 0$ and $\|V(x)\| \le C(1 + \|x\|)$ for every x, then every initial point has a unique integral curve defined for all real times. These curves give a jointly smooth flow whose time maps are inverse at opposite times.*

Proof. Fix an initial point x and a finite horizon $T > 0$. Choose $B > (1 + \|x\|)e^{CT}$ and a smooth function $\chi : E \to [0, 1]$, equal to one on the closed ball of radius B and supported in a larger ball. The field $V_B = \chi V$ is smooth and vanishes outside a compact set. Local existence on that compact set gives a uniform positive lifetime; outside the support constant curves solve the equation for every time. Thus the same positive lifetime is available at every point. Successively restarting solutions and using uniqueness gives a global curve γ_B for V_B with $\gamma_B(0) = x$. Since $\|V_B(y)\| \le C(1 + \|y\|)$, Gronwall's inequality gives

$$1 + \|\gamma_B(t)\| \le (1 + \|x\|)e^{C|t|} \qquad (|t| \le T).$$

For negative times apply the same inequality to $t \mapsto \gamma_B(-t)$, whose derivative is $-V_B(\gamma_B(-t))$ with the same norm bound. The curve stays where $\chi = 1$ throughout this interval, so it solves the original equation there. For positive integer horizons these original solutions agree on every overlap by uniqueness, and hence define one all-real integral curve through x. Uniqueness also gives the flow group law and inverse time maps. Smooth local dependence, propagated across finitely many local flow intervals near any fixed finite time, gives joint smoothness in time and initial point. This expands the source's compact-containment continuation argument using compactly supported fields, without a maximal-interval construction. □

Lemma G.16 (Both-time completeness of the original smooth Liouville flow)**.** *For every original finite input and every $K \ge 0$, the actual Liouville vector field of λ_K has a smooth flow defined on all of $\mathbb{R} \times Y$. Its time maps have the flow group law and are mutually inverse at opposite times.*

Proof. The smooth primitive and pointwise nondegenerate differential determine a smooth Liouville vector field, hence a unique smooth local flow. Lift the field through the logarithmic quotient cover to the radial and angular real vector space. The coordinate formula and positivity of $1 + Ka$ show that this lifted field is smooth. Lemma G.14 bounds its Euclidean norm by $C(1 + |u|)$, and hence by $C(1 + |(u, \theta)|)$. Equivalence of norms in finite dimension gives a linear-growth bound for any chosen vector-space norm. Lemma G.15 supplies all-real integral curves upstairs. The differential of the cover carries the lifted field to the actual Liouville field, so projecting these curves supplies an all-real integral curve through every point of Y. Uniqueness of the native manifold equation makes the result independent of the chosen lift and gives the group law. Smooth local dependence propagated over finite intervals makes the descended flow jointly smooth. The group law identifies time $-t$ as the inverse of time t. □

Lemma G.17 (Finite negative entry for the original flow)**.** *For every original input, $K \ge 0$, valid ambient end bound A_{amb} and $A \ge A_{\mathrm{amb}}$, the constructed original negative Liouville flow enters $\{\phi_0 \le A\}$ in finite time from every point. If $A > A_{\mathrm{amb}}$, it enters $\{\phi_0 < A\}$.*

Proof. The function ϕ_0 is smooth on the quotient, with compact sublevels because its radial factor is a closed bounded ball and the torus is compact; negative-height sublevels are empty. The original

derivative identity and ambient end bound give $d\phi_0(Z_K) > 0$ throughout $\{\phi_0 \geq A_{\mathrm{amb}}\}$. Apply Lemma C.19 to the constructed complete flow. For the strict assertion, first enter the sublevel at A_{amb}, which lies inside the strict sublevel at A. □

Lemma G.18 (The actual original quadratic Liouville domain)**.** *Given $K \geq 0$, a valid ambient end bound A_{amb} and $A > A_{\mathrm{amb}}$, the literal subset $\{\phi_0 \leq A\} \subset Y$ is a compact Liouville domain for λ_K. Its smoothly embedded boundary is exactly $\{\phi_0 = A\}$.*

Proof. Put $r = \sqrt{2A} > 0$. Radial scaling identifies the sublevel with the product of the closed unit Euclidean ball and the quotient torus. This product is compact and its frontier is the product of the unit sphere and the torus. The map $(v, \theta) \mapsto (rv, \theta)$ therefore embeds $S^{n-1} \times T$ with image exactly $\{\phi_0 = A\}$. The standard sphere and torus charts make it smooth, and its differential is injective because the sphere inclusion has injective differential and $r \neq 0$. The product model has dimension $(n-1) + n = 2n - 1$. For $n = 1$, the sphere is the smooth zero-dimensional two-point sphere, so the same construction applies. The function $\phi_0 - A$ defines the sublevel globally. Its differential is nonzero along the boundary because its value on the actual Liouville field is positive. The original smooth primitive has symplectic differential and its prescribed field satisfies the Liouville equation. The positivity from the valid ambient end bound makes that field strictly outward on the displayed boundary. These data give the asserted actual Liouville domain. □

Lemma G.19 (Original compact sublevels and finite Liouville type)**.** *For every original finite input and every $K \geq 0$, (Y, λ_K) is complete and of finite Liouville type. For every valid ambient end bound and every larger quadratic level, the literal sublevel is a Liouville domain whose inclusion extends to a primitive-preserving diffeomorphism from its completion onto (Y, λ_K).*

Proof. Choose the constructed ambient end bound $A_{\mathrm{amb}} > 0$ and any $A > A_{\mathrm{amb}}$. The domain $D = \{\phi_0 \leq A\}$ is compact because the angle torus is compact, and its boundary is smooth because $d\phi_0 \neq 0$ on the positive radial level. The symplectic differential is nondegenerate everywhere. The exhaustion identity gives $d\phi_0(Z_K) = (|u|^2 + KQ_W)/(1 + Ka) > 0$ on $\phi_0 \geq A_{\mathrm{amb}}$, so the Liouville field is strictly outward along ∂D. Every point outside D enters it in finite negative time. Indeed, until its first entry the complete negative orbit has decreasing potential and remains in the compact annulus $A \leq \phi_0 \leq \phi_0(y)$. On this annulus the positive derivative $d\phi_0(Z_K)$ has a positive minimum m. If the orbit never entered D, its potential would decrease at speed at least m for all negative-flow durations, contradicting its lower bound A after duration greater than $(\phi_0(y) - A)/m$. Lemma G.18 supplies the native compact domain and its exact boundary. Lemma C.21 therefore constructs the required actual inclusion-induced completion diffeomorphism, with its smooth inverse and primitive identity. Finally the ambient end bound exists from the finite input; choosing its level plus one gives a finite-type witness for each $K \geq 0$. □

Lemma G.20 (Convexity and completeness)**.** *For f as above, $Q_f > 0$ outside a common compact set. For every finite $K \geq 0$, Z_K is complete and sufficiently large potential sublevels complete to the given Y. On regular fibers over any fixed bounded base disk, the intrinsic Liouville fields are complete and outward on one common sufficiently large potential level. More precisely, for each R there is a level $A_{\mathrm{fib}}(R)$ above the common exceptional region such that $a > 0$ and $r^2 - |b|^2/a \geq r^2/2$ on $\{\phi \geq A_{\mathrm{fib}}(R),\ |f| \leq R\}$. For the original potential this level may be enlarged to be at least A_{amb}.*

Proof. The first assertion follows from Lemma G.11. The estimate $|b - hf| \leq CM$ gives, on $|f| \leq R$,

$$\frac{|b|}{r\sqrt{a}} \leq C\left(\frac{R}{M} + \frac{1}{r}\right).$$

For each fixed R, exponential growth of M makes this at most $1/\sqrt{2}$ outside a common compact set. The intrinsic exhaustion derivative is therefore at least $r^2/2 > 0$ there. When $n = 1$, on the homogeneous end with $a > 0$ one instead reads this argument as a contradiction: $r^2 = Hu^2$ and $a = |G|^2/H$ imply $|b|/(r\sqrt{a}) = 1$. Thus the set $\{|f| \le R\}$ has no points beyond the chosen radius. No positive derivative of a zero-dimensional fiber field is asserted.

On the end the actual ambient field is

$$Z_K = Z_0 + \frac{K}{1 + Ka} L^*(f/2 - b).$$

Writing $g = H(du, du) + H(d\theta, d\theta)$ for the Kähler metric of ω_0, the adjoint satisfies $\|L^*\|_{\mathbb{C}\to g} = \sqrt{a}$, hence $\|L^*\|_{\mathbb{C}\to\mathrm{Eucl}} \le C\sqrt{a}$ by metric equivalence. This estimate, $K/(1 + Ka) \le 1/a$, $|f| \le CM$, and $|b| \le r\sqrt{a}$ give $|Z_K| \le C(1 + |u|)$. The intrinsic projection has the same bound. For each fixed K the smooth field is bounded on the compact remainder. Gronwall's inequality bounds $|u|$ on every bounded time interval in both directions; compactness of the angle torus and the smooth ODE extension theorem give completeness. A regular fiber is closed in Y, so the same compactness and extension argument applies to its intrinsic field. Outside an outward potential level, negative flow reaches it in finite time: otherwise the orbit stays in a compact annulus where the exhaustion derivative has a positive minimum. Lemma C.22 gives inclusion-induced primitive-preserving completion markings, both for the ambient and for regular fibers. □

Lemma G.21 (Surjectivity of the prescribed original superpotential)**.** *For each fixed finite input, the specified function $W : Y \to \mathbb{C}$ is surjective.*

Proof. Choose $w = (1, N, \dots, N^{n-1}) \in M$ so that the integral numbers $d_v = \langle v, w\rangle$ are distinct. For each pair $v \ne v'$, the equation $\langle v - v', w\rangle = 0$ is a nonzero integral polynomial equation in N, and therefore excludes only finitely many integers. Choosing N outside their finite union gives the claim. Because zero lies in the interior of Q, this nonzero w pairs both positively and negatively with vertices. Write $\min d_v = -d < 0 < e = \max d_v$. For any desired $z \in \mathbb{C}$, the expression $\sum_v c_v t^{d_v+d} - zt^d$ is a polynomial of degree $d + e > 0$ with nonzero constant term: the extreme exponents are unique. It has a root by the fundamental theorem of algebra, and its constant term makes that root nonzero. At that root, the monomial curve $(t^{w_1}, \dots, t^{w_n}) \in (\mathbb{C}^*)^n$ satisfies $W = z$. Taking logarithmic magnitudes and quotient phases presents this point as a point of Y. □

Lemma G.22 (Nonempty fibers)**.** *Every Laurent polynomial with this support and nonzero coefficients is surjective onto $\mathbb{C}$.*

Proof. Apply Lemma G.21 to the given nonzero coefficient tuple, with the fixed lattice and polytope. This yields a preimage of every complex number for that polynomial. □

Lemma G.23 (Boundedness and nonemptiness of the original literal ratio set)**.** *There is $A_* > 0$ such that for every $A_{\mathrm{amb}} \ge A_*$ and every $R > \max(1, M(A_{\mathrm{amb}}))$, the actual ratio set S_R is nonempty and bounded above.*

Proof. Take the native gradient constants $c, L > 0$ and the exponential-growth constants $A, d > 0$. Choose $A_* = L^2/2 + 1$. The maximum of $|W|$ on $\phi_0 \le A_{\mathrm{amb}}$ is attained. Thus every point with $|W| \ge R > M(A_{\mathrm{amb}})$ has $\phi_0 > A_{\mathrm{amb}} \ge A_*$ and $r = |u| > L$. There $g = \sqrt{a} \ge cM > 0$, and $R > 1$ also gives $|W| > 0$. By Cauchy–Schwarz, $|b_0| \le rg$; hence

$$\frac{|b_0|}{a|W|} \le \frac{r}{gR} \le \frac{r}{cMR} \le \frac{1}{cAdR}.$$

The last step uses $M \geq Ae^{dr} \geq Adr$ and $r > 0$. This finite number bounds every ratio. Surjectivity gives a point with $W = R$; it belongs to the far-value region where the denominators just proved are positive, so its ratio is an element of S_R. □

Lemma G.24 (A shared original ambient bound and analytic radius threshold)**.** *For every finite input there exist an ambient end bound A_{amb} and $R_0 > \max(1, M(A_{\mathrm{amb}}))$ such that for every $R \geq R_0$, the actual S_R is nonempty and bounded above and a valid fiber end bound relative to this same A_{amb} exists.*

Proof. Choose the ambient end bound A_o from Lemma E.38 and the ratio threshold A_* from Lemma G.23. Set $A_{\mathrm{amb}} = \max(A_o, A_*)$. Enlarging an ambient end level preserves its positivity conditions, so this is still an ambient end bound. Set $R_0 = \max(1, M(A_{\mathrm{amb}})) + 1$. For every $R \geq R_0$, the strict radius inequality needed for the ratio lemma holds, proving nonemptiness and boundedness of the literal set. Lemma G.12, applied to this same ambient level and this R, gives the required fiber end bound. □

Lemma G.25 (Native regularity of the original far fiber)**.** *If A_{amb} is a valid ambient end bound and $R > \max(1, M(A_{\mathrm{amb}}))$, then R is a regular value of W: at every point with $W = R$, the actual differential $dW : T_yY \to \mathbb{C}$ is surjective.*

Proof. Such a point has $\phi_0 > A_{\mathrm{amb}}$, since otherwise the attained sublevel maximum would give $R = |W| \leq M(A_{\mathrm{amb}})$. The ambient bound therefore gives $a = \sum_j |G_j|^2 > 0$. For any $z \in \mathbb{C}$, the complex tangent vector $q_j = \overline{G_j} z/a$ satisfies $dW(q) = \sum_j G_j \overline{G_j} z/a = z$. The logarithmic cover differential is a real tangent-space isomorphism, so this preimage gives a preimage under the actual quotient-manifold differential. □

Lemma G.26 (Finite thresholds for the actual coefficient and potential path)**.** *Fix the coefficient path f_t, and a smooth potential ϕ with $cI \leq D^2\phi \leq CI$ for $c > 0$, and put $\phi_s = (1-s)|u|^2/2 + s\phi$. For every specified cylinder radius L there are $R_0, B > 0$ such that, for all $R \geq R_0$ and independent $t, s \in [0,1]$, the literal sets*

$$S_{t,s}(R) = \left\{ \frac{|b_t(y)|}{a_{t,s}(y)|f_t(y)|} : |f_t(y)| \geq R \right\}$$

are nonempty and bounded above by B. Every point in their defining region satisfies $|u| > L$ and $a_{t,s} > 0$. Consequently $1 \leq \kappa_{t,s}(R) := 1 + 2\sup S_{t,s}(R) \leq 1 + 2B$. The set of values $\{\kappa_{t,t}(R) : t \in [0,1]\}$ is nonempty and bounded; its supremum $\kappa_{\mathrm{path}}(R)$ satisfies $K_0(R) \leq \kappa_{\mathrm{path}}(R) \leq 1 + 2B$. At $t = s = 0$, the ratio set equals the original S_R literally.

Proof. The coefficient bounds give $|f_t(u,\theta)| \leq C_0 e^{U_0|u|}$, uniformly in t. Choose R_0 larger than this bound on a cylinder containing $|u| \leq L$ and the common dominance exceptional region. Thus $|f_t| \geq R \geq R_0$ implies cylinder exclusion and $a_{t,s} > 0$. Weighted Cauchy, the common inverse-Hessian dominance estimate and $D^2\phi_s \leq \max(1, C)I$ yield

$$\frac{|b_t|}{a_{t,s}|f_t|} \leq C_1 \frac{|u|}{MR} \leq C_2 \frac{|u|e^{-d|u|}}{R} \leq B$$

for a fixed positive B, using $M \geq Ae^{d|u|}$ and $e^{d|u|} \geq d|u|$. Surjectivity of each f_t supplies a point with $f_t = R$, proving nonemptiness before taking a supremum. All ratios are nonnegative. The claimed single-pair bounds now follow from the least-upper-bound property. The diagonal value set contains its value at zero and inherits the same upper bound, so its supremum is finite. At zero, $f_0 = W$ and $D^2\phi_0 = I$, identifying the set with S_R and its threshold with $K_0(R)$. □

Lemma G.27 (Numeric outward margin from the literal path threshold)**.** *For the preceding actual ratio set, suppose $R > 0$, the set is bounded above, $K \geq \kappa_{t,s}(R)$, and a point satisfies $|f_t| \geq R$ and $a_{t,s} > 0$. Writing $a = a_{t,s}$ and $b = b_t$, one has the scalar inequalities*

$$0 < \frac{a|f_t|^2}{1+Ka} \leq \frac{2\,\mathrm{Re}(\overline{f_t}b) + Ka|f_t|^2}{1+Ka}.$$

Proof. The nonnegative ratio at this point lies below the supremum, hence $K \geq 1$ and $2|b| \leq (K-1)a|f_t|$. Use $\mathrm{Re}(\overline{f_t}b) \geq -|f_t||b|$ in the numerator and divide by $1 + Ka > 0$. The lower numerator is positive since $a > 0$ and $|f_t| \geq R > 0$. □

Lemma G.28 (Literal ratio threshold and base outwardness)**.** *There is an ambient end bound A_{amb} independent of K and an excluded-value bound B_{ex}. For $R > \max(1, M(A_{\mathrm{amb}}), B_{\mathrm{ex}})$ the actual ratio set is nonempty and bounded above, its threshold is $1 + 2\sup S_R$, and every K above that threshold gives $Z_K|f|^2 > 0$ on $|f| \geq R$.*

Proof. Fix a closed radial cylinder

$$K_{\mathrm{ex}} = \{(u,\theta) : |u| \leq L_{\mathrm{ex}}\} \subset Y$$

containing the nonhomogeneous regions and every exceptional region in Lemmas E.25 and G.11, uniformly for the previously fixed unnormalized coefficient and potential paths. The radius L_{ex} is fixed independently of R, β, K and all admissible truncation choices. For the original quadratic potential choose $A_{\mathrm{amb}} > L_{\mathrm{ex}}^2/2$; then $a > 0$ and $Q_W > 0$ on $\phi_0 \geq A_{\mathrm{amb}}$, as required in (G1). Writing the unnormalized coefficient path as f_t, put

$$B_{\mathrm{ex}} = \max_{t\in[0,1],\ y\in K_{\mathrm{ex}}} |f_t(y)|.$$

Every critical value of every f_t has modulus at most B_{ex}. For the present end and threshold arguments take $R > \max(1, M(A_{\mathrm{amb}}), B_{\mathrm{ex}})$; a single threshold including all later requirements will be fixed in the final assembly. The ratio set defining

$$\kappa(f,\phi;R) = 1 + 2\sup_{|f|\geq R} \frac{|b|}{a|f|}, \qquad K_0(R) = \kappa(W,\phi_0;R) \tag{15}$$

is nonempty by Lemma G.22. Its denominators are positive, and it is bounded because on this region

$$\frac{|b|}{a|f|} \leq \frac{C|u|}{MR} \leq C'|u|e^{-c|u|}/R.$$

In particular the supremum is a genuine finite real supremum of a nonempty set, and $\kappa \geq 1$. If $K \geq \kappa(f,\phi;R)$, then

$$Z_K|f|^2 = \frac{2\,\mathrm{Re}(\overline{f}b) + Ka|f|^2}{1+Ka} \geq \frac{a|f|^2}{1+Ka} > 0 \qquad (|f| \geq R). \tag{16}$$

All critical values lie inside the excluded base disk. The end estimate in Lemma G.20 proves existence of a fiber end bound. We now verify the universal clauses for the original polynomial separately.

□

Lemma G.29 (Native base derivative and the literal path threshold)**.** *Let $f = W_t$ and ϕ_s be the fixed coefficient and potential paths. At a point where $D^2\phi_s(u)u = d\phi_s(u)$, put $H = D^2\phi_s(u)$, $a = \overline{G}^T H^{-1} G$ and $b = u \cdot G$. For $K \geq 0$ the actual native field for $\lambda_{\phi_s} + K f^* \lambda_{\mathbb{C}}$ satisfies*

$$d|f|^2(Z_K) = \frac{2\,\mathrm{Re}(\overline{f}b) + Ka|f|^2}{1 + Ka}.$$

If $R > 0$, $|f| \geq R$, $a > 0$, the actual ratio set is bounded above, and $K \geq \kappa(f, \phi_s; R)$, then $d|f|^2(Z_K) \geq a|f|^2/(1 + Ka) > 0$. The Euler condition is supplied on the common homogeneous end.

Proof. The chain rule on the logarithmic cover gives $d|f|^2(v) = 2\,\mathrm{Re}(\overline{f}\, df(v))$. The cover has a bijective differential, so this is also the identity on every native tangent space. Substitute the native holomorphic formula $df(Z_K) = (b + Kaf/2)/(1 + Ka)$ and take the real part to obtain the displayed expression. The literal-threshold numeric margin bounds exactly this expression from below by $a|f|^2/(1 + Ka)$, which is strictly positive. □

Lemma G.30 (Literal-threshold base outwardness at original points)**.** *Assume $R > 0$, the actual ratio set S_R is nonempty and bounded above, $K \geq 1 + 2\sup S_R$, and a point y satisfies $|W(y)| \geq R$ and $a(y) > 0$. Then*

$$d|W|^2(Z_K)(y) \geq \frac{a(y)|W(y)|^2}{1 + Ka(y)} > 0.$$

Proof. Each ratio in S_R is nonnegative, so its supremum is nonnegative and $K \geq 1$. The ratio at y is at most that supremum. Multiplication by its positive denominator gives $2|b_0(y)| \leq (K - 1)a(y)|W(y)|$. Since $\mathrm{Re}(\overline{W}b_0) \geq -|W||b_0|$, the numerator in Lemma E.26 is at least $a|W|^2$. Its denominator is positive, and $a|W|^2 > 0$, giving both assertions. □

Lemma G.31 (The two outward derivatives on an original corner)**.** *Suppose A_{amb} is a valid ambient end bound, the maximum $M(A_{\mathrm{amb}})$ is attained, $R > \max(1, M(A_{\mathrm{amb}}))$, S_R is nonempty and bounded above, $K \geq K_0(R)$, and $A \geq A_{\mathrm{amb}}$. At every point with $\phi_0 = A$ and $|W| = R$, both $d\phi_0(Z_K)$ and $d|W|^2(Z_K)$ are positive.*

Proof. At such a point the ambient end bound gives $a > 0$ and $Q_W > 0$. Since $A_{\mathrm{amb}} > 0$, also $|u|^2 = 2A > 0$. The literal threshold gives $K \geq 1$, so the exhaustion formula in Lemma E.26 has positive numerator and denominator. Its base derivative is positive by Lemma G.30. □

Lemma G.32 (Smooth convex maximum with exact tails)**.** *For every $\epsilon > 0$ there exists a smooth convex profile $m_\epsilon : \mathbb{R}^2 \to \mathbb{R}$ with nonnegative partial derivatives summing to one, satisfying $\max(x, y) \leq m_\epsilon(x, y) \leq \max(x, y) + \epsilon$ and $m_\epsilon(x, y) = \max(x, y)$ whenever $|x - y| \geq \epsilon$.*

Proof. Let $q(t)$ be a smooth nondecreasing function with values in $[0, 1]$, zero for $t \leq 0$ and one for $t \geq \epsilon$. Set $F(t) = \int_0^t q(s)\,ds$, $c = \epsilon - F(\epsilon)$ and $g(t) = F(t) + F(-t) + c$. The fundamental theorem of calculus makes F smooth with derivative q; it is convex since q is nondecreasing. It vanishes for $t \leq 0$, and for $t \geq \epsilon$ it equals $F(\epsilon) + t - \epsilon$. Hence g is smooth, convex and even, with $g(t) = |t|$ on $|t| \geq \epsilon$. For $0 \leq t \leq \epsilon$ one has $g(t) - t = \int_t^\epsilon (1 - q(s))\,ds$, which lies in $[0, \epsilon]$. Evenness therefore gives $|t| \leq g(t) \leq |t| + \epsilon$ everywhere. Moreover $g'(t) = q(t) - q(-t) \in [-1, 1]$. Now put $m_\epsilon(x, y) = (x + y + g(x - y))/2$. Composition with an affine map and addition of an affine function preserve convexity. Its partial derivatives are $(1 + g'(x - y))/2$ and $(1 - g'(x - y))/2$, so have the asserted signs and sum. The identity $\max(x, y) = (x + y + |x - y|)/2$ gives the bounds and exact tails. This integrated-cutoff construction replaces the source's convolution while retaining every profile condition. □

Lemma G.33 (Uniform original corner-band geometry). *Suppose A_{amb} is an ambient end bound, $R > \max(1, M(A_{\mathrm{amb}}))$, S_R is nonempty and bounded above, $K \geq K_0(R)$, and A_{fib} is any valid fiber end bound. For every $A_- > A_{\mathrm{fib}}$ and $A > A_-$ there is $0 < \delta < A - A_-$ such that, for every $\epsilon \leq \delta$, throughout B_ϵ the actual derivative $d|W|^2(Z_K)$ is positive and $d\phi_0, d|W|^2$ are linearly independent.*

Proof. Write $D = d|W|^2(Z_K)$ and consider the compact set $S = \{\phi_0 \leq A,\ |W|^2 \leq R^2,\ D \leq 0\}$. The continuous nonnegative function $g = A - \phi_0 + R^2 - |W|^2$ is strictly positive on S: a zero would be a corner point, where Lemma G.31 gives $D > 0$. If S is nonempty let $m > 0$ be the minimum of g on it and take $\delta_0 = m/2$; otherwise take $\delta_0 = 1$. A point of B_ϵ with $\epsilon \leq \delta_0$ has $g < 2\epsilon$, and thus cannot be in S. Consequently $D > 0$ on that band. Shrink to $\delta = \min(\delta_0, (A - A_-)/2)$. Then $\phi_0 > A - \epsilon > A_- > A_{\mathrm{fib}}$ and $|W| \leq R$ on the band, so the fiber bound supplies $a > 0$ and $|u|^2 - |b_0|^2/a > 0$. In complex tangent coordinates the vector $v = u - \overline{G} b_0/a$ satisfies $dW(v) = 0$ and $d\phi_0(v) = |u|^2 - |b_0|^2/a > 0$. Evaluating a relation $s\, d\phi_0 + t\, d|W|^2 = 0$ on v gives $s = 0$; evaluating it on Z_K then gives $t = 0$. This proof also covers an empty band. The compact minimum of the level deficits replaces the source's shrinking-sequence argument. □

Lemma G.34 (Existence of every original admissible rounding). *Under the hypotheses of Lemma G.33, there exists an admissible rounding for the specified R, K, A_-, A.*

Proof. Choose its positive width $\delta < A - A_-$ from that lemma and a profile m_δ from Lemma G.32. The width inequalities and both geometric conditions on B_δ are exactly the required admissibility conditions. □

Lemma G.35 (Shared bounds with all original rounding choices). *For every finite input there exist an ambient end bound A_{amb} and $R_0 > \max(1, M(A_{\mathrm{amb}}))$ such that, for every $R \geq R_0$, the literal set S_R is nonempty and bounded above and a fiber end bound exists. For every valid fiber end bound, every $K \geq K_0(R)$ and every $A_- > A_{\mathrm{fib}}$, $A > A_-$, an admissible rounding exists.*

Proof. Take the shared bounds of Lemma G.24. For every $R \geq R_0$ its strict radius inequality, ratio-set conclusions and fiber-bound existence hold. Apply Lemma G.34 to each of the universally supplied fiber bounds, parameters and levels. □

Lemma G.36 (Unchanged faces of an arbitrary smooth profile). *Let m be any rounding profile of positive width ϵ. If $m(x, y) = x$, its full differential at (x, y) is the first-coordinate projection; if $m(x, y) = y$, it is the second-coordinate projection. If $m(x, y) = 0$, $x \leq -\epsilon$ and $y \leq 0$, then $y = 0$. Symmetrically, if $m(x, y) = 0$, $x \leq 0$ and $y \leq -\epsilon$, then $x = 0$.*

Proof. The nonnegative smooth function $m(x, y) - x$ has a global minimum zero where $m(x, y) = x$, so its differential vanishes there. This proves the first projection identity, and the second is identical with y replacing x. The nonnegative profile partials imply monotonicity in each coordinate. Suppose $x \leq -\epsilon$ and $y < 0$ with $m(x, y) = 0$. Exact tails give $m(x, 0) = 0$. For every $t > y$, coordinate monotonicity gives $m(x, t) \geq 0$. Thus $t = 0$ is a local minimum of $t \mapsto m(x, t)$, so its derivative there is zero. But the second projection identity at $(x, 0)$ makes that derivative one, a contradiction. Interchanging the coordinates proves the other assertion. This argument treats flat junctions directly using the differential at a minimum. □

Lemma G.37 (Smooth rounded scalar and compact zero level). *For every original input, real R, A and rounding profile m, the function $\rho = m(\phi_0 - A, |W|^2 - R^2)$ is smooth, and both $\{\rho \leq 0\}$ and $\{\rho = 0\}$ are compact.*

Proof. Smoothness follows by composition of the smooth profile with the two smooth scalar functions. The profile lower bound gives $\phi_0 - A \le \rho$, so its closed sublevel $\{\rho \le 0\}$ lies in the compact potential sublevel $\{\phi_0 \le \max(A, 0)\}$. It is therefore compact, and its closed subset $\{\rho = 0\}$ is compact as well. □

Lemma G.38 (Actual outwardness for every admissible rounded boundary)**.** *Fix a valid ambient bound, $R > \max(1, M(A_{\mathrm{amb}}))$ with S_R nonempty and bounded above, $K \ge K_0(R)$, any valid fiber bound, and $A_- > A_{\mathrm{fib}}$. For every admissible rounding at the levels A_-, A, the actual differential satisfies $d\rho(Z_K) > 0$ at every point of $\{\rho = 0\}$. In particular $d\rho : T_y Y \to \mathbb{R}$ is surjective there.*

Proof. Put $x = \phi_0 - A$ and $y = |W|^2 - R^2$ at a zero point. The lower bound on the profile implies $x, y \le 0$. If both exceed $-\epsilon$, the point is in the admissible band. There $d|W|^2(Z_K) > 0$ by admissibility and $\phi_0 > A - \epsilon > A_- > A_{\mathrm{amb}}$ gives $d\phi_0(Z_K) > 0$. The chain rule expresses $d\rho(Z_K)$ as their weighted sum with nonnegative weights summing to one, hence it is positive. Outside the strict band, Lemma G.36 places the point on one of the unchanged faces and identifies the full profile differential with its corresponding coordinate projection. On the potential face $\phi_0 = A$ the ambient end bound gives positivity. On the base face $|W| = R$ the literal-threshold base inequality gives positivity. These cases include the flat junctions. Finally, if $L = d\rho$ and $L(Z_K) > 0$, the vector $(r/L(Z_K))Z_K$ maps to each prescribed real r, proving surjectivity. □

Lemma G.39 (An unchanged neighborhood of the entire initial ribbon)**.** *For every positive-width profile with $\epsilon < A - A_-$, there is an open neighborhood of $\{W = R, \phi_0 \le A_-\}$ on which ρ equals $|W|^2 - R^2$ literally. On that entire ribbon its differential is exactly $d|W|^2$.*

Proof. Take the open set where $(|W|^2 - R^2) - (\phi_0 - A) > \epsilon$. At every ribbon point its left side is $A - \phi_0 \ge A - A_- > \epsilon$. On this open set the profile's exact-tail condition makes it equal to its second argument, which is $|W|^2 - R^2$. Differentiating this local equality gives the stated covector identity. □

Proposition G.40 (Every admissible initial pair)**.** *For the original quadratic potential and every R as above, every valid A_{fib}, every $K \ge K_0(R)$, and every $A_- > A_{\mathrm{fib}}$, $A > A_-$ admit an admissible rounding. Every admissible rounding gives the pair and both specified markings in (G2), with nonempty ribbon.*

Proof. The existence clause follows from Lemma G.35, which constructs the shared bounds and every required admissible profile. Fix any admissible profile. Lemma C.10 supplies its literal compact ambient domain and original primitive. Lemma C.11 supplies its completion with the actual inclusion-induced marking onto the original ambient.

Lemma C.18 constructs the entire compact fiber truncation and its actual marking onto the complete full fiber. It is nonempty: surjectivity supplies a point of the full fiber, and finite negative entry carries it into the truncation. Lemma C.6 restricts only the extension neighborhood, retaining the whole compact carrier, and embeds it in the unchanged rounded boundary. Its exact primitive identity supplies the ribbon condition with constant positive witness one; its literal inclusion square and image give exactly the required full ribbon. The completion restriction preserves the specified full-fiber marking. These data assemble the asserted initial pair and both markings.

All constructions apply in rank one: the regular fiber has dimension zero, the intrinsic field is stationary, and regularity of the truncation boundary makes that boundary empty. No positive-dimensional argument or component selection is used in the assembly. □

Corollary G.41 (The complete initial geometric assertions). Every finite toric input satisfies assertions G1 and G2 of the target, with one shared ambient bound and far-radius threshold and all the prescribed parameter and admissible-choice quantifiers.

Proof. The original finite-type theorem proves G1 for every $K \geq 0$. Choose the common ambient bound and radius threshold of the initial rounding-bounds lemma. For each radius above that threshold it supplies the literal ratio-set bounds and a valid fiber-end bound. For every valid fiber-end bound, permitted parameter and strict pair of levels, it supplies an admissible rounding. For every admissible rounding, the preceding proposition constructs the required initial pair and both actual markings. This is exactly the stated quantifier order of G2. □

G.0.1 Changes of ambient truncation at a fixed full ribbon

Lemma G.42 (All ambient choices at a fixed original ribbon level)**.** *Let two supplied original initial pairs have the same R, K, A_-, but arbitrary possibly different ambient levels and admissible rounding profiles. They admit a strict completion identification with ribbon scale one. Its composite with the target inclusion marking is the source inclusion marking on the whole completed ambient. The boundary map fixes the entire marked embedded ribbon and its end shift is zero there.*

Proof. Compose the source inclusion marking with the inverse target marking. Both markings preserve λ_K, so this composite preserves the completed primitive. The boundary-coordinate theorem supplies its actual smooth boundary diffeomorphism and smooth shift, together with the all-time end equation and the contact pullback formula.

The two embedded ribbon carriers are both exactly $W^{-1}(R) \cap \{\phi_0 \leq A_-\}$. The theorem comparing full compact embedded carriers constructs a compact ribbon diffeomorphism whose ambient square holds at every compact point. Its primitive pullback is exactly the source primitive: both ribbon primitives pull back the same λ_0, and equality on a full compact carrier determines the derivatives at its boundary as well as its interior. Thus its scale is one.

At a compact ribbon point, the ambient marking square and the compact ribbon square show that the completed ambient map takes its source boundary representative to its target boundary representative. The all-time end equation at time zero and injectivity of the target end coordinates imply that the induced boundary map is this representative and its shift is zero. This proves the required ribbon square and hence the strict identification. The construction uses no nonempty boundary or positive ribbon dimension assumption. □

Lemma G.43 (Graded comparison for fixed ribbon level)**.** *For the two initial pairs of the preceding lemma, every supplied logarithmic Maslov representative on the target is the literal target representative of a length-one comparison of these initial choices. The same geometric arrow carries the prescribed logarithmic grading, and the fixed-ambient decoration uses the actual inclusion markings.*

Proof. Construct the preceding strict identification q. Pull the supplied target representative back along its actual ambient map. The equality of the two marked ambient maps and the chain rule identify its tensor with the prescribed logarithmic tensor at the source. A constant homotopy grades the actual identification from this pullback representative to the supplied target representative. Take the path consisting of this one identification and decorate it by the given inclusion markings; their composite equality makes the identification the identity on marked Y. All endpoint tuples and representatives therefore agree literally. □

Lemma G.44 (Attach all ambient choices to the convenient prefix)**.** *Let a five-arrow original marked prefix be retained. Every independently supplied initial pair at its R, K, A_-, with arbitrary*

ambient level and rounding profile, starts a six-arrow path to its exact tailoring endpoint. The initial identification and original coefficient arrow form a length-two initial segment with prescribed logarithmic source datum. The full path is exactly the concatenation of this segment, the retained original-to-holomorphic identification, and the retained three-arrow marked chain.

Proof. Identify the supplied source with the convenient original source by the fixed-ribbon comparison, and pull the retained original Maslov representative back along this map. Its equality on marked Y gives the source logarithmic tensor. The constant pullback homotopy grades this identification. Append the retained original coefficient arrow, sharing its Maslov representative literally, to form the length-two graded segment. Append the retained identification and three-arrow marked chain. Path concatenation preserves every adjoining tuple and gives the stated equality and total length six. □

G.0.2 The full original ribbon on varying levels

Lemma G.45 (A common compact carrier for varying full-ribbon levels)**.** *Fix an original ambient end bound, a radius in its stated range, any valid fiber end bound, and any two levels A, B above that bound. On the actual full-fiber model $j : H \hookrightarrow Y$, set $\eta = \phi_0 \circ j$ and $L(t) = (1-t)A + tB$. There is a nonempty compact smooth domain D, with smooth embeddings J_t of one common open extension into H, jointly smooth with extensions across parameter endpoints, such that*

$$J_t(D) = \{\eta \le L(t)\}, \qquad J_t(\partial D) = \{\eta = L(t)\}.$$

Consequently $j(J_t(D))$ is the entire prescribed original ribbon at level $L(t)$, including every component and the zero-dimensional case.

Proof. Take the actual original compact sublevel at A as the source carrier. Its carrier and frontier are respectively $\{\eta \le A\}$ and $\{\eta = A\}$, and it is nonempty. The transverse full-ribbon transport provides a jointly smooth flow Φ on $\mathbb{R} \times H$. Its first coordinate at every point of this carrier is t after time t. The compact projected-flow embedding theorem therefore supplies one open neighborhood of the carrier on which every spatial projection $x \mapsto (\Phi_t(0, x))_2$ is a smooth embedding, with a jointly smooth extension in time. Restrict the original compact domain to this common neighborhood. Its carrier and frontier keep their original images. The two-direction flow image identities give exactly the two displayed sets. Composition with j gives the prescribed entire ribbon image. □

Lemma G.46 (The actual Liouville family on the common ribbon carrier)**.** *The common carrier of the preceding lemma carries a smooth family of Liouville structures with primitive $J_t^* \mu_H$, canonical intrinsic field, and outwardness along its full boundary. Both compact primitive and field families are jointly smooth. Their localized completion family is constructed from these primitives. The entire carrier and boundary images remain the literal sublevel and level sets stated above.*

Proof. The full-fiber primitive is smooth and symplectic. Equal-dimensional smooth embeddings have invertible tangent maps, so their pullbacks are symplectic on the whole common extension. The defining function on H at time t is $q_t = \eta - L(t)$. It is nonpositive on the embedded carrier and zero on its boundary. At that boundary $\eta = L(t)$ exceeds the valid fiber end bound, so the intrinsic Liouville derivative of q_t is strictly positive. The sublevel pullback theorem transfers this inequality to the carrier's own native defining functions and constructs its Liouville structure. Joint smoothness of embeddings and the constant full-fiber primitive gives joint smoothness of the pullback primitive; inversion of its nondegenerate differential gives the field family. The Liouville-family collar and completion construction supplies the localized completed family from the same primitives. □

Lemma G.47 (Restriction of a compact embedding family to an open target)**.** *Let J_t be a smooth family of embeddings of the extension of a compact domain D into a manifold H. If $J_t(D)$ lies in an open set O for every $t \in [0,1]$, there is one open neighborhood U of D on which all J_t take values in O. Their restrictions are smooth embeddings into O, jointly smooth with extensions across both parameter endpoints.*

Proof. Choose a smooth real-parameter extension f of J_t on a neighborhood of $[0,1] \times D$. Intersect this neighborhood with $f^{-1}(O)$. This is open and contains the compact product $[0,1] \times D$. The tube lemma gives a product neighborhood $V \times U$ inside it. Restrict the source to U and the target to O. Smoothness in the open submanifold follows from smoothness after its inclusion; embedding and immersion follow from those of J_t. The same restriction of f on $V \times U$ is the required real-parameter extension. □

Lemma G.48 (The varying full ribbon in one unchanged rounded face)**.** *Choose $L > \max(A, B)$ and a geometrically admissible rounded ambient domain for ribbon level L. The varying full ribbons admit one common nonempty compact domain, smooth embeddings J_t into the actual full fiber and j_t into this fixed rounded boundary, and a native smooth Liouville family μ_t. Their entire carrier and boundary images are the literal sublevel and level sets at $(1-t)A + tB$, and on the whole common extension*

$$\iota \circ j_t = j \circ J_t, \qquad \mu_t = J_t^* \mu_H = j_t^*(\lambda_K|_{\partial E}).$$

Proof. The whole common carrier has level at most $\max(A, B) < L$. Thus every image lies in the open ribbon neighborhood on which the original rounded boundary is its unchanged base face. Apply the open-target restriction lemma and compose with the actual original boundary embedding. Restrict the existing Liouville family to this same source neighborhood. Restriction preserves its compact carrier and boundary, joint smoothness, and the primitive pullback equation. The boundary primitive equation follows by composing the fixed base-face primitive identity with the restricted family maps. □

Lemma G.49 (Completing a ribbon family with fixed ambient tuple)**.** *Fix a Liouville domain E and its completion. Let D carry a native smooth Liouville family μ_t, with constructed uniform collars and localized compact and completed presentations. Suppose j_t is a smooth family of embeddings into ∂E and $j_t^*(\lambda_E|_{\partial E}) = \mu_t$ on the whole extension. These data construct a witnessed-pair deformation with witness one, constant ambient tuple, and the same localized ribbon completion family.*

Proof. At time t, use the collar-localized ribbon domain and compose j_t with its original open inclusion. Pullback composition and literal restriction of the localized primitive prove the witnessed primitive equation. Use the generated compact and completed ribbon presentations. Their primitive and field regularity is part of the constructed completion family. The original projection formula transports the ribbon embeddings to a jointly smooth family on the source extension. For the ambient tuple use identity compact and completion presentations; its forms and fields are constant smooth families. The witness is the constant scalar one. □

Lemma G.50 (Full-fiber markings of actual sublevel presentations)**.** *Let a exceed any valid original fiber end bound. If a nonempty native Liouville domain embeds into the actual whole fiber with image exactly $\{\phi_0 \le a\}$ and primitive the pullback of the full-fiber primitive, every chosen completion has an inclusion/flow marking onto that entire fiber, extending exactly this embedded compact inclusion.*

Proof. Use the defining function $q = \phi_0 - a$ on the actual full-fiber model. Its sublevel is compact and its intrinsic Liouville derivative is strictly positive on the zero set. Every full-fiber point reaches this sublevel in finite backward time. The embedded absorbing-sublevel marking theorem therefore constructs the required primitive-preserving diffeomorphism for the supplied native completion, with its exact compact inclusion and all-real flow-induced end equation. □

Lemma G.51 (Completed pairs for the actual varying full ribbon)**.** *The preceding family extends to a smooth witnessed-pair deformation, with the rounded ambient domain and its completion fixed and witness one. Every slice has an inclusion marking of its entire completed ribbon into the actual full fiber. Its carrier image remains the whole prescribed original ribbon at that slice's level. The marking of the entire completed ambient and the compact ambient inclusion remain constant, hence jointly smooth in the specified presentations of this same deformation.*

Proof. Construct the uniform Liouville collars and localized completion family from the native ribbon primitives. Compose each boundary embedding with the collar's original open inclusion. Its pulled-back ambient boundary primitive is exactly the localized ribbon primitive. The compact and completed ribbon presentations and their smooth forms and fields are those of this constructed completion family. The ambient presentations are identities. Each localized compact carrier embeds as the entire full-fiber sublevel at its actual level. Compactness, strict intrinsic outwardness, and backward absorption of that sublevel construct the full-fiber inclusion marking of its chosen completion. □

Lemma G.52 (Grading on the same full-ribbon motion)**.** *In the constructed varying full-ribbon deformation, every Maslov datum on the fixed completed ambient extends over the same deformation, with that exact datum at both endpoints.*

Proof. The completed ambient primitive and its presentation are constant, with presentation the identity at every parameter. Extend both the compatible complex field and its nonzero squared-canonical section constantly in time. Their chart expressions are smooth on the product of the entire real time axis with the spatial chart. Compatibility, canonicity and nonvanishing are inherited pointwise. Pullback through the specified endpoint presentations is the identity, so both endpoint equations hold literally. □

Lemma G.53 (Comparison of arbitrary supplied initial choices)**.** *Fix the original end bound, radius and coefficient in their stated ranges. Fix any common valid fiber end bound. Two arbitrary supplied initial pairs, at levels above that bound and with arbitrary ambient truncations and rounding profiles, admit a three-step comparison. Every step is marked inside the same original completed ambient, and the prescribed logarithmic grading extends along this same chain. Any chosen prescribed terminal representative is retained literally.*

Proof. Choose a common ambient rounding with unchanged base face containing both full ribbons, and construct their varying-level family in that face. Identify each supplied endpoint with the corresponding constructed endpoint using the strict fixed-ribbon comparison. These maps are the identity on the marked entire ambient. Choose the prescribed logarithmic representative on the supplied terminal pair and pull it through the terminal identification to the common ambient. Extend it constantly over the constructed family using the preceding lemma, then pull it through the initial identification. Both strict steps carry the grading by their actual pullbacks. Their markings commute exactly, so the endpoint representatives have the prescribed logarithmic tensor values. Concatenating these two identifications with the same middle deformation gives all asserted data on one three-step path. □

Lemma G.54 (Comparing arbitrary initial choices)**.** *At fixed original R, K, any two admissible initial pairs are joined by a finite permitted chain whose completed ambient marking is the identity on (Y, λ_K). The chain preserves $[\Omega^{\otimes 2}]$.*

Proof. Let their ribbon levels be A^0_-, A^1_-, and take the smaller of their valid fiber end bounds. Both levels exceed it. Choose $L > \max(A^0_-, A^1_-)$, $A'' > L$, and an admissible rounding with $\epsilon'' < A'' - L$, supplied by Proposition G.40. Its domain E'' has both ribbons and every intermediate fiber sublevel in the unrounded base face. Identify each original ambient completion with $\widehat{E}''$ via the inclusion markings. The induced boundary map fixes that endpoint ribbon pointwise, because it is on both boundary sections. Use the identity ribbon map and scale one. Inside E'' vary the ribbon level linearly between A^0_- and A^1_-. The intrinsic exhaustion derivative stays positive on this compact band. A vector field tangent to the fiber, with derivative one on the exhaustion there and cut off outside a slightly larger band, trivializes these sublevel domains smoothly. This is a pair deformation with witness one. Relabeling at its ends, if needed, uses an identity ambient completion step and the specified ribbon diffeomorphism. All ambient data are restrictions or pullbacks of the fixed datum on Y. □

Lemma G.55 (Exact normalization of the marked junction)**.** *Let W_1 be the coefficient-one endpoint polynomial, $R = e^\beta$, and $f_0 = R^{-1}W_1$, the initial member of the tailoring family. For every smooth ambient one-form λ and every real K,*

$$\lambda + K f_0^* \lambda_{\mathbb{C}} = \lambda + (K/R^2) W_1^* \lambda_{\mathbb{C}}.$$

For every potential ϕ, outer level A, inner level A_-, and rounding profile m_ϵ, the normalized defining function $m_\epsilon(\phi - A, R^2(|f_0|^2 - 1))$ equals $m_\epsilon(\phi - A, |W_1|^2 - R^2)$ pointwise, and

$$\{f_0 = 1,\ \phi \le A_-\} = \{W_1 = R,\ \phi \le A_-\}.$$

Thus the compact carriers and primitive forms at this junction agree literally, with the same profile and witness one.

Proof. The coefficient path ends at one, and the initial tailoring sum has every coefficient $e^{-\beta} = R^{-1}$. Factoring this scalar out of the finite sum proves $f_0 = R^{-1}W_1$. For a real constant c, differentiation gives $d(cf) = c\,df$. Substituting into $f^*\lambda_{\mathbb{C}} = \frac{1}{2}(\operatorname{Re} f\, d \operatorname{Im} f - \operatorname{Im} f\, d \operatorname{Re} f)$ yields $(cf)^*\lambda_{\mathbb{C}} = c^2 f^*\lambda_{\mathbb{C}}$. Apply this with $c = R^{-1}$ for the primitive identity. Since $R > 0$, $R^2|R^{-1}W_1|^2 = |W_1|^2$ and $R^{-1}W_1 = 1$ is equivalent to $W_1 = R$. Substitution gives both the defining-function and truncation equalities. □

Lemma G.56 (Retaining a rounding while increasing the coefficient)**.** *Let m_ϵ be any admissible rounding at coefficient $K \ge 0$. For every $L \ge K$, the same profile is admissible at L. Under the initial end and level hypotheses, the fixed rounded defining function has strictly positive Liouville derivative on its zero level for every coefficient $(1-s)K + sL$, $0 \le s \le 1$.*

Proof. The actual base derivative is

$$\frac{2\operatorname{Re}(\overline{W} b_0) + Ka|W|^2}{1 + Ka}.$$

Its denominator is positive for every nonnegative coefficient. Its numerator has nonnegative slope $a|W|^2$, so its positivity at K persists at every larger coefficient. The corner band, the width inequalities, and the independence of its two covectors do not depend on the coefficient. Hence

the same profile remains admissible. The affine path lies in $[K, L]$ and remains above the original threshold. Apply the outward derivative assertion for an arbitrary admissible original rounding at each of these coefficients. This proves outwardness on the fixed whole zero level, including its flat junctions. □

Lemma G.57 (Joint smoothness of the original coefficient family)**.** *For any real K, L, the primitives $\lambda_{(1-t)K+tL} = \lambda_0 + ((1-t)K + tL)W^*\lambda_{\mathbb{C}}$ vary jointly smoothly, with extension across both endpoints of $[0, 1]$.*

Proof. In each fixed quotient chart the coefficients of λ_0 and $W^*\lambda_{\mathbb{C}}$ are smooth functions of the spatial coordinate. Their linear combination has coefficients equal to the sum of the first coefficient function and the second multiplied by the polynomial $(1-t)K+tL$. Products and sums of smooth functions are smooth on $\mathbb{R}$ times the open chart target. These open sets cover the whole closed parameter interval and provide the required endpoint extensions. □

Lemma G.58 (The original marked coefficient deformation)**.** *Let I be a finite toric input, A_{amb} a valid ambient end bound, and $R > \max(1, M(A_{\mathrm{amb}}))$, with S_R nonempty and bounded above. Let A_{fib} be any valid fiber end bound relative to A_{amb}, choose $A_- > A_{\mathrm{fib}}$ and $A > A_-$, and take $K \geq K_0(R)$ and any admissible profile m_ϵ at K. For every $L \geq K$, there is a smooth completed pair deformation whose actual slice at every $t \in [0, 1]$ is an original initial pair at coefficient $(1-t)K + tL$, with the same profile and levels. Every slice has its entire nonempty far-fiber ribbon and both specified inclusion-induced completion markings.*

Proof. Every affine coefficient lies in $[K, L]$, so the parameter-rounding lemma preserves strict outwardness on the entire fixed rounded zero level. Symplecticity holds at every coefficient because it is nonnegative. These properties give the Liouville structures on the same original rounded domain. Choose the smooth boundary inclusion of the fixed entire ribbon into the fixed rounded domain. Its ambient inclusion square is fixed on the whole ribbon extension. Pullback along this square shows that the ambient primitive restricts to the fixed intrinsic ribbon primitive at every parameter, since the superpotential is constant there. Give all slices witness one.

The original completion theorem applies at every affine coefficient, because the same profile remains admissible. It supplies the individual ambient completions and their inclusion-induced markings. Keep the original full-fiber ribbon completion and its marking fixed. Choose the compact and completed presentations from the marked completion-germ construction. Their boundary presentations are the identity, and their transported completed primitives and fields vary jointly smoothly. Smooth pullback along the compact presentation gives joint smoothness of the transported compact primitives. Their actual fields vary smoothly because they are the unique solutions of the nondegenerate contraction equations.

Present the fixed ribbon and its completion by identity maps. Their forms and fields are constant smooth families. Since the ambient boundary presentation is the identity, the transported ribbon inclusion is the same fixed inclusion, and the witness remains one. These presentations therefore satisfy every compact and completed regularity condition for one pair deformation. The slice tuples themselves are the original initial pairs, with the individually constructed ambient markings and fixed ribbon marking. □

Lemma G.59 (A literal first parameter comparison step)**.** *For every choice in the preceding lemma, there are original initial pairs at coefficients K and L joined by a permitted deformation step, and hence by a comparison chain of length one. Both endpoint pairs retain the original profile, levels, entire ribbon, and both markings.*

Proof. Take the actual endpoint slices of the preceding deformation. The affine coefficients at times zero and one equal K and L respectively, so these slices have the asserted endpoint parameters. Relabeling these equal real coefficient expressions changes no component of their witnessed-pair tuples. The deformation is a permitted step between precisely those tuples. Its singleton finite path has length one. □

Lemma G.60 (The logarithmic datum along an affine coefficient path)**.** *For every finite toric input and $K, L \geq 0$, the original ambient symplectic forms $d\lambda_{(1-t)K+tL}$, $t \in [0,1]$, have a smooth Maslov family with the fixed logarithmic complex structure and the fixed squared logarithmic volume tensor $\Omega^{\otimes 2}$.*

Proof. The affine coefficient is nonnegative on $[0,1]$. In the logarithmic coframe the complex structure is $J(a,b) = (-b,a)$ and $d\lambda_k = \sum_j du_j \wedge d\theta_j + 2k\, d\Re W \wedge d\Im W$. Holomorphicity of W makes this form J-invariant and gives $d\lambda_k(v, Jv) = \sum_j (a_j^2 + b_j^2) + 2k|dW(v)|^2 > 0$ for $v \neq 0$. Thus J is compatible with every form in the path. The product of the complex logarithmic coframe is a nowhere-zero canonical volume form, so its square is canonical and nowhere zero. Both the complex structure and the tensor are smooth in spatial coordinates. Their constant extensions in the real parameter are jointly smooth on the product of the whole parameter line with each open spatial chart, including both parameter endpoints. □

Lemma G.61 (Completion-map germs from compact presentations and flow)**.** *Let C be a completion of a compact Liouville domain D, and let N be a smooth manifold. Suppose maps $f_t : C \to N$, $t \in [0,1]$, satisfy $f_t(i(x)) = g(t,x)$ on the compact carrier, where g is jointly smooth for real t on the domain extension. Suppose also that for some $\delta > 0$,*

$$f_t(\mathrm{end}(x,r)) = \Phi_t^r(b(x)) \qquad (r > -\delta),$$

where $b : \partial D \to N$ is smooth. Assume that the actual flow maps $(t,r,y) \mapsto \Phi_t^r(y)$ have jointly smooth real local extensions at every (t,r,y) with $t \in [0,1]$ and $r \in \mathbb{R}$. Then the maps f_t have jointly smooth real local extensions at every $(t,z) \in [0,1] \times C$.

Proof. At a point in the image of the compact interior, choose a smooth local inverse of i. Restrict its source so that this inverse stays in the interior. The formula $g(t, i^{-1}(z))$ is then a jointly smooth extension of f_t on this neighborhood.

Every other point has end coordinates (x,r) with $r \geq 0$. Choose a smooth local inverse of the end map and an actual real flow germ at $(t,r,b(x))$. The map taking (s,z) to $(s, (\mathrm{end}^{-1} z)_2, b((\mathrm{end}^{-1} z)_1))$ is smooth near this point. Restrict its domain to the preimage of the flow germ's open domain and to radii greater than $-\delta$. This is an open neighborhood because $r \geq 0$ and $\delta > 0$. Composing with the flow germ gives the required extension. Its equality to f_s follows from the end formula at actual parameters. Both constructions apply at the two parameter endpoints as well. □

Lemma G.62 (Retained inward end square for completion families)**.** *Let D be a fixed compact domain with a jointly smooth interval family of Liouville structures, and choose an individual completion C_t for each actual parameter. There are coherent compact presentations h_t with identity boundary presentation, smooth for real t, and completed presentations $p_t : C_0 \to C_t$ with jointly smooth transported primitives and fields. They can be chosen with some $\delta > 0$ such that*

$$p_t(\mathrm{end}_0(x,r)) = \mathrm{end}_t(x,r) \qquad (t \in [0,1],\ r > -\delta).$$

Proof. Construct the smooth compact collar family and its presentation isotopy. ODE uniqueness identifies the actual negative end maps with this collar. The presentation's collar identity gives the common compact completion square, so the completion-family constructor applies. Retain its common width $\delta > 0$. For $-\delta < r < 0$, compact coverage and the compact presentation square give the completed end square. For $r \geq 0$, this is already the defining end square. The boundary presentation is the identity in both formulas. □

Lemma G.63 (Actual marked completion germs for the coefficient family)**.** *Fix a toric input I, a valid ambient end bound A_{amb}, $R > \max(1, M(A_{\mathrm{amb}}))$ with nonempty bounded ratio set, and a valid fiber end bound A_{fib}. Choose $A_- > A_{\mathrm{fib}}$, $A > A_-$, $K \geq K_0(R)$, any profile admissible at these original choices, and $L \geq K$. The actual original pairs at coefficient $(1-t)K + tL$ have compact and completed presentations such that the maps from the fixed source completion to Y, obtained by following the completed presentations with the prescribed original inclusion markings, have jointly smooth real local extensions at every point and actual parameter.*

Proof. Use the compact Liouville collar family to choose a common width $\delta > 0$ and a smooth compact presentation isotopy, with identity boundary presentation. On negative collar radii the actual complete-flow ODE identifies each end map with the collar inclusion. The collar presentation identity therefore gives the compact completion square. The completion-family constructor produces completed presentations with jointly smooth transported primitives and fields. The compact square and compact coverage imply that their end square also holds for $-\delta < r < 0$; the defining nonnegative end square handles $r \geq 0$.

For the actual maps followed by their markings, the compact inclusion equation identifies their compact formula with the smooth compact presentation. Their inward end square and each marking's flow-induced equation identify their end formula with the original Liouville flow, starting at the fixed ambient boundary inclusion. The original affine primitive family is jointly smooth. Nondegeneracy of its exterior derivative gives a smooth family of its actual Liouville fields. Local real extensions of these fields and ODE uniqueness along compact finite orbit segments provide real smooth germs of their complete flows at every finite time. The preceding completion-map lemma now gives the asserted local extensions on the full fixed completion, including its seam. No joint regularity of the individually supplied marking maps was assumed. □

Lemma G.64 (Prescribed grading of the same original coefficient step)**.** *For every original admissible choice in the preceding marked-map-germ lemma, there are original initial pairs at coefficients K and L and a comparison chain of length one between their literal tuples. Both endpoints have the prescribed logarithmic ambient grading through their actual inclusion markings, and that same geometric chain has a grading lift between these representatives.*

Proof. Choose the actual compact and completed presentations of the marked coefficient family supplied by the marked-map-germ lemma. Assemble the pair deformation using precisely those presentations, the fixed full ribbon, and its fixed completion and marking. On Y, keep the logarithmic complex field and squared volume tensor fixed while varying the symplectic form along the affine coefficient path. The affine Maslov family lemma gives compatibility and smoothness of this ambient datum.

Pull this family back by the actual completed presentations followed by the prescribed inclusion markings. Their proved local smooth germs give joint local extensions of the pulled complex field and tensor. Each marking preserves the actual primitive; composing this equation with the completed presentation and taking exterior derivatives identifies the pulled symplectic form with the deformation's completed symplectic form. Thus the local-germ Maslov pullback theorem gives a

grading on this very pair deformation. At each endpoint the chain rule for the composite map identifies its representative with the prescribed representative pulled back through the actual endpoint marking.

Take the singleton chain consisting of this graded deformation. The affine endpoint coefficients equal K and L. Relabeling by these equal real expressions changes neither endpoint tuple nor its marking or representative, so it transports the same length-one graded chain to the stated original endpoint data. □

Lemma G.65 (Nonempty tailored fibers on the positive real locus)**.** *For all sufficiently large β, every $s \in [0,1]$ has a point $(u,0)$ with $f_s(u,0) = 1$.*

Proof. On the zero-angle locus all monomials are positive real and

$$f_s = s\eta(1-\chi_0) + \sum_v \big(1 + s\eta(\chi_v - 1)\big) e^{\langle v,u\rangle - \beta}.$$

All summands are nonnegative, since each weight belongs to $[0,1]$. At $u = 0$ and $\beta \geq 0$, the constant term is maximal, so $\chi_0 = 1$ and $f_s(0,0) \leq |V| e^{-\beta} \leq 1$ once β is sufficiently large. Choose a nonzero vertex and a real vector u pairing with it to $\beta + 1$. A vertex maximizing the pairing at u has weight at least 1 in logarithmic coordinates and is maximal even among the augmented exponents. Its cutoff is one, hence its coefficient in the displayed sum is one. Therefore $f_s(u,0) \geq e > 1$. The real continuous function f_s on the connected real coordinate space takes the value one by the intermediate value theorem. □

This positive-real argument supplies nonemptiness directly from the explicit family. It replaces the source's open-and-closed parameter-image argument for this conclusion, without assuming a bundle trivialization.

H Joint local charts of marked level families

Lemma H.1 (Adding the truncation coordinate)**.** *Let $L : E \to F$ be a surjective continuous linear map and $r : E \to \mathbb{R}$ a continuous linear functional. If $z \in \ker L$ and $r(z) \neq 0$, then $(L, r) : E \to F \times \mathbb{R}$ is surjective. In particular, if $f : M \to \mathbb{C}$ is regular at $j(x)$ on a smooth embedded fiber $j : N \hookrightarrow M$, $f \circ j$ is constant, and $d(\rho \circ j)_x(v) \neq 0$, then (f, ρ) is regular at $j(x)$.*

Proof. Given (c, a), choose v with $L(v) = c$ and set $w = v + (a - r(v))z/r(z)$. Linearity gives $L(w) = c$ and $r(w) = a$. For the embedded fiber take $L = df_{j(x)}$, $r = d\rho_{j(x)}$, and $z = dj_x(v)$. Differentiating the constant function $f \circ j$ gives $L(z) = 0$; the chain rule gives $r(z) = d(\rho \circ j)_x(v) \neq 0$. □

Lemma H.2 (Joint normal forms preserving the parameter)**.** *Let M be a finite-dimensional smooth manifold without boundary, F a finite-dimensional real vector space, and $f : \mathbb{R} \times M \to F$ smooth. If the spatial differential of f is surjective at (t, x), then $(s, y) \mapsto (s, f(s, y))$ has a smooth local product normal form at (t, x) whose first coordinates are exactly $(s, f(s, y))$.*

Proof. Write the full differential of f as D. For a desired tangent target (a, c) choose a spatial vector v satisfying $D(0, v) = c - D(a, 0)$. Then the differential of $(s, f(s, y))$ sends (a, v) to (a, c), so it is surjective. The regular-level normal-form theorem applied to this actual augmented map produces the asserted smooth chart, with its first coordinate equal to the augmented map itself. □

Lemma H.3 (Smooth local motions preserving marked equations)**.** *Let e be a smooth local product chart on Q with first coordinates $(\tau(x), f(x)) \in \mathbb{R} \times F$. At every a in its source there are an open neighborhood O of $(0, a)$ in $\mathbb{R} \times Q$ and a jointly smooth motion $\Psi : O \to Q$ such that*

$$\Psi(0, x) = x, \qquad \tau(\Psi(u, x)) = \tau(x) + u, \qquad f(\Psi(u, x)) = f(x).$$

The image remains in the source of e. Consequently a smooth parameterized family with surjective spatial differential has such a local motion at each regular point, with τ its actual parameter coordinate.

Proof. Write $e(x) = ((\tau(x), f(x)), k(x))$. On the open set

$$O = \{(u, x) : x \in \text{source}(e), ((\tau(x) + u, f(x)), k(x)) \in \text{target}(e)\},$$

define $\Psi(u, x) = e^{-1}((\tau(x) + u, f(x)), k(x))$. The chart and its inverse are smooth, and coordinate translation is smooth; therefore this motion is jointly smooth. The chart target is open, so O is open and contains $(0, a)$. The inverse identity gives $\Psi(0, x) = x$. The other equations follow by applying e to the defining formula. The joint normal form of the preceding lemma supplies such an e directly from the surjective spatial differential. □

Lemma H.4 (Native joint smoothness of the tailoring family)**.** *For fixed β, the actual quotient map $(s, y) \mapsto f_{\beta,s}(y)$ is jointly smooth in the native ambient coordinates.*

Proof. The literal quotient family is already jointly smooth. The native ambient atlas is obtained by the fixed linear recharting of the quotient atlas; the identity from native to quotient coordinates is smooth. Applying this identity to the spatial coordinate and retaining the real parameter gives a smooth product map. Composition with the quotient family proves the claim. □

Lemma H.5 (Uniform local motions of the full tailoring fibers)**.** *For a smooth potential with fixed positive Hessian lower and upper bounds, there is $\beta_0 > 0$ such that for every $\beta \geq \beta_0$, $s \in [0, 1]$ and y with $f_{\beta,s}(y) = 1$, an open neighborhood of $(0, (s, y))$ has a jointly smooth local motion Ψ satisfying*

$$\Psi(0, (t, z)) = (t, z), \quad \Psi(u, (t, z))_t = t + u, \quad f_{\beta,\Psi(u,(t,z))_t}(\Psi(u, (t, z))_y) = f_{\beta,t}(z).$$

No nonempty-fiber or global-trivialization conclusion is part of this lemma.

Proof. Choose the threshold from native Poisson positivity. At every stated point the Poisson pairing is positive because the fiber value has norm one. The two Hamiltonian normal vectors therefore give surjectivity of the actual complex differential. Apply the local-motion theorem to the jointly smooth native tailoring family. Its open real domain includes parameter extensions across either interval endpoint and preserves the original level function by the exact chart equation. □

Lemma H.6 (Uniform local motions preserving the truncation boundary)**.** *Retain a smooth potential with fixed positive Hessian bounds and a C^1 degree-two homogeneous end. There are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$, $s \in [0, 1]$, and point of the full fiber $f_{\beta,s}(y) = 1$ satisfying $\phi(y) \geq F^2\beta^2/2$, the preceding local-motion equations hold together with $\phi(\Psi(u, (t, z))_y) = \phi(z)$. Thus they apply at the boundary of every truncation at a later level $A \geq F^2\beta^2/2$.*

Proof. Take the constants from the actual full-fiber truncation-marking theorem and increase the parameter threshold to the native Poisson threshold. That theorem supplies a full embedded realization of the fiber and an intrinsic Liouville vector with strictly positive potential derivative throughout the stated end. At y, choose its preimage x in this realization. Its pushed-forward

intrinsic vector is in the complex differential kernel, and its potential derivative remains positive by the chain rule. Native Poisson positivity makes the complex differential surjective. The first lemma therefore makes $(f_{\beta,s}, \phi)$ regular at y. This pair of equations is jointly smooth in the real parameter, so its joint local motion preserves both equations exactly. In dimension one the required end on the fiber is empty, consistently with the same statement. □

Lemma H.7 (Nonempty full-fiber potential truncations)**.** *For the potential in the preceding lemma there are $F, \beta_0 > 0$ such that every $\beta \geq \beta_0$, $s \in [0,1]$, and $A \geq F^2\beta^2/2$ has a nonempty literal compact ribbon $\{f_{\beta,s} = 1,\ \phi \leq A\}$. More generally, each native full-fiber potential truncation with its proved outwardness and complete intrinsic flow is nonempty whenever the whole literal fiber is nonempty.*

Proof. Choose a common threshold for the full-fiber truncation theorem and the real-positive-slice nonemptiness theorem. The latter gives a point in the whole fiber. The full embedded image equation gives a point in its native realization. The actual complete intrinsic flow absorbs this point into the potential sublevel in finite negative time. That image point belongs to the native truncation carrier and hence to its displayed literal image. This also proves the stated general nonemptiness implication. □

Lemma H.8 (Compact tailoring ribbon total space and proper projection)**.** *Let ϕ be smooth, uniformly strictly convex, and equal outside a ball to a positively degree-two homogeneous end potential that is C^1 everywhere and smooth away from the origin. For every real β, A, the literal total space*

$$T_{\beta,A} = \{(s,y) : s \in [0,1],\ f_{\beta,s}(y) = 1,\ \phi(y) \leq A\}$$

is compact. Its actual projection $(s,y) \mapsto s$ to $\mathbb{R}$ is a proper map, including when this total space is empty.

Proof. The ambient potential sublevel $\{\phi \leq A\}$ is compact by the quadratic lower bound on the homogeneous end. Its product with $[0,1]$ is compact. Joint continuity of $f_{\beta,s}$ and continuity of ϕ show that $T_{\beta,A}$ is closed in this product, hence compact. The parameter projection is continuous from a compact space to the Hausdorff real line, hence proper. □

Lemma H.9 (The compact tailoring total space hits every parameter)**.** *For a potential satisfying the preceding hypotheses and a uniform upper Hessian bound, there are $F, \beta_0 > 0$ such that every $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$ satisfy*

$$\mathrm{pr}_{\mathbb{R}}(T_{\beta,A}) = [0,1].$$

Proof. The definition of $T_{\beta,A}$ puts its projection inside $[0,1]$. For each $s \in [0,1]$, the uniform literal truncation nonemptiness lemma gives y with $f_{\beta,s}(y) = 1$ and $\phi(y) \leq A$. Then $(s,y) \in T_{\beta,A}$ projects to s, proving the reverse inclusion. □

Lemma H.10 (Actual local horizontal fields of marked levels)**.** *Let $f : \mathbb{R} \times M \to F$ be a jointly smooth map between finite-dimensional real manifolds, with F a vector space. If $d(f(a_1,\cdot))_{a_2}$ is surjective, there is an open neighborhood O of a and a smooth native tangent section V on O satisfying*

$$d\big((s,x) \mapsto (s, f(s,x))\big)_p V_p = (1,0) \qquad (p \in O).$$

In particular the actual parameter derivative is one and every actual marked-level derivative is zero. The case $F = \mathbb{C} \times \mathbb{R}$ keeps both real fiber equations and the truncation equation. More generally, if an actual partial diffeomorphism $e : Q \dashrightarrow (\mathbb{R} \times F) \times K$ has first coordinate g, its open source supports a smooth field with $dg(V) = (1,0)$; pulling any constant coordinate field through an inverse partial diffeomorphism is smooth on its source.

Proof. The joint normal-form lemma constructs e whose source contains a and whose first coordinate is the actual map $g(s,x) = (s, f(s,x))$. Put $v = ((1,0),0)$ and $V_p = d(e^{-1})_{e(p)}v$. The tangent map of the smooth inverse is smooth; compose it with the smooth constant-vector section over $e(p)$. Since $e^{-1}(e(p)) = p$ on the source, the resulting native tangent section has base point exactly p. On the open target, $g \circ e^{-1} = \mathrm{pr}_{\mathbb{R}\times F}$. Differentiating this identity and applying to v gives $dg_p(V_p) = (1,0)$. This also proves the stated general pullback-field smoothness and conditional normal-form specialization. □

I Local horizontal lifts of the actual tailoring equations

Lemma I.1 (Open lifts preserving the full fiber equation)**.** *Let ϕ be a smooth potential with fixed positive lower and upper Hessian bounds. There is $\beta_0 > 0$ such that for every $\beta \geq \beta_0$, $s \in [0,1]$, and y satisfying $f_{\beta,s}(y) = 1$, an open neighborhood O of (s,y) carries a smooth tangent field V such that*

$$d\big((t,z) \mapsto (t, f_{\beta,t}(z))\big)_p(V_p) = (1,0) \qquad (p \in O).$$

The equation holds on the open neighborhood, including its points whose parameter lies outside $[0,1]$.

Proof. Choose the uniform threshold supplied by native Poisson positivity. At the specified point the fiber value has norm one, so the native Poisson pairing is strictly positive. Its Hamiltonian normal vectors make the complex spatial differential surjective. Joint native smoothness of the tailoring map and the horizontal-field theorem therefore construct the asserted open neighborhood and tangent field. The latter theorem applies on the real parameter manifold, so it gives the displayed identity throughout that neighborhood without restricting its parameter coordinate. □

Lemma I.2 (Open lifts retaining the truncation coordinate)**.** *Retain such a potential, with a degree-two homogeneous end that is C^1 everywhere and smooth away from the origin. There are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$, $s \in [0,1]$, and point y with $f_{\beta,s}(y) = 1$ and $\phi(y) \geq F^2\beta^2/2$, there are an open neighborhood O of (s,y) and a smooth tangent field V satisfying*

$$d\big((t,z) \mapsto (t, (f_{\beta,t}(z), \phi(z)))\big)_p(V_p) = (1,(0,0)) \qquad (p \in O).$$

Proof. Take the uniform constants from the full-fiber truncation-marking theorem and enlarge the scale threshold to ensure Poisson positivity. Its embedded full-fiber realization contains y. Push the intrinsic Liouville vector at a preimage of y into the ambient tangent space. Differentiating the constant fiber equation shows that this vector belongs to the complex differential kernel. Its potential derivative is strictly positive by the whole-fiber end estimate. The complex differential is surjective by Poisson positivity. Adding the potential covector is consequently surjective by the marked-level surjectivity lemma. Apply the horizontal-field theorem to the jointly smooth map $(t,z) \mapsto (f_{\beta,t}(z), \phi(z))$. It constructs the desired field and proves both level identities on its entire open domain. In rank one the specified end in the fiber is empty; no positive derivative is imposed on a nonempty zero-dimensional end. □

J Confinement in the marked parameter slab

Lemma J.1 (Horizontal curves stay in a truncated fiber)**.** *Let Q be a topological space, E a real normed vector space, and $\gamma : \mathbb{R} \to Q$, $\tau, \rho : Q \to \mathbb{R}$, and $f : Q \to E$ continuous. Fix $w \in E$ and $A, v \in \mathbb{R}$. Write*

$$C = \{q : \tau(q) \in [0,1],\ f(q) = w,\ \rho(q) \leq A\}.$$

Suppose O, B are open, $C \subset O$, and $C \cap \{\rho = A\} \subset B$. Suppose

$$(\tau \circ \gamma)'(t) = v, \qquad (f \circ \gamma)'(t) = 0 \quad \text{when } \gamma(t) \in O, \qquad (\rho \circ \gamma)'(t) = 0 \quad \text{when } \gamma(t) \in O \cap B.$$

Let $J \subset \mathbb{R}$ be connected with $t_0 \in J$, $f(\gamma(t_0)) = w$, $\rho(\gamma(t_0)) \le A$, and $\tau(\gamma(t_0)) + v(t - t_0) \in [0, 1]$ for all $t \in J$. Then every $t \in J$ satisfies

$$\tau(\gamma(t)) = \tau(\gamma(t_0)) + v(t - t_0), \qquad f(\gamma(t)) = w, \qquad \rho(\gamma(t)) \le A.$$

In particular this statement applies to positive or negative time intervals.

Proof. Let P be the set of real times satisfying the three displayed conclusions. Continuity makes P closed, and $t_0 \in P$. We show that $P \cap J$ is open relative to J. At $t \in P \cap J$ the affine parameter bound puts $\gamma(t)$ in C, hence in O. Continuity keeps $\gamma(u)$ in O for all u in a sufficiently small open interval around t. On that interval the functions $\tau(\gamma(u)) - vu$ and $f(\gamma(u))$ have zero derivative, so the mean-value theorem makes each constant. Their values at t give the first two defining equations of P nearby. If $\rho(\gamma(t)) < A$, continuity preserves this inequality nearby. If equality holds, $\gamma(t) \in B$; shrink the interval until its image lies in $O \cap B$. There $\rho \circ \gamma$ has zero derivative and is constant, equal to A. Thus the third defining inequality also persists. Consequently $P \cap J$ is both relatively open and relatively closed in J, and is nonempty. Connectedness gives $J \subset P$, as required. The derivative-zero claim for vector-valued functions follows from the normed-space mean-value inequality on the small convex interval. □

Lemma J.2 (Preservation of the truncation boundary)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ be continuous, J a connected subset of $\mathbb{R}$, and $t_0 \in J$ with $g(t_0) = A$. If each $t \in J$ with $g(t) = A$ has a neighborhood on which g' exists and vanishes, then $g(t) = A$ for every $t \in J$.*

Proof. The level set is closed in J by continuity and contains t_0. At each of its points choose an open interval on which the derivative vanishes. The mean-value theorem makes g constant there, with value A, so the level set is also relatively open in J. Connectedness gives the assertion. Applied to the potential along a confined horizontal curve, the open boundary collar supplies exactly the derivative-zero neighborhoods. □

K Simultaneous gluing of marked horizontal fields

Lemma K.1 (Gluing all applicable affine constraints)**.** *Let Q be a finite-dimensional Hausdorff σ-compact smooth manifold without boundary and $C \subset Q$ closed. For each j let $S_j \subset Q$ be a marked locus and let $L_j(x) : T_xQ \to F$ be a continuous linear map, with prescribed value $b_j(x) \in F$. Suppose every $a \in C$ has an open neighborhood O_a and a smooth tangent field V_a on O_a such that*

$$L_j(x)V_a(x) = b_j(x) \qquad (x \in O_a \cap C \cap S_j)$$

for every j. There is a globally smooth tangent field V satisfying $L_j(x)V(x) = b_j(x)$ for every $x \in C \cap S_j$, simultaneously for all j.

Proof. At x let A_x consist of vectors satisfying every equation whose marked locus contains x, if $x \in C$, and let $A_x = T_xQ$ otherwise. Each A_x is convex: if $v, w \in A_x$ and $r, s \ge 0$ with $r + s = 1$, linearity gives $L_j(x)(rv + sw) = (r + s)b_j(x) = b_j(x)$ for each applicable j. The stated fields give local smooth sections taking values in A_x around points of C. Outside C, its open complement carries the zero field, which also takes values in A_x. Take a smooth partition of unity subordinate to

this open cover. Multiply each local field by its partition function and extend by zero; the support is contained in the corresponding open set, so the extended field is smooth. Their locally finite sum is smooth. At each point the nonzero terms belong to A_x, their coefficients are nonnegative and add to one, so their sum belongs to A_x by convexity. This is the required field. □

Lemma K.2 (Affine lift on a neighborhood of a compact family)**.** *Let Q also be normal and let C be compact. Suppose the local fields of the preceding lemma satisfy each equation throughout $O_a \cap S_j$, without restriction to C. There are an open set O containing C and a globally smooth tangent field satisfying every equation on $O \cap S_j$.*

Proof. The union $U = \bigcup_{a \in C} O_a$ is open and contains the closed set C. Normality gives an open O containing C with $\overline{O} \subset U$. Every point of $\overline{O}$ lies in some O_a, so the original field on that source gives the local solution required by the preceding lemma with closed family $\overline{O}$. Its glued field satisfies all equations on $\overline{O} \cap S_j$, and hence on $O \cap S_j$. □

Lemma K.3 (One horizontal lift with an open marked collar)**.** *Retain the normal manifold Q of the preceding lemma. Let $C \subset Q$ be compact and $T \subset Q$ closed. Suppose around each point of C there is a smooth field solving $L(x)V = b(x)$ on its whole source, and around each point of $C \cap T$ there is a smooth field solving both $L(x)V = b(x)$ and $H(x)V = c(x)$ on its whole source. There are open sets O, B and one globally smooth field V such that $C \subset O$, $C \cap T \subset B$, the first equation holds on O, and the second holds on $O \cap B$.*

Proof. Let U be the union of the marked sources. Normality gives an open B with $C \cap T \subset B$ and $\overline{B} \subset U$. At a point of $C \cap \overline{B}$, choose a marked source containing it; its field solves both equations throughout that source. At a point of $C \setminus \overline{B}$, take the first local field and intersect its source with the open complement of $\overline{B}$. It solves the first equation and imposes no extra equation there. Apply neighborhood gluing with the first equation marked on all of Q and the second marked on $\overline{B}$. The resulting open O and field solve the first equation on O and the second on $O \cap \overline{B}$, hence on $O \cap B$. □

Lemma K.4 (One field for the compact tailoring ribbon family)**.** *Let ϕ be smooth with uniform positive Hessian lower and upper bounds, and agree outside a ball with a positive degree-two homogeneous end potential that is C^1 everywhere and smooth away from zero. There are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$, the literal compact total space*

$$C = \{(s, y) : s \in [0, 1],\ f_{\beta,s}(y) = 1,\ \phi(y) \leq A\}$$

has an open neighborhood O carrying one smooth native field V with $d(s, f_{\beta,s})V = (1, 0)$ throughout O. Its boundary locus $C \cap \{\phi = A\}$ has an open neighborhood B such that $d\phi(V) = 0$ throughout $O \cap B$. The field extends smoothly to the whole real parameter ambient manifold.

Proof. Take a common scale threshold for local fiber horizontal fields and local marked fiber horizontal fields, retaining the latter's uniform end constant F. The total space C is compact. At every point of C the first local producer gives the augmented fiber equation on an open source. At every point of $C \cap \{\phi = A\}$ the level inequality permits the marked local producer, which gives $d(s, (f_{\beta,s}, \phi))V = (1, 0)$ on its source. Its components are the same augmented fiber equation and $d\phi(V) = 0$. The set $\{\phi = A\}$ is closed by continuity. Apply the preceding lemma to these two affine equations, placing the scalar potential equation in the first component of the fiber-equation target. Its field and open neighborhoods have exactly the asserted properties. □

Lemma K.5 (Supported complete flow near a compact family)**.** *Let $C \subset Q$ be compact and O an open neighborhood of C in a finite-dimensional Hausdorff second-countable smooth manifold without boundary. A smooth tangent field V on O admits a globally smooth field W, vanishing outside a compact set, that agrees with V on an open neighborhood P of C with $P \subset O$. The field W has a jointly smooth flow $\Phi : \mathbb{R} \times Q \to Q$ defined for every real time.*

Proof. Choose a smooth compactly supported function χ with closed support inside O and equal to one on an open neighborhood P of C. Set $W = \chi V$ on O and extend by zero. The support condition makes this extension smooth, and $W = V$ on P. The field vanishes outside the compact closed support of χ. Local smooth ODE existence gives a uniform integration time on this compact support. Outside it the constant curve solves the ODE. Iterate the resulting uniform time step in both directions; uniqueness makes the concatenations agree, giving integral curves for every real time. Uniqueness also gives $\Phi_{s+t} = \Phi_s \circ \Phi_t$ and $\Phi_0 = \mathrm{id}$. Smooth dependence of the ODE on its initial point gives joint smoothness on each local step, and composition gives joint smoothness about every finite time. □

Lemma K.6 (Actual supported tailoring horizontal flow)**.** *With the hypotheses and uniform constants of the compact tailoring horizontal-field lemma, there are open neighborhoods O of the actual compact ribbon total space and B of its truncation boundary, a globally smooth native tangent field W, and an all-real jointly smooth flow Φ of W such that*

$$d(s, f_{\beta,s})W = (1, 0) \text{ on } O, \qquad d\phi(W) = 0 \text{ on } O \cap B.$$

Proof. Apply the supported-flow construction to the field of the compact tailoring horizontal-field lemma, using compactness of its literal total space. Replace the first open neighborhood by the smaller neighborhood on which the supported field agrees with that field, and retain its marked collar. Both differential equations follow from the agreement. The constructed supported flow is jointly smooth for every real time. □

Lemma K.7 (Full-interval transport from the horizontal equations)**.** *Let $\tau, \rho : Q \to \mathbb{R}$ and $f : Q \to F$ be smooth, where F is a real normed vector space, and fix $w \in F$ and $A \in \mathbb{R}$. Let O, B be open neighborhoods of*

$$C = \{q : \tau(q) \in [0, 1],\ f(q) = w,\ \rho(q) \le A\}, \qquad \partial C = \{q : \tau(q) \in [0, 1],\ f(q) = w,\ \rho(q) = A\},$$

respectively. Suppose a flow Φ integrates a native tangent field W with $d(\tau, f)W = (1, 0)$ on O and $d\rho(W) = 0$ on $O \cap B$. For every $p \in C$ and every real t with $\tau(p) + t \in [0, 1]$,

$$\tau(\Phi_t p) = \tau(p) + t, \qquad f(\Phi_t p) = w, \qquad \rho(\Phi_t p) \le A.$$

If $\rho(p) = A$, then $\rho(\Phi_t p) = A$.

Proof. Along $\gamma(u) = \Phi_u p$, the chain rule and the integral-curve equation give $(\tau \circ \gamma)' = 1$ and $(f \circ \gamma)' = 0$ whenever $\gamma(u) \in O$, and $(\rho \circ \gamma)' = 0$ whenever $\gamma(u) \in O \cap B$. Apply marked-curve confinement to the connected interval $J = [-\tau(p), 1 - \tau(p)]$, starting at $u = 0$. The affine predicted parameter lies in $[0, 1]$ throughout J, so confinement gives all three assertions on J. If $\rho(p) = A$, every time in J at which the potential equals A lies in $O \cap B$. Openness and continuity give a time neighborhood on which the potential derivative is zero. The level- preservation lemma on the connected interval J then gives $\rho \circ \gamma = A$ throughout J. The desired t belongs to J. □

Lemma K.8 (Actual transport of the full compact tailoring ribbon)**.** *Under the potential hypotheses of the compact tailoring horizontal-field lemma, there are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$ one jointly smooth all-real flow Φ on the actual parameter ambient manifold has the following property. If $p = (s, y)$ satisfies $s \in [0, 1]$, $f_{\beta,s}(y) = 1$, and $\phi(y) \leq A$, then every real t with $s + t \in [0, 1]$ satisfies*

$$(\Phi_t p)_1 = s + t, \qquad f_{\beta,(\Phi_t p)_1}((\Phi_t p)_2) = 1, \qquad \phi((\Phi_t p)_2) \leq A,$$

$$\phi((\Phi_t p)_2) = A \iff \phi(y) = A, \qquad \Phi_{-t}(\Phi_t p) = p.$$

Thus times $t = r - s$ give mutually inverse maps between the literal truncated fibers at any $s, r \in [0, 1]$, preserving their boundaries.

Proof. Use the constructed tailoring horizontal flow and apply the preceding lemma with $\tau(s, y) = s$, $f(s, y) = f_{\beta,s}(y)$, and $\rho(s, y) = \phi(y)$. These are smooth in the native ambient manifold; the neighborhoods and equations are exactly those furnished by that flow construction. This gives the three transport equations and forward boundary preservation. The flow law gives $\Phi_{-t}\Phi_t p = \Phi_0 p = p$. Apply the same transport lemma to $\Phi_t p$ and time $-t$ to obtain reverse boundary preservation. The parameter condition for the reverse step is the original $s \in [0, 1]$. Replacing t by $r - s$ gives the claimed maps, and applying the inverse identity in each fiber proves they are mutually inverse. □

Lemma K.9 (Native immersion of the transported fixed carrier)**.** *Let D be a native compact domain with boundaryless extension M, $j : M \to N$ a smooth embedding, and Φ a jointly smooth all-real flow on $\mathbb{R} \times N$. Fix $t \in \mathbb{R}$. If $(\Phi_t(0, j(x)))_1 = t$ for every $x \in D$, the smooth map $E_t(x) = (\Phi_t(0, j(x)))_2$ is immersive at every point of D.*

Proof. In dimension zero the source tangent is zero. In positive dimension the native inclusion $i : D \to M$ has injective differential: an ambient extension of each native compact chart is a local left inverse. Equal source and target dimensions make di bijective. Differentiate the constant identity $(\Phi_t(0, j(i(x))))_1 = t$ on the native compact manifold. By the chain rule and surjectivity of di, the first component of the differential of $x \mapsto \Phi_t(0, j(x))$ is zero in every ambient tangent direction, including at boundary points. This total map is immersive, since j is immersive and Φ_t is a diffeomorphism with smooth inverse Φ_{-t}. Consequently equality of two spatial differential values gives equality of their total differential values, and hence equality of the source vectors. This proves immersion of E_t. □

Lemma K.10 (One smooth embedding neighborhood for the fixed carrier)**.** *With the preceding data, suppose the constant parameter identity holds for every $t \in [0, 1]$ and every $x \in D$. There is one open neighborhood $U \subset M$ of the entire carrier and a jointly smooth interval family of smooth embeddings $J_t : U \to N$ whose actual maps are $J_t(x) = (\Phi_t(0, j(x)))_2$.*

Proof. The flow formula is jointly smooth for all real times and all points of M. The preceding lemma gives immersion on the compact carrier for every interval time. On that carrier, equality of spatial images gives equality of total images because their time coordinates both equal t. Applying Φ_{-t} and injectivity of j gives equality of the original points. The uniform embedding-neighborhood theorem applied to this jointly smooth family, its compact carrier, and these injection and immersion statements supplies a common U and the asserted embedding family. Its extension retains the original flow formula. □

Lemma K.11 (One actual fixed carrier for the full tailoring ribbon family)**.** *Under the potential hypotheses of the compact tailoring transport lemma, there are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$*

and $A \geq F^2\beta^2/2$ there exist a nonempty native compact domain D of dimension $2n-2$, a smooth embedding j of its boundaryless extension onto the entire base fiber $\{f_{\beta,0}=1\}$, an all-real flow Φ, an open extension neighborhood U of the carrier, and a jointly smooth interval family of smooth embeddings $J_s : U \to$ LGSpace. They retain the actual formula

$$J_s(x) = (\Phi_s(0, j(x)))_2,$$

and satisfy

$$j(D) = \{f_{\beta,0} = 1,\ \phi \leq A\}, \qquad J_s(D) = \{f_{\beta,s} = 1,\ \phi \leq A\}, \qquad J_s(\partial D) = \{f_{\beta,s} = 1,\ \phi = A\}$$

for every $s \in [0,1]$. Here D and ∂D in the last two images mean their actual restrictions to the same open extension U.

Proof. Take the maximum of the end constants for compact transport, full-fiber markings, and nonempty truncations, and the maximum of their scale thresholds. Positivity makes the resulting level bound imply each original bound. At parameter zero the full-fiber marking producer constructs one full-fiber manifold and embedding j, with its actual complete primitive. Its proved outwardness constructs the native regular-sublevel domain $D = \{x : \phi(j(x)) \leq A\}$, whose frontier is exactly $\{x : \phi(j(x)) = A\}$. The literal image identity and the uniform nonempty truncation theorem make D nonempty.

Use the constructed compact transport flow. For every $x \in D$ and $s \in [0,1]$, its parameter equation is $(\Phi_s(0, j(x)))_1 = s$. The common embedding-neighborhood lemma now gives U and J_s with the displayed formula and genuine open smooth extensions across both parameter endpoints. Forward transport puts the whole image inside the stated truncated fiber. Conversely, for any y in that fiber, apply time $-s$ to (s, y). The result has parameter zero, satisfies the base fiber equation, and has potential at most A. The base image identity provides a point $x \in D$ with $(0, j(x)) = \Phi_{-s}(s, y)$. The flow inverse then gives $J_s(x) = y$. Boundary preservation in both directions gives the corresponding boundary image equality. Open restriction preserves the carrier and frontier, so these are exactly the asserted images in U. □

Lemma K.12 (The actual tangent kernel of an embedded compact carrier)**.** *Let $f : M^d \to F$ be smooth, with surjective differential on $f^{-1}(w)$, where F is a finite-dimensional real vector space. Let D be a native compact domain of dimension $d - \dim F$ and let j be a smooth embedding of its boundaryless extension into M. If $f(j(x)) = w$ for every point x of the actual compact carrier, then*

$$\operatorname{im}(dj_x) = \ker(df_{j(x)}) \qquad (x \in D).$$

Only the carrier equation is required; the open extension need not lie in the level set.

Proof. Write $i : D \hookrightarrow X$ for the native inclusion into the chosen boundaryless extension. In positive dimension its differential is injective, including at boundary points: in a native half-space chart, an ambient smooth chart extension composed with i has identity differential, so equality of two values of di gives equality of the vectors. Its source and target have equal dimension, hence di is surjective. The actual map $f \circ j \circ i$ is constant w. The chain rule and surjectivity of di therefore give $df \circ dj = 0$. In dimension zero the source tangent space is zero and this identity is immediate. Thus the image of dj is contained in the kernel of df. Since j is immersive, its tangent image has dimension $d - \dim F$. Surjectivity of df and rank-nullity give the same dimension for its kernel. The inclusion is consequently equality. □

Lemma K.13 (The actual fixed-carrier tailoring primitive family)**.** *Under the same potential hypotheses as the fixed-carrier embedding lemma, there are $F, \beta_0 > 0$ such that, for every $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$, one obtains a nonempty native compact domain D of dimension $2n-2$ and a jointly smooth interval family of smooth embeddings $J_s : X \to \mathrm{LGSpace}$ of its boundaryless extension, with the literal carrier and frontier images*

$$J_s(D) = \{f_{\beta,s} = 1,\ \phi \leq A\}, \qquad J_s(\partial D) = \{f_{\beta,s} = 1,\ \phi = A\}.$$

For every $s \in [0,1]$ and $x \in D$ their actual differentials satisfy

$$\mathrm{im}(dJ_s|_x) = \ker(df_{\beta,s}|_{J_s(x)}).$$

The one-forms $\mu_s = J_s^*\lambda_\phi$ form a jointly smooth interval family, with smooth extensions across the parameter endpoints. There is one open spatial neighborhood $O \supset D$ on which every $d\mu_s$ is symplectic. These conclusions concern the same transported ribbon carrier and do not assert an ambient rounding or completed pair family.

Proof. Retain the fixed-carrier embeddings, restricting the original compact domain to their common open embedding neighborhood. This preserves its nonempty carrier and frontier. Increase only the scale threshold to include the uniform positive native Poisson-pairing threshold. On the literal fiber $f_{\beta,s} = 1$, this positivity and nondegeneracy of $d\lambda_\phi$ imply surjectivity of the actual complex differential. The carrier image identity gives $f_{\beta,s} \circ J_s = 1$ on D. Apply the preceding tangent-kernel lemma with $F = \mathbb{C}$, regarded as a two-dimensional real vector space, to derive the displayed tangent image equality at every carrier point, including the boundary.

The constant family λ_ϕ is jointly smooth: in any spatial chart its smooth coefficients are simply composed with the spatial projection from the real-parameter product. Pullback by the jointly smooth actual maps J_s therefore gives the actual joint family μ_s. At a carrier point, identify its tangent image with the common kernel of the real and imaginary parts of $df_{\beta,s}$. The positive native Poisson pairing makes the restriction of $d\lambda_\phi$ to this kernel nondegenerate by the rank-two projection lemma. Exterior differentiation commutes with the actual pullback, hence $d\mu_s$ is nondegenerate there. The common symplectic-neighborhood lemma applied to this actual joint primitive and compact carrier gives one $O \supset D$ where all these differentials remain nondegenerate. Each is smooth and closed because it is the exterior derivative of a smooth primitive, hence symplectic. □

Lemma K.14 (Total-space trivialization from horizontal transport)**.** *Let X be a topological space, $S_t \subset X$, and let Φ be a continuous flow on $\mathbb{R} \times X$. Suppose that whenever $s, s+t \in [0,1]$ and $x \in S_s$, the point $\Phi_t(s,x)$ has first coordinate $s+t$ and second coordinate in S_{s+t}. Then the literal total subset $\{(s,x) : s \in [0,1], x \in S_s\}$ is homeomorphic over $[0,1]$ to $[0,1] \times S_0$.*

Proof. Map (s,x) to $(s, \mathrm{pr}_X\, \Phi_{-s}(s,x))$. The transport hypothesis places the second component in S_0. Conversely map (s,y) to $\Phi_s(0,y)$; the same hypothesis places this point in the literal total subset with parameter s. The flow law proves that the two maps are inverse, since the backward image has first coordinate zero. Joint continuity of the flow and the subspace topologies prove continuity in both directions. □

Lemma K.15 (Trivialization of the literal tailored ribbon total space)**.** *Under the potential hypotheses of Lemma K.8, there are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$, the literal total space*

$$C = \{(s,y) : s \in [0,1],\ f_{\beta,s}(y) = 1,\ \phi(y) \leq A\}$$

is homeomorphic over $[0,1]$ to $[0,1] \times \{y : f_{\beta,0}(y) = 1, \phi(y) \leq A\}$. If $e(s,y) = (s,z)$ is this homeomorphism, then $\phi(z) = A$ if and only if $\phi(y) = A$.

Proof. Take the jointly smooth flow furnished by Lemma K.8. Its parameter and truncated fiber equations satisfy the preceding lemma with $S_s = \{y : f_{\beta,s}(y) = 1, \phi(y) \le A\}$. The resulting homeomorphism has the claimed first coordinate. Its second coordinate is $\mathrm{pr}_X \Phi_{-s}(s,y)$, so the boundary equivalence is exactly the transport lemma's boundary equivalence at time $-s$. This time is allowed because its final parameter is zero. □

L Outwardness on the transported tailoring ribbon

Lemma L.1 (Sublevel derivative and the fixed boundary normal)**.** *Let D be a compact smooth domain in a boundaryless extension X. Let $\eta : X \to \mathbb{R}$ be smooth, with $\eta \le 0$ on D and $\eta = 0$ on its actual frontier. At a frontier point x, let $v \in T_xX$ satisfy $d\eta_x(v) > 0$. Then every regular local defining function ρ of the same compact domain, with sublevel convention $D = \{\rho \le 0\}$, satisfies $d\rho_x(v) > 0$.*

Proof. In dimension zero the frontier is empty. In positive dimension, let $i : D \hookrightarrow X$ be the native inclusion and choose its native half-space chart with first coordinate nonnegative. Its differential is injective and has equal-dimensional source and target, hence is bijective. Write $v = di_x(w)$. The native function $\eta \circ i$ is nonpositive and vanishes at x; its derivative is regular because its value on w is positive. In the half-space chart, this derivative is $c\,dq_0$ for a strictly negative constant c. Indeed, the local maximum on $q_0 \ge 0$ makes its derivative nonpositive on every vector with nonnegative first coordinate. Testing each vector with first coordinate zero and its negative shows that the derivative vanishes on the tangent hyperplane. Regularity makes its remaining coefficient nonzero, hence strictly negative. Thus $c\,w_0 = d\eta_x(v) > 0$ implies $w_0 < 0$. The native pullback of ρ is also nonpositive near x, vanishes there and is regular, since di_x is surjective. The same argument gives $d(\rho \circ i)_x = c'\,dq_0$ with $c' < 0$. Therefore $d\rho_x(v) = c'w_0 > 0$. □

Lemma L.2 (Outwardness of the actual fixed-carrier tailoring primitives)**.** *Under the potential hypotheses of the fixed-carrier primitive-family lemma, there are $F, \beta_0 > 0$ such that for every $\beta \ge \beta_0$ and $A \ge F^2\beta^2/2$ the same conclusion holds with, additionally, the intrinsic Liouville vector of every $\mu_s = J_s^*\lambda_\phi$ strictly outward along the actual frontier of that one fixed carrier D. Thus, for every stored regular local defining function ρ_x of D,*

$$d(\rho_x)_x(Z_{\mu_s}(x)) > 0 \qquad (s \in [0,1],\ x \in \partial D).$$

The nonempty carrier, literal carrier and frontier images, tangent-kernel identity, jointly smooth primitive family and common symplectic extension neighborhood are retained together with this outwardness conclusion.

Proof. Take the maximum of the primitive-family radius constant and the whole-end intrinsic-exhaustion radius constant. Increase the logarithmic threshold to dominate both thresholds and one. On the compact logarithmic ball $|u| \le L$ where the potential may differ from its homogeneous end, let V bound ϕ. Increase the threshold once more so that $F_{\mathrm{end}}^2\beta^2/2 > V$. Obtain D, J_s, O from the fixed-carrier primitive-family lemma; this choice is retained throughout. For $x \in \partial D$, the literal frontier image gives $f_{\beta,s}(J_s(x)) = 1$ and $\phi(J_s(x)) = A$. The inequality $A > V$ places its logarithmic component outside $|u| \le L$. Quadratic homogeneity there gives $r^2 = 2\phi(J_s(x)) = 2A$ for the Hessian radial square. Hence $r \ge F_{\mathrm{end}}\beta$. The actual tangent-kernel identity at x allows the whole-end intrinsic exhaustion estimate to be applied to the very embedding J_s, giving

$$d(\phi \circ J_s)_x(Z_{\mu_s}(x)) \ge r^2/2 = A > 0.$$

The smooth function $\eta_s = \phi \circ J_s - A$ is nonpositive on D by the literal carrier image, and is zero on the actual frontier by the literal frontier image. Its derivative is the displayed positive derivative. Apply the preceding lemma to this η_s and the stored defining function of D. This proves outwardness for every parameter on the same fixed carrier. In dimension zero the frontier is empty and the outward assertion is vacuous. □

Lemma L.3 (The ambient plane update on a fiber tangent kernel)**.** *Let $j : N \hookrightarrow M$ be a smooth embedding, α a one-form, and $f : M \to \mathbb{C}$. If $dj_x(T_xN) \subset \ker df_{j(x)}$, then for every real K,*

$$\big(j^*(\alpha + Kf^*\lambda_{\mathbb{C}})\big)_x = (j^*\alpha)_x.$$

This is the tangent-kernel version of the constant-fiber primitive identity, and does not require holomorphicity or positivity of K.

Proof. For every $v \in T_xN$, the kernel hypothesis gives $df_{j(x)}(dj_x(v)) = 0$. By the definition of pullback, the plane contribution evaluated on v is $\lambda_{\mathbb{C},f(j(x))}(df_{j(x)}(dj_x(v))) = 0$. Evaluating the displayed identity on every v proves equality of its one-form fibers. □

Lemma L.4 (The fixed tailoring ribbon Liouville and completion family)**.** *Under the same potential hypotheses, there are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$, one fixed nonempty native compact ribbon D carries a jointly smooth family of embeddings J_s with the same literal full carrier and frontier images as above. It has an actual Liouville structure L_s with primitive exactly $\mu_s = J_s^*\lambda_\phi$, and both the primitives and their canonical Liouville fields form smooth interval families. Each $d\mu_s$ is symplectic on the entire common restricted extension. These structures construct a compatible localized completion family, with jointly smooth compact and completed primitive and field families. This conclusion is a ribbon completion family; it does not assert a marked ambient pair. For every $s \in [0,1]$, every real K, and every $x \in D$, its actual primitive also satisfies the full ambient restriction identity*

$$\big(J_s^*(\lambda_\phi + Kf_{\beta,s}^*\lambda_{\mathbb{C}})\big)_x = (\mu_s)_x.$$

Proof. Obtain the domain D_0, embeddings and common symplectic neighborhood O from the outward primitive-family lemma. Restrict its extension to O. The restricted carrier is the inverse image of the entire old carrier, whose inclusion maps onto that carrier since $D_0 \subset O$. The frontier is its corresponding inverse image; it too is mapped onto the original frontier because the compact domain is closed. Thus restricting the actual embeddings preserves both literal image identities and nonemptiness. The open inclusion has identity tangent map, so it also preserves the actual tangent kernel, the primitive, and its exterior derivative. Restriction preserves the real parameter-endpoint extensions of the embedding and primitive families.

On the restricted extension every $d\mu_s$ is nondegenerate. Its unique contraction solution is therefore a smooth vector field on the entire extension, and joint primitive smoothness proves joint smoothness of these canonical fields. The identity tangent map identifies this solution with the previous canonical field wherever it is restricted. Each stored local defining function is the restriction of its original one; the chain rule therefore preserves its positive derivative on this field. Consequently the primitive, its symplectic differential, canonical contraction field and proved outwardness form the actual Liouville structure L_s.

Apply the fixed-domain localized completion-family lemma to these L_s and their proved jointly smooth primitives. That construction supplies the common positive collar width, actual flow collars, compatible compact presentations and their cylindrical completions. Its regularity conclusions give the stated compact and completed primitive and field families. Finally the retained actual tangent-kernel equality and the preceding plane-update lemma prove the ambient restriction identity at every point of the full compact carrier, including its boundary, for every real K. □

Lemma L.5 (Horizontal field for affine compact scalar sublevels)**.** *Let M be a finite-dimensional Hausdorff second-countable smooth manifold without boundary, and let $\eta : M \to \mathbb{R}$ be smooth with every sublevel compact. Fix any real A, B, and assume $d\eta_x \neq 0$ whenever $\min(A,B) \leq \eta(x) \leq \max(A,B)$. Set*

$$L(s) = (1-s)A + sB, \qquad \rho(s,x) = \eta(x) - L(s), \qquad C = \{(s,x) : s \in [0,1],\ \rho(s,x) \leq 0\}.$$

There are open neighborhoods O of C and T of its marked level $\{(s,x) : s \in [0,1], \rho(s,x) = 0\}$, and a globally smooth tangent field V on $\mathbb{R} \times M$, such that $ds(V) = 1$ on O and $d\rho(V) = 0$ on $O \cap T$. This includes dimension zero and empty marked levels.

Proof. The affine level lies in $[\min(A,B), \max(A,B)]$ for $s \in [0,1]$. Thus C is a closed subset of the compact set $[0,1] \times \{\eta \leq \max(A,B)\}$, and is compact. The marked locus $\{\rho = 0\}$ is closed. Near any point of C, the parameter normal-form construction with a zero-dimensional, constant equation gives a smooth field satisfying $ds(V) = 1$. At a point of $C \cap \{\rho = 0\}$, the spatial derivative of ρ is $d\eta$, which is nonzero by hypothesis. Its scalar target makes it surjective. The parameter normal form for ρ therefore gives a smooth field with $d(s,\rho)(V) = (1,0)$ on an open neighborhood. The simultaneous boundary gluing theorem combines these local fields. It retains the parameter equation near all of C and the scalar equation on an open neighborhood of the marked boundary. No field or equation is required on an empty marked locus. □

Lemma L.6 (Complete motion of affine compact scalar sublevels)**.** *Under the preceding hypotheses, there is an all-real flow Φ on $\mathbb{R} \times M$ with jointly smooth action. For any $s \in [0,1]$, $\eta(x) \leq L(s)$ and any real t with $s + t \in [0,1]$,*

$$(\Phi_t(s,x))_1 = s + t, \qquad \eta((\Phi_t(s,x))_2) \leq L(s+t).$$

If $\eta(x) = L(s)$, the latter inequality is equality. Consequently, for every $s, u \in [0,1]$, the spatial motion $g_{s,u}(x) = (\Phi_{u-s}(s,x))_2$ has literal images

$$g_{s,u}(\{\eta \leq L(s)\}) = \{\eta \leq L(u)\}, \qquad g_{s,u}(\{\eta = L(s)\}) = \{\eta = L(u)\}.$$

Proof. Multiply the preceding field by a compactly supported smooth cutoff equal to one on an open neighborhood of C contained in O. The supported integration theorem constructs its all-real jointly smooth flow. The parameter derivative remains one near C, and the scalar derivative remains zero on an open boundary collar. Apply the full-interval confinement theorem with the constant zero equation and truncation function ρ. It gives the stated parameter and sublevel identities, and equality on the marked boundary, in either time direction. The forward motion gives containment of each asserted image. Starting at (u,y) in the target sublevel, apply time $s-u$ to obtain a point in the source sublevel with first coordinate s. The flow group law shows that time $u-s$ returns this point to (u,y). This proves reverse containment. The same argument with equality in place of inequality proves the boundary-level image identity. □

Lemma L.7 (Whole original ribbon-level transport)**.** *Let I be a finite toric input, let A_{amb} be any valid original ambient end bound, and choose $R > \max(1, M(A_{\mathrm{amb}}))$. Let A_{fib} be any valid fiber end bound relative to these choices. For every $A, B > A_{\mathrm{fib}}$, take the constructed complete full-fiber model $j : F \hookrightarrow Y$ with image $W^{-1}(R)$ and $\eta = \phi_0 \circ j$. There is a jointly smooth all-real flow Φ on $\mathbb{R} \times F$ with the parameter, sublevel and boundary laws of the preceding lemma. At every $s, u \in [0,1]$ it satisfies*

$$(j \circ g_{s,u})(\{\eta \leq L(s)\}) = W^{-1}(R) \cap \{\phi_0 \leq L(u)\}, \qquad g_{s,u}(\{\eta = L(s)\}) = \{\eta = L(u)\}.$$

The full ribbon is retained, including all its components and the rank-one case. These motions supply the native sublevel transport used in the varying-ribbon-level stage of the comparison of initial choices.

Proof. The exhaustion η is smooth and all its sublevels are compact on the actual constructed full fiber. On every point with $\eta \geq A_{\mathrm{fib}}$, its derivative along the intrinsic Liouville field is strictly positive. Since $A_{\mathrm{fib}} < \min(A, B)$, this forces $d\eta \neq 0$ on the entire level band. Apply the preceding transport construction. Every affine level $L(u)$ exceeds A_{fib}, and the embedded compact sublevel at that level is literally $W^{-1}(R) \cap \{\phi_0 \leq L(u)\}$. Composing its native image equality with j proves the displayed literal ribbon image equality. The construction used no positive-dimensional tangent or nonempty boundary assumption, so it also applies in rank one. □

Lemma L.8 (Compact families with marked boundaries)**.** *Let a compact family of domains, together with compact embedded ribbons in their boundaries, be given over an interval by smooth level equations. Suppose the domains have smooth boundary, each ribbon is a smooth submanifold with boundary, and the equations admit local parameter lifts tangent to the ambient boundary, the ribbon and its boundary wherever those conditions apply. Then the family admits a smooth parameter lift preserving all these subsets. Its flow trivializes both bundles over the whole interval. Their actual fibers are the endpoint pairs.*

Proof. Write a lift as $\partial_t + V_t$. The tangency conditions are affine linear equations in V_t: for a moving defining function ρ_t, they read $d\rho_t(V_t) = -\partial_t \rho_t$ on its zero set. Local solutions, on charts adapted to the nested marked submanifolds, glue by a partition of unity, since affine combinations with coefficients summing to one preserve these equations. Compactness of the total family and tangency to its boundary give the flow for the entire parameter interval. The inverse is the reverse flow. □

Here the needed local lifts are explicit linear systems. Near a ribbon the equations are $f_t = R$ (or $f_s = 1$), two real equations; at its boundary include $\phi_t = A_-(t)$. The regularity of f_t and the strict intrinsic exhaustion derivative make these equations independent. The ambient boundary equation there is the unchanged base face and is already implied by the two fiber equations, so it adds no independent constraint. Away from the ribbon, the rounded ambient boundary has a nonzero defining covector. Before rounding, the two corner equations are independent by the same intrinsic derivative. These checks provide the local lifts required by Lemma L.8, including the ribbon truncation constraint. Uniform rounding in a corner collar disjoint from the ribbon preserves the checks. Thus the coefficient, potential and tailoring paths give smooth compact pair families, with the actual endpoint domains and ribbons as their endpoint tuples.

Lemma L.9 (Common logarithmic fiber levels for the quadratic coefficient path)**.** *With the quadratic potential there are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$, every $t \in [0, 1]$, and every point with $r = |u| \geq F\beta$ and $|W_t| \leq e^{\beta}$, one has*

$$a_t > 0, \qquad r^2 - |b_t|^2/a_t \geq r^2/2.$$

Given any real A_{amb}, enlarging β_0 makes $F^2\beta^2/2$ a valid fiber end bound relative to that level for every member of the coefficient path.

Proof. Choose the uniform constants $c, C, L, A, d, H > 0$ so that $g_t = \sqrt{a_t} \geq cM_t$ beyond L, $M_t \geq Ae^{dr}$, $h \leq Hr$, and $|b_t| \leq h|W_t| + CM_t$. Take $F = 2/d$. Choose $\beta_0 > 0$ so that $F\beta_0 \geq L$,

$cF\beta_0 \ge 4C$, and $e^{\beta_0} \ge 4H/(cA)$. If $r \ge F\beta$ and $\beta \ge \beta_0$, then $dr \ge 2\beta$, so $4He^{\beta} \le cAe^{dr} \le cM_t$. Also $4CM_t \le crM_t \le rg_t$. Thus

$$|b_t| \le Hre^{\beta} + CM_t \le rg_t/2, \qquad r^2 - |b_t|^2/a_t \ge 3r^2/4 \ge r^2/2.$$

Positivity of g_t proves $a_t > 0$. Increasing β_0 so that $F\beta_0 \ge \sqrt{2\max(A_{\mathrm{amb}}, 0)}$ proves the relative-level condition. The estimate is strictly positive on this end because $F\beta > 0$. For rank one the actual defect is zero, so the displayed bounded-value end is empty, as required. □

Lemma L.10 (Matching logarithmic truncations)**.** *There are fixed $C_+ > C_- > C_{\mathrm{fib}} > 0$ such that, for all sufficiently large β, the coefficient and potential paths have valid fiber bound $C_{\mathrm{fib}}^2\beta^2/2$ and strictly larger admissible ribbon and ambient levels $C_-^2\beta^2/2$ and $C_+^2\beta^2/2$.*

Proof. The radii in this construction can be chosen proportional to $\log R$, uniformly along the fixed paths. For the unnormalized polynomial, still write $h(u) = \max_v \langle v, u\rangle$. The coefficient bounds give

$$|b - hf| \le CM, \qquad \frac{|b|}{r\sqrt{a}} \le C(R/M + r^{-1}) \quad \text{on } |f| \le R.$$

The first estimate follows by summing $(h - \langle v, u\rangle)e^{\langle v,u\rangle - h} \le 1/e$, with the bounded coefficients. Write $\phi_{t,\infty} = (1-t)\phi_0 + t\phi_\infty$ for the homogeneous end of the fixed potential path, and set

$$c_* = \min_{\substack{t\in[0,1]\\ 2\phi_{t,\infty}(\xi)=1}} h(\xi) > 0.$$

The constraint set is compact and avoids zero, and h is positive away from zero. Choose C_{fib} with $c_*C_{\mathrm{fib}} > 1$. The coefficient path has a common positive lower bound $c_{\min}$, so on its homogeneous end

$$M \ge c_{\min}e^{h(u)} \ge c_{\min}e^{c_*r}, \qquad \frac{R}{M} \le c_{\min}^{-1}e^{-(c_*C_{\mathrm{fib}}-1)\beta} \quad (r \ge C_{\mathrm{fib}}\beta,\ R = e^{\beta}).$$

Together with $r^{-1} \to 0$, this gives the required intrinsic exhaustion derivative at least $r^2/2$ uniformly along the path. Choose fixed $C_+ > C_- > C_{\mathrm{fib}}$, enlarging them as needed below. Increase the large-β threshold so that $A_{\mathrm{fib}} = C_{\mathrm{fib}}^2\beta^2/2 \ge A_{\mathrm{amb}}$ and the homogeneous-end estimates apply. This is a valid original fiber end bound, strictly below $A_- = C_-^2\beta^2/2$. We take the initial inner and outer levels to be $r = C_-\beta$ and $r = C_+\beta$, uniformly for every path member. These are exactly the levels in Section M. This explicitly identifies the endpoint marked truncation with the starting truncation of the tailoring family; no comparison of two unspecified "sufficiently large" domains is being assumed.

□

M Finite tailoring with separate control of the full fiber ends

We now have coefficients one and the potential of Lemma D.33. Let $R = e^{\beta}$, $N_0 = |V|$, and

$$m_v = e^{-\beta}z^v, \quad f_0 = \sum_v m_v = W/R, \quad M = e^{h(u)-\beta}, \quad \varepsilon_\beta = e^{-\sqrt{\beta}}.$$

Definition M.1 (The finite all-pairs cutoffs)**.** *Put $E(s) = e^{-1/s}$ for $s > 0$ and $E(s) = 0$ for $s \le 0$, $S(s) = E(s)/(E(s) + E(1-s))$, and $\chi(s) = S((s+2)/2)$. Thus $\chi : \mathbb{R} \to [0,1]$ is smooth and nondecreasing, zero on $(-\infty, -2]$, one on $[0,\infty)$, and strictly between zero and one on $(-2, 0)$. For*

$\mathcal{A} = \{0\} \cup V$, *put* $\nu(0) = 0$, $\nu(v) = 1$. *For arbitrary real* β, *the symbol* $\sqrt{\beta}$ *in these formulas denotes* $\sqrt{\max(\beta, 0)}$; *for* $\beta \geq 0$ *this is the usual square root. Set*

$$\ell_a(u) = \langle a, u\rangle - \beta\nu(a), \qquad \chi_a(u) = \prod_{b \in \mathcal{A}\setminus\{a\}} \chi\big(\ell_a(u) - \ell_b(u) + \sqrt{\beta}\big). \tag{17}$$

Define

$$\mathcal{F}_\beta = -\chi_0 + \sum_v \chi_v m_v, \qquad H_\beta = \mathcal{F}_\beta^{-1}(0).$$

Lemma M.2 (Elementary cutoff estimates)**.** *Every maximal term has cutoff one. Retention* $\chi_a(u) > 0$ *implies* $\ell_a(u) > \max_b \ell_b(u) - \sqrt{\beta} - 2$. *If* $\chi_a(u) \neq 1$, *or any positive-order derivative of* χ_a *at* u *is nonzero, its uncut magnitude satisfies*

$$e^{\ell_a(u)} \leq e^{-\sqrt{\beta}} \max(1, e^{h(u)-\beta}).$$

Proof. All product factors lie in $[0, 1]$. If a is maximal, all their arguments are nonnegative, so every factor and the product equal one. If the product is positive, each factor is positive; hence its argument is greater than -2. This gives the retention inequality for $b \neq a$; for $b = a$ it holds because $\sqrt{\beta} + 2 > 0$. Taking a maximizing b proves the claimed deficit bound. Put $L = \max_b \ell_b(u) = \max(0, h(u) - \beta)$. If $\ell_a(u) \geq L - \sqrt{\beta}$, every factor argument is nonnegative. Every positive-order derivative of χ vanishes there: it vanishes on the open one tail and at its endpoint by continuity. The higher chain rule for an affine argument and the product rule therefore show that every positive-order derivative of the product vanishes at u. The product itself is one. Thus either of the hypotheses in the last claim forces $\ell_a(u) < L - \sqrt{\beta}$. Exponentiating and using $e^L = \max(1, e^{h(u)-\beta})$ proves the estimate. □

Lemma M.3 (Uniform bounds for all cutoff derivatives)**.** *Here derivative operators are measured in the coordinate maximum norm on* $M_{\mathbb{R}}$ *and its induced multilinear operator norm. For every nonnegative integer* m *there is* $C_m \geq 0$, *independent of* u *and* β, *such that* $\|D^m \chi_a(u)\| \leq C_m$ *for every* $a \in \mathcal{A}$.

Proof. The zeroth derivative of χ is bounded by one. For $m > 0$, its mth derivative vanishes outside $[-2, 0]$ and is continuous, so its absolute value has a finite bound B_m on that compact interval. The affine argument in the factor indexed by b has differential $L_{ab}(w) = \langle a - b, w\rangle$. Its mth factor derivative is bounded by $B_m \|L_{ab}\|^m$, independently of its translation and therefore of u, β. The product rule for m derivatives distributes the ordered m differentiation directions among the finitely many factors. Each of its finitely many terms is bounded by the product of these factor bounds, and the triangle inequality bounds the sum by a finite constant. Finally take the maximum over the finite set of a to obtain C_m. □

Lemma M.4 (The retained exponents belong to one simplex)**.** *For all sufficiently large* β, *at every* u *there is a facet with vertex set* T *such that the retained exponents are contained in* $\{0\} \cup T$, *an affinely independent simplex of* $0 * \partial Q$. *Their nonzero vertices are linearly independent over* $\mathbb{R}$.

Proof. Let $\eta > 0$ be the near-maximal-face constant and $C > 0$ the support Lipschitz bound in the coordinate maximum norm. Put $r = \sqrt{\beta} + 2$. Put $T = 4 + 4C/\eta$ and $\beta_0 = T^2$. For $\beta \geq \beta_0$, write $s = \sqrt{\beta} \geq T \geq 4$. Then $r = s + 2 \leq 3s/2$ and $\beta - r = s^2 - s - 2 \geq s^2/2 > 0$. Moreover $\eta(\beta - r) \geq \eta s^2/2 \geq 2Cs \geq Cr$. Let B be the retained nonzero vertices at u. For $v \in B$, the retention estimate gives

$$\langle v, u\rangle > \beta + \max(0, h(u) - \beta) - r \geq \beta - r > 0, \qquad h(u) - \langle v, u\rangle < r.$$

If B is nonempty, the first inequality and $\langle v, u\rangle \le h(u) \le C\|u\|_\infty$ show that $u \neq 0$ and $C\|u\|_\infty > \beta - r$. Thus $r \le \eta\|u\|_\infty$ and every member of B satisfies the near-maximal-face estimate. That lemma puts B in one proper face. Extend this face to a facet. The input's facet-basis condition makes its vertices linearly independent, so their union with zero is an affinely independent simplex of $0 * \partial Q$ containing all retained exponents, whether or not zero is retained. If B is empty, the retained set is contained in $\{0\}$, which lies in the cone from zero over any facet. Such a facet exists because the input polytope is full-dimensional and has positive dimension. In both cases the nonzero retained vertices are a subset of a facet basis and are linearly independent. □

Lemma M.5 (Cutoff estimates and simplicial support)**.** *Every maximal term has cutoff one. Retention implies $\ell_a > \max_b \ell_b - \sqrt{\beta} - 2$. Changed terms and terms with nonzero positive-order cutoff derivatives have uncut magnitude at most $\varepsilon_\beta \max(1, M)$. Every fixed derivative order is uniformly bounded in the coordinate maximum multilinear operator norm. For sufficiently large β, all retained exponents lie in one simplex of $0 * \partial Q$ and their nonzero vertices are linearly independent.*

Proof. Combine the elementary estimates, uniform derivative bounds and retained-simplicial-support lemma. Their constants depend only on the fixed finite input and profile, as asserted. □

Definition M.6 (The literal tailoring family)**.** *Put $c = 1/(8(N_0 + 1))$, $S = (\sum_v |m_v|^2)^{1/2}$, and $\eta(S) = \chi(2(S - c)/c - 2)$. Thus η is zero for $S \le c$ and one for $S \ge 2c$. Set*

$$f_s = f_0 + s\eta(S)\left[(1 - \chi_0) + \sum_v (\chi_v - 1)m_v\right]. \tag{18}$$

The normalized polynomial is $f_0 = \sum_v m_v$, and the tailored hypersurface is $H_\beta = \mathcal{F}_\beta^{-1}(0)$ with $\mathcal{F}_\beta = -\chi_0 + \sum_v \chi_v m_v$ as above.

Lemma M.7 (Smoothness and the entire endpoint fiber)**.** *For every real β, the family f_s is jointly smooth on $\mathbb{R} \times Y$, starts at the normalized polynomial, and has the literal full endpoint fiber equality $f_1^{-1}(1) = H_\beta$. Whenever $\eta(S) < 1$, one has $\chi_0 = 1$ and $|f_s| < 1$ for $0 \le s \le 1$.*

Proof. Every monomial is smooth on the quotient. Its squared magnitude is $e^{2\ell_v(u)}$, so S^2 is a smooth positive finite sum: the vertex set is nonempty and every summand is positive. The square root is smooth on the positive real axis, proving smoothness of S and then of $\eta(S)$. The finite formula for f_s proves joint smoothness for all real s, and substitution of zero gives f_0. If $\eta(S) < 1$, then $S < 2c$ and each $|m_v| \le S < 2c < 1$. Thus $\ell_v < 0 = \ell_0$, making zero a maximal term. Its cutoff is one by the elementary cutoff lemma; this holds for every real β. For $0 \le s \le 1$ write

$$f_s = \sum_v w_v m_v, \qquad w_v = 1 - s\eta(S) + s\eta(S)\chi_v \in [0, 1].$$

Consequently $|f_s| \le \sum_v |m_v| \le N_0 S < 2N_0 c < 1$. The same bound applies to $\sum_v \chi_v m_v$, so $\mathcal{F}_\beta = -1 + \sum_v \chi_v m_v \neq 0$ there. On the complementary region $\eta(S) = 1$, direct substitution gives $f_1 - 1 = \mathcal{F}_\beta$. On the first region neither of the two level equations has a solution; on the second they are the same equation. This proves equality of the entire fibers. □

Lemma M.8 (First derivative closeness of the actual tailoring maps)**.** *There is $C > 0$, depending only on the fixed input and profiles, such that for every real β, uniformly in $s \in [0, 1]$ and $y = (u, \theta)$,*

$$|f_s - f_0| + \|d(f_s - f_0)\| \le C\varepsilon_\beta M. \tag{19}$$

The differential norm is the operator norm for the coordinate maximum norm on the real-angle cover. For $\beta < 0$ the cutoff width uses $\sqrt{\max(\beta, 0)}$, so $\varepsilon_\beta = 1$ there.

Proof. Write $\Delta = (1-\chi_0) + \sum_v (\chi_v - 1) m_v$. The elementary exponential suppression applies wherever a cutoff is changed or its first differential is nonzero. Thus each uncut term appearing in Δ or in a differentiated cutoff has magnitude at most $\varepsilon_\beta \max(1, M)$. The monomial differential has norm at most a fixed constant times its magnitude, since its exponent is a fixed lattice vector. The cutoff derivatives have uniform bounds in the coordinate maximum norm. The product rule and finite sum therefore give $|\Delta| + \|d\Delta\| \le C_0 \varepsilon_\beta \max(1, M)$. Let $B = 1 + \sum_v \|\langle v, \cdot \rangle\|$, using the covector operator norm for the coordinate maximum norm. Since $|m_v|$ depends only on u, differentiation of the positive square gives

$$dS = \frac{1}{S} \sum_v |m_v|^2 \langle v, du \rangle, \qquad \|dS\| \le BS.$$

The scalar derivative η' is bounded and vanishes outside $[c, 2c]$. Hence $\|d(\eta(S))\|$ is uniformly bounded, using $S \le 2c$ wherever that derivative is nonzero. Where η or its differential is nonzero, $S \ge c$ and $S \le N_0 M$, so $M \ge c/N_0$. It follows that $\max(1, M) \le C_2 M$ there. The product rule applied to $f_s - f_0 = s\eta(S)\Delta$, with $0 \le s \le 1$, gives (19). Elsewhere the function and its differential are zero, so the bound holds everywhere. □

Remark M.9 (Range and constants in the closeness estimate). The source uses the estimate only for sufficiently large positive β. The same product estimates hold with the total real square root convention for every real β. The bound $S \le N_0 M$ replaces $S \le \sqrt{N_0} M$ in the support argument; only the unspecified constant C increases. The manuscript states the differential estimate in the Euclidean norm. Equivalence of these fixed finite-dimensional norms transfers the bound after increasing C; the attached statement records the coordinate norm used in the derivative estimates.

Lemma M.10 (The quotient differential in logarithmic coordinates)**.** *For the actual logarithmic cover p and $e = f_s - f_0$, $D(e \circ p)_x = (de)_{p(x)} \circ Dp_x$. Consequently the same C^1 estimate holds for the native quotient differential composed with Dp_x, with its coordinate maximum operator norm. The corresponding C^0 bound holds at every point of the quotient.*

Proof. Both family maps are smooth on the quotient. Apply the manifold chain rule to their difference and the actual smooth cover. On its vector-space domain the manifold derivative is the Fréchet derivative, so the identity is literal. Substitution in the preceding estimate proves the differential bound. For the zeroth-order conclusion choose a preimage under the surjective cover and discard the nonnegative derivative norm. □

Lemma M.11 (Tailoring support and marked fibers in the homogeneous end)**.** *There are $c_1, \beta_0 > 0$ such that for $\beta \ge \beta_0$ the modification vanishes on $|u| \le c_1 \beta$, for every real s. For $s \in [0, 1]$, on $f_s = 1$ one has $M \ge 1/(N_0 + C\varepsilon_\beta)$, where C is the closeness constant. For sufficiently large β this entire marked fiber lies in the homogeneous end of the chosen potential ϕ.*

Proof. Choose $H > 0$ with $h(u) \le H|u|$ and put $c_1 = 1/(2H)$. If $|u| \le c_1 \beta$, then $M \le e^{-\beta/2}$ and $S \le N_0 e^{-\beta/2}$. For sufficiently large β this is at most the positive small-monomial threshold c. Hence $\eta = 0$ and the literal family formula gives $f_s = f_0$ for every real s. Finally on $f_s = 1$ the estimate and $|f_0| \le N_0 M$ give $1 \le (N_0 + C\varepsilon_\beta) M$. Since $M = e^{h(u) - \beta}$, this implies $h(u) \ge \beta - \log(N_0 + C\varepsilon_\beta)$. The bound $h(u) \le H|u|$ and $\varepsilon_\beta \le 1$ give $|u| \ge (\beta - \log(N_0 + C))/H$. Increasing the threshold ensures $|u| \ge \beta/(2H)$ on the entire level-one fiber. Given any fixed radius R, increase the threshold again so that $\beta/(2H) \ge R$. Apply this to the Euclidean radius outside which the selected extension agrees with its homogeneous end. This proves the last assertion without assuming any fiber marking or completeness. □

M.1 Rank-two identities for a nonholomorphic map

Definition M.12 (The actual rank-two primitive)**.** *Let α be a smooth one-form on a finite-dimensional manifold and $f = A + iB$ a smooth complex-valued map, smooth as a real map. Set $\alpha_K = \alpha + Kf^*\lambda_{\mathbb{C}}$ and $\omega_0 = d\alpha$. At a point where ω_0 is nondegenerate, let Z_0, X_A, X_B be its unique contraction solutions $\iota_{Z_0}\omega_0 = \alpha$, $\iota_{X_A}\omega_0 = dA$ and $\iota_{X_B}\omega_0 = dB$. Put $\pi_f = dA(X_B)$, $a_f = dA(Z_0)$, and $b_f = dB(Z_0)$.*

Lemma M.13 (Exact rank-two identities on the native tangent spaces)**.** *At a point where ω_0 is nondegenerate, the actual differential $d\alpha_K = \omega_0 + K\,dA \wedge dB$ is nondegenerate exactly when $D = 1 + K\pi_f \neq 0$. Under this condition its actual Liouville vector and differentials satisfy*

$$Z_K = Z_0 + \frac{K}{D}\big[(b_f - B/2)X_A + (A/2 - a_f)X_B\big], \tag{20}$$

$$df(Z_K) = \frac{Z_0 f + (K\pi_f/2)f}{D}, \tag{21}$$

$$d\psi(Z_K) = \frac{C_0' + KB_f}{D}, \qquad C_0' = d\psi(Z_0), \tag{22}$$

$$B_f = C_0'\pi_f + (b_f - B/2)d\psi(X_A) + (A/2 - a_f)d\psi(X_B).$$

The last identity holds for any differentiable real function ψ. If ω_0 is nondegenerate and $\pi_f \geq 0$ throughout a subset, then $d\alpha_K$ is symplectic there for every $K \geq 0$.

Proof. The pulled-back plane primitive is $(A\,dB - B\,dA)/2$. The product rule and $d^2A = d^2B = 0$ give its differential $dA \wedge dB$. Alternation gives $dA(X_A) = dB(X_B) = 0$ and $dB(X_A) = -\pi_f$. Consequently contraction with the updated form gives $\iota_{X_A}d\alpha_K = D\,dA$ and $\iota_{X_B}d\alpha_K = D\,dB$. If a vector lies in its kernel, evaluation against X_A, X_B gives $D\,dA(v) = D\,dB(v) = 0$. When $D \neq 0$, these pairings vanish and the original nondegeneracy forces $v = 0$. When $D = 0$, $\pi_f \neq 0$, so $X_A \neq 0$ and the displayed contraction puts it in the kernel. This proves the criterion. Furthermore $\iota_{Z_0}d\alpha_K = \alpha + K(a_f\,dB - b_f\,dA)$. Adding the two corrections in (20) cancels these extra terms and gives $\alpha + K(A\,dB - B\,dA)/2 = \alpha_K$. Nondegeneracy identifies this vector with the actual Liouville vector. Applying dA, dB to it gives, respectively, $(a_f + K\pi_f A/2)/D$ and $(b_f + K\pi_f B/2)/D$, proving the complex identity. Applying $d\psi$ and collecting the numerator gives (22). If $K, \pi_f \geq 0$, then $D \geq 1$, so the criterion proves nondegeneracy. The form is closed because it is an exterior differential of a smooth primitive. This proves symplecticity. □

Lemma M.14 (Tailoring pairing estimate in logarithmic cover coordinates)**.** *For a complex real-linear covector L on the logarithmic cover, represent its real and imaginary parts by radial and angular Euclidean coefficient vectors $A_u, A_\theta, B_u, B_\theta$, and put*

$$P_H(L) = \langle H^{-1}B_\theta, A_u\rangle - \langle H^{-1}B_u, A_\theta\rangle.$$

Given $c > 0$, there is $C > 0$ such that, for every real β, $s \in [0,1]$, cover point x, and invertible operator H satisfying $c|v|^2 \leq \langle Hv, v\rangle$, the actual cover differentials obey

$$|P_H(df_s) - P_H(df_0)| \leq Ce^{-\sqrt{\beta}}M^2, \qquad M = e^{h(u)-\beta}.$$

Here the square root is zero for negative β, and the covector operator norms in the proof use the coordinate maximum norm of the cover.

Proof. Radial and angular restriction followed by the Euclidean Riesz inverse are fixed bounded linear operators on the cover covectors. Their norms include the fixed norm-conversion constants. Cauchy's inequality bounds the bilinear expression $P_H(A,B)$ by $C_0\|H^{-1}\|\|A\|\|B\|$. Subtracting its two products therefore gives, for complex covectors L, L_0,

$$|P_H(L) - P_H(L_0)| \le C_1\|H^{-1}\|\|L - L_0\|(\|L - L_0\| + 2\|L_0\|).$$

The actual finite monomial differential gives $\|df_0\| \le BM$, with B the sum of the norms of the exponent covectors. The proved first-derivative estimate gives $\|df_s - df_0\| \le A\epsilon M$, where $\epsilon = e^{-\sqrt{\beta}} \le 1$. The inverse operator bound gives $\|H^{-1}\| \le c^{-1}$. Substitution bounds the difference by $C_1c^{-1}A(A+2B)\epsilon M^2$. Increasing that nonnegative constant by one makes it strictly positive and proves the assertion. □

Lemma M.15 (Native Poisson error for the actual tailoring family)**.** *For the same smooth potential with global positive Hessian lower bound, the native potential form is nondegenerate everywhere and there is $C > 0$ such that for every real β, $s \in [0,1]$, and $y \in Y$,*

$$|\pi_{f_s}(y) - \pi_{f_0}(y)| \le Ce^{-\sqrt{\beta}}M(y)^2.$$

The constant is independent of β, s, y and of the later base coefficient.

Proof. The lower bound constructs the actual inverse Hessian, whose cover two-form is nondegenerate. The bijective cover tangent map transfers nondegeneracy to every quotient tangent space. The chain rule identifies each native differential pulled back by this map with its actual cover Fréchet differential. The native Hamiltonian coordinate formula then identifies the native pairing with P_H for that actual Hessian. Apply the cover pairing estimate and use surjectivity of the logarithmic cover to obtain the assertion at every y. □

N Quantitative control of the tailoring exhaustion numerator

Definition N.1 (The homogeneous exhaustion numerator)**.** *Let H be the invertible Hessian of a potential, put $R(u) = \langle Hu, u\rangle$, and let $P_H(L)$ be the inverse-Hessian Poisson pairing of the real and imaginary parts of a complex covector L. Define*

$$B_H(u,f,L) = R(u)P_H(L) - \operatorname{Im}\big(\overline{L(u,0) - f/2}\, L(0,u)\big).$$

The native numerator is

$$B_f = C_0\pi_f + (Z_0 \operatorname{Im} f - \operatorname{Im} f/2)d\phi(X_A) + (\operatorname{Re} f/2 - Z_0 \operatorname{Re} f)d\phi(X_B), \qquad C_0 = d\phi(Z_0),$$

where $\iota_{X_A}\omega_0 = d\operatorname{Re} f$ and $\iota_{X_B}\omega_0 = d\operatorname{Im} f$.

Lemma N.2 (The cover exhaustion error)**.** *Fix the toric input and $c > 0$. There is $C > 0$ such that for every real β, every $s \in [0,1]$, every u with $|u| \ge 1$, every angle, and every invertible H with $\langle Hv, v\rangle \ge c|v|^2$,*

$$|B_H(u, f_s, df_s) - B_H(u, f_0, df_0)| \le Ce^{-\sqrt{\beta}}R(u)M^2, \qquad M = e^{h(u)-\beta}.$$

Proof. Write $\epsilon = e^{-\sqrt{\beta}}$, $r = |u|$, $L = df_s$, and $L_0 = df_0$. The first derivative estimate gives $|f_s - f_0|, \|L - L_0\| \le A\epsilon M$ for fixed $A > 0$. Finite sums give $|f_0|, \|L_0\| \le BM$ for fixed $B > 0$. Choose $E \ge 1$ bounding the two fixed coordinate inclusions, so that $\|(u,0)\|, \|(0,u)\| \le Er$. Since

$\epsilon \le 1$ and $Er \ge 1$, put $z = L(u,0) - f_s/2$, $z_0 = L_0(u,0) - f_0/2$, $w = L(0,u)$, and $w_0 = L_0(0,u)$ to obtain

$$|z - z_0| \le 2AE\epsilon Mr, \quad |w| \le (A+B)EMr, \quad |z_0| \le 2BEMr, \quad |w - w_0| \le AE\epsilon Mr.$$

The identity $\bar{z}w - \bar{z}_0 w_0 = \overline{z - z_0}w + \bar{z}_0(w - w_0)$ bounds the imaginary-part difference by $2AE^2(A + 2B)\epsilon M^2 r^2$. The pairing estimate bounds the first term by $P\epsilon R(u)M^2$. Finally $cr^2 \le R(u)$ absorbs the cross term. The positive constant $1 + P + 2AE^2(A + 2B)/c$ proves the assertion. □

Lemma N.3 (The native exhaustion error)**.** *For a smooth positive-Hessian potential with lower bound $c > 0$, the same estimate holds for the actual native numerator wherever $|u| \ge 1$ and the radial Euler identity $H(u)u = \nabla\phi(u)$ holds:*

$$|B_{f_s} - Q_{f_0}| \le Ce^{-\sqrt{\beta}} r_\phi^2 M^2, \qquad r_\phi^2 = \langle H(u)u, u\rangle.$$

The constants are chosen before β, s, u and the later base coefficient.

Proof. The native Hamiltonian formula gives radial component $H^{-1}p_\theta$ for a covector with angular coefficient p_θ. Euler's identity therefore gives $d\phi(X_A) = d\,\mathrm{Re}\, f(0,u)$ and $d\phi(X_B) = d\,\mathrm{Im}\, f(0,u)$. It also gives $Z_0 = (u,0)$ and $C_0 = r_\phi^2$. Substitution into the native numerator and expansion of the imaginary part identify it with $B_H(u, f, df)$. For a holomorphic polynomial, $df(0,u) = i\,df(u,0)$ and $df(u,0) = b_0$; the pairing equals the inverse-metric scalar a. Thus its numerator is $r_\phi^2 a - |b_0|^2 + \mathrm{Re}(\bar{f}b_0)/2 = Q_f$. Apply this to the literal scalar-normalized coefficient tuple and use the preceding cover estimate. □

Lemma N.4 (Finite radial-band outwardness)**.** *Fix a smooth uniformly positive-Hessian potential agreeing with a degree-two homogeneous end outside a ball, with one fixed Hessian upper bound. For any fixed $c_2, C_+ > 0$, all sufficiently large β satisfy $B_{f_s} > 0$ and $d\phi(Z_{s,K}) > 0$ on $c_2\beta \le r_\phi \le C_+\beta$, simultaneously for all $s \in [0,1]$ and all $K \ge 0$.*

Proof. The upper Hessian bound ensures that the lower radial-band bound puts the point beyond both the fixed margin radius and the homogeneous-end radius when β is large. The normalized margin is $Q_{f_0} \ge qr_\phi M^2$ for fixed $q > 0$. The error estimate and $r_\phi \le C_+\beta$ give

$$B_{f_s} \ge (q - CC_+\beta e^{-\sqrt{\beta}})r_\phi M^2.$$

The coefficient in parentheses is at least $q/2$ eventually: with $t = \sqrt{\beta}$, $\beta e^{-\sqrt{\beta}} = t^2 e^{-t} \to 0$. The band has positive radius and $M > 0$, so this proves strict positivity. The already proved Poisson nonnegativity gives $1 + K\pi_{f_s} \ge 1$. The rank-two identity then gives $d\phi(Z_{s,K}) = (r_\phi^2 + KB_{f_s})/(1 + K\pi_{f_s}) > 0$. All constants precede the arbitrary later coefficient K. □

Lemma N.5 (The native rank-two exhaustion identity)**.** *For a smooth potential and complex map at a homogeneous-end point, with invertible actual Hessian H and $1 + K\pi_f \ne 0$, the actual rank-two-updated Liouville field satisfies*

$$d\phi(Z_K) = \frac{\langle Hu, u\rangle + KB_f}{1 + K\pi_f}.$$

Proof. The radial Euler identity determines $Z_0 = (u,0)$ and $d\phi(Z_0) = \langle Hu, u\rangle$. Substitute this identity into the general rank-two exhaustion formula. Its expanded coefficient of K is precisely the native numerator defined above. □

Lemma N.6 (The literal finite potential sublevel domains)**.** *Let ϕ be smooth with fixed Hessian bounds $cI \le D^2\phi \le UI$, where $c, U > 0$. Suppose it agrees outside $|u| \ge L > 0$ with a C^1 degree-two homogeneous function smooth away from zero. Fix $D > 0$. For all sufficiently large β, all $s \in [0,1]$ and every independently chosen $K \ge 0$, the literal nonempty compact set*

$$D_\beta = \{(u,\theta) : 2\phi(u) \le D^2\beta^2\}$$

is a Liouville domain in the logarithmic quotient ambient. Its primitive is the actual rank-two primitive of λ_ϕ and f_s with coefficient K, and its field is the Liouville field of that same ambient primitive. This assertion does not supply a smooth family of domains or a completion marking.

Proof. Choose V bounding ϕ above on the compact cylinder $|u| \le L$. Increase the common threshold for β so that $a_\beta = D^2\beta^2/2 > V$, and so that both the finite-band outwardness and global symplecticity conclusions hold. The potential sublevel is compact by Lemma F.7, and it contains a point of the inner cylinder, so it is nonempty. Every point of the level $\phi = a_\beta$ lies in $|u| > L$. The radial identity there gives $r_\phi^2 = 2\phi = D^2\beta^2$, hence $r_\phi = D\beta$. Apply the finite-band lemma with both band endpoints equal to D. It gives strict positivity of the actual tailored field on this level, simultaneously for every later $K \ge 0$. The smooth defining function $\rho = \phi - a_\beta$ has the same differential as ϕ. The actual primitive is smooth and its derivative is symplectic on the whole ambient by the Poisson positivity lemma. Thus Lemma C.9 constructs the Liouville domain with defining function ρ, the stated literal carrier, primitive, and native field. □

Definition N.7 (The native projected exhaustion derivative)**.** *For the potential primitive λ_ϕ and a smooth complex map f, write Π_f for the actual symplectic projection and Z_0 for the actual potential Liouville field. Define $S_f = d\phi(\Pi_f Z_0)$. At a point satisfying $Hu = \nabla\phi(u)$, the same scalar in fixed logarithmic covering coordinates is $d\phi(u)[(P(df)(u,0))_u]$.*

Proof. The Euler identity determines $Z_0 = (u, 0)$ in covering coordinates. The covering derivative intertwines the displayed projector and the native projector. The chain rule for the potential now identifies their exhaustion derivatives. □

Lemma N.8 (The native projected exhaustion error)**.** *Fix a smooth potential with $cI \le H(u) \le UI$ for $c, U > 0$. There are $C, R, \beta_0 > 0$ such that, for $\beta \ge \beta_0$, $s \in [0,1]$, $|u| \ge R$, and $Hu = \nabla\phi(u)$,*

$$\left|S_{f_s} - \left(r_\phi^2 - |b_0|^2/a_0\right)\right| \le Ce^{-\sqrt{\beta}}r_\phi^2, \qquad |P(df_s)(u,0)| \le Cr_\phi.$$

Here a_0, b_0 are the actual inverse-metric and radial scalars of the normalized holomorphic polynomial f_0.

Proof. Weighted Cauchy–Schwarz for H gives $|\langle Hu, v\rangle| \le r_\phi\sqrt{\langle Hv, v\rangle} \le \sqrt{U}\, r_\phi|v|$. The Euler identity identifies this pairing with $d\phi(u)[v]$. The fixed radial inclusion and projection have bounded norms, and $\sqrt{c}\,|u| \le r_\phi$, so applying this differential to $(P(df_s) - P(df_0))(u,0)$ bounds its absolute value by $C'e^{-\sqrt{\beta}}r_\phi^2$. The holomorphic projected identity identifies the unperturbed scalar with $r_\phi^2 - |b_0|^2/a_0$. The uniform bound for $P(df_s)$ also gives the stated linear field bound. A larger positive constant works for both estimates. □

Lemma N.9 (Native outwardness on the entire bounded-value end)**.** *Under the preceding Hessian bounds, suppose ϕ agrees outside a fixed ball with a degree-two homogeneous function smooth away from zero. There are $F, C, \beta_0 > 0$ such that for every $\beta \ge \beta_0$, $s \in [0,1]$, and point with $r_\phi \ge F\beta$ and $|f_s| \le 2$,*

$$\pi_{f_s} > 0, \qquad d\phi(\Pi_{f_s} Z_0) \ge r_\phi^2/2, \qquad |P(df_s)(u,0)| \le Cr_\phi.$$

There is no upper radial cutoff. The normalized holomorphic scalars also satisfy $a_0 > 0$ *and* $r_\phi^2 - |b_0|^2/a_0 \geq 3r_\phi^2/4$ *on this set.*

Proof. The support-height lower bound and $r_\phi \leq \sqrt{U}\,|u|$ give $h(u) \geq dr_\phi$ for fixed $d > 0$. Choose $F \geq 1$ with $dF \geq 2$. Then $r_\phi \geq F\beta$ gives $M^{-1} \leq e^{-\beta}$ and $r_\phi^{-1} \leq \beta^{-1}$. The almost-tangency lemma consequently bounds $|b_0|/(r_\phi\sqrt{a_0})$ by $T(e^{-\beta} + e^{-\sqrt{\beta}} + \beta^{-1})$, which tends to zero. Increase β_0 to make this ratio at most $1/2$, and make the projected error coefficient $Ce^{-\sqrt{\beta}}$ at most $1/4$. The holomorphic defect is then at least $3r_\phi^2/4$, so the actual projected derivative is at least $r_\phi^2/2$. Increase the threshold again so the entire end lies beyond the fixed projector, positivity, and homogeneous-end radii. The native Poisson lower bound gives strict positivity there, and the preceding field bound applies throughout this end. □

Lemma N.10 (Actual intrinsic outwardness of embedded tailored fibers)**.** *The same* F, β_0 *give, on any actual smooth embedded realization* $j : N \hookrightarrow Y$ *with tangent image* $\ker df_s$*,*

$$d(\phi \circ j)(Z_N) \geq r_\phi^2/2 \quad \text{when } r_\phi \geq F\beta \text{ and } |f_s \circ j| \leq 2, \qquad Z_N = Z_{j^*\lambda_\phi}.$$

This conclusion concerns the native field of the restricted primitive; it does not itself construct a completion marking.

Proof. The preceding strict Poisson positivity makes the restricted two-form nondegenerate on the actual tangent kernel. The intrinsic contraction equation therefore identifies $dj(Z_N) = \Pi_{f_s} Z_0$. Applying the chain rule to $\phi \circ j$ gives $d(\phi \circ j)(Z_N) = d\phi(\Pi_{f_s} Z_0)$. The preceding whole-end projected inequality proves the assertion. □

Lemma N.11 (The entire tailored bounded-value end is empty in rank one)**.** *In rank one, under the same potential hypotheses, there are* $F, \beta_0 > 0$ *such that for every* $\beta \geq \beta_0$ *and* $s \in [0, 1]$*,*

$$\{y : r_\phi(y) \geq F\beta,\ |f_s(y)| \leq 2\} = \varnothing.$$

The assertion includes all bounded values of f_s*, and does not assume that a fiber is nonempty.*

Proof. The preceding whole-end data give $a_0 > 0$ and a holomorphic defect at least $3r_\phi^2/4$ on the displayed set. In rank one the exact metric identity is $r_\phi^2 a_0 = |b_0|^2$, so this defect equals zero. But $r_\phi \geq F\beta \geq F\beta_0 > 0$, a contradiction. □

O A fixed outer domain for the tailoring family

Lemma O.1 (Affine plane pullbacks)**.** *Let* $f, g : X \to \mathbb{C}$ *be smooth maps on a smooth manifold, and put* $P(f) = f^*\lambda_{\mathbb{C}}$*. Then*

$$P(f + sg) = P(f) + s\big(P(f + g) - P(f) - P(g)\big) + s^2 P(g).$$

For every fixed smooth one-form α *and real* K*, the forms* $\alpha + KP(f + sg)$ *are jointly smooth for all real* s*.*

Proof. Write $f = A + iB$ and $g = C + iD$. The formula $P(f) = \frac{1}{2}(A\,dB - B\,dA)$ and linearity of differentiation give

$$P(f + sg) = P(f) + \tfrac{s}{2}(A\,dD + C\,dB - B\,dC - D\,dA) + s^2 P(g).$$

The coefficient of s is $P(f+g) - P(f) - P(g)$, proving the identity. Each coefficient is a smooth one-form. In every fixed coordinate chart the family is polynomial in s with smooth spatial coefficients, which proves joint smoothness, including extension across both interval endpoints. □

Lemma O.2 (One outer compact carrier)**.** *Fix a smooth uniformly positive-Hessian potential with homogeneous positive-Hessian end and a positive outer-radius constant D. For all sufficiently large β, the single compact domain $\{2\phi \le D^2\beta^2\}$ is nonempty and carries $\lambda_\phi + Kf_s^*\lambda_{\mathbb{C}}$ as a Liouville primitive for every $s \in [0,1]$ and $K \ge 0$. Its boundary presentation is fixed before s and K are chosen. For every fixed K, its primitives and the associated localized completions form smooth interval families.*

Proof. Quadratic growth makes the sublevel compact. Choose β large enough that the level exceeds the maximum of ϕ on the fixed exceptional cylinder; this also ensures nonemptiness. Every point on the boundary is then on the homogeneous end and has homogeneous radius $D\beta$. The finite-band inequality makes the derivative of ϕ along every one of the indicated Liouville fields positive there. At $s = K = 0$ this also proves that the level is regular, giving one fixed smooth boundary presentation. The same derivative inequality proves outwardness for all other s, K on that presentation. The rank-two positivity theorem gives symplecticity. The tailoring function is affine in s, so the preceding lemma proves smoothness of the actual primitive family. Parameter-dependent Liouville collars and the cylindrical completion construction applied to this fixed compact family give the stated smooth completed family. □

Lemma O.3 (Perturbation errors for the tailoring family)**.** *For sufficiently large β, uniformly in $s \in [0,1]$, the actual Poisson scalars satisfy $|\pi_{f_s} - a| \le C\varepsilon_\beta M^2$. On the homogeneous end the exhaustion numerators satisfy*

$$|\pi_{f_s} - a| \le C\varepsilon_\beta M^2, \qquad |B_{f_s} - Q_{f_0}| \le C\varepsilon_\beta r^2 M^2. \tag{23}$$

Here $a = \overline{G}^T H^{-1} G$ and $\psi = \phi$ in (22).

Proof. The first-derivative estimate gives $|f_s - f_0| + \|df_s - df_0\| \le C\varepsilon_\beta M$. The finite monomial formula bounds $|f_0| + \|df_0\|$ by CM. The fixed global Hessian lower bound gives a uniform bound on the inverse of the symplectic coefficient matrix. Thus the Hamiltonian vectors are bounded by CM and their changes by $C\varepsilon_\beta M$. The scalar $\pi_f = dA(X_B)$ is bilinear in these differential coefficients. Subtract its two expressions and use the bounds on each factor to obtain $|\pi_{f_s} - \pi_{f_0}| \le C\varepsilon_\beta M^2$. For the holomorphic f_0, the field identities give $\pi_{f_0} = a$. On the homogeneous end $Z_0 = u\partial_u$, $d\phi(Z_0) = r^2$ and $\|d\phi\| \le Cr$. Enlarge its fixed inner radius so $r \ge 1$. The coefficients a_f, b_f, A, B in the exact numerator are then bounded by CrM, with changes bounded by $C\varepsilon_\beta rM$. The coefficients $d\phi(X_A), d\phi(X_B)$ obey the same bounds. Subtracting the products in B_f bounds their changes by $C\varepsilon_\beta r^2 M^2$; its first term contributes $r^2|\pi_{f_s} - \pi_{f_0}|$ with that same bound. For f_0 the holomorphic identities give $B_{f_0} = Q_{f_0}$ on the end, proving the claimed errors. □

Lemma O.4 (Nonholomorphic rank-two identities and perturbation estimates)**.** *For the actual tailoring maps, nondegeneracy is equivalent to $1 + K\pi_{f_s} \ne 0$. Under this condition the Liouville, base and exhaustion identities (20)–(22) hold, with the perturbation estimates (23).*

Proof. The tailoring-family lemma supplies smoothness of the actual maps. Apply the exact rank-two identities to their primitive updates, and apply the perturbation-error lemma to compare them with the normalized holomorphic endpoint. The second error estimate is asserted on the homogeneous end, exactly as in that lemma. □

Lemma O.5 (Uniform positivity for the tailored family)**.** *Fix a smooth potential ϕ with global Euclidean Hessian bounds $cI \le D^2\phi \le UI$, where $c, U > 0$, as supplied by the constructed potential. For all sufficiently large β, uniformly in $s \in [0,1]$, one has $\pi_{f_s} \ge 0$ on Y and*

$$\pi_{f_s} \ge c_{\mathrm{dom}} M^2/2 > 0 \quad \text{on } Y \setminus K_{\mathrm{ex}}, \qquad \pi_{f_s} > 0 \quad \text{where } |f_s| \ge 1.$$

The threshold and $c_{\mathrm{dom}} > 0$ are independent of the later base coefficient K. Consequently $\lambda_{s,K} = \lambda_\phi + K f_s^* \lambda_{\mathbb{C}}$ is a primitive of a symplectic form for every $s \in [0,1]$ and $K \geq 0$.

Proof. Multiplying all monomials by the same nonzero scalar leaves each ratio m_v/M unchanged. Thus the bound $|G| \geq cM$ and its exceptional cylinder in Lemma E.25 also hold for $f_0 = W/R$, with constants independent of β. The global Hessian bounds give fixed positive upper and lower bounds for its eigenvalues and those of its inverse. Hence $a \geq c_{\mathrm{dom}} M^2$ outside K_{ex}. Lemma M.15 gives

$$\pi_{f_s} \geq (c_{\mathrm{dom}} - C\varepsilon_\beta) M^2 \geq c_{\mathrm{dom}} M^2/2$$

there once β is large. This scalar-normalization argument applies to the relative bound $|G|/M$; it does not assert a β-independent prefactor in the exponential growth bound for M.

The modification is supported in $|u| \geq c_1\beta$, so for large β it vanishes on a neighborhood of the fixed cylinder K_{ex}. There the actual differentials agree, so $\pi_{f_s} = a \geq 0$. On the entire fixed cylinder the elementary bound $|f_0| \leq |V| \exp(h(u) - \beta) < 1$ holds for sufficiently large β. Since $f_s = f_0$ there, that cylinder contains no point with $|f_s| \geq 1$. This proves strict positivity on $|f_s| \geq 1$ everywhere. Finally, $1 + K\pi_{f_s} \geq 1$, so the rank-two nondegeneracy criterion applies. □

O.1 An outward finite boundary

Lemma O.6 (Outwardness on the finite ambient band)**.** *For fixed $c_2, C_+ > 0$ and large β, $B_{f_s} > 0$ on $c_2\beta \leq r \leq C_+\beta$, uniformly in s. Consequently $D_\beta = \{2\phi \leq C_+^2\beta^2\}$ is Liouville for every s and every $K \geq 0$.*

Proof. By Lemma G.11, applied after the scalar normalization $f_0 = W/R$, one has $Q_{f_0} \geq crM^2$ for $r \geq r_0$, with constants independent of β.

Choose fixed $C_- < C_+$ sufficiently large as specified below and let

$$D_\beta = \{2\phi \leq C_+^2\beta^2\}.$$

On any fixed band $c_2\beta \leq r \leq C_+\beta$, $c_2 > 0$, (23) and Lemma G.11 give $B_{f_s} > 0$ for large β, because $\beta\varepsilon_\beta \to 0$. Thus D_β is a Liouville domain for every s, K. This is explicitly a finite-band statement; the error bound does not establish ambient convexity at every arbitrarily large radius for a fixed β. □

P The whole-end tailored fiber projection

Definition P.1 (Projection in fixed logarithmic coordinates)**.** *In fixed logarithmic coordinates, let H be the actual invertible potential Hessian. For a real covector p, let p_u, p_θ be its Euclidean Riesz coefficient vectors and set $X_p = (H^{-1}p_\theta, -H^{-1}p_u)$. For a complex differential $L = p + iq$, define*

$$\pi(L) = p(X_q), \qquad N(L)v = q(v)X_p - p(v)X_q, \qquad P(L) = \mathrm{id} + \pi(L)^{-1}N(L).$$

For the differential of any smooth complex map, the derivative of the logarithmic quotient covering carries X_p and $P(L)$ to the native Hamiltonian vector and native rank-two projection, respectively.

Proof. The potential symplectic form on cover vectors satisfies $\omega((a,b),(v,w)) = \langle Ha, w\rangle - \langle Hv, b\rangle$. Symmetry of the actual Hessian gives $\omega(X_p,(v,w)) = \langle p_u, v\rangle + \langle p_\theta, w\rangle = p(v,w)$. Nondegeneracy uniquely determines the native Hamiltonian, and the covering derivative is bijective, proving the Hamiltonian identification. The actual complex differential pulls back to L, so the native Poisson

scalar equals $p(X_q)$. Applying the linear covering derivative to $v + q(v)X_p/\pi(L) - p(v)X_q/\pi(L)$ now gives the native projection formula term by term. All norms below are in these fixed cover coordinates. □

Lemma P.2 (Uniform whole-end projector bounds)**.** *Fix a smooth potential whose actual Hessian satisfies* $c\mathrm{id} \le H(u) \le U\mathrm{id}$ *for fixed* $c, U > 0$*. There are* $C, R, \beta_0 > 0$ *such that for every* $\beta \ge \beta_0$*,* $s \in [0,1]$*, and point with Euclidean logarithmic radius* $|u| \ge R$*,*

$$\|P(df_s) - P(df_0)\| \le Ce^{-\sqrt{\beta}}, \qquad \|P(df_s)\| \le C.$$

The end has no upper radial cutoff and the constants precede the parameters and the point.

Proof. Put $\epsilon = e^{-\sqrt{\beta}}$ and $M = e^{h(u)-\beta} > 0$. The first derivative estimate and the finite monomial sum give fixed $A, B > 0$ with $\|L - L_0\| \le A\epsilon M$ and $\|L_0\| \le BM$. Since $\epsilon \le 1$, also $\|L\| \le (A+B)M$. Coercivity bounds $\|H^{-1}\| \le c^{-1}$: for $v = H^{-1}w$, $c\|v\|^2 \le \langle w, v\rangle \le \|w\|\|v\|$, and cancellation when $v \ne 0$ gives the bound; it is immediate when $v = 0$. The fixed coordinate inclusions and Riesz maps consequently give a fixed $D \ge 0$ with $\|N(L)\| \le D\|L\|^2$ and

$$\|N(L) - N(L_0)\| \le D\|L - L_0\|(\|L - L_0\| + 2\|L_0\|).$$

Indeed expand the bilinear tensor difference as $N(p - p_0, q) + N(p_0, q - q_0)$ and apply the two Hamiltonian norm bounds. Thus fixed nonnegative D_1, D_0, D_2 bound these three quantities by $D_1\epsilon M^2$, D_0M^2, and D_2M^2, respectively. The original dominance and tailored positivity estimates give a common $d > 0$ and end radius with $\pi(L), \pi(L_0) \ge dM^2$. The native pairing error, transported to the cover, gives $|\pi(L) - \pi(L_0)| \le E\epsilon M^2$ for fixed $E > 0$. The exact identity

$$P(L) - P(L_0) = \pi(L)^{-1}(N(L) - N(L_0)) + (\pi(L)^{-1} - \pi(L_0)^{-1})N(L_0)$$

and $|a^{-1} - b^{-1}| = |a - b|/(ab)$ for $a, b > 0$ yield $\|P(L) - P(L_0)\| \le (D_1/d + ED_0/d^2)\epsilon$. Likewise $\|P(L)\| \le 1 + D_2/d$. Increasing the sum of these constants by one gives the asserted positive common constant. The powers of M cancel, so no upper radial cutoff enters. □

Lemma P.3 (The full native level-one fiber and intrinsic end growth)**.** *Under the preceding Hessian hypotheses, suppose the smooth potential agrees outside a fixed Euclidean ball with a degree-two homogeneous function smooth away from zero. For all sufficiently large* β*, uniformly in* $s \in [0,1]$*, the literal full level* $f_s^{-1}(1)$ *is the image of a smooth embedding of dimension* $2n - 2$*, and the restricted potential primitive is symplectic. There are* $C, R > 0$*, chosen before* β, s*, such that its intrinsic Liouville field, pushed into fixed logarithmic coordinates, satisfies* $|Z_s^{\mathrm{fib}}| \le C|u|$ *whenever* $|u| \ge R$*. No nonemptiness of that end is asserted in rank one.*

Proof. At every point of the level, $|f_s| = 1$, so tailored positivity gives $\pi_{f_s} > 0$. The two real differential covectors are surjective onto $\mathbb{R}^2$: the vectors X_q/π_{f_s} and $-X_p/\pi_{f_s}$ map to $(1,0)$ and $(0,1)$. The regular-level construction therefore gives the full smooth level with its literal inclusion and tangent image $\ker df_s$. The rank-two kernel lemma makes the restricted form nondegenerate; smoothness and closedness follow by pullback of the smooth exact ambient form. On the open homogeneous end, Euler's identity gives $Hu = \nabla\phi(u)$, hence the actual ambient Liouville vector is $(u, 0)$. The intrinsic contraction equation and the kernel lemma identify the pushforward of the actual intrinsic field with $P(df_s)(u, 0)$. The preceding uniform projector bound and the fixed equivalence between Euclidean and logarithmic coordinate norms now give $|P(df_s)(u, 0)| \le C'|u|$. Choose a radius larger than the projector, positivity, and homogeneous-end radii, and a parameter threshold larger than both earlier thresholds. Increasing C' by one supplies a positive constant valid throughout this whole end. The construction allows dimension zero, so the argument also applies when $n = 1$. □

Lemma P.4 (Complete flow from logarithmic growth on a closed submanifold)**.** *Let $j : N \hookrightarrow (\mathbb{C}^*)^n$ be a smooth embedding with closed image, and let Z be a smooth vector field on N. Suppose there are $C, R > 0$ such that, whenever $|u(j(x))| \geq R$, the logarithmic-coordinate representative v of $dj_x Z(x)$ satisfies $\|v\| \leq C|u(j(x))|$. Then Z has a jointly smooth all-real flow.*

Proof. Set $\rho(x) = |u(j(x))|^2/2$. Closedness of the image and compactness of the ambient quadratic sublevels make ρ proper. If k is the norm of the fixed conversion from the coordinate norm to the Euclidean norm, Cauchy–Schwarz gives $|d\rho(Z)| \leq k|u|\|v\| \leq 2kC\rho$ outside the compact sublevel $\rho \leq R^2/2$. On that sublevel smoothness bounds $|d\rho(Z)|$ by a finite constant B. Thus $|d\rho(Z)| \leq D(1+\rho)$ for $D = \max(B, 2kC+1) > 0$. The proper-exhaustion complete-flow theorem applies. □

Lemma P.5 (Complete intrinsic primitives on the full tailored fibers)**.** *Retain a smooth potential with fixed bounds $cI \leq D^2\phi \leq UI$, where $c, U > 0$, and the prescribed homogeneous end. For one $\beta_0 > 0$, every $\beta \geq \beta_0$ and $s \in [0,1]$ admit a smooth embedding whose image is the entire level $\{f_{\beta,s} = 1\}$ and whose literal restricted primitive is complete.*

Proof. The full-fiber growth theorem constructs the smooth symplectic level and bounds its actual intrinsic Liouville field uniformly on the whole end. Its image is closed since $f_{\beta,s}$ is continuous. Apply the preceding completeness theorem to this actual field. The primitive is smooth and symplectic by its construction, and its native Liouville field satisfies the contraction equation. Together with the constructed all-real flow these give the asserted complete primitive. □

Q Actual completion markings of full tailored-fiber truncations

Lemma Q.1 (The native full tailored-fiber sublevel completion)**.** *Retain the smooth chosen positive-Hessian potential with its C^1 degree-two homogeneous end, and a full smooth realization $j : F \hookrightarrow Y$ of $\{f_{\beta,s} = 1\}$ with an actual complete primitive $\mu = j^*\lambda_\phi$. Suppose at a chosen level A its actual intrinsic derivative is positive throughout $\phi \circ j \geq A$. Then the entire literal truncation $D = \{\phi \circ j \leq A\}$ is a compact native Liouville domain, with frontier $\phi \circ j = A$ and image $\{f_{\beta,s} = 1,\ \phi \leq A\}$. It has a flow-built completion, and every chosen completion of this same domain has a primitive-preserving inclusion marking onto the whole F. No nonemptiness claim is made by this lemma.*

Proof. The entire fiber image is closed by continuity of $f_{\beta,s}$. Closedness of its embedding pulls the proved compact ambient potential sublevels back to compact intrinsic sublevels. The defining function $\phi \circ j - A$ has the same differential as $\phi \circ j$; its positive Liouville derivative makes its zero level regular and outward. The regular-sublevel construction gives the asserted native domain and frontier. The full-image equation gives the displayed image equality. The actual complete intrinsic flow, compact exhaustion bands and positive derivative give finite negative entry from every point of F. The absorption completion construction uses this same flow for its boundary-orbit maps, so its marking is onto the entire fiber. Strict completion comparison transports it to every chosen completion. □

Lemma Q.2 (Uniform actual markings for all large tailored truncations)**.** *For the retained smooth potential with fixed bounds $cI \leq D^2\phi \leq UI$, $c, U > 0$, and its C^1 degree-two homogeneous end, there are $F, \beta_0 > 0$ such that every $\beta \geq \beta_0$ and $s \in [0,1]$ admit an actual complete full smooth realization of $\{f_{\beta,s} = 1\}$. On that same realization, for every $A \geq F^2\beta^2/2$, the entire literal potential truncation is a compact native Liouville domain and every chosen completion has its actual inclusion marking onto that same full fiber. The assertion is pointwise in the parameters and does not assert nonemptiness of the tailored level or a parameter-smooth marked family.*

Proof. Take the common constants from actual tailored-fiber completeness, native Poisson positivity, and whole-end intrinsic exhaustion. Continuity bounds ϕ above by V on the compact cylinder $|u| \le L$. Increase β_0 so that $F^2\beta^2/2 > V$ and $\beta > 0$. Every point above the stated potential level lies on the strict homogeneous end, where $r^2 = u^T D^2\phi(u)u = 2\phi(u)$. Thus $\phi \ge F^2\beta^2/2$ implies $r \ge F\beta$. Positive native Poisson pairing over $f_{\beta,s} = 1$ supplies actual regularity, so the regular-level theorem gives the tangent kernel. The entire fiber has $|f_{\beta,s}| = 1 \le 2$, hence the whole-end intrinsic estimate gives $d\phi(Z_F) \ge r^2/2 = \phi > 0$ there. This holds above every $A \ge F^2\beta^2/2$. Apply the preceding domain and absorption completion construction to the actual complete fiber produced by the completeness theorem. Transport its inclusion marking to every chosen completion of each stated truncation. All constants precede the parameters and the truncation level. □

Q.1 Whole-fiber ends

Lemma Q.3 (Almost tangency on actual bounded-value tailored fibers)**.** *Fix positive metric bounds $cI \le H \le UI$ with H symmetric. For $f_0 = e^{-\beta}W_1$, write $G_0 = df_0/(du + i\,d\theta)$, $b_0 = u \cdot G_0$, $a_0 = \overline{G_0}^T H^{-1} G_0$, $r^2 = u^T H u$, and $M = e^{h(u)-\beta}$. There are $C, L > 0$, independent of β, s, H, such that for every real β, $s \in [0,1]$, $|u| \ge L$ and $|f_s| \le 2$ one has*

$$a_0 > 0, \qquad \frac{|b_0|}{r\sqrt{a_0}} \le C\big(M^{-1} + e^{-\sqrt{\beta}} + r^{-1}\big).$$

For negative β the square root is taken to be zero. There is no upper radial restriction.

Proof. The coefficient-one discrepancy, multiplied by $e^{-\beta}$, gives $|b_0| \le h|f_0| + BM$ with $B > 0$ fixed. Actual first-derivative closeness includes $|f_s - f_0| \le E\epsilon M$, where $\epsilon = e^{-\sqrt{\beta}}$ and $E > 0$ is fixed. Thus $|f_0| \le 2 + E\epsilon M$ on the stated fibers. Metric comparison gives $r \ge \sqrt{c}\,|u|$ and $h \le Ar$ for fixed $A > 0$. Outside a fixed positive radius, normalized dominance gives $\sqrt{a_0} \ge dM > 0$ for fixed $d > 0$: both sides of the unnormalized gradient estimate scale by $e^{-\beta}$. Consequently

$$\frac{|b_0|}{r\sqrt{a_0}} \le \frac{Ar(2 + E\epsilon M) + BM}{rdM} = \frac{2A}{dM} + \frac{AE}{d}\epsilon + \frac{B}{dr}.$$

Take $C = 2A/d + AE/d + B/d + 1$; each nonnegative summand is bounded by its corresponding summand with coefficient C. All constants precede β, s, H, as claimed. □

Lemma Q.4 (Rank-two symplectic projection)**.** *Let ω be a nondegenerate alternating form on a finite-dimensional real vector space, and let p, q be covectors. Define X_p, X_q by $\omega(X_p, \cdot) = p$, $\omega(X_q, \cdot) = q$, and put $\pi = p(X_q) \ne 0$. Then the restriction of ω to $E = \ker p \cap \ker q$ is nondegenerate, and*

$$\Pi(v) = v + \frac{q(v)}{\pi}X_p - \frac{p(v)}{\pi}X_q$$

belongs to E and fixes every vector in E. For $w \in E$ one has $\omega(\Pi(v), w) = \omega(v, w)$.

Proof. Alternation gives $p(X_p) = q(X_q) = 0$ and $q(X_p) = -\pi$. Substitution therefore gives $p(\Pi(v)) = q(\Pi(v)) = 0$. If $v \in E$, the formula gives $\Pi(v) = v$, so this is a projection onto the entire common kernel. For $w \in E$, both correction terms pair to zero with w, which proves the last identity. Suppose $v \in E$ pairs to zero with every $w \in E$. Every ambient vector w decomposes as $\Pi(w)$ plus a linear combination of X_p, X_q. The first summand pairs to zero by the hypothesis, and the remaining two pair to zero since $p(v) = q(v) = 0$. Thus v pairs to zero with every ambient vector, so $v = 0$ by nondegeneracy of ω. This proves the restricted nondegeneracy. □

Lemma Q.5 (The actual intrinsic Liouville field of a regular fiber)**.** *Let α be a smooth one-form on a finite-dimensional manifold, with $d\alpha$ symplectic, and let $f = A + iB$ be smooth. Let $j : F \hookrightarrow X$ be a smooth embedded fiber whose tangent image at each point is $\ker dA \cap \ker dB$. If $\pi_f = dA(X_B) \neq 0$ along the fiber, its restricted differential $d(j^*\alpha)$ is symplectic and the actual Liouville vector of $j^*\alpha$ satisfies*

$$dj(Z_F) = \Pi_f(Z_0).$$

Here Z_0 is the actual ambient Liouville vector and Π_f is the projection in the preceding lemma with $p = dA$, $q = dB$.

Proof. The derivative of an embedding identifies each tangent space of F with the stated common kernel. The projection lemma proves that the restriction of $d\alpha$ to this kernel is nondegenerate. Since exterior differentiation commutes with pullback, $d(j^*\alpha) = j^*(d\alpha)$; it is also closed, so is symplectic. The vector $\Pi_f(Z_0)$ belongs to the tangent image, hence has a unique tangent preimage V under dj. For every tangent vector w of F, the projection identity gives

$$d(j^*\alpha)(V, w) = d\alpha(\Pi_f(Z_0), dj(w)) = d\alpha(Z_0, dj(w)) = \alpha(dj(w)) = (j^*\alpha)(w).$$

This is the intrinsic Liouville contraction equation. Its unique solution is Z_F, proving the stated identity. □

Lemma Q.6 (A smooth symplectic fiber from nonzero Poisson pairing)**.** *Let X be a smooth d-manifold, α a smooth one-form with $d\alpha$ symplectic, $f = A + iB$ smooth, and $w \in \mathbb{C}$. Suppose $\pi_f \neq 0$ at every point of $f^{-1}(w)$. Then there is a smooth $(d-2)$-manifold F with a smooth embedding $j : F \hookrightarrow X$ whose image is literally $f^{-1}(w)$, and whose restricted primitive has symplectic differential $d(j^*\alpha)$. Empty fibers are allowed; if the fiber is nonempty then $d \geq 2$, and $d = 2$ gives dimension zero.*

Proof. At a point of the level, nonzero π_f makes the two real differentials independent. More explicitly, for $z = a + ib \in \mathbb{C}$ the vector $(a/\pi_f)X_B - (b/\pi_f)X_A$ maps under df to z, because $dA(X_A) = dB(X_B) = 0$, $dA(X_B) = \pi_f$ and $dB(X_A) = -\pi_f$. Thus df is surjective at every point of the level. The regular-level coordinates give its smooth submanifold structure and literal inclusion embedding, with tangent image $\ker df$; the dimension is $d - 2$ by rank-nullity. The equality $\ker df = \ker dA \cap \ker dB$ follows by taking real and imaginary parts. The native fiber lemma gives nondegeneracy of the restricted differential at each point. It is closed because it is an exterior differential, hence symplectic. If there are no level points, take the empty manifold of the indicated dimension; every condition holds there. The construction also applies when $d = 2$, with zero-dimensional tangent spaces. □

Lemma Q.7 (Global intrinsic tailored fiber ends)**.** *For all large β the regular fibers $f_s^{-1}(1)$ have complete intrinsic fields, and their truncations $F_s^- = \{f_s = 1, 2\phi \leq C_-^2\beta^2\}$ complete to the entire fibers by inclusion. The end exhaustion derivative is at least $r^2/2$ in positive fiber dimension; in rank one that end is empty.*

Proof. Choose C_- so that $C_- \min_{2\phi_\infty(\xi)=1} h(\xi) > 1$ with a strict margin. For $r \geq C_-\beta$, $M = e^{h-\beta}$ is exponentially large uniformly as β grows. On $f_s = w$, $|w| \leq 2$, the identity for E and (19) imply

$$\frac{|u \cdot G|}{r\sqrt{a}} \leq C\big(M^{-1} + \varepsilon_\beta + r^{-1}\big). \tag{24}$$

For $\pi_f > 0$ the rank-two projection lemma gives the symplectic projection onto $\ker df$ as

$$\Pi_f(v) = v + \frac{dB(v)}{\pi_f} X_A - \frac{dA(v)}{\pi_f} X_B.$$

For large β this end lies outside K_{ex}. Lemma O.5 bounds its denominators below by a fixed positive multiple of M^2, so (19) gives $\|\Pi_{f_s} - \Pi_{f_0}\| \le C\varepsilon_\beta$ there. The native fiber-field lemma identifies the actual intrinsic Liouville field as $Z_s^{\mathrm{fib}} = \Pi_{f_s} Z_0$. Consequently

$$d\phi(Z_s^{\mathrm{fib}}) \ge r^2 - |u \cdot G|^2/a - C\varepsilon_\beta r^2 \ge r^2/2, \qquad |Z_s^{\mathrm{fib}}| \le Cr. \tag{25}$$

For $n = 1$, the same identity $|u \cdot G|/(r\sqrt{a}) = 1$ holds on this end. Since the right side of (24) is uniformly less than one for large β, the set $\{f_s = w,\ |w| \le 2,\ r \ge C_-\beta\}$ is empty for every s. Thus the end inequality has empty domain in dimension one. The zero-dimensional fibers have zero intrinsic field, and the same actual sublevel completion construction gives their inclusion markings. The corner at the outer radius is empty. The fibers over $|w| = 1$ are regular for large β: in the modification region $\pi_{f_s} > 0$, and outside it the original critical values, scaled by $e^{-\beta}$, have modulus below one. Thus

$$F_s^- = \{f_s = 1,\ 2\phi \le C_-^2\beta^2\}$$

is a compact Liouville domain completing to the entire marked fiber. This uses a global intrinsic end estimate separate from the finite ambient band.

□

Q.2 The marked deformation

Q.2.1 Uniform coefficients and rounded finite tailoring domains

Lemma Q.8 (A positive lower bound above a positive base radius)**.** *For a fixed smooth potential with positive lower and upper Hessian bounds, and a fixed $r_b > 0$, there are $d, \beta_0 > 0$ such that for every $\beta \ge \beta_0$, $s \in [0,1]$ and $|f_s(y)| \ge r_b$,*

$$\pi_{f_s}(y) \ge de^{-2\beta}/2 > 0.$$

Proof. Choose the exceptional radius and dominant-face constant from the native Poisson positivity estimate. Increase the scale threshold so the entire exceptional cylinder maps into the open disk of radius r_b. Indeed the unchanged normalized polynomial on this cylinder is bounded by $|V|e^{HR-\beta}$, which tends to zero. A point with $|f_s| \ge r_b$ therefore lies beyond that radius, where $\pi_{f_s} \ge de^{2(h(u)-\beta)}/2$. Interior containment of zero in the polytope gives $h(u) \ge 0$, so $e^{2(h(u)-\beta)} \ge e^{-2\beta}$. Both d and the exponential are positive, proving the assertion. □

Remark Q.9. The source uses $r_b = 1$. Taking $r_b = 1/2$ gives a quantitative collar below the unit circle by the same finite-input argument.

Lemma Q.10 (One finite base coefficient on every compact carrier)**.** *Fix $r_b > 0$ and a compact subset $C \subset Y$. For every sufficiently large β there is a finite $K_* > 0$ such that for every $K \ge K_*$, $s \in [0,1]$, and $y \in C$ with $|f_s(y)| \ge r_b$,*

$$d|f_s|^2(Z_{s,K})(y) > 0.$$

The coefficient may be increased arbitrarily while preserving this conclusion.

Proof. Write $Z_0 = Z_{\lambda_\phi}$ and $a(s,y) = 2\,\mathrm{Re}(\bar{f}_s\, df_s(Z_0))$. The tailoring homotopy is affine in s, hence so is its spatial differential. Smoothness of its endpoint maps and of Z_0 shows that a is continuous on $[0,1] \times C$. Compactness bounds $|a|$ there by a finite $N \geq 0$, including when C is empty. The preceding lemma supplies $p = de^{-2\beta}/2 > 0$ with $\pi_{f_s} \geq p$ whenever $|f_s| \geq r_b$. Choose $K_* = (N+1)/(pr_b^2)$. For $K \geq K_*$, the numerator $a + K\pi_{f_s}|f_s|^2$ is at least $-N + Kpr_b^2 \geq 1$. The native rank-two formula gives the derivative as this numerator divided by $1 + K\pi_{f_s}$, whose positivity follows from global Poisson nonnegativity and $K \geq K_* > 0$. □

Remark Q.11. The compact carrier can be the literal finite potential domain $\{2\phi \leq C_+^2\beta^2\}$. This argument proves the uniform coefficient before choosing any rounding width. The following lemmas construct collars and rounded-domain outwardness. Completion absorption and the full marking remain further geometric steps.

Lemma Q.12 (A fixed potential collar inside the outer domain)**.** *Assume the potential agrees outside a fixed ball with a degree-two homogeneous potential, smooth away from zero and continuously differentiable at zero, and has positive lower and upper Hessian bounds. Fix $D > 0$. For all sufficiently large β, put $A = D^2\beta^2/2$. For every $s \in [0,1]$, $K \geq 0$, and point with $A/2 \leq \phi \leq A$, one has $d\phi(Z_{s,K}) > 0$.*

Proof. The potential has a finite upper bound V on the fixed exceptional cylinder. Increase β until $A/2 > V$, so every point in the displayed annulus lies in the homogeneous end. There the radial Euler identity gives $r_\phi^2 = 2\phi$. Hence $D\beta/2 \leq r_\phi \leq D\beta$. The finite-band outwardness estimate on this fixed radial band gives the positive native derivative for every s and K in the stated ranges. □

Lemma Q.13 (Localization of active rounded partials)**.** *For any allowed profile m_ϵ with $\epsilon > 0$ and $m_\epsilon(x,y) \geq 0$, a positive first partial implies $x \geq -2\epsilon$, and a positive second partial implies $y \geq -2\epsilon$.*

Proof. The upper profile bound gives $\max(x,y) \geq -\epsilon$. If $x < -2\epsilon$, this forces $y \geq -\epsilon$ and $y - x > \epsilon$. The profile equals the second coordinate at this point. Since the profile is bounded below by the second coordinate everywhere, their difference has a smooth local minimum of zero, whose derivative vanishes. Thus the first partial is zero, contradicting its positivity. The second assertion follows by interchanging the two coordinates; the corresponding difference with the first coordinate has a local minimum of zero and its second derivative component vanishes. □

Definition Q.14 (The literal normalized rounding)**.** *For $R > 0$, level A, scale β and homotopy parameter s, define*

$$q_s(y) = m_\epsilon\big(\phi(y) - A, R^2(|f_s(y)|^2 - 1)\big), \qquad E_s = \{q_s \leq 0\}.$$

Here f_s is the actual normalized tailoring homotopy. The source normalization is obtained by taking $R = e^\beta$.

Lemma Q.15 (One width gives positive derivative on the compact exterior)**.** *Under the same potential hypotheses and fixed $D > 0$, for all sufficiently large β there is $K_* > 0$ with the following property. For every $R > 0$ and $A_- < A = D^2\beta^2/2$, there is $\delta > 0$ with $\delta < A - A_-$ such that every allowed profile of width $0 < \epsilon \leq \delta$ satisfies, for all $K \geq K_*$, $s \in [0,1]$ and $y \in \{\phi \leq A,\ q_s \geq 0\}$,*

$$dq_s(Z_{s,K})(y) > 0.$$

Proof. Use the compact base coefficient lemma with $r_b = 1/2$ on the literal compact potential sublevel $\{\phi \leq A\}$. Choose $\delta = \min(A/4, R^2/4, (A - A_-)/2) > 0$. An active first partial implies

$\phi - A \geq -2\epsilon$, hence $\phi \geq A/2$; the potential collar lemma gives a positive derivative for that input. An active second partial implies $R^2(|f_s|^2 - 1) \geq -2\epsilon$, hence $|f_s|^2 \geq 1/2$ and $|f_s| \geq 1/2$; the base estimate gives a positive derivative for that input. The chain rule expresses $dq_s(Z_{s,K})$ as the sum of these two derivatives multiplied by the nonnegative profile partials. Their sum is one, so at least one is positive. Inactive partials are zero and contribute zero. This proves strict positivity. □

Lemma Q.16 (Actual rounded compact Liouville domains)**.** *With the same parameter order, every allowed profile below the produced width and every $s \in [0,1]$, $K \geq K_*$ gives a native compact Liouville domain with literal carrier E_s, primitive $\lambda_{s,K}$, and field $Z_{s,K}$. Its entire boundary is the literal zero level $\{q_s = 0\}$.*

Proof. The profile and the actual potential and tailoring map are smooth, so q_s is smooth. Its nonpositive sublevel is closed and lies in $\{\phi \leq A\}$, since $q_s \geq \phi - A$. The potential sublevel is compact, hence E_s is compact. Global Poisson nonnegativity gives symplecticity of $d\lambda_{s,K}$. On $q_s = 0$, the preceding strict derivative makes dq_s nonzero. The regular-sublevel construction therefore supplies a native smooth boundary, the restricted primitive and its actual outward Liouville field, with the entire zero level as boundary. □

Q.2.2 The literal rounded family equations

Lemma Q.17 (Jointly smooth rounded equation)**.** *For every fixed smooth potential ϕ, scale β, real R, A, and smooth rounding profile m, the function*

$$(s,y) \longmapsto q_s(y) = m\big(\phi(y) - A, R^2(|f_s(y)|^2 - 1)\big)$$

is jointly smooth for all real s. In particular it extends smoothly across both endpoints of the parameter interval.

Proof. The actual tailoring map $(s,y) \mapsto f_s(y)$ is smooth for all real parameters, since its formula is affine in s with smooth spatial terms. The squared complex norm is a real polynomial, hence smooth. Compose these functions with the smooth potential, multiplication by R^2, and the given smooth profile to obtain the assertion. □

Lemma Q.18 (Actual local motions of every rounded boundary)**.** *Assume the potential and end hypotheses of the uniform rounded outwardness lemma, and fix $D > 0$. For all sufficiently large β, choose its coefficient $K_* > 0$. For every $R > 0$ and $A_- < A = D^2\beta^2/2$, there is $0 < \delta < A - A_-$ such that, for every allowed profile of width $0 < \epsilon \leq \delta$, every $s \in [0,1]$, and every y with $q_s(y) = 0$, there is an open neighborhood O of $(0,s,y)$ and a jointly smooth motion $\Psi : O \to \mathbb{R} \times Y$ satisfying*

$$\Psi(0,t,z) = (t,z), \qquad \Psi(h,t,z) = (t+h, z'), \qquad q_{t+h}(z') = q_t(z).$$

Proof. At a zero of q_s, the profile lower bound gives $\phi(y) \leq A$. Uniform outwardness at K_* gives $dq_s(Z_{s,K_*})(y) > 0$. Thus the spatial derivative of q_s is nonzero and hence surjects onto $\mathbb{R}$. The parameter-dependent submersion normal form for the jointly smooth function $(s,y) \mapsto q_s(y)$ gives the stated local motion, preserving this function and translating the parameter. Its neighborhood is open in the full real parameter space, including at either endpoint of the closed interval. □

Lemma Q.19 (Compact total rounded sublevel)**.** *For a smooth uniformly positive-Hessian potential with homogeneous positive-Hessian end, the literal set*

$$\{(s,y) : 0 \leq s \leq 1,\ q_s(y) \leq 0\}$$

is compact, for every fixed β, R, A, m as above.

Proof. The lower smooth-maximum bound gives $\phi(y)-A \le \max(\phi(y)-A, R^2(|f_s(y)|^2-1)) \le q_s(y)$. Thus the indicated set lies in $[0,1]\times\{\phi \le A\}$, which is compact by quadratic growth of the potential and compactness of the angle torus. The parameter interval and the sublevel inequality of the continuous function q are closed. Their intersection is therefore a closed subset of that compact product. □

Q.2.3 One compatible lift for the rounded tailoring pair

Lemma Q.20 (Components of the parameter and level equation)**.** *Let $\tau : Q \to \mathbb{R}$ and $f : Q \to V$ be differentiable at x, where Q is a smooth manifold and V a real normed vector space. For $v \in T_xQ$, $d(\tau, f)_x(v) = (1,0)$ implies $d\tau_x(v) = 1$ and $df_x(v) = 0$.*

Proof. The differential of the product map is the product of the two differentials. Apply its two coordinate projections to the equality. □

Lemma Q.21 (Literal unrounded germ along the full ribbon)**.** *Let I be a finite toric input, ϕ a smooth potential, β, R, A, A_- real, and m_ϵ a rounding profile of positive width ϵ. Write $f_s = f_{\beta,s}$ and*

$$q(s,y) = m_\epsilon\big(\phi(y) - A, R^2(|f_s(y)|^2 - 1)\big).$$

If $A_- < A - \epsilon$, then at every (s,y) with $f_s(y) = 1$ and $\phi(y) \le A_-$ there is an open neighborhood U on which $q(s',y') = R^2(|f_{s'}(y')|^2 - 1)$ literally. In particular $q(s,y) = 0$ for every point of the full truncated fiber.

Proof. Put $x(s,y) = \phi(y) - A$ and $z(s,y) = R^2(|f_s(y)|^2 - 1)$. Both are continuous. At the specified point, $z = 0$ and $z - x \ge A - A_- > \epsilon$. Thus $U = \{z - x > \epsilon\}$ is an open neighborhood. On U, positivity of ϵ gives $x < z$ and $|x - z| > \epsilon$. Exactness of the profile away from the diagonal therefore gives $m_\epsilon(x,z) = \max(x,z) = z$. Evaluation at the original fiber point gives zero. □

Lemma Q.22 (Fiber lifts imply the ambient boundary equation)**.** *On an open set U where $q = R^2(|f|^2 - 1)$, any native vector v satisfying $d(s,f)(v) = (1,0)$ also satisfies $dq(v) = 0$.*

Proof. The component equation gives $df(v) = 0$. The chain rule gives $d|f|^2(v) = d(|\cdot|^2)_f(df(v)) = 0$. Subtracting a constant and multiplying by R^2 preserves this zero derivative. Equality of the two functions on the open set identifies their differentials at every point of U. □

Lemma Q.23 (One smooth lift preserving all rounded pair markings)**.** *Let I be a finite toric input. Suppose ϕ is smooth and its Euclidean Hessian satisfies $c\|v\|^2 \le H_\phi(v,v) \le U\|v\|^2$ for fixed $c, U > 0$. Suppose outside a positive logarithmic radius L it equals ϕ_{end}, where ϕ_{end} is globally C^1, smooth off zero and satisfies $\phi_{\mathrm{end}}(ru) = r^2\phi_{\mathrm{end}}(u)$ for every $r > 0$. Then there is $F > 0$ such that for every $D > F$ there is $\beta_0 > 0$ with the following property. For every $\beta \ge \beta_0$, every $R > 0$ and every*

$$F^2\beta^2/2 \le A_- < A := D^2\beta^2/2$$

there is $0 < \delta < A - A_-$ for which every profile of width $0 < \epsilon \le \delta$ admits one globally smooth native tangent field V on $\mathbb{R} \times Y$, open sets O, B_a, B_r, B_f, and equations

$$ds(V) = 1 \text{ on } O, \qquad dq(V) = 0 \text{ on } O \cap B_a, \qquad d(s,f)(V) = (1,0) \text{ on } O \cap B_r, \qquad d\phi(V) = 0 \text{ on } O \cap B_f.$$

Here O contains the literal compact total rounded family $C = \{0 \le s \le 1, q \le 0\}$; B_a contains its zero boundary; B_r contains the full truncated ribbon $C_r = \{0 \le s \le 1, f_s = 1, \phi \le A_-\}$; and B_f contains $\{0 \le s \le 1, f_s = 1, \phi = A_-\}$. All choices come from the finite analytic hypotheses. In particular neither a compatible lift nor a supplied family trivialization is assumed.

Proof. Choose F and the first scale threshold from the compact ribbon lift, independently of D. For each $D > F$, take the maximum of that threshold and the rounded outwardness threshold. The latter produces K_* and, after R, A_-, a width bound $0 < \delta < A - A_-$. This order ensures a nonempty interval of admissible truncations. Fix a profile of width $0 < \epsilon \le \delta$.

The existing compact ribbon lift V_r satisfies $d(s,f)(V_r) = (1,0)$ on an open set O_r and $d\phi(V_r) = 0$ on an open truncation collar. Intersect O_r with $\{z - x > \epsilon\}$ from the unrounded-germ proof. It still contains the entire compact ribbon, and the compatibility lemma gives $dq(V_r) = 0$ throughout it. By normality choose B_r around C_r whose closure lies in this intersection. Choose B_f around the compact ribbon frontier with closure contained in B_r and the original truncation collar.

On the actual rounded zero boundary, the literal carrier inclusion gives $\phi \le A$. The proved rounded derivative $dq(Z_{s,K_*}) > 0$ makes the spatial covector nonzero, hence surjective. The joint parameter-level normal form therefore constructs actual smooth local fields satisfying $d(s,q)(V) = (1,0)$. Their open sources cover the compact zero boundary. Choose B_a around that boundary with closure contained in their union.

Use four affine constraints: the parameter equation everywhere, the rounded equation on $\overline{B_a}$, the fiber equation on $\overline{B_r}$, and the fiber and potential equations on $\overline{B_f}$. At points of $\overline{B_r}$ use V_r, which already solves the rounded equation and solves the potential equation where the last constraint applies. Away from $\overline{B_r}$ but in $\overline{B_a}$ use one of the scalar local lifts, shrinking its source to avoid $\overline{B_r}$. Away from both closed collars use a local parameter-coordinate lift, shrinking its source to avoid both. The inclusion $\overline{B_f} \subset B_r$ makes the last constraint inapplicable in the latter two cases. The smooth affine gluing theorem constructs one global smooth field satisfying these constraints on an open neighborhood O of C. Project its four affine equations to obtain the displayed equations on the actual open collars. □

Q.2.4 One flow for both members of the marked pair

Lemma Q.24 (Simultaneous compact transport from compatible local equations)**.** *Let Q be a finite-dimensional Hausdorff second-countable smooth manifold, $\tau, q, \rho : Q \to \mathbb{R}$ and $f : Q \to F$ smooth, where F is a real normed vector space. Suppose*

$$C = \{0 \le \tau \le 1,\ q \le 0\} \quad \textit{is compact}, \qquad T = \{0 \le \tau \le 1,\ f = w,\ \rho \le a\} \subset C.$$

Let open sets O, B, R, J contain respectively C, its zero face $\{0 \le \tau \le 1, q = 0\}$, T, and the marked face $\{0 \le \tau \le 1, f = w, \rho = a\}$. Suppose a smooth vector field on O satisfies

$$d\tau(V) = 1 \text{ on } O, \quad dq(V) = 0 \text{ on } O \cap B, \quad d(\tau, f)(V) = (1,0) \text{ on } O \cap R, \quad d\rho(V) = 0 \text{ on } O \cap R \cap J.$$

Then there is a globally defined smooth flow Φ such that whenever $\tau(p) + t \in [0,1]$, it transports C and T with parameter $\tau(\Phi_t(p)) = \tau(p) + t$. On C it preserves the zero face in both directions; on T it preserves $f = w$ and the face $\rho = a$ in both directions. The same flow realizes all these assertions.

Proof. Choose a compactly supported smooth cutoff equal to one on an open neighborhood P of C and supported inside O. Multiplying V by this cutoff gives a complete smooth field and its all-real smooth flow. On P the parameter equation remains $d\tau(V) = 1$, while on $P \cap B$ the rounded equation has zero derivative. Apply marked-curve confinement with constant auxiliary function $f_0 = 0$, truncating function q and level zero. It proves transport of C and forward preservation of its zero face throughout every allowed parameter interval.

The same cutoff is one near T, since $T \subset C$. Apply confinement again on $P \cap R$, now with the original f, truncation ρ and level a. This proves transport of T and forward preservation

of its marked face. Applying each assertion from $\Phi_t(p)$ for time $-t$ proves the reverse boundary implications, because the flow law gives $\Phi_{-t}\Phi_t(p) = p$. Thus both carriers and both faces are transported by one smooth flow. □

Lemma Q.25 (One actual flow for the rounded tailoring pair)**.** *For a smooth potential with positive lower and upper Hessian bounds and a homogeneous end, there is $F > 0$ such that every $D > F$ admits a scale threshold $\beta_0 > 0$. For every $\beta \geq \beta_0$, every $R > 0$ and*

$$F^2\beta^2/2 \leq A_- < A = D^2\beta^2/2$$

there is $0 < \delta < A - A_-$ such that every allowed profile of width $0 < \epsilon \leq \delta$ admits one globally defined smooth flow on $\mathbb{R} \times Y$ transporting both

$$E_s = \{q_s \leq 0\}, \qquad F_s^- = \{f_s = 1,\ \phi \leq A_-\}$$

throughout $s \in [0,1]$. It translates the parameter and preserves each of the faces $q_s = 0$ and $\phi = A_-$ in both directions.

Proof. Construct the compatible native field on neighborhoods of the rounded total space, its boundary, the whole ribbon and its frontier. The strict gap $A_- < A - \epsilon$ ensures that $q_s = R^2(|f_s|^2 - 1)$ on a neighborhood of the ribbon. In particular every ribbon point lies on $q_s = 0$, so the ribbon total space lies in the compact rounded total space. Apply the preceding simultaneous transport lemma to the actual functions $\tau(s,y) = s$, $q(s,y) = q_s(y)$, $f(s,y) = f_s(y)$ and $\rho(s,y) = \phi(y)$. Its single supported flow gives both assertions and both boundary equivalences. The order $F < D$ ensures that the displayed interval of permitted inner levels is nonempty for every positive β. □

Q.2.5 The same native carriers for the rounded tailoring pair

Lemma Q.26 (Identity smooth embedding)**.** *The identity of a smooth manifold is a smooth embedding.*

Proof. It is smooth and induces the unchanged topology. Its differential is the identity linear map at every point, hence injective. □

Lemma Q.27 (One nested pair of native compact carriers)**.** *Let I be a finite toric input. Let ϕ be smooth with Hessian bounds $c\|v\|^2 \leq H_\phi(v,v) \leq U\|v\|^2$, where $c, U > 0$. Suppose outside a positive logarithmic radius L it equals a globally C^1 potential ϕ_{end} that is smooth off zero and satisfies $\phi_{\mathrm{end}}(ru) = r^2\phi_{\mathrm{end}}(u)$ for every $r > 0$. There is $F > 0$ such that for every $C_+ > F$ there is $\beta_0 > 0$ with the following property. For every $\beta \geq \beta_0$, every $R > 0$, and every $F^2\beta^2/2 \leq A_- < A = C_+^2\beta^2/2$, there is $0 < \delta < A - A_-$ such that every rounding profile of width $0 < \epsilon \leq \delta$ admits native compact domains E of dimension $2n$ and D of dimension $2n-2$, smooth embeddings*

$$\chi : E^{\mathrm{ext}} \hookrightarrow Y, \qquad i : D^{\mathrm{ext}} \hookrightarrow E^{\mathrm{ext}}, \qquad k : D^{\mathrm{ext}} \hookrightarrow \partial E,$$

one constructed flow Φ on $\mathbb{R} \times Y$, and a jointly smooth embedding family $J_s : E^{\mathrm{ext}} \hookrightarrow Y$ such that both compact carriers are nonempty, the entire range of i lies in the actual frontier of E, and

$$\iota_{\partial E} \circ k = i \quad \text{on } D^{\mathrm{ext}}, \qquad J_0 = \chi, \qquad J_s(x) = \mathrm{pr}_Y\, \Phi_s(0, \chi(x)).$$

The ribbon family is the literal composition $R_s = J_s \circ i$ on the whole common extension and is jointly smooth there. Put $f_s = f_{\beta,s}$ and $q_s(y) = m_\epsilon(\phi(y) - A, R^2(|f_s(y)|^2 - 1))$. For every $s \in [0,1]$ the four literal image equalities are

$$J_s(E) = \{q_s \leq 0\}, \qquad J_s(\partial E) = \{q_s = 0\}, \qquad R_s(D) = \{f_s = 1, \phi \leq A_-\}, \qquad R_s(\partial D) = \{f_s = 1, \phi = A_-\}.$$

The source embedding has these same ambient image equalities at $s = 0$. This statement constructs carriers and compatible maps; it does not yet assert their smoothly varying Liouville structures or completion markings.

Proof. Choose F as the maximum of the pair-transport radius, the full-fiber marking radius, and the nonempty full-fiber truncation radius. For each $C_+ > F$, take the maximum of the three relevant scale thresholds and the rounded-outwardness threshold. For β, R, A_- as stated take the minimum of the pair-transport and outwardness width bounds. It is positive and still strictly below $A - A_-$. The outwardness coefficient is used internally to prove regularity; no unspecified coefficient is returned as a Liouville threshold certificate.

At $s = 0$, the actual rounded Liouville derivative $dq_0(Z) > 0$ on the zero level makes dq_0 nonzero there. The regular scalar-sublevel construction therefore supplies a native compact domain E_0 with carrier $\{q_0 \le 0\}$ and actual frontier $\{q_0 = 0\}$. The full-fiber marking producer supplies the smooth embedding $j : N \hookrightarrow Y$ with entire image $\{f_0 = 1\}$ and its native compact potential truncation D_0. Its whole carrier image is $\{f_0 = 1, \phi \le A_-\}$ and its frontier image has $\phi = A_-$. The separate proved full-fiber nonemptiness producer shows $D_0 \neq \varnothing$.

Take the one constructed simultaneous pair flow Φ. Its parameter coordinate equals s along E_0, so the projected-flow embedding neighborhood theorem, applied to the identity ambient embedding, produces one open neighborhood U_a of E_0 and the jointly smooth embeddings J_s on that common neighborhood. Set $E = E_0|_{U_a}$ and let χ be the native open-subset inclusion into Y. Restrict the ribbon extension to

$$U_r = \{x \in N : j(x) \in U_a,\ \phi(j(x)) < A - \epsilon\}.$$

The strict gap and unchanged base face put every point of D_0 in U_r, so $D = D_0|_{U_r}$ retains its entire carrier and frontier. The restricted embedding $j|_{U_r}$ has range inside the image of χ. Smooth embedding factorization constructs i with $\chi \circ i = j|_{U_r}$. Every point of this extension still has $f_0 = 1$ and $\phi < A - \epsilon$, hence $q_0 = 0$ by the literal unrounded germ. Thus the entire range of i lies in the native frontier of E. Factor it through the native boundary inclusion to construct k and the displayed equality on the entire extension. The retained nonempty ribbon carrier also gives nonemptiness of the ambient carrier.

Restriction identities give the two literal source ambient images. Composition with the fixed smooth map i makes $R_s = J_s \circ i$ jointly smooth, and the factorization identifies its flow formula with $\mathrm{pr}_Y \Phi_s(0, j(x))$. At time zero the global flow identity gives $J_0 = \chi$ on the whole extension. Forward pair transport proves containment in all four displayed images. Conversely, flow backward for time $-s$ from an arbitrary point of any one of the four target equation sets. The same pair transport brings it into the corresponding initial full carrier, retaining the appropriate zero or truncation boundary value. The entire initial carriers lie in U_a and U_r, respectively. Their literal initial image equalities therefore give preimages in the retained native carriers. The flow identity $\Phi_s \Phi_{-s} = \mathrm{id}$ gives the required forward preimage exactly. This proves equality of the whole images and frontier images, rather than just containment of transported selected points. □

Q.2.6 Liouville structures on transported carriers

Lemma Q.28 (Actual Liouville pullback on a compact sublevel carrier)**.** *Let D be a compact smooth domain with boundary in a d-dimensional extension N, and let $j : N \hookrightarrow M$ be a smooth embedding into another d-dimensional manifold. Let α be a smooth one-form on M with $d\alpha$ everywhere nondegenerate. Suppose $q : M \to \mathbb{R}$ is smooth, $q \circ j \le 0$ on D, $q \circ j = 0$ on ∂D, and*

$$dq(Z_\alpha)(j(x)) > 0 \qquad (x \in \partial D).$$

Then D has a Liouville structure with primitive $j^*\alpha$ and its canonical field. Its exterior derivative is symplectic on all of N, and $dj(Z_{j^*\alpha}) = Z_\alpha \circ j$.

Proof. The derivative of j is injective and its source and target have equal finite dimension, so it is bijective. Naturality of the exterior derivative then makes $d(j^*\alpha) = j^*d\alpha$ nondegenerate everywhere. The contraction equation uniquely determines the Liouville fields, and applying the bijective derivative proves their displayed transport identity. The chain rule now gives $d(q \circ j)(Z_{j^*\alpha}) = dq(Z_\alpha) \circ j > 0$ on the boundary. Since $q \circ j$ is nonpositive on the carrier and zero on its boundary, outward transfer to any native boundary defining function gives the required outwardness. Smooth inversion of the nondegenerate two-form makes the canonical field smooth. These data form the claimed Liouville structure. □

Lemma Q.29 (Smooth pulled-back family and localized completions)**.** *Under the preceding hypotheses for every $t \in [0,1]$, suppose the embeddings j_t and ambient primitives α_t vary smoothly jointly with the parameter, with smooth extensions across the interval endpoints. Then the Liouville primitives $j_t^*\alpha_t$ and their canonical fields form smooth interval families on the same compact domain, and these structures admit a constructed localized completion family.*

Proof. Apply the preceding construction at each parameter. Pullback of jointly smooth map and one-form families gives smoothness of $j_t^*\alpha_t$, including at both endpoints. Nondegeneracy holds on the entire extension, so smooth matrix inversion in charts makes the canonical Liouville fields jointly smooth. The compact Liouville-family completion construction then supplies the localized completion family for these very structures. □

Lemma Q.30 (Actual ambient Liouville families on the compatible pair carriers)**.** *For a smooth uniformly positive-Hessian potential with a homogeneous end, there is $F > 0$ such that every $C_+ > F$ admits a sufficiently large scale threshold. For each such β there is $K_* > 0$ preceding all choices $R > 0$ and $F^2\beta^2/2 \le A_- < A = C_+^2\beta^2/2$. One positive width $\delta < A - A_-$ then works for every profile with $0 < \epsilon \le \delta$. The actual compatible nonempty fixed ambient and ribbon carriers E, D, native boundary embedding $k : D^{\mathrm{ext}} \hookrightarrow \partial E$, and their same-flow embeddings J_t can be chosen before every $K \ge K_*$. On that same E, every such coefficient has Liouville structures*

$$\Lambda_{K,t}.\lambda = J_t^*\big(\lambda_\phi + K f_t^*\lambda_{\mathbb{C}}\big).$$

For each fixed coefficient their primitives and canonical fields are smooth interval families, symplectic throughout the common extension, with constructed localized completion families. All literal full carrier and frontier images and the original same-flow formulas are retained.

Proof. Choose F from the compatible-carrier construction, before C_+. Increase the scale threshold to cover that construction, global tailored symplecticity and uniform rounded outwardness. The latter supplies K_* before R and A_-. Take the smaller of its admissible width and the compatible-carrier width. The carrier construction then supplies E, D, k, J_t independently of the subsequent coefficient.

For $K \ge K_*$, the literal ambient image equality implies $q_t \circ J_t \le 0$ on E, and the frontier image equality implies $q_t \circ J_t = 0$ on ∂E. The profile lower bound puts their images inside the finite outer potential sublevel. Thus the proved rounded exterior estimate gives strict outward derivative there. Global tailored symplecticity and the affine primitive-family identity supply the other hypotheses of the preceding pullback-family lemma. Applying it on the same E for every such K gives the claimed structures, fields and localized completions. The construction changes none of the carrier or embedding data. □

Q.2.7 Liouville data on the supplied tailoring ribbon carrier

Lemma Q.31 (Primitive family for prescribed ribbon embeddings)**.** *Fix a smooth potential with positive lower and upper Hessian bounds. For all sufficiently large β, let D be any compact domain of dimension $2n-2$ and J_t a smooth interval family of embeddings into Y whose compact carrier satisfies $f_t(J_t x) = 1$. Then the actual tangent image at every carrier point is* $\ker df_t$*, the pulled-back primitives $\mu_t = J_t^* \lambda_\phi$ form a smooth family, and their differentials are symplectic on one common open neighborhood of D.*

Proof. The native positive Poisson estimate on $|f_t| \geq 1$ makes df_t surjective at every point of the indicated level. The level equation on the whole native compact carrier differentiates to zero in every native tangent direction, including at the boundary. The embedding is immersive, and its dimension is exactly the dimension of the kernel, so the tangent image equals that kernel. Joint pullback smoothness gives the actual primitive family. The rank-two symplectic kernel lemma and positive Poisson scalar show that $d\mu_t$ is nondegenerate at every compact carrier point. Joint continuity of this family, compactness of the parameter interval and openness of nondegeneracy then give one common open spatial neighborhood on which all $d\mu_t$ are nondegenerate. Since each is an exterior derivative, it is closed and hence symplectic. □

Lemma Q.32 (Outwardness on the same supplied carrier)**.** *Assume additionally that the potential agrees outside a fixed ball with a degree-two homogeneous potential, continuously differentiable at zero and smooth away from zero. There are $F, \beta_0 > 0$ such that for $\beta \geq \beta_0$ and $A \geq F^2\beta^2/2$, every supplied smooth interval family J_t of compact-domain embeddings with full carrier image $\{f_t = 1, \phi \leq A\}$ and frontier image $\{f_t = 1, \phi = A\}$ has strictly outward canonical fields for the actual pulled primitives μ_t on the same carrier. The tangent, smoothness and common symplectic neighborhood conclusions above remain valid.*

Proof. Let F be the threshold in the whole-end intrinsic exhaustion estimate. The potential has a finite upper bound V on the fixed exceptional ball. Increase β_0 so that $F^2\beta^2/2 > V$; every image of a frontier point therefore lies in the homogeneous end. At such a point Euler's identity gives $r_\phi^2 = 2A$, so $r_\phi \geq F\beta$. The intrinsic estimate, using the proved tangent kernel, gives $d(\phi \circ J_t)(Z_{\mu_t}) \geq r_\phi^2/2 = A > 0$. The full carrier image gives $\phi \circ J_t - A \leq 0$ on the carrier, and the frontier image gives equality on its boundary. The native sublevel-normal comparison lemma therefore transfers this positive derivative to every stored boundary defining function of the supplied compact domain. No carrier or embedding is replaced. □

Lemma Q.33 (One common restriction gives the actual Liouville family)**.** *Under the preceding hypotheses and parameter ranges, there is one common open neighborhood O of the supplied carrier such that the same compact domain, with extension restricted to O, carries actual Liouville structures with primitive $J_t^* \lambda_\phi$. Their primitives and canonical fields are smooth interval families, their differentials are symplectic on the whole restricted extension, and their compatible localized completion family is constructed. The embeddings are literally the original J_t composed with the native open inclusion; their full carrier and frontier images and tangent kernels are retained. Every rank-two ambient tailoring update pulls back to that same primitive on the compact ribbon carrier.*

Proof. Restrict the original extension to the common symplectic neighborhood produced above. Exterior differentiation commutes with this open restriction, so the pulled primitives are globally symplectic there. The canonical contraction solution is the same native vector as on the original extension at every restricted point; its contraction and smoothness follow from nondegeneracy. The proved outwardness transfers under the identity tangent map of the native inclusion. This

constructs the actual Liouville structures, and joint pullback smoothness together with smooth inversion of contraction constructs their interval field family. The native completion-family existence theorem applies to these actual structures. Restriction projects the compact carrier and its frontier onto exactly the original carrier and frontier, preserving both literal image equations. Their tangent images remain $\ker df_t$. Finally the rank-two update term vanishes on every vector in that kernel, so every ambient coefficient yields the same pulled primitive there. □

Q.2.8 Actual completed witnessed pairs along tailoring

Lemma Q.34 (The primitive square at the native boundary)**.** *Let E be a compact smooth domain, N, M smooth manifolds, $k : N \hookrightarrow \partial E$ and $J : E^{\mathrm{ext}} \hookrightarrow M$ smooth embeddings, and α a one-form on M. Then*

$$k^*((J^*\alpha)|_{\partial E}) = (J \circ \iota_{\partial E} \circ k)^*\alpha.$$

Proof. The boundary form is the pullback by the native boundary inclusion. Functoriality of pullback first combines this inclusion with k, and then combines the resulting map with J. This gives the displayed equality at every point and on every tangent vector. □

Lemma Q.35 (The entire frontier after collar localization)**.** *For a realized Liouville flow collar of a compact domain E, let $a : E^{\mathrm{ext}}_{\mathrm{loc}} \hookrightarrow E^{\mathrm{ext}}$ be its original inclusion. Then $a(\partial E_{\mathrm{loc}}) = \partial E$, including when this frontier is empty.*

Proof. Both native boundary manifolds are the same manifold ∂E. Their native inclusions have images equal to the respective frontiers, and the original inclusion composed with the localized boundary inclusion is the original boundary inclusion. Every point of the localized frontier is therefore sent into the original frontier. Conversely write any original frontier point as the original boundary inclusion of $b \in \partial E$. The localized boundary inclusion of the same b is a preimage, proving the reverse containment. □

Let I be a finite toric input. Let ϕ be smooth with Hessian bounds $c\|v\|^2 \le H_\phi(v,v) \le U\|v\|^2$, where $c, U > 0$. Suppose outside a positive logarithmic radius L it agrees with a globally C^1 function ϕ_{end}, smooth away from zero, with $\phi_{\mathrm{end}}(ru) = r^2\phi_{\mathrm{end}}(u)$ for all $r > 0$. There is $F > 0$ such that every $C_+ > F$ admits $\beta_0 > 0$ with the following property. For every $\beta \ge \beta_0$ there is $K_* > 0$ preceding all choices of $R > 0$ and

$$F^2\beta^2/2 \le A_- < A = C_+^2\beta^2/2.$$

There is $0 < \delta < A - A_-$ such that every rounding profile of width $0 < \epsilon \le \delta$ admits nonempty native compact carriers E, D of dimensions $2n, 2n-2$, a smooth embedding $k : D^{\mathrm{ext}} \hookrightarrow \partial E$, and jointly smooth embeddings $J_t : E^{\mathrm{ext}} \hookrightarrow Y$ and $W_t = J_t \circ \iota_{\partial E} \circ k : D^{\mathrm{ext}} \hookrightarrow Y$. For every $t \in [0,1]$, writing $q_t = m_\epsilon(\phi - A, R^2(|f_t|^2 - 1))$, their literal images are

$$J_t(E) = \{q_t \le 0\}, \quad J_t(\partial E) = \{q_t = 0\}, \qquad W_t(D) = \{f_t = 1, \phi \le A_-\}, \quad W_t(\partial D) = \{f_t = 1, \phi = A_-\}.$$

These same carriers have Liouville structures $\Lambda_{K,t}$ for every $K \ge K_*$ and M_t, with exact primitives

$$\Lambda_{K,t}.\lambda = J_t^*(\lambda_\phi + K f_t^* \lambda_{\mathbb{C}}), \qquad M_t.\lambda = W_t^*\lambda_\phi.$$

For every such coefficient there are realized ambient and ribbon collar families A_t, B_t, their original inclusions a_t, b_t, and an actual pair deformation Q including both smooth completed families, whose tuple at t is exactly the localized witnessed pair

$$\big(E_{A_t}, a_t^*\Lambda_{K,t}.\lambda, D_{B_t}, k \circ b_t, 1\big).$$

The completions in this tuple are the cylindrical completions of these same collars. The four displayed full and frontier image equalities also hold for its actual localized ambient and ribbon carriers under $J_t \circ a_t$ and $W_t \circ b_t$, respectively. The compact primitives vary smoothly, and the ambient and ribbon families retain the very same compact and completion identifications used in Q. Thus subsequent maps on these completions can be expressed in these chosen family coordinates without any change of presentation.

Proof. Let F_a be the radius threshold for the compatible ambient family and F_r the threshold for the supplied-ribbon Liouville family. Take $F = \max(F_a, F_r)$. It is positive. For $C_+ > F$, take the maximum of the ambient scale threshold for C_+ and the supplied-ribbon scale threshold. The ambient theorem then supplies K_* before R, A_-, and its width δ has precisely the required strict gap. The lower bound using F implies both lower bounds using F_a, F_r, since both thresholds are positive and at most F.

For the chosen profile the ambient theorem produces the same nonempty E, D_0, boundary embedding k_0, ambient embeddings J_t and structures $\Lambda_{K,t}$ for every $K \geq K_*$. Write $R_t = J_t \circ \iota_{\partial E} \circ k_0$; its supplied whole carrier and frontier images are exactly the stated full-fiber truncations. Apply the supplied-ribbon theorem to D_0, R_t, A_-. It produces one open neighborhood O of the entire carrier and actual structures M_t on $D = D_0|_O$. Let $v : D^{\mathrm{ext}} \hookrightarrow D_0^{\mathrm{ext}}$ be the native inclusion and set $k = k_0 \circ v$. Then $W_t = R_t \circ v$ on the entire extension, by the original boundary factorization. Consequently it is jointly smooth and has exactly the retained whole and frontier images. Restriction retains nonemptiness because every point of the original carrier belongs to O. Its primitive is the exact pullback $W_t^* \lambda_\phi$.

Fix $K \geq K_*$. The actual tangent-kernel result in the supplied-ribbon theorem makes the rank-two update pull back to zero on every ribbon carrier tangent vector. The native boundary pullback square therefore gives

$$k^*(\Lambda_{K,t}.\lambda|_{\partial E}) = W_t^*(\lambda_\phi + K f_t^* \lambda_{\mathbb{C}}) = M_t.\lambda \quad \text{on } D.$$

Thus the constant function 1 is the positive witnessing factor. The fixed map k and constant witnessing function extend smoothly to all real parameters; both primitive families have the joint smooth endpoint extensions supplied by the two actual Liouville producers. Apply the localized witnessed-pair construction to these very structures, embeddings and witnesses. It constructs both collar families and both smooth completed families, and its defining pair tuple is the one displayed above.

Each original collar inclusion maps the localized compact carrier onto the entire original compact carrier. The preceding frontier lemma proves the corresponding equality for the entire frontier. Composing these four equalities with J_t, W_t proves all four asserted localized image equalities. □

Remark Q.36. This is a pair deformation in the sense of the specified compact and completed smooth data. A smooth family of inclusion completion markings onto the named outer spaces and full fibers, and strict identifications with the normalized original endpoint tuple, are additional geometric requirements. They are not asserted by this theorem.

Q.2.9 Finite negative absorption in the actual outer completion

Lemma Q.37 (Compact confinement and entry in a completion)**.** *For every actual completion of a compact Liouville domain, its compact piece is invariant under negative flow, and every completed point reaches that piece in finite nonpositive time.*

Proof. The completion is covered by its compact piece and the nonnegative boundary end, parametrized by the same complete flow. An end point at radius $r \geq 0$ reaches the boundary at time $-r$, proving entry. For confinement, suppose a compact point flowed backwards for time $t \leq 0$ becomes exterior. The covering expresses this exterior point as a boundary end point at radius $r > 0$: radius zero is in the compact boundary. Flowing forward by $-t$ then puts the original compact point at radius $r - t > 0$. The completion overlap law excludes every strictly positive end point from the compact piece, a contradiction. □

Lemma Q.38 (Absorption inside a compact negatively invariant region)**.** *Let Φ be a continuous all-real flow, C a compact set invariant under nonpositive flow, and ρ, q continuous real functions such that $\frac{d}{dt}\rho(\Phi^t x) = q(\Phi^t x)$ at every time. If $q(x) > 0$ for $x \in C$ with $\rho(x) \geq a$, then every point of C reaches $\{\rho \leq a\}$ in finite nonpositive time.*

Proof. Suppose a point $x \in C$ never reaches that sublevel in negative time. Then its negative orbit lies in the compact set $C \cap \{\rho \geq a\}$, where continuity gives a positive minimum c of q. The mean-value inequality on $[-T, 0]$ gives $cT \leq \rho(x) - \rho(\Phi^{-T}x)$. Choose $T = (\rho(x) - a + 1)/c > 0$. Since the assumed failure of entry gives $\rho(\Phi^{-T}x) > a$, the right side is strictly smaller than $\rho(x) - a = cT - 1$, a contradiction. □

Lemma Q.39 (Native cylindrical scalar absorption)**.** *Let D be a compact Liouville domain and ρ a smooth function on its extension. If $d\rho(Z_D) > 0$ on $D \cap \{\rho \geq a\}$, every point of the actual cylindrical completion constructed from any native flow collar reaches $D \cap \{\rho \leq a\}$ in finite nonpositive time. Here ρ on the completed space is its pullback by the actual native open-subset projection; the field derivative hypothesis is only on D.*

Proof. The native projection is the composition of two open-subset inclusions, so its derivative is the identity in native tangent fibers. The completed field at a compact point is exactly Z_D. Thus the pulled-back scalar has the stated positive derivative throughout the compact exterior. Its derivative along the smooth complete flow is continuous and equals its differential on that very completed field. The first lemma gives entry into and negative confinement in the compact piece; the second gives subsequent finite entry into $\{\rho \leq a\}$. Adding the two nonnegative backward times proves the conclusion. □

Lemma Q.40 (Actual uniform rounded tailoring absorption)**.** *Assume a smooth potential with positive lower and upper Hessian bounds that agrees off a fixed ball with a degree-two homogeneous potential, continuously differentiable at zero and smooth away from zero. Fix $D > 0$. For every sufficiently large β, there is a common $K_* > 0$ such that for every $R > 0$ and $A_- < A = D^2\beta^2/2$ there is $0 < \delta < A - A_-$ with the following property. For every allowed profile of width $0 < \epsilon \leq \delta$, every $K \geq K_*$ and $s \in [0, 1]$, the literal nonempty outer domain $\{\phi \leq A\}$ with actual primitive $\lambda_{s,K}$ is a compact Liouville domain. In the actual cylindrical completion constructed from any of its native flow collars, every point enters the literal rounded domain $E_s = \{q_s \leq 0\}$ in finite nonpositive time, using the native projection to the original space.*

Proof. Increase the scale threshold to satisfy both the finite outer-domain producer and the rounded exterior outwardness producer. The former provides compactness, nonemptiness, actual symplecticity and strict outer boundary derivative, which construct the regular-sublevel Liouville domain on the literal outer carrier. The latter provides K_* and δ in the displayed parameter order and proves $dq_s(Z_{s,K}) > 0$ throughout $\{\phi \leq A, q_s \geq 0\}$. Apply the preceding lemma to this produced domain, smooth scalar q_s, and level $a = 0$. Its actual primitive and field are the specified tailoring data, so the derivative hypothesis follows directly from the rounded producer. The resulting entry scalar inequality is precisely membership in the literal rounded domain. □

Remark Q.41. This proves finite entry in the completed finite ambient domain. The smooth family of primitive-preserving completion identifications, full ribbon markings, and geometric comparison chain remain additional constructions; none is an input to these absorption lemmas.

Q.2.10 Individual inclusion completion maps for rounded tailoring domains

Lemma Q.42 (Completion localization along a native collar)**.** *Any actual completion of a compact Liouville domain restricts to the native localized presentation defined by a flow collar, preserving its completed ambient, primitive, complete flow and all-real end parametrization. The compact inclusion is the composition with the native open inclusion.*

Proof. The localized primitive is the pullback of the original primitive by the two native open inclusions. Their differential is the identity in native tangent fibers, and the localized canonical field equals the original field on the compact carrier. These facts give the compact primitive and field equations after composing the original completion inclusion. The localized compact carrier projects onto exactly the original compact carrier; thus covering and positive-end overlap are unchanged. The localized boundary is the same presented boundary and its primitive is unchanged, so the original all-real end equations supply all end data. □

Lemma Q.43 (A native inclusion completion marking from absorption)**.** *Let P be a complete Liouville manifold and ρ a smooth scalar with compact nonpositive sublevel and positive Liouville derivative at every point of $\{\rho = 0\}$. Suppose every point reaches $\{\rho \le 0\}$ in finite nonpositive time. The native cylindrical completion of this regular sublevel, using a collar supplied by collar existence, admits a primitive-preserving inclusion marking onto P. Its compact formula is the literal native inclusion, and its end formula is the flow in P at every real radius.*

Proof. Strict derivative positivity makes $d\rho$ nonzero on the zero level, so the regular-sublevel theorem constructs the compact domain. Boundary outwardness and negative entry realize P itself as a completion of that domain, with identity compact inclusion and the actual boundary flow. Localize this completion to the same native collar presentation as the chosen cylindrical completion by the preceding lemma. The strict comparison theorem for two completions of one represented compact domain gives a primitive-preserving smooth equivalence between them. Its compact square is the literal native inclusion. Its all-real end square is the same boundary flow in P, giving the claimed marking. □

Lemma Q.44 (Actual individual rounded tailoring completion maps)**.** *Assume a smooth potential with positive lower and upper Hessian bounds, agreeing outside a fixed ball with a degree-two homogeneous potential that is continuously differentiable at zero and smooth away from zero. Fix $D > 0$. For every sufficiently large β there is $K_* > 0$ such that for every $R > 0$ and $A_- < A = D^2\beta^2/2$ there is $0 < \delta < A - A_-$ with the following property. For every allowed profile of width $0 < \epsilon \le \delta$, $K \ge K_*$, and $s \in [0,1]$, the actual rounded compact domain inside the constructed outer completion admits a native cylindrical completion with a primitive-preserving inclusion marking onto that outer completion. Its projected compact carrier is exactly the literal $E_s = \{q_s \le 0\}$, its compact primitive is the prescribed $\lambda_{s,K}$, and the marking equals literal inclusion on every compact point.*

Proof. Use the finite-input outer-domain and rounded exterior estimates in the displayed parameter order to construct the literal outer domain $\{\phi \le A\}$ and its native cylindrical completion P. Pull q_s back by its actual native projection. The profile lower bound implies that its nonpositive sublevel lies in the outer compact carrier; closedness therefore makes this sublevel compact. At a zero of the pulled scalar, the point lies in the compact carrier, where the completed field is the original

tailoring field. Thus the proved strict rounded exterior derivative gives regularity and outwardness there. Native cylindrical scalar absorption gives finite entry for every point of P. Apply the preceding marking lemma to this scalar and complete ambient. The resulting native compact inclusion projects onto exactly E_s: one inclusion follows from the defining scalar equation, and the converse lifts every E_s point through the outer compact homeomorphism since $E_s \subseteq \{\phi \leq A\}$. The completed primitive at every compact point is exactly the original $\lambda_{s,K}$, giving the required compact primitive identity as well as the literal inclusion formula. □

Remark Q.45. These are individual inclusion completion maps. Their jointly smooth parameter family requires compatible native presentations of the varying rounded domains and their boundaries; choosing each collar individually does not provide that compatibility.

Q.2.11 Comparing compact presentations through their embedded images

Lemma Q.46 (Compact diffeomorphism from a common embedded image)**.** *Let D, E be compact smooth domains of dimension d, let M be a smooth d-manifold, and let j, k be smooth embeddings of their respective open extension manifolds into M. Suppose D is nonempty and $j(D) = k(E)$. There is a compact diffeomorphism $h : D \to E$ with smooth extensions of it and its inverse to open neighborhoods, satisfying $k \circ h = j$ on the source neighborhood and $j \circ h^{-1} = k$ on the target neighborhood. This includes $d = 0$.*

Proof. An embedding between manifolds of equal dimension has bijective differential, so the inverse function theorem makes both images open and both inverse maps smooth there. On $U = j^{-1}(\operatorname{im} k)$ define $h = k^{-1}j$; on $V = k^{-1}(\operatorname{im} j)$ define $h^{-1} = j^{-1}k$. The image equality puts the entire compact carriers inside U, V and shows these formulas map them bijectively to each other. They are mutually inverse on these neighborhoods. In particular their differentials are inverse and h is immersive at every compact point, including boundary points. Nonemptiness of D and the image equality give points of both extension manifolds; use any such points to extend the two functions outside U, V. Their smoothness is required only on U, V, so these extensions do not affect the compact diffeomorphism or either open-neighborhood square. □

Lemma Q.47 (Differential forms on the same embedded compact domain)**.** *For any compact diffeomorphism h with $k \circ h = j$ on its open source neighborhood, and any differential form α on M, one has $h^*(k^*\alpha) = j^*\alpha$ at every point of D.*

Proof. At every point of D the map equality holds on an open neighborhood. Differentiating gives $dk \circ dh = dj$ on the full tangent space, also at a boundary point. Substituting this equality in each argument of α gives the stated equality of alternating forms. □

Q.2.12 Inverse regularity for the same parameterized maps

Lemma Q.48 (Smooth inverse germs of a family of diffeomorphisms)**.** *Let M, N be smooth manifolds of the same dimension, and let $e_t : M \to N$ be a diffeomorphism for each $t \in [0, 1]$. Suppose near every (t, x) the function $(s, z) \mapsto e_s(z)$ has a smooth extension to an open subset of $\mathbb{R} \times M$. Then the actual inverse family $(s, y) \mapsto e_s^{-1}(y)$ has the same local extension property on $\mathbb{R} \times N$.*

Proof. Fix t, y and set $x = e_t^{-1}(y)$. Choose a smooth local extension $f(s, z)$ of the forward family near (t, x), and consider $G(s, z) = (s, f(s, z))$. At (t, x) its derivative is triangular with identity parameter block and invertible spatial block $de_t|_x$. It is therefore invertible, and the inverse function theorem supplies a smooth local inverse L near (t, y). Shrink this neighborhood so that L takes

values inside the chosen extension neighborhood. The equation $GL(s, y') = (s, y')$ implies that the first component of $L(s, y')$ is exactly s. If $s \in [0, 1]$, the second component z' therefore satisfies $e_s(z') = y'$, hence is $e_s^{-1}(y')$ by injectivity. The second component of L is the desired real smooth extension. The argument works at the interval endpoints because the forward extension was defined on an open real neighborhood. □

Q.2.13 The fixed outer potential domain

Lemma Q.49 (One native domain for the varying outer primitives)**.** *Fix a smooth function ρ with compact nonpositive sublevel and regular zero level, and fix its regular-level boundary presentation. Suppose α_t are symplectic primitives whose Liouville fields are strictly outward along $\rho = 0$. The compact domains underlying the resulting Liouville domains are all this same chosen domain H. Identifying each domain with H and transporting its Liouville structure reconstructs the original Liouville domain exactly.*

Proof. The extension manifold, carrier $\{\rho \leq 0\}$, boundary $\{\rho = 0\}$, boundary embedding, and defining function are fixed by the chosen scalar sublevel presentation. None depends on the primitive. The primitive and its Liouville field define only the Liouville structure on this domain. Transporting that structure along the domain identification and pairing it with H therefore gives the original domain with its original structure. For the tailoring outer domains the scalar is the same function $\rho(y) = \varphi(y_u) - A$ for every parameter, so the assertion applies. □

Q.2.14 Strict identifications at the normalized junction

Lemma Q.50 (The full compact pair square determines a strict junction)**.** *Let P, Q be witnessed compact pairs, with their already chosen completions. Let $a : E_P \to E_Q$ and $g : F_P \to F_Q$ be compact diffeomorphisms having smooth extensions on open neighborhoods of the whole compact carriers. Suppose $a^*\lambda_Q = \lambda_P$ and $g^*\mu_Q = c\mu_P$ on these carriers, where $c > 0$ is constant, and suppose for every $x \in F_P$ that*

$$a(\iota_P j_P(x)) = \iota_Q j_Q(g(x)).$$

Then there is a permitted completion identification from P to Q with ribbon map g, scale c, and radial shift zero. Its ambient map on the entire compact piece is the prescribed map a.

Proof. The strict compact-map extension theorem constructs a primitive-preserving completed diffeomorphism f extending a, with boundary diffeomorphism b and end equation $f(y, r) = (b(y), r)$ for $r \geq 0$. Its boundary square says $a\iota_P = \iota_Q b$. Combining this square with the stated full-ribbon square and injectivity of ι_Q gives $bj_P = j_Q g$ at every ribbon point. Thus the completed trajectories of corresponding ribbon points agree at radius zero. The ribbon-trajectory identification theorem constructs a permitted identification I with ambient map f, ribbon map g and scale c. For a fixed boundary point y, its eventual end equation has the form $f(y, r) = (I_\partial(y), r + s(y))$. Choose r both nonnegative and above its eventual threshold. The already proved end equation and injectivity of the completed end coordinates give $r = r + s(y)$, hence $s(y) = 0$. The compact equation is the one supplied by the strict extension theorem. □

Remark Q.51. This lemma supplies the permitted step once the compact ambient and ribbon maps, their primitive equations, and their square on the entire ribbon have been proved. Scalar normalization alone does not supply those maps between independently selected compact presentations.

Lemma Q.52 (Independent collar presentations of the same compact pair)**.** *Let E, F be nonempty compact Liouville domains of dimensions $2n, 2n - 2$, with a fixed boundary embedding $j : F \hookrightarrow \partial E$*

and fixed positive witness b satisfying $j^*(b\alpha_E) = \mu_F$. Choose another Liouville structure on each same compact domain whose primitive equals the original primitive on its whole extension. Choose independently a realized flow collar for each of the four structures. The two localized witnessed pairs admit a permitted strict completion identification of scale one and radial shift zero. On the full ribbon its compact map preserves the original ribbon point. On the ambient compact domain its completed map carries each compact point to a compact point with the same original ambient point. The collar widths need not agree.

Proof. Each original collar inclusion maps the localized compact carrier onto the entire original compact carrier. Nonemptiness therefore passes to the localized carriers. Apply the common-embedding compact-map theorem to the two ambient collar inclusions into E^{ext}, and separately to the ribbon collar inclusions into F^{ext}. This constructs compact diffeomorphisms a, g whose original-inclusion squares hold on open neighborhoods of their full compact carriers.

The localized primitives are pullbacks of the original primitives by these inclusions. The common-embedding form-pullback lemma and the assumed primitive equalities consequently show $a^*\lambda' = \lambda$ and $g^*\mu' = \mu$ on the full localized carriers, including their boundaries. To check the full pair square, compose both sides with the target original ambient inclusion. The square for a replaces its composition by the source inclusion. The native boundary inclusions then reduce both sides to $\iota_E j$ applied to the respective original ribbon projections, which agree by the square for g. Injectivity of the target original ambient inclusion proves the pair square itself. Apply the preceding compact-pair-square lemma at scale one. Its retained compact equations give exactly the asserted original point equations, and its completed end calculation gives radial shift zero. □

Lemma Q.53 (The normalized zero endpoint has a strict collar junction)**.** *Let I be a toric input, β, K real, $R = e^{\beta}$, and W_1 the coefficient-one endpoint polynomial. Let D be a nonempty compact domain embedded in Y by J, with two Liouville structures having primitives*

$$J^*(\alpha + Kf_0^*\lambda_{\mathbb{C}}), \qquad J^*(\alpha + (K/R^2)W_1^*\lambda_{\mathbb{C}}),$$

where f_0 is the zero endpoint of the actual tailoring family and α is any ambient one-form. Suppose a fixed nonempty compact Liouville ribbon F with boundary embedding j is witnessed by the constant function one for the first structure. Independently chosen ambient and ribbon collars give two localized witnessed pairs. There is a permitted strict completion identification between these pairs, with scale one and radial shift zero, preserving original points on the full ribbon and compact ambient domain.

Proof. Exact normalization gives $\alpha + Kf_0^*\lambda_{\mathbb{C}} = \alpha + (K/R^2)W_1^*\lambda_{\mathbb{C}}$. Pull back this equality by the same embedding J to obtain equality of the actual native ambient primitives. In particular their boundary forms agree, so the given witness-one ribbon equation holds for the second structure as well. The ribbon primitive itself is the same on both sides. Apply the independent-collar-pair identification lemma to these structures, the same native domains and embedding, and the four chosen collars. It constructs the claimed permitted identification with the stated point and end equations. □

Remark Q.54. The last lemma is conditional on the two stated structures having been constructed on the same native compact domain with the same embedded ribbon. It does not assert existence of the preceding holomorphic compact family, or a strict identification of its independently chosen endpoint carrier with this common carrier.

Q.2.15 Inclusion markings on the retained rounded presentations

Lemma Q.55 (Transfer from an absorbing regular sublevel)**.** *Let D be a compact Liouville domain with a prescribed completion C, and P a complete Liouville manifold. Let $\rho : P \to \mathbb{R}$ be smooth with compact nonpositive sublevel B, positive Liouville derivative at every zero, and finite negative absorption into B. Suppose a compact diffeomorphism $h : D \to B$ has smooth local extensions and satisfies $h^*\lambda_P = \lambda_D$ on the compact carrier. Then C admits a primitive-preserving diffeomorphism $q : C \to P$, with*

$$q|_D = h, \qquad q(y, r) = \mathrm{Fl}^r_{Z_P}(h(y)) \quad (y \in \partial D,\ r \in \mathbb{R}).$$

Proof. Positive Liouville derivative on the zero level makes that level regular and makes B a Liouville domain with primitive $\lambda_P|_B$. Boundary outwardness and absorption construct P as a completion of this very domain, with identity compact inclusion and end map $(z, r) \mapsto \mathrm{Fl}^r_{Z_P}(z)$. Extend the prescribed compact diffeomorphism h to the two supplied completions using its exact primitive equality. This produces a smooth diffeomorphism q with smooth inverse, exact primitive equality, and compact formula $q|_D = h$. Primitive preservation intertwines the unique Liouville fields, and uniqueness of integral curves therefore intertwines their complete flows for every real time. The source end is the flow of the native source boundary inclusion. Applying q to this flow and using its compact formula at the boundary gives the displayed all-real end law. □

Lemma Q.56 (Common outer family from the actual slice markings)**.** *Let the retained rounded ambient completion family have nonempty original carrier, jointly smooth original embeddings, and the stated rounded carrier images. Suppose, for each parameter, actual compactness, symplecticity and outwardness witnesses for the outer potential domain are supplied, and every native outer collar completion admits an exact inclusion marking from this retained rounded completion with the original compact square and all-real flow formula. Then these actual slice data produce a common localized outer completion family and actual inclusion markings into it, with the same compact and end formulas and real smooth germs of both the markings and their actual inverses in the retained presentations.*

Proof. Select the supplied witnesses as functions of the interval parameter. Use the explicit regular scalar sublevel at parameter zero as the fixed native compact domain. The fixed-domain lemma identifies every actual outer domain with this one, without changing its extension, carrier or boundary presentation. Transport each actual Liouville structure along that equality. Its primitive is still the actual tailoring primitive, which has a smooth real parameter extension. Construct a localized completion family from this smooth primitive family. Its carrier is nonempty because a point in the nonempty rounded carrier maps into the outer potential sublevel. Each constructed outer collar is a collar of the corresponding actual outer Liouville domain, so select its supplied actual marking. The compact square and all-real flow formula are retained. Primitive preservation gives the hypotheses of the retained marking regularity lemma; applying that lemma proves real forward and inverse germs for these selected maps. No joint regularity of the slice markings was required as an input. □

Let I be a finite toric input, and let ϕ be smooth with positive uniform lower and upper Hessian bounds. Suppose ϕ agrees outside a positive radius with a degree-two homogeneous function, smooth away from the origin and continuously differentiable everywhere. There is $F > 0$ such that, for every $C^+ > F$, there is $\beta_0 > 0$ such that, for every $\beta \geq \beta_0$, there is $K_* > 0$ with the following property. For every $R > 0$ and

$$F^2\beta^2/2 \leq A^- < (C^+)^2\beta^2/2 = A,$$

there is $0 < \delta < A - A^-$ such that every admissible rounding m with $0 < \epsilon_m \le \delta$ gives the retained rounded pair family, with the stated full carrier and frontier images, for every $K \ge K_*$. Together with this very family one obtains a constructed outer completion family of $D_\beta = \{\phi \le A\}$ with primitives $\lambda_{s,K}$ and actual exact ambient inclusion markings

$$q_s : (\widehat{E_s, \lambda_{s,K}}) \longrightarrow (\widehat{D_\beta, \lambda_{s,K}}).$$

Their full compact original projections are inclusion; their native end formula is outer complete flow for every real radius. In the retained completion presentations, both q_s and their actual inverses admit real smooth germs at every completed point and every $s \in [0,1]$. The marking conclusion concerns the ambient completions. The embedded ribbon carriers and their frontiers remain the original full-fiber truncations.

Proof. Use the preceding finite marking theorem without changing any threshold, rounding quantifier, original domain, ribbon, or compact/completed presentation. Choose its actual outer symplecticity, outwardness and compactness witnesses at every parameter. The outer defining function $\phi - A$ is independent of the parameter, so these domains have one common native compact presentation. The fixed-domain lemma identifies each with the explicit regular scalar sublevel at parameter zero. Transporting the actual Liouville structures along this proved equality retains their primitives, which are the actual smooth tailoring family. The localized-family construction therefore produces a common outer collar family with its completed presentations and smooth transported primitives. It is nonempty because the nonempty rounded domain embeds inside this outer carrier. Select the actual inclusion marking supplied by the finite theorem into each of these particular outer collar completions. The selected maps retain the compact projection and all-real end formulas. Their primitive equality and compact projection square, together with the already retained smooth original embedding family, satisfy the actual marking regularity lemma. Apply it to obtain forward and inverse real germs on the full completions. The pair itself and all four original whole-carrier/frontier identities were retained throughout. □

Lemma Q.57 (Whole-map regularity by interior absorption)**.** *Let $f_t : M \to N$ be maps and $V \subset M$. Let ϕ_t and ψ_t have real smooth flow germs at every parameter in $[0,1]$, every real time, and every point. Suppose each x reaches V under ϕ_t^{-T} for some finite T, the return identity $\psi_t^r(f_t(\phi_t^{-r}(x))) = f_t(x)$ holds, and f_t has real smooth map germs at every point of V. Then it has real smooth map germs on all of M.*

Proof. Fix (t,x) and choose T with $y = \phi_t^{-T}(x) \in V$. Choose the actual source flow germ near $(t, -T, x)$, the compact map germ near (t, y), and the target flow germ near $(t, T, f_t(y))$. Successively restrict their open domains to the preimages of the next germ's domain. The resulting open neighborhood of (t,x) carries the smooth composite $\psi_s^T(f_s(\phi_s^{-T}(z)))$. The return identity identifies its values with $f_s(z)$ for every actual interval parameter. All three germs are real-open, so the argument includes both endpoints. □

Lemma Q.58 (Regularity of the actual retained marking maps and inverses)**.** *Let two constructed localized completion families have fixed original compact domains E, H, with H nonempty. Let $j_t : E \to H$ be an original smooth interval map family. Suppose actual primitive-preserving slice diffeomorphisms q_t between their completions satisfy the full compact original-inclusion square with j_t. In their retained completion presentations, q_t and its actual inverse have real smooth germs at every completed point and every interval parameter.*

Proof. The retained original presentations and their inverses are jointly smooth. Compose the source original presentation, the actual real extension of j_t, and the inverse target original presentation. On the compact carrier the given square and the retained presentation squares identify this composite with the original projection of q_t in fixed coordinates. The reference target original inclusion is a same-dimensional smooth embedding, so its image is open and its inverse is smooth there. Restrict to the preimage of that image and the source compact interior. Composing with this actual inverse gives real smooth compact map germs. The constructed completed primitive families produce actual complete-flow germs in the retained coordinates. Exact primitive preservation intertwines their unique Liouville fields and flows, giving the return identity. Finite backward interior absorption is unchanged by a retained compact presentation. The preceding lemma now supplies whole-completion forward germs, including the seam. Apply the parameter graph inverse function theorem to these germs and the actual slice diffeomorphisms to obtain germs of those same inverses. □

Lemma Q.59 (Markings of the retained rounded pair)**.** *Under the finite tailoring Hessian bounds and the stated uniform choices of β, K, R, A^-, A, m, the rounded pair deformation can be chosen with its actual ambient and ribbon collar families retained. For each parameter, every cylindrical completion of the selected outer potential domain admits an exact inclusion marking from the ambient completion of this same rounded pair. On the full compact carrier its projection to the original ambient manifold is the original inclusion, and on every native boundary ray it is the complete outer Liouville flow for all real times.*

Proof. Take the maximum of the rounded-pair and rounded-outwardness thresholds, and the minimum of their positive smoothing widths. The original finite quantifiers and four image equalities are consequently retained. At each parameter pull the rounded defining function back by the chosen outer completion's original inclusion. Its nonpositive sublevel is compact: the rounded domain is contained in the compact outer potential domain. The scalar derivative is positive on its boundary by rounded outwardness, and finite backward absorption follows from the positive derivative on the outer compact region and the cylindrical end formula. The retained localized rounded domain and this sublevel embed into the original ambient manifold with exactly equal carrier images. The common-embedding construction supplies a compact diffeomorphism between these particular presentations. Pullback composition and their original primitive identities make it strict on the carrier. Apply the preceding transfer lemma to the already retained rounded completion. Its compact and all-real flow formulas give the asserted marking without changing the pair presentation. □

Lemma Q.60 (Finite backward absorption into the compact interior)**.** *In every completion of a compact Liouville domain, positive backward time sends the compact carrier into its interior. Every point of the full completion reaches that interior in finite nonpositive time.*

Proof. Backward invariance first puts the image in the compact carrier. If this image lay on its boundary, forward flow for the positive elapsed time would represent the starting compact point on a positive end ray. This contradicts the disjointness of positive end rays and the compact carrier. For an arbitrary point, either apply this argument directly with time one, or use the completion covering to write the point on a ray of radius $r \geq 0$. Backward time $r+1$ moves it to the time-minus-one image of that ray's boundary point, which lies in the compact interior by the first assertion. This also covers the empty-boundary case. □

Q.2.16 One rounding profile along the holomorphic path

Lemma Q.61 (Uniform base outwardness above a far value)**.** *Suppose the fixed smooth potential has positive lower and finite upper Hessian bounds and agrees, outside a ball of positive radius, with a degree-two homogeneous function smooth away from zero. There is $R_0 > 0$ such that for every $R \geq R_0$ and every $K \geq \kappa_{\mathrm{path}}(R)$, the actual primitive $\lambda_{\phi_t} + KW_t^*\lambda_{\mathbb{C}}$ satisfies*

$$Z_{t,K}|W_t|^2 > 0 \quad \text{on } \{|W_t| \geq R\}, \qquad 0 \leq t \leq 1.$$

Proof. The uniform ratio-set theorem supplies a positive far-value threshold above which every such point lies strictly outside the fixed ball, has positive inverse-Hessian pairing, and belongs to a nonempty bounded ratio set. Each individual path threshold is at most the supremum of the threshold values along the path. On this homogeneous end the Euler identity and the holomorphic Liouville identity apply. The ratio threshold then bounds the actual base derivative below by $a_t|W_t|^2/(1 + Ka_t)$, which is strictly positive because $a_t > 0$, $|W_t| \geq R > 0$, and $K \geq \kappa_{\mathrm{path}}(R) \geq 1$. □

Remark Q.62. To produce a whole collar below the far circle without an additional compact-continuity argument, the subsequent coefficient choice also dominates $\kappa_{\mathrm{path}}(R/2)$. This is an additional lower bound on a later freely chosen finite coefficient. It imposes no additional condition on the supplied original coefficient.

Lemma Q.63 (Spatial derivative of the literal holomorphic rounding)**.** *Let ϕ be smooth and let W_t, ϕ_t be the literal coefficient and affine potential paths. For every real t, R, A, profile m, point y and tangent vector v, put*

$$q_t(y) = m(\phi_t(y) - A, |W_t(y)|^2 - R^2).$$

Then

$$dq_t(y)v = dm(\phi_t(y) - A, |W_t(y)|^2 - R^2)\big(d\phi_t(y)v, d|W_t|^2(y)v\big).$$

Proof. The two entries of the displayed argument map are smooth. Subtracting their respective constants does not change their spatial differentials. The product differential is the pair of their differentials. Applying the ordinary chain rule to the smooth profile composed with this product map gives the displayed identity on each tangent vector. □

Let I be a toric input. Suppose ϕ is smooth, its Euclidean Hessian has bounds $c\|v\|^2 \leq H_\phi(v, v) \leq U\|v\|^2$ with $c > 0$, and outside a ball of radius $L > 0$ it agrees with a degree-two homogeneous function ϕ_{end} smooth away from zero. There are $A_0, R_0 > 0$ such that for every $R \geq R_0$ there is a finite $K_* > 0$ with $K_* \geq \kappa_{\mathrm{path}}(R)$ and the following property. For every $A \geq 2A_0$ and $A_- < A$ there is $0 < \delta < A - A_-$ such that every allowed profile of width $0 < \epsilon \leq \delta$ satisfies, simultaneously for every $K \geq K_*$ and $t \in [0, 1]$,

$$Z_{t,K}q_t > 0 \quad \text{on } \{q_t \geq 0\}, \qquad q_t = m_\epsilon(\phi_t - A, |W_t|^2 - R^2).$$

Proof. The uniform ambient-potential theorem gives $A_0 > 0$ such that $Z_{t,K}\phi_t > 0$ when $\phi_t \geq A_0$, for every path parameter and every $K \geq 0$. Let $R_b > 0$ be the threshold of the far-base lemma, and take $R_0 = 2R_b$. For the chosen R, take

$$K_* = \max\big(1, \kappa_{\mathrm{path}}(R), \kappa_{\mathrm{path}}(R/2)\big).$$

Both path thresholds are the nonempty bounded suprema supplied by the ratio-set theorem. In particular this is a finite positive choice. It gives $Z_{t,K}|W_t|^2 > 0$ wherever $|W_t| \geq R/2$ for every $K \geq K_*$, since $R/2 \geq R_b$.

Choose

$$\delta = \min\bigl(A/4, R^2/4, (A - A_-)/2\bigr) > 0.$$

It is strictly smaller than $A - A_-$. At a point with $q_t \geq 0$, an active first profile partial forces $\phi_t - A \geq -2\epsilon$, hence $\phi_t \geq A - 2\epsilon \geq A/2 \geq A_0$. An active second partial forces $|W_t|^2 - R^2 \geq -2\epsilon$, hence $|W_t|^2 \geq R^2/2$ and therefore $|W_t| \geq R/2$. Thus every active partial multiplies a strictly positive actual input derivative. The inactive partials are zero, all partials are nonnegative, and their sum is one. Their weighted sum is consequently strictly positive. The spatial chain-rule lemma identifies that sum with the actual $Z_{t,K} q_t$, proving the assertion on the whole nonnegative region at every parameter. □

Q.2.17 Actual scalar boundary motions along the holomorphic path

Lemma Q.64 (Compact total space of the literal rounded domains)**.** *Let I be a toric input. Suppose ϕ is smooth with a positive uniform lower Hessian bound, and agrees outside a ball of radius $L > 0$ with a degree-two homogeneous function ϕ_{end} that is C^1 everywhere and smooth away from zero. For every $A \geq 0$, every real R and every allowed rounding profile m, the literal total space*

$$\{(t, y) : 0 \leq t \leq 1, \quad m(\phi_t(y) - A, |W_t(y)|^2 - R^2) \leq 0\}$$

is compact, where $\phi_t = (1 - t)\phi_0 + t\phi$.

Proof. The lower bound of the smooth maximum implies $\phi_t(y) \leq A$ on the displayed set. If both $\phi_0(y) > A$ and $\phi(y) > A$, their convex combination is strictly greater than A for every $t \in [0, 1]$, a contradiction. Thus the total space lies in

$$[0, 1] \times \bigl(\{\phi_0 \leq A\} \cup \{\phi \leq A\}\bigr).$$

Both endpoint potential sublevels are compact by the properness results. The literal rounded equation is jointly smooth, so its nonpositive set is closed, as is the parameter slab. Their intersection is a closed subset of the displayed compact cylinder and is therefore compact. □

Let I and ϕ satisfy the finite smoothness, lower and upper Hessian, homogeneous end, and end-agreement hypotheses of the whole-exterior outwardness theorem. There exist $A_0, R_0 > 0$ such that for every $R \geq R_0$ there is $K_* > 0$, dominating the literal $\kappa_{\mathrm{path}}(R)$, with the following property. For every $A \geq 2A_0$ and $A_- < A$ there is $0 < \delta < A - A_-$ such that, for every allowed profile with $0 < \epsilon \leq \delta$, every $t \in [0, 1]$, and every y with $q_t(y) = 0$, where $q_t = m(\phi_t - A, |W_t|^2 - R^2)$:

1. An open neighborhood of $(0, (t, y))$ carries a smooth local motion Ψ that is the identity at motion time zero and satisfies

$$\mathrm{time}\,\Psi(a, p) = \mathrm{time}(p) + a, \qquad q(\Psi(a, p)) = q(p).$$

2. An open neighborhood of (t, y) carries a smooth tangent field V satisfying the actual equation $d(\mathrm{time}, q)V = (1, 0)$ throughout that neighborhood.

Proof. Choose exactly the constants and profile width supplied by the whole-exterior theorem. At a zero point its actual Liouville vector Z_{t,K_*} has $dq_t(Z_{t,K_*}) > 0$. Therefore the spatial covector $D = dq_t(y)$ is surjective: for every real a, the vector $(a/DZ_{t,K_*})Z_{t,K_*}$ maps to a. Joint smoothness of the literal scalar and this spatial surjectivity give a genuine joint parameter-level normal form. Translating its parameter coordinate constructs Ψ; pulling back the constant parameter-coordinate vector constructs V. Both equations hold on actual open neighborhoods, including beyond either closed interval endpoint. □

Remark Q.65. These results construct scalar ambient boundary data. The next pair-family producer must simultaneously solve the moving equations $W_t = R$ and, at the ribbon frontier, $\phi_t = A_-$. A compatible full-ribbon lift, global simultaneous transport, and smooth completed pair family are not conclusions of these two statements. This distinction is required by Target Theorem 4.1 and the specification's Section 6.

Q.2.18 Local holomorphic ribbon equations and the unchanged rounding

Lemma Q.66 (Joint unchanged face along the full holomorphic ribbon)**.** *Let ϕ be smooth. Fix real R, A, A_- and an allowed profile of width $\epsilon > 0$ with $A_- < A - \epsilon$. At every real t and every point y satisfying $W_t(y) = R$ and $\phi_t(y) \le A_-$, there is an open neighborhood O of (t, y) such that throughout O the literal scalar is*

$$q(s, z) = m(\phi_s(z) - A, |W_s(z)|^2 - R^2) = |W_s(z)|^2 - R^2.$$

Proof. Write $x(s, z) = \phi_s(z) - A$ and $b(s, z) = |W_s(z)|^2 - R^2$. Both functions are continuous. The set $O = \{b - x > \epsilon\}$ is open. At (t, y) one has $b = 0$ and $x \le A_- - A < -\epsilon$, so this point lies in O. At every point of O, $b > x$ and $|x - b| > \epsilon$. The defining exactness of the allowed profile away from its corner band therefore gives $m(x, b) = \max(x, b) = b$ on all of O. □

Corollary Q.67 (Full truncated holomorphic fiber on the rounded zero face). Under the hypotheses of the preceding lemma, every point of the literal full truncated fiber $\{W_t = R,\ \phi_t \le A_-\}$ satisfies $q_t = 0$.

Proof. Apply the joint unchanged-face identity at the given point. Its right side is $|R|^2 - R^2 = 0$, since R is real. □

Lemma Q.68 (Fiber horizontality implies rounded boundary tangency)**.** *Suppose on an open set O the literal rounded equation is $q(t, y) = |W_t(y)|^2 - R^2$. At a point $p \in O$, every tangent vector v with $d(\mathrm{time}, W)_p(v) = (1, 0)$ satisfies $dq_p(v) = 0$.*

Proof. Taking the complex component of the horizontal equation gives $dW_p(v) = 0$. The two defining functions agree on a neighborhood of p, so their differentials there agree. The chain rule gives $dq_p(v) = d|\cdot|^2_{W(p)}(dW_p(v)) = 0$; subtracting the constant R^2 does not change the differential. □

Lemma Q.69 (Actual local horizontal lifts of every far holomorphic fiber)**.** *For every toric input there is $R_0 > 0$ such that, for every $R \ge R_0$, every $t \in [0, 1]$ and every y with $W_t(y) = R$, an open neighborhood of (t, y) carries a smooth tangent field V with $d(\mathrm{time}, W)V = (1, 0)$ throughout that neighborhood.*

Proof. The uniform critical-value bound supplies $B > 0$ such that dW_t is surjective wherever $|W_t| > B$, for all $t \in [0, 1]$. Set $R_0 = B + 1$. At a point on the specified full fiber, $|W_t(y)| = R \ge R_0 > B$. The actual map $(s, z) \mapsto W_s(z)$ is jointly smooth. The parameter-level normal form and its constructed horizontal field therefore give the claimed open neighborhood and tangent field. □

Lemma Q.70 (Actual local lifts of the moving holomorphic frontier)**.** *Let I be a toric input. Suppose ϕ is smooth with positive lower and finite upper Euclidean Hessian bounds, and agrees outside a ball of positive radius with a degree-two homogeneous function that is C^1 everywhere and smooth away from zero. There exist $F, \beta_0 > 0$ such that for every $\beta \ge \beta_0$, every $t \in [0, 1]$ and every y with*

$$W_t(y) = e^\beta, \qquad \phi_t(y) \ge F^2\beta^2/2,$$

an open neighborhood of (t, y) carries a smooth tangent field V satisfying the actual equation

$$d\big(\text{time}, (W, \phi_{\text{time}})\big)V = (1, 0)$$

throughout that neighborhood.

Proof. The proved intrinsic estimate gives $F > 0$ and a positive scale threshold B_0 such that, whenever $\phi_t \geq F^2\beta^2/2$ and $|W_t| \leq e^\beta$, the intrinsic Liouville derivative of ϕ_t on the full fiber is strictly positive. Let $B > 0$ be the uniform critical-value bound and take $\beta_0 = \max(B_0, \log(B+1))$. Then $e^\beta \geq B + 1 > B$. The entire literal fiber $W_t^{-1}(e^\beta)$ is regular and hence has its constructed smooth manifold N, inclusion j, and tangent image $\text{range}(dj) = \ker(dW_t)$.

The given point is $j(x)$ for some $x \in N$. Let Z be the actual intrinsic Liouville vector of $j^*\lambda_{\phi_t}$ at x. The intrinsic estimate implies $d(\phi_t \circ j)_x(Z) > 0$. The vector $dj_x(Z)$ therefore lies in $\ker(dW_t)$ and has nonzero $d\phi_t$ value. Together with surjectivity of dW_t, this makes $d(W_t, \phi_t)$ surjective: first lift the desired complex value; then add a scalar multiple of $dj_x(Z)$ to correct its real potential value without changing the complex value. The joint map $(s, z) \mapsto (W_s(z), \phi_s(z))$ is smooth. Its parameter-level normal form supplies the claimed actual smooth horizontal field. □

Remark Q.71. At a ribbon frontier $\phi_t = A_-$ above the stated logarithmic bound, the last lift preserves both the moving fiber and the moving frontier. Intersecting its open source with the unchanged germ also enforces the actual rounded ambient boundary equation. The same construction with the unmarked fiber lift applies away from the frontier. Compact simultaneous gluing, flow trivialization, the resulting fixed-carrier pair family, and the smooth family of completion markings remain separate producers; none is an input to these local statements. In rank one the full fiber is zero-dimensional, so the high-potential premise of the marked-lift statement has no points: a map from its two-dimensional ambient tangent space to the three-dimensional target $\mathbb{C} \times \mathbb{R}$ cannot be surjective. Thus the unmarked full-fiber lift remains substantive, while the sufficiently large truncation frontier is empty, as prescribed in the specification.

Q.2.19 The primitive along the holomorphic path

Lemma Q.72 (Joint regularity of the plane pullback)**.** *Let $f_t : X \to \mathbb{C}$ be jointly smooth for real t on a smooth manifold X. Then $f_t^*\lambda_{\mathbb{C}}$ is a jointly smooth family of one-forms.*

Proof. In a fixed chart write $g(t, x) = f_t(x)$. The coefficients of the pullback are $\lambda_{\mathbb{C}}(g(t, x)) \circ D_x g(t, x)$. The spatial derivative is the restriction of the derivative of g to vectors $(0, v)$. Smooth differentiation, composition and this fixed linear restriction show that these coefficients are jointly smooth. The calculation holds for real parameters on both sides of the interval endpoints. □

Lemma Q.73 (The prescribed holomorphic primitive family)**.** *For the prescribed coefficient path f_t and potential path $\phi_t = (1-t)\phi_0 + t\phi$, the forms $\lambda_{\phi_t} + Kf_t^*\lambda_{\mathbb{C}}$ are jointly smooth for every fixed real K.*

Proof. Differentiation of the affine potential gives $\lambda_{\phi_t} = (1-t)\lambda_{\phi_0} + t\lambda_\phi$. The two endpoint forms are smooth, so this is a smooth family. Each coefficient of f_t is $\exp((1-t)\ell_v)$ and each quotient character is smooth. Their finite sum is jointly smooth. Apply the preceding lemma and take the indicated sum of forms. □

Assume the smooth potential has positive lower and finite upper Hessian bounds and agrees outside a fixed ball with a smooth degree-two end. There are $A_0, R_0 > 0$ such that, for every $R \geq R_0$, one finite $K_* > 0$ dominates $\kappa_{\text{path}}(R)$. For every $A \geq 2A_0$ and $A_- < A$, one positive

width bound below $A - A_-$ works for every smaller positive profile width and every $K \geq K_*$ as follows. If a fixed compact smooth domain E has a smooth family of ambient embeddings J_t with entire carrier image $\{q_t \leq 0\}$ and frontier image $\{q_t = 0\}$, where $q_t = m(\phi_t - A, |W_t|^2 - R^2)$, then

$$J_t^*(\lambda_{\phi_t} + KW_t^*\lambda_{\mathbb{C}})$$

is an actual smooth Liouville family on this same E. Its fields are jointly smooth, its symplectic forms are nondegenerate on the entire extension, and it admits smooth localized completions.

Proof. Take the constants and width from the whole-exterior outwardness theorem. The affine Hessian has lower bound $\min(1, c) > 0$ on the parameter interval. Holomorphicity makes the plane update semipositive for the standard complex structure, so the ambient exterior derivative is nondegenerate for each nonnegative K. The preceding primitive-family lemma gives joint smoothness. The two image identities give $q_t \circ J_t \leq 0$ on E and equality on its boundary. The proved strict outward derivative on $q_t \geq 0$ thus applies to every image boundary point. The compact-sublevel pullback construction gives the Liouville structures, with the displayed primitive. Pullback regularity and smooth inversion of the nondegenerate form give jointly smooth primitives and fields. Parameter-dependent Liouville collars construct the localized completions of precisely this compact family. □

Q.2.20 Simultaneous holomorphic rounded transport

Lemma Q.74 (Compact moving-potential truncations)**.** *Let $\psi_t = (1 - t)\phi_0 + t\phi$, where ϕ has a positive lower Hessian bound and a positive degree-two homogeneous end. For every $A_- \geq 0$ and $R \in \mathbb{R}$, the set*

$$\{(t, y) : 0 \leq t \leq 1,\ W_t(y) = R,\ \psi_t(y) \leq A_-\}$$

is compact.

Proof. If both $\phi_0(y)$ and $\phi(y)$ exceed A_-, their convex combination exceeds A_- for every $t \in [0, 1]$. Thus the displayed set lies in the product of $[0, 1]$ with the union of the two compact endpoint sublevels. Its three defining conditions are closed by joint continuity. It is therefore a closed subset of a compact set. □

Lemma Q.75 (A field preserving the whole holomorphic ribbon)**.** *Under the same finite potential hypotheses, including fixed upper and lower Hessian bounds, there are $F, \beta_0 > 0$ such that for $\beta \geq \beta_0$ and $A_- \geq F^2\beta^2/2$, one globally smooth field V is defined on $\mathbb{R} \times Y$. On an open neighborhood of the entire compact ribbon family it satisfies $d(t, W_t)(V) = (1, 0)$, and on an open collar of the frontier $\psi_t = A_-$ it also satisfies $d\psi(V) = 0$.*

Proof. Take the maximum of the far-value regularity threshold and the logarithmic marked-lift threshold. Local fiber lifts exist at every point of the compact ribbon family. At its frontier the actual intrinsic exhaustion inequality gives local lifts satisfying both equations. Glue these local fields by the boundary horizontal gluing lemma, imposing the fiber equation everywhere and the additional potential equation on a smaller closed collar of the frontier. The affine equations survive every partition-of-unity combination. The resulting equations hold on open neighborhoods, as asserted. □

Lemma Q.76 (A simultaneous lift of the rounded holomorphic pair)**.** *There is $F > 0$ such that every $D > F$ admits $\beta_0 > 0$ with the following property. For every $\beta \geq \beta_0$, set $R = e^\beta$ and $A = D^2\beta^2/2$. For every $F^2\beta^2/2 \leq A_- < A$, there is $0 < \delta < A - A_-$ such that every permitted*

rounding profile of width $0 < \epsilon \leq \delta$ admits one globally smooth field on $\mathbb{R} \times Y$ with $dt(V) = 1$ near the rounded total space. On open collars of the rounded boundary, the entire truncated fiber $W_t = R$, and its frontier, respectively, it satisfies $dq(V) = 0$, $d(t, W_t)(V) = (1, 0)$ and $d\psi(V) = 0$, where $q(t, y) = m_\epsilon(\psi_t(y) - A, |W_t(y)|^2 - R^2)$.

Proof. Increase β_0 so the outer level exceeds the ambient lift threshold and e^β exceeds every required far-value threshold. The rounded boundary lifts give one permitted width, uniform on the compact family. Shrink an open ribbon neighborhood into the unrounded germ $q = |W_t|^2 - R^2$. There its fiber lift also satisfies $dq(V) = 0$. Choose smaller closed collars of the ambient boundary, ribbon and frontier. At points in the ribbon collar use the ribbon field; away from it use ambient boundary lifts where needed and a free parameter lift elsewhere. Their applicable equations are compatible on each local source. Glue them using the affine neighborhood gluing lemma. The closed collars contain the required actual subsets, so the resulting field has every claimed open collar identity. □

Lemma Q.77 (Transport and reverse transport of the actual pair)**.** *For precisely the choices in the preceding lemma, one globally defined smooth flow on $\mathbb{R} \times Y$ transports both $q_t \leq 0$ and $\{W_t = R, \psi_t \leq A_-\}$ between any two parameters in $[0, 1]$. It translates the parameter and preserves each of the faces $q_t = 0$ and $\psi_t = A_-$ in both directions. Its reverse-time map is the inverse of each such transport.*

Proof. The strict gap makes the whole ribbon lie on the rounded zero face, so its total space is contained in the compact rounded total space. Apply simultaneous compact transport to $\tau = t$, q, $\rho = \psi$ and $f = W_t$, using the field and open collars just constructed. The confinement theorem gives both carrier-preservation assertions throughout the closed interval. Applying them for time $-s$ from the image proves each reverse boundary implication. The flow identity $\Phi_{-s}\Phi_s = \mathrm{id}$ supplies the inverse. □

Lemma Q.78 (Nonempty native initial ribbon carriers)**.** *Under the same finite potential hypotheses there are $F, \beta_0 > 0$ such that every $\beta \geq \beta_0$ and $A_- \geq F^2\beta^2/2$ admits a nonempty native compact domain D in the actual initial full fiber $W_0 = e^\beta$. Its carrier and frontier have embedded images $\{W_0 = e^\beta, \phi_0 \leq A_-\}$ and $\{W_0 = e^\beta, \phi_0 = A_-\}$, respectively.*

Proof. Increase the threshold so that the original full-fiber model is available. The logarithmic intrinsic derivative estimate at both path parameters zero makes $\phi_0 - A_-$ a regular boundary scalar. Its sublevel is compact, and the regular-sublevel construction gives D with its exact frontier. The quadratic potential primitive is literally $\lambda_0 = \sum u_j d\theta_j$, so the original complete intrinsic flow has strictly positive exhaustion derivative above A_-. Every point therefore enters the sublevel in finite negative time. The original Laurent superpotential is surjective, so its full fiber is nonempty; negative entry proves that the compact truncation is also nonempty. The full-fiber embedding is onto the literal entire fiber, proving the two asserted images without choosing a component. □

Lemma Q.79 (One fixed pair of native carriers)**.** *For the same quantifier order $F > 0$, $C_+ > F$, sufficiently large β, every allowed A_- and every sufficiently thin supplied profile, there are nonempty native compact domains E, D, an ambient embedding family J_t, and fixed embeddings $i : D^{\mathrm{ext}} \longrightarrow E^{\mathrm{ext}}$, $k : D^{\mathrm{ext}} \longrightarrow \partial E$, with $i = \iota_{\partial E} k$ on the whole extension. For every $t \in [0, 1]$, the images of $E, \partial E, D, \partial D$ under $J_t, J_t, J_t i, J_t i$ are exactly the rounded domain, rounded boundary, entire truncated holomorphic fiber and its moving potential frontier. Both families are jointly smooth and arise from one pair flow.*

Proof. Use the regular initial rounded scalar sublevel for E and the preceding nonempty initial fiber truncation for D. Take the minimum of the outwardness and transport profile widths. The projected flow is an embedding on a common open neighborhood of the compact initial ambient carrier, uniformly for the whole interval. Restrict the ambient extension to that neighborhood. Restrict the ribbon extension further by the strict condition $\phi_0 < A - \epsilon$, keeping its entire compact carrier. The exact unrounded germ makes every point of this smaller extension lie on the initial rounded boundary. Factor its full-fiber embedding through the ambient inclusion and then its boundary inclusion to obtain i, k. Set $J_t(x) = \mathrm{pr}_Y \Phi_t(0, x)$; its composition with i gives the ribbon family on the whole extension. Forward transport proves each image inclusion, including the frontier equations. For a point of any asserted image at time t, apply Φ_{-t} to (t, y). Confinement places its inverse in the exact initial carrier, and the flow group identity sends it back to y. The frontier equivalences give the corresponding reverse inclusions for both boundaries. Nonemptiness of D, together with its inclusion into the ambient boundary, also proves nonemptiness of E. □

Q.2.21 Liouville data on the prescribed holomorphic path ribbon

Lemma Q.80 (Common restriction of an outward primitive family)**.** *Let D be a native compact smooth domain, and μ_t a smooth interval family of one-forms on its extension. Suppose $d\mu_t$ is symplectic on one open neighborhood O containing D, and the canonical Liouville field of μ_t is strictly outward at every boundary point of D. Then the same carrier, with extension restricted to O, has Liouville structures with the restricted primitives and canonical fields. These primitives and fields form smooth interval families, their differentials are symplectic on the whole restricted extension, and their localized completion family exists.*

Proof. Pull back each primitive under the open inclusion. Exterior differentiation commutes with this pullback, and the inclusion has the identity tangent map, so its differential remains nondegenerate and closed throughout the restricted extension. The contraction equation therefore uniquely determines the same canonical field in the native tangent coordinates. The identity tangent map also transfers the strict boundary-normal inequality. Smooth inversion of the nondegenerate contraction maps gives the jointly smooth canonical fields, and pullback gives joint smoothness of the primitives. These actual compact Liouville data satisfy the localized completion-family construction. □

Lemma Q.81 (The primitive on the same holomorphic ribbon embeddings)**.** *Fix the finite toric input and a smooth potential ϕ with a positive lower Hessian bound and finite upper Hessian bound. Write $W_t = W_{c(t)}$ and $\psi_t = (1-t)\phi_0 + t\phi$. For all sufficiently large β, let D be any compact domain of dimension $2n-2$ and $J_t : D^{\mathrm{ext}} \hookrightarrow Y$ a smooth interval family satisfying $W_t(J_t x) = e^\beta$ on its carrier. Then the native tangent image at every carrier point is $\ker dW_t$, the literal primitives $\mu_t = J_t^* \lambda_{\psi_t}$ are jointly smooth, and their differentials are symplectic on one common open neighborhood of the same carrier.*

Proof. The finite coefficient path admits one disk containing every critical value. Increase β so that e^β lies outside it. Differentiating the level equation on the native carrier annihilates all its native tangent directions, including at its boundary. Immersion and the regular level dimension therefore give precisely the kernel. The Hessian of ψ_t has lower bound $\min(1, c) > 0$ and upper bound $\max(1, U)$, so the ambient form is symplectic. At a regular holomorphic point its native Poisson scalar is the positive inverse-Hessian norm of the polynomial derivative. The kernel restriction lemma makes $d\mu_t$ nondegenerate on the carrier. The potential primitive is the literal affine

interpolation of its endpoint primitives; simultaneous smooth pullback by J_t gives joint smoothness. Compactness of the carrier and interval and openness of nondegeneracy give one common open neighborhood. Each $d\mu_t$ is closed, hence symplectic there. □

Lemma Q.82 (Native outwardness for the actual moving truncations)**.** *Assume additionally that ϕ agrees outside a fixed ball with a degree-two homogeneous potential, continuously differentiable at zero and smooth away from zero. There are $F, \beta_0 > 0$ such that, for $\beta \geq \beta_0$ and $A_- \geq F^2\beta^2/2$, every supplied family J_t with literal carrier image $\{W_t = e^\beta, \psi_t \leq A_-\}$ and frontier image $\{W_t = e^\beta, \psi_t = A_-\}$ has strictly outward canonical fields on the same native carrier. The preceding primitive, tangent and common-neighborhood conclusions hold for this family.*

Proof. Use the logarithmic intrinsic end estimate for the independent coefficient and potential parameters, restricting both to t. At a frontier point the potential equals A_-, so its level hypothesis holds, and the polynomial equals e^β, so its bounded-base hypothesis holds. The proved tangent kernel then gives $d(\psi_t \circ J_t)(Z_{\mu_t}) > 0$ there. The carrier image makes $\psi_t \circ J_t - A_-$ nonpositive on the carrier, and the frontier image makes it zero on the boundary. The native sublevel-normal lemma transfers the strict derivative to the stored boundary defining function. In dimension zero the native boundary is empty and this assertion imposes no positive derivative on a zero-dimensional fiber. □

Lemma Q.83 (The same holomorphic ribbon Liouville and completion family)**.** *Under the preceding hypotheses and ranges, one common open restriction of the supplied extension gives actual Liouville structures with primitive $J_t^*\lambda_{\psi_t}$, jointly smooth primitive and canonical field families, and constructed localized completions. The embeddings are the original J_t precomposed only by the native open inclusion; both literal images and the tangent kernels are retained. For every real K the ambient rank-two primitive $\lambda_{\psi_t} + KW_t^*\lambda_{\mathbb{C}}$ pulls back to that very ribbon primitive at every point of its actual compact carrier.*

Proof. Apply the common-restriction lemma to the primitive family and native outwardness just proved. The restricted inclusion projects the carrier and frontier onto exactly their originals, and has identity derivative; this retains both image equations and the kernel equality. Naturality of pullback identifies the constructed primitives with the actual restricted embeddings' pullbacks. On that kernel dW_t vanishes, so the plane pullback term vanishes for every real coefficient K. Thus its pullback is precisely the same primitive, supplying witness one for the actual ribbon in the rounded pair family. □

Q.2.22 The actual holomorphic completed pair deformation

For the fixed toric input and the prescribed smooth positive-Hessian potential with its homogeneous end, there is an inner-radius constant $F > 0$. For every larger outer-radius constant C_+ and all sufficiently large β, put $R = e^\beta$ and $A = C_+^2\beta^2/2$. There is a finite $K_* > 0$ dominating $\kappa_{\mathrm{path}}(R)$ such that, for every $F^2\beta^2/2 \leq A_- < A$, one positive width bound below $A - A_-$ has the following property. For every profile of smaller positive width and every $K \geq K_*$, the actual rounded domains

$$E_t = \{m(\phi_t - A, |W_t|^2 - R^2) \leq 0\}, \qquad F_t = \{W_t = R, \phi_t \leq A_-\}$$

form a smooth deformation of completed witnessed compact pairs with primitive $\lambda_{\phi_t} + KW_t^*\lambda_{\mathbb{C}}$ and witness one. Both carriers are nonempty. The fixed presentations and their localized collars retain the entire displayed carrier and frontier images at every parameter.

Proof. Take the maximum of the inner constants for simultaneous transport and intrinsic ribbon outwardness, and enlarge it to at least one. For any larger C_+, enlarge the lower bound on β so that all transport and ribbon estimates apply, $e^\beta \geq R_0$ and $C_+^2\beta^2/2 \geq 2A_0$, where R_0, A_0 are the ambient-family constants. Choose its later coefficient threshold K_* and take the minimum of the transport and outwardness widths.

The same simultaneous flow supplies compact ambient and ribbon presentations, with the ribbon embedding factoring through the ambient boundary on the full extension. The retained ambient-family theorem constructs the ambient Liouville structures on exactly those embeddings. For the ribbon, the actual intrinsic exhaustion estimate gives outwardness in its native defining coordinates, and positivity of the fiber symplectic form persists on a common open neighborhood of its compact carrier. Restrict to this neighborhood once. The ribbon embeddings are therefore still the restrictions of the same flow embeddings, with unchanged full carrier and frontier images.

Pulling the ambient primitive back to the ribbon removes the plane term: the derivative of W_t vanishes on every tangent vector of the full fiber. Its remaining primitive is exactly the pulled-back λ_{ϕ_t}. Thus witness one gives the required ribbon primitive on its entire carrier. Both primitive families are jointly smooth. The localized Liouville collar constructions yield smooth ambient and ribbon completions; their boundary embeddings give an actual pair deformation. The original inclusions from each localized collar map its carrier and frontier onto the original ones. Composing these inclusions with the same flow embeddings proves all four displayed image equalities, including at the two endpoints. □

Q.2.23 Ambient markings on the retained holomorphic presentations

Lemma Q.84 (Inclusion marking of an embedded absorbing sublevel)**.** *Let P be a complete Liouville manifold and q a smooth function with compact nonpositive sublevel, whose derivative along the Liouville field is positive on $q = 0$. Suppose every point enters $\{q \leq 0\}$ in finite negative time. If a nonempty compact Liouville domain S embeds in P with image exactly $\{q \leq 0\}$ and its primitive is the pullback of that of P, then every completion of this same S has an inclusion-induced strict marking onto P.*

Proof. The sublevel has its native compact regular presentation. Its identity embedding and the given embedding of S have the same image. Their local inverses give a represented compact diffeomorphism, with the same primitive by the differential embedding square. The absorption completion of the native sublevel is strictly equivalent to every prescribed completion of S. The equivalence agrees with the given embedding on the compact carrier; strictness intertwines both complete flows, giving the required formula on every boundary cylinder point. □

Lemma Q.85 (Holomorphic rounded ambient markings)**.** *For the prescribed coefficient and potential path, choose the uniform rounded outwardness bounds. For every permitted radius, levels, profile, coefficient and path parameter, every nonempty Liouville presentation of the literal rounded sublevel, equipped with the pulled-back actual primitive, has an inclusion marking from each of its prescribed completions onto the whole holomorphic ambient with its native complete Liouville flow.*

Proof. The actual potential-path primitive has a complete Liouville flow. The rounded defining function q satisfies $q \geq \phi_t - A$, so each of its sublevels is a closed subset of a compact potential sublevel. The proved rounded estimate gives $dq(Z) > 0$ throughout $q \geq 0$. Proper-below compactness and the minimum of $dq(Z)$ on a compact exterior band force finite negative entry into $q \leq 0$. Apply the preceding lemma to the supplied presentation and completion, retaining its original embedding. □

Q.2.24 Actual pointwise markings on the retained holomorphic pair

Lemma Q.86 (The whole moving fiber marks every retained ribbon completion)**.** *Let I be a finite toric input and let ϕ be smooth with a positive uniform lower Hessian bound cI and finite upper Hessian bound UI. Suppose outside a positive radius it equals a degree-two homogeneous end which is continuously differentiable everywhere and smooth away from the origin. Write $W_s = W(I_s)$ for the prescribed coefficient path and $\psi_s = (1-s)|u|^2/2 + s\phi$. There are $F, \beta_0 > 0$ such that for every $\beta \geq \beta_0$, $s \in [0,1]$ and $A^- \geq F^2\beta^2/2$, the following holds. Let (D,μ) be a nonempty retained compact Liouville domain with actual extension embedding j_D satisfying*

$$j_D(D) = \{W_s = e^\beta,\ \psi_s \leq A^-\}, \qquad \mu = j_D^*\lambda_{\psi_s} \quad \text{on } D.$$

Every supplied completion C of this very domain has a marking onto an actual complete full-fiber realization $j : F_s \hookrightarrow Y$, with

$$j(F_s) = \{W_s = e^\beta\}, \qquad \lambda_{F_s} = j^*\lambda_{\psi_s}.$$

Its compact square is $j \circ q|_D = j_D$; it preserves primitives and maps every all-real source boundary orbit to the all-real native full-fiber orbit. The assertion is pointwise in the path parameter.

Proof. Use the common intrinsic outwardness threshold for the independent coefficient and potential parameters and the uniform far-value regularity bound B. Enlarge β_0 so that $\beta \geq B$, whence $e^\beta \geq \beta + 1 > B$. The literal level of W_s is then regular. The actual path potential has Hessian bounds $\min(1,c)I \leq D^2\psi_s \leq \max(1,U)I$ and its actual interpolated homogeneous end. Apply the native regular-fiber construction and the complete intrinsic flow theorem to this potential and this full level. They produce F_s, j and the complete primitive $j^*\lambda_{\psi_s}$.

The tangent range of j is the actual kernel of dW_s. Apply the independent-path intrinsic outward estimate with both parameters equal to s and $|W_s| = e^\beta$. Conversion between the fixed Euclidean coordinates and the real logarithmic coordinates identifies its potential with the displayed ψ_s. Its derivative is therefore positive above every allowed A^-. The potential-fiber outward-sublevel theorem gives the full native truncation E of F_s and its flow-built completion whose entire ambient is F_s.

The full images of D and E in Y agree. The arbitrary-ambient compact comparison constructs a retained compact diffeomorphism $h : D \to E$ with $jh = j_D$ on every carrier point. The full-carrier form identity gives $h^*\lambda_{F_s} = \mu$, including at boundary points. Extend this compact map to the arbitrary supplied completion C and the constructed completion of E. Its compact formula is h, and its primitive preservation intertwines the actual Liouville fields and complete flows. This gives the all-real boundary-orbit formula and the asserted full-fiber marking. □

Under the preceding finite potential hypotheses there is $F > 0$ such that every $C^+ > F$ has a threshold $\beta_0 > 0$. For every $\beta \geq \beta_0$, put $R = e^\beta$ and $A = (C^+)^2\beta^2/2$. There is $K_* > 0$ dominating the prescribed path coefficient threshold such that every $F^2\beta^2/2 \leq A^- < A$ has a positive allowed rounding width below $A - A^-$. For every rounding of smaller positive width and every $K \geq K_*$, the actual completed holomorphic pair deformation retains its fixed compact domains, same embeddings, native primitive families, witness one, exact pair formulas, and all whole-carrier and frontier images. Each of its same ambient completions has an actual pointwise inclusion marking onto the whole holomorphic ambient with its actual primitive $\lambda_{\psi_s} + KW_s^*\lambda_{\mathbb{C}}$ and native complete flow. Each of its same ribbon completions has the preceding actual pointwise marking onto the whole $W_s = e^\beta$ fiber. Both compact inclusion squares are the original ones. The marking assertions are pointwise in s.

Proof. Take the maximum of the inner constants for the actual pair deformation and retained ribbon marking, and enlarge it to at least one. After C^+ is chosen, enlarge the β threshold to meet both producers and to imply $R \geq R_0$ and $A \geq 2A_0$ for the rounded ambient marking producer. Take the maximum of its coefficient threshold and that of the actual pair deformation. This still dominates the prescribed path threshold. For any permitted A^-, take the minimum of the two positive rounding widths. All original quantifier ranges are preserved.

Apply the already constructed pair deformation with its own exact domains, embeddings, structures and completion families. Restrict its coefficient-indexed structures to the enlarged range without changing any member. Retain its same Q and its literal localized pair formula. Each localized carrier has the original full image and is nonempty. Its primitive is the pullback of the original primitive under the native original inclusion. Pullback composition therefore gives exactly the actual ambient or ribbon primitive under the respective retained embedding. Apply the ambient marking producer to each same ambient cylindrical completion. Apply the preceding ribbon producer to each same ribbon cylindrical completion. These are the actual completion witnesses in Q, so both pointwise markings are obtained without changing the pair deformation or any compact or primitive square. The ribbon marking also retains its explicit all-real flow end formula. Neither pointwise construction asserts parameter-smooth marking data. □

Q.2.25 Smooth ambient markings on the retained holomorphic family

Lemma Q.87 (Real regularity of inclusion markings into a fixed ambient)**.** *Let E be a compact Liouville domain presentation with a smooth primitive family and its constructed localized completion family A_t. Let N be a fixed smooth manifold, with a smooth interval family of primitives η_t whose actual Liouville fields are complete. Suppose j_t is a smooth interval family of maps from the original extension of E to N. If primitive-preserving slice diffeomorphisms $q_t : A_t \to N$ agree with j_t on the full compact carrier, then the maps q_t and their actual inverses have real smooth germs at every completed point in the retained completion presentations, including both parameter endpoints.*

Proof. Use the retained presentation to express every source completion on the reference completion $M = A_0$. The source completed primitive family and the target primitive family give real smooth germs of their actual complete flows. Exact primitive preservation intertwines these flows. Every source point reaches the compact interior in finite backward time. On that interior, compose the smooth retained original presentation with the actual real extension of j_t. The compact square identifies this composite with the actual marking in reference coordinates. Restrict its real-open neighborhood to the compact interior; this gives a real smooth germ of the marking there. For an arbitrary point choose a finite backward time into the interior, compose the source backward flow germ, this compact map germ, and the target forward flow germ, and restrict successively to the preimages of their open domains. Flow intertwining identifies the composite with the marking. This proves forward real germs everywhere. The parameter graph inverse function theorem, applied to these germs and the actual slice diffeomorphisms, gives real germs of their same inverses. □

Under the finite potential hypotheses and the same nested parameter ranges of the preceding pointwise marking theorem, its very rounded pair family admits selected ambient inclusion markings onto the whole holomorphic ambient. In its retained completion coordinates these markings and their actual inverses have real smooth germs at every interval parameter and every completed point. All compact carrier and frontier images, both primitive pullback identities, the witnessing factor one, the literal pair formula, and both actual pointwise full-fiber ribbon markings are retained.

Proof. Apply the pointwise theorem with its exact thresholds and its selected compact domains,

embeddings, primitives, collars and pair deformation. Select one of its actual ambient inclusion markings at each parameter. The ambient primitive is the actual simultaneous coefficient and potential primitive, whose real parameter extension is smooth. The original ambient embedding family is smooth, and the inclusion property of each selected marking supplies its full compact square. The preceding lemma gives real germs of these markings and their inverses in the same retained completion presentations. Keep every other component of the pointwise theorem, including its ribbon markings. Thus the new regularity is on the same pair and requires no alteration of its thresholds or admissible choices. □

Q.2.26 Full-fiber ribbon markings on the retained tailoring pair

Lemma Q.88 (Smooth local left inverse of an embedded submanifold)**.** *Let $j : N^d \hookrightarrow M^e$ be a smooth embedding of finite-dimensional boundaryless manifolds. For every $a \in N$ there are an open neighborhood U of $j(a)$ and a smooth map $r : U \to N$ such that $r(j(y)) = y$ whenever $j(y) \in U$. The dimensions may be zero and need not be equal.*

Proof. Take a chart c of M at $j(a)$ with differential the identity in the chosen tangent coordinates. The injective derivative dj_a has a linear left inverse L. Thus the derivative of $h = Lcj$ at a is the identity. The inverse function theorem produces a smooth local inverse q^{-1} of h on a neighborhood of a. The embedding topology gives an ambient open set S whose preimage under j is the source of this inverse. Intersect S with the chart source and with $(Lc)^{-1}$ of the inverse's target. This gives U containing $j(a)$, and $r = q^{-1}Lc$ is smooth there. For any y with $j(y) \in U$, the choice of S places y in the inverse's source, so $r(j(y)) = q^{-1}(h(y)) = y$. A zero-dimensional source has zero tangent space, on which the same inverse construction applies. □

Lemma Q.89 (Smooth compact lift and detection of boundary derivatives)**.** *Let $D \subset X^d$ be a native compact smooth domain, and let $j : X \hookrightarrow M^e$ and $k : N^\ell \hookrightarrow M$ be smooth embeddings. If $j(D) \subset k(N)$, there are an open neighborhood U of D and a smooth $g : U \to N$ such that $k(g(x)) = j(x)$ for all $x \in D$. Also, if two maps $f_1, f_2 : X \to M$ are differentiable at each point of D and agree throughout D, their derivatives agree at every point of D, including its boundary.*

Proof. The embedding k makes the carrier lift $f : D \to N$ unique. At each $x \in D$, use the preceding smooth left inverse near $k(f(x))$ and compose it with j. This gives a smooth map on an open neighborhood of x which equals f at every carrier point in that neighborhood. The finite compact map extension theorem glues these germs to g while preserving their values on all of D; no agreement of the germs outside D is needed.

For the derivative assertion in positive dimension, the actual native inclusion $i : D \hookrightarrow X$ is smooth and has injective differential even at the boundary: in a half-space chart, its ambient smooth chart extension composes with i to the native coordinate map, with identity differential. Equal dimensions make di_x surjective. Differentiating $f_1 i = f_2 i$ and cancelling this surjection gives $df_{1,x} = df_{2,x}$. In dimension zero both derivatives have zero source and are equal directly. □

Lemma Q.90 (Compact comparison in an arbitrary common ambient)**.** *Let $D \subset X^d$ and $E \subset Z^d$ be compact smooth domains, D nonempty, and $j : X \hookrightarrow M^e$, $k : Z \hookrightarrow M^e$ smooth embeddings. If $j(D) = k(E)$, there is a compact diffeomorphism $h : D \to E$, represented together with its inverse by smooth maps on open neighborhoods, satisfying*

$$k(h(x)) = j(x) \quad (x \in D), \qquad j(h^{-1}(y)) = k(y) \quad (y \in E).$$

The common ambient dimension e is arbitrary. The squares are asserted on the full compact carriers.

Proof. Apply the compact lift construction in both directions. It gives smooth extensions near D and E of $k^{-1}j$ and $j^{-1}k$ on their carriers. Nonemptiness and the image equality supply points of both extension manifolds, so the maps can be defined arbitrarily elsewhere. Image equality and injectivity of j, k show the carrier maps land in E, D and are mutually inverse there. The preceding derivative assertion applied to $k \circ h = j$ on D gives $dk \circ dh = dj$ on its full tangent spaces. Since dj is injective, dh is injective at every carrier point. These are precisely the required smooth neighborhood representation, inverse, and immersion properties of a compact diffeomorphism. □

Lemma Q.91 (Exact form pullback from the full embedded carrier)**.** *Let $h : D \to E$ be any compact diffeomorphism with smooth neighborhood extensions, and j, k smooth embeddings into a common finite-dimensional manifold. If $k \circ h = j$ on the full carrier D, then for every differential form α on that manifold, $h^*(k^*\alpha) = j^*\alpha$ at every point of D.*

Proof. The full carrier derivative assertion gives $dk \circ dh = dj$ at every point of D, including its boundary. The carrier square identifies the base point at which α is evaluated. Substituting the derivative identity in each argument of the alternating form proves the equality. □

Lemma Q.92 (Full tailored-fiber marking of any retained ribbon completion)**.** *Let I be finite toric input and let ϕ be smooth with $cI \le D^2\phi \le UI$, $c, U > 0$. Suppose outside a positive radius it agrees with a degree-two homogeneous end which is continuously differentiable everywhere and smooth away from the origin. There are $F, \beta_0 > 0$ such that for every $\beta \ge \beta_0$, $s \in [0,1]$ and $A^- \ge F^2\beta^2/2$ the following holds. Let (D, μ) be any nonempty retained compact Liouville domain with an actual smooth embedding W of its extension into Y, satisfying*

$$W(D) = \{f_{\beta,s} = 1,\ \phi \le A^-\}, \qquad \mu = W^*\lambda_\phi \quad \text{on } D.$$

For every supplied completion C of this very domain there is an actual complete full-fiber realization $j : F_s \hookrightarrow Y$, with $j(F_s) = \{f_{\beta,s} = 1\}$ and complete primitive $\lambda_{F_s} = j^*\lambda_\phi$, and a marking $q : C \to F_s$ satisfying

$$q^*\lambda_{F_s} = \lambda_C, \qquad j(q(x)) = W(x)\ (x \in D), \qquad q(b, r) = \mathrm{Fl}^r_{Z_{F_s}}(q(b))\ (b \in \partial D, r \in \mathbb{R}).$$

The full-fiber assertion is pointwise in the parameters.

Proof. The uniform full-fiber theorem produces the actual complete restricted primitive and positivity above A^-. Its outward-sublevel construction therefore gives a native compact truncation $E \subset F_s$ with $j(E) = \{f_{\beta,s} = 1, \phi \le A^-\}$ and a completion whose entire ambient is F_s. Compare D and E in Y by their equal full images using the arbitrary-ambient compact comparison. This constructs h with $jh = W$ on D. The exact carrier pullback identity gives $h^*\lambda_{F_s} = W^*\lambda_\phi = \mu$ on all of D. Extend this prescribed compact diffeomorphism to the supplied completion C and the constructed completion of E. The resulting smooth diffeomorphism q preserves the primitive and restricts to h. Primitive preservation intertwines the actual Liouville fields and their complete flows. Applying this equality to the source boundary orbits gives the all-real end formula. No retained marking, fiber lift, or completion comparison was assumed. □

Under the preceding finite potential hypotheses there is $F > 0$ such that, for every $C^+ > F$, there is $\beta_0 > 0$ such that every $\beta \ge \beta_0$ admits $K_* > 0$. For every $R > 0$ and

$$F^2\beta^2/2 \le A^- < (C^+)^2\beta^2/2 = A$$

there is $0 < \delta < A - A^-$ such that every admissible rounding with $0 < \varepsilon \le \delta$ admits the retained rounded pair deformation for every $K \ge K_*$. It retains its original compact domains, embeddings,

full carrier and frontier images, native primitives, collars, completions, pair formulas and smooth ambient inclusion markings with their inverse germs. For every $s \in [0,1]$, its same ribbon completion additionally has the pointwise full-fiber marking of the preceding lemma. The compact formula uses its original ribbon inclusion into Y, and its end formula is the actual full-fiber flow for every real radius.

Proof. Take the maximum of the positive F constants for the retained rounded pair producer and full-fiber ribbon marking producer. After C^{+} is chosen, take the maximum of their β thresholds. Nonnegative square monotonicity makes the resulting lower bound on A^{-} sufficient for both producers. Invoke the existing smooth ambient marking theorem and retain its exact domains, maps, primitive families, collars and deformation Q. Its literal equality with the localized witnessed pairs is retained.

For each s, its localized ribbon image is the entire stated truncated fiber. This carrier is nonempty, since the original retained carrier is nonempty and has the same image. The localized primitive is the pullback of the original ribbon primitive under its original inclusion. The original primitive identity and pullback composition therefore identify it with the pullback of λ_{ϕ} under this same localized ribbon embedding. Apply the preceding finite full-fiber marking producer to this precise domain and its already constructed cylindrical completion. This adds its actual marking, compact square and all-real end formula without changing Q. Every earlier full image, primitive and ambient marking formula remains among the conclusions. No smoothness assertion for the new full-fiber ribbon markings is made. □

Q.2.27 The actual normalized pair junction

Lemma Q.93 (Smooth embeddings through the retained compact presentations)**.** *Let a smooth Liouville family on a fixed compact domain have its constructed localized completion family and compact presentation maps a_t. If J_t is a jointly smooth embedding family of the original extension into a fixed manifold, then $J_t \iota_t a_t$ is a jointly smooth map family on the localized extension at parameter zero.*

Proof. The original projection identity for the compact presentations writes $\iota_t a_t = h_t \iota_0$, where h_t is their jointly smooth original presentation family. Compose the joint family J_t with the joint family $h_t \iota_0$. The displayed projection identity identifies this smooth composition with the asserted family, including real extension germs across both parameter endpoints. □

Lemma Q.94 (Equality of the two retained endpoint scalars)**.** *For every supplied profile and all ϕ, β, A, y, the holomorphic rounded scalar at parameter one equals the tailoring rounded scalar at parameter zero with normalization factor $e^{2\beta}$.*

Proof. At parameter one the moving potential is ϕ and the polynomial is W_1. The normalized initial tailoring polynomial is $e^{-\beta} W_1$, so $e^{2\beta}(|f_0|^2 - 1) = |W_1|^2 - e^{2\beta}$. Both arguments of the same supplied profile are therefore identical. □

Lemma Q.95 (Strict comparison from the full embedded pair)**.** *Let P, Q be witnessed compact pairs whose nonempty ambient carriers and entire ribbon carriers have equal images in one smooth ambient manifold. Suppose their native ambient primitives are restrictions of the same ambient form and their native ribbon primitives are restrictions of the same ribbon form under these embeddings. Then there is a strict completion identification from P to Q with ribbon scale one and end shift zero. Its compact ambient and entire compact ribbon maps preserve their images in the common ambient.*

Proof. The same-dimensional ambient embeddings give a compact diffeomorphism a with its embedding square on an open neighborhood. The embedded carrier theorem gives a compact ribbon diffeomorphism g, including for zero-dimensional ribbons. Its square holds on the full compact carrier; equality there determines all boundary tangent derivatives. These squares give the exact ambient and ribbon primitive pullback identities. For a ribbon point, its ambient-boundary inclusion lies in the ambient carrier. Applying the ambient square there and the ribbon square at the original point proves the pair square by injectivity of the target ambient embedding. Extend this compact pair square to the chosen completions. The resulting strict identification has scale one and zero end shift and retains both compact image squares. □

Lemma Q.96 (The holomorphic-to-tailoring junction)**.** *Under the finite potential hypotheses there is $F > 0$ such that every $C_+ > F$, sufficiently large β, and every permitted $F^2\beta^2/2 \le A_- < A = C_+^2\beta^2/2$ admit one positive rounding width. Every supplied profile thinner than this width, and every sufficiently large normalized coefficient K, gives the actual completed holomorphic and tailoring pair deformations and a specified strict completion identification between the holomorphic final pair at coefficient $K/e^{2\beta}$ and the tailoring initial pair at K. Its ribbon scale is one and its end shift zero; its compact maps preserve the whole embedded rounded carrier and entire ribbon. The ambient and ribbon embedding families remain jointly smooth under the very compact presentation maps of these same two pair deformations.*

Proof. Take the maximum of the two finite-input truncation constants and the maximum of the two large-parameter thresholds. Choose the normalized coefficient threshold at least both the tailoring threshold and $e^{2\beta}$ times the holomorphic threshold. Take the minimum of the two rounding widths and retain the supplied profile in both constructed families. At the holomorphic final endpoint, the potential is ϕ. At tailoring parameter zero the polynomial is $e^{-\beta}W_1$. The factor $e^{2\beta}$ in its rounded scalar therefore makes that scalar exactly equal to the holomorphic final scalar. Their entire truncated fiber images agree as well. The actual plane pullback identity gives $Kf_0^*\lambda_{\mathbb{C}} = (K/e^{2\beta})W_1^*\lambda_{\mathbb{C}}$, so the native ambient primitives and the native ribbon primitives are restrictions of the same forms. Collar localization preserves their entire carriers and these pullback identities. The smooth original presentation families and their exact projection squares preserve the joint ambient and ribbon embedding families on the localized base extensions. Apply the preceding strict comparison to those two retained endpoint pairs. Their witnesses are the constant one inherited from the constructed pair families. □

Q.2.28 The finite marked holomorphic and tailoring prefix

This is the segment from the convenient original pair at coefficient $K/e^{2\beta}$ to the final tailored rounded pair, corresponding to P_5 through P_7 of the source itinerary. The two independently constructed presentations of the normalized junction are recorded as distinct vertices with their specified strict comparison between them.

Definition Q.97 (Retained marked rounded family)**.** *A retained marked rounded family consists of an actual completed pair deformation, its ambient embeddings, the entire embedded compact carriers and their frontiers, its literal ambient and ribbon primitive pullbacks, the constant witness one, and inclusion markings of each ribbon completion onto the entire regular fiber. Joint ambient and ribbon embedding maps are measured in the compact presentations of that very deformation. An ambient marking family additionally specifies its actual target completions and presentation maps, and requires real smooth germs of the marking maps and their inverses in those completed presentations. For the holomorphic family its target is the entire logarithmic ambient; for the*

tailoring family its target is the constructed outer completion. The latter retains the native compact domain, its entire image, its literal Liouville primitive family and the localized completion family.

Lemma Q.98 (The exact three-step prefix)**.** *Two retained marked rounded families H, T, together with their strict normalized junction $q : H_1 \to T_0$, determine the finite chain*

$$H_0 \xrightarrow{H} H_1 \xrightarrow{q} T_0 \xrightarrow{T} T_1.$$

It has exactly three arrows and the ordered vertex list $[H_0, H_1, T_0, T_1]$. All markings remain on these very pairs.

Proof. Start with the empty path at H_0, append its deformation arrow to H_1, append the specified identification arrow to T_0, and append the second deformation arrow to T_1. Each append increases the length by one and adds exactly its target vertex. The dependent endpoints enforce the displayed equalities at each concatenation. □

Under the finite potential hypotheses, there is $F > 0$ such that every $C_+ > F$ and sufficiently large β admit a finite normalized coefficient threshold $K_* > 0$. For every $F^2\beta^2/2 \le A_- < C_+^2\beta^2/2$ there is one positive rounding width. Every supplied profile thinner than that width and every $K \ge K_*$ produces the three-step chain above, with the holomorphic coefficient $K/e^{2\beta}$ and tailoring coefficient K. Both actual pair deformations retain their whole carrier and frontier equations, exact primitives, nonempty ribbons, smooth embedding families, and pointwise full-fiber inclusion markings. Their actual ambient marking maps and inverse maps are jointly smooth in completed coordinates. The junction has scale one and zero end shift and preserves the entire compact ambient and ribbon images pointwise.

Proof. Take the maximum of the two marked-family truncation constants, then the maximum of their large-β thresholds. The normalized coefficient threshold is the maximum of the tailoring threshold and $e^{2\beta}$ times the holomorphic threshold. For each inner level take the minimum of the two rounding widths, and retain the supplied profile in both marked-family constructions. The constructions supply nonempty native carriers and their chosen localized completions; localization preserves the whole carrier images and primitive pullback identities. Compose the native smooth embedding families with their compact presentation maps, using their exact original projection identities. This retains smooth embeddings on the actual source extensions.

At holomorphic parameter one the potential equals ϕ and the polynomial is W_1; at tailoring parameter zero it is $e^{-\beta}W_1$. The normalized rounded scalar equals the holomorphic endpoint scalar for the same profile. The entire truncated fibers agree as well. Quadratic homogeneity of the plane primitive gives $K(e^{-\beta}W_1)^*\lambda_{\mathbb{C}} = (K/e^{2\beta})W_1^*\lambda_{\mathbb{C}}$. Thus the two ambient primitives and the two ribbon primitives are pullbacks of the same forms. The full embedded-pair comparison constructs the strict junction on the chosen completions, with the claimed compact squares, scale and shift. Append precisely these three produced arrows. The markings have never been replaced: the holomorphic maps target the whole logarithmic ambient, the tailoring maps target the same outer completion family, and the ribbon maps target their whole regular fibers. Their already established smooth completed map and inverse germs remain in the retained presentations. □

Corollary Q.99 (Finite toric input supplies the marked prefix)*.* Every finite toric input admits a strongly compatible homogeneous end potential and a smooth global extension for which the marked prefix exists with the truncation, parameter, coefficient and profile quantifiers of the preceding theorem. The same output retains their agreement outside one positive radius, the corresponding agreement along the affine path, the path's smoothness, the end path's homogeneity, and positive lower and upper Hessian bounds uniformly along that path.

Proof. Construct the strongly compatible end and its global extension with uniform Hessian bounds along the affine potential path. At parameter one these are the positive lower and upper Hessian bounds on the global extension. The end is smooth away from zero and homogeneous of degree two, is continuously differentiable at zero, and equals the global extension outside the constructed positive radius. These are every potential hypothesis of the preceding theorem. Apply it and retain its output truncation constant and all subsequent choices and universal quantifiers. □

Proposition Q.100 (Marked tailoring deformation and common completions)**.** *For fixed large β and supplied K, there is finite K_* with $K_*/R^2 \geq \max(K, \kappa_{\mathrm{path}}(R))$ and a uniformly rounded pair family $(E_s, \lambda_{s,K_*}, F_s^-, \mathrm{incl}, 1)$. It has smooth inclusion-induced maps $q_s : \widehat{E}_s \to \widehat{D}_\beta$. At $s = 0$ its actual tuple equals the preceding unnormalized endpoint, including the rounding factor R^2.*

Proof. Let

$$C_\beta = \{(s, x) \in [0, 1] \times D_\beta : |f_s(x)| \geq 1\}.$$

This is compact. Lemma O.5 makes $\pi_{f_s}|f_s|^2$ strictly positive on it, hence bounded below by a positive constant when C_β is nonempty. The numerator in (21) for $Z_{s,K}|f_s|^2$ is

$$2\,\mathrm{Re}(\overline{f}_s Z_0 f_s) + K\pi_{f_s}|f_s|^2.$$

It is therefore positive on C_β for one finite $K = K_*$, uniformly in s, and remains positive when K_* increases. If C_β is empty, this condition is vacuous. The cornered domains

$$E_s^{\mathrm{corner}} = D_\beta \cap \{|f_s| \leq 1\}$$

have outward faces and transverse corners, by (25). Writing $A = C_+^2\beta^2/2$, choose one sufficiently small profile and set

$$E_s = \{m_\epsilon(\phi - A,\ R^2(|f_s|^2 - 1)) \leq 0\}. \tag{26}$$

Choose its width uniformly so the corner collar misses F_s^-. The positive derivative of $|f_s|^2$ persists on a uniform compact collar of $|f_s| = 1$ inside D_β; the potential derivative persists on a collar of ∂D_β. On the changed part of the rounded boundary, both derivatives are positive. On the unchanged faces the corresponding one is positive, so these are smooth Liouville domains.

There is an actual common completion marking. Work in $(\widehat{D_\beta, \lambda_{s,K_*}})$, whose negative flow enters D_β. Let $x = \phi - A$, $y = R^2(|f_s|^2 - 1)$ and $q = m_\epsilon(x, y)$. On $D_\beta \cap \{q \geq 0\}$, if $\partial_x m_\epsilon > 0$ then $x \geq -2\epsilon$, and if $\partial_y m_\epsilon > 0$ then $y \geq -2\epsilon$. Indeed $q \leq \max(x, y) + \epsilon$ and an active partial requires either its argument to be maximal or the arguments to differ by less than ϵ. For small enough ϵ, each active partial therefore multiplies a strictly positive derivative, using the collars just chosen and the inequality on $|f_s| \geq 1$. Thus $Z_{s,K_*}q > 0$ on this compact set. A positive minimum proves finite negative absorption into E_s. Lemma C.22 gives the specified primitive-preserving map

$$q_s : (\widehat{E_s, \lambda_{s,K_*}}) \longrightarrow (\widehat{D_\beta, \lambda_{s,K_*}}),$$

equal to inclusion on E_s. These maps form a smooth family of completion identifications. Indeed, parameter-dependent Liouville collars give smooth total spaces for both completed families, and on the source end the map is

$$q_s(y, r) = \mathrm{Fl}^r_{Z_{s,K_*}}(y), \qquad y \in \partial E_s, \quad r \geq 0,$$

with flow computed in $(\widehat{D_\beta, \lambda_{s,K_*}})$. This is smooth in (s, y, r) and agrees smoothly with inclusion across the seam by the collar construction. Absorption gives its inverse; away from E_s the inverse

is obtained by the unique backward hit on ∂E_s. The strict inequality $Z_{s,K_*}q > 0$ at that boundary makes this hitting time smooth in the point and in s by Lemma U.58. The same parameter-dependent flow argument gives smooth families of the inclusion completion markings used on the preceding holomorphic paths. This proves no global completion claim for the tailored form on all of Y. The fibers F_s^- form a smooth proper family of embedded full ribbons; their primitive is λ_ϕ. This gives a marked deformation from the normalized original fiber to H_β. Since

$$K_* f_0^* \lambda_{\mathbb{C}} = (K_*/R^2) W^* \lambda_{\mathbb{C}},$$

the normalized and unnormalized primitives agree exactly at $s = 0$. The factor R^2 in (26) makes the rounded domains agree as well when the preceding family uses the same profile on $(\phi - A, |W|^2 - R^2)$. Their ribbon and witnessing factor one also agree literally. For the previously fixed coefficient/potential path (f_t, ϕ_t), set

$$\kappa_{\mathrm{path}}(R) = \sup_{t\in[0,1]} \kappa(f_t, \phi_t; R) < \infty.$$

Uniformity in Lemma E.25 and the ratio bound above proves finiteness; the supremum needs no continuity assumption on κ. Increase K_* until $K_*/R^2 \geq \max(K, \kappa_{\mathrm{path}}(R))$ for the supplied K. All positivity inequalities remain valid under this increase. To start with any original $K \geq K_0(R)$, first vary K to K_*/R^2 with the original coefficients and potential fixed; only then traverse the chosen path at that common value.

□

R The exact retained normal of the tailored hypersurface

Lemma R.1 (An actual annihilator lies close to the retained complex normal line)**.** *There are $C, \beta_0 > 0$ such that, on every full endpoint fiber with $\beta \geq \beta_0$, any real coefficient vector p whose radial or angular covector vanishes on the actual tangent space admits $t \in \mathbb{C}$ with*

$$\|p - tG_\chi\|_\infty \leq Ce^{-\sqrt{\beta}}\|p\|_\infty.$$

Possibly increasing C, a witness also satisfies the same inequality in the Euclidean norms of the manuscript. Consequently the infimum of the Euclidean line distances is at most $Ce^{-\sqrt{\beta}}|p|$. These absolute estimates include $p = 0$ and require no division by $|p|$.

Proof. Choose a coordinate j with $|(G_\chi)_j| = \|G_\chi\|_\infty$ and set $t = p_j/(G_\chi)_j$. For every k, correct each of the real and imaginary coordinate unit vectors at k in coordinate j so that the resulting vectors belong to $\ker L_\chi$. Their coordinate product norms are at most two. The radial evaluations of these two vectors are respectively $\mathrm{Re}(p_k - t(G_\chi)_k)$ and $-\mathrm{Im}(p_k - t(G_\chi)_k)$; the angular evaluations are the imaginary and real parts, respectively. Replace these model tangent vectors by actual tangent vectors using the preceding comparison. Since the prescribed covector vanishes on the actual tangent space, each of the two evaluations has absolute value at most $2nCe^{-\sqrt{\beta}}\|p\|_\infty$. Thus $\|p - tG_\chi\|_\infty \leq 4nCe^{-\sqrt{\beta}}\|p\|_\infty$. The explicit fixed-dimensional Euclidean/supremum norm inequalities give the displayed Euclidean estimate, after increasing the constant. Taking the infimum over t completes the proof. □

Lemma R.2 (A nearby vector in the actual tangent hyperplane)**.** *For all sufficiently large β, every $v \in \ker L_\chi$ has a vector $w \in \ker df_1$ with $\|v - w\| \leq Ce^{-\sqrt{\beta}}\|v\|$, uniformly on the full endpoint fiber in fixed logarithmic coordinate norms.*

Proof. Choose a maximal normal coordinate and let R be the real linear map that sends $z \in \mathbb{C}$ to the radial and angular coordinate correction for $z/(G_\chi)_j$. Then $L_\chi R = \mathrm{id}$ and $\|Rz\| \le |z|/\|G_\chi\|$. For large β, the proved error bound and normal lower bound give $\|E_\chi\| \le \|G_\chi\|/2$. The real linear endomorphism $T = df_1 R$ of $\mathbb{C}$ consequently satisfies $|Tz - z| \le |z|/2$. The triangle inequality gives $|z| \le 2|Tz|$, so T is injective and, in real dimension two, surjective. Choose z with $Tz = df_1(v)$ and set $w = v - Rz$. Then $df_1(w) = 0$ and $\|v - w\| \le 2|df_1(v)|/\|G_\chi\|$. Since $L_\chi(v) = 0$, the exact differential gives $|df_1(v)| = |E_\chi(v_u)| \le De^{-\sqrt{\beta}} M\|v\|$. Using $cM \le \|G_\chi\|$ proves the result with $C = 2D/c$. □

Definition R.3 (Retained monomials and the radial error)**.** *Put $m_v = e^{\langle v,u\rangle - \beta + i\langle v,\theta\rangle}$, $q_v = \chi_v m_v$ and $G_\chi = \sum_v q_v v$. On logarithmic covering coordinates define the complex linear form and radial error by*

$$L_\chi(X,Y) = \sum_v q_v(\langle v, X\rangle + i\langle v, Y\rangle), \qquad E_\chi = \sum_v m_v\, d\chi_v - d\chi_0.$$

These use the literal finite cutoffs, with no choice of active support.

Lemma R.4 (Exact differential of the tailored equation)**.** *On the full endpoint fiber one has $\sum_v q_v = \chi_0$. Everywhere, the equation $\mathcal{F}_\beta = -\chi_0 + \sum_v \chi_v m_v$ satisfies $d\mathcal{F}_\beta(X,Y) = L_\chi(X,Y) + E_\chi(X)$.*

Proof. The full endpoint fiber is precisely $\mathcal{F}_\beta = 0$, so its equation gives the first identity. Each monomial has differential $dm_v = m_v(\langle v, du\rangle + i\langle v, d\theta\rangle)$. The product rule in the finite sum gives $d(\chi_v m_v) = q_v(\langle v, du\rangle + i\langle v, d\theta\rangle) + m_v d\chi_v$. Subtracting $d\chi_0$ gives the asserted differential. On the endpoint fiber the small-monomial cutoff is one. Its derivative vanishes because it attains its global maximum, so the actual family function f_1 has the same differential as $\mathcal{F}_\beta$ there. □

Lemma R.5 (Exact normal coordinate formula)**.** *The leading differential satisfies*

$$L_\chi(X,Y) = \sum_j (G_\chi)_j (X_j + iY_j).$$

Proof. Expand the lattice pairing in the defining sum for L_χ and interchange the two finite sums. The coefficient of $X_j + iY_j$ is $\sum_v q_v v_j = (G_\chi)_j$. □

Lemma R.6 (A nearby vector in the leading tangent hyperplane)**.** *There are $C, \beta_0 > 0$ such that every actual tangent vector v of a full endpoint fiber with $\beta \ge \beta_0$ has a vector $w \in \ker L_\chi$ with $\|v - w\| \le Ce^{-\sqrt{\beta}}\|v\|$ in fixed logarithmic coordinate product norms.*

Proof. The retained normal is nonzero. Choose a coordinate j with $|(G_\chi)_j| = \|G_\chi\|$ and put $z = L_\chi(v)/(G_\chi)_j$. Let e have radial component $\operatorname{Re} z$ in coordinate j, angular component $\operatorname{Im} z$ there, and all other components zero. The normal coordinate formula gives $L_\chi(e) = L_\chi(v)$, and the coordinate product norm satisfies $\|e\| \le |z|$. Thus $w = v - e$ belongs to $\ker L_\chi$ and $\|v - w\| \le |L_\chi(v)|/\|G_\chi\|$. The relative tangent defect estimate gives the desired uniform bound. □

Lemma R.7 (Maximum monomial on the entire endpoint fiber)**.** *Let $N = |V|$ and $M = e^{h(u)-\beta}$. On the entire endpoint fiber, $1/N \le M$. If $\log N \le \sqrt{\beta}$, a vertex maximizing $\langle v, u\rangle$ has cutoff one and its retained monomial has norm M.*

Proof. If $M \ge 1$, the lower bound follows from $N \ge 1$. If $M \le 1$, all vertex affine weights are nonpositive, so zero is a maximal affine exponent and $\chi_0 = 1$. The fiber equation and $0 \le \chi_v \le 1$ give $1 = |\sum_v q_v| \le \sum_v |q_v| \le NM$, hence the lower bound. Taking logarithms gives $h(u) - \beta \ge$

$-\log N$. A maximizing vertex is at least every other vertex weight and is at least the zero weight minus $\sqrt{\beta}$. Each all-pairs profile factor is consequently one, so its cutoff is one. Its monomial norm is M. □

Lemma R.8 (Uniform radial error and its tangent consequence)**.** *There is $C > 0$, depending only on the finite toric input, such that on every full endpoint fiber*

$$\|E_\chi\| \le Ce^{-\sqrt{\beta}}M.$$

For every vector (X, Y) tangent to its defining equation, $|L_\chi(X, Y)| \le Ce^{-\sqrt{\beta}}M\,\|(X, Y)\|$. The covector norms in this estimate are the fixed logarithmic coordinate operator norms.

Proof. Choose a uniform first-derivative bound $D \ge 0$ for all cutoffs. When $d\chi_v \ne 0$, the cutoff deficit estimate gives $|m_v| \le e^{-\sqrt{\beta}} \max(1, M)$; when $d\chi_v = 0$, its contribution vanishes. Thus each term has norm at most $De^{-\sqrt{\beta}} \max(1, M)$. The same inequality bounds $d\chi_0$: its nonzero derivative forces $1 \le e^{-\sqrt{\beta}} \max(1, M)$. The finite sum is bounded by $(N+1)D$ times this factor, with the fixed real-to-complex inclusion operator norm included. The preceding lemma gives $\max(1, M) \le NM$. Enlarging the resulting constant by one makes it positive. If the actual equation differential vanishes on (X, Y), its exact decomposition gives $L_\chi(X, Y) = -E_\chi(X)$; the operator norm inequality and $\|X\| \le \|(X, Y)\|$ prove the tangent estimate. □

Lemma R.9 (Uniform retained-normal noncancellation)**.** *There exist $c, C, \beta_0 > 0$ such that every full endpoint fiber with $\beta \ge \beta_0$ satisfies $cM \le \|G_\chi\| \le CM$. Here G_χ uses the supremum norm on complex coordinates.*

Proof. Choose β_0 so that the retained vertices lie in a proper nonempty face and $\log N \le \sqrt{\beta}$. Extend the retained coefficients by zero on a containing face. A maximizing vertex is retained with norm M, so the coefficient supremum norm is at least M. The uniform proper-face noncancellation estimate bounds the corresponding vertex sum below by cM. This sum is precisely G_χ, since every coefficient outside the containing face vanishes. For the upper bound, $|q_v| \le |m_v| \le M$, hence $\|G_\chi\| \le M \sum_v \|v\|$. Increasing the last constant by one makes it positive. □

Lemma R.10 (Relative tangent defect)**.** *There are $C, \beta_0 > 0$ such that for every full endpoint fiber with $\beta \ge \beta_0$ and every actual tangent vector v,*

$$|L_\chi(v)| \le Ce^{-\sqrt{\beta}}\|G_\chi\|\,\|v\|.$$

This is a relative covector defect estimate in fixed logarithmic coordinate norms.

Proof. Choose the constants D and c in the error estimate and normal lower bound. Then $M \le \|G_\chi\|/c$, and the tangent error estimate gives $|L_\chi(v)| \le De^{-\sqrt{\beta}}M\|v\| \le (D/c)e^{-\sqrt{\beta}}\|G_\chi\|\|v\|$. The constant $C = D/c$ is positive and independent of the fiber point. □

R.1 Conormal estimates and the actual critical covectors

Lemma R.11 (Euclidean and coordinate conormal norms)**.** *For every real coordinate vector p one has $\|p\|_\infty \le |p|$. There is a finite constant A_n, depending only on the fixed coordinate basis, such that every complex coordinate vector z satisfies $|z| \le A_n\|z\|_\infty$. Regarding a real vector as complex does not change its Euclidean norm.*

Proof. For each coordinate, $|p_j|^2 \le \sum_k |p_k|^2$, so $|p_j| \le |p|$; take the maximum over j. The canonical linear change from the coordinate supremum norm to the Euclidean norm is continuous in finite dimension. Its operator norm A_n gives the displayed complex inequality directly from the defining operator-norm inequality. Finally $|p_j + i0| = |p_j|$ coordinatewise, so the two sums of squared coordinate norms agree. □

Lemma R.12 (The actual native critical covectors)**.** *There are $C, \beta_0 > 0$, depending only on the toric input, such that the following holds on each full endpoint fiber with $\beta \ge \beta_0$. Let ϕ be a smooth real potential and suppose $d\lambda_\phi$ is nondegenerate at the point. If $d\phi$ annihilates the actual fiber tangent kernel, or if the native symplectic projection of Z_{λ_ϕ} to that kernel is zero, then $p = \nabla\phi(u)$ admits $t \in \mathbb{C}$ with $\|p - tG_\chi\|_\infty \le Ce^{-\sqrt{\beta}}\|p\|_\infty$.*

Proof. The logarithmic covering map pulls back the actual endpoint differential to its coordinate derivative. It pulls $d\phi$ back to $\sum_j p_j du_j$ and λ_ϕ back to $\sum_j p_j d\theta_j$. The first critical condition therefore gives radial annihilation. For the second, let PZ denote the symplectic fiber projection. Its normal correction pairs to zero under $d\lambda_\phi$ with every actual fiber tangent vector v. Thus $d\lambda_\phi(PZ, v) = d\lambda_\phi(Z, v) = \lambda_\phi(v)$. When $PZ = 0$ this gives angular annihilation. Apply the preceding annihilator-line estimate with the actual coefficient vector p. □

Lemma R.13 (The intrinsic critical conditions on an embedded fiber)**.** *Use the same uniform constants as above. Let $j : H \hookrightarrow Y$ be a smooth embedding whose derivative at $a \in H$ has image precisely the actual endpoint tangent kernel at $j(a)$. Suppose $j(a)$ belongs to the endpoint fiber, $d\lambda_\phi$ is nondegenerate there, and the Poisson pairing of the real and imaginary endpoint differentials is nonzero. If a is a critical point of $\phi \circ j$, or a zero of the intrinsic Liouville field for $j^*\lambda_\phi$, then $p = \nabla\phi(u)$ admits the same retained-line approximation. This assertion permits a zero-dimensional H.*

Proof. At a critical point, each vector in the actual tangent kernel has a preimage under dj_a. The chain rule gives vanishing of $d\phi$ on this vector. At an intrinsic field zero, nonzero Poisson pairing makes the restricted form nondegenerate, and the intrinsic field pushes forward to the symplectic projection of Z_{λ_ϕ}. Its vanishing thus gives zero of that projection. The preceding native covector argument and annihilator-line estimate apply to the two cases. □

R.2 Small conormal distance and the polar boundary

Lemma R.14 (Radial contraction of the retained normal)**.** *Write $M = e^{h(u)-\beta}$, $q_v = \chi_v m_v$, and $G_\chi = \sum_v q_v v$. On the full endpoint fiber,*

$$|u \cdot G_\chi| \le h(u)\chi_0 + |V|(\sqrt{\beta} + 2)M.$$

Proof. The exact fiber equation gives $\sum_v q_v = \chi_0$. Therefore $u \cdot G_\chi = h(u)\chi_0 - \sum_v (h(u) - \langle v, u\rangle)q_v$. If $q_v \ne 0$, its cutoff is positive and its support deficit belongs to $[0, \sqrt{\beta} + 2)$. Also $|q_v| \le M$. If the cutoff is zero, its summand is zero. The triangle inequality, the nonnegativity of h and χ_0, and summation over the finite vertex set prove the bound. □

Lemma R.15 (Positive radial angle of a homogeneous gradient)**.** *For a strongly compatible degree-two homogeneous potential ϕ with $\phi(u) \ge a\|u\|_\infty^2$ for some $a > 0$, there are $P, k > 0$ such that, for $u \ne 0$ and $p = \nabla\phi(u)$,*

$$\|p\|_\infty \le P\|u\|_\infty, \qquad \langle p, u\rangle \ge k\|p\|_\infty\|u\|_\infty > 0.$$

Proof. The gradient is continuous on the compact unit sphere and homogeneous of degree one. Its maximum norm there, increased by one, is a positive constant P, giving the first inequality by rescaling u to the unit sphere. Euler's identity gives $\langle p, u\rangle = 2\phi(u) \geq 2a\|u\|_\infty^2$. Thus the second inequality holds with $k = 2a/P$. Its strict positivity also shows that $p \neq 0$. □

Lemma R.16 (Small conormal localization)**.** *Fix a strongly compatible homogeneous potential supplied by the finite-input construction. There exist $\delta, M_*, \beta_0 > 0$ such that, for every $\beta \geq \beta_0$ and every point of the full endpoint fiber, existence of $t \in \mathbb{C}$ with $\|p - tG_\chi\|_\infty \leq \delta\|p\|_\infty$ implies*

$$|V|^{-1} \leq M \leq M_*, \qquad \chi_0 = 1, \qquad |h(u) - \beta| \leq C, \qquad d\beta \leq \|u\|_\infty \leq D\beta$$

for fixed positive C, d, D.

Proof. Choose $c > 0$ with $cM \leq \|G_\chi\|_\infty$. The triangle inequality and $\delta \leq 1$ give $|t|M \leq 2\|p\|_\infty/c$. Let H bound $h(u)/\|u\|_\infty$. Contract $p = tG_\chi + (p - tG_\chi)$ with u. The preceding radial bound and the coordinate estimate $|u \cdot e| \leq n\|u\|_\infty\|e\|_\infty$ give

$$k \leq \frac{2H\chi_0}{cM} + \frac{2|V|(\sqrt{\beta} + 2)}{c\|u\|_\infty} + n\delta.$$

The whole endpoint fiber has $\|u\|_\infty \geq d_0\beta$. Choose δ so $n\delta \leq k/4$ and increase β_0 until the middle term is at most $k/4$. Since $\chi_0 \leq 1$, this gives $M \leq 4H/(ck) = M_*$. Increase β_0 again until $\log M_* \leq \sqrt{\beta}$. The all-pairs cutoff is then exactly one for the constant exponent. The already established bound $M \geq |V|^{-1}$ and $\log M = h(u) - \beta$ give a fixed bound on the support-height error. The upper and lower linear support-height inequalities now give the two radial bounds after increasing the positive constants and threshold. □

The supremum norms in these statements are equivalent to the prescribed Euclidean norms by fixed-dimensional constants. Consequently the existential error criterion contains a sufficiently small Euclidean complex-line distance criterion as in the source lemma.

Lemma R.17 (Localization from the Euclidean conormal distance)**.** *For the same fixed potential, define*

$$\zeta = \frac{\operatorname{dist}_2(p, \mathbb{C}G_\chi)}{\|p\|_2}.$$

There exists $\zeta_* > 0$ such that $\zeta \leq \zeta_*$ on every sufficiently large full endpoint fiber implies all the conclusions of the preceding lemma, with its radial annulus expressed in the prescribed Euclidean norm.

Proof. Choose $A > 0$ such that $\|p\|_2 \leq A\|p\|_\infty$ and let $\zeta_* = \delta/(2A)$. The gradient is nonzero on the whole endpoint fiber, so its Euclidean norm is positive. The assumed normalized distance bound gives $\operatorname{dist}_2(p, \mathbb{C}G_\chi) \leq \delta\|p\|_\infty/2 < \delta\|p\|_\infty$. The complex line contains zero, so the defining infimum supplies t with $\|p - tG_\chi\|_2 < \delta\|p\|_\infty$. Since the supremum norm is at most the Euclidean norm, this t satisfies the preceding lemma's error criterion. The inequalities $\|u\|_\infty \leq \|u\|_2 \leq A\|u\|_\infty$ transfer the radial annulus, increasing its upper constant by A. □

Lemma R.18 (Bounded normalized complex-line approximants)**.** *There are $\delta, T, \beta_0 > 0$ such that every point of the full endpoint fiber with $\beta \geq \beta_0$ and every $t \in \mathbb{C}$ satisfying $\|p - tG_\chi\|_\infty \leq \delta\|p\|_\infty$ obey $|t| \leq T\beta$.*

Proof. Choose $\delta \leq 1$ sufficiently small for the localization lemma. Its lower maximum-monomial bound and normal noncancellation give $c/|V| \leq \|G_\chi\|_\infty$. The triangle inequality gives $|t|\|G_\chi\|_\infty \leq 2\|p\|_\infty$. The gradient upper bound and localized radial upper bound give $\|p\|_\infty \leq PD\beta$. Thus $|t| \leq (2PD|V|/c)\beta$, as asserted. □

R.3 Critical conormals of the full smooth endpoint fiber

Lemma R.19 (Critical conormals on every full endpoint presentation)**.** *Fix a smooth potential with positive uniform lower and upper Hessian bounds. There are $C, \beta_0 > 0$ such that the following holds for every $\beta \geq \beta_0$ and every smooth embedding $j : H \hookrightarrow Y$ with image the entire endpoint fiber. Every critical point of $\phi \circ j$, and every zero of the intrinsic field of $j^*\lambda_\phi$, admits $t \in \mathbb{C}$ with*

$$\|\nabla\phi(u) - tG_\chi\|_\infty \leq Ce^{-\sqrt{\beta}}\|\nabla\phi(u)\|_\infty.$$

No separate tangent-image or symplectic regularity assumptions are required.

Proof. The Hessian lower bound gives ambient nondegeneracy. The uniform tailoring Poisson estimate gives strictly positive real-imaginary Poisson pairing on the entire level-one fiber. Thus the endpoint differential is surjective. For every full endpoint presentation the regular-level tangent theorem identifies the image of dj with the exact differential kernel. The embedded critical-covector theorem now applies, proving the estimate for both the restricted-potential critical condition and the intrinsic field-zero condition. All thresholds are combined by taking their maximum. □

Lemma R.20 (The actual end-gradient estimate at every intrinsic critical point)**.** *Under the smooth positive Hessian hypotheses and agreement with the homogeneous end outside a fixed ball, every intrinsic critical point or field zero on every full endpoint presentation satisfies the uniform retained-line estimate with $p = \nabla\phi_\infty(u)$.*

Proof. Combine the threshold for the actual full-fiber critical estimate with the threshold for equality of the two gradients. At each full-fiber point both the coefficient vector and its norm agree, so substitution yields precisely the same error bound for the homogeneous-end gradient. □

Lemma R.21 (A full symplectic endpoint with the critical estimate)**.** *With the preceding potential, for every sufficiently large β there exists a smooth symplectic full endpoint hypersurface $j : H \hookrightarrow Y$, and every intrinsic critical point or field zero on that same hypersurface satisfies the preceding estimate. The hypersurface has dimension $2n - 2$, including dimension zero when $n = 1$.*

Proof. Positive Poisson pairing constructs the smooth symplectic regular level with image the entire fiber. Apply the preceding estimate to this constructed embedding, and retain its symplectic property in the same conclusion. □

Lemma R.22 (The homogeneous gradient on the full endpoint fiber)**.** *If ϕ and ϕ_∞ agree outside a fixed Euclidean ball, then $\nabla\phi = \nabla\phi_\infty$ on every full level-one tailoring fiber for all sufficiently large β, uniformly in the closed tailoring interval.*

Proof. The growing fiber-radius bound puts every fiber point strictly beyond the fixed ball. Continuity of the Euclidean norm supplies a neighborhood of the point still beyond the ball. The functions agree there, so their differentials, and hence each coordinate gradient coefficient, agree at the point. □

S All critical points and a global intrinsic Lyapunov function

Lemma S.1 (Tailored tangent and conormal control)**.** *The marked hypersurface H_β is smooth and symplectic, $|G_\chi|$ is comparable with M, and $|E_\chi| \leq C\varepsilon_\beta M$. Its tangent plane is uniformly $O(\varepsilon_\beta)$ close to the displayed complex hyperplane.*

Proof. All of H_β lies in the homogeneous end. Put

$$q_v = \chi_v m_v, \qquad G_\chi = \sum_v q_v v, \qquad E_\chi = \sum_v m_v \nabla\chi_v - \nabla\chi_0.$$

On H_β the exact identities are

$$\sum_v q_v = \chi_0, \qquad d\mathcal{F}_\beta = G_\chi \cdot (du + i\, d\theta) + E_\chi \cdot du. \tag{27}$$

If $M \le 1$, the constant has cutoff one, so $1 \le N_0 M$. Thus always $M \ge N_0^{-1}$ and $|u| \ge c\beta$. A maximal nonconstant term has cutoff one once $\beta \ge (\log N_0)^2$: its deficit from the constant is at most $\log N_0 \le \sqrt{\beta}$. Lemma M.5 and independence of the retained vertices give

$$cM \le |G_\chi| \le CM, \qquad |E_\chi| \le C\varepsilon_\beta M. \tag{28}$$

The constants are uniform because there are only finitely many independent support sets. Therefore H_β is smooth and symplectic, and its tangent space is uniformly $O(\varepsilon_\beta)$-close to the complex hyperplane

$$T^0 = \ker\big(G_\chi \cdot (du + i\, d\theta)\big).$$

□

S.1 A uniform exclusion of cutoff-transition critical points

Write $p = \Phi_\infty(u)$ and

$$\zeta(u,\theta) = \frac{\inf_{t\in\mathbb{C}} |p - tG_\chi|}{|p|}.$$

The Euclidean distance of either $(p, 0) = d\phi$ or $(0, p) = \lambda_\phi$ from the real conormal plane of T^0 is $\zeta|p|$. To check this, its orthogonal generators are $(\operatorname{Re} G_\chi, -\operatorname{Im} G_\chi)$ and $(\operatorname{Im} G_\chi, \operatorname{Re} G_\chi)$, of equal norm. Their two projection coefficients give the displayed complex-line distance. By (28), replacing T^0 by the actual tangent plane changes the distance by at most $C\varepsilon_\beta|p|$.

Lemma S.2 (Small conormal distance occurs near the polar boundary)**.** *There exist $\zeta_* > 0$ and $M_* < \infty$ such that, for all sufficiently large β, a point of H_β with $\zeta \le \zeta_*$ satisfies*

$$N_0^{-1} \le M \le M_*, \quad \chi_0 = 1, \quad h(u) = \beta + O(1), \quad |u| \asymp \beta.$$

Proof. Homogeneity and positivity give $|p| \asymp |u|$ and $\langle p, u\rangle \ge \kappa|p||u|$ for some $\kappa > 0$. For the minimizing t, orthogonal projection gives $|t||G_\chi| \le |p|$, hence $|t|M \le C|p|$. Each retained vertex has $0 \le h(u) - \langle v, u\rangle \le \sqrt{\beta} + 2$. Thus

$$|u \cdot G_\chi| \le h(u)\chi_0 + N_0(\sqrt{\beta} + 2)M.$$

Contract $p = tG_\chi + (p - tG_\chi)$ with u and divide by $|p||u|$ to get

$$\kappa \le C\chi_0/M + C(\sqrt{\beta} + 2)/|u| + \zeta.$$

Since $|u| \ge c\beta$, the middle term tends uniformly to zero. Choosing $\zeta_* < \kappa/4$ bounds M above. The constant can only be cut when $\log M > \sqrt{\beta}$, so it is now exactly uncut. The remaining claims follow from $M = e^{h-\beta}$ and $h(u) \asymp |u|$. □

Call a point *bad* if $0 < \chi_a < 1$ for at least one $a \in \mathcal{A}$; otherwise call it good.

S.2 Limits of transition exponents

Lemma S.3 (A transition exponent vanishes on its limiting active face)**.** *Let $a \in V$ be fixed, $\beta_j \to +\infty$, and $y_j = (u_j, \theta_j)$. Suppose the maximum augmented weight is bounded above by a constant $B \geq 0$ and $0 < \chi_a(u_j) < 1$ eventually. Then $q_a(y_j) \to 0$. If also $u_j/\beta_j \to x$, then $\langle a, x\rangle = 1$.*

Proof. The retention and change inequalities give

$$-\sqrt{\beta_j} - 2 < \ell_a(u_j) < B - \sqrt{\beta_j}.$$

The upper bound implies $|q_a(y_j)| \leq \exp(B)\exp(-\sqrt{\beta_j}) \to 0$, since $0 \leq \chi_a \leq 1$. Both endpoints of the displayed interval, divided by $\beta_j > 0$, tend to zero: use $\sqrt{\beta_j}/\beta_j = 1/\sqrt{\beta_j}$. Thus $\ell_a(u_j)/\beta_j \to 0$. As a is a vertex, its offset is one, so $\langle a, u_j/\beta_j\rangle = 1 + \ell_a(u_j)/\beta_j \to 1$. Continuity of the lattice pairing gives the asserted equality. □

S.3 The active face of a normalized limit

Lemma S.4 (Limits of retained coefficients are supported on the active face)**.** *Suppose $\beta_j \to +\infty$, $u_j/\beta_j \to x$, and the augmented maximum weights are bounded above. Then $\langle v, x\rangle \leq 1$ for every $v \in V$. If $q_v(y_j) \to q_v$, every nonzero q_v has $\langle v, x\rangle = 1$. If the points lie in the tailored endpoint fibers and $\chi_0(u_j) = 1$ eventually, then $\sum_v q_v = 1$; in particular some limiting coefficient is nonzero.*

Proof. The weight bound gives $\langle v, u_j\rangle \leq \beta_j + B$. Divide by the eventually positive β_j and pass to the limit. If $\langle v, x\rangle < 1$, choose $d > 0$ such that eventually $\langle v, u_j/\beta_j\rangle \leq 1 - d$. Then $|q_v(y_j)| \leq e^{-d\beta_j} \to 0$, so uniqueness of limits forces $q_v = 0$. Finally the exact fiber equation gives $\sum_v q_v(y_j) = \chi_0(u_j) = 1$ eventually. Finite sums commute with limits, proving the sum identity and nonemptiness of its support. □

S.4 Compactness for vanishing conormal errors

Lemma S.5 (Simultaneous normalized subsequence)**.** *Fix a strongly compatible homogeneous potential with a positive quadratic lower bound. Suppose $\beta_j \to +\infty$, $y_j = (u_j, \theta_j) \in H_{\beta_j}$, $\varepsilon_j \to 0$, and $|p_j - t_j G_{\chi,j}|_\infty \leq \varepsilon_j |p_j|_\infty$ eventually. There is a common subsequence on which $u_j/\beta_j \to x \neq 0$, every retained coefficient converges, and t_j/β_j converges. The augmented maximum weight is eventually bounded and the constant cutoff is eventually one.*

Proof. The errors are eventually below each fixed positive small-conormal threshold. Localization gives $d\beta_j \leq \|u_j\|_\infty \leq D\beta_j$, $M_j \leq M_*$, a bounded support-height error, and $\chi_0 = 1$. Noncancellation and the triangle inequality give $|t_j|\|G_{\chi,j}\|_\infty \leq 2\|p_j\|_\infty$ after shrinking the threshold to at most one. Since $M_j \geq 1/|V|$ and $\|p_j\|_\infty \leq P\|u_j\|_\infty$, this bounds $|t_j|/\beta_j$. Each retained coefficient has magnitude at most M_*. Thus the triple consisting of u_j/β_j, the finite coefficient tuple and t_j/β_j eventually belongs to one closed ball in a finite-dimensional real vector space. Its compactness supplies a common convergent subsequence. The lower bound $d \leq \|u_j/\beta_j\|_\infty$ passes to its limit and gives $x \neq 0$. □

S.5 Excluding critical conormals in the transition region

Lemma S.6 (Every active normal survives a conormal limit)**.** *Let ϕ be a strongly compatible end potential, let $x \in P$ with $h(x) = 1$, and let $A = \{v \in V : \langle v, x\rangle = 1\}$. If $p = \nabla\phi(x) = \sum_{v\in A} b_v v$, with arbitrary real coefficients, then $b_a \neq 0$ for every $a \in A$. Consequently, if $q_v \in \mathbb{C}$ vanish outside A and $p = t\sum_{v\in V} q_v v$ for $t \in \mathbb{C}$, then $q_a \neq 0$ for every $a \in A$.*

Proof. The finite set A is a nonempty proper simplicial face. Compatibility places p in the interior of the fan star of each $a \in A$. Choose a fan cone containing p and write it as $\mathrm{Cone}(R)$, where R is a proper face. The open-star characterization implies $a \in \mathrm{Cone}(R)$ for each $a \in A$. The fan intersection identity then gives $a \in R$: otherwise the cones of R and $\{a\}$ meet only at zero, whereas an active normal is nonzero. Thus $A \subset R$.

Extend R to a facet lattice basis and write $p = \sum_{v \in R} c_v v$ with $c_v \geq 0$. The basis coordinate functional at any $a \in A$ evaluates this expression to c_a and the proposed expression on A to b_a. If $b_a = 0$, then $c_a = 0$, so $p \in \mathrm{Cone}(R \setminus \{a\})$. This is another fan cone. The open-star characterization again puts a in this cone; the fan intersection identity forces $a \in R \setminus \{a\}$, a contradiction.

For the complex assertion take real parts coordinatewise: $p = \sum_{v \in A} \mathrm{Re}(t q_v) v$. A vanishing q_a would give a vanishing real coefficient, contradicting the first assertion. This derives the limiting contradiction directly from open stars and the simplicial fan, replacing the source's intermediate face-minimum argument without strengthening its potential hypotheses. □

Lemma S.7 (Uniform separation on the actual bad region)**.** *Let ϕ be a strongly compatible end potential with $a\|u\|_\infty^2 \leq \phi(u)$ for some $a > 0$. There are $\delta, \beta_0 > 0$ such that on every full endpoint fiber with $\beta \geq \beta_0$, if $0 < \chi_v(u) < 1$ for some augmented exponent $v \in \{0\} \cup V$, then $\delta\|\nabla\phi(u)\|_\infty \leq \|\nabla\phi(u) - tG_\chi(u,\theta)\|_\infty$ for every $t \in \mathbb{C}$. Consequently there are $\zeta_0, \beta_1 > 0$ such that the literal Euclidean distance ratio ζ is at least ζ_0 at every such point for $\beta \geq \beta_1$.*

Proof. If no uniform coordinate bound existed, choose $\beta_j \geq j + 1$, endpoint points with an intermediate cutoff, and scalars t_j whose relative errors are smaller than $1/(j+1)$. The small-conormal compactness lemma supplies one subsequence on which u_j/β_j, every retained coefficient, and t_j/β_j converge; it also bounds augmented maximal weights and makes $\chi_0 = 1$ eventually. Thus the intermediate cutoff eventually belongs to a vertex. Finiteness of V supplies a further subsequence with one fixed intermediate vertex. The preceding transition limit contradiction applies, proving the uniform coordinate bound.

Let $A > 0$ bound the Euclidean norm by A times the real coordinate norm. At large endpoint parameters the position and its gradient are nonzero. If the Euclidean distance ratio were smaller than $\delta/(4A)$, the infimum characterization of distance would supply a retained-line approximant whose coordinate error is at most $(\delta/2)\|\nabla\phi(u)\|_\infty$. This contradicts the coordinate lower bound and the nonzero gradient. The resulting constants depend only on the fixed toric input and the given potential and lower bound. □

Lemma S.8 (The literal normalized conormal limit)**.** *Let $\beta_j \to \infty$, $u_j/\beta_j \to x \neq 0$, $q_{v,j} = \chi_v(u_j) m_v(u_j, \theta_j) \to q_v$ for every $v \in V$, and $t_j/\beta_j \to \tau$. Suppose $\delta_j \to 0$ and eventually $\|p_j - t_j G_{\chi,j}\|_\infty \leq \delta_j \|p_j\|_\infty$, where $p_j = \nabla\phi(u_j)$ and ϕ is a strongly compatible end potential. Then $\nabla\phi(x) = \tau \sum_{v \in V} q_v v$ in complex coordinates.*

Proof. The homogeneous identity for the coordinate gradient gives $p_j/\beta_j = \nabla\phi(u_j/\beta_j)$ eventually. Smoothness off zero and $x \neq 0$ therefore give $p_j/\beta_j \to \nabla\phi(x)$. Finite summation gives $G_{\chi,j} \to \sum_v q_v v$. After division by β_j, the norm of the error is bounded by $\delta_j \|p_j/\beta_j\|_\infty \to 0$. Taking limits in this same equation proves the stated identity. □

Lemma S.9 (No transition exponent admits a vanishing conormal error)**.** *Under the preceding convergence hypotheses, suppose the augmented maximal weights stay bounded and one fixed $a \in V$ satisfies $0 < \chi_a(u_j) < 1$ eventually. These conditions are impossible.*

Proof. Bounded maximal weights imply $x \in P$ and that $q_v = 0$ whenever $\langle v, x \rangle < 1$. The fixed transition exponent has $\langle a, x \rangle = 1$, while its retained coefficient tends to zero, so $q_a = 0$. In

particular $x \neq 0$ and $h(x) = 1$. The normalized conormal equation and the active-coefficient lemma imply $q_a \neq 0$, giving the contradiction. □

Lemma S.10 (Uniform bad-region separation)**.** *For the constructed strongly compatible homogeneous potential, with its positive quadratic lower bound, there are $\delta_1 > 0$ and β_0 such that $\zeta \geq \delta_1$ at every bad point of H_β whenever $\beta \geq \beta_0$.*

Proof. Apply Lemma S.7 to the constructed potential and its positive quadratic lower bound. Its compactness argument uses the direct facet-star contradiction of Lemma S.6: a transition vertex is active at the normalized limit but has zero limiting retained coefficient, contradicting the nonvanishing of every active conormal coefficient. □

S.6 Every intrinsic endpoint critical point has constant cutoff values

Lemma S.11 (The actual critical and zero sets avoid cutoff transitions)**.** *Fix the constructed homogeneous compatible potential and a smooth extension agreeing with it outside a fixed ball, with uniform positive Hessian bounds. For all sufficiently large β, every critical point of the restricted potential and every intrinsic Liouville-field zero on every smooth presentation of the full endpoint fiber has $\chi_a \in \{0, 1\}$ for every augmented exponent $a \in \{0\} \cup V$.*

Proof. The preceding actual intrinsic critical estimate supplies t with error at most $Ce^{-\sqrt{\beta}}\|\nabla\phi_\infty(u)\|_\infty$. On the full endpoint fiber, the growing radial lower bound gives $u \neq 0$; Euler positivity therefore makes this gradient nonzero. Uniform bad-region separation gives a lower bound $\delta\|\nabla\phi_\infty(u)\|_\infty$ for every t whenever some augmented cutoff lies strictly between zero and one. Choose the common threshold so $Ce^{-\sqrt{\beta}} < \delta$. The two bounds then contradict each other at a transition point. Since each cutoff belongs to $[0, 1]$, absence of transitions gives the asserted two values. □

This conclusion specifies the cutoff values at critical points. It does not assert holomorphic neighborhoods, positive active coefficients, equality of the two critical sets, or identification of the full intrinsic core.

Every finite toric input admits a strongly compatible homogeneous potential ϕ_∞, continuously differentiable on the whole vector space, and a globally smooth extension ϕ equal to it outside a fixed Euclidean ball. There is $B > 0$ such that for every $\beta \geq B$ the entire endpoint fiber admits a smooth symplectic embedding with exactly that image. Every critical point of its restricted potential and every zero of its intrinsic Liouville field has all augmented cutoff values in $\{0, 1\}$.

Proof. The homogeneous compatible-potential construction gives $a > 0$ and $\phi_\infty(u) \geq a\|u\|_2^2$, with the corrected potential $\phi_\infty - a\|\cdot\|_2^2$ convex. The smooth convex extension therefore has Hessian bounded below by a and agrees with ϕ_∞ outside a ball. Its Hessian has a finite global upper bound: compactness controls the inner ball, while homogeneity of degree two makes the end Hessian homogeneous of degree zero and hence bounded by compactness on the unit sphere. Also $\|u\|_\infty \leq \|u\|_2$, so the same positive lower bound supplies the native-norm hypothesis of the preceding lemma. Apply that lemma to these constructed potentials. Uniform positive Poisson pairing, supplied by their Hessian bounds, constructs the full smooth symplectic endpoint fiber for every sufficiently large parameter. Taking the maximum of these two thresholds gives the stated embedding and cutoff conclusion on that same full fiber. □

S.7 The exact differential at points with plateau cutoff values

Lemma S.12 (Plateau values give a complex tangent hyperplane)**.** *Suppose every augmented cutoff at a point of the endpoint fiber has value zero or one. Then the radial error is zero and the exact endpoint differential is L_χ. Moreover $L_\chi(Jv) = iL_\chi(v)$, so its kernel is invariant under the standard complex rotation.*

Proof. Each cutoff takes values in $[0,1]$. At a point with value zero it attains a minimum, and at a point with value one it attains a maximum. Its real differential vanishes in either case. Every term defining E_χ consequently vanishes. Substituting this into the exact endpoint differential gives $df_1 = L_\chi$. The coordinate identity $L_\chi(X,Y) = \sum_j (G_\chi)_j (X_j + iY_j)$ shows directly that $L_\chi(-Y,X) = iL_\chi(X,Y)$. □

Lemma S.13 (The exact conormal equation at good critical points)**.** *Fix a smooth potential with uniformly positive and bounded Hessian. For all sufficiently large β, at any point of any smooth presentation of the actual full endpoint fiber with all augmented cutoff values in $\{0,1\}$, either intrinsic potential criticality or an intrinsic Liouville-field zero implies $\nabla\phi(u) = tG_\chi$ for some $t \in \mathbb{C}$.*

Proof. Uniform Poisson positivity gives the regular symplectic fiber and identifies the embedding tangent image with $\ker df_1$. Intrinsic criticality therefore annihilates this kernel by the radial potential covector. An intrinsic Liouville zero annihilates it by the angular primitive covector, by the Liouville equation. At a good point the preceding lemma gives $\ker df_1 = \ker L_\chi$. Apply the coordinate conormal extraction with error zero to the respective annihilating covector. Its resulting norm estimate is $\|\nabla\phi(u) - tG_\chi\| \le 0$, which is precisely the asserted equality. □

S.8 Strict cutoff plateaus at every intrinsic critical point

Lemma S.14 (Closed transition strips have positive conormal separation)**.** *Let ϕ be a strongly compatible end potential with a positive quadratic lower bound. Write $L = \max_{a\in\{0\}\cup V} \ell_a$. There are $\delta, B > 0$ such that, for $\beta \ge B$, every endpoint point having an exponent a with*

$$L - \sqrt{\beta} - 2 \le \ell_a \le L - \sqrt{\beta}$$

satisfies $\delta\|\nabla\phi\|_\infty \le \|\nabla\phi - tG_\chi\|_\infty$ for every complex t.

Proof. If no such constants existed, choose $\beta_j \ge j+1$ and endpoint points with relative conormal error less than $1/(j+1)$ and an exponent in the displayed closed strip. The small-conormal compactness lemma gives a common subsequence with bounded L_j, $u_j/\beta_j \to x \ne 0$, all retained coefficients $q_{v,j} \to q_v$, and $t_j/\beta_j \to \tau$. The strip excludes the constant exponent eventually because $L_j - \sqrt{\beta_j} < 0$. Finiteness gives a further subsequence with one fixed vertex a. The bounds $-\sqrt{\beta_j} - 2 \le \ell_{a,j} \le C - \sqrt{\beta_j}$ imply $\ell_{a,j}/\beta_j \to 0$, hence $\langle a, x\rangle = 1$. On the other hand $|q_{a,j}| \le e^{\ell_{a,j}} \le e^C e^{-\sqrt{\beta_j}} \to 0$. Bounded maximal weights give $x \in P$ and vanishing limiting coefficients off its active facets. The normalized conormal equation gives $\nabla\phi(x) = \tau \sum_v q_v v$, so the active-coefficient lemma forces $q_a \ne 0$. This contradicts $q_a = 0$. This strengthens the source's exclusion to include both plateau endpoints; it uses the same normalized-limit contradiction without a face minimum. □

Lemma S.15 (Every intrinsic critical point has a holomorphic neighborhood)**.** *Let ϕ be a smooth positive-Hessian potential with fixed positive lower and finite upper Hessian bounds, agreeing outside a compact set with a strongly compatible homogeneous end ϕ_∞ having a positive quadratic lower*

bound. There is $B > 0$ such that, for every $\beta \geq B$, every zero of the intrinsic Liouville field and every critical point of ϕ on the actual full endpoint fiber admits a nonempty proper face $s \subset V$ and an ambient open neighborhood O on which

$$\chi_0 = 1, \qquad \chi_v = \begin{cases} 1 & v \in s, \\ 0 & v \notin s, \end{cases} \qquad f_1 = \sum_{v \in s} e^{\langle v,u\rangle - \beta + i\langle v,\theta\rangle}.$$

The face s is exactly the actual retained support at the point.

Proof. The actual full-fiber critical conormal estimate supplies t with $\|\nabla\phi_\infty - tG_\chi\|_\infty \leq Ce^{-\sqrt{\beta}}\|\nabla\phi_\infty\|_\infty$. Every endpoint position and its homogeneous gradient are nonzero for large β. Increase B until $Ce^{-\sqrt{\beta}} < \delta$, where δ is the preceding separation constant. No exponent can lie in the closed strip. Thus every exponent has either $\ell_a < L - \sqrt{\beta} - 2$ or $\ell_a > L - \sqrt{\beta}$. The same error bound, now below the small-conormal localization threshold, makes $\chi_0 = 1$ at the point.

In the first case the factor associated with an attained maximal exponent has argument less than -2, making $\chi_a = 0$. In the second case every factor has positive argument, making $\chi_a = 1$. All these strict finite inequalities persist on a common real open neighborhood. Hence every cutoff keeps its exact value there. The finite retained-simplex theorem puts the retained support s inside a proper simplicial face; every subset of that face is a face. The equation $\sum_v \chi_v m_v = \chi_0 = 1$ excludes $s = \varnothing$.

Write S for the monomial Euclidean size and $N = |V|$. The equation and the triangle inequality give $1 \leq NS$. The fixed small constant satisfies $4Nc < 1$, so $S > 2c$ at the point. Intersect the cutoff neighborhood with the open set $S > 2c$. On it the small-monomial cutoff η is exactly one. The literal formula for f_1 therefore simplifies to the displayed finite holomorphic sum on this actual ambient open neighborhood. □

Lemma S.16 (The limiting face is the full retained face)**.** *Let ϕ be strongly compatible, and let $\beta_j \to \infty$ and endpoint points $y_j = (u_j, \theta_j)$ satisfy the following convergences and bounds:*

$$L_j \leq C, \quad \chi_{0,j} = 1, \quad u_j/\beta_j \to x \neq 0, \quad q_{v,j} = \chi_{v,j} m_v(y_j) \to q_v, \quad t_j/\beta_j \to \tau, \qquad \|\nabla\phi(u_j) - t_j G_{\chi,j}\|_\infty \leq \epsilon_j \|\nabla\phi(u_j$$

Here the first two conditions and the conormal estimate may hold eventually. Then $x \in P$, $\ell(x) = 1$, $\sum_{v\in V} q_v = 1$, and

$$q_v \neq 0 \iff \langle v, x\rangle = 1, \qquad \nabla\phi(x) = \tau \sum_{v \in V} q_v v.$$

The multiplier τ is positive real, as is each q_v on the active face $A(x) = \{v \in V : \langle v, x\rangle = 1\}$. Moreover the actual retained supports s_j eventually equal $A(x)$.

Proof. Bounded maximal weights imply $\langle v, x\rangle \leq 1$ for every $v \in V$. The literal fiber equation with uncut constant passes to the finite limit, giving $\sum_v q_v = 1$. Thus some $q_a \neq 0$. A coefficient off the active face tends to zero exponentially, so this a satisfies $\langle a, x\rangle = 1$ and consequently $\ell(x) = 1$. Homogeneity and the vanishing relative conormal error give the displayed limiting equation. The active-complex-conormal positivity lemma gives the positive multiplier and every positive active coefficient. This also proves both implications in the support equivalence.

For an active v, convergence to $q_v \neq 0$ makes $q_{v,j} \neq 0$ eventually, so v is retained. For an inactive v, consider

$$H_{v,j} = \langle v, u_j/\beta_j\rangle + \sqrt{\beta_j}/\beta_j + 2/\beta_j \longrightarrow \langle v, x\rangle < 1.$$

If v were retained, the actual cutoff-deficit estimate and $L_j \geq 0$ would give $\langle v, u_j\rangle - \beta_j > -\sqrt{\beta_j} - 2$, hence $H_{v,j} \geq 1$ for positive β_j. Thus inactive vertices are eventually not retained. Finiteness of V gives simultaneous equality $s_j = A(x)$. □

Lemma S.17 (Actual critical conormal subsequences converge to face minima)**.** *Assume additionally that ϕ is globally C^1, has a positive quadratic lower bound, and admits $c > 0$ such that $\phi - c\|\cdot\|_2^2$ is convex in the prescribed Euclidean coordinates. Every sequence of actual endpoint points with $\beta_j \to \infty$ and relative conormal errors tending to zero has a subsequence with an eventual fixed nonempty proper retained face s, normalized positions converging to the unique minimum x of ϕ on F_s, and retained coefficients converging to positive real q_v on s. The limits satisfy $\sum_{v\in V} q_v = 1$ and $\nabla\phi(x) = \tau \sum_{v\in V} q_v v$ with $\tau > 0$ real.*

Proof. The small-conormal compactness theorem provides bounded maximal weights, an eventually uncut constant, and a common convergent subsequence for u_j/β_j, all actual retained coefficients, and t_j/β_j; its radial lower bound makes the position limit nonzero. The preceding lemma then gives $s = A(x)$ and eventual equality with the actual retained supports. The maximizing-face theorem makes s a nonempty proper face.

Take real parts of the limiting conormal equation. The resulting real gradient combination is supported on s, since all other q_v vanish. For any $z \in F_s$, every normal $v \in s$ annihilates $z - x$, so $d\phi_x(z-x) = 0$. Convexity gives $0 = d\phi_x(z-x) \le \phi(z) - \phi(x)$, proving literal minimality on F_s. The positive quadratic convexity correction makes this minimum unique: strong gradient monotonicity applied to two face minima would give $2c\|z - x\|_2^2 \le 0$, since both normal gradients annihilate their difference. Hence $z = x$. All coefficient and multiplier conclusions and their convergences are retained from the same actual subsequence. □

Definition S.18 (Uniform approximation by the retained-face minimum)**.** *For an endpoint point $y = (u, \theta)$ let s be its actual retained support. An approximation to precision η consists of positive real parts of the current retained coefficients and a literal minimum x of ϕ on F_s, together with coefficients q_v supported on s and $\tau > 0$ real such that*

$$\sum_{v\in V} q_v = 1, \qquad q_v > 0 \text{ real for } v \in s, \qquad \nabla\phi(x) = \tau \sum_{v\in V} q_v v,$$

and, for every $v \in V$,

$$\|u/\beta - x\|_2 + |\chi_v m_v(y) - q_v| < \eta.$$

The position norm is the prescribed Euclidean norm. On the finite set of vertices these inequalities are equivalent to the bound with the maximum coefficient error; requiring all vertices also includes the vanished coefficients outside s.

Lemma S.19 (The comparison coefficients are normalized gradient coefficients)**.** *An approximation has the source's coefficient form: there are positive real a_v on s with $\nabla\phi(x) = \sum_{v\in s} a_v v$ and with the displayed error bounded by η after substituting $q_v = a_v / \sum_{w\in s} a_w$.*

Proof. Set $a_v = \tau \operatorname{Re} q_v$. Positivity follows from that of τ and the active real coefficients. The off-face coefficients vanish and $\sum_{v\in s} q_v = 1$, so $\sum_{v\in s} a_v = \tau > 0$. Taking real parts of the normal representation gives the gradient formula. Consequently $a_v / \sum_w a_w = \operatorname{Re} q_v = q_v$ on s, and the claimed estimate is the same proved approximation inequality. □

Lemma S.20 (Uniform face-minimum and coefficient approximation)**.** *Under the hypotheses of the preceding face-minimum subsequence lemma, for every $\eta > 0$ there are $\delta, B > 0$ such that any actual endpoint point with $\beta \ge B$ and relative critical conormal error at most δ admits the displayed approximation to precision η. Consequently, for a smooth positive-Hessian potential with fixed lower and upper bounds agreeing at infinity with that end potential, every intrinsic critical or Liouville-zero point on every full endpoint fiber has such an approximation, uniformly for large β.*

Proof. If the first statement failed, choose $\beta_j \geq j+1$ and actual endpoint points with conormal errors at most $1/(j+1)$ but without the approximation. The preceding subsequence lemma gives a fixed eventual retained face s, its literal minimum x, and convergent coefficients q_v with the stated positive normalized representation. Coefficients outside s vanish because the actual retained support eventually equals s. Continuity of the fixed change from native coordinates to Euclidean coordinates gives position error less than $\eta/2$ eventually. Finiteness gives every coefficient error less than $\eta/2$ simultaneously. The current real parts on s are eventually positive by convergence to the strictly positive limiting real parts. Thus every field of the approximation holds on this same subsequence, a contradiction.

For actual full-fiber critical points, the proved intrinsic conormal estimate is $Ce^{-\sqrt{\beta}}$ relative to the homogeneous end gradient. Increase B until this is below δ and apply the first statement. All choices of full-fiber smooth presentation are covered by that intrinsic estimate. □

Lemma S.21 (Positive real face monomials at every intrinsic critical point)**.** *Under the smooth potential and homogeneous-end hypotheses above, for every $\eta > 0$ and all sufficiently large β, each actual full-fiber critical or Liouville-zero point simultaneously has the constructed open proper-face holomorphic equation, strictly positive real monomials on that face at the point, and the uniform face-minimum approximation to precision η.*

Proof. Choose one threshold for the open holomorphic region, the uniform approximation, the exact good-point critical conormal equation, and end gradient agreement. On the constructed neighborhood all cutoffs are exact indicators of the actual nonempty proper face s and $\chi_0 = 1$. The exact good-point conormal equation therefore reads $\nabla\phi(u) = t\sum_{v\in s} q_v v$, where the literal fiber equation gives $\sum_{v\in s} q_v = 1$. End agreement and the radial gradient-angle bound make this real gradient nonzero. Independence of the proper-face normals then forces t and each q_v to be real. The uniform approximation makes each current q_v have positive real part. Since $\chi_v = 1$ on s, $q_v = m_v(y)$, so these actual face monomials are strictly positive real. The same point and actual support retain the open face equation and the Euclidean summed-error approximation already proved. □

S.9 Positive normalized conormal coefficients

Lemma S.22 (Strictly positive active gradient coefficients)**.** *Let ϕ_∞ be strongly compatible, $x \in P$ with $h(x) = 1$, and $A = \{v \in V : \langle v, x\rangle = 1\}$. If $\nabla\phi_\infty(x) = \sum_{v\in A} b_v v$ with real coefficients, then $b_v > 0$ for every $v \in A$.*

Proof. Choose a cone containing the gradient. Strong compatibility and the open-star criterion imply that every generator in A belongs to that cone. Extend its face generators to a facet lattice basis. The coordinate functional of each $a \in A$ reads both the coefficient b_a and a nonnegative coefficient of the gradient in this cone, so $b_a \geq 0$. If $b_a = 0$, the same cone representation puts the gradient in the cone with a erased. That cone is a fan face. Strong compatibility at the active facet a and the open-star criterion would force a to belong to this erased face, a contradiction. Thus $b_a > 0$. □

Lemma S.23 (Reality and positivity of normalized complex coefficients)**.** *With the same x, suppose complex coefficients q_v vanish off A, $\sum_{v\in V} q_v = 1$, and $\nabla\phi_\infty(x) = t\sum_{v\in V} q_v v$ for $t \in \mathbb{C}$. Then t is positive real and every q_v, $v \in A$, is positive real.*

Proof. Taking real and imaginary parts gives $\nabla\phi_\infty(x) = \sum_{v\in A} \mathrm{Re}(tq_v)v$ and $0 = \sum_{v\in A} \mathrm{Im}(tq_v)v$. The preceding lemma makes every real coefficient positive. Extend the independent active face to a

facet lattice basis; applying its coordinate functionals to the imaginary equation gives $\mathrm{Im}(tq_v) = 0$ for each active v. Since $t = \sum_{v \in A} tq_v$ and A is nonempty, t is positive real. Dividing each positive real tq_v by t proves the assertion. □

Lemma S.24 (Normal stationarity identifies a face minimum)**.** *Let ϕ be differentiable and convex. If $x \in F_s$ and $\nabla\phi(x) = \sum_{v \in s} a_v v$ with real coefficients, then x minimizes ϕ on F_s. The convexity hypothesis holds in particular if $\phi - c\|\cdot\|_2^2$ is convex for $c \geq 0$.*

Proof. For $y \in F_s$, each defining normal satisfies $\langle v, y - x\rangle = 1 - 1 = 0$. Hence $d\phi_x(y - x) = \sum_{v \in s} a_v \langle v, y - x\rangle = 0$. The differentiable convex supporting inequality gives $\phi(y) - \phi(x) \geq d\phi_x(y - x) = 0$. The additional claim follows by adding the nonnegative convex quadratic $c\|\cdot\|_2^2$ back to the corrected potential. An invertible linear change between the Euclidean and fixed coordinate presentations preserves convexity, so the same conclusion holds in the specified based space. □

Lemma S.25 (Reality of an exact face conormal)**.** *Let s be a nonempty proper face of the smooth fan polytope, $p \neq 0$ a real vector, and q_v, t complex numbers satisfying $\sum_{v \in s} q_v = 1$ and $p = t \sum_{v \in s} q_v v$. Then t is nonzero real and every q_v is real.*

Proof. Extend the generators of s to a facet lattice basis. The real extensions of the dual coordinate functionals applied to the imaginary part of the conormal equation give $\mathrm{Im}(tq_v) = 0$ for each $v \in s$. Summation and the normalization show $\mathrm{Im}(t) = 0$. The conormal equation and $p \neq 0$ give $t \neq 0$; division by this real number proves $\mathrm{Im}(q_v) = 0$. □

S.10 Quantitative positivity of the potential form

Lemma S.26 (Hessian lower bound for the complex rotation)**.** *For a smooth potential ϕ, let $H = D^2\phi$ in the fixed Euclidean coordinates and suppose $\langle Hv, v\rangle \geq c\|v\|_2^2$ with $c > 0$. For $q = (X, Y)$, $Jq = (-Y, X)$, the actual descended form satisfies*

$$d\lambda_\phi(q, Jq) = \langle HX, X\rangle + \langle HY, Y\rangle \geq c(\|X\|_2^2 + \|Y\|_2^2) \geq c\|q\|_\infty^2.$$

In particular its value is positive for $q \neq 0$.

Proof. The Hessian block formula for the exterior derivative gives $d\lambda_\phi((X, Y), (X', Y')) = \langle HX, Y'\rangle - \langle HX', Y\rangle$. Substitute $(X', Y') = (-Y, X)$ and use linearity of H to obtain the claimed identity. Apply the Hessian lower bound to each summand. Each coordinate supremum norm is bounded by the Euclidean norm of its block. The square of the maximum of the two block supremum norms is therefore at most the sum of their Euclidean squares. Multiplication by $c > 0$ proves the last inequality and strict positivity for a nonzero vector. □

Lemma S.27 (Uniform mixed bound for the potential form)**.** *Suppose in addition $\langle Hv, v\rangle \leq U\|v\|_2^2$ for $U \geq 0$. Let K be the operator norm of the identity from coordinate supremum norm to Euclidean norm. Then*

$$|\langle Hv, w\rangle| \leq U\|v\|_2\|w\|_2, \qquad |d\lambda_\phi(q, r)| \leq 2UK^2\|q\|_\infty\|r\|_\infty.$$

Proof. The Hessian is symmetric by equality of mixed second derivatives and nonnegative by its lower bound. Cauchy's inequality for its symmetric nonnegative bilinear form gives $\langle Hv, w\rangle^2 \leq \langle Hv, v\rangle\langle Hw, w\rangle \leq U^2\|v\|_2^2\|w\|_2^2$. Taking nonnegative square roots proves the mixed bound. The exterior form has two mixed Hessian terms. Bound each by that inequality and use $\|v\|_2 \leq K\|v\|_\infty$ for each block; each block supremum norm is bounded by the product supremum norm. Adding the two bounds proves the stated constant. □

S.11 The intrinsic Lyapunov identity on the good region

Lemma S.28 (Intrinsic energy at a good endpoint point)**.** *At a good endpoint point, let Z be the symplectic projection of the ambient Liouville vector onto the actual fiber tangent space. Then $d\phi(Z) = \omega(Z, JZ)$.*

Proof. The actual fiber tangent kernel is complex by the plateau differential lemma. Thus JZ lies in that kernel. Symplectic projection preserves contraction against every vector in the kernel, and the ambient Liouville equation gives $\omega(Z, JZ) = \lambda_\phi(JZ)$. In logarithmic coordinates $J(X, Y) = (-Y, X)$ and $\lambda_\phi(X, Y) = \nabla\phi \cdot Y$. Hence $\lambda_\phi(JZ) = d\phi(Z)$, proving the identity. □

Lemma S.29 (An intrinsic zero at a good point is critical)**.** *Let a smooth embedding present the actual endpoint tangent kernel at a good point, with nondegenerate ambient potential form and nonzero normal-plane pairing. If its intrinsic Liouville vector is zero there, then every intrinsic directional derivative of the restricted potential vanishes there.*

Proof. The intrinsic vector pushes forward to the symplectic projection of the ambient vector. Thus zero intrinsic vector implies that projection is zero. Contraction on the tangent kernel gives $\lambda_\phi(v) = 0$ for every actual tangent v. At a good point the actual differential is the retained complex differential, so Jv is an actual tangent whenever v is. The coordinate formulas give $d\phi(v) = \lambda_\phi(Jv) = 0$. For every intrinsic tangent, its image under the embedding derivative belongs to that kernel. The chain rule identifies its intrinsic restricted derivative with the zero ambient derivative, as required. □

Lemma S.30 (Strict intrinsic Lyapunov positivity at a good point)**.** *For a smooth potential with positive Hessian, at every good endpoint point where the actual fiber-projected Liouville vector is nonzero, $d\phi(Z) > 0$.*

Proof. The preceding identity expresses the derivative as $\omega(Z, JZ)$. In coordinates this is the sum of the Hessian quadratic forms on the radial and angular components. A nonzero vector has a nonzero component; positive definiteness makes its summand positive and the other summand nonnegative. Their sum is positive. □

Lemma S.31 (Strict positivity for the actual intrinsic field at good points)**.** *Fix a smooth potential with uniformly positive and bounded Hessian. For all sufficiently large β, every smooth presentation of the actual full endpoint fiber has strictly positive derivative of the restricted potential along its intrinsic Liouville vector at each good point where that vector is nonzero.*

Proof. Uniform Poisson positivity identifies the embedding tangent image with the actual defining-kernel and gives its restricted symplectic form. The intrinsic Liouville vector pushes forward to the symplectic projection of the ambient vector. Injectivity of the embedding derivative preserves nonzero vectors. Apply the preceding good-point positivity to its covering coordinate vector. The chain rule identifies this ambient derivative with the intrinsic derivative of the restricted potential, giving the result. □

S.12 Positive intrinsic energy in the transition region

Lemma S.32 (A nearby actual tangent for the complex rotation)**.** *There are $C, B > 0$ such that every actual endpoint tangent vector v with $\beta \geq B$ has an actual tangent vector w with $\|w - Jv\| \leq Ce^{-\sqrt{\beta}}\|v\|$ in fixed coordinate product norms.*

Proof. First choose $u \in \ker L_\chi$ with $\|u - v\| \le Ae^{-\sqrt{\beta}}\|v\|$. The complex linearity of L_χ puts Ju in its kernel. Choose an actual tangent w with $\|w - Ju\| \le De^{-\sqrt{\beta}}\|Ju\|$. The rotation preserves the coordinate norm, and $\|u\| \le (1 + A)\|v\|$, since $e^{-\sqrt{\beta}} \le 1$. The triangle inequality gives the claim with $C = D(1 + A) + A$. □

Lemma S.33 (A quantitative conormal bound from the intrinsic vector)**.** *For a smooth potential with uniformly positive and bounded Hessian there are $A, D, B > 0$ such that at every endpoint point with $\beta \ge B$, the actual projected Liouville vector Z admits $t \in \mathbb{C}$ with $\|\nabla\phi - tG_\chi\| \le Ae^{-\sqrt{\beta}}\|\nabla\phi\| + D\|Z\|$.*

Proof. Put $p = \nabla\phi$. For each $v \in \ker L_\chi$, choose actual tangent w with $\|v - w\| \le Ce^{-\sqrt{\beta}}\|v\|$ and $\|w\| \le (1+C)\|v\|$. The coordinate primitive obeys $|\lambda_\phi(v-w)| \le n\|p\|\|v-w\|$. The actual Liouville contraction gives $\lambda_\phi(w) = \omega(Z, w)$, whose norm is at most $V\|Z\|\|w\|$ for the uniform form bound V. Thus the primitive on the leading kernel has operator bound $nCe^{-\sqrt{\beta}}\|p\| + V(1 + C)\|Z\|$. The coordinate conormal extraction multiplies this bound by four. Increasing both resulting constants by one gives positive A, D. □

Lemma S.34 (Energy against a nearby rotated tangent)**.** *Let the Hessian of a smooth potential have lower bound $c > 0$ and upper bound $U > 0$. Put $V = 2U\|\iota\|^2$, where ι sends each coordinate supremum-norm block to its Euclidean block. Let Z be the actual projected Liouville vector. If an actual tangent w satisfies $\|w - JZ\| \le \eta\|Z\|$ and $\|\nabla\phi\| \le K\|Z\|$, with $K \ge 0$, then*

$$d\phi(Z) \ge \big(c - (V + nK)\eta\big)\|Z\|^2.$$

Proof. Set $e = w - JZ$. Contraction on actual tangents gives $\omega(Z, w) = \lambda_\phi(w)$, while the coordinate formula gives $\lambda_\phi(JZ) = d\phi(Z)$. Subtracting the evaluations on e gives

$$d\phi(Z) = \omega(Z, JZ) + \omega(Z, e) - \lambda_\phi(e).$$

The first term is at least $c\|Z\|^2$. The mixed form bound gives $|\omega(Z, e)| \le V\|Z\|\|e\| \le V\eta\|Z\|^2$. The primitive's coordinate pairing gives $|\lambda_\phi(e)| \le n\|\nabla\phi\|\|e\| \le nK\eta\|Z\|^2$. Combining these three estimates proves the assertion. □

Lemma S.35 (The intrinsic vector controls the gradient at bad points)**.** *Let ϕ have uniformly positive and bounded Hessian and agree outside a fixed Euclidean ball with a strongly compatible homogeneous end potential ϕ_∞ satisfying $\phi_\infty(u) \ge a\|u\|^2$ for $a > 0$. There are $K, B > 0$ such that, at every actual bad endpoint point with $\beta \ge B$, $\|\nabla\phi\| \le K\|Z\|$ for its actual projected Liouville vector.*

Proof. Write $p = \nabla\phi$ and $\epsilon_\beta = e^{-\sqrt{\beta}}$. Outside a uniform threshold, gradient agreement on the entire fiber gives $p = \nabla\phi_\infty$. Bad-region separation gives a uniform $\delta > 0$ with $\delta\|p\| \le \|p - tG_\chi\|$ for every complex t. Choose the t from the intrinsic-vector conormal bound to obtain $\delta\|p\| \le A\epsilon_\beta\|p\| + D\|Z\|$. As $\epsilon_\beta \to 0$, increase the threshold until $A\epsilon_\beta \le \delta/2$. Rearrangement gives $\|p\| \le (2D/\delta)\|Z\|$. Taking the maximum of the three thresholds proves the stated uniform bound with $K = 2D/\delta > 0$. □

Lemma S.36 (Strict intrinsic derivative in the bad region)**.** *Under the preceding potential and end hypotheses, for all sufficiently large β the actual endpoint projected Liouville vector satisfies $d\phi(Z) > 0$ at every bad point with $Z \ne 0$ and nonzero normal-plane pairing.*

Proof. Nonzero normal-plane pairing puts Z in the actual tangent kernel. Choose an actual tangent w with $\|w - JZ\| \le C\epsilon_\beta \|Z\|$. The bad gradient bound gives $\|\nabla\phi\| \le K\|Z\|$. The nearby-rotation energy bound therefore gives $d\phi(Z) \ge (c - (V + nK)C\epsilon_\beta)\|Z\|^2$. Choose a threshold at which $(V + nK)C\epsilon_\beta < c/2$, and increase it to cover the two preceding estimates. The resulting lower bound is greater than $(c/2)\|Z\|^2 > 0$ whenever $Z \neq 0$. □

This proof uses the actual intrinsic vector and its contraction equation in place of introducing the induced-metric gradient. The same tangent comparison and conormal separation control its energy directly.

S.13 The strict derivative on the entire endpoint full fiber

Throughout this subsection fix a smooth potential ϕ whose Euclidean Hessian satisfies $c\,\mathrm{id} \le D^2\phi \le U\,\mathrm{id}$ for $c, U > 0$. Assume it agrees outside a fixed Euclidean ball with a strongly compatible homogeneous end potential ϕ_∞ satisfying $\phi_\infty(u) \ge a\|u\|^2$ for $a > 0$. Every intrinsic field below is the Liouville vector determined by the pullback of λ_ϕ.

Lemma S.37 (Strict projected derivative everywhere on the endpoint)**.** *For all sufficiently large β, the actual projected Liouville vector satisfies $d\phi(Z) > 0$ at every point of the endpoint fiber where $Z \neq 0$ and the normal-plane pairing is nonzero.*

Proof. Every cutoff lies in $[0, 1]$. Either all augmented cutoffs are zero or one, giving good-point positivity, or some cutoff is strictly between zero and one, giving bad-point positivity. Use the uniform threshold from the bad estimate; the good estimate holds for every parameter at which its pointwise hypotheses hold. These two alternatives exhaust every endpoint point and prove the assertion. □

Lemma S.38 (Strict derivative for every full-fiber intrinsic field)**.** *There is $B > 0$ such that for every $\beta \ge B$ and every smooth embedding with image the entire endpoint fiber, its actual intrinsic Liouville vector Z_H satisfies $d(\phi|_H)(Z_H) > 0$ wherever $Z_H \neq 0$.*

Proof. Uniform normal-plane positivity and the full-fiber tangent theorem give the embedding tangent image equal to the actual defining kernel. They also give nondegeneracy of the pulled-back exterior form. The intrinsic Liouville contraction identity identifies the image of Z_H with the actual ambient symplectic projection. Injectivity of the embedding derivative makes that image nonzero when $Z_H \neq 0$. Lift the image to the logarithmic cover and apply the projected positivity theorem. The chain rule identifies the resulting ambient derivative with $d(\phi|_H)(Z_H)$. Take the maximum of the normal-plane and projected positivity thresholds to obtain a threshold independent of the embedding and its point. □

Lemma S.39 (Equality of the critical set and the intrinsic zero set)**.** *For all sufficiently large β, on every smooth presentation of the entire endpoint fiber the critical set of the restricted potential equals the zero set of its actual intrinsic Liouville vector.*

Proof. At a critical point every tangent derivative vanishes, including the derivative along Z_H. Strict positivity wherever $Z_H \neq 0$ therefore forces $Z_H = 0$. Conversely, the full-fiber critical-good theorem makes every intrinsic zero a good point. Uniform normal-plane positivity and the full-fiber tangent theorem give the tangent-kernel and symplectic contracts there. The good-point zero-to-critical theorem then makes every directional derivative vanish. The maximum of these three uniform thresholds proves both implications simultaneously. □

Proposition S.40 (Strict Lyapunov derivative away from the critical set)**.** *For all sufficiently large β, on every smooth presentation of the actual full endpoint fiber, the potential critical set equals the zero set of the primitive-determined intrinsic Liouville vector, and $d(\phi|_H)(Z_H) > 0$ away from that critical set.*

Proof. Choose the maximum of the thresholds for strict positivity away from the intrinsic zero set and for equality of the two sets. At a noncritical point that equality implies $Z_H \neq 0$; strict positivity follows. The same threshold retains the equality of the critical and zero sets at every point, proving both assertions for the fixed potential. □

Lemma S.41 (Localization of every critical point)**.** *Fix the constructed strongly compatible end ϕ_∞, retaining its continuously differentiable zero extension and positive Euclidean convexity correction, and its smooth extension ϕ with uniformly positive and bounded Hessian. They agree outside a fixed ball. For all sufficiently large β, the following assertions hold on every full-fiber presentation.*

Every zero of the intrinsic Liouville field Z_H and every critical point of $\phi|_{H_\beta}$ lies in an open region where the equation is exactly

$$\sum_{i=1}^{k} e^{\langle v_i, u\rangle - \beta + i\langle v_i, \theta\rangle} = 1 \tag{29}$$

for the normals of one proper nonempty face $F \subset P$. At such a point all these monomials are positive real and, with $\Phi_\infty(x_F) = \sum_i a_i v_i$,

$$u/\beta \longrightarrow x_F, \qquad q_{v_i} \longrightarrow a_i / \sum_j a_j > 0.$$

The convergence assertions are uniform over all such points as $\beta \to \infty$. The zero set and critical set coincide. Precisely, for every $\delta > 0$ there is β_δ such that every such point for $\beta \geq \beta_\delta$ admits one of these finitely many faces with

$$|u/\beta - x_F| + \max_i \left| q_{v_i} - \frac{a_i}{\sum_j a_j} \right| < \delta; \qquad q_v = 0 \quad (v \notin \{v_1, \ldots, v_k\}).$$

Proof. The positive holomorphic localization lemma constructs the actual nonempty proper retained support and an ambient open neighborhood on which the literal endpoint equation is the sum of precisely these monomials. At the point each such monomial is positive real. For every $\delta > 0$, the same lemma gives, above one threshold, a minimum x on this actual polar face and positive normalized coefficients with Euclidean position error plus each monomial error less than δ. The quantitative uniqueness lemma identifies x with x_F. The normalized-coefficient lemma writes $\Phi_\infty(x_F) = \sum_i a_i v_i$ with $a_i > 0$ and identifies each limit coefficient exactly as $a_i / \sum_j a_j$. Independence of the face normals makes this representation unique. The support is nonempty and finite, so the individual summed-error inequalities imply the displayed maximum inequality. Outside the retained support the retained monomials vanish identically; the constructed neighborhood has those cutoffs identically zero. All thresholds are uniform over every point and full-fiber presentation. Finally the full-fiber critical-zero lemma, applied to the same ϕ and ϕ_∞, identifies the two sets. Taking the maximum of its threshold and the localization threshold proves all assertions simultaneously. □

S.14 Properness and the full-core orbit bridge

In the fiber results below, fix a smooth potential ϕ with the quantitative Euclidean Hessian bound $D^2\phi \geq c\,\mathrm{id}$, $c > 0$. Its end potential ϕ_∞ is C^1 everywhere, smooth away from zero and homogeneous of degree two for every positive dilation. Fix $L > 0$ such that $\phi = \phi_\infty$ for $\|u\| \geq L$. The complete primitive and flow always mean the restriction of this same λ_ϕ to the entire literal tailored level-one fiber.

Lemma S.42 (Relative compactness characterizes the marked full core)**.** *Let C be an actual completion of a compact Liouville domain D. Then its native full core and its image in C are compact, and the image is exactly the set of points whose entire forward orbit has compact closure. Under a primitive-preserving inclusion marking onto a complete Liouville manifold H, the image of this same core in H is exactly the set of relatively compact forward orbits in H.*

Proof. The all-forward-time core is a closed subset of the compact image of D: each time map is continuous and the compact image is closed. The inclusion is an embedding, so compactness of the image also gives compactness of the native core. Every core orbit stays in that compact image, hence has compact closure.

Conversely, suppose an orbit leaves the compact image at time $t \geq 0$. The completion covering writes the point as (b, r) on the nonnegative cylinder. Its subsequent points have radius $r + s$ for $s \geq 0$. The cylinder image is closed: its complement is precisely the image of the compact-domain interior, which is open by the local diffeomorphism property of the inclusion. Thus if the forward orbit had compact closure, the preimage of that closure under the closed cylinder embedding would be compact. The continuous radius would be bounded above there, contrary to the points of radius $r + s$ for all $s \geq 0$. The orbit therefore cannot leave the compact image. Finally primitive preservation intertwines the Liouville fields and their complete flows. A homeomorphism carries the closure of each orbit onto the closure of its transported orbit, and preserves compactness. The prescribed compact inclusion square identifies the transported core with its marked image in H. □

Lemma S.43 (The retained potential is proper on the whole fiber)**.** *Let ϕ be the retained potential with the quantitative lower Hessian bound and homogeneous end specified above, and let $j : H \hookrightarrow Y$ present the entire closed tailored fiber. Then $\phi \circ j : H \to \mathbb{R}$ is proper.*

Proof. The preceding full-fiber sublevel result makes every set $\{\phi \circ j \leq A\}$ compact. A compact set $K \subset \mathbb{R}$ is closed and bounded above, say by A. Its preimage is closed by continuity of $\phi \circ j$ and is contained in the compact sublevel $\{\phi \circ j \leq A\}$. It is therefore compact. Continuity and compact preimages characterize proper maps to the locally compact Hausdorff space $\mathbb{R}$, proving the assertion. □

Lemma S.44 (Compact full skeleton on the same fiber presentation)**.** *Retain an actual complete intrinsic primitive $\mu = j^*\lambda_\phi$ on the full fiber and a potential level A such that $d(\phi \circ j)(Z_H) > 0$ throughout $\phi \circ j \geq A$. For every completion of the literal truncation $\{\phi \circ j \leq A\}$, its native full core is precisely the set of points of H with relatively compact forward intrinsic orbit. This set and its embedded image under j are compact.*

Proof. Every chosen completion has the proved inclusion marking onto this same full fiber. Apply the preceding marked orbit characterization to that marking, whose compact inclusion is the literal inclusion of the truncation into H. The subtype image of its native core is therefore the same core subset of H. Compactness follows from compactness of the completion's native core. The continuous embedding j takes this compact set to a compact embedded image. □

Proposition S.45 (Strict Lyapunov function and complete ends)**.** *Fix the retained smooth potential ϕ, with uniform Euclidean Hessian bounds $c\,\mathrm{id} \le D^2\phi \le U\,\mathrm{id}$ for $c, U > 0$. It agrees outside a fixed ball with the strongly compatible degree-two end ϕ_∞, which is C^1 globally, smooth away from zero, and satisfies $\phi_\infty(u) \ge a\|u\|^2$ for $a > 0$. For all sufficiently large β, the restriction $\phi|_{H_\beta}$ is proper and $d\phi(Z_H) > 0$ away from the critical set. The field is complete, and H_β is the completion of a compact Liouville domain with compact core. The same complete flow has nonempty forward omega-limit sets contained in the critical set for every relatively compact forward orbit.*

Proof. Choose the maximum of the thresholds for full-fiber completion markings and strict intrinsic Lyapunov positivity. The former constructs a smooth presentation of the entire fiber, its actual complete restricted primitive, and inclusion markings from every sufficiently large potential truncation. On that very presentation the strict derivative theorem applies: the good points satisfy the exact complex tangent energy identity, while conormal separation and the perturbation estimate give positivity at bad points. This uses the direct contraction-energy proof rather than an auxiliary comparison with a separately chosen induced-metric gradient.

Closedness of the full fiber and the compact potential sublevels give properness. The actual completion marking identifies its relatively compact forward orbits with the full compact-domain core. That core is compact by the all-forward-time characterization. Finally the smooth flow chain rule and strict positivity give a nonempty omega-limit set contained in the field's zero set on every relatively compact forward orbit. The proved equivalence between intrinsic zeros and potential critical points identifies the latter set with the critical set. Every step retains the same potential, fiber presentation and complete flow. □

S.15 Critical tori, uniqueness, and the full Hessian

S.16 Literal face annihilator tori

Lemma S.46 (An integral basis containing a face)**.** *Every nonempty proper face support s is contained in the vertex support t of a facet, and the vectors of t form an integral basis of N.*

Proof. The proper face supports containing s form a nonempty finite partially ordered set, since they are subsets of the finite vertex set of Q. Choose a maximal member t. It is nonempty and a maximal proper face, so is a facet. The integral-basis property of its vertices is one of the finite input conditions. □

Lemma S.47 (Integral generation and generator equations)**.** *Let s be the vertices of a nonempty proper face of Q. Every lattice point in $\mathrm{cone}(s)$ is a nonnegative integral combination of s. Consequently*

$$T_{\mathrm{cone}(s)} = \{\theta : \exp(i\langle v, \theta\rangle) = 1 \text{ for every } v \in s\}.$$

Proof. Extend s to the integral basis b given by a containing facet. The real extension of each coordinate of b reads off the coefficient in any real basis expansion. A nonnegative real expansion supported on s therefore equals the integral basis expansion coefficient by coefficient. The latter coefficients are integers, nonnegative on s, and zero elsewhere. This proves the first assertion. The integral angle pairing is additive and commutes with integral scaling. Its vanishing on s thus implies its vanishing on every lattice point of the cone; the converse follows because each generator is itself a lattice point of the cone. Finally the exponential map from $\mathbb{R}/(2\pi\mathbb{Z})$ to the unit circle is injective and sends zero to one, giving the displayed character equations. □

Lemma S.48 (Connected face annihilator)**.** *The torus $T_{\mathrm{cone}(s)}$ in the preceding lemma is compact and connected.*

Proof. The annihilator is an intersection of closed character kernels in the compact quotient torus, hence is compact. Let b be the containing integral facet basis. Prescribe arbitrary quotient angles on its vectors outside s and zero on its vectors in s. Extend these values uniquely to an integral-linear map from the lattice to $\mathbb{R}/(2\pi\mathbb{Z})$. Its values on the original coordinate basis give a point of the original quotient torus. This construction is continuous: each output coordinate is a finite integral-linear combination of the prescribed angles. Its image is precisely the face annihilator, by the preceding generator criterion and uniqueness of a linear map on a basis. Its domain is a finite product of connected circles, so its image is compact and connected. □

S.17 Affine constrained minima for the critical-torus equations

Definition S.49 (Shifted face equations)**.** *For a finite face-normal set s and levels $b : s \to \mathbb{R}$, put*

$$A_s(b) = \{x : \langle v, x\rangle = b_v \text{ for every } v \in s\}.$$

At $b_v = 1$ this is the affine subspace containing the corresponding polar face. The shifted constraints in the reduced critical equations are $A_s(\mathbf{1} + y)$.

Lemma S.50 (Every face level is attainable)**.** *For a nonempty proper face of the original finite toric input, $A_s(b)$ is nonempty for every real level vector b.*

Proof. Choose a facet containing s and its integral lattice basis. Assign value b_v to each basis vector in s, and arbitrary real values to the remaining basis vectors. Extend the resulting lattice-linear functional to the real lattice space. In the fixed Euclidean basis it is pairing with a vector x. Consequently $\langle v, x\rangle = b_v$ for every $v \in s$, so $x \in A_s(b)$. □

Lemma S.51 (Stationarity on a shifted affine slice)**.** *Let ϕ be differentiable and convex on the whole real coordinate space, and let $x \in A_s(b)$. Then x minimizes ϕ on $A_s(b)$ if and only if $d\phi_x$ vanishes on $E_s = \{d : \langle v, d\rangle = 0 \text{ for every } v \in s\}$. In particular a representation $\nabla\phi(x) = \sum_{v\in s} a_v v$ implies constrained minimality.*

Proof. For $d \in E_s$, the whole line $x + td$ lies in $A_s(b)$. A constrained minimum is therefore a local minimum of the differentiable one-variable function $t \mapsto \phi(x + td)$ at zero; its derivative $d\phi_x(d)$ is zero. Conversely, if $y \in A_s(b)$ then $y - x \in E_s$. The supporting derivative inequality for a convex differentiable function gives $0 = d\phi_x(y - x) \le \phi(y) - \phi(x)$, proving the minimum inequality. For the normal representation, $d\phi_x(d) = \sum_{v\in s} a_v\langle v, d\rangle = 0$ on E_s. □

Lemma S.52 (A unique minimum on every shifted face slice)**.** *Suppose ϕ is C^1, $\phi(x) \ge a|x|^2$ for some $a > 0$, and $\phi(x) - c|x|^2$ is convex for some $c > 0$. Every shifted slice $A_s(\mathbf{1} + y)$ associated with a nonempty proper face has exactly one global constrained minimum.*

Proof. The affine slice is a closed intersection of linear level sets and is nonempty by the preceding level-attainment lemma. Fix x_0 in it. For $|u| \ge \max(1, |\phi(x_0)|/a)$, the quadratic lower bound implies $\phi(u) \ge a|u|^2 \ge |\phi(x_0)| \ge \phi(x_0)$. Thus minimization on the closed slice reduces to minimization on a compact ball together with x_0; continuity gives an attained minimum. If x and z were two minima, both derivatives annihilate $z - x$ by the preceding stationarity lemma. Quantitative convexity gives

$$2c|z - x|^2 \le \langle \nabla\phi(z) - \nabla\phi(x), z - x\rangle = 0.$$

Since $c > 0$, this forces $z = x$. □

Remark S.53. This supplies global constrained minima for the literal shifted equations. Smooth dependence of that minimum and its multiplier on the levels, the reduced positive implicit-function branch, and the full Hessian inertia are further assertions of the critical-torus construction.

S.18 Construction of the reduced coefficients

Definition S.54 (The face constraint and normal maps)**.** *For a nonempty proper face-normal set s, use one coordinate for each $v \in s$, and set*

$$A_s x = (\langle v, x\rangle)_{v\in s}, \qquad B_s a = \sum_{v\in s} a_v v.$$

These are linear maps and satisfy $\langle B_s a, x\rangle = \sum_{v\in s} a_v (A_s x)_v$.

Lemma S.55 (Invertibility of the constrained derivative)**.** *The map A_s is surjective and B_s is injective. If H satisfies $\langle HX, X\rangle > 0$ for every $X \neq 0$, then*

$$(X, b) \longmapsto (HX - B_s b, A_s X)$$

is an invertible linear map on $\mathbb{R}^n \times \mathbb{R}^s$.

Proof. Extend any prescribed levels on s to the lattice by assigning zero outside s. Nonemptiness of the resulting affine slice gives surjectivity of A_s. If $B_s b = 0$, choose X with $A_s X = b$. Then $0 = \langle B_s b, X\rangle = \sum_v b_v^2$, so every $b_v = 0$. For a vector in the kernel of the displayed map, $A_s X = 0$ and $HX = B_s b$. Therefore $\langle HX, X\rangle = \sum_v b_v (A_s X)_v = 0$. Positive definiteness gives $X = 0$, and injectivity of B_s gives $b = 0$. The map is an injective endomorphism of a finite-dimensional vector space, and hence is bijective. □

Lemma S.56 (Smooth branch of actual constrained minima)**.** *Let ϕ be the strongly compatible homogeneous potential, globally C^1 and with the positive quantitative convexity contract. Let x_F be its polar-face minimum, and a^0 its positive normal coefficients. There are functions $x(y), a(y)$ smooth at zero in every finite order, with $x(0) = x_F$, $a(0) = a^0$, such that, for all sufficiently small y,*

$$A_s x(y) = \mathbf{1} + y, \qquad \nabla\phi(x(y)) = B_s a(y).$$

Each $x(y)$ is the unique global minimum of ϕ on its displayed affine slice.

Proof. The face equations imply $x_F \neq 0$, so ϕ is smooth at x_F. Apply the smooth implicit-function theorem to

$$(y, (x, a)) \longmapsto (\nabla\phi(x) - B_s a, A_s x - \mathbf{1} - y).$$

It vanishes at $(0, (x_F, a^0))$. Its derivative in (x, a) is the invertible map of the preceding lemma, using the positive Hessian at x_F. Thus the implicit-function theorem supplies the stated branch and the two equations. The normal equation annihilates every direction in the kernel of A_s, and convexity therefore makes $x(y)$ a global constrained minimum. Quantitative convexity supplies its uniqueness. □

Lemma S.57 (Envelope identity and the literal reduced coefficient)**.** *For $\psi(y) = \phi(x(y))$, its derivative is the coefficient $a(y)$ for all sufficiently small y. Consequently the literal coefficient $d\psi$ is smooth at zero in every finite order, and every coordinate of $d\psi(0)$ is strictly positive.*

Proof. Smoothness at zero in order one gives differentiability of x throughout a neighborhood of zero. At such a parameter y, differentiating the locally valid level equation gives $A_s Dx(y)h = h$. The chain rule and the normal equation give

$$D\psi(y)h = \langle \nabla\phi(x(y)), Dx(y)h\rangle = \langle B_s a(y), Dx(y)h\rangle = \sum_v a_v(y) h_v.$$

Thus $d\psi = a$ on a neighborhood of zero. Their local equality transfers smoothness at zero from a to $d\psi$. Their values at zero coincide with the strictly positive coefficients a^0. □

Corollary S.58 (The normalized branch from the actual reduced potential)*.* The literal coefficient $d\psi$ admits a locally unique positive branch of the normalized equations

$$d\psi(\epsilon \log q) = \mu q, \qquad \sum_v q_v = 1,$$

smooth at zero in every finite order, with $\mu(0) = \sum_v a_v^0$ and $q_v(0) = a_v^0 / \sum_w a_w^0$.

Proof. The envelope identity supplies smoothness and positivity for the literal coefficient. Apply the positive normalized reduced-branch lemma to that coefficient. Its base normalization formulas give the stated values. □

Lemma S.59 (The reconstructed radial normal equation)**.** *For all sufficiently large β, the reconstructed radial branch satisfies*

$$\nabla\phi(u_\beta) = \beta\mu(\beta^{-1}) B_s q(\beta^{-1}).$$

Proof. The positive continuous weights imply $\beta^{-1} \log q(\beta^{-1}) \longrightarrow 0$ as $\beta \to \infty$. Continuity of x and $x(0) \neq 0$ ensure that its values at these parameters are nonzero for all sufficiently large β. Degree-two homogeneity of ϕ implies $\nabla\phi(\beta X) = \beta \nabla\phi(X)$ for $X \neq 0$ and $\beta > 0$: differentiating $\phi(\beta X) = \beta^2 \phi(X)$ in a direction D gives $D\phi(\beta X)(\beta D) = \beta^2 D\phi(X) D$, and cancellation of β gives the claimed gradient scaling in every coordinate. Now apply the normal equation at the small parameter, replace its coefficient by μq using the reduced branch equation, and use linearity of B_s. This gives the displayed identity. □

Corollary S.60 (The constructed radial branch lies on the full fiber)*.* For the potential constructed from the finite toric input, every nonempty proper face-normal set s has the preceding reduced branch. For all sufficiently large β, its radial point

$$u_\beta = \beta x(\beta^{-1} \log q(\beta^{-1}))$$

satisfies $f_{1,\beta}(u_\beta, \theta) = 1$ for every angle θ whose characters at all normals in s are one.

Proof. The constructed strong potential gives its actual polar-face minimum, whose unprescribed normal levels are strictly below one and whose prescribed levels equal one. The smooth minimum and normalized branches are differentiable at zero, with positive base weights. The reconstructed fiber lemma therefore applies to their actual shifted levels and normalization. It yields exactly the asserted fiber membership. The finite-input assertion follows by first applying the proved strong potential existence theorem and then this construction to each face. □

S.19 The positive reduced Hessian

Lemma S.61 (The reduced Hessian is the positive constrained Schur form)**.** *Let $x(y)$ and $a(y) = d\psi(y)$ be the actual constrained minimizer and reduced coefficient, with $\psi(y) = \phi_\infty(x(y))$, constructed above. For every sufficiently small y and every nonzero $h \in \mathbb{R}^k$,*

$$\langle Da(y)h, h\rangle = \langle D^2\phi_\infty(x(y)) Dx(y)h, Dx(y)h\rangle > 0.$$

Proof. The local branches are differentiable on a neighborhood of zero. Differentiate $Ax(y) = \mathbf{1} + y$ and $\nabla\phi_\infty(x(y)) = A^T a(y)$ to obtain $ADx(y) = \mathrm{id}$ and $H_y Dx(y) = A^T Da(y)$. Pair the second identity with $Dx(y)h$ and use the first to get the displayed equality. For $h \neq 0$, the first identity also gives $Dx(y)h \neq 0$. The base minimum is nonzero, so continuity permits shrinking the neighborhood until $x(y) \neq 0$. The positive Hessian of the homogeneous potential at $x(y)$ then proves the strict inequality. □

S.20 The reduced positive implicit-function branch

Lemma S.62 (Smooth coefficients of the actual constrained minimum)**.** *Let $s = \{v_1, \dots, v_k\}$ be a nonempty proper face-normal set and $Ax = (\langle v_i, x\rangle)_i$. Let ϕ_∞ be the constructed strongly compatible homogeneous potential, globally C^1, with the positive quantitative convexity contract. Write x_F for its unique polar-face minimum and $a_i^0 > 0$ for its normal coefficients. The unique global minimum $x(y)$ on $Ax = \mathbf{1} + y$ and its unique coefficients $a(y)$ are smooth at zero in every finite order, and satisfy on a neighborhood of zero*

$$x(0) = x_F, \qquad a(0) = a^0, \qquad \nabla\phi_\infty(x(y)) = A^T a(y).$$

The reduced function $\psi(y) = \phi_\infty(x(y))$ has derivative $a(y)$ on that neighborhood, and its Hessian is positive definite at every sufficiently small parameter.

Proof. The normals are linearly independent because they belong to an integral facet basis. Hence A is surjective and A^T is injective. Consider the smooth equation near $(x_F, a^0, 0)$

$$(x, a, y) \longmapsto (Ax - \mathbf{1} - y, \nabla\phi_\infty(x) - A^T a).$$

It is smooth there because $x_F \neq 0$. Its derivative in (x, a) is $(X, b) \mapsto (AX, HX - A^T b)$, where $H = D^2\phi_\infty(x_F)$ is positive definite. If this derivative vanishes, then $AX = 0$ and $X^T HX = X^T A^T b = 0$, so $X = 0$. Injectivity of A^T then gives $b = 0$. This derivative has equal domain and codomain dimension, hence is invertible. The implicit-function theorem supplies smooth $x(y), a(y)$ solving the displayed equations near zero.

Every point of this branch is a constrained global minimum by the normal stationarity criterion. The proved quantitative uniqueness of that minimum identifies it with the actual global minimum on the shifted slice. Injectivity of A^T makes the coefficients unique. For a parameter direction h, differentiating $Ax(y) = \mathbf{1} + y$ gives $A(Dx(y)h) = h$. Therefore

$$D\psi(y)h = \langle A^T a(y), Dx(y)h\rangle = \langle a(y), h\rangle.$$

Differentiating the normal equation gives $H_y Dx(y)h = A^T Da(y)h$, where $H_y = D^2\phi_\infty(x(y))$. Consequently $\langle Da(y)h, h\rangle = \langle H_y Dx(y)h, Dx(y)h\rangle > 0$ for $h \neq 0$, since $ADx(y)h = h$ forces $Dx(y)h \neq 0$. This is the asserted positive Hessian of the reduced function. □

Definition S.63 (Normalized reduced critical equation)**.** *For a map $a : \mathbb{R}^k \to \mathbb{R}^k$, define*

$$\mathcal{F}(\epsilon, q, \mu) = \left(a(\epsilon \log q) - \mu q,\ \sum_i q_i - 1 \right),$$

where $\log q$ is coordinatewise. The coefficient map is an explicit argument, whose smoothness must be established before applying the next lemmas. The equation imposes both the normal critical equation and normalization.

Lemma S.64 (The normalization derivative has an explicit inverse)**.** *If $\mu_0 \neq 0$ and $\sum_i q_i^0 = 1$, the linear map*

$$J(r, s) = (-\mu_0 r - s q^0, \sum_i r_i)$$

is invertible. For an output (h, b) its inverse is

$$\tau = \sum_i h_i + \mu_0 b, \qquad J^{-1}(h, b) = (\mu_0^{-1}(\tau q^0 - h), -\tau).$$

Proof. The sum of the first component of the proposed inverse is $\mu_0^{-1}(\tau - \sum_i h_i) = b$, while its image under the first component of J is $-(\tau q^0 - h) + \tau q^0 = h$. Conversely, if $(h, b) = J(r, s)$ then $\tau = -\mu_0 \sum_i r_i - s + \mu_0 \sum_i r_i = -s$. Thus $\mu_0^{-1}(\tau q^0 - h) = r$ and $-\tau = s$. These two identities prove that the displayed linear maps are mutual inverses. □

Lemma S.65 (Derivative of the literal reduced equation)**.** *Suppose a is smooth at zero, in every finite differentiability order, and all $q_i^0 > 0$. Then $\mathcal{F}$ is smooth in every finite differentiability order at $(0, q^0, \mu_0)$ and its derivative in (q, μ) at that point is $J(r, s) = (-\mu_0 r - s q^0, \sum_i r_i)$.*

Proof. Each coordinate logarithm is smooth on the positive half-line, so the argument $\epsilon \log q$ and its composition with a are smooth at the stated point. On the slice $\epsilon = 0$ the equation is exactly $(a(0) - \mu q, \sum_i q_i - 1)$. Differentiating this slice in direction (r, s) gives $(-\mu_0 r - s q^0, \sum_i r_i)$. The chain rule for insertion of the zero parameter identifies this with the unknown-variable derivative of the full equation. □

Lemma S.66 (A locally unique positive reduced branch smooth at zero)**.** *Let $a : \mathbb{R}^k \to \mathbb{R}^k$ be smooth at zero in every finite differentiability order, with $k \geq 1$ and $a_i(0) > 0$ for every i. Set $\mu_0 = \sum_i a_i(0)$ and $q_i^0 = a_i(0)/\mu_0$. There is a function $z(\epsilon) = (q(\epsilon), \mu(\epsilon))$ smooth at zero with $z(0) = (q^0, \mu_0)$, such that for all sufficiently small ϵ all its weights and multiplier are positive and*

$$a(\epsilon \log q(\epsilon)) = \mu(\epsilon) q(\epsilon), \qquad \sum_i q_i(\epsilon) = 1.$$

On a neighborhood of $(0, q^0, \mu_0)$, a point solves the reduced equation if and only if its unknowns equal $z(\epsilon)$.

Proof. Positivity and $k \geq 1$ give $\mu_0 > 0$, $q_i^0 > 0$, $\sum_i q_i^0 = 1$, and $a(0) = \mu_0 q^0$. Hence the reduced equation vanishes at the stated base point. Its unknown-variable derivative is invertible by the preceding two lemmas. The smooth implicit-function theorem therefore supplies a branch smooth at zero and the stated local uniqueness equivalence. Continuity of the branch and the finitely many strictly positive base coordinates preserve positivity of all weights and of the multiplier on a common neighborhood of zero. □

Remark S.67. The positive-branch lemma explicitly requires its coefficient map to be smooth at zero in every finite differentiability order. The smooth-constrained-minimum and envelope lemmas furnish this hypothesis for the literal derivative of the reduced potential. The radial reconstruction, angle torus, uniqueness among all critical points, and full Hessian are further geometric assertions of the critical-torus proposition. The regularity statements here retain smoothness at the base parameter in every finite order and differentiability on a neighborhood, which suffice for this implicit equation and the positive reduced Hessian. They do not assert a common neighborhood on which the chosen branches are smooth in every order, the stronger regularity phrasing of the source.

S.21 Recovering the radial scale from the reduced branch

Lemma S.68 (Bounded radial displacement of a smooth reduced branch)**.** *Let $x : \mathbb{R}^k \to E$ be differentiable at zero, where E is a real normed vector space. Let $q : \mathbb{R} \to \mathbb{R}^k$ be differentiable at zero with every $q_i(0) > 0$. Define*

$$u(\beta) = \beta x\big(\beta^{-1} \log q(\beta^{-1})\big).$$

There is $C > 0$ such that $\|u(\beta) - \beta x(0)\| \leq C$ for every sufficiently large β. Also $x(\beta^{-1} \log q(\beta^{-1})) \to x(0)$ as $\beta \to \infty$.

Proof. Each coordinate logarithm is differentiable at the positive value $q_i(0)$. Thus $f(\epsilon) = x(\epsilon \log q(\epsilon))$ is differentiable at zero and $f(0) = x(0)$. Differentiability gives $C > 0$ and a neighborhood of zero on which $\|f(\epsilon) - f(0)\| \le C|\epsilon|$. For all sufficiently large positive β, substitute $\epsilon = \beta^{-1}$ and multiply by β to obtain the asserted uniform bound. The convergence follows from continuity of f at zero and $\beta^{-1} \to 0$. □

Lemma S.69 (Reconstruction in the full tailored fiber)**.** *Let $s \subset V$, $\beta \neq 0$, $q_v > 0$, and $\sum_{v\in s} q_v = 1$. Suppose $\langle v, x\rangle = 1 + \beta^{-1} \log q_v$ and the character of v at θ is one for every $v \in s$. Suppose at $u = \beta x$ the cutoffs are exactly one on $\{0\} \cup s$ and zero on $V \setminus s$. Then the literal normalized monomial at (u, θ) is q_v for $v \in s$, and $(u, \theta) \in f_1^{-1}(1)$.*

Proof. Linearity gives $\langle v, \beta x\rangle - \beta = \log q_v$. Exponentiating and using the trivial character proves the monomial identity. The defining tailored equation therefore equals $-1 + \sum_{v\in s} q_v = 0$. The full endpoint fiber equals the zero set of this equation, including the small-monomial cutoff in its definition, so the reconstructed point belongs to $f_1^{-1}(1)$. □

Lemma S.70 (Exact plateau neighborhoods of a reconstructed face branch)**.** *Let $s \subset V$ and let x satisfy $\langle v, x\rangle = 1$ for $v \in s$ and $\langle v, x\rangle < 1$ for $v \in V \setminus s$. If $\|u(\beta) - \beta x\|$ is bounded for all sufficiently large β, then for every sufficiently large β there is an open neighborhood of $u(\beta)$ on which $\chi_0 = 1$, $\chi_v = 1$ for $v \in s$, and $\chi_v = 0$ for $v \in V \setminus s$.*

Proof. The finitely many linear pairings of the bounded displacement are bounded in absolute value by one constant $C \ge 0$. Every active affine weight lies in $[-C, C]$, the constant weight is zero, and every inactive weight is at most $C - \beta d_v$, where $d_v = 1 - \langle v, x\rangle > 0$. Thus the maximal augmented weight lies in $[0, C]$. For large β, $\sqrt{\beta} > 2C$ and $C - \beta d_v < -\sqrt{\beta} - 2$ for every inactive vertex. The latter follows, for example, once $\sqrt{\beta} \ge 2/d_v$ and $\sqrt{\beta} > C + 2$. Every active and constant weight is strictly within $\sqrt{\beta}$ of the maximum; every inactive weight is strictly more than $\sqrt{\beta} + 2$ below it. The defining cutoff product therefore has precisely the asserted values. These finitely many strict inequalities persist on an open neighborhood by continuity of the affine weights and their maximum. □

S.22 The exact face Lagrangian Hessian

Definition S.71 (Face equation and weighted normal square)**.** *For a finite set of lattice normals s, write*

$$f_s(u, \vartheta) = \sum_{v\in s} e^{\langle v,u\rangle - \beta + i\langle v,\vartheta\rangle}, \qquad G_q(X) = \sum_{v\in s} q_v \langle v, X\rangle^2, \qquad \mathcal{L} = \phi(u) - \lambda \operatorname{Re} f_s(u, \vartheta).$$

The angular variable here is a local real lift of the period-2π torus.

Lemma S.72 (Second derivative of the literal face equation)**.** *Let $m_v = e^{\langle v,u\rangle - \beta + i\langle v,\vartheta\rangle}$ and $\ell_v(X, Y) = \langle v, X\rangle + i\langle v, Y\rangle$. Then $D^2 f_s[V, W] = \sum_{v\in s} m_v \ell_v(V)\ell_v(W)$. If every m_v is real, with $q_v = m_v$, then $D^2 \operatorname{Re} f_s[(X, Y), (X, Y)] = G_q(X) - G_q(Y)$.*

Proof. The derivative of each exponential summand is $m_v \ell_v$, because its exponent is affine linear. Differentiating this covector differentiates only m_v, and gives $m_v \ell_v(V)\ell_v(W)$. Sum over the finite set. For $V = W = (X, Y)$ and real $m_v = q_v$, the real part of the summand is $q_v(\langle v, X\rangle^2 - \langle v, Y\rangle^2)$; summing proves the formula. □

Lemma S.73 (Exact ambient Lagrangian Hessian)**.** *For a smooth real potential ϕ and real face monomials q_v, the actual second derivative of $\mathcal{L}$ is*

$$D^2\mathcal{L}[(X,Y),(X,Y)] = D^2\phi(u)[X,X] - \lambda G_q(X) + \lambda G_q(Y).$$

This is an ambient Lagrangian identity; the identification with the intrinsic constrained Hessian requires stationarity and a local fiber parametrization.

Proof. The potential depends only on u. The linear projection to u has zero second derivative, so its pullback has Hessian $D^2\phi(u)[X,X]$. Second derivatives are linear under subtraction and constant scaling. Apply the preceding identity to $-\lambda \operatorname{Re} f_s$ and expand. □

Lemma S.74 (Positive weighted normal square and its kernel)**.** *If $q_v \geq 0$ for all $v \in s$, then $G_q(X) \geq 0$. If all these weights are strictly positive, then $G_q(X) = 0$ if and only if $\langle v, X\rangle = 0$ for every $v \in s$; consequently $G_q(X) > 0$ for every direction outside that common kernel.*

Proof. Every term $q_v\langle v, X\rangle^2$ is nonnegative. A finite sum of nonnegative real terms vanishes exactly when every term vanishes. For positive q_v, this is equivalent to $\langle v, X\rangle^2 = 0$, hence to $\langle v, X\rangle = 0$. A nonnegative nonzero real number is strictly positive, giving the last assertion. □

Lemma S.75 (Stationarity of the actual face Lagrangian)**.** *Suppose ϕ is differentiable at u, all face monomials at (u,ϑ) are real with values q_v, and $\nabla\phi(u) = \lambda\sum_{v\in s} q_v v$. Then $D\mathcal{L}(u,\vartheta) = 0$.*

Proof. For a real logarithmic tangent (X,Y), differentiation of the face sum gives $D\operatorname{Re} f_s(X,Y) = \sum_v q_v\langle v, X\rangle$; the angular contribution is purely imaginary. The normal equation gives $D\phi(u)[X] = \lambda\sum_v q_v\langle v, X\rangle$. Their difference is zero on every tangent vector, proving stationarity. □

Lemma S.76 (Second derivative of a stationary pullback)**.** *Let $f : E \to \mathbb{R}$ and $g : F \to E$ be smooth maps between real normed spaces. If $Df(g(a)) = 0$, then for $V, W \in F$,*

$$D^2(f\circ g)(a)[V,W] = D^2 f(g(a))[Dg(a)V, Dg(a)W].$$

Proof. The first derivative is $D(f\circ g) = Df(g)\circ Dg$. Differentiate this composition of linear maps using the product rule. Evaluating in V, W gives $D^2 f(g(a))[Dg(a)V, Dg(a)W] + Df(g(a))[D^2 g(a)[V,W]]$. The second summand vanishes by stationarity. □

Lemma S.77 (Actual constrained Hessian in a local fiber parametrization)**.** *Let $g : F \to \mathbb{R}^n \times \mathbb{R}^n$ be smooth, with $f_s(g(z)) = 1$ for all z near a. Assume that at $g(a) = (u,\vartheta)$ the face monomials are real and the normal equation of the preceding stationarity lemma holds for a smooth potential ϕ. Then $D(\phi\circ \mathrm{pr}_u \circ g)(a) = 0$ and, writing $(X,Y) = Dg(a)V$,*

$$D^2(\phi\circ\mathrm{pr}_u\circ g)(a)[V,V] = D^2\phi(u)[X,X] - \lambda G_q(X) + \lambda G_q(Y).$$

Proof. On the stated neighborhood, $\mathcal{L}\circ g + \lambda$ equals $\phi\circ\mathrm{pr}_u\circ g$ literally, since $\operatorname{Re} f_s \circ g = 1$. Thus their first and second derivatives at a agree. Stationarity and the first-derivative chain rule give a zero first derivative. The stationary pullback identity gives the second derivative of $\mathcal{L}$ evaluated on $Dg(a)V$. Substitute the exact ambient Lagrangian Hessian formula. □

Lemma S.78 (Normal equation implies intrinsic criticality on the full fiber)**.** *Let $j : N \hookrightarrow Y$ be a smooth embedding with image the entire endpoint fiber $f_1^{-1}(1)$, and let $j(z) = (u,\theta)$ be a good point, where every augmented cutoff is zero or one. Write $q_v = \chi_v(u)e^{\langle v,u\rangle - \beta + i\langle v,\theta\rangle}$. If these retained monomials are real and a smooth potential satisfies $\nabla\phi(u) = \lambda\sum_{v\in V} q_v v$, then z is an intrinsic critical point of $\phi\circ\mathrm{pr}_u\circ j$.*

Proof. Because $f_1 \circ j = 1$, its derivative vanishes, so every vector in the image of $Dj(z)$ belongs to the actual endpoint differential kernel. At a good point the cutoff derivative terms vanish, and the endpoint differential is $\sum_v q_v \ell_v$. For a lifted tangent (X, Y) in this kernel, its real part is $\sum_v q_v \langle v, X\rangle = 0$. The normal equation therefore gives $D\phi(u)[X] = 0$. Every ambient tangent has a unique logarithmic lift, and the chain rule for j identifies this derivative with the intrinsic scalar derivative on N. It vanishes on every intrinsic tangent vector, as asserted. □

S.23 Actual critical points from the constructed reduced branch

Lemma S.79 (The reconstructed retained normal)**.** *Let $s \subset V$, $\beta \neq 0$, and $q_v > 0$. Suppose $\langle v, x\rangle = 1 + \beta^{-1} \log q_v$ and the character of v at θ is one for every $v \in s$. At $u = \beta x$, assume the cutoffs are one on $\{0\} \cup s$ and zero at the other vertices. Then the literal retained monomials equal q_v on s and zero on $V \setminus s$. They are real, and their retained real normal equals $B_s q = \sum_{v \in s} q_v v$.*

Proof. The shifted level and trivial character give the literal monomial $e^{\langle v, \beta x\rangle - \beta + i\langle v, \theta\rangle} = q_v$ on s. Multiplication by its cutoff one preserves this value. At every vertex outside s, the cutoff is zero; zero is not a vertex, so the exceptional uncut constant does not contribute to the vertex normal. Thus the retained monomials are exactly the stated real values. Terms outside s vanish in their finite normal sum; summing the remaining $q_v v$ gives $B_s q$. □

Let ϕ_∞ be a strongly compatible end potential which is globally C^1, and suppose that $\phi_\infty(u) - c\|u\|^2$ is convex for some $c > 0$, where the norm is Euclidean. Let ϕ be a globally smooth potential agreeing with ϕ_∞ outside a fixed Euclidean ball. For each nonempty proper face-normal set s, there exist maps x and $z = (q, \mu)$, smooth at zero in every finite order, such that $x(0)$ belongs to the polar face of s and minimizes ϕ_∞ there, and a number $B > 0$ with the following property. For every $\beta \geq B$, all $q_v(\beta^{-1})$ and $\mu(\beta^{-1})$ are positive, and

$$u_\beta = \beta x\big(\beta^{-1} \log q(\beta^{-1})\big)$$

satisfies $f_{1,\beta}(u_\beta, \theta) = 1$ for every angle θ whose characters at all $v \in s$ are one. For every smooth embedding $j : N^{2n-2} \longrightarrow \mathbb{R}^n \times T^n$ with image the entire fiber $\{f_{1,\beta} = 1\}$, there is $p \in N$ with $j(p) = (u_\beta, \theta)$ and

$$d\big(\phi \circ \mathrm{pr}_1 \circ j\big)_p(V) = 0 \qquad \text{for every } V \in T_p N.$$

The assertion applies to every embedding with this image; it does not assert existence of such an embedding, exhaustive classification of critical points, or the Hessian inertia.

Proof. The actual positive reduced branch supplies the polar-face minimum $x(0)$, the smooth maps x, z, positive base weights, and, for all sufficiently large β, positive weights and multiplier, the shifted levels, the radial normal equation

$$\nabla\phi_\infty(u_\beta) = \beta\mu(\beta^{-1}) B_s q(\beta^{-1}),$$

and membership of (u_β, θ) in the full fiber for every stated angle. Its strict polar-face inequalities and smoothness at zero give the cutoff plateaus: the cutoffs are one on $\{0\} \cup s$ and zero on the remaining vertices at u_β. The small parameter used for these levels tends to zero, since $q(\beta^{-1})$ tends to the positive vector $q(0)$, so its coordinate logarithms remain bounded and their products with β^{-1} tend to zero.

The preceding retained-normal lemma therefore makes every retained monomial real and identifies its real normal with $B_s q(\beta^{-1})$. On every sufficiently large full fiber the gradients of ϕ and

ϕ_∞ agree. Combining this with the radial equation gives the literal normal equation for ϕ, with multiplier $\lambda = \beta\mu(\beta^{-1})$. Choose $B > 0$ larger than all the finitely many thresholds just used.

Fix $\beta \geq B$ and one of the stated angles. Full-fiber membership and the image equality of j supply a preimage p. Lift (u_β, θ) to logarithmic angle coordinates. At this lift the cutoffs are zero or one, the retained monomials are real, and the literal normal equation holds. The full-fiber normal converse, which differentiates the constant equation $f_{1,\beta} \circ j = 1$, now gives the asserted vanishing scalar differential at p. □

S.24 Exhaustion by the actual critical face tori

Write $A_s x = (\langle v, x\rangle)_{v\in s}$ and $B_s a = \sum_{v\in s} a_v v$. For a proper nonempty face-normal set s, let $x_s(y)$ denote the actual constrained minimum on $A_s x = \mathbf{1} + y$, and let $a_s(y)$ be its actual reduced coefficient vector. The actual normalized implicit branch is $z_s(\epsilon) = (q_s(\epsilon), \mu_s(\epsilon))$. All norms on weight vectors below are the maximum coordinate norm.

Lemma S.80 (Positive monomials determine the angle and shifted level)**.** *If the literal monomial $m_v(u,\theta) = e^{\langle v,u\rangle - \beta}\exp(i\langle v,\theta\rangle)$ is positive real, then*

$$\exp(i\langle v,\theta\rangle) = 1, \qquad \log m_v(u,\theta) = \langle v,u\rangle - \beta.$$

Here the character and its equation are defined on the quotient torus.

Proof. The character has norm one, so $|m_v| = e^{\langle v,u\rangle-\beta}$. Positivity makes $|m_v| = m_v$. Cancel this positive real factor in the monomial formula to obtain the character equation. Taking its real logarithm gives the shifted level. □

Lemma S.81 (Weight proximity controls the normalized multiplier)**.** *Let $a : \mathbb{R}^s \to \mathbb{R}^s$ be continuous at zero, $q_v^0 > 0$, and $\mu^0 = \sum_v a_v(0)$. Suppose a branch z has the local uniqueness property*

$$a(\epsilon \log q) = \mu q, \quad \sum_v q_v = 1 \quad \Longleftrightarrow \quad z(\epsilon) = (q,\mu)$$

in a neighborhood of $(0, q^0, \mu^0)$. There is a neighborhood of $(0, q^0)$ such that any μ solving these equations at a pair (ϵ, q) in that neighborhood satisfies $z(\epsilon) = (q,\mu)$.

Proof. On a neighborhood of q^0 every coordinate is positive, so coordinate logarithms are continuous there. Thus the map

$$(\epsilon, q) \longmapsto \left(\epsilon, q, \sum_v a_v(\epsilon \log q)\right)$$

is continuous at $(0, q^0)$ and takes it to $(0, q^0, \mu^0)$. Choose a neighborhood mapped into the stated uniqueness neighborhood. For a solution with $\sum_v q_v = 1$, summing its coefficient equation gives $\sum_v a_v(\epsilon \log q) = \mu \sum_v q_v = \mu$. Consequently the solution belongs to that uniqueness neighborhood and equals the branch value there. □

Lemma S.82 (The approximation uses the actual base coefficients)**.** *Let ϕ_∞ be a strongly compatible, globally C^1 end potential, and suppose $\phi_\infty - d\|\cdot\|_2^2$ is convex for some $d > 0$. For a proper nonempty face s, let $x^0 \in F_s$ minimize ϕ_∞ on F_s, and suppose $\nabla\phi_\infty(x^0) = B_s a^0$. If an endpoint point has actual retained support s and has a face-minimum approximation to precision η, then*

$$\left|\mathrm{Re}(\chi_v m_v) - \frac{a_v^0}{\sum_w a_w^0}\right| < \eta \qquad (v \in s).$$

Proof. The normalized-coefficient lemma gives a face minimum b and coefficients $a_v > 0$ with $\nabla\phi_\infty(b) = B_s a$; its approximation bounds the complex difference from $a_v/\sum_w a_w$ by η after discarding the nonnegative position error. Uniqueness of the polar-face minimum gives $b = x^0$. Surjectivity of A_s makes B_s injective, so the two gradient representations give $a = a^0$. Substitute this equality in the bound. The absolute value of the real part of a complex difference is at most its norm, proving the asserted real coefficient estimate. □

In the next two theorems, ϕ_∞ is strongly compatible and globally C^1, with $\phi_\infty - d\|\cdot\|_2^2$ convex for some $d > 0$ and $a\|u\|_\infty^2 \le \phi_\infty(u)$ for some $a > 0$. The potential ϕ is globally smooth, has fixed Euclidean Hessian bounds $c\|v\|_2^2 \le D^2\phi_u(v,v) \le U\|v\|_2^2$ with $c, U > 0$, and agrees with ϕ_∞ outside a fixed Euclidean ball. The toric input and its admissible cutoff data are as in Definition A.1.

For every proper nonempty face-normal set s, there are maps x_s and $z_s = (q_s, \mu_s)$, smooth at zero, with $x_s(0)$ minimizing ϕ_∞ on F_s. Put

$$u_s(\beta) = \beta x_s\big(\beta^{-1}\log q_s(\beta^{-1})\big).$$

There is $B_s > 0$ such that, for $\beta \ge B_s$, on every smooth embedded presentation $j : N^{2n-2} \to \mathbb{R}^n \times T^n$ whose image is the entire fiber $H_\beta = \{f_{1,\beta} = 1\}$, every critical point of $\phi \circ \mathrm{pr}_1 \circ j$ or zero of its intrinsic Liouville field with actual retained support s satisfies

$$j(p) = (u_s(\beta), \theta), \qquad \exp(i\langle v, \theta\rangle) = 1 \quad (v \in s).$$

Conversely, for all sufficiently large β, every such angle gives a point $(u_s(\beta), \theta) \in H_\beta$ with exact retained support s, and in every such presentation it has a preimage which is critical.

Proof. Take the actual affine minimum and its actual normalized implicit branch. Write $a_s(y)$ for its reduced coefficients and $q^0 = a_s(0)/\sum_v a_{s,v}(0)$. The positive coefficients make every coordinate of q^0 positive. Local affine minimality and the preceding multiplier lemma give a neighborhood of $(0, q^0)$ on which the affine minimum equations hold and every normalized reduced solution is the constructed branch. Choose $\eta > 0$ so that the ball of radius η about $(0, q^0)$ lies in this neighborhood.

Uniform positive holomorphic localization at precision η gives, for every critical or Liouville-zero point of retained support s and all sufficiently large β, a cutoff neighborhood with exactly the indicators of $\{0\} \cup s$, positive real face monomials, and the face-minimum approximation. Increase the threshold until $\beta > 0$, $\beta^{-1} < \eta$, end-gradient agreement holds, the radial coordinate u is nonzero, and the exact good-stratum conormal equation holds. Put $q_v = m_v(u, \theta)$. The literal fiber equation gives $\sum_v q_v = 1$. The positive-monomial lemma gives the character equations and

$$A_s(\beta^{-1}u) = \mathbf{1} + \beta^{-1}\log q.$$

The actual-base-weight lemma gives $\|q - q^0\|_\infty < \eta$. The exact complex conormal equation, whose retained monomials are these real q_v, gives on taking real parts $\nabla\phi_\infty(u) = \lambda B_s q$ for a real λ.

Set $y = \beta^{-1}\log q$. Degree-two homogeneity gives $\nabla\phi_\infty(\beta^{-1}u) = \beta^{-1}\lambda B_s q$. For every tangent vector to $A_s x = \mathbf{1} + y$ this gradient has zero pairing. Convexity therefore makes $\beta^{-1}u$ a global minimum on that affine slice. Its unique minimum is $x_s(y)$, so $\beta^{-1}u = x_s(y)$. Comparing gradients gives $B_s a_s(y) = B_s((\lambda/\beta)q)$. Injectivity of B_s yields the literal reduced equation $a_s(y) = (\lambda/\beta)q$. The pair (β^{-1}, q) is in the chosen uniqueness neighborhood, hence $z_s(\beta^{-1}) = (q, \lambda/\beta)$. This proves $u = u_s(\beta)$.

For the converse, the actual branch supplies positive normalized weights, the fiber equation at every stated angle, and its radial normal equation. The strict face inequalities at $x_s(0)$ persist in the reconstructed large-β cutoffs, so exactly s is retained and all other vertex cutoffs vanish.

At the stated angles the retained monomials are exactly the real branch weights; their normal is $B_s q_s(\beta^{-1})$. End-gradient agreement transports the radial normal equation to ϕ. The full-fiber normal converse then makes every tangent derivative vanish at a preimage supplied by the image equality of j. Take the maximum of these finitely many thresholds. □

Let $\mathcal{F}$ be the finite set of proper nonempty face-normal sets. There are radial maps u_s for $s \in \mathcal{F}$ and one $B > 0$ such that, for every $\beta \geq B$, the literal sets

$$C_s(\beta) = \{u_s(\beta)\} \times T_{\mathrm{cone}(s)}$$

are pairwise disjoint and contained in H_β. Every point of $C_s(\beta)$ has actual retained support s. For every smooth embedded presentation $j : N^{2n-2} \to \mathbb{R}^n \times T^n$ of the entire fiber and every $p \in N$, the following are equivalent: p is a critical point of $\phi \circ \mathrm{pr}_1 \circ j$; p is an intrinsic Liouville zero; and $j(p) \in \bigcup_{s \in \mathcal{F}} C_s(\beta)$.

Proof. Every face-normal set is a subset of the finite vertex set, so $\mathcal{F}$ injects into its finite powerset. Choose the radial maps and thresholds of the preceding theorem for each face. Finiteness gives a common threshold for their two implications and the exact support claims. Increase it to include uniform full-fiber holomorphic localization and the equality of the critical and intrinsic zero sets.

At any critical or intrinsic-zero point, holomorphic localization identifies its actual support as some proper nonempty face s. The exhaustive branch theorem makes its radial coordinate $u_s(\beta)$ and every generator character one. The face-torus generator criterion puts its angle in the literal $T_{\mathrm{cone}(s)}$. Conversely this criterion turns any point of $C_s(\beta)$ into one of the branch theorem's constructed critical points. If that theorem supplies a different preimage of the same ambient point, injectivity of the embedding identifies it with the specified p. Thus membership in the union is equivalent to criticality, and the critical-zero theorem gives the intrinsic-zero equivalence.

Finally, if an ambient point belongs to both $C_s(\beta)$ and $C_t(\beta)$, the exact retained-support assertions compute its single actual retained support as both s and t. Hence $s = t$, proving pairwise disjointness. □

S.25 Orthogonal blocks of the face Hessian

Fix a finite set s of k face normals, let $A : \mathbb{R}^n \to \mathbb{R}^s$ be the face-level map, and let A^T denote its adjoint in the fixed basis. For a positive vector $q \in \mathbb{R}^s$, write

$$\ell_q(h) = \sum_{v \in s} q_v h_v, \qquad G_q(h) = \sum_{v \in s} q_v h_v^2, \qquad E = \ker A, \quad Q_q = \ker \ell_q, \quad T_q = \ker(\ell_q A).$$

For a linear map $H : \mathbb{R}^n \to \mathbb{R}^n$ and a scalar λ, put

$$\mathcal{B}((X,Y),(X',Y')) = \langle HX, X' \rangle - \lambda \sum_v q_v (AX)_v (AX')_v + \lambda \sum_v q_v (AY)_v (AY')_v, \qquad \mathcal{Q}(V) = \mathcal{B}(V,V).$$

Definition S.83 (The literal face-tangent form)**.** *The radial and angular tangent spaces are both T_q; the quadratic and bilinear forms are the explicitly displayed $\mathcal{Q}$ and $\mathcal{B}$.*

Lemma S.84 (Differentiated constrained equations)**.** *Suppose $x(y)$ and $a(y)$ are differentiable at y, the potential is smooth there, and near y they satisfy $Ax(w) = \mathbf{1} + w$ and $\nabla\phi(x(w)) = A^T a(w)$. For $B = Dx(y)$, $R = Da(y)$ and $H = D\nabla\phi(x(y))$, one has $AB = \mathrm{id}$ and $HB = A^T R$.*

Proof. Differentiate the two equalities at y. The constant $\mathbf{1}$ has zero derivative, A and A^T are linear, and the chain rule gives $ADx(y) = \mathrm{id}$ and $D\nabla\phi(x(y))Dx(y) = A^T Da(y)$. □

Lemma S.85 (Equality of end germs transfers the actual Hessian)**.** *If two potentials agree on a neighborhood of u, their coordinate gradients and coordinate Hessians at u agree. In particular, if they agree outside a closed Euclidean ball of radius R, this applies at every u with $|u| > R$.*

Proof. Differentiation is local: equality on a neighborhood gives equality of the first derivatives there, hence equality of their coordinate gradients. Differentiating that equality at u gives equality of the Hessians. The strict inequality $|u| > R$ persists on an open neighborhood by continuity of the Euclidean norm, supplying the required equality there. □

Lemma S.86 (The actual branch escapes the exceptional region)**.** *Let x and $z = (q, \mu)$ be continuous at zero, with $x(0) \neq 0$ and $q_v(0) > 0$. Then $u_\beta = \beta x(\beta^{-1} \log q(\beta^{-1}))$ eventually lies outside every fixed Euclidean ball. If ϕ and ϕ_∞ agree outside such a ball, their germs at u_β agree for every sufficiently large β.*

Proof. Positivity makes the logarithm of every q_v continuous near zero. Consequently the argument of x tends to zero. Continuity of x and $|x(0)| > 0$ give $|x(\beta^{-1} \log q(\beta^{-1}))| > |x(0)|/2$ for large β. Multiplication by $\beta \to \infty$ proves escape from every ball. Apply the preceding germ lemma at each such point. □

Lemma S.87 (The full smooth global potential has the same constrained form)**.** *Let ϕ be smooth globally and let ϕ_∞ agree with ϕ near the radial component u of $g(a)$, where g is a smooth local parametrization satisfying $f_s \circ g = 1$ near a. Suppose the face monomials at $g(a)$ are the real values q_v and $\nabla \phi_\infty(u) = \lambda A^T q$. Then the first derivative of $\phi \circ \mathrm{pr}_u \circ g$ at a vanishes and*

$$D^2(\phi \circ \mathrm{pr}_u \circ g)(a)[V, V] = \mathcal{Q}(Dg(a)V), \qquad H = D^2\phi_\infty(u).$$

No smoothness of ϕ_∞ at the origin is required.

Proof. The germ identity gives equality of gradients and Hessians at u. Thus the stated normal equation is also the normal equation of ϕ. The stationary-pullback face-Hessian identity gives $D^2\phi(u)[X, X] - \lambda \sum_v q_v \langle v, X\rangle^2 + \lambda \sum_v q_v \langle v, Y\rangle^2$, where $(X, Y) = Dg(a)V$, and gives the zero first derivative. Replace the potential Hessian by that of ϕ_∞. The identity $\sum_v q_v \langle v, X\rangle^2 = G_q(AX)$ follows term by term from the definition of the face-level map, giving precisely $\mathcal{Q}$. □

Lemma S.88 (The minimizer lift is the orthogonal complement)**.** *Assume $AB = \mathrm{id}$, $HB = A^T R$, and $\langle He, e\rangle > 0$ for nonzero e. Then*

$$\langle HBh, e\rangle = 0 \ (e \in E), \qquad \langle HBh, Bh'\rangle = \langle Rh, h'\rangle.$$

Moreover X is H-orthogonal to E if and only if $X = BAX$. The map $(e, h) \mapsto e + Bh$ is a linear isomorphism $E \times Q_q \to T_q$.

Proof. Pair $HBh = A^T Rh$ with e and with Bh' and use $Ae = 0$ and $ABh' = h'$. For arbitrary X, set $e = X - BAX$; then $Ae = 0$. If X is orthogonal to E, both $\langle HX, e\rangle$ and $\langle HBAX, e\rangle$ vanish. Subtracting shows $\langle He, e\rangle = 0$, whence $e = 0$ by positivity. The converse follows from the first identity. If $X \in T_q$, then $AX \in Q_q$, and $(X - BAX, AX)$ is the inverse of the displayed map. Both inverse identities follow directly from $AB = \mathrm{id}$. □

Remark S.89 (Derivative form of the complement estimate). The source proves uniform transversality of the moving complement by a compact unit-vector contradiction. The minimizer derivative gives the same complement explicitly: $D = T_q \cap E^{\perp_H} = B(Q_q)$. Its radial Hessian is exactly the reduced Hessian, so bounding Da directly supplies the required domination without a separate moving-complement estimate.

Lemma S.90 (Exact orthogonal four-block formula)**.** *Assume additionally that H is symmetric. For $e, f \in E$ and $h, k' \in \mathbb{R}^s$,*

$$\mathcal{Q}(e + Bh, f + Bk') = \langle He, e\rangle + \big(\langle Rh, h\rangle - \lambda G_q(h)\big) + \lambda G_q(k').$$

Proof. Expand $\langle H(e + Bh), e + Bh\rangle$. The two cross terms vanish by orthogonality and symmetry, and its last term is $\langle Rh, h\rangle$. The face levels are $A(e + Bh) = h$ and $A(f + Bk') = k'$, giving the formula. In the bilinear expansion, the same two identities make every cross term between distinct blocks vanish, and the angular E block vanishes identically. □

Lemma S.91 (Uniform domination along the actual positive branch)**.** *Let a be smooth near zero, let $z(\epsilon) = (q(\epsilon), \mu(\epsilon))$ be continuous at zero, and suppose every $q_v(0) > 0$ and $\mu(0) > 0$. Set $y_\beta = \beta^{-1} \log q(\beta^{-1})$ and $\lambda_\beta = \beta\mu(\beta^{-1})$. For all sufficiently large β and every nonzero h,*

$$\langle Da(y_\beta)h, h\rangle - \lambda_\beta \sum_v q_v(\beta^{-1})h_v^2 < 0.$$

Proof. Choose $\delta > 0$ below every $q_v(0)$ and $\mu_* > 0$ below $\mu(0)$. Finiteness and continuity give $q_v(\epsilon) > \delta$ and $\mu(\epsilon) > \mu_*$ near zero. The logarithms are continuous there, so $y_\beta \to 0$. Continuity of Da gives a finite bound $\|Da(y_\beta)\| < C$ for large β. In the maximum coordinate norm,

$$\langle Rh, h\rangle \le k\|R\|\|h\|_\infty^2 \le k\|R\| \sum_v h_v^2.$$

The first inequality follows from $|(Rh)_v| \le \|R\|\|h\|_\infty$ and $|h_v| \le \|h\|_\infty$, term by term. The second follows because every coordinate square is bounded by their sum. Choose β so large that $kC < \beta\mu_*\delta$. The negative weighted term is at least $\beta\mu_*\delta \sum_v h_v^2$, and that sum is positive for $h \neq 0$. Subtracting proves the strict inequality. □

Lemma S.92 (The full radical and block dimensions)**.** *Suppose H is positive, $q > 0$, $\lambda > 0$, and the lifted radial form is negative on every nonzero face-coordinate vector. The radical of $\mathcal{B}$ on $T_q \times T_q$ is exactly $\{0\} \times E$. If A is surjective and s is nonempty, then* $\dim E = n - k$, $\dim Q_q = k - 1$, *and* $\dim(E \times Q_q) = n - 1$.

Proof. Write $X = e + Bh$ with $e \in E$ and $h \in Q_q$. Pair a radical vector (X, Y) first with $(e, 0)$. Orthogonality reduces this to $\langle He, e\rangle = 0$, so $e = 0$. Pair next with $(Bh, 0)$; negative definiteness gives $h = 0$, hence $X = 0$. Pair finally with $(0, Y)$; positivity of λ and all weights gives $AY = 0$. Conversely $X = 0$ and $AY = 0$ make the displayed bilinear form zero against every tangent. Rank-nullity for the surjection A gives $\dim E = n - k$. The functional ℓ_q is surjective: for any scalar a, a vector with one coordinate a/q_v and all others zero maps to a. Rank-nullity therefore gives $\dim Q_q = k - 1$. Since surjectivity implies $k \le n$ and nonemptiness gives $k \ge 1$, the two dimensions add to $n - 1$. □

Lemma S.93 (The actual reconstructed Hessian blocks)**.** *For the constructed constrained minimum x, its actual reduced coefficient $a = d(\phi_\infty \circ x)$, and the positive normalized branch z, take $u_\beta = \beta x(y_\beta)$, $B = Dx(y_\beta)$, $R = Da(y_\beta)$, and $H = D^2\phi_\infty(u_\beta)$. For all sufficiently large β the identities, signs, radical and dimensions of the preceding lemmas hold. The four blocks have signs positive, negative, zero, positive; their dimensions are $n - k, k - 1, n - k, k - 1$ respectively.*

Proof. The branch parameter tends to zero. Smoothness gives differentiability on one neighborhood; nonzeroness of $x(0)$ gives $x(y_\beta) \neq 0$ there. The constrained identities can therefore be

differentiated at y_β. Degree-two homogeneity makes the gradient degree one. Differentiating $\nabla\phi_\infty(\beta x) = \beta\nabla\phi_\infty(x)$ off zero gives $D\nabla\phi_\infty(\beta x) = D\nabla\phi_\infty(x)$ by cancellation of $\beta > 0$. Thus the differentiated equations involve exactly the Hessian at u_β. Schwarz symmetry follows from local smoothness; positivity follows from the strong potential. Apply domination and the four-block identity. Angular positivity follows from positive weights and multiplier, while the angular E block vanishes identically. The radical and rank-nullity assertions now follow from the preceding lemma. □

S.26 Native logarithmic charts at a face critical point

The local Hessian in this subsection is the second derivative of the restricted potential in an actual chart of the smooth fiber. Its logarithmic lift is constructed from the quotient covering map. This supplies the chart interpretation of the formula in `prop:morsebott`, without asserting a tubular neighborhood or local trapping.

Lemma S.94 (Smooth integral coordinates on the quotient torus)**.** *Every integral basis $b_1, \dots, b_n$ of the lattice induces an actual smooth diffeomorphism*

$$T \longrightarrow (\mathbb{R}/2\pi\mathbb{Z})^n, \qquad \theta \longmapsto (\langle b_i, \theta\rangle)_i.$$

Proof. For circle-valued coordinates α_i, let L_α be the unique integer-linear map from the lattice to the circle taking b_i to α_i. Its values on the original coordinate basis define the inverse torus point. Evaluation on b_i proves one inverse identity; uniqueness of an integer-linear map with specified basis values proves the other. Every component of either map is a finite integer-linear combination of circle coordinates. Such a component is smooth because under the smooth identification of the quotient circle with the unit circle it is a finite product of integer powers. Smoothness of a map to the finite product follows componentwise. Both maps are therefore smooth. □

Lemma S.95 (Smooth embedded parametrization of the literal face torus)**.** *Let s be a nonempty proper face of the smooth toric input. There are smooth maps $f : T^{n-|s|} \to T^n$ and $r : T^n \to T^{n-|s|}$ with $r \circ f = \mathrm{id}$, and f is a topological embedding with injective differential everywhere. Its image is literally the cone annihilator $T_{\mathrm{pos}(s)}$.*

Proof. Extend the face generators to the integral basis supplied by a facet containing the face. Reindex the basis by $1, \dots, n$, and let J be the indices of the face generators. Basis injectivity and containment give $|J| = |s|$. Enumerate the complementary indices by $1, \dots, n - |s|$. In basis angle coordinates define f by inserting zero at every active index and the free angles at the complementary indices, then apply the inverse basis diffeomorphism. Define r by applying the basis diffeomorphism and extracting those free angles. Both are smooth componentwise and $rf = \mathrm{id}$. The continuous left inverse proves that f is an embedding; differentiating the same identity proves that its differential is injective. Its image consists exactly of the points where every face generator has zero quotient angle. The generator characterization of the literal cone annihilator identifies this image with $T_{\mathrm{pos}(s)}$. The argument also applies when the complement is empty. □

Lemma S.96 (Logarithmic lift of an embedded chart)**.** *Let $j : N \hookrightarrow Y$ be a smooth embedding, $a \in N$, and choose a logarithmic representative x of $j(a)$. Write c for the chart of N at a and $y = c(a)$. There exists a map g defined near y, smooth there, such that $g(y) = x$ and $\mathrm{cover} \circ g = j \circ c^{-1}$ near y. Its differential satisfies*

$$D\,\mathrm{cover}_x \; Dg_y = Dj_a \, D(c^{-1})_y.$$

In particular Dg_y is injective.

Proof. The logarithmic quotient cover is a smooth local diffeomorphism. Choose its smooth local inverse ℓ at x, and set $g = \ell \circ j \circ c^{-1}$. The point $j(a)$ lies in the domain of ℓ, and continuity of $j \circ c^{-1}$ gives the composition on a neighborhood of y. The inverse identity gives $g(y) = x$ and the stated identity of maps there. Differentiating gives the displayed equality. The derivative of c^{-1} is bijective and Dj_a is injective, so Dg_y is injective. □

Lemma S.97 (The bilinear chart Hessian and its radical)**.** *Suppose a smooth chart potential f has diagonal Hessian $D^2 f_y[v, v] = \mathcal{Q}(Dv)$, where D has image $T_q \times T_q$. Assume H is symmetric positive definite, $q > 0$, $\lambda > 0$, and there are maps B, R with $AB = \mathrm{id}$, $HB = A^T R$, and $\langle Rh, h\rangle - \lambda G_q(h) < 0$ for $h \neq 0$. Then*

$$D^2 f_y[v, w] = \mathcal{B}(Dv, Dw), \qquad \big(D^2 f_y[v, w] = 0 \text{ for all } w\big) \iff (Dv)_{\mathrm{rad}} = 0,\ A(Dv)_{\mathrm{ang}} = 0.$$

Proof. The actual Hessian is symmetric by the equality of mixed derivatives, and the displayed formula for $\mathcal{B}$ is symmetric because H is. Expand the diagonal identity at $v + w$ and subtract the identities at v and w. Dividing by two proves the mixed identity. By surjectivity of D onto $T_q \times T_q$, annihilating all chart directions is now equivalent to belonging to the bilinear radical of $\mathcal{B}$ on that tangent space. Its four orthogonal blocks are positive, negative, zero, and positive under the stated assumptions; the radical is precisely the angular copy of $\ker A$. Equivalently, split each radial vector as $e + Bh$ with $Ae = 0$. Testing first against e, then Bh, forces these two components to vanish by positivity and negativity. Testing against the angular component forces its weighted square to vanish, hence $AY = 0$ because every $q_v > 0$. The converse follows directly from the formula for $\mathcal{B}$. □

Lemma S.98 (Exact tangent image in a lifted fiber chart)**.** *Assume the native inclusion differential has image equal to the kernel of the tailored differential, the tailored function agrees locally with the face polynomial, and its monomials at the base point are the real weights q_v. Then the image of the constructed logarithmic chart differential is precisely $T_q \times T_q$.*

Proof. Differentiating the face polynomial gives $Df_s(X, Y) = \ell_q(AX) + i\ell_q(AY)$, whose kernel is exactly $T_q \times T_q$. The local equality with the tailored function identifies their differentials after pullback through the quotient cover. The differential identity of the logarithmic chart lemma and surjectivity of the inverse-chart differential show that the cover differential sends the image of Dg onto the image of the native inclusion differential. The latter is the native fiber kernel by hypothesis. The cover differential is bijective, so taking its inverse identifies the image of Dg with the displayed coordinate kernel. □

Lemma S.99 (Stationary pullback Hessian for a local chart)**.** *Let f be smooth near $g(y)$ and g smooth near y, with $Df_{g(y)} = 0$. Then*

$$D^2(f \circ g)_y[v, w] = D^2 f_{g(y)}[Dg_y v, Dg_y w].$$

For the face Lagrangian $L = \phi - \lambda \operatorname{Re} f_s$, if $f_s \circ g = 1$ near y and $DL_{g(y)} = 0$, the same formula computes the actual chart Hessian of $\phi \circ g$ using $D^2 L$.

Proof. The derivative of the composition near y is $Df_{g(z)} Dg_z$. Differentiate this product at y. The two terms are $D^2 f_{g(y)}[Dg_y v, Dg_y w]$ and $Df_{g(y)} D^2 g_y[v, w]$. The latter vanishes by stationarity. For the Lagrangian, $\phi \circ g = L \circ g + \lambda$ near y, so its first and second derivatives agree with those of $L \circ g$. □

Lemma S.100 (End-germ formula for the native fiber-chart Hessian)**.** *Let $j : N \hookrightarrow Y$ have image the entire tailored fiber, and let $a \in N$ have logarithmic representative $x = (u, \theta)$. Suppose on a neighborhood of $j(a)$ the tailored function is the face polynomial f_s. Suppose its monomials at x are the real numbers q_v, the smooth global potential ϕ agrees near u with ψ, and $\nabla\psi(u) = \lambda A^T q$. Then the logarithmic lift g of the actual chart c^{-1} exists as above, the first chart derivative of $\phi \circ j$ vanishes, and for every chart tangent v,*

$$D^2(\phi \circ j \circ c^{-1})_{c(a)}[v, v] = \mathcal{Q}(Dg_{c(a)}v),$$

where $H = D\nabla\psi(u)$ and $\mathcal{Q}$ is the literal constrained form of the face. No parametrization is part of these hypotheses.

Proof. Construct g by the logarithmic chart lemma. The image condition gives $f_s(g(y)) = 1$ near $c(a)$: the lifted chart lies in the actual fiber, and the two defining functions agree near $j(a)$. The potential germ agreement identifies both gradients and Hessians at u. The real monomial values and the normal equation imply that $L = \phi - \lambda \operatorname{Re} f_s$ is stationary at x. The preceding local composition lemma gives the first derivative zero and computes the second derivative by D^2L. Twice differentiating each exponential gives

$$D^2L_x[(X, Y), (X, Y)] = \langle HX, X\rangle - \lambda \sum_v q_v(AX)_v^2 + \lambda \sum_v q_v(AY)_v^2.$$

The lifted potential and $\phi \circ j \circ c^{-1}$ agree near $c(a)$, so their actual second derivatives agree there. This proves the formula. □

Lemma S.101 (Native angular-kernel radical on the actual fiber)**.** *Assume the global potential is smooth and has a uniform positive lower Hessian bound, and the native Poisson pairing is positive on the entire tailored fiber. Take an actual full-fiber embedding of dimension $2n - 2$. At a face critical point satisfying the hypotheses of the native Hessian formula, suppose $q_v > 0$, $\lambda > 0$, and the actual end Hessian and minimizer derivative satisfy the block hypotheses of the bilinear radical lemma. Then there is a logarithmic lift of the actual fiber chart with injective differential of image $T_q \times T_q$, and the actual chart Hessian has radical exactly the directions whose logarithmic radial component is zero and whose angular component lies in* $\ker A$.

Proof. Positive native Poisson pairing makes the fiber equation a submersion; the tangent range theorem for a full regular level embedding therefore identifies the native tangent image with its kernel. Construct the logarithmic chart by the native Hessian lemma, and use the exact tangent image lemma. The smooth global potential gives symmetry of its actual Hessian, and equality of the potential germs transfers that symmetry to the end Hessian. Apply the bilinear radical lemma to the actual chart potential. All its directions are covered by the proved tangent image. □

S.27 Uniform native Hessians on the constructed face tori

For a nonempty proper face-normal set s, write $A = A_s$ for its face-level map, $k = |s|$, and $T_q = \ker(\ell_q A)$ for the real tangent hyperplane of a positive weight vector q.

Lemma S.102 (The endpoint face equation on a plateau neighborhood)**.** *Let (u, θ) belong to the actual endpoint fiber. Suppose an open neighborhood of u has cutoff one on $\{0\} \cup s$ and zero at every other augmented exponent. Then on a neighborhood of (u, θ), $f_{1,\beta} = \sum_{v \in s} e^{\langle v,u\rangle - \beta + i\langle v,\theta\rangle}$.*

Proof. The endpoint fiber equation gives $\sum_v \chi_v m_v = 1$. If S is the Euclidean norm of the vector of monomials, the triangle inequality gives $1 \le |V|S$. The specified small constant ε_0 satisfies $4|V|\varepsilon_0 < 1$, so $S > 2\varepsilon_0$. This strict inequality persists on a neighborhood, where the small-monomial cutoff is therefore one. Intersect that neighborhood with the pullback of the given radial plateau neighborhood. The actual family formula there becomes $\sum_{v\in V} \chi_v m_v$ because its constant cutoff is one. All terms outside s vanish and each term on s has coefficient one, proving the stated identity. □

Lemma S.103 (Dimension of the native Hessian kernel)**.** *Let $D : E \to \mathbb{R}^n \times \mathbb{R}^n$ be injective with image $T_q \times T_q$, and let a bilinear form L on E have radical exactly those V for which $(DV)_u = 0$ and $A(DV)_\theta = 0$. If A is surjective, then* $\dim \ker L = n - k$.

Proof. Map a radical vector V to $(DV)_\theta \in \ker A$. This map is linear. If two radical vectors have the same angular image, their radial images are both zero and hence their full D images agree. Injectivity of D makes the vectors equal. Conversely, for $Y \in \ker A$, the vector $(0, Y)$ belongs to $T_q \times T_q$, so it equals DV for some V. The assumed radical criterion makes that V a radical vector. Thus the radical is linearly isomorphic to $\ker A$. Rank-nullity for the surjection $A : \mathbb{R}^n \to \mathbb{R}^s$ gives dimension $n - k$. □

Definition S.104 (Native face Hessian chart)**.** *For an actual embedded full fiber $j : N \to Y$ and a point $a \in N$, a native face Hessian chart consists of a smooth logarithmic lift g of $j \circ c^{-1}$ near $c(a)$, with the prescribed center, whose differential is injective and has image $T_q \times T_q$. For $f = \phi \circ \mathrm{pr}_u \circ j \circ c^{-1}$ it satisfies $df_{c(a)} = 0$ and*

$$D^2 f_{c(a)}(V, W) = \mathcal{B}(Dg_{c(a)}V, Dg_{c(a)}W), \quad \ker D^2 f_{c(a)} = \{V : (DgV)_u = 0,\ A(DgV)_\theta = 0\}, \quad \dim \ker D^2 f_{c(a)} = n-k$$

Here $\mathcal{B}$ is the displayed face form for the actual radial Hessian H, the weights q and the multiplier λ.

Let ϕ_∞ be a strongly compatible homogeneous end potential, globally C^1, with $\phi_\infty - d|u|^2$ convex for some $d > 0$. Let ϕ be globally smooth with fixed positive lower and finite upper Hessian bounds, agreeing with ϕ_∞ outside a fixed ball. For every nonempty proper face-normal set s, the actual reduced branch constructs smooth maps $x, z = (q, \mu)$ near zero and $B > 0$ such that for all $\beta \ge B$, every point of

$$C_{s,\beta} = \{\beta x(\beta^{-1} \log q(\beta^{-1}))\} \times T_{\sigma_s}$$

lies in the full endpoint fiber with retained support s. Every actual full-fiber presentation has a native face Hessian chart at each such point and each chosen logarithmic lift, with $H = D^2\phi_\infty(u_\beta)$, $q = q(\beta^{-1})$ and $\lambda = \beta\mu(\beta^{-1}) > 0$.

Proof. Construct x and z by the positive reduced-branch theorem. Their base values have positive weights and multiplier, and $x(0)$ is the nonzero minimum on the polar face. The shifted level equations and trivial face characters identify every monomial on s with its positive weight. The exact reconstructed cutoff neighborhood gives the local endpoint face equation by the plateau lemma. The branch escapes the exceptional ball, giving equality of the two potential germs at its radial point. Its normal equation is $\nabla\phi_\infty(u_\beta) = \lambda A^T q$.

Differentiate the actual constrained-minimum equations. The resulting lift B_β and reduced Hessian R_β satisfy $AB_\beta = \mathrm{id}$ and $HB_\beta = A^T R_\beta$. The actual Hessian-block theorem gives symmetry and positivity of H and strict negativity of $\langle R_\beta h, h\rangle - \lambda G_q(h)$ for every nonzero h. Positive native Poisson pairing makes the whole fiber a regular level and determines its tangent kernel. Construct

the logarithmic lift of each actual fiber chart. The local face equation and normal equation give its stationary first derivative and diagonal Hessian; polarization gives the full bilinear identity. The exact tangent-image theorem gives image $T_q \times T_q$, so the bilinear radical theorem identifies its radical with the zero radial and angular face-kernel directions. The preceding dimension lemma gives its dimension $n-k$. Take B larger than every threshold used in these finitely many arguments. None depends on the angle or on the chosen presentation, proving the uniform assertion. □

Lemma S.105 (Stationarity in the native chart is intrinsic criticality)**.** *If ϕ is smooth and a native face Hessian chart exists at a, then $d(\phi \circ \mathrm{pr}_u \circ j)_a = 0$ on the actual tangent space of N.*

Proof. For each $V \in T_aN$, the derivative in a native chart evaluated on Dc_aV equals the intrinsic derivative evaluated on V, by the chain rule and the inverse-chart differential identity. The chart derivative is zero by its stationary field, so every intrinsic derivative vanishes. □

Under the potential hypotheses of the constructed-face theorem, there are radial functions $u_s(\beta)$, positive weight vectors $q_s(\beta)$, positive multipliers $\lambda_s(\beta)$, and $B > 0$, simultaneously for all nonempty proper face-normal sets s, such that every $\beta \geq B$ has an actual smooth symplectic full-fiber presentation and all the native chart conclusions hold at every point of $\{u_s(\beta)\} \times T_{\sigma_s}$. These points belong to the actual fiber and have retained support s. The native chart conclusions hold for every smooth full-fiber presentation and every logarithmic lift.

Proof. Every face-normal set is a subset of the finite vertex set, so there are finitely many such sets. Choose the branches and thresholds supplied by the preceding theorem for each one. Their finite collection has a common upper threshold. Increase it above the uniform symplectic-level threshold, which supplies an actual smooth symplectic embedding with image the full level $f_{1,\beta} = 1$. All face assertions and the full level construction now hold at each β beyond the same threshold. □

Under the same hypotheses, assume also $\phi_\infty(u) \geq a\|u\|_\infty^2$ for some $a > 0$ in the fixed coordinate norm. There are one threshold and face-indexed positive weights and multipliers such that every critical point, and every intrinsic Liouville-zero point, on every smooth full-fiber presentation has a native face Hessian chart, for every logarithmic lift. Its face index is exactly its literal retained support, its radial Hessian is $D^2\phi_\infty$ at that actual point, and the kernel of its actual chart Hessian has dimension $n - |s|$.

Proof. Take a common threshold for the uniform native chart construction and the exhaustive critical-torus classification. Let a be an actual critical or zero point, and let s be its classified face, so $j(a) = (u_s^0, \theta)$ with $\theta \in T_{\sigma_s}$. The reconstructed point (u_s, θ) belongs to the full fiber, hence has a preimage b in the same presentation. It has a native Hessian chart, and the preceding stationarity lemma makes b an actual critical point. Classify b in turn, obtaining a face t. The reconstructed point has retained support s, while its classified point has retained support t. These are the same literal radial point, so $s = t$. Classification therefore gives $j(b)_u = u_s^0 = j(a)_u$, while reconstruction gives $j(b)_u = u_s$. Thus $u_s = j(a)_u$. The reconstructed chart theorem, applied now at a and at any chosen logarithmic lift of $j(a)$, gives the required native chart. Its kernel dimension and bilinear formula are the fields of that chart. The weights and multiplier are those of the same reconstructed face branch and are positive beyond the uniform threshold. □

Remark S.106 (Scope of the native Hessian identification). The chart witnesses identify the actual second derivative and its radical uniformly on each constructed torus. Identifying this radical with the differential of the constructed torus embedding inside the fiber, and constructing a neighborhood of its conormal branch with positive normal gradient linearization, are separate geometric steps. The result does not assert local trapping or equality of the full core.

S.28 The actual tangent of the critical face torus

Definition S.107 (Constant-radius quotient torus)**.** *If $f : T^{n-|s|} \to T^n$ is the constructed parametrization of T_{σ_s} and $u \in \mathbb{R}^n$, the constant-radius torus map is $k_u(\alpha) = (u, f(\alpha))$. It is smooth in the actual quotient atlas.*

Proof. The product of the constant radial map and the smooth quotient torus parametrization is smooth. The fixed linear Euclidean rechart of the ambient product atlas preserves smoothness. □

Lemma S.108 (The face-torus differential is angular and annihilates the face)**.** *Let k_u be a smooth constant-radius parametrization with image $\{u\} \times T_{\sigma_s}$. At every point and every logarithmic representative, each actual tangent vector of its image has coordinates $(0, Y)$ with $A_s Y = 0$.*

Proof. Each radial coordinate of k_u is constant, so differentiating it gives zero radial component. For $v \in s$, the character of v is one at every angle in T_{σ_s}. Thus the descended monomial of coefficient one restricts to the nonzero constant $e^{\langle v,u\rangle}$ on k_u. Its actual logarithmic differential is $m_v(\langle v, X\rangle + i\langle v, Y\rangle)$, obtained by the chain rule from the quotient cover and the derivative of the exponential. The tangent has $X = 0$, so its zero derivative implies $im_v\langle v, Y\rangle = 0$. Since both i and m_v are nonzero, $\langle v, Y\rangle = 0$ for every $v \in s$, which is $A_s Y = 0$. □

For every nonempty proper face-normal set s and every u, there is an actual smooth embedded immersion $k_u : T^{n-|s|} \to Y$ with literal image $\{u\} \times T_{\sigma_s}$. Its differential image at every point, in every chosen logarithmic representative, is exactly $\{0\} \times \ker A_s$.

Proof. Construct the quotient parametrization f and its global smooth retraction r. The map $(u', \theta) \mapsto r(\theta)$ is a smooth left inverse to k_u, proving that k_u is a topological embedding and its differential is injective. Pull its differential back by the invertible logarithmic-cover differential. The preceding lemma puts its image in $\{0\} \times \ker A_s$. Angular projection restricts to an injective linear map from the source tangent to $\ker A_s$. The source has dimension $n - |s|$, and surjectivity of the face-level map gives the same dimension for $\ker A_s$. Thus this linear map is surjective. Together with its zero radial component, this proves the exact displayed tangent image. The literal image follows from the range equality of f. □

Let C be a native face Hessian chart at a on an actual full-fiber presentation $j : N \to Y$. Let k_u be the constructed face-torus embedding, with $j(a) = k_u(\alpha)$ and the same chosen logarithmic representative. A vector in the actual fiber chart lies in the kernel of its actual Hessian if and only if its pushforward by the actual inverse-chart and inclusion differentials belongs to $Dk_u(T_\alpha T^{n-|s|})$.

Proof. The logarithmic lift differential, followed by the cover differential, equals the inclusion differential followed by the inverse-chart differential. The native Hessian radical criterion says that its lift has zero radial component and angular component in $\ker A_s$. The exact tangent theorem identifies precisely those lifted vectors with the differential image of k_u. Substitution in the differential identity gives both implications. □

Proposition S.109. *For each proper nonempty face $F \subset P$ of codimension k, there is exactly one critical torus*

$$C_{F,\beta} = \{u_{F,\beta}\} \times T_{\sigma_F}, \qquad u_{F,\beta} = \beta x_F + O(1), \qquad \sigma_F = \operatorname{cone}(v_1, \dots, v_k).$$

There are no other critical points. The Hessian of $\phi|_{H_\beta}$ has $k-1$ negative directions, $n-1$ positive directions, and kernel $TC_{F,\beta}$ of dimension $n-k$.

Proof. Let $A_F x = (\langle v_i, x\rangle)_i$ and $E_F = \ker A_F$. For small $y \in \mathbb{R}^k$, strict convexity and the implicit function theorem give a unique constrained minimizer $x(y)$ of ϕ_∞ on $A_F x = \mathbf{1} + y$, near x_F. Its reduced function $\psi_F(y) = \phi_\infty(x(y))$ has positive Hessian, the relevant Schur complement. Write $a(y) = d\psi_F(y)$, so $a(0) = (a_1, \dots, a_k) > 0$. For $q_i = e^{\langle v_i, u\rangle - \beta} > 0$ and $\epsilon = \beta^{-1}$, the critical equations, with normalized multiplier $\mu = \lambda/\beta$, reduce to

$$a(\epsilon \log q) = \mu q, \qquad \sum_i q_i = 1. \tag{30}$$

At $\epsilon = 0$, $\mu^0 = \sum_i a_i$ and $q_i^0 = a_i/\mu^0$. The linearization in (q, μ) has zero kernel: a kernel vector satisfies $\mu^0 \dot q + q^0 \dot\mu = 0$, $\sum_i \dot q_i = 0$, which force $\dot\mu = \dot q = 0$. It is square, hence invertible. The implicit function theorem gives a unique nearby positive solution and $u = \beta x_F + O(1)$. At this solution the cutoffs have the exact values in (29). The angle conditions give T_{σ_F}, connected because these primitive generators extend to a lattice basis. For every possible additional solution, summing (30) gives $\mu = \sum_i a_i(\epsilon \log q)$. The uniform convergence in Lemma S.41 therefore forces both $q \to q^0$ and $\mu \to \mu^0$, placing all solutions in these fixed implicit-function neighborhoods, where uniqueness excludes additional ones.

For the Hessian put $D = \operatorname{diag}(q_i)$, $m = A_F^T q$, $H = D^2\phi(u)$, and $\lambda = \beta\mu(\beta^{-1}) \asymp \beta$. Then $d\phi = \lambda m$. A real tangent vector (X, Y) obeys $m \cdot X = m \cdot Y = 0$. The exact constrained Hessian is

$$X^T H X - \lambda (A_F X)^T D (A_F X) + \lambda (A_F Y)^T D (A_F Y). \tag{31}$$

There is no mixed radial-angular term. Decompose the radial tangent as

$$m^\perp = E_F \oplus_H D_F, \qquad D_F = m^\perp \cap E_F^{\perp_H}.$$

On E_F the form is positive; on D_F, of dimension $k-1$, the negative term dominates for large β because A_F is injective there and q_i stay bounded away from zero. To make this estimate uniform, note that homogeneity gives $H(u_{F,\beta}) = D^2\phi_\infty(u_{F,\beta}/\beta) \to D^2\phi_\infty(x_F)$, while $q \to q^0$. The limiting H-orthogonal complement is transverse to $\ker A_F$. A compact unit-vector contradiction therefore gives one positive lower bound for $|A_F X|/|X|$ on all these D_F for large β; when $D_F = 0$ this assertion is vacuous. Use the same orthogonal splitting for the angular tangent. In the four metric-orthogonal blocks

$$(E_F)_{\mathrm{rad}} \oplus (D_F)_{\mathrm{rad}} \oplus (E_F)_{\mathrm{ang}} \oplus (D_F)_{\mathrm{ang}},$$

the Hessian has signs positive, negative, zero, positive, respectively. Every mixed block vanishes: the radial H cross term vanishes by orthogonality and A_F vanishes on E_F, while there is no radial-angular cross term. This gives precisely the asserted inertia and identifies the invariant eigenspaces of its metric-selfadjoint operator. □

T Identification of the full embedded skeleton

For a complete finite-type Liouville manifold H, write $\mathrm{Sk}(H)$ for the set of points with relatively compact forward orbit. Under an inclusion completion marking this is exactly the full core of the compact domain: an orbit leaving that domain enters its cylindrical end, where the end coordinate grows at speed one and cannot return. On the homogeneous end define

$$\Gamma = (\Phi_\infty \times \mathrm{id})^{-1} \left(\bigcup_{\sigma \neq 0} (\sigma \setminus \{0\}) \times T_\sigma \right), \qquad L_\beta = \Gamma \cap H_\beta.$$

T.1 The radial section of the positive Legendre conic set

Fix the original finite toric input and its actual all-pairs cutoffs χ_a. Write $N = |V|$, $h(u) = \max_{v\in V}\langle v,u\rangle$, $M_\beta(u) = e^{h(u)-\beta}$, and $\epsilon_\beta = e^{-\sqrt{\beta}}$.

Definition T.1 (The real radial equation)**.** *Define the smooth real function*

$$\mathcal{R}_\beta(u) = 1 - \chi_0(u) + \sum_{v\in V} \chi_v(u) e^{\langle v,u\rangle - \beta}.$$

Lemma T.2 (Uniform radial derivative on an unbounded region)**.** *For every $L \geq 0$ there exists B such that for every $\beta \geq B$ and every u with $h(u) \geq \beta - L$, $D\mathcal{R}_\beta(u)[u] > 0$.*

Proof. The finite input supplies $c > 0$ with $c|u| \leq h(u)$; the coordinate supremum norm is at most $|u|$. The cutoff derivative bounds supply $D \geq 0$ independent of u, β with $\|D\chi_a\| \leq D$ for every augmented exponent. If $D\chi_a(u) \neq 0$, its uncut magnitude is at most $\epsilon_\beta \max(1, M_\beta)$. Consequently the error

$$E = -D\chi_0(u)[u] + \sum_v D\chi_v(u)[u] e^{\langle v,u\rangle - \beta}$$

satisfies $|E| \leq C\|u\|\epsilon_\beta \max(1, M_\beta)$ with $C = (N+1)D$. This also covers a derivative that vanishes: its summand is zero. For the constant exponent, the same changed-term estimate reads $1 \leq \epsilon_\beta \max(1, M_\beta)$ whenever its derivative is nonzero.

Retention implies $\langle v,u\rangle > h(u) - \sqrt{\beta} - 2$. Choose B so $\sqrt{\beta} \geq L$ and $\beta - L - \sqrt{\beta} - 2 \geq 0$ for $\beta \geq B$. Every exponential-slope summand $\chi_v e^{\langle v,u\rangle - \beta}\langle v,u\rangle$ is then nonnegative. A maximizing vertex has cutoff one: it is within $\sqrt{\beta}$ of every vertex weight and, because $h \geq \beta - L$, also within that distance of the constant weight. Its slope summand is hM_β, so the sum of slopes is at least that. On the region in question, $\max(1, M_\beta) \leq e^L M_\beta$ and $\|u\| \leq h/c$. Thus

$$|E| \leq (Ce^L/c)\epsilon_\beta h M_\beta < hM_\beta$$

after increasing B, since $\epsilon_\beta \to 0$; also $h > 0$ there. Differentiating the displayed definition gives the slope sum plus E. This proves the strict inequality uniformly throughout the unbounded region. □

Lemma T.3 (A unique positive intersection on every ray)**.** *For all sufficiently large β, every u with $h(u) = 1$ admits exactly one $t > 0$ such that $\mathcal{R}_\beta(tu) = 1$.*

Proof. Put $L = \log(2N)$. For $h(w) \leq \beta$, the constant term is maximal, so $\chi_0(w) = 1$ and $\mathcal{R}_\beta(w) \leq Ne^{h(w)-\beta}$. Thus the value at $(\beta - L)u$ is at most $1/2$. At $(\beta + \log 2)u$ a maximizing vertex is uncut and contributes 2; all other terms and $1 - \chi_0$ are nonnegative, so the value is at least 2. Continuity gives an intersection between those two radii. Increase the threshold so $\beta - L > 0$. The radial-derivative lemma gives a positive derivative with respect to t on $[\beta - L, \infty)$, since that derivative is $D\mathcal{R}_\beta(tu)[tu]/t$. The mean-value theorem therefore makes the function strictly increasing there. A positive radius below $\beta - L$ has value at most $Ne^{-L} = 1/2$, so cannot be another solution. This proves existence and uniqueness on the whole positive ray. □

Lemma T.4 (Height bounds and compactness of the real level)**.** *For every real β, the level $\mathcal{R}_\beta = 1$ is compact and its points satisfy $\beta - \log N \leq h(u) \leq \beta$.*

Proof. If $h(u) > \beta$, a maximizing vertex has cutoff one and contributes $e^{h-\beta} > 1$, contradicting $\mathcal{R}_\beta = 1$. Hence $h \leq \beta$. Then $\chi_0 = 1$ and the sum is at most $Ne^{h-\beta}$, giving $1 \leq Ne^{h-\beta}$ and therefore the lower height bound. The level is closed by continuity. The bound $c|u| \leq h(u) \leq \beta$ puts it in a closed bounded ball when it is nonempty; the empty case is compact as well. Finite-dimensional compactness proves the assertion. □

Lemma T.5 (Exact exclusion throughout the outward region)**.** *Fix a strongly compatible homogeneous potential ϕ_∞ and $L \in \mathbb{R}$. For all sufficiently large β, for every native fan cone σ and every u with $\nabla\phi_\infty(u) \in \sigma$ and $h(u) \geq \beta - L$, all vertices outside σ have cutoff zero.*

Proof. The angular gap gives $h(u) - \langle v, u\rangle \geq \delta|u|$ for every outside vertex. The upper support bound $h(u) \leq H|u|$ consequently gives a lower bound $\delta(\beta - L)/H$ for this deficit. For large β that exceeds $\sqrt{\beta} + 2$: divide by β and use $\beta^{-1/2} + 2\beta^{-1} \to 0$ and $L/\beta \to 0$. Retention would require the deficit to be strictly less than $\sqrt{\beta} + 2$, a contradiction. Nonnegativity of the cutoffs then forces each outside cutoff to equal zero. The estimate is uniform in the radius, the point, the cone and the outside vertex. □

Definition T.6 (The actual Legendre conic section)**.** *For the native gradient $\Phi_\infty = \nabla\phi_\infty$, define*

$$\Gamma = \{(u, \theta) : \Phi_\infty(u) \neq 0,\ \Phi_\infty(u) \in \sigma,\ \theta \in T_\sigma \text{ for some nonzero fan cone } \sigma\}, \qquad L_\beta = \Gamma \cap \{f_1 = 1\}.$$

Define $\rho(u, \theta) = (\theta, \Phi_\infty(u)/|\Phi_\infty(u)|)$.

Lemma T.7 (The endpoint is exactly the real radial equation on the cone)**.** *For all sufficiently large β, at every point with $\Phi_\infty(u) \in \sigma$, $\theta \in T_\sigma$ and $h(u) \geq \beta - \log(2N)$, one has $f_1(u, \theta) = \mathcal{R}_\beta(u)$. Moreover $L_\beta = \Gamma \cap \{\mathcal{R}_\beta = 1\}$.*

Proof. Off-cone terms vanish by exact exclusion. For an on-cone vertex the literal annihilator condition gives $\langle v, \theta\rangle = 0$ in $\mathbb{R}/2\pi\mathbb{Z}$, so its character is exactly one. Thus $\mathcal{F}_\beta = \mathcal{R}_\beta - 1$. A largest normalized monomial has magnitude $M \geq e^{-\log(2N)} = 1/(2N)$, and the prescribed monomial size is at least that magnitude. Since the fixed small constant satisfies $4Nc < 1$, this size exceeds $2c$; hence $\eta = 1$. The exact endpoint identity $f_1 - 1 = \mathcal{F}_\beta$ proves the first assertion.

For a point in the actual endpoint fiber, its retained-monomial equation gives $M \geq 1/N$, hence $h \geq \beta - \log N$. For a point of the real radial level, the preceding height-bounds lemma gives the same inequality. Both lower bounds exceed or equal $\beta - \log(2N)$, so the first assertion applies in both directions and identifies the two intersections with Γ exactly. □

Lemma T.8 (Compactness of the actual section)**.** *Suppose Φ_∞ is the native gradient homeomorphism fixing zero of the compatible-potential construction. Then L_β is compact for all sufficiently large β.*

Proof. Every T_σ is closed, since it is an intersection of kernels of continuous integral angle characters; the quotient torus is compact. Every native cone is closed, and the native fan is finite. Therefore the finite union of the closed sets $\{(u, \theta) : \Phi_\infty(u) \in \sigma, \theta \in T_\sigma\}$ over nonzero cones is closed. Intersect it with the compact real radial level times the torus. For $\beta > \log N$, the lower height bound excludes $u = 0$; injectivity of Φ_∞ and $\Phi_\infty(0) = 0$ exclude $\Phi_\infty(u) = 0$ as well. This compact intersection is consequently exactly L_β, by the endpoint-equation lemma. □

Proposition T.9 (A bijective section over the entire literal link)**.** *For the same native gradient homeomorphism, $\rho : L_\beta \to \Lambda^\infty_\Sigma$ is bijective for all sufficiently large β. In particular its image equals the entire literal FLTZ link in the fixed Euclidean unit cotangent boundary.*

Proof. For every point of Γ, normalization of its nonzero cone vector stays in the same cone and has Euclidean squared norm one. Its angle stays in the same literal annihilator torus, so $\rho(\Gamma)$ lies in the link. Conversely take (θ, p) in the link and let $x = \Phi_\infty^{-1}(p)$. It is nonzero, and $h(x) > 0$ by the support lower bound. Normalize $u = x/h(x)$, so $h(u) = 1$, and take its unique radius $t > 0$ from the ray lemma. Degree-one homogeneity of the actual gradient gives $\Phi_\infty(tu) = (t/h(x))p$. Thus

$(tu, \theta) \in \Gamma$, its real radial value is one, and its projection is (θ, p); the endpoint equation makes it a point of L_β.

If two section points project to the same link point, their angles coincide and their gradients differ by the positive ratio of their Euclidean lengths. Homogeneity and injectivity of the native gradient then place their radial coordinates on the same positive ray. Normalize that ray by $h = 1$. Both coordinates solve the same real radial equation, so uniqueness makes them equal. This proves injectivity. The radial function contains no angle; in particular its unique radius is independent of the angle on each T_σ. □

Lemma T.10 (Positive scalar-ray derivative of the actual endpoint)**.** *For all sufficiently large β, if $u \neq 0$, $\Phi_\infty(u) \in \sigma$, $\theta \in T_\sigma$, and $h(u) > \beta - \log(2N)$, then $f_1(u, \theta)$ is positive real and the scalar-ray function $t \mapsto f_1(tu, \theta)$ has a strictly positive real derivative at $t = 1$.*

Proof. The strict height inequality persists for t in a neighborhood of one, where $t > 0$. Homogeneity keeps $\Phi_\infty(tu)$ in the same cone; the angle is fixed. Hence the endpoint-equation lemma identifies this scalar-ray function near one with the real function $t \mapsto \mathcal{R}_\beta(tu)$. Its derivative is $D\mathcal{R}_\beta(u)[u] > 0$, by the radial-derivative lemma. A maximizing uncut term contributes $M > 0$ to $\mathcal{R}_\beta(u)$, so its value is positive real. On a section point $h \geq \beta - \log N$; every outward radial continuation therefore satisfies the strict height inequality, since $\log(2N) > \log N$. □

Corollary T.11 (Construction from the finite input). The original finite toric input admits a strongly compatible homogeneous potential with its actual gradient homeomorphism fixing zero such that, for all sufficiently large β, the actual section is compact and its radial projection is bijective onto the literal FLTZ link.

Proof. Choose the potential and native homogeneous gradient homeomorphism from the proved compatible-potential construction. Apply compactness and the bijective-section proposition to that same choice. Their two thresholds have a finite maximum, proving both assertions simultaneously. □

Lemma T.12 (Smoothness on full-span conormal extensions)**.** *Near every point of L_β belonging to the σ piece, the equation $f_1 = 1$ cuts a smooth transverse section in the smooth full-span conormal extension $\mathcal{C}_\sigma = \{(p, \theta) : p \in \operatorname{span} \sigma, \theta \in T_\sigma\}$ in Legendre coordinates.*

Proof. At the point, each off-cone term has a deficit strictly exceeding the retention width. Continuity preserves these finitely many strict inequalities on an ambient neighborhood, so all these cutoffs vanish there. On $\mathcal{C}_\sigma$ every remaining character equals one; the restriction of f_1 is consequently real and smooth. The radial direction lies in this extension, and its derivative of f_1 is strictly positive by the scalar-ray derivative lemma. The real implicit-function theorem therefore makes $f_1 = 1$ a smooth transverse codimension-one section in this extension. This statement concerns the smooth extension; the union of cone strata may be singular. □

The compact section, ambient radial transversality, intrinsic flow invariance and equality with the full intrinsic core are distinct assertions.

T.2 Actual conormal differentials and Liouville directions

Lemma T.13 (Ambient absence of outside exponents)**.** *Fix a strongly compatible end potential and a real constant L. For all sufficiently large β, if $\nabla\phi_\infty(u) \in \sigma$ and $h(u) \geq \beta - L$, every cutoff whose vertex lies outside σ vanishes on a neighborhood of u in the entire real ambient space.*

Proof. Let $\delta > 0$ be the uniform angular gap and let $h(u) \le H|u|$. Choose the threshold so $\delta(\beta - L) > H(\sqrt{\beta}+2)$. Then the gap estimate gives $h(u) - \langle v, u\rangle > \sqrt{\beta}+2$ for every outside vertex. The deficit is continuous, so this strict inequality holds throughout a neighborhood of u and makes the cutoff zero there. Intersecting these finitely many neighborhoods gives one neighborhood for every outside exponent. □

Lemma T.14 (Conormal differential blocks)**.** *At a point of the actual endpoint fiber with $\theta \in T_\sigma$ and all outside cutoffs zero, the radial differential equals $D\mathcal{R}_\beta$ and the angular differential is purely imaginary. Its angular normal is the literal real vector $G = \sum_v \chi_v e^{\langle v,u\rangle - \beta} v$ and belongs to* $\operatorname{span}_{\mathbb{R}} \sigma$.

Proof. For a vertex in σ, its character is one on T_σ. For a vertex outside σ, the cutoff and its first derivative vanish; the derivative vanishes because the nonnegative smooth cutoff attains its minimum zero. Differentiating the finite monomial sum therefore gives a real radial derivative, including all cutoff derivatives, and the angular derivative $iG \cdot d\theta$. Every nonzero summand of G is a real multiple of a vector of σ, so their sum lies in its real span. □

Lemma T.15 (Native intrinsic field on a conormal piece)**.** *Let the smooth potential ϕ have invertible Hessian at u and satisfy $D(\nabla\phi)_u u = \nabla\phi(u)$. Consider an ordinary smooth regular-fiber embedding with tangent image* $\ker Df_\beta$*, at a conormal point as above, and suppose its Poisson pairing π_f is nonzero. The pushforward of the intrinsic Liouville field determined by the pulled-back primitive is*

$$\dot{u} = u - \frac{D\mathcal{R}_\beta(u)[u]}{\pi_f}(D\nabla\phi_u)^{-1}G, \qquad \dot{\theta} = 0.$$

Its native Legendre derivative is $\dot{p} = p - (D\mathcal{R}_\beta(u)[u]/\pi_f)G$. If $p \in \sigma$, this velocity belongs to $\operatorname{span}_{\mathbb{R}} \sigma$.

Proof. The preceding differential formula has zero imaginary radial block and zero real angular block. The Hamiltonian vector of its imaginary part is therefore $((D\nabla\phi_u)^{-1}G, 0)$. Applying the proved symplectic projection formula to the ambient Liouville vector $(u, 0)$ gives the displayed radial correction. The regular-fiber tangent-image equality identifies this projection with the pushforward of the intrinsic field for the literal pulled-back primitive. Differentiating the Legendre map sends the correction to G, and sends u to p by the Euler identity. Both p and G belong to the real span of σ. □

The last assertion is tangency to the full span. Preservation of the closed cone requires the smallest-face argument and two-sided ODE uniqueness of Lemma T.37.

Lemma T.16 (Smooth ambient radial extension)**.** *For a strongly compatible end potential and sufficiently large β, let (u, θ) be any point of the actual endpoint fiber with $\nabla\phi_\infty(u) \in \sigma$ and $\theta \in T_\sigma$. There is a function r, smooth at u, with $r(u) = 1$, such that throughout an ambient neighborhood of u one has $r(w) > 0$ and*

$$f_\beta(r(w)w, \vartheta) = 1 \qquad (\vartheta \in T_\sigma).$$

In particular the radial graph extends beyond the closed cone, before restricting its parameters to the full face span in Legendre coordinates.

Proof. The actual-fiber height lower bound gives $h(u) \ge \beta - \log|A| > \beta - \log(2|A|)$. The preceding cutoff germ and continuity of the height therefore imply that the actual endpoint equals $\mathcal{R}_\beta$ on an ambient neighborhood of u, uniformly for every angle in T_σ; the prescribed small-monomial cutoff is identically one there. The quantitative radial estimate gives $D\mathcal{R}_\beta(u)[u] > 0$. Apply the smooth

scalar implicit-function theorem to $(w,t) \mapsto \mathcal{R}_\beta(tw)$ at $(u,1)$. Its derivative in t is the positive displayed derivative, so the implicit function r is smooth and satisfies $\mathcal{R}_\beta(r(w)w) = 1$. Continuity, $r(u) = 1$, and the ambient equation germ give the claimed actual-fiber equation and positivity. □

Lemma T.17 (Uniform native evaluation for the actual end extension)**.** *Let ϕ be a smooth extension of the same strongly compatible homogeneous end ϕ_∞, with uniformly positive Hessian. One β threshold makes the evaluation of the actual native ambient Liouville field on df_β a positive real number at every actual conormal endpoint point. If the Hessian also has a uniform upper bound, one threshold gives the preceding intrinsic field formula on every smooth full endpoint presentation, without separately assuming its tangent-image contract, nondegeneracy, Poisson positivity, off-cone absence, or the Euler identity.*

Proof. Every full endpoint fiber lies beyond any fixed radius for large β. Choose that radius beyond the extension radius; local end agreement then gives the Euler identity for the actual smooth potential. The endpoint height lower bound places every conormal point in the uniform angular-exclusion and radial-positivity region. The native ambient field is consequently $(u,0)$, and its evaluation by the actual differential is the positive real number $D\mathcal{R}_\beta(u)[u]$. Uniform Poisson positivity on the full endpoint fiber implies its regularity and the tangent-image identity for any full-dimensional smooth endpoint embedding. The previous native projection theorem therefore applies with the Hessian equivalence of the actual potential. Taking the maximum of the finitely many thresholds proves uniformity. □

Lemma T.18 (Local integral curves on a smooth retract)**.** *Let M, N be finite-dimensional smooth manifolds without boundary, $j : M \to N$ and $r : N \to M$ smooth, and $r \circ j = \mathrm{id}_M$. Let Z be a smooth vector field on N, with $Z(j(x)) \in \operatorname{im} dj_x$ for all $x \in M$. For any integral curve γ of Z through $j(x)$ at time t_0, there is a neighborhood of t_0 on which γ belongs to $j(M)$.*

Proof. Define $W(x) = dr_{j(x)}Z(j(x))$. The tangent map of a smooth map is smooth, so this is a smooth vector field, using $r(j(x)) = x$ to identify its base point. Differentiating $r \circ j = \mathrm{id}$ gives $dr_{j(x)}dj_x = \mathrm{id}$. Write $Z(j(x)) = dj_x v$ by tangency. Then $dj_x W(x) = dj_x dr_{j(x)} dj_x v = Z(j(x))$. The local existence theorem gives an integral curve α of W through x. The chain rule shows that $j \circ \alpha$ is an integral curve of Z through $j(x)$. Local uniqueness identifies it with γ, proving the assertion. This argument constructs the restricted field and its integral curve; it does not require flow invariance as an assumption. □

Lemma T.19 (Local retraction germs suffice)**.** *Let $j : M \to N$ and $r : N \to M$ be twice continuously differentiable near x and $j(x)$, respectively, and suppose $r \circ j = \mathrm{id}$ near x. Let Z be continuously differentiable near $j(x)$, with $Z(j(y)) \in \operatorname{im} dj_y$ for all y sufficiently close to x. For every neighborhood U of x, an integral curve through $j(x)$ belongs to $j(U)$ for all times sufficiently close to its initial time.*

Proof. Twice continuous differentiability on an open neighborhood makes the tangent map once continuously differentiable there. The formula $W(y) = dr_{j(y)}Z(j(y))$ consequently gives a continuously differentiable local vector field. The local identity $r(j(y)) = y$ places this vector in the tangent fiber at y. Differentiating the identity in a neighborhood of each sufficiently close y and using tangency gives $dj_y W(y) = Z(j(y))$. Construct a local integral curve α of W through x. Its continuity keeps it in U and in the neighborhoods where these identities and derivatives hold after restricting time. The chain rule makes $j \circ \alpha$ an integral curve of Z there. Local uniqueness identifies it with the specified ambient curve. □

Lemma T.20 (Closed local invariance extends to every real time)**.** *Let A be a closed subset of a topological space X and $\gamma : \mathbb{R} \to X$ a continuous curve. Suppose that whenever $\gamma(t) \in A$, all sufficiently nearby times also map into A. If γ meets A, then its whole image lies in A. Consequently a closed set locally preserved by a continuous real flow at every one of its points is preserved for every real time.*

Proof. The inverse image $\gamma^{-1}(A)$ is closed by continuity and open by the local hypothesis. It is nonempty by assumption. Connectedness of $\mathbb{R}$ therefore makes it all of $\mathbb{R}$. For a flow Φ and $x \in A$, apply this argument to $\gamma(t) = \Phi_t(x)$. At a time t with $\Phi_t(x) \in A$, local preservation at this point and the identity $\Phi_{s-t}(\Phi_t(x)) = \Phi_s(x)$ give the required local condition. □

Lemma T.21 (Compact invariant sets have relatively compact forward orbits)**.** *Let A be a compact invariant subset of a Hausdorff space carrying a continuous real flow. Every point of A has relatively compact forward orbit.*

Proof. Invariance places the forward orbit in A. Since A is closed, the closure of the orbit is still contained in A. It is a closed subset of a compact set and is therefore compact. □

Lemma T.22 (Integral curves lift through the logarithmic cover)**.** *Let $j : N \to Y$ be smooth, let Z be a vector field on N, and let V be a vector field on the logarithmic covering space. Suppose*

$$dj_a Z(a) = d\exp_x V(x) \qquad \text{whenever } j(a) = \exp(x).$$

Every local integral curve γ of Z, together with a chosen logarithmic representative x_0 of $j(\gamma(t_0))$, has a local lift ξ with $\xi(t_0) = x_0$, $\exp \circ \xi = j \circ \gamma$ near t_0, and $\dot{\xi} = V(\xi)$ near t_0.

Proof. Choose the smooth local inverse l of the covering at x_0 and put $\xi = l \circ j \circ \gamma$. Continuity keeps $j \circ \gamma$ in the domain of l near t_0, proving the lifting equation and initial value. Differentiating $l \circ \exp = \mathrm{id}$ on the open inverse chart gives $dl_{\exp(x)} d\exp_x = \mathrm{id}$. The chain rule and the assumed field identity therefore give $\dot{\xi} = dl\, dj\, Z = dl\, d\exp\, V(\xi) = V(\xi)$ throughout a neighborhood of t_0. □

Lemma T.23 (A positive face combination has a full-span neighborhood)**.** *If s is a nonempty proper face support and $p = \sum_{v \in s} a_v v$ with every $a_v > 0$, there is an ambient neighborhood O of p such that $O \cap \operatorname{span}\operatorname{cone}(s) \subset \operatorname{cone}(s)$.*

Proof. The face generators are independent. The linear map $A : (b_v) \mapsto \sum b_v v$ therefore has a linear left inverse B on the ambient finite-dimensional vector space. All linear maps here are continuous. Each coefficient $(Bq)_v$ remains positive for q sufficiently close to p, since $Bp = a$. The span of the cone is the range of A: the cone lies in that range, and every generator belongs to the cone. For q in that range, $ABq = q$. Thus q close to p in the span is a positive combination of the same generators. □

Lemma T.24 (A local cone branch at every conic point)**.** *For $0 \neq p \in \sigma$ in the face fan, there is a nonempty proper face support s such that $p \in \operatorname{cone}(s) \subset \sigma$ and a neighborhood O of p with $O \cap \operatorname{span}\operatorname{cone}(s) \subset \operatorname{cone}(s)$. Moreover $T_\sigma \subset T_{\operatorname{cone}(s)}$.*

Proof. Present σ as the cone on a proper support t, and write $p = \sum_{v \in t} a_v v$ with $a_v \geq 0$. Retain precisely the indices with $a_v > 0$. Their support s is nonempty because $p \neq 0$; it is a proper face because every subset of a proper simplicial face is a face. Removing zero terms leaves the same sum, so the preceding lemma applies. Cone inclusion follows from $s \subset t$. Every integral vector of the smaller cone belongs to the larger one, so its character vanishes on T_σ, proving the torus inclusion. □

Lemma T.25 (A common smooth implicit neighborhood)**.** *Let $f : E \times \mathbb{R} \to \mathbb{R}$ be smooth, where E is a real Banach space, and suppose $D_2 f(a,b)$ is invertible. There are open neighborhoods U of a and O of (a,b) and a function r, smooth throughout U, such that $r(a) = b$, $(x, r(x)) \in O$ and $f(x, r(x)) = f(a,b)$ for every $x \in U$, and*

$$f(x,t) = f(a,b) \quad \Longleftrightarrow \quad t = r(x) \qquad ((x,t) \in O).$$

Proof. The implicit-function theorem supplies r, smooth at a in every finite order, its equation, and its uniqueness equation near (a,b). Continuity of $D_2 f$ and openness of the invertible operators permit an open neighborhood O on which both uniqueness and invertibility hold. Continuity of r gives an open neighborhood U of a such that its graph lies in O; shrink U also so that r is continuously differentiable there. Fix $x \in U$. Apply the implicit-function theorem at $(x, r(x))$, obtaining a local function r_x smooth at x in every finite order, with $r_x(x) = r(x)$. Its graph remains in O near x by continuity. Its defining equation equals $f(a,b)$, and uniqueness on O gives $r_x = r$ near x. Consequently r is smooth at every point of this single U. □

Definition T.26 (Transverse Legendre parameter space)**.** *Let V be the real span of a proper face, let $p_0 \neq 0$ lie in V, and let ℓ be a real linear functional satisfying $\ell(p_0) = 1$. The transverse parameter space is $K = V \cap \ker \ell$. Its affine inclusion is $q \mapsto p_0 + q$.*

Lemma T.27 (Radial tangent from the differentiated section equations)**.** *Let $K = V \cap \ker \ell$, let $p \in V$ satisfy $\ell(p) = 1$, and let $H : E \to E$ be a linear isomorphism with $Hu = p$. Let d be a linear functional with $d(u) \neq 0$. If $L : K \to E$ satisfies*

$$HL(a) - \ell(HL(a))p = a, \qquad d(L(a)) = 0 \quad (a \in K),$$

then $\operatorname{range} L = \{v : Hv \in V,\ d(v) = 0\}$.

Proof. The displayed identity expresses $HL(a)$ as a plus a multiple of p, hence places it in V; the second identity gives the forward inclusion. Conversely suppose $Hv \in V$ and $d(v) = 0$. Put $a = Hv - \ell(Hv)p \in K$ and $c = \ell(Hv) - \ell(HL(a))$. The first identity gives $H(v - L(a)) = cp = cHu$, so injectivity of H implies $v - L(a) = cu$. Applying d gives $0 = cd(u)$, hence $c = 0$ and $v = L(a)$. □

Lemma T.28 (The differentiated central radial graph)**.** *Let $K = V \cap \ker \ell$, $p \in V$ and $\ell(p) = 1$. Suppose $g(0) = u$, $\alpha(0) = 1$, and g and α are differentiable at zero. Suppose, near zero,*

$$\Phi(g(a)) - \ell(\Phi(g(a)))p = \alpha(a)a, \qquad \rho(g(a)) = 1.$$

If $H = D\Phi_u$ is invertible, $Hu = p$, and $d = D\rho_u$ satisfies $d(u) \neq 0$, then $\operatorname{range} Dg_0 = \{v : Hv \in V,\ d(v) = 0\}$.

Proof. Differentiate the first identity at zero. Its right side has derivative the inclusion of K because $\alpha(0) = 1$ and its factor a vanishes at zero. Thus $HDg_0(a) - \ell(HDg_0(a))p = a$. The derivative of the second identity gives $d(Dg_0(a)) = 0$. The preceding linear lemma applied to $L = Dg_0$ proves the assertion. □

Lemma T.29 (Radial tangent throughout an open branch)**.** *Let $K = V \cap \ker \ell$, $p \in V$ and $\ell(p) = 1$. On an open set $B \subset K$, suppose the differentiable maps g and α satisfy*

$$\Phi(g(a)) = \alpha(a)(p + a), \qquad \rho(g(a)) = 1.$$

At $q \in B$, suppose $\alpha(q) > 0$, $H = D\Phi_{g(q)}$ is invertible, $Hg(q) = \Phi(g(q))$, and $d = D\rho_{g(q)}$ satisfies $d(g(q)) \neq 0$. Then

$$\operatorname{range} Dg_q = \{v : Hv \in V,\ d(v) = 0\}.$$

Proof. Put $c = \alpha(q) > 0$, $\ell' = \ell/c$, $G(a) = g(q + a/c)$, and $A(a) = \alpha(q + a/c)/c$. The new radial point is $g(q)$ with momentum $\Phi(g(q)) = c(p + q)$, and $\ell'(\Phi(g(q))) = 1$. The kernel of ℓ' equals that of ℓ. On a neighborhood of zero, the momentum and level identities give

$$\Phi(G(a)) - \ell'(\Phi(G(a)))\Phi(g(q)) = A(a)a, \quad A(0) = 1, \quad \rho(G(a)) = 1.$$

The preceding central lemma applies to G. Since $DG_0 = Dg_q/c$ and multiplication by c bijects K, these derivatives have the same range. □

Lemma T.30 (Embedded full-span conormal section)**.** *Suppose Φ is the degree-one Legendre homeomorphism of a strongly compatible homogeneous potential, smooth with smooth inverse off zero. At an actual conormal endpoint point $u_0 = \Phi^{-1}(p_0)$, write r for the local radial implicit function. Let $f : T^{n-k} \to T_\sigma$ be the integral-basis torus parametrization, with smooth retraction a. There is an open neighborhood B of zero in K on which*

$$j(q, \vartheta) = \big(r(\Phi^{-1}(p_0 + q))\Phi^{-1}(p_0 + q), f(\vartheta)\big)$$

is a smooth embedded immersion, belongs to the actual endpoint fiber, and has momentum in V. It contains the entire torus $\{u_0\} \times T_\sigma$. Uniformly over $\theta \in T$, its image near that torus is exactly

$$\{(u, \theta) : f_\beta(u, \theta) = 1,\ \Phi(u) \in V,\ \theta \in T_\sigma\}.$$

At every $q \in B$, the radial differential has the exact range

$$\operatorname{range} Dg_q = \{v : D\Phi_{g(q)}v \in V,\ D\mathcal{R}_\beta(g(q))[v] = 0\}.$$

Proof. The affine slice avoids zero because $\ell(p_0 + q) = 1$. Smoothness of Φ^{-1} and the preceding common smooth implicit neighborhood make the radial component smooth after shrinking B. Shrink it so that $r > 1/2$ and the actual endpoint equation holds uniformly at every angle of T_σ. Homogeneity gives the literal momentum $p = r(\Phi^{-1}(p_0 + q))(p_0 + q)$, which lies in V and has $\ell(p) = r(\Phi^{-1}(p_0 + q))$. The inverse parameters are therefore

$$q = p/\ell(p) - p_0, \qquad \vartheta = a(\theta).$$

These are smooth near the image. A continuous linear projection onto K extends the first inverse parameter to the ambient momentum space; replacing its denominator away from the image by $\max(1/2, \ell(p))$ makes a continuous ambient retraction. This proves that j is a topological embedding. On the image the maximum agrees locally with $\ell(p)$, so the same retraction is smooth there. Differentiating its composition with j, which is the identity, proves injectivity of the differential. Finally $r(u_0) = 1$ and $\operatorname{range} f = T_\sigma$ give the entire specified torus. For the local image assertion, put $t = \ell(\Phi(u))$ and $w = u/t$. Near u_0, $t > 1/2$ and (w, t) lies in the uniqueness neighborhood of the radial implicit equation. If $\Phi(u) \in V$, then $q = \Phi(u)/t - p_0$ belongs to K and tends to zero, so belongs to B. Homogeneity gives $w = \Phi^{-1}(p_0 + q)$. On the uniform ambient cutoff germ, the actual endpoint equation at an angle in T_σ is the scalar radial equation. Uniqueness therefore gives $r(w) = t$, and hence $j(q, \vartheta) = (u, \theta)$ whenever $f(\vartheta) = \theta$. The converse follows from the endpoint and span assertions already proved. Finally shrink B so that the positive radial derivative remains positive at $g(q)$ throughout it. Differentiating the inverse identities of the punctured smooth Legendre homeomorphism constructs the inverse of $D\Phi_{g(q)}$. Differentiated homogeneity gives $D\Phi_{g(q)}g(q) = \Phi(g(q))$. The actual radial graph equation and the already proved momentum identity therefore satisfy all hypotheses of the preceding tangent lemma, giving the displayed exact range. □

Lemma T.31 (Smoothness of the actual inverse Hessian)**.** *For a smooth potential whose Hessian has a positive uniform lower bound, the inverse Hessian is smooth as a function of the base point.*

Proof. The derivative of the smooth gradient is smooth. The lower bound makes each Hessian injective and hence invertible in finite dimension. Inversion on the open set of invertible linear operators is smooth, so composition with the Hessian gives the assertion. □

Lemma T.32 (The smooth projected field is the intrinsic field)**.** *Fix a smooth potential with uniform positive lower and finite upper Hessian bounds, agreeing outside a positive radius with the prescribed strongly compatible homogeneous end. For all sufficiently large β, on every smooth presentation of the entire endpoint fiber, the actual intrinsic Liouville field pushes forward to the logarithmic-cover differential applied to the smooth projected radial field. This holds at every fiber point and every lift, without any conormal restriction.*

Proof. The whole endpoint fiber lies beyond the radius of agreement. Euler's identity there identifies the ambient Liouville field with $(u, 0)$ in logarithmic coordinates. The native Poisson bound makes the level regular; its embedding tangent image is exactly the complex differential kernel. The intrinsic contraction equation thus identifies the pushed field with the symplectic projection of the ambient field. The covering differential identifies this projection with its inverse-Hessian formula. The preceding uniform smoothness result applies since the endpoint value is one. □

Lemma T.33 (Smooth logarithmic symplectic projection)**.** *For the actual smooth tailored complex map and a smooth uniformly convex potential, applying the symplectic projection in logarithmic coordinates to any smooth vector field gives a smooth field near every point with nonzero Poisson pairing. In particular its application to the radial vector $(u, 0)$ is smooth there.*

Proof. The actual differential of the tailored map is smooth. Its real and imaginary covectors, their Riesz representatives, and their images under the actual inverse Hessian are smooth. Thus both Hamiltonian vectors and their Poisson pairing are smooth. Nonvanishing persists near the specified point, so division by the pairing in the explicit rank-two projection preserves smoothness. Evaluation on $(u, 0)$ is a smooth bilinear operation. □

Lemma T.34 (One threshold for local projected-field smoothness)**.** *For a smooth potential with uniform positive lower and finite upper Hessian bounds, one positive threshold B ensures the preceding projected radial field is smooth near every logarithmic lift x with $|f_{\beta,s}(x)| \geq 1$, for every $\beta \geq B$ and $0 \leq s \leq 1$.*

Proof. Use the uniform positive native Poisson bound on $|f_{\beta,s}| \geq 1$. The covering differential identifies that scalar with the actual inverse-Hessian pairing in logarithmic coordinates. Its nonvanishing therefore supplies the preceding lemma at every specified point. □

T.3 Local invariance of the literal conormal section

Lemma T.35 (The conormal Poisson scalar)**.** *At an endpoint-fiber point (u, θ) with $\theta \in T_\sigma$ and all outside cutoffs zero, let H be an invertible real Hessian, let $d = D\mathcal{R}_\beta(u)$, and let $G = \sum_v \chi_v e^{\langle v,u\rangle - \beta} v$. The actual logarithmic Poisson scalar is*

$$\pi_f = d(H^{-1}G).$$

Proof. The real angular differential and imaginary radial differential are zero. The imaginary angular covector has Euclidean representative G, while the real radial covector has representative $d^\sharp$. The two-block inverse-Hessian pairing therefore reduces to $\langle H^{-1}G, d^\sharp\rangle = d(H^{-1}G)$. □

Lemma T.36 (Tangency of the actual projected radial velocity)**.** *Suppose the smooth potential has Hessian H at u and satisfies $Hu = \nabla\phi(u)$. Suppose the outside cutoffs vanish, $\theta \in T_\sigma$, $f_\beta(u,\theta) = 1$, and $\pi_f \neq 0$. If $\nabla\phi(u) \in V = \operatorname{span}_{\mathbb{R}} \sigma$, then the actual projected radial velocity v has*

$$Hv_u \in V, \qquad D\mathcal{R}_\beta(u)[v_u] = 0, \qquad v_\theta = 0.$$

Consequently, if a differentiable radial branch g has exact tangent range $\{w : Hw \in V,\ D\mathcal{R}_\beta(u)[w] = 0\}$ at $g(q) = u$, then v belongs to the differential image of $q \mapsto (g(q), \theta)$ at q.

Proof. The actual projection formula gives $v_u = u - (d(u)/\pi_f)H^{-1}G$ and $v_\theta = 0$. Applying H gives $Hv_u = \nabla\phi(u) - (d(u)/\pi_f)G$. Both terms belong to V, because every retained exponent belongs to σ. The preceding pairing identity gives $d(v_u) = d(u) - (d(u)/\pi_f)\pi_f = 0$. The exact radial tangent range therefore provides a parameter vector a with $Dg_q(a) = v_u$. The derivative of the fixed-angle branch sends a to $(Dg_q(a), 0) = v$. □

Lemma T.37 (Two-sided local invariance at every cone stratum)**.** *Let ϕ be smooth with uniform positive lower and finite upper Hessian bounds, agreeing outside a positive radius with a strongly compatible homogeneous end ϕ_∞. Suppose $\Phi = \nabla\phi_\infty$ is a homeomorphism fixing zero, smooth with smooth inverse off zero. For all sufficiently large β, let V_β be the actual logarithmic symplectic projection of $(u, 0)$ to the endpoint-fiber tangent kernel. If a local integral curve γ of V_β has logarithmic quotient image in the literal section L_β at time t_0, then throughout a neighborhood of t_0 its quotient image remains in L_β and its lifted angular coordinate remains equal to its angular coordinate at t_0.*

Proof. Choose one threshold for the smoothness and positive Poisson estimates, the actual full-span branch construction, the ambient outside-cutoff germs, and the escape of the entire endpoint fiber beyond the extension radius. All these estimates are uniform in the fan cones and endpoint points. Thus, along every constructed branch, the Hessian and gradient of the smooth extension agree with those of the homogeneous end. Differentiated homogeneity supplies the Euler identity for that same Hessian.

Write $x = \gamma(t_0)$ and $p = \Phi(x_u) \neq 0$. Choose a smallest face cone τ containing p in its relative interior. Its full span has a neighborhood of p contained in τ, and the starting angle belongs to T_τ. Choose $\ell(p) = 1$ and the constructed radial branch $g : B \subset K \to M_{\mathbb{R}}$, where $K = \operatorname{span}\tau \cap \ker\ell$ and $g(0) = x_u$. The integral torus parametrization provides a parameter for the starting quotient angle. Fix the original lifted angle x_θ and set $j(q) = (g(q), x_\theta)$. Composing the ambient branch retraction with the logarithmic quotient map and taking its radial parameter gives a retraction r with $r \circ j = \mathrm{id}$ near zero. Both maps are smooth there.

The genuine ambient cutoff germ at x_u pulls back under g to a parameter neighborhood on which every outside cutoff vanishes. Each branch point belongs to the actual endpoint fiber. Hence Poisson positivity, the end Euler identity, and the exact radial tangent range allow the preceding tangency lemma to place $V_\beta(j(q))$ in $\operatorname{im} Dj_q$ throughout that neighborhood. The projected field is smooth near x.

Restrict the parameter neighborhood further so that $\Phi(g(q)) \neq 0$ and every momentum in $\operatorname{span}\tau$ lies in τ there. The local retraction theorem constructs the restricted integral curve and uses two-sided uniqueness to show that γ belongs to the image of this parameter neighborhood near t_0. Every image point has momentum in τ, angle in T_τ, and actual endpoint value one, so belongs to L_β. Its lifted angle is exactly x_θ. The time neighborhood contains times on both sides of t_0, including when p lies on a face of the original starting cone. □

Proposition T.38 (An invariant complete radial section from finite input)**.** *From the finite toric input alone, there exist a strongly compatible degree-two end potential ϕ_∞, with $\phi_\infty(0) = 0$ and*

$\phi_\infty \in C^1$, and a smooth global potential ϕ, with positive uniform lower and finite upper Euclidean Hessian bounds. The global potential is quadratic plus one near zero and agrees literally with ϕ_∞ outside a positive radius. Their actual gradient maps are homeomorphisms: the end map fixes zero and is smooth with smooth inverse off zero, and the global map and its inverse are smooth everywhere.

There is a single $B > 0$ such that, for every $\beta \geq B$, the literal section L_β for this same ϕ_∞ is compact and its radial projection is a bijection onto the entire literal FLTZ link, with image equal to that link. There is a smooth embedded presentation $j : N \to (\mathbb{C}^*)^n$ of the entire endpoint fiber, equipped with the complete restricted primitive of this same ϕ. The inverse image $j^{-1}(L_\beta)$ is compact and invariant under its intrinsic Liouville flow for every real time. Every orbit starting there has constant angle and relatively compact forward orbit.

Proof. The proved finite-input potential construction supplies $\phi_\infty \in C^1$, $\phi_\infty(0) = 0$, and $\delta > 0$ such that $\phi_\infty - \delta\|u\|^2$ is convex, $\delta\|u\|^2 \leq \phi_\infty(u)$, and its punctured Hessian has lower bound 2δ. The smooth convex extension construction gives ϕ with Hessian lower bound δ, quadratic germ $(\delta/2)\|u\|^2 + 1$, literal agreement with ϕ_∞ outside some $R > 0$, and its actual smooth gradient homeomorphism with smooth inverse.

The global Hessian has a finite operator bound: compactness bounds it inside the radius R, and outside that radius it equals the homogeneous end Hessian, whose degree-zero homogeneity bounds it by compactness of the unit sphere. The operator bound gives the claimed quadratic-form upper bound by Cauchy–Schwarz. The lower quadratic bound makes zero a minimum of ϕ_∞, so its derivative at zero vanishes. Strong convexity and coercivity construct its actual gradient homeomorphism fixing zero. Punctured smoothness and the lower bound 2δ give smoothness of this map and its inverse away from zero.

Apply the constructed-complete-fiber invariance corollary to precisely these potentials, bounds, end radius, and end gradient homeomorphism. Apply radial compactness and the radial bijection theorem to the same end potential and map. Take the maximum of their three thresholds. For each parameter beyond this maximum, retain the complete fiber and primitive constructed by the corollary. All stated flow properties then hold on its actual section, and the bijection supplies the literal radial image equality as well. □

Lemma T.39 (Local preservation by the intrinsic flow)**.** *Fix a smooth strictly convex potential with uniform positive lower and finite upper Hessian bounds, agreeing outside a ball with the chosen strongly compatible homogeneous potential. Fix its homogeneous Legendre homeomorphism, smooth with smooth inverse away from zero. For all sufficiently large β, every local intrinsic Liouville integral curve on a smooth embedded presentation of the full endpoint fiber, starting in L_β, remains in L_β for nearby times and keeps its angle.*

Proof. The uniform projected-field identity identifies the pushforward of the intrinsic Liouville field at every fiber point with the logarithmic projected field. Lift the given curve through a smooth local inverse of the logarithmic cover. Its derivative is the projected field. Local preservation of the actual conormal section by this field puts the lift in the section and keeps its real angle coordinate fixed. The lifting identity transfers both conclusions to the original curve. Taking the maximum of the two uniform thresholds makes the conclusion simultaneous. □

Proposition T.40 (Two-sided intrinsic invariance of the entire section)**.** *Under the same potential and Legendre hypotheses, for all sufficiently large β and every embedded full-fiber presentation carrying its complete intrinsic Liouville flow, the inverse image of L_β is compact and invariant for every real time. The angle is constant on each of its orbits. Every such point has relatively compact forward orbit.*

Proof. The section is compact. The full-fiber embedding has closed image, since the endpoint fiber is a level set of a continuous function. It therefore pulls the section back to a compact set A. Local intrinsic preservation and closedness of A extend to all real times by the connected-time argument. For a fixed initial angle θ, the subset of A with that angle is closed, and the local theorem preserves it as well. The same argument shows that the angle remains θ for all time. The closure of any forward orbit is a closed subset of the compact invariant set A and hence is compact. □

Corollary T.41 (A constructed complete fiber with invariant section). Under the same potential hypotheses, for every sufficiently large β there exists an embedded presentation of the entire endpoint fiber with its complete restricted Liouville primitive. Its inverse image of L_β is compact and invariant under this constructed flow, its orbits keep their angle, and its points have relatively compact forward orbits.

Proof. Construct the whole smooth fiber and its complete intrinsic flow from the proved global intrinsic growth estimates for the same potential. Take the maximum of the completeness and invariance thresholds, then apply the preceding proposition to this very presentation and flow. □

Proposition T.42 (The conormal section lies in every marked full core)**.** *Fix the same smooth uniformly convex potential and strongly compatible homogeneous end, with the end potential C^1 at zero. There are $F > 0$ and $B > 0$ such that, for every $\beta \geq B$, one can construct a complete embedded presentation $j : H \hookrightarrow Y$ of the entire endpoint fiber with the following property. For every $A \geq F^2\beta^2/2$, the literal truncation $D_A = \{\phi \circ j \leq A\}$ has a completion, and every completion C has its prescribed inclusion marking onto H and satisfies*

$$L_\beta \subset j(\mathrm{Core}(C)).$$

Proof. Take the maximum of the invariance and full-fiber marking thresholds. The marking construction gives the complete fiber and every indicated truncation and completion. For $y \in L_\beta$, the level-one condition places y in the image of j, say $y = j(x)$. Compact invariance puts the entire forward orbit of x in a compact subset of H. For every completion of D_A, its inclusion marking identifies its full core with the points of H having relatively compact forward orbit. Thus x belongs to that native full core and y belongs to its literal embedded image. This proves the inclusion simultaneously for all the stated levels and every completion of each truncation. □

T.4 The metric normal of the smooth conormal branch

Write A for the face-level map, $V = \operatorname{span}\operatorname{pos}(s)$, $E = \ker A$, and $T_q = \ker(\ell_q A)$, with the notation of the face Hessian. The metric on logarithmic tangent coordinates is

$$g_H((X, Y), (X', Y')) = \langle HX, X' \rangle + \langle HY, Y' \rangle.$$

Lemma T.43 (Face span and the Hessian complement)**.** *The range of A^T is exactly V. Suppose $AB = \mathrm{id}$, $HB = A^T R$, and H is positive definite. Then*

$$HX \in V \quad \Longleftrightarrow \quad X = BAX.$$

Proof. Every positive combination lies in the range of A^T, so its real span does too. Conversely write any coefficient vector as its nonnegative part minus its nonpositive part. Their images are in the positive cone, so the image of their difference is in its span. If $HX \in V$, express it as $A^T a$. For $e \in E$, $\langle HX, e \rangle = \langle a, Ae \rangle = 0$. The orthogonal-lift lemma gives $X = BAX$. In the reverse direction, $HX = HBAX = A^T RAX$ lies in V. □

Definition T.44 (Conormal tangent and its metric)**.** *The conormal tangent is*

$$S_H = \{(X, Y) : X \in T_q,\ HX \in V,\ AY = 0\}.$$

The metric g_H is the displayed direct sum of the radial and angular Hessian metrics.

Lemma T.45 (The literal potential induces the displayed metric)**.** *For a smooth potential ϕ, $H = D\nabla\phi(u)$, and logarithmic representative $\xi = (u, \theta)$, the native descended form satisfies*

$$d\lambda_\phi(D\operatorname{cover}_\xi V, D\operatorname{cover}_\xi JW) = g_H(V, W).$$

Proof. The descended Hessian-form formula gives $\sum_j((HX)_j(JW)_{\theta,j} - (H(JW)_u)_j Y_j)$. Since $JW = (-Y', X')$, linearity gives $\sum_j((HX)_j X'_j + (HY')_j Y_j)$. The coordinate Hessian is symmetric, so the angular summand equals $\langle HY, Y'\rangle$. This is g_H. □

Lemma T.46 (The radial level derivative in the actual face germ)**.** *Suppose the actual endpoint family agrees near* $\operatorname{cover}(u, \theta)$ *with the holomorphic face sum, and at the fixed angle θ it agrees near u with the real radial level function $\mathcal{R}_\beta$. If the face monomials there are the real weights w_v, then*

$$D\mathcal{R}_\beta(u)[X] = \ell_w(AX).$$

Proof. Compose both germ equalities with $u' \mapsto \operatorname{cover}(u', \theta)$ and take real parts. The scalar functions then have the same germ at u, hence the same differential. The differential of the face sum in a radial direction is $\sum_v w_v\langle v, X\rangle$ by the monomial derivative formula. All the weights are real, so taking real parts leaves this expression unchanged. It is $\ell_w(AX)$. □

Lemma T.47 (The full native differential of a conormal branch)**.** *Let $g : K \to \mathbb{R}^n$ be smooth at q, and let $f : T^{n-|s|} \to T^n$ be a smooth parametrization with image $T_{\operatorname{pos}(s)}$ and injective ambient constant-radius differential. Suppose the actual radial differential satisfies*

$$\operatorname{range} Dg_q = \{X : X \in T_w,\ HX \in V\}.$$

For the actual ambient map $j(a, \vartheta) = (g(a), f(\vartheta))$, and any logarithmic representative ξ of $j(q, \vartheta)$,

$$D\operatorname{cover}_\xi Z \in \operatorname{range} Dj_{(q,\vartheta)} \quad\Longleftrightarrow\quad Z \in S_H.$$

Proof. The differential along the first source factor is $Dj(a, 0) = D\operatorname{cover}_\xi(Dg_q a, 0)$, by differentiating the equality $j(a, \vartheta) = \operatorname{cover}(g(a), \xi_\theta)$. Along the second source factor it is the differential of the actual constant-radius torus. Its logarithmic range is exactly $\{0\} \times E$ by the true face-torus tangent theorem. Linearity gives the sum of these two differential images. The cover differential is bijective, so its inverse identifies this sum with $\operatorname{range} Dg_q \times E = S_H$, as asserted. □

Lemma T.48 (The constructed conormal branch has the exact native tangent)**.** *Let ϕ be strongly compatible and let its Legendre homeomorphism $\Phi = \nabla\phi$, with $\Phi(0) = 0$, be smooth with smooth inverse off zero. For all sufficiently large β, uniformly over nonempty proper faces s, consider an actual endpoint point $\xi = (u, \theta)$ with $\Phi(u) \in \operatorname{pos}(s)$ and $\theta \in T_{\operatorname{pos}(s)}$. Suppose its actual family germ is the holomorphic face sum, whose monomials at ξ are real weights w_v. For any linear ℓ with $\ell(\Phi(u)) = 1$, put $K = V \cap \ker \ell$. There exist an open $B \subset K$ containing zero, a map $g : K \to \mathbb{R}^n$ smooth at zero, and a smooth torus immersion $f : T^{n-|s|} \to T^n$ with image $T_{\operatorname{pos}(s)}$, such that $g(0) = u$, $j(q, \vartheta) = (g(q), f(\vartheta))$ is embedded on $B \times T^{n-|s|}$ and lies in the actual endpoint fiber. Moreover,*

$$\operatorname{range} Dg_0 = \{X : X \in T_w,\ D\Phi_u X \in V\},$$

and, for every ϑ and every logarithmic representative η of $j(0, \vartheta)$,

$$D\operatorname{cover}_\eta Z \in \operatorname{range} Dj_{(0,\vartheta)} \quad\Longleftrightarrow\quad Z \in S_{D\Phi_u}.$$

Proof. The embedded-branch construction gives B, g, f, the exact literal torus image, the endpoint equation and an injective product differential. Composing its smooth map with $q \mapsto (q, 0)$ and $\vartheta \mapsto (0, \vartheta)$ proves the stated factor smoothness. If $Df_\vartheta a = Df_\vartheta b$, the product differential takes $(0, a)$ and $(0, b)$ to the same vector. Its injectivity implies $a = b$, so f is an immersion. The constructed radial range is $\{X : D\Phi_u X \in V,\ D\mathcal{R}_\beta(u)[X] = 0\}$. Strong compatibility gives the required radial germ at the actual conormal point. The radial face-constraint lemma identifies its second condition with $X \in T_w$. Apply the full native branch-differential lemma at each torus point. No branch parametrization is part of the hypotheses. □

Lemma T.49 (Positive Hessian on the conormal metric normal)**.** *Suppose H is symmetric positive definite, $AB = \mathrm{id}$, $HB = A^T R$, every $q_v > 0$, and $\lambda > 0$. For every nonzero $V = (X, Y)$ with $X \in T_q$ which is g_H-orthogonal to S_H, $\mathcal{Q}(V) > 0$.*

Proof. Put $h = AX$ and $e = X - Bh$. Then $Ae = 0$ and $\ell_q(h) = 0$. The vector $(Bh, 0)$ is in S_H. Orthogonality gives $\langle HX, Bh\rangle = 0$. Symmetry and the orthogonal-lift identity give $\langle He, Bh\rangle = 0$, hence $\langle HBh, Bh\rangle = 0$. Positive definiteness implies $Bh = 0$; applying A gives $h = 0$. Testing against every $(0, f)$ with $Af = 0$ shows that Y is H-orthogonal to E. Thus $Y = BAY$ by the orthogonal-lift lemma. The constrained Hessian now equals $\langle HX, X\rangle + \lambda G_q(AY)$. Both summands are nonnegative. If $X \neq 0$ the first is positive. Otherwise $Y \neq 0$; the equality $Y = BAY$ implies $AY \neq 0$, and the positive weights make the second summand positive. □

Lemma T.50 (The actual fiber Hessian on the actual branch normal)**.** *Let C be an actual fiber Hessian chart at ξ, and let D be the differential of its logarithmic lift. Suppose the positive-normal block hypotheses above hold, $H = D\nabla\phi(\xi_u)$ for the smooth actual potential ϕ, and j_B is an actual conormal branch as in the full native differential lemma. For any nonzero fiber-chart vector v, if*

$$d\lambda_\phi\big(D\,\mathrm{cover}_\xi\, Dv, D\,\mathrm{cover}_\xi\, JW\big) = 0 \quad \textit{whenever } D\,\mathrm{cover}_\xi\, W \in \mathrm{range}\, Dj_B,$$

then the actual restricted potential has $D^2(\phi \circ j \circ c^{-1})[v, v] > 0$.

Proof. The full native branch differential identifies the quantified branch directions with S_H. The literal descended-form identity changes the displayed orthogonality condition into $g_H(Dv, W) = 0$ for every $W \in S_H$. Positivity of the actual fiber-chart Hessian applies. The constructed-branch lemma supplies the radial range and torus immersion hypotheses when j_B is its constructed branch. □

Lemma T.51 (Positivity of the actual fiber-chart Hessian)**.** *Let D be the differential of the constructed logarithmic lift of an actual fiber chart, and suppose its native Hessian chart satisfies the preceding metric hypotheses. For every nonzero chart tangent v, if Dv is orthogonal to S_H, then the actual restricted potential has $D^2(\phi \circ j \circ c^{-1})[v, v] > 0$.*

Proof. The constructed lift is an immersion, so $Dv \neq 0$. Its exact tangent range gives $Dv \in T_q \times T_q$. The native Hessian formula identifies the actual second derivative with $\mathcal{B}(Dv, Dv) = \mathcal{Q}(Dv)$. Apply the preceding positive-normal lemma. □

T.5 The actual native metric gradient and its linearization

Definition T.52 (The field and metric in a native lifted fiber chart)**.** *Let $j : N \hookrightarrow Y$ present the actual full endpoint fiber, let c be a chart at $a \in N$, and let L be its smooth logarithmic*

lift, so $\operatorname{cover} \circ L = j \circ c^{-1}$ *near* $c(a)$. *Set* $\mu = j^*\lambda_\phi$, $Z(y) = Dc_{c^{-1}(y)} Z_\mu(c^{-1}(y))$, $D_y = DL_y$, $H_y = D\nabla\phi(L(y)_u)$, *and*

$$G(y)(v, w) = g_{H_y}(D_y v, D_y w), \qquad f(y) = \phi(j(c^{-1}(y))_u).$$

These are the actual native coordinate field and descended induced metric.

Lemma T.53 (Local native gradient identity)**.** *Suppose* ϕ *is smooth with a positive uniform Hessian lower bound, the actual fiber has positive Poisson plane pairing, and the tailoring cutoffs are each zero or one on a neighborhood of* $j(a)$. *Then* $G(y)(Z(y), \cdot) = Df_y$ *on a neighborhood of* $c(a)$.

Proof. Shrink so the lift is continuously differentiable, its covering equality holds as a germ at every point, and y lies in the chart target. Put $b = c^{-1}(y)$. Differentiating the covering equality gives $D \operatorname{cover}_{L(y)} D_y = Dj_b Dc_y^{-1}$. The inverse-chart identity consequently transports $D_y Z(y)$ to $Dj_b Z_\mu(b)$. The actual regular-level tangent theorem identifies $\operatorname{range} Dj_b$ with the kernel of the family differential. The actual contraction equation therefore identifies $Dj_b Z_\mu(b)$ with the symplectic fiber projection of Z_{λ_ϕ}. For each w, the vector $D_y w$ is tangent to the fiber, so its family differential vanishes. Apply the good-point projected gradient identity to $D_y Z(y)$ and $D_y w$. The descended metric formula changes its left side to $G(y)(Z(y), w)$. The covering chain rule and the actual equality $f = \phi \circ \operatorname{pr}_u \circ L$ change its right side to $Df_y[w]$. Equality for every w is equality of covectors. □

Definition T.54 (The fixed cover-plane transverse metric)**.** *Write* A *for the face level map,* $F_q = \ker(q^T A)$, *and* $P_q = F_q \times F_q$. *Let* Q *be a continuous linear radial quotient with kernel* $\operatorname{span}(\operatorname{pos}(s))$, *let* $v_0 \in s$, *and let* $r_v > 0$. *The transverse differential on this fixed plane is*

$$D_0(X, Y) = \left(QHX, (r_v \langle v, Y\rangle)_{v \in s \setminus \{v_0\}}\right).$$

Restrict the actual Hessian metric g_H *to* P_q, *writing this restriction as* G_0. *For a continuous linear map* L *from the transverse target to* P_q, *set* $M_L(z, w) = G_0(Lz, Lw)$. *These definitions have no torus-angle or native-chart parameter.*

Lemma T.55 (The unique fixed metric-normal lift)**.** *Assume* Q, H, A *are surjective,* H *is symmetric positive definite,* $q_v, r_v > 0$, *and the actual face lift matrices satisfy* $AB = \mathrm{id}$ *and* $HB = A^T R$. *There exists a unique continuous linear map* L *with* $D_0 L = \mathrm{id}$ *and* $G_0(Lz, W) = 0$ *for* $W \in \ker D_0$. *The form* M_L *is symmetric positive definite. Moreover, for all* $V \in P_q$,

$$G_0(LD_0 V, Lw) = G_0(V, Lw).$$

Proof. The actual transverse submersion lemma makes D_0 surjective. The metric G_0 is symmetric and positive definite because H is. Let $K = \ker D_0$ and $W = K^{\perp_{G_0}}$. If $V \in K \cap W$, then $G_0(V, V) = 0$, hence $V = 0$. Finite-dimensional bilinear orthogonality therefore gives $P_q = K \oplus W$. The restriction $D_0|_W$ is injective by this zero intersection and surjective by this decomposition and the surjectivity of D_0. Its linear inverse is continuous in finite dimension. Compose that inverse with $W \hookrightarrow P_q$ to obtain L. For another normal lift of the same target vector, their difference belongs to K and is orthogonal to K. Positivity makes that difference zero, proving uniqueness. If $z \neq 0$, then $Lz \neq 0$ because $D_0 Lz = z$, so $M_L(z, z) > 0$; symmetry is inherited from G_0. Finally, $LD_0 V - V \in K$, and symmetry and normality of Lw give $G_0(LD_0 V - V, Lw) = 0$, which is the claimed pairing identity. □

Lemma T.56 (The actual transverse rate derivative)**.** *Use the actual full-fiber chart, field and metric of the native gradient lemma. Let Φ be smooth at the radial critical point, let δ_c be the literal transverse map composed with the full-fiber chart inverse, and set $R_c(y) = D\delta_c(y)Z(y)$. At the critical center b,*

$$DR_c(b) = D\delta_c(b)DZ(b).$$

Thus on any constructed normal lift L_c the induced operator $D\delta_c(b)DZ(b)L_c$ equals $DR_c(b)L_c$.

Proof. The actual transverse map is smooth in the native ambient manifold. Its covering composite is smooth at the logarithmic representative. Compose with the smooth actual lift and use its covering germ to obtain smoothness of δ_c. The actual native Liouville field is smooth by nondegeneracy of the actual restricted symplectic form. The native gradient lemma proves $Z(b) = 0$. Differentiating the continuous-linear-map evaluation $D\delta_c(y)Z(y)$ gives, in direction v,

$$D^2\delta_c(b)(v, Z(b)) + D\delta_c(b)DZ(b)v.$$

The first term vanishes. Equality on all directions proves the asserted continuous linear map identity, and composing with L_c proves the last claim. □

Assume the actual native gradient hypotheses and the fixed normal-lift hypotheses above, with positive multiplier λ. Assume also $H = D\nabla\phi(x_u) = D\Phi(x_u)$ and the actual monomial values at the critical point are the positive real weights r_v. Let e be the true lift differential from native chart coordinates to P_q, let L be the constructed fixed normal lift, and put $L_c = e^{-1}L$. Then $D\delta_c(b)L_c = \mathrm{id}$ and the actual transverse rate operator

$$T = DR_c(b)L_c = D\delta_c(b)DZ(b)L_c$$

is selfadjoint and strictly positive in M_L:

$$M_L(Tz, w) = M_L(z, Tw), \qquad M_L(Tz, z) > 0 \quad (z \neq 0).$$

For any family of such critical points with the same radial H, q, r, Q, v_0, this construction uses the same fixed transverse metric M_L, and

$$M_L(Tz, w) = \mathrm{Hess}_{s,H,q,\lambda}(Lz, Lw).$$

Thus the actual transverse linearization pairings are also independent of angle and chart when λ agrees.

Proof. The actual lift differential is injective and has range exactly P_q, so it gives the continuous linear equivalence e. The actual descended metric formula and the Hessian equality identify the chart metric with the pullback of G_0 under e. The actual transverse differential factors as $D\delta_c(b) = D_0e$. Hence $D\delta_c(b)L_c = D_0L = \mathrm{id}$. Its kernel is the full-span smooth conormal branch tangent. Thus the normality of L transports to actual conormal normality of L_c.

Apply the fixed normal-lift pairing identity to $V = eDZ(b)L_cz$. It gives the exact equality

$$M_L(Tz, w) = G(b)(DZ(b)L_cz, L_cw).$$

Selfadjointness now follows from the proved actual native selfadjointness and the symmetry of the metrics. For $z \neq 0$, the right-inverse equation implies $L_cz \neq 0$. Its actual conormal normality allows application of the proved positive native linearization, giving the strict inequality. The actual rate-derivative lemma identifies T with $DR_c(b)L_c$. The actual chart Hessian formula further identifies its metric pairing with the literal constrained cover Hessian on Lz, Lw. Finally D_0, G_0, L and M_L involve only the stated fixed radial data; the uniqueness of L prevents any angle-dependent complement choice. □

Let ϕ_∞ be strongly compatible, continuously differentiable, and strongly convex with a positive constant d. Let ϕ be a smooth extension equal to ϕ_∞ outside a fixed radius, with positive uniform lower and upper Hessian bounds. For each nonempty proper face s, there exist smooth functions x and $z(\varepsilon) = (q(\varepsilon), \mu(\varepsilon))$ constructed from the constrained-minimum and reduced branches, and a finite positive threshold B such that $x(0)$ minimizes ϕ_∞ on the literal polar face. For every $\beta \geq B$, the actual full endpoint fiber admits a smooth symplectic presentation, $q_\beta = q(\beta^{-1})$ is positive, and $\lambda_\beta = \beta\mu(\beta^{-1}) > 0$. At every such radius there is an open radial neighborhood O_β containing u_β on which the augmented cutoffs are one on $\{0\} \cup s$ and zero elsewhere. At every $\theta \in T_{\mathrm{pos}(s)}$ the point (u_β, θ), with

$$u_\beta = \beta x\big(\beta^{-1} \log q_\beta\big),$$

lies in the actual endpoint fiber and has retained support s. For every full-fiber presentation and same-point logarithmic representative, there exists a native Hessian chart C with $H = D\nabla\phi_\infty(u_\beta)$, weights q_β, and multiplier λ_β, whose actual coordinate field vanishes at the center, satisfies its metric-gradient identity on a neighborhood, and has a selfadjoint derivative. For every nonzero chart vector v whose lift differential is g_H-orthogonal to S_H, this derivative satisfies $G(z)(DZ_z v, v) > 0$.

Proof. Construct x, z by the positive reduced-branch theorem. Write $a(y)$ for the coefficients of the constrained minimum and $y_\beta = \beta^{-1} \log q_\beta$. The exact level and normal equations are $Ax(y) = \mathbf{1} + y$ and $\nabla\phi_\infty(x(y)) = A^T a(y)$. The actual four-block theorem consequently supplies $B_\beta = Dx_{y_\beta}$ and $R_\beta = Da_{y_\beta}$ with $AB_\beta = \mathrm{id}$ and $HB_\beta = A^T R_\beta$, symmetric positive H, and the negative reduced radial block. Positive limits of the reduced weights and multiplier give $q_\beta > 0$ and $\lambda_\beta > 0$ for large β.

The shifted face levels and strict gaps at the polar-face minimum give an open radial plateau neighborhood: its augmented cutoffs are one on $\{0\} \cup s$ and zero elsewhere. These same data reconstruct the exact fiber equation and retained support at every stated angle. The radius escapes to infinity, so the actual global potential has the same germ as ϕ_∞ there. Choose the maximum of the finitely many thresholds for these assertions, positive plane pairing, and existence of the actual smooth symplectic full fiber.

At any point on that torus and in any full-fiber presentation, the native chart-Hessian constructor applies to the exact fiber germ, positive weights, multiplier, and the produced B_β, R_β, H. It constructs the logarithmic lift and the actual stationary Hessian. Pull back the radial plateau neighborhood along the first projection to obtain the good ambient germ, independently of angle. The native gradient lemma then gives the actual field zero, neighborhood covector identity and selfadjoint linearization. Apply the positive native normal lemma using the produced block data to obtain the last strict inequality. Every threshold was chosen before the angle and the fiber presentation. □

Lemma T.57 (Actual field zero and metric linearization)**.** *In the preceding setting, at a native critical Hessian chart center $z = c(a)$, both G and Z are smooth, $Z(z) = 0$, and*

$$G(z)(DZ_z v, w) = D^2 f_z(v, w).$$

The derivative DZ_z is selfadjoint for $G(z)$.

Proof. The actual fiber primitive is smooth, and its differential is nondegenerate by the actual tangent-kernel and positive plane-pairing theorems. Smoothness of the Liouville field then gives smoothness of its chart field. Smoothness of L and $\nabla\phi$, followed by taking derivatives and finite sums of compositions, gives smoothness of G. At the center $Df_z = 0$. The gradient identity gives

$G(z)(Z(z), Z(z)) = 0$. The lift differential is injective and the ambient Hessian metric is positive definite, so $Z(z) = 0$. Differentiate $G(y)Z(y) = Df_y$. The derivative of G is multiplied by $Z(z) = 0$, leaving the displayed identity. The metric is symmetric because the potential Hessian is symmetric. Symmetry of $D^2 f_z$ therefore gives $G(z)(DZ_z v, w) = G(z)(v, DZ_z w)$. □

Lemma T.58 (Positive actual normal linearization)**.** *Under the actual face Hessian hypotheses, if $v \neq 0$ and $D_z v$ is orthogonal to the conormal tangent S_H, then $G(z)(DZ_z v, v) > 0$.*

Proof. The preceding identity identifies the claimed quadratic expression with $D^2 f_z(v, v)$. Positivity of the actual fiber Hessian on the conormal metric normal proves the strict inequality. □

Lemma T.59 (Positive normal to the literal native branch differential)**.** *Let $j_B(b, \vartheta) = (g(b), f(\vartheta))$ be a radial-angular branch with smooth radial factor at b, smooth angular factor with image $T_{\mathrm{pos}(s)}$, and injective constant-radius angular differential. Suppose the literal radial differential image is $\{X : X \in T_q,\ HX \in \operatorname{span} \mathrm{pos}(s)\}$, where $H = D\nabla\phi(u)$. At a same-point native critical chart, assume the face Hessian hypotheses of the preceding lemma. Then for every $v \neq 0$ such that*

$$d\lambda_\phi(D \operatorname{cover} D_z v, D \operatorname{cover} JW) = 0 \quad \text{for every } D \operatorname{cover} W \in \operatorname{range} Dj_B,$$

$G(z)(DZ_z v, v) > 0$.

Proof. The actual branch-Hessian theorem gives $D^2 f_z(v, v) > 0$ under precisely this literal differential-image orthogonality. The native gradient linearization identifies that expression with $G(z)(DZ_z v, v)$. □

Lemma T.60 (Uniform native gradient data on all critical charts)**.** *Let ϕ_∞ be strongly compatible and satisfy $\phi_\infty(u) \geq a_0|u|^2$ for some $a_0 > 0$. Let ϕ be a smooth extension equal to ϕ_∞ outside a fixed radius, with positive uniform lower and upper Hessian bounds. One finite positive threshold in β ensures, for every presentation of the full endpoint fiber and every native critical Hessian chart, that its actual coordinate field is smooth at the center, vanishes there, satisfies the gradient identity on a neighborhood, and has the preceding Hessian linearization and selfadjointness identities.*

Proof. Take the maximum of the uniform threshold for positive fiber plane pairing and that for the open holomorphic critical neighborhoods. Stationarity of the native Hessian chart implies actual intrinsic criticality. The latter theorem therefore supplies an open neighborhood of its image where the constant cutoff is one, and every vertex cutoff is the indicator of the retained face. Thus every augmented cutoff is zero or one there. The chart center maps to its logarithmic representative by the covering identity and the inverse-chart identity. Pull the open neighborhood back to this representative and apply the local native gradient and linearization lemmas. Both thresholds are independent of the angle and of the chosen full-fiber presentation. □

Fix critical normal data at the specified u, q, λ, B, R, with $H = D\nabla\phi_\infty(u)$, a strongly compatible globally C^1 end potential, and a nonempty proper face. Let ϕ be globally smooth with a positive uniform Hessian lower bound and positive actual plane pairing on the full endpoint fiber. Fix an actual full-fiber presentation $j : N \hookrightarrow Y$ and $v_0 \in s$. There exist one defining neighborhood from these data and one metric-normal lift L for its radial quotient Q, chosen before all torus angles, using the fixed weights $r_v = e^\beta q_v$. The resulting M_L is symmetric positive definite. Every face-torus angle θ has a point $a \in N$ with $j(a) = (u, \theta)$. For every such native center and logarithmic representative, the same data construct a native Hessian chart whose actual field and transverse rate vanish at the center. Its rate derivative on L_c is selfadjoint and strictly positive in this same fixed M_L, and satisfies

$$M_L\big(DR_c(b)L_c z, w\big) = \mathrm{Hess}_{s,H,q,\lambda}(Lz, Lw).$$

Proof. Select the defining neighborhood supplied by the fixed datum for v_0. Its quotient is surjective with kernel $\operatorname{span}\operatorname{pos}(s)$. The same datum proves symmetry, positivity and surjectivity of H, and $AB = \mathrm{id}$ makes A surjective. The normalized monomial identity in the datum, multiplied by e^{β}, gives the unnormalized values $r_v = e^{\beta} q_v$ at every angle. These are strictly positive. Apply the fixed cover-plane normal-lift theorem once using these data, before selecting an angle. It supplies L and the fixed positive metric.

For an arbitrary face-torus angle, the datum proves actual endpoint membership of (u, θ). The full image equality for j therefore supplies a native preimage. At every stated center and representative, the same-data native-chart theorem constructs the actual logarithmic lift and Hessian chart. The common radial plateau gives its good-cutoff germ. The end gradient is smooth at $u \neq 0$, and the potential germ identifies its Hessian with that of the actual global potential. Apply the native gradient theorem to obtain the actual field zero; the literal transverse rate consequently vanishes as well. Apply the actual rate-derivative identity, fixed constrained pairing identity, selfadjointness and positive normal transport using the already selected Q, L and the same H, B, R, q, λ. All these identities hold for the arbitrary angle and center, establishing the assertions without making another critical-branch or metric choice. □

T.6 Transverse defining coordinates over a whole face torus

Fix a nonempty proper face s, put $V = \operatorname{span}\operatorname{pos}(s)$, and choose one vertex $v_0 \in s$. Write $s' = s \setminus \{v_0\}$. The defining coordinates below are single quotient-defined functions near the whole face torus. They do not choose real representatives of its free angles. They give the transverse defining map for the normal-coordinate construction in the trapping argument; a product neighborhood and its uniform expansion estimate remain additional assertions.

Lemma T.61 (A complement-valued radial quotient)**.** *For each linear subspace $V \subset \mathbb{R}^n$ there are a linear retraction $P : \mathbb{R}^n \to V$ and a surjective linear map $Q : \mathbb{R}^n \to \ker P$ with $\ker Q = V$.*

Proof. Extend a basis of V to a basis of $\mathbb{R}^n$ and project onto the coordinates of the first basis. This defines P with $P|_V = \mathrm{id}$. Put $Qx = x - Px$. Then $P(Qx) = Px - P(Px) = 0$, so Q has values in $\ker P$. For $x \in \ker P$, $Qx = x$, proving surjectivity. Finally $Qx = 0$ implies $x = Px \in V$, and $x \in V$ implies $Px = x$, hence $Qx = 0$. □

Definition T.62 (The positive phase neighborhood and transverse map)**.** *Write $m_v(u, \theta) = e^{\langle v, u \rangle} \chi_v(\theta)$ and let Φ be the actual Legendre map. Define*

$$O_s = \{(u, \theta) : \operatorname{Re} m_v(u, \theta) > 0 \ (v \in s)\}, \qquad \delta(u, \theta) = \big(Q\Phi(u), (\operatorname{Im} m_v(u, \theta))_{v \in s'}\big).$$

Lemma T.63 (A whole-torus exact zero set)**.** *The set O_s is open and contains $\mathbb{R}^n \times T_{\operatorname{pos}(s)}$. On $O_s \cap \{\sum_{v \in s} c m_v = 1\}$, for any positive real constant c,*

$$\delta(u, \theta) = 0 \quad \Longleftrightarrow \quad \Phi(u) \in V \ \text{ and } \ \theta \in T_{\operatorname{pos}(s)}.$$

Proof. The real parts of the monomials are continuous, so their finitely many strict positive inequalities define an open set. On the face torus all characters equal one; the monomials are positive exponentials, proving the inclusion. The radial coordinate vanishes exactly when $\Phi(u) \in V$. If the retained imaginary coordinates vanish, the imaginary part of the fiber equation gives $c \operatorname{Im} m_{v_0} = 0$. Since $c > 0$, the omitted imaginary part also vanishes. For each $v \in s$, divide m_v by its positive exponential factor. The resulting unit complex number has zero imaginary part and positive real part, hence equals one. The character criterion for the face torus now applies. Conversely the face-torus characters are all one, making every retained imaginary coordinate zero; $\Phi(u) \in V$ makes the radial coordinate zero as well. □

Lemma T.64 (The actual branch as a common transverse zero set)**.** *Let Φ be the Legendre homeomorphism of a strongly compatible end potential, with its punctured smooth inverse. For all sufficiently large β, suppose $y = (u_*, \theta_*)$ is an actual conormal endpoint point, $\Phi(u_*) = A^T a$ with every $a_v > 0$, and $\ell\Phi(u_*) = 1$. Suppose also that one radial open neighborhood U of u_* has the literal cutoff plateau one on $\{0\} \cup s$ and zero off this set. Then there are a smooth embedded immersion $j_B(q, \vartheta) = (g(q), f(\vartheta))$, a quotient Q as above, and an open ambient neighborhood O containing the entire torus $\{u_*\} \times T_{\mathrm{pos}(s)}$, on which δ is smooth. Every point of that torus belongs to the actual endpoint fiber and has $\delta = 0$. On the actual endpoint fiber in O,*

$$\delta(u, \theta) = 0 \quad \Longleftrightarrow \quad (u, \theta) \in \operatorname{range} j_B \quad \Longleftrightarrow \quad \Phi(u) \in \mathrm{pos}(s), \quad \theta \in T_{\mathrm{pos}(s)}.$$

Proof. The embedded-branch lemma constructs B, g, f and gives one radial neighborhood on which its image is exactly the full-span conormal locus, uniformly over every angle. The positive-face-span lemma supplies a momentum neighborhood of $A^T a$ in which membership in V implies membership in $\mathrm{pos}(s)$. Pull it back by Φ, intersect with U, the preceding radial neighborhood, and the complement of zero. The last set contains u_* because $\ell\Phi(u_*) = 1$ and $\Phi(0) = 0$. Choose a smaller radial open neighborhood W of u_* in this intersection, and set $O = (W \times T^n) \cap O_s$. It is open and contains the entire face torus. Smoothness of Φ off zero proves smoothness of δ on O. On the actual endpoint fiber the literal family equation is $\sum_{v \in \mathrm{Vert}(P)} \chi_v m_v = \chi_0$; the plateau makes $\chi_0 = 1$, removes every nonface term, and makes each face coefficient one. Thus the face equation holds, including the specified small-monomial cutoff through the proved endpoint-fiber identity. The preceding exact zero-set lemma identifies $\delta = 0$ with the full-span conormal locus, which is the branch image by the chosen radial neighborhood. The relative interior neighborhood identifies the span and cone conditions there. Finally $g(0) = u_*$, $0 \in B$, and $\operatorname{range} f = T_{\mathrm{pos}(s)}$; the branch's actual fiber identity proves the asserted fiber membership for every angle, and its span condition proves $\delta = 0$. □

Lemma T.65 (Constructed critical tori have the common zero neighborhood)**.** *Let ϕ be a strongly compatible end potential, globally C^1 with $\phi(0) = 0$, and suppose that for some $c > 0$ the function $\phi(u) - c\|u\|^2$ is convex, $\phi(u) \geq c\|u\|^2$, and its punctured Euclidean Hessian is bounded below by $2c$ times the identity. For each nonempty proper face s, there are functions x and $z = (q, \mu)$, smooth at zero and constructed from the constrained-minimum and normalized reduced branches, with $x(0)$ the minimum of ϕ on the polar face. For every sufficiently large β and every choice $v_0 \in s$, the critical radius*

$$u_\beta = \beta x(\beta^{-1} \log q(\beta^{-1}))$$

has the embedded branch, quotient Q, and common open neighborhood O of the preceding lemma. Here $\Phi = \nabla\phi$, all points of $\{u_\beta\} \times T_{\mathrm{pos}(s)}$ belong to the actual endpoint fiber and have $\delta = 0$, and the two exact zero-set equivalences hold on that fiber in O.

Proof. The quantitative convexity and quadratic lower bound construct the actual gradient homeomorphism Φ; its punctured smooth inverse follows from the positive Hessian bound. The lower bound makes zero a local minimum, so $\Phi(0) = 0$. The actual reduced-branch producer constructs x, z and gives, for sufficiently large β,

$$q_v(\beta^{-1}) > 0, \quad \mu(\beta^{-1}) > 0, \quad \Phi(u_\beta) = \beta\mu(\beta^{-1}) A^T q(\beta^{-1}).$$

Strong compatibility makes the polar minimum lie strictly below the support height of every off-face vertex. Smoothness at zero of x and q bounds the radial displacement from $\beta x(0)$; the proved cutoff plateau theorem therefore supplies one actual radial plateau neighborhood of u_β for all sufficiently

large β. The reduced-branch fiber identity puts every face-torus angle on the actual endpoint fiber. Set $a = \beta\mu(\beta^{-1})q(\beta^{-1})$; every coordinate is positive. Since A is surjective, A^T is injective, so $A^T a \neq 0$. Choose a nonzero coordinate of this momentum and divide its coordinate functional by that coordinate to obtain $\ell\Phi(u_\beta) = 1$. All hypotheses of the preceding neighborhood lemma are now proved by these producers. Applying it at the zero torus angle constructs the claimed objects and gives its conclusions simultaneously for all angles. □

Lemma T.66 (The literal transverse differential)**.** *At a face-torus point $\xi = (u, \theta)$ write $H = D\Phi_u$, $r_v = e^{\langle v,u\rangle} > 0$, and let $q_v > 0$ be the normalized face weights. The transverse map is smooth wherever Φ is smooth, and its logarithmic differential is*

$$L(X, Y) = \big(QHX, (r_v\langle v, Y\rangle)_{v\in s'}\big).$$

For the actual fiber-chart lift g at its center a, its native coordinate differential is $L \circ Dg_a$. On the fiber tangent $T_q \times T_q$, where $T_q = \ker(\ell_q A)$, the kernel of L is exactly $\{(X, Y) : X \in T_q,\ HX \in V,\ AY = 0\}$.

Proof. The radial component is the composition of the smooth Legendre map with the linear map Q. The angular components are imaginary parts of smooth descended monomials, so they are smooth. The chain rule gives QHX for the radial derivative. The monomial derivative is $m_v(\xi)(\langle v, X\rangle + i\langle v, Y\rangle)$; at the face torus $m_v(\xi) = r_v$ is real, so its imaginary part is $r_v\langle v, Y\rangle$. Composing the identity of the actual logarithmic lift with the transverse map gives the native factorization. Finally, vanishing of the radial part is equivalent to $HX \in V$. Positivity of the retained r_v makes all retained angular pairings vanish. The fiber constraint $\sum_{v\in s} q_v\langle v, Y\rangle = 0$ then gives $q_{v_0}\langle v_0, Y\rangle = 0$; positivity of q_{v_0} recovers the omitted pairing. This proves $AY = 0$. The reverse implication follows by substitution. □

Lemma T.67 (Surjectivity after removing the redundant angle)**.** *Suppose Q and H are surjective, $AB = \mathrm{id}$, $HB = A^T R$, and $\ker Q = V$. For positive face weights q, r, the restriction of L to $T_q \times T_q$ is surjective. Consequently the actual native fiber differential of δ is surjective.*

Proof. Given a target (z, b), choose X_0 with $QHX_0 = z$. Put e_0 equal to the coordinate vector of v_0. Since $ABe_0 = e_0$, the vector Be_0 has $\ell_q ABe_0 = q_{v_0} > 0$; also $HBe_0 \in V$, so $QHBe_0 = 0$. Thus

$$X = X_0 - \frac{\ell_q AX_0}{q_{v_0}} Be_0$$

lies in T_q and satisfies $QHX = z$. Extend b_v/r_v from s' to a vector a_0 on s by setting its v_0 coordinate to zero, then put $a = a_0 - (\ell_q a_0/q_{v_0})e_0$. This has $\ell_q a = 0$ and $r_v a_v = b_v$ for every retained vertex. Surjectivity of A gives Y with $AY = a$, hence $Y \in T_q$ and the angular part of $L(X, Y)$ equals b. This proves surjectivity. The logarithmic differential of the actual fiber chart has image exactly $T_q \times T_q$, so each such (X, Y) is the image of a native tangent. Composing with L proves the final assertion. □

Lemma T.68 (The native transverse orbit equation)**.** *Let $j : N \hookrightarrow Y$ be the actual smooth fiber embedding and Z_μ the native Liouville field of its specified primitive. Set $b(a) = D(\delta \circ j)_a Z_\mu(a)$. Every native integral curve γ satisfies, at each time when Φ is smooth at its radial coordinate,*

$$\frac{d}{dt}\delta(j\gamma(t)) = b(\gamma(t)).$$

Proof. The transverse map is smooth at $j\gamma(t)$ and the embedding is smooth, so their composition is differentiable at $\gamma(t)$. The derivative of the native integral curve is $Z_\mu(\gamma(t))$. Applying the manifold chain rule to these two differentials gives the displayed derivative. The definition of b identifies its right-hand side. □

T.7 Shared critical data for the conormal neighborhood and native normal metric

Definition T.69 (Critical normal data at a specified radius)**.** *Fix the face s, parameter β, end potential ϕ_∞, global potential ϕ, radial point $u \neq 0$, weights q, multiplier λ, and linear maps B, R. Write $A = A_s$ and $H = D(\nabla\phi_\infty)_u$. Critical normal data at these specified objects consist of the following statements and geometric objects:*

$$q_v > 0, \qquad \sum_v q_v = 1, \qquad \lambda > 0, \qquad \nabla\phi_\infty(u) = \lambda A^T q, \qquad AB = \mathrm{id}, \qquad HB = A^T R.$$

The actual form H is symmetric and positive definite, while $\langle Rh, h\rangle - \lambda \sum_v q_v h_v^2 < 0$ for every $h \neq 0$. The potentials have the same germ at u. There is one radial open plateau neighborhood U of u, with cutoff one on $\{0\} \cup s$ and zero elsewhere. At every $\theta \in T_{\mathrm{pos}(s)}$, the normalized monomial $m_v(u, \theta) = q_v$ for $v \in s$ and the actual endpoint fiber equation holds. Finally, for every choice $v_0 \in s$ there is an explicit common transverse neighborhood: a radial quotient Q with kernel $\operatorname{span}\mathrm{pos}(s)$*, a smooth embedded conormal immersion (g, f) through the whole torus at u, and one open ambient neighborhood O of that torus, with smooth defining map δ. On the actual endpoint fiber in O, $\delta = 0$ is exactly the branch image and exactly the literal positive-cone conormal locus. These objects are a defining neighborhood; a product inverse is an additional assertion.*

Lemma T.70 (The same data construct the actual native Hessian chart)**.** *Suppose ϕ_∞ is a strongly compatible end potential of class C^1. The critical normal data at u imply that H is surjective, that $\nabla\phi_\infty$ is smooth at u, and that $H = D(\nabla\phi)_u$. They give a good-cutoff germ at (u, θ) for every angle θ. Suppose ϕ is globally smooth with a positive uniform Hessian lower bound, and its actual plane pairing is positive on the whole endpoint fiber. For every actual smooth full-fiber embedding $j : N \hookrightarrow Y$, every face-torus angle θ, and every $a \in N$, ξ with $j(a) = \mathrm{cover}(\xi) = (u, \theta)$, these same data construct a native face Hessian chart with the specified H, q, λ.*

Proof. If $HX = HY$ with $X \neq Y$, positivity applied to $X - Y$ contradicts $H(X - Y) = 0$. Thus H is injective and, in finite dimension, surjective. Smoothness of the end potential off zero gives smoothness of its gradient at u. Differentiating the equality of potential germs twice gives the literal equality of their coordinate Hessians. The pullback of U by the radial projection is a neighborhood of (u, θ) for every angle; its cutoffs are all zero or one, proving the good germ. For a face-torus angle, the data give actual endpoint membership, the normalized monomial weights, and the normal equation. The plateau lemma makes the actual endpoint equation the finite face sum on a neighborhood. The native full-fiber Hessian-chart construction now applies with the potential germ, $AB = \mathrm{id}$, $HB = A^T R$, symmetry and positivity of H, positive q, λ, and the reduced negative form supplied by the same data. It constructs a genuine logarithmic lift with full fiber tangent image and the second derivative of the actual restricted global potential. All parameters remain the specified u, H, q, λ. □

Lemma T.71 (One reduced branch supplies both geometric contracts)**.** *Let ϕ_∞ be a strongly compatible end potential, globally C^1 with $\phi_\infty(0) = 0$. Suppose for some $c > 0$ that $\phi_\infty - c\|u\|^2$ is convex, $\phi_\infty \geq c\|u\|^2$, and its punctured Hessian is bounded below by $2c$ times the identity. Let ϕ agree with ϕ_∞ outside a fixed ball. For each nonempty proper face s there are functions $x, z = (q, \mu)$, smooth at zero and constructed from the constrained-minimum and normalized reduced branches, with $x(0)$ the minimum of ϕ_∞ on the polar face. For every sufficiently large β, critical normal data exist at the single specified radius and actual branch derivatives*

$$t_\beta = \beta^{-1} \log q(\beta^{-1}), \qquad u_\beta = \beta x(t_\beta), \qquad q_\beta = q(\beta^{-1}), \qquad \lambda_\beta = \beta\mu(\beta^{-1}),$$
$$B_\beta = Dx_{t_\beta}, \qquad R_\beta = D(a_{\phi_\infty, x})_{t_\beta}, \qquad H_\beta = D(\nabla\phi_\infty)_{u_\beta}.$$

Here $a_{\phi_\infty,x}$ is the actual reduced coefficient map. In particular the conormal neighborhood and all native-chart normal metric inputs refer to this same radius and these same weights and multiplier.

Proof. The quantitative convexity constructs the actual Legendre homeomorphism and its smooth punctured inverse. The quadratic lower bound makes zero a local minimum, so its gradient is zero. Choose the actual positive reduced branch once. Its normalized reduced equation gives positive $q_\beta, \mu(\beta^{-1})$ and $\sum_v q_{\beta,v} = 1$ for sufficiently large β. Its reconstructed normal equation is exactly $\nabla\phi_\infty(u_\beta) = \lambda_\beta A^T q_\beta$. The branch's shifted face levels give the normalized monomial $q_{\beta,v}$ at every face-torus angle; its actual fiber producer gives endpoint membership for all these angles. Differentiating the face-level and normal identities gives $AB_\beta = \mathrm{id}$ and $H_\beta B_\beta = A^T R_\beta$. The actual face-block theorem gives symmetry and positivity of H_β and negativity of the reduced radial form. All these assertions concern the already chosen branch. The positive base weights and nonzero polar minimum imply that u_β escapes every fixed ball, so the potentials have the same germ there. Strong compatibility supplies strict off-face inequalities at the polar minimum; smoothness at zero of this same branch bounds its displacement from $\beta x(0)$. The cutoff plateau producer therefore gives one radial open plateau U_β at u_β. Set $a = \lambda_\beta q_\beta$, which has strictly positive coordinates. Injectivity of A^T gives nonzero momentum; hence $u_\beta \neq 0$. Normalize a nonzero momentum coordinate to choose a functional $\ell\nabla\phi_\infty(u_\beta) = 1$. The common-neighborhood theorem then constructs the embedded branch, quotient and open O at this same u_β, using the just-proved positive coefficients and plateau. Intersecting the finitely many eventual parameter conditions gives one threshold. Recording their witnesses gives precisely the stated critical normal data, with no second choice of x, z. □

T.8 Vanishing transverse velocity on the actual smooth branch

Lemma T.72 (The native rate in logarithmic coordinates)**.** *Let $j : N \hookrightarrow Y$ be a smooth fiber embedding, let μ be its primitive, and let $j(a) = \mathrm{Exp}(x)$. Suppose the native Liouville vector pushes forward to $D\,\mathrm{Exp}_x[V]$, and that the Legendre map Φ is smooth at x_u. Then the actual native rate is*

$$D(\delta \circ j)_a Z_\mu(a) = \Big(QD\Phi_{x_u}[V_u], \big(\mathrm{Im}(m_v(\mathrm{Exp}\,x)(\langle v, V_u\rangle + i\langle v, V_\theta\rangle))\big)_{v\neq v_0}\Big).$$

Proof. The transverse map is smooth at $\mathrm{Exp}(x)$. The chain rule for $\delta \circ j$ and the assumed pushed vector give $D\delta_{\mathrm{Exp}(x)} D\,\mathrm{Exp}_x[V]$. The chain rule for $\delta \circ \mathrm{Exp}$ identifies this with $D(\delta \circ \mathrm{Exp})_x[V]$. The literal logarithmic differential of δ gives the displayed two components. □

Lemma T.73 (The intrinsic transverse rate vanishes on the branch)**.** *Let ϕ be a smooth global potential with positive uniform lower and finite upper Hessian bounds, agreeing outside a positive radius with a strongly compatible degree-two end potential ϕ_∞. There is one threshold such that, for every later endpoint parameter and every smooth presentation $j : N \hookrightarrow Y$ of its full fiber, the following holds. Fix a nonempty proper face s, a vertex $v_0 \in s$, and a radial quotient Q with kernel $V = \mathrm{span\,pos}(s)$. At any fiber point in the positive phase neighborhood with literal cutoff plateau one on $\{0\} \cup s$ and zero elsewhere, if $\delta = 0$, then*

$$D(\delta \circ j)_a Z_{j^*\lambda_\phi}(a) = 0, \qquad \Phi = \nabla\phi_\infty.$$

This applies at every point of the smooth branch in the common defining neighborhood, after intersecting that neighborhood with its radial plateau.

Proof. Choose one threshold for the native projected-field identity, Poisson positivity, and escape of the whole fiber beyond the end-agreement radius. At the chosen point the literal plateau reduces the

actual fiber equation to $\sum_{v\in s} e^{-\beta} m_v = 1$. The exact zero-set lemma therefore gives $\nabla\phi_\infty(u) \in V$ and $\theta \in T_{\mathrm{pos}(s)}$. Every off-cone vertex lies outside s, so its cutoff vanishes.

Take any logarithmic representative $x = (u, \widetilde{\theta})$. Fiber escape puts u strictly beyond the agreement radius; the potentials, gradients, and Hessians consequently agree as germs there. The homogeneous Euler identity gives $D\nabla\phi_u[u] = \nabla\phi(u)$, and the Poisson scalar is nonzero. The actual projected-radial tangency lemma gives $D\nabla\phi_u[V_u] \in V$ and $V_\theta = 0$ for the pushed intrinsic vector. The germ equality replaces the gradient differential by $D\Phi_u$. Thus its image under Q is zero. On the face torus every m_v is a real positive exponential; because $V_\theta = 0$, every retained imaginary differential is also zero. The preceding native-to-logarithmic rate identity proves the conclusion. No vanishing of the vector field at the chosen branch point is used. □

Corollary T.74 (A common open neighborhood with zero transverse rate). Under the same potential hypotheses and for every sufficiently large endpoint parameter, let O be an open set containing a whole radial face torus $\{u_*\} \times T_{\mathrm{pos}(s)}$. Let U be an open radial neighborhood of u_* with the literal face cutoff plateau. There is an open $W \subset O$ containing that whole torus on which, at every actual fiber point, $\delta = 0$ implies that the intrinsic transverse rate is zero. In particular, the actual smooth branch and its exact transverse zero-set description restrict to this same neighborhood.

Proof. Put $W = O \cap \mathrm{pr}_u^{-1}(U) \cap O_s$, where O_s is the positive phase neighborhood. Every set in this intersection is open. Every point of the indicated torus belongs to all three sets: the first two by hypothesis, and the third because its monomials are positive real exponentials. At a fiber point of W the actual radial plateau and the positive phase hypotheses of the preceding lemma hold, so $\delta = 0$ forces the native rate to vanish. Since $W \subset O$, every branch image or exact zero-set equivalence already established on O remains true after restriction to W. □

Lemma T.75 (The whole literal outward radial continuation)**.** *For a strongly compatible homogeneous end potential, for all sufficiently large β, every $(u, \theta) \in L_\beta$ and every $t \geq 1$ satisfy*

$$(tu, \theta) \in \Gamma, \qquad f_1(tu, \theta) = \mathcal{R}_\beta(tu) \in \mathbb{R}_{>0}.$$

There is $d > 0$ such that the actual complex-valued scalar-ray maps have

$$\frac{d}{dr} f_1(rtu, \theta)\Big|_{r=1} = d, \qquad \frac{d}{dr} f_1(ru, \theta)\Big|_{r=t} = d/t.$$

In particular both derivatives are positive real numbers.

Proof. Take the finite intersection of the thresholds for the endpoint radial equation, positive scalar-ray derivative, and $\beta \geq \log N_0 + 1$. The exact height bound gives $h(u) \geq \beta - \log N_0 \geq 1$, so $u \neq 0$. If the starting point belongs to the σ piece, degree-one homogeneity of the actual end gradient gives $\Phi_\infty(tu) = t\Phi_\infty(u) \in \sigma \setminus \{0\}$. The angle is unchanged and remains in T_σ, proving membership in Γ.

Positive homogeneity of the support height and $t \geq 1$ give

$$h(tu) = th(u) \geq h(u) \geq \beta - \log N_0 > \beta - \log(2N_0).$$

Thus the actual endpoint equals the real radial function at (tu, θ), and the region-wide positive scalar-ray derivative theorem applies there. It supplies the positive value and the first derivative $d > 0$. Composing that actual scalar-ray map with $r \mapsto r/t$ at $r = t$ gives exactly $r \mapsto f_1(ru, \theta)$, with derivative d/t. Since $t \geq 1$, this derivative is positive as well. □

Lemma T.76 (The actual ambient Liouville derivative on every outward ray)**.** *Let ϕ be smooth with a uniform positive lower Euclidean Hessian bound, and suppose it agrees outside a radius $R > 0$ with the chosen strongly compatible homogeneous end ϕ_∞. There is $B > 0$ such that, for every $\beta \geq B$, every $(u, \theta) \in L_\beta$, and every $t \geq 1$, the literal ambient Liouville field Z_0 determined by λ_ϕ satisfies*

$$Df_1(tu, \theta)[Z_0(tu, \theta)] \in \mathbb{R}_{>0}.$$

Proof. Increase the preceding threshold so that the entire endpoint fiber is outside radius $R + 1$. Then every starting section point has $\|u\| \geq R + 1$. Since $t \geq 1$, its entire outward continuation satisfies $\|tu\| = t\|u\| > R$. Literal agreement of the potentials on the open exterior transfers the homogeneous Euler identity to the actual smooth potential: $D(\nabla\phi)_{tu}(tu) = \nabla\phi(tu)$. The positive lower Hessian bound makes the actual potential two-form nondegenerate. Its radial contraction equation therefore identifies the primitive-determined ambient field with the logarithmic vector $(tu, 0)$ at the continued point.

Choose a logarithmic lift of (tu, θ). The actual native complex differential pulls back to the actual cover differential. Evaluating that differential on the lifted radial vector is the derivative at one of $r \mapsto f_1(rtu, \theta)$. Uniqueness of the derivative identifies this value with the preceding $d > 0$. Hence the displayed evaluation is positive real for every outward point, as asserted. □

Lemma T.77 (Increasing base value on each outward ray)**.** *For a strongly compatible homogeneous end potential and all sufficiently large β, each $(u, \theta) \in L_\beta$ has strictly increasing real base value $t \mapsto \operatorname{Re} f_1(tu, \theta)$ on $[1, \infty)$. In particular this value is at least one throughout that interval.*

Proof. The outward-ray derivative theorem gives a positive derivative at every $t \geq 1$. Taking real parts preserves that positive real derivative. The mean value theorem on the convex interval gives strict monotonicity. At $t = 1$ the fiber equation is $f_1(u, \theta) = 1$, proving the lower bound. □

Lemma T.78 (Ambient transversality for every base coefficient)**.** *Let ϕ be a smooth potential with uniform positive lower and finite upper Hessian bounds, equal outside a positive radius to its strongly compatible homogeneous end ϕ_∞. There is a threshold $B > 0$ such that for every $\beta \geq B$, $(u, \theta) \in L_\beta$, $t \geq 1$ and $K \geq 0$, the actual field of $\lambda_\phi + Kf_1^*\lambda_{\mathbb{C}}$ satisfies*

$$Df_1(tu, \theta)[Z_{1,K}(tu, \theta)] \in \mathbb{R}_{>0}.$$

The threshold is independent of K.

Proof. Take the maximum of the thresholds for outward radial positivity, the actual unmodified Liouville derivative, and global nonnegativity of the tailored Poisson pairing. At any of the stated points write $b = f_1(tu, \theta) > 0$, $a = Df_1[Z_0] > 0$, and $\pi = \pi_{f_1} \geq 0$; the first two numbers are real by the outward-ray statements. The rank-two formula for the actual updated primitive gives

$$Df_1[Z_{1,K}] = \frac{a + (K\pi/2)b}{1 + K\pi}.$$

Its denominator is at least one and its numerator is at least $a > 0$. Thus it is a positive real number for every nonnegative K. □

Lemma T.79 (A positive base margin on each compact outward band)**.** *Under the preceding potential hypotheses, suppose the homogeneous gradient extends to a homeomorphism of the real vector spaces fixing zero. There is $B > 0$ such that for every $\beta \geq B$ and finite $T \in \mathbb{R}$ there is $\varepsilon > 0$ satisfying*

$$\operatorname{Re} Df_1(tu, \theta)[Z_{1,K}(tu, \theta)] \geq \varepsilon \quad ((u, \theta) \in L_\beta,\ 1 \leq t \leq T,\ K \geq 0).$$

In particular the lower bound does not require an upper bound on K.

Proof. The section L_β is compact by the radial-section theorem. On the compact parameter set $[1,T]\times L_\beta$, put $x=(tu,\theta)$, $a(x)=d(\operatorname{Re} f_1)_x[Z_0(x)]$ and $b(x)=\operatorname{Re} f_1(x)$. The smoothness of f_1 and of the actual unmodified Liouville field makes a and b continuous. The preceding ray theorems say that both are strictly positive. Thus the continuous function $\min(a,b/2)$ has a positive lower bound ε on this compact set; if it is empty take any positive bound. The actual complex derivative of f_1 along Z_0 equals the real number a, by the chain rule for the real part. For $\pi\ge 0$ and $K\ge 0$ the rank-two identity gives

$$\operatorname{Re} Df_1[Z_{1,K}]=\frac{a+K\pi\, b/2}{1+K\pi}\ge\frac{\varepsilon+K\pi\varepsilon}{1+K\pi}=\varepsilon.$$

This proves the claim even for an unbounded range of nonnegative K. □

Lemma T.80 (Finite base bands have bounded radial parameter)**.** *For all sufficiently large β, if $y=(u,\theta)\in L_\beta$, $t\ge 1$ and $\operatorname{Re} f_1(tu,\theta)\le M$, then $t\le\beta+\max(M,0)$. Under the smooth potential and gradient homeomorphism hypotheses of the preceding lemma, each such finite base band has a positive lower bound for $\operatorname{Re} Df_1[Z_{1,K}]$ independent of $K\ge 0$. With the gradient homeomorphism hypothesis, the actual band $\{(tu,\theta):(u,\theta)\in L_\beta,\ t\ge 1,\operatorname{Re} f_1(tu,\theta)\le M\}$ is compact.*

Proof. The level-one height estimate gives $H(u)\ge\beta-\log|A|\ge 1$. Write $h=H(tu)=tH(u)\ge t$. If $h<\beta-\sqrt{\beta}$, then $t\le\beta$. Otherwise the maximal retained monomial yields $e^{h-\beta}\le\operatorname{Re} f_1(tu,\theta)\le M$. Since $h-\beta\le e^{h-\beta}$, this implies $t\le\beta+M$. Both cases give the stated bound. Apply the preceding compact-parameter margin with $T=\beta+\max(M,0)$. The band is the continuous image of the closed base sublevel subset of $[1,T]\times L_\beta$ and hence is compact. □

Lemma T.81 (A compact radial section)**.** *For large β, L_β is a compact radial section over the entire FLTZ link. Its radial coordinate is independent of the angle on each T_σ, and*

$$\beta-\log N_0\le h(u)\le\beta.$$

On its whole outward radial continuation in Γ, f_1 is real positive and $Z_0f_1>0$.

Proof. Any point of H_β has $h\ge\beta-\log N_0$. On Γ in the slightly larger region $h\ge\beta-\log(2N_0)$, every exponent outside the indicated cone has a deficit at least $\delta_0|u|$ by (12). For large β this exceeds $\sqrt{\beta}+2$, so its cutoff vanishes identically on an ambient neighborhood. The retained characters are one on T_σ, and $\eta=1$ there. Hence f_1 restricts to the real function

$$\mathcal{R}_\beta(u)=1-\chi_0+\sum_v\chi_v e^{\langle v,u\rangle-\beta}.$$

At $h=\beta-\log(2N_0)$ this is at most $1/2$, and at $h=\beta+\log 2$ it is at least 2 because a largest nonconstant monomial has cutoff one. Every retained exponential has positive radial derivative: its exponent slope is at least $h-\sqrt{\beta}-2>0$. A largest one contributes hM. All cutoff-derivative terms total at most $C|u|\varepsilon_\beta\max(1,M)$, strictly smaller than hM on this region. Thus $Z_0\mathcal{R}_\beta>0$, including at all larger radii. There is exactly one transverse solution of $\mathcal{R}_\beta=1$ on every ray. At it $M\le 1$, since a retained largest term alone would otherwise exceed $\chi_0\le 1$; also $\chi_0=1$ and $M\ge 1/N_0$. This proves the bounds and compactness. Smoothness means smoothness on each full-span conormal extension; the union can of course be singular along its cone strata. □

Lemma T.82 (Two-sided invariance, including the cone faces)**.** *For the chosen smooth uniformly convex potential, agreeing with its strongly compatible homogeneous end, and all sufficiently large β, the intrinsic Liouville flow preserves L_β in both time directions and fixes the angle.*

Proof. We use the minimal-face neighborhood argument in the local preservation lemma to treat points on every cone stratum. In Legendre coordinates, the actual projected Liouville field on the conormal extension satisfies

$$\dot{\theta} = 0, \qquad \dot{p} = p - (a_f/\pi_f)G, \tag{32}$$

where $G = \partial_\theta \operatorname{Im} f_1$, $a_f = p \cdot \partial_p \operatorname{Re} f_1$, and $\pi_f = \partial_p \operatorname{Re} f_1 \cdot G > 0$. At a given point choose the minimal face τ containing p in its relative interior. The off-face exclusion implies $G \in \operatorname{span}_{\mathbb{R}} \tau$, and the exact differential range of the smooth branch contains this field. The local retraction and ODE uniqueness therefore preserve the branch. Shrink the neighborhood so that its intersection with $\operatorname{span}_{\mathbb{R}} \tau$ lies in τ; the resulting local orbit lies in L_β. The logarithmic covering transfers this preservation and the fixed angle to the intrinsic flow on the entire fiber. The section is compact and hence closed. For an orbit starting in the section, the set of times at which it lies there is closed and, by local preservation, open. It is nonempty and therefore all of $\mathbb{R}$. Applying the same argument with the angle fixed proves the angular assertion. The constructed complete fiber supplies the flow for every real time. □

T.9 Normal expansion in a fixed normal-coordinate metric

The following argument treats a trivialized normal bundle with its fixed Euclidean metric. It supplies the analytic trapping mechanism in [1, `lem:trapping`], lines 1661–1694. Passing from an embedded invariant submanifold to these coordinates, including the varying-metric terms, is a separate geometric assertion.

Lemma T.83 (A bounded expanding scalar starts at zero)**.** *Let $r : \mathbb{R} \to \mathbb{R}$ be differentiable, and suppose that for $t \geq 0$ one has $0 \leq r(t) \leq B$ and $r'(t) \geq a r(t)$, where $a > 0$. Then $r(0) = 0$.*

Proof. The derivative is nonnegative on $[0, \infty)$, so $r(t) \geq r(0)$ there. If $r(0) > 0$, the derivative is bounded below by the positive constant $ar(0)$. The mean value inequality gives $r(t) \geq r(0) + ar(0)t$ for $t \geq 0$. Taking $t = (B - r(0) + 1)/(ar(0)) > 0$ contradicts $r(t) \leq B$. Nonnegativity rules out $r(0) < 0$, proving the claim. □

Lemma T.84 (The normal derivative gives the radial estimate)**.** *Let E be a real inner-product space, $w : E \to E$ differentiable, $w(0) = 0$, and $v \in E$. Suppose that along the segment from zero to v,*

$$\langle Dw(tv)v, v\rangle \geq a\|v\|^2 \qquad (0 \leq t \leq 1).$$

Then $\langle w(v), v\rangle \geq a\|v\|^2$.

Proof. Differentiate $h(t) = \langle w(tv), v\rangle$. Its derivative is $\langle Dw(tv)v, v\rangle$. The mean value inequality on $[0, 1]$ gives $h(1) - h(0) \geq a\|v\|^2$. Since $w(0) = 0$, this is the conclusion. □

Let X be a topological space, $C \subset X$ compact, and E a finite-dimensional real inner-product space. Let $w : X \times E \to E$ vanish on $X \times \{0\}$. Suppose that its derivatives in the second variable exist and define a continuous function $D : X \times E \to \operatorname{Hom}(E, E)$. Assume $\langle D(c, 0)v, v\rangle > 0$ for $c \in C$ and $v \neq 0$. Then there are $a > 0$ and an open neighborhood U of $C \times \{0\}$, contained in $X \times \{\|v\| < 1\}$, on which

$$\langle w(s, v), v\rangle \geq a\|v\|^2.$$

Proof. The product of C and the unit sphere is compact. The continuous function $\langle D(c, 0)v, v\rangle$ is positive on this product, so it has a uniform lower bound $b > 0$. This statement also holds when the product is empty, by choosing any $b > 0$. For each $c \in C$, continuity and compactness of the

unit sphere give a neighborhood of $(c,0)$ where $\langle D(s,y)v,v\rangle > b/2$ for every unit vector v. Choose inside it a product $V_c \times \{\|y\| < r_c\}$, with V_c open containing c and $0 < r_c \le 1$. Homogeneity extends the bound to all vectors with the factor $\|v\|^2$. Every segment tv with $0 \le t \le 1$ remains in this ball. The radial estimate gives $\langle w(s,v),v\rangle \ge (b/2)\|v\|^2$ throughout the product. Its union over $c \in C$ is the required open neighborhood, with $a = b/2$. □

Under the hypotheses of the preceding theorem, let F be a real flow on $X \times E$ whose normal coordinate satisfies

$$\frac{d}{dt}(F^t(s,v))_E = w(F^t(s,v)).$$

There is an open neighborhood U of $C \times \{0\}$ such that the full forward-time core of U is contained in $X \times \{0\}$.

Proof. Use the constructed normal tube. Along any orbit remaining there, $r(t) = \|(F^t(s,v))_E\|^2$ lies between zero and one, and differentiation gives $r'(t) = 2\langle (F^t(s,v))_E, w(F^t(s,v))\rangle \ge 2ar(t)$. The bounded expanding scalar lemma gives $r(0) = 0$. The flow is the identity at time zero, so $v = 0$, proving the stated containment. □

Lemma T.85 (Native trapping from an actual scalar derivative)**.** *Let F be a complete flow solving a smooth vector field Z on a smooth manifold, and let ρ be smooth. Suppose $0 \le \rho \le B$ on U and $d\rho(Z) \ge a\rho$ there, with $a > 0$. Then the full forward-time core of U is contained in $\{\rho = 0\}$.*

Proof. For a point of this core the orbit stays in U at every nonnegative real time. The chain rule for this very field and flow gives $(\rho \circ F^\bullet)' = d\rho(Z) \circ F^\bullet$. Apply the bounded expanding scalar lemma and evaluate the flow at zero. □

T.10 Trapping in local normal coordinates

The normal expansion argument can be performed on an open coordinate domain. No extension of its normal field or conjugation of the complete flow to a global product is needed. The metric in these coordinates is fixed; construction of the geometric coordinates and the corresponding positive normal linearization remains a separate assertion.

Lemma T.86 (Expansion with derivatives only at forward times)**.** *Let $r : \mathbb{R} \to \mathbb{R}$ have a derivative at every $t \ge 0$. If $0 \le r(t) \le B$ and $r'(t) \ge ar(t)$ for all $t \ge 0$, where $a > 0$, then $r(0) = 0$.*

Proof. The derivative is nonnegative on the interior of $[0,\infty)$, and r is continuous on that interval, so r is nondecreasing there. If $r(0) > 0$, the mean value inequality gives $r(t) \ge r(0) + ar(0)t$ for $t \ge 0$. At $t = (B - r(0) + 1)/(ar(0)) > 0$ this contradicts the upper bound. Nonnegativity gives the other inequality. □

Lemma T.87 (Integration on a single normal segment)**.** *Let E be a real inner-product space, $v \in E$ and $w : E \to E$. Suppose $w(0) = 0$, w is differentiable at each tv, $0 \le t \le 1$, and $\langle Dw(tv)v, v\rangle \ge a\|v\|^2$ there. Then $\langle w(v), v\rangle \ge a\|v\|^2$.*

Proof. The function $f(t) = \langle w(tv), v\rangle$ is continuous on $[0,1]$ and differentiable there, with $f'(t) = \langle Dw(tv)v, v\rangle \ge a\|v\|^2$. The mean value inequality between zero and one gives the result. □

Let X be a topological space, $C \subset X$ compact, and E a finite-dimensional real inner-product space. Let $O \subset X \times E$ be open and contain $C \times \{0\}$. Suppose $w : X \times E \to E$ has normal derivative $D(s,v)$ at every $(s,v) \in O$ and vanishes at $(s,0) \in O$. Assume D is continuous at each $(c,0)$, $c \in C$,

and $\langle D(c,0)v, v\rangle > 0$ whenever $v \neq 0$. There exist $a > 0$ and an open U with $C \times \{0\} \subset U \subset O$ such that, for $(s, v) \in U$,

$$\|v\| < 1, \qquad \langle w(s,v), v\rangle \geq a\|v\|^2.$$

Proof. On the compact product of C and the unit sphere, the continuous positive function $\langle D(c,0)v, v\rangle$ has a uniform lower bound $b > 0$. If that product is empty, any positive b suffices. For each $c \in C$, continuity at $(c, 0)$ and compactness of the unit sphere give a neighborhood where all its values exceed $b/2$. Intersect with O and choose a product neighborhood $V_c \times B(0, r_c)$ inside this intersection with $0 < r_c \leq 1$. For (s, v) in this product, the segment (s, tv) stays in the product. Homogeneity extends the unit-sphere bound to $\langle D(s,tv)v, v\rangle \geq (b/2)\|v\|^2$. The segment lemma therefore gives the same lower bound for $\langle w(s,v), v\rangle$. The union of these product neighborhoods is the desired U, with $a = b/2$. This proof also covers zero-dimensional E. □

Under the hypotheses of the expanding-tube theorem there is an open U with $C \times \{0\} \subset U \subset O$ such that every curve $\gamma : \mathbb{R} \to X \times E$ satisfying

$$\gamma(t) \in U, \qquad \frac{d}{dt}\gamma_E(t) = w(\gamma(t)) \qquad (t \geq 0)$$

has $\gamma_E(0) = 0$.

Proof. Use the tube already constructed and set $r(t) = \|\gamma_E(t)\|^2$. For $t \geq 0$ it is differentiable, lies between zero and one, and satisfies $r'(t) = 2\langle \gamma_E(t), w(\gamma(t))\rangle \geq 2ar(t)$. The forward scalar expansion lemma gives $r(0) = 0$. Positive definiteness of the norm gives $\gamma_E(0) = 0$. □

Lemma T.88 (Linearization of a local metric gradient at a zero)**.** *Let E be a real normed vector space, $G(z)$ a continuous bilinear form depending differentiably on z at c, and $Z : E \to E$ differentiable at c. Suppose $Z(c) = 0$ and $G(z)(Z(z), \cdot) = Df_z$ near c. Then for all $v, w \in E$,*

$$G(c)(DZ_c v, w) = D^2 f_c[v, w].$$

If f is twice continuously differentiable near c, the left side is symmetric in v, w. If also $G(c)$ is symmetric, DZ_c is selfadjoint with respect to that form.

Proof. Differentiate the displayed identity of covectors and evaluate on v, w. The product rule gives the sum $DG_c[v](Z(c), w) + G(c)(DZ_c v, w)$. Its first summand is zero since $Z(c) = 0$, proving the formula. The second derivative of a twice continuously differentiable function is symmetric. Symmetry of $G(c)$ then gives $G(c)(DZ_c v, w) = G(c)(v, DZ_c w)$. □

Lemma T.89 (The actual intrinsic gradient identity at good points)**.** *At a good point of the endpoint fiber, let q be the logarithmic representative of the actual symplectic projection of the ambient Liouville vector onto the fiber. For every fiber tangent v,*

$$d\lambda_\phi(q, Jv) = d\phi(v).$$

Here the potential is smooth, its ambient two-form is nondegenerate, and $J(X, Y) = (-Y, X)$ is the prescribed complex structure.

Proof. At a good point every cutoff is at zero or one and its derivative vanishes. The actual defining differential is therefore the complex linear retained-monomial differential. Thus Jv is tangent whenever v is. The symplectic projection equation and the ambient Liouville equation give $d\lambda_\phi(q, Jv) = \lambda_\phi(Jv)$. In logarithmic coordinates the right side is $\sum_j \partial_j \phi \, v_{u,j} = d\phi(v)$. □

Corollary T.90 (Local coordinate application to a native flow). Let F be a real flow on a topological space M, and let $\kappa : M \to X \times E$ be a coordinate map. Suppose U has the forward-curve trapping property just stated, and that whenever $\kappa(F^t x) \in U$ the normal coordinate satisfies

$$\frac{d}{dt}\kappa_E(F^t x) = w(\kappa(F^t x)).$$

Then every point whose full forward orbit lies in $\kappa^{-1}(U)$ belongs to $\{x : \kappa_E(x) = 0\}$.

Proof. For such a point x, apply the forward-curve trapping property to $\gamma(t) = \kappa(F^t x)$. Both hypotheses hold for every $t \geq 0$. Since $F^0 x = x$, the conclusion is precisely $\kappa_E(x) = 0$. □

Lemma T.91 (Local expansion in a fixed bilinear metric)**.** *Let E be a finite-dimensional real normed space and G a continuous bilinear form on E. Let C be a compact subset of a topological space X, and let $O \subset X \times E$ be open and contain $C \times \{0\}$. Suppose $w : X \times E \to E$ has derivative $D(s, v)$ in the second variable on O, that D is continuous at every $(c, 0)$ with $c \in C$, and that $w(s, 0) = 0$ whenever $(s, 0) \in O$. If $G(D(c, 0)v, v) > 0$ for every $c \in C$ and nonzero v, there exist $a > 0$ and an open neighborhood $U \subset O$ of $C \times \{0\}$ such that $\|v\| < 1$ and $G(w(s, v), v) \geq a\|v\|^2$ on U.*

Proof. The function $G(D(c, 0)v, v)$ is continuous and strictly positive on the compact product of C and the unit sphere. Choose a positive lower bound b there. If that product is empty, any positive b has the required property. Continuity and compactness of the unit sphere give, around each $(c, 0)$, a product neighborhood $V_c \times B(0, r_c) \subset O$, with $0 < r_c \leq 1$, where $G(D(s, y)v, v) \geq b/2$ for unit v. Bilinearity and normalization give the bound $(b/2)\|v\|^2$ for every v. For (s, v) in this product, its entire radial segment is in the product. The derivative of $t \mapsto G(w(s, tv), v)$ is $G(D(s, tv)v, v)$. The mean-value inequality between zero and one, and $w(s, 0) = 0$, prove the desired bound with $a = b/2$. Take the union of these product neighborhoods. This uses the fixed transverse metric of the critical torus; it does not identify it with the pre-existing norm. □

Lemma T.92 (Trapping in a fixed positive bilinear metric)**.** *Suppose additionally that G is symmetric and positive definite. The neighborhood U above can be chosen so that every curve $\gamma(t) = (s(t), v(t))$ remaining in U for $t \geq 0$ and satisfying $v'(t) = w(\gamma(t))$ has $v(0) = 0$.*

Proof. Put $r(t) = G(v(t), v(t))$. Symmetry gives $r'(t) = 2G(w(\gamma(t)), v(t)) \geq 2a\|v(t)\|^2$. The operator norm bound gives $0 \leq r(t) \leq \|G\|\|v(t)\|^2 \leq \|G\|$, so $r'(t) \geq [2a/(\|G\| + 1)]r(t)$. The bounded scalar expansion lemma gives $r(0) = 0$, and positive definiteness implies $v(0) = 0$. □

Lemma T.93 (A common expansion coefficient near a compact set)**.** *Let C be compact in a topological space M, and let $\rho, Q : M \to \mathbb{R}$ with $\rho \geq 0$. If every $c \in C$ has an open neighborhood U_c and a constant $a_c > 0$ with $a_c\rho \leq Q$ on U_c, there are $a > 0$ and an open neighborhood U of C with $a\rho \leq Q$ on U.*

Proof. Choose a finite subcover of the U_c and a positive lower bound a for its finitely many constants. The union of this subcover is open and contains C. At any point of that union choose one covering member; nonnegativity of ρ gives $a\rho \leq a_c\rho \leq Q$. If the cover is empty, use any positive a. This applies to one fixed transverse quadratic scalar and its rate, so local inverse coordinates need not agree on overlaps. □

Lemma T.94 (A single trapping neighborhood from local expansion)**.** *Let $C \subset M$ be compact, G a continuous symmetric positive definite bilinear form on a real normed space E, and $\delta, R : M \to E$.*

Suppose δ is continuous at C and zero there. If every point of C has an open neighborhood with an estimate $a_c\|\delta(x)\|^2 \le G(R(x),\delta(x))$, $a_c > 0$, then there is one open neighborhood U of C such that any curve γ remaining in U for nonnegative time and satisfying $(\delta\circ\gamma)' = R\circ\gamma$ has $\delta(\gamma(0)) = 0$.

Proof. The preceding compactness lemma gives a common positive coefficient on an open neighborhood. Intersect with the interior of $\{\|\delta\| < 1\}$; continuity and vanishing on C ensure that this still contains C. Apply the bounded bilinear expansion argument to $v = \delta\circ\gamma$ and $w = R\circ\gamma$. □

T.11 Product coordinates for the actual transverse defining map

Lemma T.95 (Product inverse preserving a specified submersion)**.** *Let E, F be real normed spaces, with E complete. Let $\delta : E \to F$ be smooth at a, with $\delta(a) = 0$ and actual derivative $A = D\delta_a$. Suppose a specified continuous linear map $L : F \to E$ satisfies $AL = \mathrm{id}$. Put $K = \ker A$ and $P = \mathrm{id} - LA$, regarded as a map into K. There is a local homeomorphism e from a neighborhood of a onto a neighborhood of zero in $K \times F$, with*

$$e(y) = (P(y-a), \delta(y)), \qquad \delta(e^{-1}(s,v)) = v.$$

Its inverse is smooth at zero, its derivative there is $D(e^{-1})_0(s,v) = s + Lv$, and its derivative in the normal variable is exactly L. This includes the case $F = 0$.

Proof. The identity $AL = \mathrm{id}$ gives $AP = 0$, so P takes values in K. The linear map $J : E \to K \times F$, $Jx = (Px, Ax)$, has continuous inverse $J^{-1}(s,v) = s + Lv$: indeed $As = 0$, $PL = 0$, and $Px + LAx = x$. Thus J is a continuous linear equivalence. The map $\kappa(y) = (P(y-a), \delta(y))$ is smooth at a, takes a to zero, and has derivative J. The inverse function theorem constructs the stated local homeomorphism with forward map κ and a smooth inverse at zero. The chain rule for its inverse gives derivative J^{-1}. Applying this derivative to $(0,v)$ gives Lv. Applying the second projection to $e(e^{-1}(s,v)) = (s,v)$ proves the literal defining-map identity throughout the target neighborhood. □

Remark T.96. This construction replaces the tubular coordinates of the cited argument by product coordinates of the actual transverse submersion. It preserves one fixed normal right inverse and uses the literal transverse value as its normal coordinate; the neighborhood expansion estimate is an additional assertion.

Lemma T.97 (Expansion from the derivative transverse to an actual zero set)**.** *Let E be a complete real normed space, F finite-dimensional, and G a continuous bilinear form on F. Suppose $\delta : E \to F$ is smooth at a, $\delta(a) = 0$, and $D\delta_a L = \mathrm{id}$ for a continuous linear map $L : F \to E$. Let $R : E \to F$ be continuously differentiable at a, vanishing near a wherever $\delta = 0$. If $G(DR_aLv, v) > 0$ for every nonzero v, then there are $b > 0$ and an open neighborhood U of a such that $b\|\delta(y)\|^2 \le G(R(y),\delta(y))$ for every $y \in U$.*

Proof. Use the product inverse e^{-1} constructed from δ and L. Set $w(s,v) = R(e^{-1}(s,v))$. On a sufficiently small open neighborhood of zero, w is continuously differentiable, $w(s,0) = 0$, and its normal derivative at zero is DR_aL. Apply the local bilinear expansion lemma with the compact base consisting of the single zero vector. Pull its open neighborhood back by the local homeomorphism e; the identity $\delta(e^{-1}(s,v)) = v$ gives the asserted inequality. □

T.12 Actual native transverse products and nonlinear expansion

Lemma T.98 (Smooth actual defining map and rate in a fiber chart)**.** *Let $j : N \hookrightarrow Y$ be the actual smooth fiber embedding, c its chart at a, and $b = c(a)$. Let a native face Hessian chart have logarithmic lift g with $g(b) = x$ and $\operatorname{cover} \circ g = j \circ c^{-1}$ near b. If Φ is smooth at x_{rad}, the actual defining map*

$$\delta_c(y) = \delta(j(c^{-1}y)), \qquad \delta_c(b) = \delta(\operatorname{cover} x)$$

is smooth at b. If the global potential is smooth and its pulled-back fiber primitive is nondegenerate everywhere, the actual coordinate field Z_c and rate $R_c(y) = D\delta_c(y)Z_c(y)$ are smooth at b.

Proof. The actual conormal transverse map composed with the logarithmic cover is smooth at x. Composing with g and using its covering identity gives smoothness of δ_c at b. Evaluating the same covering identity at b proves the center formula. Smoothness and nondegeneracy of the actual pulled-back primitive imply smoothness of its uniquely determined intrinsic Liouville field in the chart. Differentiating the smooth map δ_c gives a smooth continuous-linear-map-valued function; evaluation on the smooth field Z_c proves smoothness of R_c. □

Lemma T.99 (Product inverse for the actual native defining map)**.** *In the preceding chart suppose $\delta(\operatorname{cover} x) = 0$ and the actual transverse differential has the specified normal right inverse L_c induced from the fixed cover-plane normal lift L. Put $A = D\delta_c(b)$ and $K = \ker A$. There is a local homeomorphism e from a neighborhood of b onto a neighborhood of zero in $K \times F$ such that $e(b) = 0$, $e^{-1}(0) = b$,*

$$(e(y))_2 = \delta_c(y), \qquad \delta_c(e^{-1}(s, v)) = v, \qquad D(e^{-1})_0(0, v) = L_c v.$$

Its inverse is smooth at zero.

Proof. The actual derivative identity for the native defining map and the specified cover-plane normal lift gives $AL_c = \mathrm{id}$. The preceding lemma gives smoothness and the zero center value. Apply the submersion product-coordinate lemma to δ_c, A, L_c. Its forward map has literal second coordinate δ_c, and its inverse has normal derivative L_c. The inverse identity at b gives $e^{-1}(0) = b$. □

Lemma T.100 (Chart transport of the actual native transverse rate)**.** *At a point y in the actual chart target where $\delta \circ j$ is differentiable, the chart rate equals the native rate:*

$$R_c(y) = D(\delta \circ j)_{c^{-1}(y)} Z(c^{-1}(y)).$$

Proof. The chain rule gives $D\delta_c(y) = D(\delta \circ j)_{c^{-1}(y)} D(c^{-1})_y$. By definition $Z_c(y) = Dc_{c^{-1}(y)} Z(c^{-1}(y))$. Differentiating $c^{-1} \circ c = \mathrm{id}$ on the chart source gives $D(c^{-1})_y Dc_{c^{-1}(y)} = \mathrm{id}$. Applying these identities to $Z(c^{-1}(y))$ proves the formula. □

Under the hypotheses of the same-data fixed-transverse-linearization theorem, one defining neighborhood and one cover-plane normal lift L are chosen before all face-torus angles. Their fixed metric G is symmetric positive definite. At every actual native preimage of a point (u, θ) of that torus and every logarithmic representative, the same datum constructs the actual chart and the product inverse of the preceding lemma. Its rate R_c is smooth at the center, and

$$G\big(D(R_c \circ e^{-1})_0(0, z), z\big) > 0 \qquad (z \neq 0).$$

The original defining map is the normal coordinate of every such product.

Proof. Choose the common defining neighborhood and fixed metric-normal lift using the same-data fixed-linearization theorem. For an arbitrary angle and actual native center, its chart has the specified actual end Hessian, weights, and multiplier. The common datum supplies the transverse zero value, smooth gradient, and unnormalized monomial values. The product lemma therefore constructs e using this same lift. The smooth-rate lemma gives smoothness of R_c. By the chain rule and the normal derivative identity, $D(R_c \circ e^{-1})_0(0,z) = DR_c(b)L_c z$. The existing positive actual transverse-linearization identity is precisely strict positivity of the latter vector in the already fixed metric G. □

Lemma T.101 (Native nonlinear expansion on an actual branch neighborhood)**.** *In the actual native chart above, assume finite-dimensional transverse space F and continuous bilinear G, and $G(DR_c(b)L_c z, z) > 0$ for $z \neq 0$. Let $O \subset N$ be open containing a, with $\delta \circ j$ smooth on O, and suppose its actual native rate vanishes at every point of O where $\delta \circ j = 0$. Then some $k > 0$ and open $V \subset O$ containing a satisfy*

$$k\|\delta(j(y))\|^2 \le G\big(D(\delta \circ j)_y Z(y), \delta(j(y))\big) \qquad (y \in V).$$

Proof. Restrict the chart target to points whose inverse lies in O. The rate-transport lemma and the native zero-section equation imply that $R_c = 0$ wherever $\delta_c = 0$ there. The actual defining-map derivative has right inverse L_c by the preceding product construction. Apply the submersion expansion lemma to the smooth δ_c, smooth R_c, and specified G, L_c. It gives a positive k and open chart neighborhood with $k\|\delta_c\|^2 \le G(R_c, \delta_c)$. Pull this neighborhood back to the chart source and intersect with O, then take an open neighborhood of a in this intersection. The chart inverse identity identifies $\delta_c(c(y))$ with $\delta(j(y))$; the rate-transport lemma identifies $R_c(c(y))$ with the actual native rate. Substitution proves the assertion. □

T.13 One trapping neighborhood for a whole critical torus

Lemma T.102 (Compactness of the actual native face torus)**.** *Let $j : N \hookrightarrow Y$ be a topological embedding and suppose $\{u\} \times T_\sigma$ lies in its image. Its inverse image in N is compact. In particular, the whole face torus supplied by one critical normal datum has compact inverse image under every actual full-fiber embedding.*

Proof. The annihilator torus T_σ is closed in the compact quotient torus, hence compact. The product $\{u\} \times T_\sigma$ is compact as well. The embedding identifies its inverse image homeomorphically with this product, because the assumed image inclusion makes the image of that inverse image equal to the whole product. Compactness follows. For the critical datum, the fiber equation at every annihilator angle and the prescribed full image of the embedding give that image inclusion. □

Lemma T.103 (The defining map along an intrinsic integral curve)**.** *Let W be the constructed defining neighborhood with defining map δ, let $j : N \hookrightarrow Y$ be a smooth embedding, and let Z be the intrinsic Liouville vector field for a one-form on N. The map $\delta \circ j$ is smooth at every point whose image is in W. For an intrinsic integral curve γ with $j\gamma(t) \in W$,*

$$\frac{d}{dt}\delta(j\gamma(t)) = D(\delta \circ j)_{\gamma(t)} Z(\gamma(t)).$$

Proof. Smoothness follows by composing the defining map's smoothness on W with the smooth embedding. Apply the chain rule to this composition and the integral-curve equation $\gamma' = Z \circ \gamma$. Its differential is the intrinsic transverse rate by its definition. □

Let the smooth global potential have uniform positive lower and finite upper Hessian bounds and agree outside a positive radius with the strongly compatible homogeneous end potential, which is globally C^1. There is a positive threshold, independent of the face, the chosen critical datum, and the embedded full-fiber presentation, such that the following holds for every endpoint parameter at or above it. Fix a nonempty proper face, one constructed critical face datum at u, any actual smooth embedded presentation $j : N \hookrightarrow Y$ of the whole endpoint fiber, and any $v_0 \in s$. There are a defining neighborhood W from the same datum and one open native neighborhood V containing the entire native critical torus, with $j(V) \subset W$, such that every intrinsic forward integral curve remaining in V starts in the same smooth conormal branch and in the literal core section Γ_β. Consequently, for every complete intrinsic Liouville flow F,

$$\mathrm{Core}(F, V) \subset j^{-1}(\Gamma_\beta).$$

Proof. Use the fixed transverse quotient and metric produced from the same critical datum. Intersect the branch neighborhood with the datum's radial plateau and the positive phase neighborhood. The actual transverse rate vanishes on $\delta = 0$ there, by the native conormal vanishing theorem and the plateau cutoff identities. At every point of the face torus, the same datum supplies the native Hessian chart and the positive rate derivative in the same fixed metric. The actual native product-coordinate expansion theorem therefore gives an open neighborhood with positive lower bound for that same transverse quadratic rate. The native critical torus is compact by the preceding lemma. A finite subcover and the minimum of its positive coefficients give one positive coefficient on their union; each coefficient may be decreased since the squared transverse norm is nonnegative. Intersect with the open set where that norm is less than one and with the common branch neighborhood.

For a forward intrinsic curve γ in this intersection, put $v(t) = \delta(j\gamma(t))$ and $b(t) = D(\delta \circ j)Z(\gamma(t))$. The native orbit equation gives $v' = b$. Symmetry of the same fixed metric G gives $(G(v, v))' = 2G(b, v)$. Its operator bound and the local estimate give a positive scalar expansion coefficient, while $\|v(t)\| < 1$ bounds $G(v(t), v(t))$. The bounded scalar expansion lemma therefore forces $G(v(0), v(0)) = 0$. Positive definiteness gives $v(0) = 0$. The exact zero-set equivalence in the common branch neighborhood now places $j\gamma(0)$ in the actual smooth conormal branch.

The defining neighborhood's normalizing functional takes value one on $\nabla\phi_\infty(u)$, so this gradient is nonzero. Its continuity at u, supplied by the datum, shows that the interior of $\{a : \nabla\phi_\infty(j(a)_1) \neq 0\}$ contains the entire native torus. Intersect V with this open set. The zero-set equivalence now gives a nonzero gradient in $\mathrm{pos}(s)$ and an angle in its annihilator torus. The cone cannot be $\{0\}$, since it contains that nonzero gradient. The native full-fiber equation supplies the remaining condition defining Γ_β. For a point of the flow core, its forward orbit remains in V and is an intrinsic integral curve; apply the assertion and evaluate the orbit at time zero. □

Lemma T.104 (Normal expansion forces local trapping)**.** *Let X be a smooth manifold, Z a smooth vector field, S an embedded smooth submanifold locally invariant under its flow, and $C \subset S$ a compact set of zeros of Z. Let g be a smooth Riemannian metric. Suppose that, at every $c \in C$, the linearization DZ_c is g_c-selfadjoint and is positive definite on $(T_cS)^{\perp_{g_c}}$. Then there is a neighborhood U of C such that every forward orbit remaining in U for all nonnegative time starts in S.*

Proof. If S has codimension zero near C, choose $U \subset S$. Otherwise use a smooth normal tubular neighborhood over a relatively compact portion of S around C, with coordinates (s, v) and $\rho(s, v) = g_s(v, v)$. Choose a tube of bounded radius, so ρ is bounded. Write $Q_Z = Z\rho$ in these coordinates. Tangency to S gives, for every base point s,

$$Q_Z(s, 0) = 0, \qquad D_v Q_Z(s, 0) = 0.$$

At $c \in C$, invariance and selfadjointness preserve the normal subspace, and direct Taylor expansion gives

$$D_v^2 Q_Z(c,0)[v,v] = 4g_c(DZ_c v, v).$$

Terms from differentiating the varying normal metric contain $Z(c) = 0$. Positivity on the compact unit normal bundle over C, continuity, and finitely many bundle charts give $a > 0$ and a smaller tube such that

$$D_v^2 Q_Z(s,tv)[v,v] \geq 2ag_s(v,v) \qquad (0 \leq t \leq 1).$$

This step controls the varying-metric terms at nearby s as well; no equality of the last Hessian with its value at c is asserted there. Taylor's integral remainder now gives

$$Z\rho(s,v) = \int_0^1 (1-t) D_v^2 Q_Z(s,tv)[v,v]\, dt \geq a\rho(s,v).$$

Along a forward orbit staying in this tube, $\rho(t) \geq e^{at}\rho(0)$. Boundedness of ρ forces $\rho(0) = 0$, which means the starting point lies in S. □

Lemma T.105 (Trapping at each critical torus)**.** *For each $C_{F,\beta}$ there is a neighborhood in H_β such that any forward intrinsic orbit remaining there belongs to the smooth σ_F branch of L_β.*

Proof. In the genuine holomorphic neighborhood (29), use the branch S given by $p \in \operatorname{relint}\sigma_F$, $\theta \in T_{\sigma_F}$, intersected with the fiber. This is one smooth submanifold near the compact torus, not the full singular union of cone pieces. It is invariant by Lemma T.82. Its radial tangent at the torus is

$$\{X : m \cdot X = 0,\ HX \in \operatorname{span}_{\mathbb{R}}(v_1, \ldots, v_k)\} = D_F,$$

and its angular tangent is E_F. The induced Kähler metric is H on each radial and angular block. The orthogonal block decomposition of Proposition S.109 therefore identifies TS with the negative and zero eigenspaces of the gradient linearization, and its normal with the positive eigenspace of dimension $n-1$. The linearization is selfadjoint since Z_H is the gradient of the restricted potential on this neighborhood. Lemma T.104 applies. When $n = 1$, the fiber is zero-dimensional and its codimension-zero clause applies. □

T.14 Lyapunov values on forward limit sets

Lemma T.106 (A monotone orbit has constant potential on its limit set)**.** *Let φ be a continuous real flow on a topological space X, and let $V : X \to \mathbb{R}$ be continuous. Suppose $t \mapsto V(\varphi_t(x))$ is nondecreasing and bounded above. Every point y of the forward omega-limit set of x satisfies*

$$V(y) = \sup_{t \in \mathbb{R}} V(\varphi_t(x)).$$

Proof. Write L for the displayed supremum. The entire orbit lies in the closed set $\{V \leq L\}$, so every forward limit point lies there. For each real s, the orbit tail with $t \geq s$ lies in the closed set $\{V \geq V(\varphi_s(x))\}$ by monotonicity. A forward limit point lies in the closure of each such tail, hence $V(y) \geq V(\varphi_s(x))$ for every s. Taking the supremum proves the reverse inequality. □

Lemma T.107 (Vanishing Lyapunov derivative on the limit set)**.** *Let X be Hausdorff, let φ be a continuous real flow, and let $V : X \to \mathbb{R}$ be continuous. Suppose a function $d : X \to \mathbb{R}$ satisfies*

$$\frac{d}{dt} V(\varphi_t(y)) = d(\varphi_t(y)) \geq 0 \quad (t \in \mathbb{R},\ y \in X).$$

If the forward orbit of x has compact closure, its forward omega-limit set is nonempty and is contained in $\{d = 0\}$.

Proof. The derivative inequality makes every orbit potential nondecreasing. Compactness and continuity bound V above on the forward orbit closure; at negative times monotonicity bounds it above by $V(x)$. Thus the preceding lemma makes V constant on the omega-limit set. This set is flow invariant: translation by any fixed real time preserves the filter of future tails. Continuity of each time map sends the closure of a tail into the closure of its translated tail; intersecting these closures proves invariance. Consequently $t \mapsto V(\varphi_t(y))$ is constant for each point y in that set. Its derivative at zero is $d(y)$ and also zero. Finally the closures of future tails form a nested family of nonempty compact subsets of the forward orbit closure, so their intersection, the omega-limit set, is nonempty. □

Lemma T.108 (Forward limit points of a strict smooth Lyapunov flow)**.** *Let Z be a smooth complete vector field on a smooth manifold X, and let V be smooth with $dV(Z) > 0$ wherever $Z \neq 0$. Every relatively compact forward orbit has a nonempty forward omega-limit set contained in the zero set of Z.*

Proof. The chain rule along the actual integral curves gives $\frac{d}{dt}V(\varphi_t(y)) = dV_{\varphi_t(y)}(Z_{\varphi_t(y)})$. This derivative is positive at nonzeros and zero at zeros by linearity, so is nonnegative everywhere. The preceding lemma makes it zero on the nonempty omega-limit set. Strict positivity there would contradict this vanishing unless the vector field itself is zero. □

Lemma T.109 (Connected compact forward limit set)**.** *Let φ be a continuous real flow on a Hausdorff space and suppose the forward orbit of x has compact closure. Its omega-limit set is nonempty, compact, and connected. For every open set containing that limit set, all sufficiently late points of the orbit lie in the open set.*

Proof. The omega-limit set is the intersection of the closures of the forward tails. These sets are nonempty, closed, nested, and contained in the compact forward-orbit closure, so their intersection is nonempty and compact. If an open neighborhood failed to contain a whole tail, compactness would give a cluster point of arbitrarily late points in its closed complement. That point would belong to the omega-limit set, a contradiction.

To prove connectedness, suppose two disjoint closed sets cover the limit set. Their intersections with it are disjoint compact sets, so have disjoint open neighborhoods. Their union contains the limit set and therefore contains a whole forward tail. The tail is a continuous image of a real half-line, hence is connected and lies in one of these two open neighborhoods. Its closure contains the limit set and misses the other open neighborhood. Consequently the limit set lies in just one member of the original closed cover. This is the closed-cover criterion for connectedness. □

Lemma T.110 (Forward limits of the actual tailored fiber)**.** *Fix a strongly compatible homogeneous end potential with positive quadratic lower bound, global C^1 regularity and quantitative strict convexity, and a smooth extension with uniform positive lower and finite upper Hessian bounds. For all sufficiently large β, on every smooth embedded presentation of the whole tailored fiber, each relatively compact orbit of its actual complete intrinsic Liouville flow has omega-limit set contained in one of the literal critical face tori. Every open neighborhood of that torus in the fiber contains the entire sufficiently late forward orbit.*

Proof. Use one threshold for the exhaustive critical-torus classification and the strict intrinsic Lyapunov theorem for the same potential. The latter places the nonempty omega-limit set in the intrinsic zero locus. Classification identifies that zero locus with the finite disjoint union of the embedded face tori. Each is closed: its ambient expression is a singleton radial coordinate times a closed annihilator torus, pulled back by the continuous fiber embedding. Pullback preserves

disjointness. The preceding finite-piece lemma therefore selects one torus and gives eventual neighborhood containment. Completeness refers to the flow of this very pulled-back primitive. □

Lemma T.111 (One closed critical piece captures a forward limit set)**.** *Suppose the omega-limit set above is contained in a finite union of pairwise disjoint closed sets C_i. Then it is contained in one C_i, and every open neighborhood of that C_i contains the entire sufficiently late forward orbit.*

Proof. Choose a limit point and a member C_i containing it. The union of the other members is closed and disjoint from C_i. Connectedness of the limit set forces it to lie in one side of this closed cover, and the chosen point excludes the other side. Apply eventual neighborhood containment from the preceding lemma. □

T.15 Identical critical radii for trapping and omega limits

Lemma T.112 (Retained support of the specified critical datum)**.** *Let critical normal data be given at the specified radial point u and face $s \subset \mathrm{Vert}(Q)$. The actual retained vertex set at u is exactly s.*

Proof. The specified plateau contains u and its cutoff is one on $\{0\} \cup s$ and zero elsewhere. A retained vertex has nonzero cutoff. Every vertex is nonzero, so a vertex outside s has cutoff zero and is not retained. Conversely every element of s is a vertex with cutoff one, hence retained. These two inclusions give equality. □

Lemma T.113 (Identification with the exhaustive classifier on the same face)**.** *Let the global potential be smooth with a uniform positive Hessian lower bound, and let its actual plane pairing be positive on the full endpoint fiber. Let $j : N \hookrightarrow Y$ present that full fiber. For every nonempty proper face t, suppose a radial point r_t has retained support t, and every native critical point lies in one of the tori $\{r_t\} \times T_{\mathrm{pos}(t)}$. If critical normal data exist at a specified u for face s, then $u = r_s$.*

Proof. The zero quotient angle belongs to the face torus, so the data put $(u, 0)$ in the actual endpoint fiber. The full image condition gives a native preimage a. Choose a logarithmic representative and construct the native Hessian chart from the same data. Its actual restricted potential derivative vanishes at a, so a is a native critical point. The assumed exhaustive classifier therefore places $j(a)$ in $\{r_t\} \times T_{\mathrm{pos}(t)}$ for some t, giving $u = r_t$. The preceding lemma says the retained support at u is s, while the classifier support at r_t is t. Equality of the radial points gives $s = t$, and hence $u = r_s$. The argument uses actual criticality and exact supports, so it does not require uniqueness of a separately chosen local implicit branch. □

Assume the strongly compatible end potential is globally C^1, admits a positive quadratic lower bound, and becomes convex after subtraction of a positive multiple of the squared norm. Let the smooth global potential have uniform positive lower and finite upper Hessian bounds and agree with the end potential outside a fixed radius. There are radial functions $r_s(\beta)$ for every nonempty proper face and one positive threshold such that, for every parameter above it, their literal face tori lie in the endpoint fiber, have their prescribed retained supports, and are pairwise disjoint.

For every actual full-fiber presentation, its native critical or Liouville-zero locus is exactly the union of these tori. Every critical normal datum at a specified u for face s satisfies $u = r_s(\beta)$. Moreover every complete intrinsic forward orbit with compact closure has its omega-limit set in one such torus and eventually lies in every open neighborhood of that entire native torus. These assertions use the same functions r_s.

Proof. Choose the radial functions and threshold from the exhaustive critical face-torus classification. Increase the threshold to satisfy the uniform plane-pairing positivity and strict Lyapunov results. The classifier gives the stated fiber membership, supports, disjointness, and native critical-or-zero characterization. For an arbitrary constructed datum, apply the preceding identification lemma to this very classifier. This proves $u = r_s(\beta)$ without making a second choice of radial functions.

For a complete intrinsic flow and orbit with compact closure, the same-potential Lyapunov theorem says the smooth restricted potential has positive derivative wherever the actual intrinsic field is nonzero. The omega-limit Lyapunov theorem therefore places the omega-limit set in the actual zero locus. The classifier covers that locus by the finitely many native face tori. Each such torus is closed, because it is the preimage under the continuous embedding of a radial singleton times a closed cone-annihilator torus, and they are pairwise disjoint by the classifier. The connected omega-limit theorem for compact forward orbits places the whole nonempty limit set in one torus and proves eventual membership in every open neighborhood of that torus. All tori in this application have the already fixed classifier radii, which were just identified with every constructed critical-normal datum. □

T.16 From critical-torus trapping to the actual full core

Let the strongly compatible end potential and its smooth strongly convex extension satisfy the preceding quantitative Hessian and end-agreement hypotheses. For all sufficiently large β, every actual native presentation of the entire endpoint fiber, with its complete intrinsic Liouville flow, satisfies

$$\{x : \overline{\{\varphi_t(x) : t \geq 0\}} \text{ is compact}\} = j^{-1}(L_\beta).$$

Proof. There are finitely many nonempty proper faces. For each, the reduced branch construction supplies its actual critical normal data for all sufficiently large β; take one threshold for all faces. Enlarge it to the thresholds for classification, strict Lyapunov increase, trapping, and two-sided conormal invariance. For a relatively compact forward orbit, its omega-limit lies in one classified critical torus and the orbit eventually lies in every open neighborhood of that torus. The critical normal data for the same face have the same radius: their actual critical point lies in a classified torus, and equality of the retained supports forces equality of the faces and then of the radii. Hence the constructed trapping neighborhood applies to this very torus. Choose a late time T after which the orbit stays in that neighborhood. Applying trapping to the orbit starting at $\varphi_T(x)$ places this point in L_β. Two-sided invariance, at time $-T$, places x in L_β. Conversely the conormal section is compact and invariant, so every forward orbit starting there has compact closure. □

T.17 The literal full endpoint core from finite input

For every finite toric input there are a strongly compatible, degree-two homogeneous end potential ϕ_{end}, continuously differentiable everywhere and smooth off the origin, and a smooth potential ϕ with uniform positive lower and finite upper Hessian bounds. The latter is a positive quadratic plus 1 near the origin and agrees with ϕ_{end} outside a fixed compact set. Their coordinate gradients are homeomorphisms, smooth with smooth inverse on the respective smooth loci. There are $F, B > 0$ such that, for every $\beta \geq B$, the entire endpoint fiber has an actual complete native presentation $j : N \hookrightarrow Y$ with primitive $j^*\lambda_\phi$ and flow φ. Its forward-orbit criterion is

$$\overline{\{\varphi_t(p) : t \geq 0\}} \text{ compact} \quad \Longleftrightarrow \quad j(p) \in L_\beta.$$

For every $A \geq F^2\beta^2/2$, the actual truncation $D_A = \{\phi \circ j \leq A\}$ is a compact outward Liouville domain, with literal image $\{f_{\beta,1} = 1,\ \phi \leq A\}$ and at least one completion. Every chosen completion C of D_A has its inclusion marking onto the same entire N and satisfies

$$j(\operatorname{Core} C) = L_\beta, \qquad \operatorname{Rad}_{\phi_{\mathrm{end}}}(j(\operatorname{Core} C)) = \operatorname{Link}(\Lambda_\Sigma).$$

The section L_β is compact and its stated radial projection is bijective onto this literal link.

Proof. Construct the strongly compatible end with a fixed positive convexity correction $d|u|^2$. The smooth convex extension construction retains that end, a positive Hessian lower bound, and the prescribed quadratic germ. Homogeneity bounds the end Hessian on the unit sphere; compactness bounds the extension Hessian on the remaining compact ball. This gives one finite upper Hessian bound. The positive convexity correction gives the global Legendre homeomorphism of the end, smooth with smooth inverse off the origin; the extension has its global smooth Legendre inverse. The Euclidean quadratic lower bound also bounds the square of the coordinate maximum norm, since that norm is no larger than the Euclidean norm.

Apply the actual full-core forward-orbit criterion with these constructed potentials. Apply the full-fiber truncation-marking theorem to the same pair of potentials, obtaining $F > 0$. Increase B to all thresholds for that marking theorem, the forward-orbit criterion, compactness of L_β, and the literal radial bijection. For each $\beta \geq B$, choose the complete entire endpoint fiber supplied by the marking theorem. Its image is exactly $f_{\beta,1}^{-1}(1)$.

For each admissible A, positive intrinsic derivative on $\phi \circ j \geq A$ gives the stated actual outward truncation, its nonempty completion space, and an inclusion marking for every completion. The core characterization under that marking identifies the core with the points of the whole fiber whose forward orbits have compact closure. The already proved forward-orbit criterion makes its embedded image a subset of L_β. Conversely every point of L_β lies on the literal endpoint fiber, hence has a native preimage; that criterion places the preimage in the core. Thus the displayed embedded equality is literal. Apply the already proved radial-image equality to obtain the second equality. The choices of potentials and F, B precede β, and the choices of the full fiber and primitive precede both A and the arbitrary completion. □

T.17.1 The full core on the retained marked ribbon

Lemma T.114 (The retained outer boundary is the literal level)**.** *Suppose a retained outer completion family has compact carrier image $\{\rho \leq 0\}$ under its native ambient embedding, where ρ is smooth with regular zero level. Its actual boundary image is exactly $\{\rho = 0\}$.*

Proof. The ambient embedding has equal source and target dimensions, so the inverse function theorem makes it open. It maps the compact carrier's frontier onto the frontier of its image: openness gives the local frontier identity, and compactness prevents additional frontier points outside its open image. The compact-domain boundary parametrizes that frontier. Finally the regular-level theorem identifies the frontier of the closed sublevel with the zero level, proving the asserted literal image equation. □

Lemma T.115 (Full-fiber markings preserve the embedded full core)**.** *Let a compact ribbon have an inclusion completion marking onto an embedded complete fiber H, with the primitive induced from the ambient potential. Its embedded full core is the image in the logarithmic ambient of the points of H with relatively compact forward orbit.*

Proof. The inclusion marking intertwines the complete Liouville flows. The full core of a compact domain consists exactly of the points with relatively compact forward orbit in its completion. Indeed an orbit leaving the compact domain enters a cylindrical end and has unbounded end coordinate. Apply the marking to this equality, then the fiber embedding. The marking agrees with the compact ribbon embedding on the entire compact carrier, so the resulting image is precisely the stated embedded full core. □

Lemma T.116 (Changing a full-fiber presentation)**.** *Two smooth embedded presentations of the same full fiber, equipped with the pullback of the same ambient primitive and complete Liouville flows, have the same embedded set of relatively compact forward orbits.*

Proof. Factor each embedding through the other. The induced maps are smooth because both presentations are embedded submanifolds. Injectivity of the ambient embeddings makes these maps inverse, giving a diffeomorphism. Pullback composition shows that it preserves the primitives. Consequently it intertwines the Liouville flows and maps each forward orbit onto the corresponding forward orbit. A homeomorphism commutes with closure and preserves compactness in both directions. Applying the equal ambient embeddings gives the asserted equality. □

Finite toric input supplies one strongly compatible end potential and smooth uniformly convex extension, one ambient bound, convenient inner and outer radius constants, and a threshold in β. For every larger β and every supplied original coefficient above its literal threshold, there is a common admissible rounding and an actual marked five-arrow prefix ending at the tailored pair whose full embedded ribbon core is exactly L_β for that same potential. The original ratio bounds and a valid fiber end bound below the convenient inner level hold throughout this range.

Proof. First construct the potentials and full-fiber core classification from Theorem T.17. Apply the original marked-prefix construction to these very potentials, using their smoothness, common end and uniform Hessian bounds. If F is its lower radius constant, take $C_- = F+1$ and $C_+ = F+2$. Enlarge the parameter threshold to the maximum of the core and prefix thresholds. The convenient levels now strictly exceed the original fiber bound and have positive separation, so the common-width result gives an admissible profile. Choose the normalized coefficient as the maximum of the tailoring threshold and the supplied original coefficient times $e^{2\beta}$. This constructs the actual marked prefix.

Its terminal ribbon marking and the full-fiber core theorem may present the same complete fiber by different manifolds. Both have exactly the fiber $f_1 = 1$ as their ambient image and pull back the identical potential primitive. The preceding presentation lemma transfers the proved orbit classification to the terminal marking. The inclusion-marking core lemma then identifies the full core of this actual terminal compact ribbon with L_β. Thus no identification of the separately chosen potential witnesses is needed. □

For the strongly compatible end potential and smooth strongly convex extension constructed from the finite toric input, there are $F, B > 0$ such that for every $\beta \geq B$,

$$\mathrm{Sk}(H_\beta, \lambda_\phi|_{H_\beta}) = L_\beta$$

as embedded subsets. More precisely, every constructed full-fiber truncation at level $A \geq F^2\beta^2/2$ admits a completion, and every such completion retains an inclusion marking whose embedded full core is L_β. Its radial image under the Legendre map is exactly the full FLTZ link.

Proof. Construct the potentials and marked full-fiber truncations by Theorem T.17. Compact two-sided invariance implies $L_\beta \subset \mathrm{Sk}(H_\beta)$. Conversely a core point has a relatively compact forward orbit. Its omega-limit set is nonempty, compact, connected and invariant. The connectedness assertion follows by intersecting the nested compact closures of connected orbit tails. The proper Lyapunov function of Proposition S.45 is increasing and bounded on the orbit, so it is constant on the limit set. That set lies in the critical locus, hence in one of the finitely many disjoint critical tori. The distance to this torus tends to zero. A sufficiently late portion of the orbit stays in its trapping neighborhood for all future time, so the actual whole-torus trapping theorem places it in the relevant smooth branch of L_β. Backward invariance puts the original point in L_β. No distinct-critical-value or Morse–Smale assertion is required. □

U Removing the base term with embedded full ribbons

Use the convenient potentials and radii constructed from the finite toric input in Theorem U.6. Fix any β above its common threshold and any original coefficient above its literal threshold, and let K_* be the normalized coefficient of the resulting retained marked prefix. Consider

$$\lambda_{1,K} = \lambda_\phi + K f_1^* \lambda_{\mathbb{C}}, \qquad 0 \le K \le K_*.$$

Every one makes D_β a Liouville domain. Its restriction to H_β is always exactly λ_ϕ.

U.1 Actual conormal differential on the ambient cone band

Lemma U.1 (The auxiliary cutoff is flat on the norm-one and larger region)**.** *For every real β and every point y of the logarithmic ambient space with $|f_1(y)| \ge 1$, the actual auxiliary cutoff is $\eta(y) = 1$. Its differential vanishes there. On any logarithmic representative x of y, the actual endpoint differential therefore equals the differential of the defining equation $\mathcal{E}_\beta$:*

$$D(f_1 \circ \mathrm{cover})_x = D(\mathcal{E}_\beta \circ \mathrm{cover})_x.$$

This holds also when $|f_1(y)| = 1$.

Proof. If $\eta(y) \ne 1$, the normalized-family estimate gives $|f_1(y)| < 1$, contradicting the hypothesis. Since η is smooth, bounded above by one, and takes that value at y, its derivative on the logarithmic cover vanishes at that point. Write the actual family as $f_1 = O + \eta C$, where O is the original normalized polynomial and $C = 1 + \mathcal{E}_\beta - O$. The product rule, $\eta = 1$ and $D\eta = 0$ give $Df_1 = DO + DC = D\mathcal{E}_\beta$. □

Lemma U.2 (Conormal differential and imaginary Hamiltonian on the band)**.** *Let σ be a subset of momentum space, let $x = (u, \vartheta)$ be a logarithmic representative, and suppose $|f_1(\mathrm{cover}(x))| \ge 1$, its angle is in T_σ, and $\chi_v(u) = 0$ for every vertex $v \notin \sigma$. Write*

$$G = \sum_{v \in \mathrm{Vert}(P)} \chi_v(u) e^{\langle v, u\rangle - \beta} v.$$

The actual endpoint is real and equals $\mathcal{R}_\beta(u)$, and its full differential, including transition-cutoff terms, is

$$D(f_1 \circ \mathrm{cover})_x(w_u, w_\vartheta) = D\mathcal{R}_\beta(u) w_u + i\langle G, w_\vartheta\rangle.$$

For an invertible Hessian operator H, the corresponding cover Hamiltonian of the imaginary differential is $(H^{-1}G, 0)$.

Proof. The auxiliary cutoff result identifies the differential with that of $\mathcal{E}_\beta$. The character of every vertex in σ is one. Each outside cutoff is zero and has zero derivative because it is nonnegative and takes its minimum there. Differentiating the finite sum leaves the stated real radial derivative and imaginary angular block. The identity $f_1 - 1 = \mathcal{E}_\beta$ and the real radial equation give $f_1 = \mathcal{R}_\beta$. The imaginary covector has zero radial coefficient and angular coefficient G; the inverse-Hessian Hamiltonian formula therefore gives $(H^{-1}G, 0)$. □

Lemma U.3 (Actual ambient rank-two field and its momentum-span direction)**.** *Keep the preceding pointwise norm, annihilator and cutoff assumptions. Let ϕ be globally smooth. At u suppose its actual Euclidean Hessian is an invertible operator H, and suppose its actual coordinate Hessian satisfies Euler's identity*

$$D(\nabla\phi)_u u = \nabla\phi(u).$$

For any real K, let π_f be the actual potential-form Poisson scalar and assume $1 + K\pi_f \neq 0$. Put

$$p = \nabla\phi(u), \qquad a_f = D\mathcal{R}_\beta(u)u, \qquad \kappa = \frac{K}{1 + K\pi_f}\left(\frac{\operatorname{Re} f_1(y)}{2} - a_f\right).$$

The native Liouville vector of the actual primitive $\lambda_\phi + Kf_1^*\lambda_{\mathbb{C}}$ is the covering derivative of

$$(u + \kappa H^{-1}G, 0).$$

Its actual Legendre differential is the cotangent-cover derivative of $(0, p+\kappa G)$. If additionally $p \in \sigma$, then $p + \kappa G \in \operatorname{span}_{\mathbb{R}} \sigma$. These conclusions assert pointwise angular vanishing and span tangency; they do not assert positive-cone preservation by the ambient flow.

Proof. The actual Hessian's invertibility gives nondegeneracy of the native potential form. Euler's identity then identifies its Liouville vector with the covering derivative of $(u, 0)$. The preceding differential calculation gives $\operatorname{Im} f_1 = 0$, $d(\operatorname{Im} f_1)(Z_0) = 0$, $d(\operatorname{Re} f_1)(Z_0) = a_f$, and $X_{\operatorname{Im} f_1} = \operatorname{cover}_*(H^{-1}G, 0)$. The actual native and cover Poisson scalars agree. Substitute these identities into the proved native rank-two Liouville formula: its real Hamiltonian term vanishes and its imaginary Hamiltonian term has coefficient κ, giving the displayed vector.

The Legendre differential applies the actual Hessian to the radial component and retains the angular component. Applying it to this vector and using $Hu = p$ and $HH^{-1}G = G$ gives $(0, p+\kappa G)$ in the stated cotangent order. Every nonzero summand of G is a real multiple of a vertex in σ, so G lies in its span. The additional hypothesis $p \in \sigma$ places p in that span as well, which is closed under addition and real scalar multiplication. □

U.2 Ambient cone preservation on the norm-one and larger region

Lemma U.4 (A neighborhood with constant auxiliary cutoff)**.** *If the actual endpoint satisfies $|f_1(y)| \geq 1$, then $\eta = 1$ on a neighborhood of y in logarithmic representative coordinates.*

Proof. Put $S = (\sum_v |m_v|^2)^{1/2}$ and $c = 1/(8(|V| + 1))$. If $S \leq 2c$, every nonconstant monomial has magnitude less than one, so the constant cutoff is one. The finite weighted-sum formula for f_1 has coefficients between zero and one. Hence $|f_1| \leq |V|S \leq 2|V|c < 1$, a contradiction. Thus $S > 2c$. The function S is continuous, so the same strict inequality holds on a neighborhood, where the defining scalar transition equals one. □

Lemma U.5 (The support height on the entire norm-one region)**.** *At every actual endpoint point with $|f_1(y)| \geq 1$, $h(u) \geq \beta - \log|V|$.*

Proof. Write $M = e^{h(u)-\beta}$. If $M \geq 1$, then $M \geq 1/|V|$. Otherwise every nonconstant weight is at most zero, so the constant cutoff is one. The auxiliary cutoff is one by the norm hypothesis, and f_1 is the sum of retained monomials. Each retained monomial has norm at most M, hence $1 \leq |f_1| \leq |V|M$. In both cases $M \geq 1/|V|$; taking logarithms gives the assertion. □

Lemma U.6 (The actual ambient direction throughout the cutoff-one locus)**.** *Let the auxiliary cutoff be one at a conormal point with angle in T_σ and all off-cone cutoffs zero. For a smooth potential whose actual Hessian there is an invertible operator H, assume the Euler identity $Hu = p = \nabla\phi(u)$. For any real K with $1 + K\pi_f \neq 0$, put $\kappa = K(A/2 - a_f)/(1 + K\pi_f)$. The actual native ambient vector is the covering derivative of $(u + \kappa H^{-1}G, 0)$; its actual Legendre derivative is the cotangent-cover derivative of $(0, p + \kappa G)$. If $p \in \operatorname{span}\sigma$, then $p + \kappa G \in \operatorname{span}\sigma$. No lower bound on $|f_1|$ is needed for this cutoff-one statement.*

Proof. The auxiliary cutoff takes its maximum one, so its differential is zero. The endpoint's product formula therefore gives the same differential as its defining equation. At a zero nonnegative off-cone cutoff its derivative is zero as well. The surviving characters are one, giving $df_1(w_u, w_\theta) = d\mathcal{R}_\beta(w_u) + i\langle G, w_\theta\rangle$ and a real endpoint value. Thus $X_{\operatorname{Im} f_1} = (H^{-1}G, 0)$, while $d(\operatorname{Im} f_1)(Z_0) = \operatorname{Im} f_1 = 0$ and $d(\operatorname{Re} f_1)(Z_0) = a_f$. The native rank-two formula gives the stated vector. Apply the actual Hessian to its radial component: $H(u + \kappa H^{-1}G) = p + \kappa G$. Each retained summand of G is in the face span, so addition and real scaling prove span membership. □

Lemma U.7 (The full-span branch is locally invariant)**.** *For a strongly compatible homogeneous end potential and a smooth uniformly convex extension with uniformly bounded Hessian agreeing with it outside a ball, sufficiently large β has the following property, for every $K \geq 0$. At every point of the actual conic set with $|f_1| \geq 1$, every integral curve of the native Liouville field remains in that conic set for sufficiently small positive and negative times, and its angle remains fixed. If its initial momentum belongs to a specified fan cone and its angle annihilates that cone, its momentum remains in that same cone locally.*

Proof. Choose the minimal face containing the nonzero momentum of the point with strictly positive face coefficients. A neighborhood in its full span is contained in the positive cone. In Legendre coordinates use the branch $q \mapsto (\Phi_\infty^{-1}(q), \theta_0)$, with q in that span, and retract by applying Φ_∞ to the radial coordinate and then a linear projection onto the span. Their composition is the identity. The norm lower bound puts the point beyond the extension radius and above the exact-exclusion height. The off-face cutoffs therefore vanish on an ambient neighborhood. The preceding lemma supplies a neighborhood where $\eta = 1$. On the branch the actual ambient differential calculation gives $\dot\theta = 0$ and $\dot p = p + \kappa G$, with G in the face span. Thus the native field is tangent to the branch near the point. Differentiating the retraction constructs a smooth restricted field. Its local integral curve, composed with the branch map, solves the ambient equation and has the same initial value. Local uniqueness identifies it with the given ambient curve. Continuity keeps the restricted curve in the relative positive neighborhood and away from zero. The fixed branch angle proves the angle assertion. □

Lemma U.8 (Cone preservation throughout a connected time band)**.** *In the same regime, an ambient integral curve on a connected time interval whose actual endpoint norm is at least one remains in the actual conic set and retains its initial angle if it starts there. Every specified starting cone piece is preserved on the same interval. Integral curves are required to solve the native ambient equation on neighborhoods of every time in the interval, including any interval endpoints.*

Proof. Fix a starting fan cone σ. Its time preimage with fixed angle is closed relative to the interval, since σ is closed and the Legendre map and the curve are continuous. At every time in the norm-one region the radial coordinate is nonzero: the actual function has norm less than one at zero for sufficiently large β. The Legendre homeomorphism fixes zero, so the momentum is nonzero there as well. The local cone-piece invariance lemma consequently applies at every time in this closed preimage and makes it relatively open. It contains the initial time. Connectedness makes it the entire interval. The momentum never vanishes, and the fixed angle annihilates the same cone; thus the curve remains in the actual conic set. □

Lemma U.9 (Forward retention of the actual conormal superlevel)**.** *For the same potentials and sufficiently large β, every native ambient trajectory starting in a specified conormal cone piece with* $\operatorname{Re} f_1 \geq 1$ *remains in that same cone piece, with its original angle and* $\operatorname{Re} f_1 \geq 1$*, throughout every forward interval on which the native equation is solved near every time. In particular this holds for trajectories starting on the actual level-one core section. The base value is strictly increasing on each such interval.*

Proof. Fix the starting cone σ and angle. The times for which momentum belongs to this closed cone, angle is fixed, and $\operatorname{Re} f_1 \geq 1$ form a closed subset of the time interval. At any such time the momentum is nonzero, since the actual endpoint has norm less than one at radial zero. It is consequently a point of the actual conormal set. The local cone-piece lemma preserves the cone and angle on a neighborhood. If the base value is greater than one, continuity preserves this inequality locally. If it equals one, the actual base derivative is positive by the conormal finite-band margin at upper level one. The defining positive limit of the difference quotient then implies that the base value is greater than one on a sufficiently small right neighborhood. Thus the closed time subset is right open at each of its points before the endpoint. Continuous induction on a closed interval makes it the entire interval. At every retained point apply the finite-band margin with upper level chosen to be that point's base value. Its derivative is positive. The mean value theorem therefore gives strict increase on the interval. □

U.2.1 Finite scalar crossing and local smooth hitting times

Lemma U.10 (Finite crossing from a lower speed before the target)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ be differentiable, $g(0) < b$, and $\varepsilon > 0$. Set $T = (b - g(0) + 1)/\varepsilon$. If $g'(t) \geq \varepsilon$ whenever $0 < t < T$ and $g(t) \leq b$, then $g(\tau) = b$ for some $0 < \tau \leq T$.*

Proof. The definition gives $T > 0$. Suppose no time in $(0, T]$ has value b. For any $0 \leq t \leq T$, continuity and $g(0) < b$ imply $g(t) < b$: otherwise the intermediate value theorem on $[0, t]$ supplies such a time, which cannot be zero. Thus the derivative bound holds on $(0, T)$. The mean value inequality gives $g(T) - g(0) \geq \varepsilon T = b - g(0) + 1$, contradicting $g(T) < b$. □

Lemma U.11 (The entire superlevel conic region is the outward section)**.** *Let ϕ_∞ be a strongly compatible homogeneous end potential, whose gradient is a homeomorphism fixing zero. For all sufficiently large β, a point y belongs to the actual conic set Γ and satisfies* $\operatorname{Re} f_1(y) \geq 1$ *if and only if $y = (ru, \theta)$ for some $(u, \theta) \in L_\beta$ and $r \geq 1$.*

Proof. Let $y \in \Gamma$ have the stated value. Project its nonzero momentum radially onto the literal FLTZ link and choose the section point z with that projection, using surjectivity of the radial section. Equality of unit momentum directions says that their gradients differ by a positive scalar s. Homogeneity of the gradient and its injectivity give $y_1 = sz_1$; equality of the torus coordinates gives $y_2 = z_2$. The norm of $f_1(y)$ is at least its real part, hence at least one. The superlevel height

bound gives $h(y_1) \geq \beta - \log|V|$. The exact cone equation therefore identifies $f_1(y)$ with the real radial function at y_1. That radial function is strictly increasing along the ray from y_1 outward: throughout this ray the height remains above $\beta - \log(2|V|)$, where its positive radial derivative applies. If $s < 1$, strict increase from s to 1 would give $\operatorname{Re} f_1(y) < f_1(z) = 1$, a contradiction. Thus $s \geq 1$. Conversely, every outward continuation of a section point remains in Γ and satisfies the exact real radial equation. Its positive radial derivative on parameters at least one makes its value nondecreasing, so its value is at least its initial value one. □

Lemma U.12 (Unique crossing for a nondecreasing orbit)**.** *In the situation of Lemma U.10, suppose also that g is nondecreasing on $[0,T]$. There is exactly one $\tau \in (0,T]$ with $g(\tau) = b$.*

Proof. Existence is the preceding lemma. If $s < t$ were two such times, monotonicity would give $g = b$ on $[s,t]$. The lower derivative bound there gives $g(t) - g(s) \geq \varepsilon(t-s) > 0$, contradicting their equal values. Interchanging s, t handles either order. □

Lemma U.13 (A smooth local time at a transverse hit)**.** *Let E be a real Banach space, $H : E \times \mathbb{R} \to \mathbb{R}$ be smooth near (p_0, t_0), and let $H(p_0, t_0) = b$. If $\partial_t H(p_0, t_0) \neq 0$, there is a function τ of class C^∞ at p_0, with $\tau(p_0) = t_0$ and $H(p, \tau(p)) = b$ near p_0.*

Proof. The partial differential on the time factor is multiplication by the nonzero number $\partial_t H(p_0, t_0)$, hence is invertible. The implicit function theorem at the smoothness order ∞ applies to the literal equation $H(p,t) = H(p_0, t_0)$ and gives a function of that order at p_0, with the stated value and local identity. The same argument applies when p includes an extended family parameter, so it includes the endpoints of any closed interval. □

Lemma U.14 (Finite crossing with smooth local dependence)**.** *Let $H : E \times \mathbb{R} \to \mathbb{R}$ be smooth and $H(p_0, 0) < b$, $\varepsilon > 0$. If $\partial_t H(p_0, t) \geq \varepsilon$ for $0 < t < (b - H(p_0, 0) + 2)/\varepsilon$ with $H(p_0, t) \leq b$, then there is a hit $0 < t_0 \leq (b - H(p_0, 0) + 1)/\varepsilon$ and a time function τ of class C^∞ at p_0, with $\tau(p_0) = t_0$ and $H(p, \tau(p)) = b$ near p_0.*

Proof. Restrict the bound to the shorter interval and apply the finite-crossing lemma. Since the resulting hit is strictly inside the longer interval, its time derivative is at least $\varepsilon > 0$. The preceding implicit-time lemma supplies τ of class C^∞ at the base point. □

Lemma U.15 (A smooth time on an open neighborhood)**.** *Let E be a finite-dimensional real normed space and $H : E \times \mathbb{R} \to \mathbb{R}$ be smooth. Suppose $H(p_0, t_0) = b$ and $\partial_t H(p_0, t_0) \neq 0$. There is an open neighborhood U of p_0 and a function τ smooth on all of U, with $\tau(p_0) = t_0$ and $H(p, \tau(p)) = b$ for every $p \in U$.*

Proof. The smooth map $S(p,t) = (p, H(p,t))$ has derivative $(v, r) \mapsto (v, DH(v,0) + r\partial_t H)$ at (p_0, t_0). It is bijective: its inverse sends (v, s) to $(v, (s - DH(v,0))/\partial_t H)$. The inverse function theorem supplies a smooth local inverse l on an open neighborhood of (p_0, b). Set $U = \{p : (p, b) \in \operatorname{dom} l\}$ and $\tau(p) = l(p, b)_2$. The identity $S(l(p,b)) = (p,b)$ identifies its first coordinate with p and its second equation with $H(p, \tau(p)) = b$. The inverse identity at (p_0, t_0) gives $\tau(p_0) = t_0$. Composition and projection preserve smoothness on the stated open set. □

Lemma U.16 (A finite hit with a smooth neighborhood time)**.** *Under the finite-crossing smooth-time hypotheses with E finite-dimensional, the constructed hit has a time function smooth on one open neighborhood of p_0, satisfying the literal level equation throughout that neighborhood.*

Proof. Construct the finite hit by the lower speed bound. Its time lies strictly inside the longer speed interval, so the actual derivative there is at least $\varepsilon > 0$. Apply the preceding open-neighborhood time lemma to that hit and its actual derivative. □

Lemma U.17 (The actual tailored band constructs a finite hit)**.** *Assume the smooth uniformly convex potential and homogeneous end-potential hypotheses of the finite-base-band margin. For all sufficiently large β and every $M > 1$, there is $\varepsilon > 0$ with this property. Let F be a jointly smooth auxiliary action on parameter times ambient space. Fix $K \geq 0$ and $y \in L_\beta$. Whenever $t \geq 0$ and $\operatorname{Re} f_1(F^t(K, y)_2) \leq M$, suppose its spatial curve solves the actual native $Z_{1,K}$ equation, belongs to Γ, and has $\operatorname{Re} f_1 \geq 1$. Then $\operatorname{Re} f_1(F^t(K, y)_2) = M$ for some $0 < t \leq M/\varepsilon$.*

Proof. The actual finite-base-band margin gives a single positive lower bound ε on the real differential of f_1 along $Z_{1,K}$ on the entire actual conormal band. For $g(t) = \operatorname{Re} f_1(F^t(K, y)_2)$, joint smoothness makes g differentiable, and the action identity gives $g(0) = 1$. At every positive time with $g(t) \leq M$, the native chain rule and the explicit band retention identify $g'(t)$ with that actual differential, hence $g'(t) \geq \varepsilon$. The finite-crossing lemma supplies $0 < t \leq (M - 1 + 1)/\varepsilon = M/\varepsilon$ with $g(t) = M$. □

Lemma U.18 (Finite unique base hitting from the actual core)**.** *Assume the smooth uniformly convex potential and homogeneous end-potential hypotheses, including smoothness of the gradient homeomorphism and its inverse off zero. For all sufficiently large β and every $M > 1$, there is $\varepsilon > 0$ with this property. Let F be a jointly smooth action on $\mathbb{R}_K \times T^*T^n$, fix $K \geq 0$ and $y \in L_\beta$, and suppose its spatial curve solves the actual native $Z_{1,K}$ equation locally at every time $t \geq 0$. There is a hit $0 < t \leq M/\varepsilon$ with $F^t(K, y)_2 \in \Gamma$ and $\operatorname{Re} f_1(F^t(K, y)_2) = M$, and any nonnegative time with this value equals t.*

Proof. Choose a parameter threshold at least those for forward retention and finite native base hitting. The action starts at the stated core point. The actual forward theorem supplies membership in Γ and the lower base bound one at every nonnegative time, so it discharges the retention hypotheses of finite native hitting. The local native ODE supplies the derivative equation there, and that lemma constructs the bounded positive hit. The forward theorem also makes the base coordinate strictly increasing on $[0, \infty)$; hence equality of base values at two nonnegative times forces equality of the times. □

Lemma U.19 (The native hitting map is smooth near a transverse hit)**.** *Let F be a jointly smooth action on $\mathbb{R}_K \times T^*T^n$, and let $\ell : \mathbb{R}^{2n} \to T^*T^n$ be the logarithmic covering. Suppose the spatial curve $s \mapsto F^s(K_0, \ell(x_0))_2$ reaches $\operatorname{Re} f_1 = M$ at time t_0. Suppose its derivative at that time is Z_{1,K_0} and $\operatorname{Re} Df_1[Z_{1,K_0}] \neq 0$ at the hit. There is an open neighborhood V of (K_0, x_0) and a smooth function $\tau : V \to \mathbb{R}$ with $\tau(K_0, x_0) = t_0$ such that*

$$\operatorname{Re} f_1\big(F^{\tau(K,x)}(K, \ell(x))_2\big) = M \qquad ((K, x) \in V).$$

The map $(K, x) \mapsto F^{\tau(K,x)}(K, \ell(x))_2$ is smooth on V.

Proof. Set $H((K, x), s) = \operatorname{Re} f_1(F^s(K, \ell(x))_2)$. Smoothness of the action, the logarithmic covering, and f_1 makes H smooth. The native equation and the chain rule identify $\partial_s H((K_0, x_0), t_0)$ with the stated nonzero differential. The open-neighborhood time lemma therefore supplies V and τ with the exact level identity. Composing the smooth map $(K, x) \mapsto (\tau(K, x), (K, \ell(x)))$ with the action and taking its spatial component proves smoothness of the resulting hitting map. □

Lemma U.20 (The constructed core hit has a smooth time and map)**.** *Under the hypotheses of finite unique core hitting, let $y = \ell(x_0) \in L_\beta$. Retain the hypothesis that the spatial curve solves the actual native equation locally at every time $s \geq 0$. For all sufficiently large β, the constructed unique hit $0 < t_0 \leq M/\varepsilon$ has an open neighborhood V of (K_0, x_0) and a smooth time function τ on V, with $\tau(K_0, x_0) = t_0$, such that $\operatorname{Re} f_1(F^{\tau(K,x)}(K, \ell(x))_2) = M$ throughout V. Its literal spatial hitting map is smooth on V.*

Proof. Take a parameter threshold at least those for finite unique core hitting and the uniform conormal-band margin. The finite hitting theorem constructs t_0 and puts its spatial endpoint in Γ with base value M. The margin supplies a positive lower bound on the actual real differential there, so that differential is nonzero. The native ODE at $t_0 \geq 0$ gives the required curve derivative. Apply the native smooth hitting-map lemma to this constructed hit and its proved transversality. It supplies the exact level equation and smooth time and spatial map on one open parameter and logarithmic-coordinate neighborhood. □

Lemma U.21 (A positive speed on the entire conormal band)**.** *Under the smooth uniformly convex potential and homogeneous end-potential hypotheses of the finite-base-band margin, for all sufficiently large β and every finite M, there is $\varepsilon > 0$ such that $\operatorname{Re} Df_1(y)[Z_{1,K}(y)] \geq \varepsilon$ for all $y \in \Gamma$ with $1 \leq \operatorname{Re} f_1(y) \leq M$ and $K \geq 0$.*

Proof. Every point in this band is an outward continuation of some point of L_β, by the preceding superlevel classification. The already proved finite-base-band margin applies to all such continuations with base value at most M. Its bound is independent of the point and of the nonnegative coefficient K, giving the assertion. □

U.3 A native flow on the finite comparison region

Lemma U.22 (A parameter preserving cutoff action)**.** *Let Q be a finite dimensional smooth manifold without boundary, let $\Omega \subset \mathbb{R} \times Q$ be open, and let Z_k be jointly smooth on Ω. For every compact $C \subset \Omega$, there is an open V containing C, with $V \subset \Omega$, and a jointly smooth action $F : \mathbb{R} \times (\mathbb{R} \times Q) \to \mathbb{R} \times Q$ preserving the first coordinate. Whenever $F^t(k, x) \in V$, its spatial curve solves Z_k on a neighborhood of t.*

Proof. Choose a smooth compactly supported function χ with closed support in Ω, equal to one on an open neighborhood V_0 of C. Set $V = V_0 \cap \Omega$ and extend $\chi(k, x)(0, Z_k(x))$ by zero outside Ω. This is smooth because the closed support of χ lies in Ω. Its compact support gives a complete smooth flow: a common local existence interval on the compact support, together with constant solutions off the support, permits successive continuation in both time directions. Uniqueness gives the action law and smooth dependence gives joint smoothness. Its first velocity component is zero, so the first coordinate is constant. If $F^t(k, x) \in V$, continuity gives a time neighborhood staying in V, where $\chi = 1$. Projecting the flow equation there gives precisely the native spatial equation. □

Lemma U.23 (A smooth native affine primitive neighborhood)**.** *Let α_0, α_1 be smooth one forms on a finite dimensional smooth manifold without boundary and put $\alpha_k = \alpha_0 + k\alpha_1$. Suppose $d\alpha_k$ is nondegenerate at every point of a compact $C \subset \mathbb{R} \times Q$. There is a jointly smooth parameter preserving cutoff action solving the actual Liouville field of α_k near every time at which its current point belongs to one open neighborhood of C.*

Proof. In each spatial chart the coefficients of α_k are smooth in (k, x), as are their spatial exterior derivatives. The contraction matrix of $d\alpha_k$ is invertible on an open neighborhood of every point

of C. Matrix inversion and evaluation on the coefficients of α_k give a jointly smooth canonical Liouville field there. The union of these neighborhoods is an open Ω containing C; the local fields agree because the contraction solution is unique. Apply the preceding cutoff action construction on Ω. □

Lemma U.24 (Native flow near the literal compact tailoring sublevel)**.** *Assume the smooth uniformly convex potential hypotheses used for native tailoring symplecticity. For all sufficiently large β, every compact spatial set C and finite interval $[0, K_*]$ have a common jointly smooth parameter preserving cutoff action solving $Z_{1,K}$ near every time when $K \in [0, K_*]$ and its spatial point lies in C. In particular one may use any compact literal potential sublevel.*

Proof. The actual primitive is the affine family $\lambda_\phi + K f_1^* \lambda_{\mathbb{C}}$. Both one forms are smooth. Native tailoring symplecticity makes its differential nondegenerate for every $K \geq 0$, so in particular on $[0, K_*] \times C$, which is compact. The preceding lemma constructs the required action and an open neighborhood of this entire compact product. Membership of the current point in that product implies membership in the native neighborhood. □

The action is complete for its cutoff field. Its agreement with $Z_{1,K}$ is a regional assertion; finite crossing arguments must keep their trajectories in the stated region until the target is reached.

U.4 Finite hitting of the literal outer potential boundary

In the tailoring statements, ϕ is smooth with Hessian bounded between positive constants c and U, and agrees outside a ball of radius $R > 0$ with a strongly compatible homogeneous end potential ϕ_{end}. The latter is globally continuously differentiable. Its gradient Φ is a homeomorphism fixing zero, and Φ and its inverse are smooth away from zero. The actual conic set is denoted by Γ and its level-one section by L_β.

Lemma U.25 (A unique outward level crossing)**.** *Let $g : \mathbb{R} \to \mathbb{R}$ be differentiable. If $g'(t) > 0$ whenever $g(t) = A$, then $g(s) = g(t) = A$ implies $s = t$.*

Proof. Boundary positivity implies that once $g(s) \geq A$, every later value is strictly greater than A. Applying this assertion to either ordered pair of distinct crossing times contradicts equality to A at the later time. □

Lemma U.26 (Finite forward hitting of the outer potential boundary)**.** *Fix a smooth uniformly convex potential with bounded Hessian, agreeing outside a ball with a strongly compatible homogeneous end potential. For sufficiently large β, let D_β be the literal compact sublevel $\{2\phi \leq D^2\beta^2\}$, for fixed $D > 0$. Let a smooth flow agree with the native ambient Liouville equation whenever its trajectory is in D_β. Every point of L_β strictly inside D_β has a unique positive-time hit of $\{2\phi = D^2\beta^2\}$. Up to that hit its trajectory stays in the actual conic set and satisfies $f_1 \geq 1$.*

Proof. Continuity bounds $\operatorname{Re} f_1$ by some finite M on the compact sublevel. Suppose a starting trajectory never hits its boundary at a positive time. The intermediate value theorem, starting from its strict interior point, keeps it strictly inside for all positive times. It therefore solves the native equation there. The forward cone and base retention theorem applies, and the positive base-speed margin on $1 \leq f_1 \leq M + 1$ forces a finite hit of $f_1 = M + 1$. This contradicts the compact sublevel bound. At each potential-boundary crossing the native potential derivative is positive. The preceding lemma makes the crossing unique. Any earlier value on or outside the boundary would force a strictly greater potential value at the crossing, so all earlier times remain inside. Apply native forward cone and base retention on this closed finite interval to obtain the final assertion. □

Lemma U.27 (A fixed outer radius encloses the whole core section)**.** *There is a fixed $D > 0$ such that, for all sufficiently large β, every $y \in L_\beta$ satisfies $\phi(y) < D^2\beta^2/2$.*

Proof. Choose $a, H > 0$ with $a|u| \le h(u) \le H|u|$. On the actual radial core section, $\beta - \log|V| \le h(u) \le \beta$. The lower inequality puts the section outside the extension radius for sufficiently large β; the upper one gives $|u| \le \beta/a$. Euler's identity and the Hessian upper bound give $2\phi(u) = \langle H_u u, u\rangle \le U|u|^2$ there. Taking $D = (U+1)/a + 1$ yields $\sqrt{U}|u| < (U+1)|u| \le (U+1)\beta/a < D\beta$; squaring proves the assertion. □

Lemma U.28 (The conic outer boundary is beyond base one)**.** *There is $D_0 > 0$ such that every fixed $D \ge D_0$ has, for sufficiently large β, the following properties: the whole core section is strictly below the outer potential level, and every point of the actual conic set on that outer level has* $\operatorname{Re} f_1 > 1$.

Proof. The radial projection bijection writes every conic point uniquely in the form (su_z, θ_z) with $z \in L_\beta$ and $s > 0$. The fixed-radius bound for the core persists for every larger radius. For large β, both the core and the outer potential level are outside the extension ball; the latter assertion follows by bounding ϕ on the compact ball. Consequently homogeneity gives $D^2\beta^2/2 = s^2\phi(z)$. The left side is positive, so $\phi(z) > 0$. The strict core bound forces $s > 1$. Along every outward core ray the actual real base value is strictly increasing and equals one at $s = 1$. Thus its value at the outer point is strictly greater than one. □

Lemma U.29 (The last crossing in a compact time interval)**.** *For a continuous scalar g on $[a, 0]$, with $a < 0$, $g(a) = A$ and $g(0) > A$, there is a largest crossing time $t < 0$, and $g \ge A$ on $[t, 0]$.*

Proof. The crossing times form a nonempty closed subset of the compact interval, so they have a maximum t. It is negative because $g(0) > A$. If $g(s) < A$ at a later time s, the intermediate value theorem between s and zero gives a crossing strictly later than t, a contradiction. □

Lemma U.30 (Backward hitting from the literal conic outer boundary)**.** *For sufficiently large β, a conic outer-boundary point with* $\operatorname{Re} f_1 > 1$ *has a finite negative-time native trajectory to the actual core section, for any smooth flow agreeing with the native equation throughout the compact outer potential sublevel.*

Proof. Boundary positivity places every negative-time point strictly inside the outer potential sublevel. Suppose the real base value never reaches one at negative time. Continuity keeps it greater than one there. Connected cone invariance on every finite negative interval keeps the trajectory conormal. Compactness bounds its base above by some M, and the native base derivative has a positive lower bound ε on $1 \le f_1 \le M$. The mean value inequality on $[-T, 0]$ gives $\varepsilon T \le f_1(0) - f_1(-T)$. For $T = (f_1(0) + 1)/\varepsilon$ this contradicts $f_1(-T) > 1$. Thus a negative crossing exists. Choose the last crossing on the compact interval from one such crossing to zero. The base is at least one on this interval, so connected cone invariance puts the crossing in the actual conic set. The exact real radial formula there turns its real-base equality into the complex equation $f_1 = 1$, hence the crossing belongs to the actual core section. □

Proposition U.31 (The native outer potential boundary bijection)**.** *Under the stated potential and gradient hypotheses, there is $D_0 > 0$ such that every fixed $D \ge D_0$ has a threshold $B > 0$ with the following property. For every $\beta \ge B$ and every K_* there is a jointly smooth parameter preserving action F on $\mathbb{R} \times Y$ whose spatial curves solve the actual endpoint Liouville equation whenever $K \in [0, K_*]$ and their current potential is at most $A = D^2\beta^2/2$. For each such K there*

is a time function $\tau_K : L_\beta \to \mathbb{R}_{>0}$ giving unique hits, among all real times of F, of $\phi = A$. The literal map

$$\pi_K(y) = F^{\tau_K(y)}(K, y)_2$$

is a bijection from L_β onto $\Gamma \cap \{\phi = A\}$. Up to this hit, the spatial trajectory stays in Γ with real base at least one. At every logarithmic representative x of a core point, there is an open neighborhood of (K, x) with a smooth local time centered at $\tau_K(\ell(x))$, a literal potential hit throughout that neighborhood, and a smooth spatial hitting map.

Proof. Choose D_0 by the outer conic separation lemma and fix $D \geq D_0$. Increase the parameter threshold to cover that separation, forward and backward hitting, finite-region native flow, forward strict base increase, potential-sublevel compactness and outwardness, and smooth outer hitting. The potential sublevel $C = \{\phi \leq A\}$ is compact. Apply the finite-region flow construction to $[0, K_*] \times C$. It supplies one jointly smooth parameter preserving action with the required regional native equation. Every core point lies strictly inside C, so unique forward hitting defines τ_K and proves the trajectory assertion. In particular $\pi_K(L_\beta)$ is contained in the stated outer conic level.

For surjectivity, take $w \in \Gamma \cap \{\phi = A\}$. Its real base is strictly greater than one. Backward hitting gives $t < 0$ with $z = F^t(K, w)_2 \in L_\beta$. Parameter preservation and the action law give $F^{-t}(K, z)_2 = w$. Uniqueness of the potential hit from z forces $-t = \tau_K(z)$, so $\pi_K(z) = w$.

For injectivity, suppose $\pi_K(y) = \pi_K(z) = w$. The inverse action identifies y and z with the points at times $a = -\tau_K(y)$ and $b = -\tau_K(z)$ on the spatial curve $\gamma(s) = F^s(K, w)_2$. Native potential outwardness at every level crossing implies $\phi(\gamma(s)) \leq A$ for all $s \leq 0$: any earlier value on or above A would force its value at zero to be strictly above A. Thus this entire negative part solves the native equation. If $a < b$, forward cone retention and strict real-base increase on $[a, b]$, from the core point y, imply $f_1(y) < f_1(z)$. Both values are one, a contradiction. The case $b < a$ is identical with the two points interchanged. Hence $a = b$ and $y = z$.

Finally, smooth outer hitting provides a time and spatial map on an open coefficient and logarithmic-coordinate neighborhood of each core point. Its center time equals τ_K there by unique potential hitting. This gives the stated local smooth extension of the chosen hit. □

Finite toric input constructs one strongly compatible end potential and one smooth extension, together with fixed radii $0 < C_- < C_+$ and a parameter threshold. For every larger β and every original coefficient in its specified range, there are an admissible convenient rounding, a finite normalized coefficient $K > 0$, and an actual marked prefix P with these same potentials and radii. The full core of its terminal retained ribbon, in its actual logarithmic embedding, is L_β. A constructed parameter preserving action solves the actual native endpoint equation throughout the compact sublevel $\phi \leq C_+^2\beta^2/2$ for all coefficients in $[0, K]$. For every coefficient $k \in [0, K]$, its unique positive hitting map is a bijection

$$L_\beta \longrightarrow \Gamma \cap \{\phi = C_+^2\beta^2/2\} = \Gamma \cap j_P(\partial D_P),$$

where D_P and j_P are the very outer compact domain and native embedding of P's retained ambient completion marking. Every chosen hit has a smooth time and spatial map on an open coefficient and logarithmic-coordinate neighborhood centered at that hit.

Proof. Take the extension and end-gradient homeomorphism from the finite-input full-core construction. Apply both the original marked-prefix producer and native outer-boundary comparison to these same potential witnesses. Let $F > 0$ be the prefix's fiber-radius bound and let $D_0 > 0$ be the comparison's lower outer radius. Choose $C_- = F + 1$ and $C_+ = \max(F + 2, D_0)$. Then

$F < C_- < C_+$ and $C_+ \geq D_0$, so both producers apply at this same radius. Increase the parameter threshold to cover the prefix, full-core classification, native comparison, and regular outer potential-sublevel data.

For each permitted original coefficient choose the normalized $K = \max(K_*, K_{\text{original}} e^{2\beta})$, where K_* is the prefix threshold. Its positivity and the normalization inequality give the actual prefix with the convenient inner level and a common admissible rounding. Transfer the full-fiber compact-forward-orbit classification through this prefix's own ribbon marking; its ambient primitive uses exactly the extension already selected. Thus its actual embedded full core is L_β.

Apply the native comparison at the finite coefficient endpoint K. It supplies the action, unique times, bijection onto the literal potential level, and the stated local smooth data. Native outwardness on that level makes its defining differential nonzero. The retained ambient marking has carrier image exactly the potential sublevel. Its native embedding is an open embedding, so it maps the frontier of its compact domain to the frontier of that sublevel. Regularity identifies the latter frontier with the literal potential level. Consequently the same bijection targets the conic intersection of the actual retained marked boundary. □

U.5 Smooth hitting of the literal outer potential boundary

Lemma U.32 (A smooth native potential hitting map)**.** *Let F be a jointly smooth action on coefficient times the logarithmic ambient, let $\ell : \mathbb{R}^{2n} \to Y$ be the logarithmic covering, and let ϕ be smooth. Suppose its spatial curve from $(K_0, \ell(x_0))$ solves the actual native Z_{1,K_0} equation near a time t_0, hits $\phi = A$ there, and $d(\phi - A)(Z_{1,K_0}) \neq 0$ at that hit. There is an actual open neighborhood V of (K_0, x_0) and a time τ smooth throughout V, with $\tau(K_0, x_0) = t_0$, satisfying*

$$\phi\big(F^{\tau(K,x)}(K, \ell(x))_2\big) = A \qquad ((K, x) \in V).$$

The literal spatial hitting map in this equation is smooth on V.

Proof. The function $H((K, x), t) = \phi(F^t(K, \ell(x))_2) - A$ is smooth in all its Euclidean parameters. At the specified hit, the native chain rule identifies its time derivative with $d(\phi - A)(Z_{1,K_0})$. This is nonzero, so the scalar implicit function theorem on a smooth open inverse neighborhood supplies V and τ, with the exact equation $H((K, x), \tau(K, x)) = 0$ throughout V. Composing the smooth flow with $(\tau(K, x), (K, \ell(x)))$ and projecting to the spatial component gives the asserted smooth map. □

Lemma U.33 (Constructed outer hits have smooth ambient neighborhoods)**.** *Under the hypotheses of native outer potential hitting, let F be jointly smooth and solve $Z_{1,K}$ locally whenever its current spatial point is in the literal potential sublevel $\phi \leq A = D^2\beta^2/2$. For sufficiently large β, every logarithmic representative of a core point strictly inside this sublevel has a unique positive hit of $\phi = A$. That constructed hit has a time smooth on an actual open coefficient and logarithmic-coordinate neighborhood, and its literal spatial hitting map is smooth there.*

Proof. Increase the parameter threshold to cover both finite unique outer hitting and native outwardness of the literal potential boundary. The former constructs the positive hit and its uniqueness. Its potential equals A, so regional native agreement supplies the native equation at the hit. The latter gives $d(\phi - A)(Z_{1,K}) > 0$ there, hence transversality. Apply the preceding smooth potential hitting lemma to this actual hit and derivative. It supplies the time, level equation, and literal spatial map on one open neighborhood. □

Lemma U.34 (Open parameter extension of transverse flow hits)**.** *Let Y be the logarithmic quotient ambient and let F be a smooth flow on $\mathbb{R} \times Y$ preserving the first coordinate, and let $\rho : Y \to \mathbb{R}$ be smooth. Suppose the level $S = \{\rho = A\}$ is compact and the time-zero derivative of $\rho(\mathrm{pr}_Y F_t(K, y))$ is positive on $P \times S$, for a compact coefficient set P. There is an open set $J \supset P$ on which every trajectory has at most one hit of S.*

Proof. Differentiation in the time coordinate of the smooth flow gives a continuous time-zero derivative on $\mathbb{R} \times Y$. This can be checked in logarithmic coordinates and descended through the quotient cover, which is an open surjection. Positivity is an open condition. The tube lemma applied to $P \times S$ therefore supplies J such that the derivative remains positive on $J \times S$. The flow identity $F_{t+s} = F_s \circ F_t$ transports this derivative to every hitting time; preservation of the coefficient keeps it in J. A differentiable real function whose derivative is positive at every crossing of a fixed level crosses that level at most once. Apply this to the potential along each trajectory. □

Lemma U.35 (A local transverse time with uniqueness)**.** *Let $H : E \times \mathbb{R} \to \mathbb{R}$ be smooth, with E a finite dimensional real vector space. If $H(p, t) = b$ and $\partial_t H(p, t) > 0$, there are an open neighborhood U of p, an interval $(a, c) \ni t$, and a smooth $\tau : U \to (a, c)$ such that $H(x, \tau(x)) = b$. For every $x \in U$, this is the unique solution in (a, c).*

Proof. The smooth implicit function theorem gives a local solution τ. Continuity of the time derivative gives product neighborhoods on which that derivative stays positive. Choose (a, c) inside the time neighborhood and shrink U so that $\tau(U) \subset (a, c)$. For each fixed $x \in U$, the mean value theorem makes $H(x, -)$ strictly increasing on (a, c), giving uniqueness there. □

U.6 One smooth hitting neighborhood in the quotient ambient

Lemma U.36 (Descent of a local logarithmic hitting time)**.** *Let F be a jointly smooth action on coefficient times the logarithmic ambient Y. Suppose a smooth time function on an open neighborhood of (K, x) in coefficient times logarithmic covering coordinates gives literal potential hits. It gives a smooth time and spatial hitting map on an actual open neighborhood of $(K, \ell(x))$ in $\mathbb{R} \times Y$, with the same center time and the same literal potential equation.*

Proof. Choose a smooth local inverse l of the logarithmic covering near $\ell(x)$, with $l(\ell(x)) = x$. On $S = \mathbb{R} \times \mathrm{source}(l)$ the map $\psi(K, y) = (K, l(y))$ is smooth. Intersect S with the preimage under ψ of the given logarithmic neighborhood. This intersection is open and contains $(K, \ell(x))$. Compose the given time with ψ. The right inverse identity $\ell(l(y)) = y$ on S proves the literal hit equation there. Smoothness of the time and of its spatial hitting map follow by composition with the smooth action. □

The same finite-input open hitting construction also gives, for its chosen action and potentials, a positive time-zero derivative of $\mathrm{Re}\, f_1$ at every point of $[0, K] \times L_\beta$.

Proof. Enlarge the common parameter threshold by the uniform native base transversality threshold on the outward continuation. A point of L_β belongs to the actual embedded full core of the retained ribbon. That core is contained in its compact ribbon carrier, whose literal image equation gives $\phi \le C_-^2 \beta^2/2$ there. Since $0 < C_- < C_+$ and $\beta > 0$, the core lies strictly below the same outer potential level. The constructed action therefore satisfies the native equation at time zero there. The native base differential is a positive real number by outward transversality at radial parameter one. The chain rule along the integral curve identifies its real part with the asserted derivative of the actual action. Retain the entire open hitting construction and its same marked prefix. The preceding theorem is the corollary forgetting this extra derivative. □

Lemma U.37 (Gluing unique native hitting times)**.** *Suppose a smooth action has at most one hit of a fixed potential level for every initial point at each coefficient in an open set J. Let $S \subset J \times Y$, and suppose every point of S has an open native neighborhood with a smooth time giving that level hit. There is one open W with $S \subset W \subset J \times Y$ and one smooth time function on W giving the literal hit throughout W. Its spatial hitting map is smooth on W.*

Proof. Intersect each local neighborhood with $J \times Y$ and take their union W. Every point of W has a hit, so define its time to be that unique hit. On any constituent neighborhood, uniqueness identifies this function with the prescribed smooth local time. Thus it is smooth at every point of W. Composition with the action proves smoothness of the literal spatial hitting map. The union is open, contains S, and is contained in $J \times Y$ by construction. □

Finite toric input constructs the same convenient retained marked prefix, potential pair and radii as in the marked native outer comparison. For every permitted parameter and original coefficient, its constructed finite normalized coefficient $K > 0$ has one actual open $W \subset \mathbb{R} \times Y$ containing $[0, K] \times L_\beta$, with one positive smooth time τ and one smooth spatial map $\pi(k, y) = F^{\tau(k,y)}(k, y)_2$ satisfying the literal outer potential equation throughout W. For every $k \in [0, K]$, the time on L_β is the unique constructed hit, its preceding trajectory is conormal with base at least one, and $\pi(k, \cdot)$ is the bijection onto the conic intersection of that same prefix's actual marked outer boundary. Its retained ribbon still has the literal full embedded core L_β.

Proof. Use the common-witness finite-input comparison and enlarge its parameter threshold to cover compact potential-sublevel data at its chosen outer radius. Keep the action and the retained prefix furnished by that comparison. The outer level is closed and is contained in the compact sublevel, hence compact. On this level, native agreement at time zero and native outwardness give a positive derivative of the actual flow's potential. Continuity of this derivative and compactness of the level times $[0, K]$ give an open coefficient neighborhood J on which the time-zero derivative remains positive on the whole level. The action law and coefficient preservation identify the derivative at any later crossing with this time-zero derivative at its current boundary point. The scalar outward-crossing lemma therefore makes all level hits unique for every initial point throughout J.

At each point of $[0, K] \times L_\beta$, the previously constructed smooth logarithmic local time descends by the local inverse of the covering. Apply the gluing lemma to these native neighborhoods and the open coefficient uniqueness just established. It produces an open union containing the entire coefficient/core product, with one smooth literal hitting time and spatial map. Unique hitting identifies the time there with the original chosen positive core time. Restrict the union to the open set where this smooth time is positive; it still contains the entire product. The same identification transfers conormal retention and both literal and actual marked-boundary bijections to the restriction. No new potential, radius, prefix, or completion marking is selected in this step. □

U.7 Immersion and embedding near the actual retained full core

Lemma U.38 (A locally unique native scalar crossing)**.** *For a smooth coefficient-preserving action on the logarithmic ambient and a smooth scalar function, a scalar hit with positive time derivative has a smooth local hitting time on an actual open neighborhood of its initial coefficient and quotient point. There is a fixed open time interval containing the center time on which this is the unique hit.*

Proof. Choose a logarithmic representative of the initial point. Apply the locally unique transverse-time lemma to the smooth scalar function of coefficient, representative, and time. Its fixed open

time interval and its unique root descend by a smooth local inverse of the covering. The covering's right inverse identity transfers both the equation and uniqueness to the quotient neighborhood. □

Lemma U.39 (Transverse projection has a local inverse on the full fiber)**.** *Let $j : H \hookrightarrow Y$ be a smooth embedding with $f \circ j = 1$, let F be a jointly smooth coefficient-preserving action, and let τ be smooth on an open neighborhood in $\mathbb{R} \times H$. At a point where the time-zero derivative of f along F is positive, the graph projection $\Pi(k, x) = (k, F^{\tau(k,x)}(k, j(x))_2)$ is injective and immersive on some open neighborhood of that point.*

Proof. Write $z_0 = F^{\tau(k_0,x_0)}(k_0, j(x_0))_2$. The backward time $-\tau(k_0, x_0)$ is a hit of $f = 1$ from (k_0, z_0), with the same positive derivative as the time-zero derivative at $j(x_0)$. The preceding lemma supplies a smooth backward hitting time σ and a fixed open interval on which its roots are unique. Shrink the source neighborhood so that its projection lies in the inverse neighborhood and $-\tau$ lies in this interval. The action law and $f \circ j = 1$ show that $-\tau$ is a root, so uniqueness identifies it with $\sigma \circ \Pi$. Thus the smooth map $R(k, z) = (k, F^{\sigma(k,z)}(k, z)_2)$ satisfies $R \circ \Pi(k, x) = (k, j(x))$ there. The right side is injective with injective differential, so the chain rule makes Π immersive; the same identity makes it injective. □

Lemma U.40 (Common embeddings of the full-fiber hitting neighborhood)**.** *Let $j : H \hookrightarrow Y$ be a smooth embedding in a scalar level $f = 1$, and let $P \subset \mathbb{R}$ and $C \subset H$ be compact. Suppose a coefficient-preserving smooth flow and a smooth time on an open neighborhood of $P \times C$ give maps π_k which are injective on C. Suppose the time-zero derivative of f along the flow is positive at $(k, j(x))$ for every $(k, x) \in P \times C$. There are open neighborhoods $V \supset P$, $U \supset C$ such that each $\pi_k|_U$ for $k \in V$ is a smooth embedding, with its literal flow formula and joint smoothness on $V \times U$. For an inclusion-marked full ribbon whose embedded core is L, one may take $C = j^{-1}(L)$; its compactness follows from the given marking and the compactness of the specified ribbon core.*

Proof. The locally unique backward scalar hit constructs a local inverse to $(k, x) \mapsto (k, \pi_k(x))$ near each point of $P \times C$. Consequently this graph is locally injective and has injective differential there. Take the union of neighborhoods where its differential remains injective. Compact injectivity and local injectivity give an embedding neighborhood of $P \times C$ within this union. The tube lemma gives a product $V \times U$ inside that neighborhood. Each slice is an embedding, and injectivity of the graph differential implies injectivity of its spatial differential. Restricting the original smooth formula gives the asserted smooth embeddings and joint smoothness.

For the marked ribbon, map its compact core into the full fiber by the specified completion inclusion followed by the inclusion marking. This image is compact. The full compact marking square identifies its j-image with L. Injectivity of j identifies this compact image with $j^{-1}(L)$, so the preceding argument applies without a new compactness assumption about the native fiber core. □

U.8 Actual hitting embeddings for the retained marked full fiber

Finite toric input constructs the same potential pair, radii, retained marked prefix, full embedded core and literal marked outer boundary as the native open transverse hitting producer. Let $j : H_\beta \hookrightarrow Y$ be that prefix's actual full-fiber completion inclusion marking. There are open sets $V \subset \mathbb{R}$ and $U \subset H_\beta$, with $[0, K] \subset V$ and $j^{-1}(L_\beta) \subset U$, such that the same constructed flow projection gives an actual smooth embedding of U for every $k \in V$. These maps are jointly smooth on $V \times U$, and that product maps into the previously constructed native open hitting neighborhood. The time stays positive and the literal outer potential equation holds there. On $[0, K] \times L_\beta$ the

maps remain the proved bijections onto the conic intersection of the same retained marked outer boundary.

Proof. Retain every witness from the open transverse hitting producer. The actual full-fiber inclusion marking identifies the intrinsic compact completion core with the native preimage $j^{-1}(L_\beta)$. Its compact square and the literal retained-core equality prove this identification, and injectivity of j makes it an equality of full sets. Pull back the native open time through $(k, x) \mapsto (k, j(x))$. The full-fiber image equation gives $\operatorname{Re} f_1 \circ j = 1$. Native core transversality and the constructed backward scalar inverse therefore give local graph injectivity and immersion at every point of the compact parameter/core product. The proved core bijections give its fiberwise injectivity. Apply the compact common-neighborhood embedding construction to obtain V and U and the actual smooth embeddings on U. Their maps are the restrictions of the same native flow projection, so smoothness, positive times, the literal potential hit and the core bijections all retain the original witnesses. □

Lemma U.41 (Hitting embeddings on the whole shrinking extension)**.** *Let U be an open neighborhood in the actual marked full fiber H, and let $e_k : U \hookrightarrow Y$ be smooth embeddings for k in an open coefficient set V. Choose a concrete shrinking whose entire open endpoint extension E maps into U under $a = q \circ \Phi_C^{-T} \circ i_C$. Then $e_k \circ a : E \hookrightarrow Y$ are smooth embeddings with that literal formula. More generally, every jointly smooth map $g : V \times U \to N$ pulls back to the jointly smooth map $(k, x) \mapsto g(k, a(x))$ on $V \times E$.*

Proof. The negative-time flow and the actual completion marking are smooth diffeomorphisms; the completion inclusion is a smooth embedding. Their composition a is therefore a smooth embedding into H. Because U is open and contains its whole image, the same map with codomain U is smooth. Its induced topology is unchanged; composing its differential with the open inclusion gives the injective differential of a, so its differential is injective as well. Composing with e_k proves the first claim. The map $(k, x) \mapsto (k, a(x))$ is jointly smooth and maps $V \times E$ into $V \times U$. The chain rule gives the final claim. □

U.9 Positive witnesses on the actual outer boundary

Lemma U.42 (Scalar extension along a compact immersed carrier)**.** *Let X, Y be finite dimensional smooth manifolds without boundary, with X Hausdorff and Y Hausdorff and second countable. Let $A \subset X$ be compact, $W \subset X$ open containing A, and $j : W \to Y$ smooth, injective on A, and immersive there. Every smooth real function f on W has a smooth extension $g : Y \to \mathbb{R}$ with $g(j(x)) = f(x)$ for $x \in A$.*

Proof. An injective derivative has a linear left inverse. Compose j with a target chart and this projection. The inverse function theorem yields local injectivity. Compact injectivity then gives an open neighborhood O of A on which j is an embedding and remains immersive. At $a \in O$ apply the same projected inverse function theorem. The induced topology of the embedding allows its target neighborhood to be restricted so that every point of $j(O)$ in it comes from the local source inverse domain. The projected local inverse is therefore a smooth left inverse of j on that target neighborhood. Composing it with f gives a smooth scalar germ matching f at all points of $j(A)$ in the neighborhood. The compact set $j(A)$ is closed. A smooth partition of unity subordinate to these target neighborhoods and its complement combines the scalar germs and zero on the complement. Every germ present at a point of $j(A)$ takes the same value there, and the weights sum to one, proving exact agreement. □

Lemma U.43 (Witness of the actual full-ribbon hitting family)**.** *Let F be the same compact full ribbon after intrinsic shrinking, with smooth extension X. Let B be the native outer boundary and $i : B \hookrightarrow Y$ its actual inclusion embedding. Suppose the actual hitting formula $h(k, x)$ is smooth on an open product $V \times X$, with $[0, K_*] \subset V$, and each spatial slice is a smooth embedding. Suppose $h(k, x) \in i(B)$ throughout that product, and its actual time $t(k, x)$ is jointly smooth. Then the hitting maps factor as smooth embeddings $q_k : X \hookrightarrow B$, jointly smooth in k, with $iq_k = h(k, -)$. There is a jointly smooth positive function $w : \mathbb{R} \times B \to \mathbb{R}_{>0}$ such that $w(k, q_k(x)) = e^{-t(k,x)}$ for every $k \in [0, K_*]$ and $x \in F$.*

Proof. Injectivity of i defines each range lift q_k. Its induced topology makes the lift continuous. Local smooth left inverses of i make the joint lift smooth. The differential chain rule and injectivity of the hitting differential prove immersion of each lift; its induced topology follows from that of the two ambient embeddings. The parameter graph $(k, x) \mapsto (k, h(k, x))$ is injective on the compact product $[0, K_*] \times F$. Its derivative is triangular; a vector in its kernel has zero parameter component and then zero spatial component by slice immersion. Apply the preceding scalar extension lemma to $-t$ on this graph to obtain a on $\mathbb{R} \times Y$ with $a(k, h(k, x)) = -t(k, x)$ there. Set $w(k, b) = \exp a(k, i(b))$. Composition proves joint smoothness, exponentiation proves positivity, and the graph equation proves every prescribed value on the full ribbon, including its boundary and both parameter endpoints. □

Remark U.44 (Extension by scalar germs). The source uses a tubular projection and a cutoff to extend $-t$. Here local projected inverses of the same embedded graph and a smooth partition of unity give its scalar extension. Both constructions retain exactly the values on the full compact parameter/ribbon product. The positive witness is a function on the same boundary section.

Finite toric input constructs all the same potentials, radii, thresholds, original marked prefix, native action, hitting time, core bijections and common open full-fiber hitting embeddings as the preceding retained producer. In addition it constructs a finite intrinsic shrinking of that same entire full ribbon. Write a_T for the negative-flow embedding of its fixed source presentation into its actual full marked fiber. The whole open source extension lies in the common hitting neighborhood. On the actual native outer boundary of the same retained outer domain, there are smooth embeddings j_k of the source extension, jointly smooth for k in an open neighborhood V of $[0, K_*]$, whose ambient images are exactly $\pi_k a_T$. There is a globally positive jointly smooth real-parameter function w on this native boundary satisfying

$$w(k, j_k(x)) = \exp\big(-t_k(a_T(x))\big) \quad (0 \le k \le K_*,\ x \in F_0).$$

Every point of the full compact source ribbon, including its boundary, is covered by this equation. Its intrinsic completion and literal full core remain the marked ones of the shrinking construction.

Proof. Retain the witnesses of the preceding hitting-embedding producer. Its full-core equality identifies the actual native fiber preimage of the core with the marked compact ribbon core. The common native full-fiber open neighborhood therefore admits the finite shrinking selector. This gives a_T with its entire open extension inside that same neighborhood, while retaining the full-fiber marking. Precompose each previously constructed hitting embedding with a_T. The negative-flow embedding is smooth and independent of k, so the resulting embeddings and times are jointly smooth on V times the whole source extension. Their literal potential hit lies in the native outer boundary because the marked boundary image is exactly that potential level. Apply the preceding boundary hitting witness lemma to these actual maps and times. Its factorization square and prescribed-value identity give all the asserted equations without altering any previously retained potential, prefix, native flow, core bijection or outer marking. □

U.10 The primitive of the retained outer hitting map

Lemma U.45 (Regional primitive scaling up to a transverse hit)**.** *Let α be a smooth one form on a smooth manifold M, and let $F : \mathbb{R} \times N \to M$ be smooth. Let $S \subset M$ be open and suppose $\partial_t F = Z \circ F$ and $\iota_Z d\alpha = \alpha$ wherever F takes values in S. Fix $x \in N$ and $T > 0$, with $F(t, x) \in S$ for every $t < T$. Then $(F^T)^*\alpha|_x = e^T (F^0)^*\alpha|_x$. If a differentiable time function τ satisfies $\tau(x) = T$, the time equation also holds at (T, x), and $\alpha(Z) = 0$ there, then $[y \mapsto F(\tau(y), y)]^*\alpha|_x = e^T (F^0)^*\alpha|_x$.*

Proof. Choose a chart about x and a source tangent v. Pull α back by the resulting spacetime map. On the open preimage of S, its time coefficient is $\alpha(Z) = d\alpha(Z, Z) = 0$. Exterior differentiation on (∂_t, v) therefore shows that its spatial coefficient $q(t)$ satisfies $q'(t) = q(t)$ for every $t < T$. Thus $e^{-t} q(t)$ has zero derivative on the connected interval $(-\infty, T)$ and equals $q(0)$ there. Smoothness of the spacetime map and the form makes q continuous at T, giving $q(T) = e^T q(0)$. This proves the one-form equality on every tangent. The chain rule for variable time adds $d\tau(v)Z$ to the spatial differential. Its primitive evaluation vanishes by the endpoint hypothesis, proving the second assertion. □

Lemma U.46 (The actual cutoff hitting primitive)**.** *Let a smooth parameter preserving action solve the native Liouville equation of $\alpha_k = \lambda_\phi + k f_1^* \lambda_{\mathbb{C}}$ whenever its spatial point lies in $\{\phi \le A\}$. Suppose $d\alpha_k$ is nondegenerate throughout $\{\phi \le A\}$ and $d\phi(Z_k) > 0$ on $\{\phi = A\}$. For the retained full-fiber inclusion j into $f_1^{-1}(1)$ and differentiable positive hitting time τ_k, with $\phi(F_k^{\tau_k(x)}(j(x))) = A$, the actual map $\pi_k(x) = F_k^{\tau_k(x)}(j(x))$ satisfies $\pi_k^* \alpha_k|_x = e^{\tau_k(x)} j^* \lambda_\phi|_x$.*

Proof. The scalar trajectory has positive derivative whenever it equals A. If at some earlier time it were at least A, positivity would force it to be strictly above A at the prescribed hit: positive crossing prevents a downward crossing, and a return to A would be a local minimum with positive derivative. Consequently every pre-hit time lies in the open sublevel. Nondegeneracy gives the actual native contraction equation, including at the endpoint. Apply regional primitive scaling and its variable-time chain rule. Since $f_1 \circ j = 1$, differentiating this identity annihilates the plane pullback on every source tangent; therefore $j^* \alpha_k = j^* \lambda_\phi$. □

For finite toric input, the potentials, radii and parameter threshold of the marked native outer-comparison construction can be chosen with the following additional properties. For every allowed β and original coefficient, retain its selected admissible convenient rounding, normalized $K > 0$, actual marked prefix P, smooth action G, native open hitting time τ and literal marked outer boundary. Let $j : H_\beta \hookrightarrow Y$ be this same prefix's full-fiber inclusion marking. There are common open sets $V \supset [0, K]$ and $U \supset j^{-1}(L_\beta)$ on which the same hitting maps are smoothly embedded and jointly smooth. For every $k \ge 0$, the native form $d\alpha_k$ is globally nondegenerate and $d\phi(Z_k) > 0$ on the literal outer level. For every $k \in [0, K]$ and $x \in U$,

$$\pi_k^* \alpha_k|_x = e^{\tau(k, j(x))}\, j^* \lambda_\phi|_x, \qquad \pi_k(x) = \mathrm{pr}_Y\, G^{\tau(k, j(x))}(k, j(x)).$$

The core bijections and full-fiber marking belong to these same choices.

Proof. Keep the smooth uniformly convex potential, its Hessian bounds and its end agreement from the marked native outer-comparison producer. Increase its parameter threshold to cover native symplecticity, potential-sublevel outwardness and core transversality. These analytic results apply to the same retained potential, so their global nondegeneracy and literal outer-level positivity hold for every nonnegative coefficient. Differentiating the scalar defining function $\phi - A$ gives the same differential as ϕ.

For each permitted original coefficient retain the prefix and cutoff action from that producer. Core transversality follows by applying the native derivative formula to its same core trajectories. Outer-level positivity and the action law give uniqueness of hits throughout an open coefficient neighborhood. Descend the existing local times through the logarithmic cover and glue them by this uniqueness. Restrict their open union to positive time; every compact parameter/core point remains in it because its selected time is positive. Uniqueness also identifies the glued time with every previously selected core time, preserving the core bijections.

Apply the common embedding-neighborhood theorem to the retained full-fiber marking, this glued time and the same action. It gives V, U and the literal hitting embeddings, while $V \times U$ maps into the positive open hitting domain. For $k \in [0, K]$, apply the preceding actual cutoff primitive lemma at every $x \in U$. All its hypotheses are the nondegeneracy, outwardness, regional equation, smooth time and literal hit just constructed. The full-fiber image equation cancels the plane contribution. This proves the displayed identity without changing any of the selected witnesses. □

Lemma U.47 (Base-coefficient family on the retained outer domain)**.** *Let D be the retained compact outer domain, whose native embedding i maps its carrier to $\{\phi \le A\}$ and its boundary to $\{\phi = A\}$. Suppose for $0 \le k \le K_*$ the actual primitive $\alpha_k = \lambda_\phi + k f_1^* \lambda_{\mathbb{C}}$ has nondegenerate differential everywhere, and its Liouville field has $d\phi(Z_k) > 0$ on that level. The fixed domain D then carries a smooth Liouville family with primitive exactly $i^* \alpha_{K_*(1-t)}$, including a smooth family of actual localized completions.*

Proof. The coefficient path stays in $[0, K_*]$. In each fixed chart the form coefficients are the sum of the fixed smooth coefficients of λ_ϕ and the fixed smooth coefficients of $f_1^* \lambda_{\mathbb{C}}$ multiplied by $K_*(1-t)$; hence they are jointly smooth on an open real-parameter extension. The native embedding is fixed and smooth. The carrier and boundary equations transfer the sublevel inequality and boundary equality to D. Nondegeneracy transfers through its invertible differential, and the native outward inequality transfers to the stored defining functions of D. The sublevel-pullback family theorem consequently constructs the Liouville structures, their jointly smooth fields, and the smooth localized collar and completion family, with the stated literal primitive. □

U.11 The exact primitive of the retained hitting ribbon

Lemma U.48 (Transport through the same intrinsic shrinking)**.** *Let $a_T : X \hookrightarrow H$ be the negative intrinsic-flow embedding of the same full compact ribbon F into its marked full fiber H. Write μ_T for its actual compact Liouville primitive. Suppose the whole open extension X maps into an open hitting neighborhood U, and a smooth hitting embedding $\pi_k : U \hookrightarrow Y$ satisfies $\pi_k^* \alpha_k = e^{t_k} \lambda_H$ on U. Then*

$$(\pi_k a_T)^* \alpha_k|_x = e^{t_k(a_T(x))} \mu_T|_x \quad (x \in F).$$

Proof. The native open inclusion $U \hookrightarrow H$ has identity tangent map in the inherited charts. Restrict the given primitive identity to U and pull it back by a_T with its range in U. The differential chain rule composes these two maps. Scalar multiplication pulls through the tangent pullback with its coefficient evaluated at $a_T(x)$. The same intrinsic shrinking marking gives $a_T^* \lambda_H|_x = \mu_T|_x$ on the entire compact carrier, proving the displayed equation. □

Lemma U.49 (Exact cancellation by the boundary witness)**.** *Let $i : B \hookrightarrow Y$ be the native outer boundary embedding, and $j : X \hookrightarrow B$ a ribbon embedding. If $(ij)^* \alpha|_x = e^{t(x)} \mu|_x$ and $w(j(x)) = e^{-t(x)}$, then $j^*(w\, i^* \alpha)|_x = \mu|_x$.*

Proof. Scalar multiplication and pullback composition give $j^*(w\, i^* \alpha)|_x = w(j(x))(ij)^* \alpha|_x$. Insert both hypotheses. The remaining scalar is $e^{-t(x)} e^{t(x)} = e^0 = 1$. □

U.12 Completed families of the same retained hitting ribbon

Lemma U.50 (Smooth restriction along the coefficient path)**.** *Let $V \subset \mathbb{R}$ be open, $c : \mathbb{R} \to \mathbb{R}$ smooth with $c([0,1]) \subset V$, and $j_k : X \hookrightarrow B$ jointly smooth on $V \times X$. Then $j_{c(t)}$ is a smooth interval family with an actual open extension across both endpoints, without a nonemptiness assumption on X or B.*

Proof. On the open set $\Omega = c^{-1}(V)$ use the coefficient $c(s)$; outside it use the fixed permitted coefficient $c(0)$. Composing this parameter selection with j defines a map for every real time. At any $s \in \Omega$ the selection agrees locally with c, so the map is jointly smooth on $\Omega \times X$. This open set contains $[0,1] \times X$, and restriction has the required values. □

Definition U.51 (Completed hitting family on specified compact data)**.** *For a fixed native outer compact domain D, a fixed Liouville ribbon F, primitives L_t, embeddings $j_t : F \hookrightarrow \partial D$ and witnesses b_t, completed hitting-family data consist of their joint smoothness, positivity, exact full compact primitive law and both actual localized completion families A_t, B_t. The pair deformation is computed from these substantive data; its slice is exactly the localized witnessed pair formed with A_t, B_t, j_t, b_t. In particular its primitive, embedding, witness and completions are the specified ones of these same compact data.*

Lemma U.52 (Constructed completed hitting family)**.** *Let D be the retained marked native outer domain and F the same intrinsically shrunk Liouville ribbon. Suppose the retained native forms $\alpha_k = \lambda_\phi + k f_1^* \lambda_{\mathbb{C}}$ are nondegenerate and outward at the literal outer level for $0 \le k \le K$. Suppose the jointly smooth hitting embeddings and positive boundary witness satisfy the exact compact ribbon law there. Then the coefficient path $k(t) = K(1-t)$ constructs completed hitting family data on the same D, F, with actual ambient primitive $i_D^* \alpha_{k(t)}$, embedding $j_{k(t)}$ and witness $w(k(t), -)$.*

Proof. The retained outer coefficient construction supplies L_t on D with precisely this native pulled-back primitive and joint smoothness. The ribbon primitive is constant, hence jointly smooth. The preceding open-parameter lemma makes $j_{K(1-t)}$ jointly smooth through the parameter endpoints. Compose the globally smooth witness with this affine coefficient path; it remains jointly smooth and positive. The coefficient lies in $[0, K]$, so the assumed compact law and the literal formula for L_t give the required weighted ribbon law at all times. The localized completion-family constructor produces both actual collar/completion families. Applying its pair constructor to these very data gives the computed deformation and its stated slice formula. No completed family or pair deformation is assumed. □

Lemma U.53 (The same full-fiber inclusion marking survives localization)**.** *Every computed slice of such a hitting family has an actual completion inclusion marking onto the same entire native full fiber of its selected intrinsic shrinking. Its compact map is exactly the negative intrinsic flow embedding composed with the actual original localized inclusion.*

Proof. An original completion localizes to the collar presentation with its complete ambient, primitive and all-real end flow unchanged. Thus the original marking applies to this localized completion, with its compact map precomposed by the original collar inclusion. Compare the collar's constructed cylindrical completion with this localized original completion. The comparison preserves primitives, its compact map agrees with the prescribed original inclusion, and its end map follows the same all-real flow. Composing it with the original marking proves all marking equations. Apply this construction to the actual shrinking endpoint marking retained on the same full native fiber. □

For finite toric input there are uniform allowed potentials and thresholds such that, for every allowed radius and original coefficient, one may retain all the same admissible convenient rounding, marked prefix, native action, hitting time, intrinsic shrinking, boundary embeddings and positive witness of the exact ribbon producer. On the same marked outer domain and the same compact shrinking endpoint, these choices construct actual completed hitting-family data with ambient primitive $i_D^*\alpha_{K(1-t)}$, embedding $j_{K(1-t)}$ and witness $w(K(1-t),-)$, including genuine endpoint extensions and both actual localized completion families. The parameter begins at K and ends at zero. At every parameter the image of its full embedded core under the literal native outer-boundary inclusion is exactly $\Gamma \cap \partial D_\beta$.

Proof. Use one output of the retained exact hitting-ribbon producer. Its nondegeneracy and outwardness statements hold for every nonnegative coefficient, hence in particular on $[0,K]$. Apply the preceding completed-family construction to its same native outer domain, shrinking endpoint, jointly smooth boundary hitting embeddings and positive witness. Its compact law is exactly the equation already proved on these selected data. Retain its ambient primitive formula and the resulting completed hitting-family data. This uses no second existential selection of a potential, prefix, flow, time or shrink. The collar localization core identity reduces the new core to the specified shrinking completion core. The actual hitting formula and the preserved full-fiber marking identify its native image with the image of L_β under the retained hitting map. Its proved core bijection gives exactly $\Gamma \cap \partial D_\beta$. □

Lemma U.54 (The full core through localization and hitting)**.** *Let F be the same intrinsically shrunk ribbon with its specified completion. Localizing its open presentation by any constructed collar preserves its entire intrinsic core under the original inclusion. If a hitting embedding on the full-fiber neighborhood agrees with a map g on native points, its image of this core is exactly $g(L)$, where L is the original full embedded core before shrinking. Consequently the actual completed hitting pair has this same image under its native outer-boundary embedding.*

Proof. Localize the original completion itself to the collar presentation. Its completed flow and compact image are unchanged, so the all-forward-time core condition is precisely the preimage of the original core condition. The full core is independent of the chosen completion of this fixed domain, by strict comparison intertwining the completed flows. The collar inclusion maps its entire carrier onto the original carrier; restricting that surjection to the core gives equality of the core images.

On the shrinking core every point belongs to the compact ribbon. The negative-flow fiber embedding therefore agrees there with the actual ambient ribbon embedding. The full-fiber marking proves that shrinking preserves its entire embedded core. Composing this equality with g and using the prescribed hitting formula gives exactly $g(L)$. The localized witnessed pair uses the same hitting embedding precomposed with the collar inclusion, so the preceding localization identity proves the final assertion. □

U.13 The finite hitting trajectory in the chosen completion

Lemma U.55 (Finite native trajectories lift to the actual completed flow)**.** *Let X be a complete Liouville manifold with field Z_X, let M have the same dimension, and let $e : X \hookrightarrow M$ be a smooth embedding. Fix $K \subset X$ and a native field Z on M such that $de_x(Z_X(x)) = Z(e(x))$ for $x \in K$. Let $T > 0$ and let $\gamma : [0,T] \to M$ solve the native integral-curve equation, including the one-sided derivatives at the endpoints. Suppose $\gamma(t) \in e(K)$ throughout the interval and $\gamma(0) = e(x)$. Then*

$$\mathrm{Fl}^t_{Z_X}(x) \in K, \qquad e(\mathrm{Fl}^t_{Z_X}(x)) = \gamma(t) \quad (0 \le t \le T).$$

Proof. Equal dimension and injectivity of the differential make e a local diffeomorphism. Its injectivity therefore supplies a smooth inverse g on its open image. Put $\xi = g \circ \gamma$. Containment gives $\xi(t) \in K$ and $e(\xi(t)) = \gamma(t)$. Differentiating $g \circ e = \mathrm{id}$ and the native equation gives

$$\dot{\xi}(t) = dg_{\gamma(t)} Z(\gamma(t)) = dg_{e(\xi(t))} de_{\xi(t)} Z_X(\xi(t)) = Z_X(\xi(t)).$$

Thus ξ and the completed orbit solve the same smooth equation with the same initial value. Right interval uniqueness gives equality on $[0, T)$; continuity of both curves extends it to T. The two asserted conclusions follow from the defining properties of ξ. □

Lemma U.56 (The same native boundary hit is a completed end point)**.** *Let D be a compact Liouville domain, C its specified completion constructed from a flow collar, and $a : D_C \to D$ its actual localized compact projection. Let e be an equal-dimensional smooth embedding of the native extension of D into M, and let α be a smooth native one form with $\lambda_D = e^*\alpha$ on that extension. Assume $d\alpha$ is nondegenerate on the compact image $e(D)$. Let γ solve Z_α on $[0, T]$, where $T > 0$, with $\gamma([0, T]) \subset e(D)$, $\gamma(0) = e(a(x))$, and $\gamma(T) = e(b)$ for a literal boundary point $b \in \partial D$. Then*

$$\mathrm{Fl}^T_{Z_C}(x) = \mathrm{end}_C(b, 0), \qquad x = \mathrm{end}_C(b, -T).$$

More generally, the completed orbit projects to γ throughout $[0, T]$, staying in the localized compact carrier. The same assertions hold if the carrier field square is given directly instead of the primitive hypothesis.

Proof. Primitive naturality, nondegeneracy and uniqueness of the Liouville contraction vector give $de(Z_D) = Z_\alpha \circ e$ on the compact carrier. The localized projection has identity differential, carries its compact carrier onto D, and the completed field there is the original field. Hence $e \circ a$ transports the completed field on that carrier and the preceding lemma applies. At time T its native image is $e(b)$. The time-zero completed end has the same native image; injectivity of $e \circ a$ identifies the two completed points. Apply the completed flow for time $-T$ and use the all-real end-flow equation to obtain the second identity. □

Lemma U.57 (A regional cutoff hit follows the actual completed trajectory)**.** *In the preceding native compact presentation, suppose $e(D) = \{\rho \le A\}$ for a smooth ρ and $d\alpha$ is nondegenerate on this sublevel. Let a smooth cutoff action G on the parameter space times M solve the native Z_α equation whenever its current spatial point lies in that sublevel. Assume $d\rho(Z_\alpha) > 0$ at every point of $\{\rho = A\}$. If the literal spatial trajectory from $e(a(x))$ hits $e(b)$ at time $T > 0$, with $\rho(e(b)) = A$, then it has the same completed endpoint and negative end-coordinate identities above.*

Proof. For the scalar trajectory $f(t) = \rho(\mathrm{pr}_M\, G^t(k, e(a(x))))$, the regional equation and outwardness give $f'(t) > 0$ whenever $f(t) = A$. If $f(s) \ge A$ for any $s < T$, strict positive crossing forces $f(T) > A$, contradicting the prescribed hit. Thus every earlier real time satisfies $f(s) < A$. Together with $f(T) = A$ this places the entire interval $[0, T]$ inside $e(D)$. The regional native equation therefore supplies its integral-curve equation on that whole finite interval. Apply the preceding native completed-hit lemma. Neither the native field outside the sublevel nor a complete native ambient flow is used. □

Lemma U.58 (Smooth transverse hitting)**.** *Suppose smooth parameter-dependent vector fields and hypersurfaces are given, and every point in a compact parameterized set has a unique transverse hit at a finite time. The hitting time and hitting map are smooth on a neighborhood of that set. The neighborhood can be chosen uniformly for a compact parameter interval.*

Proof. In a local defining function ρ_a for the target hypersurface, solve $\rho_a(\mathrm{Fl}^{\tau}_{Z_a}(x)) = 0$. Its derivative in τ is $d\rho_a(Z_a) \neq 0$ at the specified hit, so the parameter-dependent implicit-function theorem gives a smooth local time. Uniqueness makes these local functions agree on overlaps. Finitely many neighborhoods cover the compact set. Smooth dependence of the flow gives the hitting map. Smoothness at interval endpoints uses the prescribed extensions of the data across them. □

Proposition U.59 (Transport through a finite invariant cone band)**.** *For each $K \in [0, K_*]$, ambient forward Liouville flow carries L_β bijectively to $\Gamma \cap \partial D_\beta$. The hitting maps are smooth on neighborhoods of the core and smoothly parameterized by K.*

Proof. The common construction puts the entire core strictly inside the compact potential sublevel D_β and puts its outer conormal section in $\{f_1 > 1\}$. A smooth coefficient-preserving cutoff action agrees with the native Liouville equation whenever the current point is in this sublevel. At every outer-level crossing the potential derivative is positive, so a trajectory can cross that level at most once.

Before its first outer hit a trajectory from the core stays conormal with real base at least one. If it never hits, it remains in the compact sublevel at all positive times. The positive uniform base-speed bound then contradicts boundedness of the base on that sublevel. Thus the positive outer hit exists and is unique. A backward trajectory from the outer conormal section stays below the outer potential level. The last-crossing argument and the same positive base-speed bound force it to reach the actual complex level $f_1 = 1$ in finite negative time while remaining conormal. This inner section is exactly L_β. The flow law and strict base increase prove injectivity and surjectivity. The regular-level boundary identity identifies this target with the native boundary of the same retained outer domain.

Transversality on the entire compact outer level persists on an open coefficient neighborhood of $[0, K_*]$. The flow law gives uniqueness of all potential hits there. Smooth local implicit times descend through the logarithmic covering's local inverses; uniqueness makes them agree on overlaps. Their open union contains $[0, K_*] \times L_\beta$ and carries one smooth time and its smooth literal hitting map. All portions of the trajectories used before their outer hits satisfy the native equation; only the auxiliary cutoff action is asserted to be globally complete. □

Proposition U.60 (One full ribbon and exact positive family witnesses)**.** *For the same finite-input potentials and thresholds fixed above, there is one full-fiber neighborhood U of L_β on which every hitting map π_K is an embedding. One finite negative intrinsic flow embeds the entire original compact ribbon into U; denote this presentation by $a_T : X \to H_\beta$ and its image by F. Its specified completion is marked onto the same full native fiber, and its primitive is $\mu_T = a_T^* \lambda_\phi$ on the compact carrier. There is a globally positive jointly smooth witness w on the whole native outer boundary such that, for $0 \leq K \leq K_*$,*

$$(\pi_K a_T)^*(w_K \alpha_K) = \mu_T$$

on that entire carrier. Equivalently, on the embedded ribbon F, $\pi_K^*(w_K \alpha_K) = \lambda_\phi|_F$.

Proof. Choose one output of the retained native primitive producer. Its common embedding neighborhood contains the full core. The nested negative intrinsic-flow images of the original compact ribbon have intersection that core, so compactness supplies a finite shrinking into this neighborhood. The shrinking selector also puts its entire open source extension in U. Its negative-flow completion equivalence retains the same full native fiber and proves the compact primitive formula above.

Regional Cartan scaling, including its variable-time chain rule, gives

$$(\pi_K a_T)^* \alpha_K = e^{t_K \circ a_T} \mu_T. \tag{33}$$

Indeed every pre-hit point lies below the outer potential level, where the constructed action satisfies the actual native Liouville equation; the additional time derivative vanishes because $\alpha_K(Z_K) = 0$.

The graph $(K, x) \mapsto (K, \pi_K a_T(x))$ is injective on the compact parameter/ribbon product and is an immersion on an open neighborhood. The native boundary inclusion has a smooth local inverse along this image, yielding actual jointly smooth boundary embeddings j_K with $i j_K = \pi_K a_T$. Local inverses to the graph extend the scalar $-t_K(a_T(x))$ near every point of its compact image. Choose finitely many such neighborhoods and a smooth partition of unity, with weights supported there and summing to one near the compact image. Their weighted extensions agree with the scalar on that image; extend the result by zero outside their supports. Exponentiation produces a globally smooth positive w on the full real-parameter/boundary product, with $w(K, j_K(x)) = e^{-t_K(a_T(x))}$ on the whole compact ribbon. Pullback composition and cancellation now give

$$j_K^*(w_K i^* \alpha_K) = \mu_T. \tag{34}$$

All maps, times and markings in this argument belong to the same chosen prefix and shrinking. Transporting through a_T gives the stated identity on F. □

Lemma U.61 (Changing a witness without changing its ribbon primitive)**.** *Let (D, λ, F, j, b) be a witnessed pair. If a smooth positive function c on ∂D satisfies $c(j(x)) = b(j(x))$ for every $x \in F$, then (D, λ, F, j, c) is a witnessed pair with the same compact domains, completions, primitive on F, and full embedded core.*

Proof. At $x \in F$ the pullback of the one-form $c\alpha$ depends on the value of c at $j(x)$ and on dj_x. Substitution of the assumed equality gives $j^*(c\alpha)_x = j^*(b\alpha)_x = \mu_x$. All other data defining the pair and its full core are unchanged. □

Lemma U.62 (A positive witness interpolation with fixed completed geometry)**.** *In the preceding setting, the witnesses*

$$b_t = \exp((1-t)\log b + t \log c), \qquad 0 \le t \le 1,$$

define a permitted pair deformation between the original pair and the pair with witness c. Both completed manifolds, primitives and fields are fixed, and every embedded full core is the original embedded full core.

Proof. Positivity of b and c makes their logarithms smooth. The displayed formula is jointly smooth for all real t, is everywhere positive, and equals b at $t = 0$ and c at $t = 1$. On $j(F)$ both logarithms coincide, so $b_t = b$ there at every parameter. The preceding lemma therefore keeps the ribbon primitive fixed. Use the identity trivializations on the compact domains and their given completions. Their primitives, fields and ribbon embeddings are constant smooth families; the only varying datum is the displayed smooth witness. This proves all requirements of a pair deformation and preserves the full intrinsic and embedded cores. □

Corollary U.63 (The witness deformation fixes the ambient Maslov datum). The same witness deformation carries every ambient Maslov representative constantly from source to target.

Proof. The ambient completion and its primitive are fixed, and its completion presentations are identities. The constant compatible complex structure and constant nowhere-zero squared canonical section therefore form a smooth compatible family with the given representative at both ends. □

V The completed intrinsic shrinking family

Lemma V.1 (Constant scaling retains the completed flow)**.** *If (F,μ) is a compact Liouville domain with a specified completion, then for every constant $c>0$, $(F,c\mu)$ has the same Liouville field and the same complete flow. Its completion has primitive $c\widehat{\mu}$ and the same compact inclusion and cylindrical end parametrization.*

Proof. The identity $d(c\mu)=c\,d\mu$ preserves nondegeneracy. Moreover, $\iota_Z d(c\mu)=c\iota_Z d\mu=c\mu$, so the field and its strict outward derivative are unchanged. On the specified cylindrical end, $c\widehat{\mu}=ce^r\alpha=e^r(c\alpha)$. The compact primitive equation and contraction equation scale by c. The covering, overlap, smoothness and end-flow equations use the unchanged maps and field and therefore still hold. These observations verify every completion condition. □

Lemma V.2 (A permitted shrinking family from the retained pair)**.** *Let (D,F,j,b) be a pair with its specified completions, and let $T\geq 0$. Choose the common open extension V from the preceding neighborhood lemma. If V is nonempty, the retained pair admits a finite permitted comparison to the pair with fixed ambient D, ribbon carrier F, ribbon primitive $e^{-T}\mu$, and embedding $j\circ q_1$. The comparison is the scale-one localization of the source extension followed by a deformation with fixed ambient and witness, embedding $j\circ q_t$, and primitive $e^{-Tt}\mu$. The complete ribbon primitive is $e^{-Tt}\widehat{\mu}$ and its complete flow is unchanged. Every prescribed ambient Maslov datum is transported constantly along this same finite comparison, and the embedded full core is unchanged throughout.*

Proof. Restrict the ribbon extension to V, retaining its compact carrier, boundary inclusion and completed manifold. The identity on the compact carrier extends smoothly into V on a neighborhood of that carrier; its inverse is the inclusion $V\hookrightarrow M$. Extend the former arbitrarily outside V, using a chosen point of V. Both compact inverse equations hold, their differentials are inverse, and the primitive pullback on the carrier is the identity. Together with identity maps on the completed ambient and its boundary this is a scale-one permitted identification to the initial family member.

The preceding primitive lemma gives $(j\circ q_t)^*(b\alpha)=e^{-Tt}\mu$. Positive constant scaling supplies its specified completion with the same complete flow. The common neighborhood lemma gives joint smoothness of the embeddings. In local charts the remaining primitive coefficients are products of fixed smooth coefficients with the smooth scalar e^{-Tt}; all fields, ambient forms and the witness are constant. These are exactly the required regularity conditions for the deformation. Negative invariance keeps its entire moving ribbon in the original compact ribbon. Its completed ambient, symplectic form and presentations are constant, so the constant homotopy of the prescribed Maslov datum gives the claimed transport on each arrow. For a full-core point x, commutation of the complete flow and negative invariance show that $q_t(x)$ is again a full-core point. Conversely the positive-time image of any full-core point remains in the compact ribbon and in its full core, and its negative-time image is the original point. Thus q_t maps the entire full core onto itself, proving the stated equality. □

The same finite-input choices and parameter ranges of Theorem T.17.1 supply the actual retained marked prefix P. For every open neighborhood W of its literal full core L_β, its terminal pair has a finite permitted comparison to a pair whose entire embedded compact ribbon lies in W and whose embedded full core is literally L_β. Every prescribed ambient Maslov datum lifts along that exact comparison. The initial potential and all original-K quantifiers are retained.

Proof. Let C be the terminal ribbon's specified completion and M its produced full-fiber marking. The continuous map $g=M.\text{embedding}\circ M.\text{marking}$ maps the completed ribbon to the same full fiber and agrees with the actual ambient embedding on its compact carrier. The inverse image

$g^{-1}(W)$ is therefore an open neighborhood of the completed full core. The negative images of the compact carrier are compact, nested, and intersect in that core. Compactness gives $T \geq 0$ with $\mathrm{Fl}^{-T}(i(F)) \subset g^{-1}(W)$: otherwise the closed complements of the neighborhood inside these nested compact sets would have the finite intersection property and a point in their intersection, a contradiction. The retained ribbon is nonempty, as proved by the prefix producer, so the source localization of the preceding lemma is available. Apply its actual shrinking comparison at this T. The completion inclusion equation for q_1 and the compact marking square imply that its whole ribbon lies in W. The preceding full-core equality and the produced equality with L_β give the literal endpoint core equality. The constant grading lifts on the localization and deformation are indexed by these same two arrows. Applying this argument to each prefix supplied by the previous finite-input theorem preserves its potential, all sufficiently large β, and every prescribed admissible original K. □

Lemma V.3 (A common neighborhood for a finite shrinking interval)**.** *Let $i : F_0 \hookrightarrow H$ be the compact inclusion of a specified completion, extended to a smooth embedding on an open extension M. For every finite $T \geq 0$, there is an open neighborhood V of F_0 in M such that $\mathrm{Fl}^{-Tt}(i(V)) \subset i(M)$ for $0 \leq t \leq 1$. The induced maps $V \to M$ are smooth jointly on an open neighborhood of $[0,1] \times V$ and satisfy $i(q_t(x)) = \mathrm{Fl}^{-Tt}(i(x))$.*

Proof. The equal-dimensional embedding i has invertible differential at every point, hence is an open embedding with a smooth inverse on its image. Negative invariance sends $i(F_0)$ into itself throughout the interval. The inverse image of $i(M)$ under $(t,x) \mapsto \mathrm{Fl}^{-Tt}(i(x))$ is an open set containing $[0,1] \times F_0$. Compactness of $[0,1]$ gives an open neighborhood V of F_0 whose product with the interval lies in this set. Composing the flow there with the inverse of i gives the claimed maps and their joint smooth extension. The same identity makes each map an embedding and an immersion, since the flow at a fixed time is a diffeomorphism. □

Lemma V.4 (The actual moving primitive)**.** *On the common extension in the preceding lemma, $q_t^* \mu = e^{-Tt} \mu$ at every point of F_0.*

Proof. The contraction equation and alternation give $\mu(Z) = 0$. Cartan's formula therefore gives $\mathcal{L}_Z \mu = \mu$. Integrating its scalar coefficient equation along the actual complete flow gives $(\mathrm{Fl}^s)^* \widehat{\mu} = e^s \widehat{\mu}$ for all real s. At a point of F_0, both its inclusion and its image under q_t lie in the compact domain, by negative invariance. The compact inclusion primitive equation thus applies at both points. Pulling back $i \circ q_t = \mathrm{Fl}^{-Tt} \circ i$ and substituting these two compact primitive equations gives the stated equality. □

Lemma V.5 (The same full-fiber marking after intrinsic shrinking)**.** *Let a retained pair have an actual full-fiber inclusion marking $q : \widehat{F}_0 \to H$ and fiber embedding $e : H \hookrightarrow Y$. For the concrete shrinking pair at time t, its specified completion with primitive $e^{-Tt} \widehat{\mu}_0$ has a constructed primitive-preserving marking onto the same entire H, with the same fiber embedding and potential primitive. On the compact carrier this marking agrees with its actual moved ribbon embedding.*

Given any open neighborhood W in Y of the embedded full core, one can choose the finite shrinking time and common source extension so that the entire open endpoint extension, under its constructed native fiber embedding, maps into W. The source localization and shrinking comparison retain the prescribed ambient grading on that exact comparison.

Proof. Write Φ_C for the complete flow on the original specified ribbon completion. Define

$$q_t = q \circ \Phi_C^{-Tt} : \widehat{F}_0 \longrightarrow H.$$

Both factors are actual smooth diffeomorphisms. Primitive preservation of q and the complete-flow identity give

$$q_t^* \mu_H = (\Phi_C^{-Tt})^* \widehat{\mu}_0 = e^{-Tt} \widehat{\mu}_0,$$

which is exactly the primitive of the supplied scaled completion. On its open source extension, the map $q_t \circ i_C$ is a smooth embedding into H. On the compact carrier the flow-lift equation, original inclusion-marking square and original compact fiber square give

$$e \circ q_t \circ i_C = R \circ q_t^{\mathrm{lift}}.$$

Consequently the full-fiber marking conditions hold for the concrete shrinking pair. Its end equation follows from flow equivariance of this actual primitive-preserving marking. Its complete fiber, fiber embedding and native potential primitive are literally the original ones. The full-core orbit characterization therefore gives equality of the original and moved embedded full cores.

The inverse image of W under $e \circ q$ contains the completed full core. Compactness and nested negative images give a finite time T placing the compact carrier in that inverse image. Choose the common extension for the shrinking interval, then intersect it with the open inverse image of W under $e \circ q \circ \Phi_C^{-T} \circ i_C$. This still contains the entire compact carrier and retains all common-interval flow-lift conditions. Its whole endpoint extension now maps into W. Nonemptiness of the retained ribbon supplies a source point for the explicit localization, and the preceding constant ambient grading construction lifts the same comparison. □

The fixed source carrier and its primitive $e^{-T}\mu_0$ are retained for subsequent hitting maps. This uses the fixed trivializations permitted in Definition 2.2; the target requires an existential finite comparison rather than the proof's particular list of presentations. Thus the optional image-domain relabeling in lines 1886–1889 is omitted. Every subsequent spatial map is instead precomposed with the constructed native negative-flow embedding. In particular, if $\pi_K^*(w_K \alpha_K) = \mu_H$, then $(\pi_K \circ q_1 \circ i_C)^*(w_K \alpha_K) = e^{-T}\mu_0$ on the compact carrier. This retains the entire ribbon, its actual primitive and its specified full-fiber marking at the junction.

Lemma V.6 (Shrinking directly into a full-fiber neighborhood)**.** *Let $q : \widehat{F}_0 \to H$ be the actual full-fiber marking, and $e : H \hookrightarrow Y$ its embedding. If an open subset $U \subset H$ contains $e^{-1}(R(\mathrm{Core}(F_0)))$, a finite intrinsic shrinking can be chosen whose entire open endpoint extension maps into U under $q \circ \Phi_C^{-T} \circ i_C$.*

Proof. The inverse image $q^{-1}(U)$ is open in the specified completion. For a point of its full core, write it as $i_C(x)$ with $x \in \mathrm{Core}(F_0)$. The actual compact inclusion square gives $e(q(i_C(x))) = R(x)$, so this inverse image contains the full core. Nested compact negative-flow images therefore give a finite $T \geq 0$ with $\Phi_C^{-T}(i_C(F_0)) \subset q^{-1}(U)$. Intersect the common shrinking source extension with the open inverse image of U under $q \circ \Phi_C^{-T} \circ i_C$. This contains the compact carrier, preserves the common flow-lift conditions and has the asserted whole-extension property. □

Lemma V.7 (The actual variable-time primitive pullback)**.** *Let (X, λ) be a complete Liouville manifold with its actual smooth complete flow Φ. For any smooth manifold N, smooth map $j : N \to X$ and smooth real function τ on N, define $h(x) = \Phi^{\tau(x)}(j(x))$. Then*

$$h^* \lambda = e^{\tau}\, j^* \lambda.$$

At a specified point, differentiability of j and τ at that point already suffices for the one-form identity there. No immersion or hitting condition is required.

Proof. For $v \in T_xN$, the chain rule for the jointly smooth complete flow gives

$$dh_x(v) = d\Phi_{j(x)}^{\tau(x)}(dj_x(v)) + d\tau_x(v)Z_\lambda(h(x)).$$

The time partial derivative here is the actual integral-curve derivative, not a prescribed primitive identity. The contraction equation and alternation give $\lambda(Z_\lambda) = d\lambda(Z_\lambda, Z_\lambda) = 0$. Consequently the second term has zero primitive evaluation. The constant-time primitive law from the earlier lemma gives

$$\lambda_{h(x)}(dh_x(v)) = e^{\tau(x)}\lambda_{j(x)}(dj_x(v)).$$

This holds for every source tangent and hence is the claimed one-form identity. The same calculation uses only the two derivatives at x for the pointwise version. □

Application to a separately constructed regional hitting action still requires proving that it agrees with the actual completed Liouville flow on the relevant trajectory tube. The lemma supplies the variable-time primitive identity once that comparison is available; it does not supply a boundary hitting family or its global positive witness.

V.1 The strict comparison to the retained hitting family

Lemma V.8 (The same retained ribbon has a scale-one collar comparison)**.** *Let F be a Liouville domain with nonempty compact carrier and let B be a chosen flow collar. There exists an actual compact-domain diffeomorphism $h : F \to F_B$ whose smooth extension satisfies $\iota_B \circ h = \mathrm{id}$ on its source neighborhood and $h^*\mu_B = \mu_F$ on the whole compact carrier. In particular its ribbon comparison scale is exactly one.*

Proof. The identity embedding of the original extension and the native projection of the localized extension have identical full compact images, namely the carrier of F. The common-embedding compact-map lemma gives h and its projection square on an open neighborhood. The localized primitive is $\iota_B^*\mu_F$. Differentiating the projection square on that neighborhood gives $h^*\iota_B^*\mu_F = \mu_F$ on the carrier. □

Lemma V.9 (Strict comparison of two outer collar completions)**.** *Let D be a nonempty compact smooth domain and let L, L' be Liouville structures on its same native extension with equal primitives. Choose flow collars C, C' and completions X, X' of their localized domains. There exist a compact-domain diffeomorphism $h : D_C \to D_{C'}$ and a primitive-preserving completed diffeomorphism $e : X \to X'$ such that, for every point x of the full compact carrier,*

$$\iota_{C'}(h(x)) = \iota_C(x), \qquad e(\mathrm{inc}_X(x)) = \mathrm{inc}_{X'}(h(x)).$$

Proof. Both localized compact domains project by smooth embeddings onto exactly D. Equality of these full compact images gives h on open neighborhoods, with the native projection square holding there. The localized primitives are pullbacks of the same native primitive. Differentiating the projection square gives equality of compact primitives under h, including on the boundary. The prescribed compact-map completion theorem at scale one constructs e, with exact primitive equality and the displayed completed compact square. Restricting the neighborhood projection square to the compact carrier gives its displayed native square. □

Lemma V.10 (The retained rounded marking reaches the chosen hitting collar)**.** *Fix a retained marked tailoring prefix for $(I, \phi, \beta, C^+, A^-, K, m)$. Write E_1 for its last rounded domain, D_β for*

its retained outer domain, $a = (C^+)^2\beta^2/2$, and $e_D : D_\beta \to (\mathbb{C}^*)^n$ for its native embedding. Let L be a Liouville structure on this same outer domain with primitive

$$\lambda_L = e_D^*\lambda_{1,K}, \qquad \lambda_{1,K} = \lambda_\phi + K\, f_1^*\lambda_{\mathbb{C}},$$

where $\lambda_{\mathbb{C}}$ is the prescribed plane Liouville primitive, and let C be any chosen flow collar of (D_β, L). There is an actual strict inclusion marking $q : \widehat{E}_1 \to X_C$, with compact map i, for which

$$(e_D \circ \iota_C)(i(x)) = J_1(x) \qquad (x \in E_1).$$

Here J_1 is the original retained native rounded embedding.

Proof. The retained rounded carrier is a subset of $\{\phi \le a\}$: the rounded maximum dominates its potential coordinate, so its defining nonpositive inequality forces $\phi - a \le 0$. The retained outer carrier has precisely this native image. A point of the nonempty rounded carrier therefore proves nonemptiness of the outer carrier. Its old outer primitive at parameter one and L both pull back the displayed native primitive. Apply the preceding strict collar comparison to the old outer collar and C, and compose the resulting strict map with the retained marking of $\widehat{E}_1$. For each compact rounded point, containment in the native outer carrier and injectivity of its native embedding show that its old marked image lies in the old localized compact carrier. The compact comparison formula thus applies to that point. Composing its native square with the original marking square gives the displayed equality. Exact primitive preservation follows by composition, and intertwining the Liouville flows supplies the inclusion marking's all-real flow formula. □

Lemma V.11 (The strict boundary map agrees with every ribbon hit)**.** *Let P be a witnessed pair, D an outer Liouville domain, and C its chosen collar completion. Let $q : \widehat{P} \to X_C$ be an actual inclusion marking, with compact map i. Suppose an equal-dimensional native embedding e identifies D with $\{\rho \le A\}$ and pulls a smooth native primitive α back to the primitive on D. Assume $d\alpha$ is nondegenerate there and $d\rho(Z_\alpha) > 0$ on $\{\rho = A\}$. Let a smooth parameter action G solve the native Liouville equation whenever its current point lies in this sublevel. For every compact ribbon point x, suppose the native projection of $i(j_P(x))$ equals a specified starting point, whose actual G trajectory hits $e(j(x))$ at time $\tau(x) > 0$, with $j(x) \in \partial D$. Then the smooth induced boundary map ψ and shift s of q satisfy*

$$\psi(j_P(x)) = j(x), \qquad s(j_P(x)) = -\tau(x)$$

on the entire compact ribbon. Their contact primitive law is $\psi^*\alpha_D = e^{-s}\alpha_P$, and their end equation holds at every real radius.

Proof. The strict completed marking constructs ψ, s and their all-real end equation, as well as the contact primitive law. Fix a compact ribbon point and write $z = i(j_P(x))$. The regional completed-hit lemma applied to its specified native trajectory gives $z = \mathrm{end}_C(j(x), -\tau(x))$. The compact inclusion formula of the same marking gives $q(\mathrm{end}_P(j_P(x), 0)) = z$. The induced end equation therefore equates $\mathrm{end}_C(\psi(j_P(x)), s(j_P(x)))$ with $\mathrm{end}_C(j(x), -\tau(x))$. The all-real completed end map is injective, so its boundary and radial coordinates agree separately. This argument applies to every compact ribbon point. □

Lemma V.12 (The actual strict shift constructs the incoming witness interpolation)**.** *Let P, Q be witnessed pairs, let $\psi : \partial P \to \partial Q$ be a smooth boundary diffeomorphism, and let s be a smooth real shift. Let $h : F_P \to F_Q$ be an actual compact-domain diffeomorphism and suppose, for every $x \in F_P$,*

$$\psi(j_P(x)) = j_Q(h(x)), \quad s(j_P(x)) = -\tau(x), \quad b_Q(j_Q(h(x))) = e^{-\tau(x)}.$$

Then $b_{\mathrm{in}} = \exp(s \circ \psi^{-1})$ is smooth and positive and agrees with b_Q on the whole compact ribbon $j_Q(F_Q)$. Consequently the logarithmic interpolation from b_Q to b_{in} is an actual permitted pair deformation with both given completions and the entire ribbon primitive fixed. Its reverse has the incoming-to-hitting direction of the junction.

Proof. Smoothness follows by composition with the smooth inverse of ψ; the exponential is positive everywhere. Given $y \in F_Q$, put $x = h^{-1}(y) \in F_P$. Its compact inverse law gives $h(x) = y$. The ribbon square implies $\psi^{-1}(j_Q(y)) = j_P(x)$. Therefore $b_{\mathrm{in}}(j_Q(y)) = e^{s(j_P(x))} = e^{-\tau(x)} = b_Q(j_Q(y))$. The positive witness interpolation lemma now gives the asserted deformation on precisely Q's existing compact domains and completions, with primitive fixed because the equality just proved holds at every compact ribbon point. Reversal is permitted by the definition of a geometric comparison chain. □

V.2 The strict geometric ribbon junction

Proposition V.13 (The retained shrinking and hitting families join literally)**.** *Fix the retained marked tailoring prefix, its finite intrinsic shrinking family, and the completed decreasing-coefficient hitting family supplied together by the retained hitting-family construction, on the same outer domain and shrunk ribbon and with its primitive, regional hitting and whole-ribbon witness equations. The endpoint of intrinsic shrinking is joined to the initial hitting pair by a scale-one completion identification and an actual positive-witness deformation. Appending the hitting deformation yields a finite comparison to its zero-coefficient endpoint. Prepending the constructed shrinking chain and original marked prefix gives a finite comparison from the retained original pair. The ribbon remains in its original carrier presentation throughout shrinking; all later native spatial maps include its negative-flow map.*

Proof. Use the initial hitting slice's ambient flow collar. Its native primitive at coefficient K_* equals that of the retained outer marking, so the retained-marking lemma constructs a strict marking of the tailoring endpoint into this actual collar completion. Intrinsic shrinking changes neither that ambient domain nor its completion. Its compact native square and the full-fiber marking's compact negative-flow square identify the starting point of every shrunk ribbon trajectory. The actual regional hitting equations therefore give the induced boundary map and negative hitting-time shift on the whole ribbon. The initial hitting slice also localizes the shrunk ribbon by its chosen ribbon collar. The original-to-localized compact comparison has projection square and preserves the primitive at scale one. Composing the whole-ribbon boundary equation with this projection square gives the required literal ribbon square in the localized hitting pair. Its witness has value the exponential of the negative hitting time. Apply the incoming-identification lemma. Reverse its witness deformation, and append the already constructed decreasing-coefficient hitting deformation. Their endpoint tuples agree exactly. The shrinking chain ends at the source of this identification, and the original prefix ends at the source of shrinking, so concatenation gives the asserted finite comparison. □

V.3 The literal cotangent endpoint

Lemma V.14 (The standard cotangent disk completion)**.** *For the canonical primitive on T^*T, the Liouville flow is $\Phi_t(\theta, p) = (\theta, e^t p)$. The unit Euclidean cotangent disk is a compact smooth Liouville domain, and its inclusion completion is all of T^*T with end map $(x, r) \mapsto \Phi_r(x)$.*

Proof. The equation $\iota_{p\partial_p}(\sum dp_j \wedge d\theta_j) = \sum p_j d\theta_j$ determines the Liouville field. Exponential scaling gives its smooth flow for all real times. The defining function $\rho = |p|^2 - 1$ has derivative $d\rho(p\partial_p) =$

$2|p|^2$, equal to two on its zero level. Thus that level is regular and the field is strictly outward. The disk is compact because the base torus and the closed Euclidean momentum ball are compact. For every momentum p, negative flow reaches the disk: choose $T \geq 0$ with $e^{-2T}|p|^2 \leq 1$. On the unit sphere a trajectory crosses the boundary only at time zero, and its primitive is multiplied by e^r. The smooth boundary-flow completion construction therefore gives the claimed inclusion realization and standard end map. □

Lemma V.15 (Transport of the retained ribbon to a strict endpoint)**.** *Let P be a witnessed pair and let Ψ strictly identify its ambient completion with the standard cotangent disk completion. Write its induced boundary map and shift as ψ, s, so that $\psi^*\alpha_{\mathrm{can}} = e^{-s}\alpha_P$. On the same compact ribbon, embedding $j' = \psi \circ j$ and witness $b' = \big(e^s b\big) \circ \psi^{-1}$ give a witnessed cotangent pair and a permitted completion identification with ribbon map the identity and scale one. Its full embedded core is the image under ψ of the original full embedded core.*

Proof. The constructed strict boundary map is a smooth diffeomorphism, so j' is a smooth embedding and b' is smooth and positive. Pullback commutes with multiplication by scalar functions; hence $\psi^*(b'\alpha_{\mathrm{can}}) = e^s b\, e^{-s}\alpha_P = b\alpha_P$. Pulling this identity back by j gives exactly the original ribbon primitive. Keep its original Liouville domain and completion. The identity ribbon diffeomorphism then preserves the primitive with scale one and the boundary square holds by construction. All ambient identification equations are those of Ψ and its induced boundary data, establishing a permitted step. Since the ribbon and its completion are unchanged, its full intrinsic core is unchanged; composing its embedding with ψ gives the final core-image equality. □

Lemma V.16 (Boundary normalization and witness factor of the same arrow)**.** *Let Ψ be the strict completion arrow of a witnessed pair to a cotangent pair, expressed in the standard cotangent marking, and let ψ, s be its induced boundary map and shift. If x denotes a point of the source boundary included into its completion, then*

$$\psi(x) = \left(\Psi(x)_\theta, \frac{\Psi(x)_p}{|\Psi(x)_p|}\right), \qquad e^{s(x)} = |\Psi(x)_p|.$$

Here the target boundary point is expressed through its actual cotangent inclusion marking.

Proof. The strict primitive equality intertwines the complete Liouville flows. Consequently the induced end equation holds at every real radius. At radius zero it gives $\Psi(x) = (\psi(x)_\theta, e^{s(x)}\psi(x)_p)$, with $|\psi(x)_p| = 1$ because the marked target boundary is the unit cosphere. Taking momentum norms gives the second identity, and dividing by that positive norm gives the first. Both identities therefore concern the same strict completion arrow and its uniquely determined end data. □

Lemma V.17 (The entire homogeneous conic level projects onto the link)**.** *Let φ_∞ be a strongly compatible degree-two homogeneous potential, positive away from zero, whose gradient is a homeomorphism Φ_∞ fixing zero. For every $A > 0$, the radial map $(u, \theta) \mapsto (\theta, \Phi_\infty(u)/|\Phi_\infty(u)|)$ takes the whole positive Legendre conic set at potential level A onto the full positive-cone FLTZ link.*

Proof. For a conic point, its gradient is nonzero and lies in a nonzero fan cone, while its angular point lies in that cone's annihilator. Normalization preserves cone membership and gives a unit covector, proving inclusion. Conversely take a link point (θ, p), with $|p| = 1$ in its specified nonzero cone. Put $u = \Phi_\infty^{-1}(p)$; it is nonzero since the homeomorphism fixes zero. Set $r = \sqrt{A/\varphi_\infty(u)} > 0$. Degree-two homogeneity gives $\varphi_\infty(ru) = A$ and degree-one gradient homogeneity gives $\Phi_\infty(ru) = rp$. Thus (ru, θ) belongs to the same conic set and projects exactly to (θ, p). This supplies every link point. □

Lemma V.18 (Literal endpoint from the same potential and native core)**.** *Retain a witnessed pair P, its full ribbon, and its witness b. Let q strictly mark its ambient completion into the logarithmic ambient with primitive λ_φ, agreeing on the included boundary with the native map i. Assume the gradient of this same smooth potential φ is a global smooth diffeomorphism and agrees along i with the homogeneous gradient Φ_∞ of the previous lemma. Assume*

$$i(\mathrm{Core}_{\mathrm{emb}}\, P) = \Gamma_\infty \cap i(\partial D), \qquad i(\partial D) = \{(u,\theta) : \varphi_\infty(u) = A\}, \quad A > 0.$$

Then the composition of q with the Legendre map of φ gives a permitted strict identification to a cotangent witnessed pair on the same ribbon, with identity ribbon map and scale one. Its boundary map and positive witness satisfy

$$\chi_\infty(x) = \left((i(x))_\theta, \frac{\Phi_\infty((i(x))_u)}{|\Phi_\infty((i(x))_u)|}\right), \qquad b_\Sigma(\chi_\infty(x)) = |\Phi_\infty((i(x))_u)|\, b(x),$$

and its actual embedded full core is literally the FLTZ link.

Proof. The literal Legendre primitive identity is $\mathcal{L}_\varphi^* \lambda_{\mathrm{can}} = \lambda_\varphi$. Composing it with the assumed strict marking q therefore constructs the strict cotangent marking using the same selected potential. Apply retained-ribbon transport. The boundary normalization lemma and the gradient agreement give the displayed χ_∞ formula and the exact witness factor. Since that construction retains the original ribbon completion, its embedded full core is the normalized image of $i(\mathrm{Core}_{\mathrm{emb}}\, P)$. Substitute the assumed native core and boundary-level equations; the conic-level image lemma then identifies that whole image with the full FLTZ link. □

Proposition V.19 (The literal retained cotangent endpoint)**.** *Fix the retained strongly compatible end potential, its selected smooth extension and their Legendre data, and the terminal slice of the completed hitting family. Suppose its nonempty native outer compact image is the selected potential sublevel at a positive homogeneous level, its native boundary is that full level, and its native full core is the Legendre-conic subset of that boundary. The zero-coefficient primitive is the native potential primitive. Assume $A > 0$, $\{\phi = A\} = \{\phi_\infty = A\}$, equality of their gradients on this level, and $d\phi(Z_\phi) > 0$ throughout $\{\phi \geq A\}$. Then this exact slice has a permitted completion identification to a standard cotangent pair with literal full embedded core equal to the FLTZ link. Its ribbon and ribbon completion are unchanged, its ribbon map is the identity and its scale is one. The boundary map is radial Legendre normalization, and the witness is multiplied by the norm of the homogeneous gradient. The native complete marking agrees with the retained native embedding on the whole compact carrier, as well as on its boundary.*

Proof. Collar localization preserves the compact carrier and its primitive under the original inclusion. Composing that inclusion with the retained native embedding gives the stated potential sublevel image and native primitive on the entire compact carrier. The compact carrier is nonempty by hypothesis on the retained outer domain. Its boundary and full-core images are the displayed native level and Legendre-conic section. The native literal endpoint lemma therefore constructs a complete native marking, composes it with the selected Legendre diffeomorphism, and identifies the result with the standard cotangent completion. It retains the entire ribbon and its completion with identity ribbon map and scale one. Its radial boundary map and gradient-norm witness formula give the exact ribbon primitive. The conic-level projection sends the entire native full core onto the fixed FLTZ link, proving literal equality. The finite-input cotangent theorem below supplies the required high level and native equations for every sufficiently large radius; they are not additional toric-input hypotheses. □

Compatible structures and change of representative.

Lemma V.20 (Compatible structures and section-class transport)**.** *Compatible complex structures form a contractible space. Transport of squared canonical sections through compatible paths identifies their homotopy classes independently of the positive metric, path and connection choices.*

Proof. For a fixed symplectic form ω, the polar construction $P_\omega(g) = A(-A^2)^{-1/2}$, with $g(Av, w) = \omega(v, w)$, depends smoothly on positive metrics and is compatible with ω. If J is already compatible, then for $g_J(v, w) = \omega(v, Jw)$ it gives $P_\omega(g_J) = J$. Fixing a compatible J_*, the family

$$P_\omega\big((1-t)g_J + tg_{J_*}\big), \qquad 0 \le t \le 1,$$

contracts the space of compatible structures to J_* while fixing J_*. A complex connection transports squared canonical sections along these paths. Changing the connection or the compatible path gives a homotopy of such transports: connections form an affine space, and the displayed contraction supplies the required homotopies of compatible paths. Hence the induced identification of section homotopy classes is independent of these choices. This proves the change-of-J foundation used in Definition 2.4 without a toric grading input.

□

Lemma V.21 (An explicit symplectic identification of the rank-two update)**.** *Let ω be a non-degenerate alternating form, let p, q be real covectors, and let X_q satisfy $\omega(X_q, \cdot) = q$. The map $T_K v = v + Kp(v)X_q$ satisfies*

$$T_K^*\omega = \omega + Kp \wedge q.$$

If $1 + Kp(X_q) \neq 0$, its inverse is $v \mapsto v - Kp(v)X_q/(1 + Kp(X_q))$. For a smooth primitive α and smooth map $f = A + iB$, this identifies the actual forms $d\alpha$ and $d(\alpha + Kf^*\lambda_{\mathbb{C}})$ at each point, with $p = dA$ and $q = dB$.

Proof. Alternation gives $q(X_q) = 0$ and $\omega(v, X_q) = -q(v)$. Expanding $\omega(T_K v, T_K w)$ leaves the original form and $K(p(v)q(w) - q(v)p(w))$; its quadratic term vanishes. Composing the proposed inverse with T_K in either order leaves v plus a multiple of X_q whose coefficient is zero after multiplication by $1 + Kp(X_q)$. Both maps are continuous and linear. Finally $d(f^*\lambda_{\mathbb{C}}) = dA \wedge dB$ by the actual pullback and exterior-derivative identities, giving the native statement. □

Lemma V.22 (Complex structures and canonical squares under the shear)**.** *Suppose the denominator in the preceding lemma is nonzero and J is compatible with ω. Then $J_K = T_K^{-1} J T_K$ is compatible with $\omega + Kp \wedge q$. If s is a nonzero squared canonical tensor for J, its pullback in every tangent slot by T_K is a nonzero squared canonical tensor for J_K. These constructions are smooth for smooth data on any open parameter set where the denominator is nonzero. At $K = 0$ they recover J and s.*

Proof. The equation $T_K J_K = J T_K$ and invertibility give $J_K^2 = -1$. The form identity transports J-invariance. For $v \neq 0$ it gives

$$(\omega + Kp \wedge q)(v, J_K v) = \omega(T_K v, J T_K v) > 0,$$

since $T_K v \neq 0$. Alternation in each tensor block is preserved by pullback, and the intertwining equation proves complex linearity in every slot. If the pulled-back tensor vanished, evaluating it on the inverse images of arbitrary vectors would force $s = 0$. The formulas for T_K and its inverse use addition, composition, scalar multiplication and division by a nonzero smooth scalar. Pullback of the tensor is multilinear in its slot maps, so all the asserted smoothness follows. Substitution of $K = 0$ gives the identity map. □

Lemma V.23 (Native smooth Maslov data for nonholomorphic updates)**.** *Let α be a smooth primitive with nondegenerate differential on a smooth even-dimensional manifold, and suppose a smooth Maslov representative for $d\alpha$ is given. For every smooth map $f = A + iB$ and real K with $1 + K\,dA(X_B) \neq 0$ everywhere, the shear constructs a smooth Maslov representative for $d(\alpha + Kf^*\lambda_{\mathbb{C}})$. In particular nonnegative pairing $dA(X_B)$ permits every $K \geq 0$.*

Proof. In a coordinate chart, the Hamiltonian vector is the inverse of the contraction matrix of $d\alpha$ applied to the covector dB. Nondegeneracy makes this matrix invertible; smooth matrix inversion proves the smoothness of X_B. If G is an inverse chart derivative, the covector changes to $dA \circ G$, the vector to $G^{-1}X_B$, and the complex structure by conjugation. The shear formula commutes with these changes since $G(G^{-1}X_B) = X_B$. Consequently the preceding smoothness result applies in every actual chart to both the conjugated complex structure and the pulled-back squared tensor. The compatibility and nonvanishing statements were proved in the preceding lemma. The full exterior derivative equals $d\alpha + K\,dA \wedge dB$, so these are representatives for the asserted native primitive. When K and the pairing are nonnegative their denominator is at least one. □

Lemma V.24 (Native Maslov transport along the coefficient-removal edge)**.** *For fixed native primitive, smooth map and initial Maslov representative as above, let $k(t)$ be a smooth real coefficient with $1 + k(t)dA(X_B) \neq 0$ for $0 \leq t \leq 1$. The shear representatives form a jointly smooth Maslov family for the actual forms $d(\alpha + k(t)f^*\lambda_{\mathbb{C}})$, with smooth extension across both parameter endpoints. At a zero coefficient its complex structure and squared tensor are exactly the original ones. Nonnegative pairing therefore gives this family for $k(t) = K(1 - t)$, for every $K \geq 0$.*

Proof. In each spatial chart, use the product of its target with the real parameter line. The nonzero-denominator locus is open and contains the whole interval over that chart. The rational shear formula is smooth on this open locus. The squared tensor formula is polynomial in the shear and is smooth even without the denominator restriction. The native chart identities in the preceding proof show that these are genuine extensions of the actual fields. Compatibility, canonicity and nonvanishing hold at each interval point by the preceding construction. At $k(t) = 0$ the shear is the identity. Finally $K(1 - t) \geq 0$ on the interval, so the denominator is at least one when the pairing is nonnegative. □

Remark V.25. This is an explicit tangent construction for the rank-two forms in the grading argument. It does not identify the completions or extend a section over their cylindrical ends. Those operations, and the prescribed agreement at the marked parameter edges, remain part of the completed family grading construction.

Lemma V.26 (Uniform graded convenient comparison)**.** *For each finite toric input there are a smooth potential φ, an ambient end bound A_{amb}, constants $0 < C_- < C_+$ and $\beta_0 > 0$ with the following properties. For every $\beta \geq \beta_0$, $e^\beta > \max(1, M(A_{\mathrm{amb}}))$, the ratio set is nonempty and bounded above, and a valid fiber bound lies below $C_-^2\beta^2/2$. For every original coefficient $K \geq K_0(e^\beta)$ there is a retained convenient original pair at levels $C_-^2\beta^2/2, C_+^2\beta^2/2$ and an eleven-arrow comparison to a cotangent pair. Its entire embedded core is literally the FLTZ link, and the same eleven arrows carry the prescribed original datum to the prescribed vertical representative.*

Proof. Select the strongly compatible end potential, its smooth retained Legendre extension, and the finite native hitting construction together. Increase its threshold to include the uniform native plane-pairing positivity threshold, and to ensure that $C_+^2\beta^2/2$ exceeds the homogeneous endpoint threshold. For example, if this last threshold is $A_0 > 0$, it suffices to impose $\beta \geq 1$ and $\beta \geq 2A_0/C_+^2$.

These choices all precede K and the convenient rounding. The retained construction supplies the full fiber, its actual markings, and the regional hitting identities for the selected base coefficient.

The original coefficient arrow and the holomorphic coefficient/potential family carry the logarithmic section through their actual markings. The normalized junction is graded by the native holomorphic shear homotopy, its fixed-collar extension, and compact tensor detection. The same completed native tailoring family then grades the tailoring arrow, giving the five-arrow original prefix. Its terminal native tensor is the initial native tensor of coefficient removal, since both use φ, f_1, K_* and the same rank-two shear. Pullback through the retained comparison maps preserves this agreement on the whole compact carrier, including its boundary; compact detection identifies the actual ambient classes. Intrinsic shrinking uses two arrows with fixed ambient datum. The constructed incoming strict identification, the reversed witness interpolation, and the actual completed hitting deformation therefore form a graded three-arrow continuation. Finally the retained cotangent identification carries the completed terminal datum to the prescribed vertical datum and supplies literal entire-core equality. Concatenation gives $5+2+3+1=11$ arrows with those exact endpoints.

This proof uses explicit native shear families on the two parameter edges and actual homotopies at their junctions. It supplies the same-chain conclusion of the cited source without requiring an extension of the ribbon or its grading over the interior of a parameter square. □

There are a common ambient end bound and radius $R_0 > \max(1, M(A_{\mathrm{amb}}))$ such that for every $R \geq R_0$ the ratio set is nonempty and bounded above and a valid fiber bound exists. For every valid fiber bound, every $K \geq K_0(R)$, every A_- above that bound, and every supplied initial pair with its allowed level and rounding, there is a fourteen-arrow endpoint comparison: literal whole-core FLTZ equality, prescribed source grading, and the prescribed vertical datum on that same chain.

Proof. Take $R_0 = e^{\beta_0}$ from the preceding lemma and set $\beta = \log R$. Its convenient comparison is available for the supplied K. Use the minimum of the supplied and convenient valid fiber bounds; it is still valid and lies below both fiber levels. The comparison-of-choices construction gives three graded arrows from the supplied initial pair to the convenient pair, with terminal representative exactly its prescribed representative. Concatenate with its eleven-arrow graded comparison. Both endpoints and the full core are unchanged by this concatenation. The terminal representative already is the vertical representative, so the required endpoint homotopy can be constant. All supplied choices remained arbitrary throughout. □

V.3.1 A common rounding for the original coefficient prefix

The prefix below starts with a constructed convenient original pair. Comparing a supplied arbitrary admissible tuple to that pair requires the initial-choice comparison of Lemma 4.2. The logarithmic grading is carried by the same first coefficient arrow; grading the later arrows remains a separate requirement.

Lemma V.27 (The original endpoint has a logarithmic fiber bound)**.** *Let the smooth uniformly convex potential have a degree-two homogeneous end, and fix an ambient level A_{amb}. There are $F > 0$ and $\beta_0 > 0$ such that for every $\beta \geq \beta_0$, $F^2\beta^2/2$ is a valid original fiber end bound over $R = e^{\beta}$.*

Proof. Apply the logarithmic potential-level estimate to the same potential path and specialize both the coefficient and potential parameters to zero. The coefficient tuple is then the original one, and the inverse Hessian is the identity. Its metric fiber defect is therefore $|u|^2 - |b_0|^2/a$. The

estimate bounds this defect below by $\phi_0 \geq F^2\beta^2/2 > 0$ and also gives $a > 0$. The level is at least A_{amb}, as required. □

Fix the finite toric input and the smooth uniformly convex potential with its homogeneous end. There are an original ambient end bound, a positive truncation constant F, and, for every $C_+ > F$, a positive large-β threshold. For every larger β, the original radius $R = e^\beta$ is in the valid analytic range, $F^2\beta^2/2$ is a valid fiber bound, and there is a finite normalized coefficient threshold $K_* > 0$. For every $F^2\beta^2/2 < A_- < C_+^2\beta^2/2$, there is a width $\delta > 0$ below the level gap. Every smooth-max profile of positive width at most δ, every $K_o \geq K_0(R)$, and every $K \geq K_*$ with $K_o \leq K/R^2$ give the actual five-arrow marked prefix from the convenient original pair at K_o through the original pair at K/R^2 and the retained holomorphic and tailoring families. The same original coefficient arrow carries the prescribed logarithmic grading at its two endpoints. Both original endpoints retain their full-fiber and ambient inclusion markings.

Proof. Choose the original ambient bound and radius threshold from the finite input. Take F to be the maximum of the marked-family truncation constant and the original logarithmic fiber constant. Increasing a fiber bound preserves its inequalities by restricting its end region. Take the maximum of the two large-β thresholds and $\log R_0 + 1$; then $R_0 < e^\beta$.

The marked-family construction supplies K_* and its rounding width. At the original threshold $K_0(R)$ the original corner estimates supply another positive width below the same level gap. Take their minimum. Every supplied thinner profile is admissible at $K_0(R)$ and hence at K_o, since the numerator of the base derivative is affine with nonnegative slope in the coefficient. The original coefficient-family construction supplies its actual graded arrow from K_o to K/R^2. The common-embedded-pair comparison identifies its target with the exact retained holomorphic source, with scale one and zero end shift. Append that identification and the retained three-arrow chain. The adjoining tuples agree literally and the resulting length is five. □

Corollary V.28 (Every supplied original coefficient has a convenient prefix). Every finite toric input admits a strongly compatible homogeneous end potential and a smooth global extension, constants $0 < C_- < C_+$, and $\beta_0 > 0$ such that for every $\beta \geq \beta_0$ and every supplied original coefficient $K_o \geq K_0(e^\beta)$ there exist a finite $K > 0$ with $K_o \leq K/e^{2\beta}$ and an admissible convenient rounding. The convenient levels are $A_- = C_-^2\beta^2/2$ and $A = C_+^2\beta^2/2$. Their actual original pair at K_o starts the five-arrow marked prefix of the preceding theorem, with the original coefficient arrow graded by the prescribed logarithmic datum.

Proof. Construct the strongly compatible potential and its smooth extension, including their homogeneous-end agreement and uniform positive Hessian bounds. Apply the common-rounding theorem to this same potential. For its output $F > 0$, take $C_- = F + 1$ and $C_+ = F + 2$. For every permitted positive β these choices give the strict level inequalities. Take the common width δ and its constructed smooth convolution profile. Given K_o, set $K = \max(K_*, K_o e^{2\beta})$. Then $K > 0$, $K \geq K_*$ and $K_o \leq K/e^{2\beta}$. The preceding theorem supplies both original admissibility and the retained five-arrow prefix for this exact profile. □

Lemma V.29 (Two presentations at identical original choices)**.** *Two supplied initial pairs at the same R, K, A_-, A and rounding profile admit a strict completion identification with scale one, zero end shift, and equal full compact ambient and ribbon images. Its composite with the target inclusion marking equals the source inclusion marking on the entire completed ambient. Thus it is the identity on marked Y.*

Proof. The two ambient compact images are the same literal rounded domain, and the two ribbon images are the same literal full fiber truncation. Their inclusion markings determine their compact primitives, including the boundary derivatives: maps equal on the entire compact carrier have equal differentials there. Pull the common forms λ_K and λ_0 back through these actual embeddings. The common-embedded pair comparison constructs the strict identification, including its compact squares, scale one, and zero shift.

Composing its ambient map with the target marking preserves the same primitive as the source marking, and the two composites agree on the entire compact carrier. Primitive preservation intertwines Liouville flows. Every noncompact point is represented by a forward flow from the compact boundary. The two maps therefore agree there as well, which proves equality on the whole completion. □

Lemma V.30 (Attach a supplied presentation to the original coefficient leg)**.** *Let a five-arrow original marked prefix be supplied by the common-rounding construction. Any independently supplied initial-pair presentation at its same convenient choices admits a six-arrow chain to that exact tailoring endpoint. The initial presentation identification and the original coefficient arrow form a length-two initial segment carrying the prescribed logarithmic grading through the same actual markings.*

Proof. Construct the preceding strict identification into the retained convenient source. Pull its original Maslov representative back by that actual map. The chain rule, together with equality of the whole marked ambient maps, gives precisely the logarithmic squared tensor at the supplied source. The constant homotopy of this pulled representative grades the identification. Append the original graded coefficient arrow, sharing its representative literally at the junction. Append the existing original-to-holomorphic identification and the three-arrow marked path. The resulting lengths are respectively two and six; every adjoining endpoint tuple agrees literally. □

V.3.2 Attaching every admissible initial choice to the retained prefix

Proposition V.31 (An arbitrary source and the retained eight-arrow prefix)**.** *Fix a retained original-coefficient, holomorphic and tailoring prefix at $R = e^{\beta}$. Let S be any supplied initial pair at the same R, K_o, with arbitrary valid fiber bound, admissible levels and rounding. Assume the retained convenient ribbon level also exceeds a valid fiber bound relative to the same ambient bound. There is a three-arrow comparison from S to the retained original pair whose terminal Maslov representative is precisely the retained original representative. Concatenating it with the original-coefficient arrow gives a four-arrow graded path to the retained coefficient endpoint. Concatenating with the entire retained prefix gives an eight-arrow path to the actual tailored pair, retaining every later endpoint and marking.*

Proof. Take the smaller of the supplied and convenient fiber bounds. It is one of the valid bounds and is strictly below both ribbon levels. Apply the full-ribbon varying-level comparison in a common rounded domain, choosing its terminal representative to equal the retained original representative. Its source representative is logarithmic under the supplied inclusion marking. The three-arrow comparison and the retained original-coefficient arrow therefore have literally equal intermediate representative and pair. Their graded concatenation has length four. Append the retained presentation identification and the retained three-arrow holomorphic and tailoring prefix. This has length eight and the stated terminal pair. Associativity of path concatenation gives both descriptions of the same path; no change of ribbon presentation is omitted. □

V.4 Matching representatives with the same squared canonical tensor

Lemma V.32 (Normalized transport fixes a common canonical tensor)**.** *Let J_0, J_1 be real linear endomorphisms of a finite-dimensional real vector space, and let τ be a squared canonical tensor for both structures. Transporting every tangent slot by $A = (\mathrm{id} - J_0 J_1)/2$ leaves τ unchanged.*

Proof. Fix all slots except a slot with entry v. Complex linearity for each structure gives

$$\tau(\dots, J_0 J_1 v, \dots) = i\tau(\dots, J_1 v, \dots) = i^2 \tau(\dots, v, \dots) = -\tau(\dots, v, \dots).$$

Real multilinearity now gives $\tau(\dots, Av, \dots) = \tau(\dots, v, \dots)$. Replace the entries one at a time. Induction on the finite set of replaced slots proves that any such set of replacements preserves the value. Taking the set of all slots proves the assertion. The same argument includes an empty set of slots. □

Lemma V.33 (An explicit smooth compatible-structure path)**.** *Two smooth almost complex structures J_0, J_1 compatible with the same two-form ω admit a smooth compatible path with precisely those endpoints, smooth on an open neighborhood of the parameter interval in every spatial chart.*

Proof. For $0 \le t \le 1$, set $a = 1 - t/2$, $b = t/2$ and $T_t = a\,\mathrm{id} - bJ_1J_0$. Using $J_i^2 = -\mathrm{id}$ and preservation of ω by each J_i, expansion gives

$$\omega(T_t u, T_t v) = (a^2 + b^2)\omega(u, v) + ab\big(\omega(J_1 u, J_0 v) + \omega(J_0 u, J_1 v)\big).$$

Replacing both u, v by $J_0 u, J_0 v$ preserves this expression: $\omega(J_0 u, J_0 v) = \omega(u, v)$, and the two cross terms are interchanged because $\omega(J_1 J_0 u, v) = -\omega(J_0 u, J_1 v)$ and $\omega(u, J_1 J_0 v) = -\omega(J_1 u, J_0 v)$. Moreover direct expansion gives

$$\omega(T_t u, T_t J_0 u) = (a^2 + b^2)\omega(u, J_0 u) + ab\big(\omega(u, J_1 u) + \omega(J_0 u, J_1 J_0 u)\big) > 0 \quad (u \neq 0).$$

Here $a > 0$, $b \ge 0$, and each displayed symplectic pairing is positive by compatibility; $J_0 u \neq 0$ follows from $J_0^2 = -\mathrm{id}$. Thus T_t has zero kernel and is invertible. Define $J_t = T_t J_0 T_t^{-1}$. Its square is $-\mathrm{id}$, the first identity gives preservation of ω, and the second gives positivity for every nonzero tangent vector, after writing that vector as $T_t u$. At $t = 0$, T_t is the identity. At $t = 1$, $T_t = (\mathrm{id} - J_1 J_0)/2$ satisfies $T_t J_0 = J_1 T_t$, so $J_t = J_1$. The inversion locus of a matrix is open. In each spatial chart it contains the full parameter interval, and on this open locus T_t^{-1}, hence J_t, depends smoothly on time and space. This supplies the asserted smooth endpoint extensions. □

Lemma V.34 (Equal squared tensors give homotopic Maslov representatives)**.** *Two smooth nonvanishing Maslov representatives on the same symplectic manifold, compatible with the same two-form, and having equal squared canonical tensor sections are connected by an actual smooth Maslov homotopy. Consequently they determine the same Maslov class.*

Proof. Let the representatives have structures J_0, J_1 and common section τ. Choose the constructed smooth path of compatible complex structures from J_0 to J_1. Transport τ along this path by the normalized intertwiner $A_t = (\mathrm{id} - J_0 J_t)/2$. If $A_t v = 0$, then $v = J_0 J_t v$ and consequently $J_0 v = -J_t v$; positivity of both $\omega(v, J_0 v)$ and $\omega(v, J_t v)$ forces $v = 0$. Thus A_t is invertible and preserves nonvanishing. The equation $A_t J_t = J_0 A_t$ preserves the canonical tensor condition in every slot, and alternation is preserved within each block. Polynomial dependence on the structures gives the joint smoothness of the tensor family on open neighborhoods of both parameter endpoints. At the initial endpoint the intertwiner is the identity. At the final endpoint the preceding lemma shows that the transported tensor is the original common tensor, since it is canonical for both endpoint structures. The structures at the endpoints are precisely J_0, J_1, so this family has the two specified representatives as literal endpoints. It is the required Maslov homotopy, and its quotient classes therefore agree. □

V.4.1 The prescribed grading on the holomorphic potential path

Lemma V.35 (Logarithmic compatibility for a positive Hessian)**.** *Let ϕ be smooth with a uniformly positive Hessian and let f be holomorphic. For every $K \geq 0$, the standard logarithmic complex structure is compatible with $d(\lambda_\phi + Kf^*\lambda_{\mathbb{C}})$. The squared logarithmic volume form is a smooth nowhere-zero squared canonical section for this structure.*

Proof. In radial and angular coordinates the first two-form evaluates to $H(X, Y') - H(X', Y)$, where $H = D^2\phi$ is symmetric. Simultaneously replacing (X, Y) by $(-Y, X)$ leaves this expression unchanged, and its evaluation on (v, Jv) is $H(X, X) + H(Y, Y) > 0$ when $v \neq 0$. Holomorphicity gives $df(Jv) = i\,df(v)$. Thus the added two-form is also J-invariant and its evaluation on (v, Jv) is $K|df(v)|^2 \geq 0$. The standard logarithmic coframe is smooth on the quotient torus and is complex-linear for J. Its top exterior power and square are therefore smooth and nowhere zero, with no choice of global real angles. □

Proposition V.36 (Grading of the retained holomorphic completed family)**.** *Let the retained holomorphic family interpolate the original coefficients and quadratic potential to coefficients one and the chosen uniformly convex potential, at a fixed nonnegative base coefficient. Its actual inclusion completion markings induce Maslov representatives at both ends and a graded deformation on that very completed family. Both endpoint tensors equal the prescribed logarithmic squared volume under their respective markings.*

Proof. The interpolated Hessian has lower bound $\min(1, c) > 0$, where c is the lower bound for the chosen potential. The previous lemma therefore gives the constant logarithmic complex field and tensor as a native compatible family. Compose each completed-family presentation with its inclusion marking. These maps have smooth local extensions in the parameter and point. Pullback through their spatial derivatives gives smooth complex fields and squared canonical sections. Exterior differentiation of the primitive-preserving equation identifies their symplectic form with the completed-family form. Invertibility of the markings preserves nonvanishing. At each endpoint, associativity of pullback identifies the family restriction with the representative pulled back through that endpoint's actual marking. Naturality of the logarithmic coframe gives the asserted prescribed tensor values. □

Lemma V.37 (The retained original junction in logarithmic coordinates)**.** *The retained original-to-holomorphic identification is the identity in logarithmic coordinates on the entire completion. Pullback of any prescribed holomorphic-source Maslov representative through this identification gives a prescribed representative on the matching original pair, together with a graded lift of that exact identification.*

Proof. At the initial parameter the potential is the quadratic potential and the coefficient path is the original coefficient vector. Thus the holomorphic marking pulls the original primitive back to its completed primitive. Its composition with the retained strict identification and the original marking preserve the same primitive. The retained compact square and the inclusion equations show that they agree on the entire compact carrier. Primitive preservation intertwines complete Liouville flows, so compact/end coverage makes the two maps equal everywhere. Pulling back the prescribed tensor through this equality makes it precisely the prescribed original tensor. Choosing the pulled-back representative makes the graded identification lift reflexive. This does not identify that representative with a separately chosen representative on an earlier deformation. □

Proposition V.38 (Grading the literal original and holomorphic prefix)**.** *Under the positive-Hessian potential and nonnegative coefficient conditions, the retained original-coefficient deformation, original presentation junction and holomorphic deformation form a graded three-arrow prefix. Its source is exactly the retained original representative, and its terminal representative is prescribed by the holomorphic endpoint marking. This path is literally the prefix of the retained geometric comparison; its remaining two arrows are normalization and tailoring.*

Proof. The retained original deformation already carries prescribed representatives. The holomorphic marking family constructs prescribed representatives and a graded deformation on its actual completions. Pull its initial datum through the retained junction. The preceding marking equality proves it is prescribed under the original marking. It has the same tensor as the retained original endpoint representative. The same-tensor homotopy lemma therefore makes their Maslov classes equal, giving a graded lift of the exact junction. Concatenate the three lifts with their literal shared representatives. The existing original path has length one, so the result has length three. Expanding the retained path and associating its concatenations proves the asserted prefix identity. □

Corollary V.39 (The prescribed six-arrow prefix for every initial choice)*.* Every supplied initial pair at the same radius and original coefficient, with its own valid fiber bound and permitted ribbon level, admits a prescribed graded six-arrow comparison to the retained holomorphic endpoint. The convenient ribbon level must exceed its own valid fiber bound for the same ambient bound. Both fiber bounds and the supplied initial presentation remain arbitrary.

Proof. Use the three-arrow initial-choice comparison whose terminal datum is exactly the retained original datum. Concatenate its grading with the preceding three-arrow graded original and holomorphic prefix. The common representative and pair are literally equal. Lengths add to six, and the endpoint remains prescribed under the retained holomorphic marking. □

W Assembly of the geometric chain

Proposition W.1 (Quantifier assembly and the geometric chain)**.** *For every finite input the assertions of Theorem 4.1 hold with its stated nested quantifiers, both inclusion markings, the literal endpoint core equality, and grading on that same finite chain.*

Proof. Fix the finite toric input. The original end estimates prove completeness and finite type for every real $K \geq 0$, including coefficients below the ribbon threshold. Theorem V.3 supplies a common ambient bound and radius and, at every larger radius, nonemptiness and boundedness of the ratio set and existence of a valid fiber bound. Its thresholds were chosen before the supplied coefficient, fiber bound, levels, or rounding.

Now take any valid fiber bound, any $K \geq K_0(R)$, and any levels $A_- > A_{\mathrm{fib}}$, $A > A_-$. The original rounding theorem produces an admissible profile. For every admissible profile, the original initial-pair theorem constructs the witnessed pair, nonempty entire fiber truncation, and both inclusion-induced completion markings. Apply the arbitrary-initial-pair theorem to that supplied pair. It returns a single endpoint comparison with fourteen arrows, literal whole embedded-core FLTZ equality, and graded lifts on those very arrows. The initial datum is prescribed by the supplied ambient marking; the terminal datum is already the prescribed vertical representative, with the constant homotopy as endpoint comparison.

These constructions prove all nested existential and universal clauses of the target for the chosen input. Quantifying over its rank and finite toric data proves the stated geometric theorem. The categorical GPS deduction is separate and is not used in this proof. □

Dimension one. No positive-dimensional fiber argument has been used to bypass $n = 1$. Its initial ribbon is a finite regular fiber, as checked in Proposition G.40. Regularity makes the tailored fiber have dimension $2n - 2 = 0$; the critical-torus calculation and radial-section argument identify its entire core with the two positive-cone link points. Its intrinsic flow is zero, so shrinking is the identity and the local trapping argument is the codimension-zero case. Hitting times, positive witnesses and the ambient two-dimensional grading construction still apply. Empty corner bands impose no spurious positivity requirement.

References

[1] *A geometric proof of the superpotential–FLTZ comparison.*

W.1 Assembly of the literal retained strict junction

Lemma W.2 (A strict map and whole-ribbon hit give the incoming pair)**.** *Let P, Q be witnessed pairs. Suppose a primitive-preserving completed diffeomorphism e induces the smooth boundary map ψ and shift s, including the end equation and $\psi^*\alpha_Q = e^{-s}\alpha_P$. Suppose a compact ribbon diffeomorphism h satisfies $h^*\mu_Q = \mu_P$ and, at every compact ribbon point x,*

$$\psi(j_P(x)) = j_Q(h(x)), \quad s(j_P(x)) = -\tau(x), \quad b_Q(j_Q(h(x))) = e^{-\tau(x)}.$$

There is an incoming pair Q_{in} with precisely Q's compact domains, completions, embedding and ribbon primitive, a completion identification $P \to Q_{\mathrm{in}}$ with ambient map e, ribbon map h and scale one, and a permitted witness deformation from Q to Q_{in}.

Proof. Set $b_{\mathrm{in}} = \exp(s \circ \psi^{-1})$. The preceding interpolation lemma proves that it is smooth and positive and agrees with b_Q on the entire ribbon. Replace Q's witness by this function. Its ribbon primitive remains μ_Q, since the witness values on every ribbon point are unchanged. Use e, ψ, s, h as the identification maps. The assumed strict primitive, end, contact and ribbon-square equations give their defining properties. The assumed compact ribbon primitive equation is exactly the required scale-one equation. Logarithmic interpolation gives the deformation with the stated literal endpoints. □

For finite toric input there are a strongly compatible end potential, its chosen smooth potential with the retained global Legendre data, an ambient end bound, constants $0 < C^- < C^+$ and $\beta_0 > 0$ such that, for every $\beta \geq \beta_0$ and every supplied original coefficient $K_o \geq K_0(e^\beta)$, there are a finite $K_* > 0$, an admissible rounding at the convenient levels $(C^-)^2\beta^2/2, (C^+)^2\beta^2/2$, and a marked original prefix with $K_o \leq K_*/e^{2\beta}$. This prefix extends by its actual intrinsic shrinking and strict hitting junction to a ten-step comparison from its original pair to its completed hitting endpoint at coefficient zero. The outer boundary has precisely the native potential level $(C^+)^2\beta^2/2$, and for every slice of the retained hitting family the native image of the full embedded ribbon core equals the intersection of the retained Legendre conic subset with that same outer boundary.

Proof. The retained native hitting-family theorem constructs the potentials and constants before the supplied coefficient is chosen, and supplies its admissible rounding, marked prefix, finite shrinking data, and completed hitting family for every coefficient in the stated range. Choose one point of the nonempty original ribbon; its inclusion in the shrinking extension gives the basepoint needed by the source localization map. At the initial hitting slice the coefficient is exactly K_*. The primitive equation therefore identifies its ambient primitive with the retained outer primitive. Its hitting

time is positive by the common open-neighborhood hitting construction, and its native boundary embedding is precisely the regional flow at that time. Its witness value is the exponential of the negative hitting time. Apply the preceding strict-junction proposition to these supplied analytic equations. Concatenate the five-step marked original prefix, the two-step source-localization and shrinking comparison, and the resulting three-step strict hitting comparison. Their lengths add to ten. Their shared endpoint tuples are those of the constructed families. The native outer-boundary level, every slice's primitive formula, the full-core image equality, and the same chosen Legendre data are retained from the hitting-family theorem. □

W.2 The selected potential at the retained cotangent endpoint

Definition W.3 (Analytic data of a retained Legendre potential)**.** *For functions ϕ_∞, ϕ on the fixed real lattice coordinates, retained Legendre data consist of a smooth ϕ, a continuously differentiable ϕ_∞, positive uniform lower and finite upper Hessian bounds for ϕ, equality $\phi = \phi_\infty$ outside a finite Euclidean ball, and homeomorphisms equal to their respective coordinate gradients. The homogeneous gradient fixes zero; the inverse of the global gradient is smooth.*

Lemma W.4 (Positivity of the homogeneous potential)**.** *If ϕ_∞ is strongly compatible and continuously differentiable, then $\phi_\infty(u) > 0$ for every $u \neq 0$.*

Proof. Use the fixed Euclidean coordinates and write x for u in those coordinates. Degree-two homogeneity gives $d\phi_\infty(x)[x] = 2\phi_\infty(x)$ and $D(\nabla\phi_\infty)(x)x = \nabla\phi_\infty(x)$. Taking the inner product of the latter equality with x gives $\langle D^2\phi_\infty(x)x, x\rangle = 2\phi_\infty(x)$. The left side is strictly positive by the positive-definite Hessian and $x \neq 0$, proving the assertion. □

Lemma W.5 (High levels of the selected potential)**.** *For retained Legendre data of a strongly compatible potential there is $A_0 > 0$ such that, for every $A \geq A_0$, the full level sets $\{\phi = A\}$ and $\{\phi_\infty = A\}$ coincide, their gradients agree there, and the native Liouville field of λ_ϕ is strictly outward on the entire end $\{\phi \geq A\}$.*

Proof. Bound each of ϕ and ϕ_∞ on the closed Euclidean ball of their agreement radius. Take A_0 above both bounds and above the native ambient outwardness threshold. Every point in either level lies strictly outside the ball. Equality of the two functions there proves both inclusions of the level sets. Their equality on an open neighborhood of each such point implies equality of their differentials and coordinate gradients. Specialize the ambient outwardness inequality to base coefficient zero to obtain the claimed native inequality. □

Lemma W.6 (Native marking of a retained zero-coefficient pair)**.** *Suppose the compact ambient of a witnessed pair has a smooth native embedding with nonempty image exactly $\{\phi \leq A\}$ and primitive the pullback of λ_ϕ. If $d\phi(Z_\phi) > 0$ on $\{\phi \geq A\}$, then every chosen completion of that compact ambient has a primitive-preserving diffeomorphism to (Y, λ_ϕ) agreeing with the specified native embedding on the compact piece.*

Proof. At coefficient zero the complete holomorphic ambient associated to the retained potential has primitive λ_ϕ and its native complete Liouville field. All potential sublevels are compact. Strict positivity above A therefore implies finite negative entry into $\{\phi \leq A\}$ by the scalar absorption theorem. The function $\phi - A$ is smooth and its zero level is regular and outward. The boundary-flow completion gives the full native sublevel its inclusion marking. The common native embedding identifies the supplied compact presentation with that sublevel, preserving its primitive. Completion comparison then transfers the marking to the supplied chosen completion and retains its compact inclusion equation. □

Lemma W.7 (Literal cotangent endpoint of the retained native pair)**.** *Let a retained pair satisfy the native marking hypotheses above on a positive homogeneous level. Suppose its native full embedded core is* $\Gamma \cap \{\phi_\infty = A\}$*. Then it has a permitted completion identification to a standard cotangent pair retaining the identical ribbon domain and completion, with ribbon map the identity and scale one. The completed ambient map is the native marking followed by the Legendre map of the same* ϕ*. On the boundary its coordinate map is* $\chi(u,\theta) = (\theta, \nabla\phi_\infty(u)/|\nabla\phi_\infty(u)|)$ *and its witness satisfies* $b_\Sigma \circ \chi = |\nabla\phi_\infty| b$*. Its entire embedded core is literally* Λ^∞_Σ*.*

Proof. Construct the native marking by the preceding lemma and compose with the global Legendre diffeomorphism. Its pullback of the canonical primitive is λ_ϕ, so this is a strict ambient completion map. Gradient agreement on the native boundary identifies its value with the homogeneous gradient. The strict-map boundary normalization and witness formulas give exactly χ and the displayed witness. The ribbon primitive is unchanged, so the ribbon map is the identity with scale one and its full intrinsic core is retained. Every nonzero target momentum has a unique homogeneous-gradient preimage. Strict positivity and degree-two homogeneity scale that preimage uniquely onto $\{\phi_\infty = A\}$. The corresponding degree-one gradient scaling preserves its positive cone and normalizes to the original unit momentum. Thus normalization of the entire Legendre-conic level is the entire positive-cone FLTZ link, proving the literal core equality. □

Lemma W.8 (Legendre data of the retained hitting family)**.** *The finite-input retained hitting-family construction can be chosen with retained Legendre data for its same selected functions* ϕ_∞, ϕ*. All prefix, full-fiber, shrinking and native-core conclusions of the retained family hold for those same functions.*

Proof. The full-core construction selects the strongly compatible homogeneous potential and its smooth extension together with uniform Hessian bounds, a radius of exact end agreement, and both actual gradient homeomorphisms. Retain these functions and these analytic witnesses while choosing the common outer radius for the marked prefix and native hitting construction. The finite maxima of the prefix, full-core, outer-hitting and outer-domain thresholds do not change either function. The common smooth hitting neighborhood, exponential primitive law, finite intrinsic shrinking, positive witness extension and the two localized completion families are then constructed for these fixed functions. These operations introduce no replacement potential. Consequently all their geometric conclusions hold together with the original analytic Legendre witnesses. □

For every finite toric input there are fixed selected potentials ϕ_∞, ϕ, an ambient end bound A_{amb}, constants $0 < C_- < C_+$ and $\beta_0 > 0$ with the following property. For every $\beta \geq \beta_0$, the original ratio set at $R = e^\beta$ is well-defined and a valid fiber bound lies strictly below $C_-^2\beta^2/2$. For every original coefficient $K_o \geq K_0(e^\beta)$ there are a finite normalized $K_* > 0$, a convenient original admissible rounding and its retained marked prefix, with $K_o \leq K_*/e^{2\beta}$. Its retained strict comparison to the zero-coefficient hitting pair has ten steps. That same hitting pair admits the cotangent identification of the preceding lemma, with the identical ribbon and completion and scale one. Appending that identification gives an eleven-step comparison from the convenient original pair to a standard cotangent pair whose full embedded core equals the entire fixed FLTZ link literally. The native marking, boundary normalization and witness satisfy the explicit Legendre formulas of the preceding lemma for these same potentials.

Proof. Choose the retained strict comparison with its selected analytic Legendre data. Let $A_0 > 0$ be the high-level threshold. Increase its radius threshold to $\max\{\beta_0, 1, 2A_0/C_+^2\}$, without changing the selected potentials or either radius constant. For every such β, $\beta^2 \geq \beta$ and $C_+^2\beta \geq 2A_0$, so

$C_+^2\beta^2/2 \geq A_0$. All original admissibility and fiber-bound conclusions persist under this increase. Choose the normalized coefficient, convenient rounding, marked prefix, finite shrinking and strict hitting comparison as before.

At the terminal hitting slice its native primitive is λ_ϕ, because its base coefficient is zero. Collar localization preserves its full native compact image $\{\phi \leq C_+^2\beta^2/2\}$ and preserves that primitive by pullback through the original collar inclusion. Its native boundary is the full displayed level and its native full embedded core is the Legendre-conic subset of that level. The compact domain is nonempty: the retained rounded domain is nonempty and lies in it. Apply the native marking and literal cotangent endpoint lemmas on this level. They construct the required strict terminal identification and retain the full intrinsic ribbon core with scale one. Its source is exactly the terminal pair of the ten-step comparison. Append this one permitted identification; the resulting path has eleven steps and the displayed literal FLTZ core equality. □

Lemma W.9 (Canonical squares under positive primitive stretching)**.** *Let $d\alpha = \omega$ be symplectic, let Z satisfy $\iota_Z\omega = \alpha$, and let (J, τ) be a smooth Maslov representative. For a smooth positive function w, suppose $1 + w^{-1}dw(Z) \neq 0$. Put $p = w^{-1}dw$ and $Tv = v + p(v)Z$. Then $J' = T^{-1}JT$ and $\tau' = T^*\tau$ are a smooth Maslov representative for $d(w\alpha)$. Wherever $dw = 0$, their values equal J and τ.*

Proof. The rank-two shear identity gives $T^*\omega = \omega + p \wedge \alpha = w^{-1}d(w\alpha)$. Its denominator is $1 + p(Z)$, so T is invertible and smooth with smooth inverse. Consequently $J'^2 = -1$ and $d(w\alpha)$ is J'-invariant. For a nonzero vector v, $d(w\alpha)(v, J'v) = w\omega(Tv, JTv) > 0$. Pullback in every slot preserves separate alternation and the squared canonical equations by $TJ' = JT$. Invertibility preserves nonvanishing. All operations are smooth. If $dw = 0$, then $p = 0$ and T is the identity. □

Lemma W.10 (Maslov representatives on the actual cylindrical completion)**.** *Suppose a Maslov representative is given for the original primitive on the localized open neighborhood of a Liouville domain furnished by its flow collar. It determines a Maslov representative on that collar's actual cylindrical completion, equal to the original representative on the whole compact carrier. A representative given on the original domain's extension also supplies this input by restriction to the localized neighborhood.*

Proof. Write r for the original collar coordinate and $h(r)$ for its smooth increasing stretch to infinite radius, with $h(r) = r$ for $r \leq \delta/3$. The completed primitive is $w\alpha$, where $w = \exp(h(r) - r)$ on the collar and $w = 1$ on the compact interior. These formulas agree near the seam. Since $dr(Z) = 1$, $w^{-1}dw(Z) = h'(r) - 1$, so the denominator in the preceding lemma is $h'(r) > 0$ on the collar and 1 in the interior. The preceding construction therefore applies everywhere on the actual completed manifold. The weight is locally constant one on a neighborhood of every compact point, including boundary points because $0 < \delta/3$; there $dw = 0$. Thus its derivative vanishes at every compact point. The resulting complex field and squared tensor agree there with the original ones. Open-subset inclusions have identity tangent maps in the native models; the restricted charts are restrictions of the original charts. Restricting the original fields therefore preserves smoothness, compatibility, canonicity and nonvanishing, providing the stated native input. □

Lemma W.11 (Uniform native and completed tailoring representatives)**.** *Let ϕ be a smooth potential with global Hessian bounds $cI \leq D^2\phi \leq UI$, where $c, U > 0$. Suppose that outside a compact set it equals a degree-two homogeneous function that is C^1 everywhere and smooth off zero. Fix $D > 0$. There is $\beta_0 > 0$ such that, for every $\beta \geq \beta_0$, $s \in [0, 1]$ and $K \geq 0$, the potential sublevel $\{\phi \leq D^2\beta^2/2\}$, with primitive $\lambda_{s,K} = \lambda_\phi + Kf_s^*\lambda_{\mathbb{C}}$, is a compact Liouville domain and admits an*

explicit native Maslov representative $\tau_{s,K}$. For every flow collar of this domain, the actual cylindrical completion admits a Maslov representative equal to $\tau_{s,K}$ on its whole compact carrier. At $K = 0$, the native complex field and tensor are exactly the logarithmic complex structure and $\Omega_{\log}^{\otimes 2}$.

Proof. The Hessian lower bound makes the logarithmic complex structure compatible with $d\lambda_\phi$; its prescribed logarithmic canonical square is smooth and nowhere zero. The uniform Poisson estimate supplies a threshold, independent of s and K, beyond which $\pi_{f_s} \geq 0$. The native rank-two shear therefore has denominator $1+K\pi_{f_s} > 0$; conjugating the logarithmic complex structure and pulling back its tensor constructs $\tau_{s,K}$. For $K = 0$ the shear is the identity. The uniform sublevel estimate supplies a second threshold guaranteeing compactness and strict outwardness on $\phi = D^2\beta^2/2$, simultaneously for all $s \in [0,1]$ and $K \geq 0$. Take the maximum of these two thresholds. Restrict the native representative and apply the preceding cylindrical completion construction to any chosen flow collar. Its compact-carrier identity gives exactly the asserted recovery of $\tau_{s,K}$. □

Lemma W.12 (Joint native tailoring grading)**.** *Under the Hessian hypotheses of the preceding lemma, there is a threshold $\beta_0 > 0$ such that for every $\beta \geq \beta_0$ and $K \geq 0$ the native representatives $\tau_{s,K}$ form a smooth Maslov family for $d\lambda_{s,K}$, $s \in [0,1]$. Both the complex field and the canonical square have smooth extensions to a real open neighborhood of the closed parameter interval times each spatial chart. Every interval slice is exactly the native representative used in the preceding completion construction.*

Proof. The complex map $f_s = f_0 + s(f_1 - f_0)$ is affine in s. Its real and imaginary differentials are consequently jointly smooth covector families. In any fixed spatial chart, the Hamiltonian vector of the imaginary covector is the inverse of the smooth, everywhere invertible contraction matrix of $d\lambda_\phi$, applied to that covector. This proves joint smoothness. The shear, its conjugated complex field, and its pulled tensor therefore depend smoothly on both variables wherever $1 + K\pi_{f_s} \neq 0$. This nonzero locus is open and contains the whole required interval by uniform Poisson positivity. It supplies the asserted real open extension, including both endpoints, without a uniform time collar at spatial infinity. The pointwise formulas are the same shear formulas used above, so their complex fields and tensors agree on every slice. □

Lemma W.13 (Joint positive-weight canonical transport)**.** *Let α_t have a smooth Maslov family (J_t, τ_t), and let Z_t be a jointly smooth family with $\iota_{Z_t} d\alpha_t = \alpha_t$. Fix a smooth positive function w. If $1 + w^{-1}dw(Z_t) \neq 0$ on the closed parameter interval, the explicit shear $T_t v = v + w^{-1}dw(v)Z_t$ constructs a Maslov family for $d(w\alpha_t)$. The denominator condition follows whenever both $d\alpha_t$ and $d(w\alpha_t)$ are nondegenerate. Restriction to open submanifolds and pullback through a fixed same-dimensional smooth embedding also construct Maslov families for the actual restricted or pulled primitive.*

Proof. The pointwise positive-weight identity proved above supplies compatibility, canonicity and nonvanishing. In fixed charts all shear coefficients, complex fields and tensors vary jointly smoothly; division occurs only by the denominator. Its nonzero locus is an open neighborhood of the required interval times the chart. Restricting to that neighborhood proves joint smoothness including real extensions across both parameter endpoints. For the asserted denominator criterion, set $p = w^{-1}dw$ and $q = \alpha_t$. The two-form $d(w\alpha_t)$ equals $w(d\alpha_t + p \wedge q)$. Its nondegeneracy therefore implies that of $d\alpha_t + p \wedge q$. The rank-two criterion gives $1 + p(X_q) \neq 0$, and contraction uniqueness identifies $X_q = Z_t$. Open restriction preserves the native tangent fields and their chart formulas. For an embedding, first restrict to its actual open image and then pull back through its constructed range diffeomorphism. Exterior-derivative naturality identifies the resulting two-form with the derivative of the actual pulled primitive. □

Lemma W.14 (Completed Maslov family in the retained presentation)**.** *A native Maslov family for the original primitives of a retained localized completion family determines a Maslov family for its actual completed primitives in the specified zero-slice presentation. Write h_t for the original presentation, χ_t for the restriction of $h_t^*\alpha_t$, and w for the zero slice's radial weight. Its fields are exactly $T_t^{-1}(h_t^*J_t)T_t$ and $T_t^*(h_t^*\tau_t)$, restricted to that localized presentation, where $T_tv = v + w^{-1}dw(v)Z_{\chi_t}$. On the whole compact carrier these fields equal the restricted native pulled fields, including at its boundary.*

Proof. The retained original presentations are jointly smooth diffeomorphisms. Pull back the native family through them and restrict to the two open submanifolds defining the localized zero slice. This constructs the family for $d\chi_t$. The fixed-weight identity identifies its completed primitive with $w\chi_t$ on the entire completed manifold. The canonical Liouville vector Z_{χ_t} varies jointly smoothly: in each chart it is the inverse of the smooth contraction matrix applied to the primitive covector, on the open invertibility locus containing every required point. Completed symplecticity and the preceding denominator criterion permit the positive-weight family construction. The construction retains the displayed formulas at every interval time. Near every compact point the weight is identically one, so $dw = 0$; the shear is the identity there. All smoothness statements are on genuine open neighborhoods in parameter and space, including the compact/end seam. □

Lemma W.15 (Grading the same retained tailoring arrow)**.** *Under the positive upper and lower Hessian bounds, one threshold for β suffices, independently of $K \geq 0$ and all retained rounded-domain choices, to grade the tailoring arrow of every retained marked prefix. The resulting family is on that very pair deformation. Its fields in the source presentation are the pullbacks of the preceding explicit outer fields through the actual retained marking and inverse outer presentation. Both endpoint representatives are those of this same family.*

Proof. Uniform Poisson positivity constructs the native tailoring Maslov family. Pull it through the retained native outer embedding using its actual open-image diffeomorphism. The preceding lemma then constructs the outer completed family. At each time compose the pair's completion presentation, its given outer inclusion marking, and the inverse outer completion presentation. The retained marking regularity supplies actual real smooth map germs for this family of diffeomorphisms. Pullback of the completed primitive is exactly the pair's specified completed primitive: the outer presentation cancels its inverse, and the inclusion marking preserves the primitive. Exterior-derivative naturality gives the required two-form identity. Pull back the constructed Maslov family through these smooth germs, retaining its full field and tensor formulas at every time. Its inverse-presented endpoint representatives define the graded arrow on the original pair deformation. No adjacent-arrow class matching is used in this construction. □

Lemma W.16 (Native homotopies extend over the same collar)**.** *Fix a Liouville domain and one of its flow collars. A smooth native Maslov homotopy for its original primitive extends to a smooth Maslov homotopy between the two actual cylindrical extensions of its endpoint representatives. The compatible fields and squared tensors at both endpoints agree literally with those specified cylindrical extensions.*

Proof. Restrict the native homotopy to the symplectic neighborhood and then to the collar's localized open neighborhood. Both restrictions preserve the complex fields and tensor sections at the two endpoints. Apply the positive-weight family construction with this collar's fixed radial weight and fixed original Liouville vector field. The denominator is positive everywhere by the stretch derivative computation. The resulting family has the completed primitive as its fixed two-form

index. Evaluating the same explicit shear formula at each endpoint gives exactly the cylindrical extension of that endpoint's original field and tensor. □

Lemma W.17 (Grading the same retained coefficient-removal arrow)**.** *Let a retained hitting ribbon family have native outer primitive $\iota^*(\lambda_\phi + K(1-t)f_1^*\lambda_{\mathbb{C}})$, with $K \geq 0$, a smooth positive-Hessian potential and $\pi_{f_1} \geq 0$. Its actual completed hitting deformation admits a Maslov family and a graded arrow. More generally, any native Maslov family for its original outer primitives extends in its specified completed presentation, with the full complex-field and tensor formulas of the retained-completion lemma.*

Proof. The fixed covectors $d\operatorname{Re} f_1$ and $d\operatorname{Im} f_1$ and coefficient $K(1-t)$ give the native shear family. Its denominator is $1 + K(1-t)\pi_{f_1} > 0$ for $t \in [0,1]$, and its smooth real extensions are supplied by the nonzero-denominator locus. Restrict and pull back through the actual outer embedding. Apply the retained completed-family construction to the hitting family's own ambient collars and presentations. By definition these are exactly the ambient completed presentations of its pair deformation. The endpoint representatives of this constructed family therefore give a graded arrow on that same hitting deformation, with its original ribbon and witnessing functions unchanged. □

Lemma W.18 (Native tensor recovered in each actual compact slice)**.** *At each parameter, undoing the specified completion presentation in the constructed completed tensor recovers the cylindrical extension of that parameter's native tensor on the entire compact carrier. In particular, the actual slice representative of the retained hitting deformation has the original native squared canonical tensor there. At the zero parameter the complex fields also agree on the compact carrier.*

Proof. Both native projections of the localized collar neighborhoods have identity differential. Differentiate their commuting square with the compact and original presentations. It follows that the compact presentation's differential equals the original presentation's differential. On the compact carrier the radial weight is one on an open neighborhood, so its differential vanishes and both cylindrical tensor shears are the identity. The projection square and its differentiated equation now identify every tensor slot in the completed formula with the pullback of the target cylindrical tensor. The compact presentation carries the compact carrier onto the target carrier, and equals the completion presentation. Pull back through its inverse; the two differential actions cancel. The hitting slice representative is defined by precisely this inverse presentation, giving the claimed actual-slice formula. At zero the original presentation is the identity, giving the additional complex-field agreement. For an outer-marked deformation, cancel its pair completion presentation against the same presentation in the outer comparison. The remaining map is the actual outer marking followed by the inverse outer presentation. At every point marked into the outer compact carrier, the preceding tensor identity therefore gives the pullback of the native cylindrical tensor by that actual marking. □

Remark W.19. This explicit stretch replaces the polar construction only for the cylindrical extension of already constructed native data. Agreement with the two prescribed marked edges and transport along the full comparison chain require their own arguments.

W.3 The geometric chain for every initial choice

For every finite toric input, there exist a common ambient bound and a radius R_0 such that the initial assertions hold for every $R \geq R_0$, every valid fiber bound, every $K \geq K_0(R)$, every allowed pair of levels, and every admissible rounding. Every supplied initial pair in this range admits a fourteen-arrow geometric comparison to a cotangent pair whose entire embedded core equals the

FLTZ link literally. In particular the initial pair, full fiber and both completion markings exist for every admissible choice, and such a comparison exists for them. This assertion does not include transport of the prescribed grading.

Proof. Use the retained finite-input cotangent comparison to choose the common ambient bound and exponent threshold β_0, and put $R_0 = e^{\beta_0}$. For $R \geq R_0$, set $\beta = \log R$. Positivity of R gives $e^{\beta} = R$ and $\beta \geq \beta_0$. The retained construction gives a valid convenient fiber bound, a convenient initial pair and its eleven-arrow cotangent comparison with literal full-core equality. For the supplied initial pair use the arbitrary-choice comparison, taking the smaller of its fiber bound and the convenient fiber bound. Its three-arrow path terminates at that exact convenient pair. Concatenate it with the retained eleven-arrow path. The endpoint and its embedded core have not changed, and path lengths add to fourteen. Finally the initial-domain construction supplies a pair and its full fiber and markings for each admissible rounding; the admissible-rounding existence theorem supplies such a rounding at every allowed level pair. Completeness for all nonnegative coefficients is the previously proved initial assertion. These facts establish the stated quantifiers. □

W.4 The fixed radial weight in the actual completion family

Lemma W.20 (Weight on the specified cylindrical end)**.** *Let a Liouville collar have positive finite width δ and let σ_δ be its chosen radius stretch. At the point of its actual completion end with radius $r \geq 0$, the constructed radial multiplier is* $\exp(r - \sigma_\delta^{-1}(r))$.

Proof. Uniqueness of integral curves identifies the completion boundary flow with the constructed stretched collar end. That end is the original collar at radius $\sigma_\delta^{-1}(r)$. By definition its radial multiplier is $\exp(\sigma_\delta(\sigma_\delta^{-1}(r)) - \sigma_\delta^{-1}(r))$. The inverse stretch identity gives the displayed formula. □

Lemma W.21 (The actual native embedding identifies an open image)**.** *Let $j : M \to N$ be a smooth embedding of manifolds of the same finite dimension. Its actual image $U = j(M)$ is open and the given map induces a diffeomorphism $M \to U$. Composition with $U \hookrightarrow N$ equals the given embedding, and composition of the inverse with the embedding equals the inclusion of U.*

Proof. The derivative of j is injective. Equal finite dimensions make it bijective, so the inverse function theorem makes j a local diffeomorphism and its image open. Injectivity gives a bijection onto that image. Its inverse is continuous because the embedding induces the subspace topology. Its composition with j is the smooth open inclusion; smooth factorization through an actual smooth embedding therefore proves that inverse smooth. The forward map is smooth as a map into the open submanifold. The two stated composition identities are the original bijection identities. □

Lemma W.22 (Covariance of the actual common-width radial weight)**.** *Consider the constructed localized completion family of a fixed compact Liouville family, with one common collar width and its specified compact and completion presentations. Pullback of every slice's radial multiplier by its compact presentation equals the zero slice's radial multiplier w. Consequently pullback of its completed primitive by its completion presentation equals w times pullback of its original localized primitive by its compact presentation, on the entire zero-slice completed manifold. The latter compact pullback is exactly the restriction of the original family primitive pulled through its specified original presentation.*

Proof. The actual compact piece and the nonnegative cylindrical end cover the completion. On the compact piece both multipliers are one, since the compact presentation carries compact carrier to compact carrier. On the nonnegative end the completion presentation retains the radius exactly.

The completion presentation is the same actual diffeomorphism as the compact presentation, and all slices have the same collar width. The preceding end formula therefore gives the same multiplier there. This proves weight covariance everywhere. The completed primitive is, by construction, the slice's radial multiplier times its localized primitive. Pullback in every covector slot factors out that multiplier; weight covariance and equality of the two presentation maps give the claimed fixed-weight formula. Finally the actual original-projection square equates the two compositions into the original extension. Pullback composition and the constructed localized-primitive equation then identify the compact pullback with the stated restriction of the original transported primitive. □

W.5 Grading at the selected Legendre endpoint

Lemma W.23 (The same selected native and Legendre representatives)**.** *The retained smooth positive-Hessian potential has the prescribed native logarithmic Maslov representative for $d\lambda_\phi$. Its same selected gradient defines a global smooth Legendre diffeomorphism L satisfying $L^*\lambda_{\mathrm{can}} = \lambda_\phi$. Pullback by L of the prescribed vertical cotangent representative is an actual smooth nowhere-zero Maslov representative for that same native two-form.*

Proof. The logarithmic complex structure preserves the Hessian block two-form, and evaluation of the form on $(v, J_0 v)$ is strictly positive by the retained positive lower Hessian bound. The fixed logarithmic squared volume form is smooth, canonical and nowhere zero. The retained gradient homeomorphism and its smooth inverse give $L(u,\theta) = (\theta, \nabla\phi(u))$. The canonical primitive pulls back to $\sum_j \partial_j\phi\, d\theta_j$, which is exactly λ_ϕ. Pullback through the two inverse actual derivatives gives compatibility, smoothness, canonicality and nonvanishing of the cotangent representative. □

Lemma W.24 (Positive ratios give concrete section homotopies)**.** *Two smooth nowhere-zero squared canonical representatives for the same compatible complex structure are smoothly homotopic when the second is a pointwise positive real multiple of the first.*

Proof. Keep the complex structure fixed and interpolate the actual sections by $\tau_t = (1-t)\tau_0 + t\tau_1$. The two smooth sections make this formula jointly smooth for all real parameters. Canonicality and separate alternation are linear conditions. At each point write $\tau_1 = a\tau_0$, where $a > 0$. For $0 \le t \le 1$, $\tau_t = ((1-t)+ta)\tau_0$ and its coefficient is strictly positive, so τ_t never vanishes. Its endpoint values are exactly the two given sections. This is a concrete smooth Maslov homotopy. □

Lemma W.25 (The actual Legendre section ratio is positive)**.** *Let L be the same selected Legendre map and let (J_L, τ_L) be pullback of the prescribed cotangent representative. Normalize it to the native logarithmic structure J_0 through the compatible intertwiner $T = (I - J_L J_0)/2$. Its squared canonical section is $\det((I+H)/2)^2 \Omega^{\otimes 2}$, where $H = D(\nabla\phi)$ is the actual selected coordinate Hessian. This pointwise real coefficient is strictly positive.*

Proof. The actual quotient derivative of L has coframe blocks $(H\,du, d\theta)$. Its intertwining equation with the pulled complex structure gives, for (a,b) in logarithmic tangent coordinates, $(dp, d\theta)(dL\,T(a,b)) = ((I+H)a/2, (I+H)b/2)$. Evaluate the squared tensor on the two ordered logarithmic real frames. Both determinant factors are the complex image of the real determinant $\det((I+H)/2)$, so their product is its real square. Reconstruction of a squared canonical tensor from its ordered-frame value proves the entire tensor identity, including every tangent slot. The compatible intertwiner is invertible, so its transported section is nowhere zero. The determinant cannot vanish, since the tensor identity would then give a zero section at that point. Its real square is therefore strictly positive. The empty frame and determinant one also cover rank zero. □

The native logarithmic potential datum is smoothly homotopic to pullback of the prescribed vertical cotangent datum through the same selected Legendre map. For the actual retained zero marking q and the actual retained cotangent identification C, pullback of this native datum by q has a constructed grading lift along C to the prescribed vertical datum of its cotangent endpoint. The retained endpoint producer returns this lift together with its same ribbon, scale-one identification, exact boundary and witness formulas and literal entire FLTZ core equality.

Proof. Normalize the Legendre-pulled datum to J_0. The preceding positive-ratio identity gives a concrete straight section homotopy from the logarithmic datum to the normalized datum. The explicit compatible-structure path homotopes the Legendre-pulled datum to its normalization. Reverse that path and concatenate smoothly, producing the desired native homotopy. This proves the endpoint homotopy conclusion of the source; the path chosen here uses compatible normalization and positive section interpolation in place of the source's displayed Hessian interpolation.

Pull this actual homotopy back by q, using its primitive equation to identify the source two-form. The retained endpoint has $E.\mathrm{marking} \circ C = L \circ q$ literally. The derivative chain rule in every tangent slot identifies the homotopy's terminal structure and section with pullback by C of the endpoint's prescribed vertical structure and section. These are exactly the endpoint equations of a grading lift on C. Apply this constructor to the same q, E, C returned by the retained endpoint producer; all other endpoint conclusions are retained without a new potential or a new identification. □

Lemma W.26 (Smooth compression into the actual compact carrier)**.** *For the cylindrical completion constructed from any flow collar, there is a jointly smooth map $H : \mathbb{R} \times \widehat{D} \to \widehat{D}$ with $H(0, x) = x$ and $H(1, x)$ in its actual compact carrier for every x. The map need not fix the compact carrier pointwise.*

Proof. Let $\delta > 0$ be the collar width, w its positive radial weight, and Φ the actual complete Liouville flow. Define

$$H(t, x) = \Phi^{-t(\log w(x) + \delta)}(x).$$

Smoothness of $w > 0$ and of the complete action gives joint smoothness for every real t, and the zero-time flow gives $H(0, x) = x$. The compact/end coverage gives two cases. On the compact carrier, $w = 1$, so $H(1, x) = \Phi^{-\delta}(x)$ remains in that carrier by negative-flow invariance. On the end at radius $r \geq 0$, the radial weight formula is $w = \exp(r - \sigma^{-1}(r))$, where σ is the collar stretch. The final radius is therefore $r - (r - \sigma^{-1}(r) + \delta) = \sigma^{-1}(r) - \delta < 0$. It is a negative-time image of the boundary point and again lies in the compact carrier. Both cases concern the actual completion flow. □

Lemma W.27 (Compact detection for an arbitrary completion)**.** *Let D be a Liouville domain with nonempty carrier and let C be any of its completions, with compact inclusion j. There is a jointly smooth compression $H : \mathbb{R} \times C \to C$ with $H(0, x) = x$ and $H(1, x) \in j(D)$. Two ambient Maslov representatives whose tensor sections agree on $j(D)$ have the same homotopy class.*

Proof. Choose the constructed flow collar B of D. The identity on the native domain induces a compact primitive-preserving diffeomorphism $h : D \to B_{\mathrm{loc}}$. The completed compact-map comparison gives a strict ambient diffeomorphism $e : C \to C_B$ satisfying $e \circ j = j_B \circ h$ on all of D. Transport the preceding compression by e, namely $H(t, x) = e^{-1} H_B(t, e(x))$. Its smoothness and zero-time identity follow by composition. Since h maps the whole compact carrier onto that of B_{loc}, its time-one image is contained in $j(D)$. Apply compact tensor detection: normalize the second compatible structure to the first. Transport fixes their common compact canonical tensor. The resulting smooth nonzero scalar ratio equals one on $j(D)$; composition with $H(1 - t, x)$ gives the required homotopy. Concatenate with the compatible-structure normalization homotopy. □

W.6 Detecting a completed grading from its compact section

Lemma W.28 (The smooth ratio of actual nonzero canonical sections)**.** *Let s, r be actual smooth nowhere-zero sections of the square of the canonical line for one compatible complex structure on a smooth $2n$-manifold. There is a unique smooth function $g : M \to \mathbb{C}^*$ with $r(x) = g(x)s(x)$. Wherever the sections agree, $g = 1$.*

Proof. At each point the complex coframe identifies the actual squared canonical fiber with $\mathbb{C}$ by two-frame evaluation and determinant-square reconstruction. The coordinate of s is nonzero, so division gives the unique coordinate ratio; nonvanishing of r makes this ratio nonzero. Around each point, project that point's complex coframe against the smooth complex structure. The projection is invertible on an open neighborhood and its inverse is smooth. Evaluation of either given section on the inverse coframe is therefore smooth. Their quotient equals the unique pointwise ratio on that neighborhood, proving global smoothness. Uniqueness forces the ratio to be one at equal-section points. This also covers $n = 0$, where the two-frame evaluation has empty slots. □

Lemma W.29 (Section homotopy from a smooth compression)**.** *Let s, r be compatible Maslov representatives for the same symplectic form on M, and suppose their actual tensor sections agree on $A \subset M$. Suppose $H : \mathbb{R} \times M \to M$ is jointly smooth, $H(0, x) = x$, and $H(1, x) \in A$. Then s and r are homotopic through actual smooth nowhere-zero squared canonical sections and compatible complex structures.*

Proof. Normalize the structure of r to that of s by the compatible-structure path. At a point of A, the common tensor is complex linear for both structures. In each tensor slot the normalized map $(\mathrm{id} - J_r J_s)/2$ fixes its evaluation: the $J_r J_s$ term evaluates as -1 times the original term by complex linearity for both structures. Applying this to every slot proves that the normalized tensor still equals s on A. Let g be the preceding derived ratio of this normalized tensor to s. The section $g(H(1 - t, x))s(x)$ is jointly smooth and nowhere zero for $0 \le t \le 1$, stays canonical for the fixed structure of s, starts at s, and ends at the normalized section. The same formula is smooth on an open neighborhood of the entire interval. Concatenate with the reverse compatible-structure normalization homotopy to reach r. □

For the actual cylindrical completion constructed from a Liouville collar, two compatible ambient Maslov representatives whose actual tensor sections agree on its full compact carrier determine the same completed ambient grading class.

Proof. Let w be this collar's constructed radial weight and $\delta > 0$ its width. Its completed Liouville flow gives the jointly smooth compression

$$H(t, z) = \Phi_{-t(\log w(z) + \delta)}(z).$$

At $t = 0$ this is the identity. On the compact carrier $w = 1$, so at $t = 1$ negative-flow invariance keeps the point in the carrier. On the nonnegative cylindrical end of radius r, the actual weight is $\exp(r - \sigma_\delta^{-1}(r))$. The final flow radius is therefore $\sigma_\delta^{-1}(r) - \delta < 0$, which lies in the compact carrier by negative-flow invariance from the boundary. The compact carrier and this end cover the actual completion. Thus $H(1, z)$ always lies in the carrier, and the preceding compression-detection lemma applies. □

W.7 The prescribed holomorphic grading and the native rank-two shear

Lemma W.30 (Complex-linear normalization of a holomorphic rank-two shear)**.** *Let (ω, J) be a compatible tangent symplectic space, let p, q be real covectors satisfying $pJ = -q$, $qJ = p$, and let X satisfy $\omega(X, v) = q(v)$. For $K \geq 0$, set $A = \mathrm{id} + KX \otimes p$ and $J_A = A^{-1}JA$. Then*

$$A\frac{\mathrm{id} - J_A J}{2} = \mathrm{id} + \frac{K}{2}(X \otimes p + JX \otimes q).$$

The right side is complex linear for J, and is joined to the identity through invertible complex-linear maps by replacing $K/2$ by $tK/2$.

Proof. Alternation gives $q(X) = \omega(X, X) = 0$, so $p(JX) = -q(X) = 0$. Moreover $q(JX) = p(X) = \omega(X, JX) \geq 0$, with equality permitted when $X = 0$. Therefore A is invertible, with inverse coefficient $-K/(1 + Kp(X))$. The equation $AJ_A = JA$ gives the displayed identity after expanding $A - JAJ$. For a real coefficient c the map on the right with $K/2$ replaced by c equals the composition

$$(\mathrm{id} + cX \otimes p)(\mathrm{id} + cJX \otimes q),$$

since $p(JX) = 0$. The Cauchy–Riemann identities and $J^2 = -\mathrm{id}$ show directly that it commutes with J. For $c = tK/2$, $0 \leq t \leq 1$, both real shears are invertible because their denominators are $1 + cp(X) > 0$. Their composition is therefore invertible throughout the path, and its dependence on t, p, q, X, J is polynomial. □

Lemma W.31 (Native shear grading matches the prescribed holomorphic grading)**.** *Suppose f is smooth and holomorphic for a smooth almost complex structure J compatible with $d\alpha$. Let a smooth nowhere-zero squared canonical tensor τ give the prescribed datum. For $K \geq 0$, suppose that J is also compatible with $d(\alpha + Kf^*\lambda_{\mathbb{C}})$. The actual native rank-two shear representative for this updated primitive is Maslov homotopic to the prescribed representative (J, τ) for that same form.*

Proof. Put $p = d\,\mathrm{Re}\, f$, $q = d\,\mathrm{Im}\, f$, and let X be the $d\alpha$-Hamiltonian vector of q. Holomorphicity gives the two Cauchy–Riemann covector identities in the preceding lemma. The native shear representative is $(J_A, A^*\tau)$. Use the smooth compatible path to normalize its complex structure to J; its endpoint tensor is $C_1^*\tau$, where $C_1 = A(\mathrm{id} - J_A J)/2$. The preceding lemma supplies the path $C_t = \mathrm{id} + (tK/2)(X \otimes p + JX \otimes q)$. Keep the complex structure J fixed and take tensor $C_t^*\tau$. Complex linearity gives the canonical condition in every slot and preserves alternation in both blocks. Invertibility preserves nonvanishing. Smoothness of the Hamiltonian vector follows by inversion of the smooth contraction operator for the nondegenerate form $d\alpha$. Polynomial dependence gives joint smoothness of the section family in each spatial chart on all real times, in particular across both endpoints. This is a homotopy from the prescribed tensor to the normalized shear tensor for the fixed updated symplectic form. Reverse it and concatenate with the normalization homotopy. The shared normalized representative agrees literally, so the result has the required native shear and prescribed endpoints. □

Proposition W.32 (Actual zero-time tailoring grading match)**.** *Let φ be a smooth real logarithmic potential with Hessian bounded below by $c\,\mathrm{id}$ for $c > 0$. For every real β and $K \geq 0$, put $f_0 = W_{\beta,0}$ and $\alpha_0 = \lambda_\varphi + Kf_0^*\lambda_{\mathbb{C}}$. The datum with logarithmic complex structure and prescribed squared logarithmic volume tensor is a constructed Maslov representative for $d\alpha_0$. The actual native rank-two representative for this identical primitive is Maslov homotopic to that datum. The nonnegative native scalar needed to construct the rank-two representative is itself obtained from the Hessian bound. The conclusion applies to every selection of its proof.*

Proof. The retained normalization identity is $f_0 = e^{-\beta} W_{\mathrm{coeff},1}$. The Laurent differential intertwines the logarithmic structure with multiplication by i. Differentiating this normalization preserves that equation, so f_0 is holomorphic for the logarithmic structure. In logarithmic coordinates the potential two-form has Hessian block matrix; symmetry gives its invariance under that complex structure, and the lower bound gives strict positivity on every nonzero tangent vector. The updated two-form has, on a vector v and its complex rotation, the extra summand $K|df_0(v)|^2 \geq 0$. Therefore the same smooth logarithmic structure remains compatible with the actual updated form. Its squared volume tensor remains canonical, smooth and nowhere zero, constructing the stated prescribed representative on $d\alpha_0$.

For the $d\lambda_\varphi$-Hamiltonian vector X of $d \operatorname{Im} f_0$, the Cauchy–Riemann equation gives $d \operatorname{Re} f_0(X) = d\lambda_\varphi(X, JX) \geq 0$. Thus the native denominator is positive for every $K \geq 0$, including critical points where $X = 0$. Apply the preceding actual homotopy theorem to the potential representative and the just-constructed prescribed updated representative. Its complex fields and tensors are literally the common prescribed logarithmic fields before and after the update, as required by that theorem. This gives the asserted native homotopy without a representative, ratio, or homotopy supplied as an input. □

W.8 Retaining the actual grading map through the incoming junction

Lemma W.33 (The actual strict maps grade their constructed incoming junction)**.** *Let P be a witnessed pair and let $\mathcal{H}$ be an actual pair deformation starting at Q. Suppose a strict primitive-preserving completed map $e : P \to Q$ has boundary map ψ, smooth end shift s, and a primitive-preserving full ribbon comparison h. Suppose on every point x of the compact ribbon the boundary square holds, the shift is $s(\iota_P x) = -\tau(x)$, and the target witness at $\iota_Q h(x)$ equals $\exp(-\tau(x))$. Given an actual grading lift of $\mathcal{H}$ from u to v, these data construct a strict incoming junction whose three arrows carry the source datum e^*u to v. Its incoming identification uses the given map e and has scale one.*

Proof. The function $b(y) = \exp(s(\psi^{-1}y))$ is smooth and positive. The whole-ribbon square, inverse property of h, and shift formula show that b equals the target witness on the whole compact ribbon image. Replace Q's witness by b to obtain the actual incoming pair R. The specified maps define a scale-one completion identification $P \to R$: its ambient map is e, its boundary shift is s, and its ribbon map is h, with all required primitive and square equations given. Its grading is the constant homotopy on the actual pulled datum e^*u. The logarithmic positive-witness interpolation $Q \to R$ keeps both completed ambients fixed and carries u constantly; reverse this very graded deformation. Append the given graded arrow $\mathcal{H} : Q \to \mathcal{H}_1$. The three intermediate tuples and representatives agree literally, giving the claimed grading of the actual three-arrow chain. □

Lemma W.34 (Regional hitting data retain the same graded comparison map)**.** *Consider the actual completed decreasing-coefficient hitting family, a strict inclusion marking q of an incoming compact pair into its zero-slice collar completion, and the regional cutoff-flow hypotheses: an actual smooth native embedding and cutoff sublevel, its native primitive, nondegeneracy and outward transversality, the smooth regional Liouville flow, and the positive whole-ribbon hitting times and witnesses. Assume the compact ribbon is nonempty. Given an actual grading lift of this hitting deformation from u to v, these data construct its strict incoming three-arrow junction, graded from q^*u to v. The actual ambient map of the incoming identification is q.*

Proof. The regional cutoff-flow construction produces the actual boundary diffeomorphism, smooth shift, strict end formula, and contact primitive identity, including its whole-ribbon square and

negative hitting-time shift. The ribbon's collar localization gives the actual compact ribbon diffeomorphism and primitive identity. Its compact projection equation transfers the given hitting witness to the localized ribbon. The localized boundary primitive is the original boundary primitive, so these are precisely the raw hypotheses of the preceding lemma, with ambient map q. That lemma constructs the strict junction and the grading of its actual three arrows, retaining the tensor $q^{*}u$. □

For a retained marked prefix, its actual intrinsic ribbon shrink, and its actual completed hitting family, suppose the zero-slice native primitive, outer boundary level, regional Liouville equations and full ribbon hitting data hold. Given the actual completed hitting grading from u to v, the prefix constructs the strict incoming marking q and the strict junction graded from $q^{*}u$ to v. The compact native square of q and the actual tensor $q^{*}u$ are retained along with the grading of this same junction's three-arrow chain.

Proof. The retained rounded marking and the constructed comparison between the two collars with the same outer compact primitive give q and its native compact square. The rounded compact domain lies in the outer cutoff sublevel, and its nonemptiness gives outer nonemptiness; neither containment nor a completed comparison is added as input. The intrinsic shrink preserves the ambient data. Its full compact fiber square combines with the native marking square to identify the start of every hitting trajectory. The retained outer boundary level gives the required endpoint cutoff value; the retained native image gives the full cutoff sublevel. The actual ribbon is nonempty by the retained shrinking construction. The preceding regional lemma applies to these exact maps and hitting times and the given completed-family lift, retaining the constructed q, its compact native square, and the source section pulled through that same map. □

Lemma W.35 (The actual compact native square detects completed grading)**.** *Let C be the actual completion of a nonempty compact Liouville domain, $q : C \to N$ a smooth equivalence satisfying $q^{*}\eta = \omega_C$, with source and target Maslov representatives compatible with ω_C and η, respectively. Let $f : D \to X$, $g : N \to X$ be smooth maps such that $gq\iota_C = f$ on the entire compact carrier. Suppose the source section pulled through ι_C equals $f^{*}\tau$ there, and the target section at $q\iota_C x$ equals $g^{*}\tau$ there. Then the source section is homotopic to the target section pulled through this same q.*

Proof. Regard the full compact carrier as its smooth manifold with boundary. In positive dimension its inclusion into the ambient extension has injective differential and hence surjective differential by equality of dimensions. The two maps composed with this inclusion are identical on the entire manifold with boundary. Their derivatives are equal; the chain rule and surjectivity identify their ambient derivatives. In dimension zero the tangent space is zero and this equality is immediate. The chain rule therefore identifies $dg\,dq\,d\iota_C = df$. The inclusion is an equal-dimensional immersion, so its derivative is surjective. Evaluating the two tensor formulas on arbitrary preimages of every frame proves literal equality of the source and pulled target tensors throughout the compact image. Actual compact detection for the completion then supplies their global homotopy. No derivative equation or grading match is assumed. □

Lemma W.36 (The rounded incoming marking lands in the actual compact carrier)**.** *For the retained prefix and any actual outer flow collar, a marking whose compact native square equals the retained rounded embedding maps every rounded compact point into that collar's compact carrier.*

Proof. The rounded cutoff domain lies in the retained outer native sublevel. The native image description gives a preimage in the original compact carrier, and the collar's original inclusion carrier-image theorem gives a preimage in its localized carrier. Its native image equals that of the

specified marked point by the compact square. Injectivity of the native embedding composed with the original inclusion identifies the two points, proving that the specified marked point belongs to the carrier. □

Lemma W.37 (The incoming chain accepts the prescribed homotopic source)**.** *If the actual three-arrow strict incoming junction is graded from d to v, and the prescribed source s is homotopic to d, then this same junction chain is graded from s to v.*

Proof. Extract the three graded arrows and their two intermediate representatives. The first arrow is the actual strict completion identification, whose graded lift is equivalent to equality between its source Maslov class and the class of the pulled target. Homotopy $s \sim d$ changes the source class by equality, yielding a lift for s through the very same identification. Retain the extracted reversed witness and hitting lifts unchanged and compose these three lifts at their original intermediate representatives. □

Lemma W.38 (The same native section occurs at the adjacent parameters)**.** *For the same retained potential, cutoff parameter, tailored function and base coefficient K, the native tailoring tensor at parameter one equals the native decreasing-coefficient tensor at parameter zero.*

Proof. Both are the actual rank-two shear tensor of the potential section by the Hamiltonian data of the fully tailored function. The first coefficient is K, and the second is $K(1-0) = K$. The functions and all derivatives are identical, so the tensors agree pointwise. □

Lemma W.39 (The same marked completed sections recover their native tensors)**.** *The actual marked tailoring section, evaluated at the completed inclusion of any compact point and on included tangent frames, equals the native section pulled through the retained rounded embedding. Every actual hitting slice section equals the native section pulled through its retained native embedding composed with the collar's original inclusion, throughout the actual localized compact carrier.*

Proof. The completed-family native tensor recovery equations leave precisely the actual slice outer marking for tailoring and the actual cylindrical section for hitting. In the compact carrier the cylindrical section equals the original native-domain section. Its embedding pullback formula retains the derivative of the actual native embedding in every slot. The original collar inclusion has identity derivative in the common tangent model. For tailoring, the full compact square identifies the composite map with the retained rounded embedding. Equality on the full compact carrier gives equality of their derivatives, including the boundary. The chain rule and these tensor equations therefore give both stated native formulas. □

For the same retained prefix, intrinsic shrink, hitting family, selected potential and coefficient, the constructed tailoring endpoint section is homotopic to the constructed hitting source section pulled through the actual incoming marking. Assume only the all-slice native formulas proved by the completed-family producers and the actual compact native square.

Proof. The retained rounded sublevel containment puts both the old tailoring outer marking and the incoming hitting marking in their actual compact carriers. The preceding lemma identifies their completed tensors with the native tailoring tensor at one and native hitting tensor at zero. These native tensors are equal by the common rank-two shear formula and $K(1-0) = K$. The actual incoming compact square identifies their native maps. Consequently compact native-square detection gives a homotopy between the prescribed completed source and the actual pulled hitting source on the entire completed ambient. Compatible structures need not agree. □

Given the actual completed tailoring and hitting families with the producer formulas, and the regional hitting geometry of the retained prefix, the constructed strict incoming junction's three-arrow chain carries the actual tailoring endpoint section to the actual hitting endpoint section. The grading uses this same constructed junction, full ribbon comparison, positive witness interpolation and completed hitting arrow.

Proof. The retained regional construction produces the incoming marking and its full compact native square, the actual three-arrow junction and its lift from the pulled hitting source. The preceding theorem matches the actual tailoring endpoint with that pulled source. Literal equality of the constructed induced tensor with the pullback tensor allows compatible structures to vary by their canonical compatible-structure homotopy. Thus the source is homotopic to the induced datum. Replace only the source of the actual first identification using class equality; retain the witness and hitting lifts and their original intermediate representatives. The resulting lift is on the very same three-arrow chain. □

Lemma W.40 (Compact tensor criterion at the retained endpoint)**.** *Let H be the retained hitting deformation and B its final cylindrical collar. Let φ be a retained Legendre potential and let $e : B_{\mathrm{loc}} \to (\mathbb{C}^*)^n$ be a smooth embedding. Suppose the strict potential marking q agrees with e on the whole compact carrier. If an ambient representative r has, on that carrier, the literal tensor $e^*\Omega_{\log}^{\otimes 2}$, then r is homotopic to the native potential representative pulled back by q. Consequently any retained cotangent identification with marking equation $m \circ C = \mathcal{L}_\varphi \circ q$ carries r to its prescribed vertical datum.*

Proof. Equality of the two smooth maps q and e on the full compact carrier implies equality of their derivatives there, including at its boundary: differentiate their common restriction to the manifold with boundary; its ambient inclusion has surjective derivative. Thus the tensor of $q^*\Omega_{\log}^{\otimes 2}$ equals the stipulated tensor of r at every compact point. Compact detection gives their homotopy, without requiring their compatible complex structures to agree. The retained Legendre grading homotopy takes the native representative to the pullback of the vertical representative under $\mathcal{L}_\varphi$. Pull it back through q, concatenate, and use the literal marking equation to identify its target with pullback through the same C. □

Proposition W.41 (The completed hitting datum reaches the vertical endpoint)**.** *Use the retained endpoint hypotheses of Proposition V.19. Let the hitting primitives be the pullbacks of $\lambda_\varphi + K(1-t)f_1^*\lambda_{\mathbb{C}}$, where $\lambda_{\mathbb{C}} = \frac{1}{2}(x\,dy - y\,dx)$, where $K \geq 0$ and the native plane pairing is nonnegative. Take the completed Maslov family constructed from the native rank-two shear family and the same hitting collars. Its actual terminal representative has a graded lift along a retained cotangent identification to the prescribed vertical datum, with literal full FLTZ core equality.*

Proof. At $t = 1$ the native coefficient is zero, so the shear is the identity in every tensor slot and its section is literally $\Omega_{\log}^{\otimes 2}$. Pullback through the native embedding uses that embedding's actual derivative. At the completed terminal slice, the recovery formula for the same collar gives precisely this native tensor on every compact point. Composing with the original collar inclusion gives the stated native embedded tensor; its derivative is the composition of the native embedding derivative with the identity derivative of the collar inclusion. The retained endpoint construction has a complete potential marking agreeing with this native map on the whole compact carrier. The preceding criterion supplies a grading on that very endpoint identification. Its whole-core equality is the retained endpoint equality. □

W.9 The prescribed grading across normalization

Lemma W.42 (Native tangent formula for embedding pullback)**.** *Let $j : M \to N$ be a smooth embedding between manifolds of the same dimension $2n$. Let (J_t, τ_t) be a smooth Maslov family for $d\alpha_t$. Restricting to the actual open image and pulling back by the image diffeomorphism gives a family for $d(j^*\alpha_t)$ whose squared tensor at x is*

$$(j^*\tau_t)_x(v_1, \dots, v_{2n}) = \tau_t(j(x))(dj_x v_1, \dots, dj_x v_{2n}).$$

The image diffeomorphism has exactly dj_x as its derivative in the common tangent models.

Proof. The embedding derivative is injective and, in equal finite dimensions, surjective. The inverse function theorem therefore makes its image open and gives the inverse smooth charts defining its image diffeomorphism $\bar{j} : M \to j(M)$. The inclusion $\iota : j(M) \to N$ has identity derivative in the restricted charts. Differentiate $\iota\bar{j} = j$: $d\iota_{\bar{j}(x)} d\bar{j}_x = dj_x$, hence $d\bar{j}_x = dj_x$. The restricted tensor evaluates by the original tensor at $j(x)$. Substituting the derivative identity in each slot of its pullback proves the formula. Exterior differentiation commutes with smooth pullback, so the primitive index of this constructed family is $d(j^*\alpha_t)$. □

Lemma W.43 (Grading the retained normalization identification)**.** *Consider a retained marked holomorphic-to-tailoring prefix with smooth potential φ, Hessian lower bound $D^2\varphi \geq cI$, $c > 0$, and $K \geq 0$. Suppose the native plane pairing is nonnegative for every tailoring parameter. Write Q for its tailoring deformation and $L : Q_{\mathrm{hol},1} \to Q_0$ for its literal normalization identification. For the fixed native outer embedding $j : D \to Y$, let (J_s^N, τ_s^N) be the constructed rank-two Maslov family for*

$$\alpha_s^N = \lambda_\varphi + K f_s^* \lambda_{\mathbb{C}}.$$

Pull this family through j and apply the retained completed-family construction. Denote its tensor in the outer zero-slice presentation by $\widehat{\tau}_s$, and the actual comparison from the pair source presentation to that outer presentation by b_s.

Let F be any completed Maslov family on Q satisfying the full native formula $F_s^{\mathrm{tensor}} = b_s^* \widehat{\tau}_s$ at every interval parameter. If u is any holomorphic terminal representative prescribed by its logarithmic completion marking, then L has a graded lift from u to the actual zero-slice representative of this same F.

Proof. Let C be the fixed zero-time outer flow collar and q_0 the retained marking from Q_0 to its cylindrical completion. Set $e = q_0 L$. Both factors preserve the completed primitive, so exterior-derivative naturality identifies the pullback by e of the cylindrical symplectic form with the holomorphic terminal symplectic form.

The zero-time native theorem gives a homotopy from the rank-two representative s^N to the constructed prescribed logarithmic representative r^N, for the identical primitive α_0^N. Restrict this homotopy to the open image of j and pull it back by its image diffeomorphism. The native primitive equality identifies its two-form with that of the actual outer zero-time domain. Call its endpoints s^D, r^D. Extend this homotopy over the fixed collar C, then pull it back by e. This gives a homotopy from $s^c = e^* \operatorname{cyl}_C(s^D)$ to $r^c = e^* \operatorname{cyl}_C(r^D)$ on the actual holomorphic terminal completion.

Every compact point of that completion maps by e into the actual outer compact carrier. Indeed, the normalization compact square first identifies it with a point y of the zero-time rounded tailoring domain. That domain is contained in the potential sublevel defining the outer domain. The exact outer image supplies a native point z in the outer carrier with the same image as y. The retained marking compact square identifies the native image of $q_0(y)$ with that image. Injectivity of j gives that its underlying original point is z, establishing the asserted compact-carrier membership.

At such points, the native recovery formula for F cancels the pair presentation and gives the tensor of $\mathrm{cyl}_C(s^D)$ pulled back by q_0. Pulling through L identifies its tensor with that of s^c on the whole holomorphic compact carrier. This carrier is nonempty, so compact detection gives a homotopy from s^c to L^*F_0.

It remains to identify r^c with u in homotopy class. Let $g = j\,\iota_C e$, where ι_C is the collar's original inclusion, and let h be the holomorphic terminal logarithmic marking. The normalization, outer-marking, and holomorphic-marking compact squares give $g = h$ on the entire holomorphic compact carrier. Their derivatives agree there as well, including on the boundary: smooth equality on the carrier gives equality on its interior; every boundary point is a limit of interior points in a boundary chart, so continuity gives equality of derivatives after the compact inclusion at the boundary. The compact inclusion has invertible derivative, which cancels to give $dg = dh$ at every included compact point.

On the outer compact carrier the cylindrical shear is the identity. The embedding tensor formula, the identity derivative of ι_C, and the chain rule therefore identify the tensor of r^c there with $g^*\Omega^{\otimes 2}$. The equality of values and derivatives just proved changes this to $h^*\Omega^{\otimes 2}$, which is exactly the tensor of u by its prescribed logarithmic condition. Compact detection consequently gives $u \simeq r^c$. Concatenate

$$u \simeq r^c \simeq s^c \simeq L^*F_0.$$

Their Maslov classes are equal, which is precisely the graded-lift criterion for the actual identification L. □

Proposition W.44 (The same selected family grades the full retained prefix)**.** *Under the conditions of the preceding lemma, the actual retained five-arrow prefix is graded from its specified original representative to the terminal representative of one selected completed tailoring family. The family has the full native tensor formula at every interval parameter. More precisely, any family with that formula and its graded tailoring arrow gives a graded lift of the whole retained prefix with these exact endpoints; one such family and arrow are constructed from the retained markings and native family.*

Proof. The graded original-holomorphic prefix supplies its prescribed terminal representative u and the lifts of its three literal arrows. Apply the preceding lemma to this u and the selected tailoring family F to grade the normalization identification. Its terminal representative is literally F_0, the source of the selected graded tailoring arrow. Concatenate that identification and arrow with the three preceding lifts. The retained chain identity identifies this concatenation with the full original prefix, and its terminal representative is exactly F_1.

For existence, apply the retained completed-family producer once to the actual native rank-two family pulled through the fixed native outer embedding. It yields one family F, its full tensor formula, and its graded tailoring arrow. Applying the preceding argument to those same outputs gives the asserted graded prefix without selecting another endpoint representative or discarding the formula. □

Remark W.45. This construction realizes the required normalization grading match by the native holomorphic shear homotopy, fixed-collar extension, and compact detection. It uses equality of Maslov classes for the literal identification rather than a relative metric and connection over the two-parameter rectangle. The completed family and its terminal datum remain the same ones used for the subsequent comparison arrows.

References

[1] S. Ganatra, J. Pardon, and V. Shende, *Covariantly functorial wrapped Floer theory on Liouville sectors*, arXiv:1706.03152v3. https://arxiv.org/html/1706.03152v3.

[2] S. Ganatra, J. Pardon, and V. Shende, *Sectorial descent for wrapped Fukaya categories*, arXiv:1809.03427v4. https://arxiv.org/html/1809.03427v4.

[3] B. Fang, C.-C. M. Liu, D. Treumann, and E. Zaslow, *T-duality and homological mirror symmetry of toric varieties*, arXiv:0811.1228v4. https://arxiv.org/html/0811.1228v4.

[4] *Fukaya categories of toric superpotentials and the FLTZ stop: An explicit far-fiber comparison for smooth Fano fans*, manuscript.